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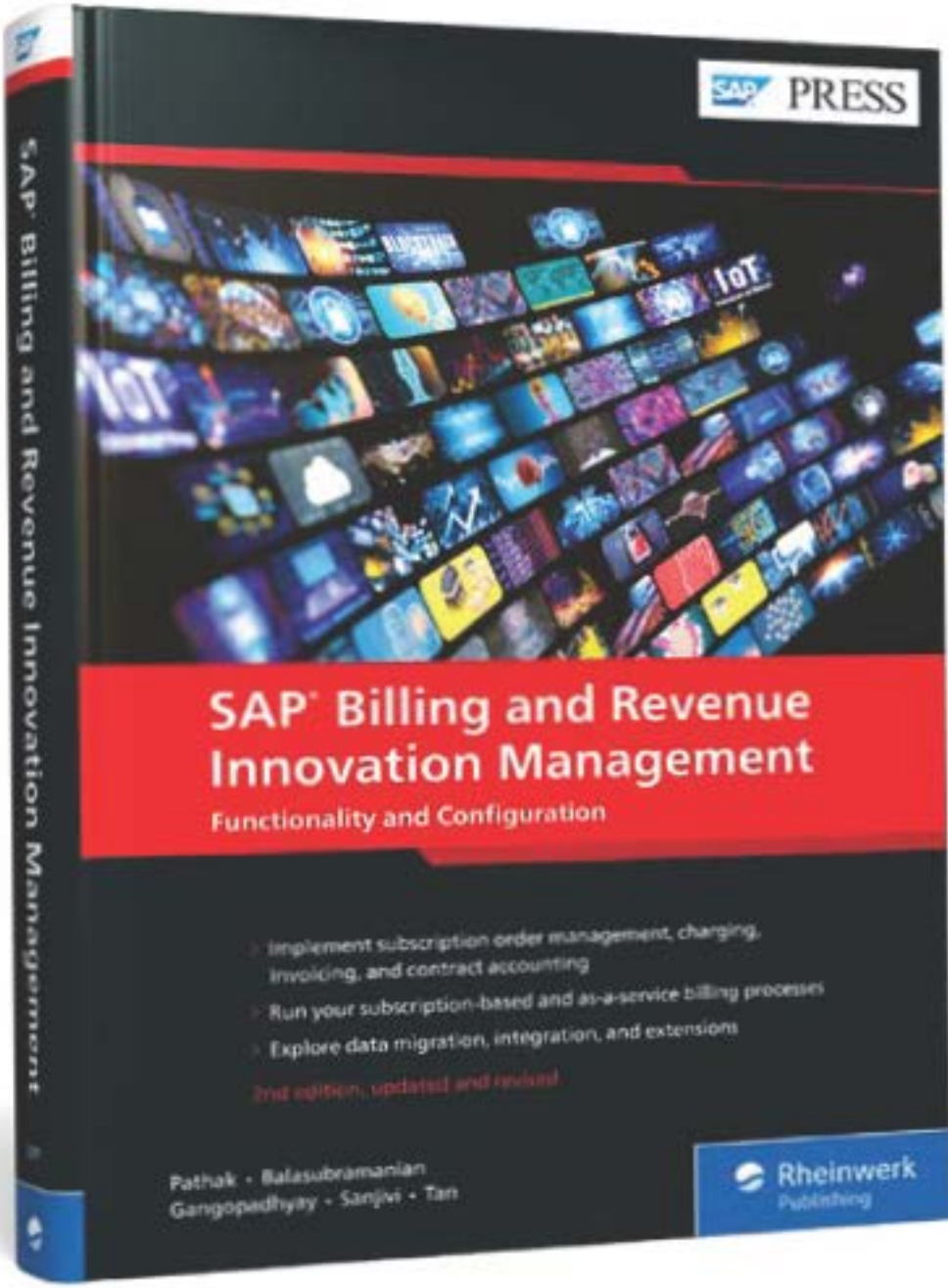

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SAP® Billing and Revenue Innovation Management

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Chapter 1

Introduction to SAP Billing and Revenue Innovation Management

This chapter highlights the shift in how institutions sell and deliver their products and services in a subscription economy. It discusses key concepts, such as the definition of flexible consumption, the driving factors behind the shift to flexible consumption, and the SAP Billing and Revenue Innovation Management solution that can help businesses adopt this new trend. This chapter also introduces all SAP Billing and Revenue Innovation Management solution components and a case study about a fictitious company shifting to a subscription economy, which will be the main theme of discussion for the following chapters.

Technology is advancing at an unprecedented rate, driving a whole new age of digital transformation and innovation. To stay ahead of these technology disruptions and to gain competitive advantages in the new economy, many companies are undergoing major revolutions. These revolutions will profoundly change their business strategies. Among these is the switch to a new billing and revenue management solution: flexible consumption. Companies that can shift to the flexible consumption model will reap great benefits—from a predictable revenue stream with higher profit margins to higher customer loyalty, resulting from the ongoing customer relationship—along with a higher cost of switching.

This first chapter begins by discussing the concept of flexible consumption and the factors driving companies to adopt this new business model, especially in the technology, media, and telecom (TMT) industry. We'll also take a quick look at SAP Billing and Revenue Innovation Management (previously SAP Hybris Billing) along with other peripheral solutions. Finally, the chapter closes with a case study about a fictitious company, Martex Corp., which serves as the main baseline for the following chapters in terms of how to enable and utilize the SAP Billing and Revenue Innovation Management solution (configurations, master data setup, etc.).

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1.1 Flexible Consumption, the Subscription Economy, and Business Model Transformation

Let's begin by examining some of the key terminology that will be used throughout the book:

- **Cloud computing**
  The delivery of on-demand computing resources and hosted services—everything from applications to data centers, including the storage and access of data—over the internet.
- **Infrastructure as a service (IaaS)**
  Providers supply a virtual server instance and various storage capacities via IaaS services. Depending on customer needs, there is a specific allocation of storage capacity and access to particular virtual machine (VM) configurations. IaaS providers can offer instances in all sizes and customizations as needed.
- **Platform as a service (PaaS)**
  Providers host development tools on their hardware and infrastructure. Web portals, gateway software, or application programming interfaces (APIs) are used to access the tools. These services are generally leveraged for software development.
- **Software as a service (SaaS)**
  Software applications are made available on the internet for users in this distribution model. These are also commonly known as web services. These services are also mobile-friendly and can be accessed from any computer or device with an internet connection.
- **Flexible consumption**
  This new business model is enabled by new trends in technology. This model affords customers flexibility in how they consume and pay based on their needs and usage. In essence, flexible consumption transforms a company's existing products or services by packaging them into different layers of use and selling these packages to the end consumer, then, depending on the amount of service consumption, customers are billed accordingly. The switch to this new business model is the main driving force behind companies seeking out a new billing solution that can support complex billing processes.
- **Subscription and subscription economy**
  A *subscription* is a business model in which customers pay a recurring fee at a regular interval to use a product or services. The *subscription economy* refers to the many companies offering subscriptions to customers and how customers consume these subscriptions.

  If you've ever used any of the following products or services, you're a participant in the subscription economy:
  - Netflix
  - Dollar Shave Club

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- Blue Apron
- Spotify
- Amazon products or services, such as Amazon Prime or Amazon Web Services

By participating in this new business model, companies and customers enter into a more long-term and intimate relationship, which requires the companies to better manage this relationship through various channels. The subscription economy pushes these companies to refocus and redesign their products/service portfolios to fit this demand, as well as the consumption style of the customers. As a result, many companies must reinvent both their offerings and their enterprise infrastructure, which in turn gives rise to the demand for the SAP Billing and Revenue Innovation Management solution.

- **Business model transformation**
  Business model transformation, in this context, describes the process in which companies develop a business case for the shift to an *everything as a service* (XaaS) model and offer constructs and use cases to explain what is sold to whom and provide a clear articulation of a strategic vision and value proposition. The transformation journey is grounded in customer experience and business simplification, as well as upgrading the billing solution. In addition, a new go-to market and new buying models also are developed to fit the redesigned business architecture and business requirements. All these activities are carried out with a focus on delivering the desired experience and a focus on lead to cash, operational capabilities, and processes.

Now that you have an understanding of the key concepts for the chapter, let's dive into the next section and explore the business drivers that fuel the growth of SAP Billing and Revenue Innovation Management.

1.2 Business Drivers for SAP Billing and Revenue Innovation Management

Let's take a detailed look at why there is such a strong demand for the switch to SAP Billing and Revenue Innovation Management by going through the key business drivers. First, we'll explore the business model changes and the benefits brought about by pursuing these changes for many companies. Then, we'll provide an overview of the three main business models that are used by companies in their journey to reinvent themselves in this new subscription economy.

1.2.1 Business Model Changes: From Products to Subscriptions

In a time of constant technological and model disruption, there is a fundamental shift occurring to flexible consumption across multiple industries. Flexible consumption

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isn't a new idea. Traditionally, this model is offered by firms that are part of the utilities industry. However, it's now expanding to businesses in a wide range of industries, creating value for both customers and the companies that adopt them (see Figure 1.1). For example, Amazon primarily used to sell books, but now it offers the Prime Reading subscription membership, via which subscribers can have unlimited access to e-books.

Benefits of flexible consumption for the consumer include the following:

- Flexibility to scale
- Asset-light model
- Lower initial investment

Benefits of flexible consumption for companies include the following:

- Predictable revenue stream
- Deeper customer engagement
- Lower unit costs from aggregation

Increasing demand for consumption flexibility is allowing companies to shift the dial on multiple internal levels.

The TMT industry may be the one that experiences the strongest impact with regard to the shift to the new model based on consumption.

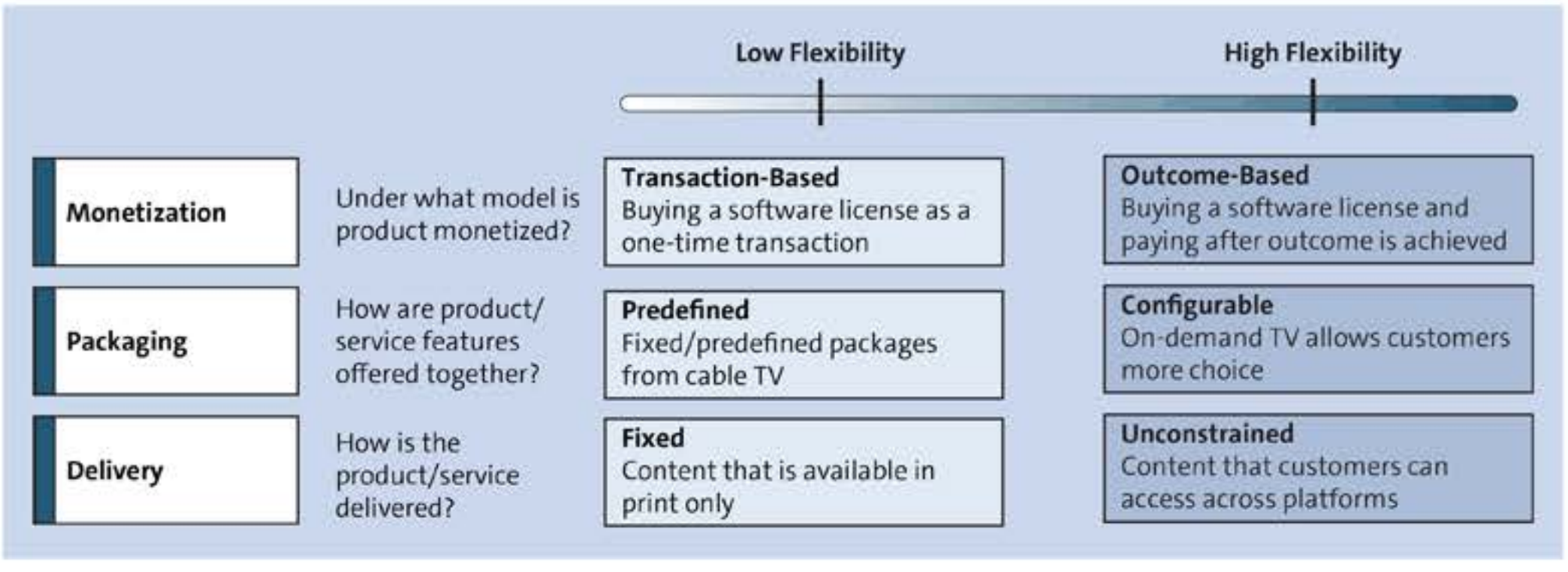

Figure 1.1 Traditional versus Flexible Model

Table 1.1 provides a list of flexible consumption models offered by many TMT companies as part of the effort to diversify their products.

Consumption Model	Description	Examples
Subscription (Unlimited)	Access to unlimited quantity of features for a predefined period	Kindle Unlimited Netflix Amazon Prime

Table 1.1 Flexible Consumption Models

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Consumption Model	Description	Examples
Subscription (Pre-defined)	Access to a predefined quantity of features for a predefined period, generally with a minimum commitment	Salesforce *The Economist*
Freemium	Access to basic services for free while charging a premium for advanced or special features	*The New York Times* Box LinkedIn
Subscription and overage	Subscription offering with additional overage charges based on actual usage	HP Ink Cisco Webex
Pay per use	Access on a pay-per-use basis, generally without a minimum commitment	Amazon Web Services (AWS) Twilio
Outcome-based	Monetization based on the value delivered to the customer, measured as a quantifiable outcome	Xerox

Table 1.1 Flexible Consumption Models (Cont.)

Many trends fuel the demand for a flexible consumption model. Some of the main trends are as follows:

- Rise in cloud product offerings
- Growth in use of mobile devices
- Increase in social networking access
- Big data
- Internet of Things (IoT)

By taking advantage of the advances in cloud-based computing, many IT companies have increased their offerings in a more flexible model based on shared hosting solutions. These solutions can range from SaaS to PaaS to IaaS, and even to XaaS. In addition, the exponential growth in mobile device usage and the expansion of broadband and Wi-Fi access points have created an expectation that customers should be able to conduct business from anywhere, at any time, from any device.

The next sections provide an overview of the three main models for flexible consumption: the consumption-based model, subscription-based model, and event-based model. These aren't exclusive and can be combined to create an even more flexible model to accommodate business needs

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1.2.2 Consumption-Based Model

The consumption-based model, also known as the usage-based model, is built on the idea that customers will pay accordingly for the amount of services or products that they consume. The philosophy for this model is pretty straightforward: First, the business will quantify the services that it provides and determine a pricing model based on actual usage. Then, at the end of every billing cycle, the customer is billed according to what it used. In one example of such a model, many network providers offer out-of-country roaming plans to their customers and charge them based on the number of messages sent or received. Another good example is the availability of many internet cafés in developing countries, which charge per minute or hour for the use of their computers or for internet access.

As shown in Figure 1.2, the invoice amount billed to customers depends on the customer usage. For example, during the first period, the customer consumes the most compared to the second and third periods. As a result, billing event 1 has the highest amount (amount 1) billed.

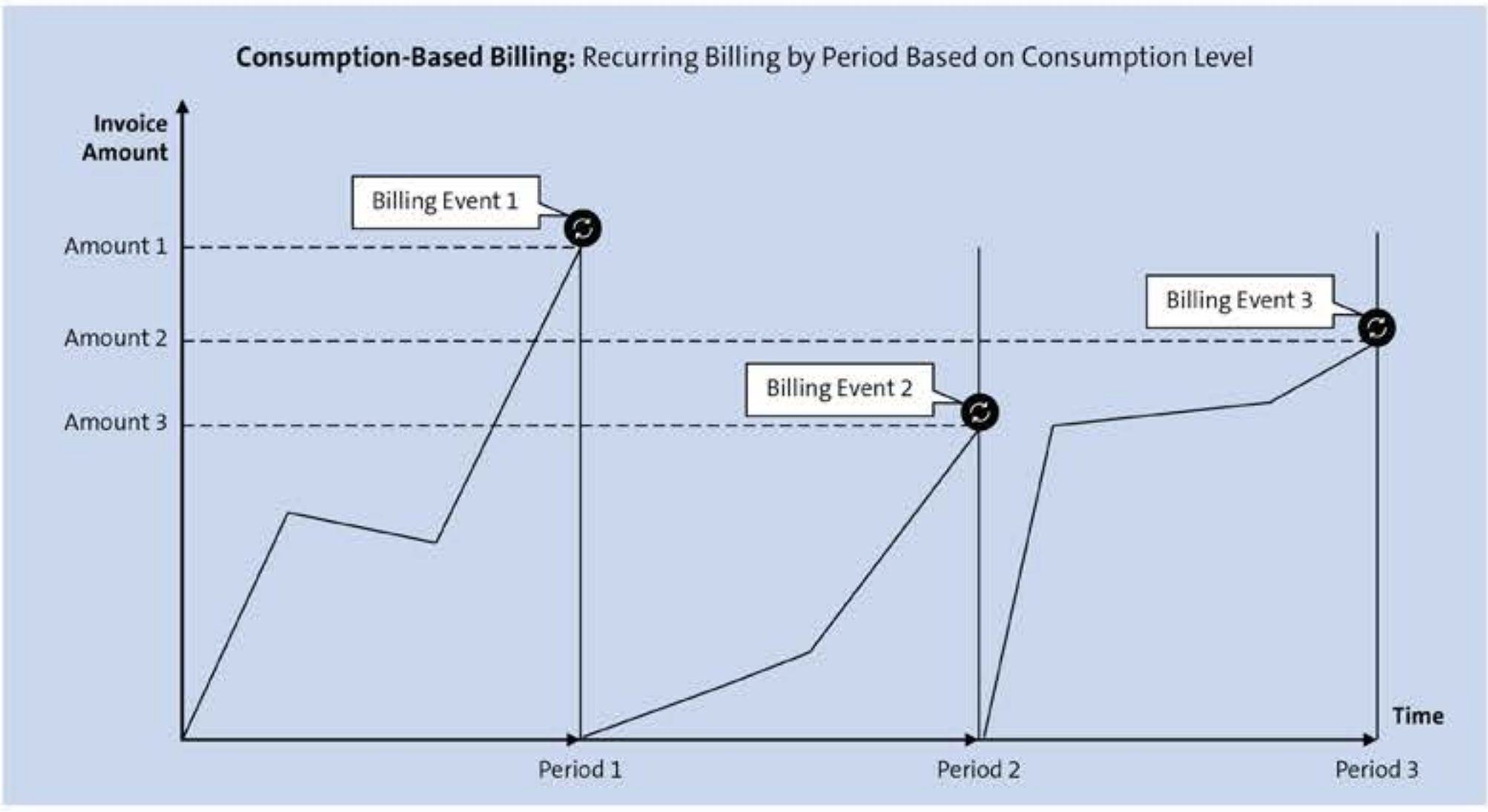

Figure 1.2 Consumption-Based Billing

Besides being simple, another advantage of this model is that it offers a linkage between the amount the customers have to pay and the actual usage and value delivered to the customers. If, for whatever reason, the customers deem that the services or products don't offer the expected return on investment, they can stop using them and aren't liable for the bill. Nonetheless, a potential drawback to this model is that there can be surprises for the customer at the end of the billing cycle. Consider again the network roaming model. The use of out-of-country roaming can be reckless, and if not

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monitored properly, this behavior can run up an outrageous bill. An additional downside can be the effect of seasonality on the revenue stream.

1.2.3 Subscription-Based Model

In contrast to the consumption-based model, companies that employ a subscription-based model offer access to their products and services for a fixed fee, and this fee is billed periodically (weekly, monthly, quarterly, annually, etc.). This model was first presented by many publishers of books and magazines. However, almost any company can adopt this model today.

Figure 1.3 provides a simple diagram of the subscription-based model. Looking at the diagram, you can see that the customer's consumption levels aren't considered. Unlike the consumption-based model, subscription-based model billable amounts and functionalities depend on subscription levels. For example, during the first and second period, the customer subscribed to the basic subscription level; as a result, the amounts charged are the same (amount 1 = amount 2). Then, during the third period, the customer decided to switch to a higher subscription level. This led to an increase in the amount billed (amount 3) and accessible functionalities for the customer.

Subscription-Based Billing: Recurring Billing by Period Based on Subscription Level

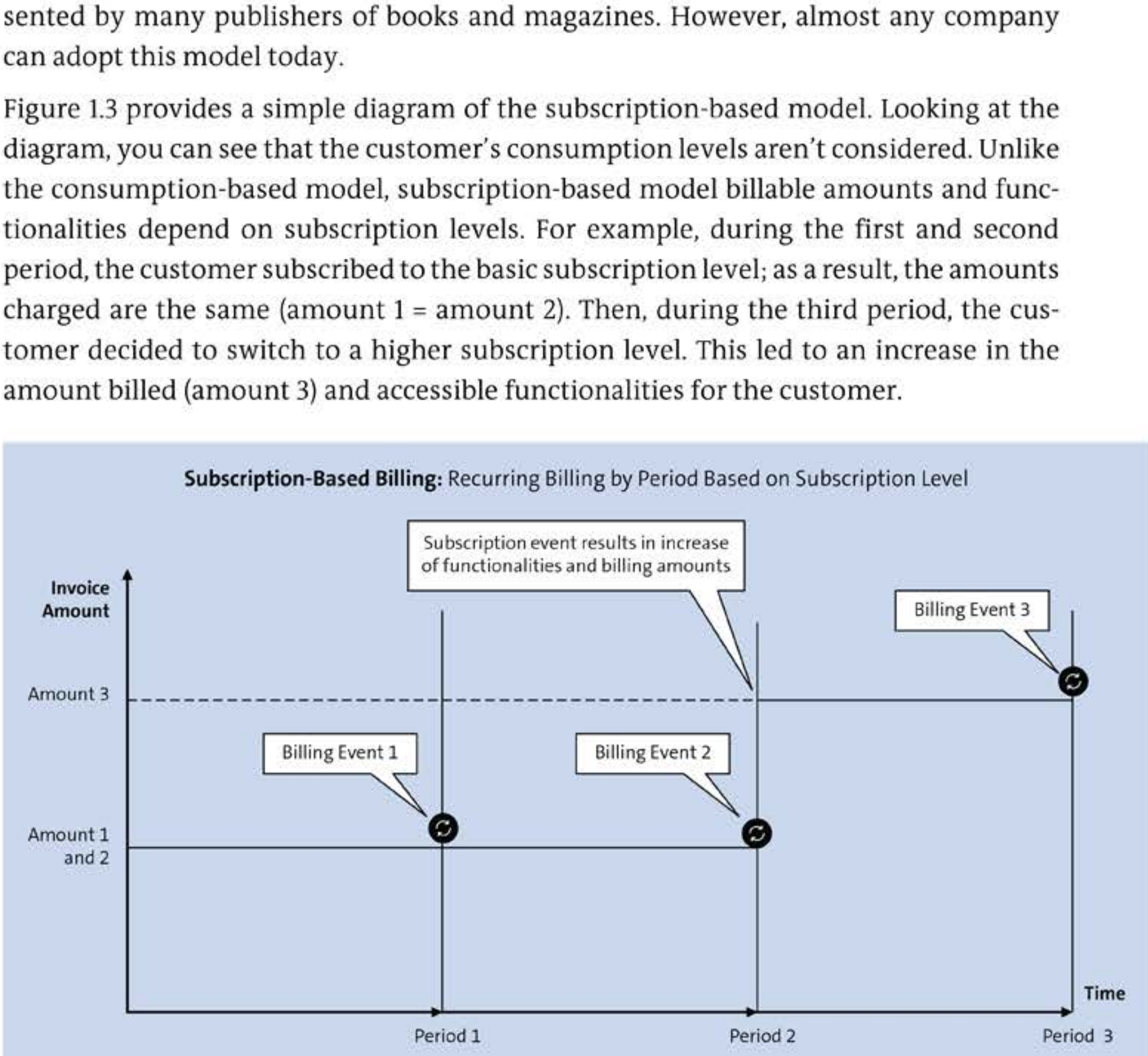

Figure 1.3 Subscription-Based Model

The key difference between consumption-based and subscription-based models is that with a subscription, customers need only to pay a fee and have virtually unlimited access to services (although limitations can sometimes apply depending on how the tier pricing is structured). In the flexible consumption economy, this model has become common thanks to the rise of companies such as Netflix, Shopify, and Dollar

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Shave Club. The subscription model is a win/win for customers and businesses. For businesses, this model is the nearest the business can get to having a crystal ball: the revenue stream is recurrent, predictable, and often not subject to aspects such as seasonality.

The following are some of the most popular ways for companies to implement this model:

- **Unlimited subscription**
  Customers have access to products and services without any limitation for a predefined period.
- **Predefined subscription**
  Customers have access to up to a specific quantity of products and amount of services for a predefined period.
- **Subscription plus coverage**
  Customers have access to up to a specific quantity of products and amount of services for a predefined period. Any additional usage that goes beyond this limit will be billed based on actual usage.
- **Freemium**
  Customers have access to the most basic features of the products and services for free. If customers want access to the premium functionalities, they will have to pay more for these features.

1.2.4 Event-Based Model

The event-based model is a hybrid model that uses the consumption-based model as the starting point and adds different events, which influence the lifecycle of the products, to charge customers. These events can range from simple events, such as the activation/deactivation of the service/device, to more complex events, such as suspension of the device usage and charging customers a prorated rate.

Figure 1.4 provides a diagram of a simple event-based model. Unlike a subscription- or consumption-based model, which has billing events triggered at regular intervals, event-based model billing events are triggered based on events (or milestones). These events are typically decided as part of the contract and will be configured into the system. These events can be automatically triggered based on the lifecycle or manually triggered by either the customer or the company. As a result, as soon as the contract reaches a certain milestone of its lifecycle, the billing event will trigger.

With all that said, you might think that switching to this new model is a no-brainer and that many companies should already be getting on board with this business model transformation.

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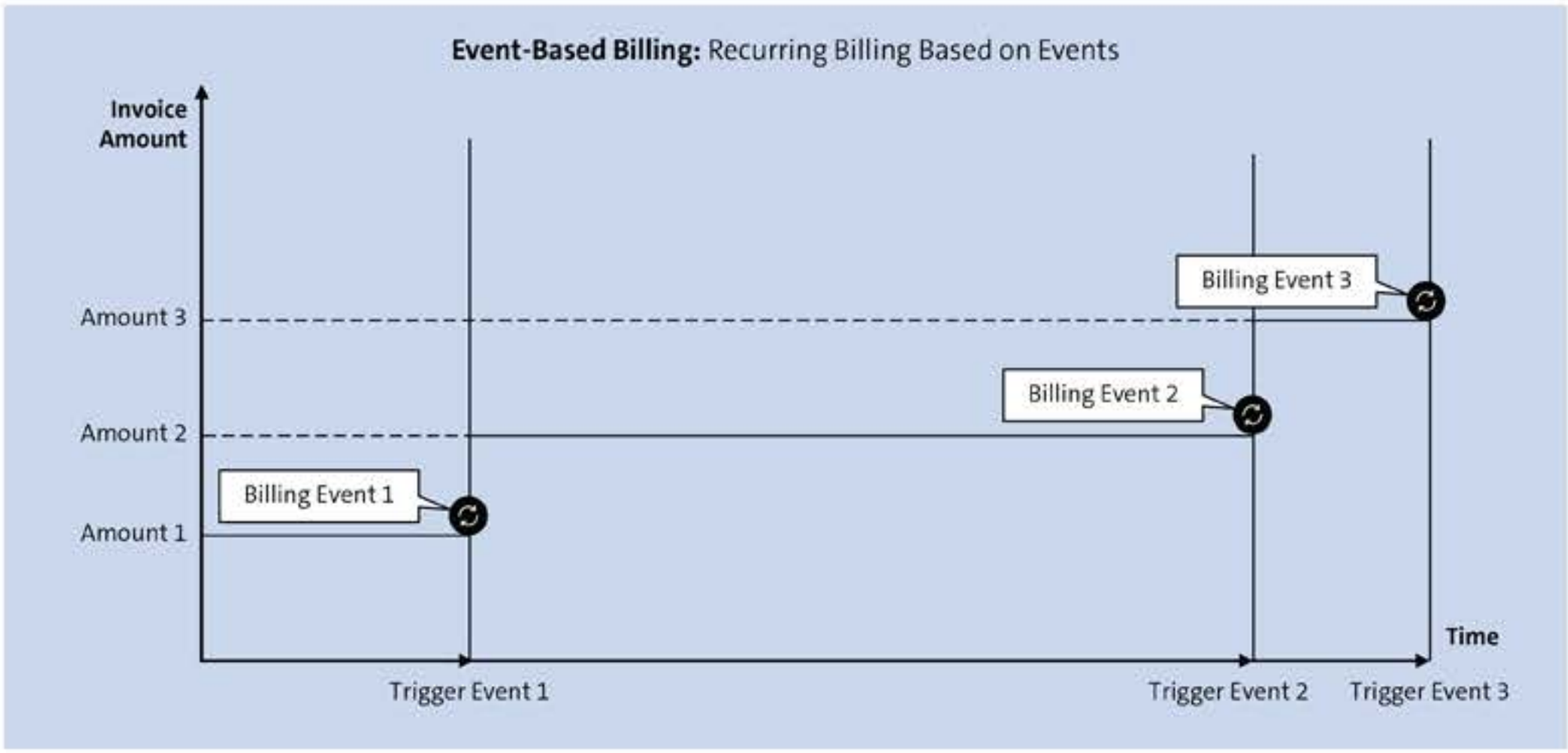

Figure 1.4 Event-Based Billing

Table 1.2 lists some of the key companies that were initially created with a flexible consumption model in mind and some of the key players that are trying to transition their current models to flexible consumption models.

	Consumer and Enterprise Software (SaaS)	Cloud Computing Services (IaaS/PaaS)	Streaming Audiovisual Content	Other Products and Services
Born Flexible Consumption Model	■ Salesforce ■ Workday ■ Box	■ Amazon Web Services (AWS) ■ Microsoft Azure	■ Netflix ■ Spotify	■ Dollar Shave Club ■ Birchbox
Transition to Flexible Consumption Model	■ Adobe ■ Intuit	■ HPE ■ VMware vCloud Air	■ Electronic Arts (EA) ■ Home Box Office (HBO)	■ Amazon Prime ■ Toyota ■ General Electric

Table 1.2 Companies Originating with or Transitioning to Flexible Consumption Models

Note that most of the companies that are in the transition process are big and established companies; not many are small or midsized.

However, this transition is easier said than done and involves many challenges. A successful transformation to a flexible consumption model requires strategic thinking and high levels of organizational coordination and alignment and is driven by visionary leadership. For instance, current organizational capabilities (e.g., billing and support) may not support the new model, or risk-averse stakeholders may be resistant to

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the change. Furthermore, the short-term impact on financial performance may have significant implications, both internally and externally.

The following are some of the challenges involved that act as deterrents when a company considers implementing an SAP Billing and Revenue Innovation Management solution:

- **The stakes are high and rising.**
  With traditional business models under threat, the pressure to offer flexibility in consumption is rising.
- **The challenge is complex.**
  The scope of disruption requires a complete reconsideration, recalibration, and redefinition of operations.
- **There is little room for error.**
  Failure to act, execute, and transform in a timely manner may threaten the viability of the business.

To solve these challenges, SAP offers businesses SAP Billing and Revenue Innovation Management—an end-to-end cross-industry solution that comprises SAP Billing and Revenue Innovation Management, subscription order management ; SAP Convergent Charging; SAP Convergent Invoicing; and contract accounting. In the next few sections, we'll take a quick look at each of these solutions individually.

1.3 SAP Billing and Revenue Innovation Management Components

SAP Billing and Revenue Innovation Management, previously known as SAP Hybris Billing, is a comprehensive solution for high-volume consumption businesses. SAP Billing and Revenue Innovation Management enables the various flexible consumption models that we discussed in Section 1.2. It's a modular solution designed to optimize the business lifecycle processes of design, sales, delivery, and billing, as depicted in Figure 1.5, which illustrates the lifecycle of the SAP Billing and Revenue Innovation Management solution based on its business benefits.

The first phase is *design*, in which SAP Billing and Revenue Innovation Management assists with the creation of a uniform design for a business and pricing model, along with the master agreement that will be used as the base for other downstream processes.

Once this is done, the next phase, *solution sales*, kicks in. In this phase, companies define how they will sell their products with the assistance of the contract management and/or quote management functionalities offered by SAP Billing and Revenue Innovation Management.

The third phase is *solution delivery*. In this phase, companies define the charging methodology for their portfolios. In other words, this is when the companies decide how and

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how much their customers will be charged based on the business models they pursue. For example, if a company follows an event-based model, it will define the business/transaction events that trigger charges for customers. If a company chooses a consumption-based model, it will calculate the tiers of consumption in which customers will be charged.

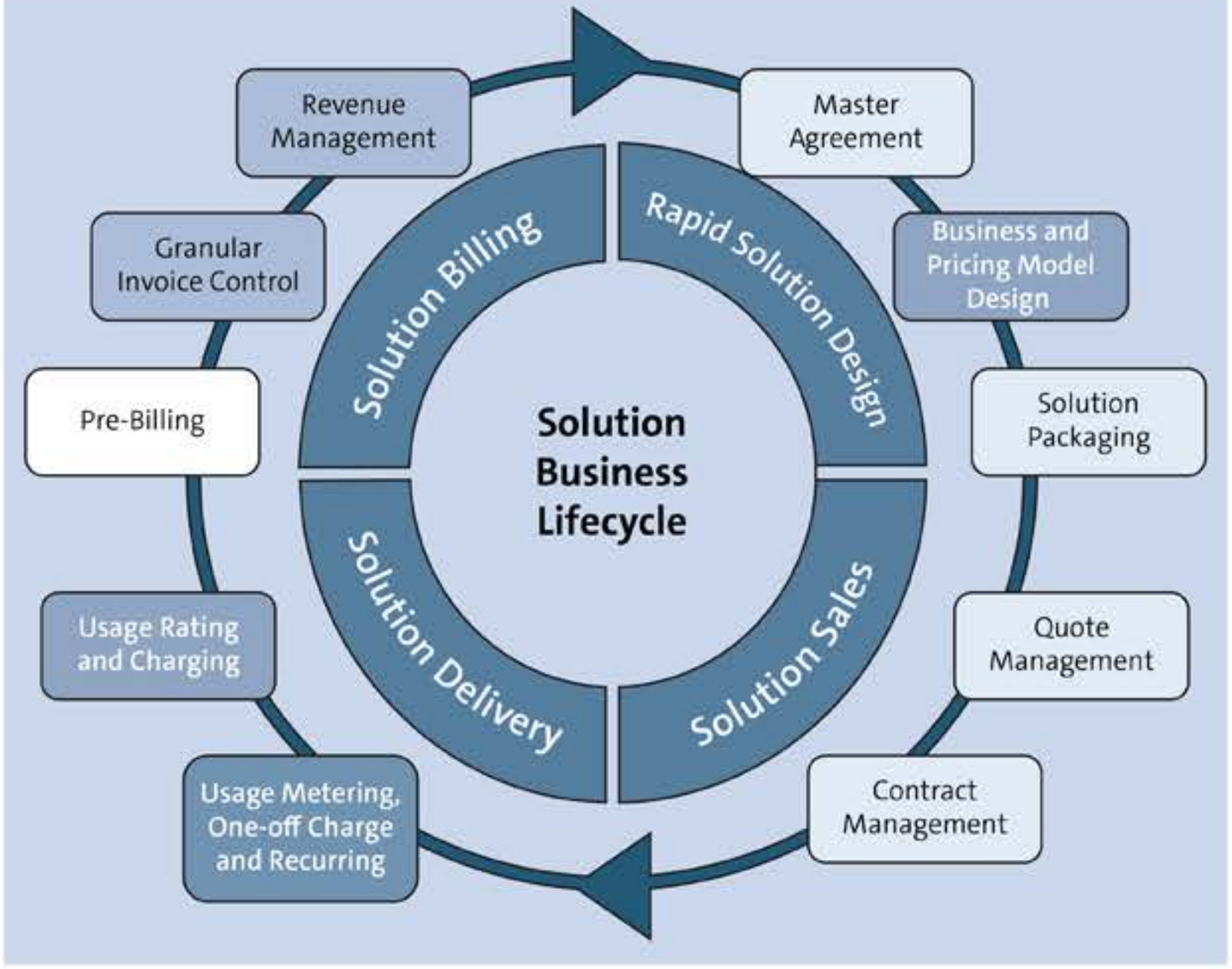

Figure 1.5 Solution Business Lifecycle

The final phase is *solution billing*. In this phase, SAP Billing and Revenue Innovation Management lets companies manage their billing and invoicing mechanisms for consumers based on what has been finalized in previous phases. Furthermore, SAP Billing and Revenue Innovation Management aids in the management of revenue recognition and management with various tools for revenue analysis.

The solution comprises four core components:

- SAP Billing and Revenue Innovation Management, subscription order management
- SAP Convergent Charging
- SAP Convergent Invoicing
- Contract accounting

Together these components help in the management of business model design; perform order management; track usage metering and transaction pricing, billing, and invoice processing; manage receivables and payables, royalties, and commissions; and provide support for financial customer care (Figure 1.6 and Figure 1.7).

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SAP Billing and Revenue Innovation Management also offers peripheral solutions such as SAP Convergent Mediation by DigitalRoute, SAP Flexible Solution Billing, SAP Revenue Accounting and Reporting, and SAP Entitlement Management,, which are implemented as needed per business requirements and complement the core SAP Billing and Revenue Innovation Management solution. Table 1.5 shows a breakdown of the SAP Billing and Revenue Innovation Management components.

Area	SAP Billing and Revenue Innovation Management Module	SAP Solution	Description
Table 1.3 Core Components	SAP Billing and Revenue Innovation Management, Subscription Order Management	■ SAP S/4HANA 1909 onward – Included with the SAP Billing and Revenue Innovation Management license	■ Manages product catalogs, including bundles of services and physical goods ■ Manages contract lifecycles
	SAP Convergent Charging	■ High Deal – Included with the SAP Billing and Revenue Innovation Management license	■ One-time, recurring, and usage-based pricing can all be integrated together ■ Pricing rate changes can be easily applied with increased flexibility
	SAP Convergent Invoicing	■ SAP S/4HANA – Included with the SAP Billing and Revenue Innovation Management license	■ Consolidates multiple billing sources into a single, simplified invoice from multiple sources ■ Supports high volumes, custom pricing, and third-party revenue sharing and partner settlement
	Contract accounting	■ SAP S/4HANA – Included with the SAP Billing and Revenue Innovation Management license	■ Posts financial documents and receives/makes payments ■ Manages disputes, credits, and collections

Table 1.5 SAP Billing and Revenue Innovation Management Core Components and Peripheral Solutions

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Area	SAP Billing and Revenue Innovation Management Module	SAP Solution	Description
Table 1.4 Peripheral Solutions	SAP Flexible Solution Billing	■ SAP S/4HANA – Included with the SAP Billing and Revenue Innovation Management license	■ Integrates easily with existing accounts receivable and with sales and distribution ■ Billing customizability decreases customer inquiries and reduces time to payment ■ Consolidated invoice
	SAP Convergent Mediation by DigitalRoute	■ Digital Route – Additional license required	■ Flexible online and batch mediation tool ■ Aggregation, filtration, and data enrichment
	SAP Entitlement Management	■ Additional license required	■ Entitlement management solution from SAP ■ Standard integration with SAP Billing and Revenue Innovation Management
	SAP Subscription Billing	■ Additional license required	■ Subscription billing solution on SAP Cloud Platform

Table 1.5 SAP Billing and Revenue Innovation Management Core Components and Peripheral Solutions (Cont.)

SAP Billing and Revenue Innovation Management solution components reside both as areas within SAP S/4HANA and as standalone modules that integrate with SAP S/4HANA. Figure 1.6 illustrates how each core component of SAP Billing and Revenue Innovation Management resides within the end-to-end solution landscape. SAP Billing and Revenue Innovation Management, subscription order management, and SAP Convergent Charging reside in different instances because they are standalone applications. This is denoted by ❶ for SAP Billing and Revenue Innovation Management, subscription order management, and ❷ for SAP Convergent Charging. On the other hand, SAP Convergent Invoicing and contract accounting are part of the SAP S/4HANA solution and come out of the box with SAP S/4HANA. This is denoted in the diagram with ❸, used for SAP S/4HANA, contract accounting, and SAP Convergent Invoicing.

As part of the new architecture of SAP S/4HANA 1909, SAP Billing and Revenue Innovation Management, subscription order management, is now part of the SAP S/4HANA solution, as shown in Figure 1.7. This significantly reduces the complexity in the

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implementation and integration when companies implement the SAP Billing and Revenue Innovation Management solution.

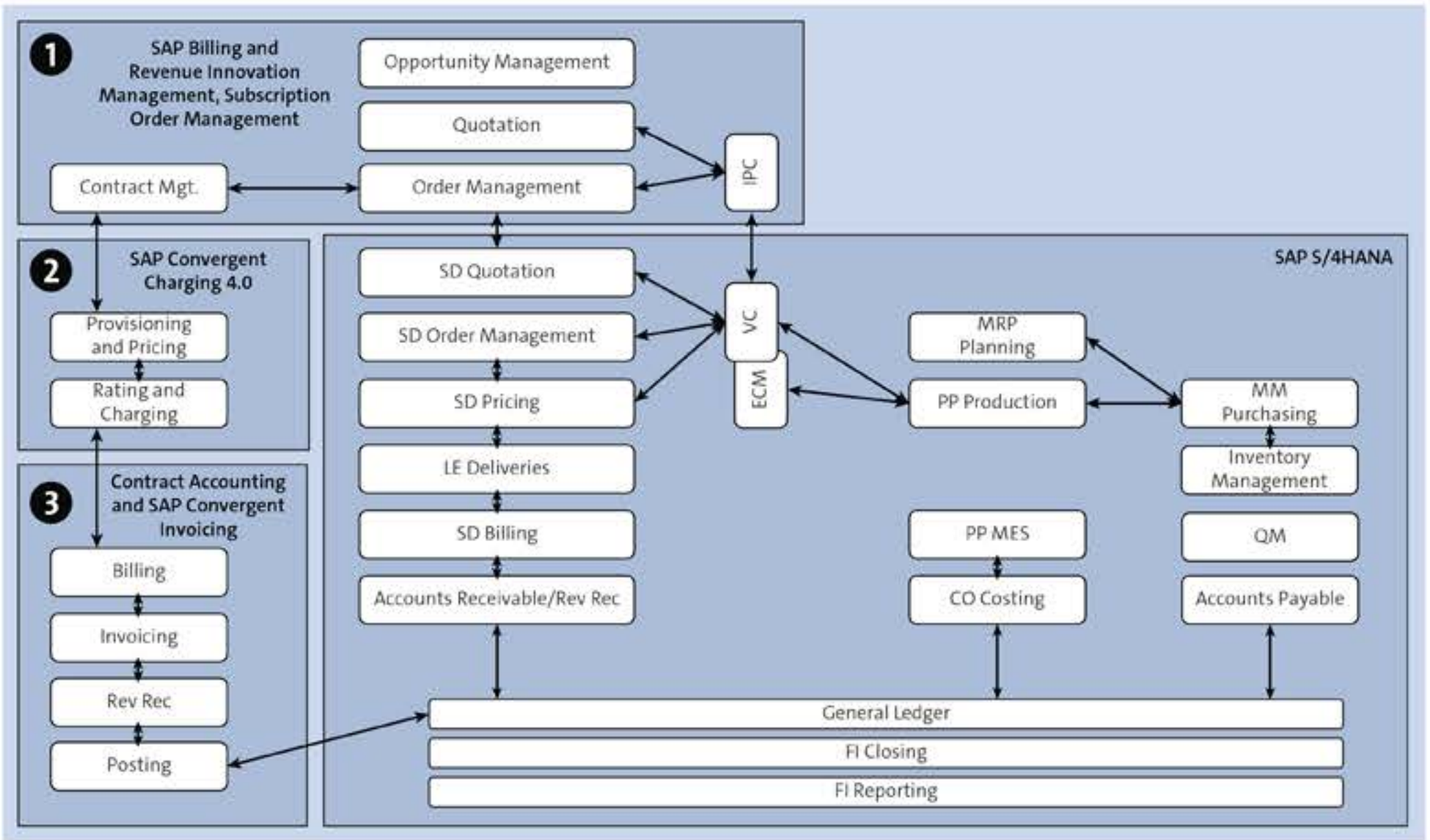

Figure 1.6 SAP Billing and Revenue Innovation Management Solution Landscape

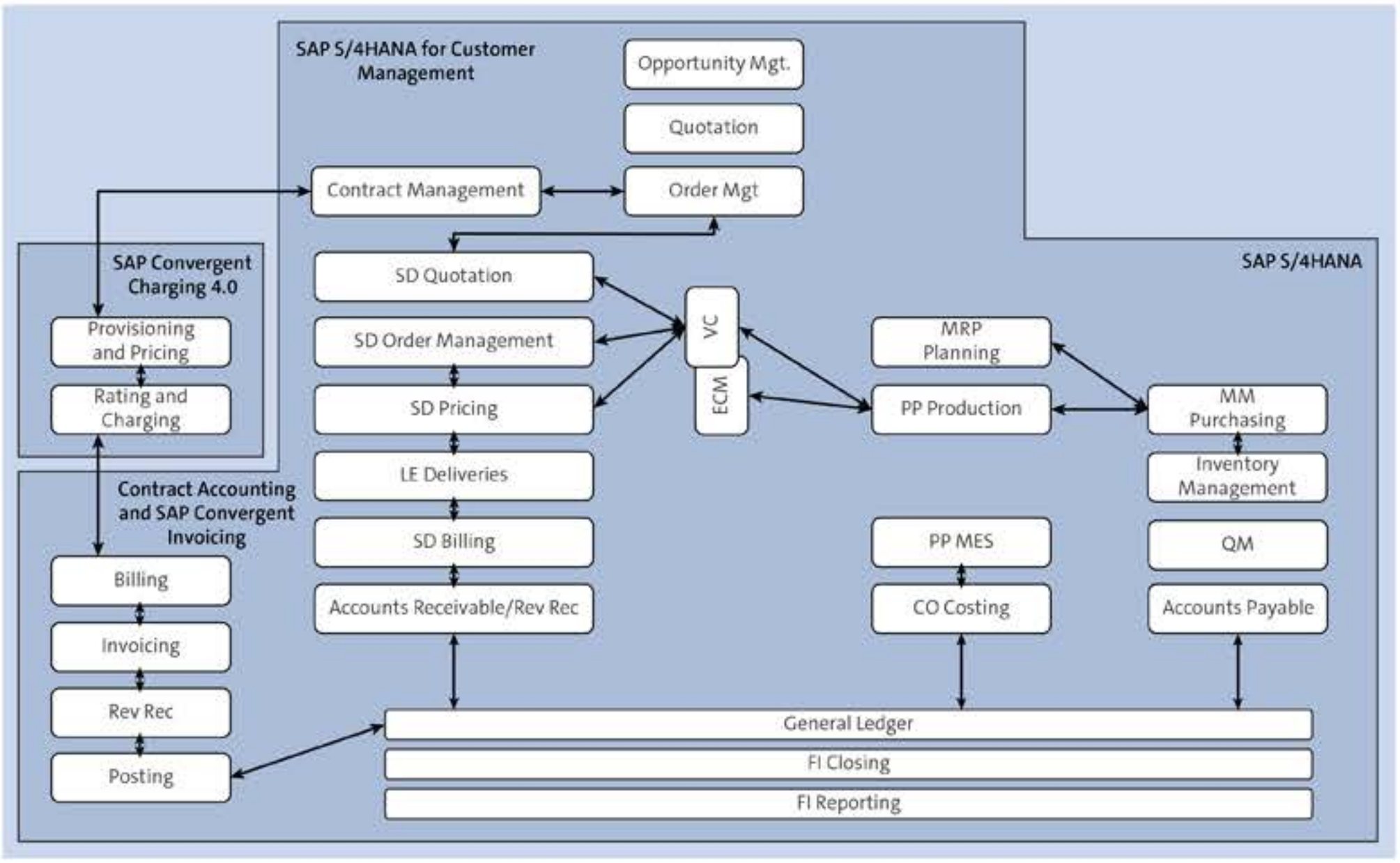

Figure 1.7 SAP Billing and Revenue Innovation Management Solution Landscape as of SAP S/4HANA 1909

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Figure 1.8 shows the flow of master data and transactional data between the various SAP Billing and Revenue Innovation Management modules and external applications.

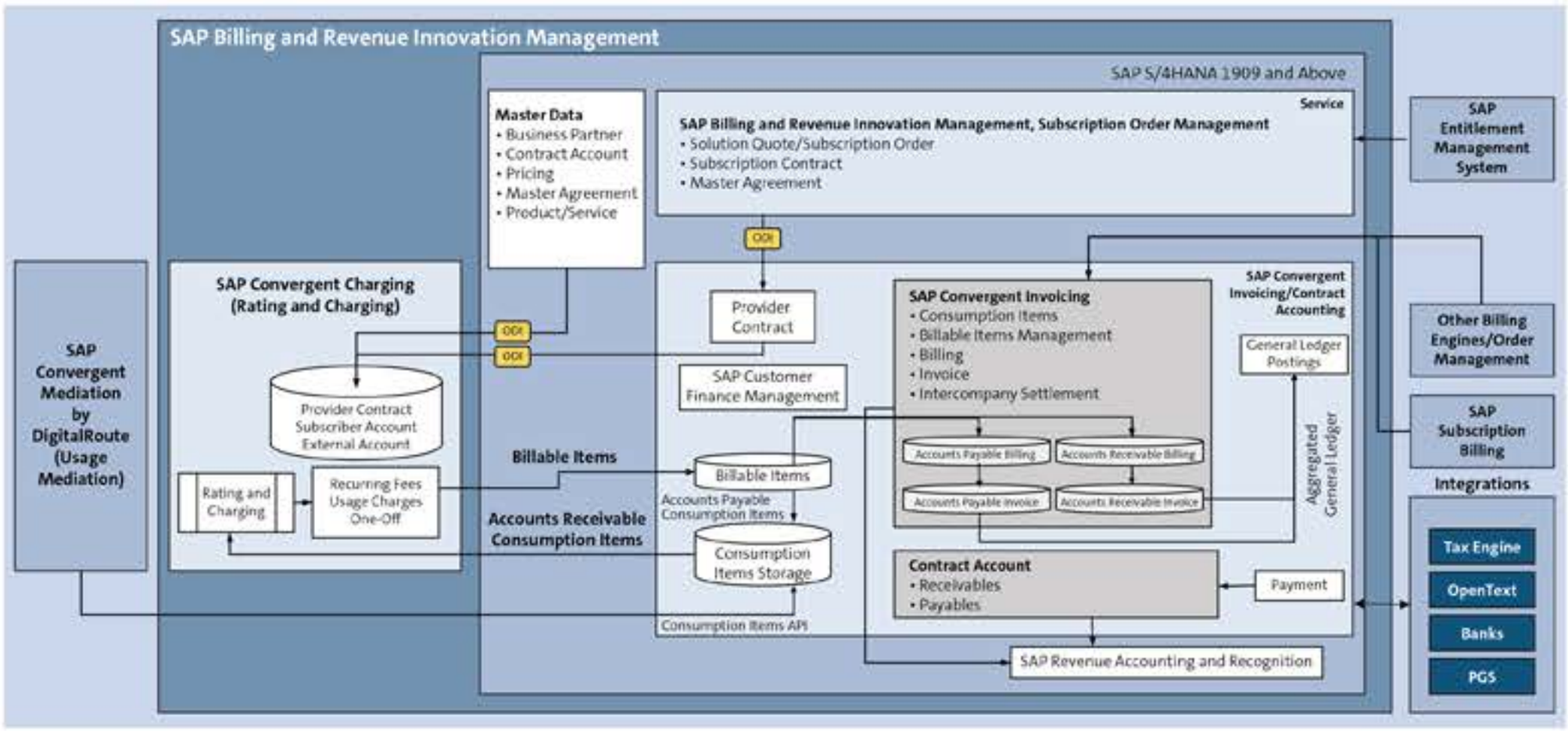

Figure 1.8 SAP Billing and Revenue Innovation Management Data Flow

1.3.1 SAP Billing and Revenue Innovation Management, Subscription Order Management

SAP Billing and Revenue Innovation Management, subscription order management, is an integral part of the end-to-end SAP Billing and Revenue Innovation Management solution. It's a business process developed in SAP for the purposes of addressing generation of fees for periodic usage of rights and the consumption of fees for the actual amount of measurable usage. As of SAP S/4HANA 1809, it has been folded into SAP S/4HANA.

SAP Billing and Revenue Innovation Management, subscription order management is the starting point for SAP Billing and Revenue Innovation Management as it manages contract lifecycles and product catalogs, including bundles of services and physical goods. Once these activities are done, SAP Billing and Revenue Innovation Management, subscription order management, creates orders that are then fulfilled by the core SAP ERP system. The order information is then passed to the next component, SAP Convergent Charging, for charging/rating defining and processing (as shown in Figure 1.9).

Equipped with the following specialized features, SAP Billing and Revenue Innovation Management, subscription order management, allows a company to effectively manage its customers by creating flexible ways to meet its customers' needs:

- Product catalog and offer management
- Order capture

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- Order distribution
- Contract management
- Customer care

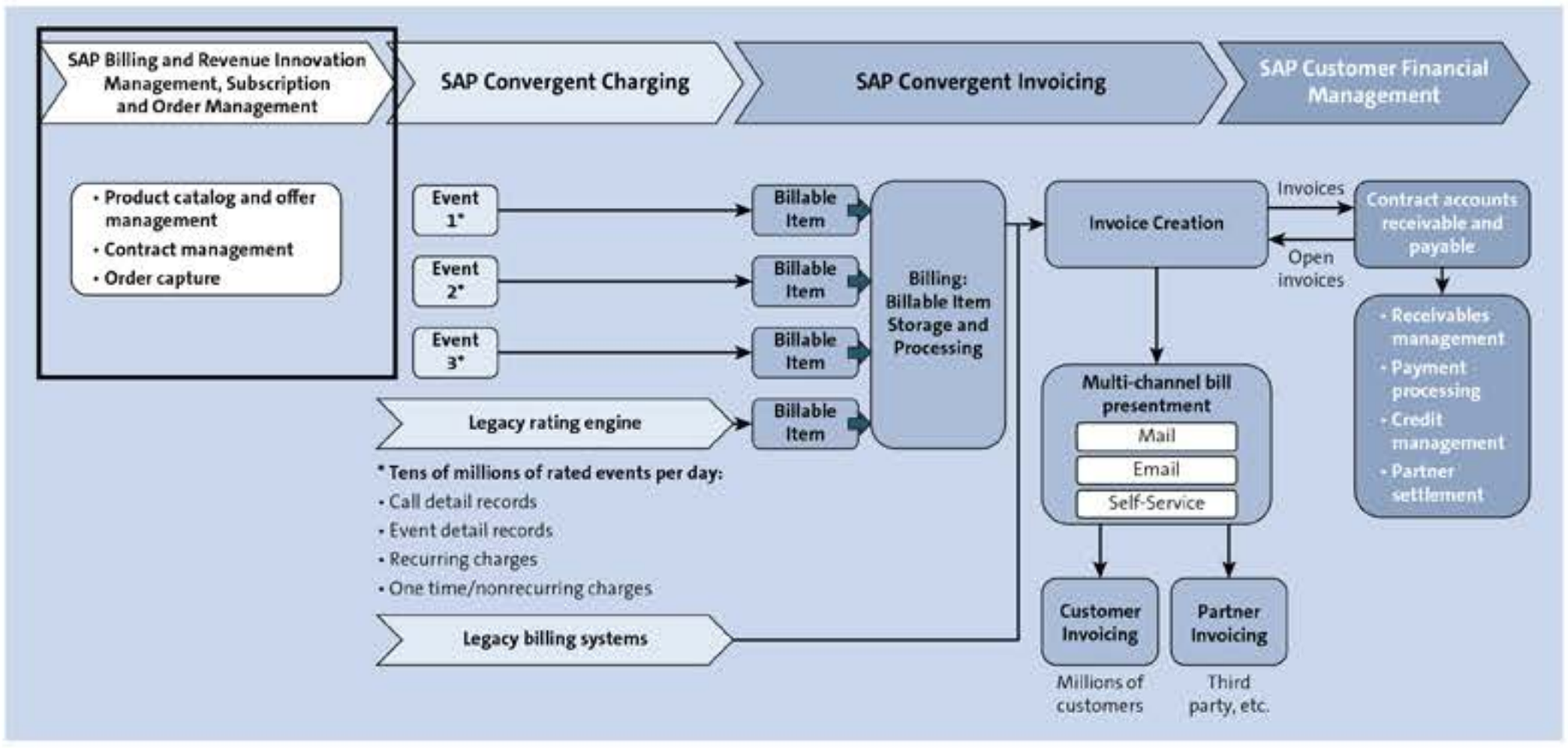

Figure 1.9 SAP Billing and Revenue Innovation Management, Subscription Order Management

Chapter 2 will cover these features in depth.

1.3.2 SAP Convergent Charging

SAP Convergent Charging is designed to deliver pricing flexibility when charging customers. In SAP Convergent Charging, a *charge* is a price generated from a rating and charging event that is charged to the customer (or subscriber). In other words, SAP Convergent Charging is a calculation engine tool used to determine the type of charges, such as recurring, one-time, and usage-based charges.

Figure 1.10 illustrates how SAP Convergent Charging fits into the SAP Billing and Revenue Innovation Management solution. There are two main components in SAP Convergent Charging: charging and rating. Charging is how SAP Convergent Charging determines which exact customer account will be charged for a specific customer. On the other hand, rating focuses on the calculation mechanism that derives the total amount to be charged in the billing document.

As part of the SAP Billing and Revenue Innovation Management solution, SAP Convergent Charging comes with an intuitive graphical user interface (GUI), which allows for visual programming (Figure 1.11). This, in turn, virtually eliminates functional and technical limitations to developing pricing offers and revenue sharing plans for different types of services.

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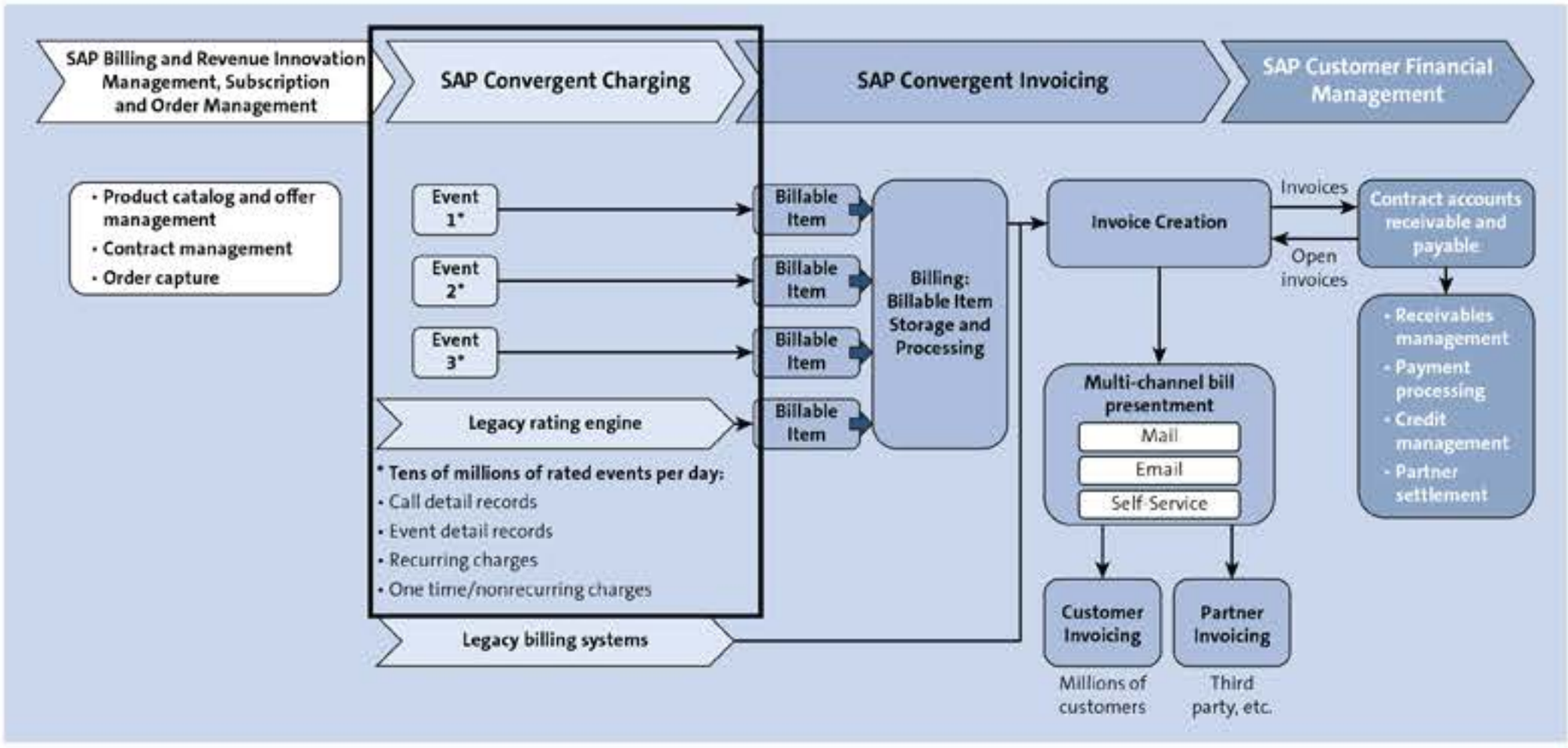

Figure 1.10 SAP Convergent Charging in SAP Billing and Revenue Innovation Management

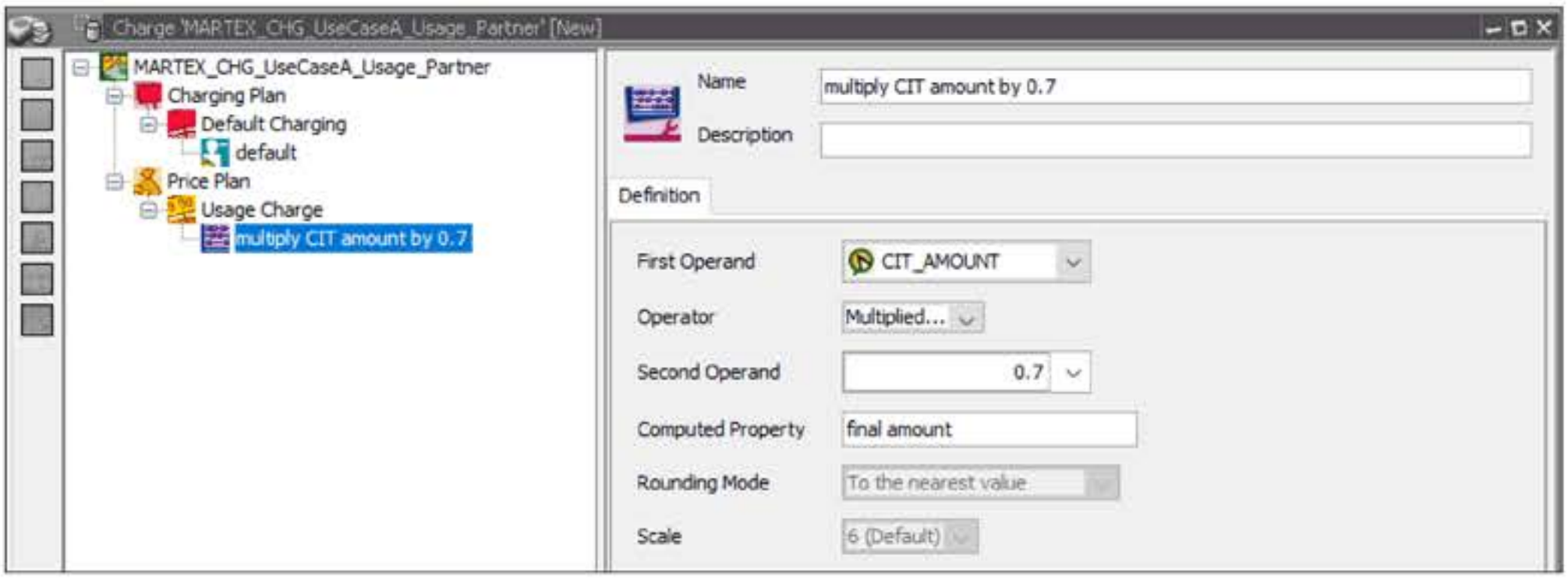

Figure 1.11 SAP Convergent Charging Interface

1.3.3 SAP Convergent Invoicing

SAP Convergent Invoicing's role in SAP Billing and Revenue Innovation Management is to consume the billable items once they are passed down from SAP Convergent Charging. Once these items reach SAP Convergent Invoicing, the application will handle the process of uploading consumption detail records, storing billable items, aggregating billable items, creating the invoice, and invoicing the customer.

SAP Convergent Invoicing stores and manages rated consumption data records from the SAP Convergent Charging system (see Figure 1.12). The rated consumption data records are also known as billable items and undergo a billing and invoicing process as part of posting receivables in contract accounting.

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Unlike SAP Billing and Revenue Innovation Management, subscription order management, and SAP Convergent Charging, which come as separate applications, SAP Convergent Invoicing isn't a separate application and is a part of contract accounting, which is included as part of the core SAP S/4HANA portfolio. This means that SAP Convergent Invoicing is natively integrated with all SAP S/4HANA areas out of the box, especially SAP S/4HANA Cloud for contract accounting and invoicing.

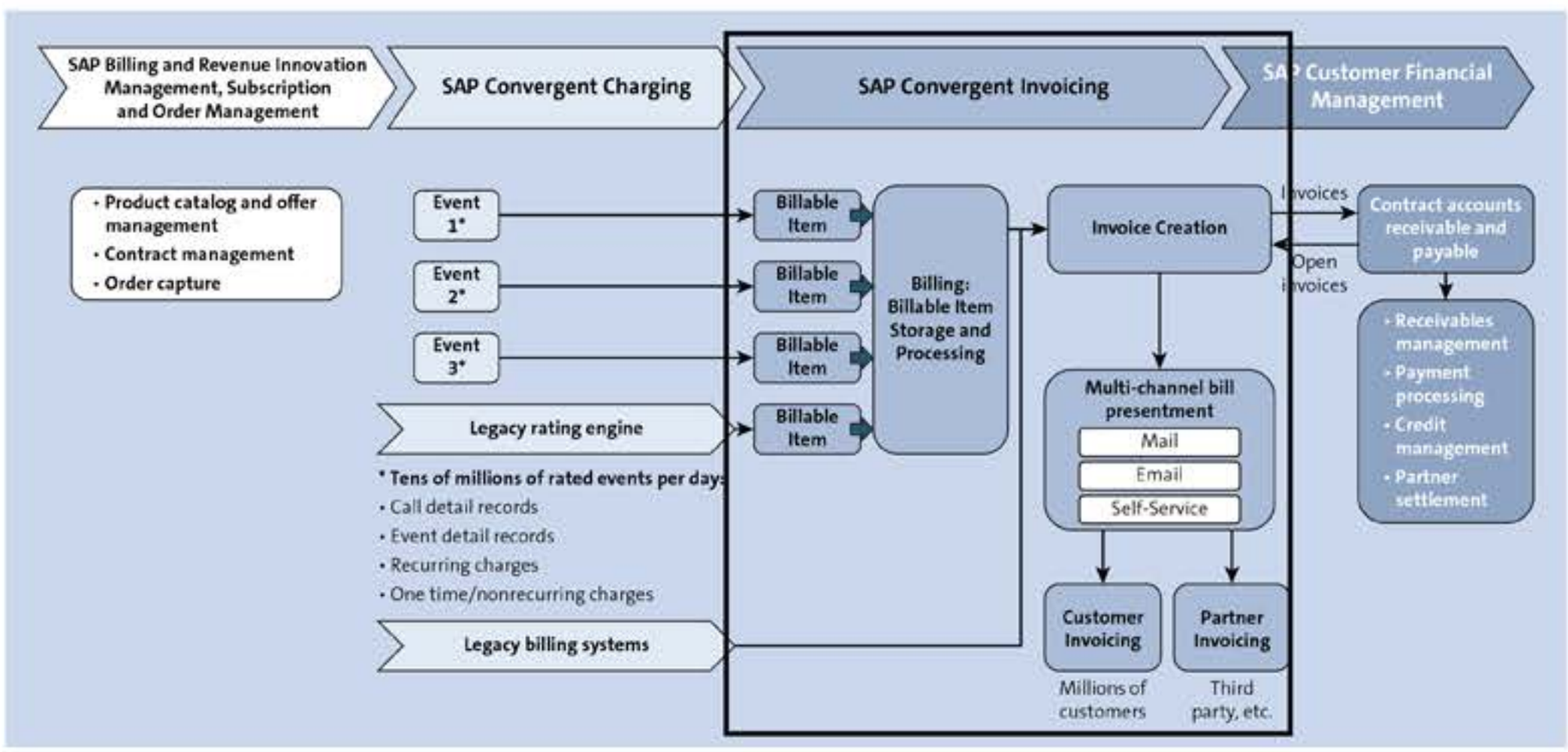

Figure 1.12 SAP Convergent Invoicing in SAP Billing and Revenue Innovation Management

1.3.4 Contract Accounting

Contract accounting is the last component of SAP Billing and Revenue Innovation Management. It's the financial backend that provides subledger accounting postings tailored to accommodate the requirements of industry sectors in which companies must interact with multiple customers (in this case, subscribers) and a large number of transactions.

As the financial backend of a robust solution in a dynamic business model, contract accounting also offers a wide range of functionalities. As shown in Figure 1.13, contract accounting offers taxation, tax calculations, and country-specific processes, such as accounting principles, payment processing, collection management, partner settlement, and so on. Because of these requirements, contract accounting is highly automated after its initial setup and requires little intervention.

To summarize at a high level, you create orders in SAP Billing and Revenue Innovation Management, subscription order management, perform calculations in SAP Convergent Charging for an order based on usage, and get the final charge amounts in SAP Convergent Charging. Bills and invoices are sent to the customer using SAP Convergent Invoicing. Finally, in contract accounting, you post the financial documents and receive/make payments. You also use contract accounting for dispute management, credit management, and collections.

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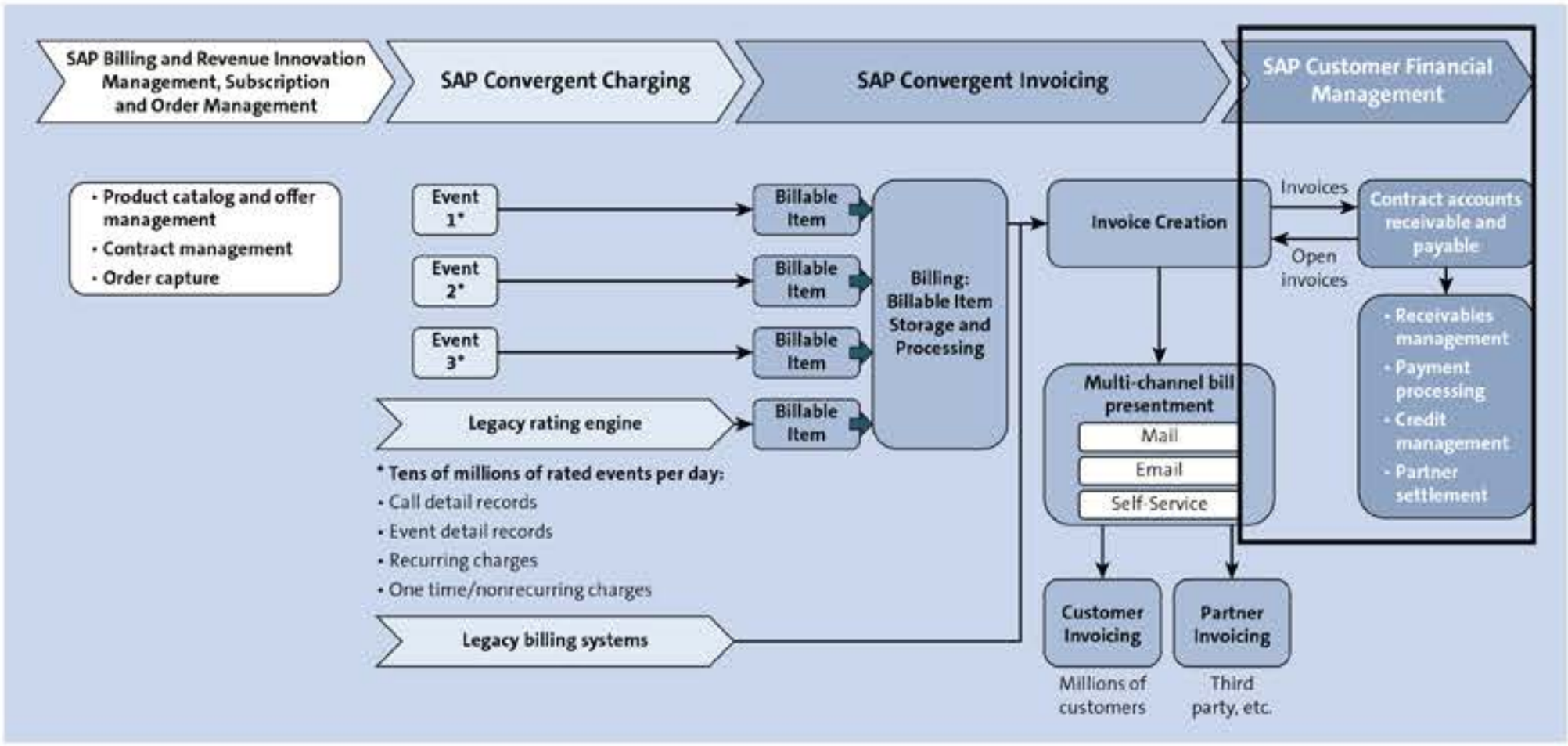

Figure 1.13 Contract Accounting in SAP Billing and Revenue Innovation Management

1.3.5 Peripheral Solutions

For the most part, SAP Billing and Revenue Innovation Management's four core components within the core SAP ERP system is well-suited to meet most business requirements. However, in certain cases, companies have the option to create even more robust and flexible solutions by integrating with other peripheral solutions offered on the market. In the following list, we discuss some of more popular peripheral solutions that can complement SAP Billing and Revenue Innovation Management:

- **SAP Convergent Mediation by DigitalRoute**
  SAP Convergent Mediation is an SAP partner product developed by DigitalRoute. Its role is to enable a standardized approach to data mediation. This means this solution first gathers data from multiple sources, then the data is filtered and converted into a format that can be consumed by other SAP Billing and Revenue Innovation Management components. The advantage of SAP Convergent Mediation is that even though it's developed by another party, it's pre-integrated into SAP Billing and Revenue Innovation Management and can be deployed quickly. The benefits that come with this solution include the reduction of multiple integration points required to manage data, reduction in infrastructure costs, and increased efficiency in data processing, which leads to higher scalability.
- **OpenText**
  OpenText is developed by the company of the same name. OpenText complements the SAP Billing and Revenue Innovation Management solution by extending the document management capabilities of all SAP modules. It enables businesses to have a comprehensive view of all scanned business documents and all related data across systems and applications. OpenText's advantage lays in its pre-integration

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with SAP modules and its ability to optimize and standardize document management through automation to ensure productivity and compliance.

- **Payment gateways**
  The *payment gateways* generic term refers to a set of applications offered by many vendors to provide portals for payment processing. These applications typically offer functionality for secured payment processing through major financial methods (credit cards, PayPal, checks, etc.), payment and personal data tokenization, and fraud monitoring,
- **SAP Customer Financial Management**
  SAP Customer Financial Management is a set of solutions created to streamline analyzing and optimizing financial processes. They provide solutions for cash management, investment and debt management, and financial risk management. In most cases, SAP Customer Financial Management is used in conjunction with SAP Billing and Revenue Innovation Management to measure, analyze, and control the risk of a business partner defaulting on financial obligations.
- **SAP Revenue Accounting and Reporting**
  SAP Revenue Accounting and Reporting lets businesses manage their revenue and income recognition cycles in compliance with various accounting policies. The solution comes with many accounting standards and rules preconfigured and automated (e.g., ASC 606 or IFRS 15). In addition, SAP Revenue Accounting and Reporting also provides a wide range of configurable reporting capabilities for cases in which the standard reports can't meet a company's requirements.
- **SAP Entitlement Management**
  SAP Entitlement Management streamlines and automates the process of entitlement management for companies. The solution manages the entitlement lifecycle (i.e., manages what the customer is entitled to). This ensures transparency into what customers are entitled to for both the customer and the sales rep (alerts, reminders, etc.). This, in turn, increases customer satisfaction and provides more opportunities for upselling or cross-selling.

1.4 Case Study: Martex Corp.

Now that you have a basic understanding of SAP Billing and Revenue Innovation Management, its driving force, and its components, let's look at our case study. At this point, you should have a high-level understanding of the benefits SAP Billing and Revenue Innovation Management offers and the role of each SAP Billing and Revenue Innovation Management component in an end-to-end solution. Moreover, you've also learned about the auxiliary solutions that can enhance the SAP Billing and Revenue Innovation Management functionalities so that the entire solution can meet your needs and deliver the best performance.

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The next several chapters will take deep dives into each component of SAP Billing and Revenue Innovation Management to further explain its features and the configurations to enable them. However, as mentioned previously, SAP Billing and Revenue Innovation Management is very comprehensive and highly configurable, which means that if we jump directly into the master data and the setup of the components, it will be nearly impossible to explore all possibilities. With this in mind, similar to many other projects and applications, SAP Billing and Revenue Innovation Management works best when it's tailored to specific use cases—like the one in our case study. We'll examine a fictional company called Martex and its requirements to transform its business model to a flexible consumption model. This case study will play a vital role in the rest of the book and will be the focal point for all the master data and configuration setup.

1.4.1 Background

Martex Corp. is a distribution company that distributes an office software product, Martex Workplace Solution (MWS), and desktop computers and laptops. MWS is a suite of workplace productivity software that includes an office suite, along with other specific applications aimed toward the content-development market. Martex doesn't develop MWS from scratch but instead purchases/licenses the rights to the MWS applications from its vendors. Martex also procures desktop computers and laptops from original equipment manufacturers (OEMs), bundles the sales with MWS, and resells the suite. The company has been operating under the traditional model, in which customers purchase the desktop/laptop computer along with a license key or a physical copy (CD) of the MWS software. Martex Corp. generates revenue from the margin between the purchase price from the vendors and the selling price for its customers.

1.4.2 Concerns

Recently, Martex has been experiencing a decline in its sales numbers; growth has been disappointing. Sales have stagnated: many of the company's clients' laptops no longer have CD drives. The drop in the number of laptops with CD drives hammers the company's ability to sell physical copies of its software. Furthermore, the license key distribution model is falling short because the price is fixed, whereas many competitors can now offer more flexible functionalities, bundled offerings, and pricing.

Martex also operates with a legacy manually managed in-house ERP solution, developed to manage order lifecycles, customer and product details, along with financials. The ERP solution also lacks the standard ERP template to onboard new mergers and acquisitions (M&A) and doesn't have revenue-driven business insights across domestic and international entities.

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1.4.3 New Business Model

After thorough reviews of its business model and customers' survey feedback, Martex's leadership realize that they must change their business model to a subscription and flexible consumption model to survive. Martex will bundle its MWS product and desktop/laptop sales as an XaaS offering through the cloud and provide hosting services for files created from the MWS suite.

Martex Corp. will transform itself to offer the XaaS business model and the new subscription economy. Under its new strategy, it will procure products from different vendors and sell them to its customers on behalf of those vendors while procuring desktops/laptops and bundling the hardware and software. Each product is sold via different models—subscription-based, flexible-consumption-based and pay-per-use models. Based on the business model offered by Martex Corp., the following use case leverages SAP Billing and Revenue Innovation Management functions to process the company's end-to-end billing requirements.

1.4.4 Use Case

The purpose of this use case is to help Martex offer its customers the ability to combine charges (hence the name "combo charge"). In this use case, we'll look at service A and service B, and each has a different chargeable rate based on whether the customer opts for a reserved instance (minimum commit). A reserved instance is an option provided to customers in which they pay a recurring charge in exchange for lower usage rates. X is a monthly recurring reserved instance membership charge of $200.00 per month to the customer. Customers with membership X can purchase additional services at lower prices, as follows:

- Service A
  - $300/GB without a reserved instance
  - $250/GB with a reserved instance
- Service B
  - $20/hour without a reserved instance
  - $10/hour with a reserved instance

In-addition, the subscription is activated based on a license key that is required to be installed on the desktop/laptop in order to initiate subscription charges. Customers can also terminate their contracts after 30 days, which will result in a prorated refund of the up-front charge.

The first month of the subscription is regarded as a trial period with the following stipulations:

- If a customer chooses to suspend their contract in the first month, there should be no charge.

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- If a customer suspends and resumes their contract in the first month, they should be charged the full price of $200.
- After the first month, a customer will only be charged based on days when contracts aren't in a suspended state.
- They also have one-time charges related to activation, installation, and termination fees.

The customer is also allowed an allowance of $2,000 toward the use of their instances only. This allowance is valid for a period of two months. If $2,000 isn't consumed within two months, the balance lapses.

Vendor A doesn't sell its software directly to customers, but rather through multiple distributors in different regions. Distributors acquire and invoice customers on behalf of vendor A. In return, vendor A provides 10% off the MSRP cost to all distributors.

Some of the end customers for Martex, can contract with Martex with a centrally negotiated price and a common pool for consumption of services that can be used for all their business units.

Now we'll list the various recurring charges and one-time fees for this case study. To better understand this case study, Table 1.6 has a representation of how the monthly invoice value would look for this use case.

- Fixed recurring charge: $200
- Reserved instance: Yes
- Reserved instance charge: $250
- Activation fee: $50
- Allowance: $2,000 (valid for 60 days)

Month	Instances Consumed (A)	Value (B) (Instances Consumed)	Value (C) (including Recurring Charge)	Invoice Value (D) (Allowance Consumed)	Remaining Allowance Value
Month 1	2	$500 (A × $250)	$750 (including activation fee) (B + $200 + $50)	$250 (B consumed from $2,000)	$1,500 ($2,000 – B)
Month 2	4	$1,000 (A × $250)	$1,200 (B + $200)	$200 (B consumed from $1,500)	$500 ($1,500 – B)
Month 3	3	$750 (A × $250)	$950 (B + $200)	$950 (Allowance expired)	$500 (Lapsed)

Table 1.6 Representation of a Sample Charge Calculation for the Case Study

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In summary, let's briefly revisit the background of Martex Corp. As sales struggle, the company is trying to bring about a new business model to replace its old one. The company wants to offer a flexible consumption model by offering XaaS via subscription billing. There can be many different use cases to consider, depending on the specific requirements from Martex drawn from its pain points and vision. However, this won't be the focus of this book. The combo charge and prorated charge use cases are just two examples of the most common use cases in which SAP Billing and Revenue Innovation Management can help Martex revamp its business model.

1.5 Summary

This chapter focused on introducing and familiarizing you with SAP Billing and Revenue Innovation Management. We first discussed the key terminology frequently used and the business drivers that lead the transformation from product-based to subscription-based business models. Then we talked about each of the core SAP Billing and Revenue Innovation Management components at a high level, in addition to the peripheral applications that enhance SAP Billing and Revenue Innovation Management functionalities. Finally, we created a case study with a fictious company, Martex Corp., with two use cases to act as the baseline for all the master data and configuration in coming chapters.

Now that you have a grasp of the basics, it's time to take a closer look at the core components of SAP Billing and Revenue Innovation Management. We'll start with Chapter 2, in which we'll begin our discussion with SAP Billing and Revenue Innovation Management, subscription order management.

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Chapter 2

Subscription Order Management

In this chapter, we'll introduce SAP Billing and Revenue Innovation Management, subscription order management, and its functions using SAP S/4HANA 1909 and above. A lot of new features have been enabled as part of recent releases, but we'll focus on some of the interesting features introduced to handle anything as a service (XaaS) transformation. We'll describe and execute the necessary configuration steps to build our Martex Inc. use case and also explore the native integration with SAP Convergent Charging and SAP Convergent Invoicing.

SAP Billing and Revenue Innovation Management, subscription order management, is a business process developed solely for the purposes of addressing the subscription model businesses that are prevailing today. In these businesses, products aren't sold outright; instead, the right to use them is sold.

Initially, SAP Billing and Revenue Innovation Management, subscription order management, was called Provider Order Management and was managed outside of SAP S/4HANA using SAP Customer Relationship Management (SAP CRM). The previous edition of this book addresses the implementation of SAP Billing and Revenue Innovation Management, subscription order management, using SAP CRM. However, as of SAP S/4HANA 1809 FPS01, SAP Billing and Revenue Innovation Management, subscription order management, now resides in SAP S/4HANA. This is analogous to the business process in the telecom industry in which a consumer is charged a monthly fee for the right to use a telephone service for a month, regardless of the number of calls or amount of data used during that month. This fee is usually charged in advance. In addition, the actual usage—that is, calls, SMS, and data—is measured and billed in the next billing cycle in arrears. This same business and billing model can be leveraged for other industries that employ this concept.

SAP Billing and Revenue Innovation Management, subscription order management, uses SAP S/4HANA as its frontend system to connect customers with services they buy and consume. The configurations from SAP CRM–based SAP Billing and Revenue Innovation Management, subscription order management, have been leveraged to a great extent in SAP S/4HANA–based SAP Billing and Revenue Innovation Management, subscription order management, as the underlying business process and concepts remain mostly the same. There are continuous innovations in every release of SAP S/4HANA to add new capabilities as the solution evolves to keep up with the changing business models, including device as a service (DaaS), anything as a service (XaaS), and so on.

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In this chapter, we'll guide you through how to configure SAP Billing and Revenue Innovation Management, subscription order management, from beginning to end. We'll begin with setting up the required master data, such as business partners, contract accounts, and subscription-specific products. Then we'll continue with the configurations of the various transaction types and item categories, required profiles and settings within transaction types, item category determination, and document distribution steps. We'll also focus on some of the exciting features and components that are used to handle critical business use cases regarding subscription lifecycle management, revenue share, and integration of SAP Billing and Revenue Innovation Management, subscription order management, with other SAP Billing and Revenue Innovation Management components.

2.1 Master Data

Master data in SAP Billing and Revenue Innovation Management is any data that is used in a business transaction in SAP Billing and Revenue Innovation Management and mainly managed using SAP Billing and Revenue Innovation Management, subscription order management. It includes data about customers, vendors, contact persons, billable accounts, and subscription products that don't change frequently throughout the lifecycle of SAP Billing and Revenue Innovation Management, subscription order management. If there is a change in the master data, then usually a governance process follows to ensure traceability and proper validation. Because master data isn't changed regularly, the benefit is that you don't have to repeatedly enter the same relevant data because SAP Billing and Revenue Innovation Management copies it automatically into related business functions.

We'll discuss setting up the following master data in detail in SAP Billing and Revenue Innovation Management, subscription order management:

- Business partner
- Contract account
- Subscription products

2.1.1 Business Partner

In this section, you'll learn about the concept of a business partner and the configuration setup of a business partner for SAP Billing and Revenue Innovation Management. A *business partner* is a master data object such as a customer, vendor, contact person, employee, organizational unit, and so on. SAP has divided the business partner into different roles to manage various business functions and features independently. The unique and usually nonmodifiable partner in any transaction is the sold-to party. Likewise, there can be different mandatory partners in any transaction such as ship-to party, bill to party, and payer.

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A single business partner in SAP Billing and Revenue Innovation Management can have more than one role, such as customer, vendor, and contract partner, depending on the interaction of the entity in a transaction. This enables a legal entity to be represented by a single business partner in SAP Billing and Revenue Innovation Management. A business partner has general data, such as their legal entity name, address, communication details, and more, stored at the global level. The concept of business partner roles was introduced to facilitate these functions.

A business partner role is an SAP technical term to define the set of characteristics a business partner can take on. It defines how the business partner will interact in each transaction.

For example:

- The sold-to party is the entity for which the transaction is placed and maintained.
- An end customer is the entity the service or product on the transaction is for.
- A vendor is a supplier of the goods.
- An employee is person involved or responsible for the transaction at the company.

Before going into the configuration of the business partner roles, we first need to understand the role of the sold-to party, which is required for business transactions. The sold-to party functions as the customer or legal entity of a transaction and has three main areas: general, sales and company code data. The basic data of a customer is saved in the general business partner role, whereas the sales and company code data is stored in the sales and distribution section.

Before you can use business partner functions, you first need to activate them. Once these functions are activated, the required fields in the business partner views are activated. To activate business partner functions (see Figure 2.1), follow menu path **Cross Application Components • SAP Business Partner • Activation Status for Functions**.

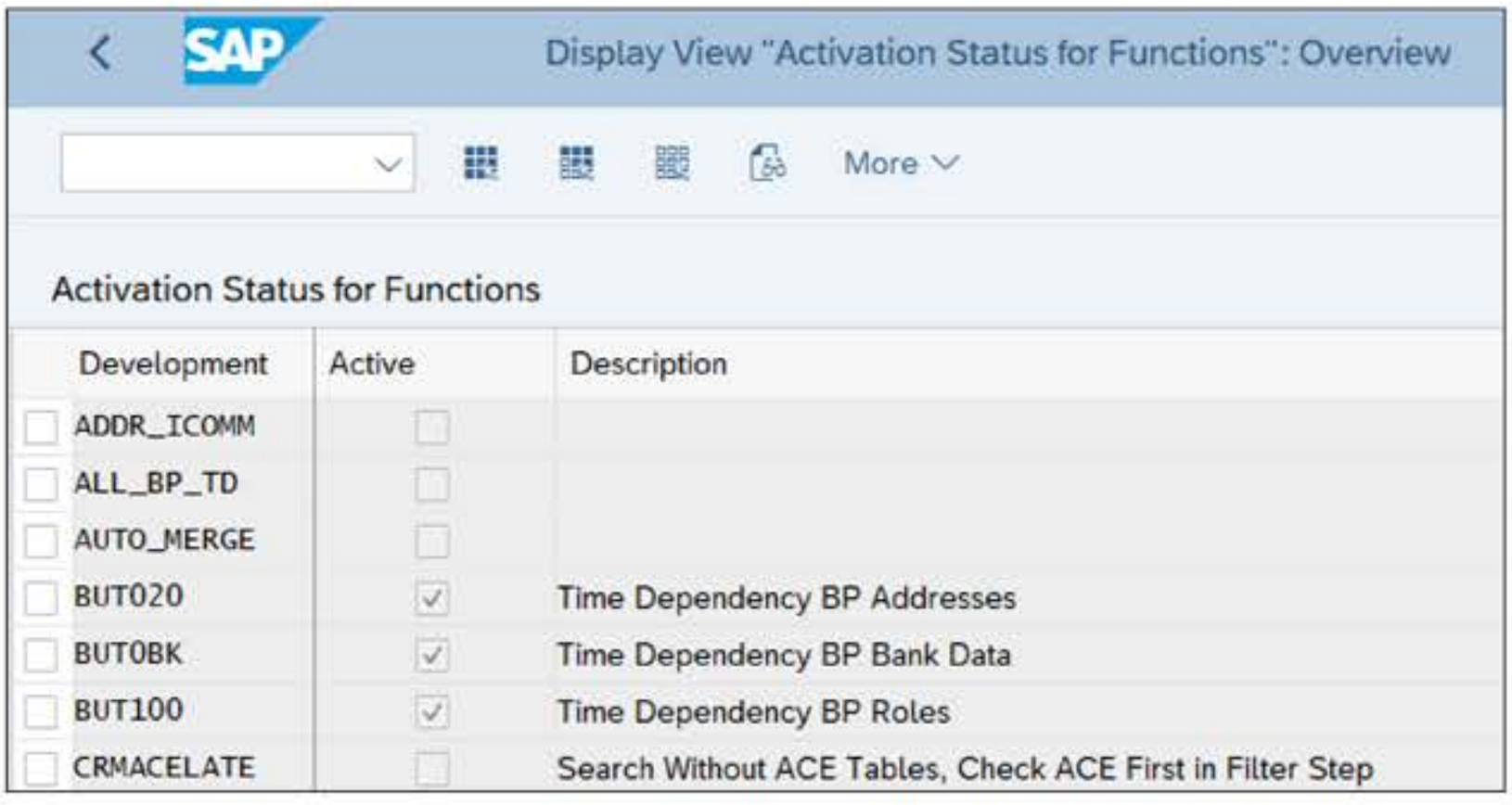

Figure 2.1 Activation Status for Business Partners Functions

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Once the business partner functions are set up, the first configuration required for the business partners is to define business partner roles. As mentioned, the roles define the function that a business partner takes on during a business transaction. Each business role has a set of related information that will be populated when setting up a new business partner. While developing the use case for Martex Inc., you won't need to configure any new business partner roles because you'll be using the set of standard roles provided by SAP. To view standard roles or define new roles, follow menu path **Cross Application Components • SAP Business Partner • Business Partner • Basic Settings • Define BP Roles.**

Figure 2.2 shows the various business partner roles defined in the system, which correspond to the different role categories.

Figure 2.2 Business Partner Roles

Each business partner role is assigned to a role category. *Role category* is a technical term to group together specific types of business partner roles. This enables the same characteristics to be assigned to the business partner roles within the same category and enables programs to use the groupings internally. Each business partner role category can be defined as an organization, person, and/or group. For example, there can be a business partner role for each different type of contact, but all are assigned to the same role category for the company's contacts.

The next step is to define the number ranges and groupings for a business partner. A unique number range should be defined for each set of business partners to easily identify different categories of business partners. A number range can be defined as numerical or alphanumerical to provide flexibility. Each number range can be set to internal or external. If the number range is set to internal, then the system will generate the next number to assign a new business partner. If the number range is set to external, then the user or an external system can enter the number if it falls within the number range and isn't already used in the system.

Each number range interval defined is assigned to a business partner grouping. A business partner grouping is assigned to each business partner created and determines the number range. The business partner grouping can't be change after it's assigned. When creating a business partner, the grouping assignment will determine how the number range field is populated—manually if an external grouping was selected or automatically

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if an internal grouping was selected. The different groups help keep the different business partners separated.

To create a number range, follow menu path **Cross Application Components • SAP Business Partner • Business Partner • Business Partner • Basic Settings • Number Ranges and Groupings • Define Number Ranges**. You can then select the **Define Number Ranges** link, click the pencil to change the intervals, and select the + button to create a new number range. Alternatively, you can also use Transaction SNRO (Number Range Object maintenance) with object BU_PARTNER, as shown in Figure 2.3.

The defined fields are as follows:

- **Number Range No.**
  Two-character number to define a number range.
- **From No.**
  Lower limit of the number range.
- **To Number**
  Higher limit of the number range.
- **NR Status**
  The number that was last used in the number range in the system.
- **External**
  Whether the number range is an internal or external interval. If the **External** check box is selected, then the number range is an external interval.

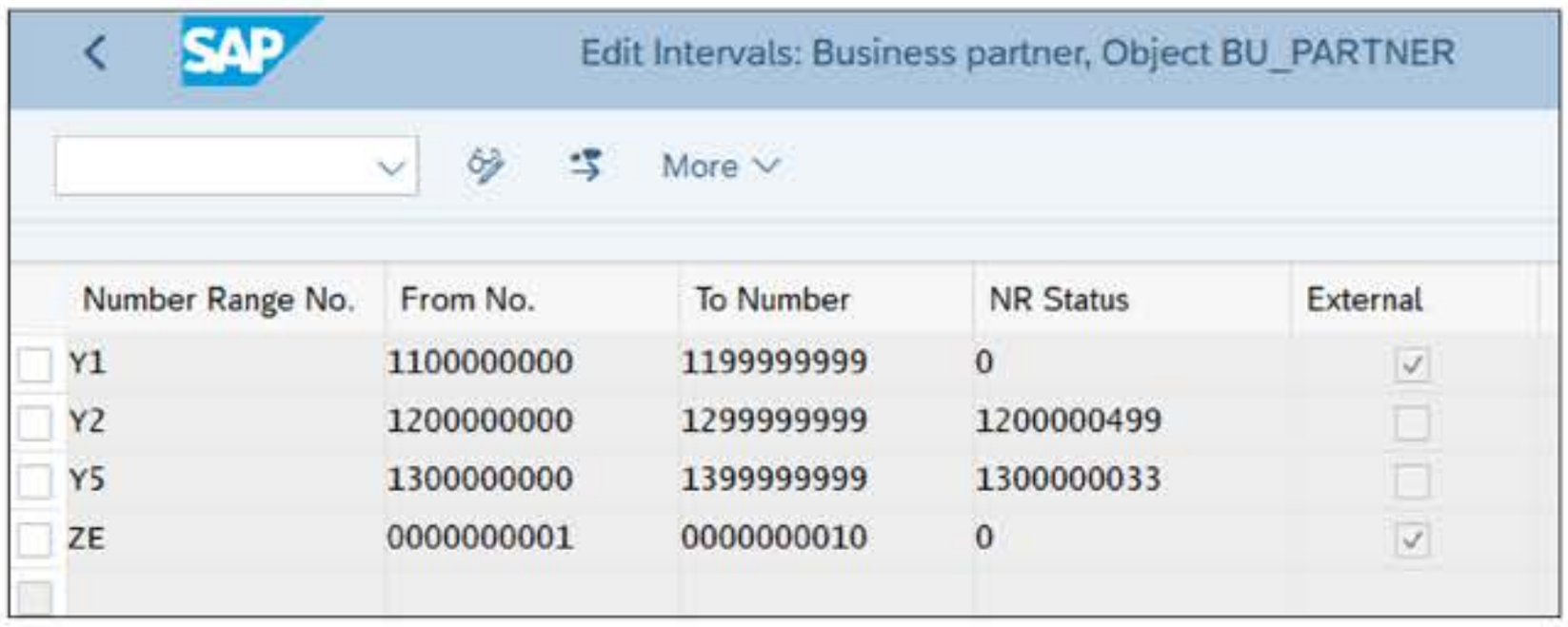

Figure 2.3 Define Business Partner Intervals

Next, you can define the groupings and assign a number range that was defined in the previous step. Grouping is a concept to enable you to organize different sets of business partners together and follow the same numbering scheme.

To define a grouping and assign a number range, follow menu path **Cross Application Components • SAP Business Partner • Business Partner • Business Partner • Basic Settings • Number Ranges and Groupings • Define Groupings and Assign Number Ranges**.

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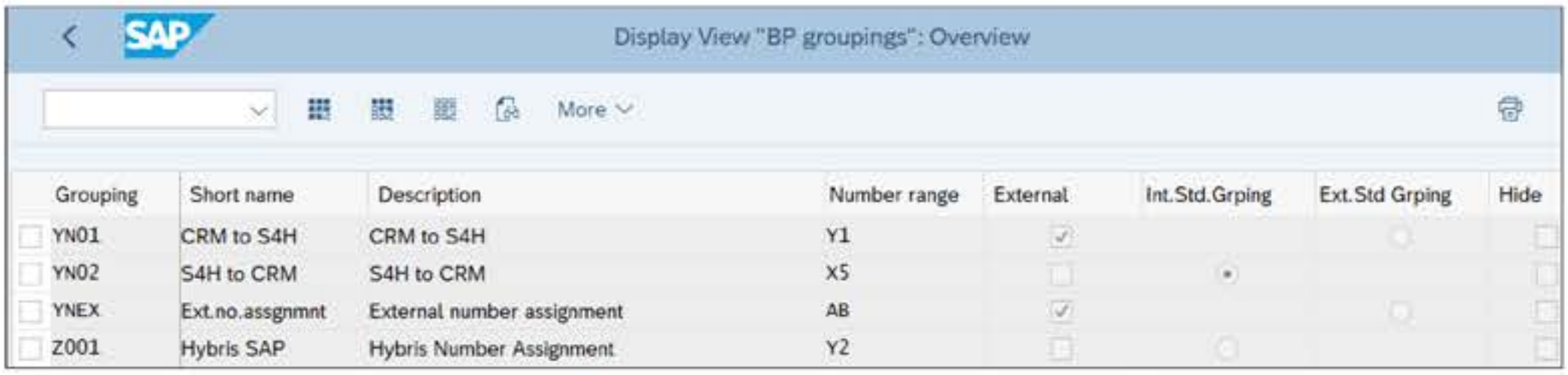
Define a new four-character **Grouping** with a **Short name** and **Description**. Assign the **Number range** defined for this grouping. The business partners created with this grouping will inherit the number range assigned during business partner creation. Figure 2.4 shows the groupings that are created and the number range assignment for the number range intervals defined previously in Figure 2.3.



Grouping	Short name	Description	Number range	External	Int.Std.Grping	Ext.Std Grping	Hide
YN01	CRM to S4H	CRM to S4H	Y1	✓			
YN02	S4H to CRM	S4H to CRM	X5		•		
YNEX	Ext.no.assgnmnt	External number assignment	AB	✓			
Z001	Hybris SAP	Hybris Number Assignment	Y2				

Figure 2.4 Define Business Partner Grouping and Assign Number Range

The defined fields are as follows:

- **Grouping**
  Defines a set of business partners with the same numbering scheme.
- **Short name**
  The grouping displayed during business partner creation.
- **Description**
  Long description of the grouping to provider more context.
- **Number range**
  Two-character number range interval defined in previous step and assigned to the corresponding group. Here, a specific number range can be assigned to many groups, which allows the sharing of the number range.
- **External**
  Indicates if the number assigned to the business partner will be internal or external. This is inherited from the number range.
- **Int.Std.Grping**
  Indicates that the internal number range will be used as the standard if no specific number range is used.
- **Ext.Std Grping**
  Indicates that the external number range will be used as the standard if no specific number range is used.

Next, customer account groups need to be defined. A *customer account* group is an SAP technical term for a classifying feature used to group together customer accounts. An account group is a required entry for organizational units, excluding persons.

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An account group determines the following:

- Necessary screens and fields for entering master data
- Required fields for each screen
- Number range assigned to each account group and how they are assigned internally or externally (like business partner number range assignment, a number range is assigned for customer accounts)
- Valid partner functions that will be populated for each account group

An account group should be defined for each type of customer that needs to be classified separately, for example, different sets of general ledger accounts and one-time accounts, or different sets of required fields.

To define an account group, follow menu path **Financial Accounting • Contract Accounts Receivable and Payable • Customer Accounts • Master Data • Preparation for Creating Customer Master Data • Define Account Groups with Screen Layouts (Customers)**. Several standard SAP account groups are already defined in the system (e.g., 0001 as sold-to-party). Here, you can also define the screen layout for each account group. The screen layout means you can indicate which fields are required, optional, displayed, or suppressed for each account group. It's recommended to define the screen layout at the customer account group level to provide the most flexibility, but you can also define the screen layout at the company code or as dependent on the transaction.

Next, you need to define the number ranges for the customer accounts. This is like the number ranges for the business partners. A number range can be defined to uniquely identify the different customer account groups. The number range intervals can be set to either internal or external. To define a number range, follow menu path **Financial Accounting • Contract Accounts Receivable and Payable • Customer Accounts • Master Data • Preparation for Creating Customer Master Data • Create Number Ranges for Customer Accounts**. Click **Change Intervals** and the **+** button to maintain entries.

Once the number ranges are defined, the unique intervals can be assigned to the customer account groups. Here, the number ranges can be shared across multiple different account groups. For example, there can be an account group for sold-to parties and bill-to parties where each have a different number range to easily identify the different customer groups, or they can have the same number range.

To assign a number range, follow menu path **Financial Accounting • Contract Accounts Receivable and Payable • Customer Accounts • Master Data • Preparation for Creating Customer Master Data • Assign Number Ranges to Customer Account Groups**. The account groups will automatically be populated, and the system will prompt you to assign a number range. The customers created with this grouping will inherit the number range assigned during creation. Figure 2.5 shows the accounts groups and the number range assignment for the intervals.

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Figure 2.5 Assigning Number Ranges to Customer Account Groups

At this point, you might be asking what the difference is between the business partner setup and the customer setup. The business partner is a single entity with multiple business partner roles such as the customer roles. The customer roles require an account with the attributes described earlier. However, now that the business partner groupings and number ranges are set up as well as the customer account groups and number ranges defined, you need to complete the master data synchronization configuration. This step ensures that the customer is updated as the same time the business partner is created. At this step, you can decide to use the same number range as the business partner that is currently being processed.

2.1.2 Contract Account

A *contract account* is another core master data entity in SAP Billing and Revenue Innovation Management in SAP S/4HANA that is used to generate invoicing at the contract account level; therefore, a contract account represents a billable account for which the posting data for contracts take place. Billing and payments require at least one contract account for a business partner, although a business partner can have multiple contract accounts.

The following attributes determine the number of contract accounts for a business partner:

- **Company code**
  A separate contract account by legal entity.
- **Group company code**
  The groups determine which company code is allowed to post to a contract account.
- **Payment terms**
  Terms used for payment, such as net 0, net 10, and so on.
- **Auto pay**
  This is controlled via the payment method information available at the contract account.

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- **Account determination ID**
  This attribute determines the general ledger account to which the transactions is to be posted.
- **Tolerance group**
  This attribute is used for defining tolerance values for payment differences.

With all the previously mentioned attributes being the same, if the contract with the customer requires different billable accounts (e.g., a company has four regions in the US: North, South, East, and West), the customer wants this to be managed via four distinct billable accounts and expects four bills. In this case, you'll need four contract accounts.

For a business partner to have a contract account created, the business partner must have the business partner role `MKK` (Contract Partner). Without this role, a contract account can't be created for the business partner. From the business partner view for the `MKK` contract partner role, you can see the contract account(s) linked to the business partner.

The following data can be defined at the contract account level.

- General data:
  - Cross partner data
  - Account management data
  - Invoicing settings
  - General remarks
- Payments and taxes:
  - Payment data
  - Taxes and tax exemption
- Dunning and correspondence:
  - Dunning control
  - Correspondence control
  - Correspondence to other partners

To configure a contract account, let's first define the contract account class that is required to process business relationships. A *contract account class* determines a collection of predefined values that can be set during replication as a contract account. A contract account has a limited number of fields compared to a business partner. The contract account class provides a template while creating a new contract account. The correct maintenance of internal and external number range assignments in the class helps to determine the correct contract account category during replication. This class is maintained to differentiate various types of business agreements/contract accounts for various business purposes, such as maintaining customer business agreements for

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incoming payments and vendor business agreements for outgoing payments. However, contract accounts can be used for both incoming and outgoing payments.

Next, you need to maintain the number range. The number range can be numerical or alphanumerical in nature and will be defined as internal or external. This setting is critical because the contract account will be used across both SAP Billing and Revenue Innovation Management, subscription order management, and SAP Convergent Invoicing. To create a number range, follow menu path **Financial Accounting • Contract Accounts Receivable and Payable • Basic Functions • Contract Accounts • Number Ranges and Contract Account Categories • Define Number Ranges**.

To define the contract account class, follow menu path **Service • Master Data • Contract Account • Define Contract Account Class**, and add a description (**Text** field) and country (Country/Region field). Define the internal and external number ranges, as shown in Figure 2.6.

Figure 2.6 Define Contract Account Class

The defined fields are as follows:

- **Contract Acct Class**
  Defines the name of the class.
- **Text**
  Description of the business agreement class.
- **Country/Region**
  Setting where you define the country for which this process is set up.
- **Int. No. Range**
  The internal number range assigned to this class.
- **Ext. No. Range**
  The external number range assigned to this class.
- **CA is usable for collective bills**
  Indicates that the definition of the class is set to a collective bill business agreement, which means the account can be assigned to an alternative account for collective bills.

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Next, you need to define the basic settings. In this activity, you define the form of open item accounting that is active in the system from the following: **FI-AR**, **FI-CA Active**, **Mixed**, or undefined. If **FI-CA Active** is selected, then the contract account is activated automatically. Because this example is using business agreements, **FI-CA Active** is selected.

These basic setting are defined via the following menu path: **Service** • **Contract Account** • **Define Basic Settings**. As shown in Figure 2.7, the contract account active indicator is determined.

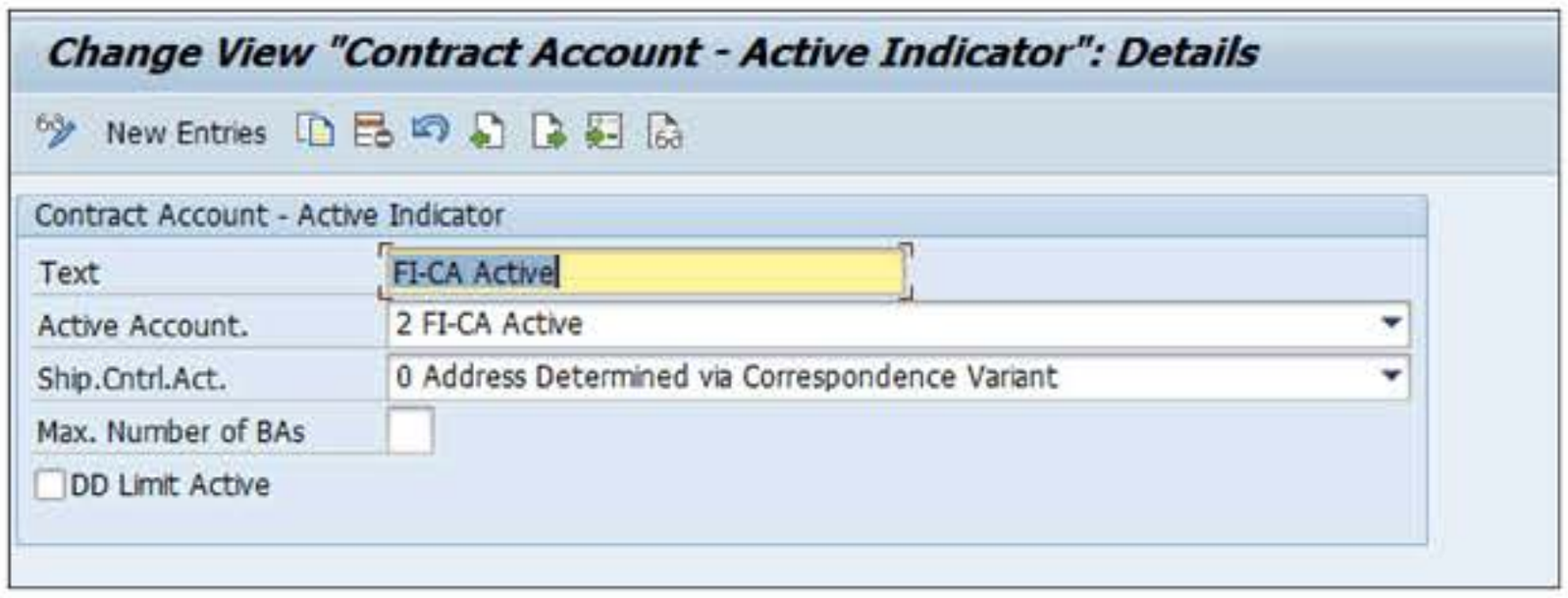

Figure 2.7 Contract Account Basic Settings: Active Indicator

The defined fields are as follows:

- **Text**
  Description of the contract account active indicator.
- **Active Account.**
  Dropdown values providing options for the method of open item accounting that is active in the system.
- **Ship.Cntrl.Act.**
  Dropdown values providing options for shipping address derivation.
- **Max. Number of BAs**
  Defines the maximum number of contract accounts displayed in the table.
- **DD Limit Active**
  Indicates that the direct debit limit is activated for a specific period.

Next, let's define the *correspondence variant*, which defines the types that contain the control parameters, such as execution intervals and the charges schema. The functions of the correspondence variant are primarily used in SAP Convergent Invoicing. You maintain the corresponding values in SAP Billing and Revenue Innovation Management, subscription order management, as defined in SAP Convergent Invoicing. You need to set this value so the replication of the contract accounts from SAP Convergent Invoicing won't fail.

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The correspondence variant can be set in the following menu path: **Service • Master Data • Contract Account • Define Parameters for Contract Account • Define Correspondence Variant**. Click the **Define Correspondence Variant** link, click the **New Entries** button, and create the entries, as shown in Figure 2.8.

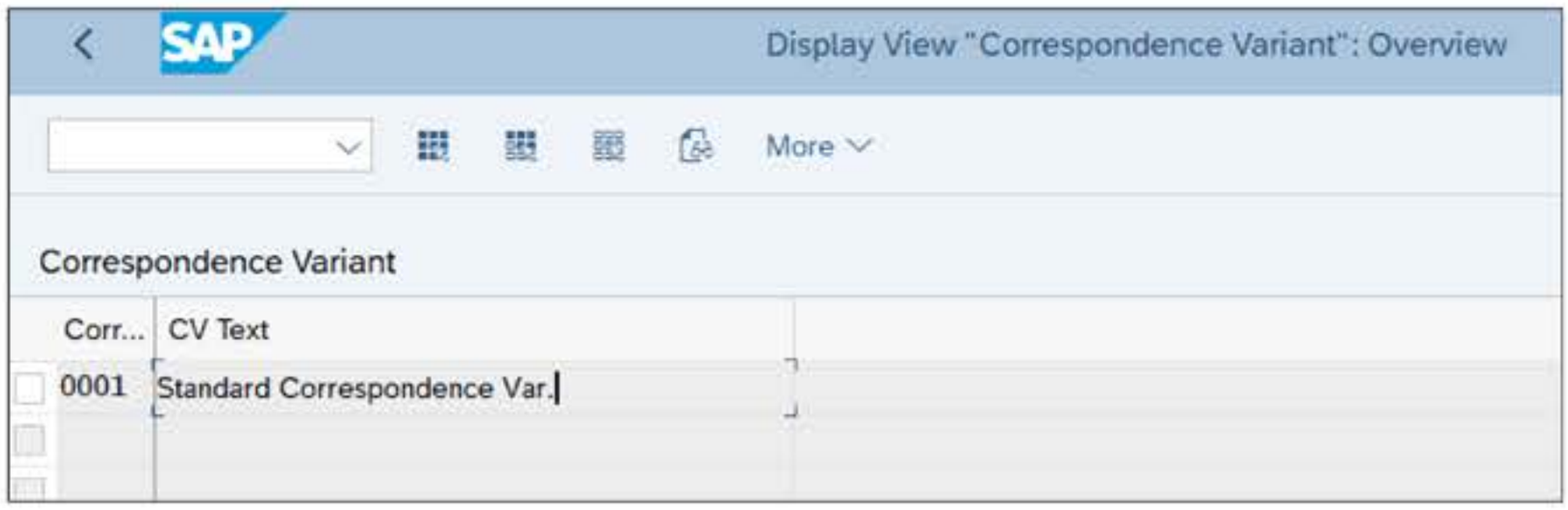

Figure 2.8 Maintain Correspondence Variant

The defined fields are as follows:

- **Correspondence Var**
  Defines the value for the correspondence variant.
- **CV Text**
  Description of correspondence variant.

Next, let's define the *payment method*, which is a value that defines the types of payment methods allowed for incoming and outgoing payments for each country. These values are used during the creation of business agreements and need to be aligned with corresponding values defined in SAP Convergent Invoicing.

Payment methods can be set in the following menu path: **Service • Master Data • Contract Account • Define Parameters for Contract Account • Define Payment Methods**. Click the **Define Payment Methods** link, and then click the **New Entries** button to create the entries shown in Figure 2.9.

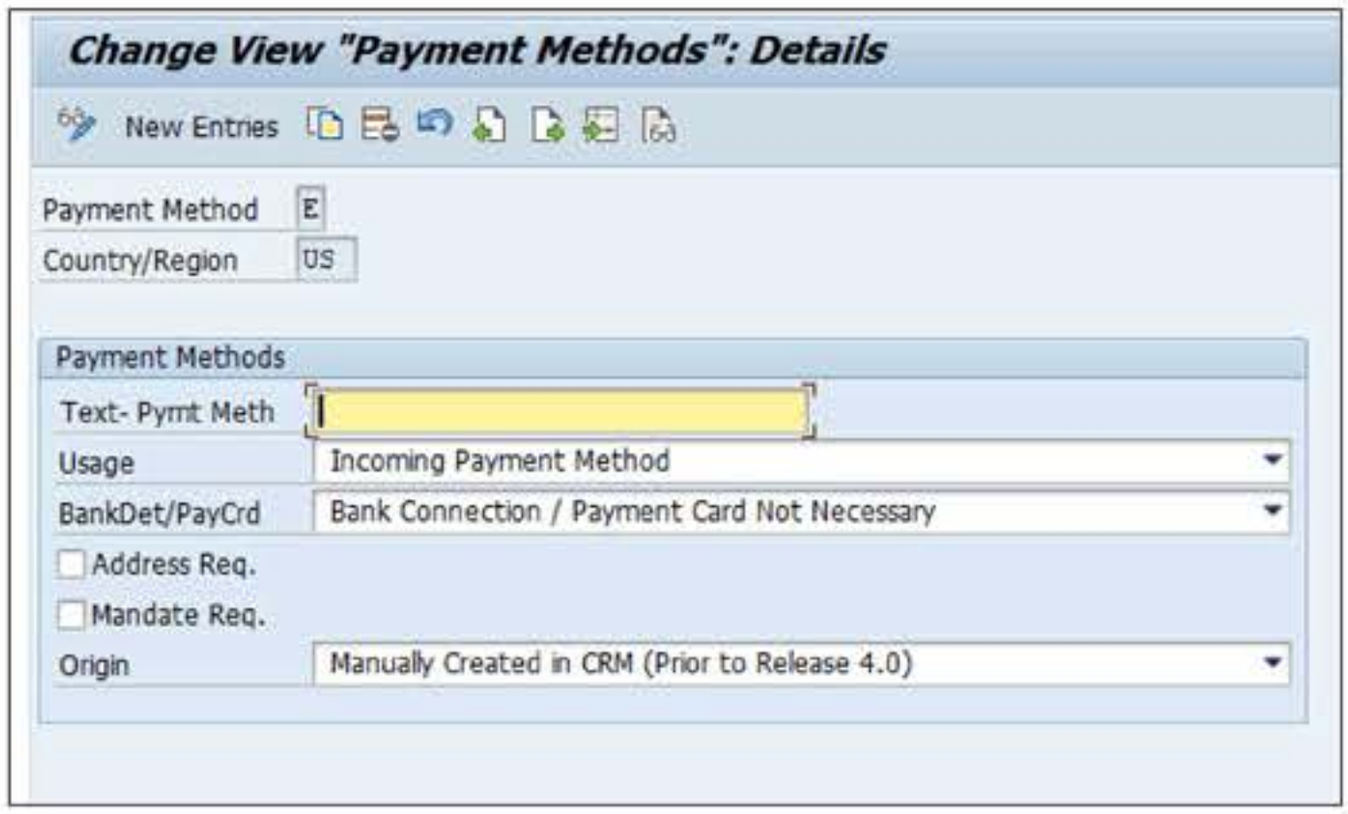

Figure 2.9 Payment Methods

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The defined fields are as follows:

- **Payment Method**
  Defines the key value for the payment method.
- **Country/Region**
  Country definition for which this value is valid.
- **Text-Pymt Meth**
  Description of the payment method.
- **Usage**
  Dropdown value defining the direction of the payment method (incoming or outgoing).
- **BankDet/PayCrd**
  Defines if the payment is via a bank or a credit card and if either is mandatory.
- **Address Req.**
  Mandates if the actual address needs to be present in the business partner master record to use the payment method.
- **Mandate Req.**
  Indicates if mandates are required (mandatory for direct debit).
- **Origin**
  Dropdown value determining the origination of the values (manually created in SAP CRM or replicated from SAP ERP).

Next, you can define the *shipping control*, which sets the basic options for shipping used for flexible control to send documents in different ways. This indicates information such as dispatch method (letter, fax, email), number of copies, archive method, and copy indicator. The shipping indictor can be set in the following menu path: **Service • Master Data • Contract Account • Define Parameters for Contract Account • Define Shipping Control**. Select **Define Shipping Control link**, and click **New Entries** to see the screen shown in Figure 2.10.

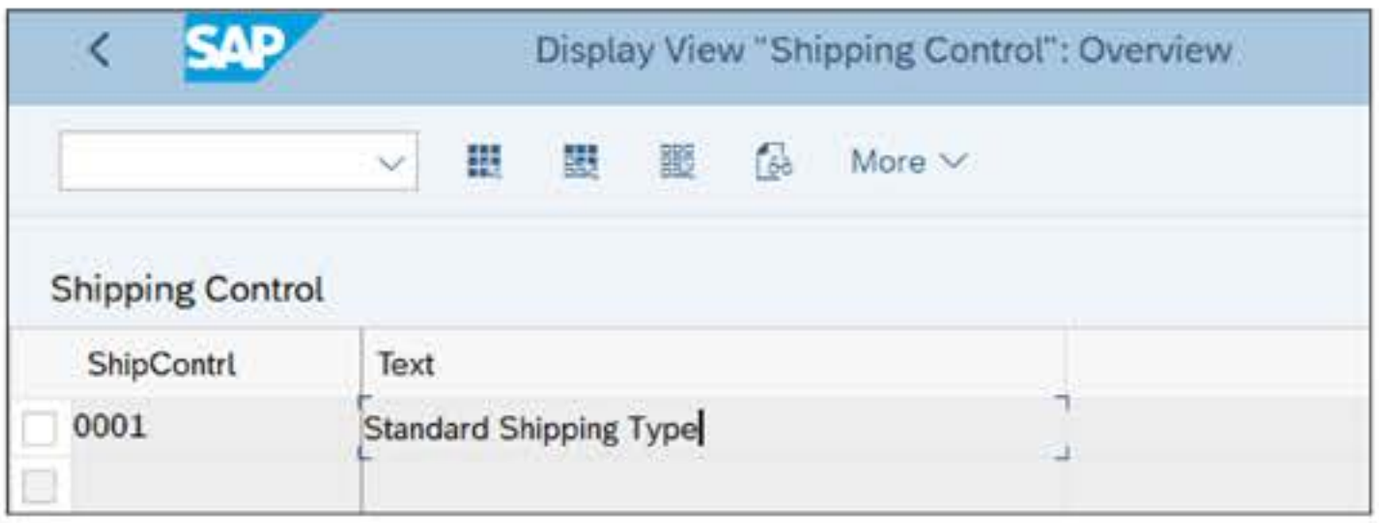

Figure 2.10 Define Shipping Control

The defined fields are as follows:

- **ShipContrl**
  Defines the value for the shipping control.

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- **Text**
  Description of the shipping control.

Next, you can define the *tax type*, which can be defaulted during contract account creation. The tax type will be used in determining the correct tax. Tax type can be set in the menu path **Service • Master Data • Contract Account • Define Parameters for Contract Account • Define Tax Types**. Select the **New Entries** button to maintain the entries shown in Figure 2.11.

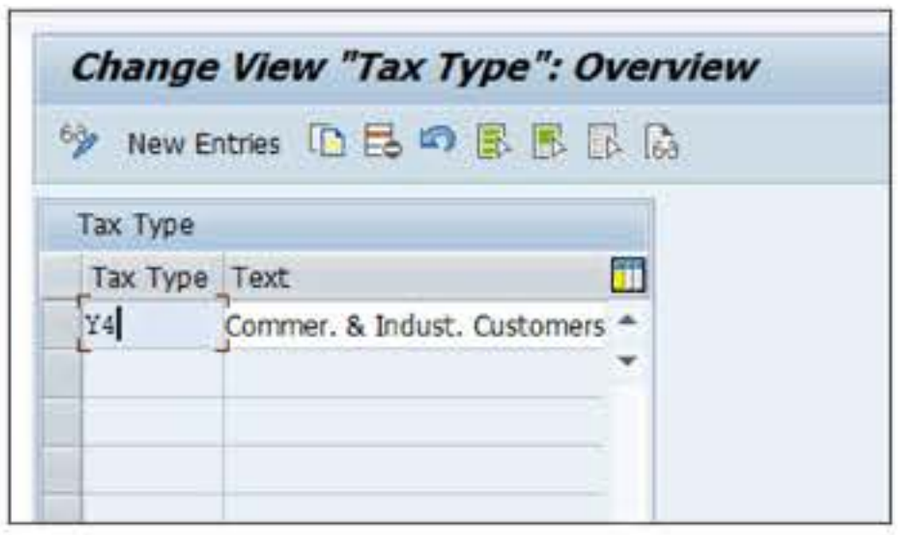

Figure 2.11 Maintaining the Tax Type

The defined fields are as follows:

- **Tax Type**
  Defines the value for the tax type.
- **Text**
  Description of the tax type.

You can also define the tax characteristics for the contract accounts, including control characteristics, which means the account determination ID for the contract account can be derived from the tax characteristics and contract account class. The tax characteristic can be used to group the various groups of customers based on tax requirements. Tax characteristics can be set in the following menu path: **Service • Master Data • Contract Account • Define Parameters for Contract Account • Define Tax Features**. Click **New Entries** to maintain the entries shown in Figure 2.12.

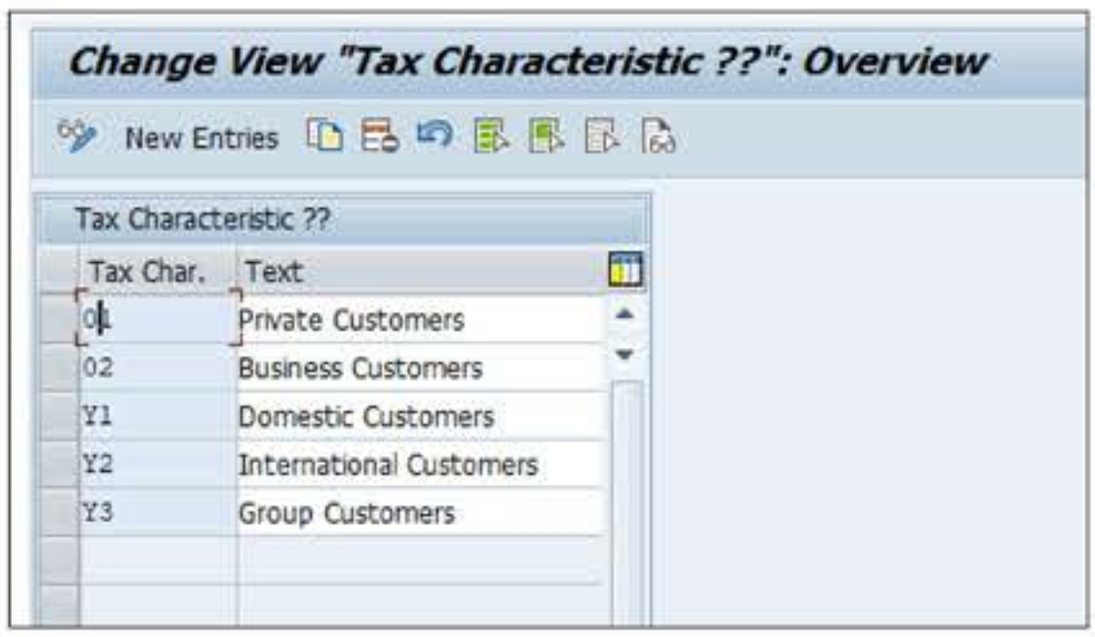

Figure 2.12 Tax Characteristic

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The defined fields are as follows:

- **Tax Char.**
  Defines the value for the tax characteristic.
- **Text**
  Description of the tax characteristic.

Now that the contract account parameters are defined, they can be assigned to the contract account class so that these values are defaulted when the contract account is created with the corresponding class. You can assign the correspondence variant and shipping control to the class defined. To assign the general parameters for a contract account class, use menu path **Service • Master Data • Contract Account • Creation of Contract Account • Allocate General Parameters to New Contract Accounts**. Click the **Define Allocate General Parameters to New Contract Accounts** link, click the **New Entries** button, and create the entries shown in Figure 2.13.

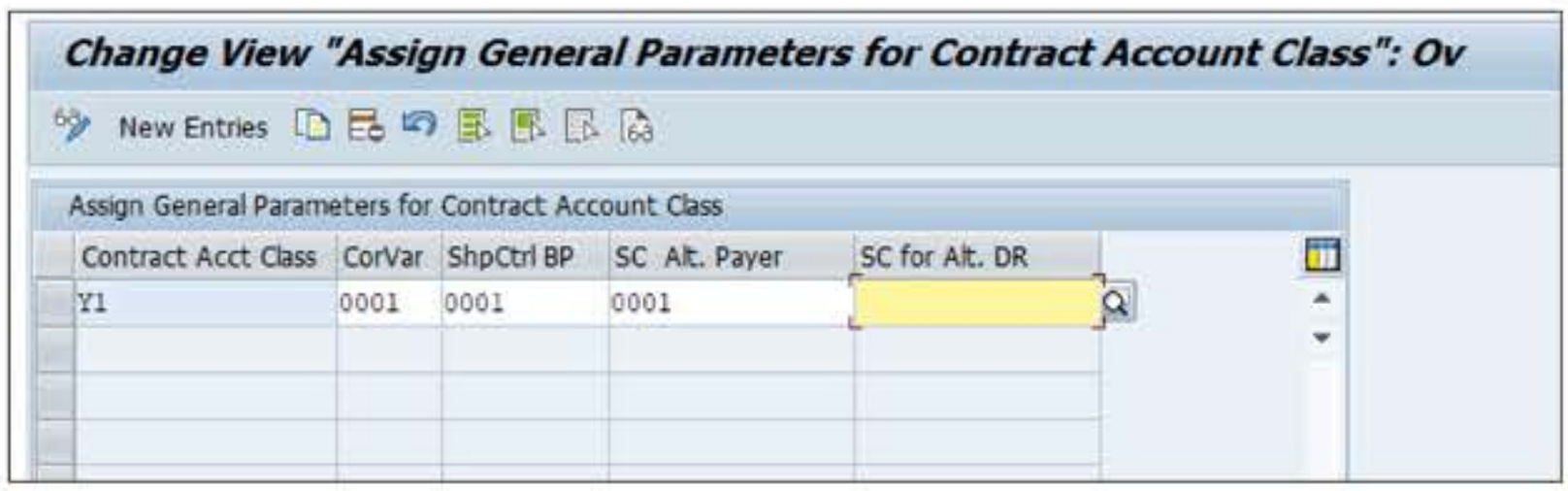

Figure 2.13 Assign General Parameters for Contract Account Class

The defined fields are as follows:

- **Contract Acct Class**
  Select the contract account class to assign general parameters.
- **CorVar**
  Assign the correspondence variant to be determined based on the selected contract account class for the business partner.
- **ShpCtrl BP**
  Assign the shipping control to be determined based on the selected contract account class for the business partner.
- **SC Alt. Payer**
  Assign the shipping control to be determined based on the selected contract account class for the alternative payer.
- **SC for Alt. DR**
  Assign the shipping control to be determined based on the selected contract account class for the alternative dunning recipient.

Lastly, you can define the *usage type* for a contract account, which determines whether an item uses standard payment data or alternative payment data. When a provider

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order is saved, based on these settings, the system will either determine the payment data for the ordered items or have the payment data valid for certain items only. The contract account class specifies the values for the contract account to be created with the **Item Uses Standard Payment Data** or **Item Uses Alternative Payment Data** option. The alternative payment data usage type is only relevant when alternative payment data is entered during order entry.

The usage type can be set in menu path **Service • Master Data • Contract Account • Creation of Contract Account • Allocate Contract Account Class** to maintain the entries shown in Figure 2.14.

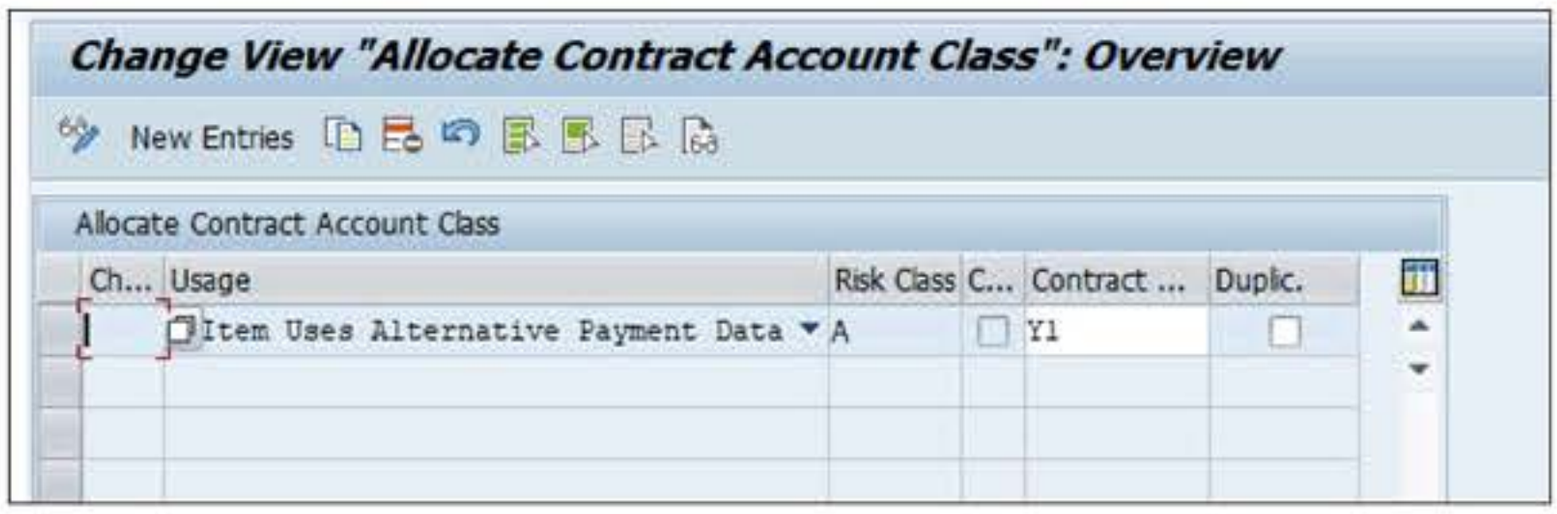

Figure 2.14 Allocate Contract Account Class to Define Usage Type

The defined fields are as follows:

- **Channel**
  Select the sales channel used in company.
- **Usage**
  Dropdown value defining the usage type for the contract account class.
- **Risk Class**
  Define the risk class used by the business partner to determine the payment-related business parameters of the contract account.
- **Contract Account Class**
  Define the contract account class for which the payment-related parameters are defined.
- **Duplicate**
  Check whether a business partner with identical values already exists to prevent a duplicate contract account from being created.

2.1.3 Subscription Materials

The material master contains all the necessary information about the service being sold in the transaction to enable the subscription selling process. It's still modeled as a traditional material, using the material master concepts from materials management. Products/materials can be tangible, such as hardware goods, or intangible, such as services or software goods. Subscription orders can handle both hardware and software

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goods for sales and service. In the context of this book, we'll focus on software goods, which will be modeled as services. Service goods aren't deliverable, so they don't require delivery processing to complete billing. The typical service process requires service confirmation to trigger the billing process. In SAP Billing and Revenue Innovation Management, subscription order management, no confirmation is involved. Although service goods are used, they are only to be offered as a subscription service to a customer to fulfill the sales order process in SAP Billing and Revenue Innovation Management, subscription order management.

The product master (aka material master) provides you with all information for the products your company purchases or sells. It's the central source from which product-specific data can be retrieved. For subscription products, you can manage the terms of duration, billing cycles, discounts, and charges, among other core attributes. This section will cover the features available as part of subscription-specific products, as well as provide an overview of bundle and sharing products, product replication, and so on.

Subscription-Specific Product Data

Subscription products need to be classified according to business purposes. This supports the use of the products in a subscription order, master agreement, or partner agreement order. The system validates the purpose of the product in the transaction in which it's used. For example, a standard product can't be used in a partner agreement order, and a sharing product can't be used in a subscription order or master agreement. Because transaction types are designed to drive different purposes, the product purpose needs to be aligned with those purposes, as follows:

- **Standard products**
  Used in subscription orders and subscription contracts. If an explicit definition isn't made, then the products take the standard product as the default assignment. However, this can't be used in partner agreements, sharing orders, or sharing purposes.
- **Partner agreement products**
  Used in partner agreements and partner agreement orders, which are associated with vendors or partners. These can't be used in other scenarios.
- **Master agreement products**
  Used in master agreements or in subscription orders and subscription contracts. They can be used as long as the subscription order is created from a master agreement or references the master agreement.
- **Sharing products**
  Used in sharing orders or sharing contracts. These can't be used in master agreements.

Settings for subscription product purposes can be made in the following menu path: **SAP Customizing Implementation Guide • Service • Master Data • Settings for Subscription Master Data • Subscription Product-Specific Data**. Click the **Assign Product Purpose to Product Types** link, click the **New Entries** button, and create the entries as shown in Figure 2.15.

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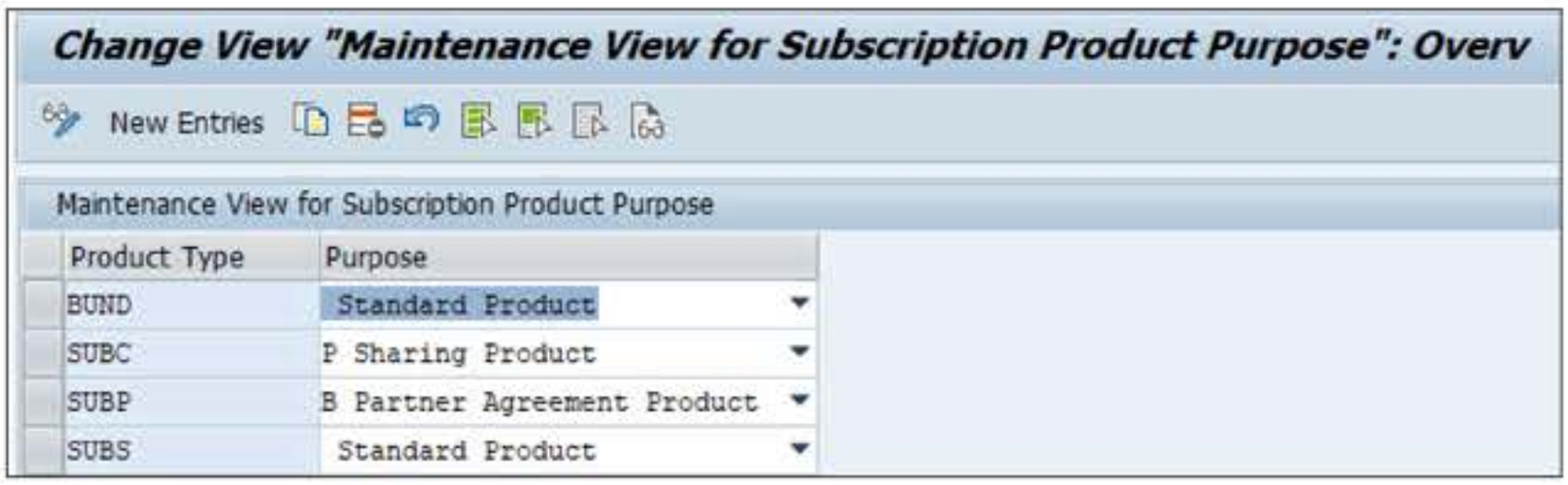

Figure 2.15 Setting for Product Purpose

The defined fields are as follows:

- **Product Type**
  Dropdown list of the created product types.
- **Purpose**
  Dropdown list of predefined values defining the product purpose.

Bundle Products

Products can be combined and sold together as an offering (e.g., subscription software can be sold with support), or they can be offered as an option. For our Martex Inc. use case, let's assume that a new bundle product called Martex Premium Software Package is offered that includes Martex Workplace Solution and Martex Activation Charge. Bundle products can be maintained by using the SAP Fiori app called Manage Product Bundles. The products to be bundled should already be created and should exist as individual products. The product bundle needs to have a Product Bundle ID, and then the individual products are added to the bundle.

Click on the **New** button, as shown in Figure 2.16, to create a new product bundle.

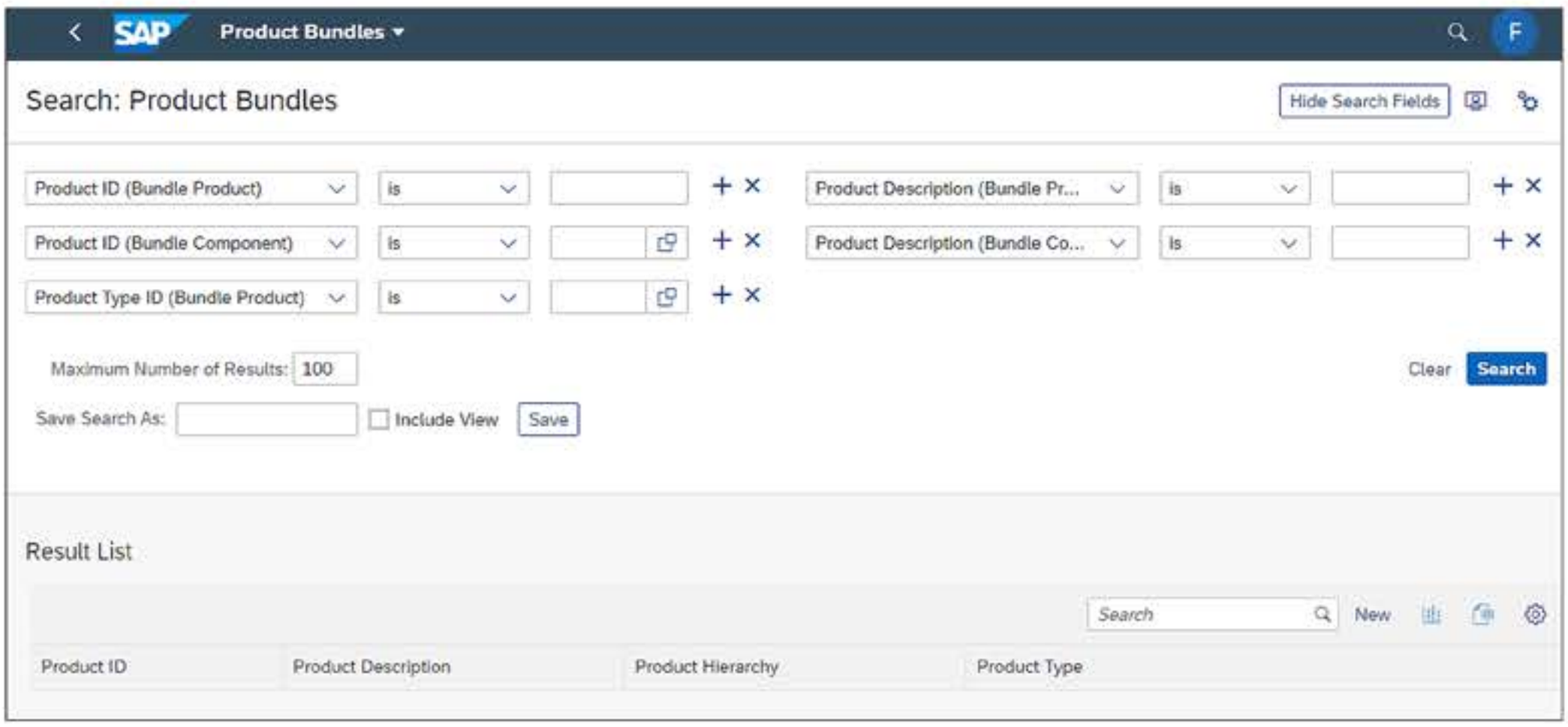

Figure 2.16 Creating New Product Bundles

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Click on the **Add Product** button, as shown in Figure 2.17, to add products to the bundle.

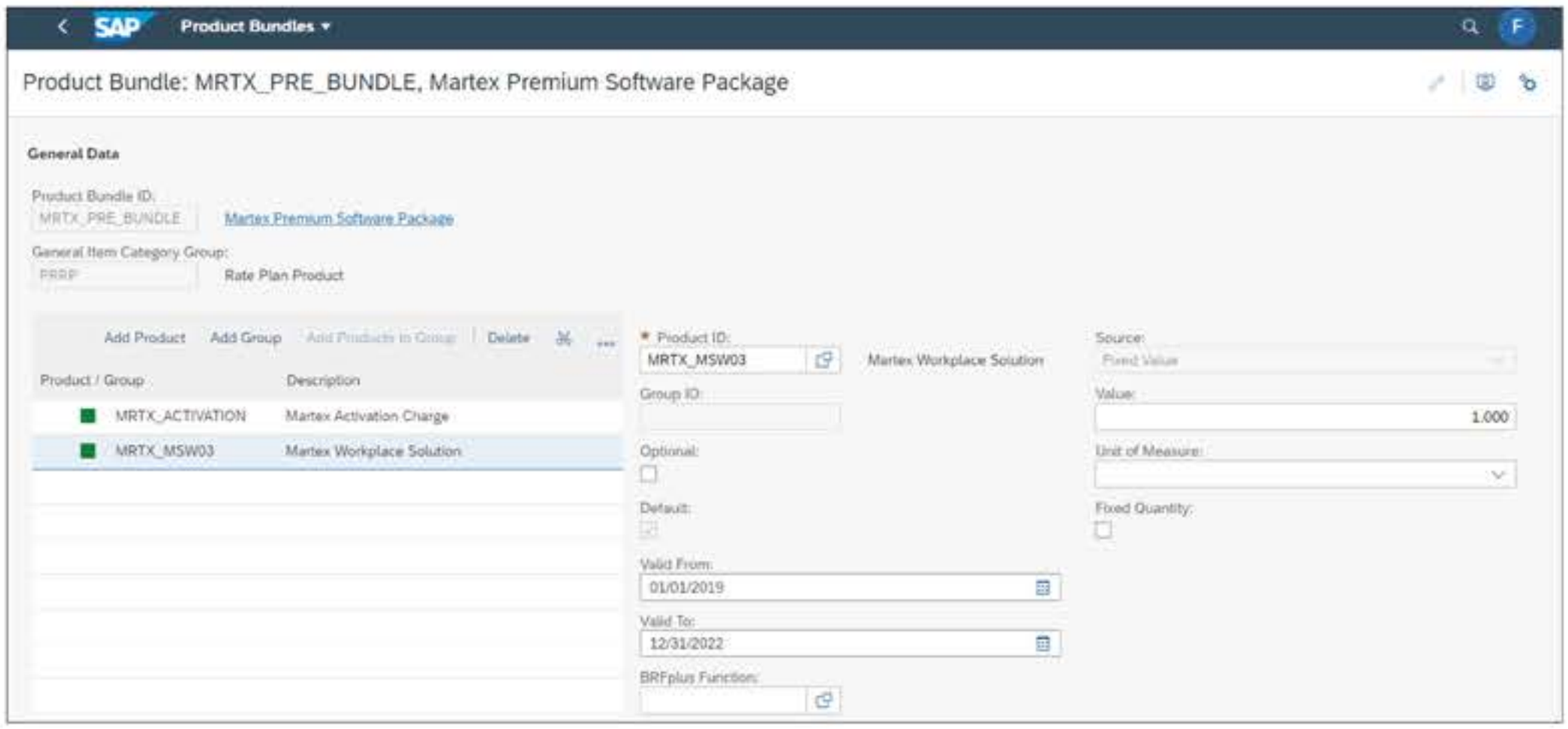

Figure 2.17 Maintaining Bundle Products

If the products always need to be sold together, select the **Default** checkbox. If it's an optional component of the bundle, select the **Optional** checkbox.

Before maintaining product bundles, you need to define products belonging to which product types are allowed to be used as product bundles. This is done in the Customizing activity **Define Product Types for Bundle Products** in the following menu path: **SAP Customizing Implementation Guide • Service • Master Data • Products • Product Bundles**. Click **Define Product Types for Bundle Products**, click the **New Entries** button, and create the entries, as shown in Figure 2.18.

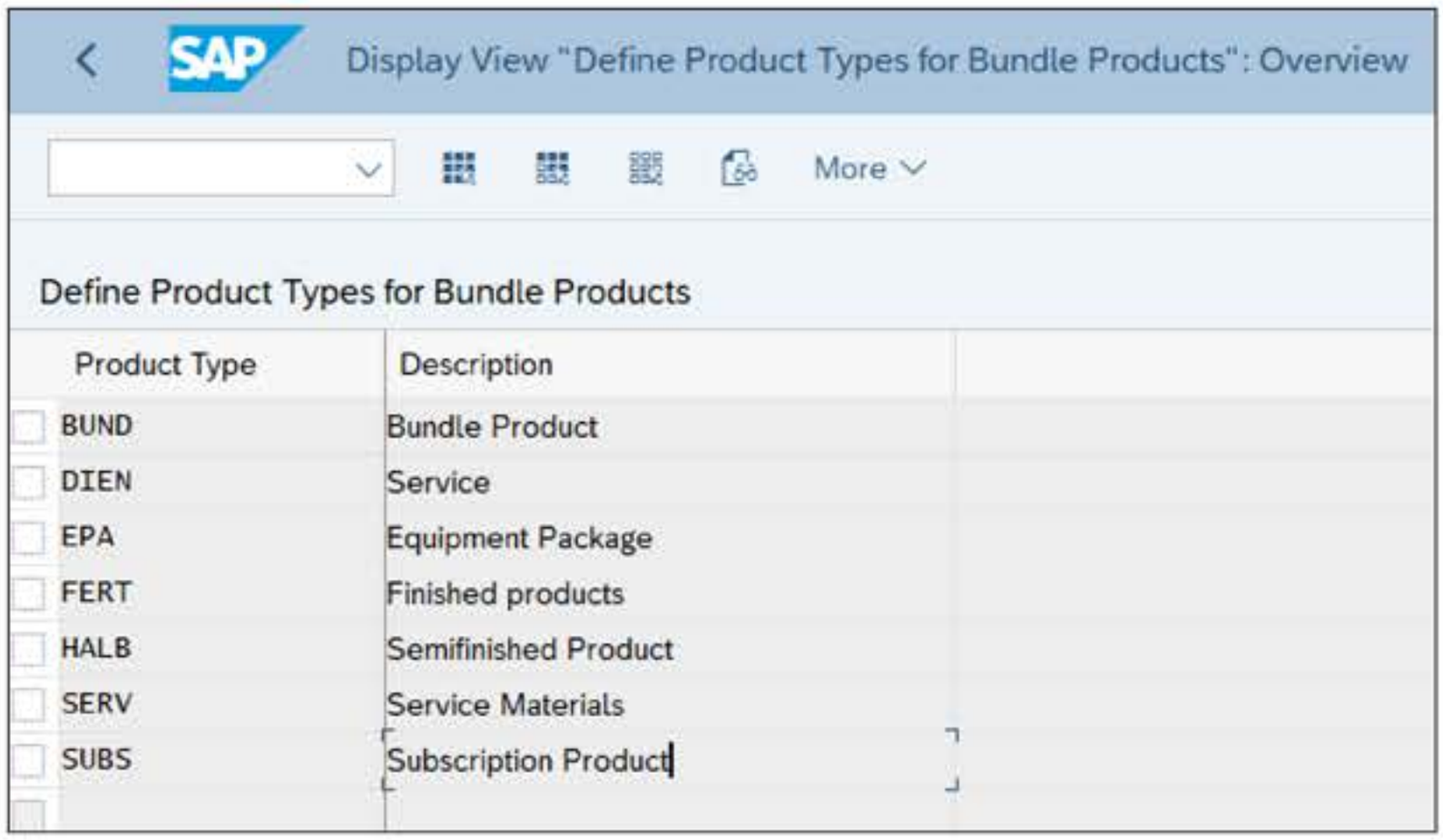

Figure 2.18 Define Product Types for Bundle Products

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The defined fields are as follows:

- **Product Type**
  Dropdown list of the created product types.
- **Description**
  Description of product types.

Sharing Products

SAP Billing and Revenue Innovation Management, subscription order management, offers a standard process to handle usage sharing scenarios by using sharing products. For example, let's assume that Martex Corp. has a customer called Active Solution Limited with two subsidiaries. Based on the agreement with Active Solution Limited, Martex will offer storage space of 800 GB each month. Now, in each scenario, if Active Solution Limited looks for flexibility to allow its subsidiaries to also share the storage space, the sharing concept can be used to manage the process. Sharing is a process that allows you to share counters or allowances during the rating process in SAP Convergent Charging within one or multiple contracts. To allow those counters or allowances are shared across contracts during rating in SAP Convergent Charging you must create a sharing contract (i.e., the root contract) and shared contracts (i.e., dependent contracts).

In a sharing contract, you can only use a subscription product with the product purpose sharing product. In a shared contract, you could assign the reference to a sharing contract only if the product used in the shared contract and the product used in the sharing contract are allowed for sharing. Sharing products are defined as fees for the usage of counter sharing over several subscription orders with or without reference to a single master agreement. Sharing products are also used in sharing orders or sharing contracts. SAP provides a standard material type SUBC (Subscription Sharing Product) to manage sharing scenarios.

In the Manage Subscription Product-Specific Data app, you define which products can be shared under **Shared Products**, as shown in Figure 2.19. Alternatively, to define which products can be shared, you can use the Business Add-In (BAdI) `CRM_PROVIDER_DET_SHARING_PROD`.

Different types of sharing options are allowed, as follows:

- Share counters and allowances across sharing contracts and their shared contracts
- Share allowances between different contracts
- Share allowances and counters within one contract.

Note

Please note that sharing contracts and shared allowances can't be used together. In addition, products that support shared allowances can't be used in a sharing contract or a shared contract or vice versa.

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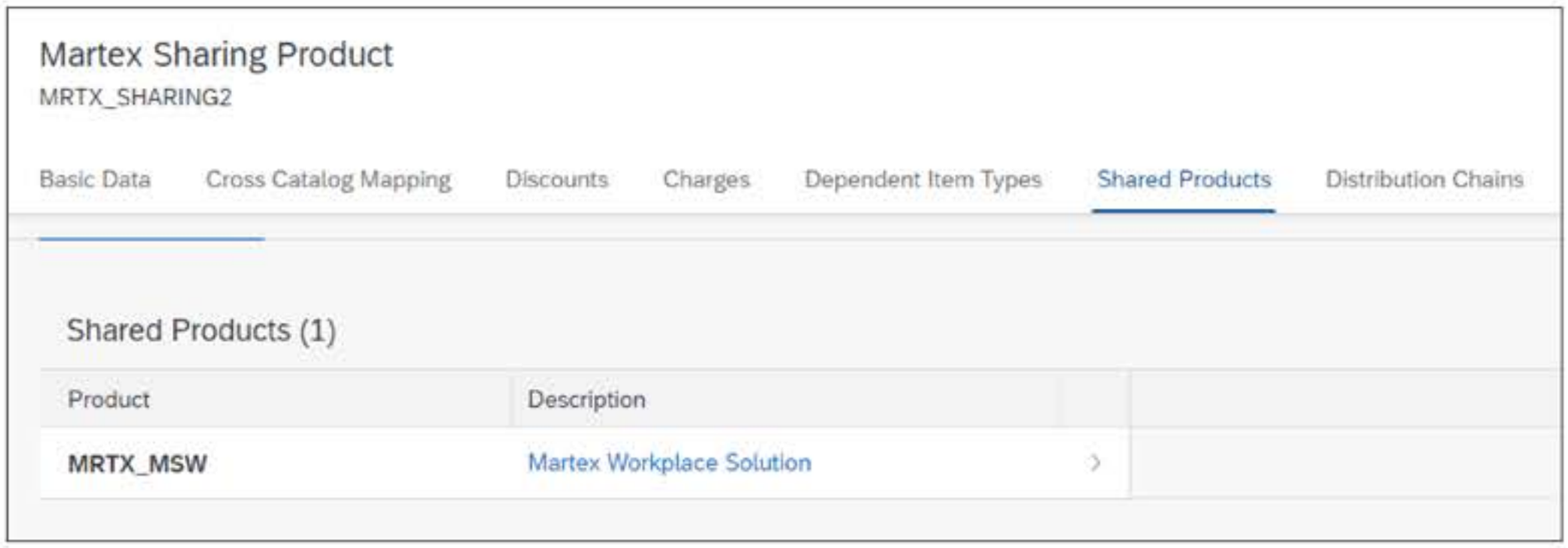

Figure 2.19 Sharing Products in SAP Billing and Revenue Innovation Management, Subscription Order Management

Product Replication Using the Data Replication Framework

Subscription product specific data can be replicated to external systems using the Data Replication Framework (DRF). The subscription product itself must exist in the target system before the data specific to the subscription product can be replicated.

The communication channel specifies how replication can be done. The standard communication channels available are services, IDocs, files, and remote function calls (RFCs).

Replication models determine which outbound implementations should be used for which target systems. To replicate data specific to a subscription product, the standard replication model and outbound implementation is `SOM_PROD`. The business object to be used in the outbound implementation is `DRF_0073`.

To configure the DRF and replication model, use Transaction DRFIMG.

2.2 Organizational Units

In this section, we'll discuss the essential activity of setting up your organizational units to support your company's structure and business functions. An organizational model is composed of the organizational units defined in the service configuration and the sales and distribution configuration. The organizational model is at the forefront of your master data setup, transactions, reporting, and finance structure.

Organizational units, such as departments, groups, or teams, define the various functions of a company and are linked as a hierarchy to display how each unit fits together. This hierarchy then creates the organizational model of your company.

There are several options to link your organizational units in SAP Billing and Revenue Innovation Management:

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- The organizational units—such as sales organization, service organization, and sales office—aren't already defined and can be defined in configuration. Not all levels are required, and new levels can be created.
- Organizational units are defined in SAP S/4HANA Sales. For example, sales organization, distribution channel, and division can be assigned to the SAP Billing and Revenue Innovation Management organizational units in SAP S/4HANA, which is required for automatically determining organizational data in transactions.
- An organization can be activated for several scenarios, enabling it to be a sales organization and a service organization at the same time.
- The organizational model is time-dependent, and validity dates can be set to allow organizational changes in the future.
- A business partner can be automatically created for an organizational unit with the organizational unit role.

The organization model in SAP Billing and Revenue Innovation Management in SAP S/4HANA is based on the organizational management set up in the sales area of SAP S/4HANA. The organization data from SAP S/4HANA Sales such as sales office, sales area, distribution channel, and so on can be copied to SAP Billing and Revenue Innovation Management.

The data can be transferred to create the organizational units in SAP Billing and Revenue Innovation Management from SAP S/4HANA by running the program in Customizing. The data transfer is helpful if you already have a system with the organizational model set up in sales, or you can manually set up the organizational model.

To generate the organizational units in SAP Billing and Revenue Innovation Management, you can access the program in menu path **Service • Master Data • Organizational Management • Data Transfer • Copy SD Sales Structure**. Click the **Copy SD Sales Structure** link, select the row of organizational units from sales and distribution to copy to the service organizational units, and then select **Generate**. The service organizational units are created as shown in Figure 2.20 and Figure 2.21.

Figure 2.20 Sales and Distribution: Enterprise/Organizational Structures

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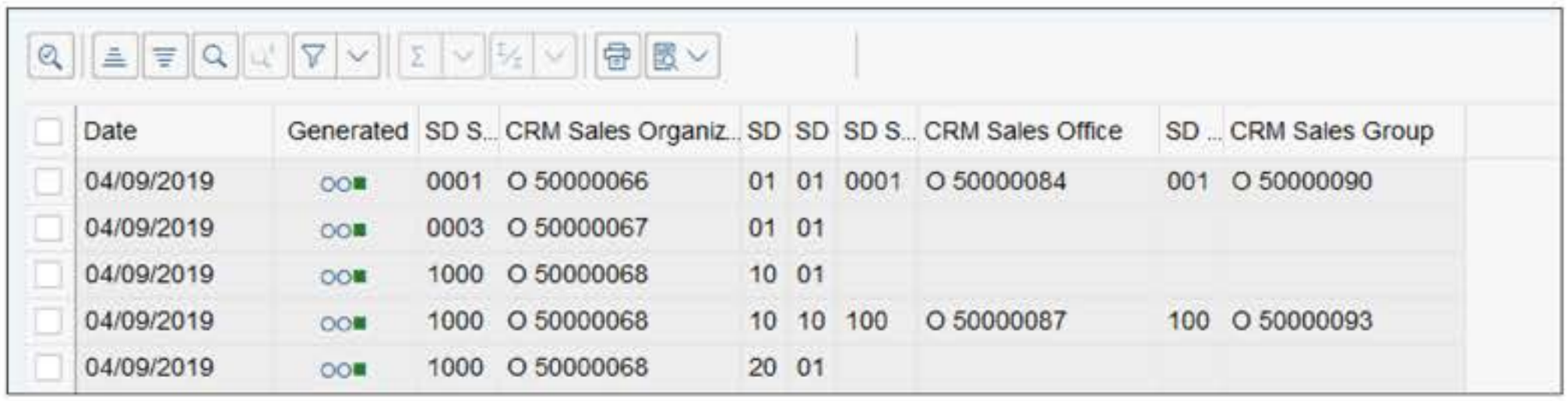

Date	Generated	SD S...	CRM Sales Organiz...	SD	SD	SD S...	CRM Sales Office	SD ...	CRM Sales Group
04/09/2019		0001	O 50000066	01	01	0001	O 50000084	001	O 50000090
04/09/2019		0003	O 50000067	01	01				
04/09/2019		1000	O 50000068	10	01				
04/09/2019		1000	O 50000068	10	10	100	O 50000087	100	O 50000093
04/09/2019		1000	O 50000068	20	01				

Figure 2.21 Generation of Service Organization Using Data Transfer

As shown in Table 2.1, organizational units are available in SAP S/4HANA for you to enter the corresponding SAP Billing and Revenue Innovation Management organizational unit.

Scenario	SAP S/4HANA Organizational Attributes	SAP Billing and Revenue Innovation Management Organizational Unit
Sales	Sales Organization	Sales Organization
Sales	Sales Office	Sales Office
Sales	Sales Group	Sales Group
Service	Planning Plant	Service Organization

Table 2.1 SAP S/4HANA Organizational Attribute Mapped to SAP Billing and Revenue Innovation Management Organizational Unit

The SAP S/4HANA organizational attributes can be mapped to the organizational units in SAP Billing and Revenue Innovation Management. You can map more than one attribute to an organizational unit. For example, you could create an organizational unit that represents both the sales office and the sales group. Additionally, you can create additional organizational units, such as a regional branch, but only the organizational units shown in Table 2.1 can have a counterpart in SAP S/4HANA. This can be defined in customizing.

First, you can assign the sales organization in SAP Billing and Revenue Innovation Management to the sales and distribution sales organization. This will link the sales organizations so that the sales organization can be automatically displayed on the service transactions.

Assignments of sales organizations the **Service** node can be set in menu path **Service • Master Data • Organizational Management • Assignment of Organizational Units to SD • Assign Sales Organizations in Service to SD Sales Organizations**. Click the **Assign Sales Organizations in Service to SD Sales Organizations** link, click **New Entries**, and create the entries as shown in Figure 2.22. For our example, you make entries for **Sales Org. 1000**.

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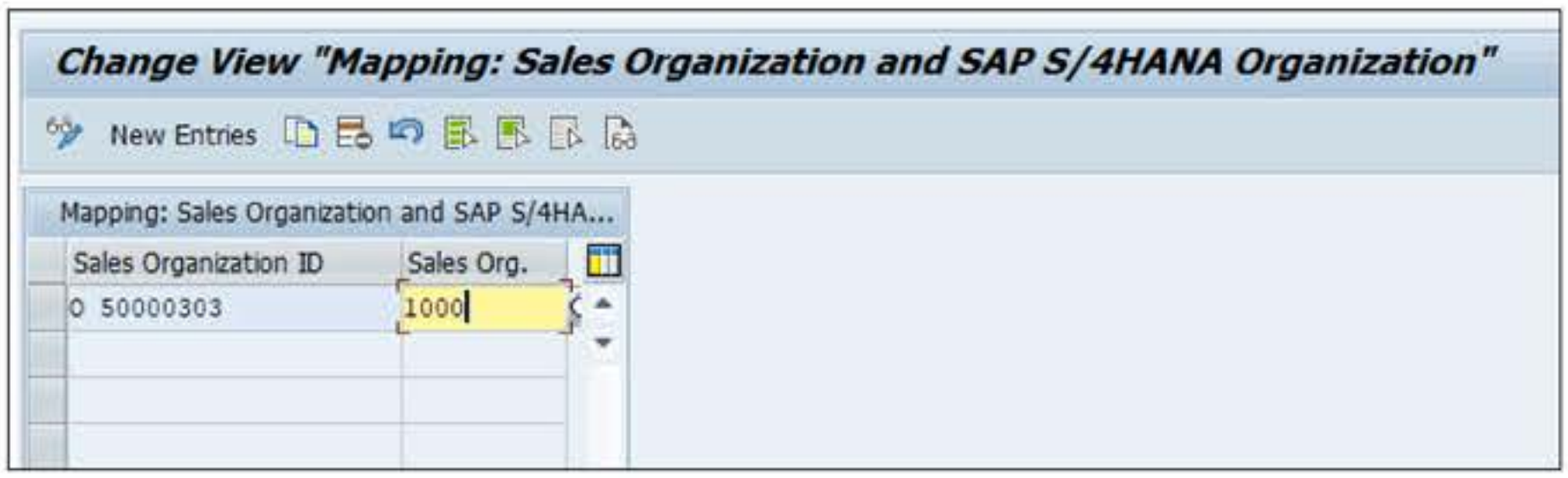

Figure 2.22 Mapping Service Sales Organization and Sales and Distribution Sales Organization

You can assign the sales office in SAP Billing and Revenue Innovation Management to the sales and distribution sales office. This will link the sales offices for population on the service transactions.

Assignments of sales organizations in the **Service** node can be set in menu path **Service • Master Data • Organizational Management • Assignment of Organizational Units to SD • Assign Sales Offices in Service to SD Sales Offices**. Click the **Assign Sales Offices in Service to SD Sales Offices** link, click **New Entries**, and create the entries as shown in Figure 2.23. For our example, you make entries for **Sls Office 100**.

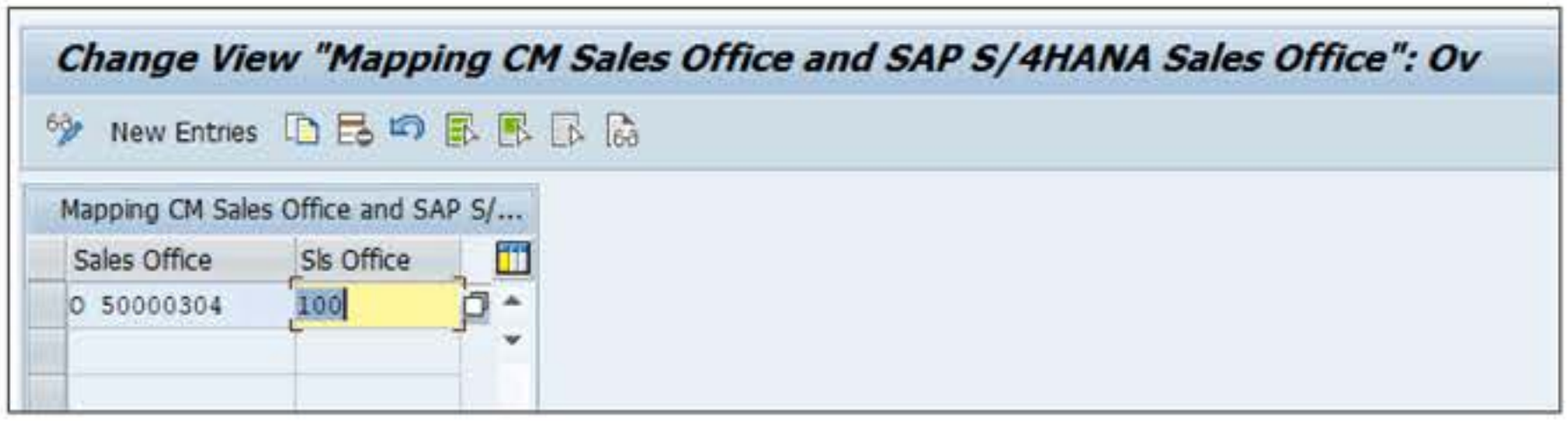

Figure 2.23 Mapping Service Sales Office and Sales and Distribution Sales Office

You can assign the sales group in SAP Billing and Revenue Innovation Management to the sales and distribution sales group. This will link the sales groups for population on the service transactions.

Assignments of sales organizations in the **Service** node can be set in menu path **Service • Master Data • Organizational Management • Assignment of Organizational Units to SD • Assign Sales Group in Service to SD Sales Group**. Click the **Assign Sales Group in Service to SD Sales Group** link, click **New Entries**, and create the entries as shown in Figure 2.24. For our example, you make entries for **SD Group 100**.

Additionally, you can assign the plant and storage location to the service organizational units. This assignment is used to determine the corresponding plant and storage locations for items on service transactions.

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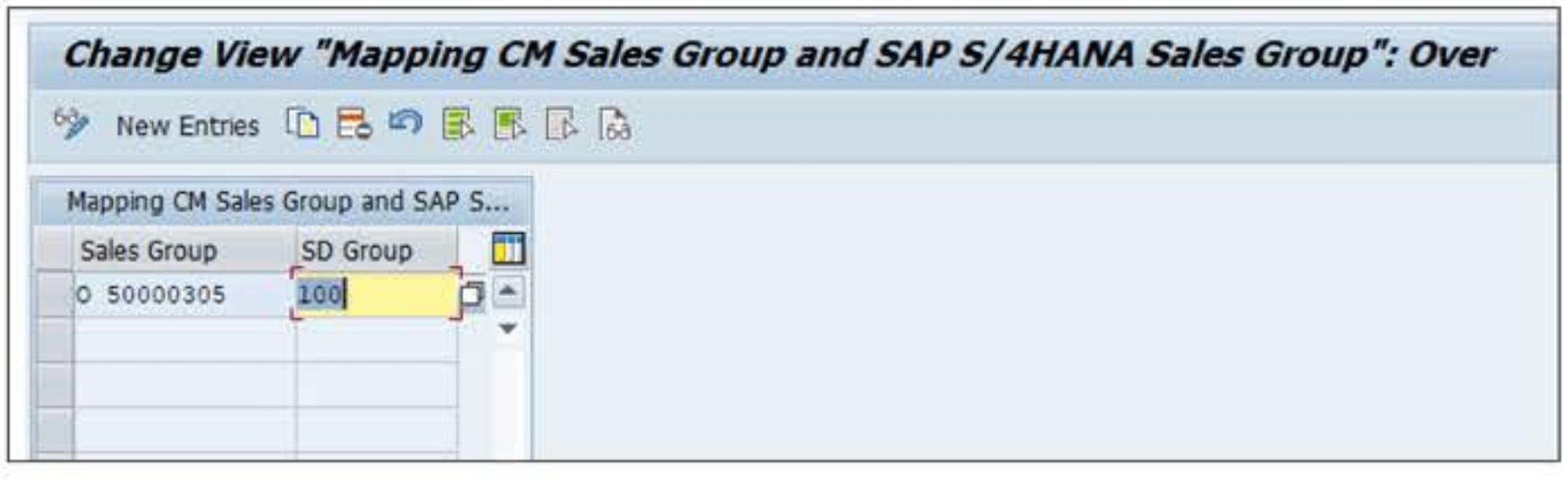

Figure 2.24 Mapping Service Sales Group and Sales and Distribution Sales Group

For assignments of sales organizations, the **Service** node can be set in menu path **Service • Master Data • Organizational Management • Cross-Application Assignment of Organizational Units • Assign Plant and Storage Location to Service Organizational Units**. Click the **Assign Plant and Storage Location to Service Organizational Units** link, click **New Entries**, and create the entries as shown in Figure 2.25. For our example, you make entries for the **Service Organization 1000**, **Plnt 3001**, and **SLoc** (storage location) **0002**.

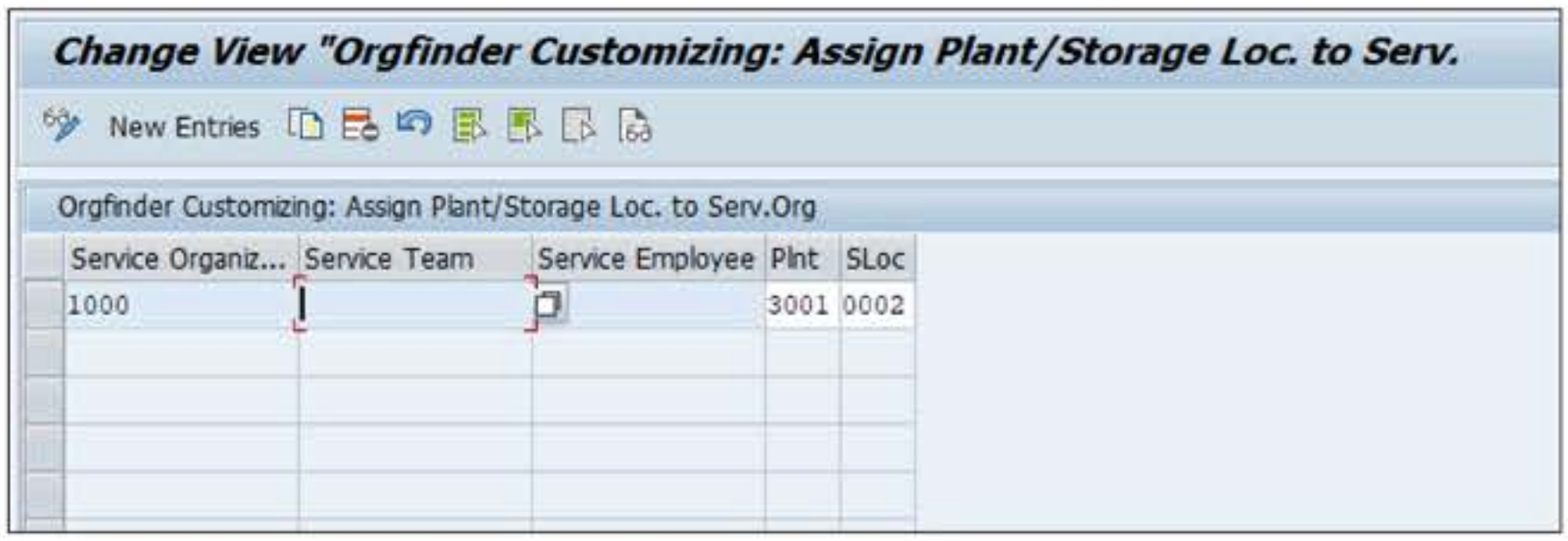

Figure 2.25 Assigning Plant and Storage Location to Service Organizational Units

You'll see how to create master data with the service organizational units in Section 2.4.1.

2.3 Configuring Subscription Order Management

In the first edition of this book, this chapter was described using SAP CRM, while a separate chapter was provided to offer a quick overview of the differences between subscription order management using SAP CRM versus SAP S/4HANA. As explained in the introduction of this chapter, Subscription Order Management has been first made available in the SAP S/4HANA 1809 version. In this edition, we'll describe configuring the SAP Billing and Revenue Innovation Management, subscription order management, using the latest version of SAP S/4HANA 2021.

In the following sections, we'll discuss the various steps for configuring the system as needed for our use case scenario. These steps range from setting up the products to adopt and the relevant transaction types to process the relevant orders.

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2.3.1 Product Modeling

While the material master in SAP S/4HANA provides the ability to define and capture the basic details of a go-to-market offering, most subscription offerings in business-to-business (B2B) scenarios require the capability to configure the offering and price it accordingly on a business transaction and pass this information to downstream systems. To enable this process on a business transaction such as a subscription order, a go-to-market subscription offering in SAP Billing and Revenue Innovation Management requires the following:

- Defining the product/material master using the Manage Product Master Data app
- Marking the product as configurable in the Manage Product Master Data app
- Performing Advanced Variant Configuration (AVC) using characteristics, classes, dependencies, configuration profiles, and so on
- Cross-catalog mapping using the Manage Subscription Product-Specific Data app
- Enabling the bundling of different offers as part of a solution via the Manage Product Bundles app

These facets are specifically required if the go-to-market offering will use SAP Convergent Charging to originate the charges or the fees. It further enables SAP Convergent Invoicing with the capability to read and use these configured product values.

SAP Billing and Revenue Innovation Management also provides an option to originate one-time and recurring charges directly in SAP Convergent Invoicing from SAP Billing and Revenue Innovation Management, subscription order management. This option could be used if the business requirements for charging the fees are basic and predictable. Such simple requirements are most likely in business-to-consumer (B2C) scenarios. In such cases, cross-catalog mapping may not be required, and this may eliminate the need to create configurable products using AVC.

In addition, it's important to note that the primary objective of product modeling is to eliminate creating multiple materials in the system to represent variants of the same product during the order management process. With the AVC capability in SAP S/4HANA, a material can be marked as a configurable product and associated with the knowledge base comprising the characteristics and their values, classes, dependencies, and so on, thus eliminating the need to proliferate the number of products maintained in the system.

In the predecessor versions of SAP Billing and Revenue Innovation Management, which used SAP CRM with SAP ERP, there were three options to make a product configurable:

- Simplified configuration
- The product modeling environment in SAP CRM with the Internet Pricing and Configurator

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- The Variant Configuration Engine in SAP ERP, with knowledge bases replicated to SAP CRM with the Internet Pricing and Configurator

Because the focus of this edition is primarily SAP Billing and Revenue Innovation Management using SAP S/4HANA 2021, in the subsequent sections of this chapter, we'll primarily discuss AVC. In addition, it's important to note that the predecessor of AVC is widely known as the Variant Configuration Engine from SAP ERP. At the same time, since subscription order management had moved from SAP CRM to SAP S/4HANA, the Java-based Internet Pricing and Configurator isn't used anymore. SAP S/4HANA–based SAP Billing and Revenue Innovation Management, subscription order management, only supports AVC and doesn't support the traditional Variant Configuration.

2.3.2 Product Configuration Using Advanced Variant Configuration

Variant configuration as a topic is a vast area with a lot of scope. The objective of this section is to help you get a basic configurable product set up in SAP S/4HANA.

All configurable products are set up using AVC. Configurable products should have the **Product is Configurable** checkbox checked in the product master, as shown in Figure 2.26.

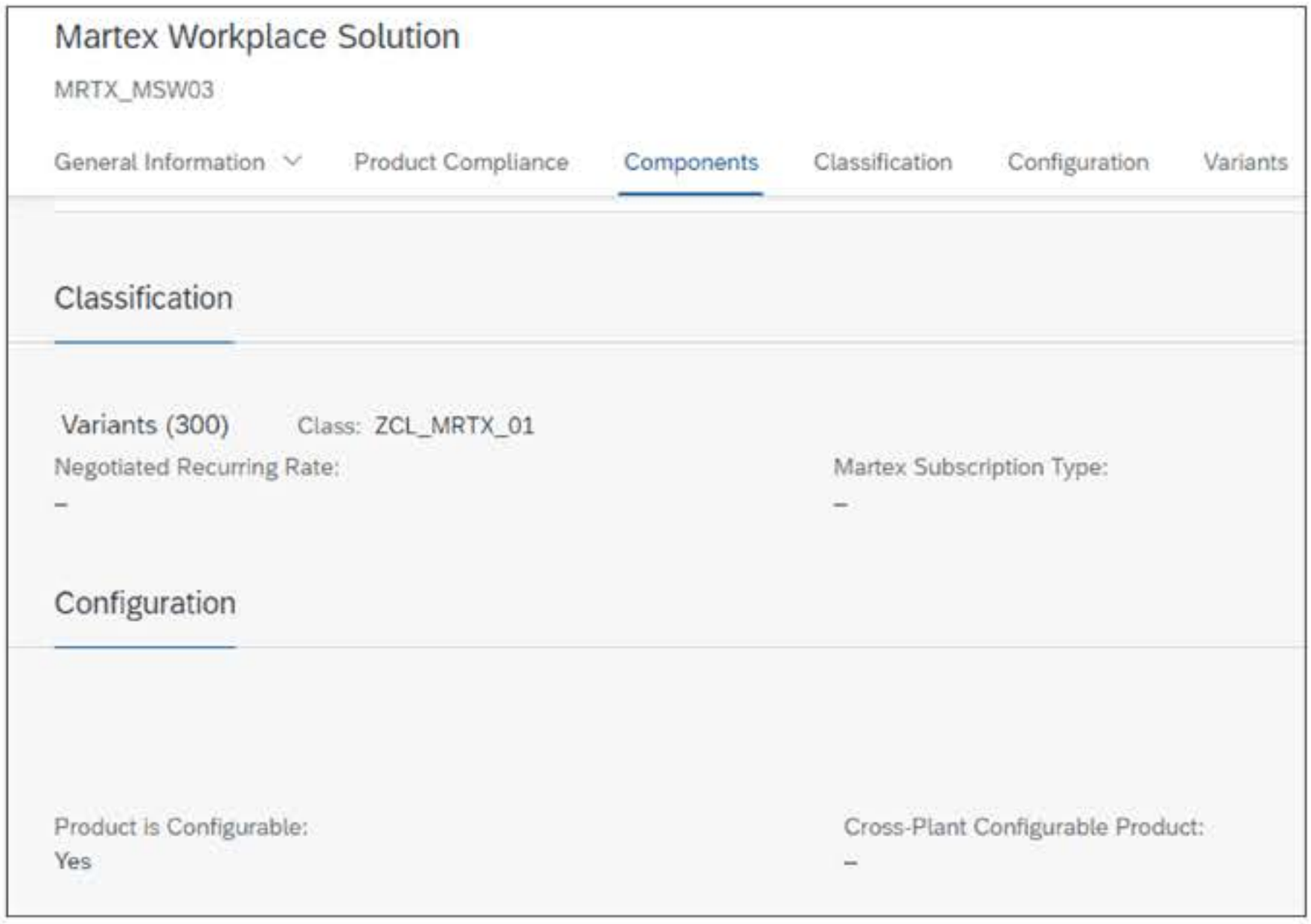

Figure 2.26 Configurable Indicator in the Subscription Product

The following sections describe the steps to set up a configurable product with attributes for selection.

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Definition of Advanced Variant Configuration Characteristics

Characteristics are attributes of the configurable product, such as type of subscription, cloud storage, and so on, which are created or modified using the Manage Characteristics app or Transaction CT04 (Characteristics), which is shown in Figure 2.27. Values for the characteristics are then created. For example, the **Martex Subscription Type** characteristic can have values such as **Base Subscription**, **Premium Subscription**, and so on.

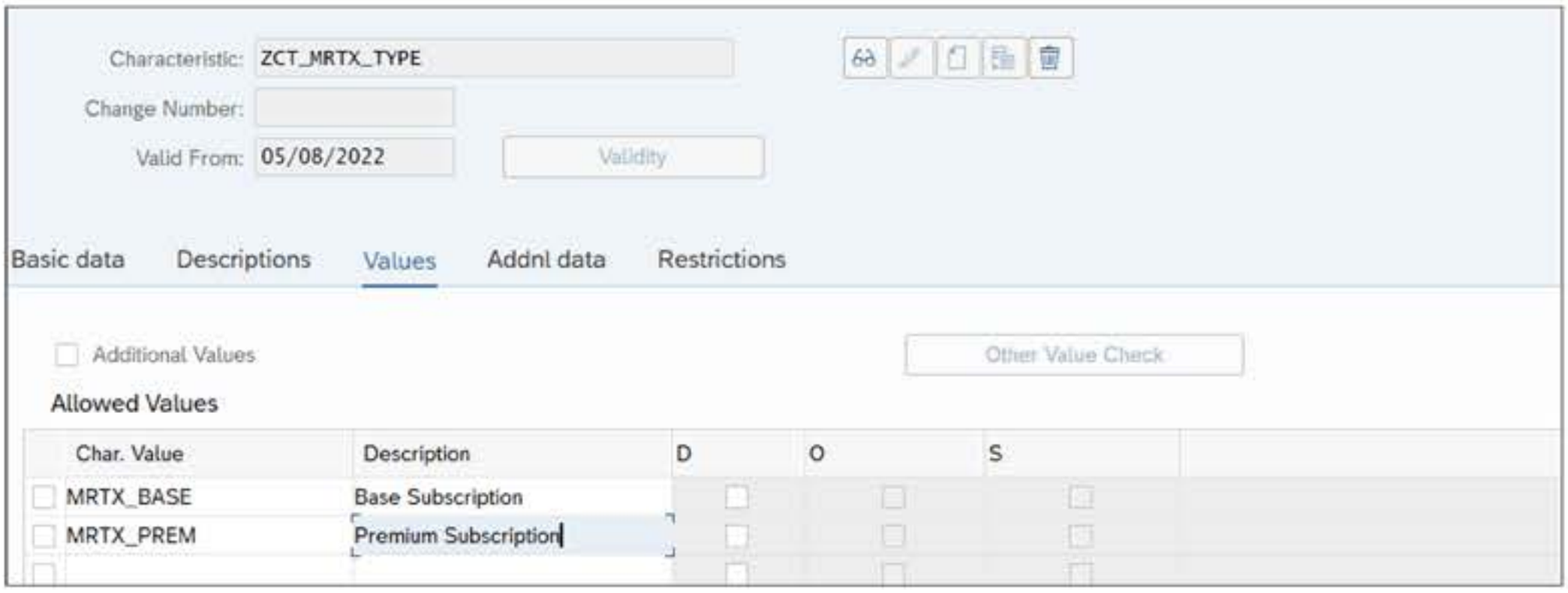

Figure 2.27 Setting Up Characteristics

Definition of Advanced Variant Configuration Classes

Configurable products have variant class type 300 maintained. Classes are created and modified using the Manage Classes app. The transaction codes to create a class is Transaction CL01 (to manage a class, you can use Transaction CL02 and to display a class, use Transaction CL03). Class is a container which is required to hold related characteristics. Once the class is created and updated with relevant information, it can then be assigned to products. In the following Figure 2.28, the characteristics created for Martex offerings are assigned to the class.

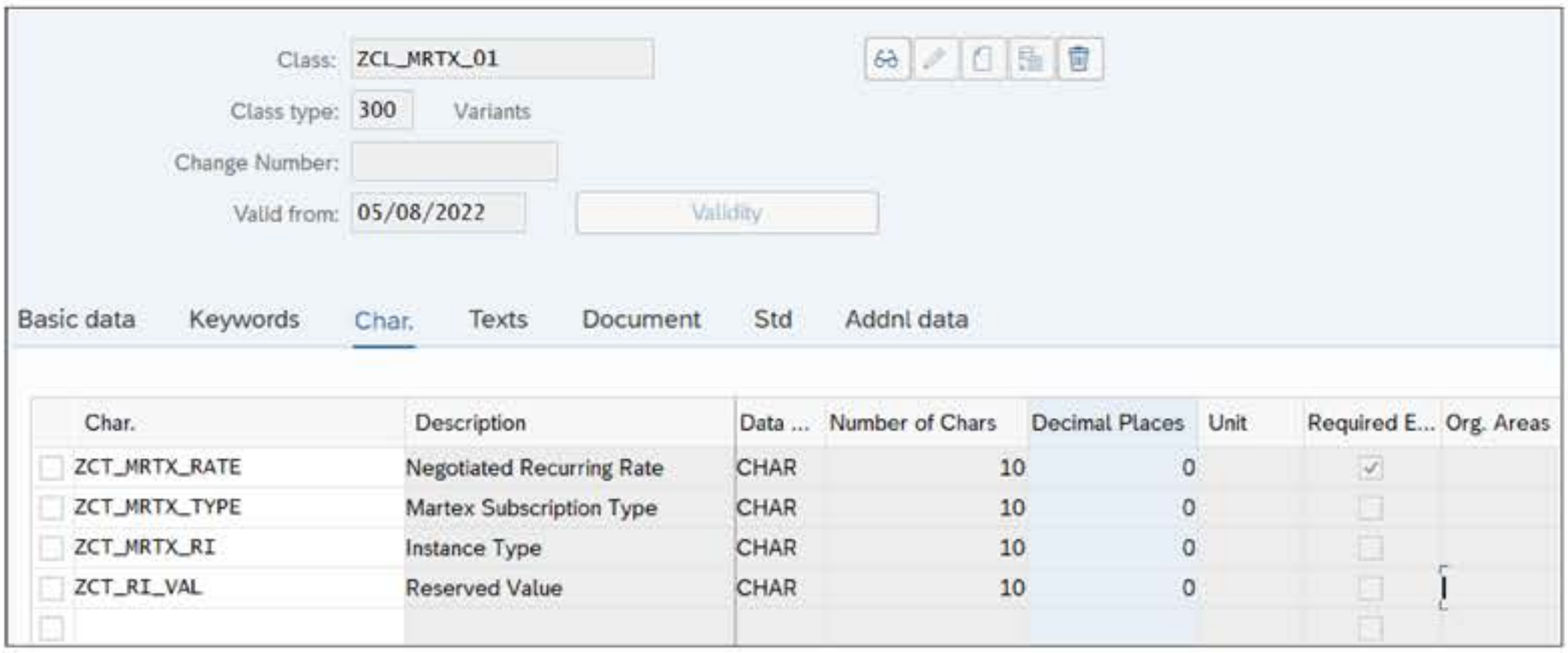

Figure 2.28 Maintenance of Classes

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Object Dependencies

Dependencies let you describe the interdependencies between characteristics and characteristic values, change the values of fields during configuration, maintain visibility control, maintain defaults, and calculate logic based on input fields.

Dependencies can have the following types:

- **Precondition**
  Preconditions can be assigned to a characteristic or a characteristic value that needs to be hidden from display, for example.
- **Selection condition**
  Selection conditions determine which variants require a specific component or operation. For example, a certain characteristic can only be presented for selection if another characteristic has a specific value.
- **Procedure**
  Procedures can be used to infer values for characteristics. It can be assigned to a characteristic or a configuration profile.
- **Constraints**
  Constraints are used to describe the dependencies between completely different objects and their characteristics. Constraints are used to store information on which conditions must be fulfilled if the configuration is to be consistent. They are grouped together in dependency nets. Dependency nets are created using Transaction CU21 (Create Dependency Net) or the Create Dependency Net app.

Dependencies can also be maintained using the Create Dependency app. You can also use Transactions CU02 and CU03 to change or display Dependency respectively.

Variant Tables

Variant tables are used to store combinations of characteristic values. They can be referenced in dependencies to infer, check, and restrict values. More specifically, they are used in the following scenarios:

- Selection conditions, preconditions, constraints, and procedures to check the consistency of the values entered
- Procedures and constraints to infer values
- Constraints to restrict values for a characteristic

One advantage of using them is that you only need to change the variant table, if the interdependencies between characteristics change, and not the dependencies. This greatly simplifies the maintenance tasks when a model changes.

The transaction codes for variant tables can be found in Table 2.2.

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Transaction Code	Description
CU61	Create variant table
CU62	Change variant table
CU63	Display variant table
CU64	Variant table structure list
CU60	Variant table maintenance
CU59	Transfer variant table contents to database table (report RCCUVTDB)
CU60E	Microsoft Excel upload for variant table contents

Table 2.2 SAP GUI Transactions for Advanced Variant Configuration

Managing Characteristic Groups

With this feature, a group of characteristics can be displayed together. You can create, update, and delete new groups of characteristics, and then maintain descriptions in different languages for each group, which is displayed on the user interface (UI).

Characteristics can be assigned and unassigned to a characteristic group and put in a specified order. Characteristics groups can be assigned or unassigned to a configuration profile, as well as be used in multiple configuration profiles.

The assigned characteristics groups for a configuration profile can be sorted to influence in which sequence they are shown in the UI. For example, you can create a characteristic group called Renewal Terms to store renewal-specific attributes (i.e., characteristics) such as renewal term, renewal opt-out period, renewal type, and so on.

Configuration Profile

The classes and various dependencies created are assigned to the configuration profile, and the configuration profile gets assigned to a configurable product to bring all the product attributes and the various rules driving the attributes and their values together. Only one configuration profile can be assigned to a product at a given time. The processing mode selected should be **AVC**. You can use Transaction CU41 to create a configuration profile for material. Additionally, you can use Transaction CU42 and CU43 to change and display the configuration profile for material, respectively.

Simulate Configuration Models

After all the steps to model a configurable product are complete, you use the Simulate Configuration Models app to check whether the characteristics and dependencies work as expected. If there are any inconsistencies, this feature flags the inconsistent

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configuration. The final configured product can also be displayed as a result of the characteristics selected, as shown in Figure 2.29.

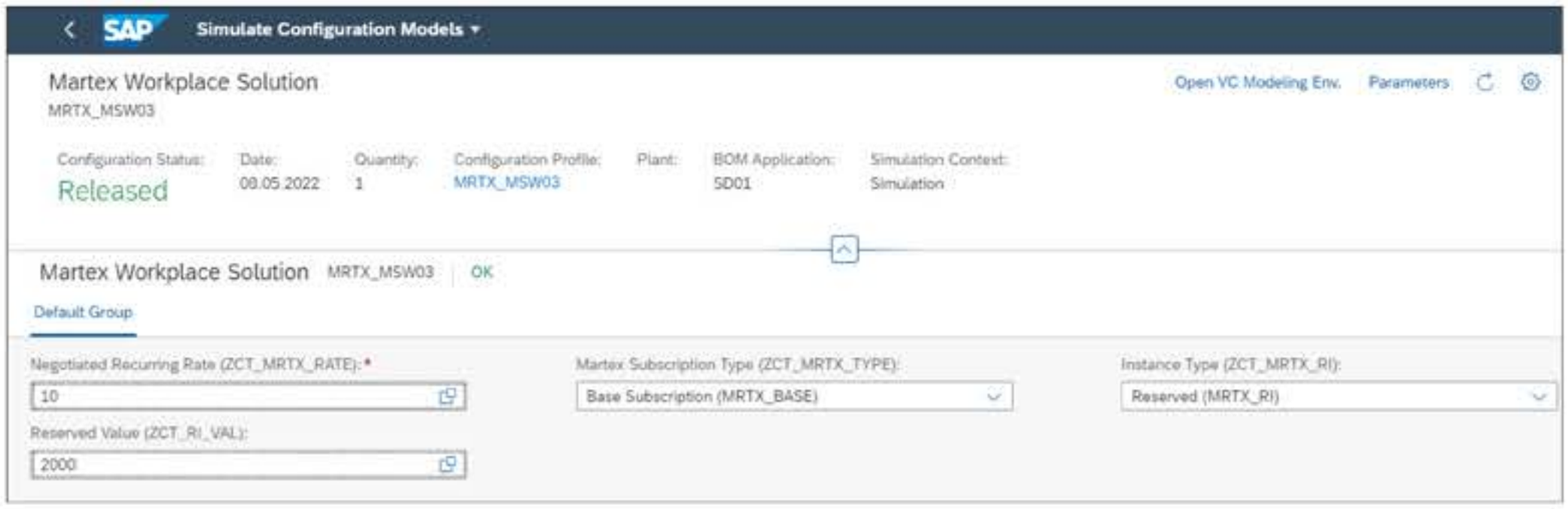

Figure 2.29 Configuration Simulation

Creating Preconfigured Products Using Material Variants

Preconfigured products are where attributes or characteristics values have already been set up per customer requirements. These attributes can't be changed by the user while placing quotes and orders.

Preconfigured products are linked to configurable products, which in turn have all the AVC rules and pricing functionality implemented. Using the Manage Product Master Data app, you enter the configurable product in the **Cross Plant Configurable Product** field under the **Configuration** tab.

2.3.3 Cross-Catalog Mapping

Cross-catalog mapping is a master data setup and maintenance activity that sets key linkages between charge plan objects created in SAP Convergent Charging (see Chapter 3 for more details) and the product setup. Within SAP Billing and Revenue Innovation Management, SAP Convergent Charging is where the rating and charging logic is defined using the charge plans for subscription products. Cross-catalog mapping between the products and its respective charge plans becomes the master data foundation on which the business transactions can be processed seamlessly. In other words, this mapping allows both SAP Billing and Revenue Innovation Management, subscription order management, and SAP Convergent Charging to understand how a subscription product will be rated and charged. Further, when the same subscription product is placed on a business transaction, it allows SAP Convergent Charging to understand whether one-time, recurring, or usage-based charges are involved.

Charge plans are grouped under file folders called catalogs in SAP Convergent Charging, allowing you to link one or many charge plans to a subscription product. Note that it's a prerequisite to have a product created using the Manage Product Master Data app before performing cross-catalog mapping.

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To perform cross-catalog mapping, you must first create a product. For our example, create the product as follows:

- Log in to the SAP Fiori launchpad in SAP Billing and Revenue Innovation Management, subscription order management, using the URL provided by your system administrator. Navigate to **Manage Subscription Product App**.
- Click **Create** in the app and navigate to the subscription product create screen.
- Enter the **Product ID** created as a prerequisite and click **Save**.
- Click **Edit**, scroll down to **Cross Catalog Mapping** section, and then click **Create**.
- A new row is added. Enter the **Description**, **Price Version**, and **Valid From** information.

Mapping versions helps to identify the mapping rules set forth during the life of the product with their own validity dates. Only the mapping versions are visible in the cross-catalog mapping assignment block. You can click a specific version to view the mapping details. Mapping version IDs are internally assigned and can't be changed.

The numbering starts with 0001 and is ascending. Unless they are released, they can't be changed or deleted. If deleted, the numbering starts with the next version after the released one or the one in processing. You can maintain multiple released versions: the system selects the version that is applicable for a specific validity period. It's recommended to maintain only one released version for the product to avoid validity issues. Only released versions are used in the subscription order and subscription contract. Versions that aren't applicable are set to obsolete and can't be changed or used in a transaction. When creating a new mapping version, the system copies the recent version and sets the highest version number in processing status.

You can also copy the mapping version from another product and adjust it, if necessary, to suit your business needs. To do so, follow these steps:

- Click on **Copy from Product**, select the product and the version using the value help, and click **OK**. A row for cross-catalog mapping is created.
- Click on the row to navigate into it. Scroll down to the **Charge Plans** section and click **Create**.
- From the value help of **Charge Plan ID**, select an appropriate charge plan.
- Select an activation option: **Characteristic**, **BRF Plus**, or **Always**.
- Enter a **Description** and click on the row to navigate into it.

Figure 2.30 shows the charge plan selection window and the mapping version screen for our product: Martex Workplace Solution (Product ID - MRTX_USW03).

You can search for a charge plan and assign it to your subscription product as part of the cross-catalog mapping (see Figure 2.31 and Figure 2.32). Once assigned, you can release the charge plan to make it active (refer to Figure 2.33).

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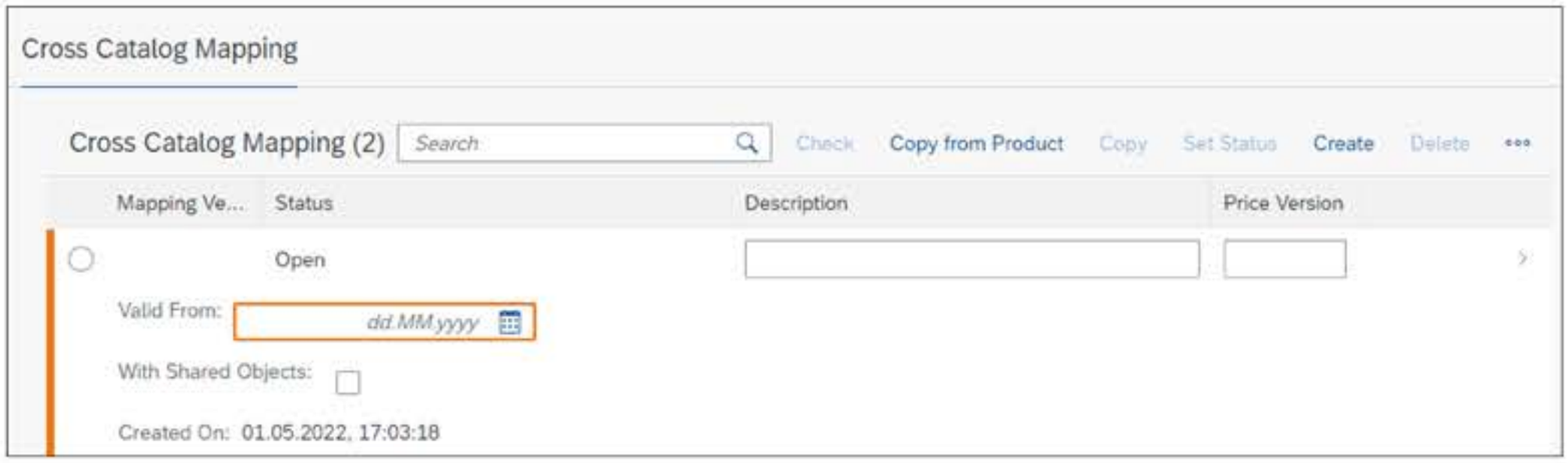

Figure 2.30 Cross-Catalog Mapping for Subscription Products

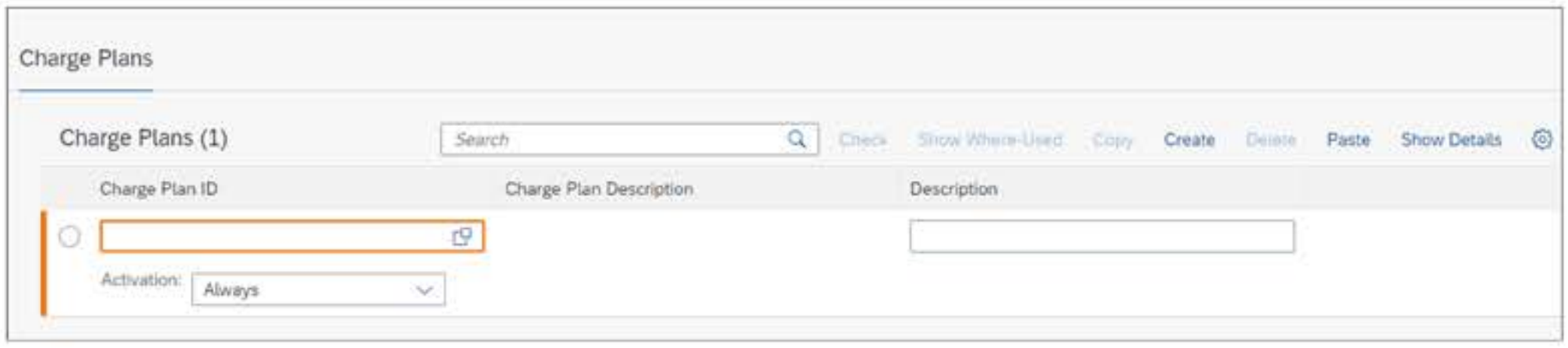

Figure 2.31 Search Charge Plan in Cross-Catalog Mapping

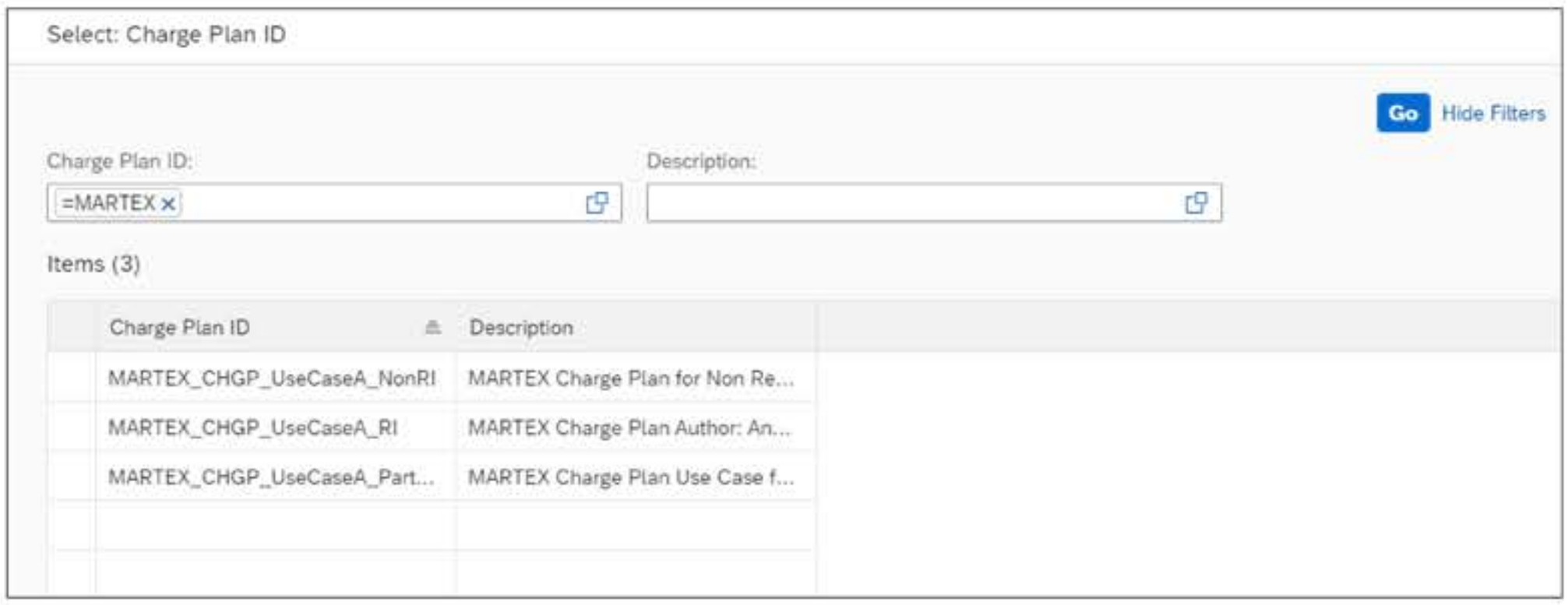

Figure 2.32 Search Charge Plan ID

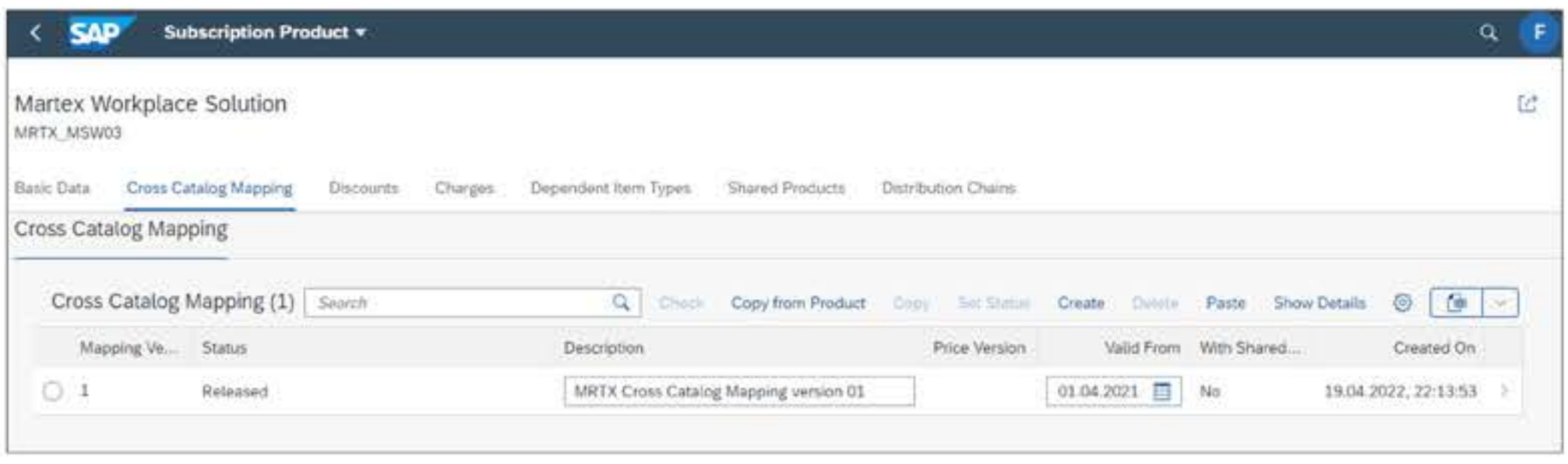

Figure 2.33 Release Charge Plan

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The charge plan assignment window shows the actual assignment, the charge plan assigned, the charge plan description, and the active status.

Go to the **Parameters** window shown in Figure 2.34. *Parameters* are considered external variables, which are used to carry values from outside systems to SAP Convergent Charging. These parameters are used flexibly to inherit default values, redefined values, characteristics/attributes, or Business Rules Framework (BRF) events. Only external variables are exposed to Subscription Order management. Internal Parameters aren't exposed to Subscription Order Management, and thus won't carry any values to SAP Convergent Charging.

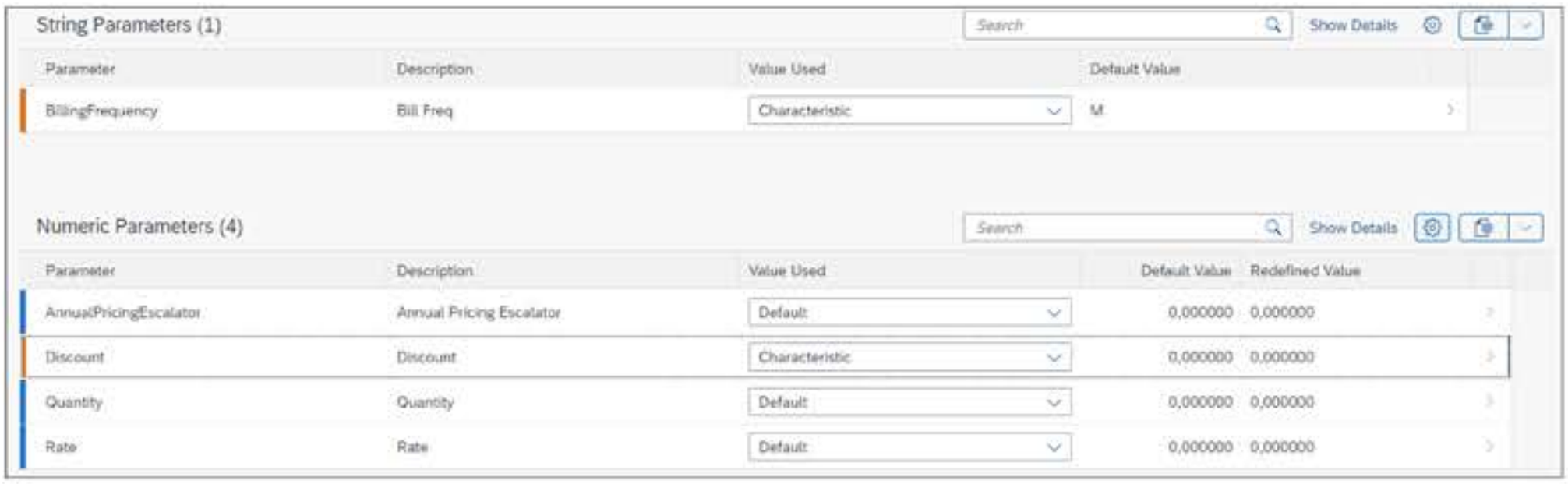

Figure 2.34 Parameter in Cross Catalog Mapping

The parameter **ID** and the **Description** are auto populated. Under the **Value Used** column, you can select one of the pre-defined options such as **BRFplus Function**, **Characteristic**, **Default**, **Redefined** and **Share ID**. In our use-case, let's use one of the parameter **Discount** which will be updated based on characteristics. Select the **Discount** row and then set the **Characteristic** as **Value Used**. We will map our existing characteristic's **Subscription Discount (in %)** as shown in Figure 2.35.

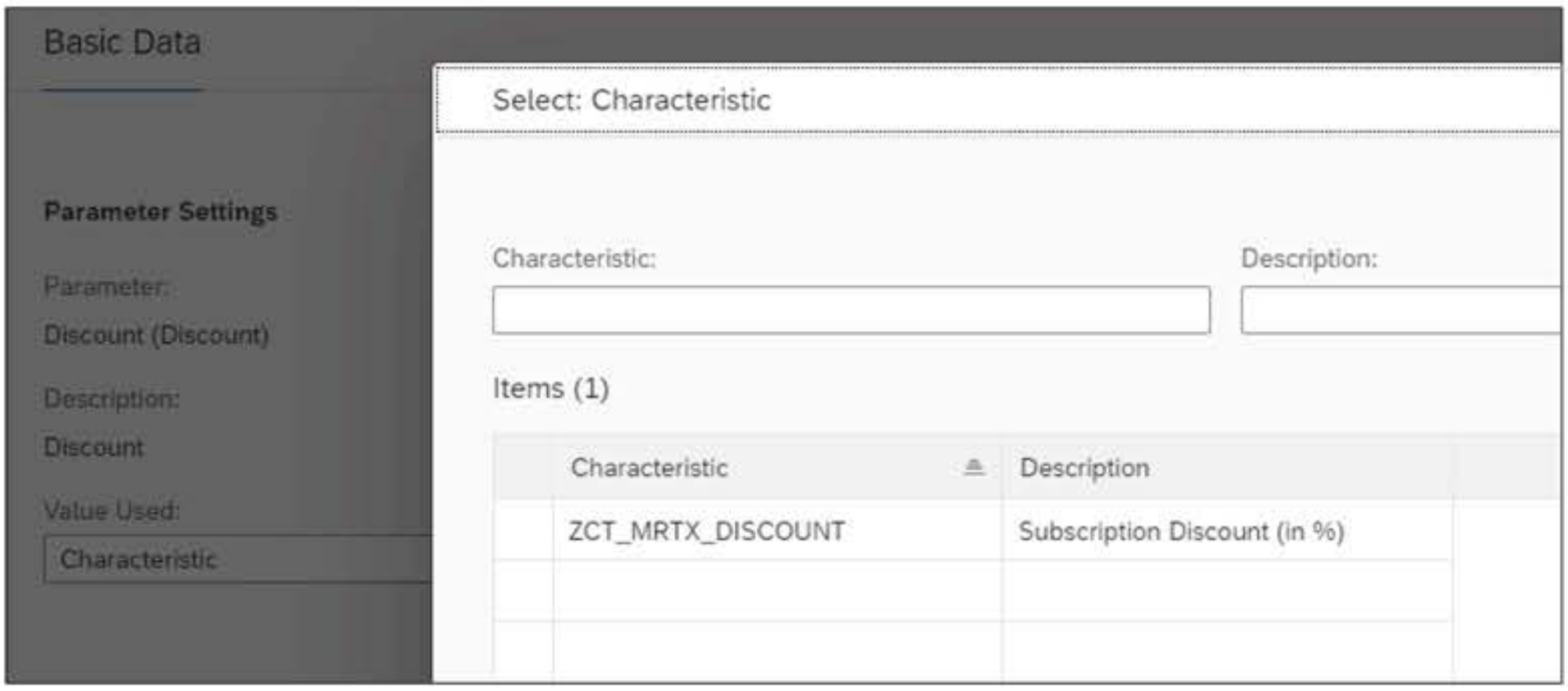

Figure 2.35 Assignment of Characteristics with Parameter

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Now, go to the **Technical Resources** screen. **Technical Resource** is a key field that helps to link the subscription contract within the SAP Billing and Revenue Innovation Management landscape that is applicable in the billing context. It can be a Booking ID, customer ID, phone number, equipment ID, device serial number, and so on based on your business requirements.

The ID and the description are auto populated. Under the **Assigned Contract Item** column, select **Own Item**. Then, under the **Technical Identifier** column, select the appropriate technical identifier from pre-defined options. The service ID in the **Service IDs** column is inherited from the charge plan in SAP Convergent Charging as shown in Figure 2.36.

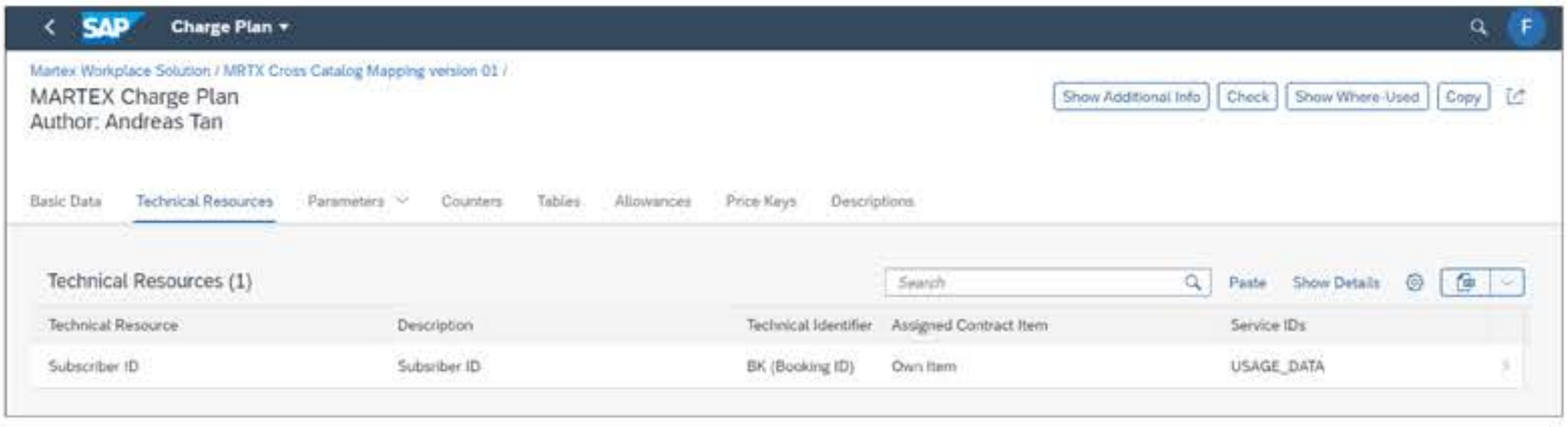

Figure 2.36 Technical Resource Assignment

2.3.4 Setting Up One-Off Charges

In general, one-off charges (previously known as one-order objects) are referred to as one-time billable items for customers. Typically, one-off charges are payable for additional services that aren't covered under subscription fees and usually aren't defined in the terms and conditions. Some examples of one-off charges include installation fees, activation charges, site visits, and so on. There are different ways of handling such one-time charges in SAP Billing and Revenue Innovation Management. These charges can be managed via SAP Convergent Charging by using one-shot rates (one-off tariffs) configured in the pricing catalog. However, there is a provision for managing one-off charges within SAP S/4HANA by configuring SAP Billing and Revenue Innovation Management, subscription order management, and SAP Convergent Charging. The one-off charges are recorded in SAP Convergent Charging in the form of billable items and can be consolidated with the related subscription or usage charges or as a separate invoice.

For our Martex use case, let's use a one-off product called MWS Activation Charge (Product ID: `MSW_ACTIVATION`) to generate a one-time charge in advance for customers opting for the Martex Workplace Solution.

As part of the standard SAP Billing and Revenue Innovation Management, subscription order management, item category PROF (Subscription One-Off Item) is provided to manage one-time charges. The settings for item categories can be accessed in **Service** •

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Transactions • Settings for Subscription Transactions • Define Settings for Item Categories, as shown in Figure 2.37.

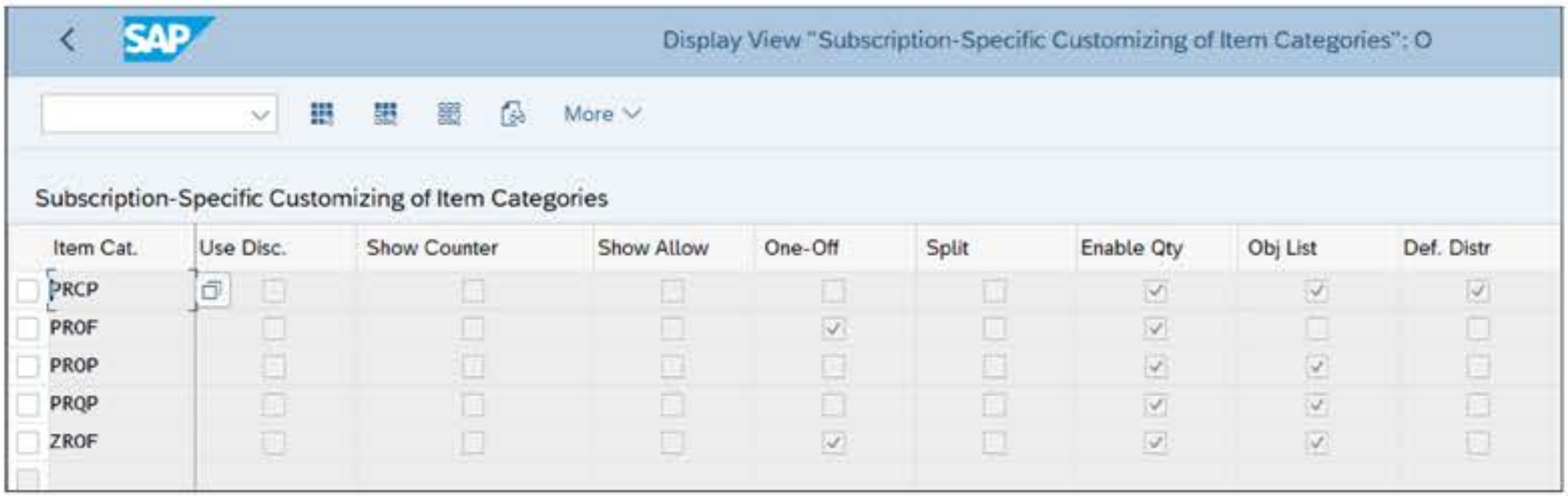

Figure 2.37 One-Off Charge Item Category

To support one-time charges in SAP Convergent Invoicing, the following settings must be executed in the configuration for contract accounting under **Financial Accounting • Contract Account Receivable and Payable • Integration • SAP Customer Relationship Management • Additional Entries for One-Off Charges from CRM**, as shown in Figure 2.38.

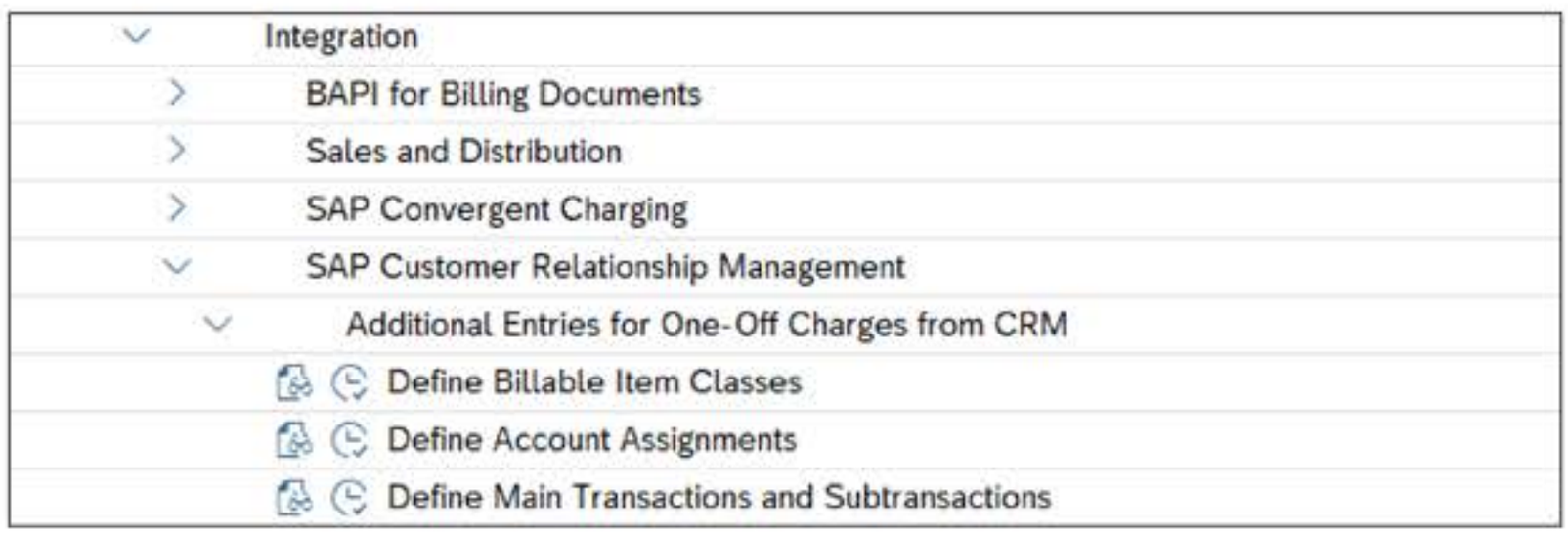

Figure 2.38 Configuration Node for One-Off Charges

In **Define Billable Item Classes**, you can specify which billable item classes, billable item types, and billing subprocesses that you want to use for one-off charges from SAP CRM (i.e., SAP Billing and Revenue Innovation Management, subscription order management). The system makes this derivation based on the current posting area. The account assignment derivation is made only for billable item classes for which the interface component **CRM Order** is active.

One-off charges are distributed from SAP Billing and Revenue Innovation Management, subscription order management, to SAP Convergent Invoicing using document distribution (Order Distribution Infrastructure [ODI]). The distribution step types ONEF (Distribute One-Off Items from Order) and PCOF (Distribute One-Off Items) are prepared to create billable items in SAP Convergent Invoicing for one-offs. The settings

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can be accessed in **Service • Transaction • Settings for Subscription Transaction • Document Distribution • Define Settings for Document Distribution**.

2.3.5 Subscription Transactions

Following are the different transaction types available to address various scenarios in SAP Billing and Revenue Innovation Management, subscription order management:

- Solution quotation
- Subscription order and subscription contracts
- Master agreement
- Partner agreement order and partner agreement

Most B2B transaction flows start with a master agreement. Note that some of the basic configuration activities required to set up these transaction types are mostly the same, but there are configuration activities specific to each of the transactions. In the following sections, we'll describe the common configuration steps needed in a detailed manner using the subscription transactions. Further, we'll transition into describing the specific configurations as required for master agreements, solution quotations, partner agreement orders, and partner agreements.

In the following sections, we'll describe the detailed configuration steps required to complete the configuration for these objects.

Solution Quotation

A solution quotation is a transactional document used for selling different solutions, such as physical goods, services, or subscriptions, to customers. These different types of product offerings are set up as a product bundle in the master data. This is increasingly common in different industries as companies aim to be a "one-stop shop" for different customer needs, as well as to lock in the customers for future recurring revenue streams. For example, think of buying a car that comes with an option to get it periodically serviced for a nominal recurring monthly charge.

When a product bundle is sold to a customer through the solution quotation, subsequent documents are automatically generated through system configuration. The subsequent documents could be a service contract for services, a sales order to deliver the physical goods, or a subscription contract for the subscription offering part of the bundle. The solution quotation framework provides a very sophisticated order distribution/orchestration capability to route the appropriate line items to different systems or modules within SAP.

The solution quotation name for this document can be misleading and cause confusion. A solution quotation once accepted by the customer is a legally binding document and not to be mistaken for a quotation.

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Subscription Order and Subscription Contract

A subscription order is a transaction used for the order capture in the process of selling, which enables you to add physical goods, services, and subscription products as line items within it. The subscription order also enables you to sell these products separately or bundle them up together. Subscription orders can also support one-time fees, which can include installation fees, activation fees, cancellation fees, and so on. Along with the product or line-item details, subscription orders also enable you to capture involved customer parties, dates, pricing, and other relevant order information.

After finalizing the order capture process, a subscription order is completed and then submitted. As a result, a new, linked document known as a subscription contract is created. This document serves to represent and maintain the long-term agreements that are made between the service provider and the end customer. This system-generated subscription contract can't be created manually by the user. When a subscription contract change is required, a linked and a follow-up subscription order known as a change order is created to capture the necessary changes being requested to the subscription contract. When this change order is submitted, a new version of the existing subscription contract will be created with the changes requested. We'll discuss the changes to an existing subscription in detail in subsequent subsections of this chapter.

Partner Provider Order and Partner Agreement

Provider orders and provider contacts are used for handling accounts receivable in a subscription-based business model, but in a typical SAP Billing and Revenue Innovation Management implementation, you need to manage accounts payable as well. SAP has provided the partner provider order and partner agreement for handling this task. A partner provider order is equivalent to a provider order, and a partner agreement is like a subscription contract for calculating the payout to vendors.

Once a partner provider order is submitted and released, it will be replicated to SAP Convergent Invoicing and SAP Convergent Charging via ODI, like the replication of provider contracts while submitting provider orders. The standard transaction types for partner provider orders and partner agreements are listed in Table 2.3.

Transaction Type	Description
PRPO	Partner Provider Order
PRPA	Partner Agreement

Table 2.3 Partner Provider Order and Partner Agreement Transaction Types

All the configurations for Subscription orders and Subscription contracts need to be maintained for the partner provider order and partner agreement transaction types as well. You create custom transaction types by copying the PRPO (Partner Provider

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Order) and PRPA (Partner Agreement) transaction types. Refer section Section 2.3.7 for more details.

Common Configurations for Subscription Order Management

Any transactions or orders that are expected to be created in the system must be assigned a number from the set number range. Like for business partners and business agreements, you define the number range for subscription transactions by using Transactions CRMC_NR_RA_PRVORD and CRMC_NR_RA_PRVCTR. The number range objects for subscription orders and subscription contracts are CRM_PRVORD and CRM_PRVCTR, respectively.

Figure 2.39 shows the number range assignment for subscription contracts.

Maintain Intervals: Provider Contract

No	From No.	To Number	NR Status	Ext
01	0000000000000000001	0000000000000009999	350	
02	0000000000000010000	0000009999999999999	10329	

Figure 2.39 Define Number Ranges for Subscription Contracts

By now, you should be familiar with the meaning of the required fields in this setup. For a quick refresher, see the number range for business partners explanation in Section 2.1.1.

Transaction Types

The next step is to configure the transaction types. A *transaction type* defines the characteristics and features of a business transaction. Examples include the subscription order, subscription contract, sales order, service request, activity, and more. Each transaction type is defined for a specific purpose and is associated with certain control attributes, such as the partner determination procedure, text determination procedure, status profile, organization data profile, and so on. The transaction type with its configuration of the control attributes drives the business transaction and its behavior.

SAP has provided a standard transaction type as a reference for subscription orders: PRVO. In an SAP Billing and Revenue Innovation Management implementation for a client, you usually make a copy of this standard transaction type to create your own custom transaction types and further make customer-specific changes to the copied transaction type.

To configure the transaction types via menu path, go to **Service • Transactions • Basic Settings • Define Transaction Types**. Figure 2.40 shows standard subscription order transaction.

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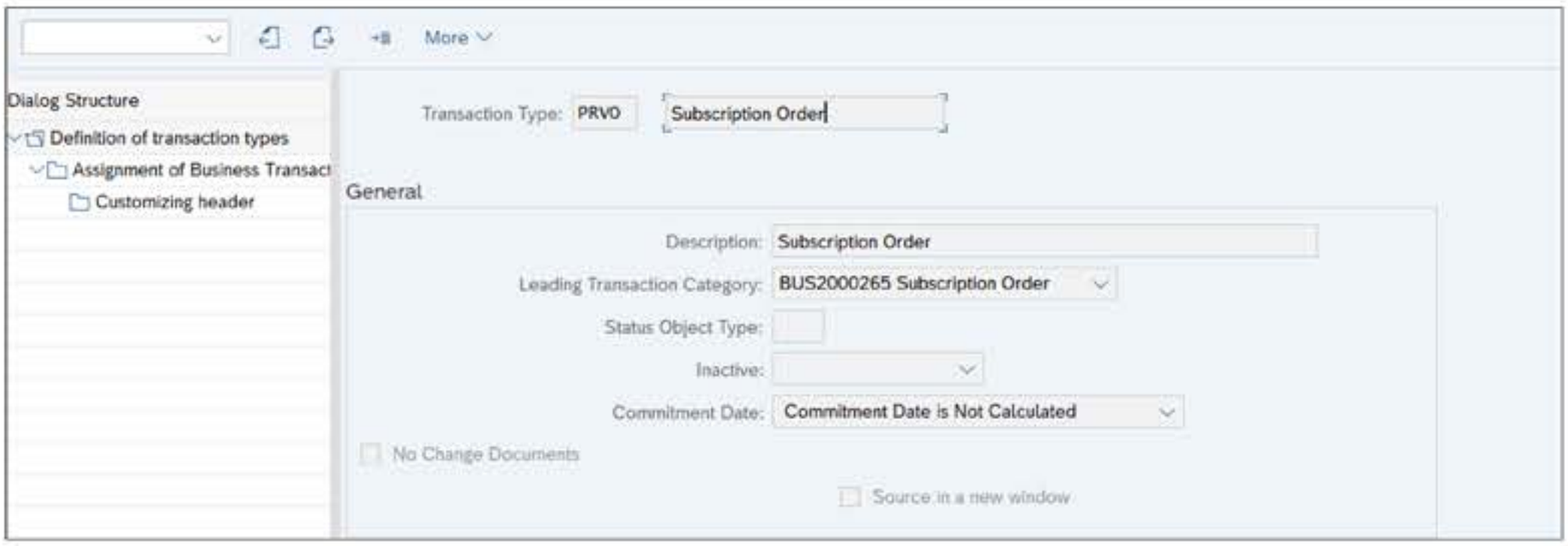

Figure 2.40 Transaction Type: Subscription Order

SAP provides standard control attributes in the standard transaction type. If you want to maintain your own control attributes, such as text determination procedure, partner determination procedure, and so on, you can copy the standard ones and assign the custom control attributes to the custom subscription order transaction type (see Figure 2.41).

Figure 2.41 Profile Customization for Transaction Types

The defined fields are as follows:

- **Text Det. Procedure**

  For every business transaction, there is a set of text types with their own sets of rules to determine the text. These text types and the rules to determine the texts are defined in the text determination profile. Text types are optional in any transaction. You define a date profile by following menu path **Service • Basic Functions • Text Management • Text Determination Procedure**.

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- **Partner Determ.Proc.**
  This is a set of rules that you define to determine how business partners are assigned during transaction processing. The system automatically determines the business partners involved in a business transaction. Partner determination procedure can be assigned to a transaction type or item category. You define this procedure by following menu path **Service • Basic Functions • Partner Processing • Define Partner Determination Procedure**.
- **Status Profile**
  SAP has provided the flexibility to create and use your own statuses along with the system statuses. These statuses can be used for accommodating additional information. You can create any number of statuses according to your requirements by following menu path **Service • Transactions • Basic Settings • Status Management • Define Status Profile for User Status**.
- **Date Profile**
  For every business transaction, there is a set of date types with its own set of rules to determine the dates. These date types and the rules to determine the dates are defined in the date profile. The date profile is important and required for defining the validity of a transaction. SAP has provided different date profiles, and you can define a date profile by following menu path **Service • Basic Functions • Date Management • Define Date Profile**.
- **Action Profile**
  SAP has provided an option to trigger the subsequent events of a transaction using actions (see Figure 2.42). Actions can automatically trigger output, follow-up activities, or workflows and can be triggered with the help of user-definable conditions in transactions. You define action profiles by following menu path **Service • Basic Functions • Actions • Actions in Transaction • Create Actions with Wizard**.

Figure 2.42 Maintenance of Action Definition

Similarly, you can also create a subscription contract transaction type by copying it from Transaction PRVC, as provided by SAP.

You define additional settings for subscription contract transaction types to enable the proper distribution of the subscription contracts into SAP Convergent Invoicing and SAP Convergent Charging. While submitting the subscription order, a subscription

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contract needs to be created asynchronously. You can make these settings at the following menu path: **Service • Transactions • Settings for Subscription Transactions • Define Settings for Transaction Types**.

Figure 2.43 shows the standard transaction of the **Subscription Contract Document**.

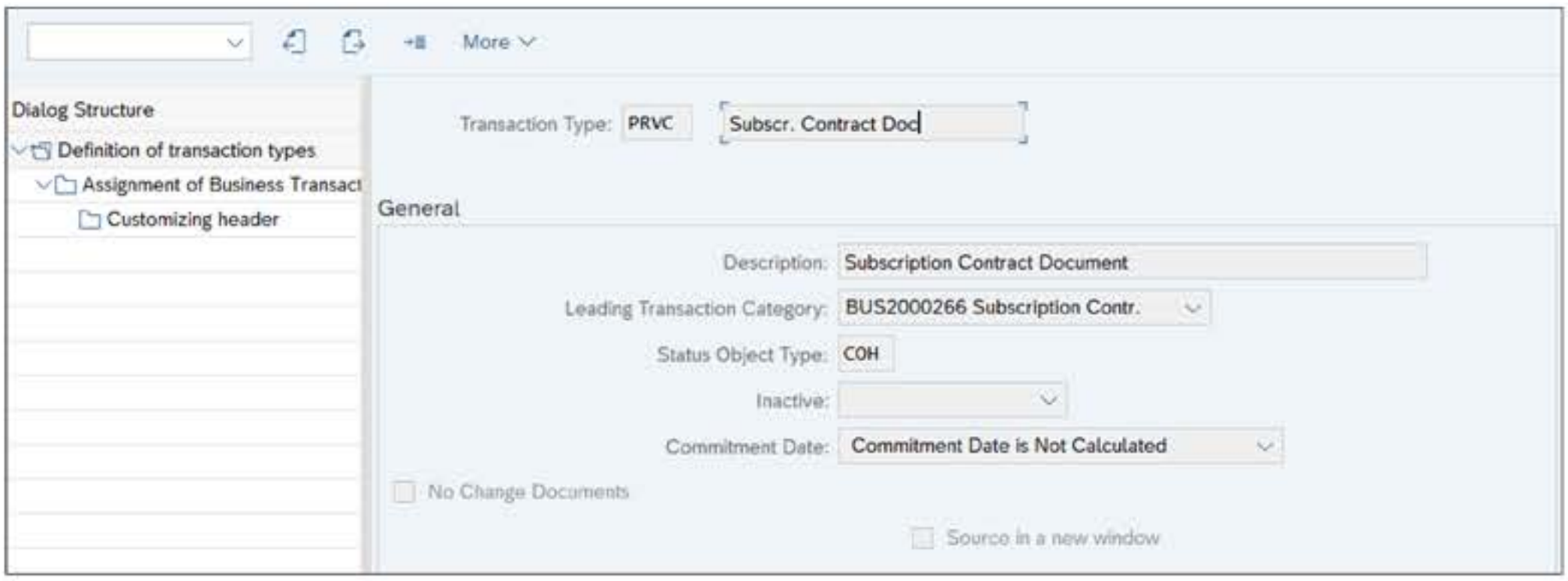

Figure 2.43 Maintenance of Subscription Contract Document

Item Category Group

Next, you would need to make the settings for the item category group (see Figure 2.44). Products are managed from a business view with the help of the item category group, which is an arbitrary setting to make the product adaptable for subscription processes. You can make the settings via menu path **Service • Transactions • Basic Settings • Define Item Category Group**.

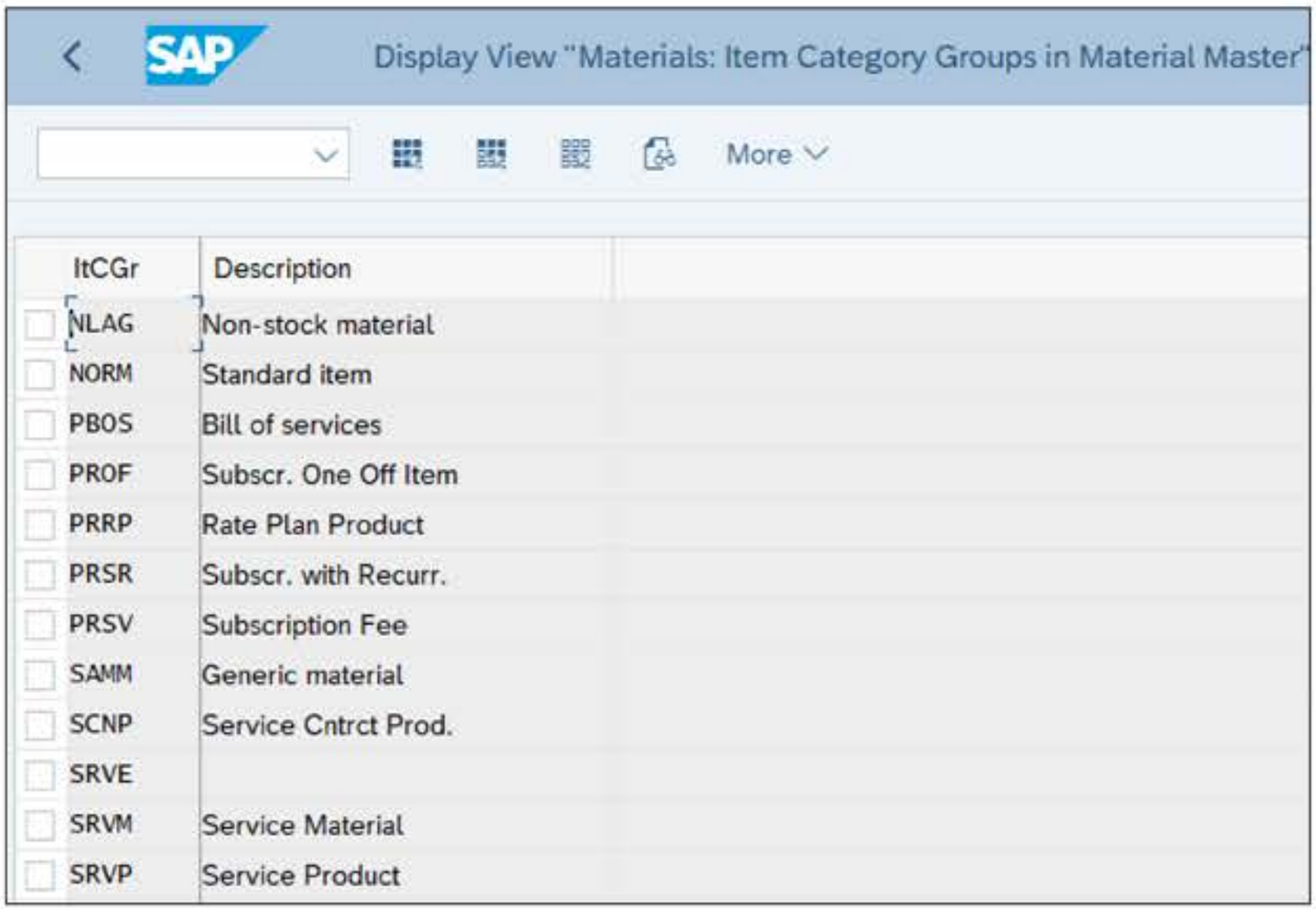

Figure 2.44 Item Category Group Settings

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The most used item category groups for SAP Billing and Revenue, subscription order management implementations are listed in Table 2.4. You can create your own item category groups by copying these standard ones.

Item Category Group	Description
PRRP	Rate Plan Product
PRSR	Subscription with Recurring
PROF	Subscription One Off Item

Table 2.4 Item Category Groups

Once you have defined the item category, you need to start configuring item categories. Item categories are settings in the system that enable the line items in transactions to behave in a certain way. You can make the settings for item categories for business transactions via menu path **Service • Transactions • Basic Settings • Define Item Categories** as shown in Figure 2.45.

The item categories for subscription order and contract are listed in Table 2.5. You can create custom item categories by copying these standard ones.

Item Category	Description
PRCP	Subscription Contract
PRCR	Subscription Contract Recurring
PRMP	Master Agreement Item
PROF	Subscription Order One Off Item
PROP	Subscription Order Item
PROR	Subscription Order Item Recurring
PROS	Subscription Order Sales Item
PRPC	Subscription Partner Contract
PRPO	Subscription Partner Order Item
PRQF	Subscription One Off Item
PRQP	Subscription Item
PRQR	Subscription Item Recurring
PRSP	Sales Package with Supply at Item Level

Table 2.5 Item Categories for Subscription Order and Contract Categories

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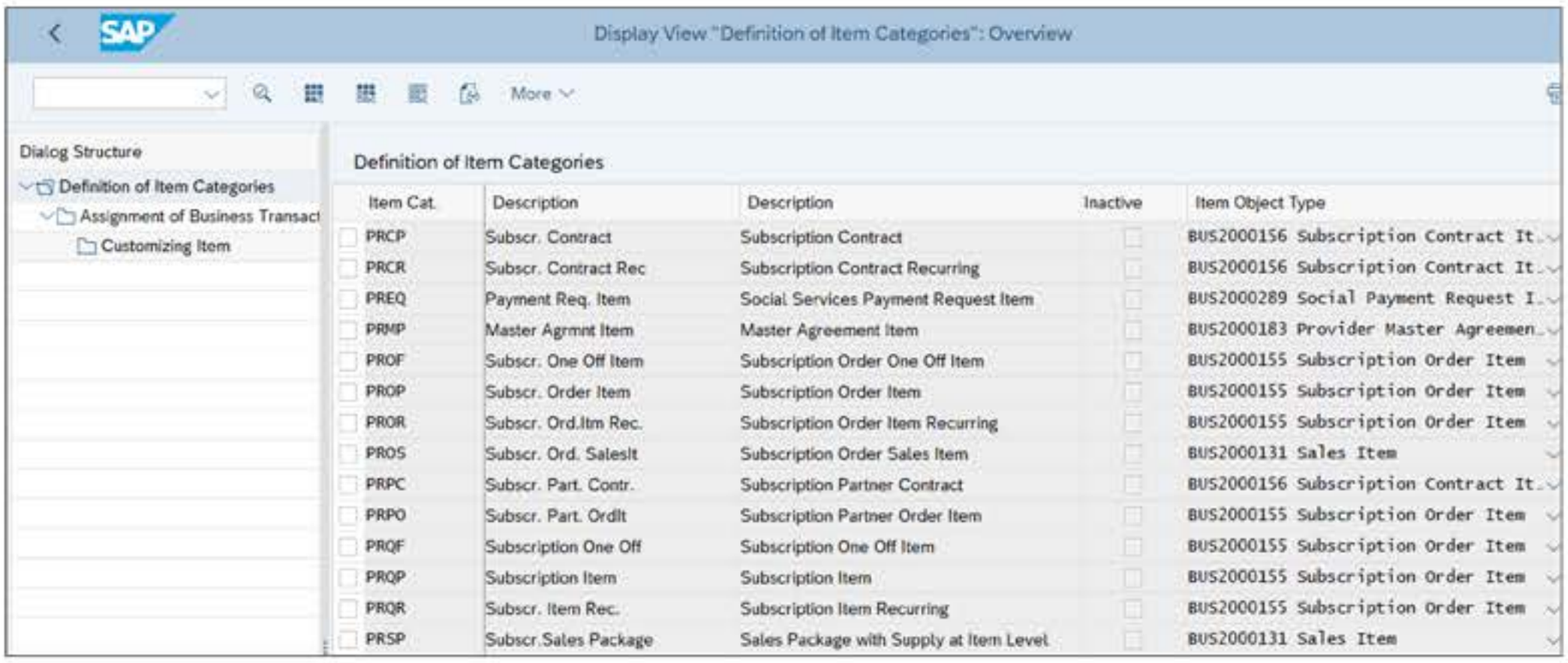

Figure 2.45 Item Categories for Subscription Order Management

After configuring the item category group and item categories, you need to define the item category determination rules. This setting enables the system to determine the correct item category for the line item automatically based on the combination of certain parameters. This helps you avoid manually assigning the item categories at the line-item level.

The item category is determined using the following parameters:

- Transaction type
- Item category group (product master)
- Item category usage
- Main item category

You make the settings for the item category determination by following menu path **Service • Transactions • Basic Settings • Define Item Category Determination**.

Figure 2.46 shows the item category determination for the standard subscription order transaction type PRVO.

Another key configuration for SAP Billing and Revenue Innovation Management, subscription order management is to define the copy control for transaction types. Copy control enables the system to copy details from the source transaction to the target transaction. You define rules to determine what details are copied and how they are copied.

To create the subscription contract while submitting the subscription order, the copy control configuration needs to be maintained for the subscription order and subscription contract transaction types. You make the copy control settings by following menu path **Service • Transactions • Basic Settings • Copying Control for Business Transactions • Define Copy Control for Transaction Types**.

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Item Category Determination

Trans.Type	Desc.TransTyp	ItCGr	ItmCat.Usg	Main Item cat.	Desc. Itm cat.	Item Cat.	Desc.ItmCty
PRVO	Subscription Order	0002 Configuration				TAC	Config.at Mat.Level
PRVO	Subscription Order	0002 Configuration		TAC	Config.at Mat.	TAE	
PRVO	Subscription Order	GG04 Config. Bundle				PROP	Subscr. Order Item
PRVO	Subscription Order	GG04 Config. Bundle		GG04	Subscr.Sales P	GG04	Subscr.Sales Package
PRVO	Subscription Order	GG04 Config. Bundle		PROP	Subscr. Order	PRSP	Subscr.Sales Package
PRVO	Subscription Order	GG04 Config. Bundle		PROS	Subscr. Ord. S	PRSP	Subscr.Sales Package
PRVO	Subscription Order	GG04 Config. Bundle		PRSP	Subscr.Sales P	PRSP	Subscr.Sales Package
PRVO	Subscription Order	LUMF Structure/mat. below				PRSP	Subscr.Sales Package
PRVO	Subscription Order	LUMF Structure/mat. below		PROP	Subscr. Order	PRSP	Subscr.Sales Package
PRVO	Subscription Order	NORM Standard item				PROS	Subscr. Ord. SalesIt
PRVO	Subscription Order	NORM Standard item		GG04	Subscr.Sales P	ZPRS	Subscr. Ord. SalesIt
PRVO	Subscription Order	NORM Standard item		PROP	Subscr. Order	PROF	Subscr. One Off Item
PRVO	Subscription Order	NORM Standard item		PROR	Subscr. Ord.It	PROS	Subscr. Ord. SalesIt
PRVO	Subscription Order	NORM Standard item		PRSP	Subscr.Sales P	PROS	Subscr. Ord. SalesIt
PRVO	Subscription Order	NORM Standard item		ZPRS	Subscr. Ord. S	ZPRS	Subscr. Ord. SalesIt
PRVO	Subscription Order	PROF Subscr. One Off Item				PROF	Subscr. One Off Item
PRVO	Subscription Order	PRRP Rate Plan Product				PROP	Subscr. Order Item
PRVO	Subscription Order	PRRP Rate Plan Product		GG04	Subscr.Sales P	PROP	Subscr. Order Item
PRVO	Subscription Order	PRRP Rate Plan Product		PROP	Subscr. Order	PROP	Subscr. Order Item
PRVO	Subscription Order	PRRP Rate Plan Product		PROR	Subscr. Ord.It	PROP	Subscr. Order Item
PRVO	Subscription Order	PRRP Rate Plan Product		PRSP	Subscr.Sales P	PROP	Subscr. Order Item

Figure 2.46 Item Category Determination

Figure 2.47 shows the copy control configurations maintained for the standard subscription order and contract transaction types.

All steps mentioned thus far need to be repeated to maintain the configurations for the subscription contract. SAP has provided the standard transaction type PRVC for the subscription contract, and you can refer to this transaction type and copy it to create your own custom subscription contract transaction.

Copy Transaction Types - General Control Data

Srce Trans. Type	Tgt Trans. Type	Short Description	Copy item no.	ComplRef	Copying routine	Copy PO data	Copy sls cycle	CopyFore	CopyClass	Copy Price Agreement
OPSM	OPSM	Sales Methodology					✓	✓	✓	
PRCC	PRCO	Sharing Order								✓
PRCO	PRCC	Sharing Contract								✓
PRMA	SRVP	Solution Quotation		✓						
PRPA	PRPO	Partner Agrmnt. Ord.					✓	✓	✓	✓
PRPO	PRPA	Partner Agreement					✓	✓	✓	✓
PRVC	PRVO	Subscription Order					✓	✓	✓	✓
PRVC	ZBSO						✓	✓	✓	✓
PRVO	PRVC	Subscr. Contract Doc				✓	✓	✓	✓	✓
REPA	REPO	Repair Order								
REPA	REPQ	Repair Quotation								
REPO	REPC	Repair Confirmation								
REPQ	REPO	Repair Order								
SC	REPO	Repair Order								
SC	REPQ	Repair Quotation								
SC	SC	Service Contract	✓							
SC	SRVO	Service Order	✓							
SC	SRVP	Solution Quotation								

Figure 2.47 Item Category Determination

Change Process

If you want to make further changes on an active subscription contract, SAP provides the change process framework to accomplish this. You either need to use a standard or a custom maintenance framework for business transactions (Business Transactions Maintenance Framework, also known as BTMF) change process to modify the contract line item. When a change process is executed, the system copies the latest technically

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active contract line item to a change order. According to the configuration, you can modify the change order and submit. Once the change order is submitted and released, it will be copied back to the subscription contract as a new technically active line item. The previous line-item status will be changed to **Technically Inactive**. This framework helps to maintain the change history of the subscription contracts.

The process types for executing the change processes on the subscription contract are listed in Table 2.6.

Process Type	Description
ISTA	Contract Change
ISTB	Change Technical Data
ISTC	Lock/Unlock
ISTF	Change without Order
ISTG	Create Repair Order

Table 2.6 Process Types

Next, you define the subscription change processes. You can define the change processes via menu path **Service • Transactions • Settings for Subscription Transactions • Contract Changes • Settings for the Maintenance Framework (BTMF)**. Figure 2.48 shows the standard change processes provided by SAP.

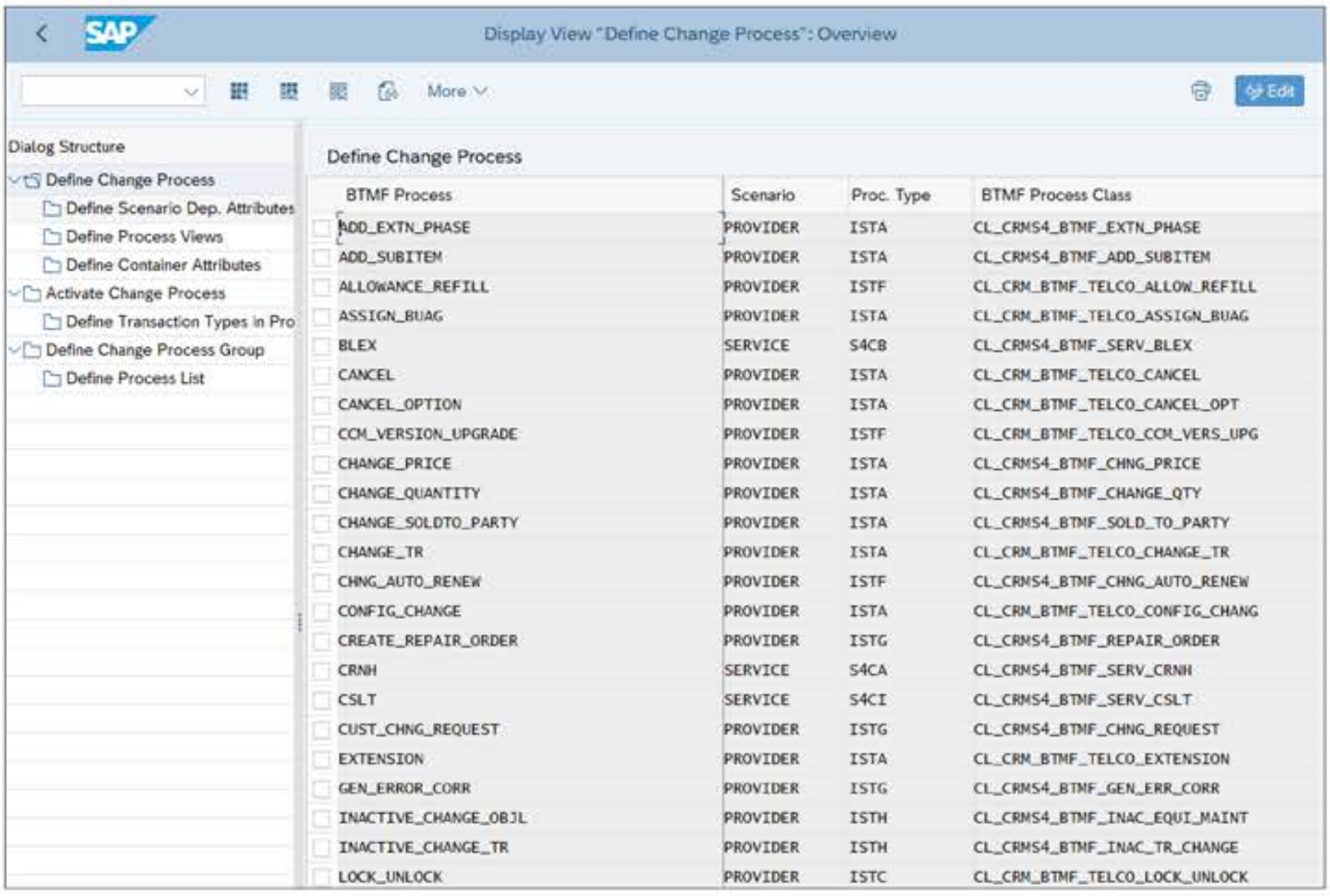

Figure 2.48 Change Process definitions

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Every change process will be associated with a process type and a process class. Process types are predefined in the system because they follow certain rules. You assign each of the change processes to a process view and define the Business Server Pages (BSPs) application view name that can be assigned to the change processes. Once the view is assigned, while executing the change process on a subscription contract, the corresponding view will be loaded, which gives you an option to make changes on the contract. You can maintain the configuration using the same menu path mentioned in the previous step. Figure 2.49 shows the view configuration for **BTMF Process: EXTENSION** provided by SAP.

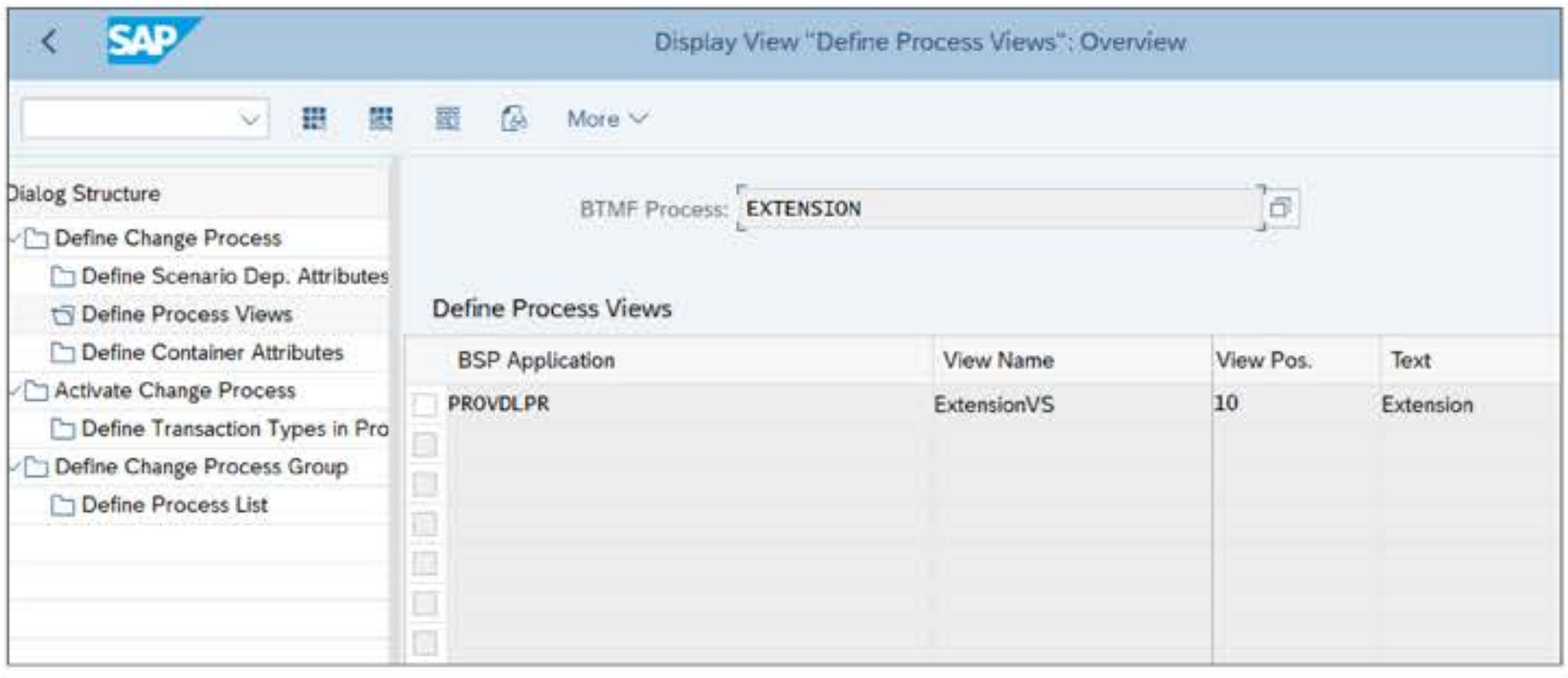

Figure 2.49 Process view definition for Change Process Extension

Next, you need to activate the change process by adding the change process defined in the previous step in the same configuration node (see Figure 2.50).

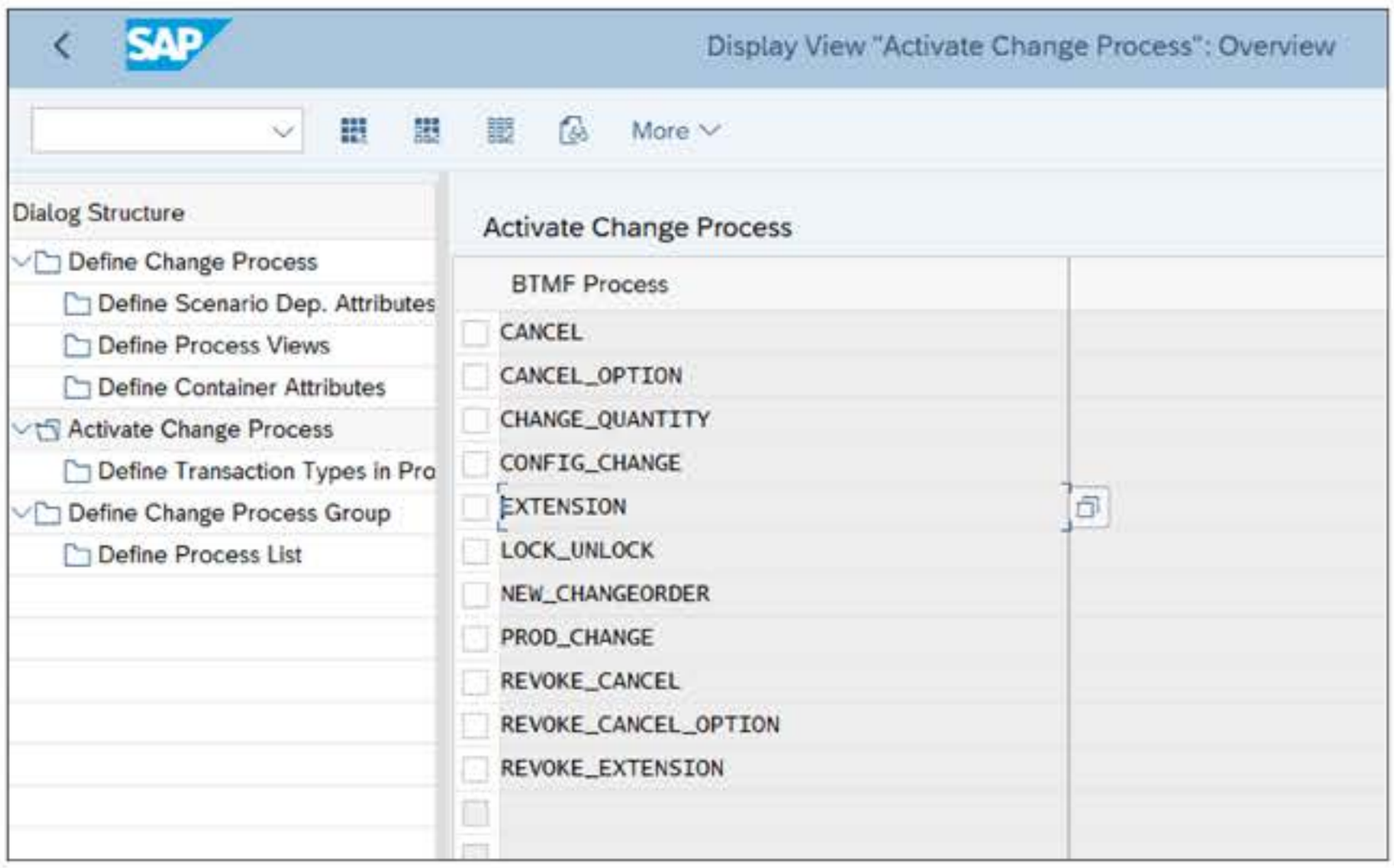

Figure 2.50 Activating the Change Process

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Next, you assign the change process to a transaction type. You do this in the same configuration node as defined in the previous step. Figure 2.51 shows the change process assigned to transaction types. Similarly, you can select the different change process created and assign it to the applicable transaction types.

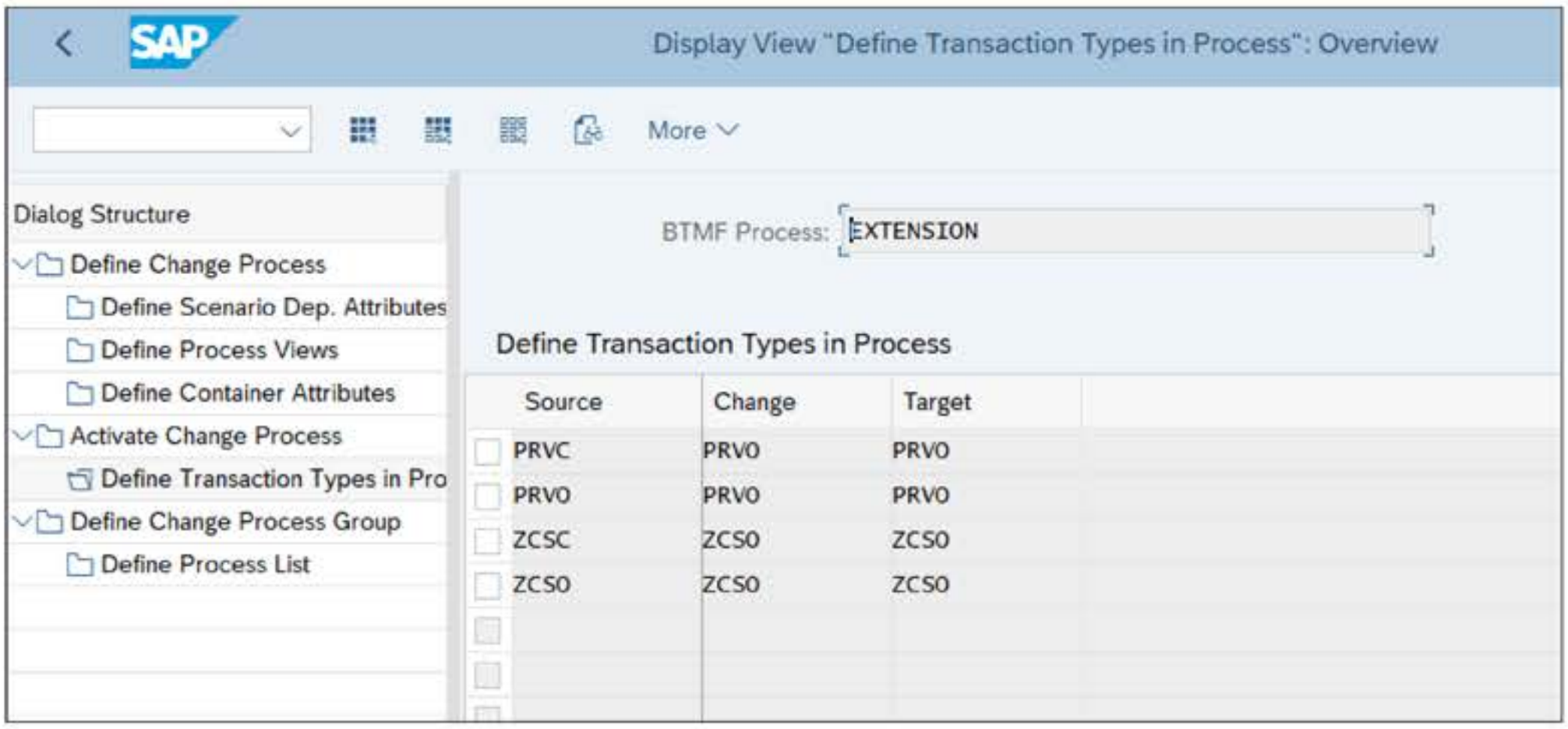

Figure 2.51 Assign Change process to Transaction Types

Next, you assign the function profile to a business role. Because users are assigned to business roles through the organizational structure assignment, this process will be inherited for the business role. You make this setting in the following menu path: **Service • UI Framework • Business Roles • Define Business Roles** (see Figure 2.52).

Figure 2.52 Assign the Function Profile to the Business Role

Select the business role and click **Assign Functional Profiles** in the left navigation bar. Search for the "ISTPROCESS" function profile ID, and assign the DEFAULT_SALES process profile, as shown earlier in Figure 2.49.

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In the recent version of SAP S/4HANA, you can also process multiple changes together using *change process groups*. This is currently enabled only through the application programming interface (API) update method. You can use this capability, for example, in a scenario where an external system such as a customer-facing self-service portal wants to allow the customer to change the configuration of their existing subscription as well as renew their subscription.

To activate this feature, you need to define the change process group. You can do so by going to menu path **Service • Transactions • Settings for Subscription Transactions • Contract Changes • Settings for the Maintenance Framework (BTMF)** (see Figure 2.53).

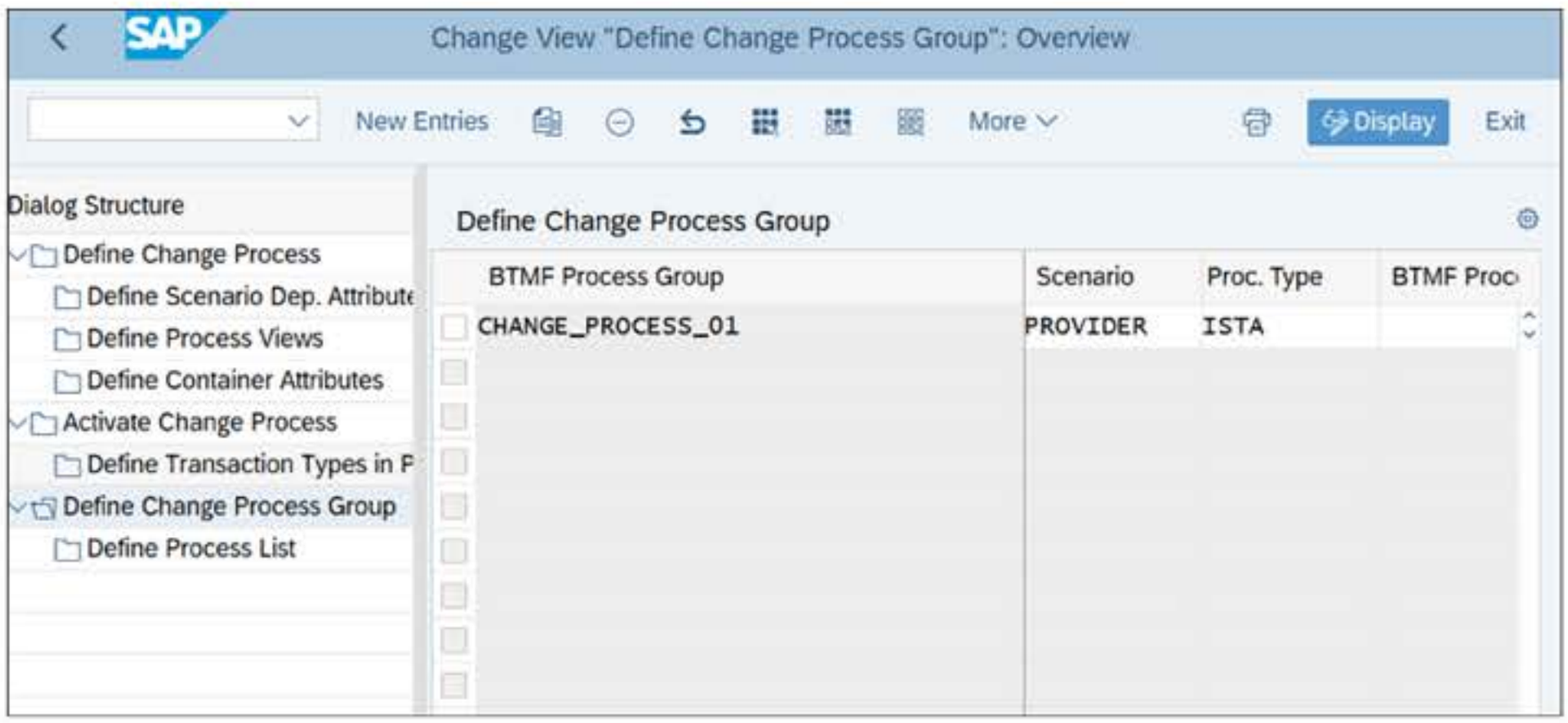

Figure 2.53 Define Change Process Group

Next, you need to assign the different change processes that are to be processed together as a part of this change process group. You do so by using the same configuration node as the previous step (see Figure 2.54).

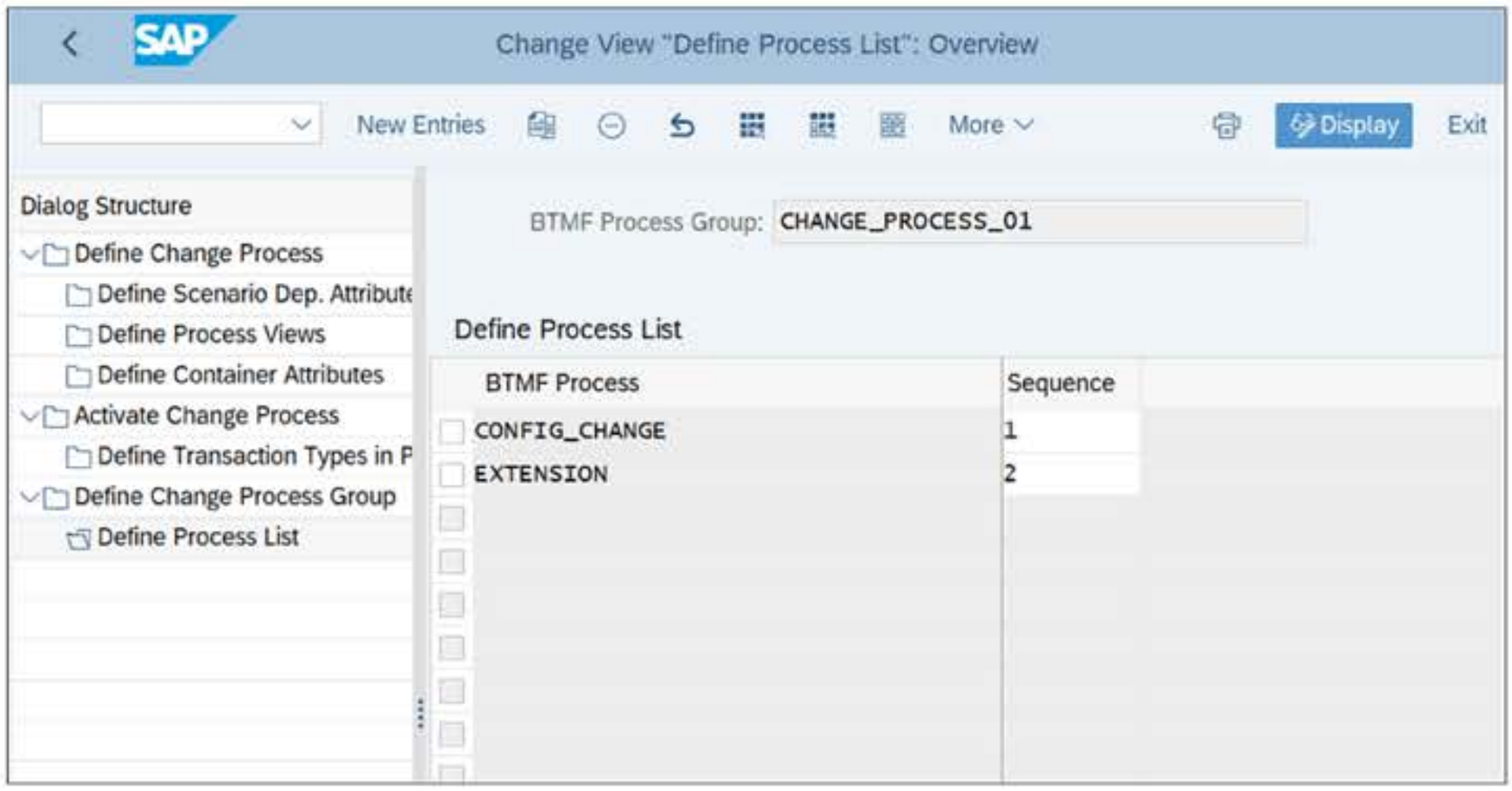

Figure 2.54 Define Process List for Change Process Group

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You can refer to section Section 2.6 to get more details on the most used change process for business use cases.

2.3.6 Master Agreements

A master agreement is an agreement between a company and its customer that governs the agreed-to commercial terms and conditions between the parties to sell different services. This serves as a template from which you create subscription orders or solution quotations as a call-off document. You define the authorized groups, products, configurations, and prices/discounts with set validity periods. The validity periods define the periods for which agreements are valid. You can only use valid master agreements from which you can create call-off documents. Authorized groups are customers that are allowed to use the agreement as a template and inherit the agreed-upon products and values in the call-off document. The system performs this validation in the background. This agreement can contain many products, each with its own set of agreed-upon values. You can import all the products or choose the ones that are relevant to the current order.

A master agreement can be saved with its line items in an **In Processing** or **Released** status. The master agreement must be in a **Released** status to be used in the subsequent call-off document. After you set the values for the product during configuration, you can set this configuration as **Fixed** or **Not Fixed**. Leaving it in a not fixed state enables the product configuration to be changed in the call-off document. The fixed state locks the product for configuration in the call-off document. Depending on the business scenario, you'll need to choose an option to leave the configuration as either **Fixed** or **Not Fixed**. The benefit of using a master agreement is that it reduces the workload of repetitive data entry during subscription order creation.

You define a master agreement in a similar manner as you defined the subscription order and subscription contract. Because you create the subsequent call-off documents from a master agreement, you set the copy controls to enable copying from master agreement to the applicable transaction types.

Discount Agreements

As part of master agreements, you can also enter into a *discount agreement* with a customer regarding the subsequent discounts. This contains the rules for calculating the discounts for the subsequent call-off documents created from the master agreement. These discounts typically aren't calculated for individual usage but for the subsequent discounting at a later point of time. An example is a volume-based discount that is calculated and applied at the end of a period. The calculation rules are contained in a discount key. You create a discount agreement in SAP S/4HANA and assign the discount keys to the discount agreement. In SAP Billing and Revenue Innovation Management, subscription order management, at the time of creating the master agreement, you have the option to assign the discount agreements to the master agreement.

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Invoicing Agreements

During the master agreement creation in SAP Billing and Revenue Innovation Management, subscription order management, you can also add invoicing agreements. An invoicing agreement can be set up to define invoicing rules related to the billing cycle, invoice recipient, and contract account to be used at the time of invoicing the customer as part of the subsequent call-off documents created from the master agreement. You can assign one or more invoicing agreements to a master agreement.

2.3.7 Partner Agreements

As traditional industries increasingly change their business model to a service-based model, a revenue-sharing business model becomes more and more important. In such a model, the service provider, who has the contractual agreement with the end customer, shares parts of the revenue with their content providers or must pay some royalties. When an end customer is buying a service, the content partner should get part of the revenue that the services provider makes with the end customer. For this reason, a service provider closes a partner agreement with the content provider. The partner agreement contains the conditions that were negotiated between the service providers and the content partner, such as the service and the revenue split. The provider that acts as a billing operator, including invoicing and collections toward the end consumer, must forward a certain amount of money to the corresponding partners or third parties.

A partner agreement order is created for a partner containing the partner product(s) for its business. When the partner agreement order is submitted, the partner agreement is created and replicated to the partner agreement and the SAP Convergent Charging provider contract for the partner. Furthermore, a customer contract is required for actual consumption. If usage is sent from SAP Convergent Charging to SAP Convergent Invoicing, this billable item is identified as relevant for partner settlement. In addition to the customer billable item, it also creates a partner chargeable item. This partner chargeable item carries a technical ID that identifies the partner provider contract in SAP Convergent Charging during charging and rating and then sends a billable item for the partner back to SAP Convergent Invoicing for further processing.

For the purposes of our example, Loren Systems partners with Martex to provide anything as a service (XaaS) services for a cost. Martex resells these services to McFarland Systems and generates revenue based on the services used. Martex pays a portion of the revenue to Loren. For this scenario to function, Martex first creates a provider partner agreement detailing the revenue-sharing conditions.

In this section, you'll create a partner provider order and customer subscription order. They both functions similarly, but a partner provider order is linked to the partner/vendor, and a customer subscription order is linked to the customer.

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Prerequisites and Setup

To set up products that can be used for partner agreements, you need to map the product types with business purpose in SAP Billing and Revenue Innovation Management, subscription order management. To maintain the setting as shown in Figure 2.55, navigate to **Service • Master Data • Settings for Subscription Master Data • Subscription Product-Specific Data • Assign Product Purpose to Product Types**.

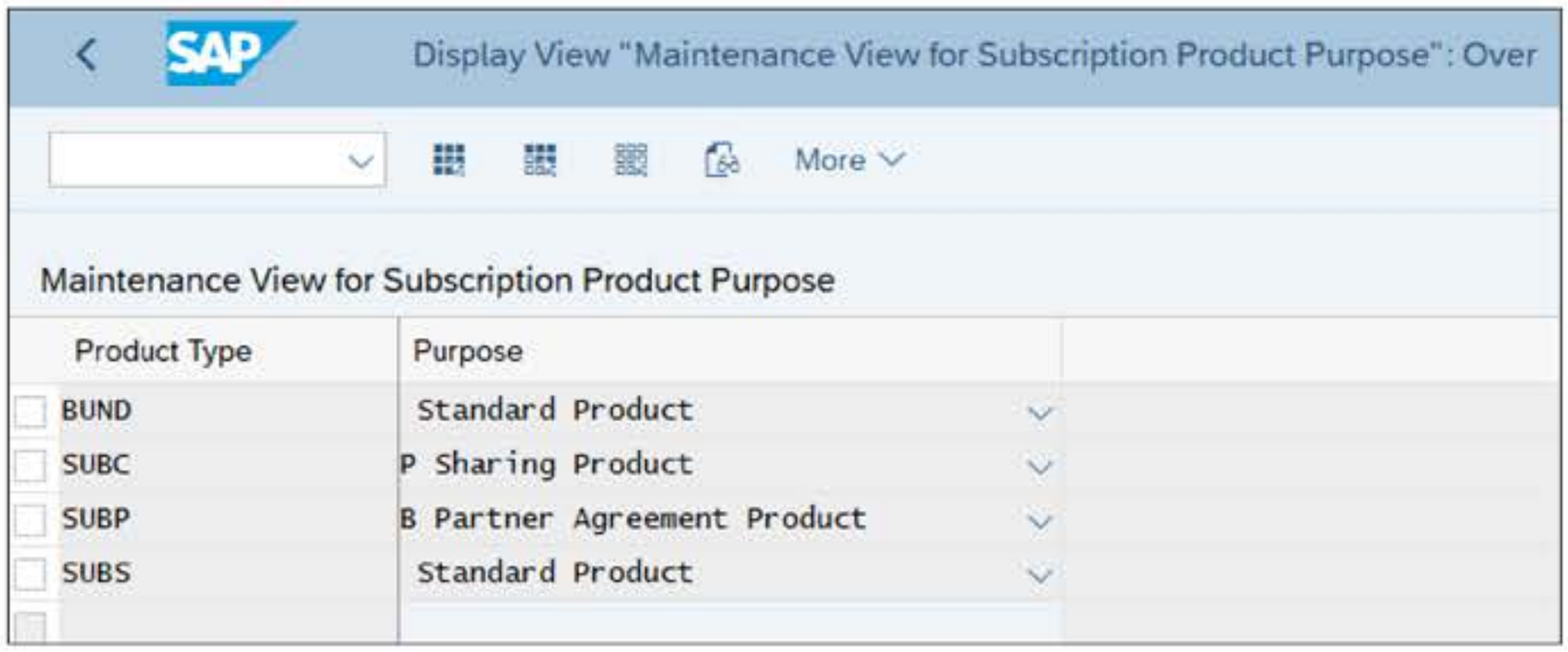

Figure 2.55 Define Product Types Applicable for the Partner Agreement Product

As part of our use case, we've created Codec 360 Solution Package (Product ID: CODEC_360_05) with **Product Type SUBP** (Subscription Partner Product). The remaining product step will be similar to setting up the subscription product described in Section 2.3.

You also must maintain a settlement rule for partner agreement orders (see Figure 2.56). To maintain the setting, navigate to **Contract Account Receivable and Payable • Convergent Invoicing • Partner Settlement • Define Settlement Rules for Partner**.

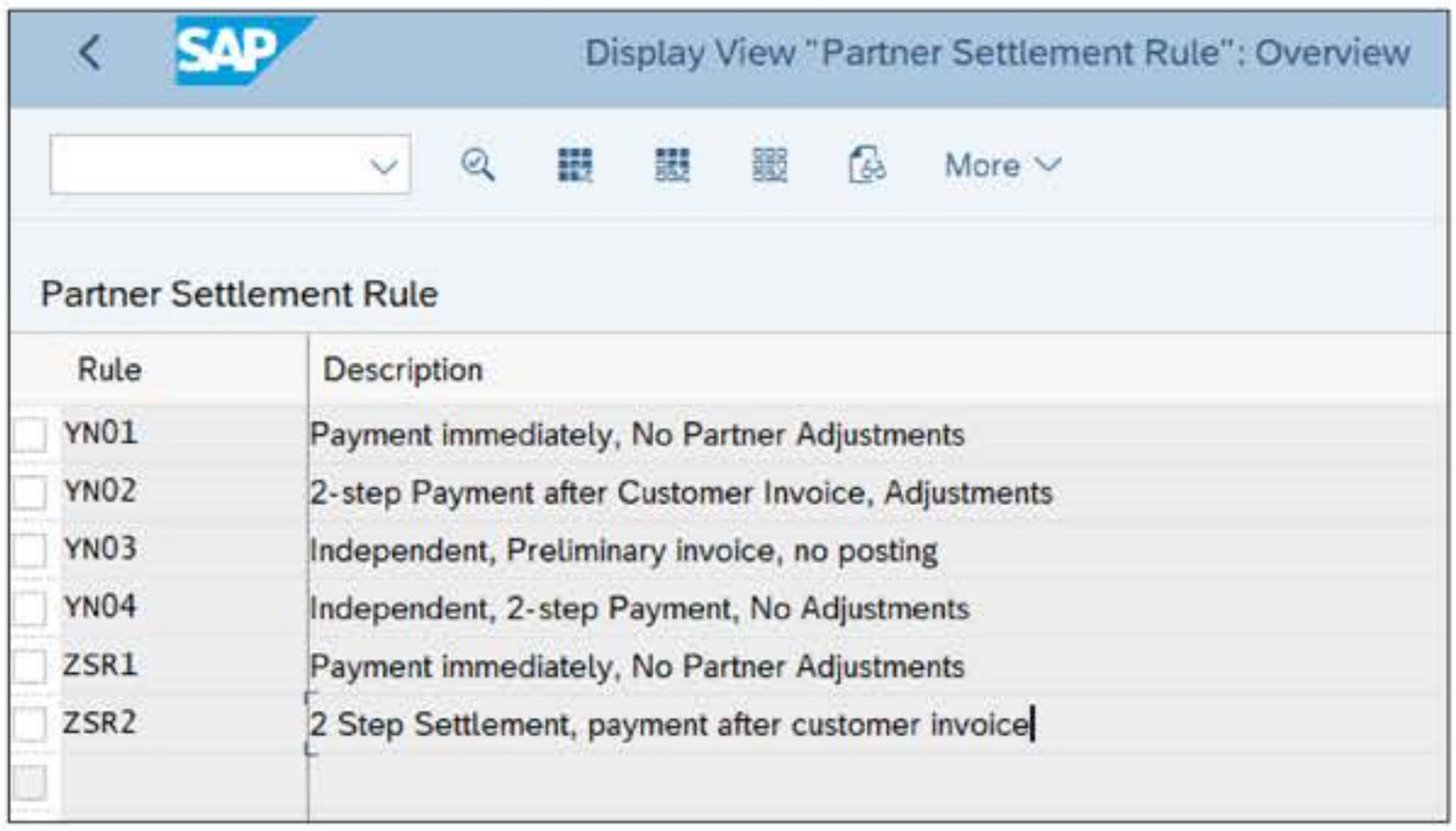

Figure 2.56 Maintaining Partner Settlement Rules in SAP Convergent Invoicing

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With all the prerequisite configuration completed, let's start with subscription partner order creation

Creating a Business Partner for the Vendor

Loren Systems will be the partner/vendor that supplies services to Martex, which in turn sells those services to McFarland Systems. You create business partners directly in SAP S/4HANA using the Manage Business Partner Master Data app. Follow these steps:

1. Log in to the SAP Fiori app using the URL provided by your system administrator.
2. Select the **Manage Business Partner Master Data** tile.
3. Click on **Create • Organization**.
4. Enter the relevant details as shown in Figure 2.57 and save.

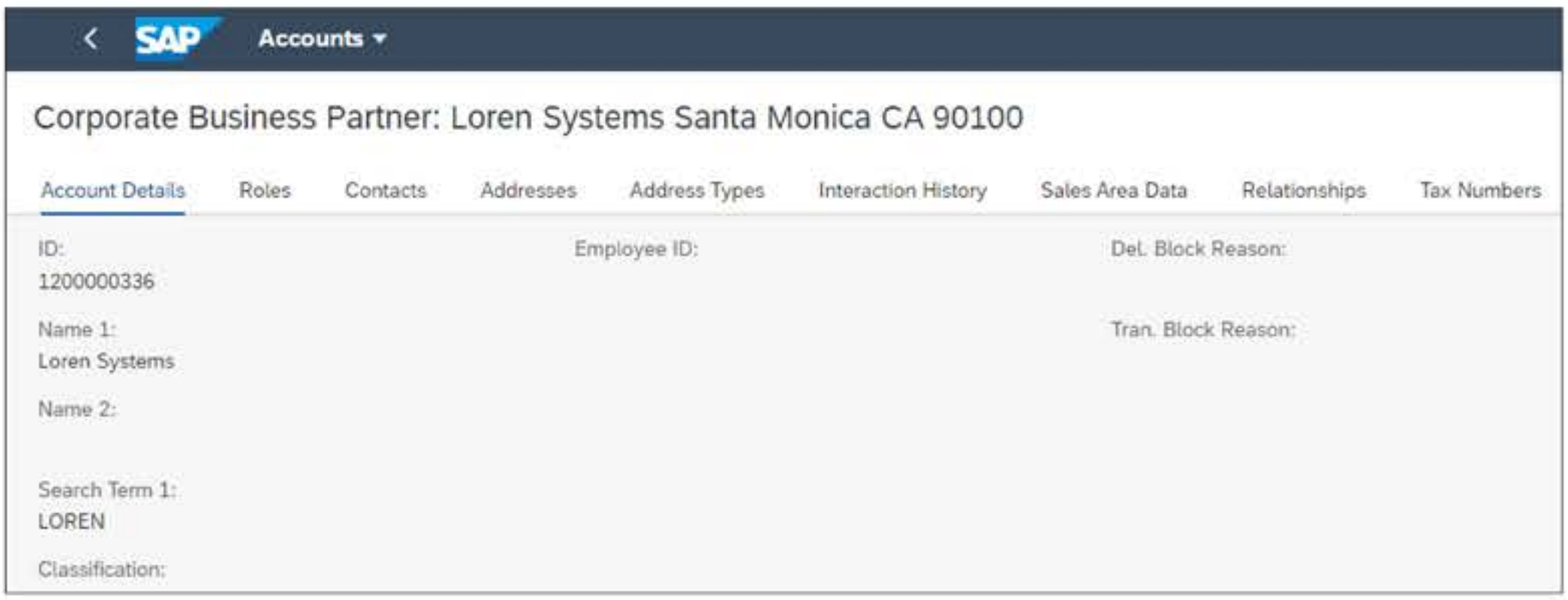

Figure 2.57 Vendor Account Creation: Loren Systems

To support SAP Billing and Revenue Innovation Management processes, the business partner should have the business partner roles to support transaction processing (see Figure 2.58). In the **Role** column, the **Contract Partner** role is mandatory to manage contract account creation for the business partner.

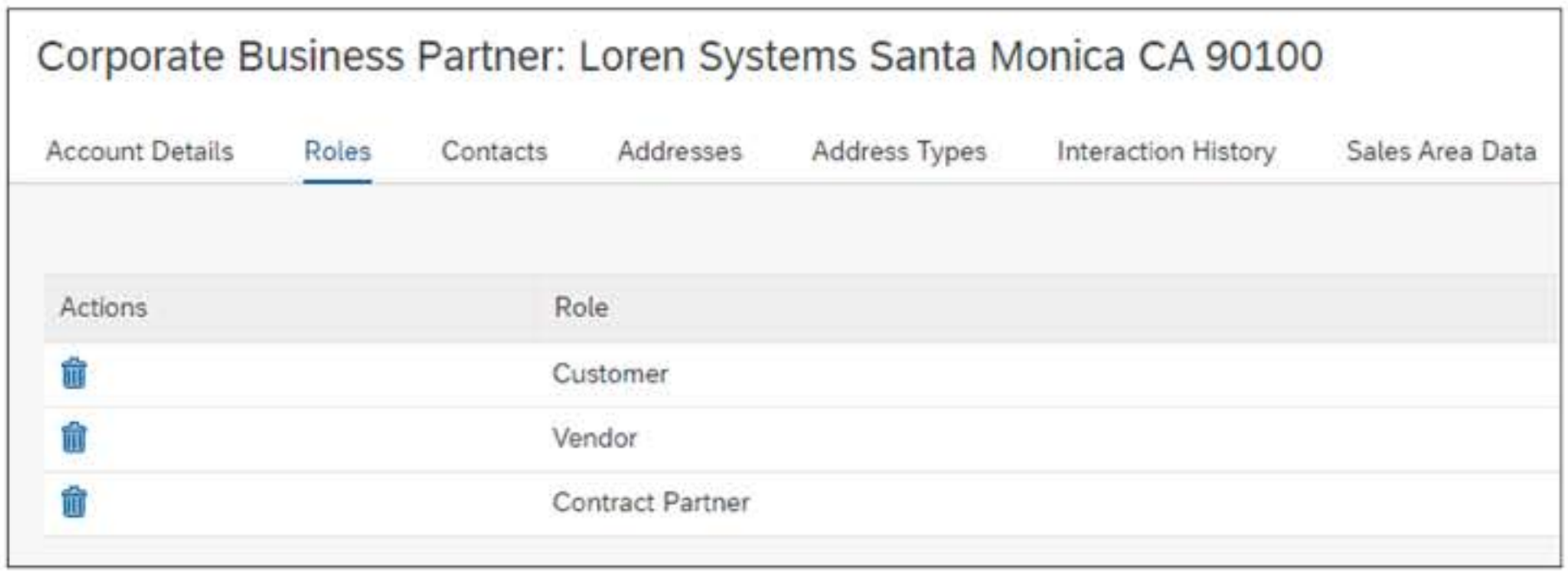

Figure 2.58 Maintain Roles for Loren Systems

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Creating Vendor Contract Accounts

In this section, let's create contract accounts for business partner Loren Systems created earlier. Contract accounts can't be created as standalone objects; they are always created for a business partner.

You create contract accounts directly in SAP S/4HANA using the Manage Contract Account app, as follows (see Figure 2.59):

1. Log in to the SAP Fiori application using the URL provided by your system administrator.
2. Select the **Manage Contract Account** tile.
3. Click on **Create** in the menu.
4. Select the **Loren Systems** business partner.
5. Select the Vendor Account class (**Customer**) and payment terms.
6. Enter all the necessary details and save.

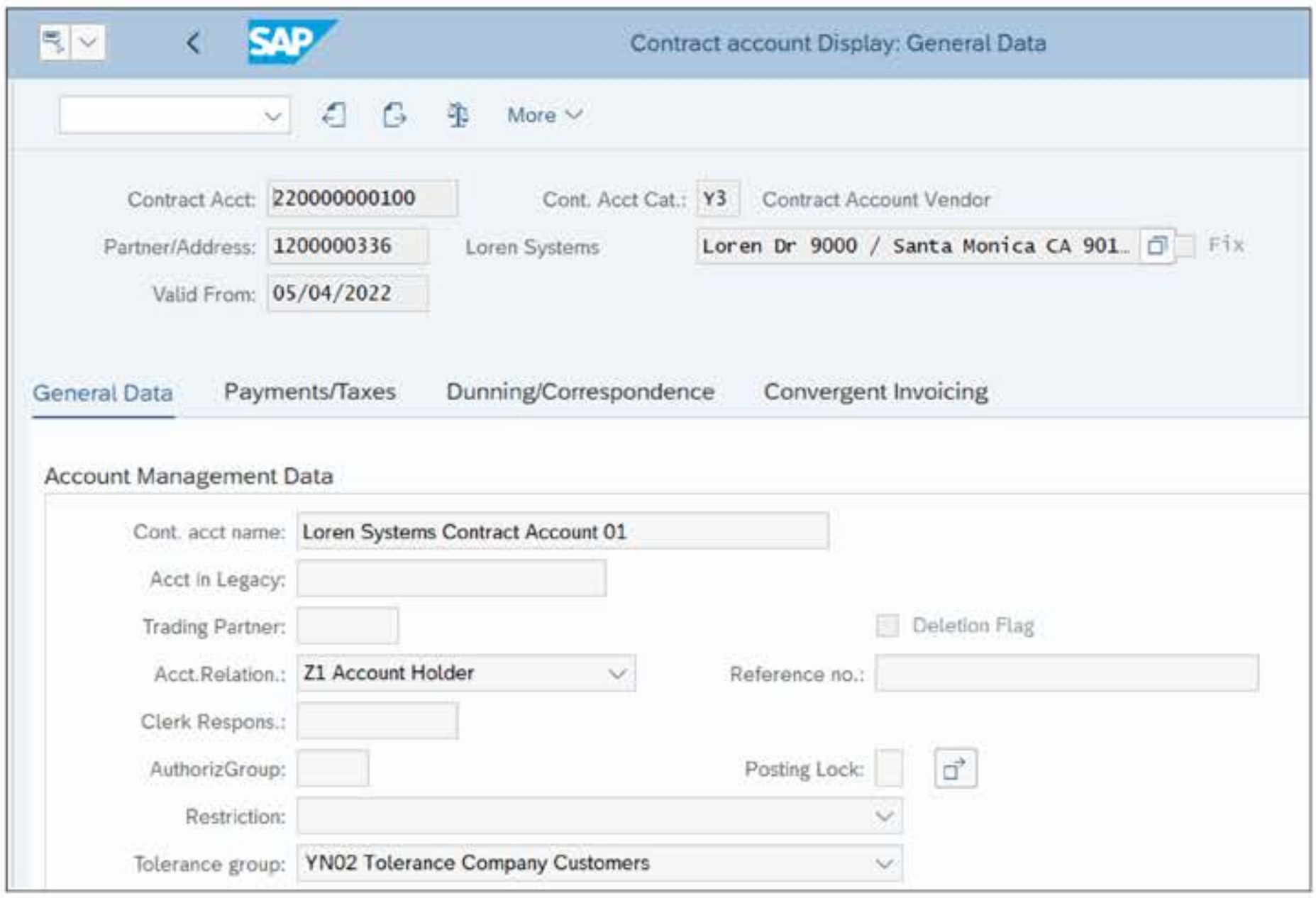

Figure 2.59 Contract Account for Loren Systems

Creating a Subscription Partner Order

Now that the master data setup is complete, you create the subscription partner order by following these steps:

1. Log in to the SAP S/4HANA Fiori application using the URL provided by the system administrator.
2. Select the **Manage Subscription Order** role.

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3. Click **New** in the **Create** window.
4. In the popup window, select **PRPO**.
5. Enter the following details:
   - **Sold-To Party**: "Loren Systems"
   - **Employee Responsible**: This is usually autopopulated based on predefined configuration. However, you can overwrite the value based on available and existing employee responsible business partners if needed.
   - **Organizational Data**: "Sales Organization"
6. In the **Items** section, enter details for the product **CODEC_360_05**, as shown in Figure 2.60.

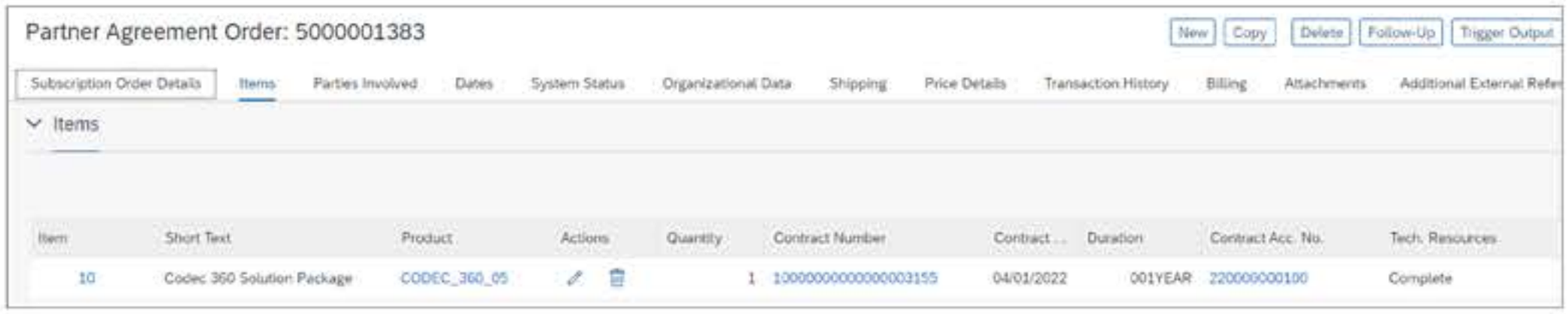

Figure 2.60 Maintain Partner Product

7. Click the **Edit** button in line item **10**.
8. Open the **Subscription-Specific Data** block (see Figure 2.61).
9. Maintain the **Settlement Rule** and save.

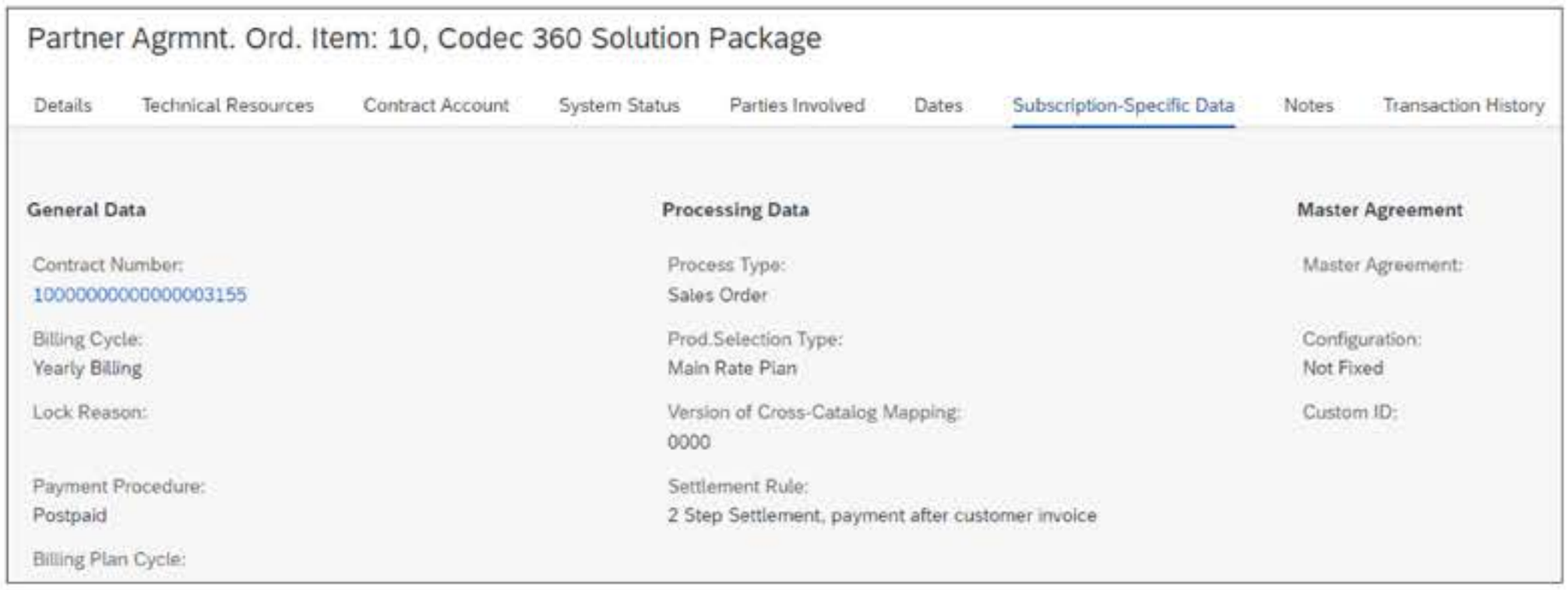

Figure 2.61 Maintain the Settlement Rule in Subscription-Specific Data

10. Click the **Save and Submit** button.
11. After clicking the **Save and Submit** button, the subscription partner agreement is created. The agreement is available in the **Items** block as a contract.
12. Click the contract number hyperlink, and you'll be taken to the partner agreement screen (Figure 2.62).

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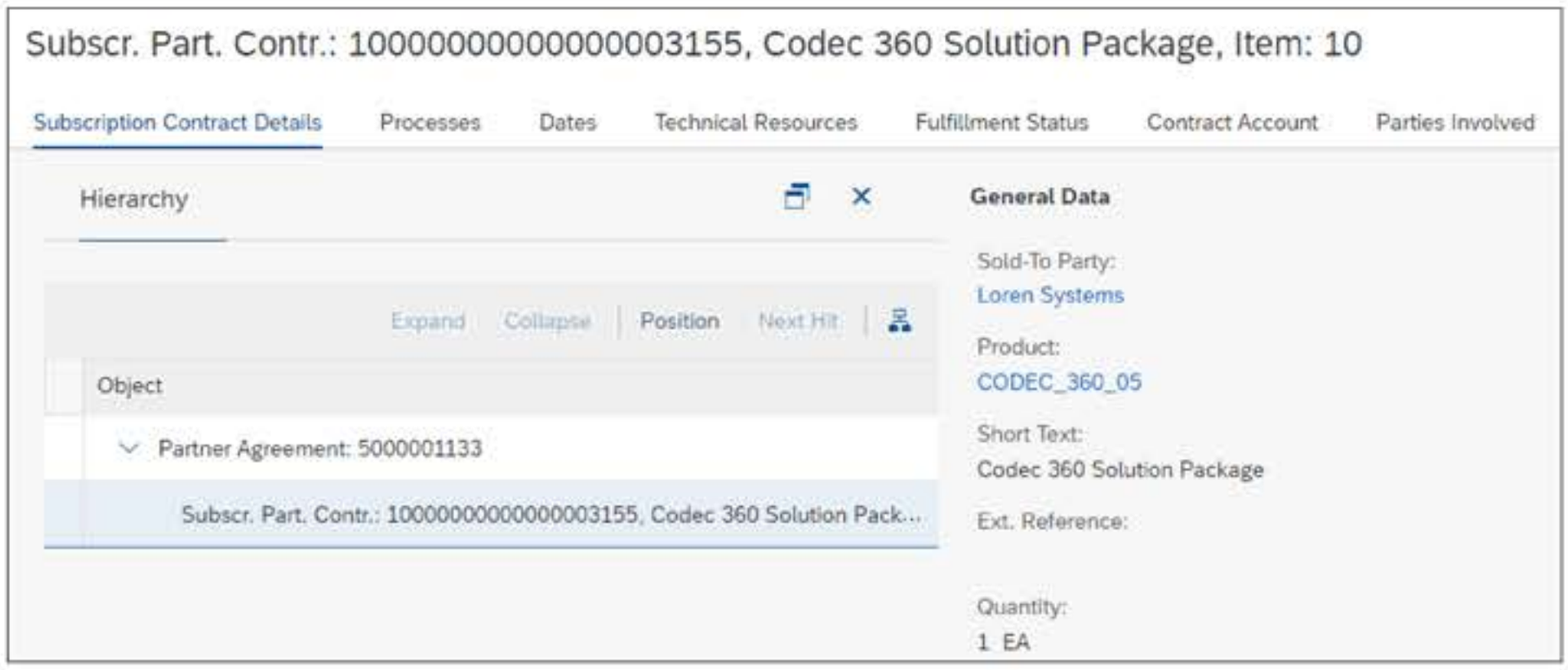

Figure 2.62 Partner Agreement for Loren Systems

The partner agreement is a secondary linked document created by the system. It can contain many vendor subscription contracts.

The **Fulfillment Status** tab provides the status of the individual distribution steps. If the status is set to **Technically Active**, the contract is successfully distributed in SAP Convergent Invoicing and SAP Convergent Charging.

2.3.8 Device as a Service: Equipment Master Integration

The device as a service (DaaS) model allows customers to have additional flexibility in the subscription- and usage-based economy and integrates the billing model with supply chain model. This also helps clients enable unique revenue streams and have better control of billing triggers. As one integrated experience, DaaS represents a step-change in supply chain management and offers significant benefits over a typical leasing model. The DaaS model is a highly desirable model for both customers and providers. Traditional hardware suppliers have a better option to enable a different monetization route and streamline leasing models. In a typical DaaS scenario, service providers will bundle hardware devices with subscription charges. The integration also helps bypass the manual approach to trigger billing for customers based on device fulfillment status to restrict revenue leakage.

Some of the important features for DaaS are as follows:

- Integrating equipment/device serial numbers with subscriptions
- Controlling customer billing based on device status/equipment activation
- Replacing and scaling equipment during the contract lifecycle
- Using equipment IDs/device serial numbers to track and bill usage

Now that you have a good understanding of the purpose and the functionality of DaaS, let's look into how Martex uses this functionality for their use case. In the subsequent section, you'll also see how to configure and enable this functionality in the system.

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Martex Device as a Service Use Case

Because Martex has adapted the DaaS process for their customer business, we'll discuss how the process can be managed using SAP Billing and Revenue Innovation Management. Remember that the DaaS process is available as part of the SAP Billing and Revenue Innovation Management standard offering since the 2020 release.

Martex has started selling a compact disc as part of their Enterprise version of Martex Workspace Solution. The disc, which is a hardware material, contains a serial number that enables the Martex Enterprise Workplace Solution once connected to the customer desktop/laptop. Because the customers won't be able to access the workplace solution until the disc is shipped to their address, the billing for the subscription will be dependent on the fulfillment of the hardware device. This would also help to bypass the current process of manual confirmation from customers and schedulers on device fulfillment to activate the subscription and would avoid any revenue leakage due to delays or errors occurring due to a manual process.

To set up this use case, we'll use the Martex Enterprise Workplace Solution subscription product (Material ID: `MRTX_MSW_ENT01`), which is a bundle product combining compact disc (Material ID: `MRTX_CD_001`) and SaaS product Martex Enterprise Service (Material ID: `MRTX_ENT_SAAS`). We also need to set up the technical resource type with category **Equipment**. To define the technical resource, navigate to **Service • Transactions • Settings for Subscription Transactions • Technical Resources • Define Types of Identification**. For the **Type of ID** field, enter "ZE", which is Martex's Equipment ID (see Figure 2.63). The **Check Time** is set as **BA Before Activation** to suppress the check during the subscription order or solution quote capturing. The **Unique** checkbox is used to check whether resources (i.e., equipment IDs) are unique for a point in time.

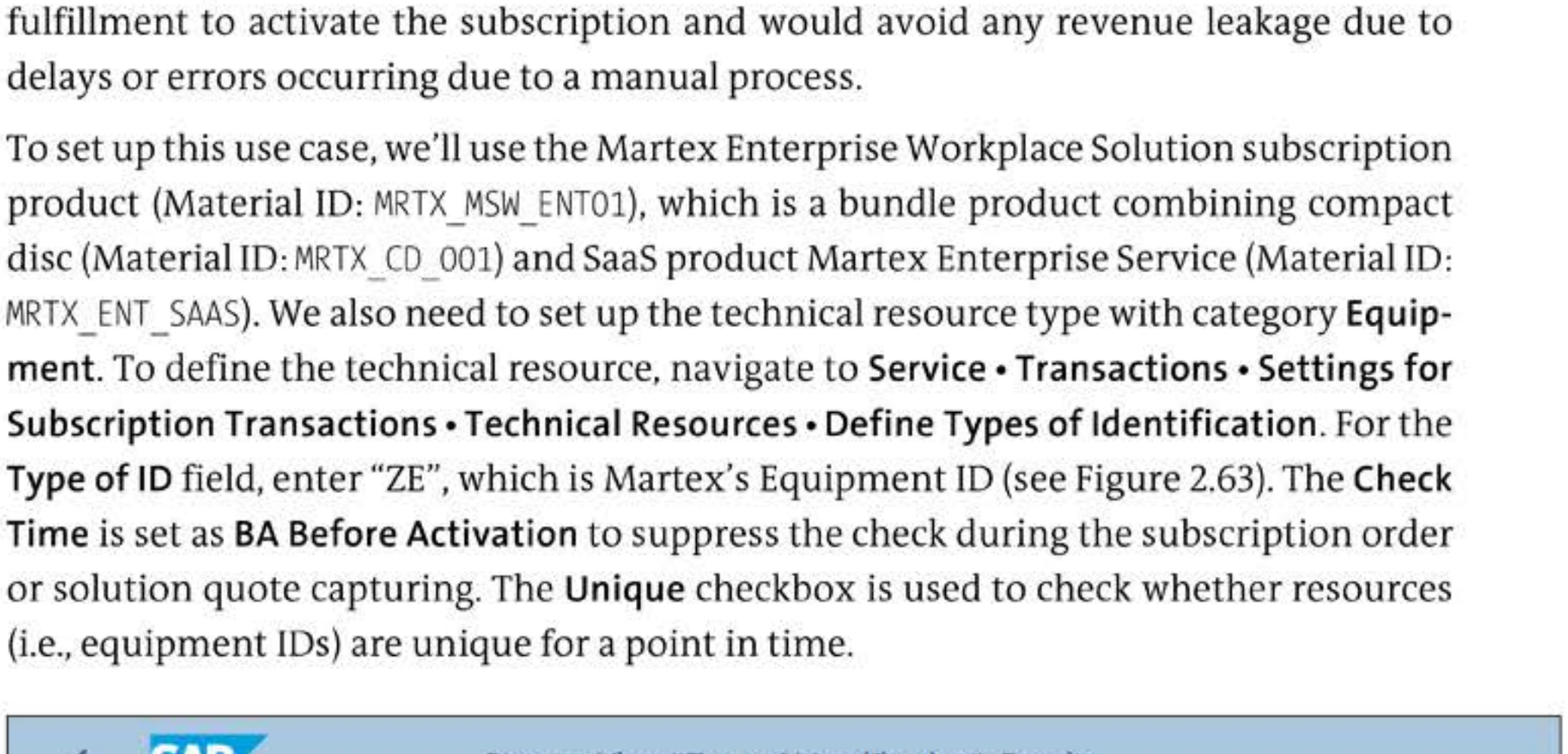

Figure 2.63 Defining Technical Resource for Martex Equipment ID

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You also need to configure the subscription-specific item categories to mark categories relevant for the object list (which is maintenance of equipment IDs). Navigate to **Service • Transactions • Settings for Subscription Transactions • Define Settings for Item Categories** to manage subscription-specific settings for item categories. In addition, **Obj List** should be checked here (see Figure 2.64).

For our Martex use case, because the contract should be active when the equipment ID is generated and shipped to the customer location, there is an additional checkbox called **Def. Distr** (deferred distribution), which allows deferred distribution even when the start date is immediate.

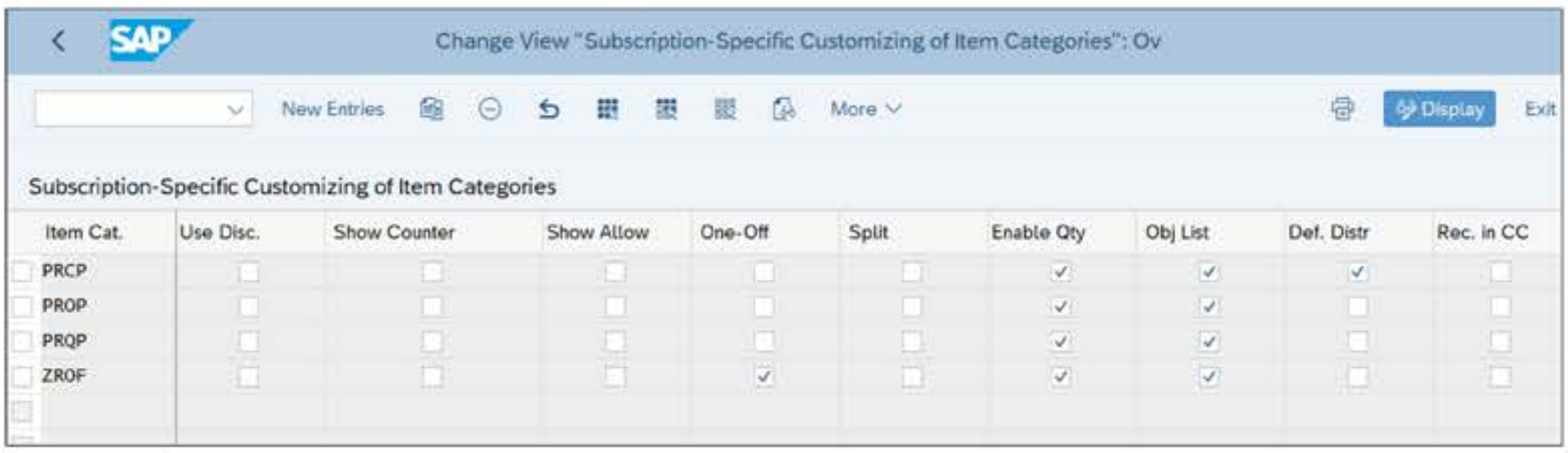

Figure 2.64 Subscription-Specific Configuration for Item Categories

The preceding settings will enable the option to define the object list in SAP Fiori for subscription items. The **Object List** option is shown in Figure 2.65. It also offers an option to search for and assign equipment numbers available for **Equipment ID** (see Figure 2.66).

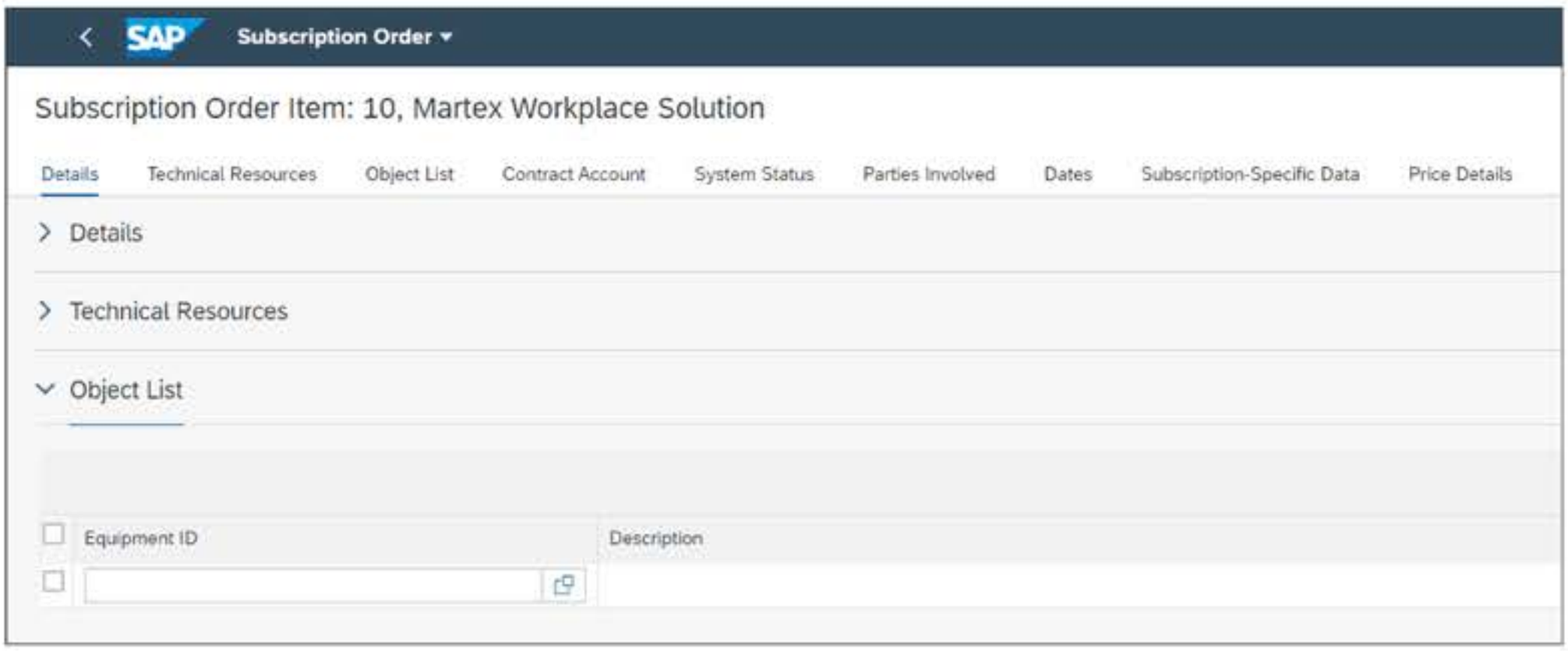

Figure 2.65 Subscription Item Object List in Transaction

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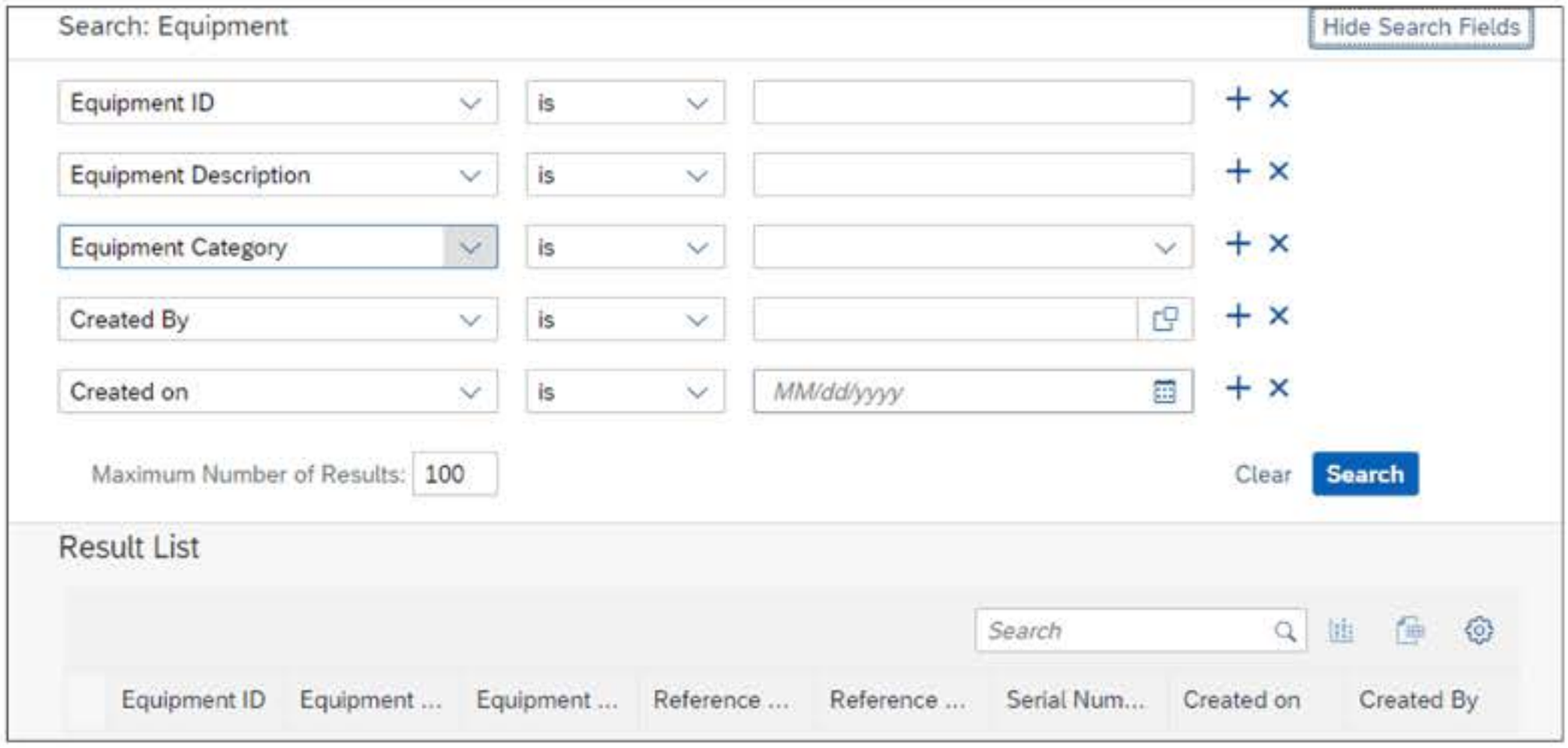

Figure 2.66 Search for Equipment in SAP Fiori

For some of the clients, where the usage is dependent on the device for tracking, equipment IDs must be distributed to SAP Convergent Charging provider contracts. This is used for calculation of usage-based charges. There is a standard feature where equipment IDs maintained in the object list are copied and distributed to the SAP Convergent Charging contract.

In addition, for our Martex use case, because equipment IDs aren't available when the order/quote is recorded, and the customer should not bill until the time fulfillment is complete, it's important to keep the subscription contract as **Technically Inactive**. Usually, the equipment ID (or device serial number) is determined in the supply chain process, and once the equipment ID is generated, it needs to be maintained in the subscription contract to process activation.

To ensure that the subscription contract is inactive until the hardware item of the order is fulfilled, you must configure the ODI framework (see Figure 2.67). The hardware device would be delivered to the customer, and the activation of subscription contract depends on the equipment delivery status. Example implementation has been provided considering this scenario where the physical device is a sales subitem of the contract main item. The sales item is processed for delivery, whereas the equipment ID, linked to the serial number of the sales product, is assigned in the delivery document. On post goods issue of the delivery document, the equipment status is updated to **ECUS**.

As part of the standard offering, ODI schema EQUI has been delivered with the steps shown in Table 2.7, which get updated based on certain events when the delivery document is created and processed.

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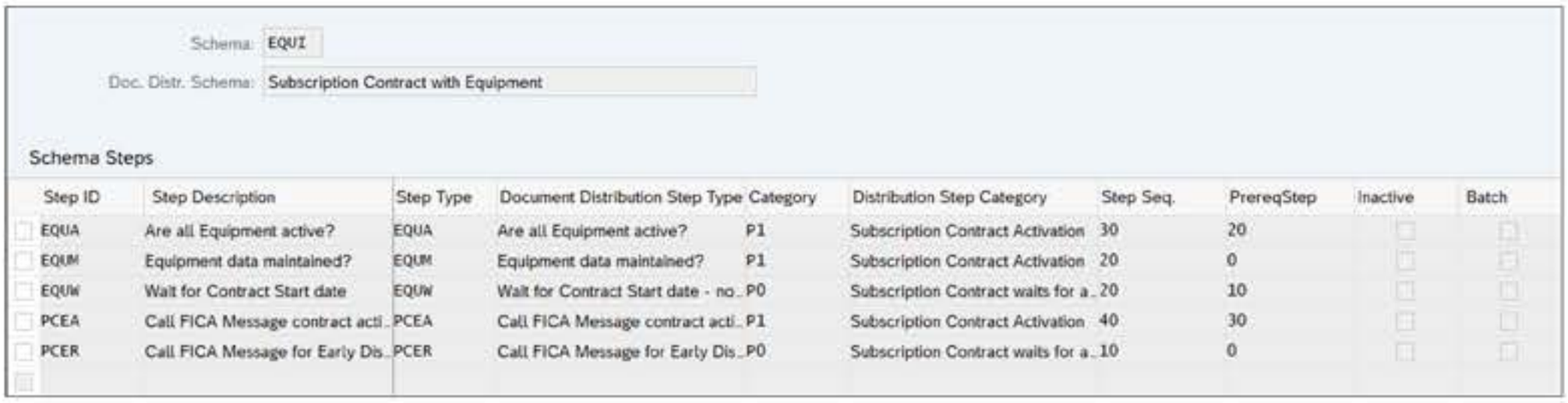

Figure 2.67 ODI Settings for Subscription Contract with Equipment

Step ID	Step Description
EQUA	*Are all Equipment active?*
EQUM	*Equipment data maintained?*
EQUW	*Wait for Contract Start date*

Table 2.7 ODI Distribution Steps Relevant for Equipment Linkage

As part of our Martex use case, the customer billing should start once the compact disc is installed in the laptop/desktop, which can be different from the standard contract start date. Or there could be some additional terms and conditions where the customer billing should start after "X" number of days after the contract start date. To support such scenarios, a new date type called Start of Billing (`CONTBILS`) was introduced. The date is synced with the bill plan "Term from," which can be used to trigger recurring billing.

Integration of Equipment ID and Status with SAP Billing and Revenue Innovation Management

When serial numbers are assigned in a delivery and the document is saved, then the event `CREATED/CHANGED` for the object LIKP is raised. If equipment is created and activated when serial numbers for a product are created, then during delivery, it's possible to identify the relevant equipment IDs. By tracking back from the delivery, it should be possible to identify the relevant sales item in the subscription order/solution quotation. By registering an event receiver and leveraging the subscription order that is still open for change or the new Maintain Equipment IDs change process, equipment IDs can be maintained on the relevant subscription contract.

When the equipment status is changed, then the event `CHANGED` for the object EQUI is raised. If this equipment, which is changed, is maintained in the object list of a subscription contract, then different processes can be triggered on the subscription contract depending on the status.

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Available Change Process for Device as a Service

Table 2.8 shows the change processes available as part of the DaaS process. The details are available in **Service • Transactions • Settings for Subscription Transactions • Contract Changes**, as shown in Figure 2.68.

BTMF Process	Process Details
INACTIVE_CHANGE_OBJL	Maintain Equipment IDs
REPLACE_EQUIPMENT	Replace Equipment
SCALE_EQUIPMENT	Scale Up/Down Equipment

Table 2.8 DaaS Change Processes

Figure 2.68 Change Processes for DaaS

2.3.9 Order Distribution Infrastructure

The ODI is an order/document distribution framework used in the SAP Billing and Revenue Innovation Management processes to distribute Subscription contracts from Subscription order management to SAP Convergent Charging and SAP Convergent Invoicing. It's highly customizable and uses the actions framework to fulfill services that are part of the subscription order and subscription contract. It can be set up as synchronous or asynchronous.

Based on your business requirements, you can adopt multiple steps during the distribution. Subscription orders aren't distributed to SAP Convergent Invoicing as a standard practice. Order distribution settings can also be designed to replicate a sales order if this contains a sales item. Only the sales item is replicated as the sales order in SAP S/4HANA. The main elements of ODI include the following:

- Document distribution categories
- Document distribution step type
- Document distribution schema
- Document distribution schema determination

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The implementation class controls the behavior of each of the elements. These classes can be adopted to meet customer-specific requirements. It's essential that you create the main element settings in the correct sequence to set up the correct distribution method.

First, you create the settings for the document distribution categories. These are the ABAP class methods that are called during runtime to execute the distribution. Although you can flexibly create your own categories, it's mandatory that you maintain the six categories listed in Table 2.9. You can make the settings for the ODI via the following menu path: **Service • Transactions • Settings for Subscription Transactions • Document Distribution • Define Settings for Document Distribution.**

Step Category	Purpose of Category
C1	Update subscription contract without using a change order
O1	Activate subscription order until a required step is completed, such as an external provisioning step
P0	Wait for activation if the contract start date is in the future
P1	Activate subscription contract if contract start date is immediate or in the past
P5	Wait for deactivation if the contract end date is in the future
P6	Deactivate subscription contract if the contract end date is immediate

Table 2.9 Settings for Document Distribution Categories

Figure 2.69 shows the step categories and the respective implementation classes assigned to them. Each step performs a specific function.

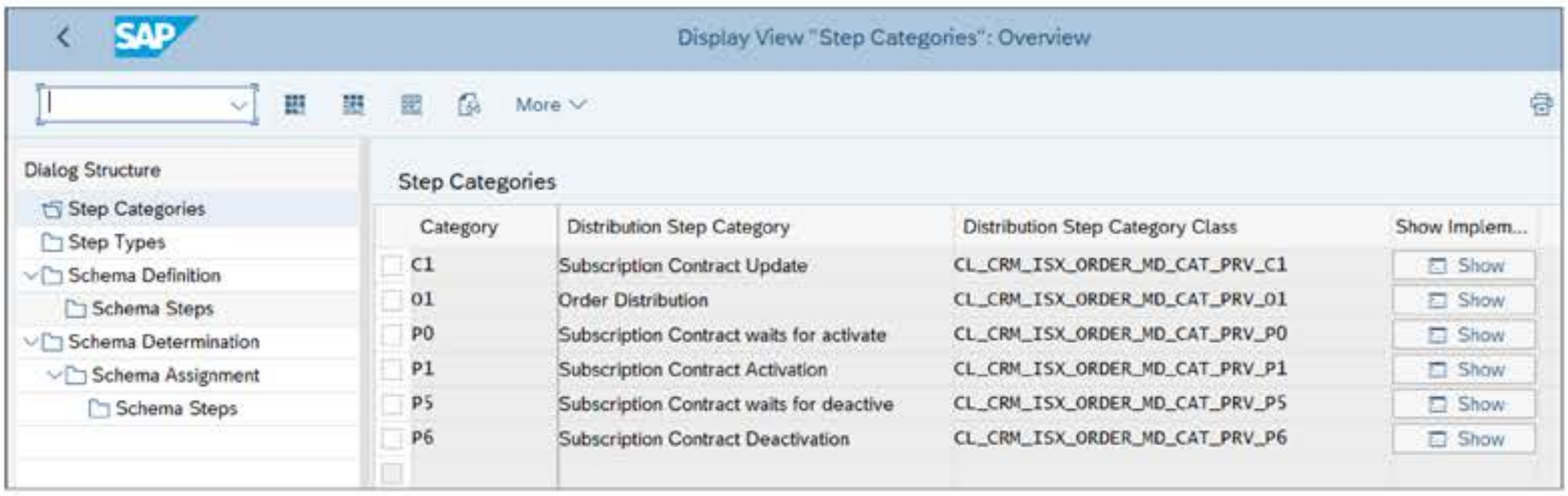

Figure 2.69 Document Distributions Step Categories

Next, you configure the step type, which defines a single step in the entire document distribution. This is driven by the combination of the distribution category and

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document distribution step type class. This class determines how the distribution category is executed.

Figure 2.70 shows the step type settings that define the document category and the step type class.

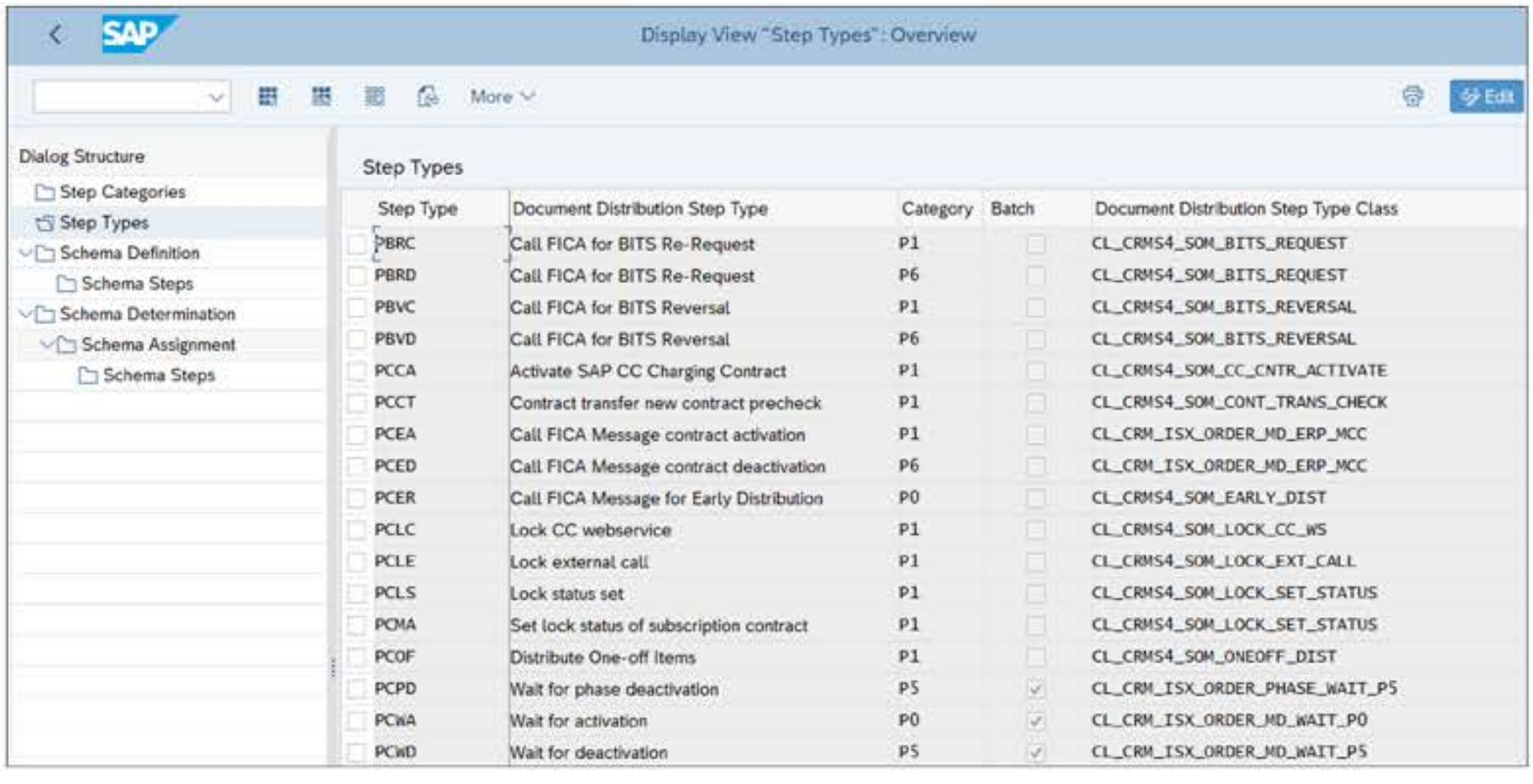

Figure 2.70 Document Distributions Step Types

Next, you define the distribution schema. The schema contains the distribution steps that are assigned to it. When called during runtime, the schema executes the steps within it. The schema is executed as a distribution schema class, which is an ABAP function.

Figure 2.71 shows the distribution schema definition that has the class assigned.

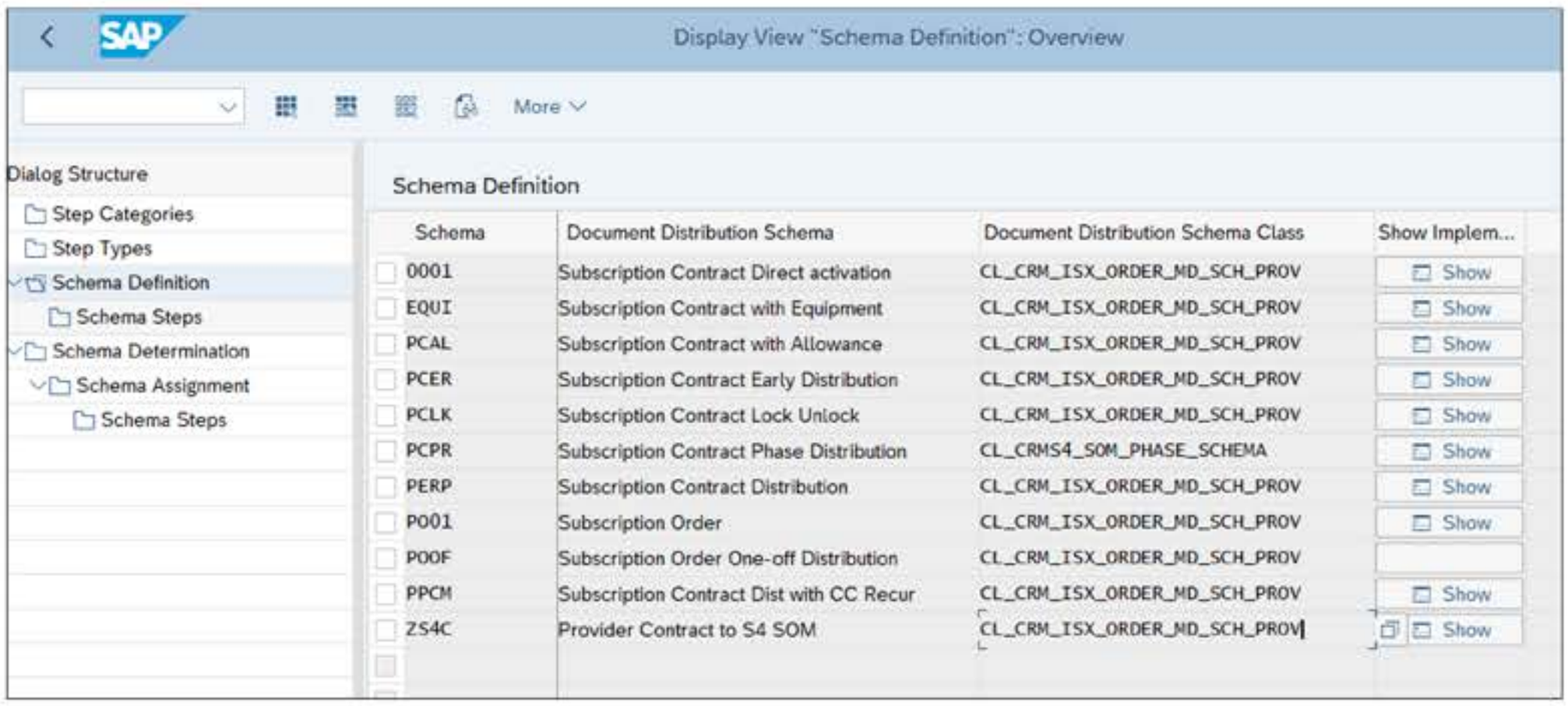

Figure 2.71 Document Distributions Schema Definition

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Next, you assign the previously defined step types to the schema definition. Here you assign the ones that are relevant for your distribution schema. Select the schema and add the step types. Figure 2.72 shows the schema definition with the assigned step types.

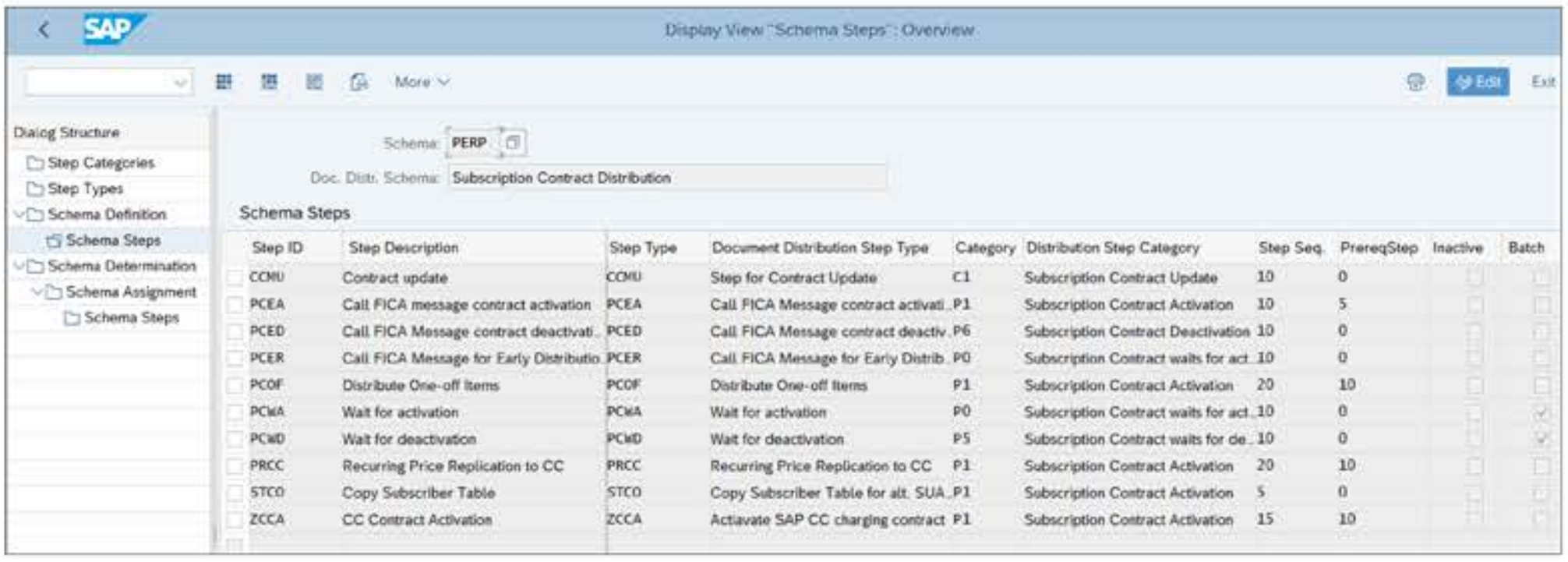

Figure 2.72 Document Distributions Schema Definition with Step Types

Next, you define the schema determination. The culmination of creating the previous steps is to assign the schema to the contract distribution. The schema determination drives the step types for the transaction type and item category combination. Figure 2.73 shows the schema determination for standard transaction type and item categories.

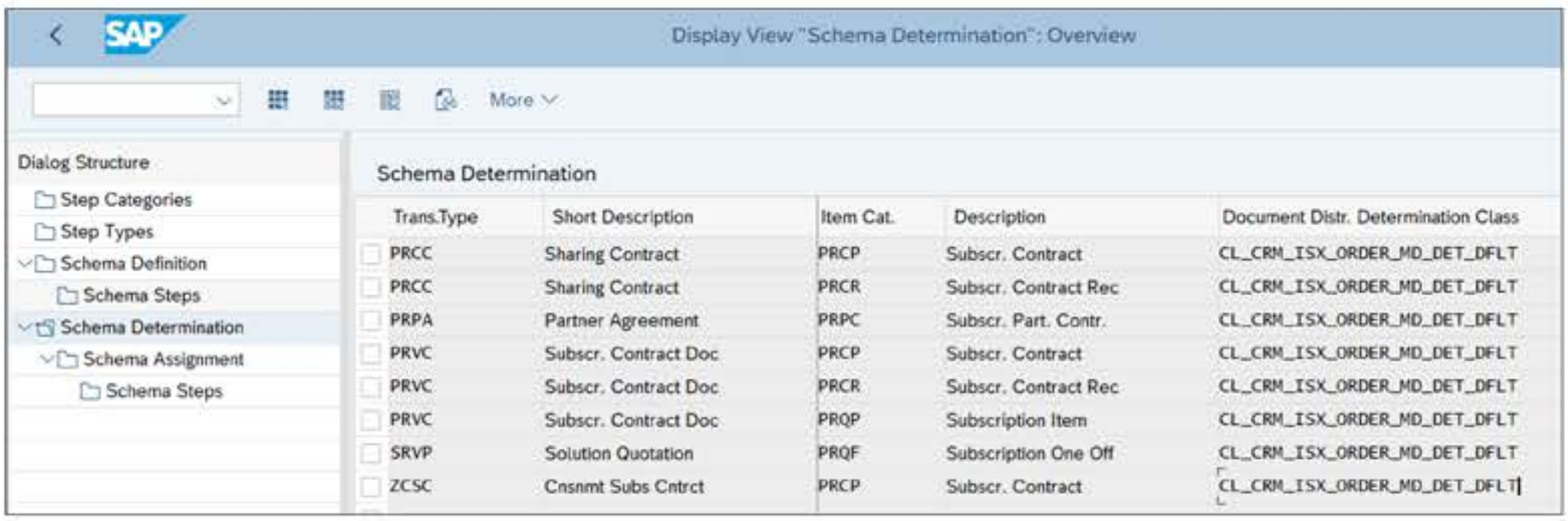

Figure 2.73 Document Distributions Schema Determination

Next, you need to assign the schemas to the schema determination. Select the combination of transaction type and item category and then double-click **Schema Assignment**. You can assign multiple schemas to one schema determination. Assign the **Default** flag to the schema you want to be the default during runtime. Figure 2.74 shows the assigned schema for the schema determination for the **PRVC/PRCR** combination.

Next, select **PERP** and double-click **Schema Steps**. The steps assigned for this schema are automatically derived for the schema determination. Figure 2.75 shows the assigned schema steps for the schema determination for the **PRVC/PRCR/PERP** combination.

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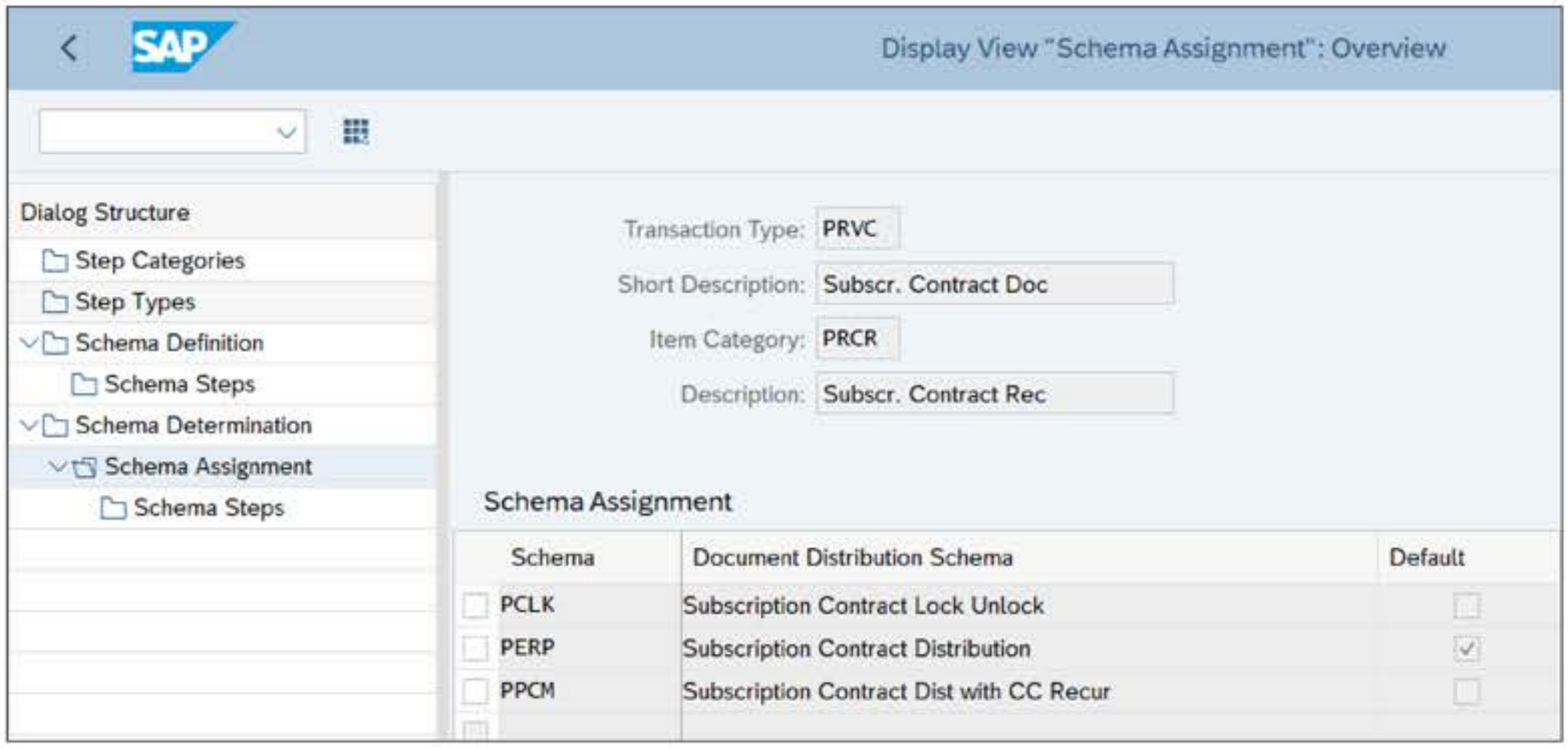

Figure 2.74 Document Distributions Schema Determination/Schema Assignment

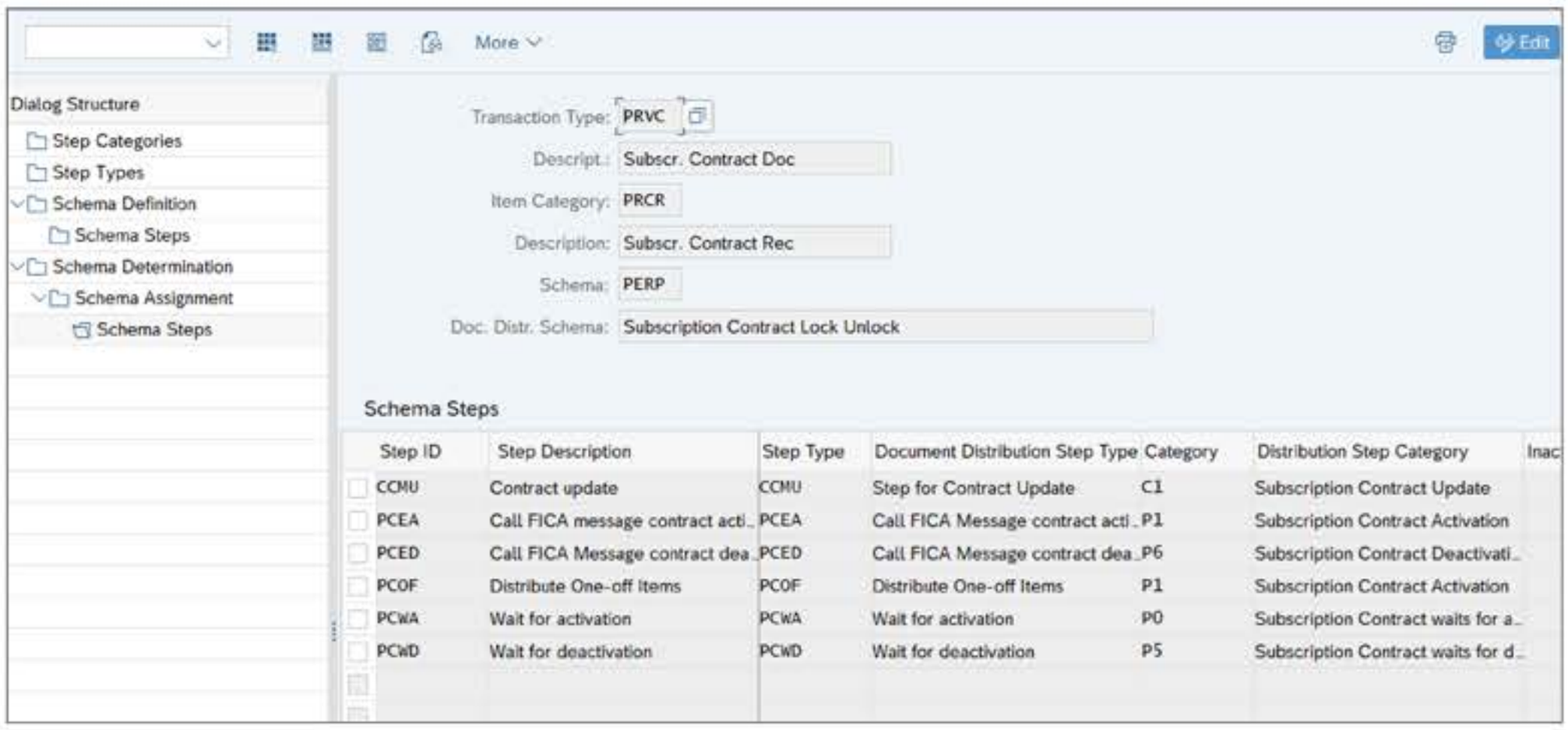

Figure 2.75 Document Distributions Schema Determination/Schema Assignment/Schema Steps

2.4 Using Subscription Order Management

In this section, we'll walk through the steps to use SAP Billing and Revenue Innovation Management, subscription order management. Using the configuration steps that we've completed so far, you can use the master data required for SAP Billing and Revenue Innovation Management, subscription order management, and create transactional data such as solution quotations, subscription orders, and subscription contracts. These contracts will replicate to SAP Convergent Invoicing and SAP Convergent Charging and thus be in sync across the three distributed but natively integrated systems.

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2.4.1 Creating Master Data

As a preliminary step, create the following master data objects, as described in the following sections:

- Organizational structure
- Business partner
- Contract account

Creating the Organizational Structure

The organizational unit is a hierarchical structure within an organization set up with different levels, each with its own functionality. As a preliminary step to create any transactions, it's mandatory that the organizational units are created in SAP Billing and Revenue Innovation Management, subscription order management. Organizational units can also be replicated from the enterprise structure so that they are in sync or can be created manually and aligned to each other as a post activity.

For our example, we'll create an organizational structure manually and align it to the organizational unit in SAP S/4HANA. First, create the root organization directly using Transaction PPOMA_CRM. Once the root organization is created, you can add additional organizational units as needed. To maintain organizational units without using SAP S/4HANA Sales integration, follow these steps:

1. Log in to the SAP S/4HANA GUI using Transaction – PPOCA_CRM.
2. In the **Create Root Organizational Object** popup that appears, define the validity period of **Root Organizational Unit**.
3. Enter the name of the root organizational unit and the following details, and then select **Save**. Because it's a root organization, leave the functions unassigned (see Figure 2.76):
   - **Basic Data**
   - **Address**

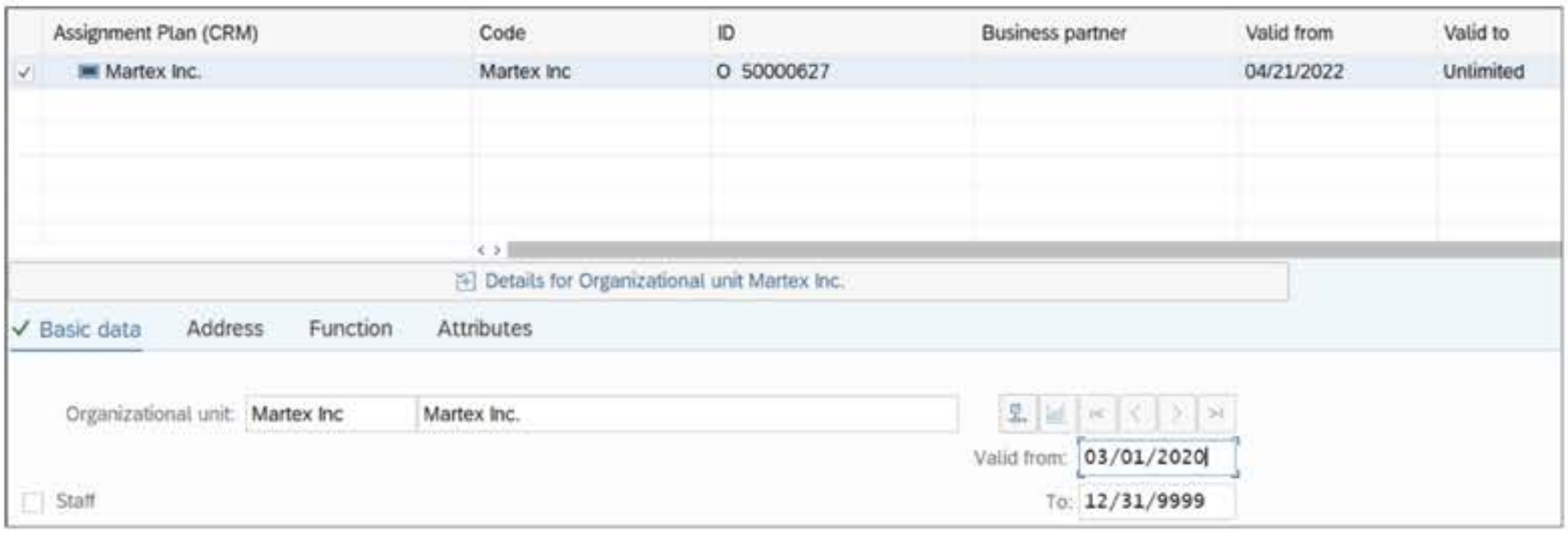

Figure 2.76 Defining the Martex Root Organization

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4. Select the newly created root organizational unit, and then click **Organizational Unit** to create a new organizational unit.
5. Enter the name of the new organizational unit and the following details, and save to add a subordinate organization to the root organization:
   - **Basic Data**
   - **Address**
   - **Valid from/To**
   - **Function**
   - **Attributes**

   The function activation ensures the type of organization this unit is set to function as. For this example, set the type as a **Sales Organization**. If the organization will be used to manage solution quotations with service processes (service order, service contract), then use the service function.

   In addition, make sure to check **Respect in Determination** in the **Attribute** tab after all attribute information is maintained. The attributes define the specific values that this sales organization will possess during runtime.

   Enter the attributes for the sales organization as shown in Figure 2.77.

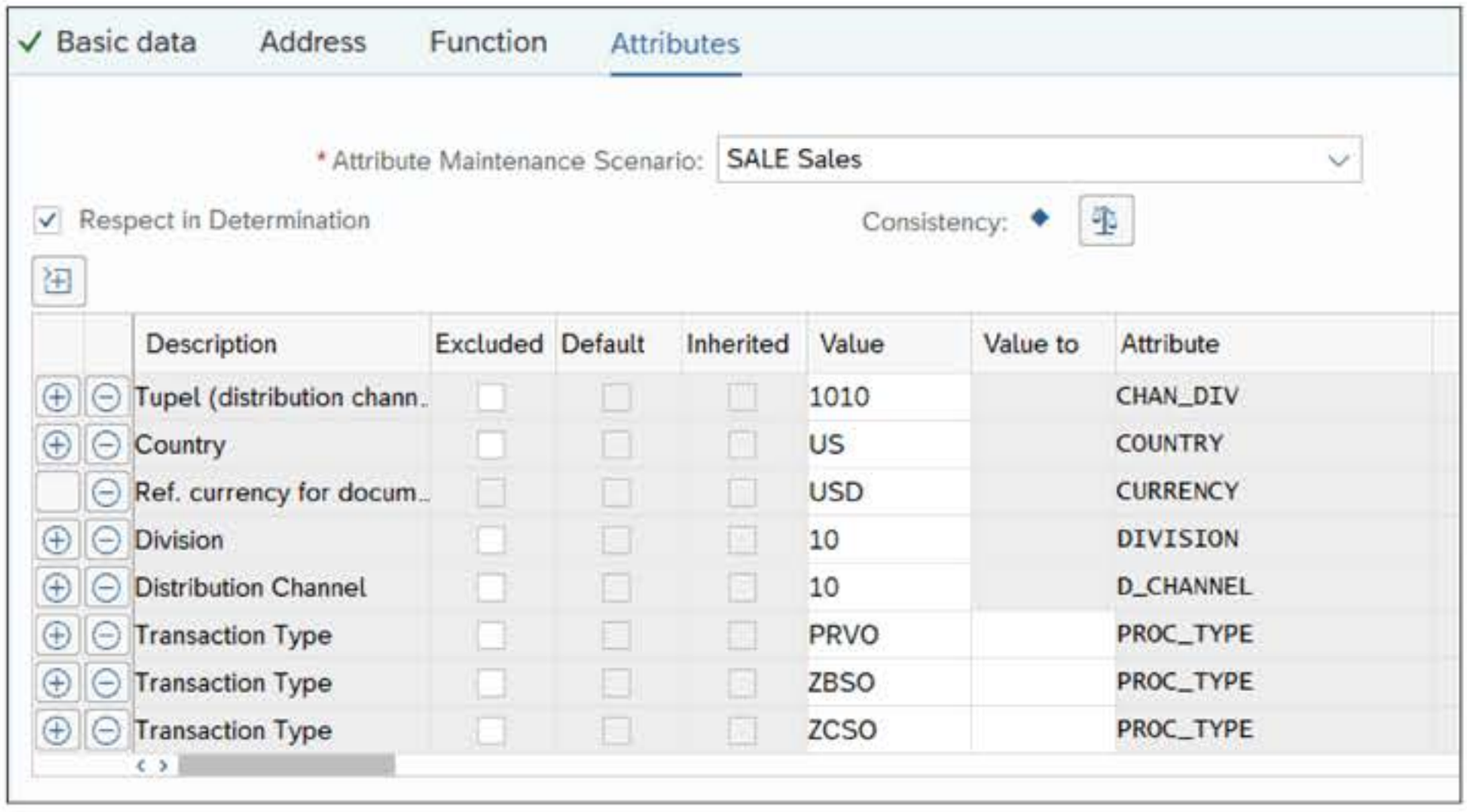

Figure 2.77 Martex Sales Organization Attributes

6. Assign the Martex sales organization to the SAP S/4HANA sales organization manually.

Based on your organizational hierarchy, you can define the structure under the root organization. For Martex, we've defined our root organization as **Martex Inc.** and

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defined three Martex organizational units as **Martex US Inc.**, **Martex EMEA Inc**, and **Martex APAC Inc.**, as shown in Figure 2.78.

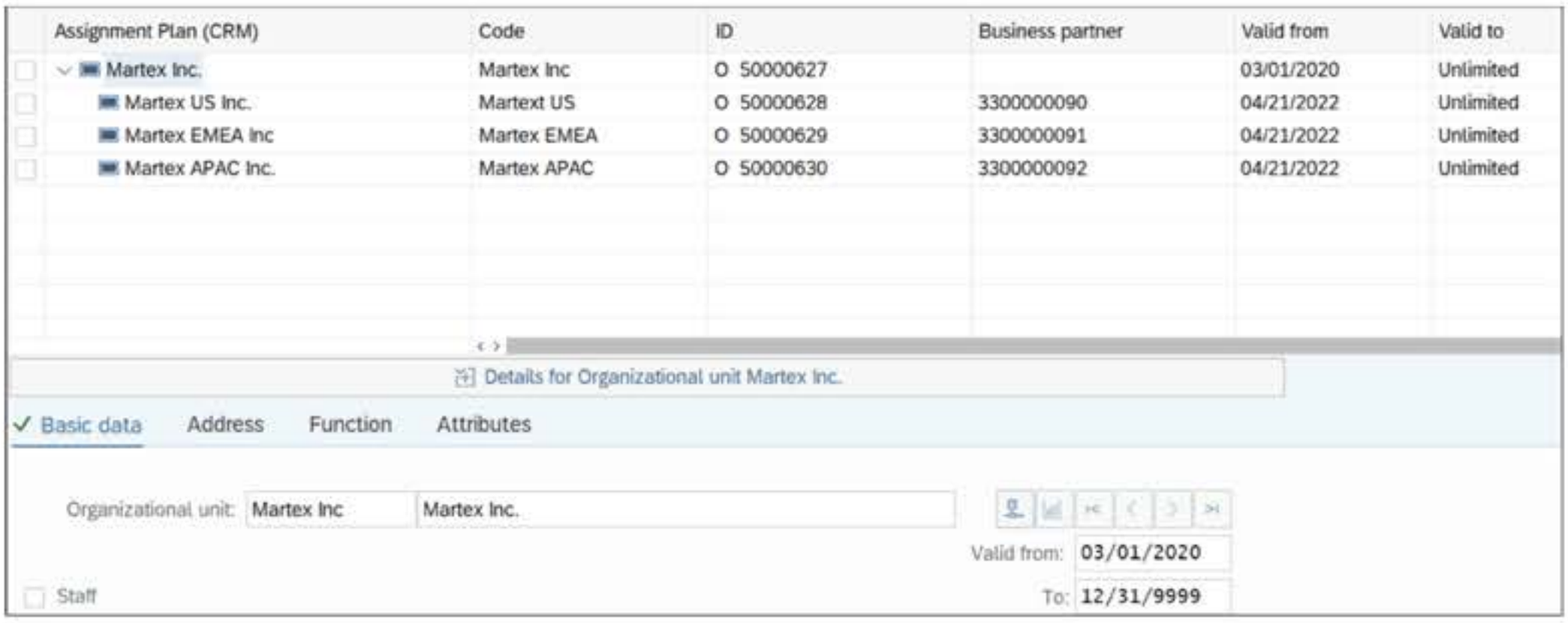

Figure 2.78 Martex Organizational Hierarchy

You can also copy organizational data such as distribution channels, divisions, and sales areas with the relevant text from SAP S/4HANA Sales to SAP S/4HANA for Service using ALE. You can display the sales structure for SAP S/4HANA Sales using a report in SAP S/4HANA Service. The details of the process are maintained under **Master Data • Organizational Management • Data Transfer • Copy SD Sales Structure**.

Creating Business Partners

In this section, you'll create two business partners to serve as your end customer and your vendor. McFarland Systems will be your customer (sold-to party), a consumer of services from Martex.

You create business partners directly in SAP S/4HANA using the Manage Business Partner Master Data app. Follow these steps:

1. Log in to the SAP Fiori application using the URL provided by your system administrator.
2. Select the **Manage Business Partner Master Data** tile.
3. Click on **Create • Organization**.
4. Enter information such as name, address, and contact details as shown in Figure 2.79, and save.

To support SAP Billing and Revenue Innovation Management transaction processes, the business partner should have the following business partner roles (see Figure 2.80). The **Contract Partner** role is mandatory to manage contract account creation for the business partner.

Be sure to extend the business partner to the relevant sales area to support transaction processing (see Figure 2.81).

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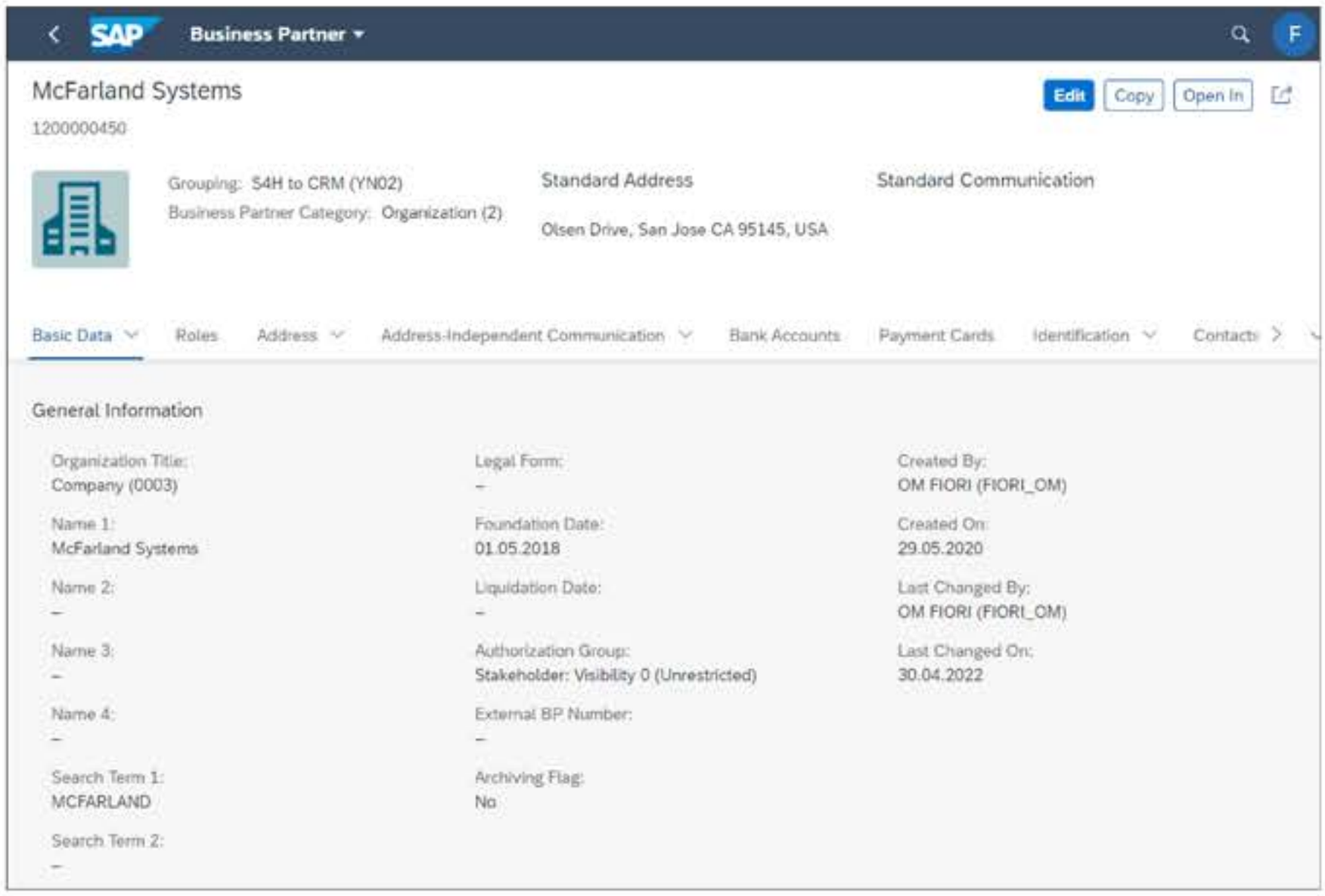

Figure 2.79 Business Partner Details: McFarland Systems

Figure 2.80 Business Partner Roles: McFarland Systems

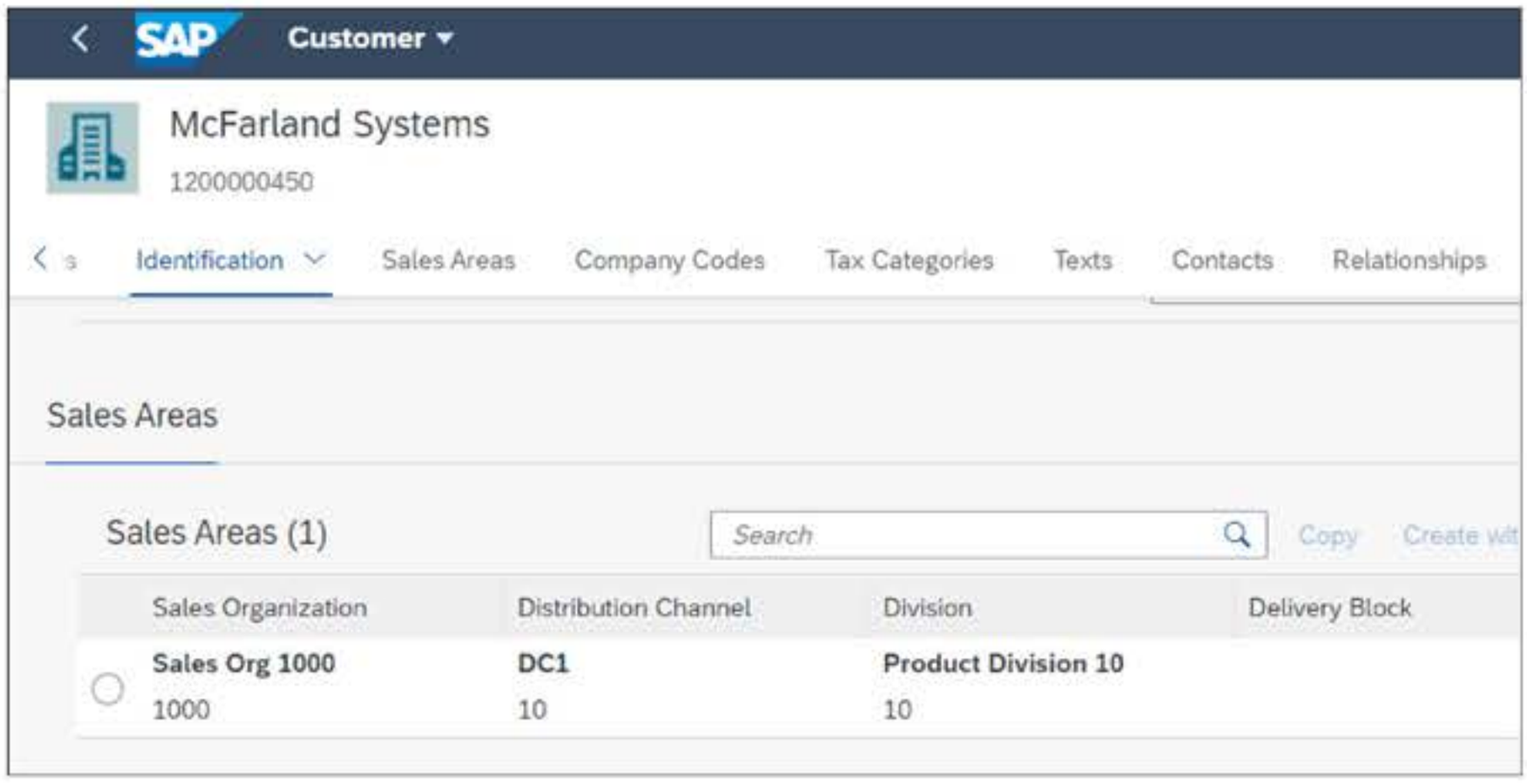

Figure 2.81 Business Partner Sales Area: McFarland Systems

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Creating Contract Accounts

In this section, we'll create contract accounts for the McFarland Systems business partner created earlier. Contract accounts can't be created as standalone objects; they are always created for a business partner.

You create contract accounts directly in SAP S/4HANA using the Manage Contract Account app, as follows:

1. Log in to SAP Fiori using the URL provided by your system administrator.
2. Select the **Manage Contract Account** tile.
3. Click on **Create** in the menu.
4. Select the **McFarland Systems** business partner.
5. Select the contract account class (**Contract Account Customer**) and **Payment Terms**.
6. Enter the **Cont. acct name**, **Tolerance group**, **Account determ. ID**, and **Company Code Grp**, and then save (see Figure 2.82, Figure 2.83, and Figure 2.84).

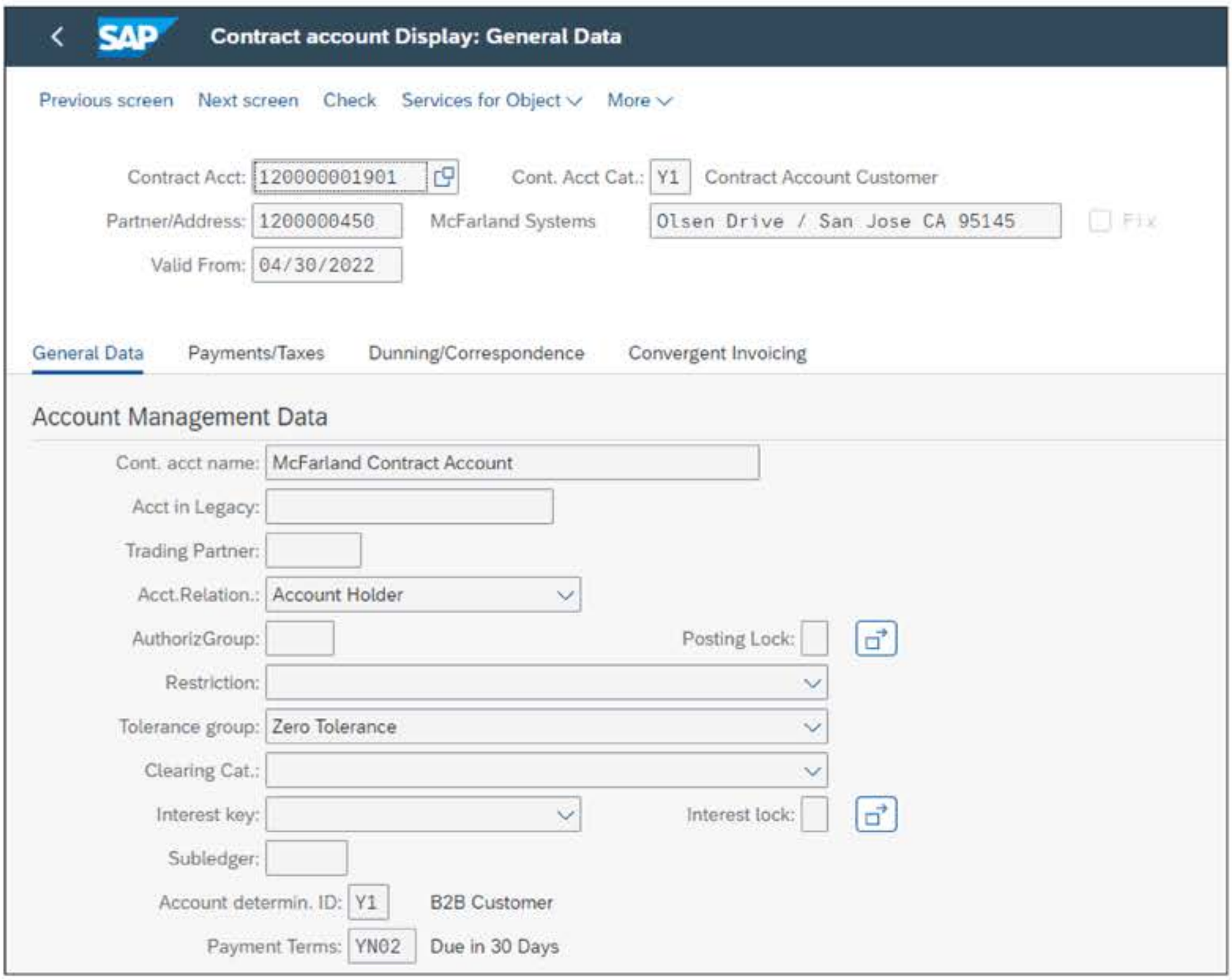

Figure 2.82 Contract Account: McFarland Systems

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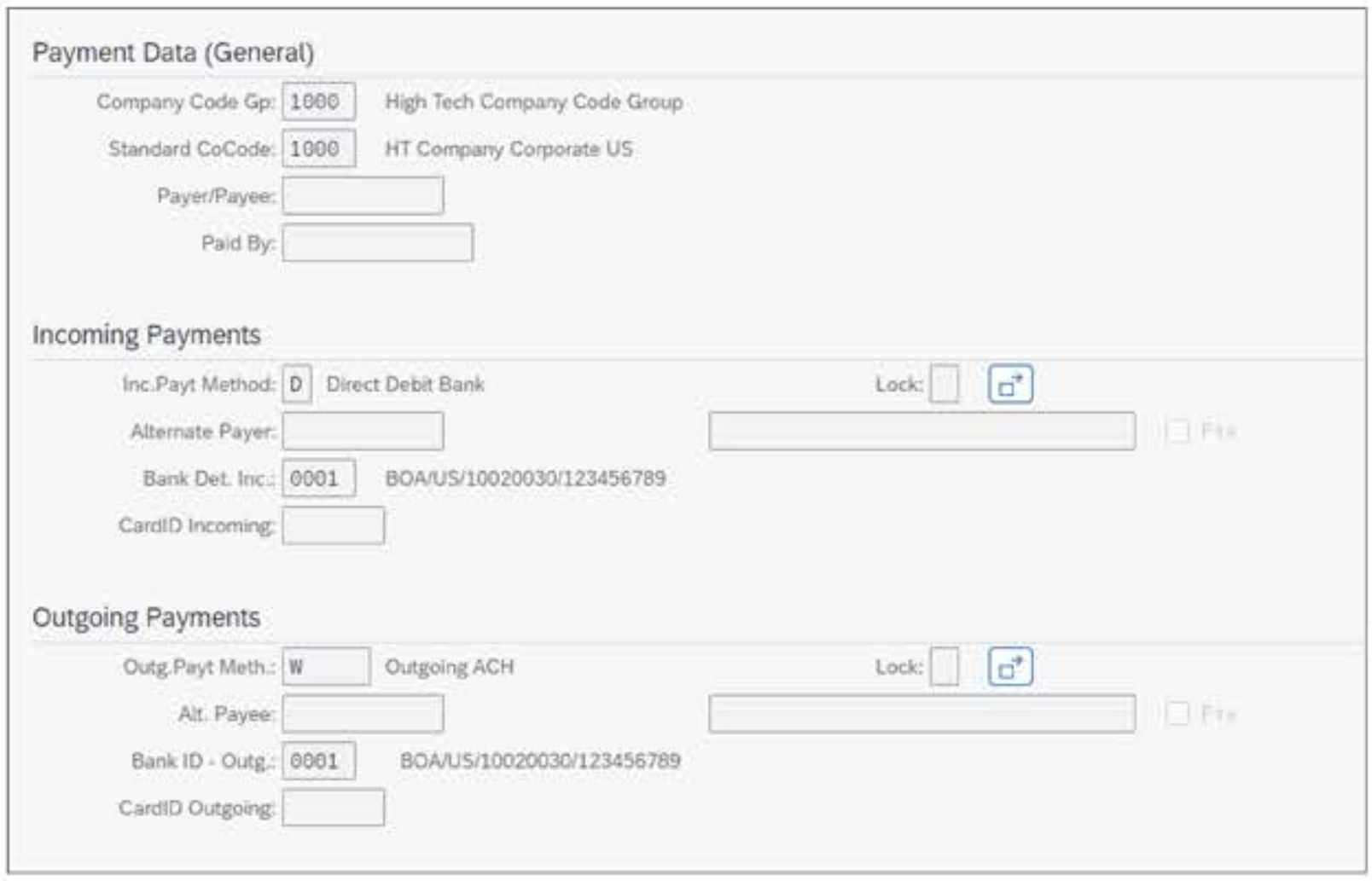

Figure 2.83 Contract Account Payments Management: McFarland Systems

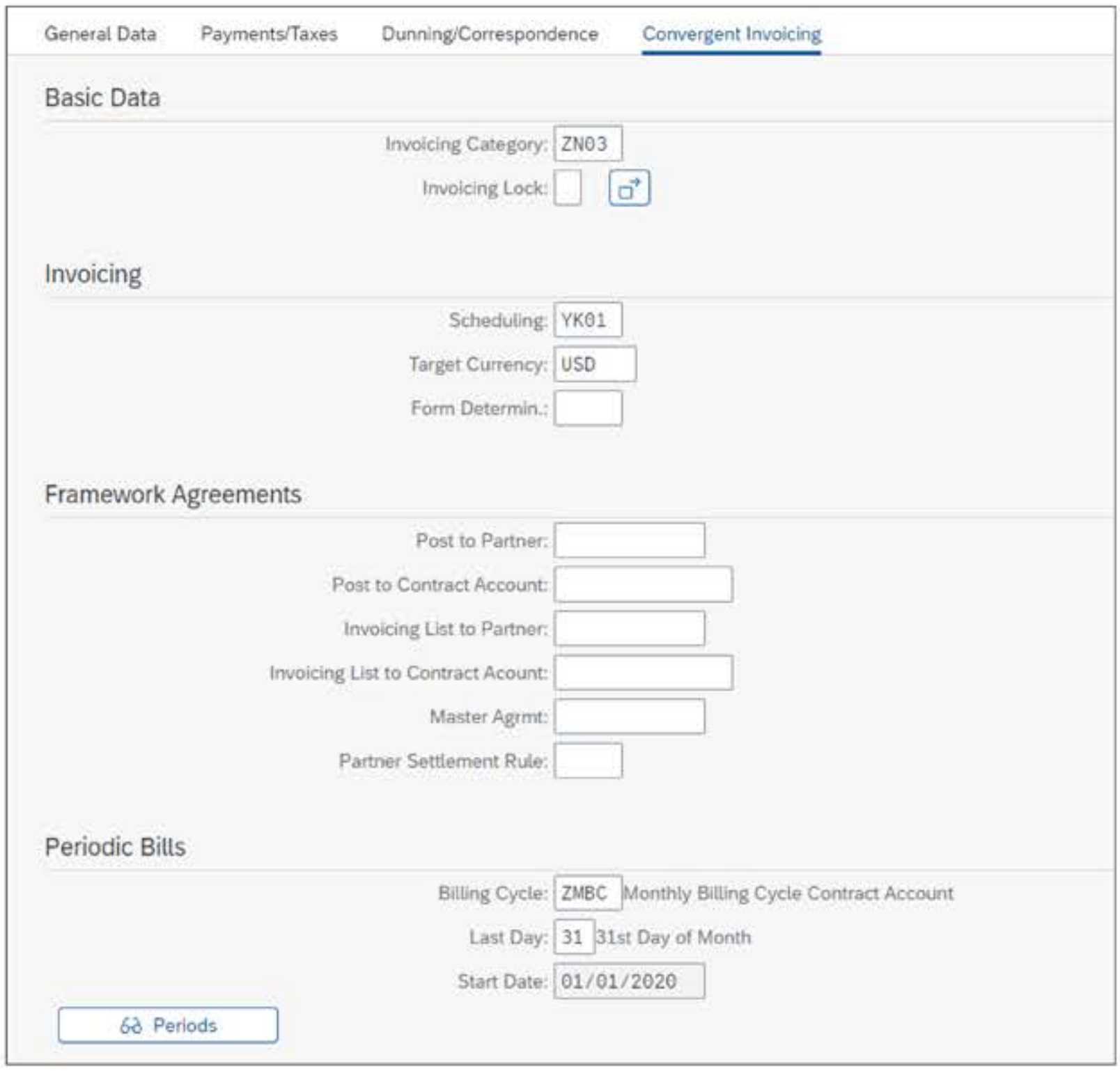

Figure 2.84 Settings Related to SAP Convergent Invoicing in Contract Accounts: McFarland Systems

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2.4.2 Creating and Mapping Subscription Products

A product is considered master data. You use it in the line item in the subscription order or solution quote, and it behaves similarly to a regular sales and distribution order. Although your product creation follows a standard process, you adapt it to function as a subscription product. This adaption is crucial to how this product will work. This step was detailed in Section 2.3.3.

2.4.3 Managing Subscription Orders and Subscription Contracts

For the purposes of our example, McFarland Systems and Loren Systems partners with Martex to provide XaaS services for a cost. Martex resells these services to McFarland Systems and generates revenue based on the services used. Martex pays a portion of the revenue to Loren Systems. For this scenario to function, Martex first creates a provider partner agreement detailing the revenue-sharing conditions.

In this section, you'll create a partner provider order and customer subscription order. They both function similarly, but a partner provider order is linked to the partner/vendor, and a customer subscription order is linked to the customer.

You create subscription orders directly in SAP S/4HANA and SAP Fiori as follows:

1. Log in to SAP S/4HANA Fiori using the URL provided by the system administrator.
2. Select the **Manage Subscription Order** tile.
3. Click **New** in the **Create** window.
4. In the popup window, select **PRVO**.
5. Enter the following details:
   - **Sold-To Party**: "McFarland Systems"
   - **Employee Responsible**: This is usually auto populated based on predefined configuration. However, you can overwrite the value based on available and existing employee responsible business partners if needed.
   - **Organizational Data**: "Sales Organization"
6. In the **Items** section, enter "MRTX_MSW03" ("Martex Workplace Solution") for the product (see Figure 2.85).

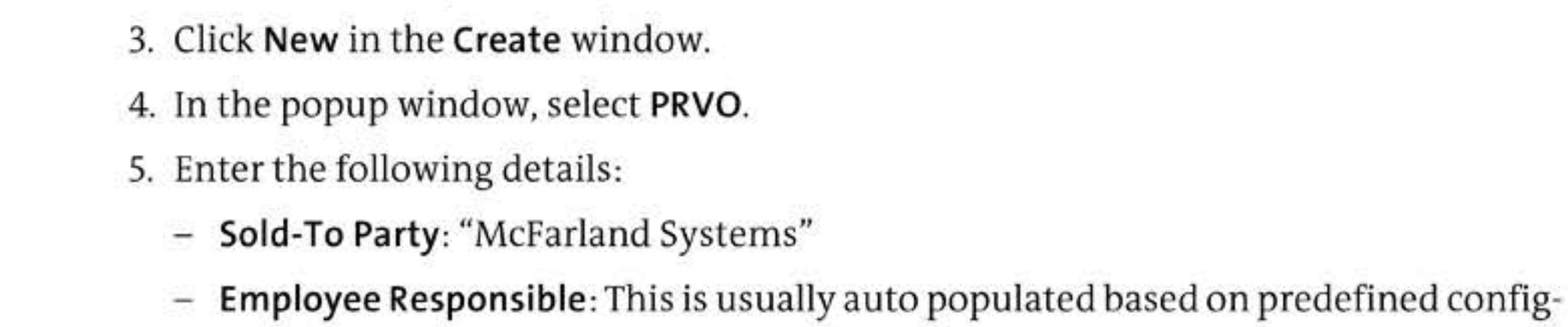

Figure 2.85 Add Product in the Subscription Order

7. Click the **Edit** button in line item 10.

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8. Open the **Configuration** block, and click the **Edit** button (see Figure 2.86).
9. Maintain values for each variant. Once maintained, save and close the configuration.

Figure 2.86 Maintain Product Configuration

10. Open the **Technical Resources** tab, and click the **Edit** button.
11. Under the **ID** column, enter a unique identifying value, and close the window (see Figure 2.87).

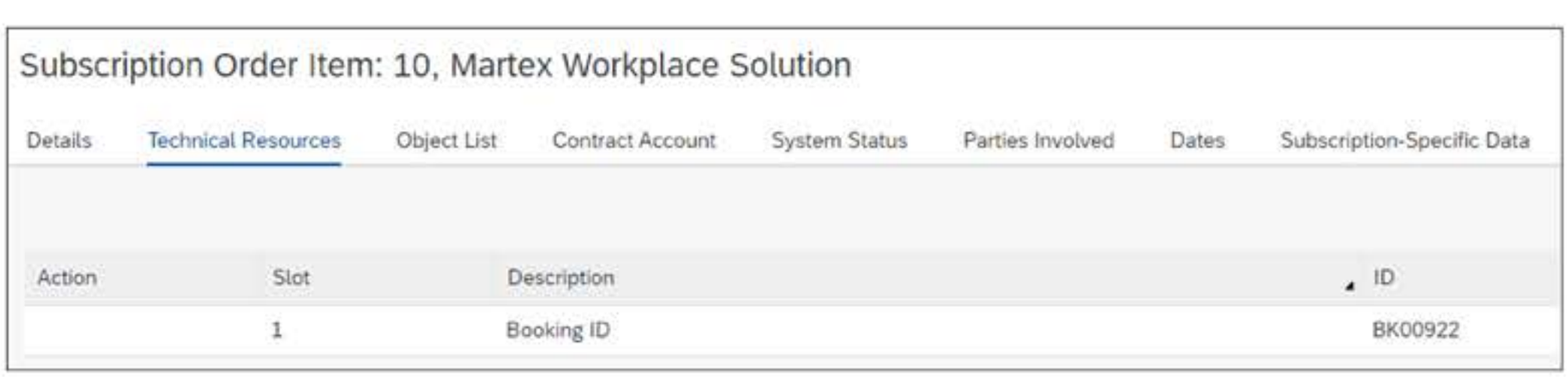

Figure 2.87 Technical Resource Maintenance for a Subscription Item

The contract account assigned to the business partner is derived by default. If no contract account is set as default, you can assign one by clicking the **Assign** button (see Figure 2.88).

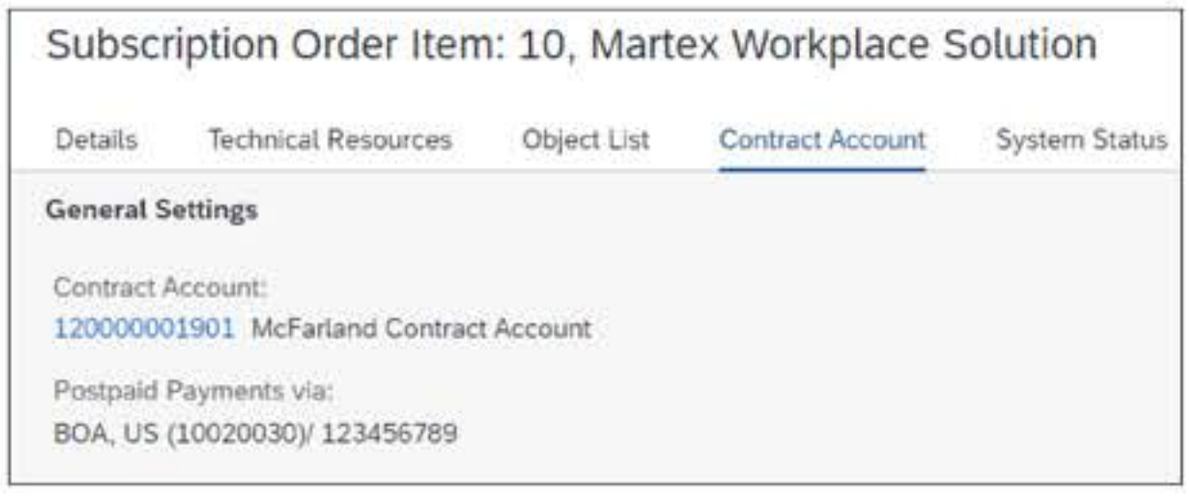

Figure 2.88 Contract Account Maintenance in Subscription Items

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12. Click **Back** to return to the **Subscription Order** screen.
13. Click the **Save and Submit** button (see Figure 2.89), and the subscription contract is created. The contract number is available in the **Item** tab (see Figure 2.90).
14. Click the contract number hyperlink, and you'll be taken to the **Subscription Contract** screen (Figure 2.90).
15. You can check the distribution of the subscription contract created to the SAP Convergent Charging and SAP Convergent Invoicing modules by clicking on the **Fulfillment Status** tab (see Figure 2.91).

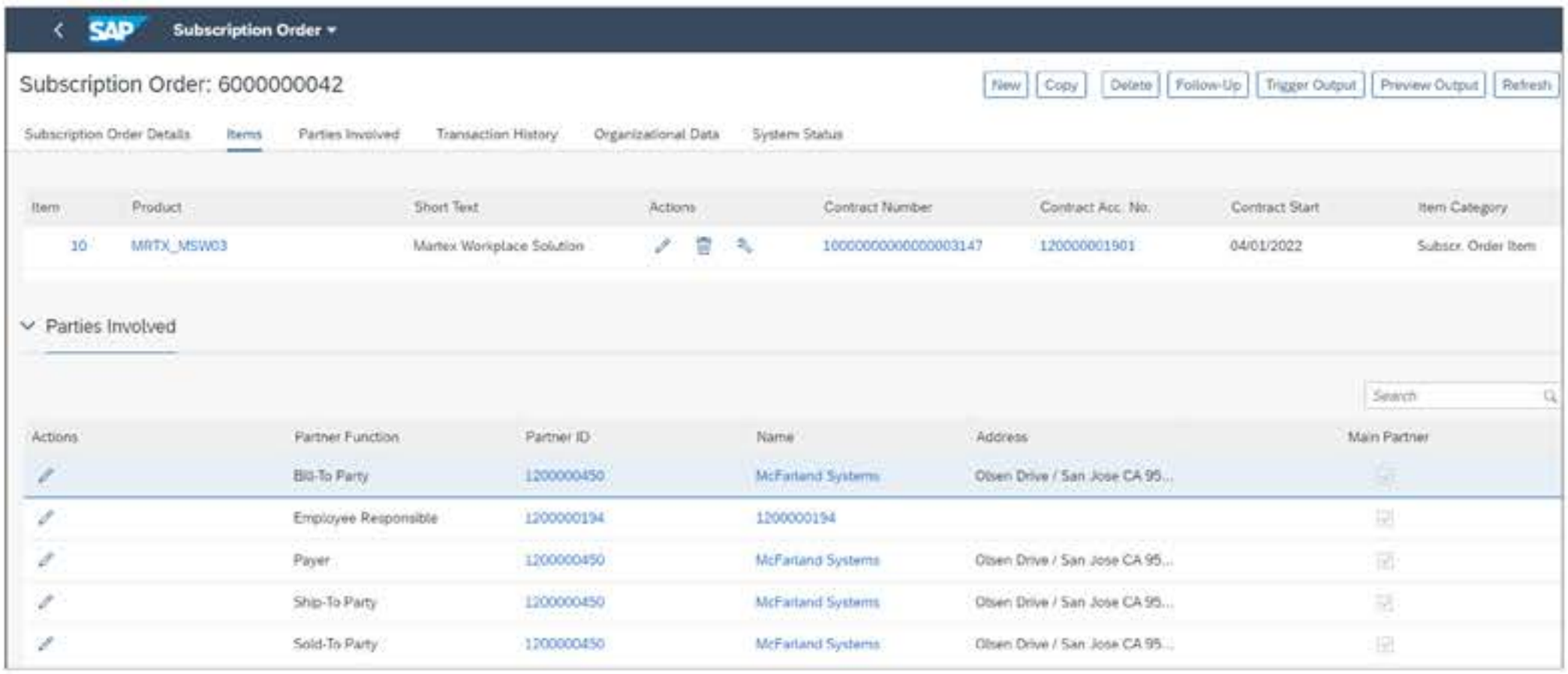

Figure 2.89 Subscription Order for McFarland Systems

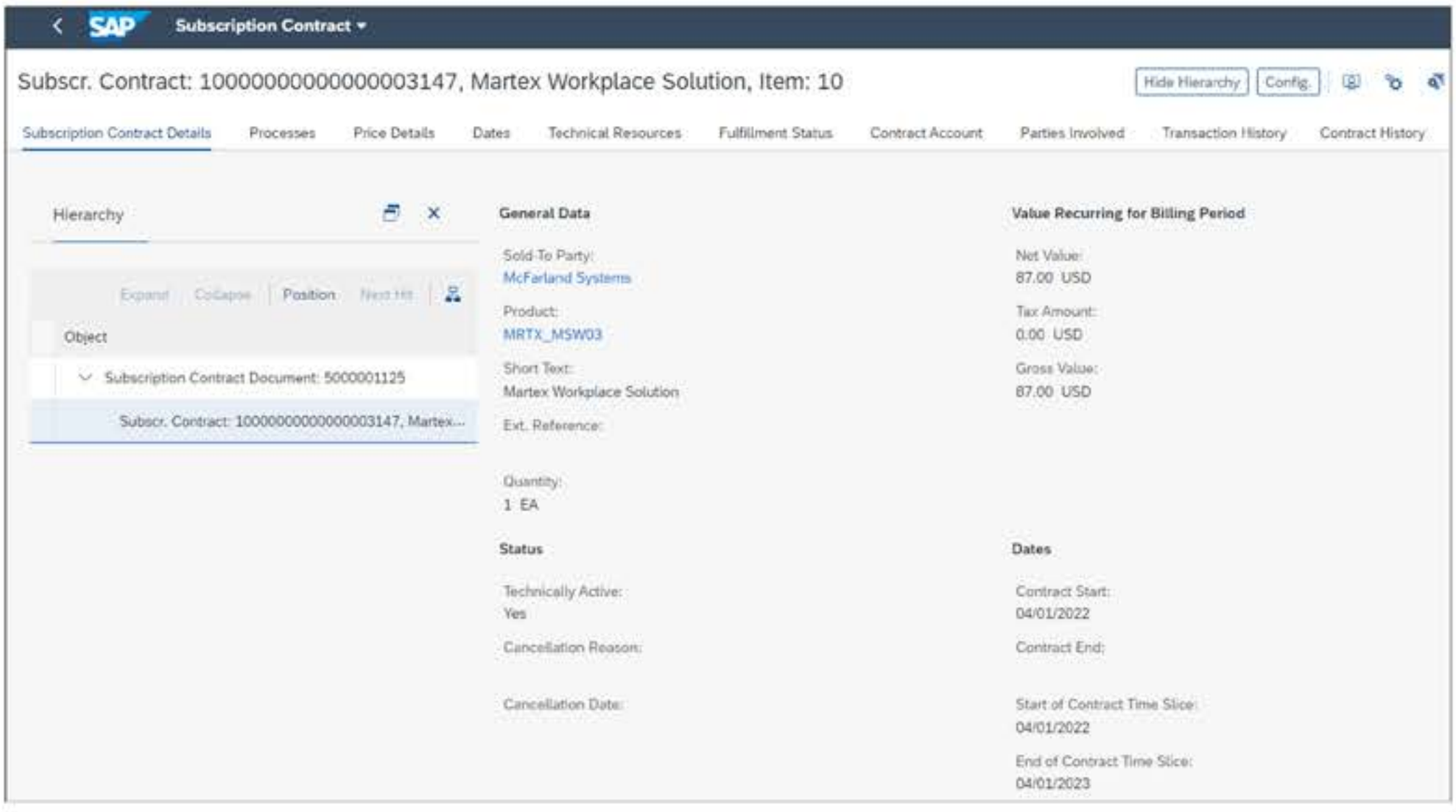

Figure 2.90 Subscription Contract for McFarland Systems

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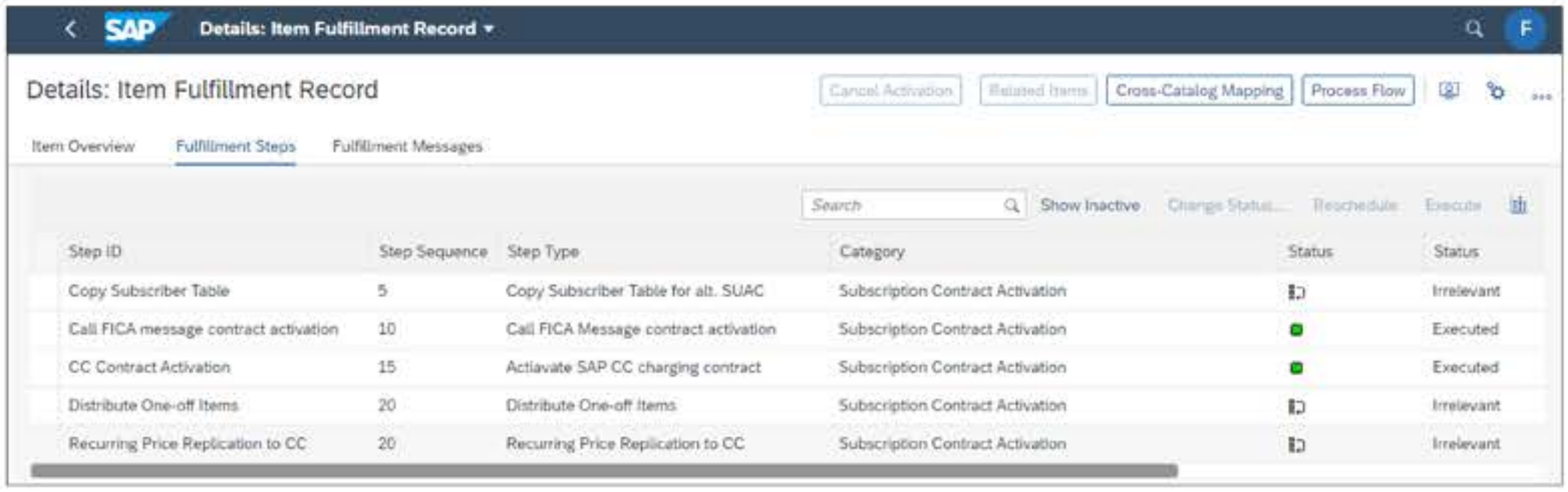

Figure 2.91 ODI Distribution Status

2.4.4 Modifying Subscription Contracts

A subscription contract can't be changed directly. This functionality is restricted to protect the data integrity across the entire system landscape. Any change must be made via a standard or custom maintenance framework for business transactions (BTMF) change process (See Section 2.6 for further details). Based on the process type, some changes are executed via a change order. Depending on the type of change, most of the changes are captured as time slices in the contract. All the changes are captured as various time slices with the effected change (see Figure 2.92).

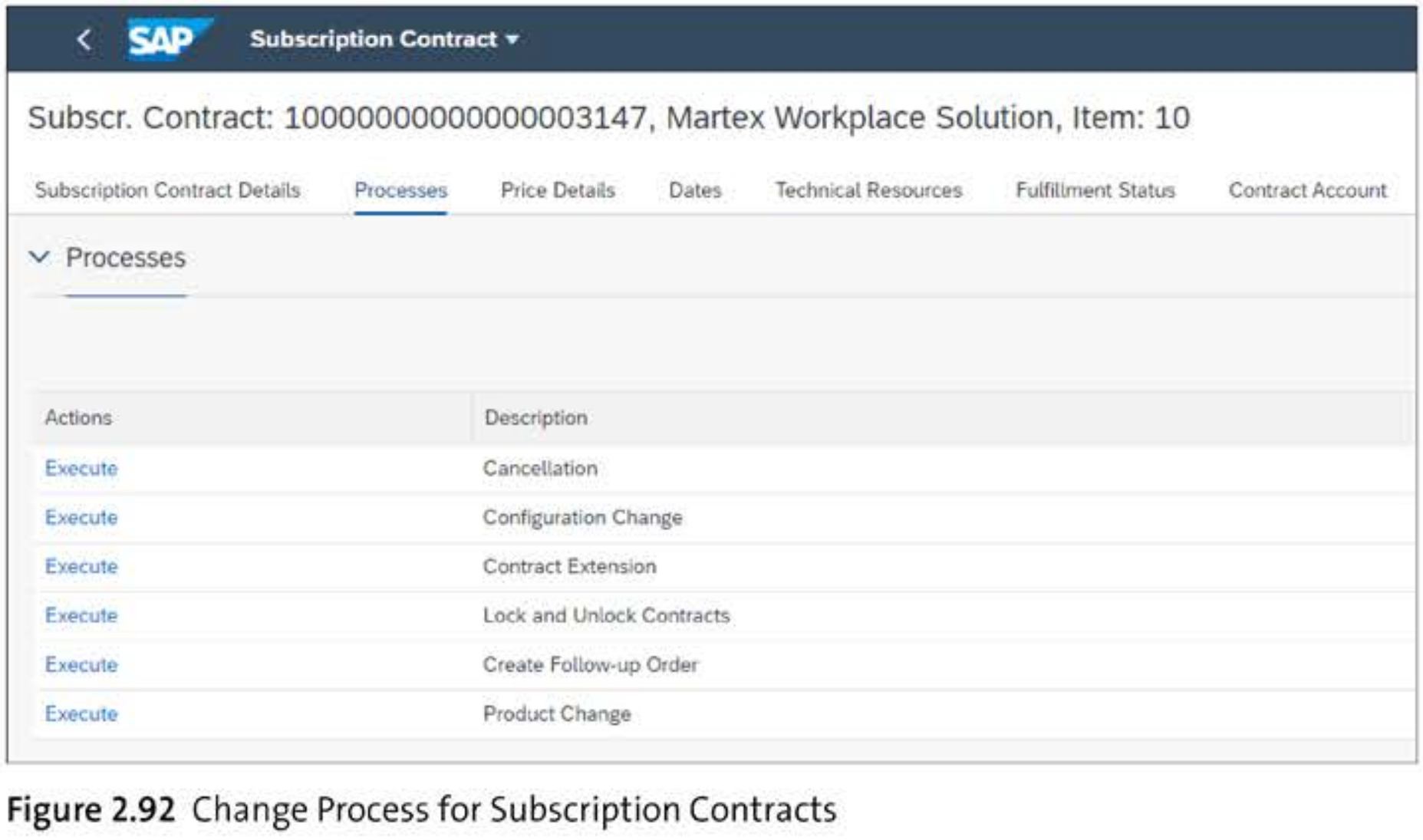

Figure 2.92 Change Process for Subscription Contracts

2.5 SAP Fiori Applications for Subscription Order Management

Because SAP Billing and Revenue Innovation Management, subscription order management, is now part of SAP S/4HANA, the traditional SAP CRM Web UI view is now

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replaced with SAP Fiori apps (see Figure 2.93). SAP Fiori is built with SAPUI5 Technology. Most of the apps are based on the Web Client UI application, and the transactions and master data are available in the SAP Fiori theme to support a seamless user experience across the SAP Fiori launchpad. The SAP Fiori launchpad uses OData services to fetch SAP S/4HANA backend information. This has helped to further enhance the UI experience as the apps can now be accessed via desktop, laptop, smartphone, and tablet.

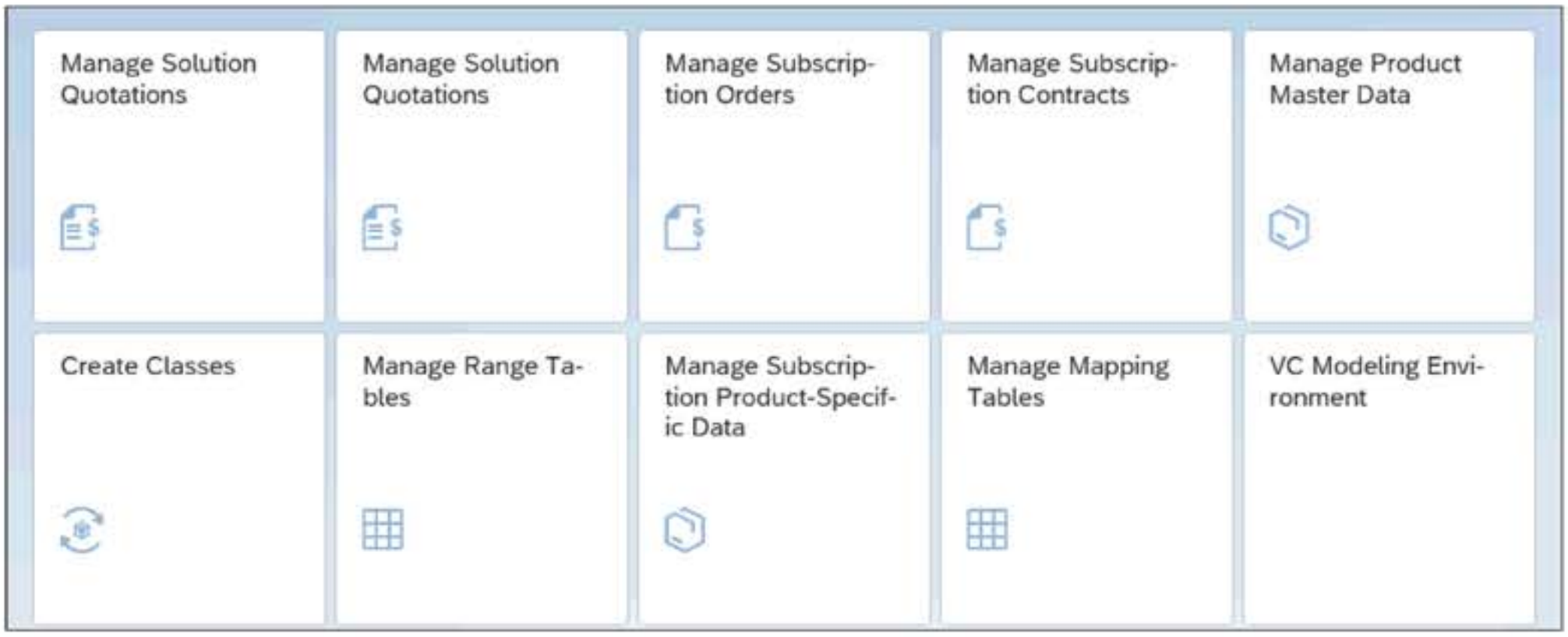

Figure 2.93 SAP Billing and Revenue Innovation Management, Subscription Order Management: SAP Fiori Apps

Following are some of the SAP Fiori apps for SAP Billing and Revenue Innovation Management, subscription order management, with brief feature descriptions:

- **Manage Subscription Order**
  This app is used to manage, search, display, edit, create, and copy subscription orders.
- **Manage Subscription Contract**
  This app is used to manage, search, display, and apply the change process for subscription contracts.
- **Manage Solution Quotation**
  This app is used to manage, search, display, edit, create, and copy solution quotations.
- **Manage Subscription Product-Specific Data**
  This app is used to create and manage subscription product master data that is required for managing the subscription contract lifecycle. This includes complex price configurations, discounts, and charges, as well as sharing the products between multiple contracts or allowances. You can also use this app to set up subscription products for use in SAP Billing and Revenue Innovation Management, subscription order management. This application also supports cross-catalog mapping either to charge plans from SAP Convergent Charging or to external objects.

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- **Manage Product Bundles**
  This app is used to define the subscription product bundle dependencies, bundle rules, and so on.
- **Manage Master Agreement**
  This app is used to manage, search, display, edit, create, and copy master agreements.

As part of standard offering, SAP_SOM_BC_SUBSCR_ORDER and SAP_SOM_BC_SUBSCR_PROD SAP Fiori catalogs are shared. SAP_SOM_BC_SUBSCR_ORDER consists of transactional applications from SAP Billing and Revenue Innovation Management, subscription order management (e.g., Create Subscription Order, Create Solution Quotation, Execute Mass Run, and Manage Sharing Groups), and SAP_SOM_BC_SUBSCR_PROD consists of applications managing subscription products and SAP Convergent Charging tables (e.g., Manage Product Master Data, Manage Subscription Product-Specific Data, and Manage Range Tables). You can create a custom catalog with applications to manage business-specific roles. In addition, to manage configurable products, you can use SAP_BR_PRODUCT_CONFIG_MODELER.

2.6 Subscription Lifecycle Management

So far, we've looked at the ability to place an initial order and convert it to a subscription. On an ongoing basis, customers commonly require the ability to modify the existing subscriptions as they keep adjusting to the changing demands of their business. As discussed earlier in this chapter, various changes can be made to a subscription. We'll look at some of the most common change processes in this section in more detail.

2.6.1 Subscription Renewal Process

Renewal process refers to extension of customer contract with either existing terms or condition or modified terms and condition. It's one of the most used post-sale processes in subscription business. Renewal ensures continuation of the business relationship with existing customers and helps streamlining revenue stream for a company. Renewals are broadly classified into two segments: auto renewal and manual renewal (or manual extension).

Auto renewal refers to a process of extension where the subscription should be extended for a predefined renewal period with existing terms of condition (like price, billing cycle, services, etc.). The feature ensures that the extension process is automated without any manual intervention from sales or IT representatives. However, if the customer requires modified terms and condition (such as a change in subscription price) or the renewal period is based on ad-hoc customer request, then the process is termed as manual renewal or extension.

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In the upcoming sections, we'll describe how SAP Billing and Revenue Innovation Management, subscription order management can manage auto renewal as well as manual Renewal (or Extension) with examples.

Auto Renewal

Auto renewal provides the capability to extend the contracts automatically without manual intervention. In our use case, let's assume that the customer MacFarland Systems require their subscription contracts to be automatically renewed for a duration of two years before 10 days of expiry of their existing contract. You can achieve this requirement by using the auto renewal feature to extend the contract for two years.

If a specific product or service should always be considered for auto renewal, you can use the **Contract Auto Renewal Indicator** field, which is provided in the product master data (see Figure 2.94). However, please note that this setting can also be overridden in the subscription order or solution quotation.

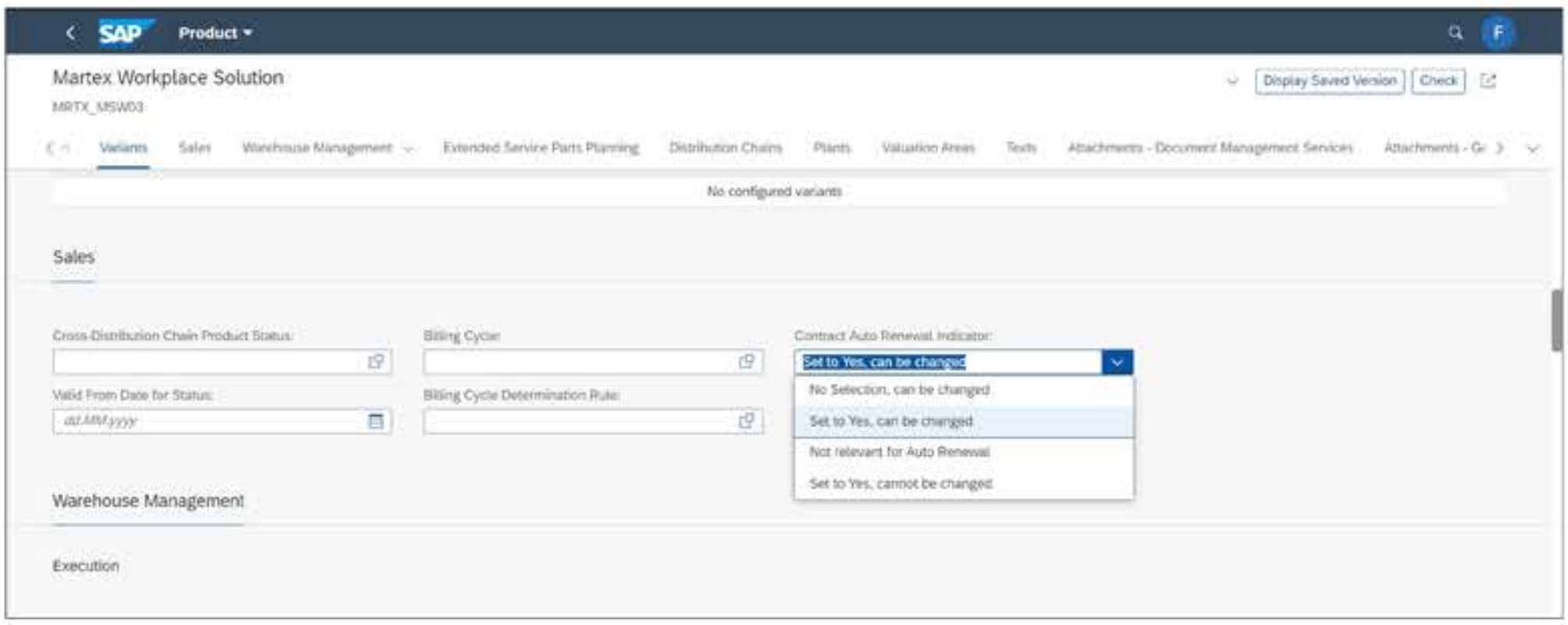

Figure 2.94 Product Master Data Sales View

The **Contract Auto Renewal Indicator** dropdown list provides different options for you to decide whether auto renewal for a given product is to be made mandatory, optional, or defaulted. You can also define whether this indicator can be overwritten at the time of subscription order or solution quotation creation. The default number of years to extend the contract is maintained at the product master data level (see Figure 2.95).

You can update the default period for extension and the unit for extension in the product master data. These fields are also visible in the solution quotation or subscription order and can be updated at the time of order entry.

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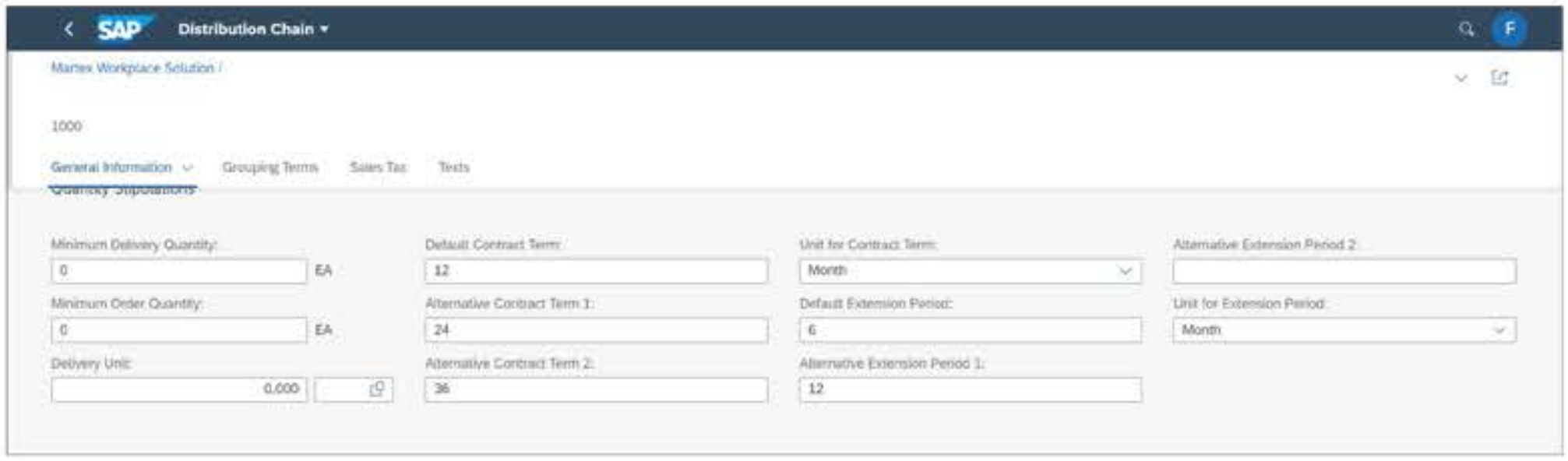

Figure 2.95 Default Extension Period and Unit in Product Master Data

To enable the auto renewal process, the following prerequisite configurations must be made in the system. You can configure the date profile by going to menu path **Service • Basic Functions • Date Management • Define Date Profile for Service Transactions**. The standard date profile for subscription orders, solution quotations, and subscription contracts has Date Types **AR_START** (see Figure 2.96) and this date type has date rule **AUTORENSTART** (see Figure 2.97). This date rule calculates the date on which auto renewal needs to be triggered with reference to the existing contract end date.

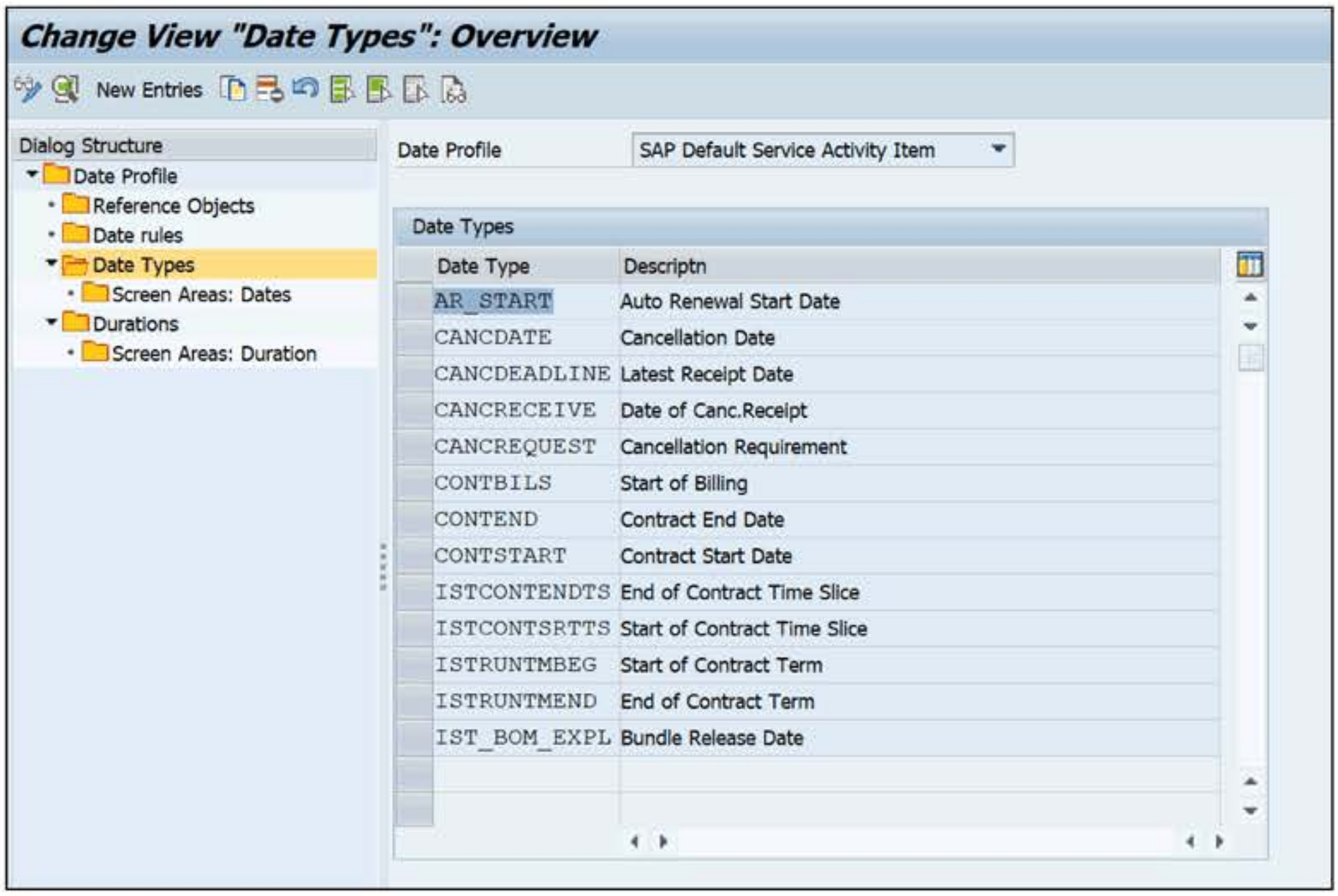

Figure 2.96 Date Type Used in Auto Renewal: New Scheduled Action Triggered Based on This Date

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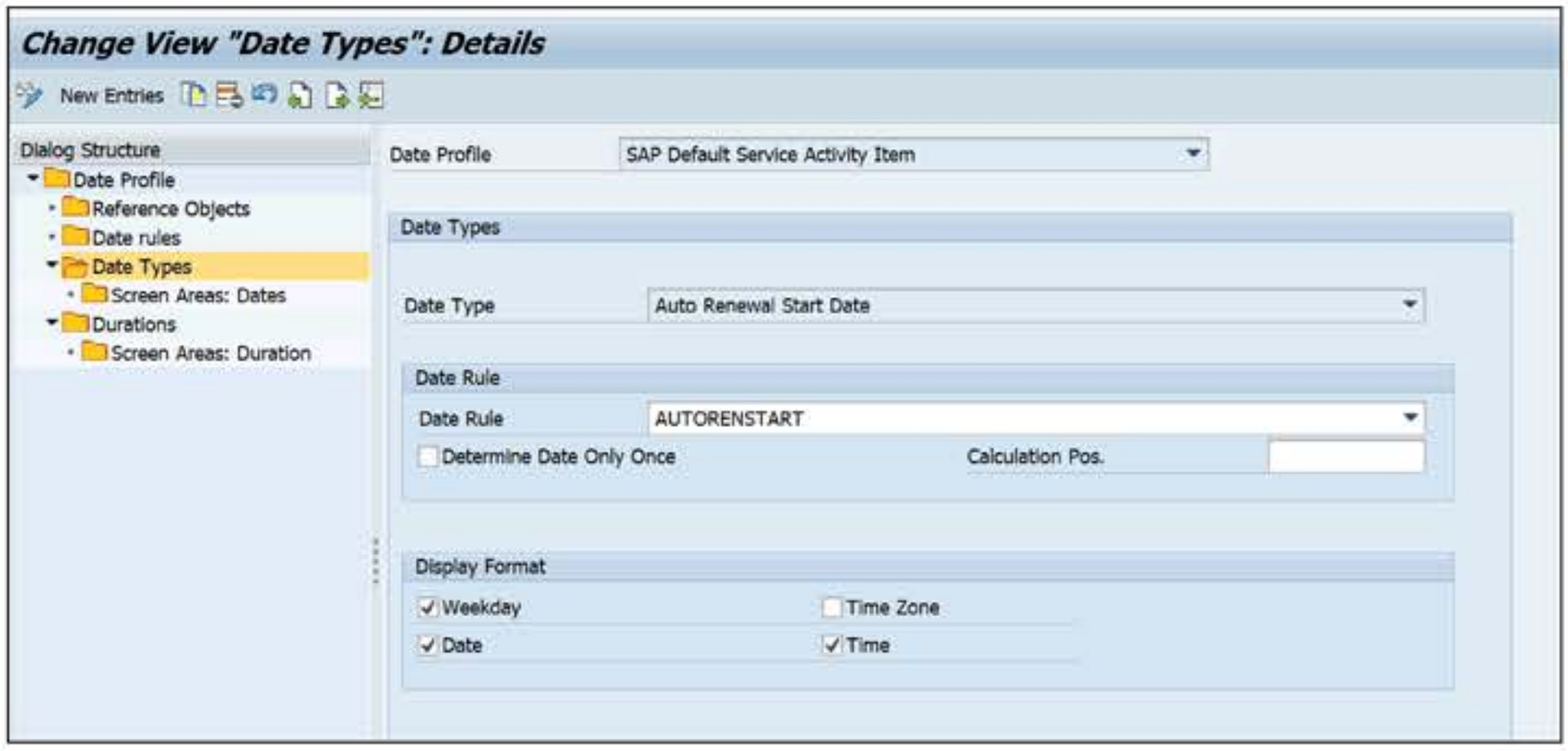

Figure 2.97 Date Rule AUTORENSTART Assigned to the Date Type

You can configure the actions by using menu path **Service • Basic Functions • Actions • Actions in Transaction • Change Actions and Conditions • Define Action Profiles and Actions** as well as maintaining the action conditions through **Service • Basic Functions • Actions • Actions in Transaction • Change Actions and Conditions • Define Condition**, as shown in Figure 2.98.

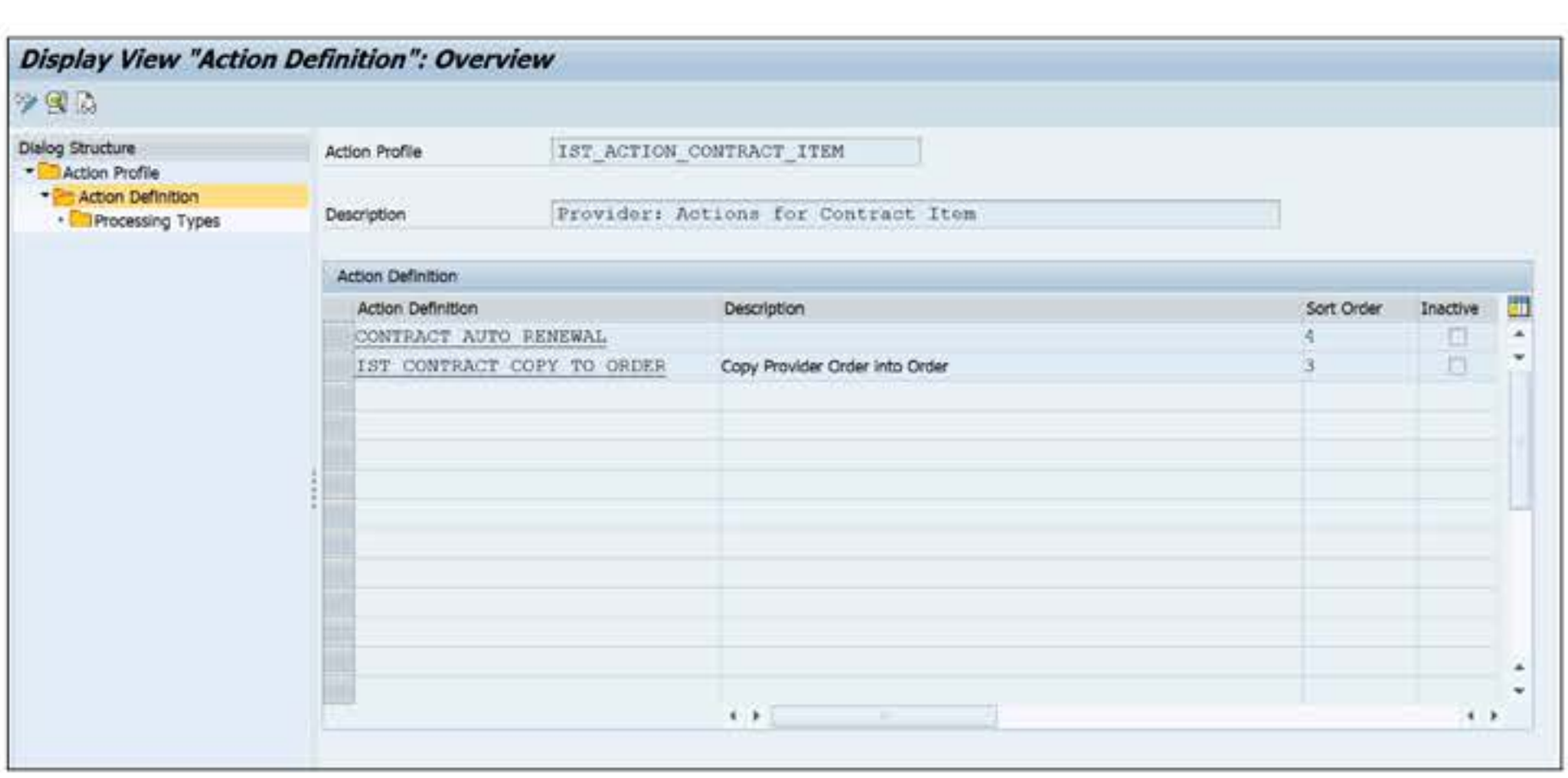

Figure 2.98 Action Profile and Action Definition for Auto Renewal

SAP has provided standard action profile **Provider: Actions for Contract Item**, which has an action for auto renewal (see Figure 2.99). When conditions for start and schedule are met, these actions are triggered, which creates a new time slice.

The start and schedule condition triggers the scheduled action and creates a new time slice in the subscription contract. If the conditions aren't maintained, then the actions get triggered immediately.

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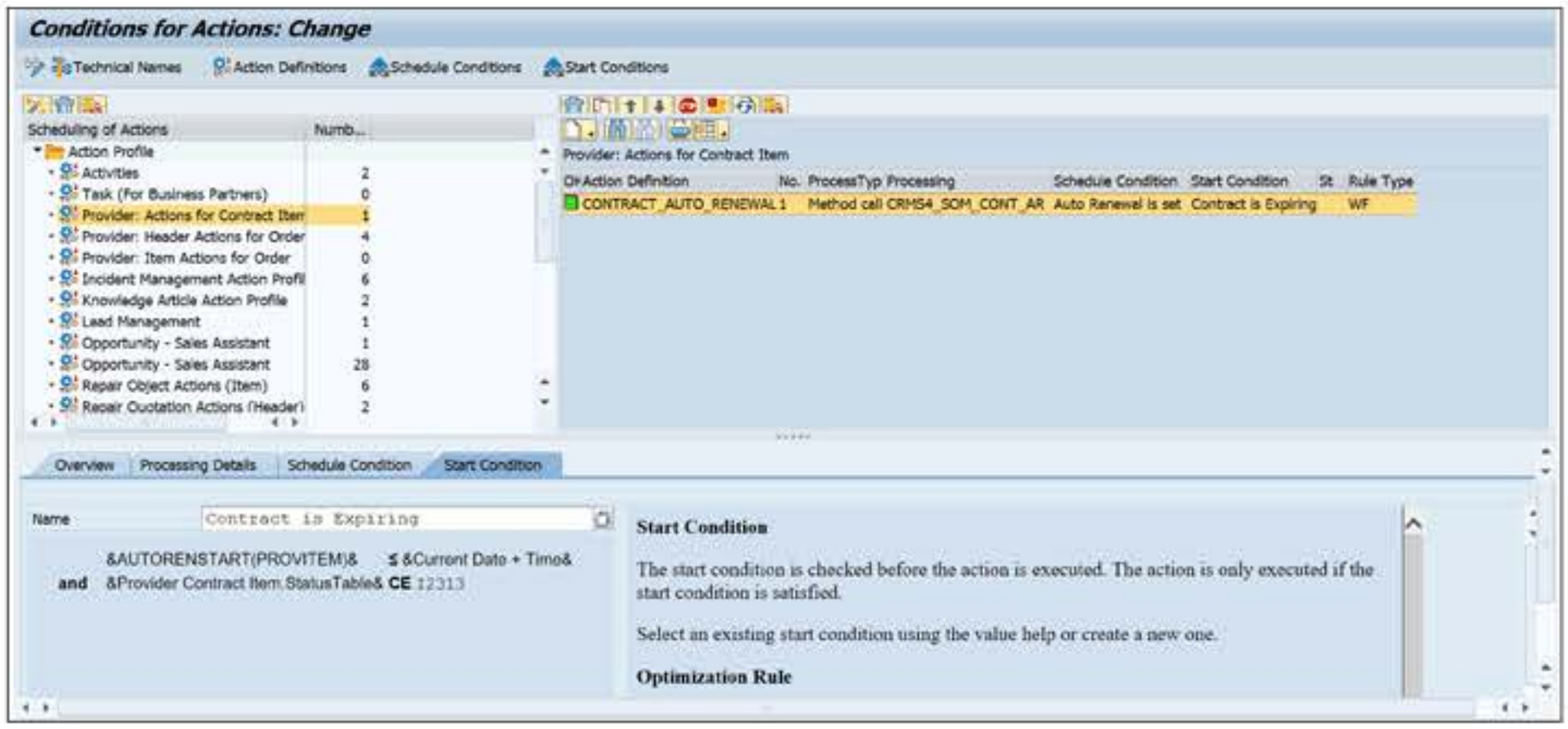

Figure 2.99 Start and Schedule Condition

The action profile configured in the previous step will need to be assigned to the item category. You can do so by selecting **Service • Basic Functions • Actions • Actions in Transaction • Assign Action Profile to Item Category**. When these action profiles are assigned to the item category of the products and the schedule conditions are met, they trigger actions (see Figure 2.100).

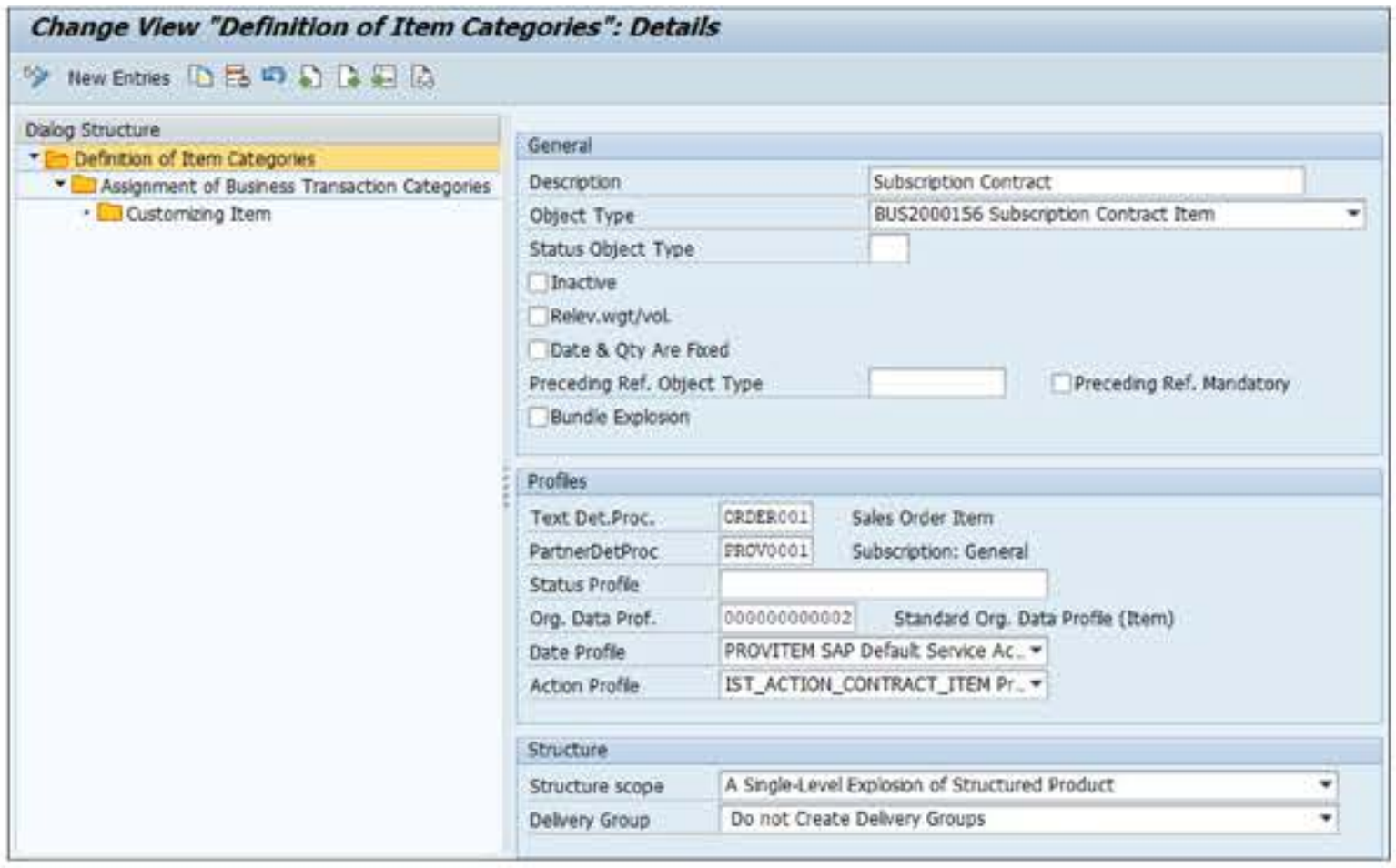

Figure 2.100 Assigning the Action Profile to Item Category

As a final step, you'll need to set up a batch job for program RSPPFPROCESS for action profile **Provider: Actions for Contract item**. When schedule conditions are satisfied, a schedule action is triggered. Another batch job for CRM_ISX_MSG_STRAT_PLANNED executes the scheduled action, which creates new actions and activates the contract.

Once these steps are complete, you can execute a transaction to execute the auto-renewal functionality with the following steps:

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1. Log in to SAP Fiori Portal with the URL shared by your administrator. Search for the Manage Subscription Order app and click on the **New** button. Select the order type PRVO.
2. Enter details for customer, contract start date, contract end, price details and product details. Navigate to the **Automatic Renewal** section (see Figure 2.101).

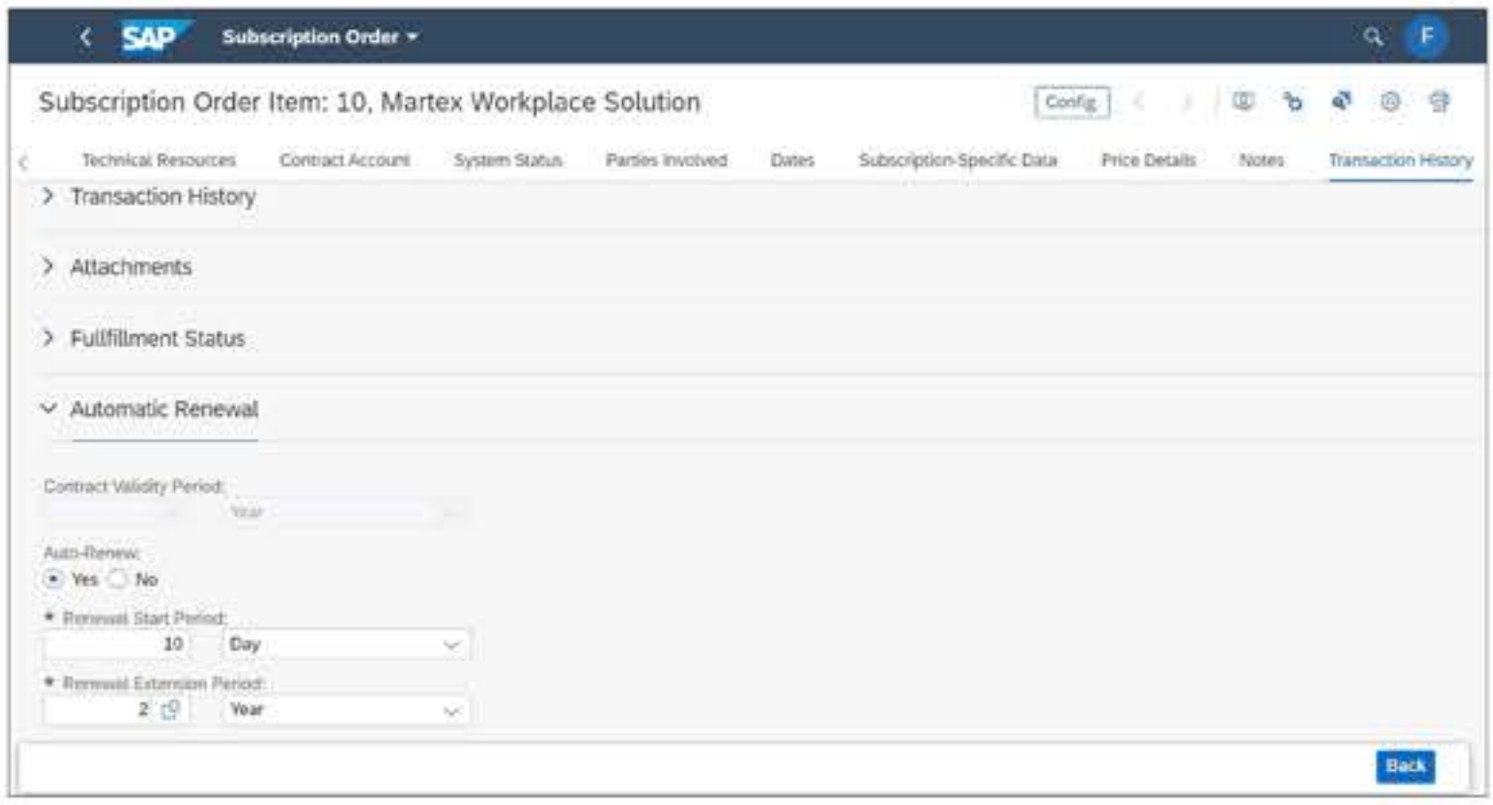

Figure 2.101 Automatic Renewal Section

3. The default **Auto–Renew** radio button indicator is derived from the product master data. Depending on the setting provided, you can edit **Renewal Start Period** (how many days before the action for renewal needs to be triggered) and **Renewal Extension Period** (the year the contract will be extended to).
4. The new scheduled action is triggered 10 days before by the scheduled batch job RSPFPROCESS and CRM_ISX_MSG_START_PLANNED. On the last day of the contract, a new time slice is triggered, extending the contract for two years, and the old time slice becomes technically inactive (see Figure 2.102).

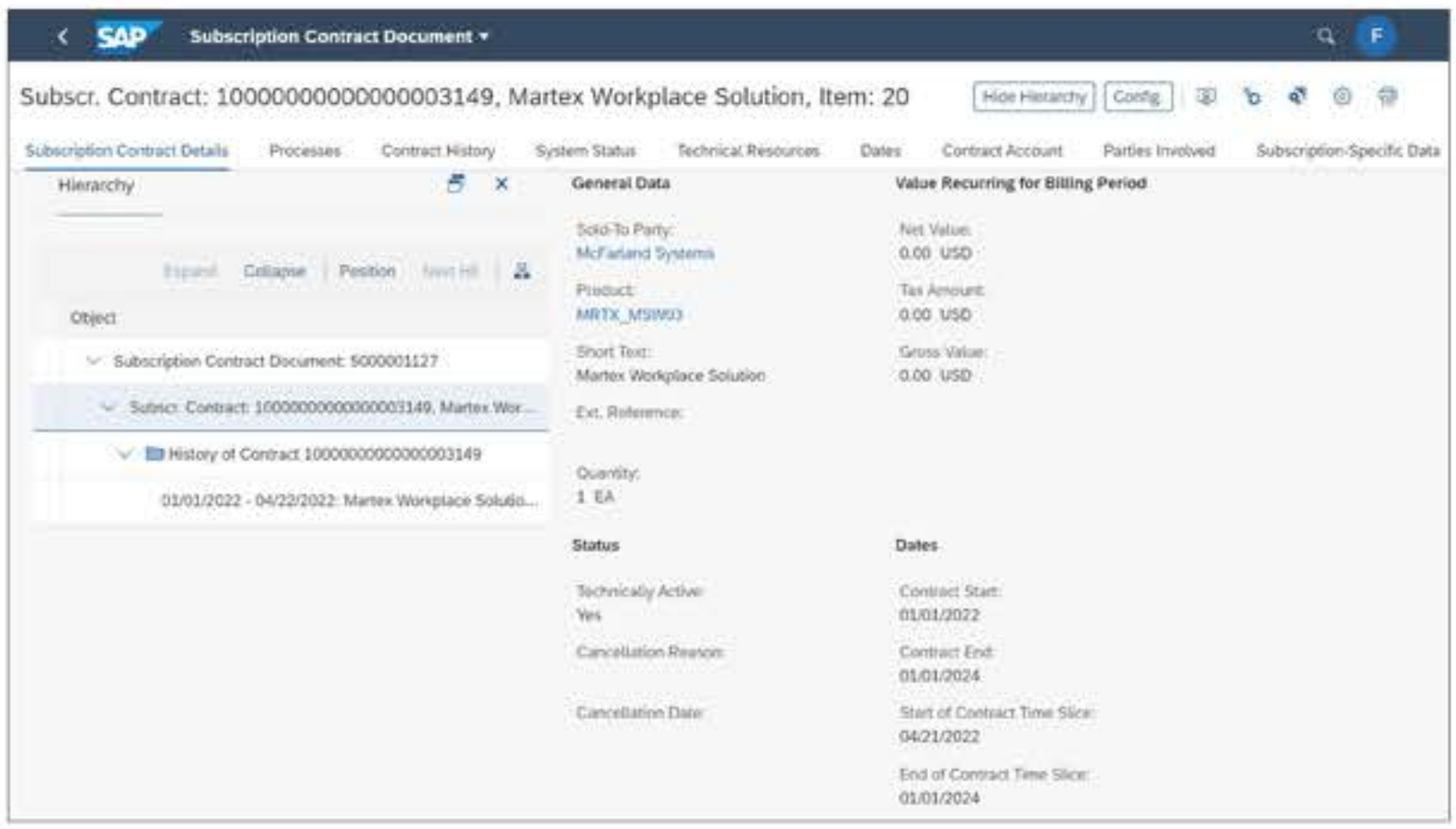

Figure 2.102 Subscription Contract Extended by Two Years

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Manual Extension

The manual extension is generally triggered when there is an add-on request to extend the contracts manually due to various reasons such as the continued negotiation for contract terms (e.g., pricing). In our use case, you'll execute the contract extension process with revised Renewal period. Per the customer's request, the contract needs to be extended by one year at the end of the current subscription period. The number of years to be extended (i.e., extension terms) resides in the product master data in the **Sales Area** data tab (refer to Figure 2.95). Once you execute the extension process, it will create a new time slice and become active at the activation date you enter. In the system, you'll be able to see the same subscription contract number with a different line-item number, and the newly extended time slice will display **Technically Active** as **Yes**, whereas the old time slice will display as **No**.

You can activate the manual extension change process via menu path **Service • Transactions • Settings for Subscription transactions • Contract Changes •Settings for Maintenance Framework (BTMF) • Activate Change Process**.

To configure the extension process, you need to add the BTMF process in the **Activate Change Process** node (see Figure 2.103). All possible change processes are defined in **Define Change Process**.

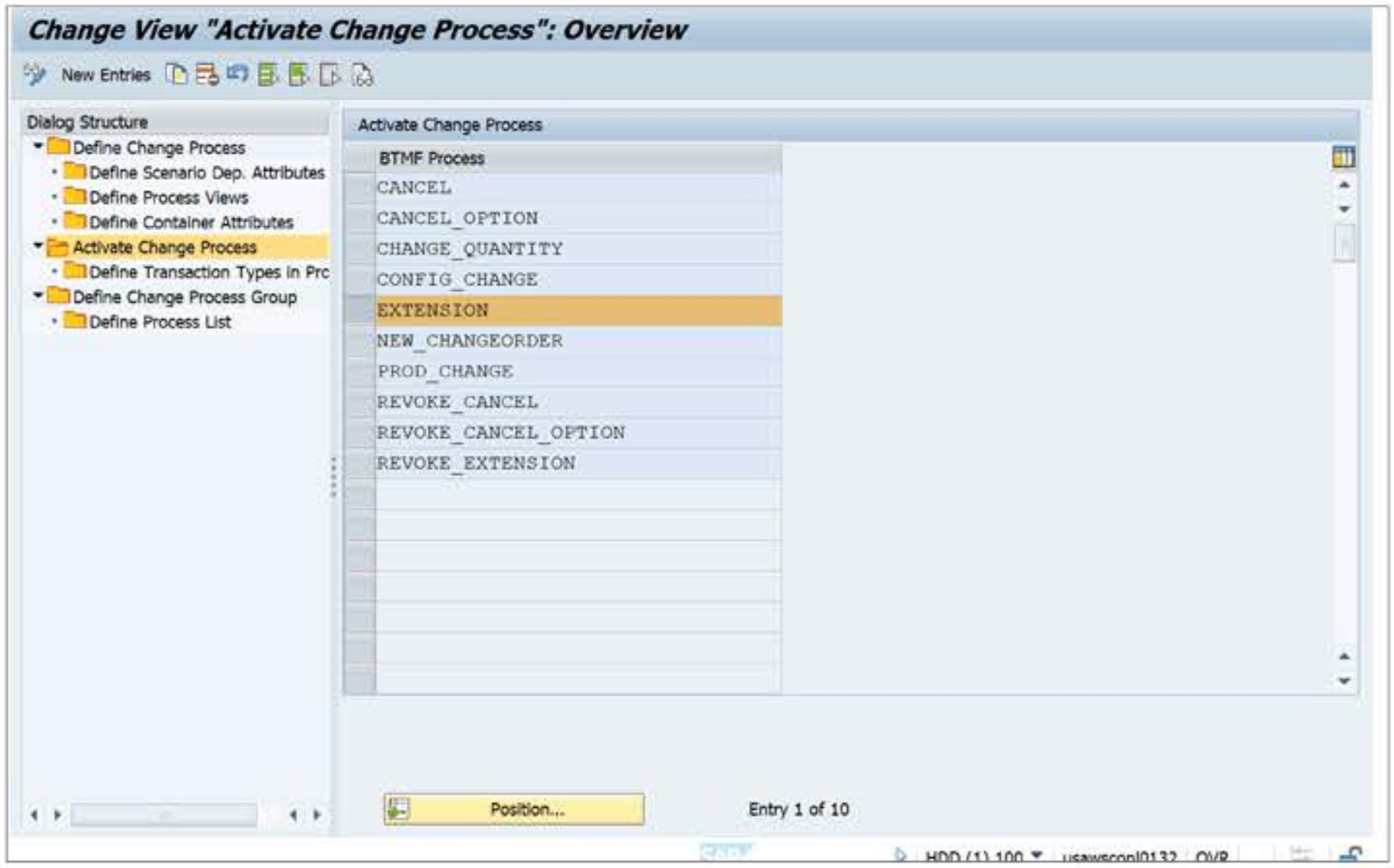

Figure 2.103 Configuration for BTMF Change Processes

To execute the extension for immediate processing, follow these steps:

1. Log in to the SAP Billing and Revenue Innovation Management and SAP Fiori using the URL shared by your system administrator. Check if the role is assigned to your ID.

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2. Select the **Manage Subscription Contracts** app tile.
3. Enter the contract number in the search box, and execute **Search**. When you see the contract number, click on its hyperlink to go to the **Subscription Contract Details** tab on the **Subscription Contract Document** screen.
4. Select line item **10** in the contract hierarchy, and you'll see **Contract General Data** (see Figure 2.104).

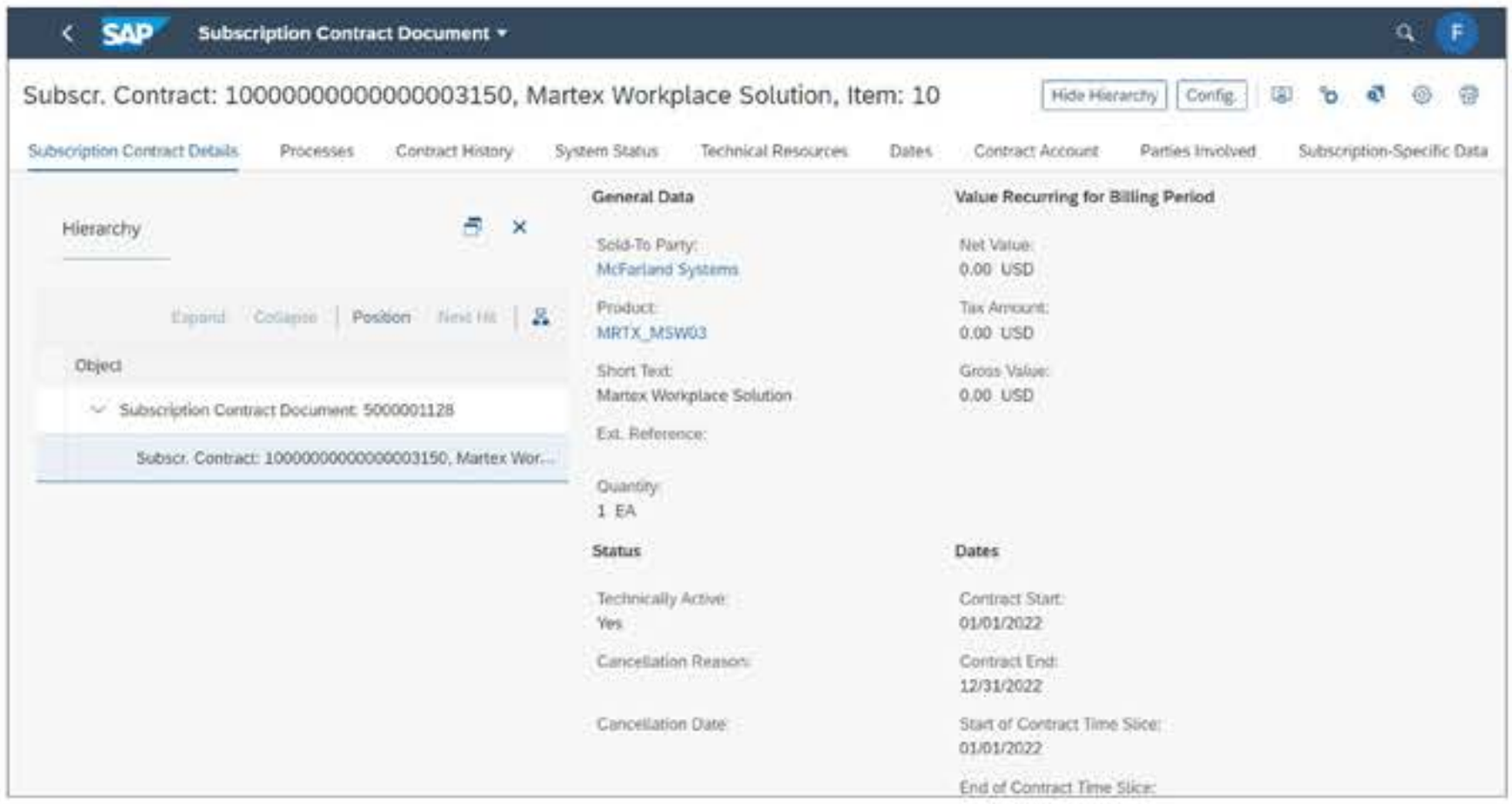

Figure 2.104 Subscription Contract: General Data

5. Navigate to the **Processes** tab, and you'll see a list of applicable change processes (see Figure 2.105).

   The **General Data** area displays the contract's start date, end date, and status (e.g., **Technically Active**: **Yes** or **No**), as shown in Figure 2.104.

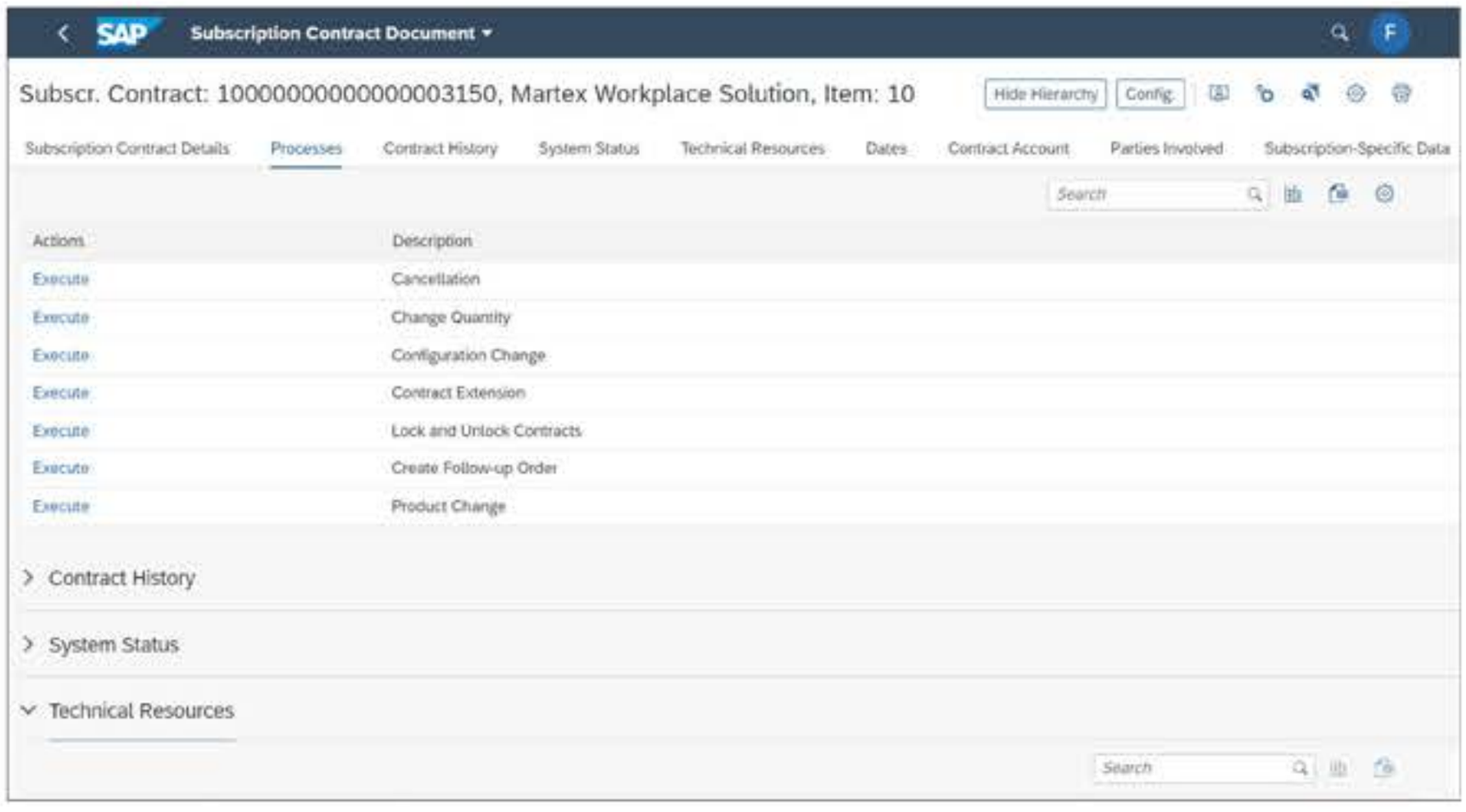

Figure 2.105 List of All Change Processes for Subscription Contracts

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All configured change processes are displayed when the contract status is **Technically Active**.

6. Click **Execute** in the row for the **Contract Extension** change process. A new popup window opens. Figure 2.106 shows the details of the change process entry fields.

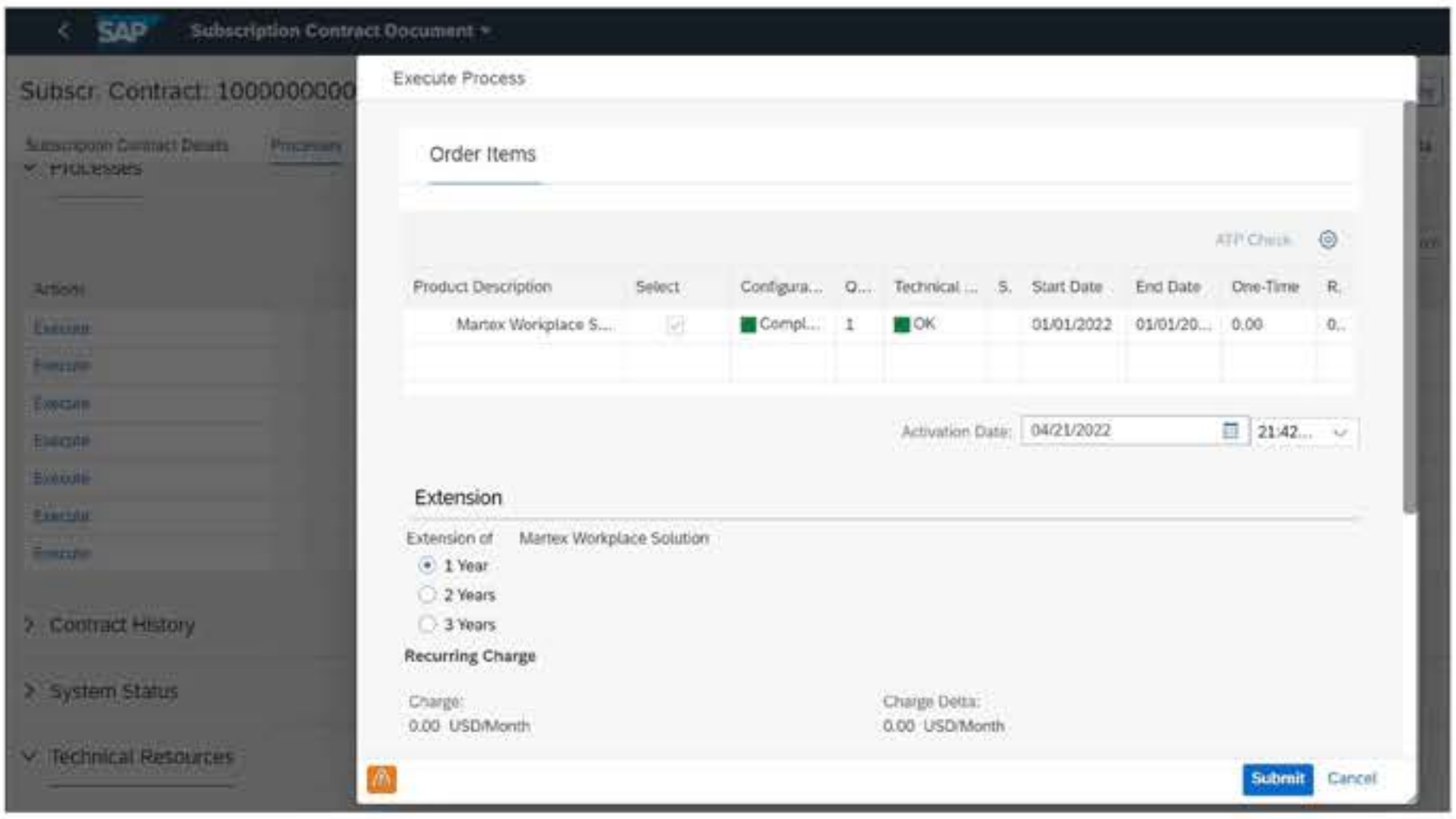

Figure 2.106 Contract Extension Screen Change Process

7. Update the contract **Activation Date** and time, select the number of years the contract needs to be extended for, and click **Submit** to go to the change order that needs to be submitted (see Figure 2.107).

   The activation date and time refers to when the extension needs to be triggered, and this date can be the current date or a future date.

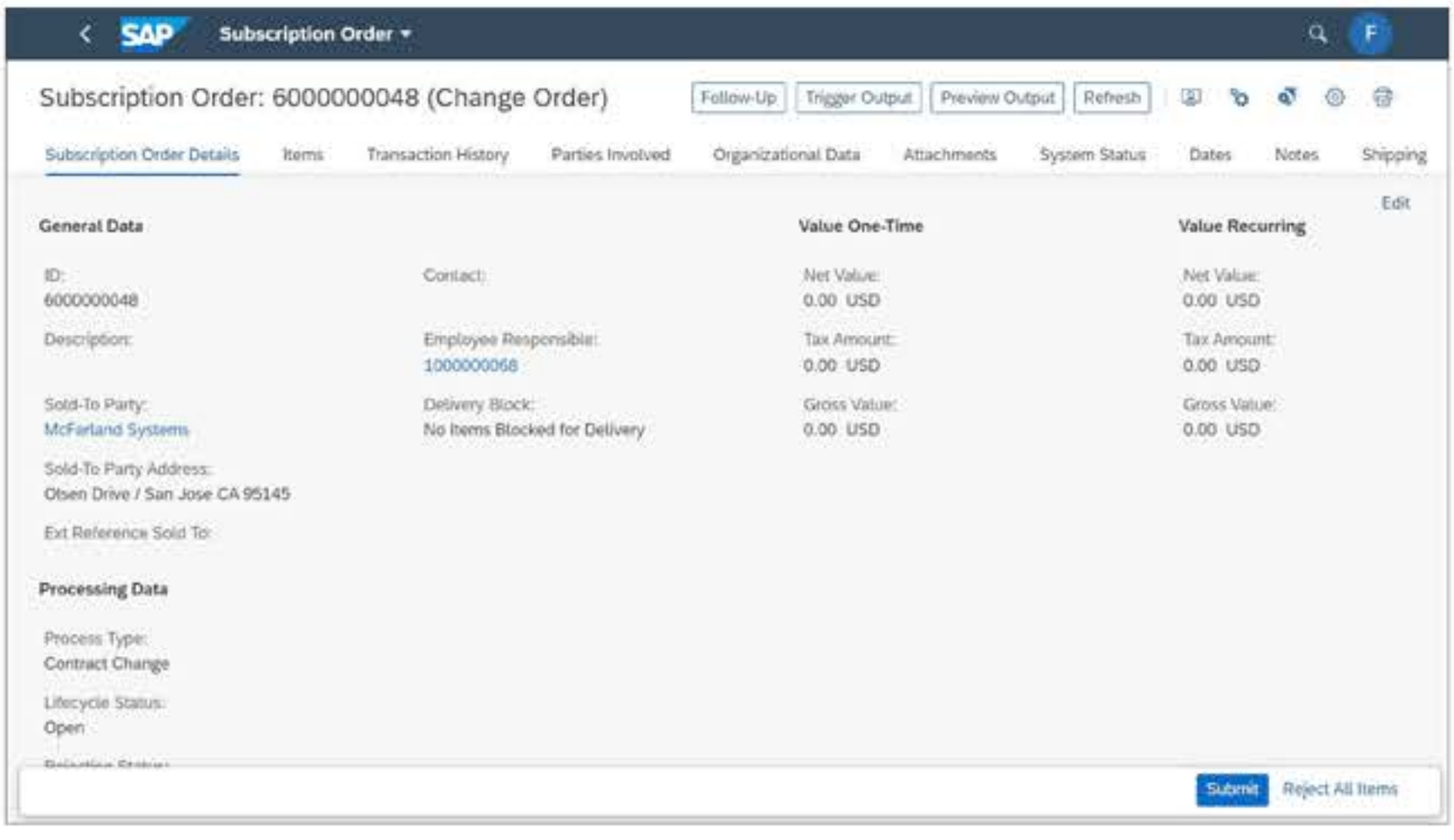

Figure 2.107 Change Order

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8. Click the **Transaction History** tab in the change order, and click on the **Subscription Contracts**. You can see a new time slice created for the same subscription contract with updated dates. The contract is **Technically Active** as well (see Figure 2.108).

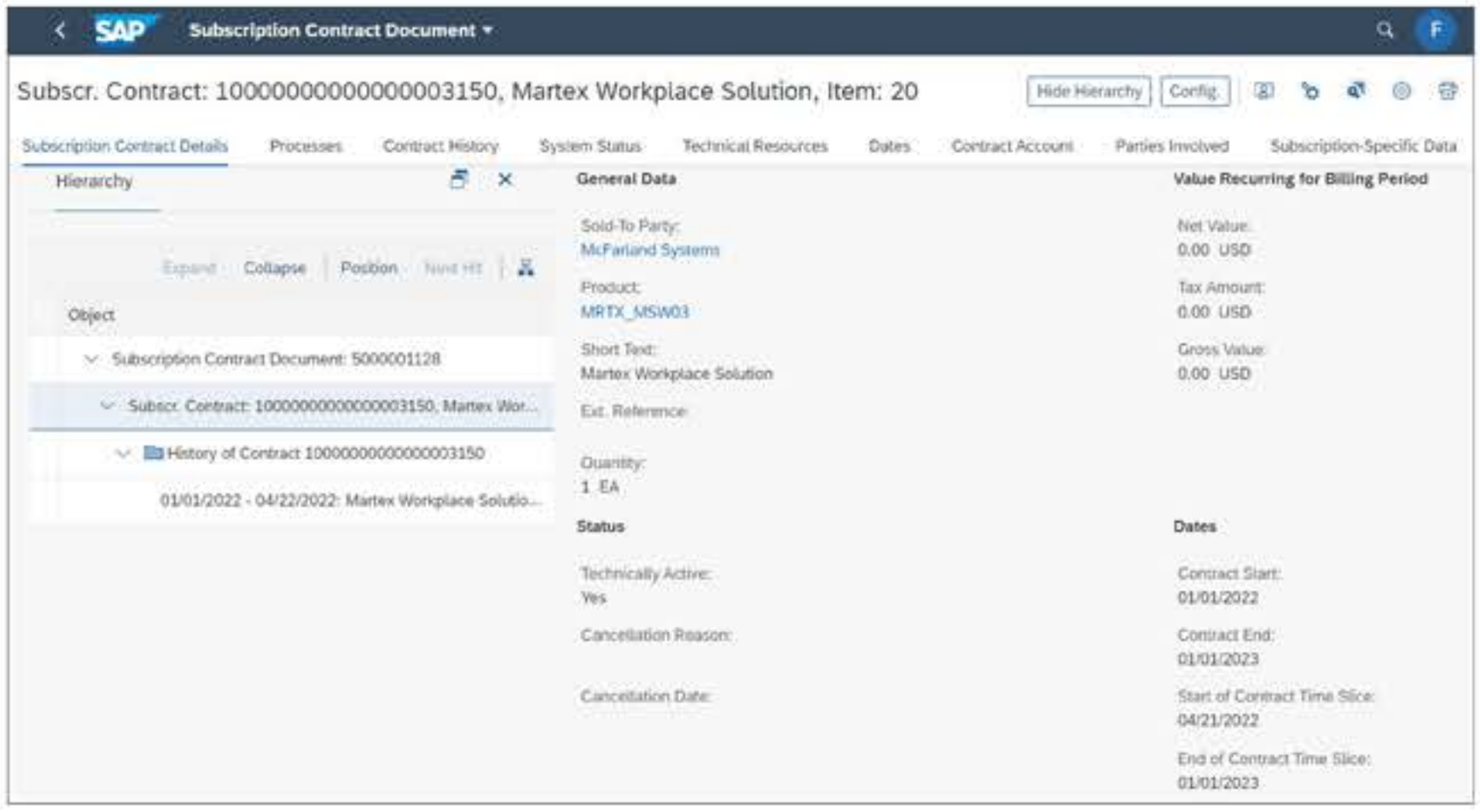

Figure 2.108 Extended Subscription Contract

2.6.2 Cancellation

Customers often request to terminate their contract before the actual contract end date. The cancellation of contracts can be achieved in mass as well as by selecting group of subscription contracts. In our use case, you execute the cancellation process using the Mass Run app as well as the individual cancellation of the contract using the standard change process, as described in the following sections. The contract cancellation can be executed immediately or can be scheduled in the future. Immediate execution will happen without any delays. Cancellation in the future will be captured in the system and executed by a standard batch process.

Mass Run Cancellation

The Mass Run app provides flexibility to find all contracts for various parameters; for example, it can find all the contracts for a given sold-to party, all contracts for a given product, and so on. When a product is supposed to be discontinued, for example, you can use the search feature of Mass Run to find all contracts with the product and trigger immediate cancellation or future date cancellation. It also provides an option to upload contracts via template.

To configure mass runs, follow these steps:

1. **Define the number range.**
   The mass run number range needs to be defined via menu path **Service • Transactions • Settings for Subscription Transactions • Mass Run •Define Number Range for Mass Run** (see Figure 2.109).

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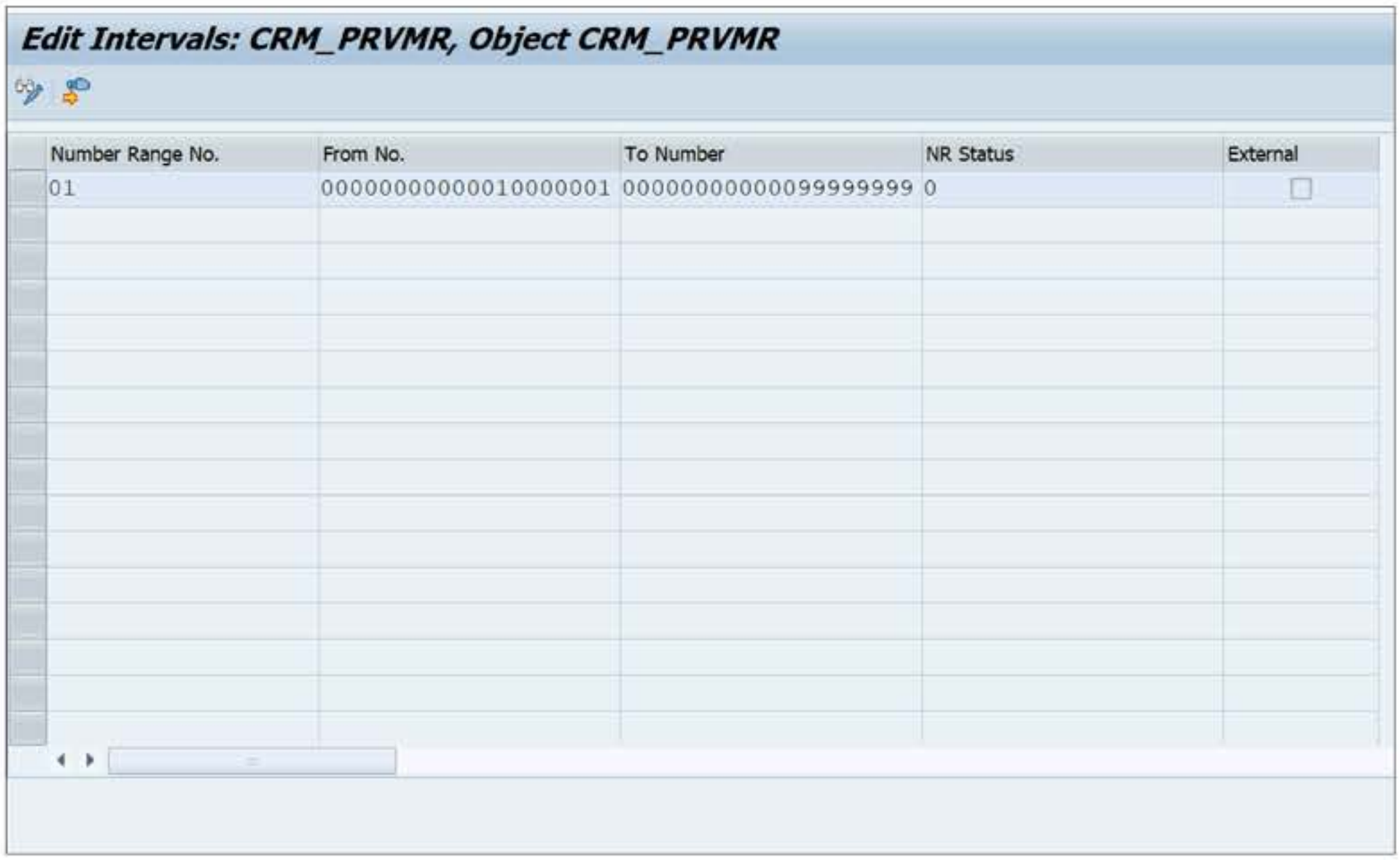

Figure 2.109 Number Range Creation and Update

2. **Set up the mass run.**

The setup for the mass run consists of defining the mass run type, assigning the change process category, and assigning the event handler class on menu path **Service • Transactions • Settings for Subscription Transactions • Mass Run** (see Figure 2.110). In addition, if you need to default fields on a mass run, then you can configure these in **Field Settings** (see Figure 2.111).

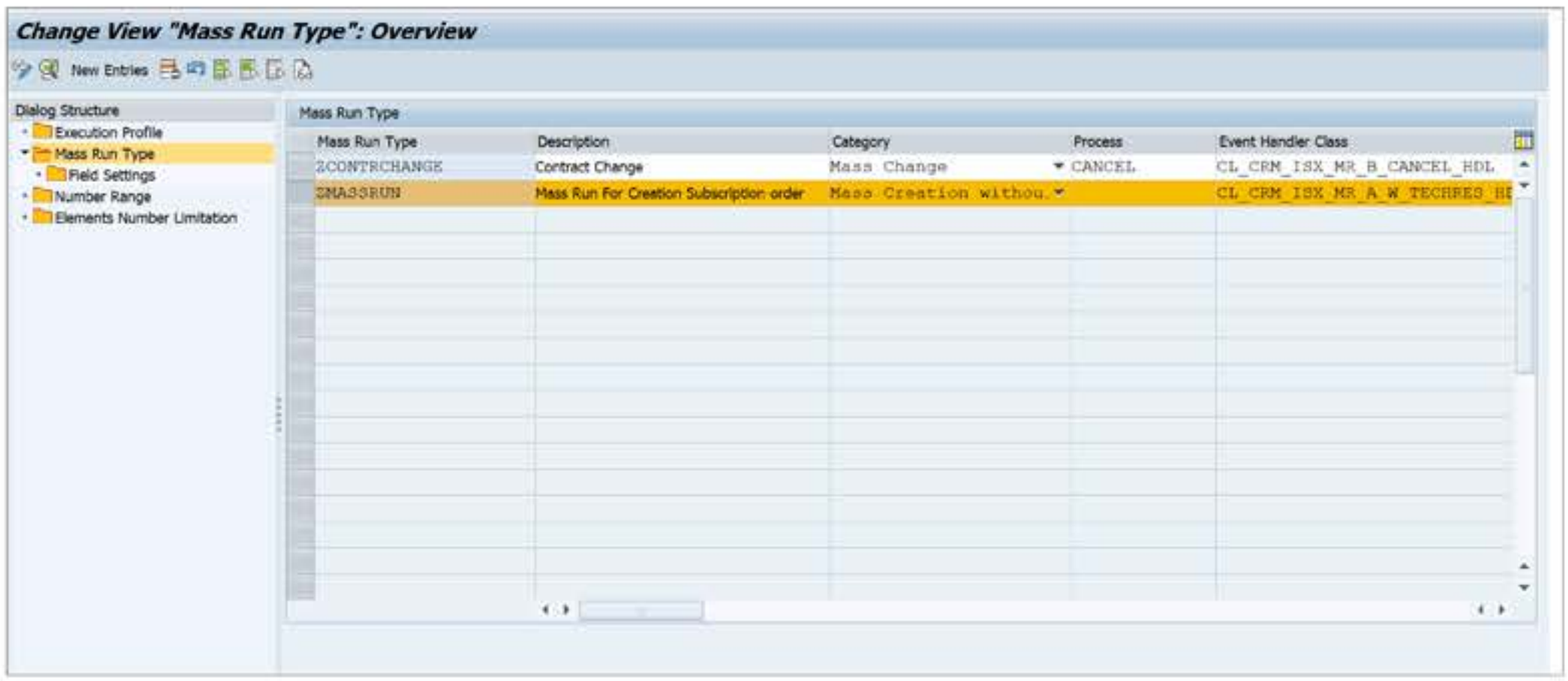

Figure 2.110 Mass Run Setup

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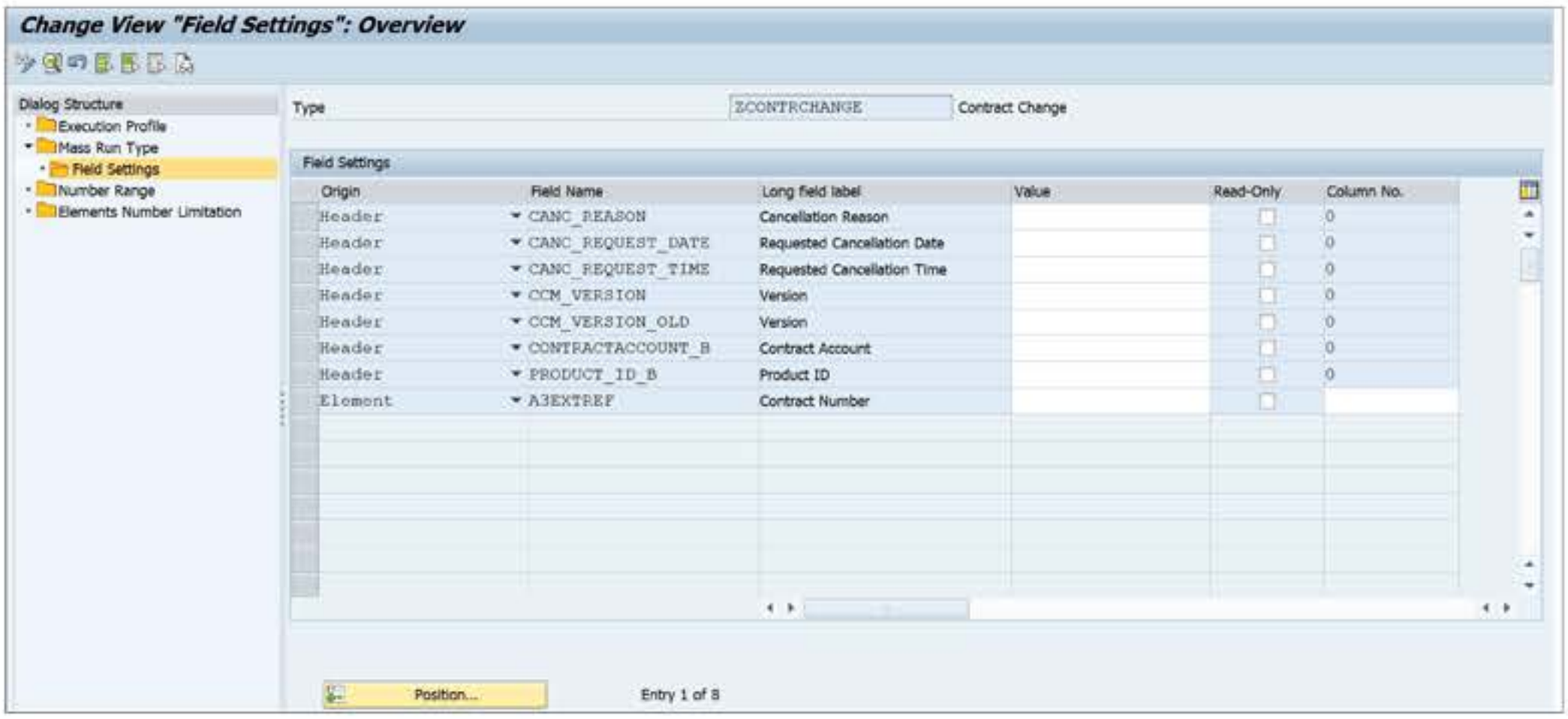

Figure 2.111 Default Values for a Mass Run

In the **Mass Run Type** area of the screen, **Mass Run Type** and **Description** are free-text fields, which you see in the dropdown when you trigger the mass run process. **Category** provides an option to select if mass runs need to be triggered with or without master agreements. **Process** is the standard BTMF process for which you would be configuring the mass run. **Event Handler Class** refers to the out-of-the-box event handler classes that SAP has provided to carry a mass run (e.g., **Cancellation**). For other processes, you need to create custom classes and assign them. You also need to assign the number for the mass run type.

In this use case, one of the products is being discontinued, so all contracts for that product will be canceled with a cancellation date in the future.

For each mass run process, SAP has provided mandatory default fields for which you must set up values.

3. **Execute the mass run.**
   Execute the mass run by following these steps:
   - Log in to SAP Billing and Revenue Innovation Management via the URL shared by the administrator, and open the Mass Run app.
   - Click on **New**, enter a description, and select the **Cancel Mass Run** type from the dropdown.
   - Enter **Activation Date**, **Req. Canc. Date**, **Req. Canc. Time**, and **Cancell. Reason** (see Figure 2.112). You can enter the activation date and cancellation date as the current or future date. You can also select the reason for cancellation manually.
   - From the **Elements** tab, you can search for contracts by entering the product number and choosing all the contracts you want to terminate (see Figure 2.113). You can find contracts via multiple search criteria provided such as product, sold-to party, and master agreement number; this search criteria can also be modified

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for custom requirements. There is also an option to upload the contracts using a Microsoft Excel template.

– The total number of contracts selected is shown under **Total**, and you can view the details by clicking on **Details**.

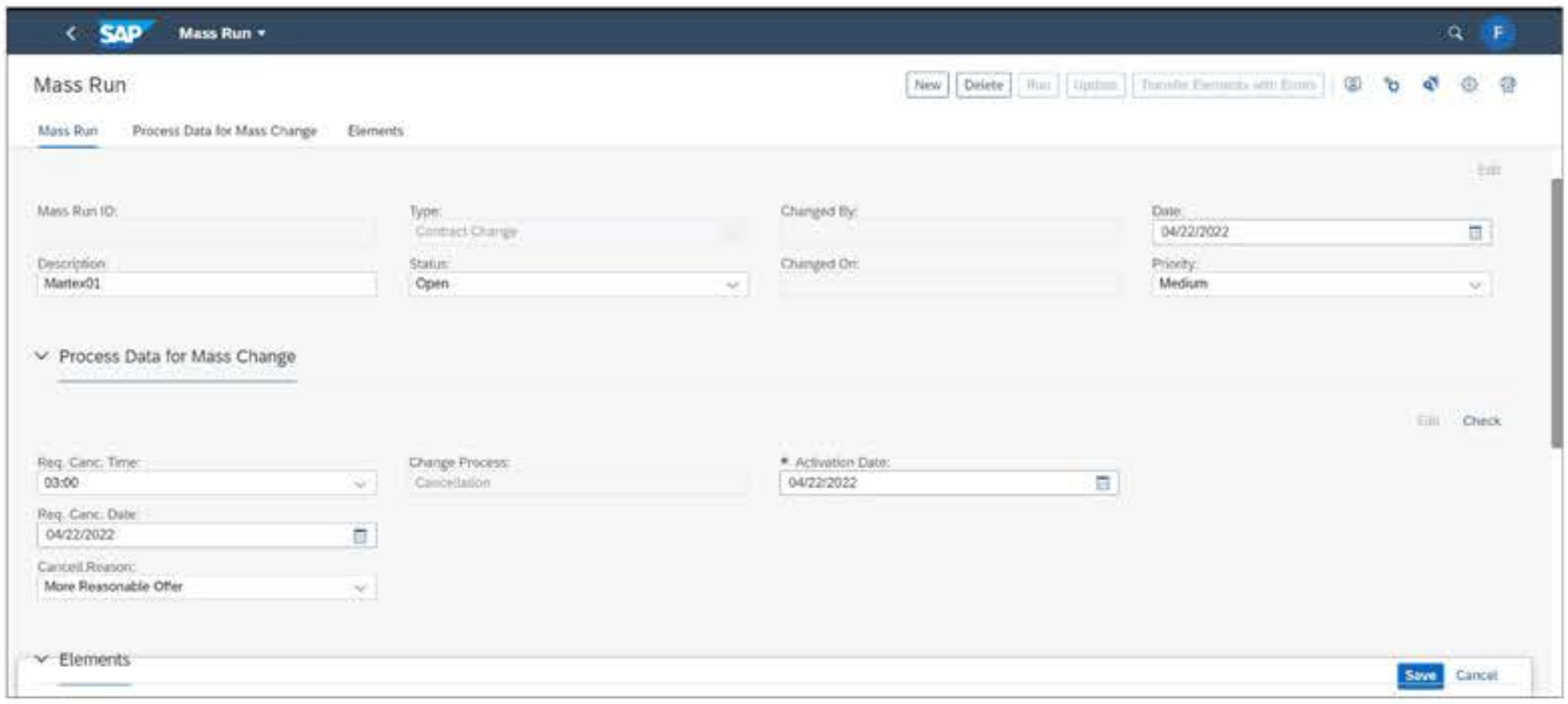

Figure 2.112 Mass Run App: Input Screen

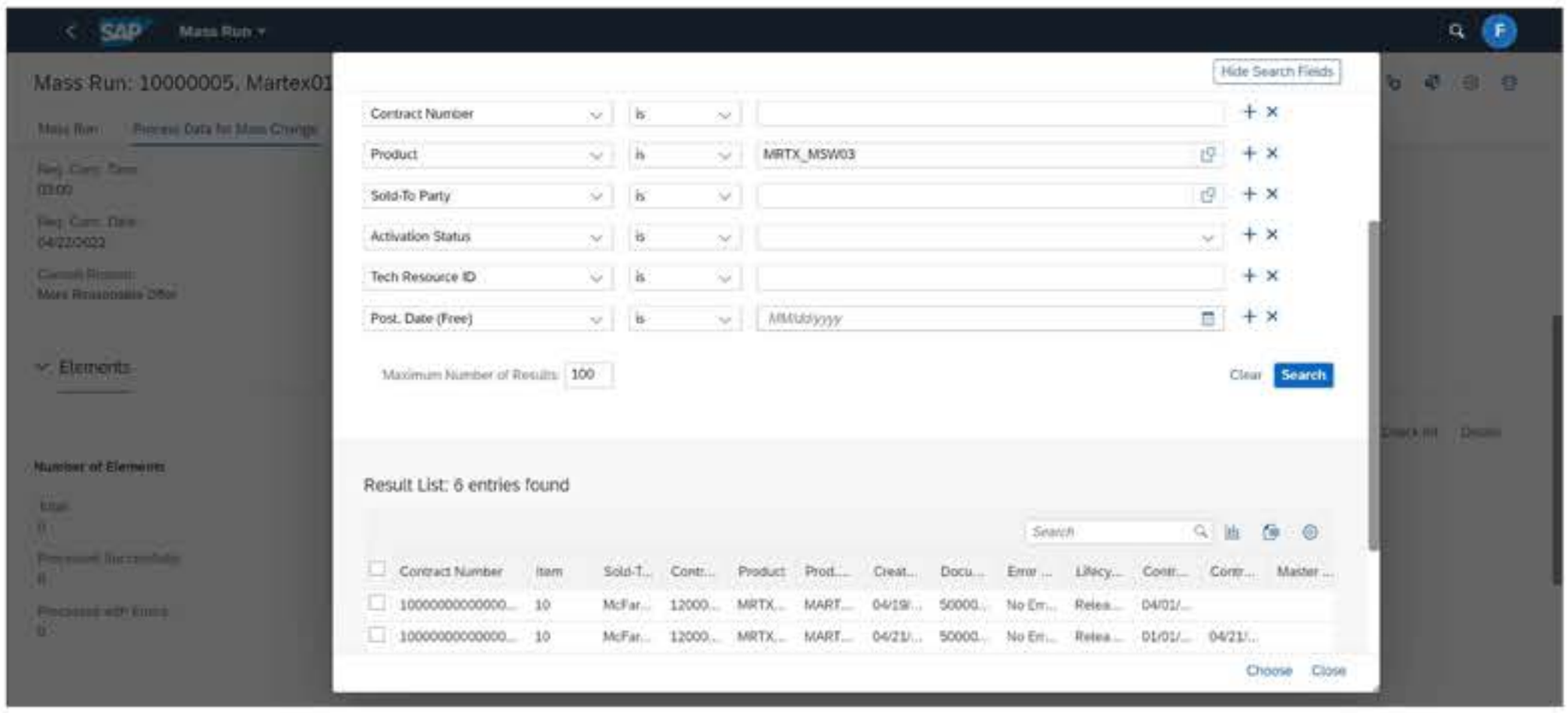

Figure 2.113 Search Screen to Find Contracts

4. Once the number of contracts is selected, click the **Edit** button on the **Mass Run** screen, which changes the status from **Open** to **Released for Processing**, and save (see Figure 2.114).
5. Once saved, the **Run** button gets enabled, which you need to execute, and the status of **Mass Run** changes to **In Process** (see Figure 2.115). Once the update is triggered, you can see the status of the mass run to be **Success** or **Error**.

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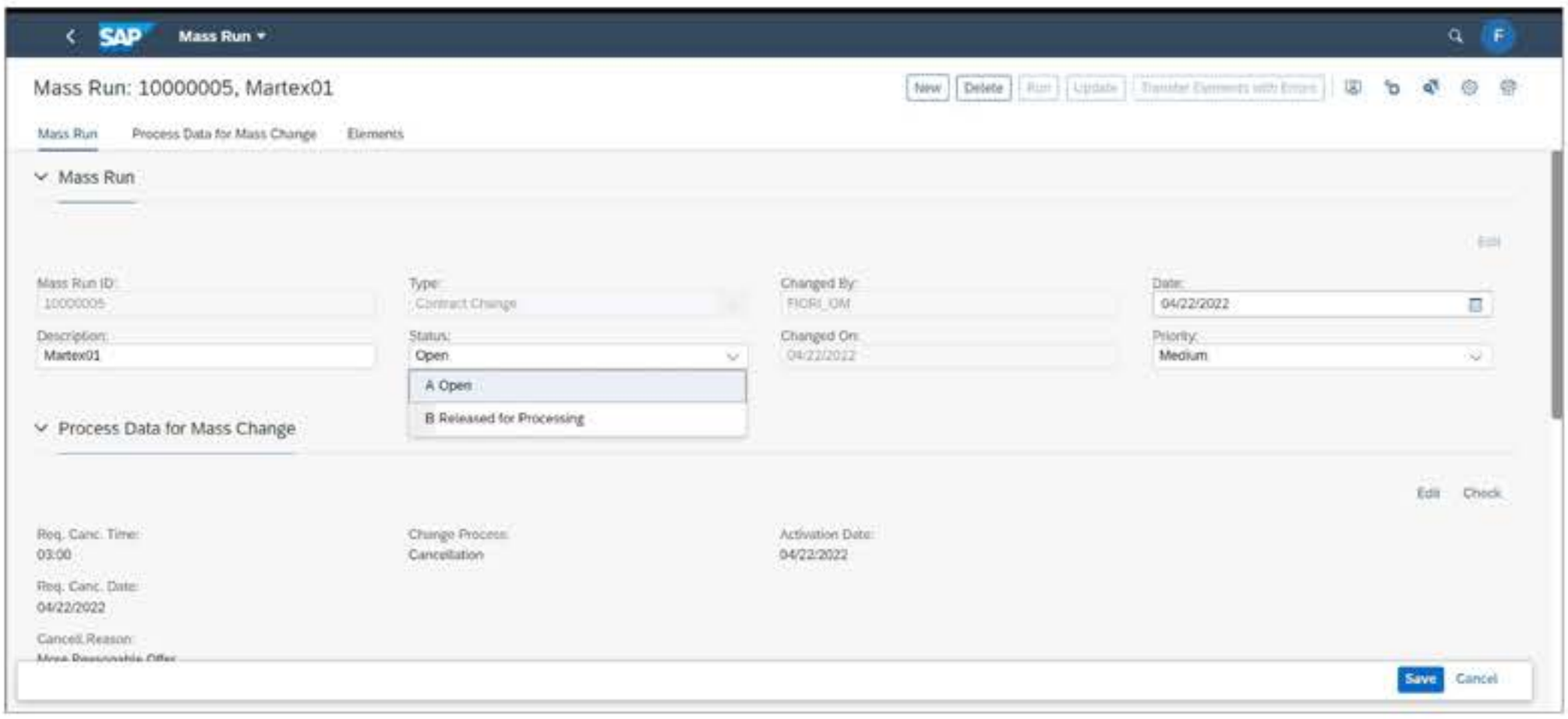

Figure 2.114 Changing the Status of the Mass Run

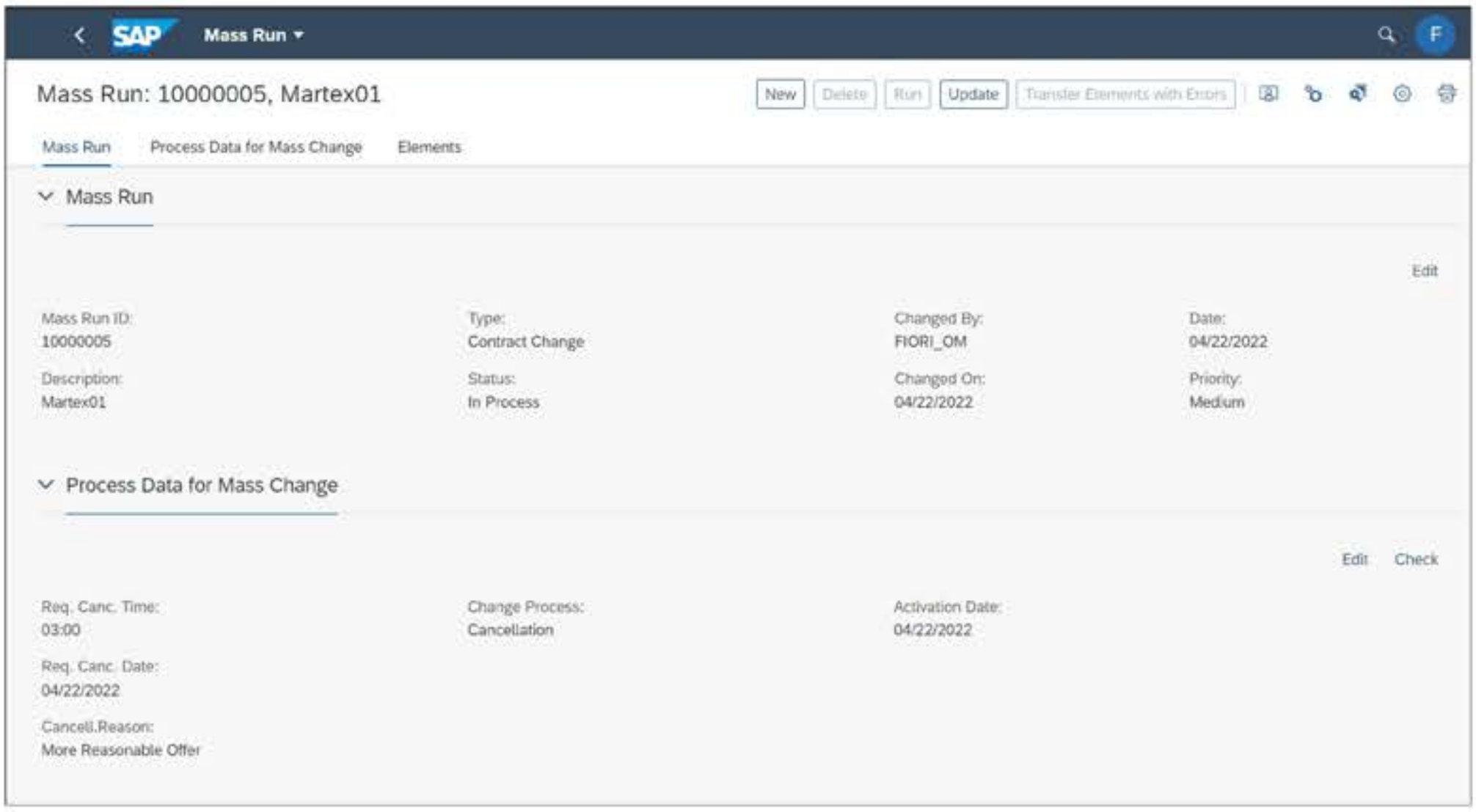

Figure 2.115 Executing the Mass Run Update

6. Click the **Update** button, which will change the status of the **Mass Run** to **Success**.
7. In the **Elements section**, you can see the total number of contracts executed successfully (see Figure 2.116). Select the **Details** button, and enter the **Mass Run ID** and the **Find Contract Number**, and check the status of each contract by clicking on **Details**. You can enter the **Mass Run ID** and find details for each contract, and you can click on the hyperlink to check if a new time slice was created.

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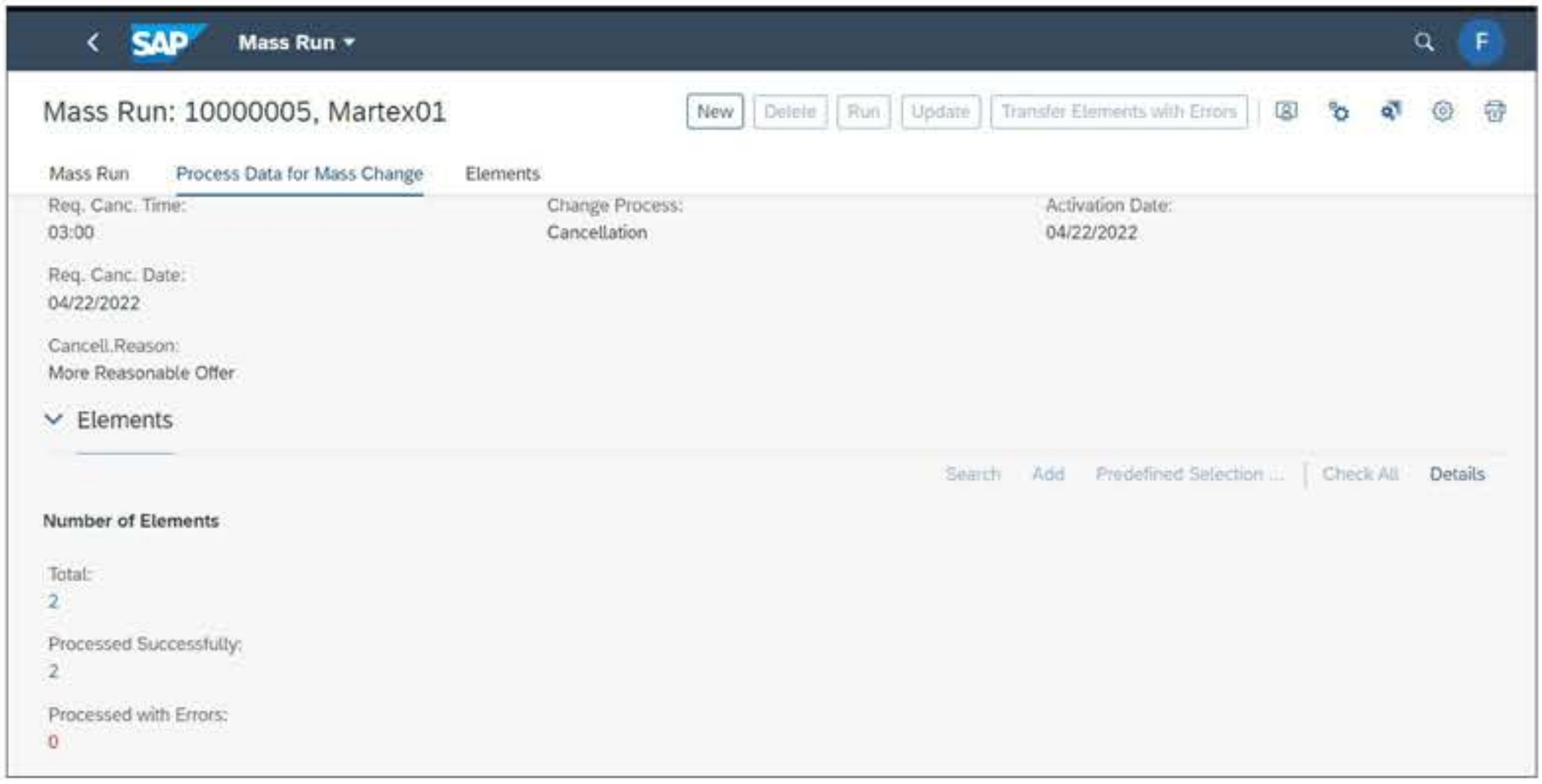

Figure 2.116 Details after the Processing

Individual Cancellation

Due to business dynamics, a customer may decide to terminate a specific contract. To terminate such individual contracts, you can leverage the standard BTMF change process, offered by Subscription Order Management. The contract cancellation can be executed immediately or scheduled in the future. Immediate execution will happen without any delays.

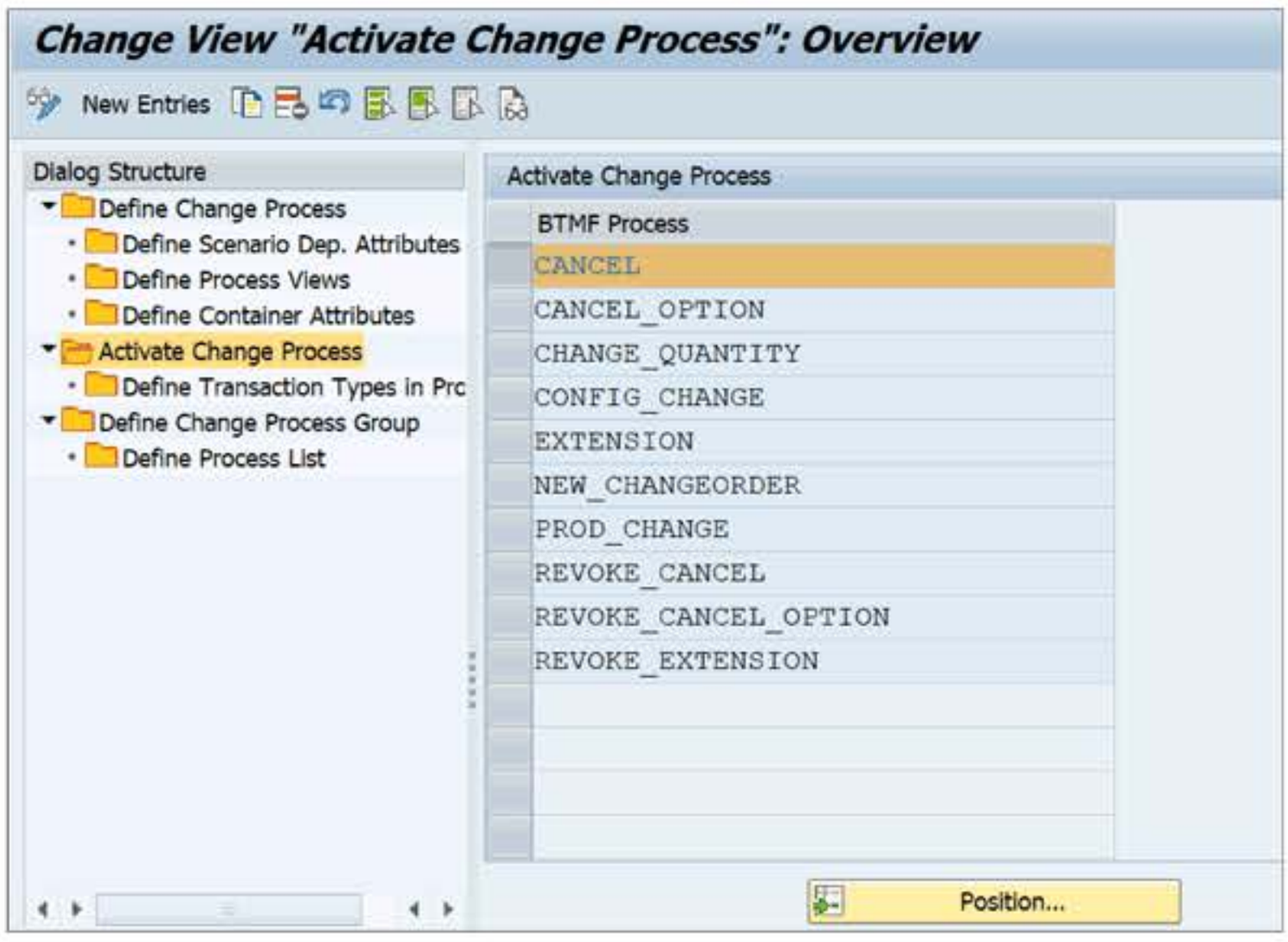

Figure 2.117 Configuration for Cancellation Change Process

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Cancellation in the future will be captured in the system and executed by a batch process. For this use case, you'll cancel the contract created for the MacFarland customer.

You need to configure the cancel process by adding the BTMF process in the **Activate Change Process** node, which can be found at the following menu path: **Service • Transactions • Settings for Subscription Transactions • Contract Changes • Settings for Maintenance Framework (BTMF) • Activate Change Process** (see Figure 2.117). All possible change processes are defined in **Define Change Process**.

To set the cancellation for immediate processing, follow these steps:

1. Log in to the SAP Billing and Revenue Innovation Management in SAP Fiori using the URL shared by your system administrator. Check if the role is assigned to your ID.
2. Select the **Manage Subscription Contracts** app tile. Enter the contract number in the search box, and click **Search**.
3. Click on the contract number hyperlink to go to the **Subscription Contract Document** screen.
4. In the **Subscription Contract Details** tab, select line item **10** in the contract hierarchy, and you'll see the contract's **General Data** (see Figure 2.118). The subscription data shows details such as contract start and end date and contract status (**Technically Active**, **Yes** or **No**). If the contract is Canceled, then the cancellation reason and cancellation date will also be displayed.

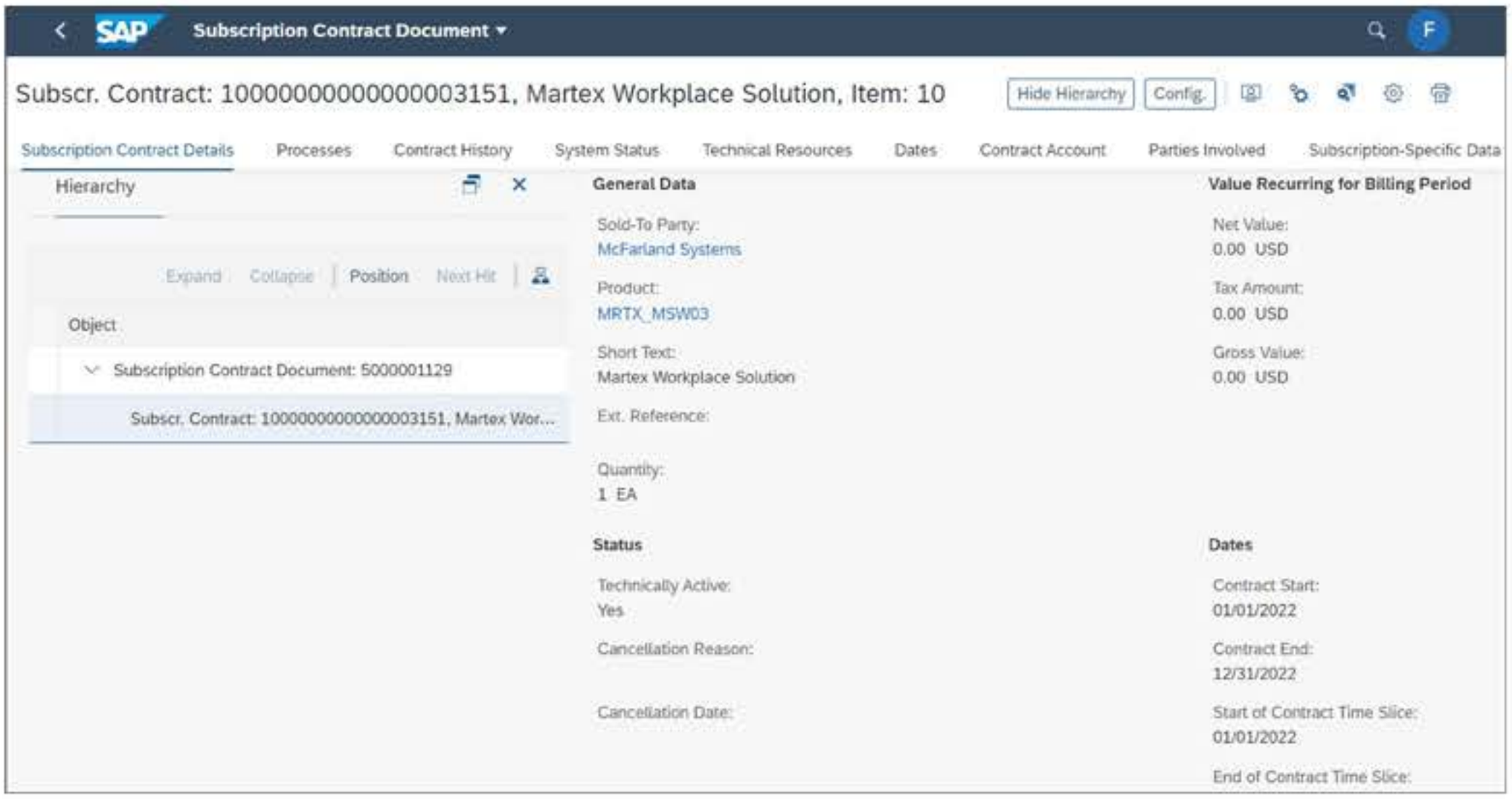

Figure 2.118 Subscription Contract: General Data

5. Navigate to the **Processes** tab, and you'll see a list of applicable and configured change processes (Figure 2.119).

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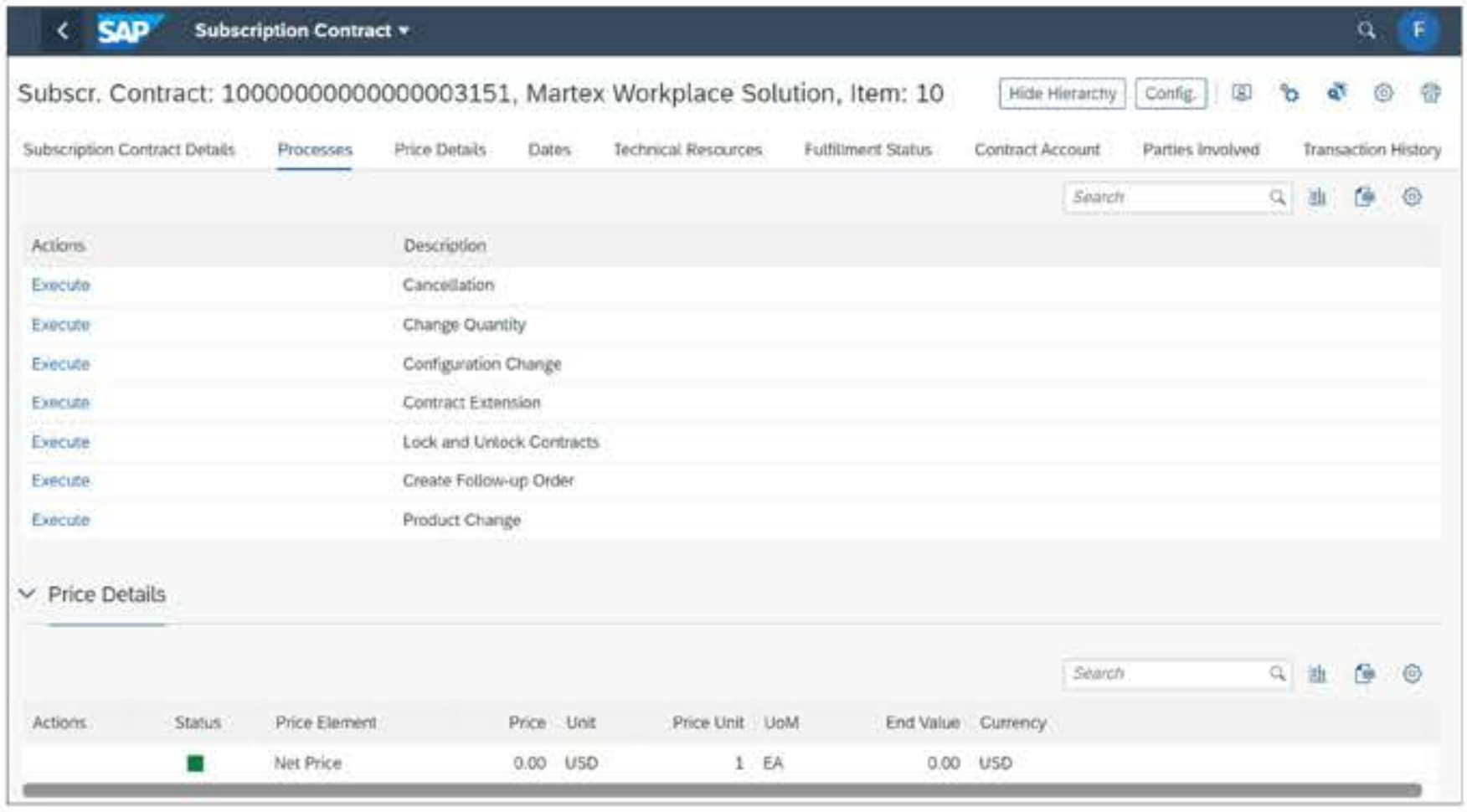

Figure 2.119 Subscription Contract: Processes Tab

6. Click **Execute** in the row for the **Cancellation** change process. A new popup window opens.
7. Update the **Activation date** and time, the **Requested Cancellation Date** and **Requested Cancellation Time** as today's date, and **Cancellation Reason** code (see Figure 2.120) and click **Submit**. The **Activation date** and time is the actual date/time the contract is slated for cancellation. **Requested Cancellation Date/Requested Cancellation Time** can be the same as or after the activation date and time. When the change order is triggered, click **Submit** (see Figure 2.121). The change order is automatically created with reference to the Subscription contract that is being changed.

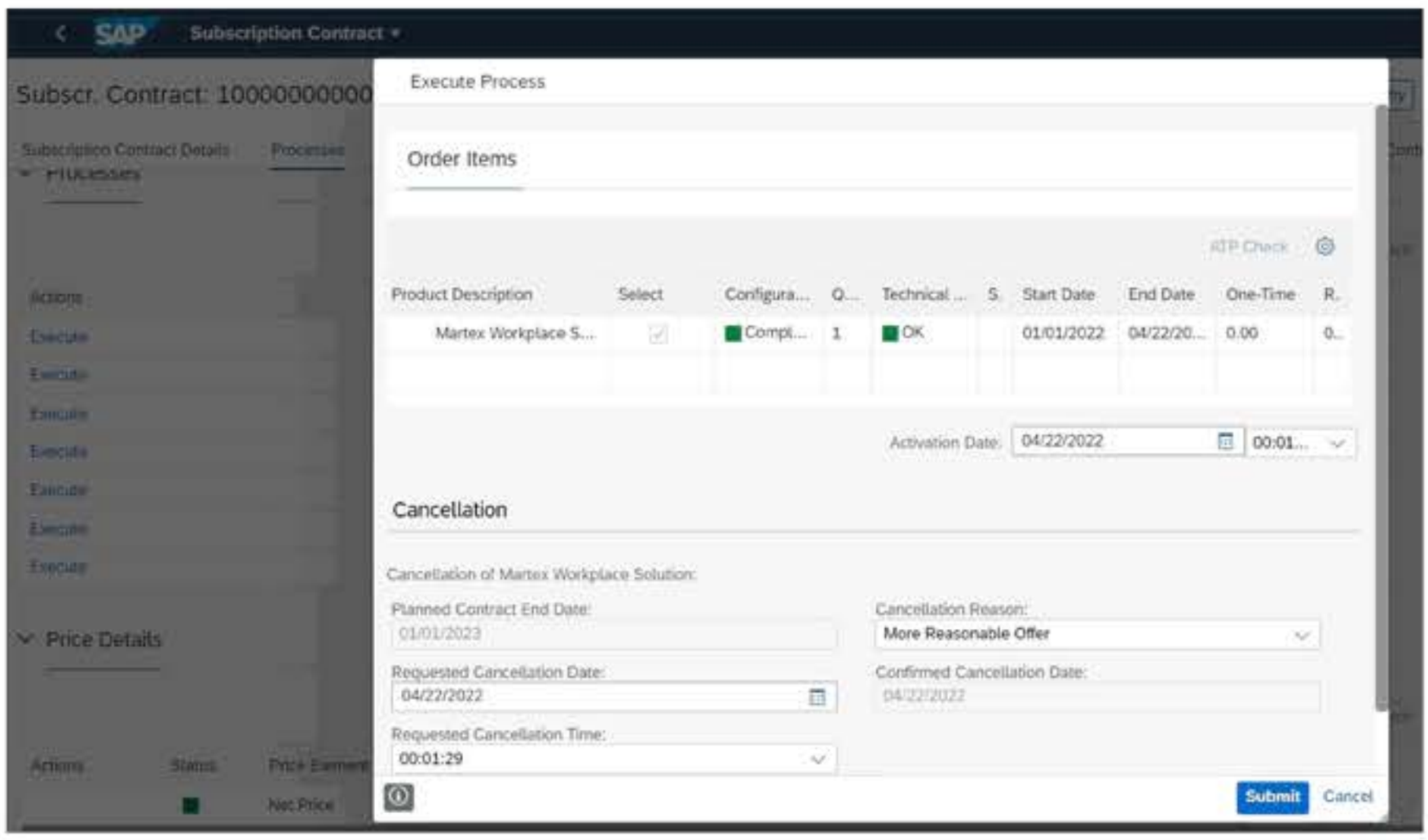

Figure 2.120 Cancellation Change Process Details

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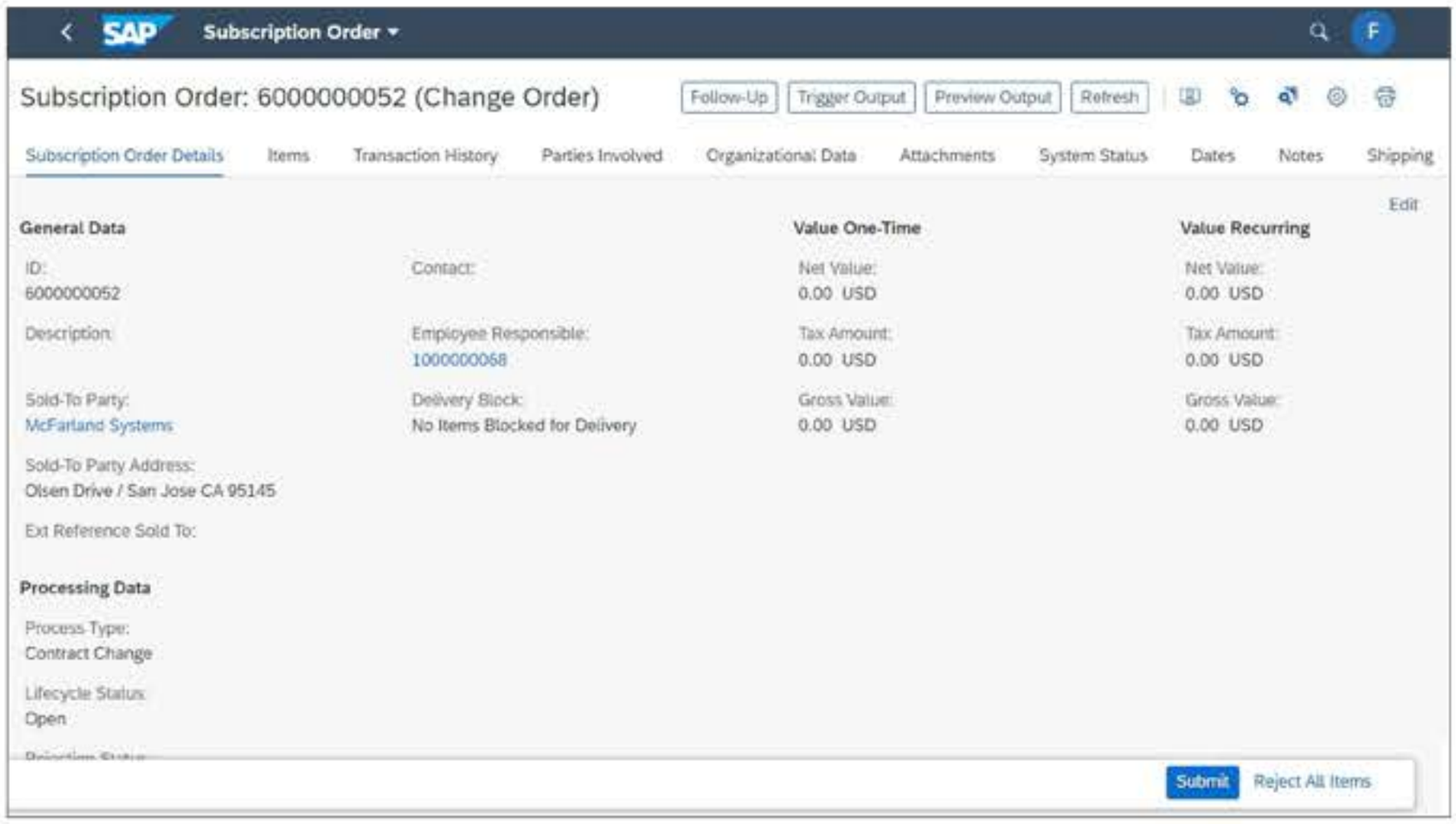

Figure 2.121 Change Order

8. Navigate to the **Transaction History** tab and click on the subscription contract hyperlink. The contract **Status** will be **Canceled**, and **Contract End** date will be the cancellation date, as shown in Figure 2.122. Note that the contract history is created with time slices if the cancellation date is in future. In such cases, a new time slice gets created with **Technically Active** status as **No**.

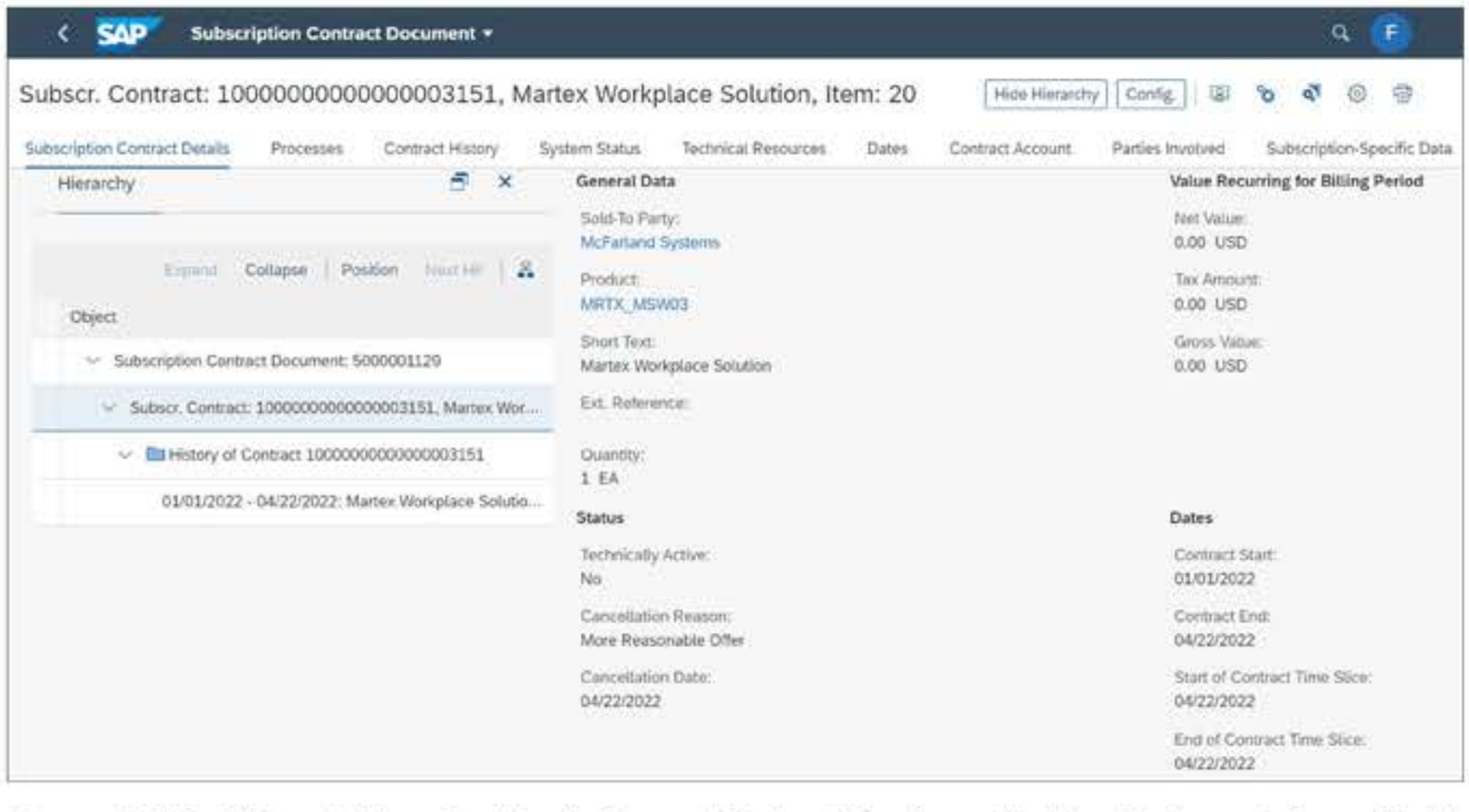

Figure 2.122 Subscription Contract General Data Tab: Cancellation Date and Cancellation Reason

2.6.3 Service Upgrade and Downgrade

Sometimes customers decide to upgrade/downgrade their product bundle and service in between live contracts. The upgrade and downgrade change process provides flexi-

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bility to achieve same on same subscription contracts. In our use case, we'll change the service from Product A to Product B. In this example, the subscription contract has an end date. In our use case, we'll replace the MWS product - Gold option with the MWS product - Silver option for our customer MacFarland Systems. After executing the change process, a new time slice will be created with a new MWS - Silver option, and the contract will have **Technically Active** set to **Yes**, whereas the old time slice will display as **No**.

You need to configure the extension process by adding the BTMF process in the **Activate Change Process** node, which can be found at the following menu path: **Service • Transactions • Settings for Subscription transactions • Contract Changes • Settings for Maintenance Framework (BTMF) • Activate Change Process** (see Figure 2.123). All possible change processes are defined in **Define Change Process**.

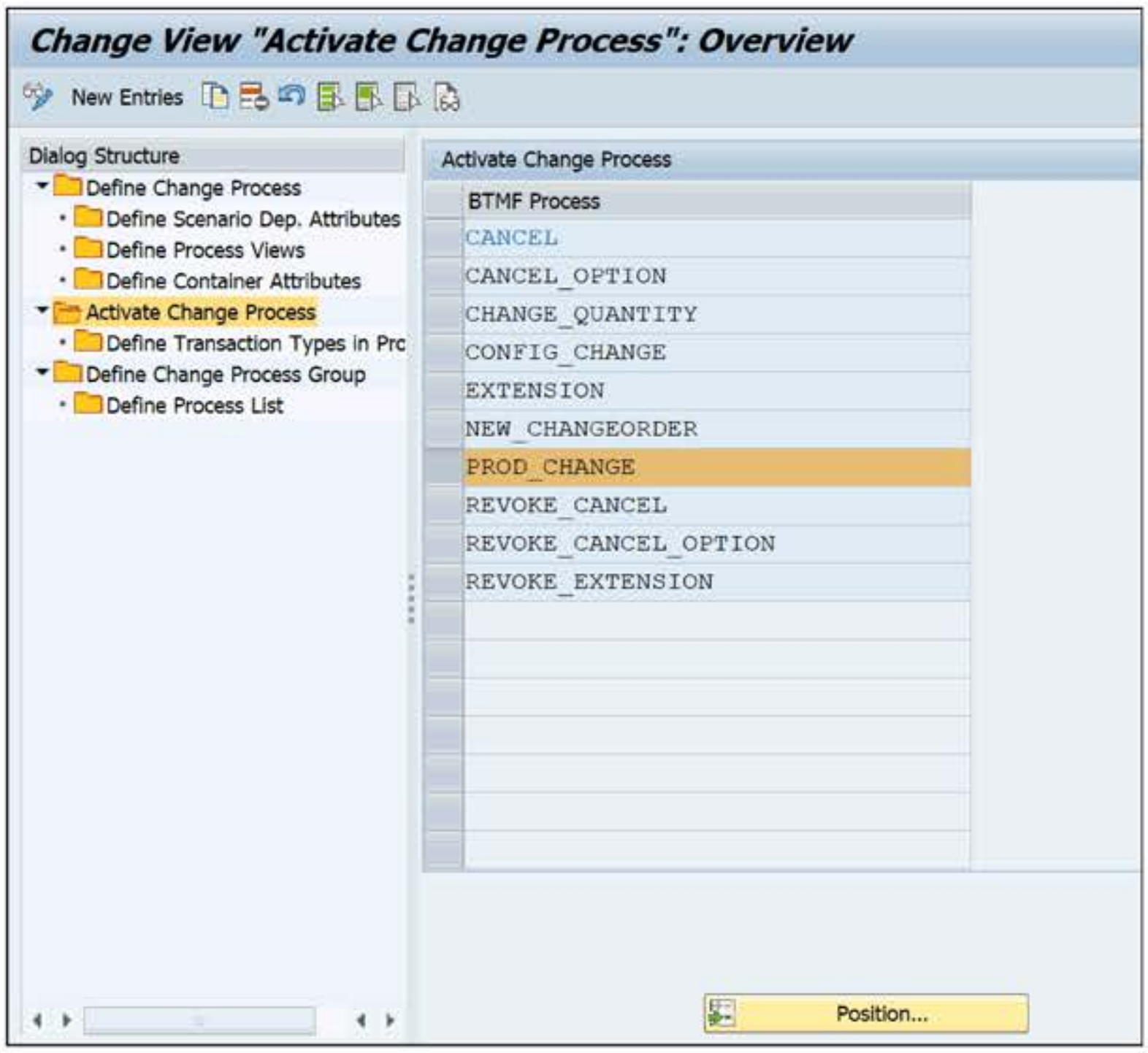

Figure 2.123 BTMF Product Change Configuration

To execute the upgrade/downgrade for immediate processing, follow these steps:

1. Log in to the SAP Billing and Revenue Innovation Management in SAP Fiori using the URL shared by your system administrator. Check if the role is assigned to your ID.
2. Select the **Manage Subscription Contracts** tile. Enter the contract number in the search box and click **Search**.

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3. Click on the hyperlink of the contract number.
4. You'll be navigated to the **Subscription Contract Details**. Select line item **10** in the contract hierarchy, and you'll see **General Data**, as shown in Figure 2.124. The subscription data shows details such as contract start/end date and contract status (**Technically Active** – **Yes** or **No**).
5. Navigate to the **Processes** tab, and you'll see a list of applicable change processes (see Figure 2.125).

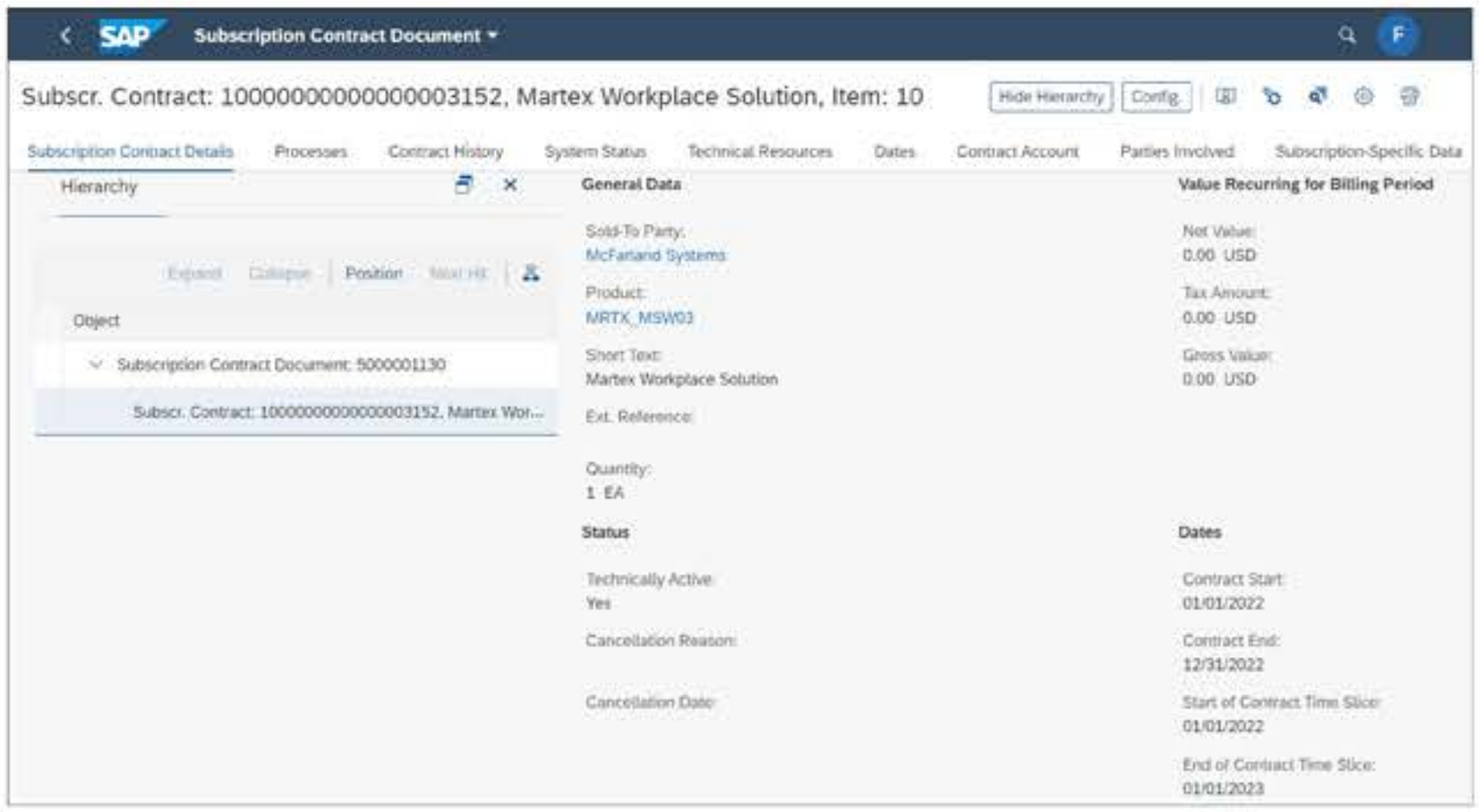

Figure 2.124 Subscription Contract: General Data Tab

Figure 2.125 Subscription Contract: Processes Tab

6. Click **Execute** in the row for the **Product Change** process. A new popup window opens. Update **Activation Date** and time to today's date, select the required product, and click **Submit** (see Figure 2.126). Activation date can be entered as a future date. Because you're changing the product, you can change the contracts account and billing plan if required. If the product is configurable, then you need to enter characteristic values.

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7. Because the product has variant characteristics, on the next screen, click on **Change Configuration**, update the **Characteristic** values, and click **Submit** (see Figure 2.127). A change order is triggered when you click **Submit**. The **Activation Date** and time is the actual date/time the contract is slated to have the new product. After maintaining the values for the characteristics, you can view them.

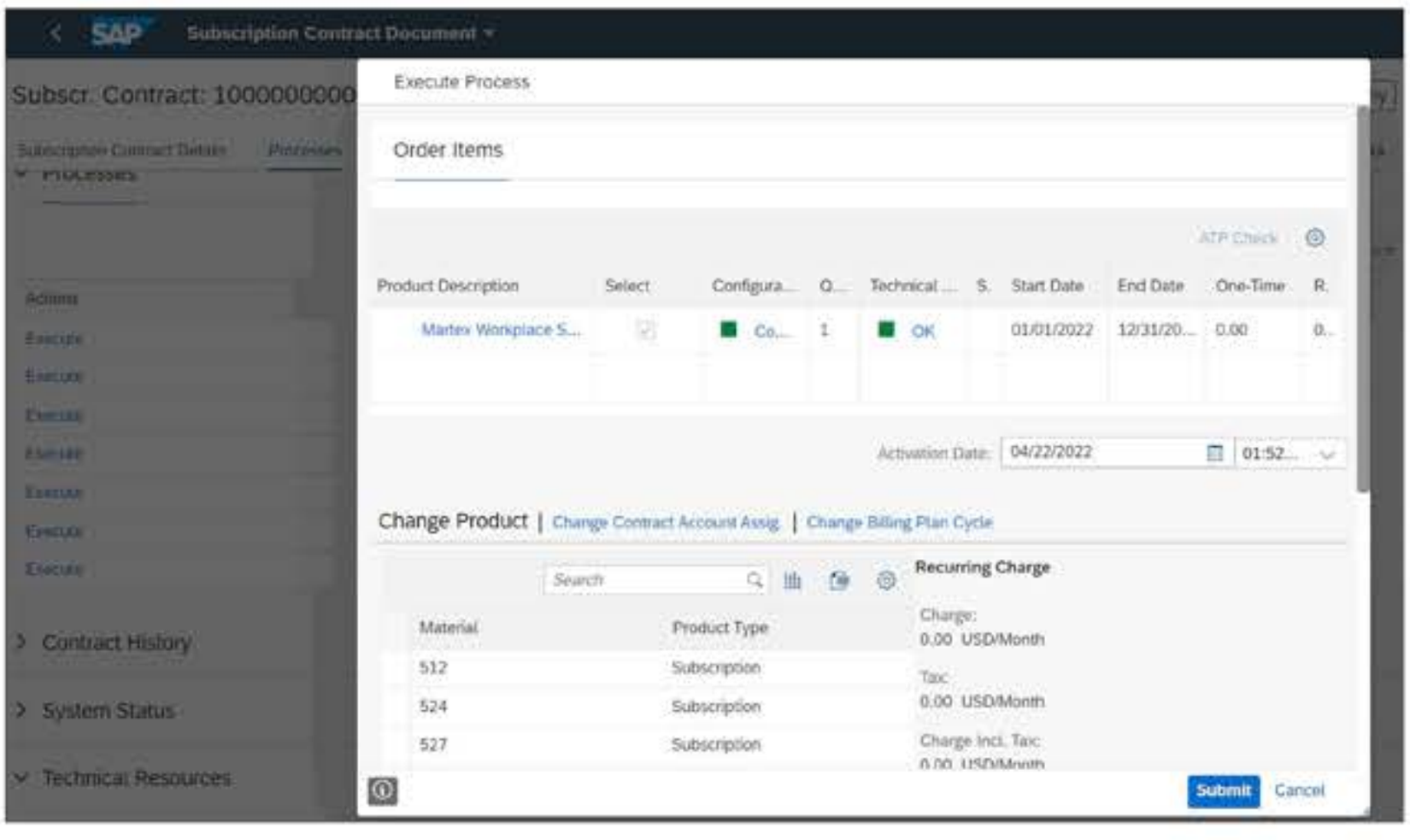

Figure 2.126 Change Product Process Screen

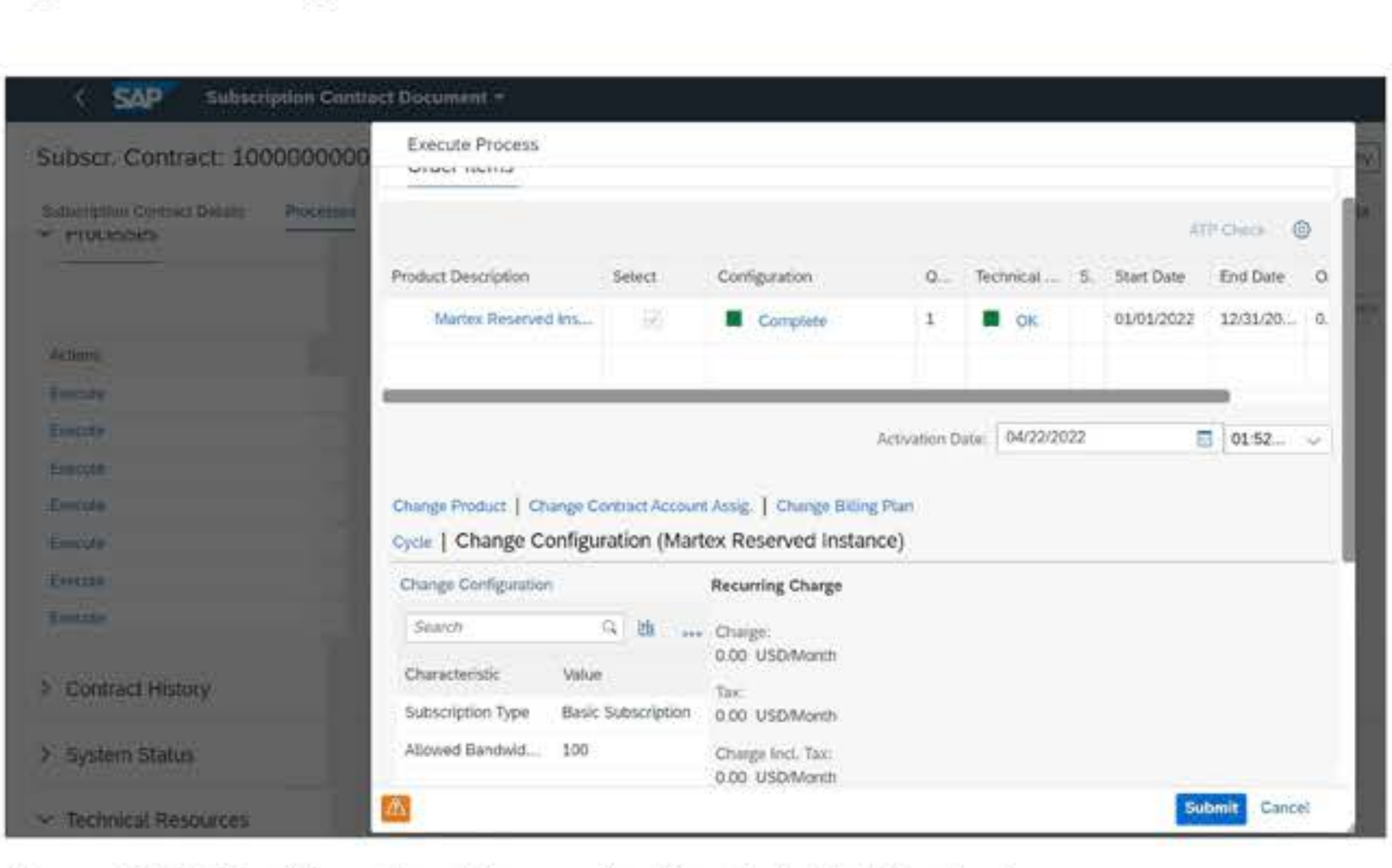

Figure 2.127 Configuration Change for Newly Added Product

8. Navigate to the **Transaction History** tab and click on the hyperlink for **Subscription Contract**. The contract will have a new time slice with a new product, and the old time slice will be technically inactive with the old product details (see Figure 2.128). Note that the contract history is created with time slices with **Technically Active** status as **No**.

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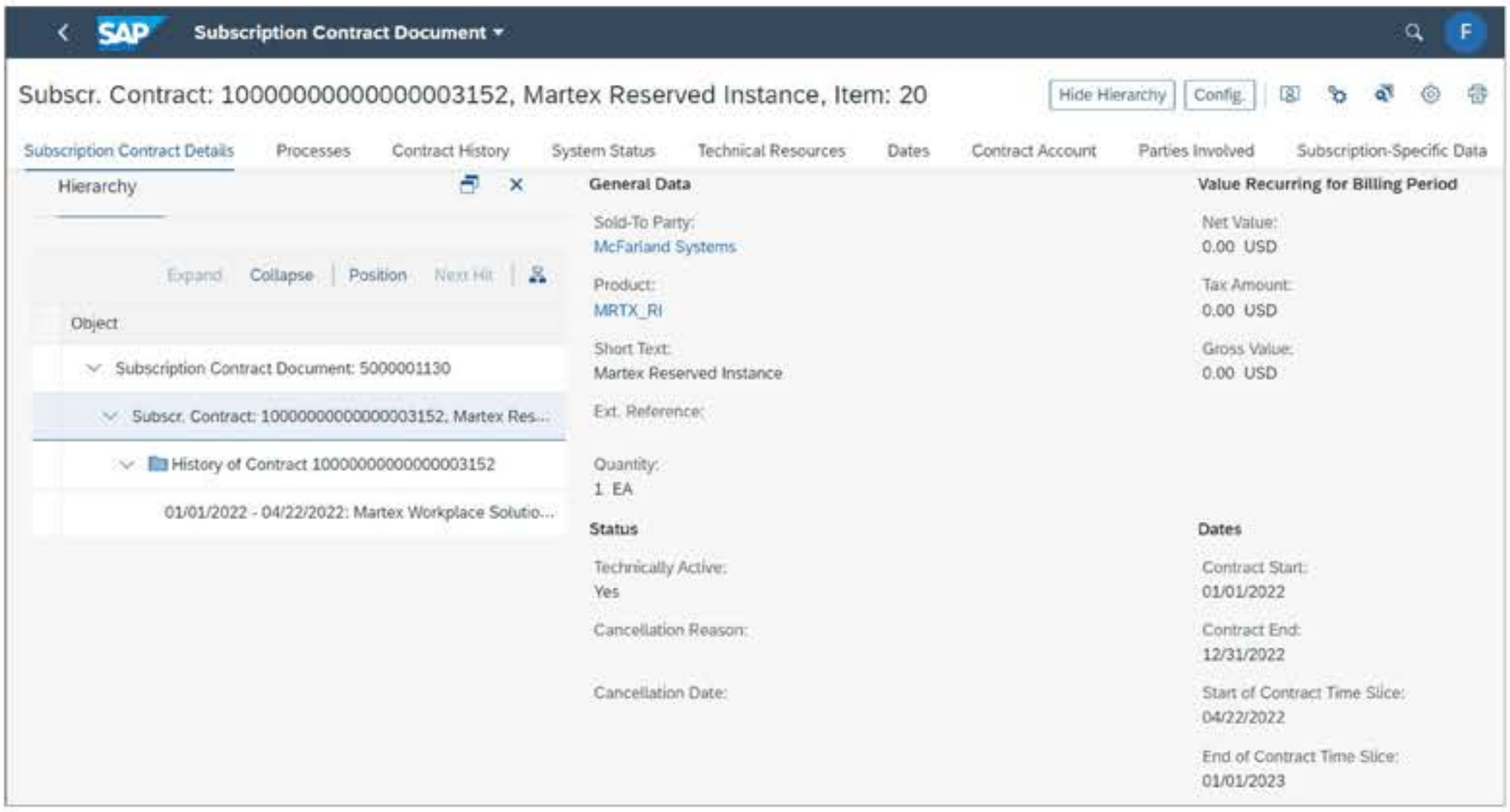

Figure 2.128 Subscription Contract with New Product in the General Data Tab

2.6.4 Change of Contract Terms

Usually the core contract terms, which consists of customer billing and pricing information, are captured in subscription configuration. Over the period of subscription life cycles, it may require an update based on customers' request or company's guidelines. Also, if the Subscription Order Management is integrated with SAP Convergent Charging and the configuration values are used to derive subscription pricing and billing related rules, it is important to capture the change with change effective period to avoid billing discrepancies.

The configuration change process provides the flexibility to achieve this for existing subscription contracts.

In our use case, we'll change the **Martex Subscription Type** from **Base Subscription** (which charges $20/month) to **Advanced Bandwidth** (which charges $30/month) for the existing contract for our customer MacFarland Systems. After executing the change process, a new time slice will be created with advance bandwidth, and the contract will have **Technically Active** set to **Yes**, whereas the old time slice will display as **No**.

To do this, you need to configure the extension process by adding the BTMF process in the **Activate Change Process** node, which can be found at menu path **Service • Transactions • Settings for Subscription Transactions • Contract Changes • Settings for Maintenance Framework (BTMF) • Activate Change Process** (see Figure 2.129). All possible change processes are defined in **Define Change Process**.

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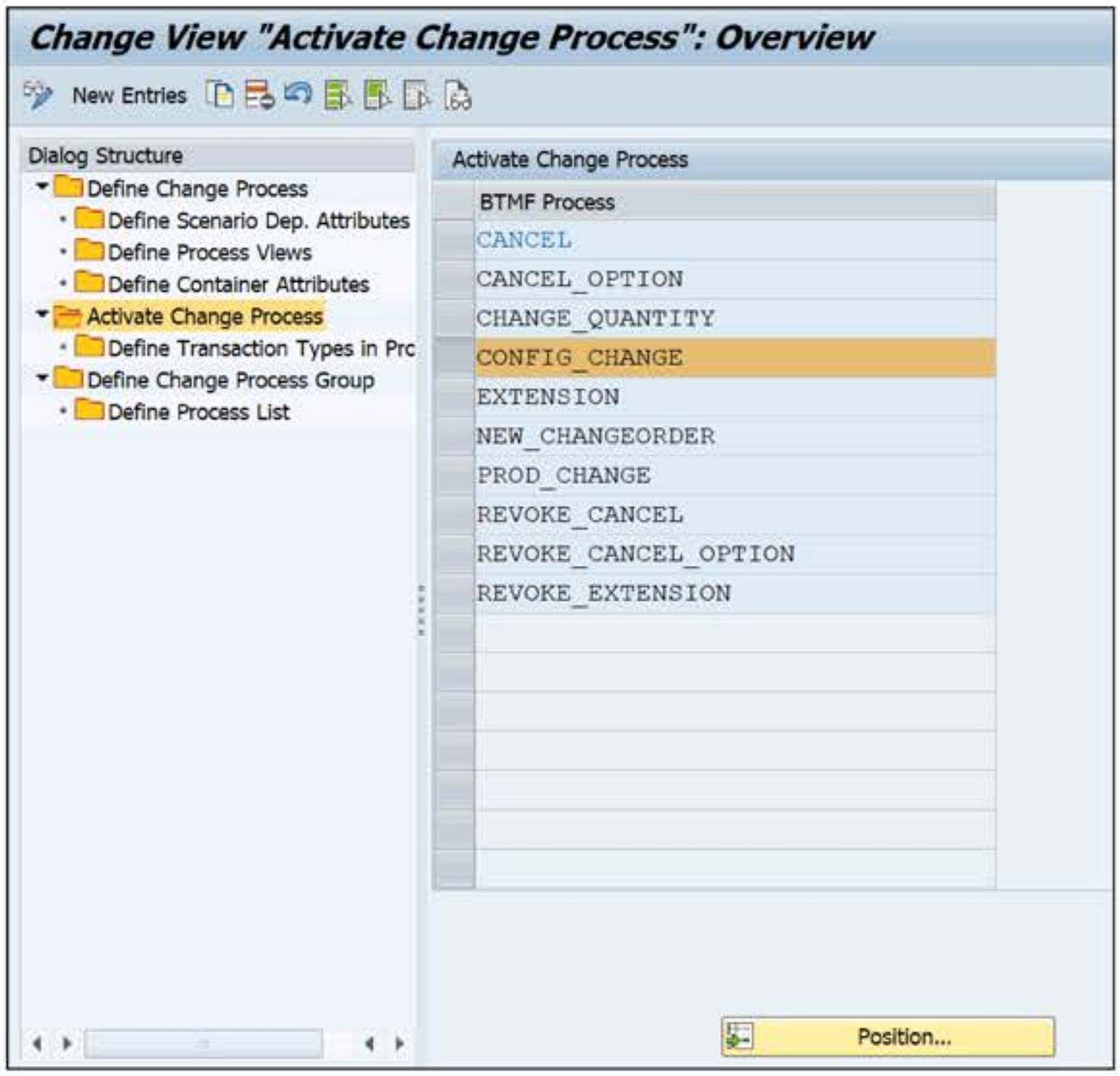

Figure 2.129 CONFIG_CHANGE Process Configuration

To execute the add-on service, that is, the configuration change process, follow these steps:

1. Log in to SAP Billing and Revenue Innovation Management in SAP Fiori using the URL shared by your system administrator. Check if the role is assigned to your ID.
2. Select the **Manage Subscription Contracts** tile. Enter the contract number in the search box and click **Search**.
3. Click on the hyperlink of the contract number. You'll be navigated to the **Subscription Contract Details**.
4. Select line item **10** in the contract hierarchy, and you'll see the contract **General Data** (see Figure 2.130). The subscription data shows details such as contract start/end date and contract **Status** (**Technically Active** – **Yes** or **No**).
5. Navigate to the **Processes** tab, and you'll see list of applicable change processes, as shown in Figure 2.131.
6. Click **Execute** in the row for the **Configuration Change** process. A new popup window opens. Update the **Activation date** and time as today's date (see Figure 2.132) and click **Change Configuration**. When this screen is loaded, it displays existing characteristics of the product; after you change the characteristic values, it displays changed values.

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2 Subscription Order Management

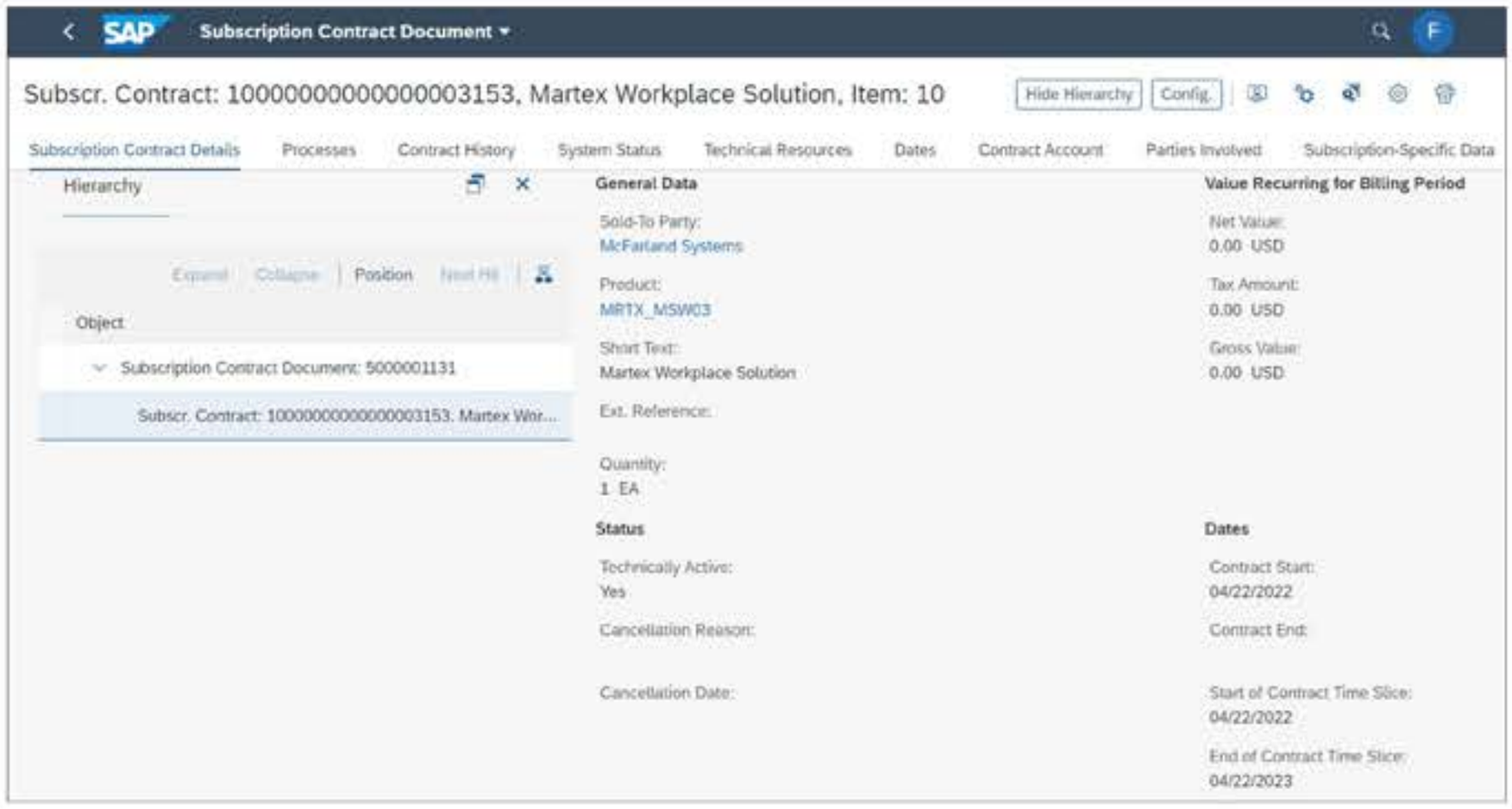

Figure 2.130 Subscription Contract: General Data Tab

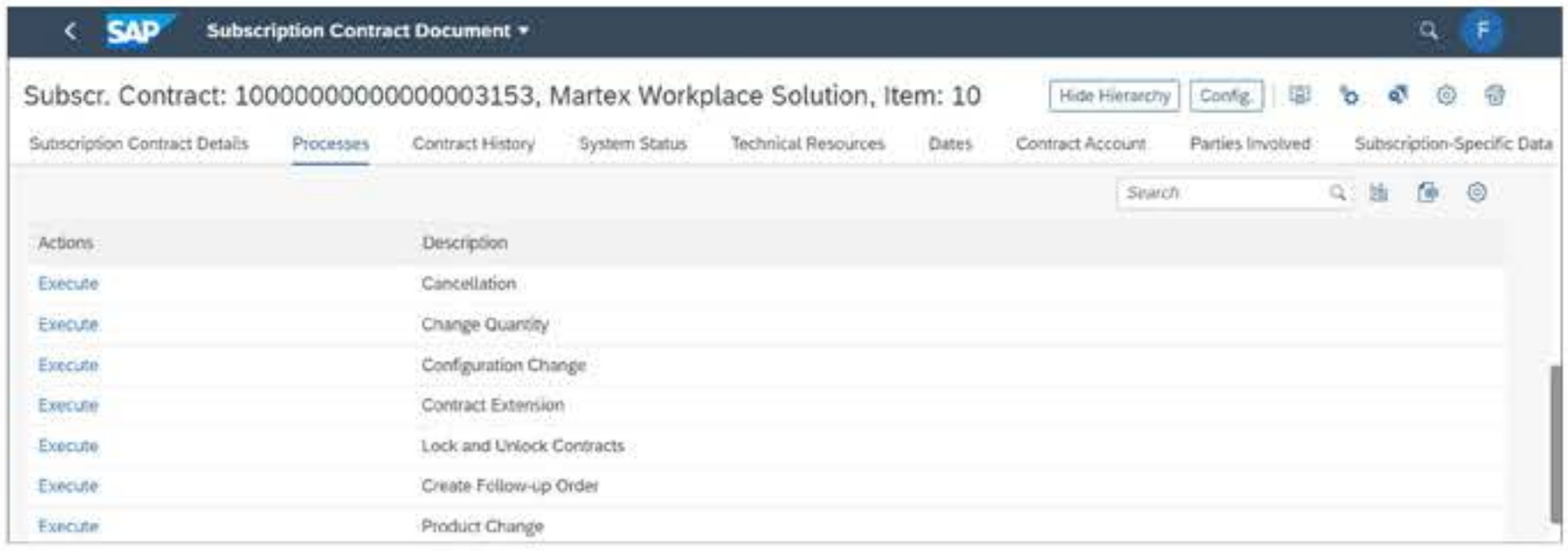

Figure 2.131 Subscription Contract: Processes Tab

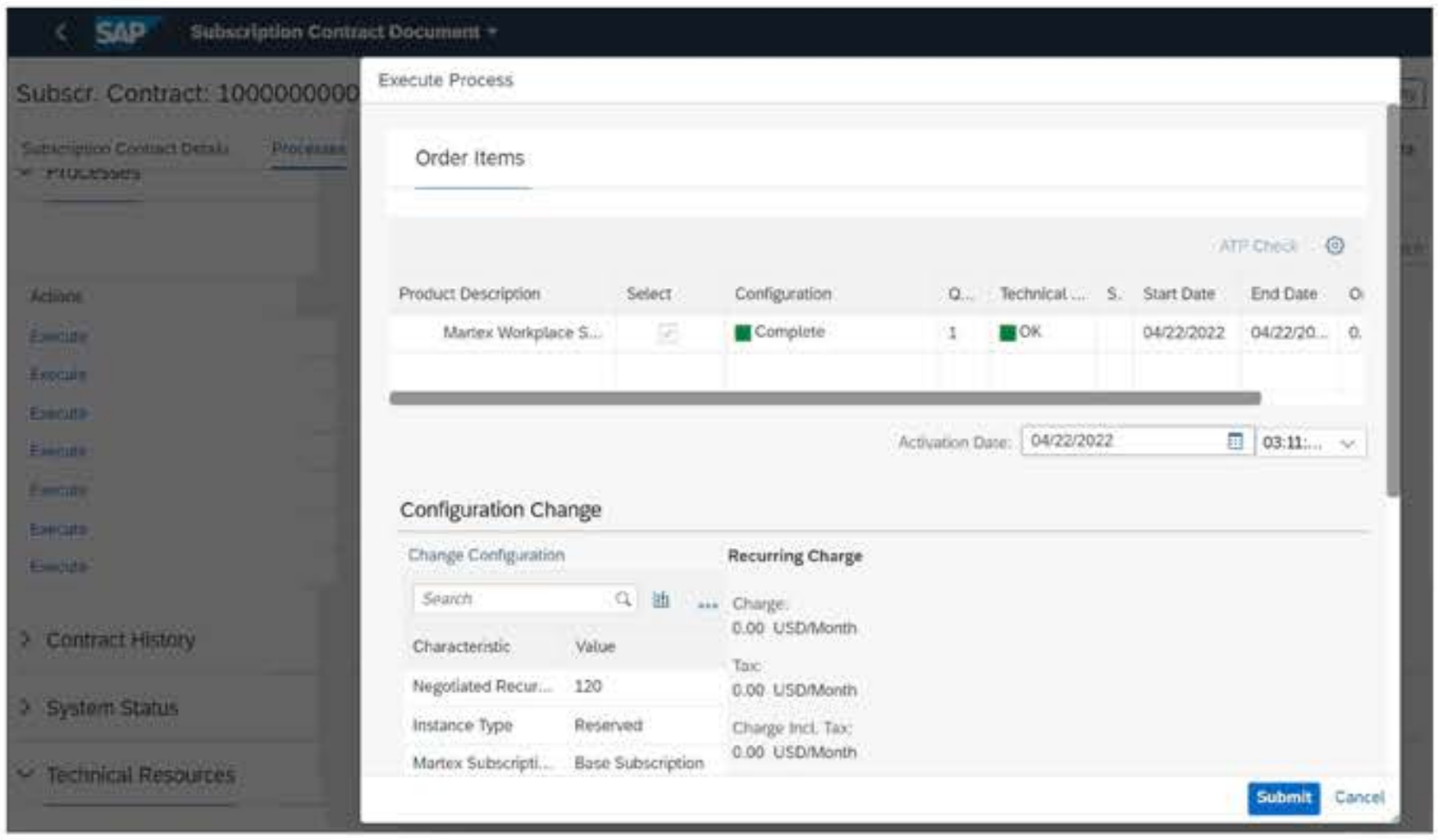

Figure 2.132 Configuration Change Process

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7. In the new pop-up screen, enter **Characteristic** values, and click **Done** (see Figure 2.133). In this screen, you can select additional services, if required, and even select other parameters.

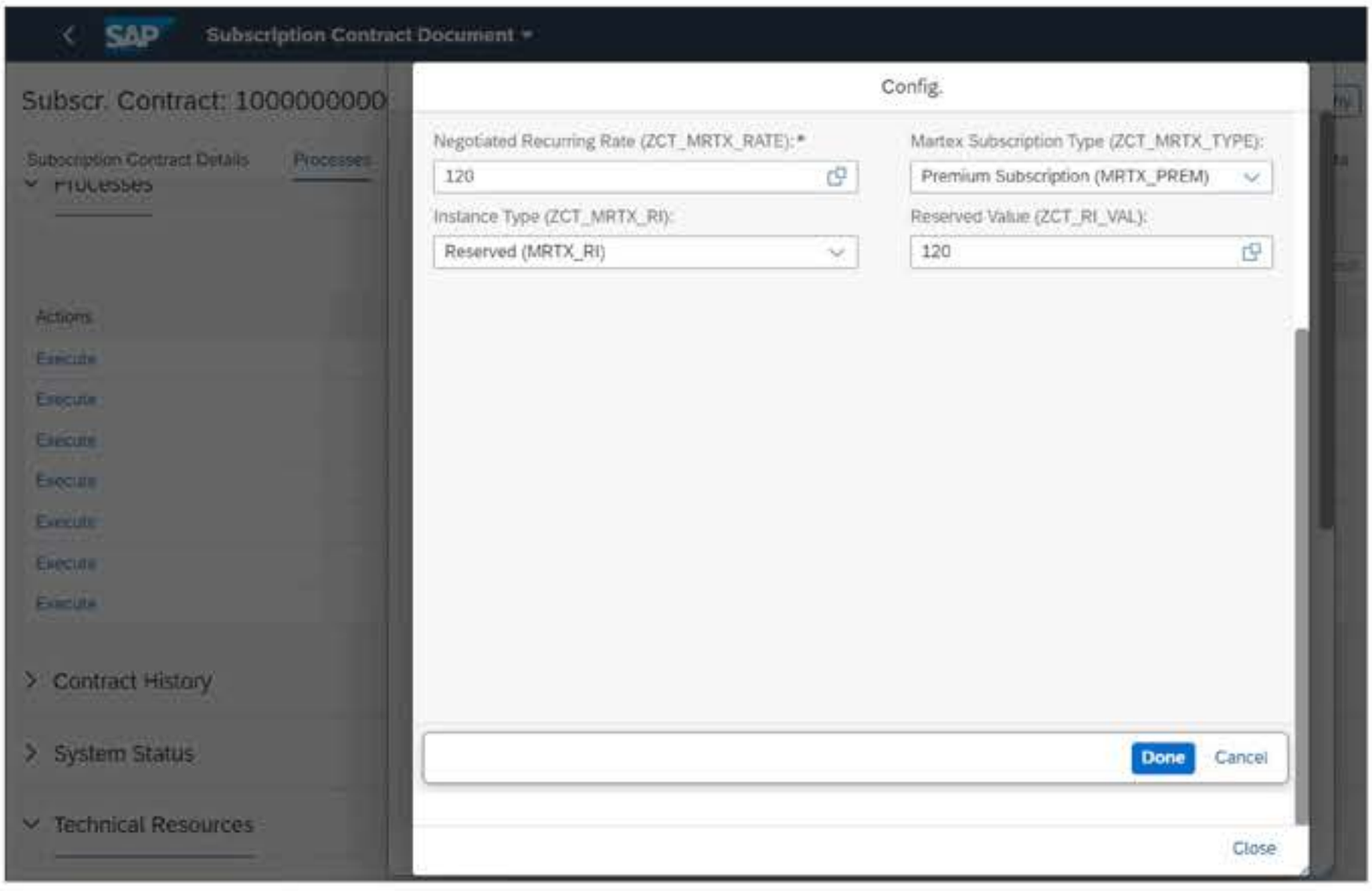

Figure 2.133 Configuration Characteristic Value Change Screen

8. Once done, click **Submit** to trigger the change order (see Figure 2.134).
9. Navigate to the **Transaction History** tab and click on the hyperlink for the subscription contract. A new contract time slice is created with the updated add-on service that you updated as a characteristic value (see Figure 2.135).

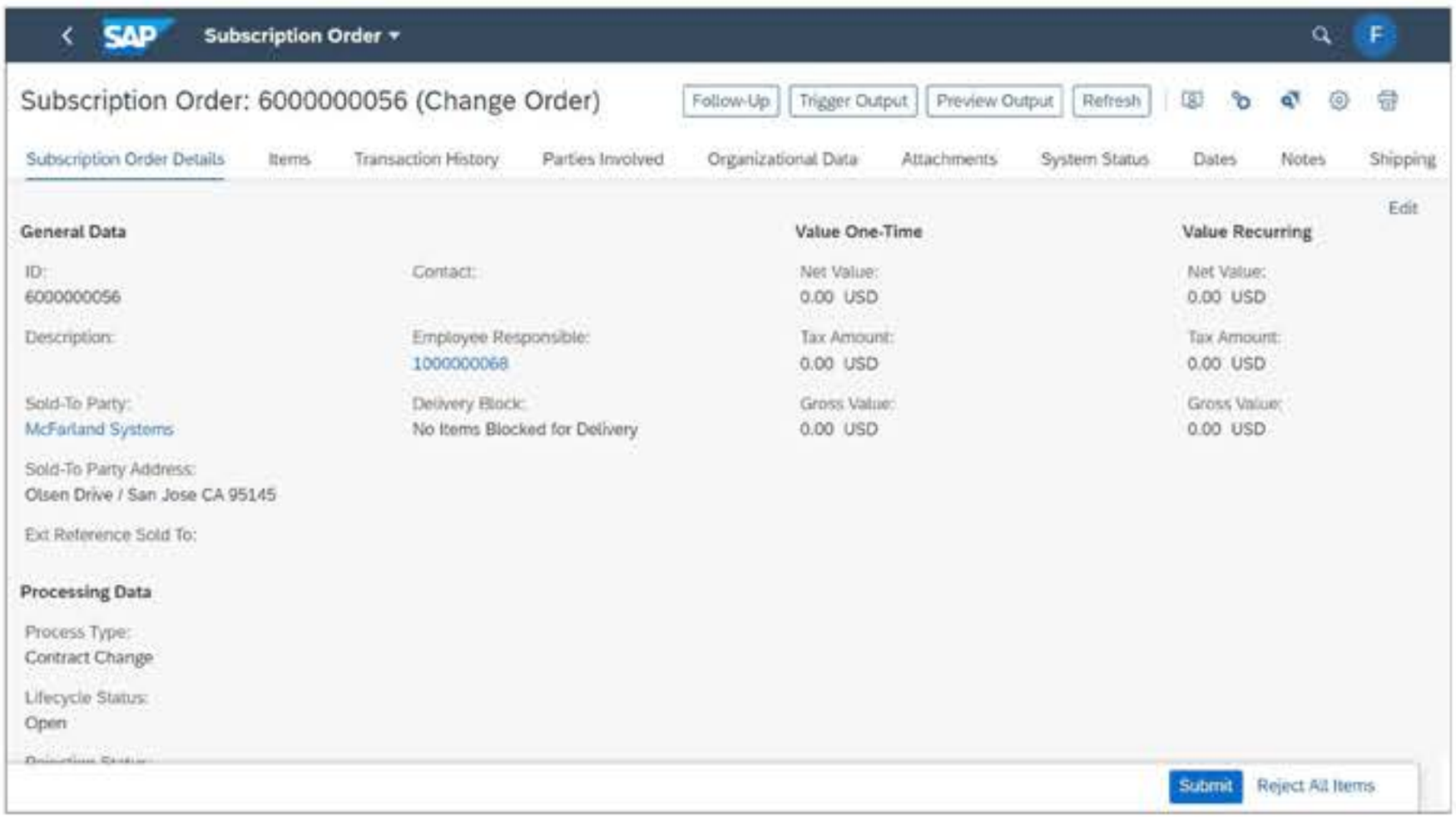

Figure 2.134 Change Order

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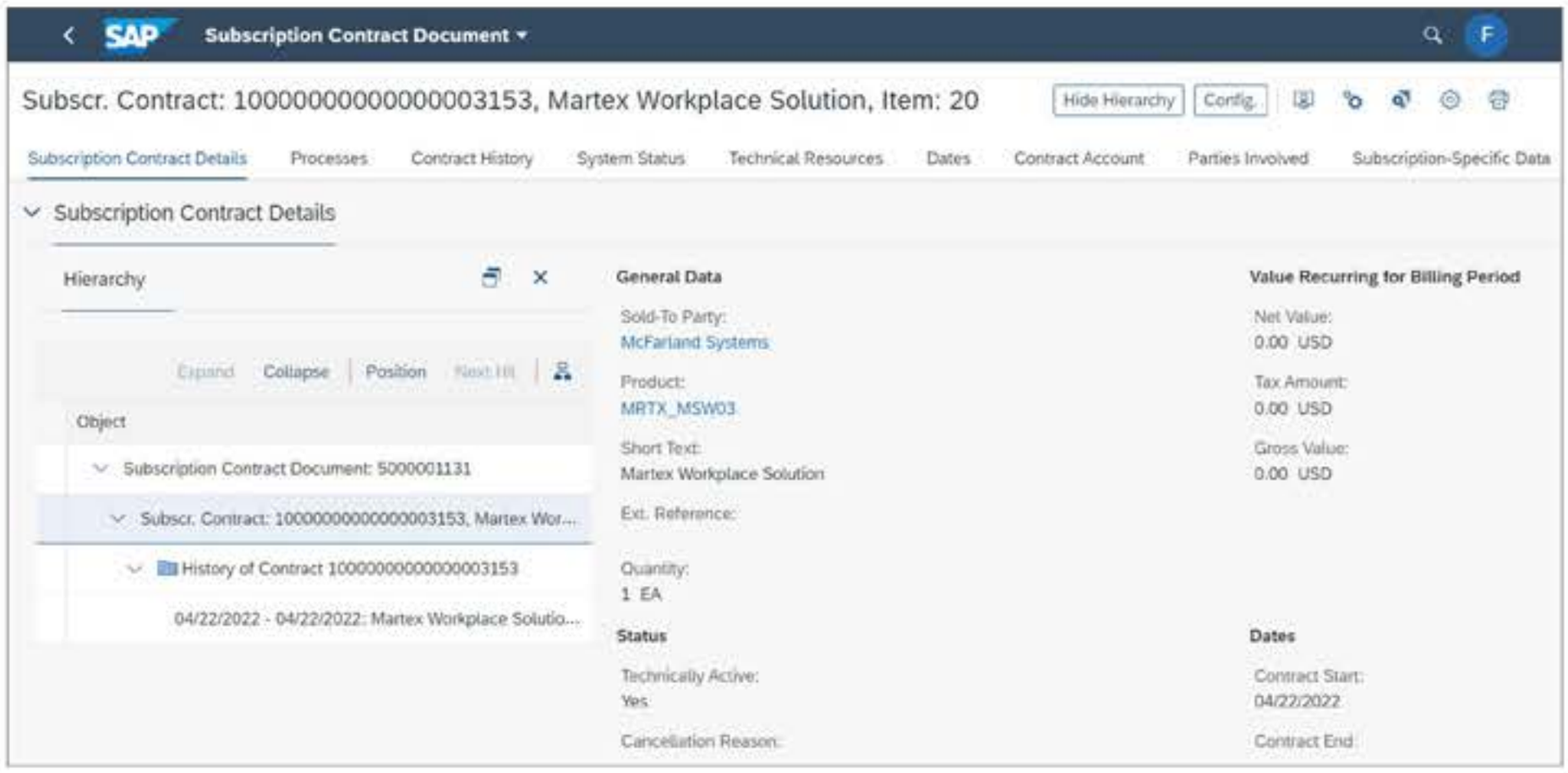

Figure 2.135 Subscription Contract with New Time Slice and Change in Contract Terms

2.7 Reporting and Analytics

Let's look at the reporting and analytics elements of SAP Billing and Revenue Innovation Management, subscription order management. We'll start by exploring standard out-of-the-box reporting features available via SAP Fiori apps in SAP Billing and Revenue Innovation Management, subscription order management, and then we'll move on to the ODI distribution monitor, which can help track transaction distribution status to natively integrated SAP Convergent Charging and SAP Convergent Invoicing.

2.7.1 SAP Fiori Applications for Subscription Order Management

SAP Billing and Revenue Innovation Management, subscription order management, has the capability to create ad hoc reports without using an SAP Business Warehouse (SAP BW) system. These reports, often called interactive reports, are used to report on limited data volumes very quickly and are very useful for business users. We'll go through some of the standard reports available in SAP Fiori apps; however, setting up these reports or adding custom attribute is out of the scope of this book.

There are different ways to search with the help of multiple search and filter criteria. The search windows for different objects already come with a limited number of commonly used criteria. You can add other search criteria fields as well.

The following objects are commonly searched for in the scope of SAP Billing and Revenue Innovation Management, subscription order management:

- Solution quotes
- Subscription orders
- Partner provider orders
- Subscription contracts
- Partner agreements
- Master agreements

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- Business partners
- Contract accounts
- Products
- SAP Convergent Charging tables

For this example, you'll search for a list of Martex subscription orders for McFarland Systems, as follows:

1. Log in to SAP Billing and Revenue Innovation Management, subscription order management, as well as the Manage Subscription Order app by using the SAP Fiori URL provided by your system administrator.
2. The landing page will have available standard provided search parameters (e.g., **Transaction ID**, **Product**, **Sold-To Party**, etc.). Let's search using the **Sold-To Party** information for McFarland Systems.
3. Enter the SAP S/4HANA business partner ID for McFarland Systems for the **Sold-To Party** number, and enter "Subscription Order" for the **Transaction Type**. If you don't have the business partner ID of McFarland Systems, you can search using the input help option using **Sold-To Party First Name** or **Last Name** and other related information.
4. Click the **Search** button, and the results will be displayed in the **Results** window in a tabular form (see Figure 2.136).

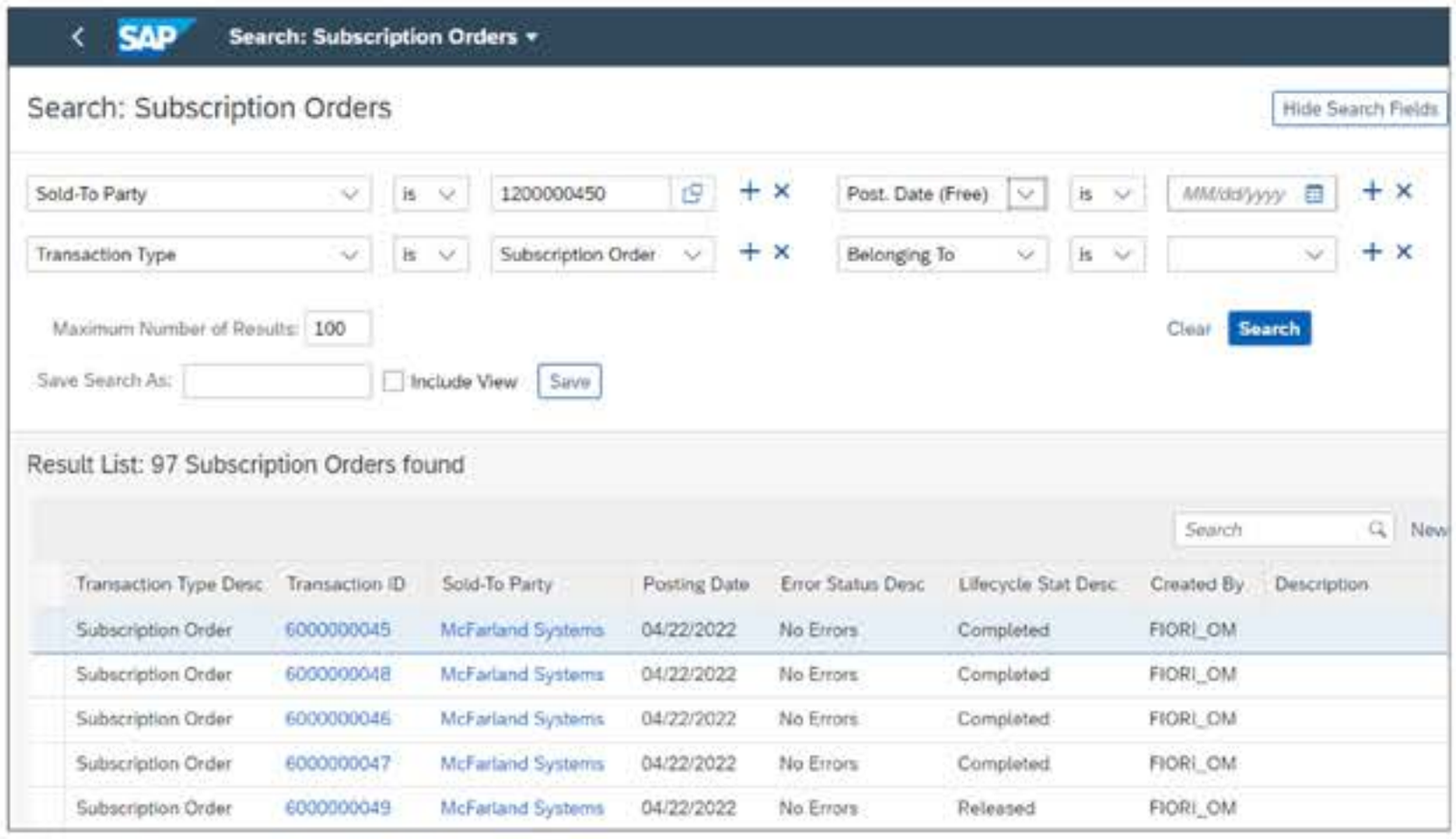

Figure 2.136 Search Subscription Orders in the Manage Subscription Order App

You can search for other objects in a similar way, adapting the filters as needed. There are also provisions to adjust the screen layout per business requirement and save the search criteria to minimize manual effort, which helps in daily monitoring. There are also options to export the result data in a spreadsheet.

Another out-of-the-box analytics view, called **Chart**, is available for monitoring. This can be used to generate a quick chart with available data sets. Figure 2.137 shows a simple example where the number of subscription orders created over a period of time is visualized.

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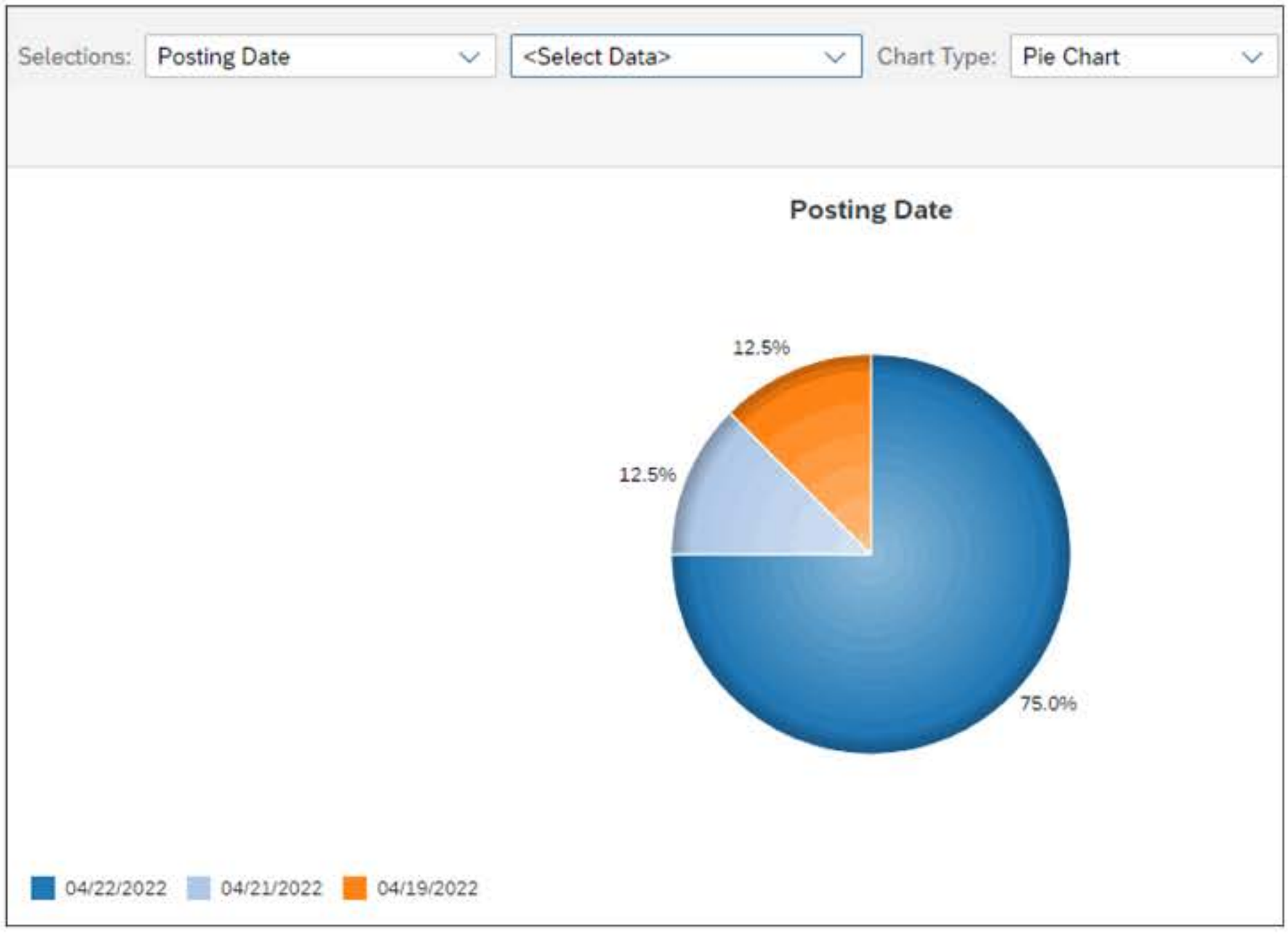

Figure 2.137 Analytics View of Manage Subscription Order

2.7.2 Order Distribution Infrastructure Distribution Monitor

One of the essential tasks for following up on the creation of subscription contracts or partner agreements is to confirm the status of the replication of these objects into SAP Convergent Invoicing and SAP Convergent Charging. If the replication isn't successful, you can't use these contracts for billing and invoicing in the downstream processes. You can view and analyze the status in the **Fulfillment** status block. You can reach this window by selecting the subscription contract line-item screen and then opening the **Fulfillment** status block to view the status. If the status is successful, the distribution steps are assigned a green light. Otherwise, the system provides error messages so that you can resolve the error, in which case the light is either yellow or red. With this option, you're limited to analyzing the particular contract you're working on.

For our example, follow these steps:

1. Log in to SAP Billing and Revenue Innovation Management, subscription order management, and the Manage Subscription Contract app using the SAP Fiori URL provided by your system administrator.
2. Search for the Subscription Contract for which you want to check the distribution status. Enter the contract number you're searching for, and click **Search** (see Figure 2.138). Alternatively, you can also search with distribution status to review and reprocess failed contracts by using the **Activation Status** filter.

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3. The **Results** window displays the status of each line item in the contract (see Figure 2.139). Click each line item, **10** and **20**, to view the statuses. Navigate to **Fulfillment Status**, and then select **Monitor**. Figure 2.140 shows the overall status of each of the line items.

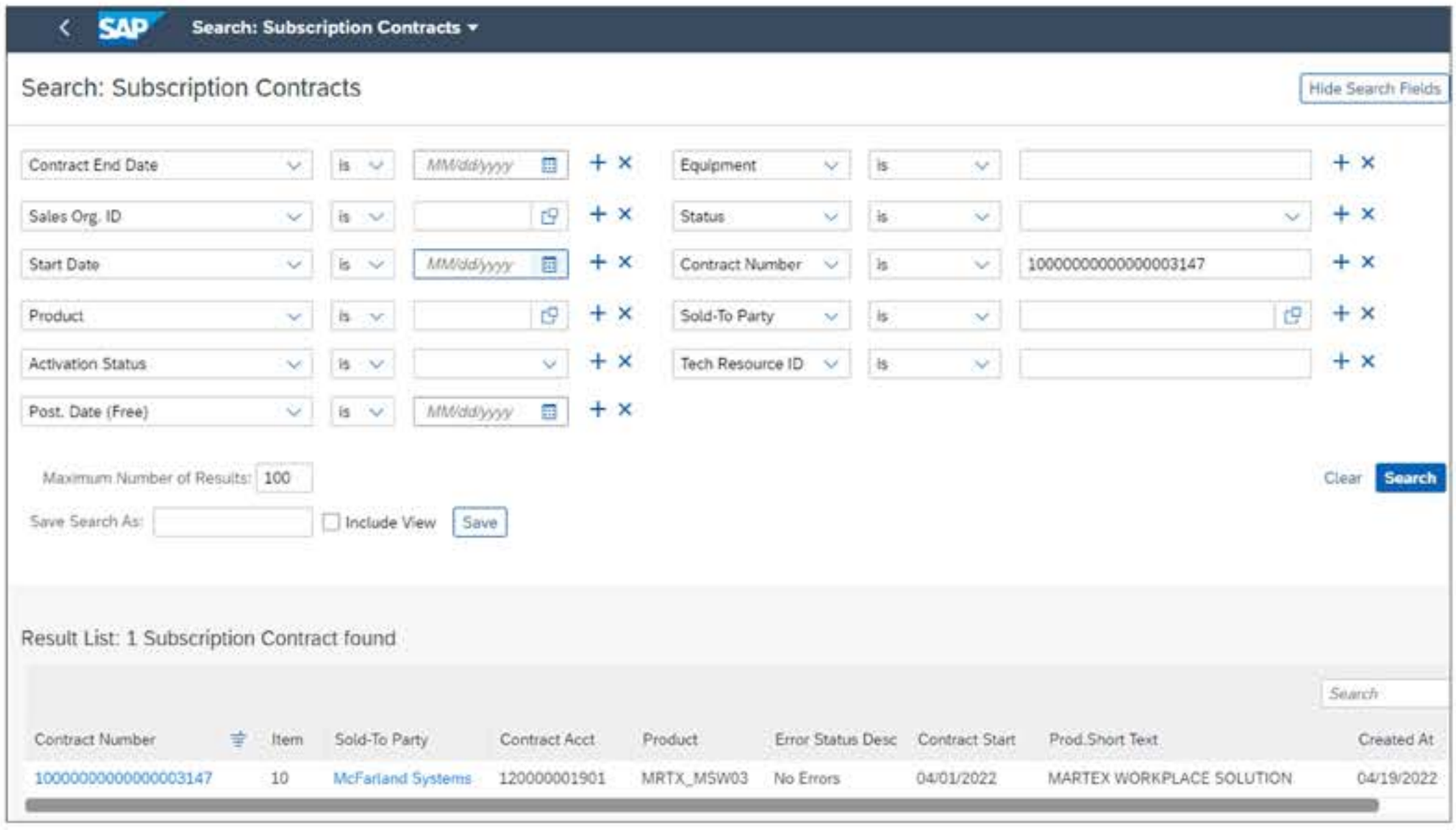

Figure 2.138 Search Subscription Contracts

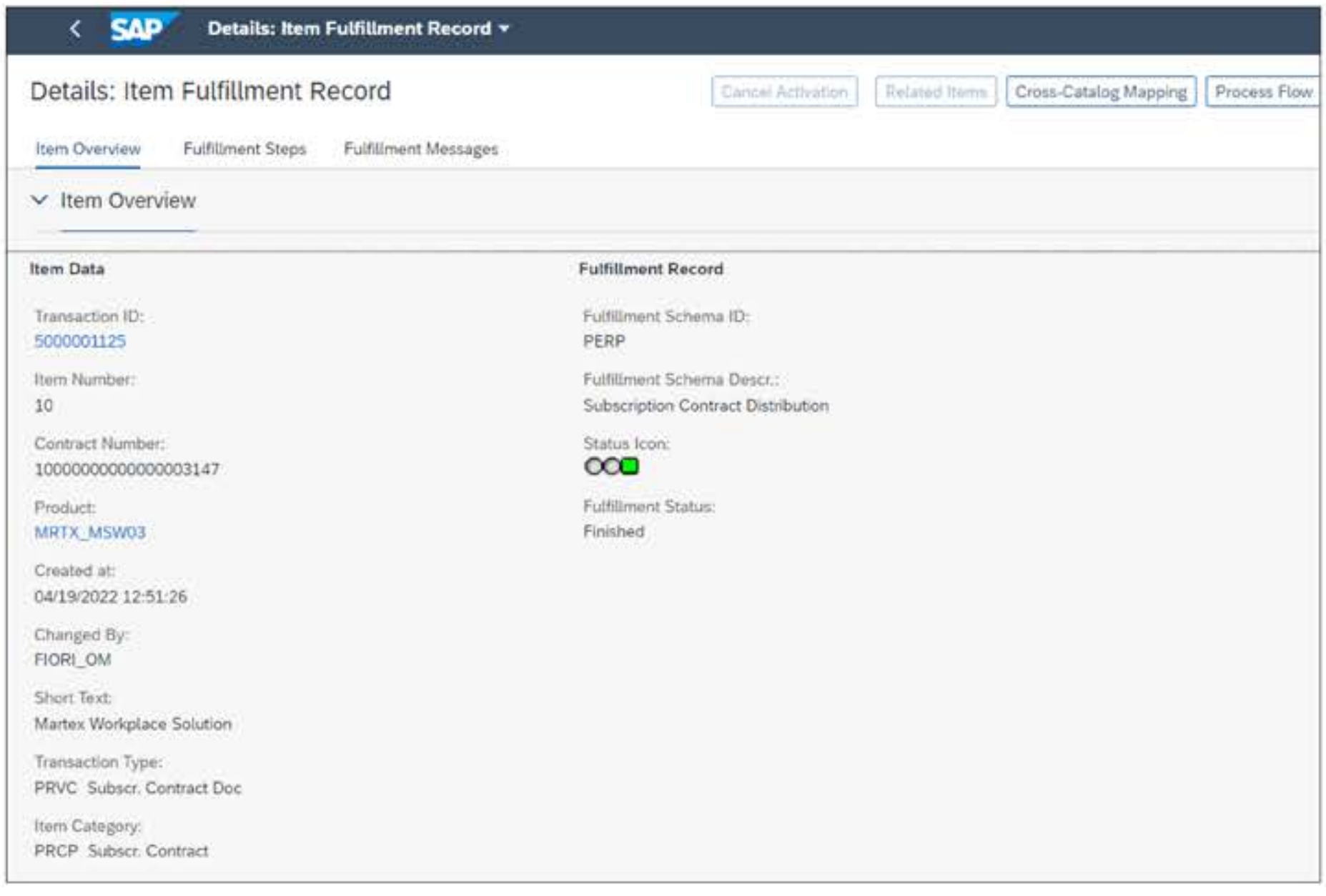

Figure 2.139 Subscription Contract Status Overview

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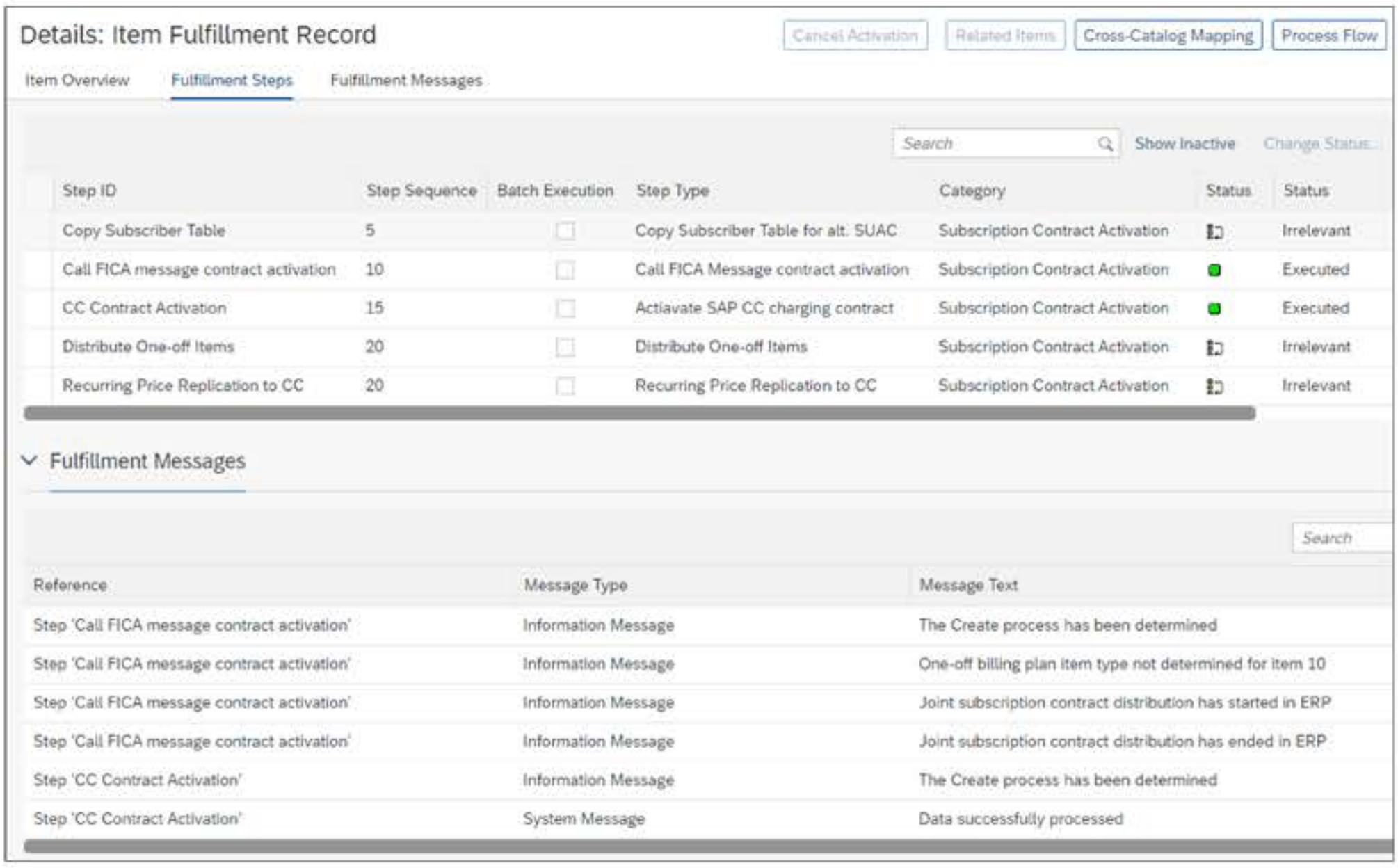

Figure 2.140 Fulfillment Records

Figure 2.139 and Figure 2.140 show the detailed statuses of each of the distribution steps, along with the distribution messages.

The overall subscription contract status can either be set to **Finished**, **In Process**, or **Failed** according to the status of the related steps. There is an option to **Refresh** subscription contract distribution, which then fetches all the distribution data for the current document reloaded from the database and displayed in monitor views. This is relevant if distribution is still ongoing, and the results of the step, which is running asynchronously, are expected.

If a step fails due to a technical failure (indicated by the status **Technically Failed**), the distribution steps must be executed again using **Rerun** option after the technical issue has been resolved.

In addition, selecting **Cross-Catalog Mapping** takes you to a dialog box where the actual data for this mapping taken from the current item is displayed.

The ODI monitor in the subscription contract provides the **Process Flow** button to display single fulfillment steps when a contract has been changed. There are also additional features, such as rescheduling of distribution steps, that are only available in Expert mode.

If any step with the category O1 (Order Distribution), P0 (Wait for Activation), or P1 (Activation) is still to be executed (status isn't **Executed**, **Failed**, **Canceled**, or **Irrelevant**), the **Cancel Activation** button is active. If you choose this, a dialog box appears in which you're asked which of the following two options is to be executed:

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- **Cancellation with Redetermination of Steps**
  After canceling all steps with the category PO and P1, new steps are determined and available with the status **Open**. This should be done if a complete restart of the activation process is required, without adaption of the subscription document. Choosing **Rerun** starts execution of the steps.
- **Cancellation without Redetermination of Steps**
  All steps that have not yet been executed (status isn't **Executed**, **Failed**, **Canceled**, or **Irrelevant**) are set to the status **Failed**. This causes the distribution process to be aborted, and control is returned to the subscription document, which can be edited again. The required changes can be made here, and when they are submitted, this triggers a new execution of the distribution process in which new steps are generated.

If any step with the category P5 (Wait for Deactivation) or P6 (Deactivation) is still to be executed (status isn't **Executed**, **Failed**, **Canceled**, or **Irrelevant**), the **Cancel Deactivation** button is active. If you choose **Cancel Deactivation**, all steps in this category are canceled, and new steps are determined and available with the status **Open**. Choosing **Rerun** starts execution of the steps.

2.8 Solving Common Order and Contract Management Issues

So far in this chapter, we've discussed in detail how SAP Billing and Revenue Innovation Management, subscription order management, can be configured as part of SAP S/4HANA to enable customers to sell their goods and services in a flexible consumption model. However, because SAP Billing and Revenue Innovation Management, subscription order management, is natively integrated with other SAP Billing and Revenue Innovation Management modules—SAP Convergent Charging, SAP Convergent Invoicing, and so on—it's important to ensure that the order/contract information is distributed to all systems to minimize disruptions for business.

Here we'll discuss some of our commonly encountered issues and troubleshooting approach, which might help determine the root cause and apply a fix as applicable. We'll limit our discussion to issues related to standard integrations only. Additional troubleshooting may be required if any of the standard integration is enhanced to support business use cases. The issues are encountered in a solution where SAP Billing and Revenue Innovation Management, subscription order management; SAP Convergent Charging; and SAP Convergent Invoicing are integrated, and no custom enhancement is introduced in the integration.

Table 2.10 lists common issues that may occur as well as their corresponding troubleshooting steps.

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Issue	Troubleshooting Steps
Sales Org Data not available after manual creation of Organization model or importing transports from lower environment.	Run report HRBCI_ATTRIBUTES_BUFFER_UPDATE using GUI transaction code SE38. You can also use the report CRM_COM_ORGMAN_BUPA_INTEGRATE to check for additional errors and correct missing integration between business partners and organizational objects.
The item configuration page isn't loading to maintain variant values in the Order/Quote Management app in SAP Billing and Revenue Innovation Management, subscription order management.	This usually happens during the very early stage of implementation when the system setup is in progress. While working with a AVC products, it's very important that AVC packages are installed in the SAP S/4HANA system. The following SAP Notes are relevant for AVC installation: ■ SAP Note 2811164 ■ SAP Note 2598627 ■ SAP Note 2551305 In addition, be sure to maintain the SAP Fiori launchpad URL for AVC in the table CRMV_IC_BORADM view for attribute S4C_FIORI.
The subscription contract is **Technically Inactive** after submission in SAP Billing and Revenue Innovation Management, subscription order management.	There are several reasons why a subscription contract is marked as **Technically Inactive** once it's submitted for distribution in SAP Billing and Revenue Innovation Management, subscription order management. **Technically Active** status works as a flag to indicate whether the subscription contract/provider contract replication was successful based on ODI settings. The first step is to verify the Contract Fulfillment status using the ODI distribution monitor in the SAP Fiori application. If the contract is distributed to contract accounting, check Transaction FKKCC_PC_MON. You can check the following steps based on the error message displayed in the transaction code: ■ **Error Message "Processing error in Internet Communication Framework" ("Direct connect to "xxx" failed: NIECONN_REFUSED (-10)"):** This error message indicates that the SAP Convergent Charging URL is incorrectly set up in Transaction SOAMANAGER, there are firewall issues to connect SAP Convergent Charging server, or the SAP Convergent Charging server isn't reachable. Troubleshoot your Transaction SOAMANAGER web services or reach out to your SAP Basis or infrastructure team to identify the root cause and fix the issue. ■ **Error Message "FailedAuthentication | The Security code could not be authenticated":** This error message indicates that the web service username and password are incorrect or expired. Reach out to your SAP Basis or security team to identify the root cause and fix the issue.

Table 2.10 Common Issues and Troubleshooting Steps

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Issue	Troubleshooting Steps
The following error occurs while submitting a subscription order: "Item 10: You can't use product xxxx" (see Figure 2.141).	This happens if the **Processing** mode of the product is selected as **Classic** instead of **Advanced Variant Configuration**. SAP Billing and Revenue Innovation Management, subscription order management, only works with AVC. You can define the processing while setting up the configuration profile (see Figure 2.142).

Table 2.10 Common Issues and Troubleshooting Steps (Cont.)

Figure 2.141 Error Message While Submitting a Subscription Order

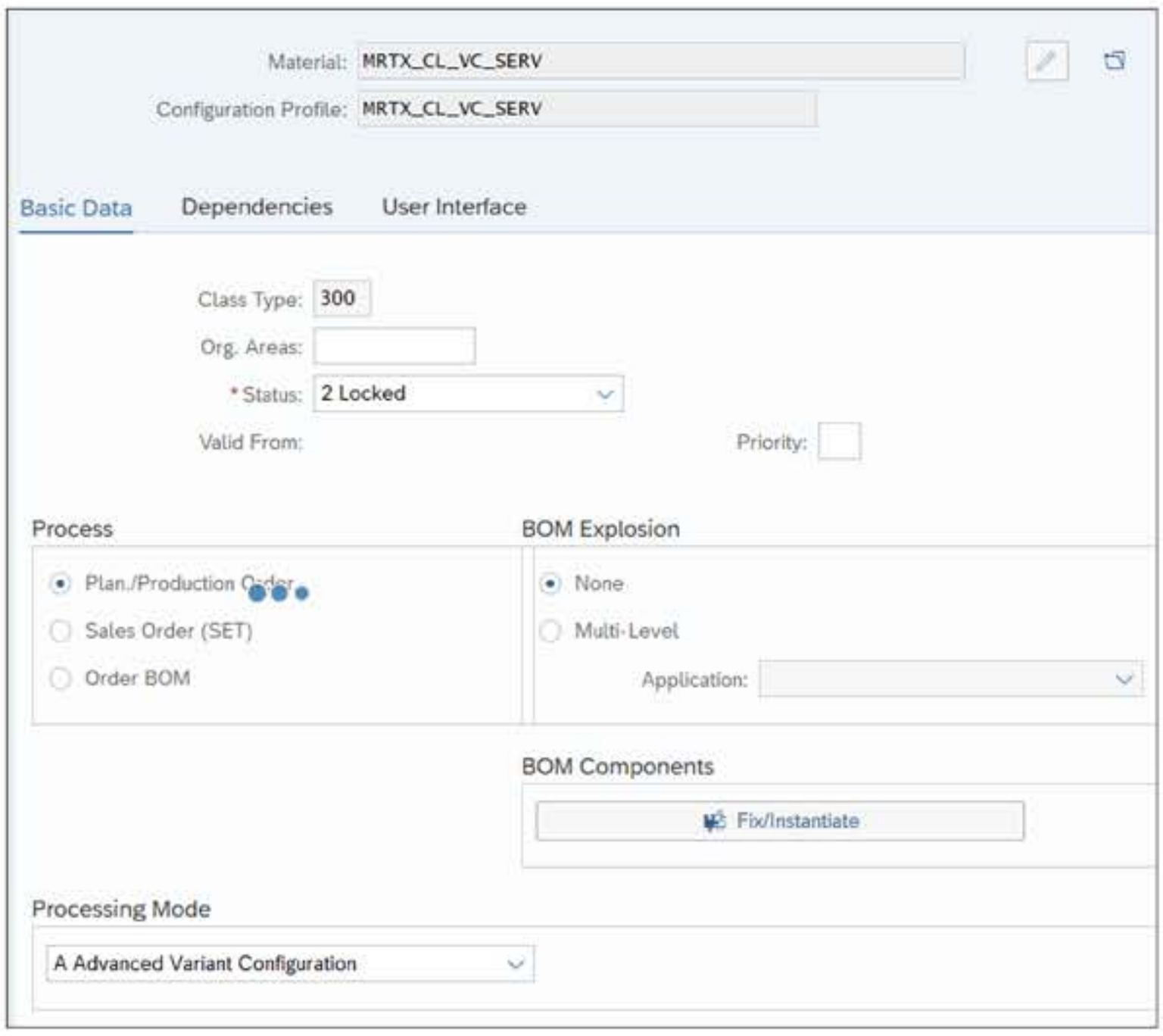

Figure 2.142 Selecting the Processing Mode in the Configuration Profile

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2.9 Summary

By now, you know that SAP Billing and Revenue Innovation Management, subscription order management, uses SAP S/4HANA as the backbone upon which its functionalities are built. Often, the functionalities cross over into other SAP modules, such as sales and distribution, materials management, variant configuration, and Internet Pricing and Configurator. You don't need to have expertise in all these areas but having functional knowledge in these areas is certainly helpful. You now have enough information to complete the necessary steps in SAP Billing and Revenue Innovation Management, subscription order management, to build the example that runs through this book. This is an extensive chapter to cover in its entirety. As you gain more knowledge by building more scenarios, you'll master the functionalities in SAP Billing and Revenue Innovation Management, subscription order management.

You've learned to set up and create all the necessary master data objects and all the necessary transactional data, and you now understand the purpose and functions of these objects and how they play their part in this process.

SAP Billing and Revenue Innovation Management is an integrated solution. We therefore recommend working through the other chapters in this book as well, which may prove helpful for reaching your goals. The next chapter on SAP Convergent Charging will take you through the functionalities and steps to set up the charge plan for rating and charging the subscription contracts created in this chapter.

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Chapter 3
Charging

This chapter first explains charging master data and tables, followed by step-by-step instructions for configuring SAP Convergent Charging. It provides an overview of SAP Convergent Charging objects creation, such as charge plan, charge, and price plan (recurring, one-time, usage), and interface objects, such as chargeable item classes and charged item classes using Core Tool.

This chapter explains how to set up the SAP Convergent Charging modeling process using the case study established in Chapter 1. Modeling in SAP Convergent Charging is typically performed using the SAP Convergent Charging Core Tool. The Core Tool is a graphical user interface that allows you to create and maintain different types of objects such as provider contracts, subscriber accounts, pricing macros, charge plans, charges, allowances, and more. The tool provides predefined components and functions to build the objects; no coding is required.

SAP Convergent Charging is an extremely robust and versatile system whose sole purpose is to rate, charge, and generate rated transactions. A rated (calculated) transaction generated out of SAP Convergent Charging is called a charged item. Charged items are stored in the file system where the SAP Convergent Charging server is installed. By default, the charged item is transferred hourly to SAP Convergent Invoicing as a billable item and is used to bill a customer. In SAP Convergent Charging, a customer is called a *subscriber account*. In contract accounting, a customer is known as a *business partner*.

As of SAP Convergent Charging 2020 FPS02, generated billable items by SAP Convergent Charging can be transferred and loaded into SAP Convergent Invoicing immediately instead of waiting for the bulkloader instance to process. By default, this option is disabled.

3.1 Overview

SAP Convergent Charging provides three types of charges: one-time, recurring, and usage. For a one-time charge, the charge is triggered based on an external event, such as contract activation, suspension, or termination. For a recurring charge, SAP Convergent Charging generates the charges based on internal scheduling determined during

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the build such as monthly, quarterly, and yearly. For a usage charge, the consumption data records (CDRs), originating from the multiple source systems or network elements, are used to perform rating and charging by SAP Convergent Charging. CDRs can be acquired through mediation systems. One of the meditation systems that is commonly used in SAP Billing and Revenue Innovation Management implementations is Mediation Zone, also known as SAP Convergent Mediation by DigitalRoute, which is also covered at a high level in this book.

For example, in the telecommunications industry, a one-time charge can be an activation charge, and a recurring charge can be a fixed monthly fee, such as unlimited voice and text messages. A usage charge can be a data plan where CDRs are generated based on a customer's consumption. In an integrated environment, CDRs are sent from multiple sources and directed to SAP Convergent Charging to perform rating and charging. SAP Convergent Charging then generates billable items in contract accounting for further billing and invoice processing. We'll talk more about billable items in later chapters.

When first deployed, SAP Convergent Charging is an empty box offering a variety of creative solutioning possibilities, and customers can create their own unique rating and charging requirements. Unlike other traditional suites of SAP products, SAP Convergent Charging doesn't have many custom objects; the customizable object is the interface design, such as chargeable item class and charged item class, which receives usage information.

SAP Convergent Charging integrates with SAP Billing and Revenue Innovation Management, subscription order management, and contract accounting systems via web services. It also can integrate with non-SAP systems using either web services or Java application programming interfaces (APIs) for rating usage events. To configure the SAP Convergent Charging system, you follow a define, assign, apply, and process (DAAP) methodology.

SAP Convergent Charging takes a decision tree approach to perform rating and charging of subscribers. There are many configuration elements provided to carry out the rating and charging tasks, which will be discussed with examples in further chapters, but, in a nutshell, they are rates, comparators, splitters, operators, and functions.

Even though SAP Convergent Invoicing has functionality to create one-time and recurring charges, the focus of this chapter is to leverage SAP Convergent Charging to handle those two types of charges. If your implementation leverages SAP Convergent Invoicing for recurring and one-time charges, the setup is done in SAP S/4HANA, and no setup in Convergent Charging is required. However, for recurring charges, the decision to use SAP Billing and Revenue Innovation Management, subscription order management, versus SAP Convergent Charging is mostly driven by the complexity of calculation, such as proration requirements during contract cancellation. Other requirements, such as minimum threshold which usually leverages the use of a counter or allowance, also require the use of SAP Convergent Charging.

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Note

The chapter doesn't include any instruction sections because SAP Convergent Charging is a pricing/rating engine that doesn't require user involvement in day-to-day operation once it has been set up.

3.2 Flexible Rating and Charging Model

The ability to adapt to the continuously changing market conditions—shifting in product offerings and go-to market strategy to support flexible consumption services—requires retooling or new platforms to support it. SAP Convergent Charging is one of the platforms of the SAP Billing and Revenue Innovation Management suite to support capability.

SAP Convergent Charging is the engine used to perform rating and charging. *Rating* is the process used to calculate transactions that are generated from usage consumption and recurring or one-off charges, while *charging* refers to the process of determining which customer account should be charged.

3.3 Master Data

In most online transaction processing (OLTP) systems, data is separated into two main categories: master data and transactional data. Any data generated from a transaction is transactional data (e.g., quotations, orders, contracts); everything else is master data (e.g., business partners). In SAP Convergent Charging terms, a provider contract is a type of transactional data, and subscriber accounts, external accounts, and charge objects are examples of master data.

Before rating and charging can be carried out in SAP Convergent Charging, master data elements need to be set up/configured based on the requirements of the use case. We'll discuss key master data elements such as subscriber accounts, external accounts, range tables, subscriber range tables, mapping tables, and translation tables within the SAP Convergent Charging system in detail, and we'll explain how they can be defined and created.

To access SAP Convergent Charging, first the Core Tool needs to be installed. The Core Tool is the user interface (UI) to perform charge modeling using visual programming in the form of a decision tree. The Core Tool is provided in the installation file of SAP Convergent Charging, which can be found by accessing the SAP Support Portal (*http://s-prs.co/v554202*) and navigating to the **Software Downloads** page, as shown in Figure 3.1.

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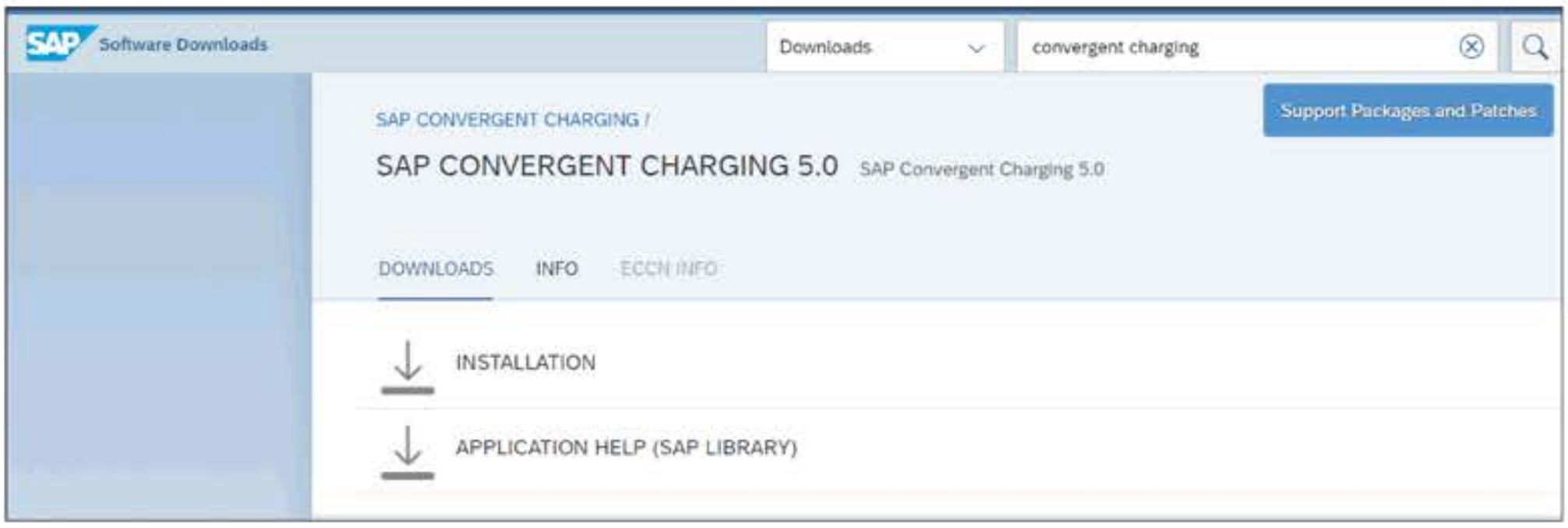

Figure 3.1 SAP Software Downloads Center

As a Java-based tool, the Core Tool will execute when the correct version of Java Virtual Machine is installed on the computer.

The Core Tool is available as part of the installation ZIP file and is in the *DATA_UNITS/CCXX_TOOLS_CONTENT_UC_OSIND/core_tool.zip* archive. After extracting the files in *core_tool.zip*, execute either *core_tool.bat* or *core_tool.sh*, depending on your operating system.

On the login screen, provide the username (**Log On**) and **Password** created by your system administrator and the **Server URL** (see Figure 3.2).

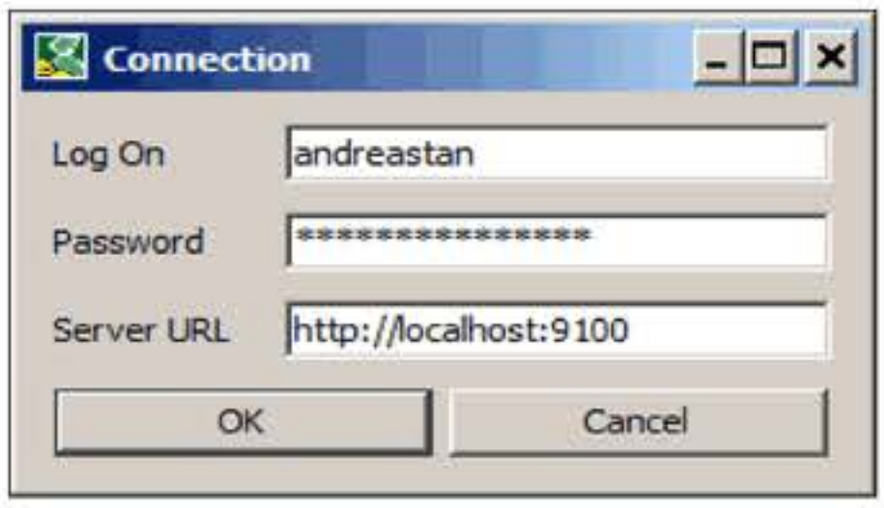

Figure 3.2 Core Tool Login Screen

3.3.1 Catalog

A catalog is a placeholder for all configuration objects. This is similar to the concept of packages in other programming languages. In an integration scenario for SAP Billing and Revenue Innovation Management, it's a best practice to only have one catalog per landscape to simplify integration. However, it's common to see multiple catalogs created in lower environments for testing purposes.

To create a catalog, navigate to **File • New • Catalog**. Once a catalog is created, you're shown a series of folders. These folders represent different types of development objects within SAP Convergent Charging. As objects are created in SAP Convergent Charging, they will be displayed in their respective folders, as shown in Figure 3.3.

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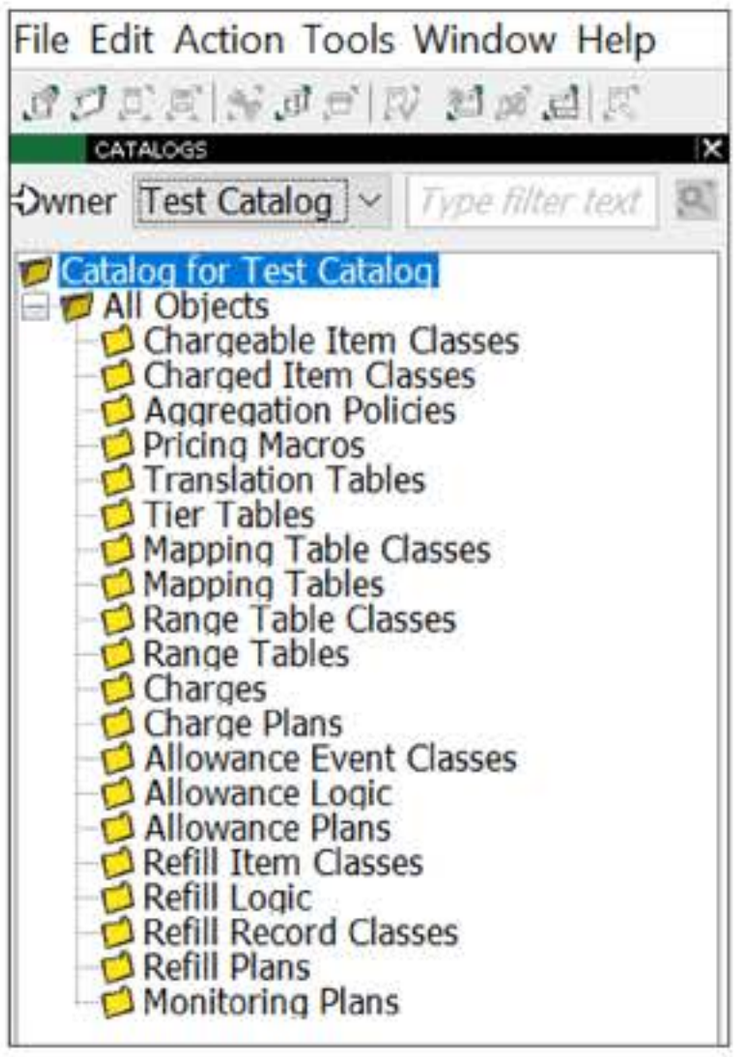

Figure 3.3 Object Categories with Catalog

3.3.2 Subscriber Account and External Account

A subscriber account within SAP Convergent Charging is a unique identifier for the end customer of the service user. Subscriber accounts can have one or many external accounts assigned to them, as shown in Figure 3.4.

In an integrated SAP Billing and Revenue Innovation Management scenario, a subscriber account is automatically created in SAP Convergent Charging via replication of a business partner from SAP Convergent Invoicing. Any changes to the subscriber account should be initiated from the provisioning system for it to be consistent across all SAP Billing and Revenue Innovation Management systems.

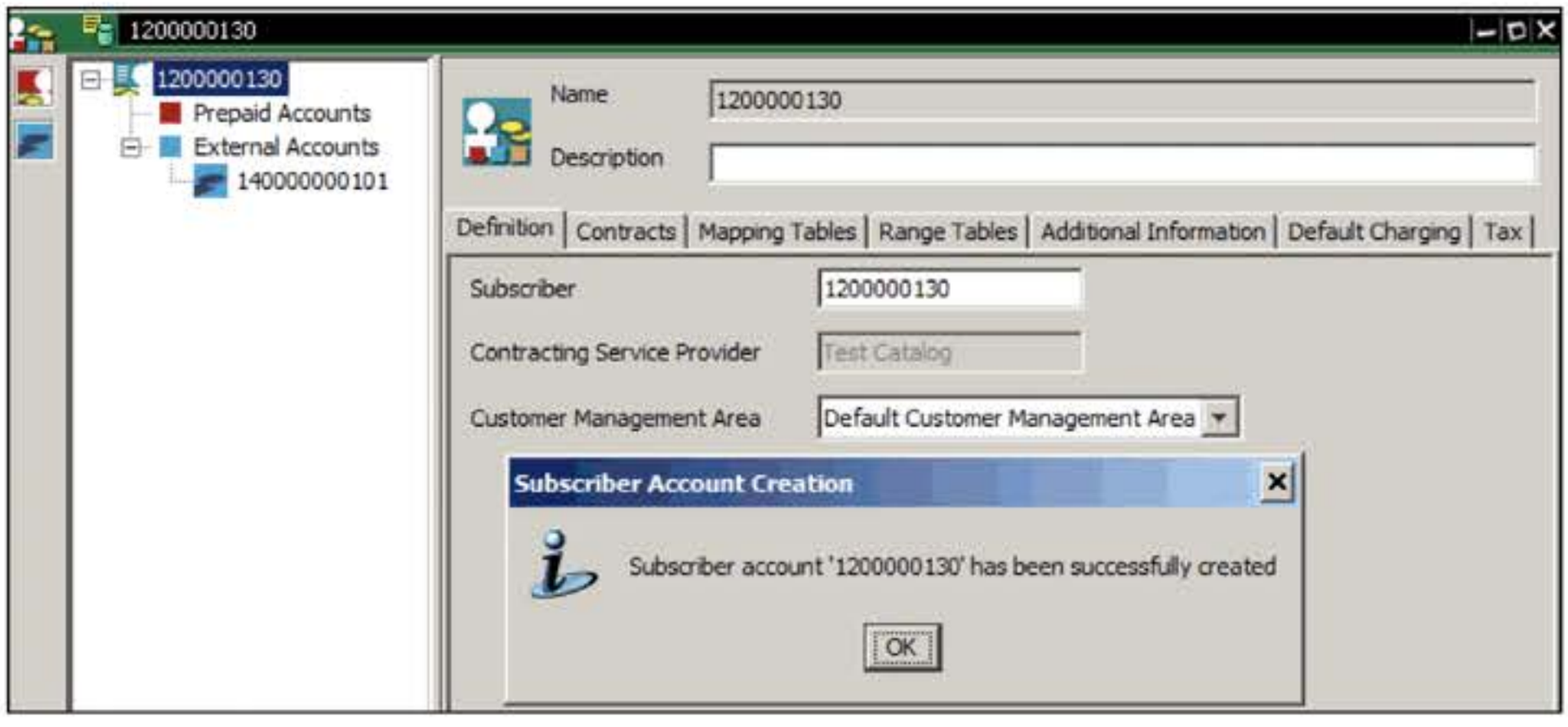

Figure 3.4 Subscriber Account and External Account

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A subscriber account contains the prepaid account and external account subobjects. An external account is also called a postpaid account. The concept of a prepaid subscriber account is analogous to a prepaid customer in the telecommunications industry, in which a pay-as-you-go charging model is used. The postpaid subscriber account is similar to most subscription models today that have different flavors such as advanced charging or arrears charging.

In SAP Convergent Charging, information about a subscriber account is lean: although they represent a customer, they often are only presented with an ID. Information such as names and addresses are typically stored in other SAP Billing and Revenue Innovation Management areas.

Subscriber accounts and external accounts are terms used exclusively in SAP Convergent Charging and are often referred to with other terminology in different SAP Billing and Revenue Innovation Management modules. Table 3.1 shows these differences in terminology.

SAP Convergent Charging	SAP Billing and Revenue Innovation Management, Subscription Order Management	Contract Accounting
Subscriber account	Business partner	Business partner
External account	Business agreement	Contract account

Table 3.1 Subscriber Account and External Account Terminology in Other Systems

3.3.3 Provider Contract

In SAP Convergent Charging, a provider contract represents a business arrangement between the provider of a service and the consumer of that service. A provider contract is generally a long-term business agreement that establishes certain agreed-upon terms and conditions.

You don't need to manually create a provider contract in SAP Convergent Charging because it will be replicated from SAP Convergent Invoicing. In an integrated scenario with SAP Billing and Revenue Innovation Management, a subscription contract is created by the provisioning system (SAP Billing and Revenue Innovation Management, subscription order management).

A provider contract is usually composed of a subscriber account, external account, and charge plan. Charge plans will be covered extensively in Section 3.6. For now, think of a charge plan as a collection of formulas that is required to calculate what to charge customers. An example of a provider contract is shown in Figure 3.5.

In an integrated scenario, SAP Billing and Revenue Innovation Management, subscription order management, can be used to update the provider contract through change processes. Whenever a change process is executed, a new version of the provider contract will be created in SAP Convergent Charging.

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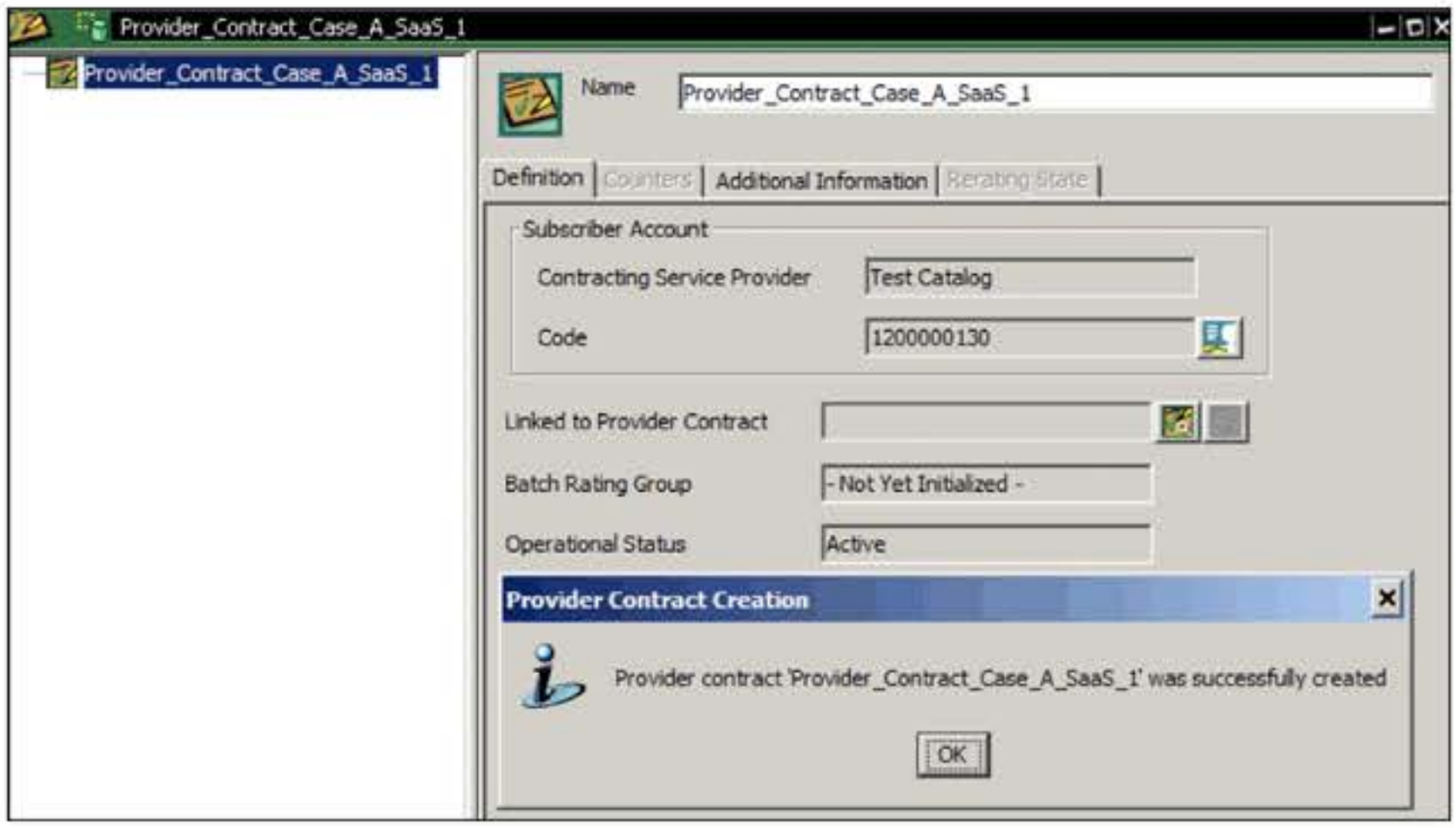

Figure 3.5 Provider Contract

3.4 Organizational Units

Unlike SAP Billing and Revenue Innovation Management, subscription order management, SAP Convergent Charging doesn't require an organizational structure setup. Organizational structures such as company code, sales organization, distribution channel, and division are needed to process invoices and financial postings. As described in Section 3.3, the required objects are subscriber account, prepaid account, external account, and provider contract, as shown in Figure 3.6.

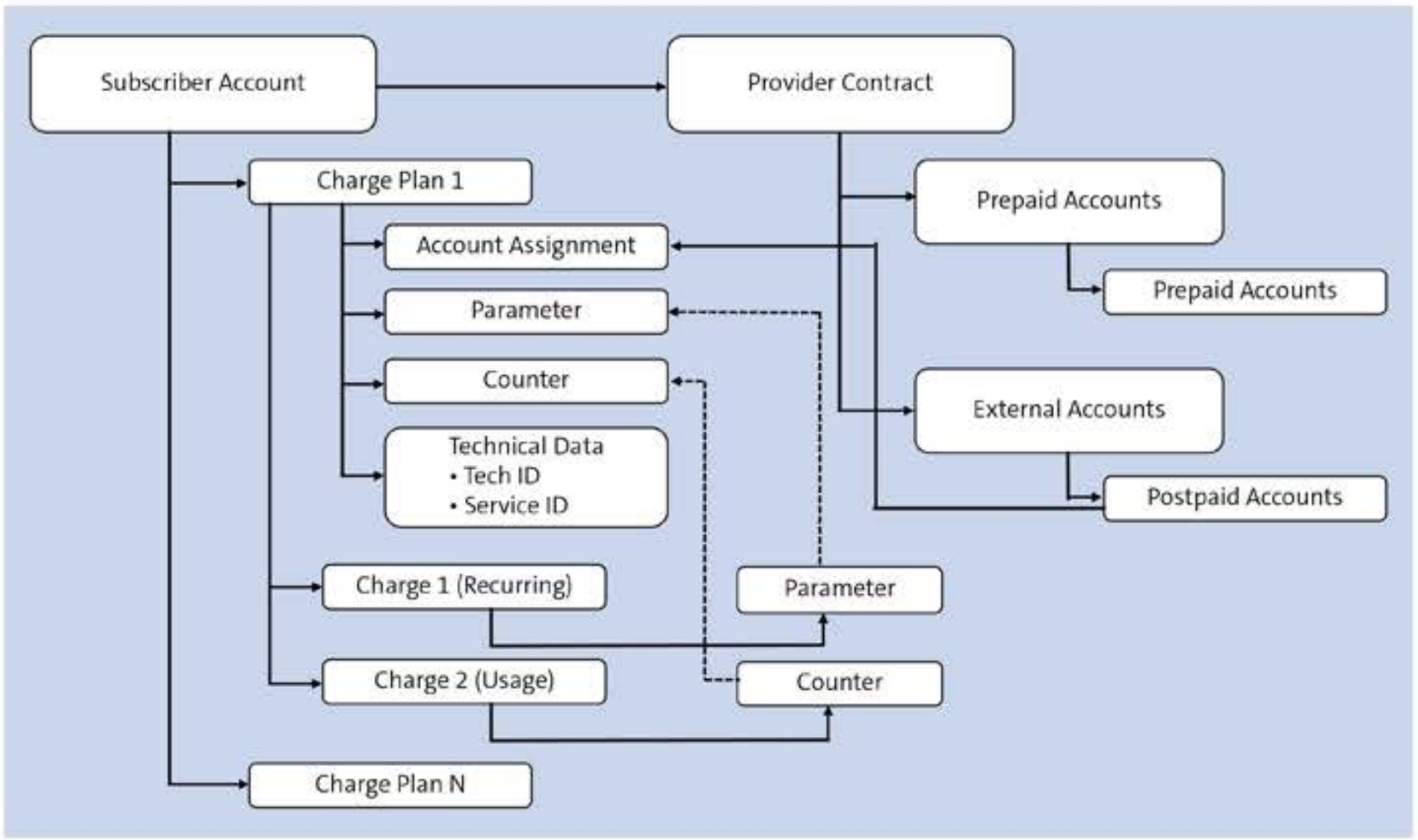

Figure 3.6 Convergent Charging Objects Hierarchy

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3.5 Managing Tables

Tables are dynamic data structures within SAP Convergent Charging used to store information inside or outside the SAP Convergent Charging system. This information is later used for designing charge logic. This section describes the various tables available in SAP Convergent Charging, and Section 3.6 shows how tables can be incorporated within charge logic.

3.5.1 Tier Table

A *tier table* is used internally within SAP Convergent Charging to calculate pricing requirements that involve tiers.

To create a tier table, follow these steps:

1. Navigate to **File • New • Tier Table**, as shown in Figure 3.7.

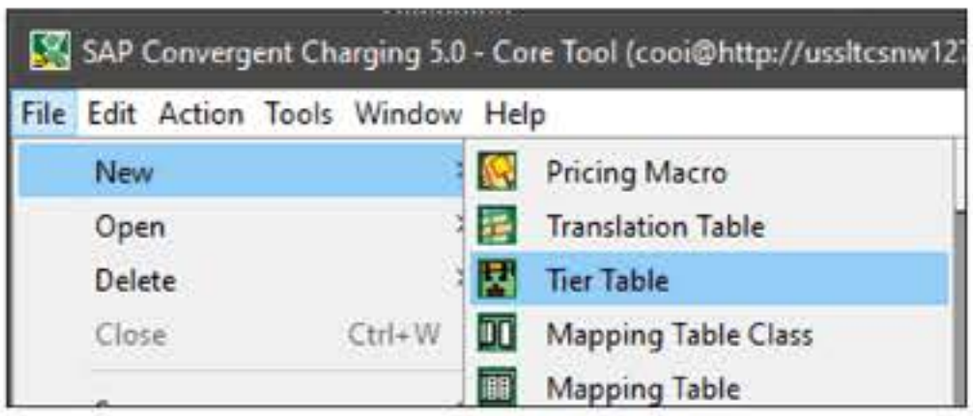

Figure 3.7 Creating a Tier Table

2. In the tier table definition window, click the **Table Schema** tab (see Figure 3.8), and then enter the **Name** and **Description**.

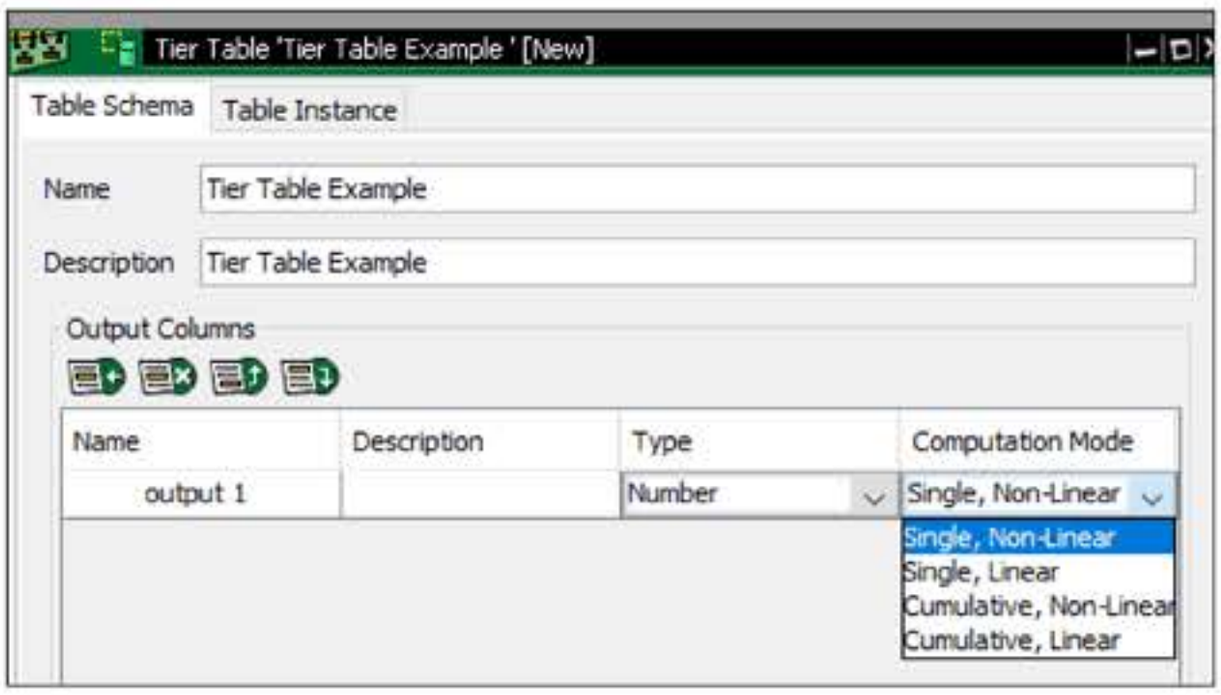

Figure 3.8 Tier Table Output Computation Modes

3. Then you must define an output column. There are four computation modes to pick from depending on your business requirements:
   - **Single, Non-Linear**
   - **Single, Linear**
   - **Cumulative, Non-Linear**
   - **Cumulative, Linear**

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Now that you've configured the table schema, let's maintain the ranges and their corresponding output values in the **Table Instance** tab. In the tier table definition window, click the **Table Instance** tab, shown in Figure 3.9.

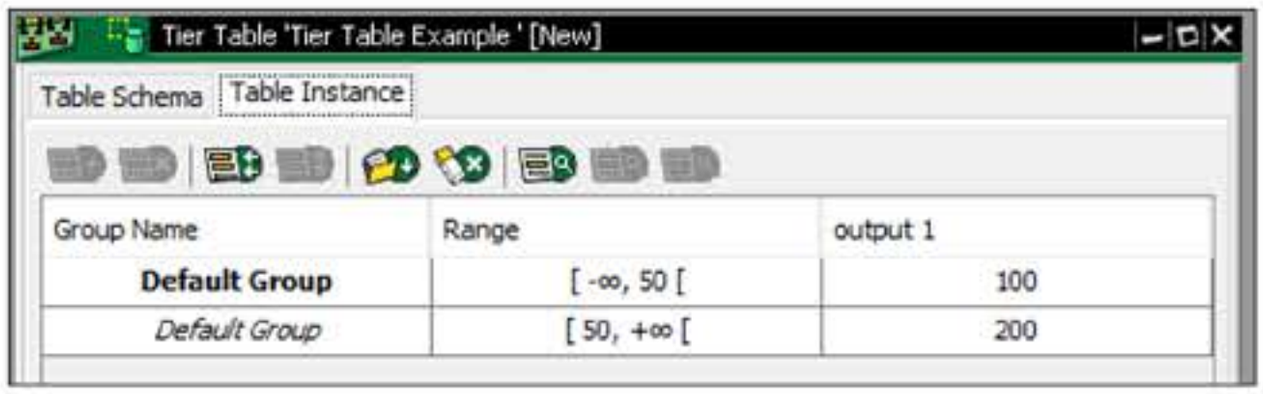

Figure 3.9 Tier Table Instance Definition

To save the tier table, follow menu path **File • Save • In the Database**. Once saved, the tier table is ready to be used in charge logic, which will be covered in Section 3.6.

3.5.2 Translation Table

A *translation table* is used internally within SAP Convergent Charging as a lookup table, in which a set of keys is passed in as input columns and a set of values is returned as output columns. To create a translation table, follow these steps:

1. Navigate to **File • New • Translation Table**.
2. Select the **Table Schema** tab. Here, specify a set of input columns by clicking in the **Input Columns** area and a set of corresponding output columns by clicking in the **Output Columns** area, as shown in Figure 3.10.

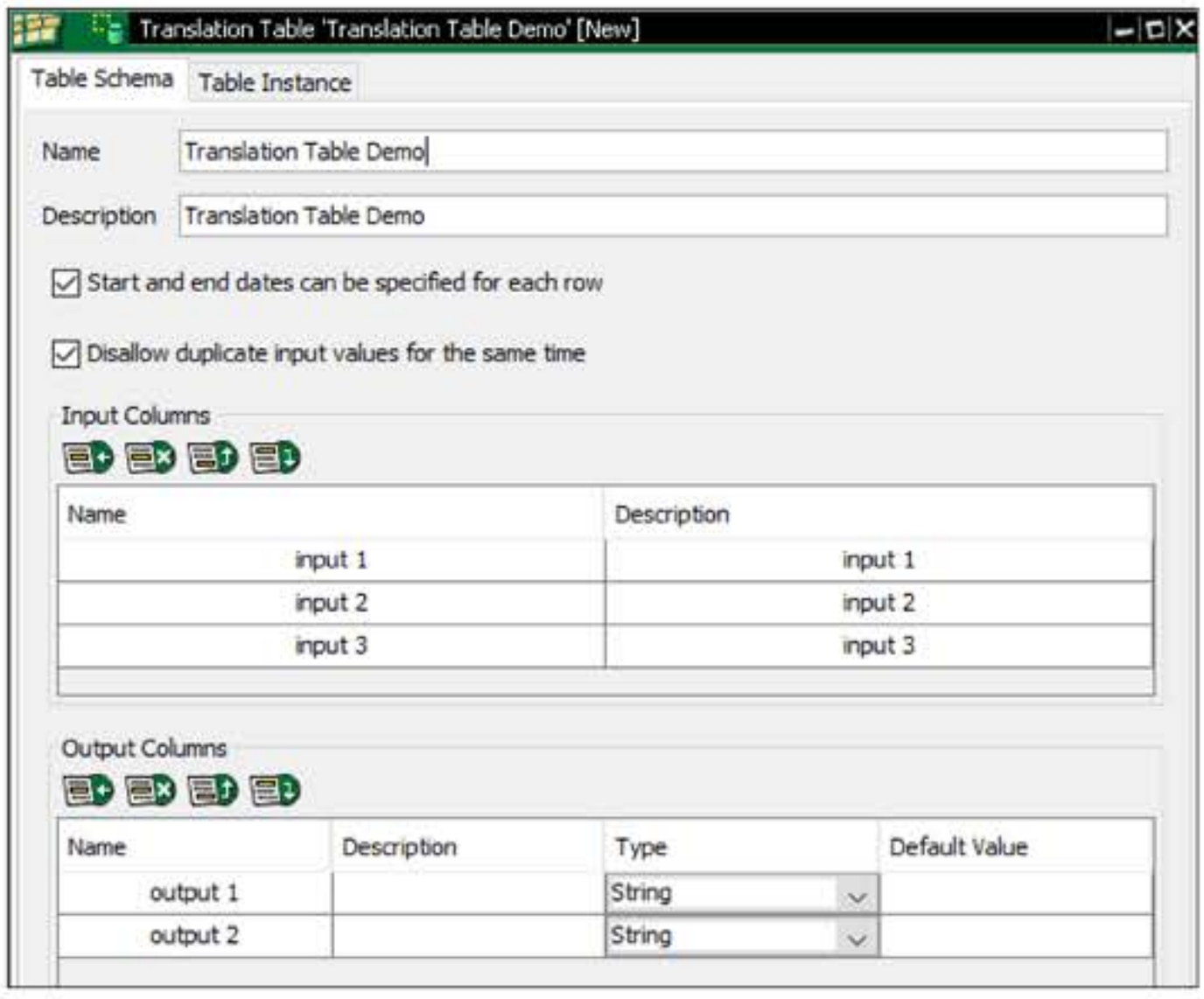

Figure 3.10 Definition View of a Translation Table

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3. Next, select the **Table Instance** tab. Here, row values for input columns and their corresponding output columns can be specified depending on your business requirements by clicking the button, as shown in Figure 3.11.

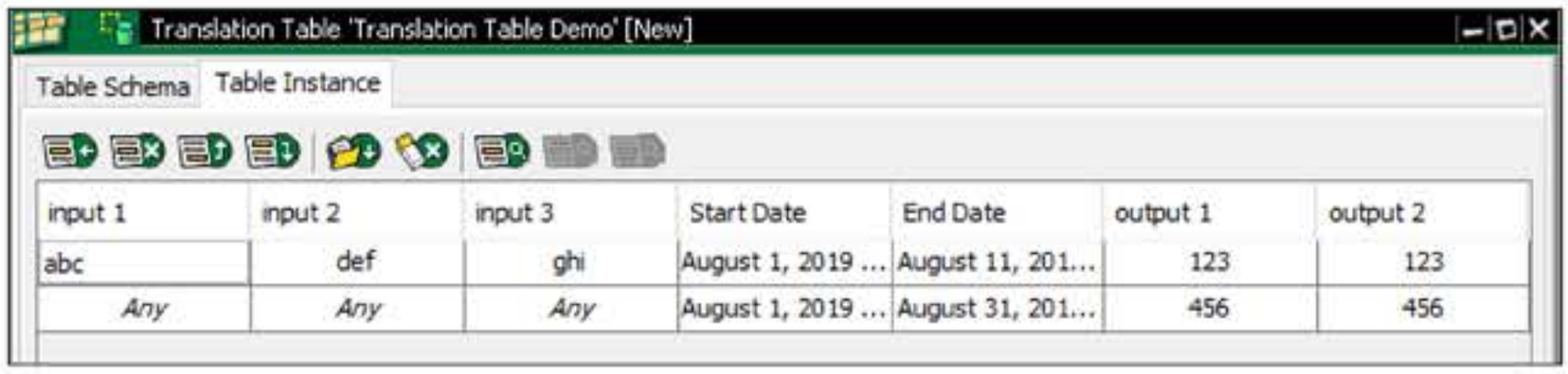

Figure 3.11 Values in a Translation Table

To save the translation table, follow menu path **File • Save • In the Database**. Once saved, the translation table is now ready to be used in charge logic, which will be covered in Section 3.6.

3.5.3 Mapping Table

A *mapping table* is used to build and store business data by validity for rating and charging purposes. Unlike their counterpart translation tables, mapping tables can be defined internally within SAP Convergent Charging or externally redefined with the help of a provisioning system contract. Because they can be redefined from a provisioning system, it's possible to have contract-specific and subscriber-specific mapping table data. Mapping tables can support up to 20 input and 30 output columns.

To define values in the mapping table, first create a mapping table class object via menu path **File • New • Mapping Table Class**. The next step is to define input and output columns and their respective data type: string or number.

In Figure 3.12, **Membership Level** and **Service Option** are set as input columns, and **Price** is set as an output column; all three have **String** as the **Type**.

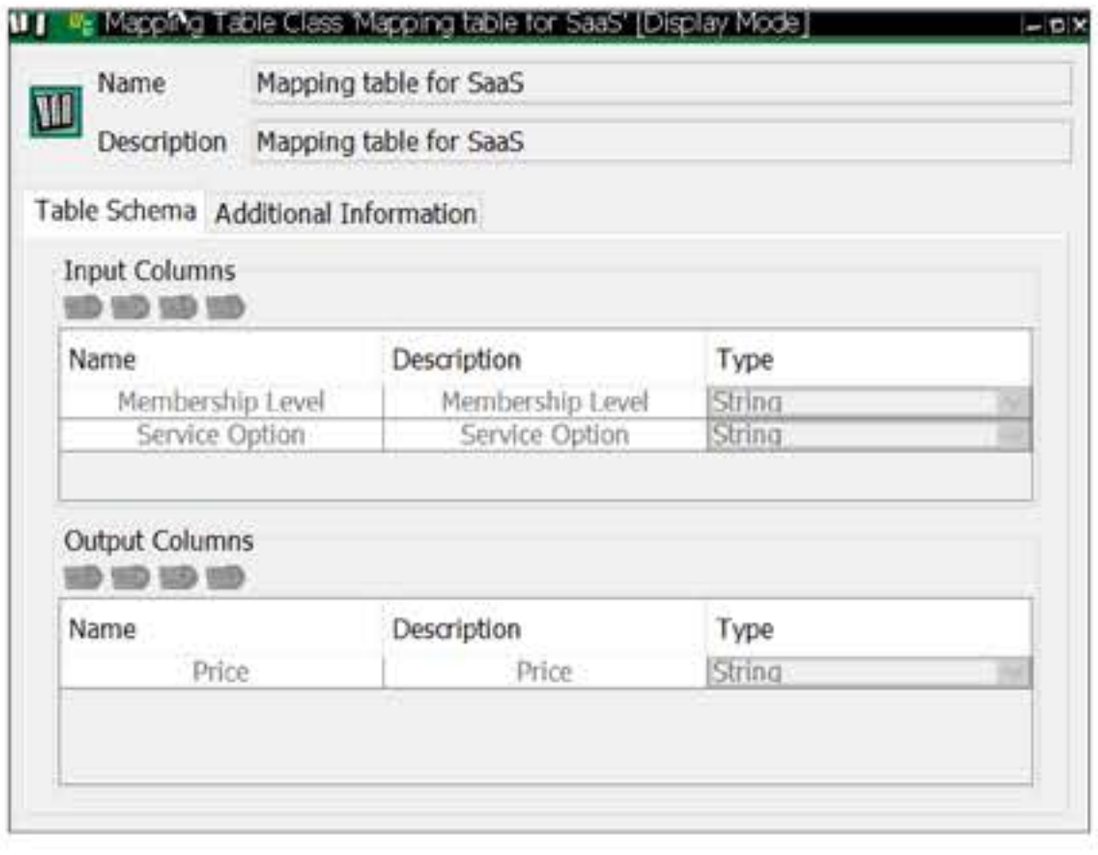

Figure 3.12 Mapping Table Class Definition

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The mapping table class can be saved via the **File • Save** shortcut. Next, create a mapping table based on the mapping table class via **File • New • Mapping Table**, as shown in Figure 3.13. To save the mapping table, follow menu path **File • Save • In the Database**. Once saved, the mapping table is ready to be used in charge logic, which will be covered in Section 3.6.

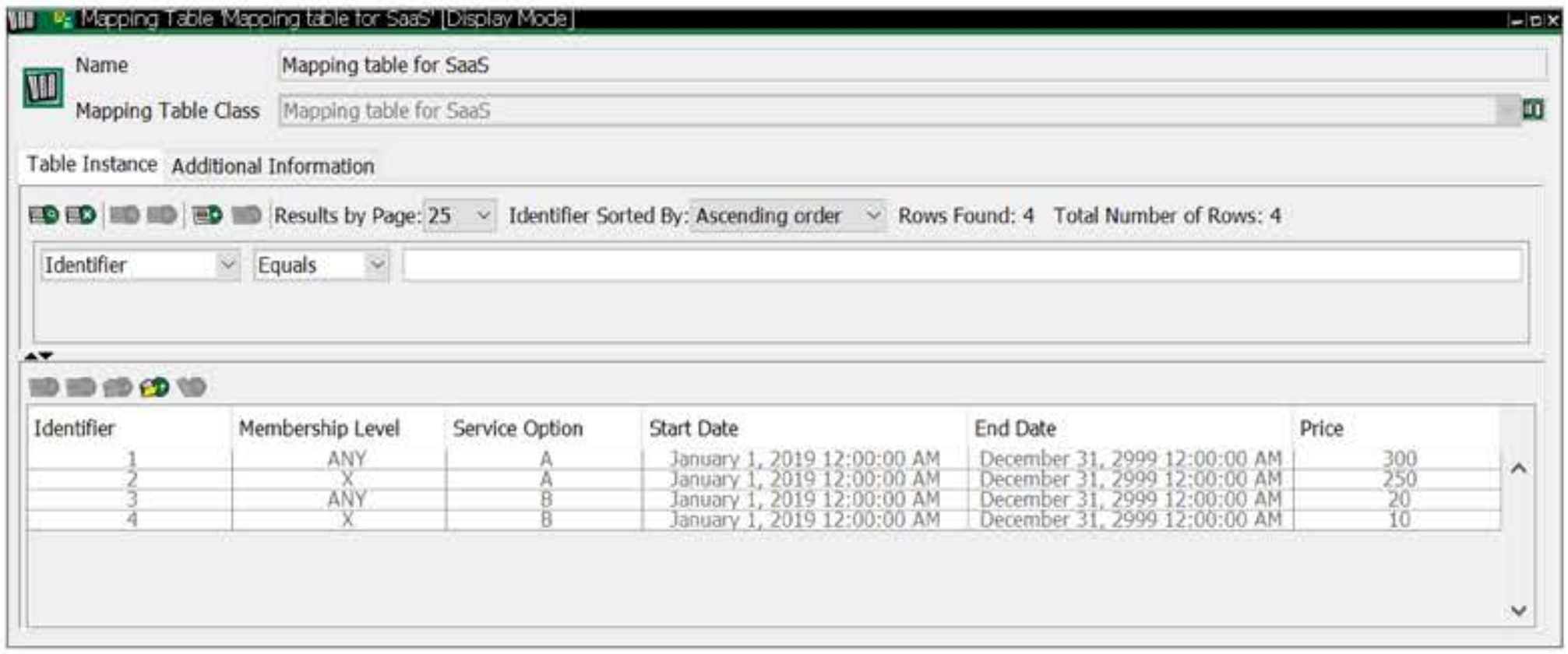

Figure 3.13 Mapping Table

3.5.4 Range Table

The *range table* is like the tier table, but with one main difference. In addition to internal SAP Convergent Charging maintenance, a range table can also be maintained via web services or directly from SAP Billing and Revenue Innovation Management, subscription order management. To create a range table, a range table class needs to be created first via menu path **File • New • Range Table Class** (see Figure 3.14).

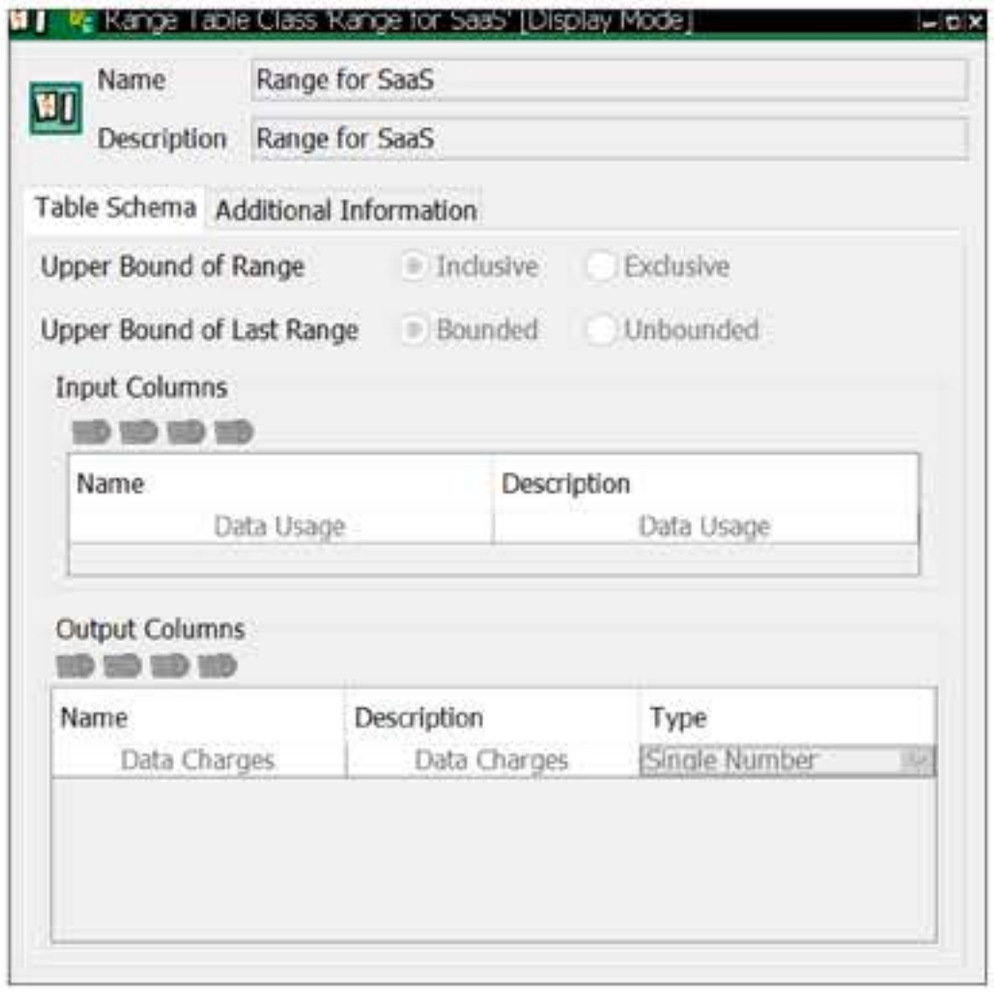

Figure 3.14 Range Table Class

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When defining a range table class, the upper bound range needs to be defined as either upper bound or lower bound. The upper bound of the last range also needs to be defined. Then, you must define the input and output columns.

The range table class can be saved using the **File • Save • In the Database** shortcut. Next, create a range table based on the range table class via **File • New • Range Table** (see Figure 3.15).

Figure 3.15 Range Table

To save the range table, follow menu path **File • Save • In the Database**. Once saved, the range table is ready to be used in charge logic, which will be covered in Section 3.6.

In this section, we've covered the building blocks of SAP Convergent Charging. In the next sections, we'll explore how these building blocks can be weaved together to design charging logic.

3.6 Configuring SAP Convergent Charging

Now let's look at the technical aspects of product modeling and how to configure SAP Convergent Charging using the Core Tool. This is arguably the most important area of SAP Billing and Revenue Innovation Management because SAP Convergent Charging plays the role of the rating and pricing engine. We'll continue to build on the Martex Corp. example to show how you can translate business requirements into SAP Convergent Charging objects.

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3.6.1 Chargeable Item Class

A chargeable item class defines the structure or format for how usage charges are passed into SAP Convergent Charging. Usage charges could be measured by, say, the number of songs downloaded, the amount of computation power consumed, or the gigabytes of disk space used.

A chargeable item class can be created in two different ways:

- Manual definition using the Core Tool
- Consumption item mapping

> **Note**
>
> A consumption item class is created within the SAP Convergent Invoicing module of SAP Billing and Revenue Innovation Management.

Let's begin by looking at manual definition within the Core Tool. The manual definition option is commonly used when SAP Convergent Charging is being used as a standalone solution. Figure 3.16 shows how the chargeable item class can be defined using the Core Tool. Follow these steps:

1. Navigate to **File • New • Chargeable Item Class**.
2. During the definition of a chargeable item class, custom fields of type date, number, and string can be defined in addition to the three mandatory fields required by SAP Convergent Charging (**Consumption Date**, **User Identifier**, and **Service Identifier**), as shown in Figure 3.16.
3. Additional user property fields can be added by clicking the button. The data type of the field can be changed by clicking the type column of the corresponding row. Subsequently, fill in the name and description and save using the following navigation path: **File • Save • In the Database** (see Figure 3.17).

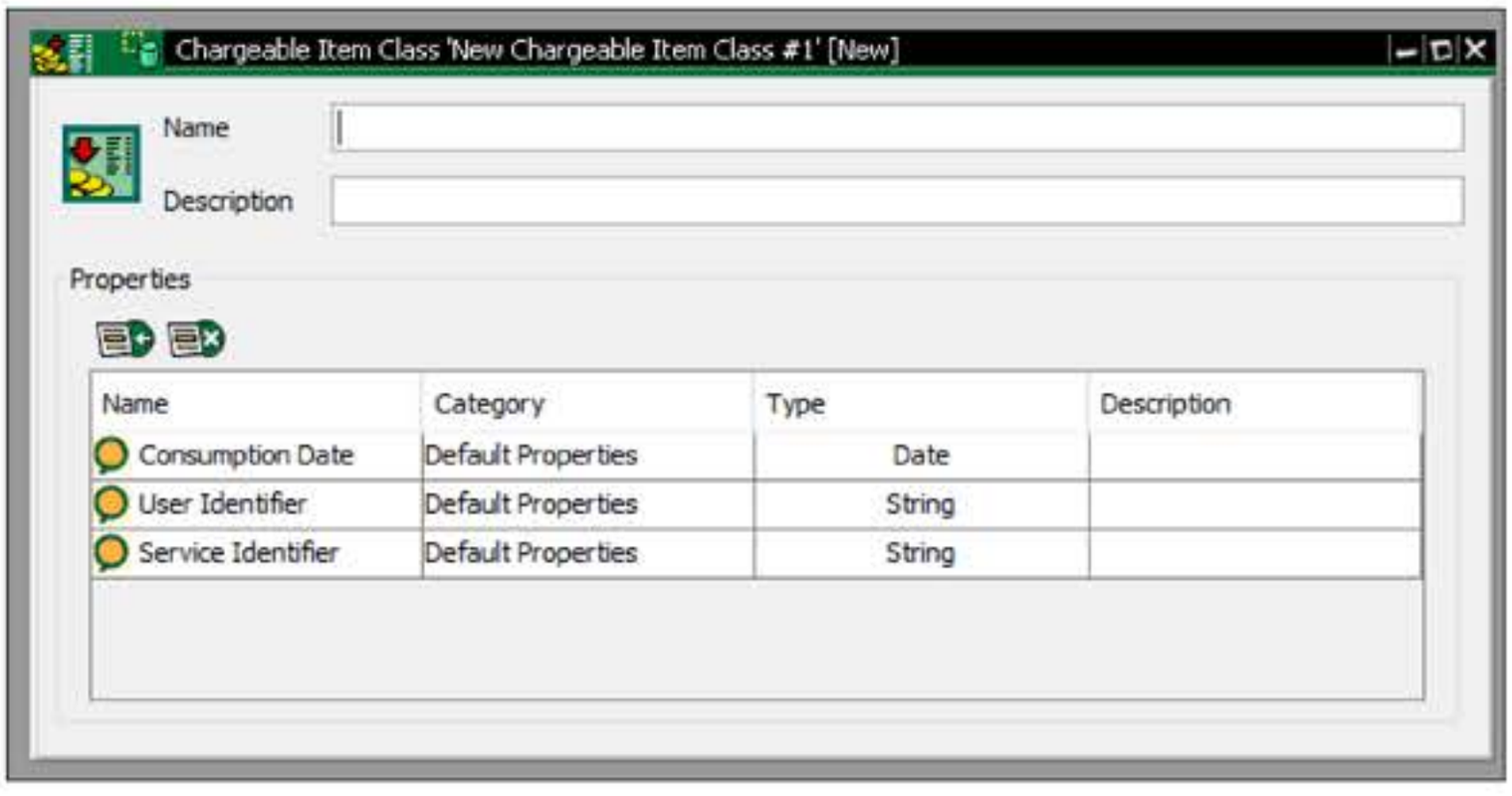

Figure 3.16 Defining the Chargeable Item Class

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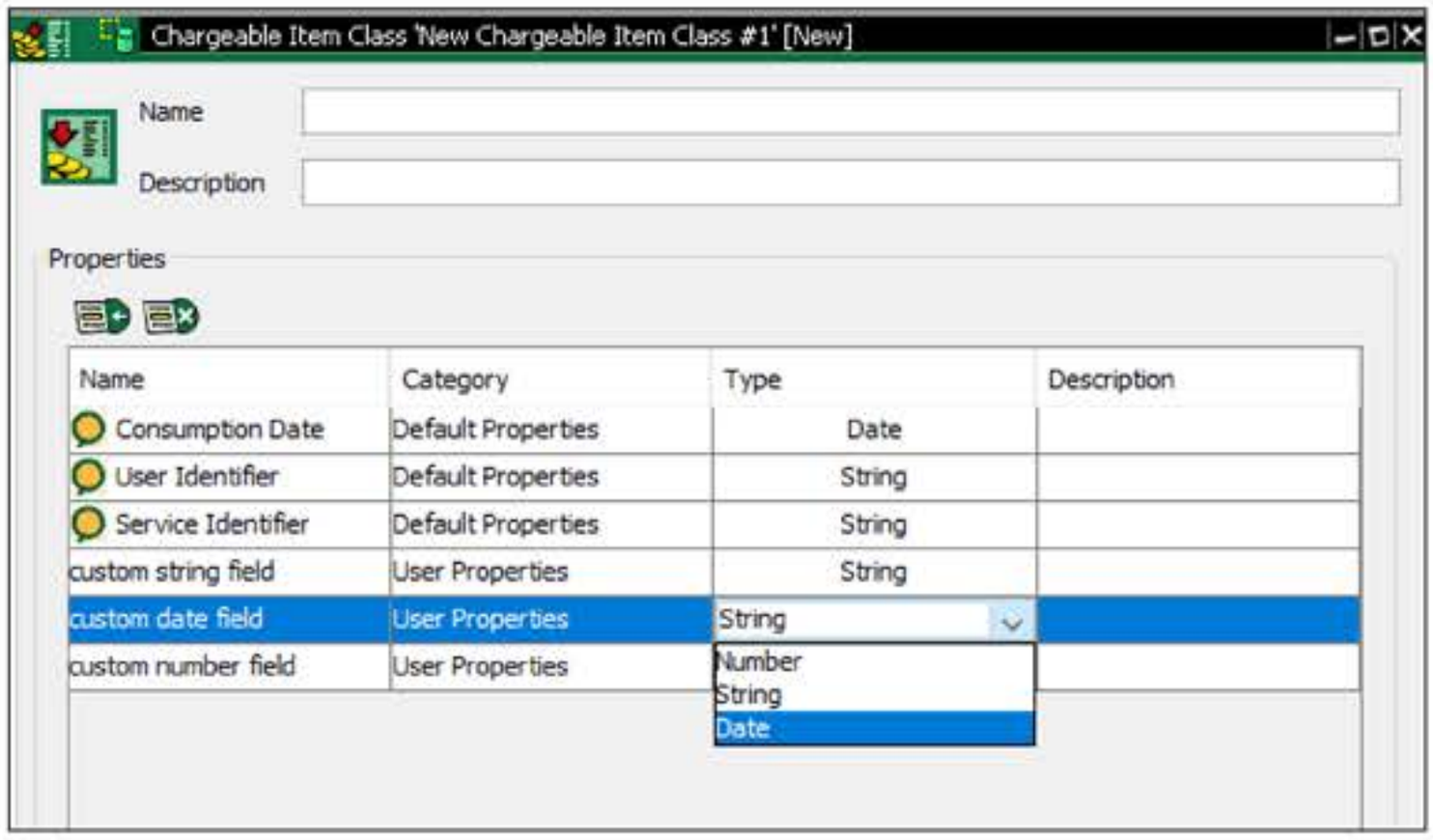

Figure 3.17 Custom Attributes in the Chargeable Item Class

Consumption item mapping is used to define chargeable item classes in an integrated scenario with SAP Convergent Invoicing. Because the objective of this book is to cover SAP Billing and Revenue Innovation Management as an end-to-end solution, chargeable item class examples will be created using the consumption item mapping approach. To create a chargeable item class using the consumption item mapping approach, navigate to **File • New • Consumption Item Mapping**.

Before you can create consumption item mapping, a consumption item class must be created first in SAP Convergent Invoicing. The consumption item classes **HCOS** and **REVE** are predefined in SAP Convergent Invoicing as consumption item classes. The purpose of having separate HCOS and REVE consumption item classes is to distinguish between consumption items for revenue (customer) and cost (vendor), as shown in Figure 3.18.

The steps and configuration to define consumption item classes in SAP Convergent Invoicing will be covered in Chapter 4.

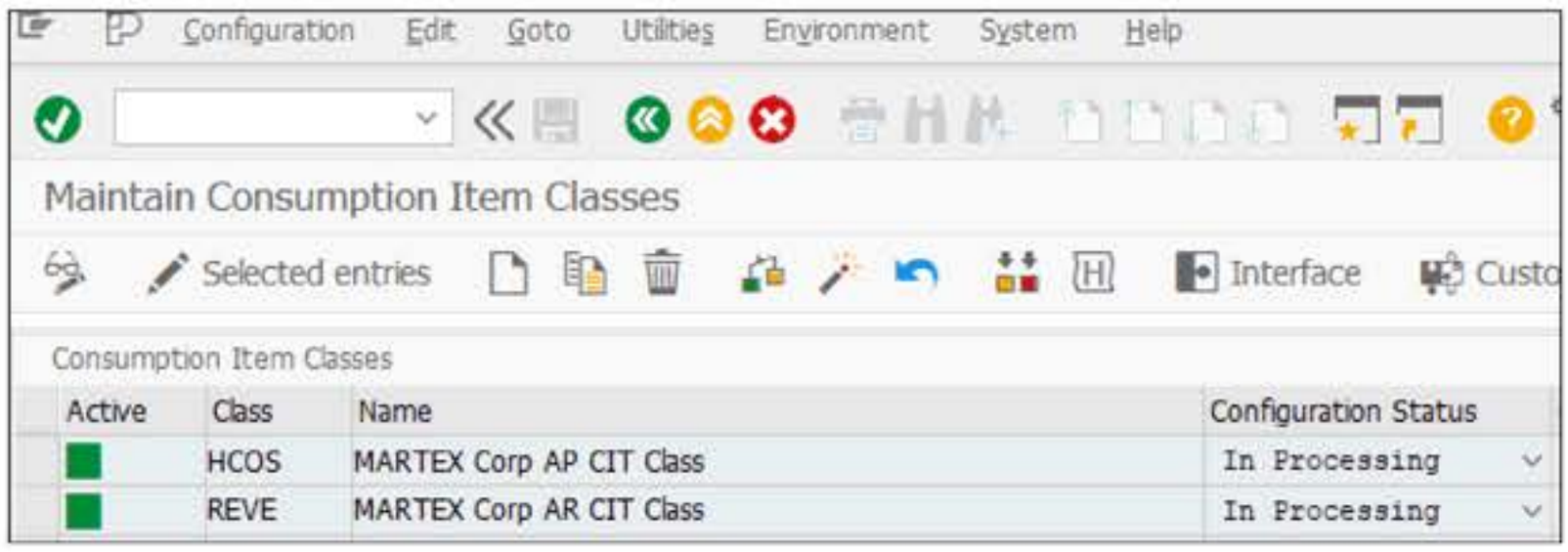

Figure 3.18 Consumption Item Classes

Figure 3.19 shows the steps to create HCOS consumption item mapping; the same steps apply when creating REVE consumption item mapping.

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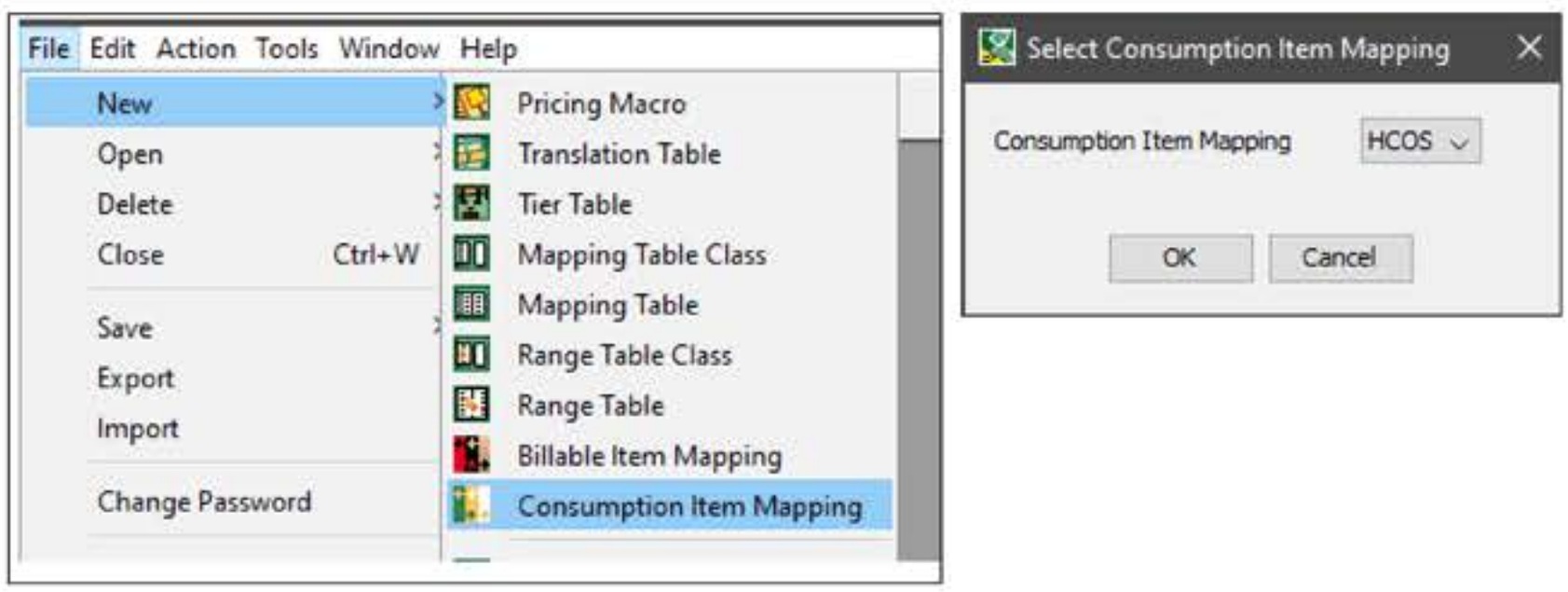

Figure 3.19 Creating Consumption Item Classes: HCOS

The creation of consumption item mapping will automatically create a chargeable item class in the background. Figure 3.20 shows the consumption item mapping class created on the left and the corresponding chargeable item class **HCOS** created together. Note that all the fields of both objects are the same, except that the chargeable item class has a consumption date, user identifier, and service identifier.

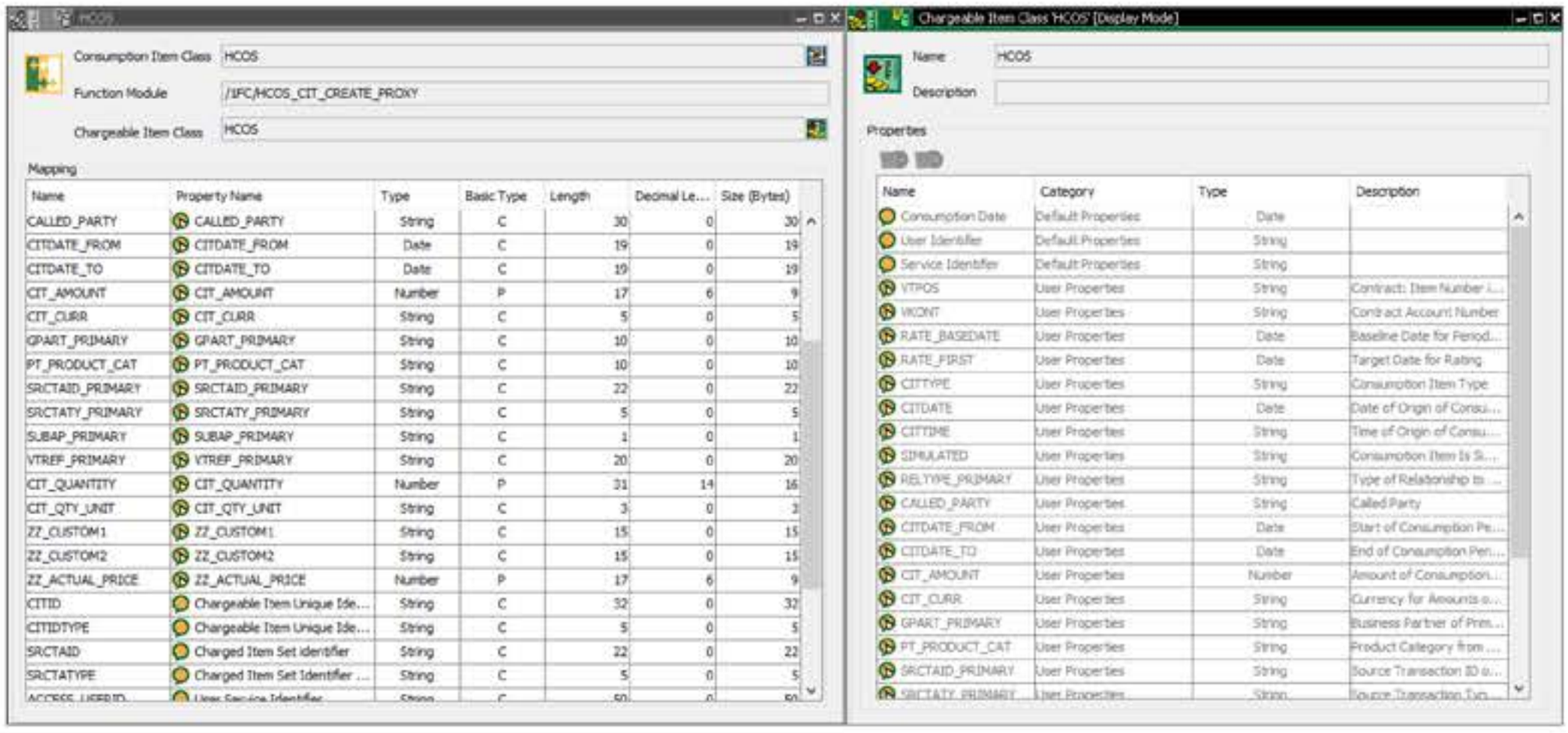

Figure 3.20 HCOS Consumption Item Mapping

3.6.2 Charged Item Class

Whereas chargeable item classes handle how usage charges are passed into SAP Convergent Charging, charged item classes handle how results from rating and charging generated by SAP Convergent Charging for usage-based or recurring subscriptions and one-off charges are passed from SAP Convergent Charging into subsequent systems for billing and invoicing. A charged item class can also be created in two different ways:

- Manual definition using the Core Tool
- Billable item mapping

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We'll discuss each of these in the following sections, along with steps for modeling monthly recurring charges, one-off charges, and usage charges.

> **Note**
>
> Billable item classes are created in the SAP Convergent Invoicing module in SAP Billing and Revenue Innovation Management.

Manual Definition

Let's begin with manual definition. The manual definition option is commonly used when SAP Convergent Charging isn't integrated with SAP Convergent Invoicing. To begin, navigate to **File • New • Charged Item Class**, as shown in Figure 3.21.

SAP Convergent Charging supports four different data types for charged items (number, string, date, and Boolean). To define a charged item class, select elements with the respective data type. The icon denotes string, denotes number, denotes date, and denotes Boolean (see Figure 3.22).

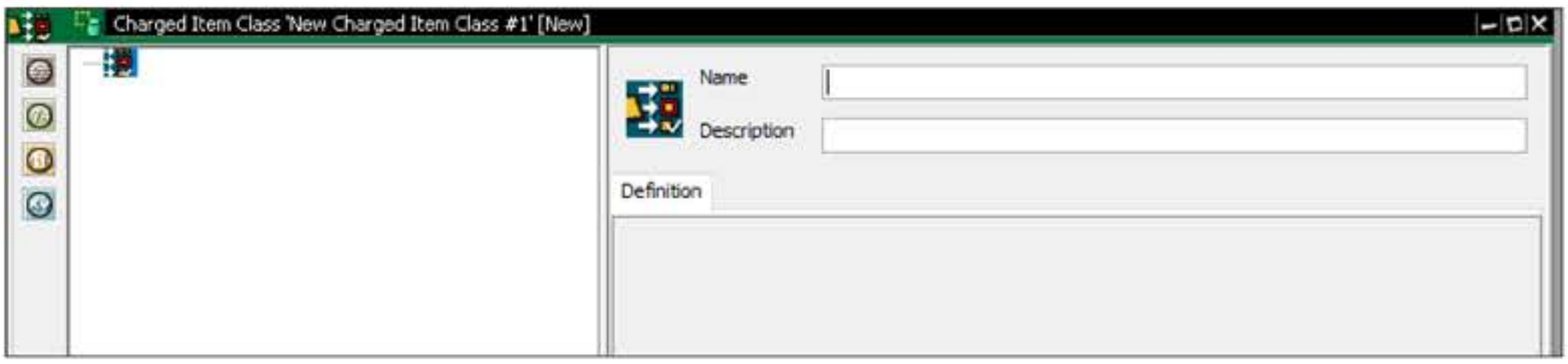

Figure 3.21 Defining a Charged Item Class

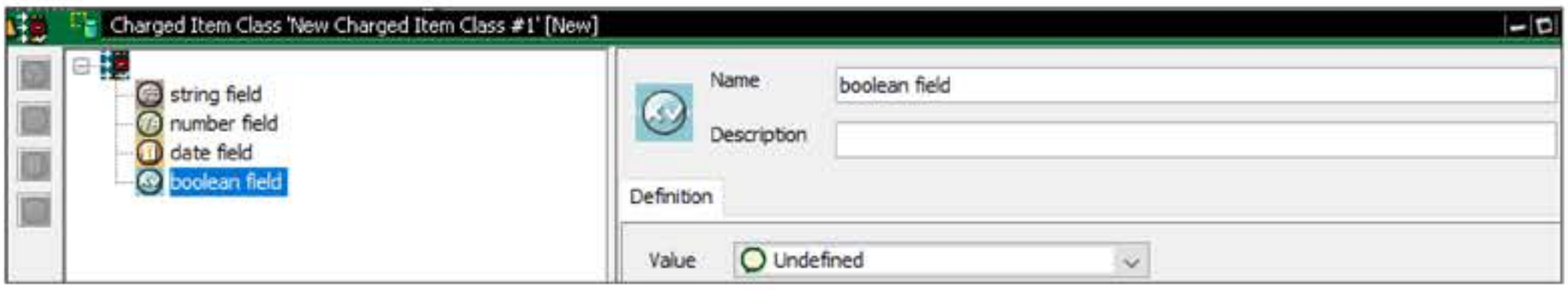

Figure 3.22 Adding Fields in a Charged Item Class

In an integrated scenario with SAP Convergent Invoicing, creation of a charged item class will be done using billable item mapping. The prerequisite for billable item mapping is to first create a billable item class in SAP Convergent Invoicing. This step will be covered in detail in Chapter 4. Figure 3.23 shows a sample of billable item classes **HCOS** and **HREV** in SAP Convergent Invoicing. The purpose of having separate HCOS and HREV billable item classes is to distinguish between billable items for revenue (customer) and costs (vendor).

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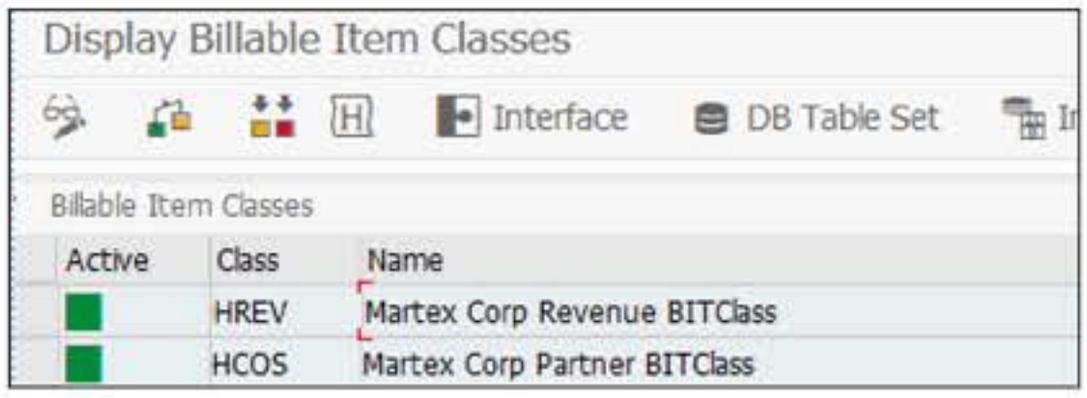

Figure 3.23 Billable Item Classes

To create billable item mapping, use navigation path **File • New • Billable Item Mapping**. The creation of billable mapping will automatically create a charged item class in the background. Figure 3.24 shows the billable item mapping created on the left and the corresponding charged item class **HCOS** created together.

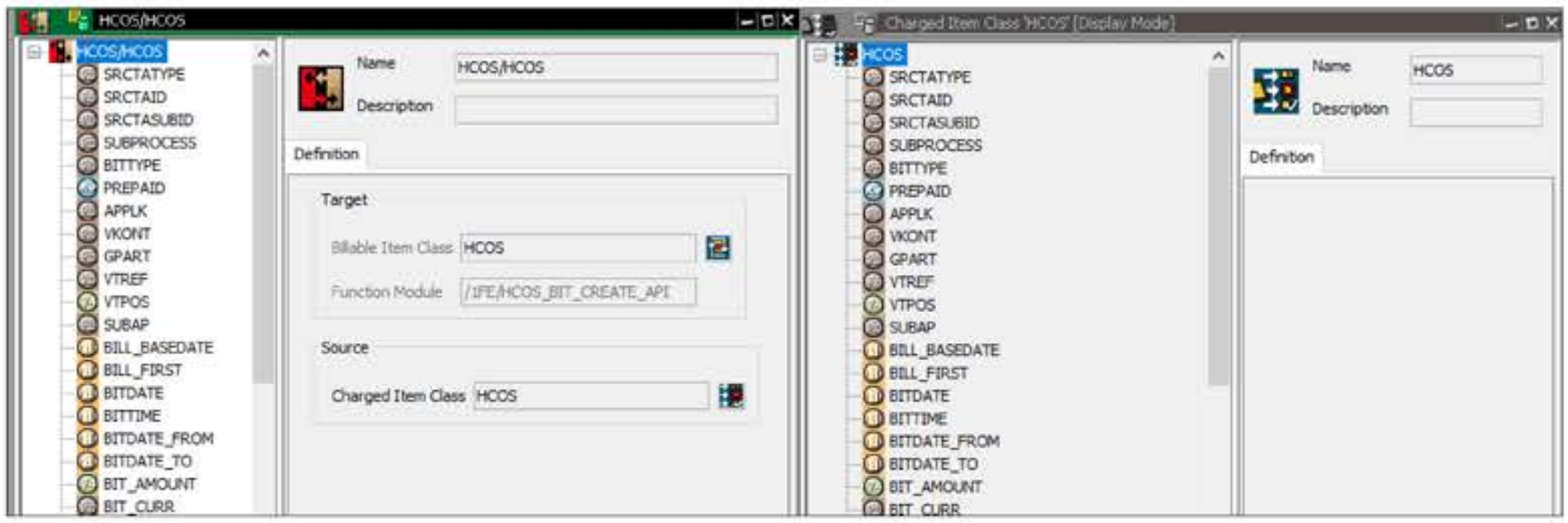

Figure 3.24 HCOS Billable Item Mapping

Now that you've defined the input/output formats for SAP Convergent Charging, the next step is to create charges that hold the calculation logic that will be used during the rating process.

Creating Charges

In SAP Convergent Charging, the charge object contains the rating and pricing logic that is triggered during the rating process. Charges are configured using the Core Tool in a decision tree structure. A charge consists of a charging plan and a price plan. The charging plan handles the determination of which reference account to charge, and the pricing plan contains the pricing and rating logic.

In this section, we'll walk through how the charges in the Martex Corp. example are being created step by step from scratch. Our example describes a few business requirements: a recurring membership cost and usage charges for service A and service B. When performing charge modeling in SAP Convergent Charging, it's important to be able to decipher the number of charges needed by understanding the business requirements. As a rule of thumb, this is usually determined by billing frequency (one-time, usage-based, or recurring).

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Modeling the Monthly Recurring Charge

The first charge to model is the recurring cost of $200 every month, and the first step is to define the charge attributes. For now, the default will be used, where **Type of Charge** is set to **Master** and **Fixed Currency** is set to **USD**. If the contracts with your customers are multicurrency, you'll need to select the **Multicurrency** option. This will allow the charge object to be used for different currency codes. The other tabs—**Parameters**, **Persistent Counters**, and **Legal Text**—will be updated as needed per business requirements. To start, navigate to **File • New • Charge**, and fill in the **Name** field, as shown in Figure 3.25.

Figure 3.25 Defining Monthly Recurring Charge

The next step is to define the charge object for the example: MARTEX_CHG_UseCaseA_RI_Recurring. Our business requirements don't prompt a need for different charging references, so default charging will be used. Figure 3.26 shows the three steps required to create a default charging reference. In the **Charging Plan** node of the tree, add a new charging reference called "default" with the **Type** set as **Postpaid** by using the button (see Figure 3.27).

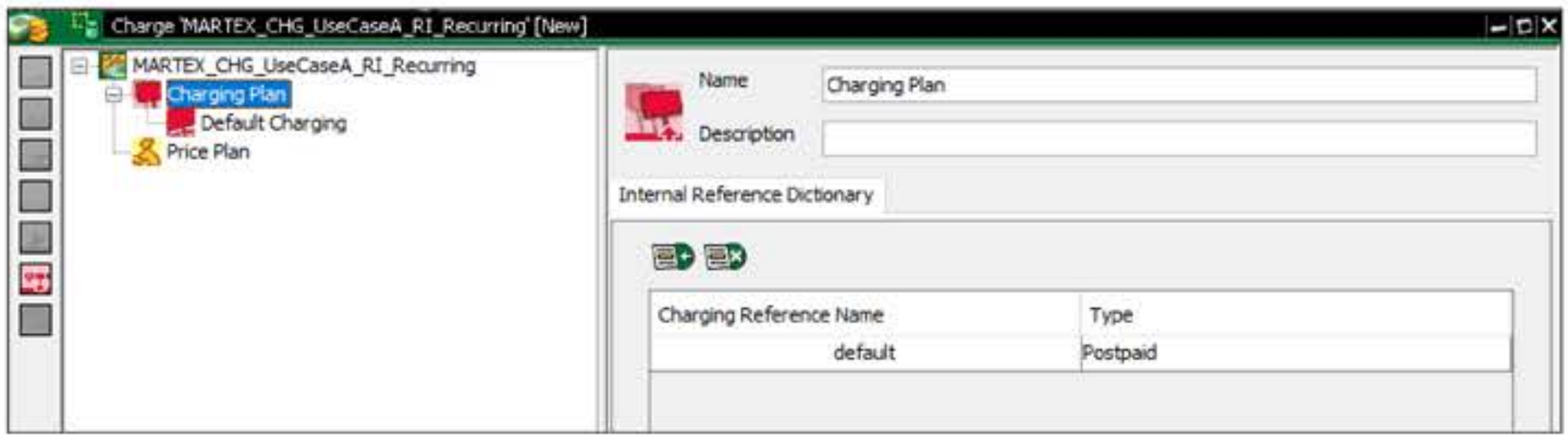

Figure 3.26 Adding a Default Charging Reference in the Account

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Next, in the **Default Charging** node, right-click and select **Add Component • References • Internal Reference** from the context menu. Next, enter the **Name** as "default", and select **default** from the **Internal Reference** dropdown list, as shown in Figure 3.27.

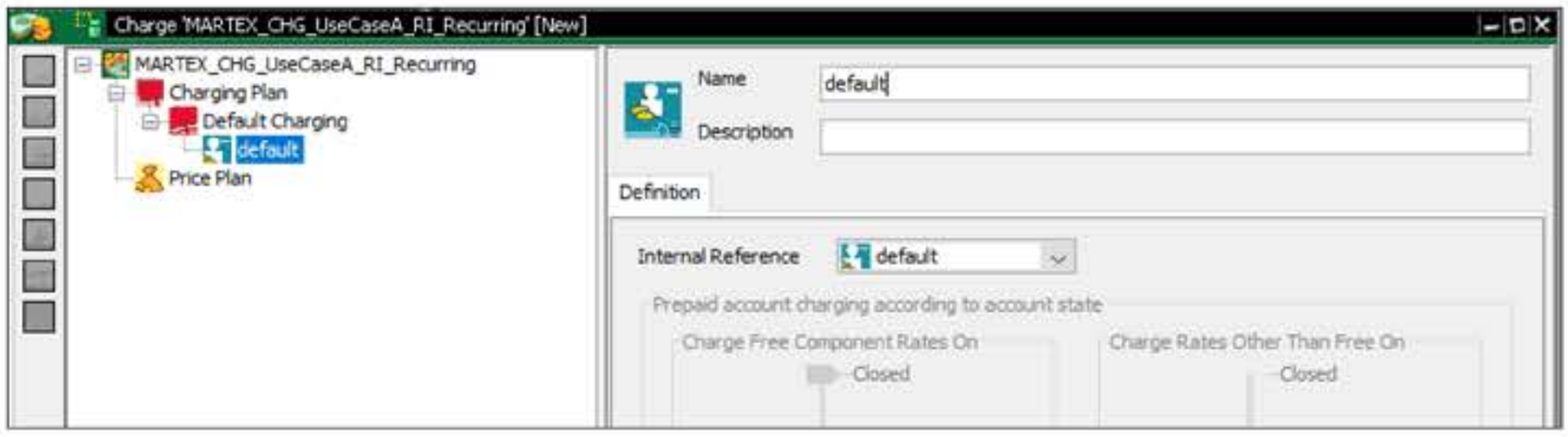

Figure 3.27 Defining the Internal References in the Charging Plan

After a charging plan is defined, the second component of the charge, price plan, needs to be defined according to the business requirements to generate a $200 charge every month for reserved instance enrollment as described in the use case. In the **Pricing Plan** node, right-click and select **Add Component • Rates • Recurring Rate**. Next, in the **Recurring** node, set the **Name** and **Description** fields to "Monthly Charge". Set **Reference Date** to **Charge Effective Date**, and set **This recurring rate is triggered** to **Every Month**, as shown in Figure 3.28.

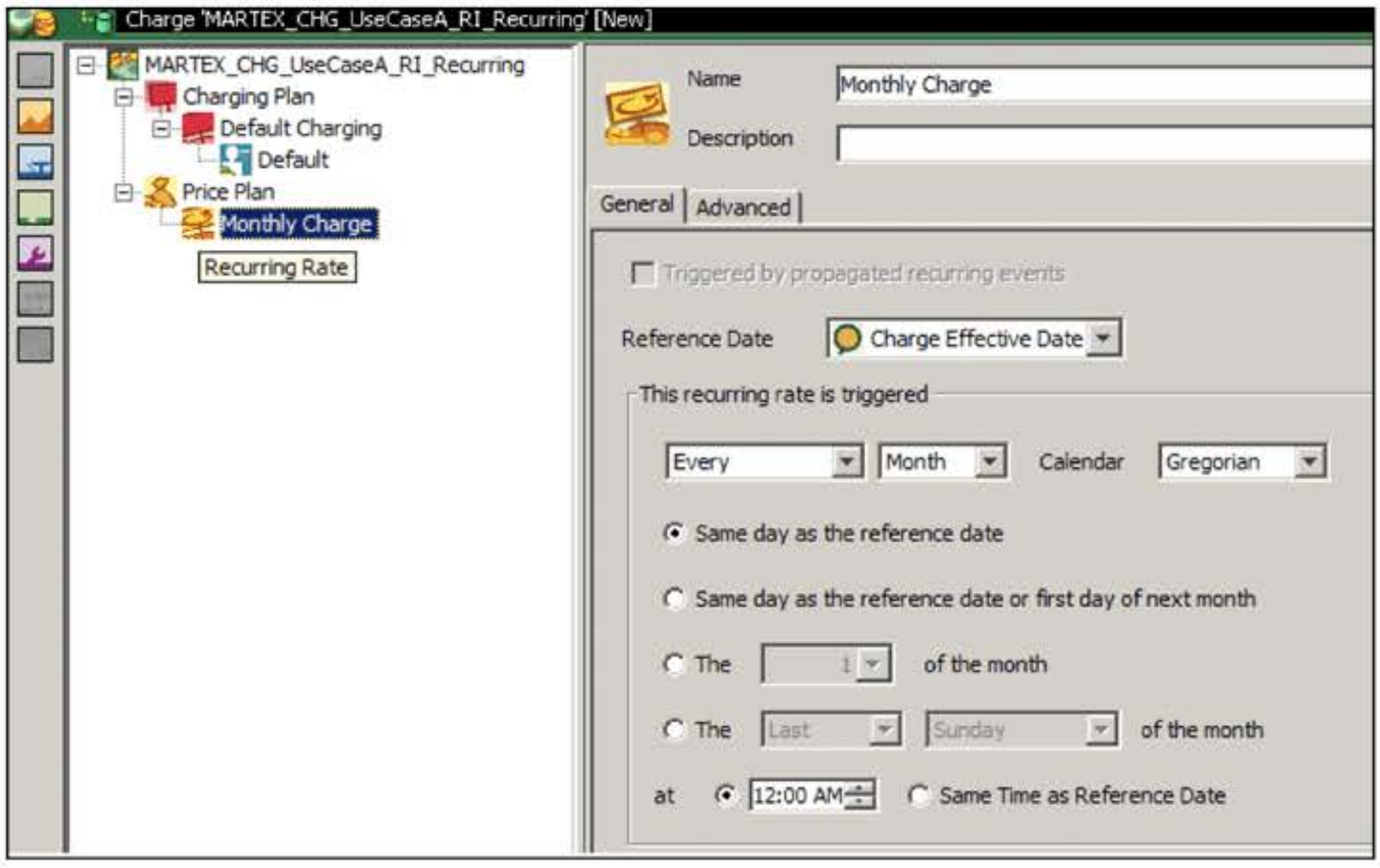

Figure 3.28 Configuring the Monthly Recurring Rate

After the recurring rate component is defined, the next step is to create the charging logic beneath it. As a rule of thumb, all paths in the charging logic tree need to end with

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a function. Based on the example requirements so far, the charge only needs to return a flat fee amount of $100. In the **Monthly Charge** node, right-click, select **Add Component • Function • Flat**, and enter a **Fixed Amount** of $100, as shown in Figure 3.29.

Note

In a real implementation, the amount should not be hard-coded. You need to create a parameter in a charge object and pass the amount from the provider contract.

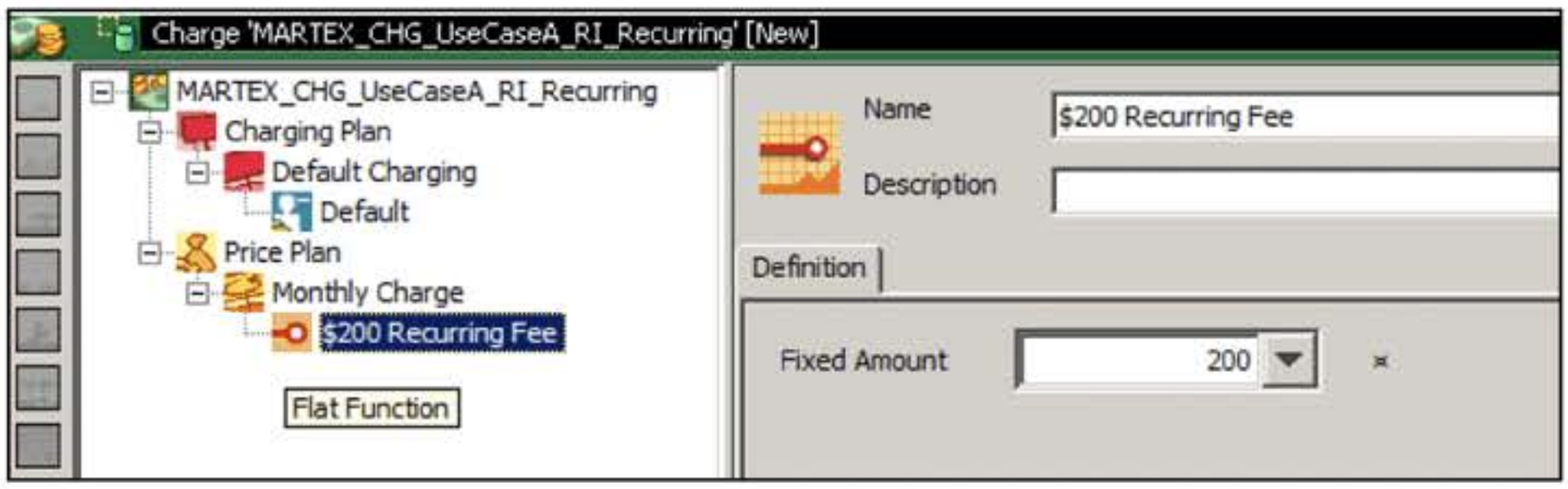

Figure 3.29 Configuring the Fixed Flat Fee Charge of $200

The charge is now configured to be $200 every month. However, there are additional rules for Martex Corp. A Martex Corp. customer will be refunded the entire up-front fee if the contract is canceled within the first month. To enable this requirement, add a persistent counter to the recurring reserved instance charge that will increment every time the charge is triggered. The purpose of the counter is to keep track of the current month of the contract. Figure 3.30 shows how a month counter is added. In the **Recurring Reserved Instance** node, select the **Persistent Counters** tab, and click **Add Counter**.

Figure 3.30 Defining Persistent Counters

To increment the counter, right-click the **Monthly Charge** node, and select **Add Component • Operators • Update Counter**. The update counter operator is configured to add the value of 1 to the month counter every time the month counter is triggered, as shown in Figure 3.31.

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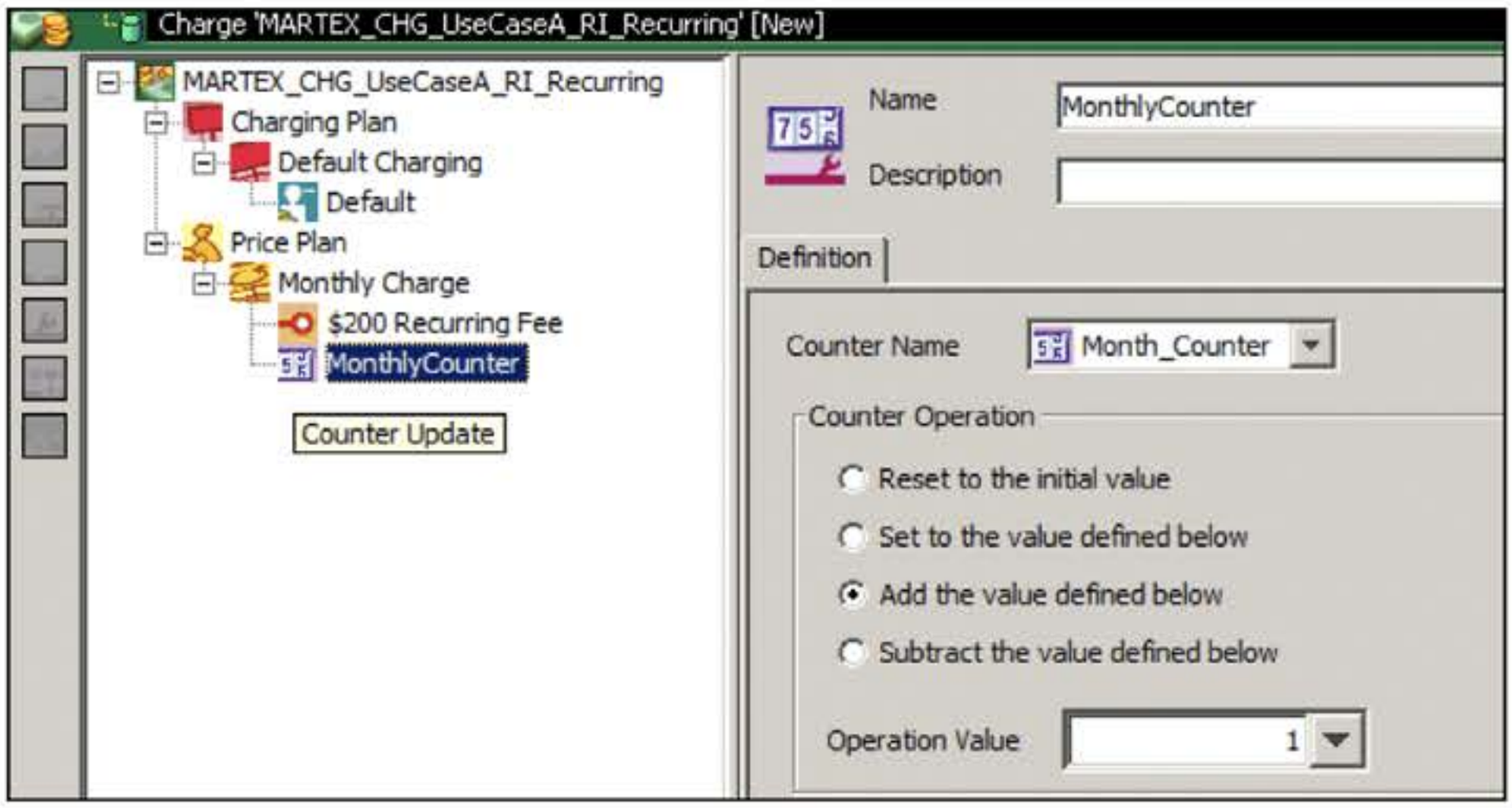

Figure 3.31 Update the Monthly Counter

Finally, save the monthly charge by following menu path **File • Save • In the Database**.

Modeling the Cancellation Charge

Now that we have a way of tracking the current contract month, the next step is to create another charge called MARTEX_CHG_UseCaseA_RI_Cancellation to calculate the cancellation fee. With the exception of the price plan, all other attributes of the charge remain the same as the recurring charge, as shown in Figure 3.32.

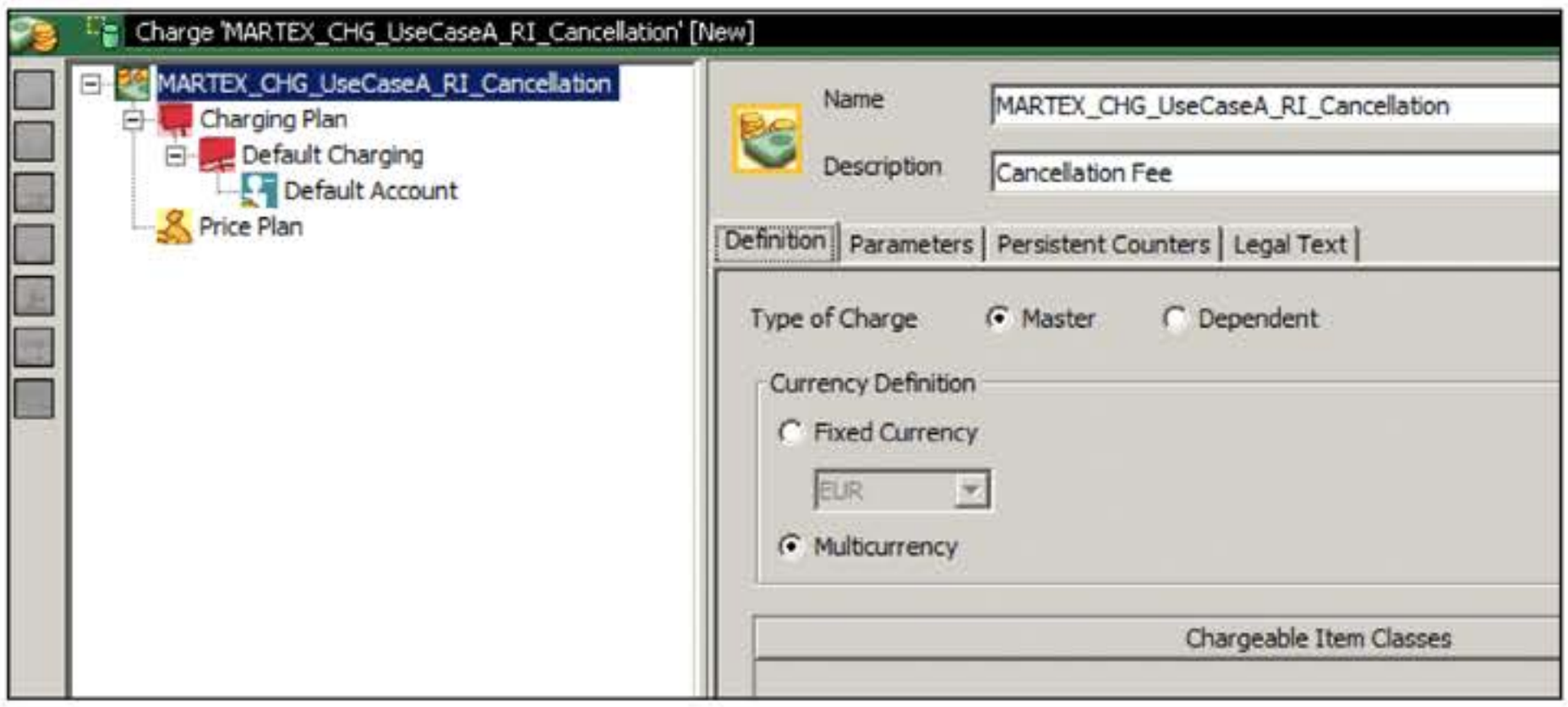

Figure 3.32 Cancellation Reserved Instance

First, add a rate component of the one-shot rate type by right-clicking the price plan, selecting **Add Component • Rates • One-Shot Rate**, and setting the **Event Name** to

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Suspension. This indicates that the rating will only happen during cancellation, as shown in Figure 3.33.

Figure 3.33 Configuring the Event Type for One-Time Charge

The next step is to add special logic for first-month cancellation by right-clicking the **Cancellation Charge** node and selecting **Add Component • Comparator • Numbers**. Because the comparator object behaves as an `if` function, adding the comparator object to check for a month counter value of 1 will trigger a fork in the tree where the month counter equals 1 and the month counter doesn't equal 1, as shown in Figure 3.34.

From here on, the logic for a condition month counter equal to 1 is to return –$200 as a refund to the customer. This step is the same as adding a flat fee charge earlier, as shown in Figure 3.35.

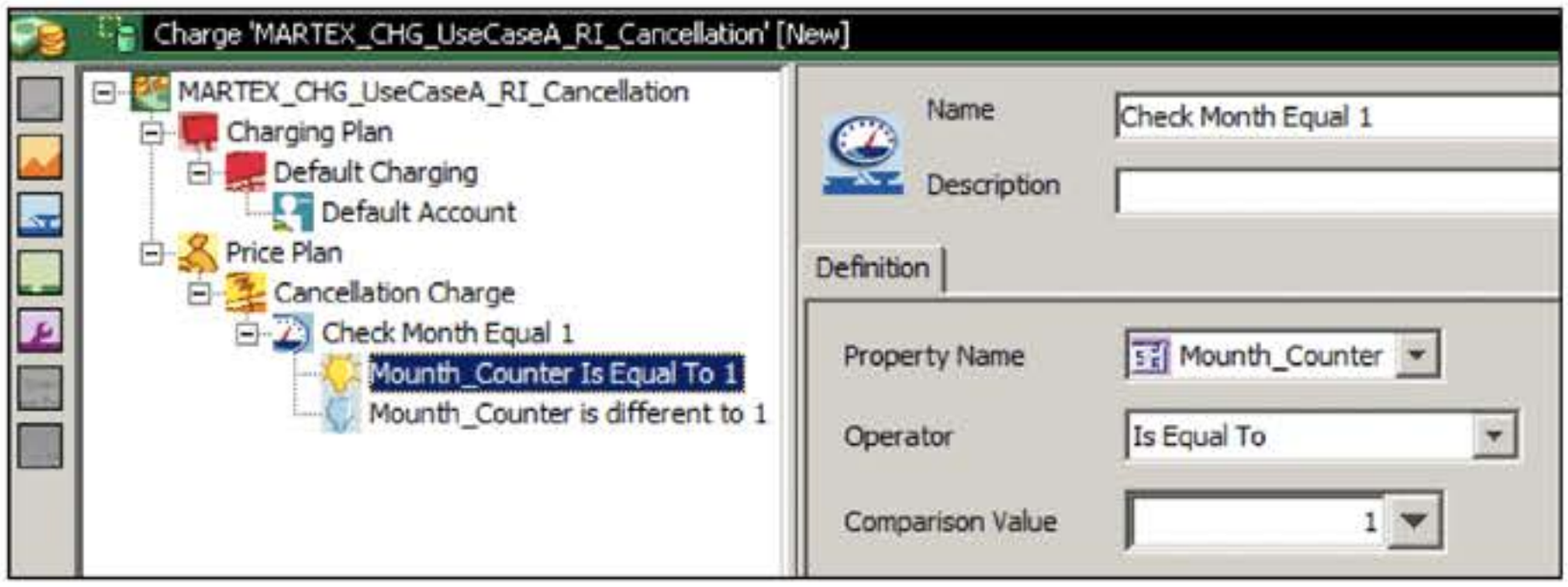

Figure 3.34 Using the Numbers Comparator in Charge Logic

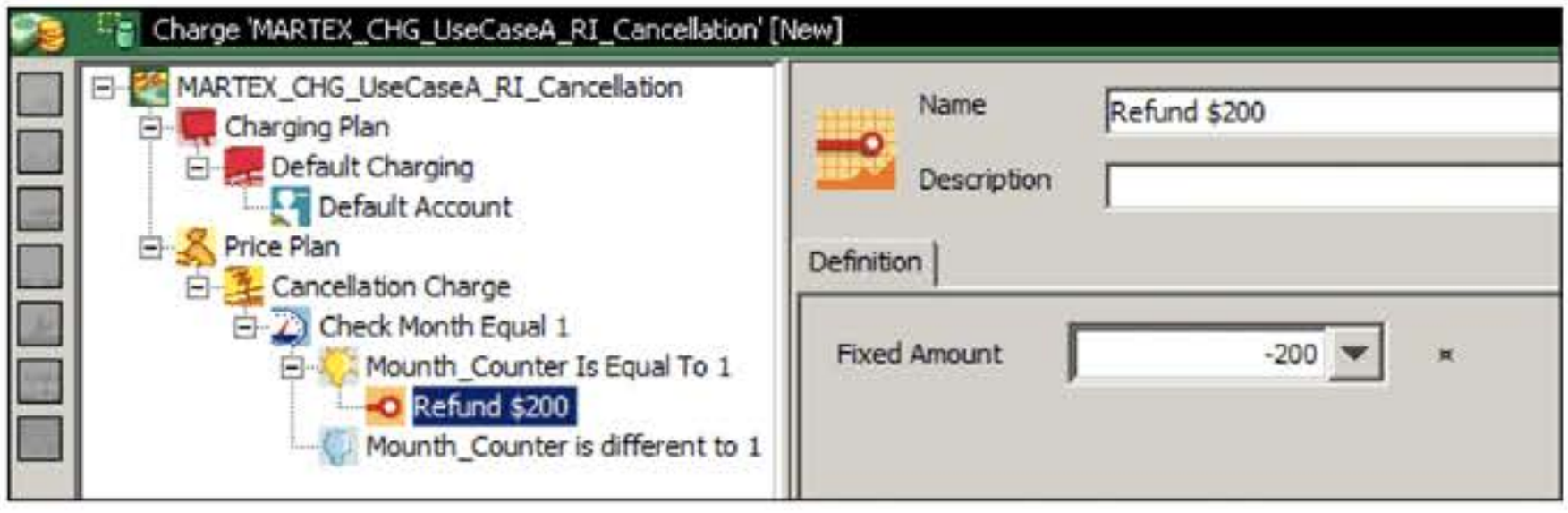

Figure 3.35 Refund $200

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If the contract is past the trial period, the month counter will have a value greater than 1. Based on our example business requirements, only the value of the unused period will be refunded to the customer. The Prorata Operator can be used to calculate the value to be returned to the customer.

The Prorata Operator has several mandatory values to be populated such as the recurring rate component. One of the properties is the **Property to Prorate** field, which needs to be populated by a number type parameter. This can be added using the **Property Introducer** operator by choosing **Add Component • Operators • Property Introducer**. The next step is to add a Prorata Operator by right-clicking the **month_counter is different to 1** node and selecting **Add Component • Operators • Prorata**.

Figure 3.36 shows the values that need to be populated. To obtain the remaining unused portion to refund, select the **Fraction from Prorata Date** radio button, and the result of the calculation will be stored in a variable called **ProratedRefundAmount**, as shown in Figure 3.37.

The prorated amount will be refunded by using the flat fee function, as shown in Figure 3.38.

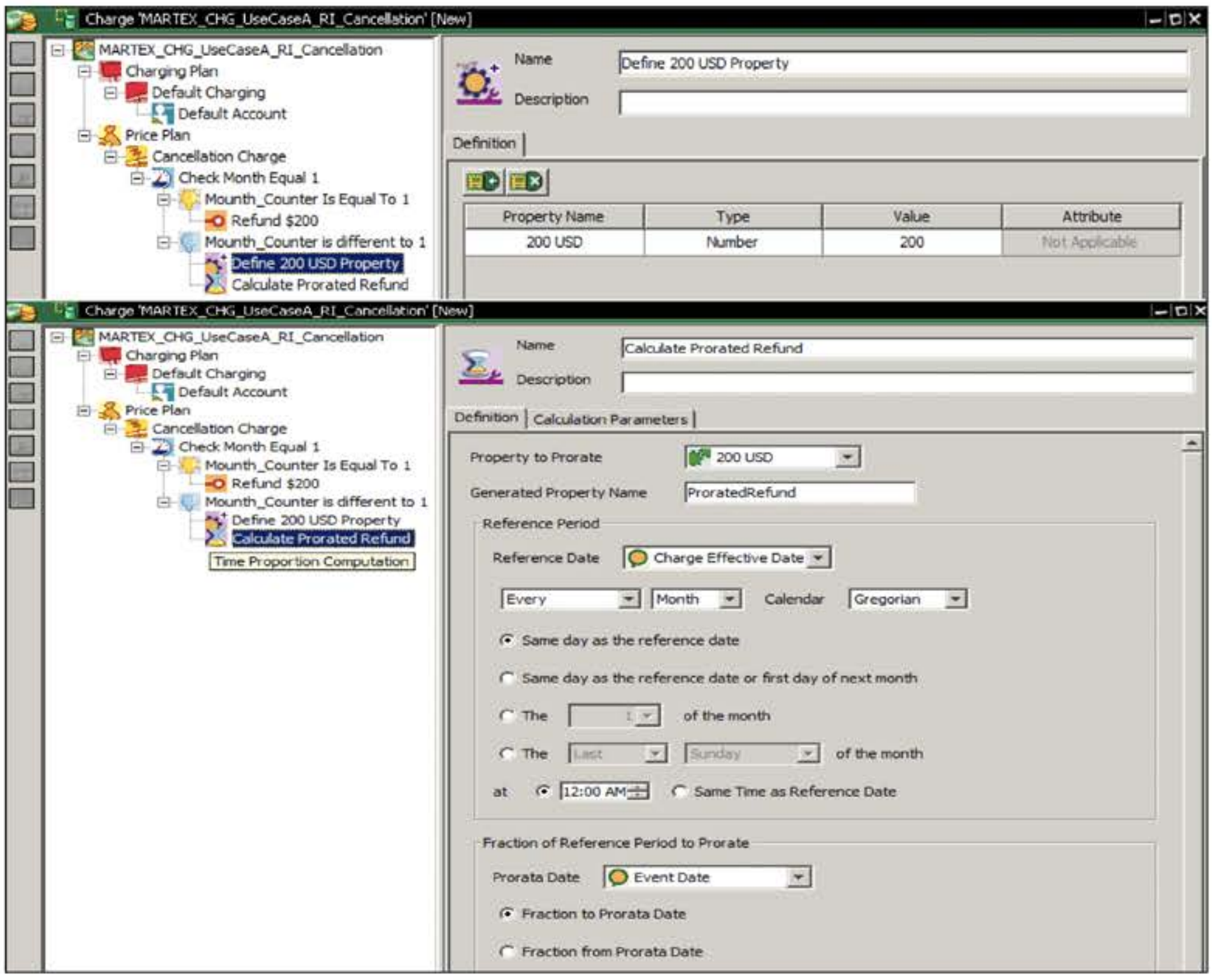

Figure 3.36 Prorate Monthly

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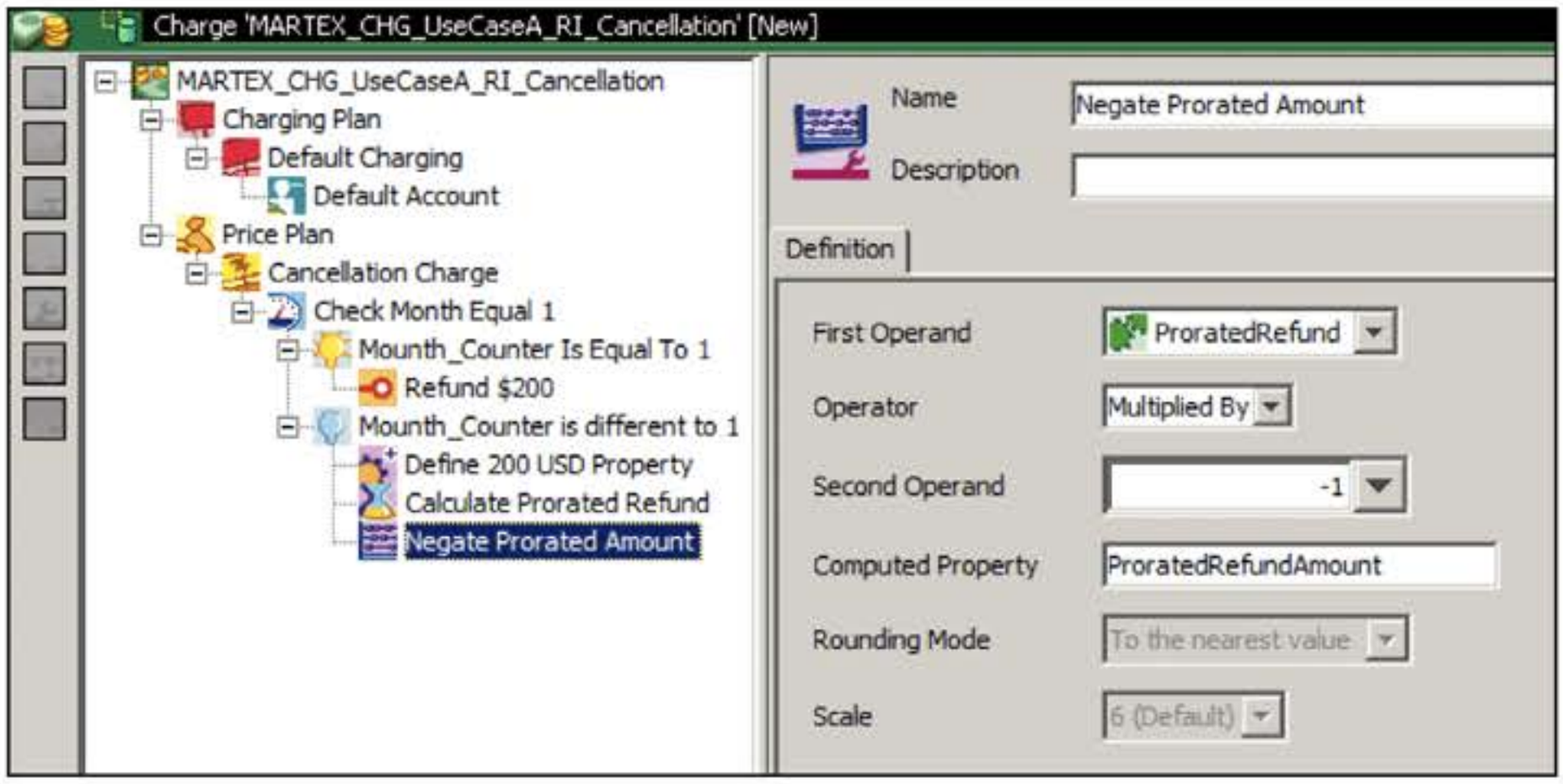

Figure 3.37 Arithmetic Computation to Negate prorated_amount via Multiplication with -1

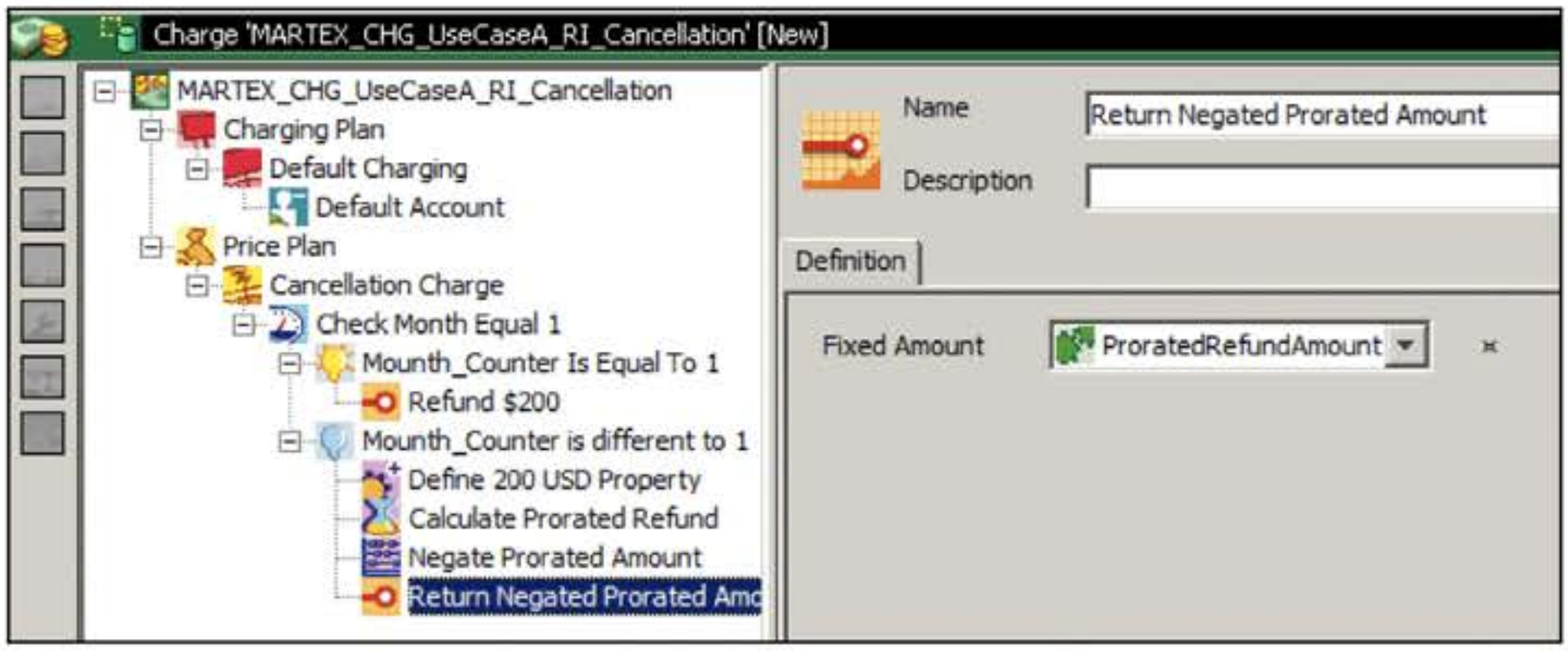

Figure 3.38 Flat Fee Function

Finally, save the cancellation charge by following menu path **File • Save • In the Database**.

Modeling the Activation Charge

As part of the use case requirement for reserved instance, a $50 one-time fee for activation needs to be created. The activation charge is called MARTEX_CHG_UseCaseA_RI_Activation. The process of creating an activation charge is the same as the cancellation charge created earlier. However, the trigger event is activation, as shown in Figure 3.39.

In addition to generating an activation fee of $50, the activation charge also creates an allowance object. Because each allowance has a validity period, you need to create a property to calculate the allowance end date, as shown in Figure 3.40.

Next is to create the allowance object as shown in Figure 3.41. The details of allowance setup are discussed in Section 3.7.

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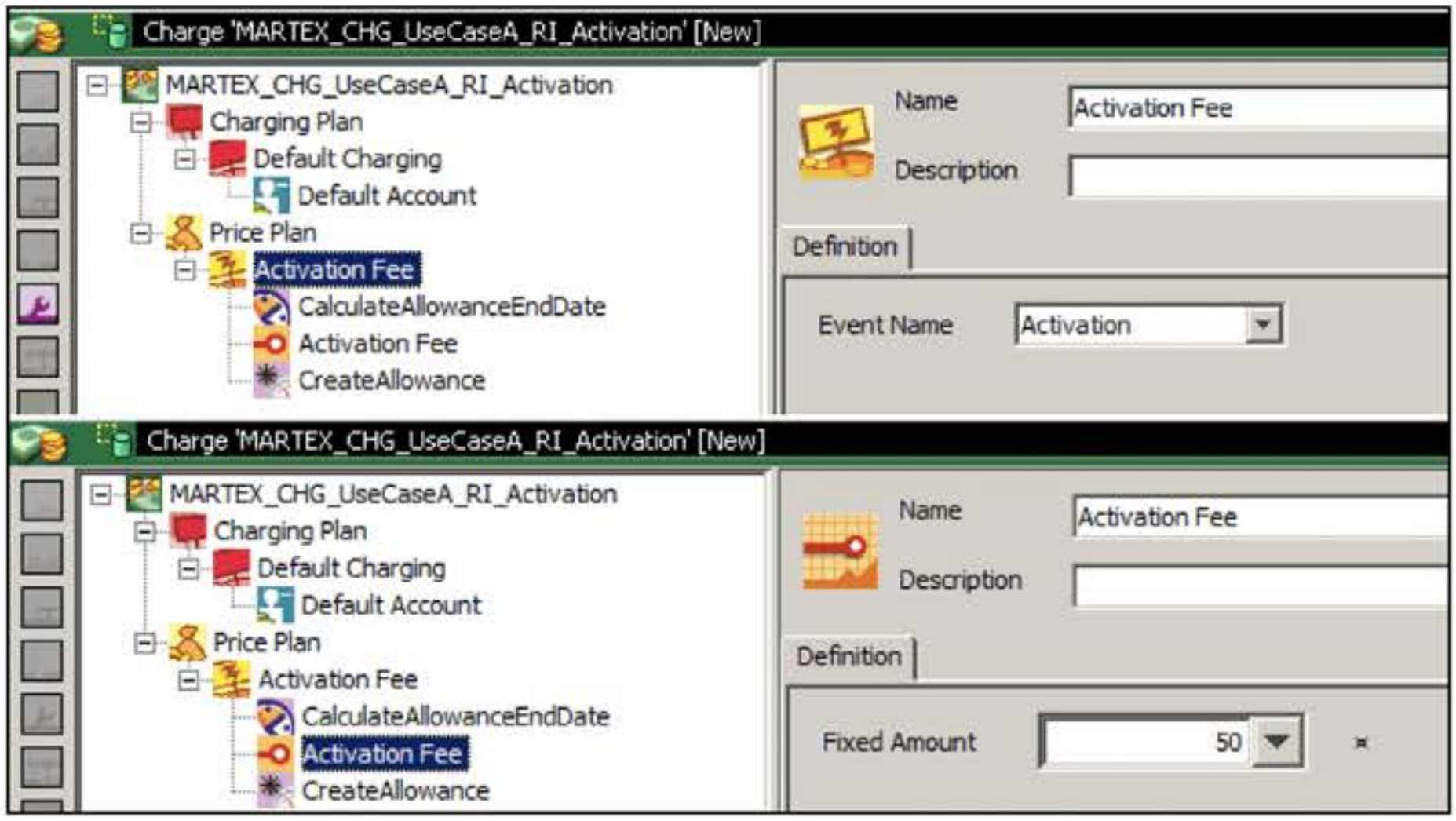

Figure 3.39 Activation Charge of $50

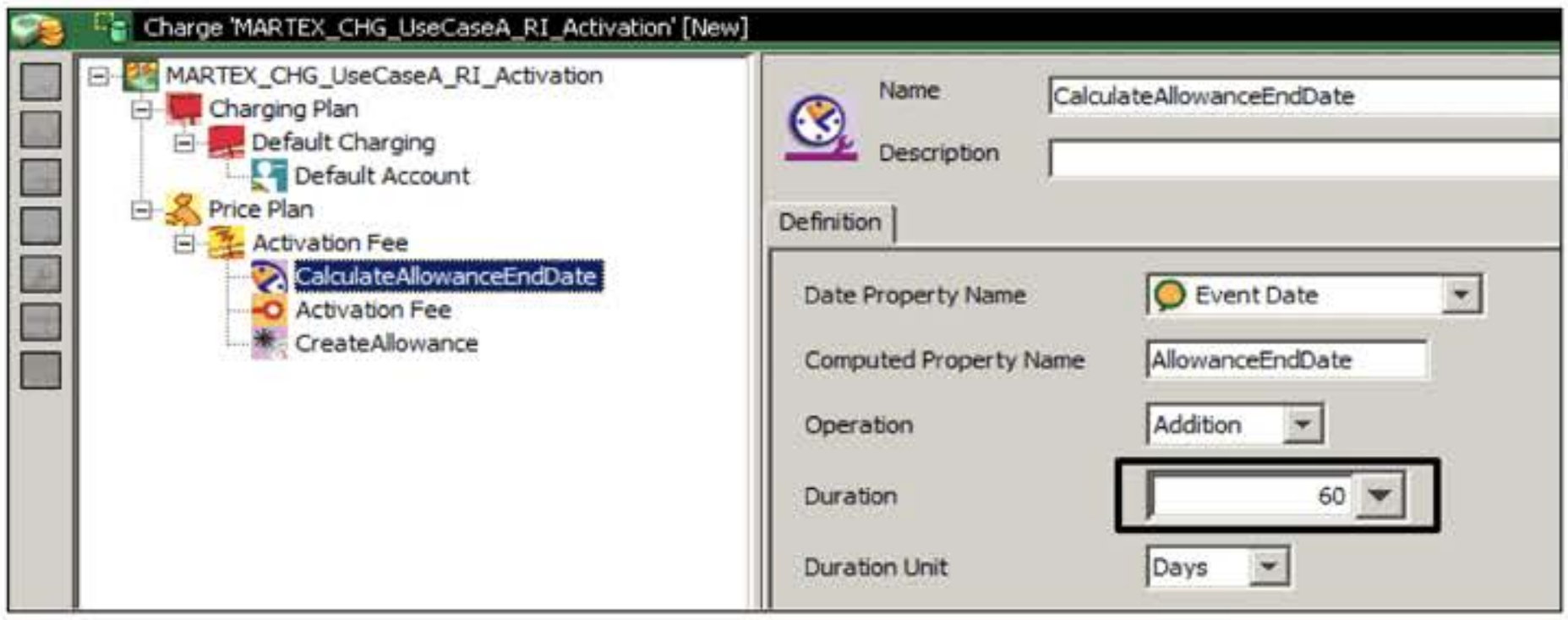

Figure 3.40 Set Allowance Duration for 60 Days

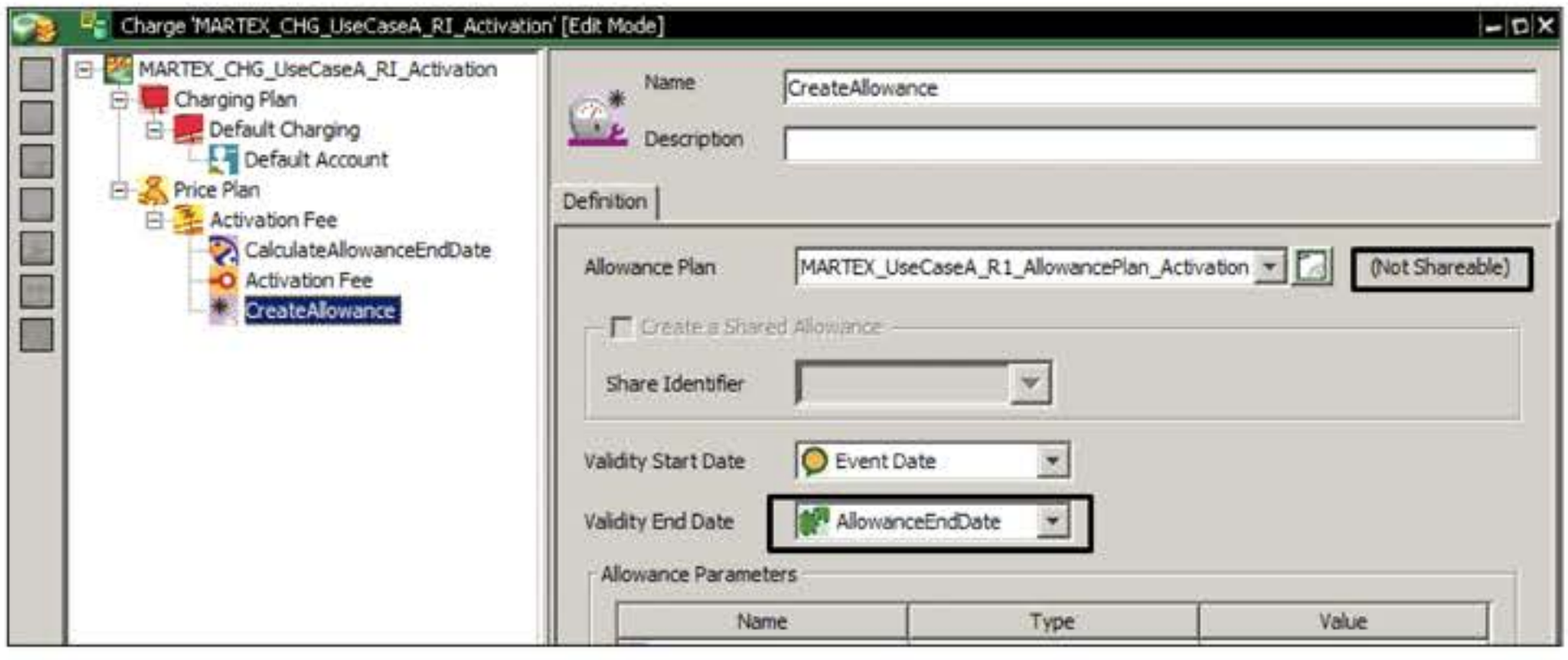

Figure 3.41 Create Allowance

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Modeling the Usage Charges for Service A and Service B

Now that the activation fee, recurring fee, and cancellation fee have been created, the next part of the requirement is to create a charge that will support incoming usage charges for both services.

The first step is to create a mapping table class structure that can be used to store the prices of both service SKUs by navigating to **File • New • Mapping Table Class**.

The next step is to define the structure of the mapping table class. For our example, the mapping table class should have one input column and two output columns. Save the mapping table class in the same catalog as all the other objects. The input column consists of microservice SKU ID, and the output columns contain the regular amount and the reserved instance amount, as shown in Figure 3.42.

Once the mapping table class has been created, the next step is to create the mapping table *instance* for the mapping table class. Prices of service SKUs are stored in the mapping table. Navigate to **File • New • Mapping Table**.

For the next steps, enter the name of the mapping table in the **Name** field, and select the mapping table class created in the previous step. Next, manually insert two entries into the mapping table by using the button, as shown in Figure 3.43. A mapping table can be populated in different ways. For example, in an SAP Billing and Revenue Innovation Management integrated scenario, the mapping table can be populated from SAP Billing and Revenue Innovation Management, subscription order management, via web services.

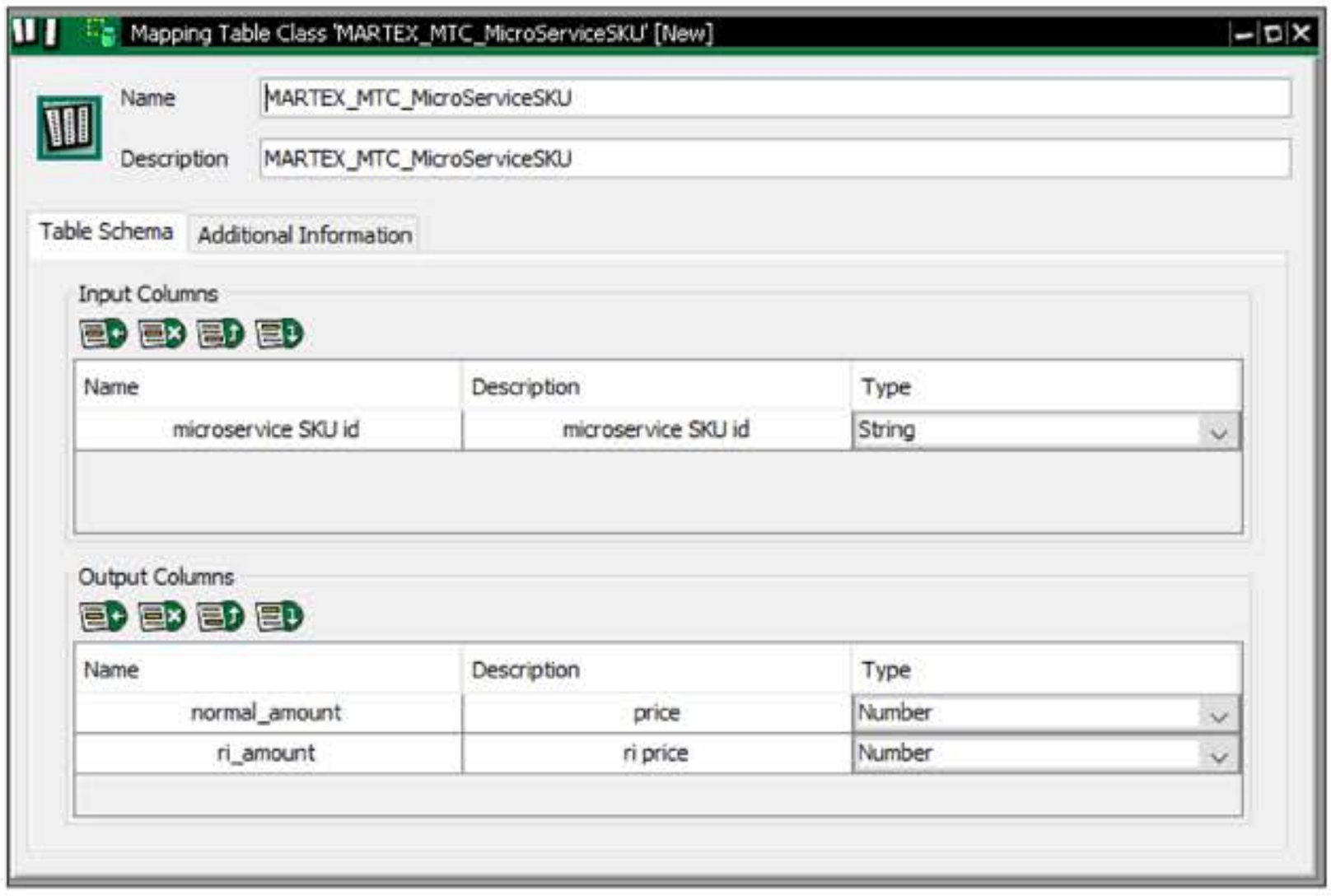

Figure 3.42 Mapping Table Class for Microservice SKU

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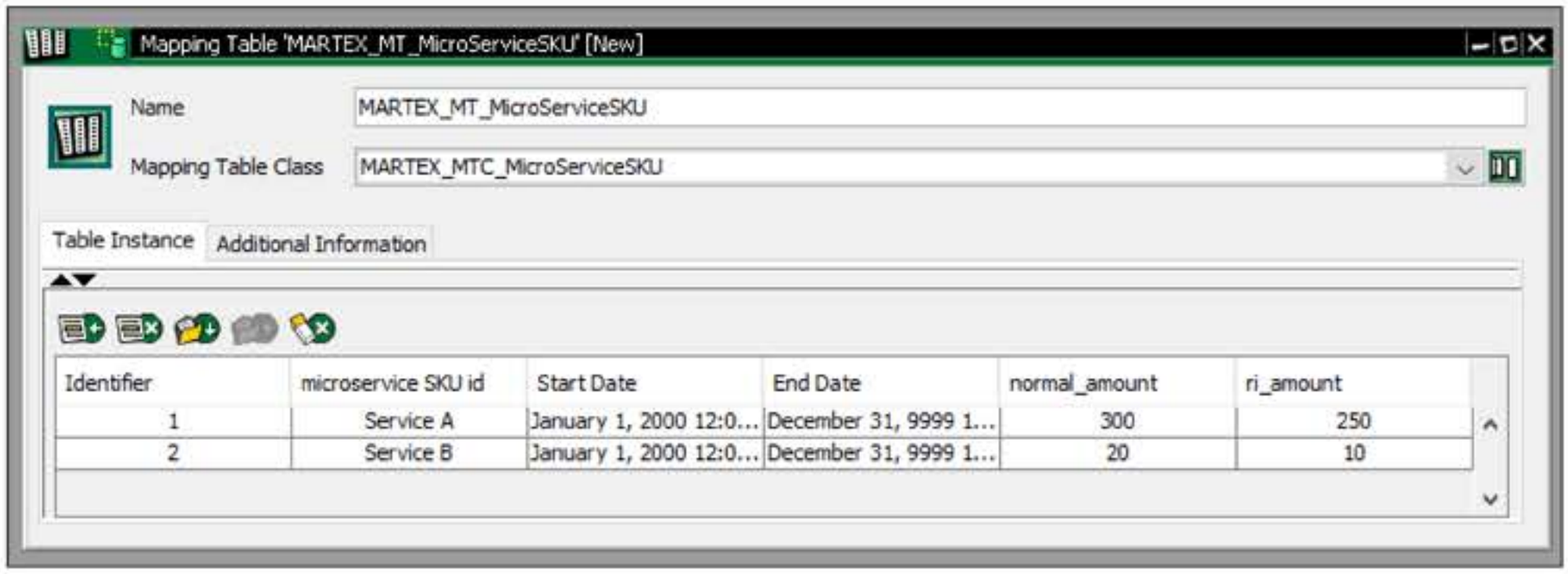

Figure 3.43 Martex_MT_MicroServiceSKU Mapping Table Defined

Finally, save the mapping table by following navigation path **File • Save • In the Database**.

Once the pricing master data for the services has been created in SAP Convergent Charging, the usage charge can be created. Based on our requirements, the price that needs to be charged when a customer is on a reserved instance is $250 for service A and $10 for service B, respectively, instead of the regular rates. To create a usage charge, proceed with the steps shown in Figure 3.44. Most of the steps are similar to those used for creating the cancellation, activation, and recurring charges earlier. Navigate to the following path in the Core Tool: **File • New • Charge**.

With the exception of the price plan, other attributes of the charge remain the same as the recurring charge. Enter "MARTEX_CHG_UseCaseA_Usage" as the **Name** for the charge, as shown in Figure 3.44.

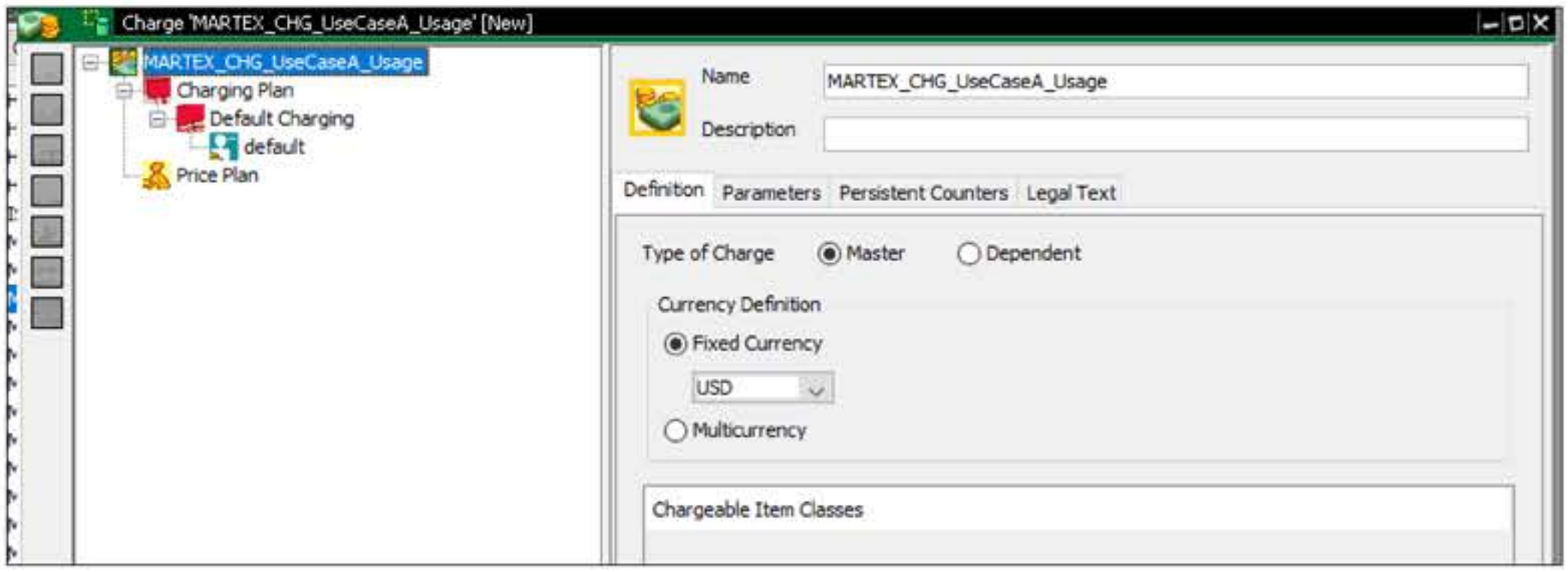

Figure 3.44 Creating Usage Type Charge

Because the scenario is based on usage consumption, a usage-based rate component needs to be added to the price plan. To create the usage rate component, navigate to **Add Component • Rates • Usage Rate**.

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Next, fill in the **Name** and **Description**, and then select a chargeable item class. The chargeable item class defines the format that SAP Convergent Charging will be expecting as an input for the charge. Pick the consumption item class **REVE** that you created earlier, as shown in Figure 3.45.

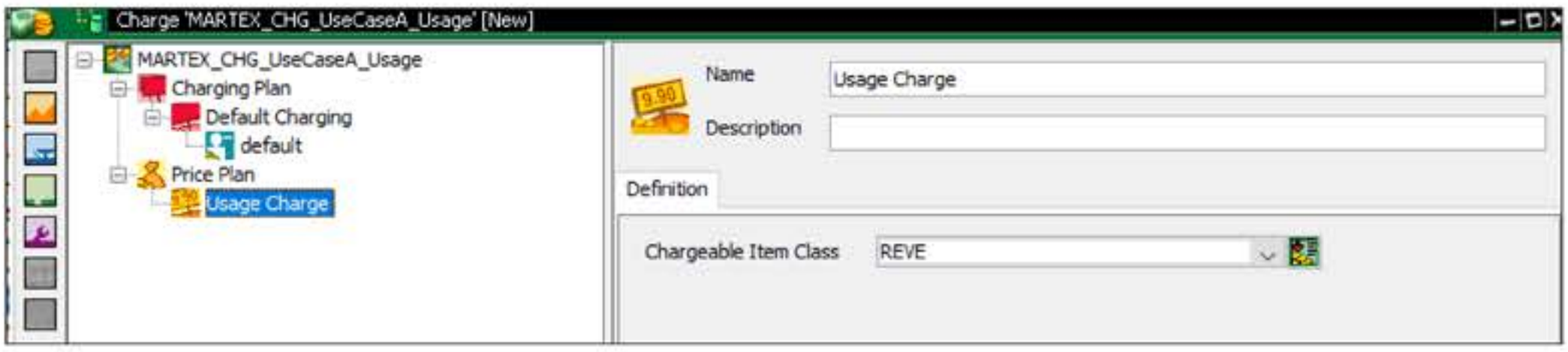

Figure 3.45 Define Usage Rate

After the usage rate component is created, an indicator is needed to let the system know that a reserved instance charge is being used. To do that, a parameter needs to be defined at the charge level: this denotes that an indicator will be passed to the charge from a provider contract. Click on the topmost level of the charge tree, select the **Parameters** tab, and then add a new parameter using the button, as shown in Figure 3.46.

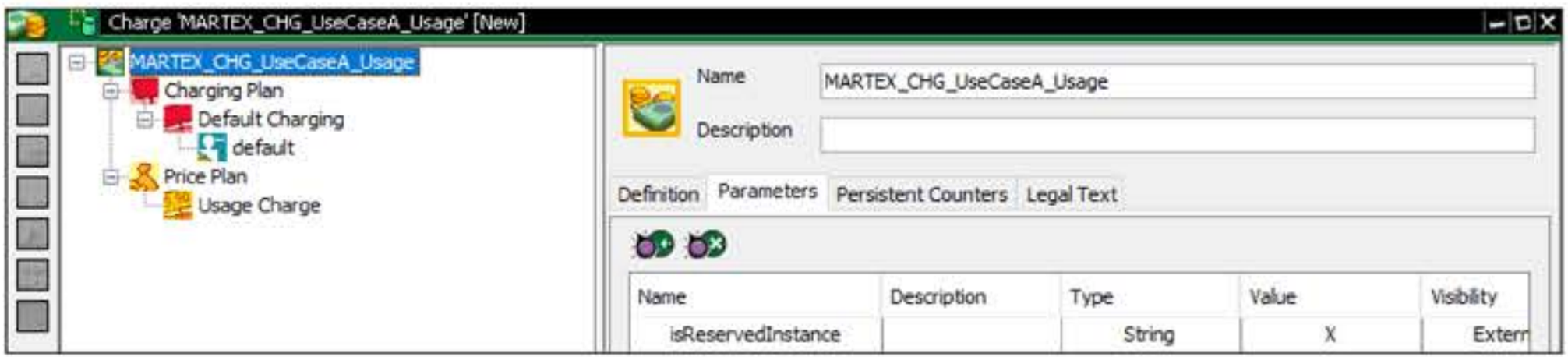

Figure 3.46 Define Parameters for Usage Charge

The next step is to read the mapping table created earlier using information from the chargeable item and reserved instance parameter. The mapping table introducer comparator will be used to read the mapping table via menu path **Add Component • Comparator • Mapping Table Introducer**. After the mapping table introducer component is added, set the **Mapping Table Class**, **Mapping Table**, and **Reference Date** to **MARTEX_MTC_MicroServiceSKU**, **MARTEX_MT_MicroServiceSKU**, and **Consumption Date**, respectively, as shown in Figure 3.47.

In the same component, set the microservice SKU id **Mapping Property** column to the **ZZ_SRVCE_SKU** field from the chargeable item class REVE, as shown in Figure 3.48. By performing these settings, the charge logic will read the mapping table based on the service SKU of the chargeable item and store the output results in the `normal_amount` and `ri_amount` variables, as shown in Figure 3.48.

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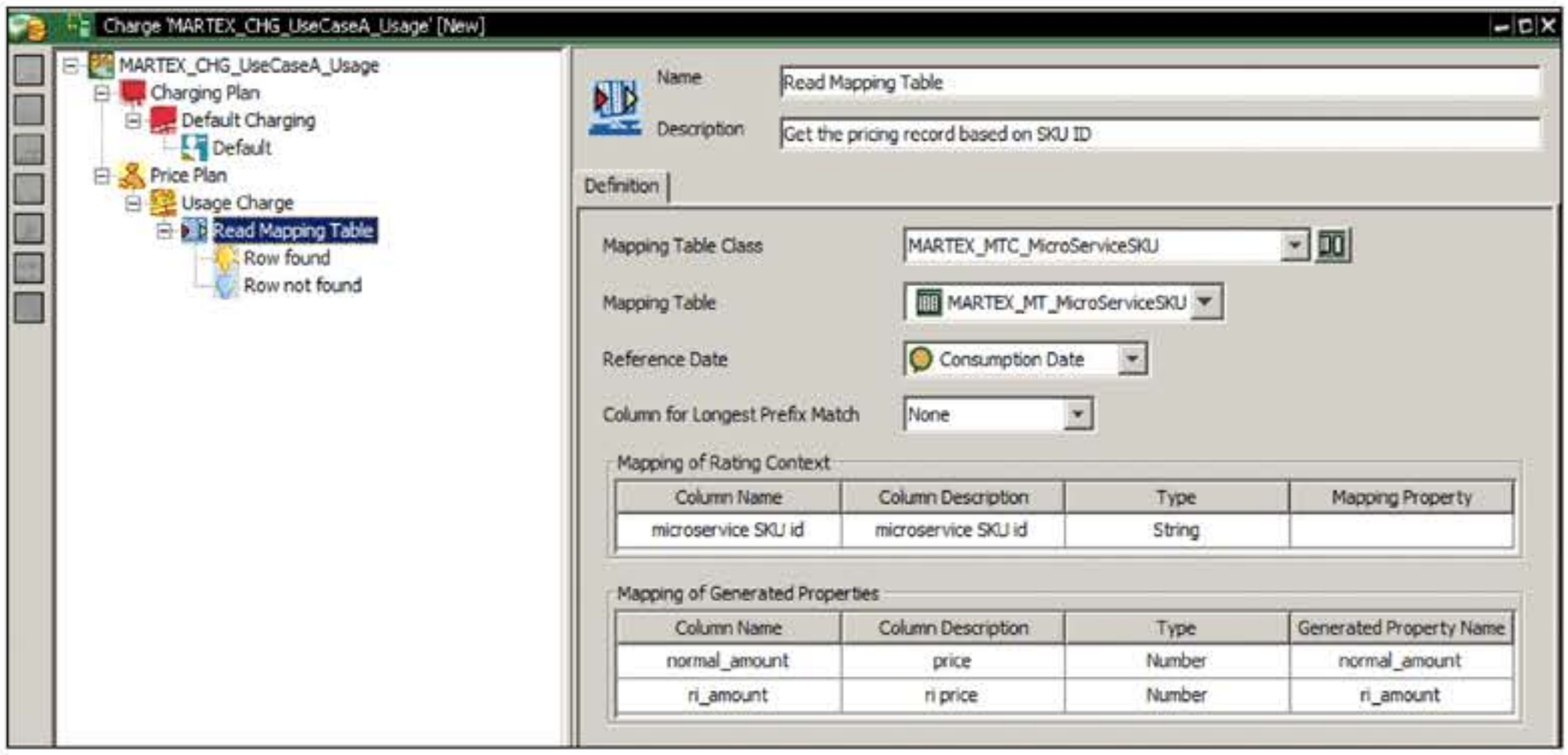

Figure 3.47 Using the Mapping Table Introducer Component

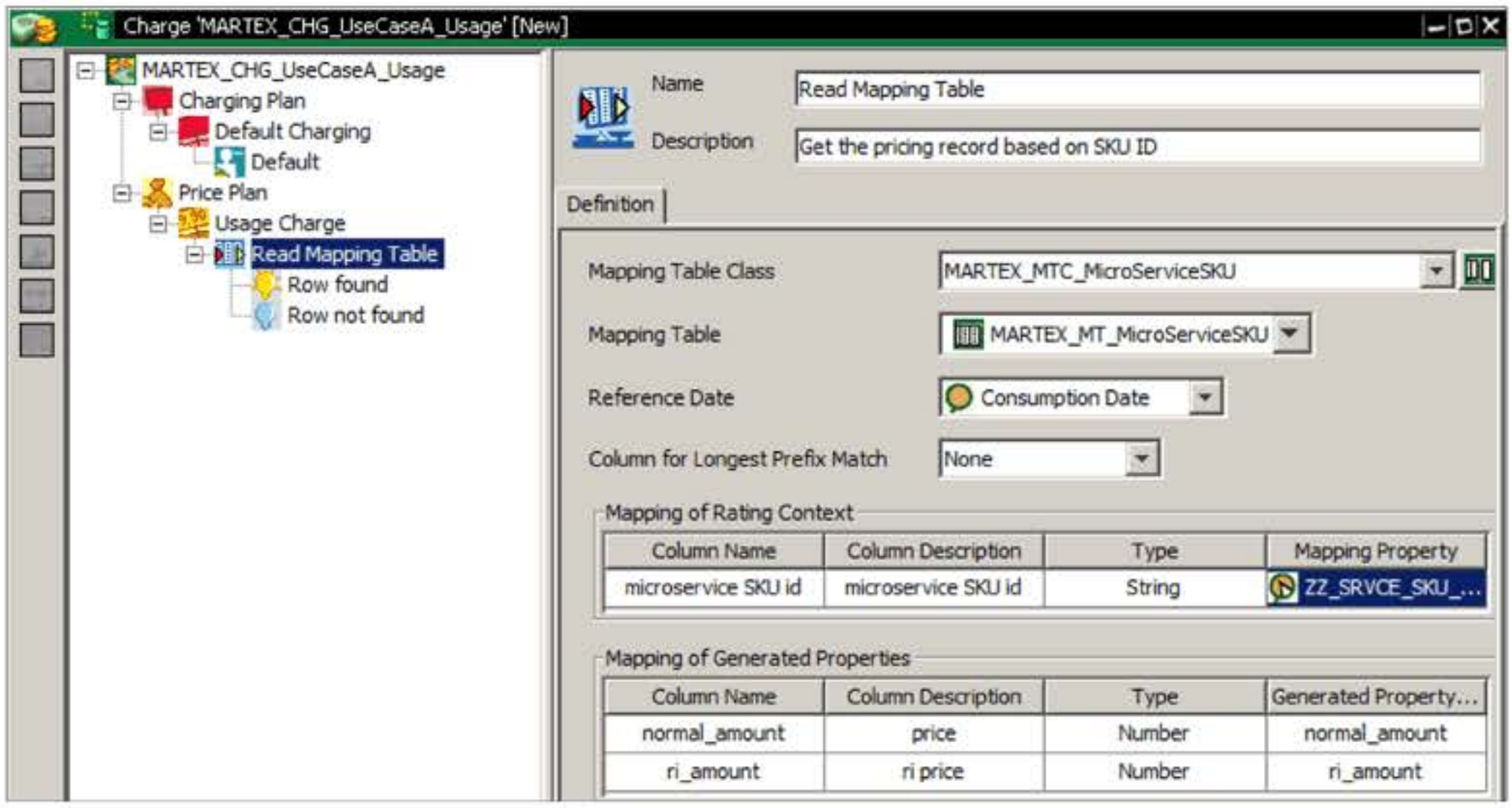

Figure 3.48 Set Mapping Property in Mapping Table Introducer

The output mapping table read will result in a fork in the logic tree, with **Row found** and **Row not found**. Under **Row found**, the first step is to check if the reserved instance is applicable for the contract by using the string comparator found at **Add Component • Comparators • Strings**. Set **Property Name** to **isReservedInstance** and **Comparison Value** to **X**, as shown in Figure 3.49.

Under the **isReservedInstance Is Equal To X** node, add arithmetic operations from navigation path **Add Component • Operators • Arithmetic Computation** to perform multiplication of **ri_amount** and **CIT_QUANTITY**, as shown in Figure 3.50.

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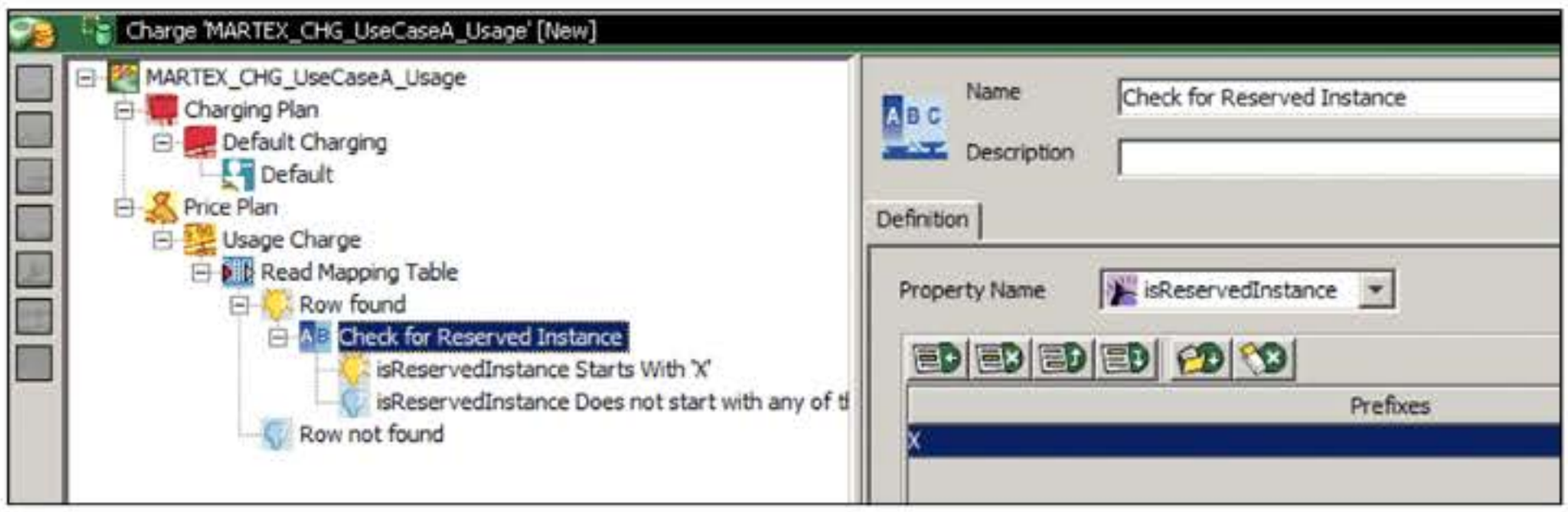

Figure 3.49 Using String Comparators

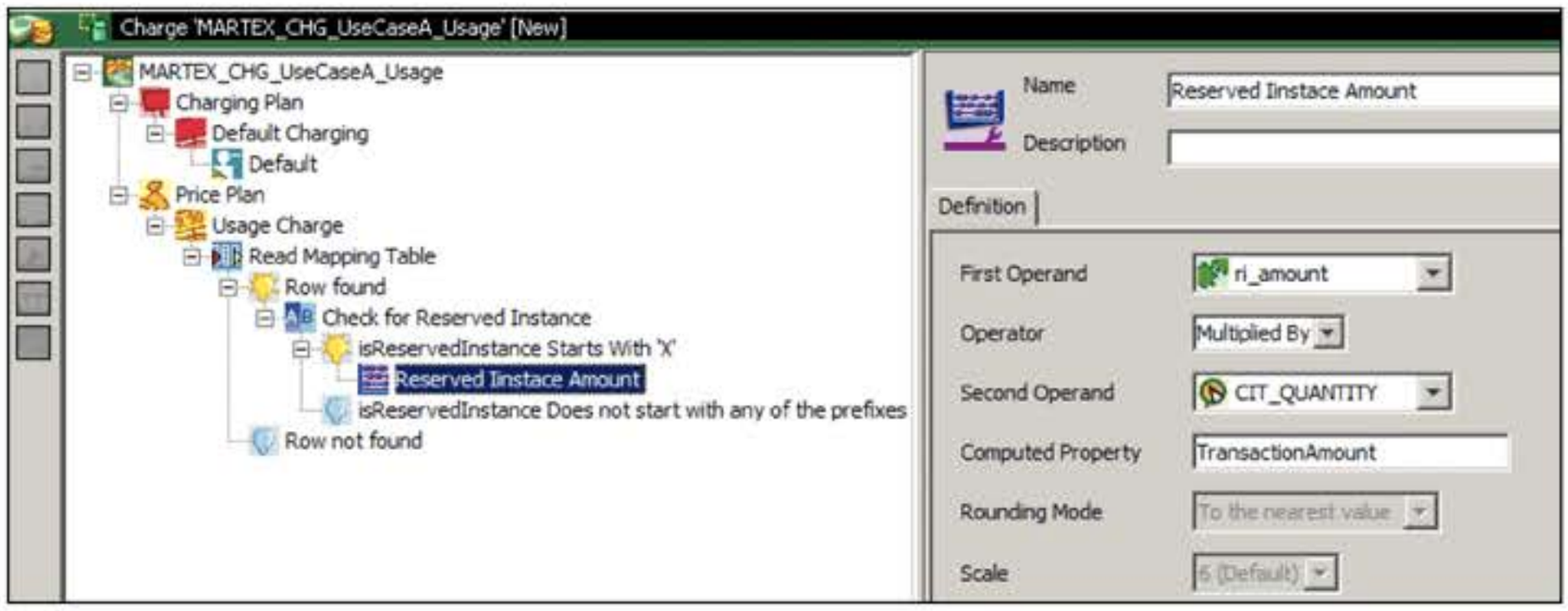

Figure 3.50 Using the Arithmetic Computation Operator to Calculate Reserved Instance

Now add an operator for the Allowance Property Introducer to get the allowance amount. To do so, you need to provide start and end dates as inputs. The output is stored in the `AmountFromAllowanceCounter` property as shown in Figure 3.51 for further use.

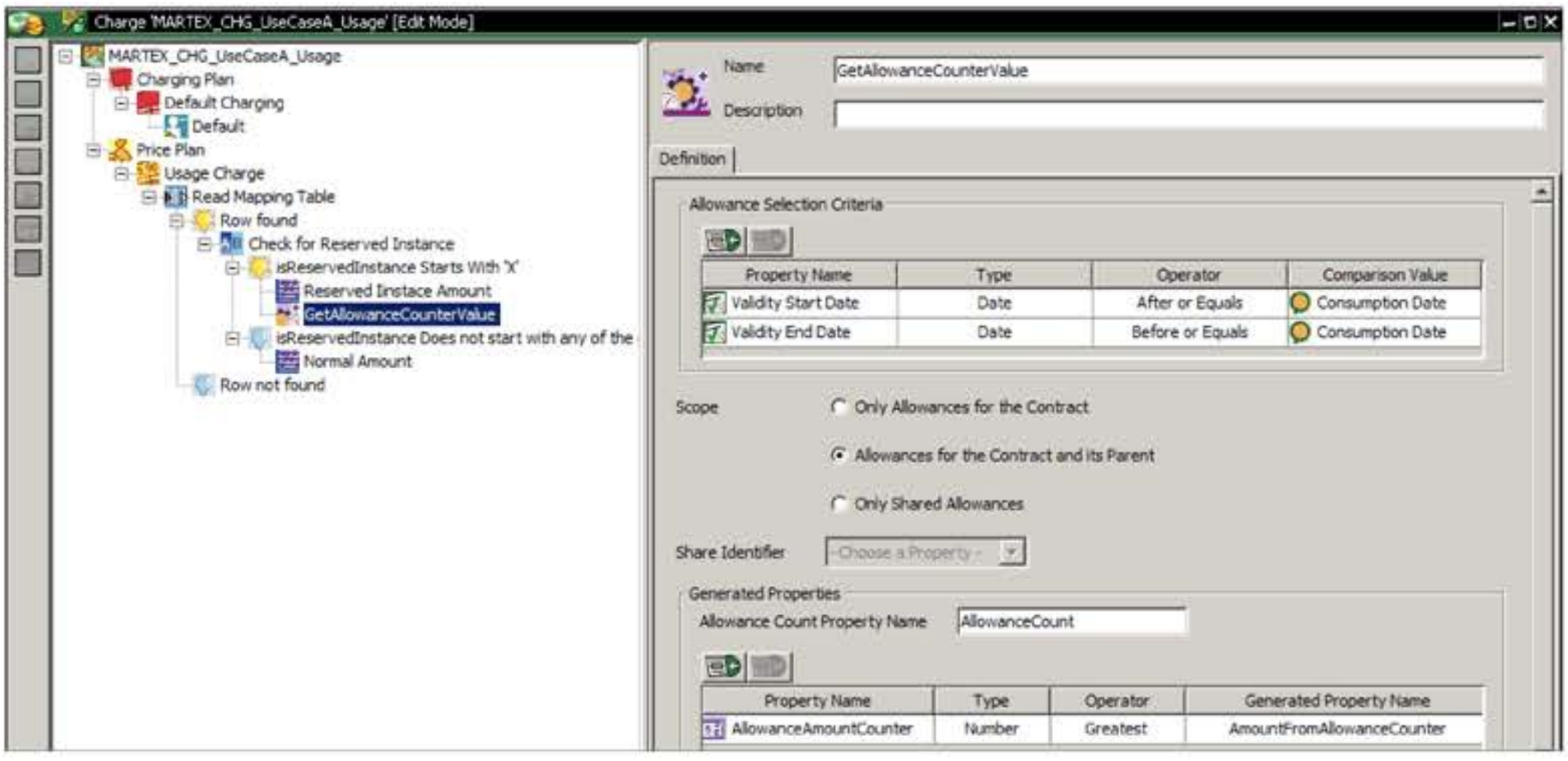

Figure 3.51 Get Allowance Amount

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As a refresher, the initial allowance object was created in the activation charge object defined earlier. The details of allowance logic to set up the initial $2,000 is covered in Section 3.7.

Next, calculate the remainder of allowance amount by using the arithmetic operation to subtract `TransactionAmount` from `AmountfromAllowanceCounter`, and store the result in `AllowanceRemainderAmount`, as shown in Figure 3.52.

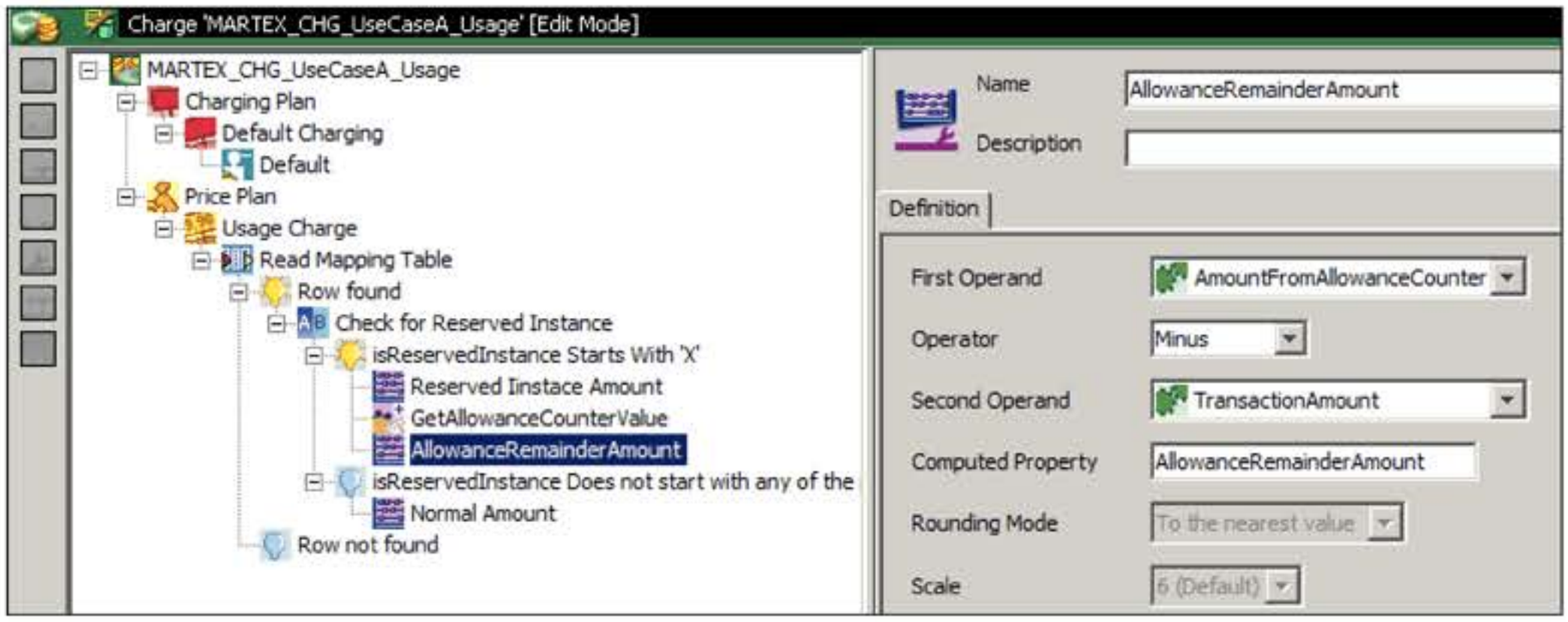

Figure 3.52 Calculate the Remainder of the Allowance Amount

Next, update the allowance amount counter. To do so, use Allowance Event Sender operator, as shown in Figure 3.53, and use the computed property `AllowanceRemainder-Amount` defined previously to update the `AllowanceAmountCounter`. The Allowance Event Sender operator executes the allowance. The details for allowance setup are discussed in Section 3.7.

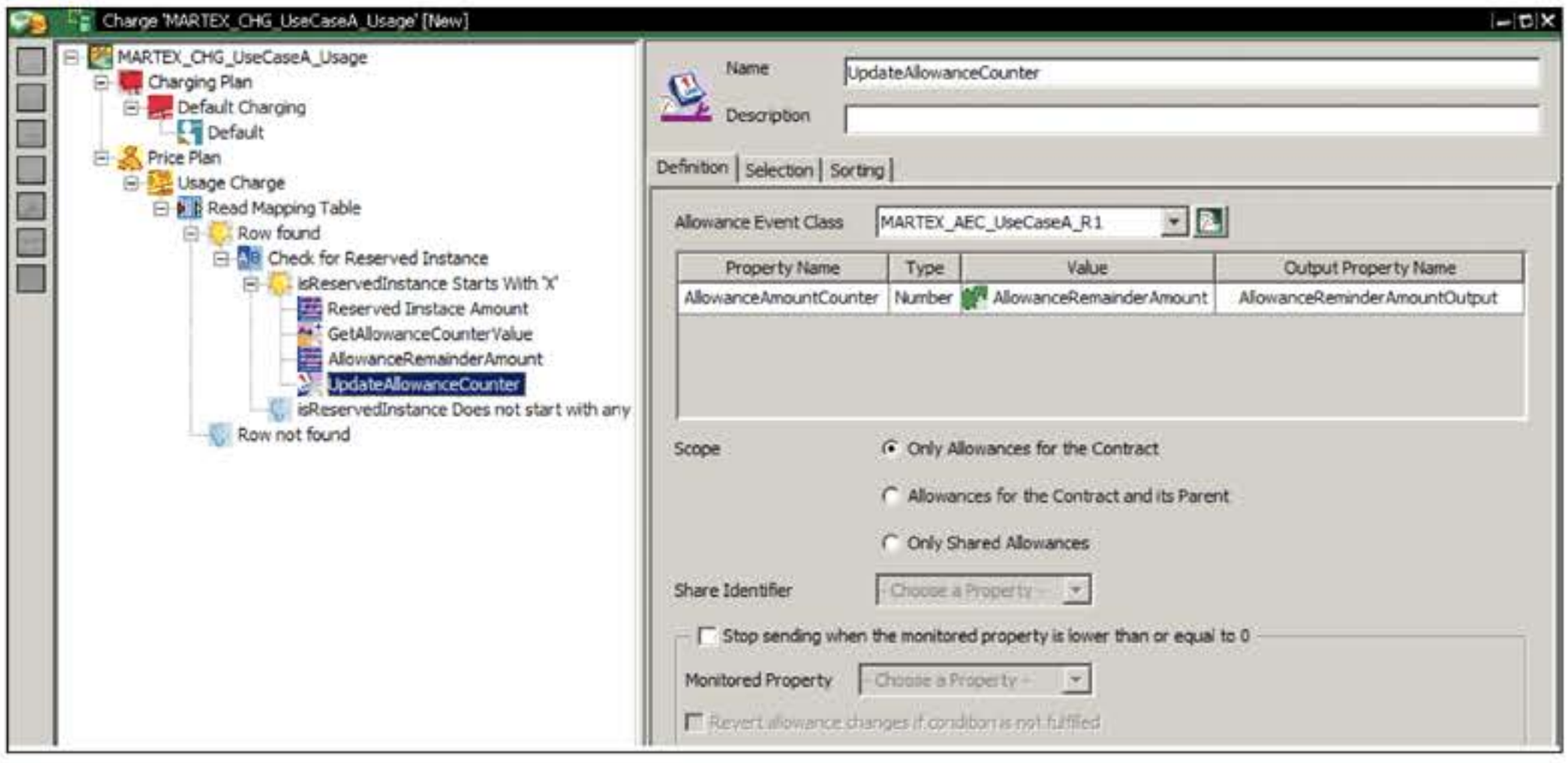

Figure 3.53 Update Allowance Counter Amount

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The next step is to generate an output for the final charge amount. However, customers who purchase the option for reserved instance will receive allowance for $2,000 in the first 60 days. For that reason, we need to compare the amounts between `AllowanceRemainderAmount` and `TransactionAmount`. If `AllowanceRemainderAmount Is Greater Than or Equal to the TransactionAmount`, then no output charge will be created, as shown in Figure 3.54. Depending on the business requirement, you can create a charge with $0. In this case, a billable item will be created with $0 in SAP Convergent Invoicing.

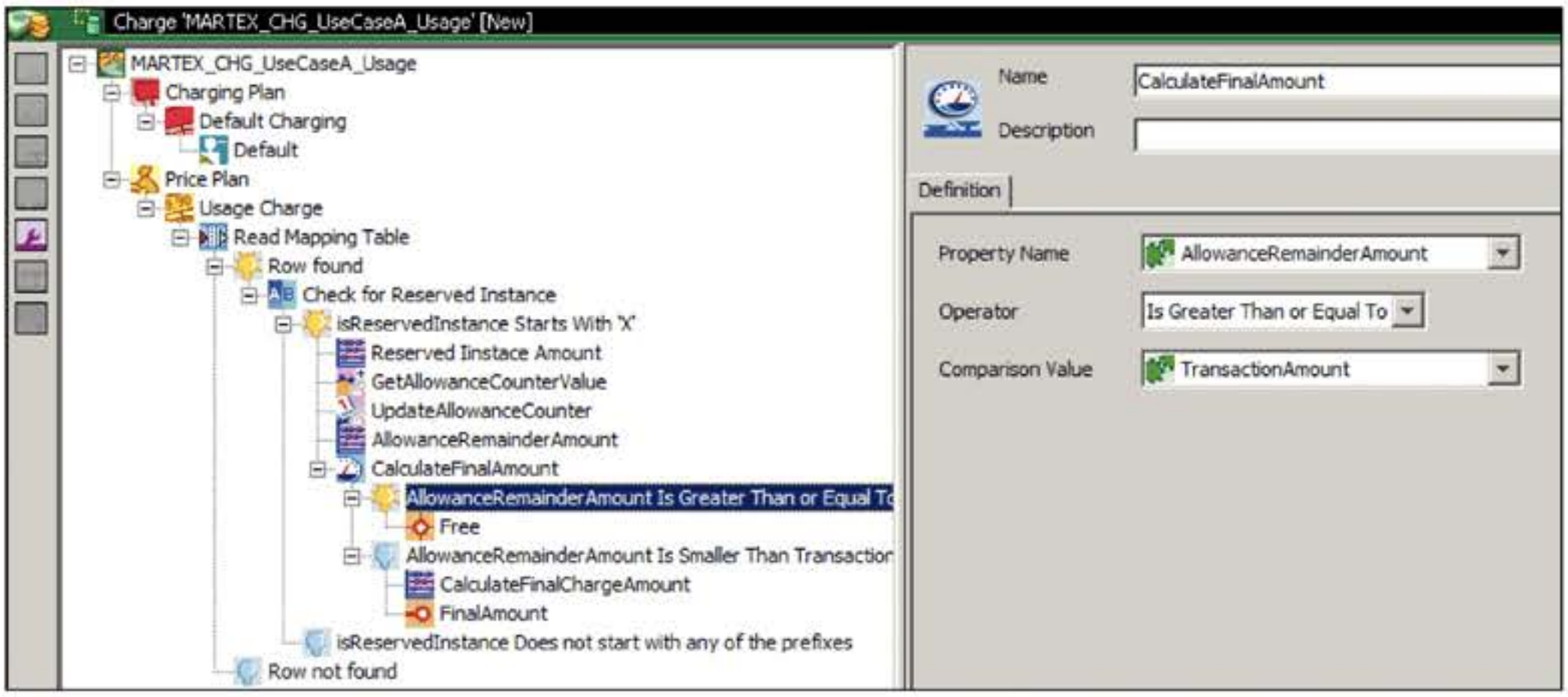

Figure 3.54 Compare Allowance Remainder Amount against Transaction Amount

Next, under the branch `AllowanceRemainderAmount Is Smaller Than TransactionAmount`, calculate the charge amount using the **Flat** function found at **Add Component • Function • Flat**, as shown in Figure 3.55.

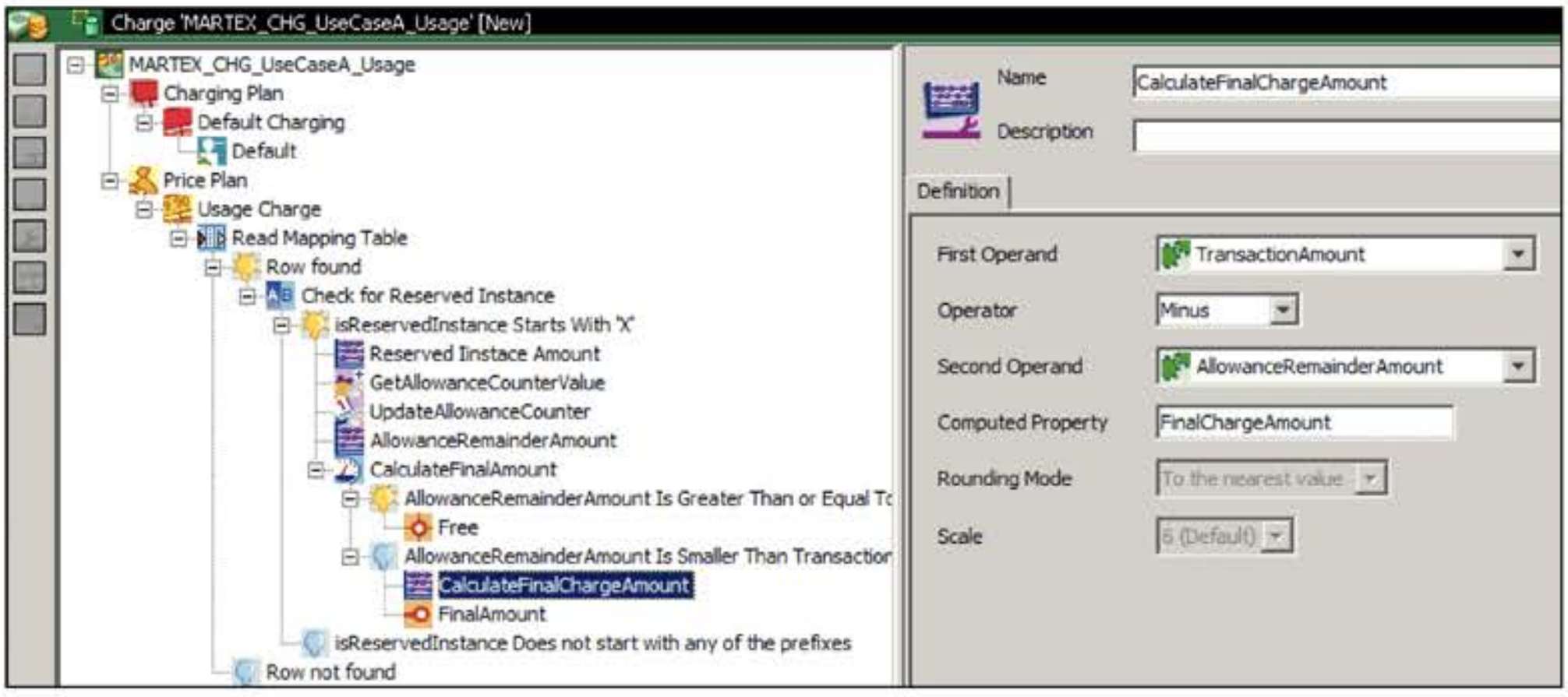

Figure 3.55 Using Arithmetic Computation Operators to Calculate Reserved Instance

Under the **isReservedInstance Does not start with any of the prefixes** node, add arithmetic operations from navigation path **Add Component • Operators • Arithmetic Computation** to

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perform multiplication of **normal_amount** and **CIT_QUANTITY**. Return a final amount as a result using a flat fee component found at **Add Component • Function • Flat**, as shown in Figure 3.56.

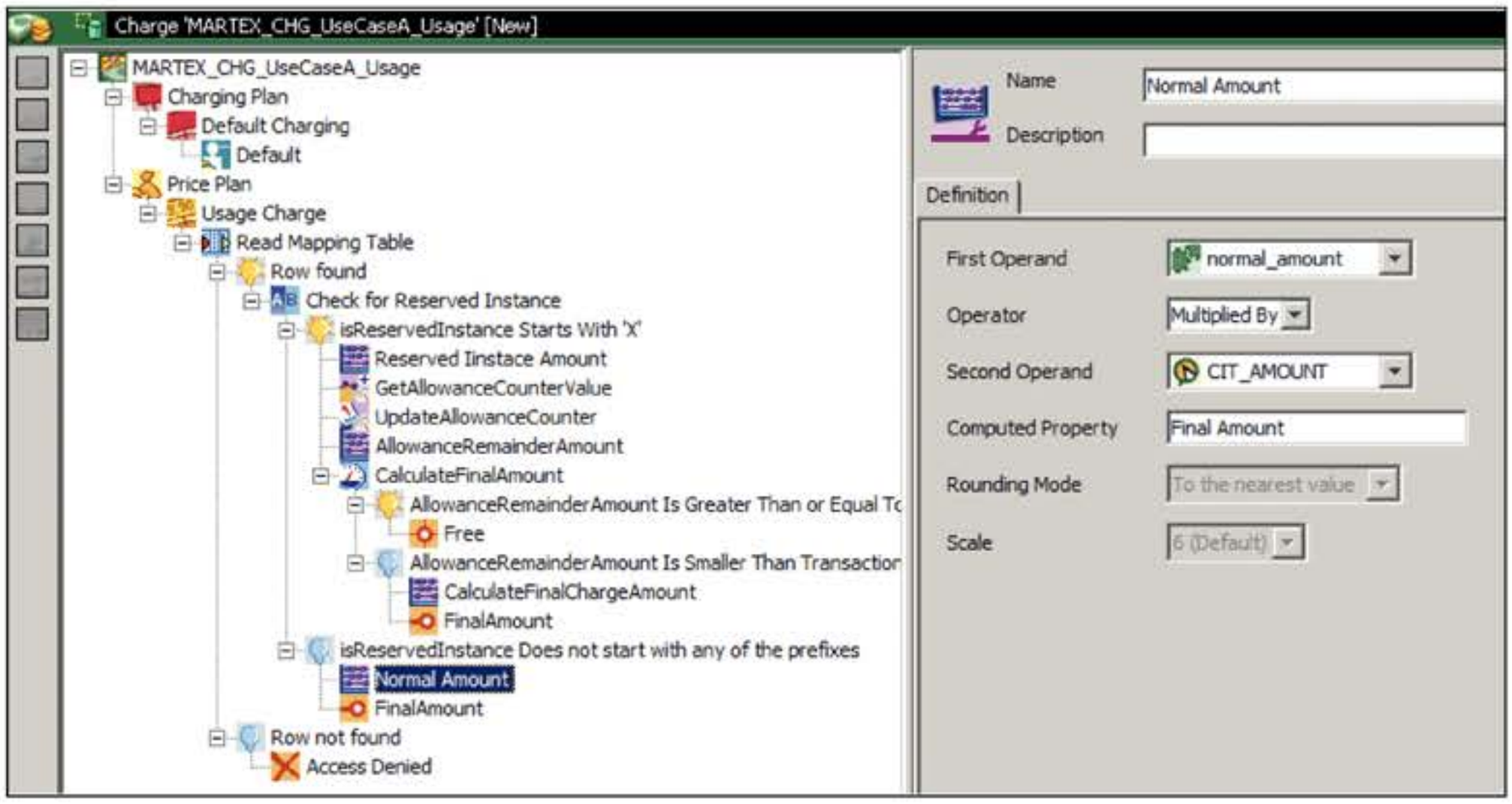

Figure 3.56 Using the Arithmetic Computation Operator to Calculate Nonreserved Instance

Next, the **Row not found** logic branch needs to be completed to handle exceptions in the event that a corresponding row isn't found. In SAP Convergent Charging, the *no access* function is used to throw exceptions. To add the no access component, navigate to **Add Component • Functions • No Access**.

The purpose of the no access function is to generate exceptions in the log files. The message in the log file, and the field values to be displayed can be configured during the creation of the object, as shown in Figure 3.57.

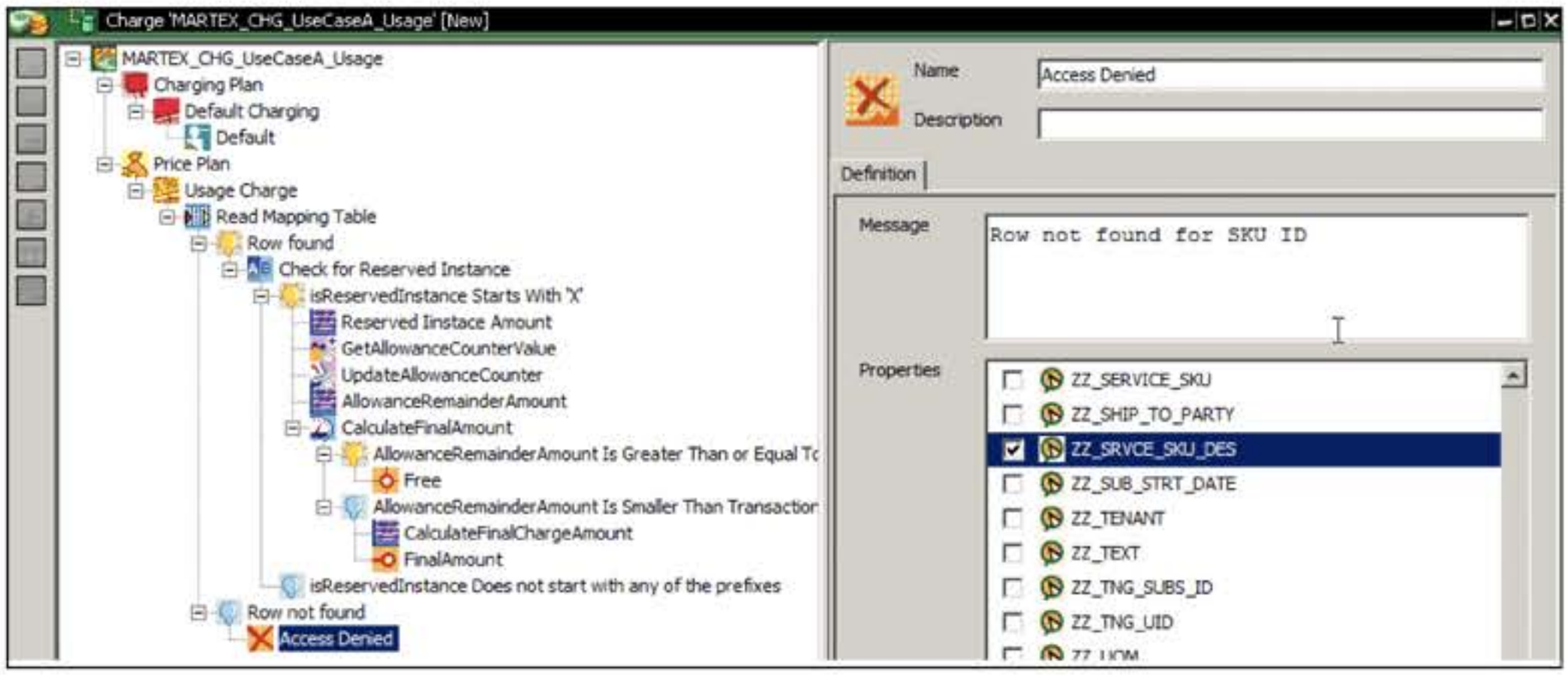

Figure 3.57 Access Denied Rate Type

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Finally, save the mapping table by following menu path **File • Save • In the Database**.

Now that the required charges are created, the next step is to combine these separate logic elements into a charge plan.

3.6.3 Creating a Charge Plan (Reserved Instance)

In the previous section, four charges were created to meet the needs of our example: activation charges, recurring charges, cancellation charges, and usage charges. Because charges are internal to SAP Convergent Charging, the next step is to group and expose the charges as an interface to other components of the SAP Billing and Revenue Innovation Management system. The component that is responsible for performing this task is the charge plan. In addition to exposing charges to external systems, the charge plan also allows for counter-sharing across all charges.

In this section, two charge plans will be created from the four charges, one for a reserved instance, which provides a reduced consumption rate with additional recurring charges, and another for regular usage, which charges for consumption at the normal rate. Let's start by creating the reserved instance charge plan:

1. Navigate to **File • New • Charge Plan**.
2. Select the catalog, and provide a **Name** and **Description** for the charge plan, as shown in Figure 3.58.

Figure 3.58 Creating Charge Plan

3. To add charges to the charge plan, right-click the **Charge Plan** object in the tree structure on the left, and select **Add Item** from the context menu, as shown in Figure 3.59. This enables the association between charges and the charge plan.
4. Add all the previously defined charges to the charge plan by repeating steps 1–3 four times. When you're finished, four charges will be assigned to the charge plan, as shown in Figure 3.60.

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Figure 3.59 Adding Charges to the Charge Plan

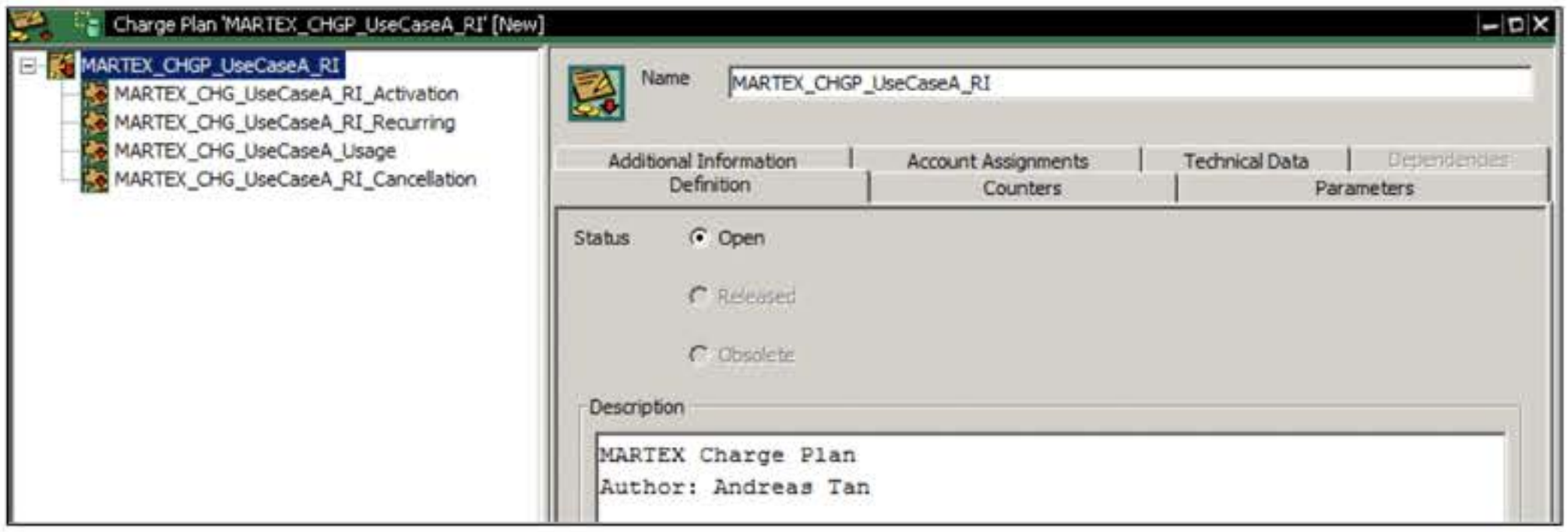

Figure 3.60 Charge Plan with Four Charges Added

Once the charges are assigned to the charge plan, the parameters of the charge plan need to be defined.

Configuring Charge Plan Parameters

The purpose of the parameters is to receive input values from order provisioning or other SAP CRM systems. In an integrated system, the order provisioning system is SAP Billing and Revenue Innovation Management, subscription order management.

For our example, the only parameter that is required is a reserved instance flag. To add the parameter, click the **Parameters** tab in the charge plan.

In an integrated SAP Billing and Revenue Innovation Management scenario, an **SAP_BILL_CYCLE** parameter is added by default, as shown in Figure 3.61. Because we don't need it for our example, we'll ignore this field.

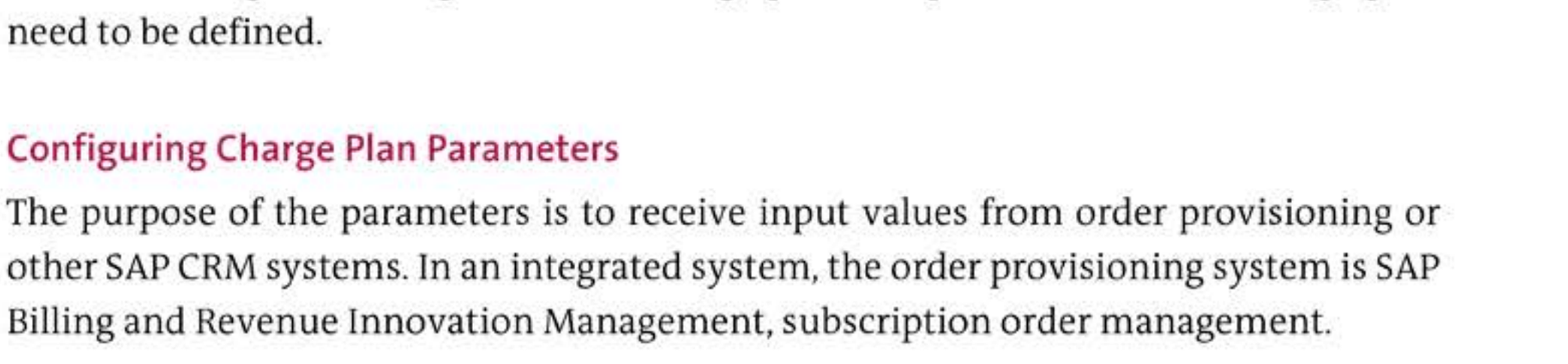

Figure 3.61 Default SAP_BILL_CYCLE Parameter

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To configure charge plan parameters, follow these steps:

1. Add a new parameter by clicking the button, as shown in Figure 3.62.

> **Note**
>
> A parameter with the internal visibility level is added because the charge plan is specifically designed for a reserved instance, so external input isn't required.

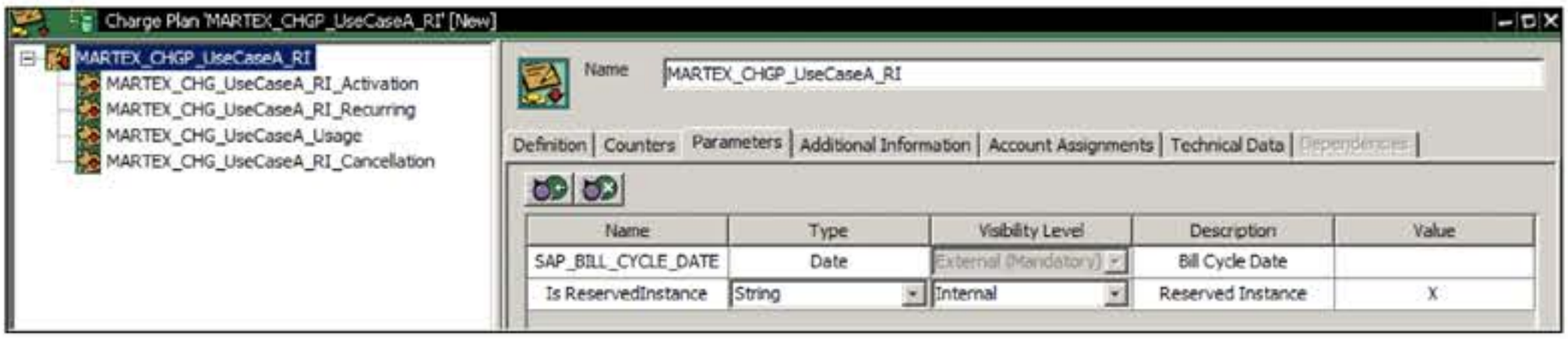

Figure 3.62 Adding Parameters to Charge Plan

2. Map the parameter defined to the specific charge that requires its value; in this case, that is the usage charge. To do so, select the charge in the tree navigation on the left, and then select the **Parameters** tab, as shown in Figure 3.63.

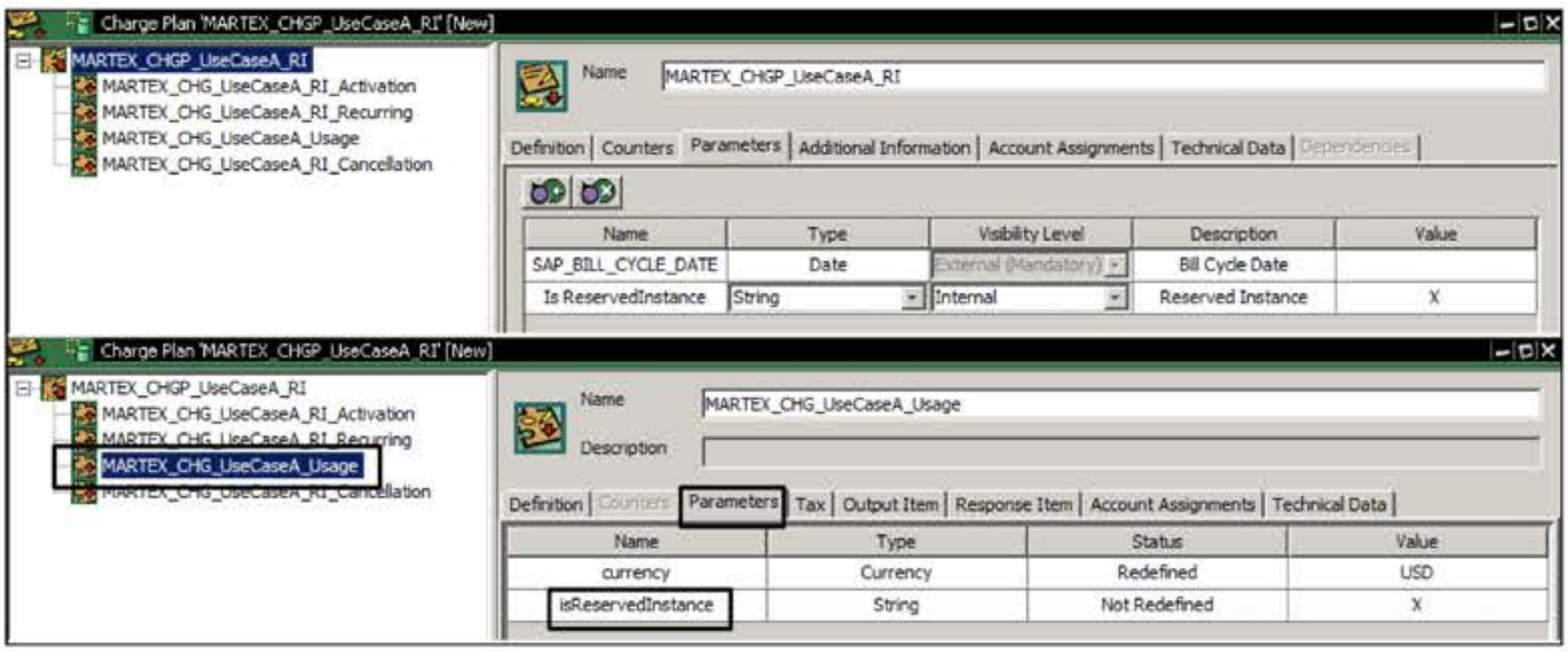

Figure 3.63 Parameters of Charges and Charge Plan

3. Change the **isReservedInstance** parameter from **Not Redefined** to **Linked** and the **Value** to **Reserved Instance**. The purpose of this action is to link the parameter from the charge plan to the charge level (see Figure 3.64).

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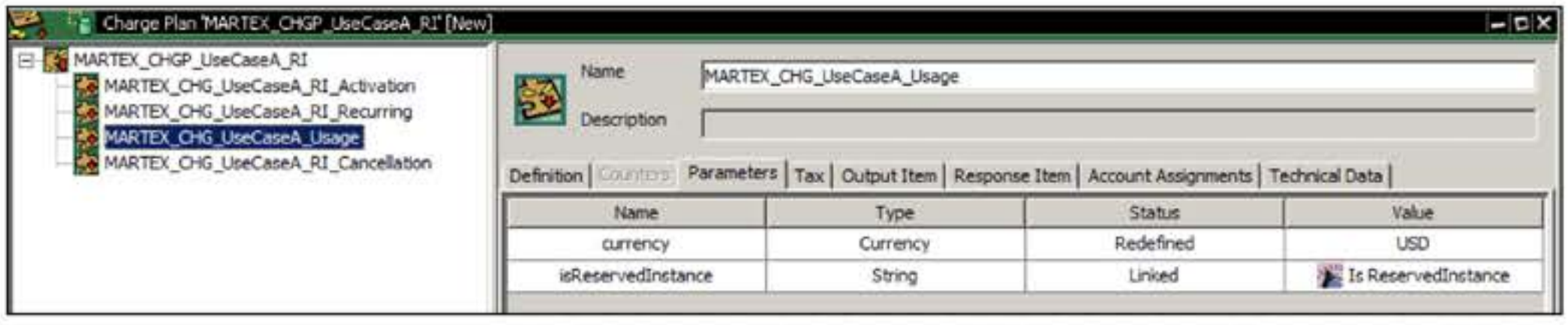

Figure 3.64 Linking Charge Parameter to Charge Plan

In summary, we've grouped all the charges that were defined separately in the previous section within a single charge plan that will be exposed to the provisioning system.

Configuring Technical Data

Whenever a charge with a usage rate is defined, SAP Convergent Charging requires technical data to be defined. The purpose of the technical data is to uniquely identify which provider contract to process CDRs or usage data. Technical data comprises a Service Identifier and User Technical Identifier. Those two identifiers are part of field values in the CDRs. For example, in the telecom industry, service identifiers can be voice, data, or text message, and a user technical identifier is normally a phone number.

To configure technical data, follow these steps:

1. To configure the technical data, first navigate to the charge plan level via the tree in the left panel, and select the **Technical Data** tab, as shown in Figure 3.65

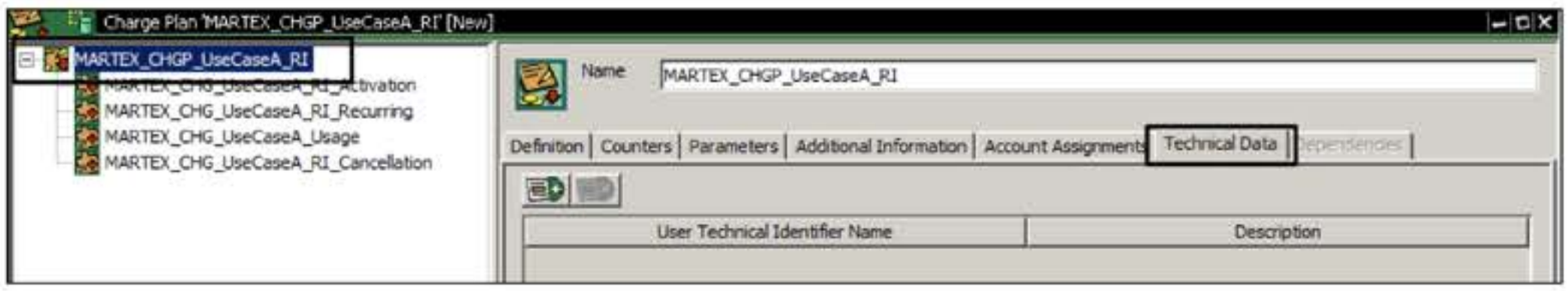

Figure 3.65 Configuring Technical Data

2. Add a new user technical identifier called "subscriber id" by clicking the button, as shown in Figure 3.66.

Figure 3.66 Adding New User Technical Identifiers

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Once the user technical identifier has been defined, the next step is to define a service identifier. The purpose of the service identifier is to determine the type of service that is consumed. For example, in the telecommunications industry, service identifiers could include data, voice, and message, which are calculated differently. However, in our example, a generic service identifier will be used.

To set up a service identifier, follow these steps:

1. Select the usage charge using the tree in the left panel, and then select the **Technical Data** tab (see Figure 3.67).

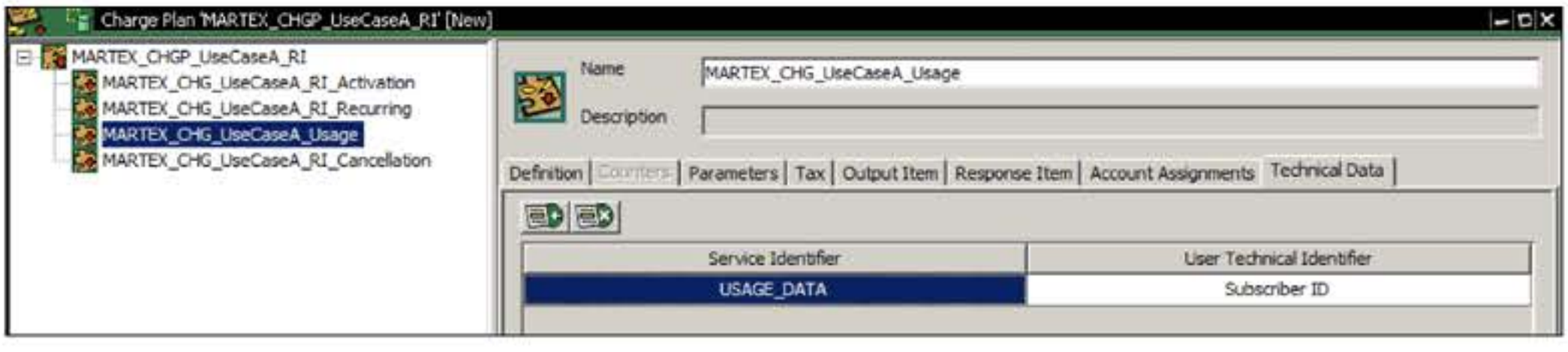

Figure 3.67 Technical Data

2. Add a new service identifier called "USAGE_DATA" by clicking the button. For the corresponding user technical identifier, SAP Convergent Charging will provide a dropdown with the user technical identifier defined in the charge plan.

Now that you've defined the technical data, the next step is to configure the account assignment.

Configuring Account Assignment

To determine which external account to use for a contract, account assignment needs to be set up during charge plan creation. However, our example requires only one default postpaid account.

To configure account assignment, follow these steps:

1. Select the change plan using the tree in the left panel. Select the **Account Assignments** tab (as see Figure 3.68).

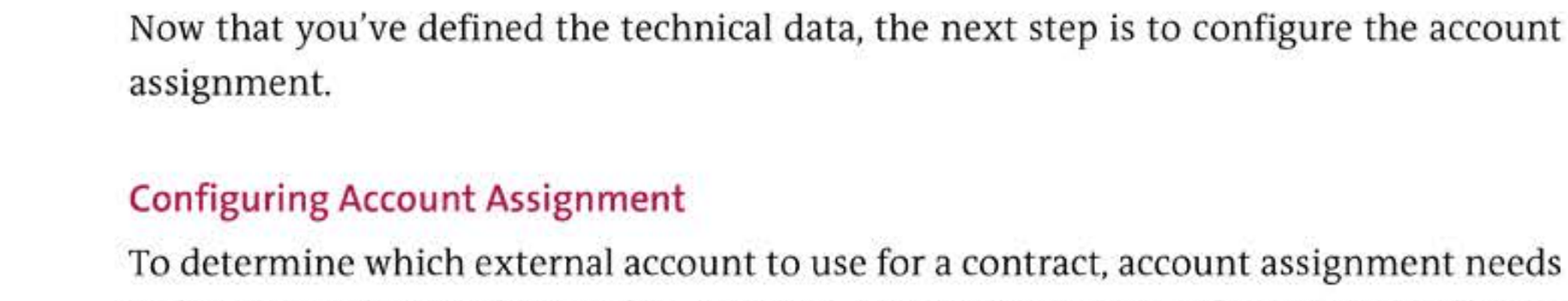

Figure 3.68 Account Assignments Tab

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2. Insert a new entry by clicking the button. Use "default" for the name and description, and set the type to **Postpaid**.
3. Account assignment needs to be performed at each individual charge level. To do that, select each charge at the charge level, and navigate to each one's **Account Assignment** tab.
4. Select the **default** value from the **Linked Value** column, as shown in Figure 3.69.

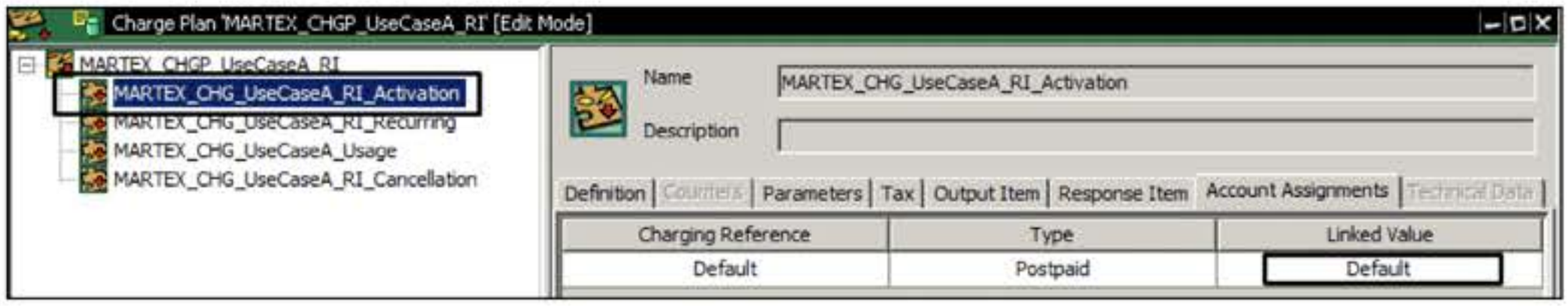

Figure 3.69 Linking Account Assignments between Charge and Charge Plan

After repeating the steps for each charge, the account assignment is considered completed. The next step is to configure the charge plan counter.

Configuring the Charge Plan Counter

When creating charges, a month counter was defined at the charge level for the recurring and cancellation charges. Assignment of the charges to the charge plan doesn't automatically link the counter. This has to be done explicitly at the charge plan level.

The first step is to navigate to the charge plan level by selecting the charge plan from the tree in the left panel, and then select the **Counters** tab, as shown in Figure 3.70.

Figure 3.70 Adding Counters in a Charge Plan

To configure the charge plan counter, follow these steps:

1. Create a new counter by clicking the button.
2. Enter "month counter" for a **Name** and **Description**, and set the **Visibility Level** to **Internal**, as shown in Figure 3.71.

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Figure 3.71 Adding a Counter in a Charge Plan

The next step is to link the counters at the charge level to the charge plan level. To do so, select the cancellation and recurring charges from the tree in the left panel, then select the **Counter** tab, and perform linking as shown in Figure 3.72 and Figure 3.73.

Figure 3.72 Linking Charge Counter and Charge Plan Counter for Recurring Charge

Figure 3.73 Linking Charge Counter and Charge Plan Counter for Cancellation Charge

This step allows for month counter values to be shared across both recurring and cancellation charges.

Disabling Tax on a Charge Plan

SAP Convergent Charging supports tax calculations. However, in an SAP Billing and Revenue Innovation Management end-to-end integration scenario, tax calculations usually are deferred to the time of invoicing and performed by the SAP Convergent Invoicing module. To disable tax on a charge plan, follow these steps:

1. To disable tax calculations in SAP Convergent Charging, navigate to each charge, and select the **Tax** tab.
2. Set the **Tax System** value to **Disabled** for each charge, as shown in Figure 3.74.

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Now that we've disabled tax configuration on the charge level of the charge plan, let's continue with the charge plan creation by configuring output items.

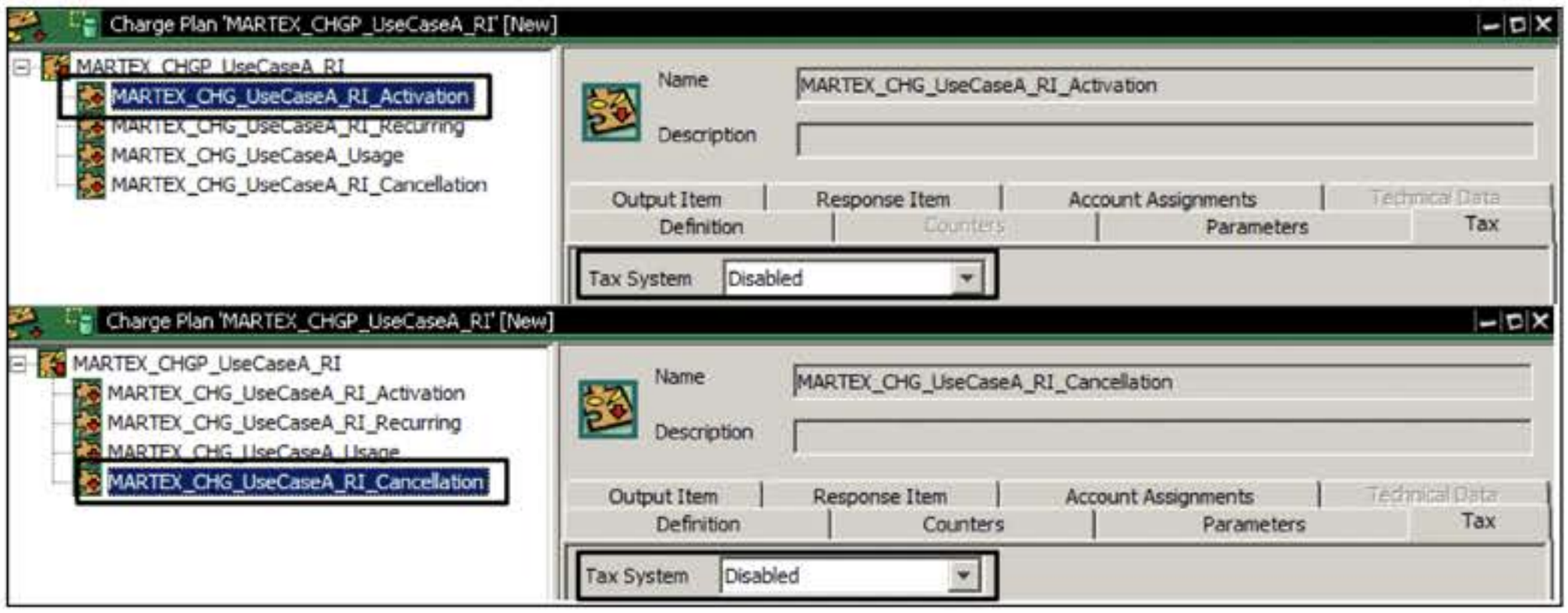

Figure 3.74 Disable Tax Configuration in the Charge Plan

Configuring Output Items

The charge plan is also responsible for how charging results are conveyed to external systems. In Section 3.6.2, billable item mapping has already been completed, resulting in the creation of charged item classes HREV_3 and HCOS.

The next step is to configure the charges for output item HREV_3 for each charge. This action can be performed by clicking the **Output Item** tab, as shown in Figure 3.75.

When the HREV_3 charged item class is picked, a list of fields belonging to the item class will be populated. In an SAP Billing and Revenue Innovation Management integrated scenario, the fields BITTYPE and SUBPROCESS are mandatory. Fields PT_USERID_REF and PT_USERID_REFTY are applicable if partner settlement scenarios are populated. The values that need to be populated in the BITTYPE and SUBPROCESS fields will be communicated by the SAP Convergent Invoicing team because it will drive billing behavior.

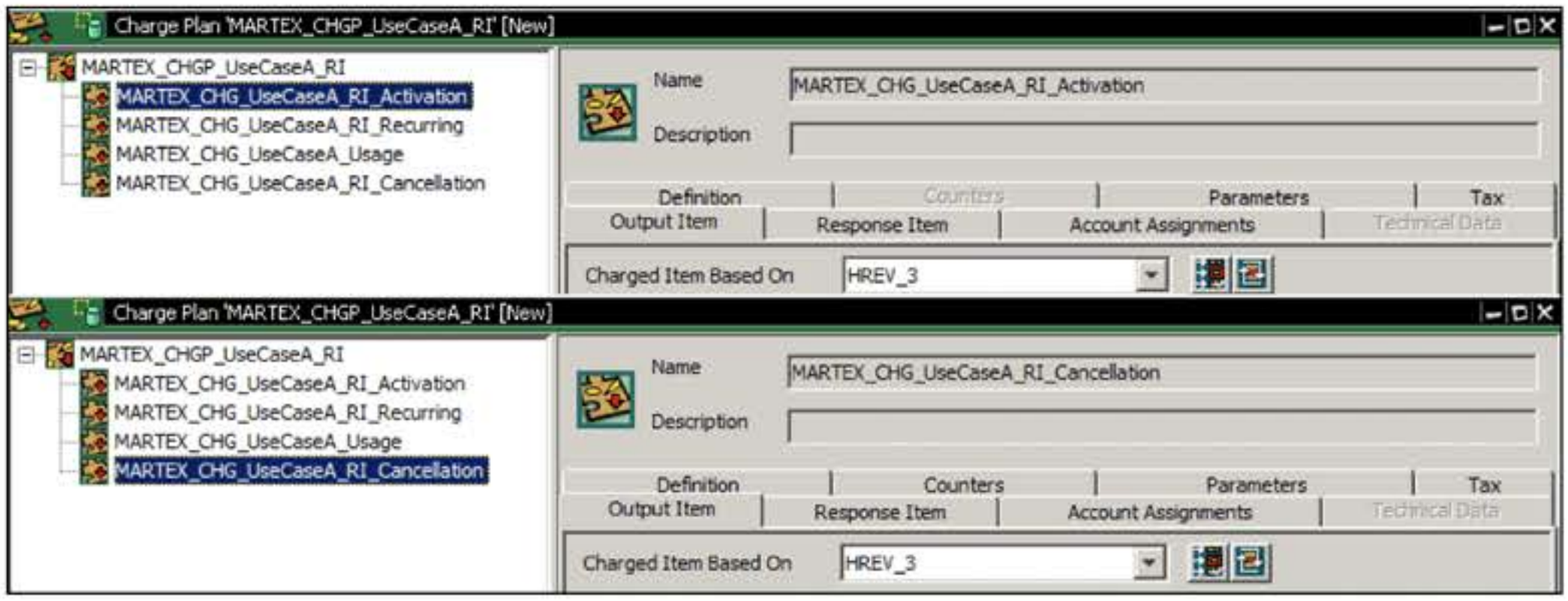

Figure 3.75 Configuring a Charged Item Class on Charges

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For our example scenario, the values defined in the SAP Convergent Invoicing chapter need to be populated for each charge, as listed in Table 3.2.

MARTEX_CHG_UseCaseA_RI_Recurring	
BITTYPE	ZBT2
SUBPROCESS	ZHR1
BIT_QTY_UNIT	EA
BIT_QTY_QUANTITY	1
MARTEX_CHG_UseCaseA_RI_Cancellation	
BITTYPE	ZBT6
SUBPROCESS	ZHR1
BIT_QTY_UNIT	EA
BIT_QTY_QUANTITY	1
MARTEX_CHG_UseCaseA_RI_Usage	
BITTYPE	ZBT3
SUBPROCESS	ZHR1
PS_CAT	RVHV
BIT_QTY_UNIT	EA
BIT_QTY_QUANTITY	CIT_QUANTITY
PT_USERID_REF	I
PT_USERID_REFTY	PARTNER TECH_RES

Table 3.2 SAP Convergent Invoicing Martex Values

After the output values for SAP Convergent Invoicing have been defined for each charge, the next step is to release the charge plan.

Releasing the Charge Plan

The release status of a charge plan enables it to be used by external provisioning systems such as SAP Billing and Revenue Innovation Management, subscription order management. By default, when a charge plan is created, it will be in **Open** status, as shown in Figure 3.76.

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Figure 3.76 Charge Plan Status

To enable the **Released** status radio button, first save the charge plan using menu path **File • Save • In the Database** (see Figure 3.77).

SAP Convergent Charging will display a prompt for catalog selection and subsequently save the charge plan and go into display mode. Next, toggle to edit mode, select the **Released** radio button by using menu path **File • Edit Mode**, and then save again.

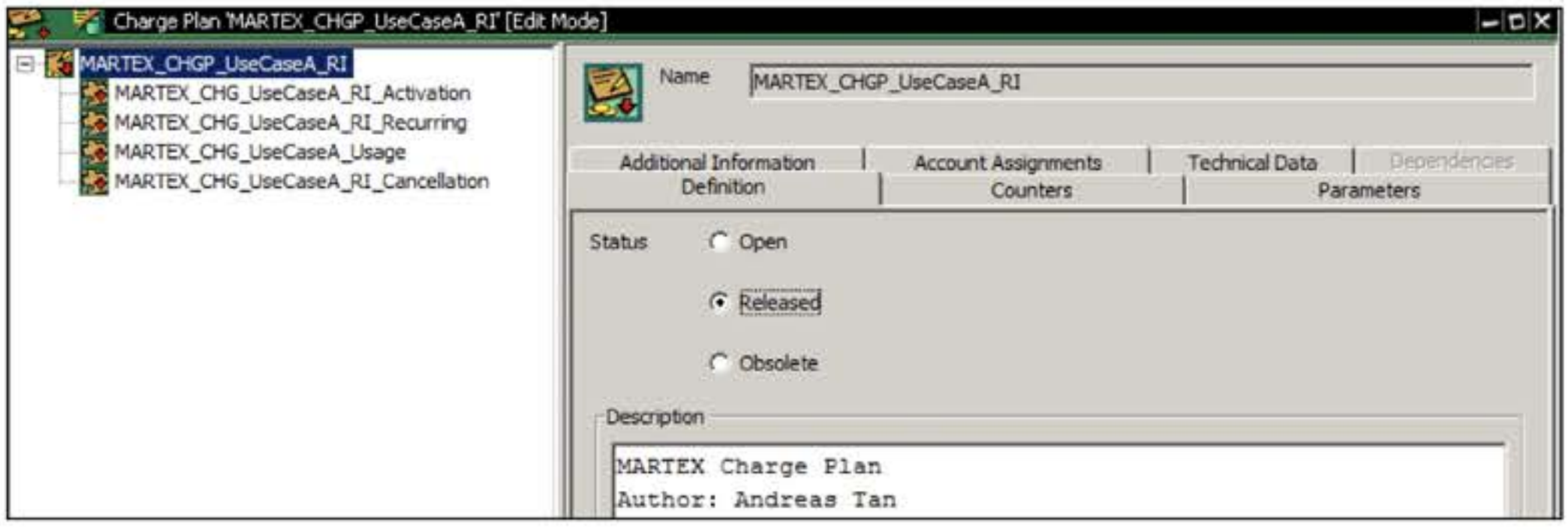

Figure 3.77 Releasing the Charge Plan

The charge plan for a reserved instanced is now created, so the next step is to create the charge plan for a nonreserved instance for those customers who did not pick the reserved instance option.

3.6.4 Creating Charge Plan (Nonreserved Instance)

Another business requirement for Martex Corp. is to support customers who don't subscribe to a reserved instance. These nonreserved-instance customers will be charged higher rates whenever they consume service A or service B. In return, they don't have to pay a monthly recurring fee of $200. As a result of that, activation and cancellation fees aren't applicable either.

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The customer choice of a reserved instance or a nonreserved instance is captured in order management systems such as SAP Billing and Revenue Innovation Management, subscription order management, in an integrated scenario. Based on the customer decision, SAP Billing and Revenue Innovation Management, subscription order management, will inform SAP Convergent Charging how to create the corresponding provider contract.

Because the nonreserved instances have the same usage charging behavior as their reserved instance counterparts, the same charge will be reused. When designing charges for SAP Convergent Charging, it's good practice to encapsulate charging logic for reuse. To create a charge plan, follow these steps:

1. First make a copy of the existing charge plan. To access the catalog, follow menu path **Window • View Catalog**. In the catalog, locate the charge plan, right-click it, and select **Open as Copy**.
2. Remove the activation charge, recurring charge, and cancellation charge from the copied charge plan by using the **Remove** option in the context menu of each respective charge, as shown in Figure 3.78. You want to remove these options because they aren't applicable to a nonreserved instance scenario.

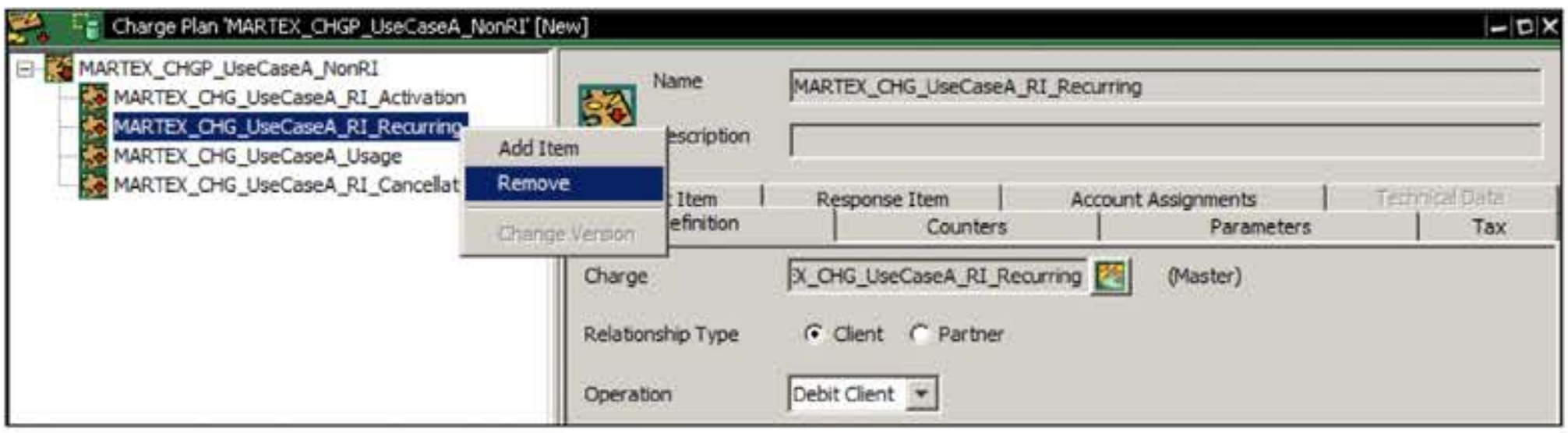

Figure 3.78 Removing Activation, Recurring, and Cancellation Charge Plan

Counters also should be deleted as they are irrelevant in a nonreserved instance charge plan. Select the month counter, and click the button (see Figure 3.79).

Figure 3.79 Removing All Counters for Nonreserved Instance Charge Plan

3. Finally, change the name and description to "Nonreserved Instance" in the **Name** and **Description** fields, and save the charge plan via **File • Save • In the Database**, as

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shown in Figure 3.80. To release the charge plan, select **File • Edit Mode** to enter edit mode, and then select the **Released** radio button.

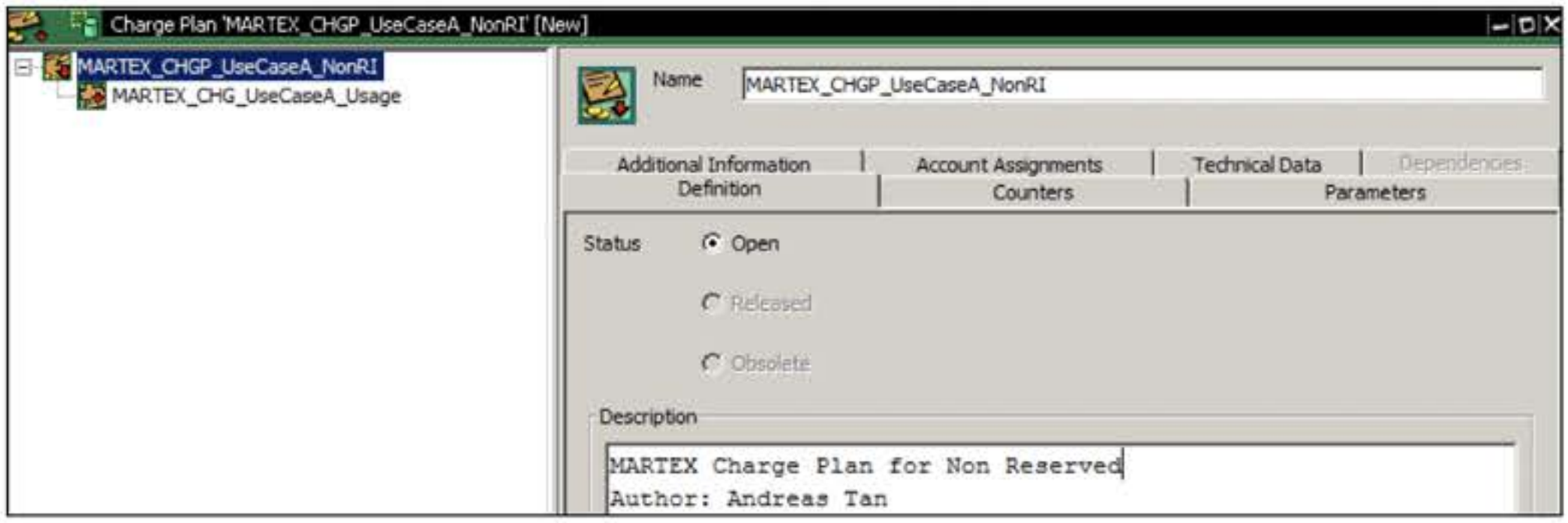

Figure 3.80 Nonreserved Instance Charge Plan Open Status

To recap, two charge plans have been created, for customers subscribing to a nonreserved instance and for customers subscribing to a reserved instance. Because Martex Corp. is a software distributor, it needs to pay vendor X a certain percentage of what it receives from its customers. In SAP Billing and Revenue Innovation Management, this can be accomplished using the partner settlement functionality, which is covered in detail in Chapter 4, Section 4.3.7.

The next section of this chapter explains the setup that is required in SAP Convergent Charging to support partner settlement capabilities.

Creating Partner Charges

As we'll discuss in Chapter 4, with the configuration of partner settlement, the creation of an accounts receivable billable item in the system leads to the creation of an accounts payable consumption item.

During the rating process of the accounts payable consumption item, the charge plan associated with the partner agreement created in SAP Billing and Revenue Innovation Management, subscription order management, will be triggered. Now, let's create a partner-related charge and charge plan:

1. Follow menu path **File • New • Charge** and define the charging plan for the `MARTEX_CHG_UseCaseA_Usage_Partner` charge. For our example, we don't need to have different charging references, so default charging will be used, as shown in Figure 3.81.
2. Figure 3.82 shows the three steps required to create a default charging reference. In the **Charging Plan** node of the tree, add a new charging reference with the name "default" with the **Type** set as **Postpaid** by using the button.

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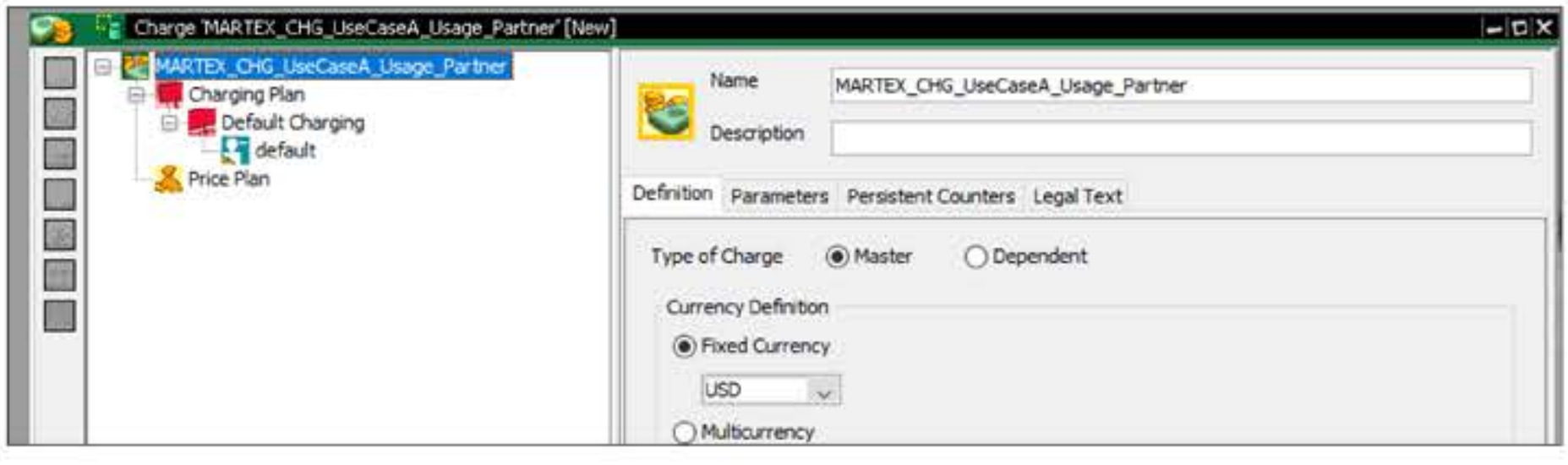

Figure 3.81 Creating Partner Charges

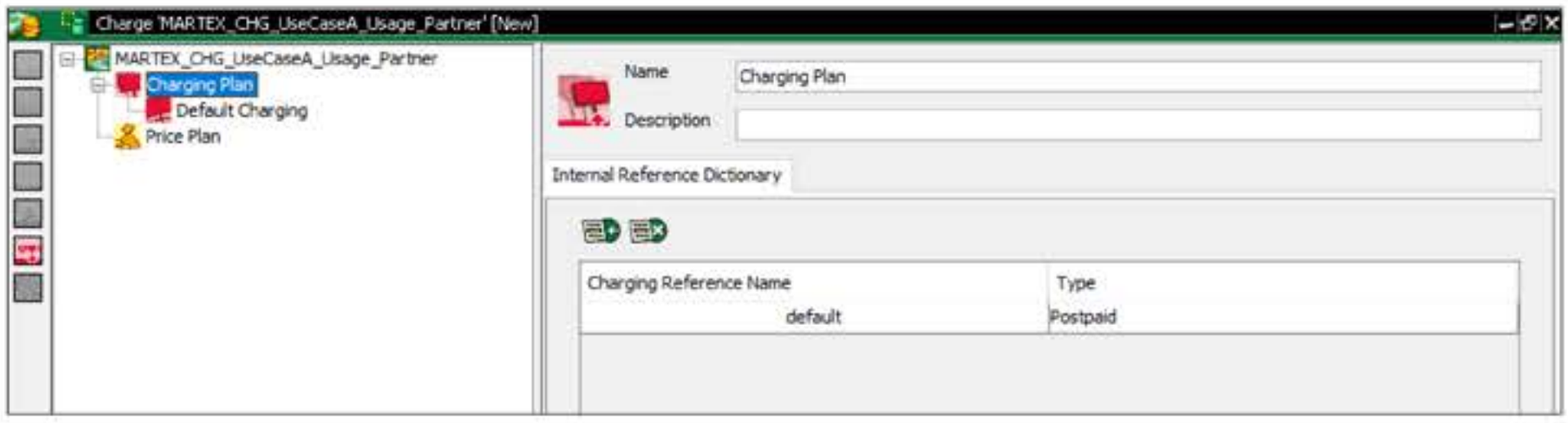

Figure 3.82 Add Charging Reference to Partner Charge

3. In the **Default Charging** node, right-click to launch the context menu, and add **Internal Reference**. Set the name to "default", and select **Default** from the **Internal References** dropdown list, as shown in Figure 3.83.

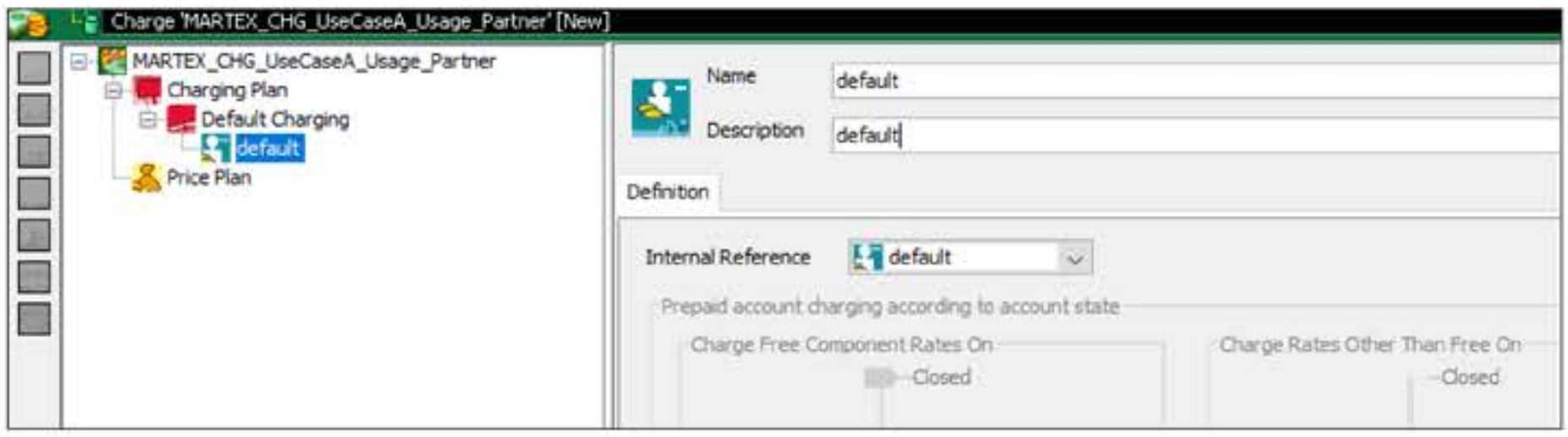

Figure 3.83 Add Default Internal Reference to Partner Charge

The next step is to model the price plan. For our example scenario, Martex Corp. needs to pay vendor X 70% of each transaction. This includes any usage for service A or service B, recurring reserved instance charges, and cancellation charges.

Partner settlement is based on usage consumption, so a usage-based rate component needs to be added to the price plan by right-clicking and selecting **Add Component • Rates • Usage Rate**.

Next, fill in the **Name** and **Description**, and select a chargeable item class. The chargeable item class defines the format that SAP Convergent Charging will be expecting as an

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input to the charge. Pick the consumption item class **HCOS** that you created earlier, as shown in Figure 3.84.

Figure 3.84 Add Usage Rate to Partner Charge

Because Martex Corp. needs to pay 70% of its revenue back to vendor X, you need to add an arithmetic operation by right-clicking and selecting **Add Component • Operators • Arithmetic Computation**, as shown in Figure 3.85.

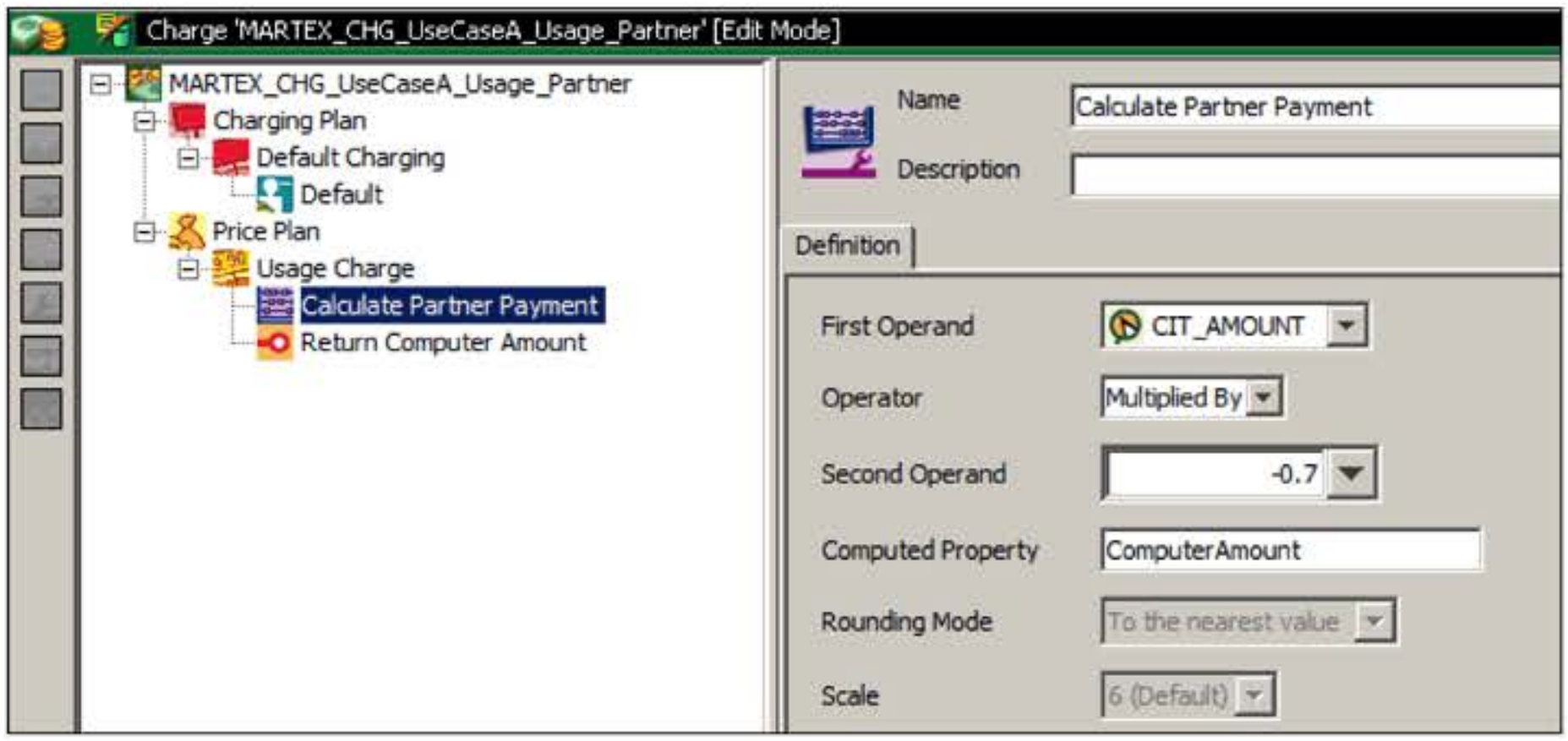

Figure 3.85 Calculating Partner Charges

Next, use a flat function to return the final amount variable by right-clicking and selecting **Add Component • Function • Flat**, as shown in Figure 3.86.

To save the charge, use menu path **File • Save • In the Database**.

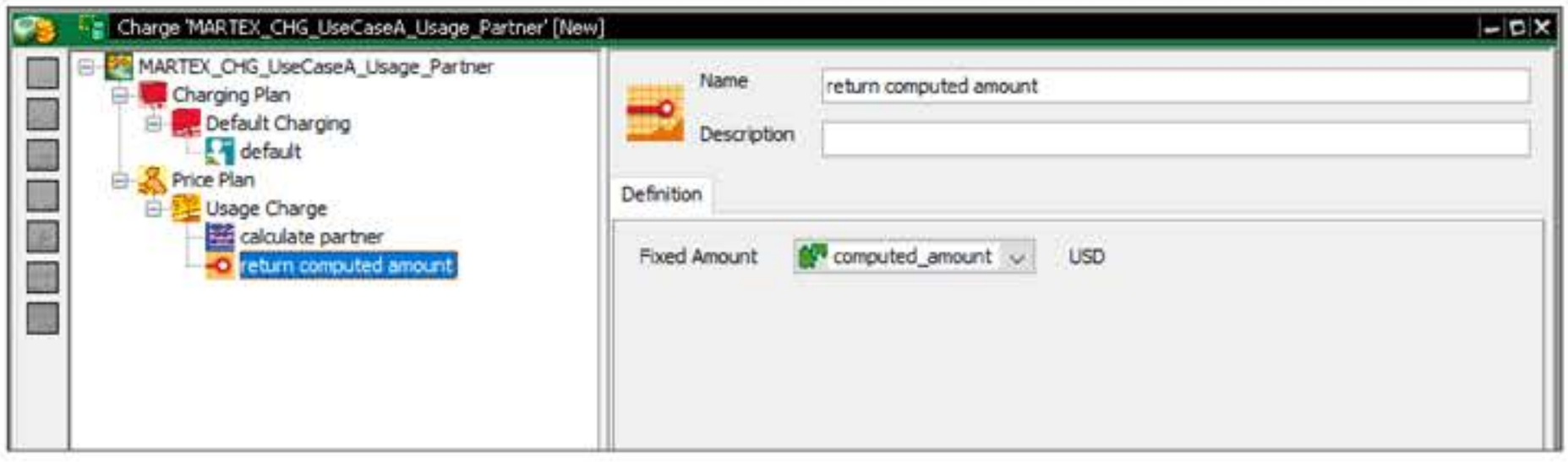

Figure 3.86 Configuring the Flat Rate Item for the Partner Charge

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Creating Partner Charge Plans

After the charge is created, the next step is to create a charge plan for SAP Billing and Revenue Innovation Management, subscription order management, to perform cross-catalog mapping. The steps to create a partner charge plan are the same as for the customer charge plan, so only the final output for each tab will be shown. Follow these steps:

1. Navigate to **File • New • Charge Plan**, and enter "MARTEX_CHGP_UseCaseA_Partner" for the **Name** and **Description**, as shown in Figure 3.87.

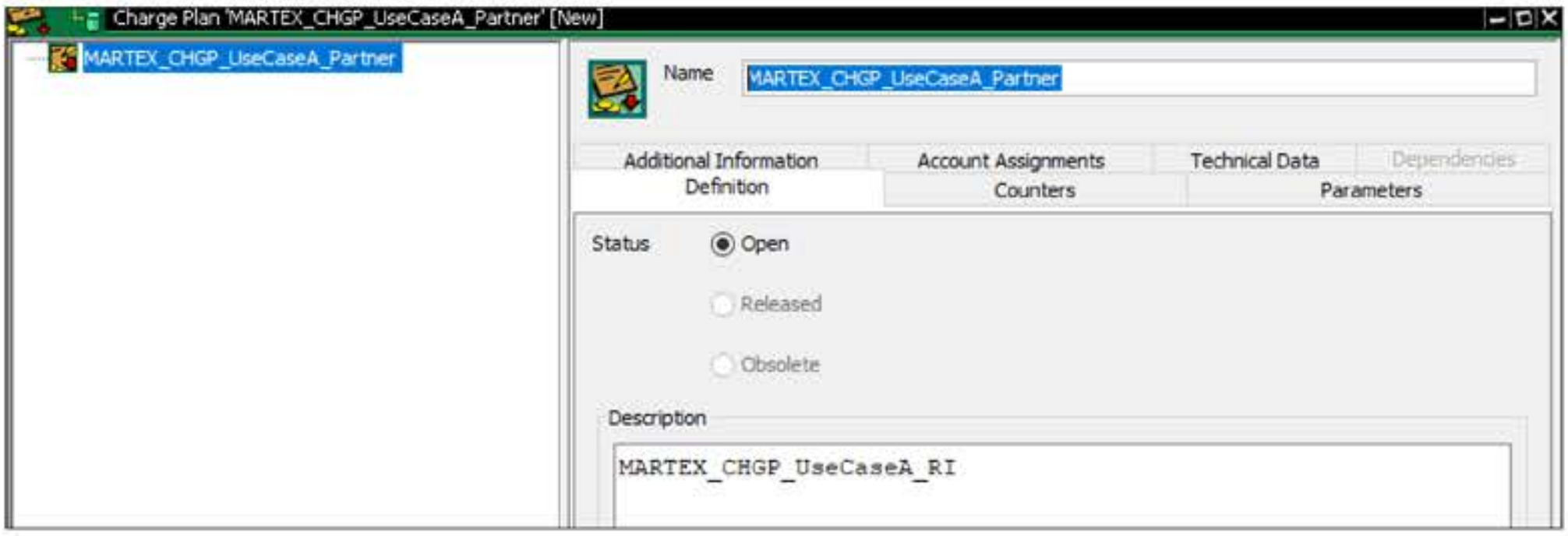

Figure 3.87 Creating the Partner Charge Plan

2. Right-click the **Charge Plan** object in the tree structure on the left, and select **Add Item** from the context menu. This setup enables the association between charges and the charge plan, as shown in Figure 3.88.
3. On the **Account Assignment** tab on the charge plan level, define the default postpaid account by clicking the button, as shown in Figure 3.89.

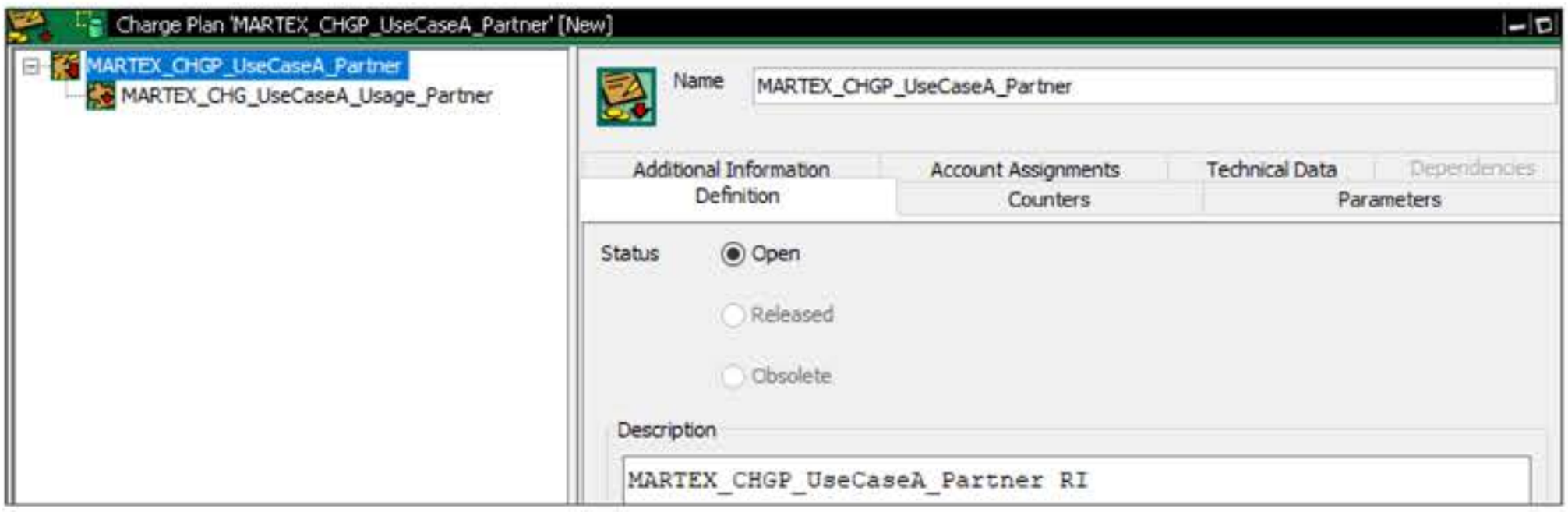

Figure 3.88 Adding a Charge to the Partner Charge Plan

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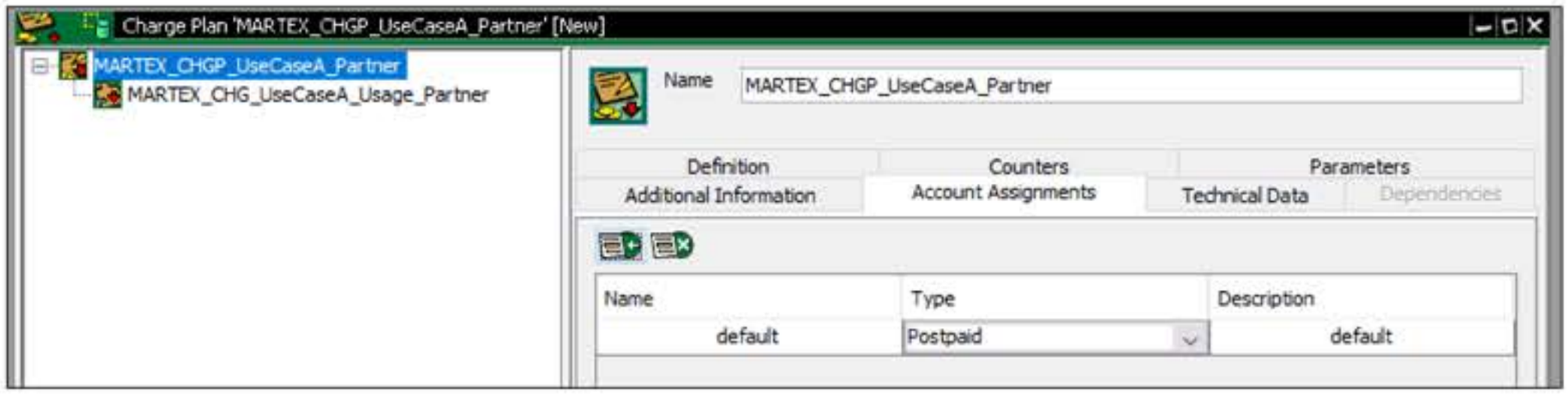

Figure 3.89 Adding an Account Assignment to the Partner Charge Plan

On the **Technical Data** tab on the charge plan level, add a user technical identifier called "subscriber id" by clicking the button (see Figure 3.90). No action is required for the **Counter** and **Parameter** tabs.

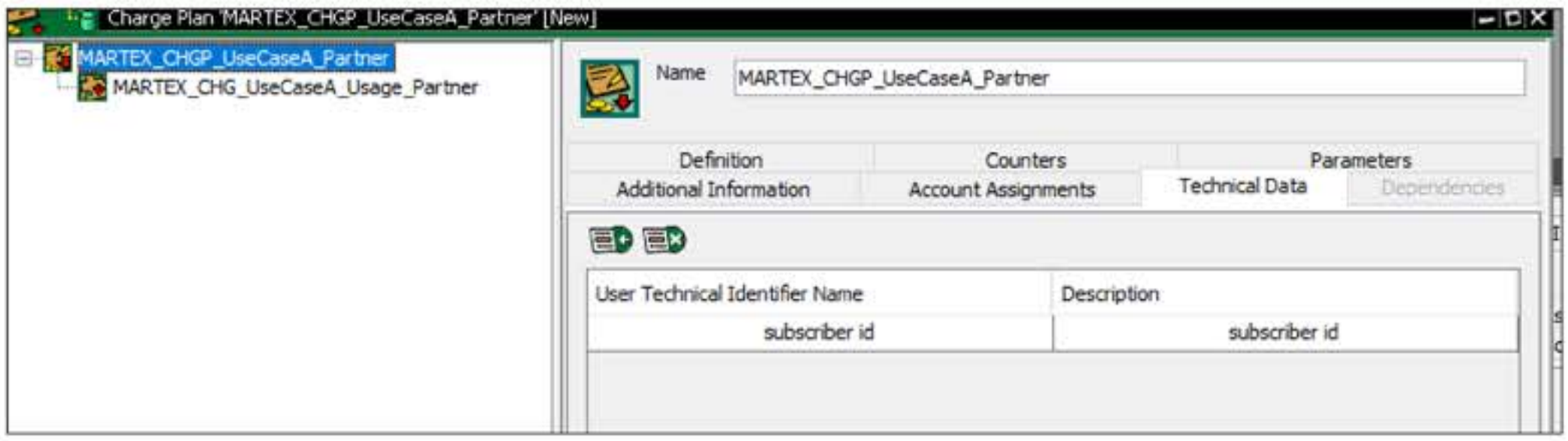

Figure 3.90 Technical Data Tab

Once the change plan level configuration is done, the next step is to configure the change level information on the charge plan:

1. On the **Tax** tab at the charge level, set the **Tax System** to **Disabled**, as shown in Figure 3.91.

Figure 3.91 Disable Tax Configuration for Partner Charge Plan

2. On the **Technical Data** tab, enter a **Service Identifier** of "HVRVSH", and map it to the **subscriber ID**, as shown in Figure 3.92.

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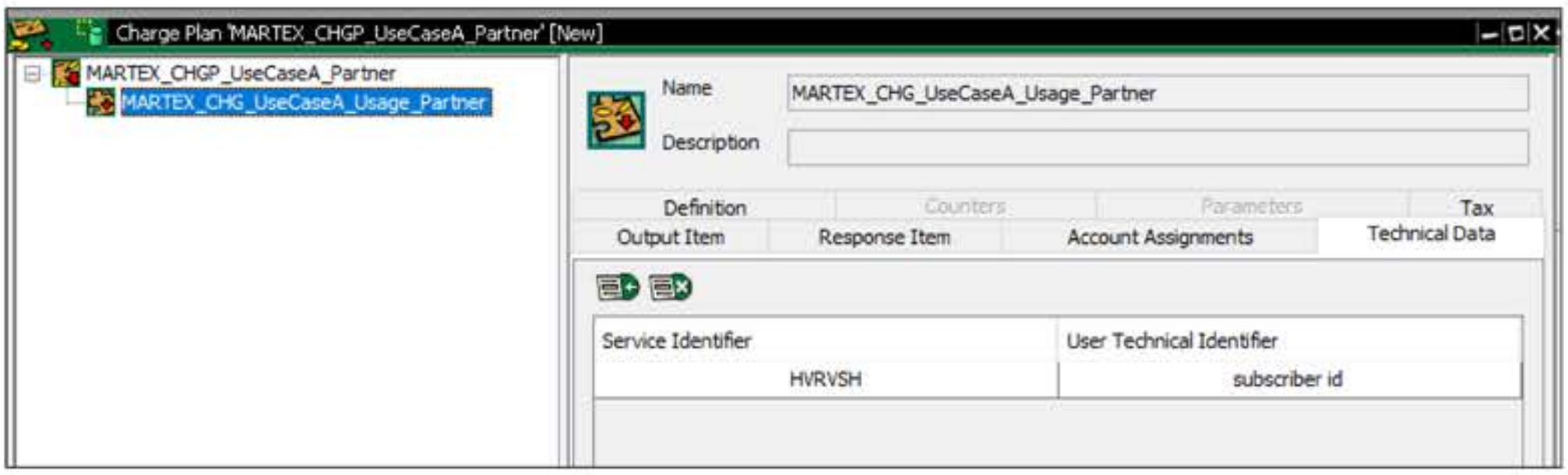

Figure 3.92 Linking Charge User Technical Identifier to Charge Plan User Technical Identifier

3. On the **Account Assignments** tab, set the charging reference **Type** as **Postpaid**, and link it to the charge plan level. This allows the account information to be passed on from the charge plan to the charge level, as shown in Figure 3.93.

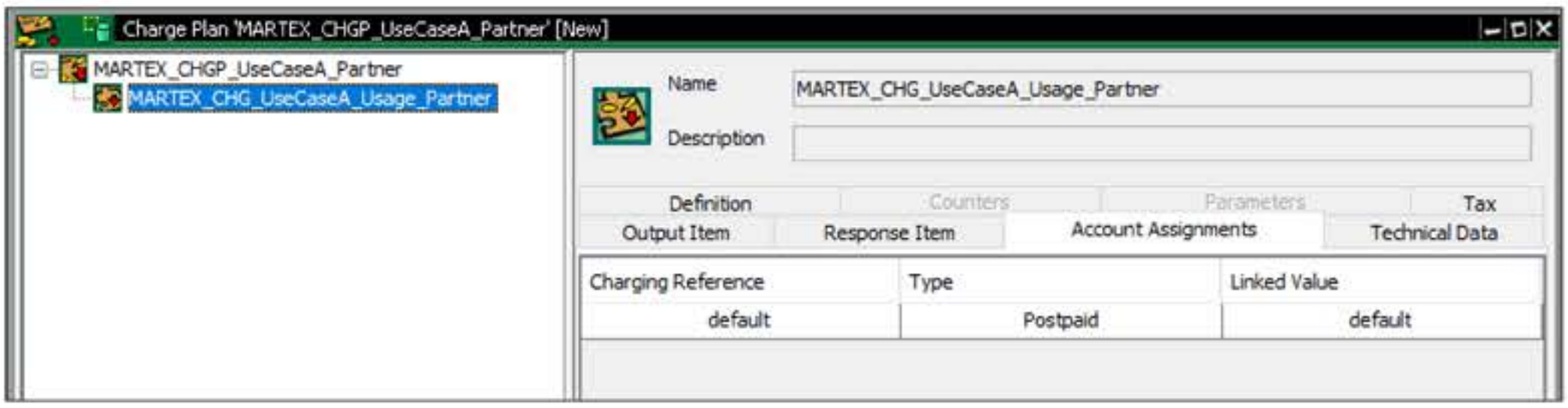

Figure 3.93 Linking Charge Account Assignments to Charge Plan Account Assignments

4. For the output item assignment, click on the **Output Item** tab, and then select **HCOS** as the output format (see Figure 3.94).

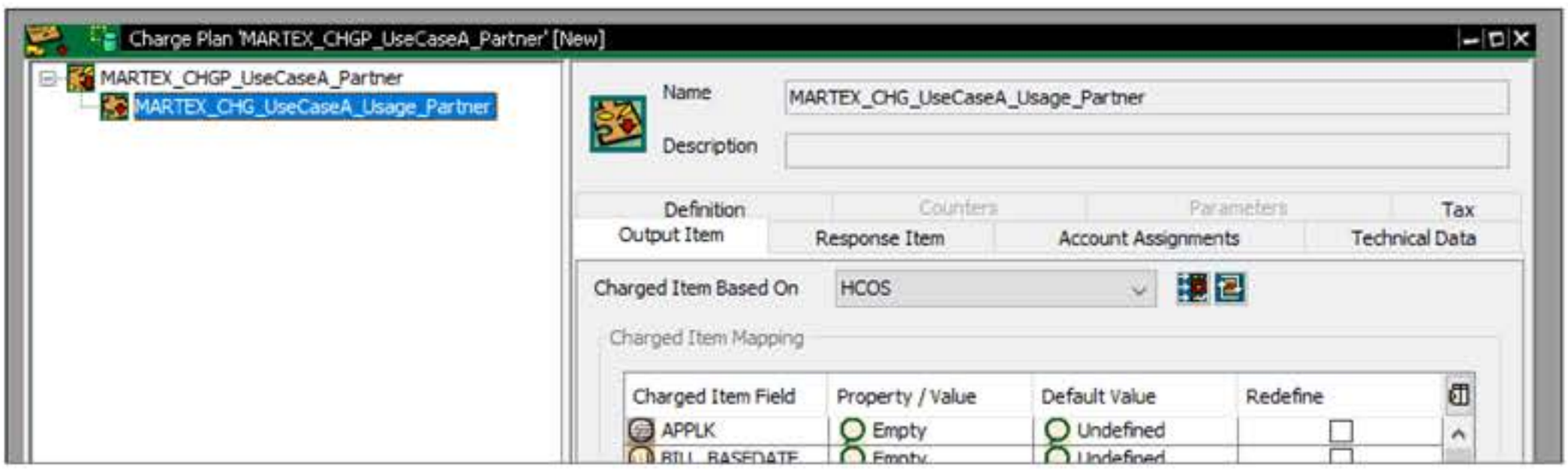

Figure 3.94 Configuring the Charged Item for the Partner Charge Plan

When the **HCOS** charged item class is selected, a list of fields belonging to the item class will be populated. In an SAP Billing and Revenue Innovation Management integrated scenario, the fields `BITTYPE` and `SUBPROCESS` are mandatory. The values that need to be

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populated in the BITTYPE and SUBPROCESS fields will be communicated by the SAP Convergent Invoicing team because they will drive billing behavior.

For our example, the values in Table 3.3 need to be populated for each charge.

MARTEX_CHG_UseCaseA_Usage Partner	
BITTYPE	ZBC1
SUBPROCESS	ZHC1

Table 3.3 Partner Charge Plan Values

Finally, release the charge plan for SAP Billing and Revenue Innovation Management subscription order management, to perform cross-catalog mapping by saving the charge plan via menu path **File • Save • In the Database**. SAP Convergent Charging will display a prompt for catalog selection and subsequently save the charge plan and go into display mode. Next, toggle to edit mode by selecting **File • Edit Mode**, and select the **Released** radio button, then save again, as shown in Figure 3.95.

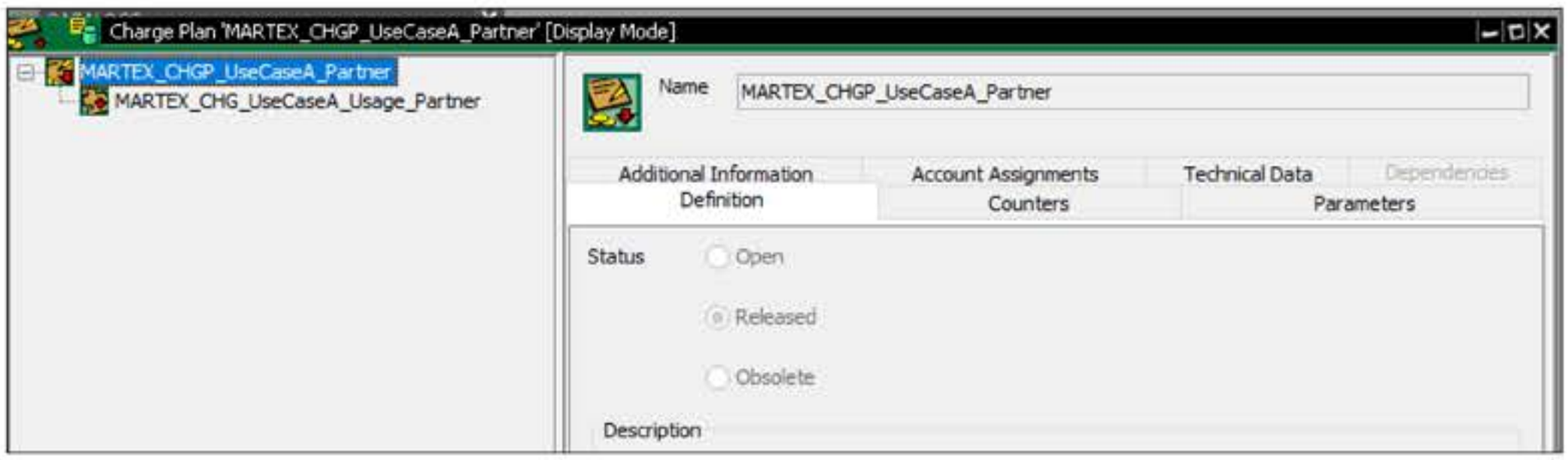

Figure 3.95 Partner Charge Plan in Released Mode

In Chapter 2, you saw how SAP Billing and Revenue Innovation Management, subscription order management, behaves as the system of engagement. Martex Corp. employees can log in to create and maintain master data elements such as the business partner (both customer and vendor), business agreements, and products, as well as transactional data such as provider contracts. You saw how entities such as business partners, business agreements, products, and provider contracts are distributed automatically to SAP Convergent Charging and SAP Convergent Invoicing.

Provider contracts in SAP Convergent Charging can be accessed by navigating to **File • Open • Provider Contract**. Figure 3.96 shows provider contract 10333 (created for reserved instance), contract 10335 (created for nonreserved instance), and contract 10337 (for partner charge plan).

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Note

There is no provider agreement object in SAP Convergent Charging. The partner agreement is created as a provider contract in SAP Convergent Charging.

Once the provider contracts are created in SAP Convergent Charging, the provider contracts containing one-time and recurring charges will be rated, and billable items will be created in SAP Convergent Invoicing. In Chapter 4, Section 4.4.4, we'll explore how usage charges for both accounts receivable and accounts payable items are triggered by the rating process of consumption items.

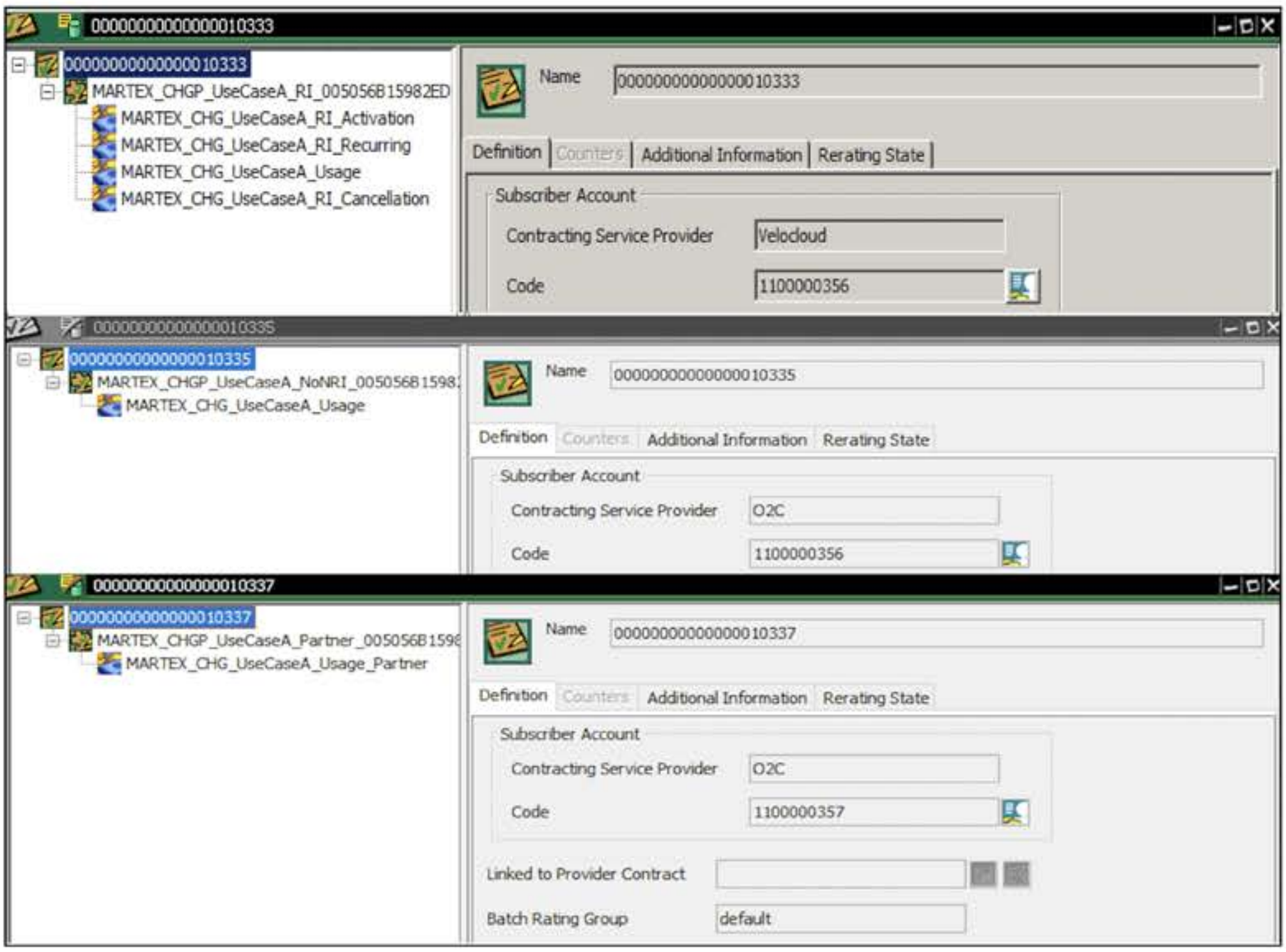

Figure 3.96 Martex Corp. Provider Contract

3.6.5 Creating Pricing Macros

Even though the configurations to complete the preceding use case don't require you to create or use a pricing macro, we'll explore how to create and work with a pricing macro in this section. A *pricing macro* is a reusable module. It's like an SAP ABAP subroutine if you're familiar with ABAP programing. The purpose of a pricing macro is reusability. If you have common calculation formulas such as proration or date calculations in your price plan, allowance logic, or refill logic, you can create a pricing macro

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and call it from these objects. A pricing macro can also be called within other pricing macros.

To create a pricing macro, select **File • New • Pricing Macro**, as shown in Figure 3.97.

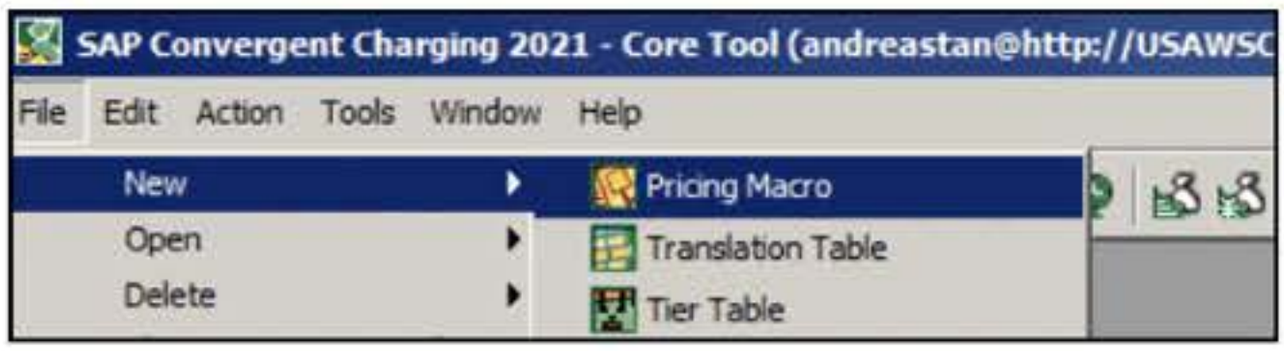

Figure 3.97 Create Pricing Macro

The sample price macro in Figure 3.98 checks if the consumption year is a leap year. It accepts a date as an input and returns output **IsLeapYear** value **Y** or **N**, as shown in Figure 3.99.

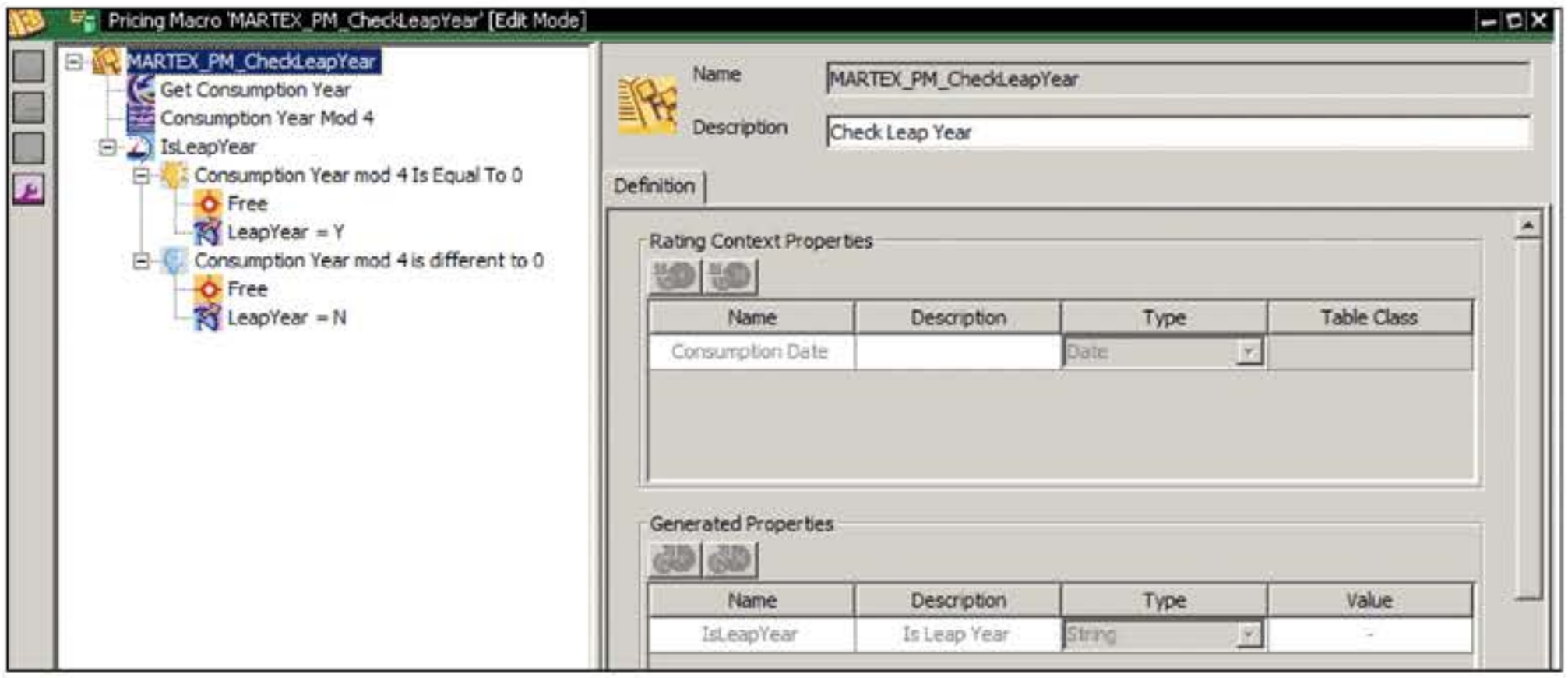

Figure 3.98 Pricing Macro Logic

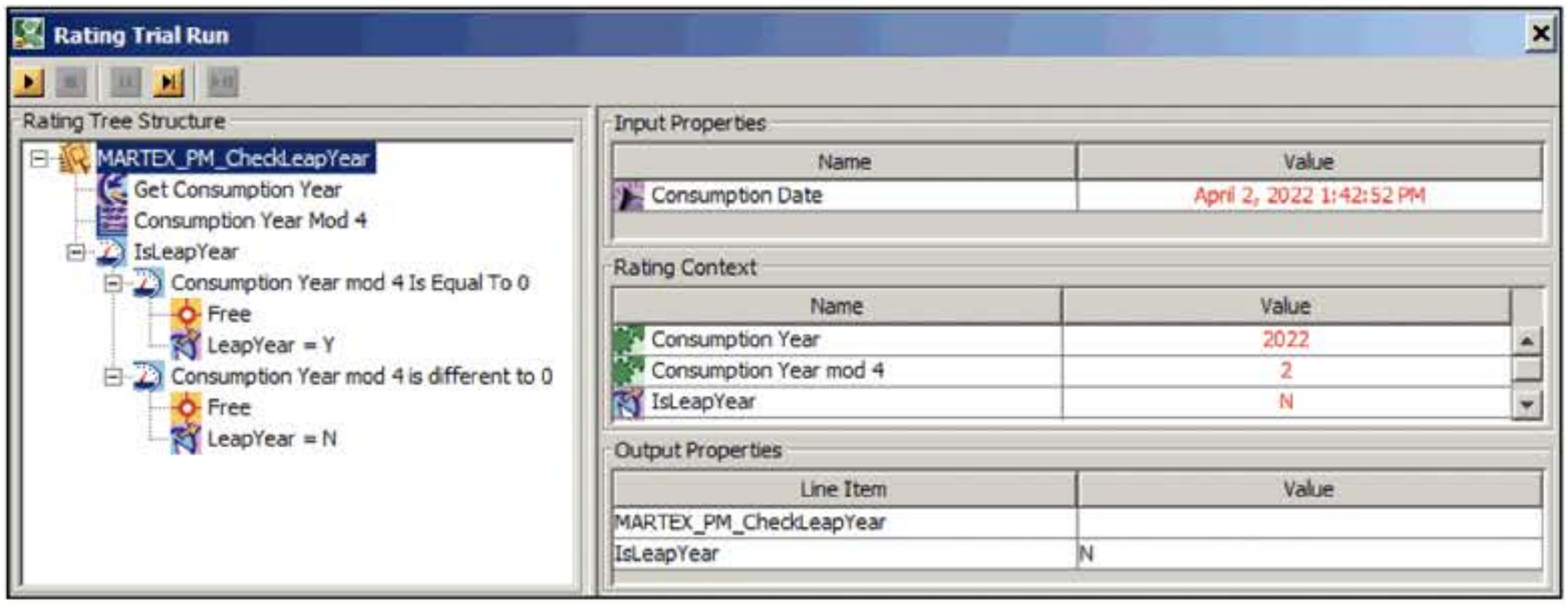

Figure 3.99 Pricing Macro Result

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3.7 Managing Allowances

An *allowance* is a credit for a customer to redeem for one or more services. Allowances can be based on quantity or amount and for a specific duration of time. Similar to charge plans and charges, allowances consist of an allowance plan and allowance logic. Allowance plans can have one or multiple allowance logic, counters, and parameters linked to the allowance logic. Allowances also have allowance event classes such as the chargeable item class; however, allowance event classes don't have default properties.

Note

The allowance topic in this section pertains only to SAP Convergent Charging. It doesn't include the integration with SAP Billing and Revenue Innovation Management, subscription order management.

The allowance logic used in the Martex use case uses a counter to represent promotions for customers who purchase the reserved instance option to receive a $2,000 allowance.

In this section, we'll cover the allowance lifecycle and necessary steps to build it.

3.7.1 Allowance Lifecycles

Allowances are similar to charge objects. They have their own set of properties, such as counters and parameters. Allowance data is created dynamically upon charging events (i.e., contract activation or consumption charging). Allowances provide features to manage how one of several services is consumed within a period of time. Each allowance has a validity period and expires if the current date/time is after its validity end date/time.

3.7.2 Define Allowances

In this section, we'll cover the type of allowance objects required and how to set up each type.

Creating Allowance Event Class

An allowance event class is similar to a chargeable item class used by a charge object with a usage rate price plan. An allowance event class doesn't have default properties, but you need to define properties, as shown in Figure 3.100. The class holds transaction data records to pass between convergent charging objects, such as from a charge object to an allowance object.

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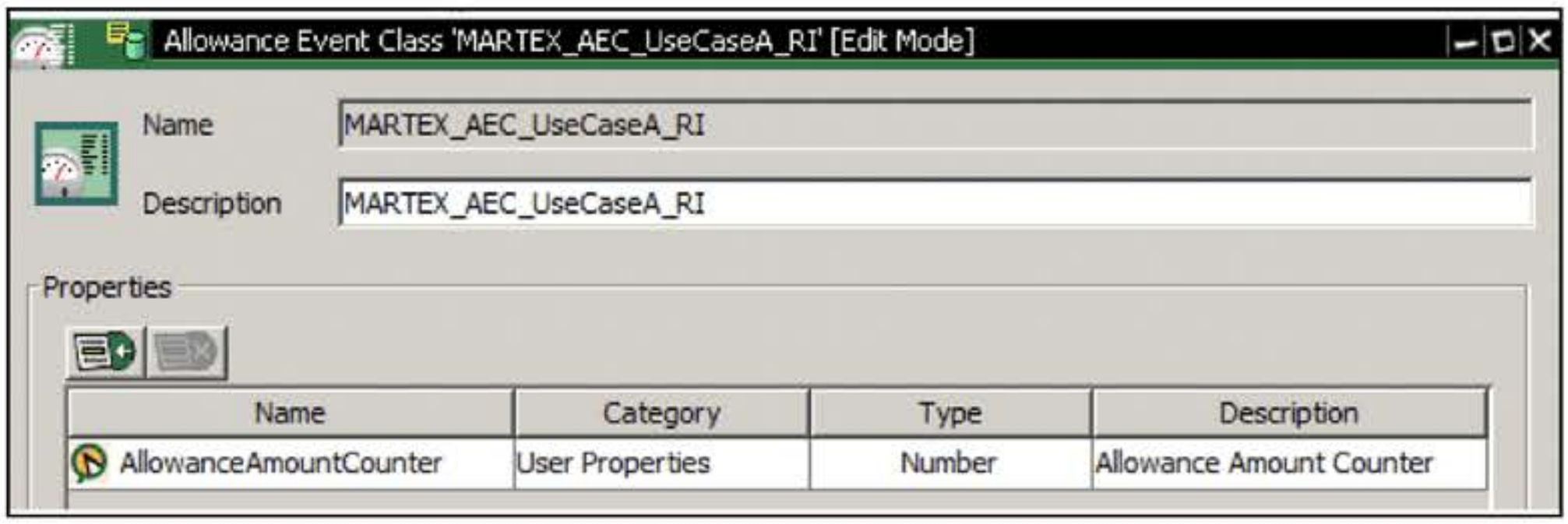

Figure 3.100 Allowance Event Class

Creating an Allowance Logic

The first allowance logic is based on an **Event-Based Trigger**, as shown in Figure 3.101. This allowance logic object is triggered by an event sent from usage charge `MARTEX_CHG_UseCaseA_Usage`, as shown in Section 3.6.

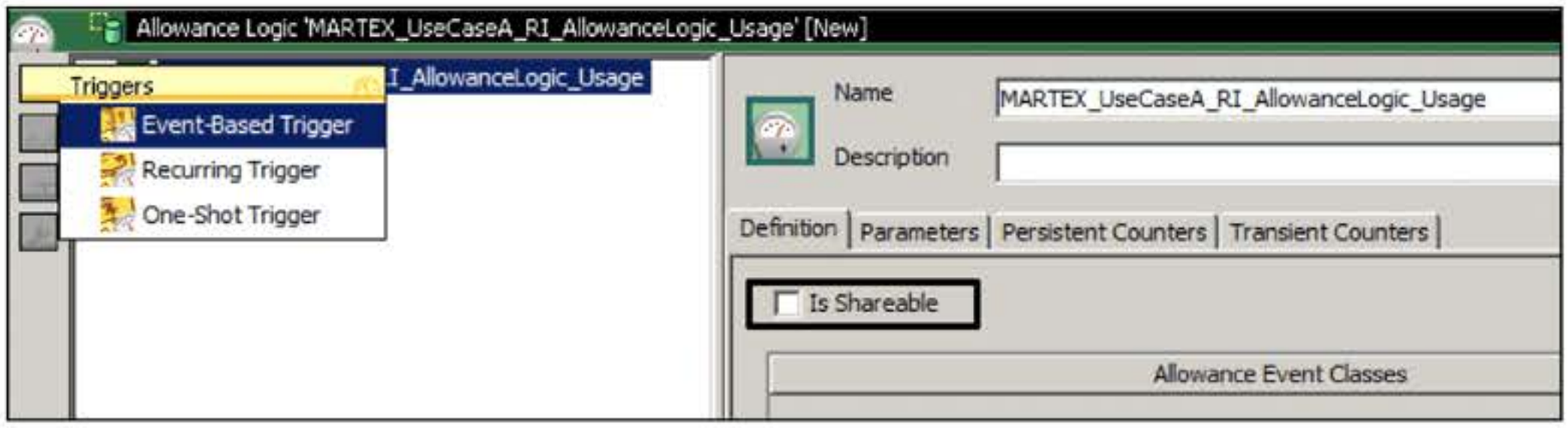

Figure 3.101 Creating Event-Based Trigger Allowance Logic

Next you need to add an event-based trigger and assign the **Allowance Event Class**, as shown in Figure 3.102.

Figure 3.102 Assigning the Allowance Event Class

Now, you add a persistent counter in the allowance logic and call it "AmountCounter", as shown in Figure 3.103. This counter is used to keep track of the allowance amount each time usage data is processed by charge object `MARTEX_CHG_UseCaseA_Usage`, shown in Section 3.6.

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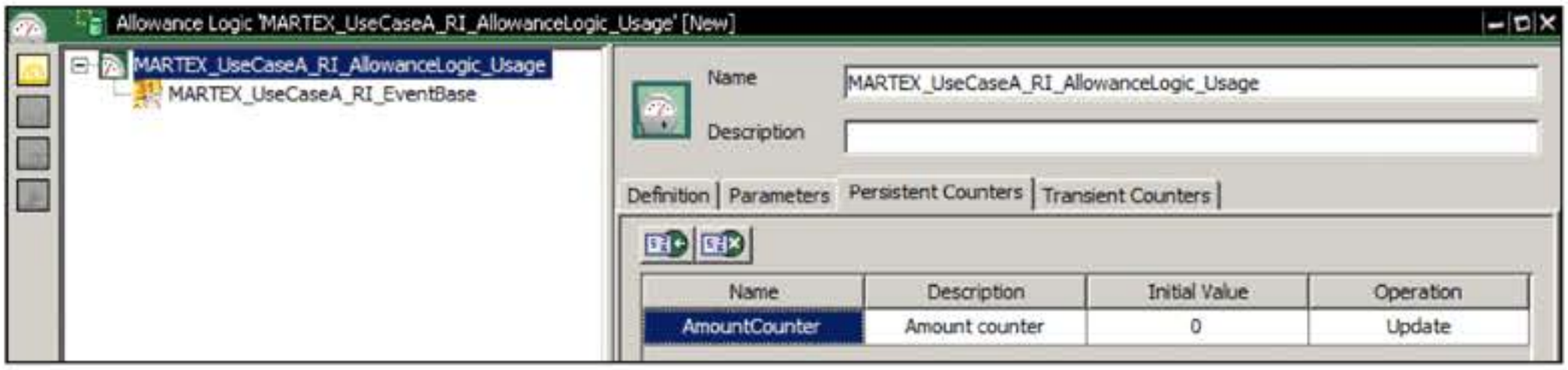

Figure 3.103 Defining the Persistent Counter

Next, add the **Allow** function. Allowance logic is only used to keep track of the allowance amount, not to generate allowance charged items; therefore, leave **Generate Charged Item** unchecked, as shown in Figure 3.104.

Figure 3.104 Adding the Allow Function

Next, add the counter update operator to update the `AmountCounter` value with `AllowanceAmountCounter`, as shown in Figure 3.105. The `AllowanceAmountCounter` value is calculated and sent by `MARTEX_UseCaseA_Usage`, as shown in Section 3.6.

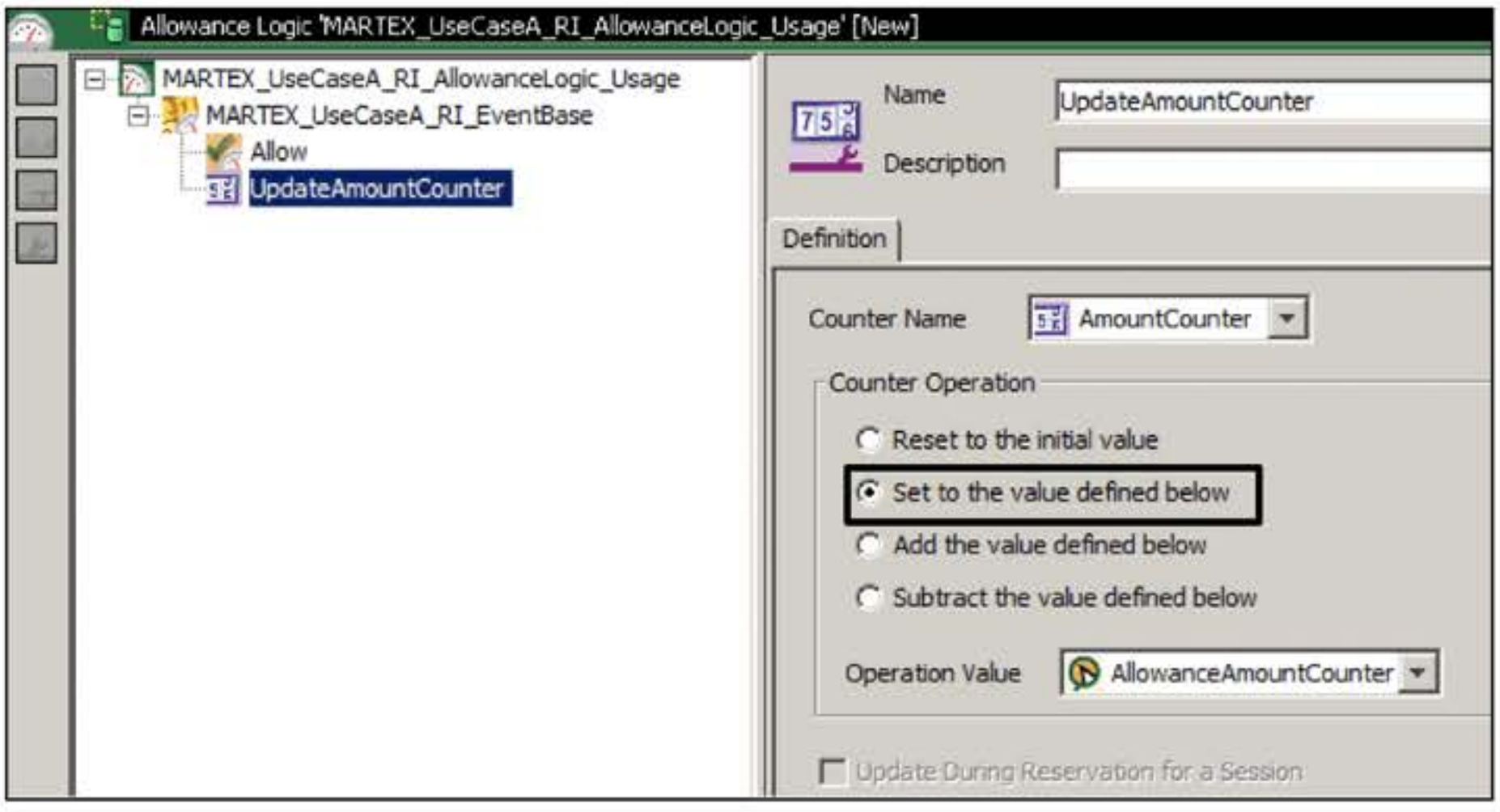

Figure 3.105 Updating AmountCounter

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The following allowance logic is based on a one-shot trigger. It's used to set the initial allowance amount to $2,000, as shown in Figure 3.106. This object is triggered by activation charge `MARTEX_CHG_UseCase_RI_Activation` in Section 3.6.

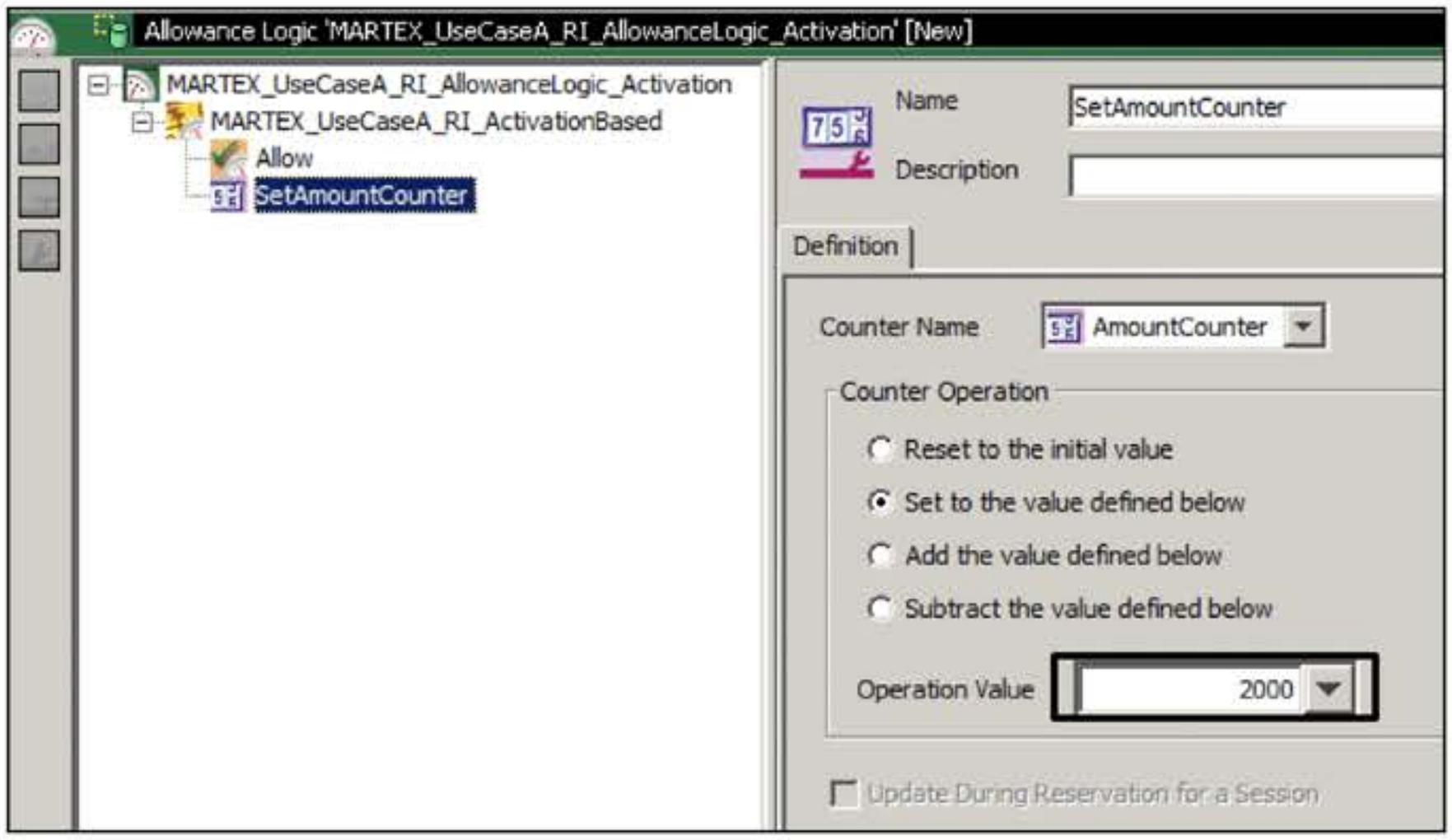

Figure 3.106 Set Initial Value $2,000 to AmountCounter

Creating an Allowance Plan

Similar to a charge plan, an allowance plan is also needed for allowance logic. Navigate to **File • New • Allowance Plan** to create a new allowance plan, as shown in Figure 3.107. The allowance plan defines the behavior of allowance logic during charging execution. Two allowance plans are created. The first one is used by allowance logic for usage, and the second is for allowance logic for activation.

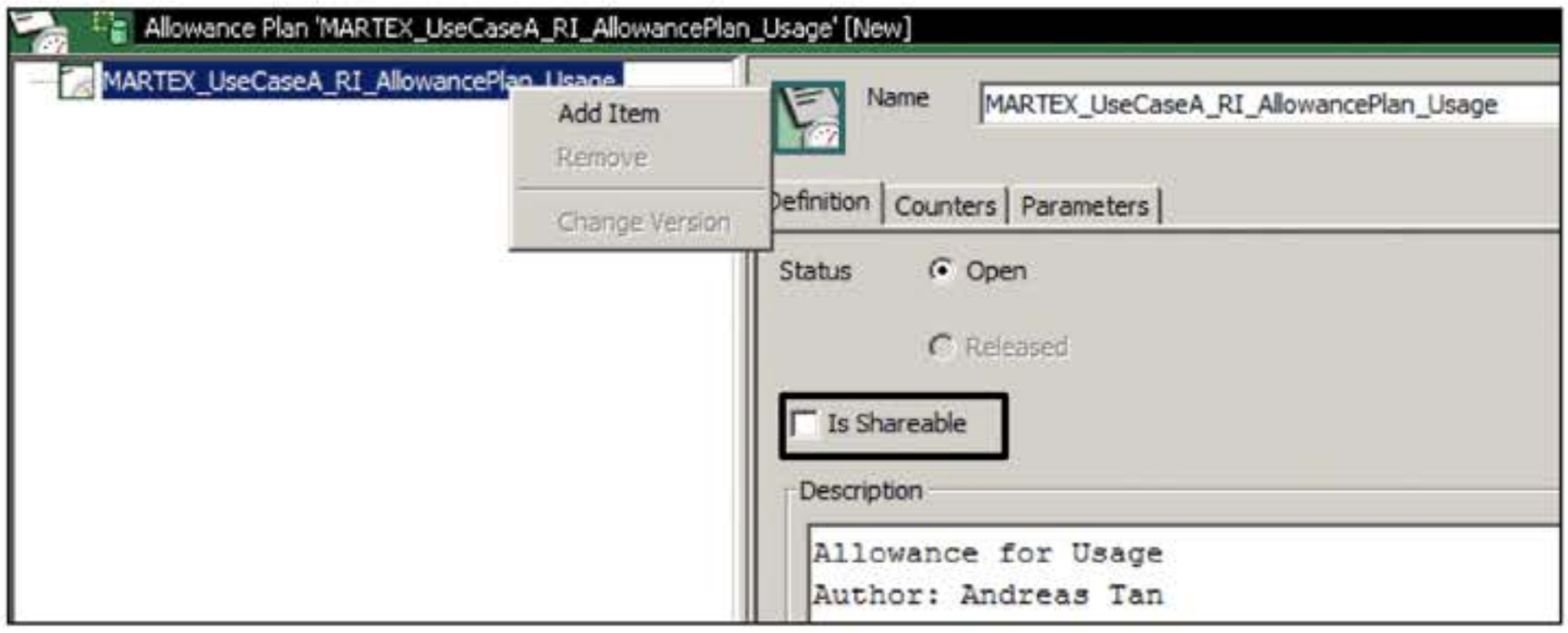

Figure 3.107 Usage Allowance Plan

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You can create an allowance plan using the following steps:

1. Add the allowance logic to the allowance plan for usage allowance (Figure 3.108).

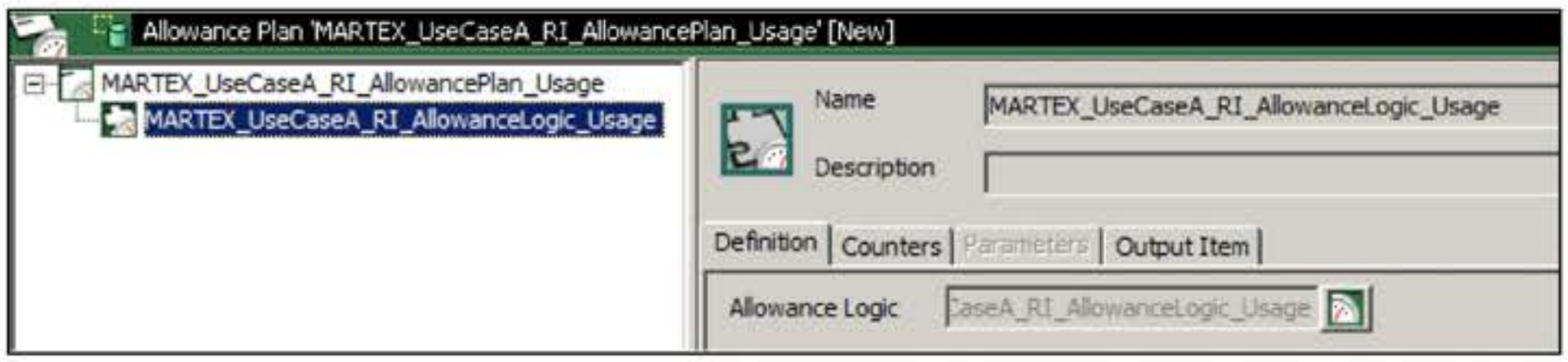

Figure 3.108 Adding the Allowance Logic to the Allowance Plan

2. Next, add a counter called "AmountCounter" by selecting the highlighted node and adding the counter, as shown in Figure 3.109.

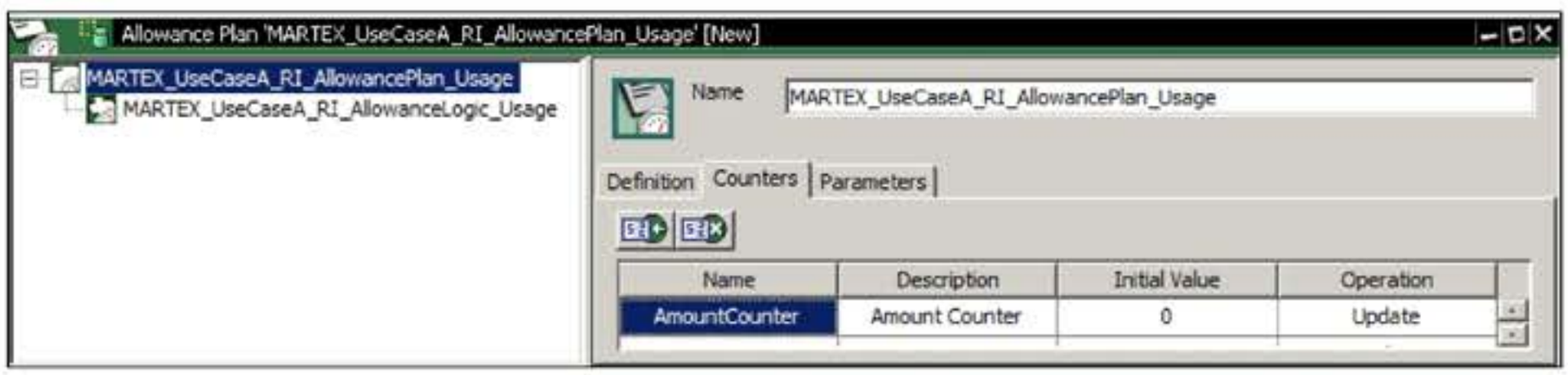

Figure 3.109 Adding AmountCounter to the Allowance Plan

3. Next, link the counter at the allowance logic level to the allowance plan level, as shown in Figure 3.110.

Figure 3.110 Linking Allowance Logic AmountCounter to the Allowance Plan

4. Because the allowance logic is only used to keep track of the allowance amount, you can select **None** for the **Charge Item Based On** checkbox, as shown in Figure 3.111.

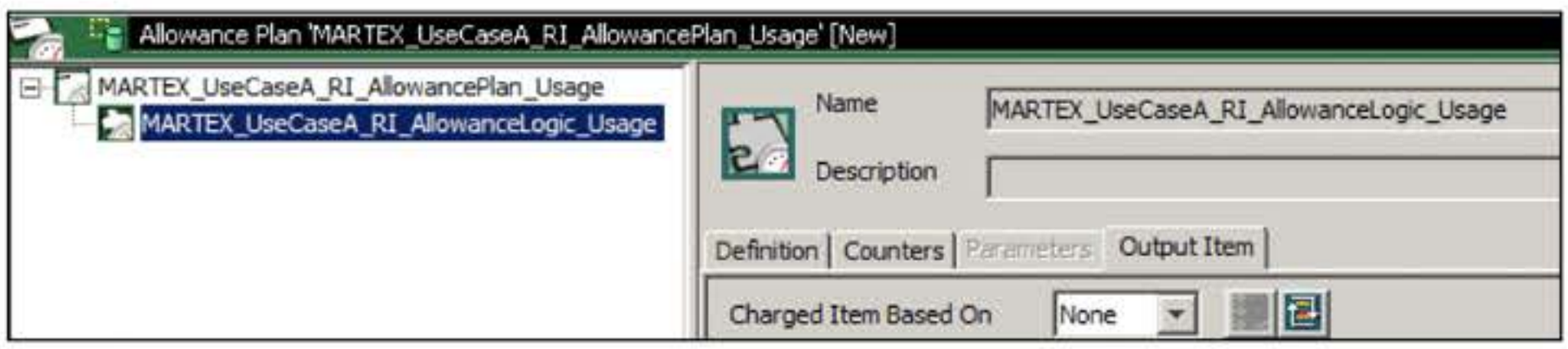

Figure 3.111 Setting the Charged Item Based On Field

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5. Repeat the same steps to create the allowance plan for activation as you've done for usage allowance plan, as shown in Figure 3.112 and Figure 3.113.

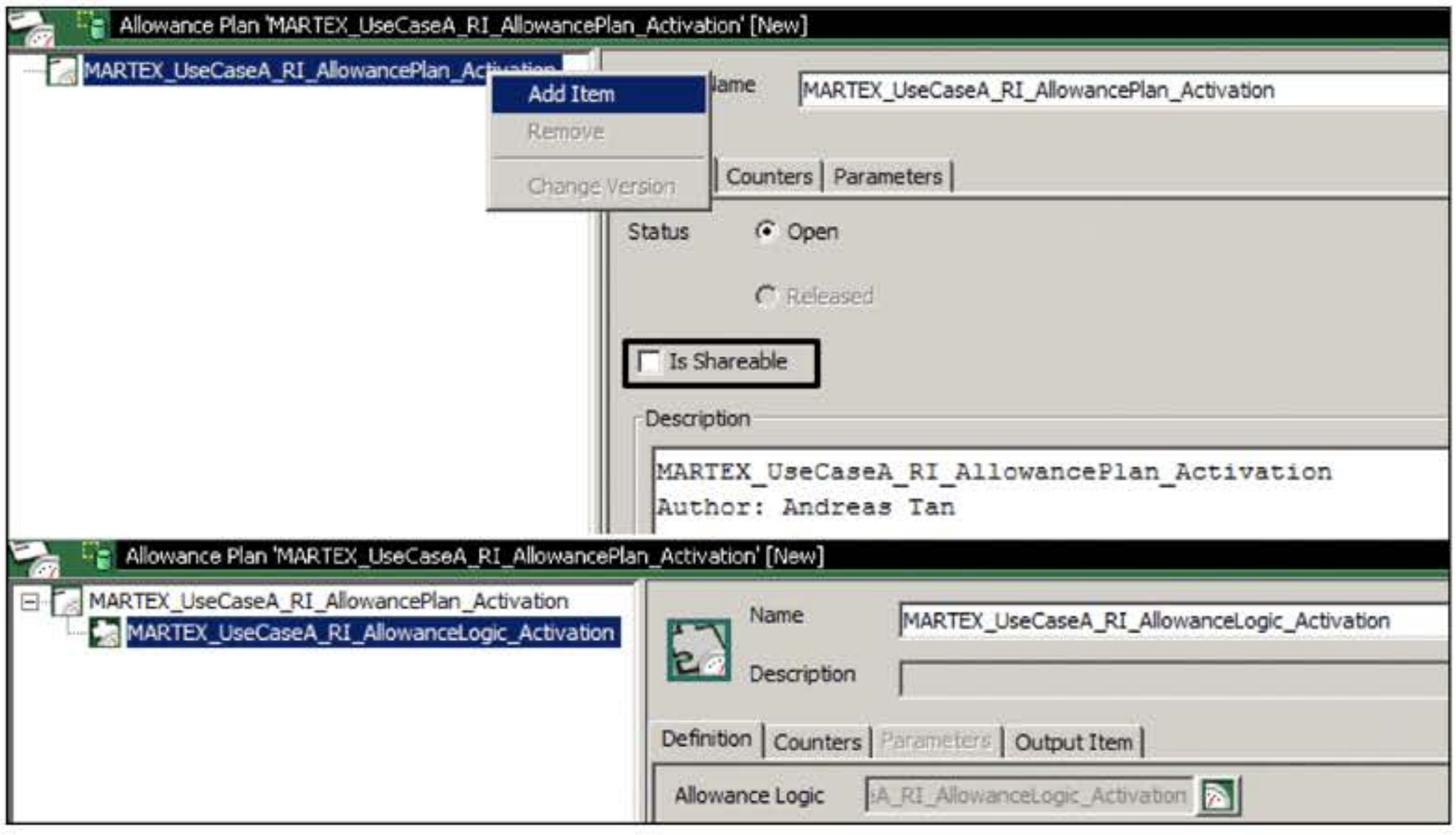

Figure 3.112 Adding Allowance Logic to the Allowance Plan

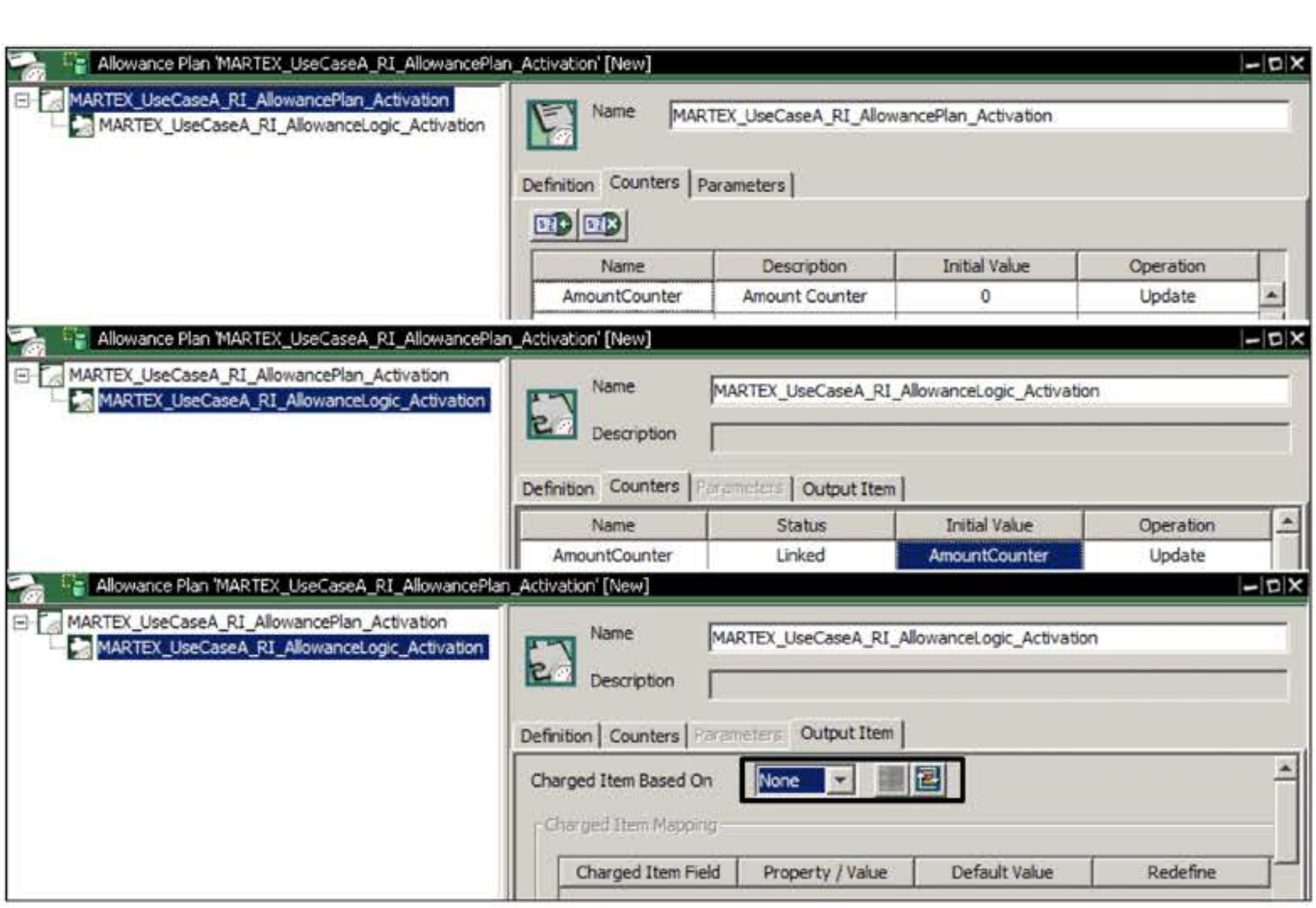

Figure 3.113 Linking the Counter and Setting the Charged Item Based On Field

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Processing Allowances

Now that the allowance setups are completed, they can be used by the usage and activation charge objects as discussed in detail in Section 3.6.

3.7.3 Shared Allowances

Allowances can be used in a single subscriber account or in different subscriber accounts (shared allowance). Shared allowances are used to share between provider contracts that belong to different subscriber accounts. For example, a company has a contract with a telco provider for their employees for the 1,000 GB of data plan per month. The 1,000 GB data pool is decremented through the subtract counter operation every time an employee consumes data. Once the counter reaches 0 bytes, the company will receive a notification that the pool of data is completely consumed. To share an allowance, the owner of the provider contract that owns the allowance specifies a unique share identifier at the creation of the shared allowance. The other provider contracts or members use this share identifier to reference the shared allowance.

3.7.4 Allowance Dashboard

In an integrated scenario with SAP Billing and Revenue Innovation Management, subscription order management, you can display active or expired allowances and check counter values for active allowances in subscription contracts. For subscription contracts that use allowances, the allowance assignment block is available in the contract UI screen in SAP Fiori. The information is pulled from SAP Convergent Charging. You can see all currently active allowances and allowances to be activated for this contract. To do so, you need to configure the item category in a Customizing form via **Service • Transaction • Setting for Subscription Transactions • Define Settings for Item Categories**.

3.8 Managing Counters

In Section 3.6, we demonstrated how to use a persistent counter to build the use case for Martex Corp. Now let's discuss in more detail how to set up counters and counter operations. The two types of counters used in SAP Convergent Charging are persistent counter and transient counter. Persistent counters are defined at the charge object definition whereas transient counters are part of the price plan definition. A counter is a numeric variable you can define and declare in a reusable charge. The following presents the different behaviors between transient and persistent counters:

- Transient counter
  - Is not persisted in the database; it's only stored in internal memory during the charging process

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  - Is not visible and can't be redefined in charge plans
  - Allows values to be passed from master to dependent charges
- Persistent counter
  - Persisted in the database during the charging process
  - Visible and can be redefined in charge plans
  - Can be shared with other charges in a charge plan
  - Is a property of a provider contract
  - Can be updated with counter operation, as shown in Figure 3.114

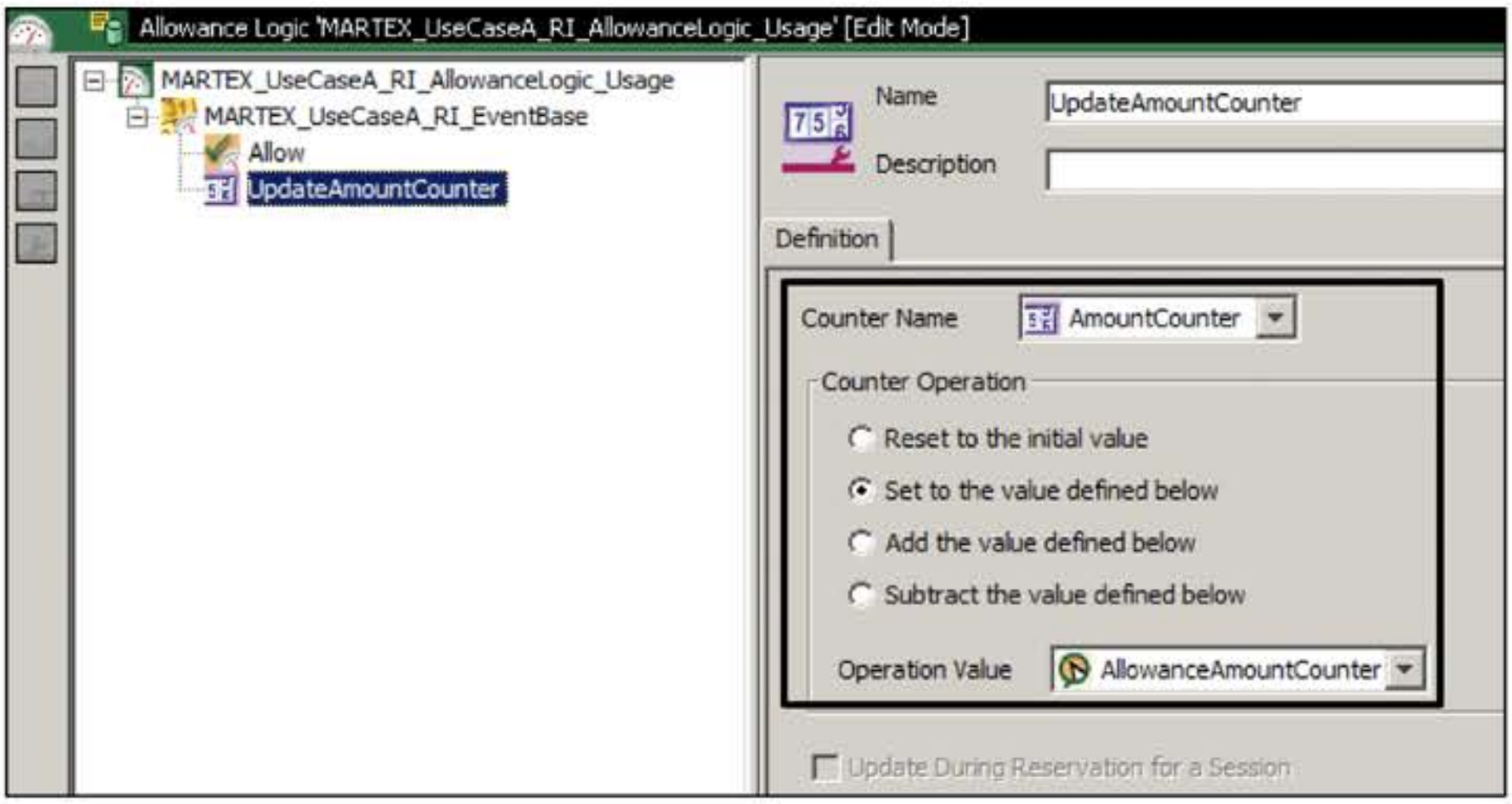

Figure 3.114 Counter Operations

3.9 Managing Integrations

In an end-to-end integrated scenario with SAP Billing and Revenue Innovation Management where SAP Billing and Revenue Innovation Management, subscription order management, and SAP Convergent Invoicing are involved, you need to configure the integration between SAP Convergent Charging and SAP S/4HANA systems. The standard integration between these two systems is connected through SOAP API web services provided by SAP Convergent Charging out of the box. SAP Convergent Charging web services make up the technical interface delivered by SAP Convergent Charging Core Server with their accessible endpoints. This set of services is used to perform your business operations such as management of master data and other operations such as rating and charging. The web services configuration is performed using Transaction SOAMANAGER.

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3.9.1 Managing Integration with SAP Billing and Revenue Innovation Management, Subscription Order Management

Most of the business services used by SAP Billing and Revenue Innovation Management, subscription order management, are related to master data such as management for customer master, product modeling, and contract activation. Depending on what type of services are required, you need to activate the necessary services to support your business operations.

Following are samples of some services required to integrate with SAP Billing and Revenue Innovation Management, subscription order management. For a complete list of services, go to *http://s-prs.co/v554203*.

- *http://Your_CC_Upadater_Instance_HostName:9081/v2/contractProvisioning?wsdl*
- *http://Your_CC_Upadater_Instance_HostName:9081/v1/suacProvisioning?wsdl*
- *http://Your_CC_Dispatcher_Instance_HostName:9180/activation?wsdl*
- *http://Your_CC_Dispatcher_Instance_HostName:9180/v1/allowanceManagement?wsdl*

Note

The preceding examples use two different port numbers: 9081 and 9180. Port 9081 is for the updater instance, and port 9080 is for the dispatcher instance. These port numbers are default values from SAP, but you can change them as needed.

3.9.2 Managing Integration with SAP Convergent Invoicing

Similar to SAP Billing and Revenue Innovation Management, subscription order management, SAP Convergent Charging also uses web services to perform business operations such as rating and rerating. However, SAP Convergent Charging also requires a different type of connection to perform consumption item mapping, billable item mapping, and charged item records sending (rated transactions) to SAP Convergent Invoicing. The connection is based on the SAP Java connector. The SAP Java connection setup is done during the SAP Convergent Charging software installation, but you can change the connection properties using the **Setup** command after installation if needed. Changing the connection is usually performed by the Basis team.

3.10 Solving Common Rating and Charging Issues

As discussed at the beginning of this chapter, SAP Convergent Charging provides three types of charges: one-time, recurring, and usage. For one-time charges, the trigger is based on an external event, such as contract activation, suspension, or cancellation. For recurring charges, SAP Convergent Charging generates the charges based on internal

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schedules such as monthly, quarterly, and yearly frequency, and for usage charges, the process is triggered by ingesting CDRs.

Even though issues can happen during rating for all these types of charges, the most common ones usually happen during the processing of usage charges, as follows:

- **Invalid technical identifier**
  This issue is caused by an incorrect technical ID value from the data source. When this happens, SAP Convergent Charging will send error message **"Contract not found."**
- **Consumption date is outside the contract effective dates**
  Like the invalid technical identifier, SAP Convergent Charging will trigger error message **"Contract not found."**
- **Duplicate user technical identifier**
  This issue usually happens during provider contract replication from SAP Billing and Revenue Innovation Management, subscription order management, to SAP Convergent Charging.

Another rating issue is related to accessing mapping tables and range tables. This issue happens when the record in those tables isn't found based on the given input key(s). When this happens, the SAP Convergent Charging server will send error **"No access found."**

In addition to these problems, issues can happen during data transfer between SAP Convergent Charging and SAP Convergent Invoicing. There are two types of transactional data that SAP Convergent Charging sends to SAP Convergent Invoicing, consumption item and billable item. As a refresher, a consumption item is transactional data to be rated by SAP Convergent Charging, and a billable item is transactional data after being rated by SAP Convergent Charging. The issue is usually caused by data type mismatch between the chargeable item class in SAP Convergent Charging and the consumption item class in SAP Convergent Invoicing or the charged item class in SAP Convergent Charging and the billable item class in SAP Convergent Invoicing. When this issue happens, the errors are logged in the working directly at the operating system level. The data type mismatch is usually caused by someone making changes in either the consumption item class or billable item class.

3.11 Summary

In this chapter, we discussed key SAP Convergent Charging components and how they interact with other components such as SAP Billing and Revenue Innovation Management, subscription order management, and SAP Convergent Invoicing. In an integration scenario, SAP Convergent Charging is used primarily for modeling logic and performing rating and charging. There are two aspects of SAP Convergent Charging

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covered in this chapter: master data objects and transactional data. Master data object, such as subscriber accounts, external accounts, and provider contracts, are replicated in SAP Convergent Charging and originated from SAP Billing and Revenue Innovation Management, subscription order management, while processing the charges are done within the SAP Convergent Charging engine itself.

As a refresher, there are three types of charges (price plans): one-off charge, recurring charge, and usage-based charge. A one-off charge is triggered by an external event such as activation, suspension, and cancellation; a recurring charge is based on a schedule set up in SAP Convergent Charging; and a usage-based charge is ingested from the external system as CDR or consumption items from SAP Convergent Invoicing. Some items, such as billable item class and consumption item class, used in this chapter are further discussed in the next chapter in more detail.

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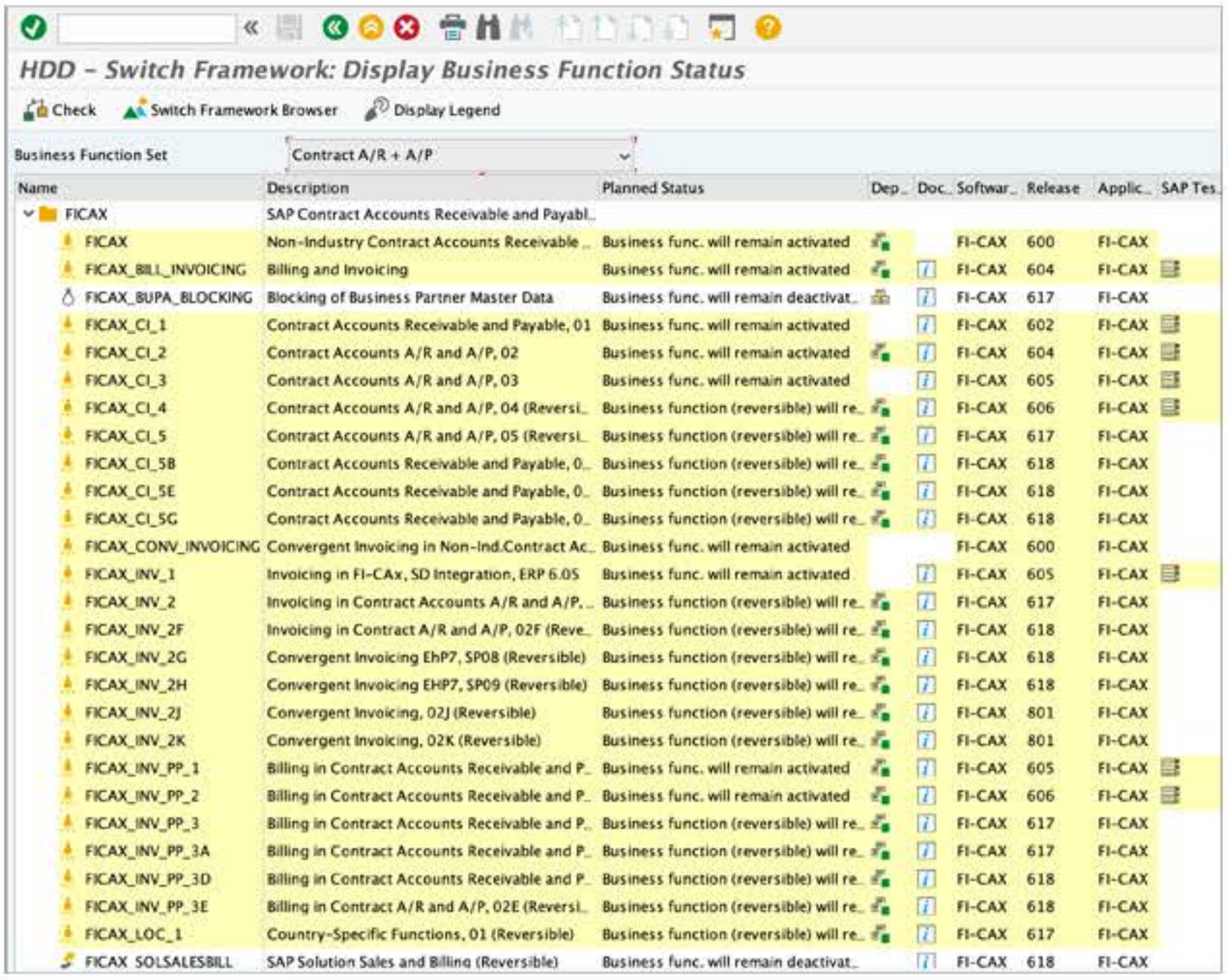
Chapter 4

Invoicing

This chapter explains how to set up and use SAP Convergent Invoicing. It also provides a brief overview of the master data and transactional data needed for the case study described in Chapter 1. We'll also discuss billable items, consumption items, rating, billing, invoicing, dependent items, partner settlement, and reporting.

SAP Convergent Invoicing is a component of SAP Billing and Revenue Innovation Management, which provides comprehensive capabilities to support business processes from inception and storage of usage/consumption records through to billing and invoicing. SAP Convergent Invoicing is thoroughly integrated with contract accounting.

Configuration nodes for SAP Convergent Invoicing are available in Transaction SPRO under **SAP Customizing Implementation Guide • Financial Accounting • Contract Accounts Receivable and Payable**. The business functions associated with SAP Convergent Invoicing, shown in Figure 4.1, must be activated to enable SAP Convergent Invoicing functionalities.



Name	Description	Planned Status	Dep...	Doc...	Softwar...	Release	Applic...	SAP Tes...
FICAX	SAP Contract Accounts Receivable and Payabl...							
FICAX	Non-Industry Contract Accounts Receivable ...	Business func. will remain activated			FI-CAX	600	FI-CAX	
FICAX_BILL_INVOICING	Billing and Invoicing	Business func. will remain activated			FI-CAX	604	FI-CAX	
FICAX_BUPA_BLOCKING	Blocking of Business Partner Master Data	Business func. will remain deactivat...			FI-CAX	617	FI-CAX	
FICAX_CI_1	Contract Accounts Receivable and Payable, 01	Business func. will remain activated			FI-CAX	602	FI-CAX	
FICAX_CI_2	Contract Accounts A/R and A/P, 02	Business func. will remain activated			FI-CAX	604	FI-CAX	
FICAX_CI_3	Contract Accounts A/R and A/P, 03	Business func. will remain activated			FI-CAX	605	FI-CAX	
FICAX_CI_4	Contract Accounts A/R and A/P, 04 (Reversi...	Business function (reversible) will re...			FI-CAX	606	FI-CAX	
FICAX_CI_5	Contract Accounts A/R and A/P, 05 (Reversi...	Business function (reversible) will re...			FI-CAX	617	FI-CAX	
FICAX_CI_5B	Contract Accounts Receivable and Payable, 0...	Business function (reversible) will re...			FI-CAX	618	FI-CAX	
FICAX_CI_5E	Contract Accounts Receivable and Payable, 0...	Business function (reversible) will re...			FI-CAX	618	FI-CAX	
FICAX_CI_5G	Contract Accounts Receivable and Payable, 0...	Business function (reversible) will re...			FI-CAX	618	FI-CAX	
FICAX_CONV_INVOICING	Convergent Invoicing in Non-Ind.Contract Ac...	Business func. will remain activated			FI-CAX	600	FI-CAX	
FICAX_INV_1	Invoicing in FI-CAx, SD Integration, ERP 6.05	Business func. will remain activated			FI-CAX	605	FI-CAX	
FICAX_INV_2	Invoicing in Contract Accounts A/R and A/P, ...	Business function (reversible) will re...			FI-CAX	617	FI-CAX	
FICAX_INV_2F	Invoicing in Contract A/R and A/P, 02F (Reve...	Business function (reversible) will re...			FI-CAX	618	FI-CAX	
FICAX_INV_2G	Convergent Invoicing EhP7, SP08 (Reversible)	Business function (reversible) will re...			FI-CAX	618	FI-CAX	
FICAX_INV_2H	Convergent Invoicing EHP7, SP09 (Reversible)	Business function (reversible) will re...			FI-CAX	618	FI-CAX	
FICAX_INV_2J	Convergent Invoicing, 02J (Reversible)	Business function (reversible) will re...			FI-CAX	801	FI-CAX	
FICAX_INV_2K	Convergent Invoicing, 02K (Reversible)	Business function (reversible) will re...			FI-CAX	801	FI-CAX	
FICAX_INV_PP_1	Billing in Contract Accounts Receivable and P...	Business func. will remain activated			FI-CAX	605	FI-CAX	
FICAX_INV_PP_2	Billing in Contract Accounts Receivable and P...	Business func. will remain activated			FI-CAX	606	FI-CAX	
FICAX_INV_PP_3	Billing in Contract Accounts Receivable and P...	Business function (reversible) will re...			FI-CAX	617	FI-CAX	
FICAX_INV_PP_3A	Billing in Contract Accounts Receivable and P...	Business function (reversible) will re...			FI-CAX	617	FI-CAX	
FICAX_INV_PP_3D	Billing in Contract Accounts Receivable and P...	Business function (reversible) will re...			FI-CAX	618	FI-CAX	
FICAX_INV_PP_3E	Billing in Contract A/R and A/P, 02E (Reversi...	Business function (reversible) will re...			FI-CAX	618	FI-CAX	
FICAX_LOC_1	Country-Specific Functions, 01 (Reversible)	Business function (reversible) will re...			FI-CAX	617	FI-CAX	
FICAX_SOLSALESBILL	SAP Solution Sales and Billing (Reversible)	Business func. will remain deactivat...			FI-CAX	618	FI-CAX	

Figure 4.1 Business Functions

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SAP Convergent Invoicing is part of the consume-to-cash process (see Figure 4.2). It provides the ability to integrate processes enabled using SAP Convergent Charging; SAP Billing and Revenue Innovation Management, subscription order management; and contract accounting. SAP Convergent Invoicing primarily focuses on capabilities involving grouping and aggregation of business transactions (billable items), creation of billing documents (including billing documents from external sources), and creation of consolidated invoice documents. It provides the ability to store and manage rated consumption data records from the SAP Convergent Charging system natively. Recurring billable items can also be created in SAP Convergent Invoicing using billing plans. Rated consumption data records are also known as billable items and undergo a billing and invoicing process with posting of receivables in contract accounting.

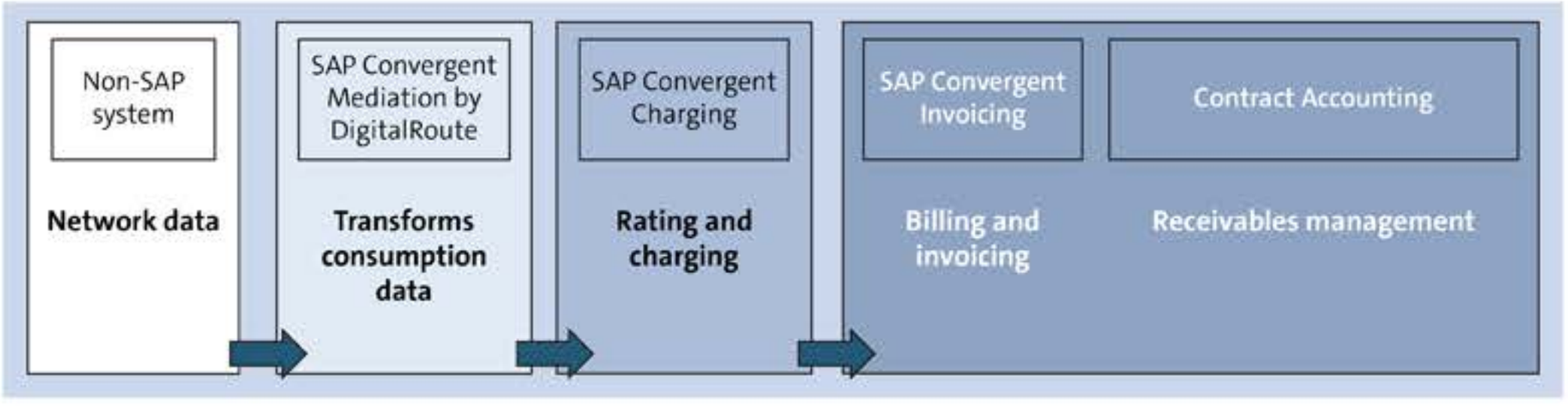

Figure 4.2 Consume-to-Cash Process

SAP Convergent Invoicing can perform grouping and aggregation at various stages of a transaction record, from rating of consumption data records to the invoicing process, which significantly reduces the storage of data. It can also be used to integrate with other non-SAP applications to receive billable transactions and perform grouping and aggregation per business requirements and present a consolidated invoice to the customer. You can invoice billing documents originating from multiple external systems, including sales and distribution and billing documents generated within SAP Convergent Invoicing.

4.1 Simplified Billing and Invoicing

SAP Convergent Invoicing integrates with SAP Billing and Revenue Innovation Management, subscription order management, and SAP Convergent Charging to enable billing and invoicing capabilities. SAP Convergent Invoicing enables grouping and aggregation of charges typically triggered for usage (from SAP Convergent Charging) and recurring charges (from SAP Convergent Invoicing) using configurable business rules. SAP Convergent Invoicing can also consolidate billing documents from sales and distribution and billing documents from SAP Convergent Invoicing to provide a unified invoicing experience to the customer. SAP Convergent Invoicing can also combine billing documents from different billing systems and consolidate them into a single customer

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invoice for a better customer experience. As shown in Figure 4.3, some of the key capabilities of SAP Convergent Invoicing include consolidated billing and invoicing, partner settlement statements, aggregation and grouping, and invoice localization.

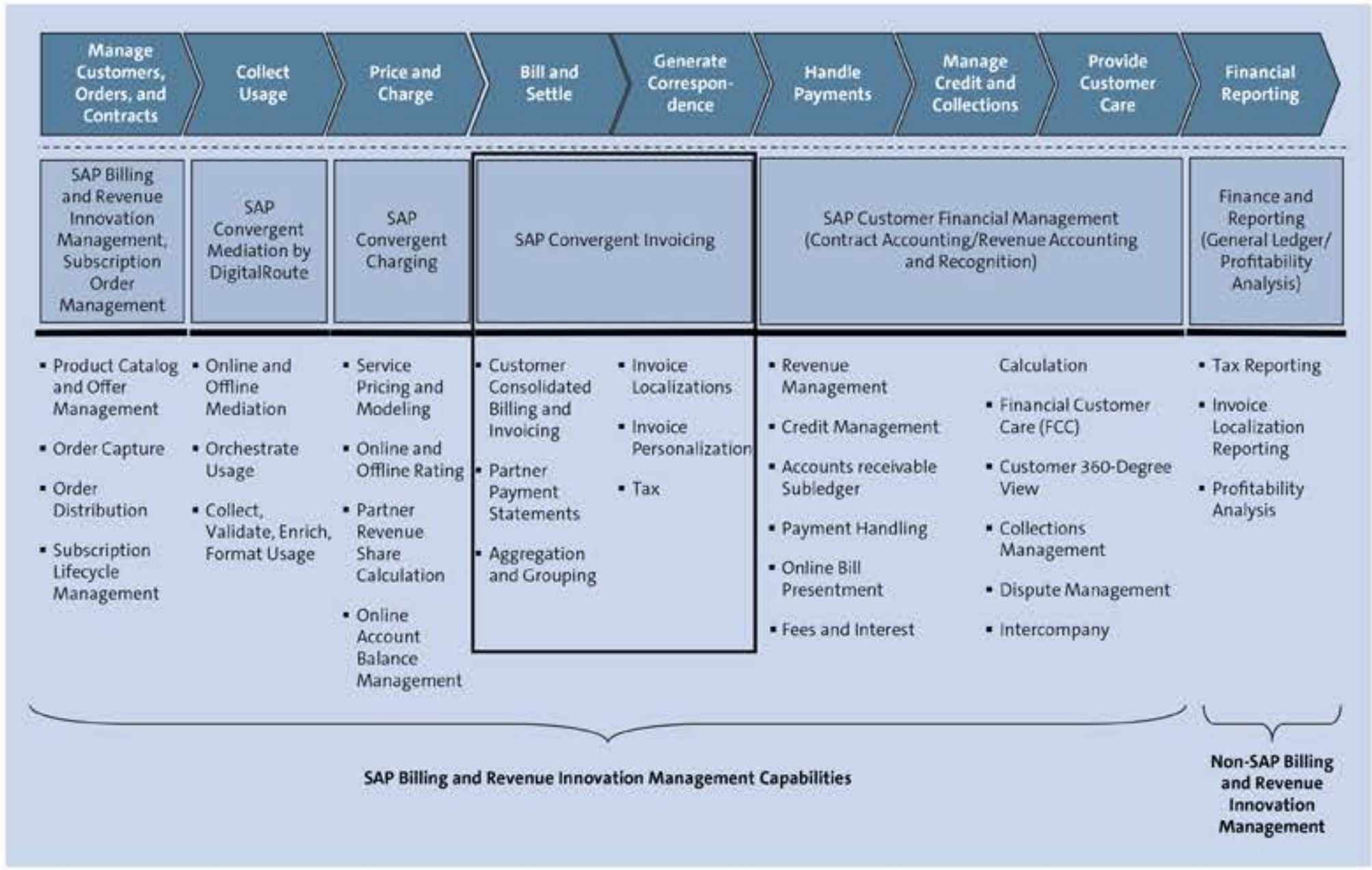

Figure 4.3 SAP Billing and Revenue Innovation Management Capabilities: SAP Convergent Invoicing

SAP Convergent Invoicing enables the following key capabilities:

- Store rated consumption data.
- Store billable transactions.
- Bill usage-based charges, recurring charges, and one-time charges.
- Generate consolidated invoices.
- Post accounting documents to contract accounting.
- Perform exception handling.

4.2 Master Data

As part of this chapter, you'll learn about various master data elements—including business partners, products, and contract accounts. Business partners contain details maintained at the SAP client level, and contract accounts have attributes maintained at the company code level. With SAP S/4HANA, customer, vendor, and business partner master data are now maintained centrally as business partner data.

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4.2.1 Business Partner

Any organization, group, or individual person with a business relationship with the company is represented as a business partner in SAP Billing and Revenue Innovation Management. Details such as names, correspondence addresses, fax numbers, telephone numbers, email addresses, bank information, and payment card information are stored in the business partner master data.

Configurations associated with business partner setup are maintained under **Cross-Application Components • SAP Business Partner • Business Partner**.

Business partner master data is created in contract accounting and can have various roles based on its usage. Business partner roles are assigned to specify their functions (see Figure 4.4). These roles are defined in Customizing and are assigned to a business partner at the time of its creation.

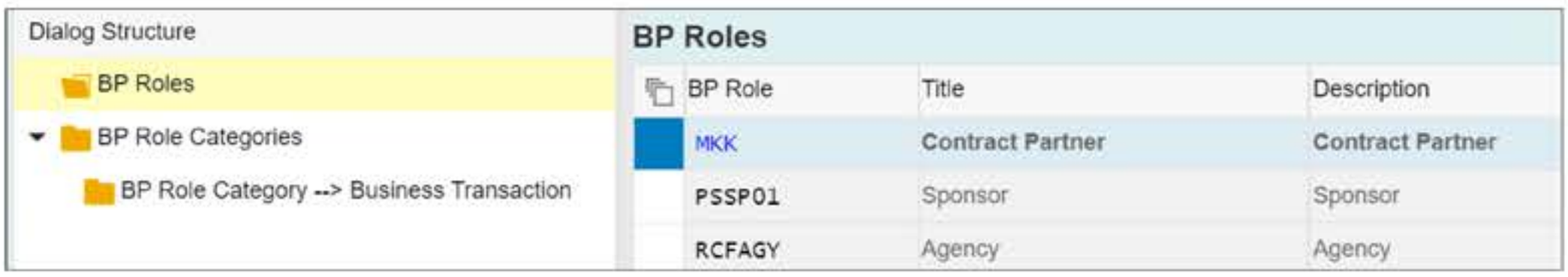

Figure 4.4 Business Partner Roles

Business partner roles are maintained under **Cross-Application Components • SAP Business Partner • Business Partner • Basic Settings • Business Partner Roles • Define BP Roles**.

A business partner can have role `FLCU000` to integrate sales and distribution with contract accounting. A business partner must have role `MKK` (Contract Partner) to post business transactions to it.

Business partners either can be created internally within SAP S/4HANA and replicated to other SAP systems such as SAP Customer Relationship Management (SAP CRM) or can be created externally from SAP CRM and replicated to SAP S/4HANA systems.

Each business partner receives a number on creation. The business partner number assignment is controlled by maintaining the number ranges for the business partner. Business partners created through an external system such as SAP CRM will have number ranges created as external in SAP S/4HANA, and those created internally within SAP S/4HANA will have internal number ranges (see Figure 4.5).

The number ranges created are assigned to the business partner grouping and are maintained under **Cross-Application Components • SAP Business Partner • Business Partner • Basic Settings • Number Ranges and Grouping** (see Figure 4.6).

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Edit Intervals: Business partner, Object BU_PARTNER

Menu | Back | Exit | Cancel | System | Display <-> Change | Free Intervals

No	From No.	To Number	NR Status	Ext
01	0000130000	0000199999	0	☑
03	0000500000	0000599999	500019	☐
AB	A	ZZZZZZZZZZ	0	☑
MD	9000000000	9999999999	0	☑
TE	1000000000	1099999999	1000000319	☐
Y1	1100000000	1199999999	0	☑
Y2	1200000000	1299999999	1200000079	☐
Y5	1300000000	1399999999	1300000033	☐
ZE	0000000001	0000000010	0	☑

Figure 4.5 Business Partner Number Ranges with 01 as External and 03 as Internal

Grouping	Short name	Description	Number ran...	External	Int.Std.Grping	Ext.Std Grping	H
0001	Int. No.Assgnmt	Internal number assignment	TE	☐	◉		
0002	Ext.No.Assgnmnt	External Number Assignment	AB	☑		◉	

Figure 4.6 Business Partner Grouping and Number Range Assignment

During the time of creation of a business partner, the business partner **Grouping** field value is assigned. A business partner number is generated based on the assigned number range for the business partner group.

4.2.2 Product

Products are goods and services sold to the customers or subscribed to by customers, through which a company generates its revenue. Products can be tangible, such as cell phones or PlayStation systems, or they can be intangible, such as software subscriptions, maintenance services, or bandwidth usage. Intangible products or subscription-based products are created in contract accounting, and tangible products are maintained in materials management, with sales handled in sales and distribution.

4.2.3 Contract Account

A contract account is another component of business master data. It's used to group together the business partner postings. All accounting transaction data associated with subscription services, payments, returns, interest calculations, and dunning is recorded in the contract account.

The contract account contains the parameters that determine the business partner posting details, such as payment terms, incoming/outgoing payment methods, taxes,

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and discounts. The contract account provides a business partner's account management functionality. One business partner can hold multiple contract accounts according to business needs.

Configurations for contract accounts are maintained under **Contract Account Receivables and Payables • Basic Functions • Contract Accounts**.

Contract Account Number Ranges

Each contract account created receives a unique two-digit number based on the number ranges maintained in the configuration. Each number range is maintained as either internal or external. Contract account creation or business agreement creation from SAP CRM uses external number ranges for SAP S/4HANA, and direct creation of a contract account in SAP S/4HANA uses internal number ranges (see Figure 4.7).

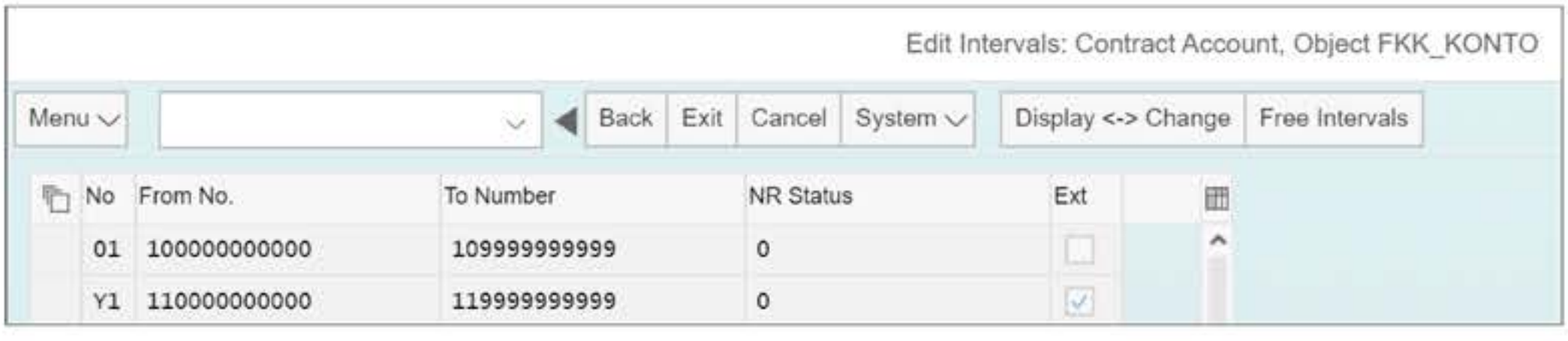

Figure 4.7 Number Ranges for Contract Account Creation

Contract Account Categories

Contract account categories are used to categorize the contract accounts. Each contract account created is assigned a contract account category. Number ranges created for the contract accounts are assigned to the contract account categories. Contract account categories control the number range assignment, specify whether the account is one time, determine if one or more business partners can be assigned, indicate whether the account is a collective account, denote whether it can hold one or more contracts, and mark if the account is a customer or vendor type (see Figure 4.8).

Figure 4.8 Contract Account Categories

Contract Account Relationship

A contract account relationship describes if the assigned business partner of the contract account is the account holder or not. The contract account relationship is assigned at the time of creation of the contract account (see Figure 4.9).

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Figure 4.9 Contract Account Relationship

Account Determination Characteristics

Account determination IDs, along with company code, division, main transaction, and sub-transactions, are used for general ledger account determination for contract accounting document postings (see Figure 4.10).

Figure 4.10 Account Determination IDs for Contract Account

4.3 Configuring SAP Convergent Invoicing

This section will cover the concept of billable items, the configuration required for their setup, and how to manage them. When you finish this section, you'll understand the definition and use of the various billable items objects and will be able to configure SAP Convergent Invoicing mappings to meet business requirements.

4.3.1 Billable Items

Billable items are rated consumption data records. Billable items hold the information for the sale of a product or service and are used to post the revenue generated by the service provider for the use of a service by a customer. These items can originate from SAP Convergent Charging, an external rating system, or even from within SAP Convergent Invoicing. Billable items can be recurring charges from SAP Convergent Charging; one-time charges originating from SAP Billing and Revenue Innovation Management, subscription order management; or rated usage consumption items.

Billable items record the details of the services used by a customer based on the contractual agreement. They are the input for the billing process for the generation of billing documents, which are inputs for the invoicing process. Billable items hold information such as product details, amount of usage, quantity, period of usage, and other details that determine the accounting posting during the invoicing process.

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This section covers the various configurations required to set up the SAP Convergent Invoicing system, such as billable item/consumption item class creation, billable item types, subprocesses, and various other configurations required for billing and invoicing processes.

Billable Item Statuses

Billable items obtain different statuses, starting with creation and going to the invoicing process, as shown in Table 4.1.

Billable Item Status	Description
0	Raw
1	Raw-Excepted
2	Billable
3	Billable-Excepted
4	Billed

Table 4.1 Billable Item Statuses

Let's take a deeper look at each status:

- **Raw**
  The billable item doesn't contain complete information, and values in the fields can be changed. Raw billable items aren't eligible for the billing process. Raw billable items should be transferred to status **2** (**Billable**) to be considered for the billing process.
- **Raw-Excepted**
  The billable items aren't considered during transfer or billing. Such items should be restored and then transferred to **Billable** status to be considered for further processing.
- **Billable**
  The billable items only are considered for the billing process. The **Billable** status indicates the information contained in the billable items is complete and consistent.
- **Billable-Excepted**
  The billable items get this status when a billing document is reversed. Billable items then should be restored if they need to be considered in the billing process.
- **Billed**
  The billable items get this status when the billing process is performed successfully on them.

Billable items are stored in separate database tables based on their statuses. There are four types of billable items:

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- **Main items (IT)**
  Main items are mandatory and contain the basic data for billing. Other record types are dependent on the main items, without which the system won't process the other record types.
- **Payment data (PY)**
  Payment data record types contain the details associated with payments and are linked to the corresponding main item by the **Payment Group** field.
- **Tax item (TX)**
  Tax items contain data associated with taxes and are linked by the **Tax Group** field with the main item.
- **Text items (TT)**
  Text items contain the test fields for the main items and are linked by the **Text Group** field.

Different record types are stored in different database tables. While creating the billable item class, the interface components are selected for the class, and upon generation and activation of the interface for the billable item class, the database table is created for each record type and for each status of the billable items. The naming conventions for the tables are described in in Figure 4.25 in the "Billable Item Management" subsection later in this chapter.

Billable Item Class

The billable item class specifies the technical attributes for the billable items. SAP doesn't provide standard classes; they are created based on your business requirements. Let's look at an example.

Martex Corp. sells software as a service (SaaS) and infrastructure as a service (IaaS) offering to customers obtained from multiple vendors. The company provides two types of services, A and B, for SaaS for cloud computing. Martex provides a reserved instance option to its customers, in which they can pay a recurring charge in exchange for lower usage rates. The services details are as follows:

- Service A:
  - $300/GB without reserved instance
  - $250/GB with reserved instance
- Service B:
  - $20/hour without reserved instance
  - $10/hour with reserved instance

Let's set up the configuration for Martex to record the service usage to perform billing and invoicing for a customer. To record the revenue from the sales of service and cost of products obtained from the vendors, let's create the billable item class `HREV` for revenue and billable item class `HCOS` for the product cost.

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Billable item classes are created under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Item Classes • Maintain Billable Item Classes** (see Figure 4.11).

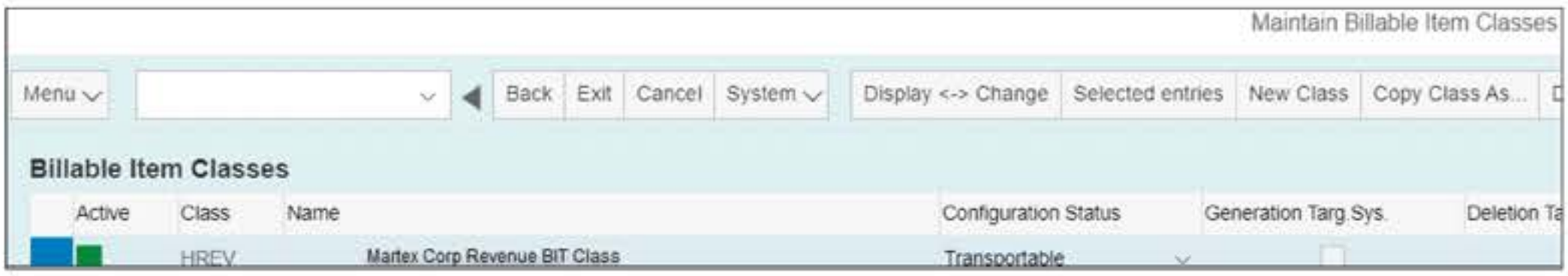

Figure 4.11 Billable Item Class Creation

SAP provides a set of standard interface components, which are assigned to a class at the time of its creation. Interface components contain technical attributes such as fields; structure definitions, which upon class activation and generation create the associated function modules for receiving data and storing data; database tables according to record type and billable item statuses; and standard fields for storing data in the billable items of the associated class. Custom fields can also be added to the class and will be extended to the billable items of the class upon creation.

Interfaces can be viewed under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Item Classes • Define Interface Components**. Interfaces selected for the billable item class, as shown in Figure 4.12, provide the fields for storing the data for the billable item class.

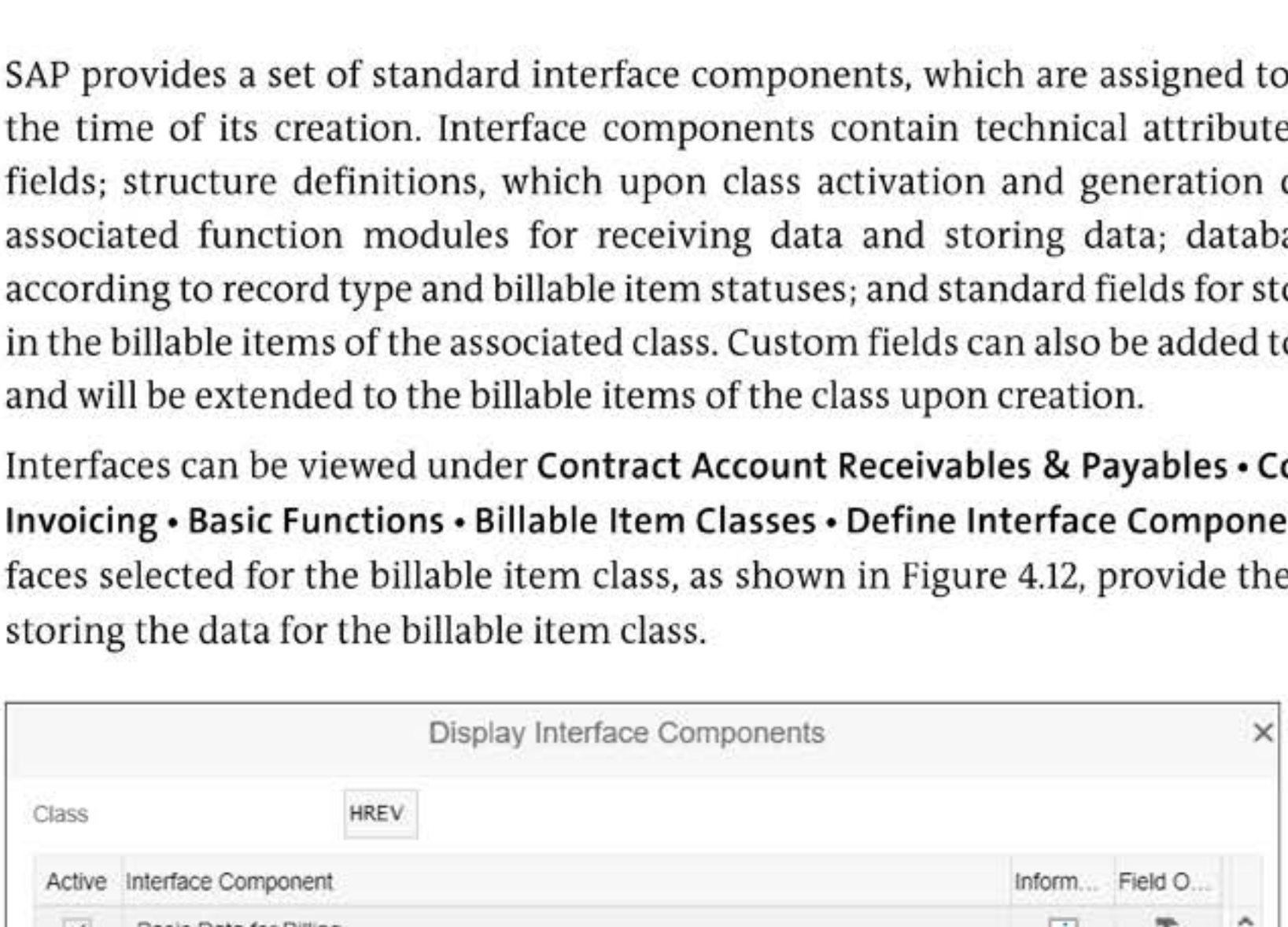

Figure 4.12 Interface Components Selected for the HREV Billable Item Class

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Upon creation of the billable item class and interface assignment, the billable item class needs to be activated, and then generation of the interface components of the class needs to be performed to generate the database tables, billable item creation application programming interface (API), and structures for the billable item class, which then can be used for storing the transactional data. Billable item classes can be activated under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Item Classes • Maintain Billable Item Classes** (see Figure 4.13).

Figure 4.13 Billable Item Class Activation

The generation of the billable item class interface components is performed under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Item Classes • Generate Interfaces for the Billable Item Classes** (see Figure 4.14).

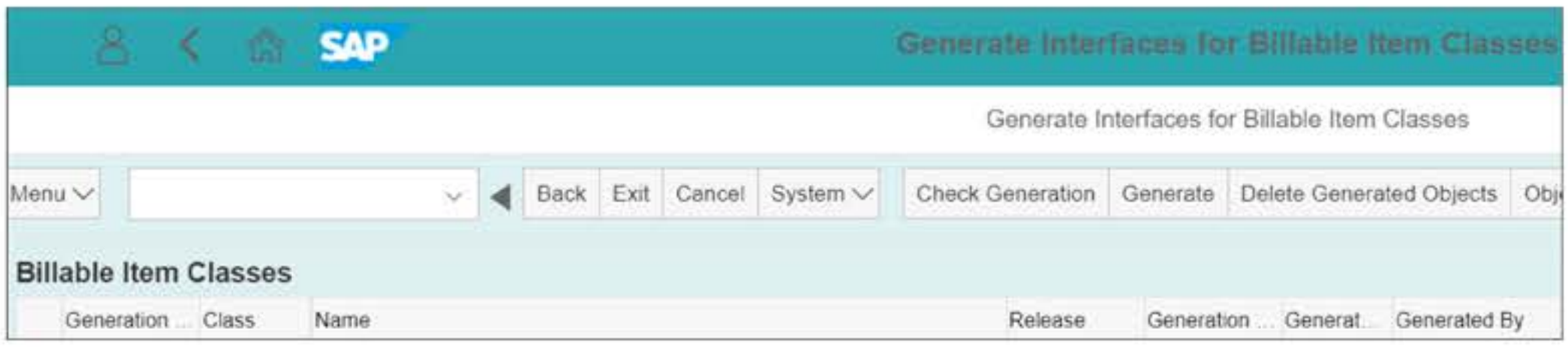

Figure 4.14 Billable Item Class Generation

Upon generation of the interfaces of the billable item class, the tables, function module, tables for billable items for each record type and status, and table for simulation records and structures will be generated (Figure 4.15).

Similarly, class `HCOS` is also created to store the cost details of products obtained from the vendors. Different classes are generally created according to business requirements, keeping in mind that if the same class is used for several products and services, then billable items might end up with a very long list of fields that aren't required for certain billable items for the products.

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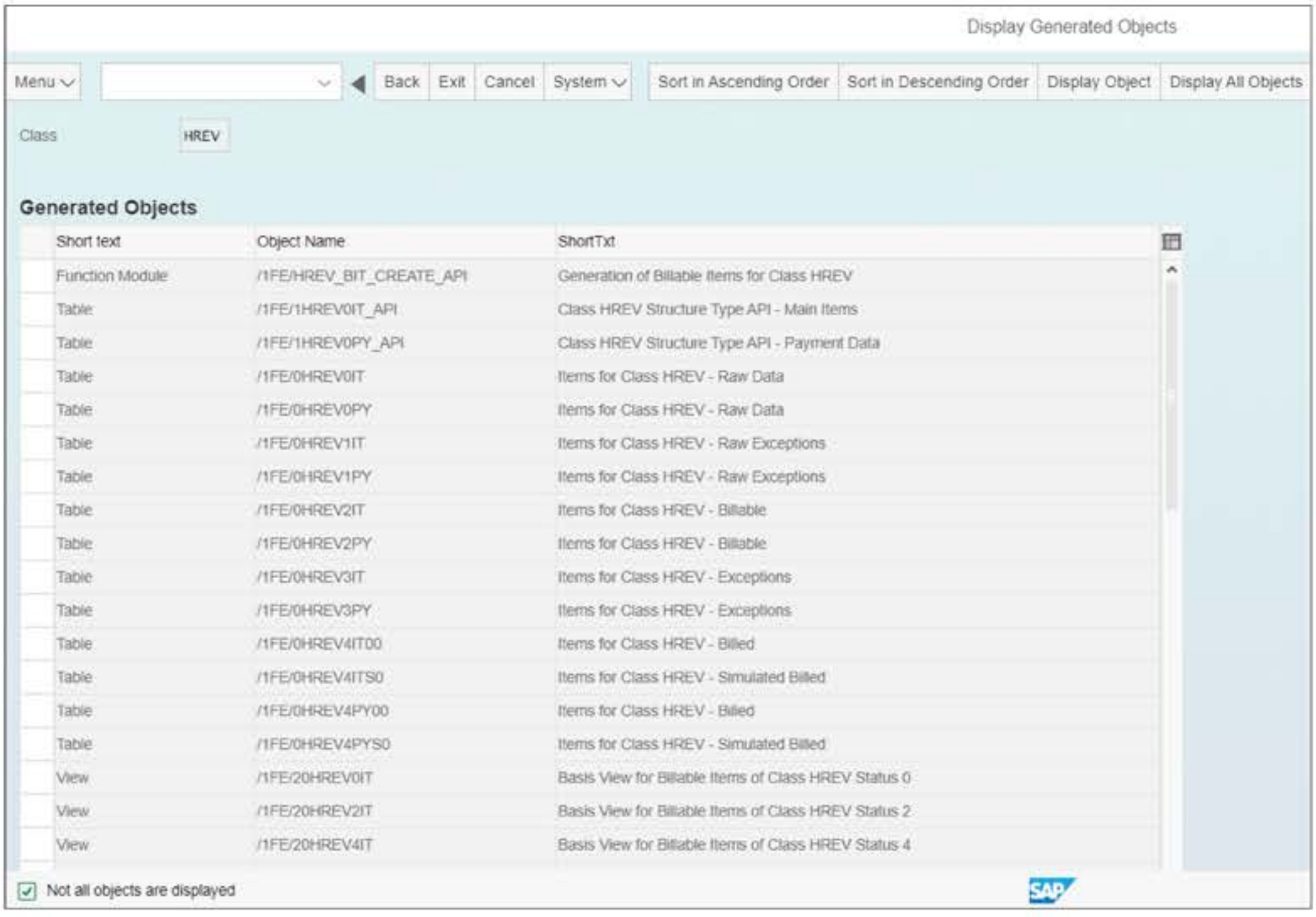

Figure 4.15 Objects Generated for the HREV Billable Item Class

A billable item class can be created or modified under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Item Classes • Maintain Billable Item Classes** (see Figure 4.16).

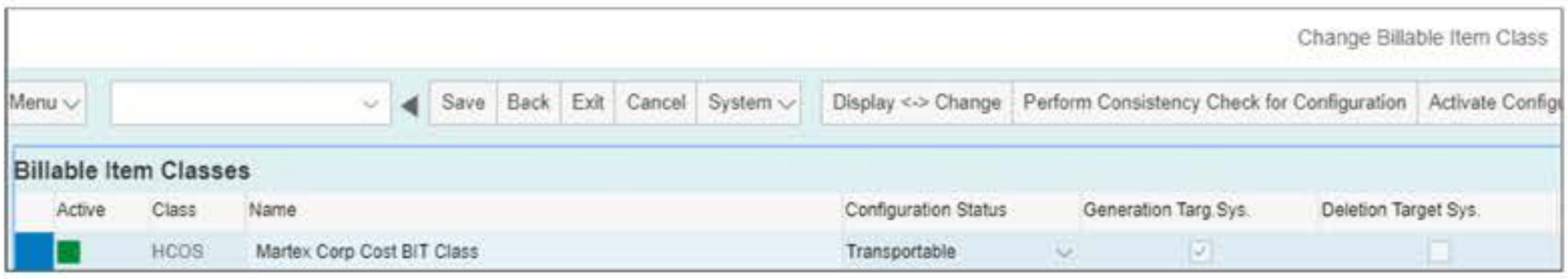

Figure 4.16 HCOS Billable Item Class for Cost

Billable Item Class Revenue

A billable item class is created to store the transaction details for the usage or sale of a service. The business requirement mapping determines the creation and use of a billable item class.

Billable item class revenue signifies that the created class will be used to record the details of business transactions relevant for the revenue generated by the sale or use of a service. The billable items described in Section 4.3.1 hold the information for the sale transaction of the service for the associated class.

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In our business example, to record the revenue, billable item class HREV is created.

Billable Item Class Cost

A billable item class is created to store the cost to the company that occurs when a product is obtained from the vendor and eventually sold to the customer. In the business example for Martex, the company purchases products from a vendor and sells them to customers. The cost incurred in obtaining the product from the vendor can be stored in the billable item class cost. For Martex, the billable item class cost created is HCOS.

Billable Item Management

Billable item management covers the management of rated consumption records, also known as billable items, and how the billable items are configured and used. Billable item storage, enrichment, and processing are covered under billable item management.

Billable item types are configured in the system to record the business transaction. Based on the type of transaction or charges, billable item types are created for a class.

After a class is activated and generated as mentioned earlier, the billable item types are created for the class. Billable item types hold business transaction details and differentiate business transactions such as sales, returns, royalty sharing, and so on.

Business Example

Martex, as described in earlier, provides services A and B with and without a reserved instance. For Martex, in a reserved instance scenario, it receives the up-front charge, monthly recurring charge, and usage charge. These three different types of charges need to be stored separately. To achieve this, three different billable item types can be created.

Let's create the three different billable item types for the billable item class HREV. Billable item type ZBT1 will be created to record the up-front one-time charge for the reserved instance scenario, billable item type ZBT2 will be created to record the recurring charges, and billable item type ZBT3 will be created for usage-based charges. Billable item types created for the class receive the business transaction details, which will be used in billing and invoicing.

Before billable item types can be created, a subprocess is created. A subprocess is a self-contained processing operation for a billing process. Subprocesses are assigned to a billable item class, and billable item types, once created, are assigned to subprocesses, thus linking the billable item types to a billable item class. Subprocesses select the billable items during billing process execution. They are created based on the business requirements for how a charge, a billable item type, should be processed during billing.

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Business Example

For Martex, create a subprocess ZHR1, and assign it to the billable item class HREV.

Subprocesses are maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Item Billable Item Management • Define Subprocess for Billing** (see Figure 4.17).

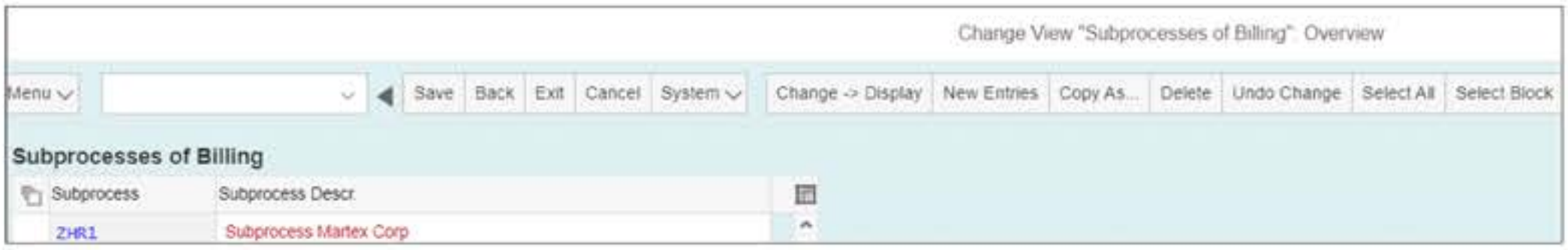

Figure 4.17 Subprocess Creation Screen

Once a subprocess is created, it's assigned to the billable item class. Assignment of a subprocess to a billable item class is done under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Item Billable Item Management • Assign Subprocesses to Classes** (see Figure 4.18).

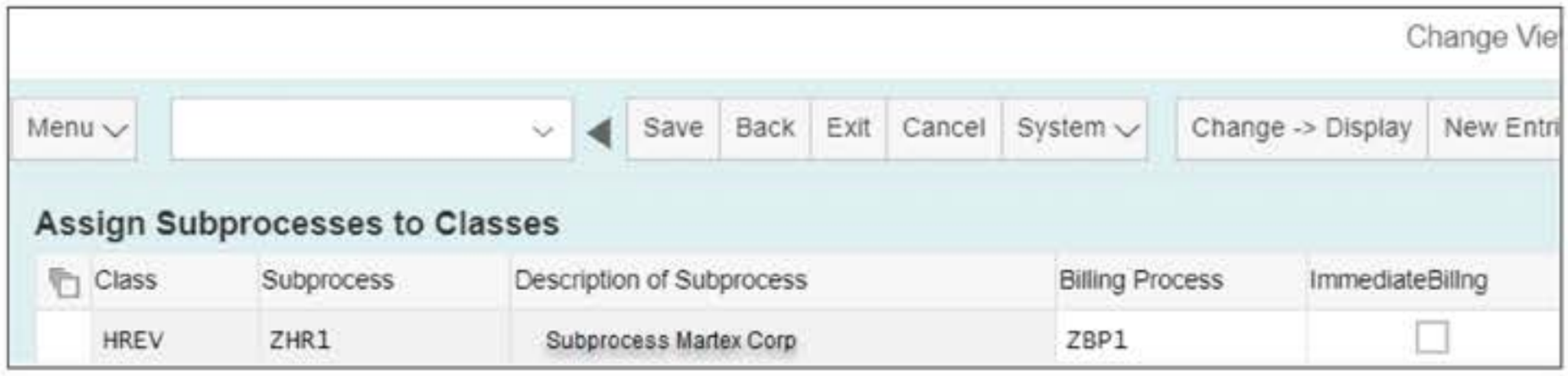

Figure 4.18 Subprocess Assignment to Billable Item Class

Once a subprocess is created and assigned to a billable item class, billable item types are created and assigned to the subprocess. Billable items are created under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Item Billable Item Management • Define Item Types** (see Figure 4.19).

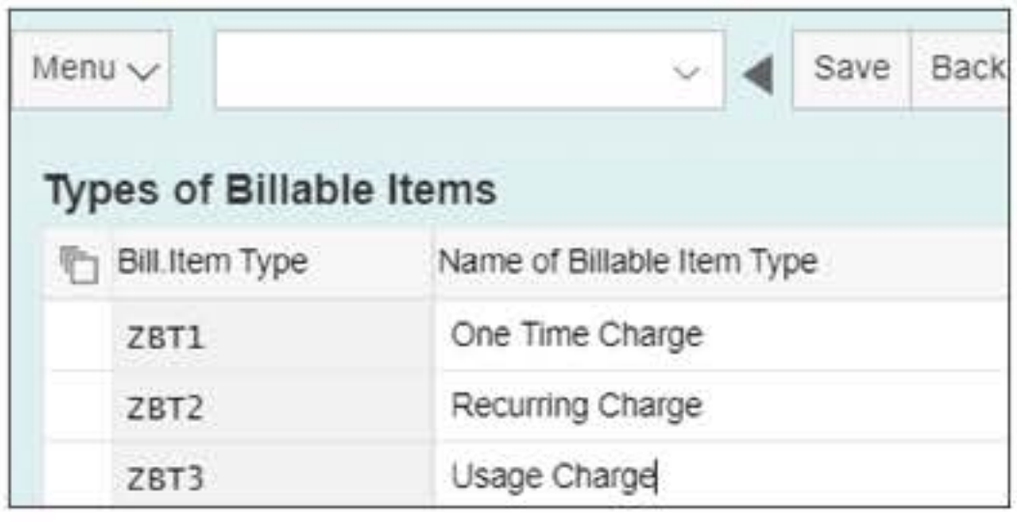

Figure 4.19 Billable Item Types

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Billable item type creation is followed by assignment of the billable items to the subprocess. Assignment of billable item types to a subprocess is done under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Item Billable Item Management • Assign Item Types to Subprocesses** (see Figure 4.20).

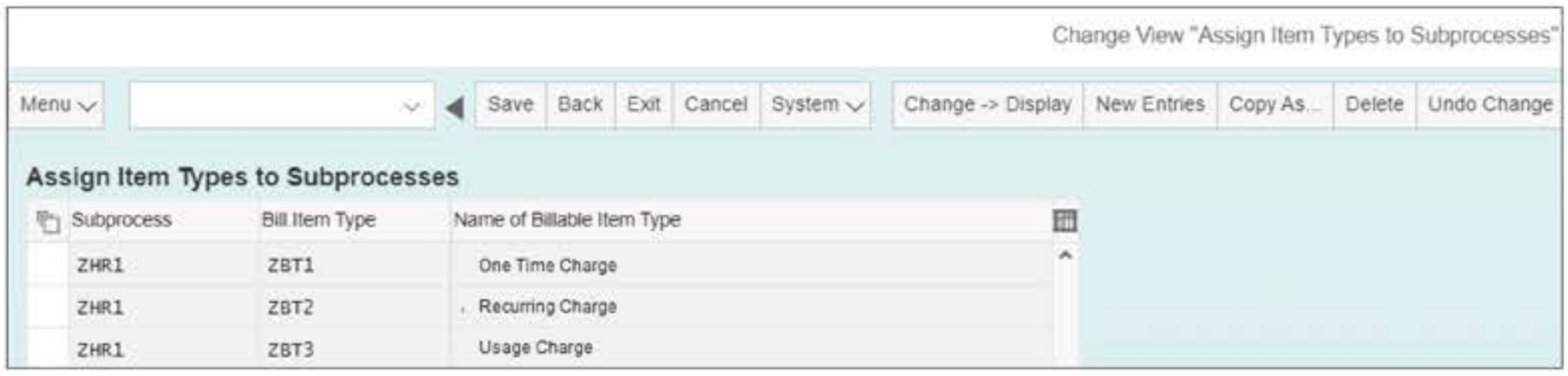

Figure 4.20 Billable Items Assignment to Subprocess

SAP Convergent Invoicing provides the functionality to receive the rated events from SAP Convergent Charging or from other external systems, store them in the form of billable items, and perform billing and invoicing. The API created at the time of generation of the billable item class is used to import data from the external systems and store it in the database tables generated as part of class generation.

Although interface components selected for the class generally provide all the necessary fields, you may need to create certain custom fields as well to meet specific business needs. The class also provides the option to add these custom fields. To assign the custom fields to the class, fields must be added to the Customizing includes before they can be added to the class (see Figure 4.21). The custom includes need to be generated before they can be used (see Table 4.2).

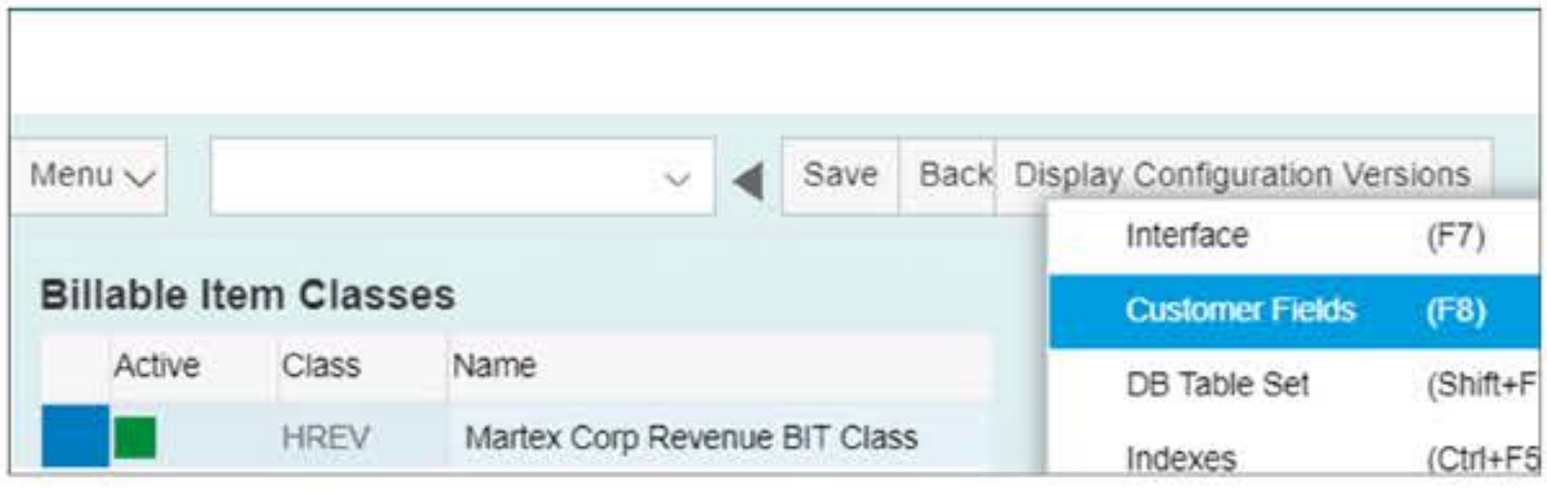

Figure 4.21 Custom Fields Option during Class Creation

Customizing Include	Description
CI_FKKBIXBIT_IT	Include for main items
CI_FKKBIXBIT_PY	Include for payment items
CI_FKKBIXBIT_TX	Include for tax items

Table 4.2 Customizing Includes for Billable Item Classes

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Billable item classes follow a sequence of creation, activation, and generation, but they can't be used to store business transaction data until they are made active for use. Activation of a billable item class for which billable items can be created is done via **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Item • Billable Item Management • Assign Item Types to Subprocesses • Activate Billable Item Classes** (see Figure 4.22).

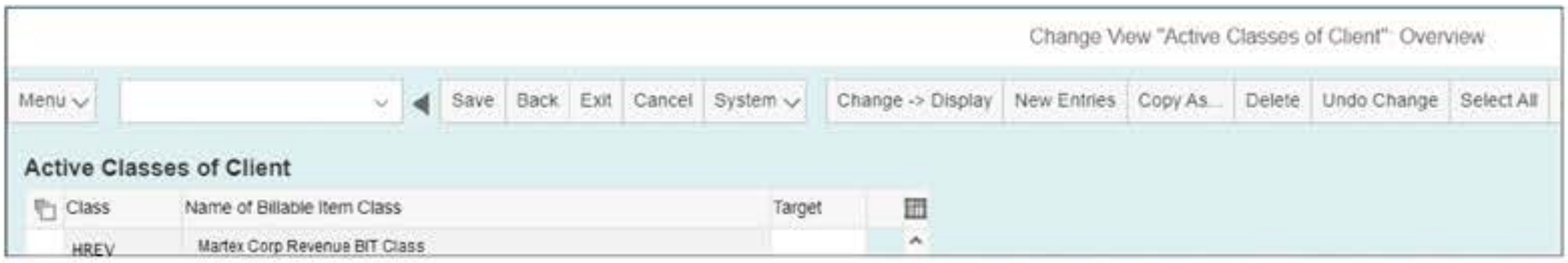

Figure 4.22 Billable ItemClass Activation for Use in Client

Billable items originating from SAP Convergent Charging or other external systems can be stored based on business needs. SAP provides various upload options to store the input billable items. In addition, SAP provides various standard events that can be used to perform checks or even perform enrichment of billable items, from receiving to saving the billable items in SAP Convergent Invoicing.

Storing and processing the input billable items is handled via **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Item Classes • Billable Item Management • Define Processing Rules and Program Enhancements** (see Figure 4.23).

Billable items received in SAP Convergent Invoicing can be stored based on the upload rule for the billable item class. All billable items can be stored in raw status; items with errors can be stored in raw status or in billable status. In addition, you could return the received data to the originating system if the data received is incomplete.

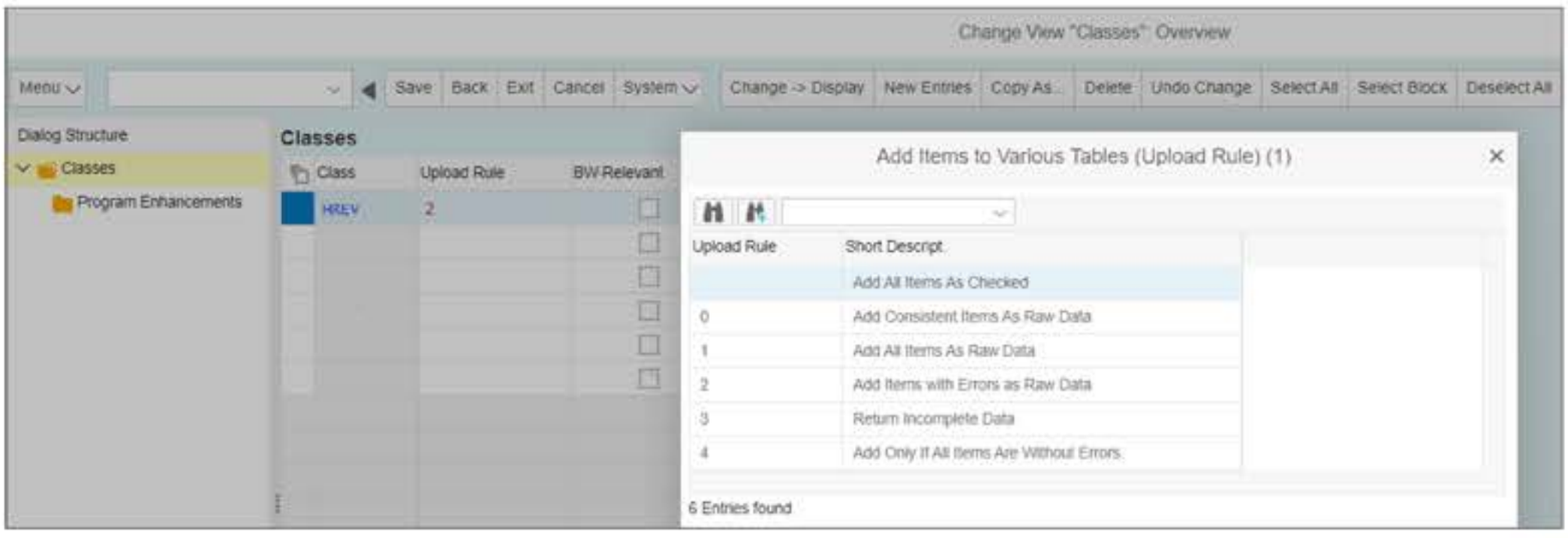

Figure 4.23 Billable Items: Processing Rules

In certain business scenarios, you must put a check on the billable item's field for validation of data. There can be scenarios in which the billable item fields may require data

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to be populated at the time of storing because the data required in the fields can only be obtained within SAP Convergent Invoicing; in such scenarios, program enhancements come into play.

SAP has provided 17 events to handle the various steps of transferring, storing, reversal, and exceptions for billable items. For each associated event, standard sample function modules are provided that can be used to create the custom function modules to code the requirements (see Figure 4.24).

For each billable item class activation and generation, database tables are created for the class for storing data. Tables are created for each record type, status, and simulation. The tables generated follow a pattern in their naming (see Figure 4.25).

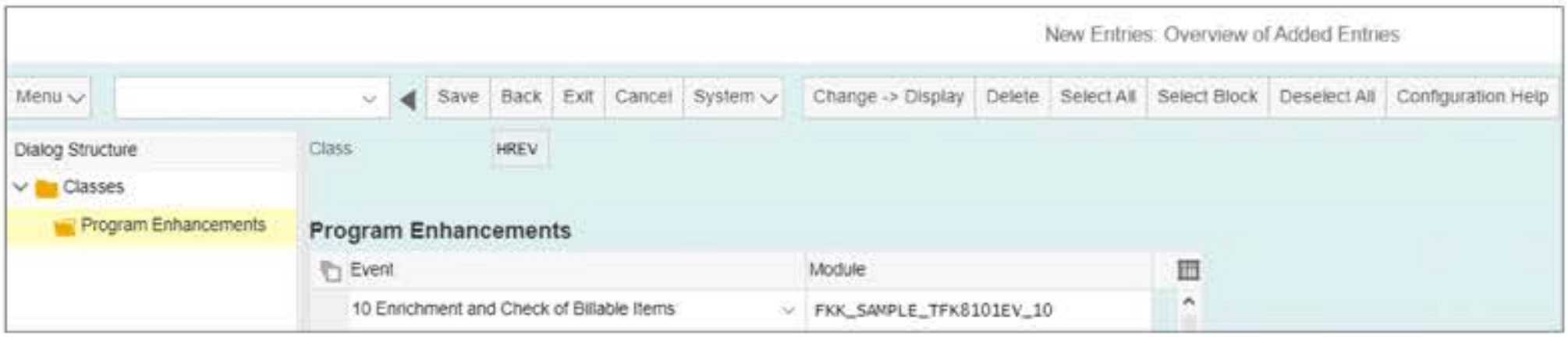

Figure 4.24 Billable Items Program Enrichment

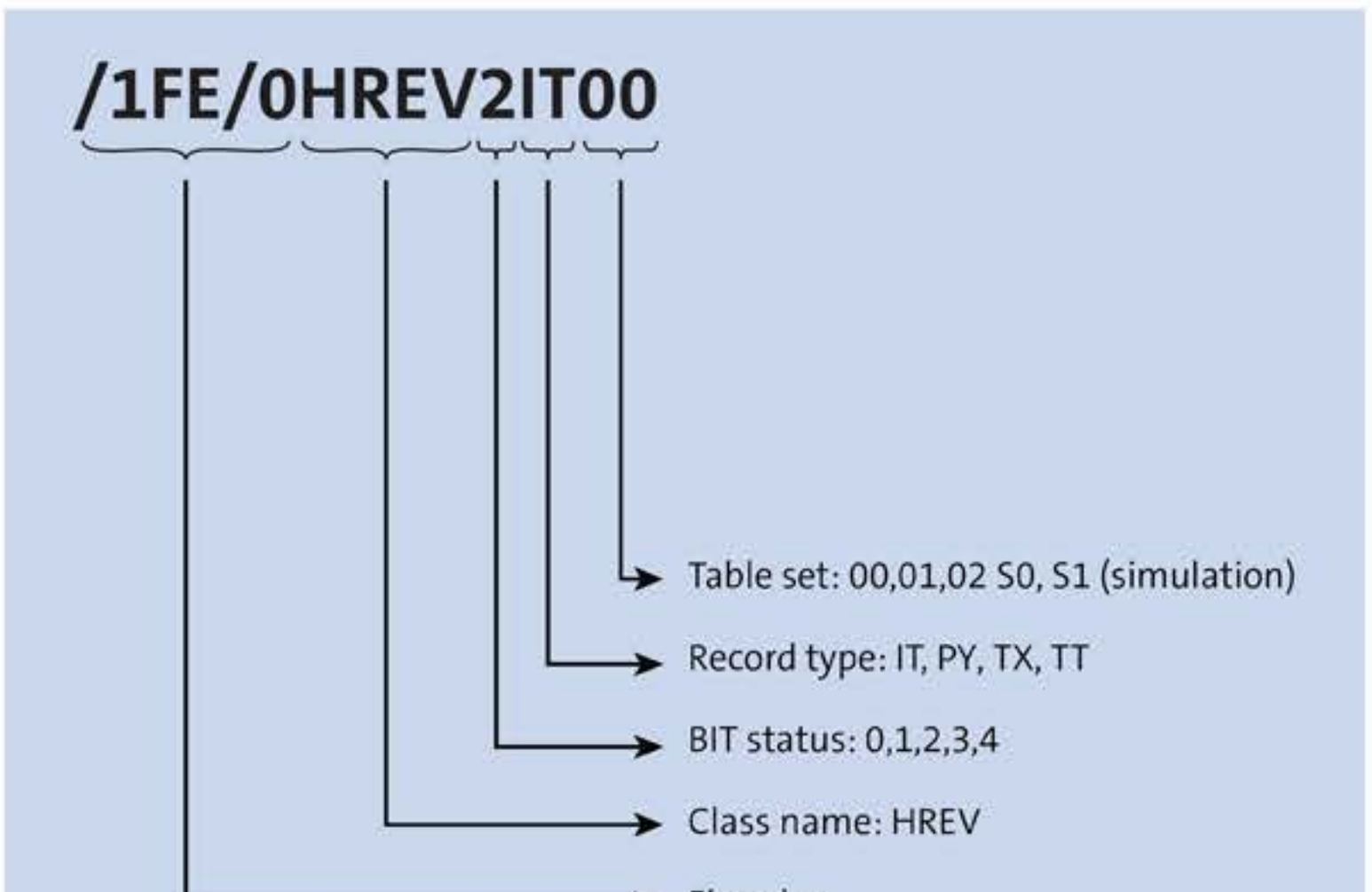

Figure 4.25 Billable Items Table Naming Convention

Billable items move from one table to another based on their status. Additional table sets can also be assigned for billed items and simulated items, and those table sets will become applicable based on the validity period of each table. A table set should always be assigned for both simulated and actual billed items.

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The additional tables are created in the billable item class in the creation screen. These tables are then further assigned to the billable item classes to be used based on the validity period (see Figure 4.26).

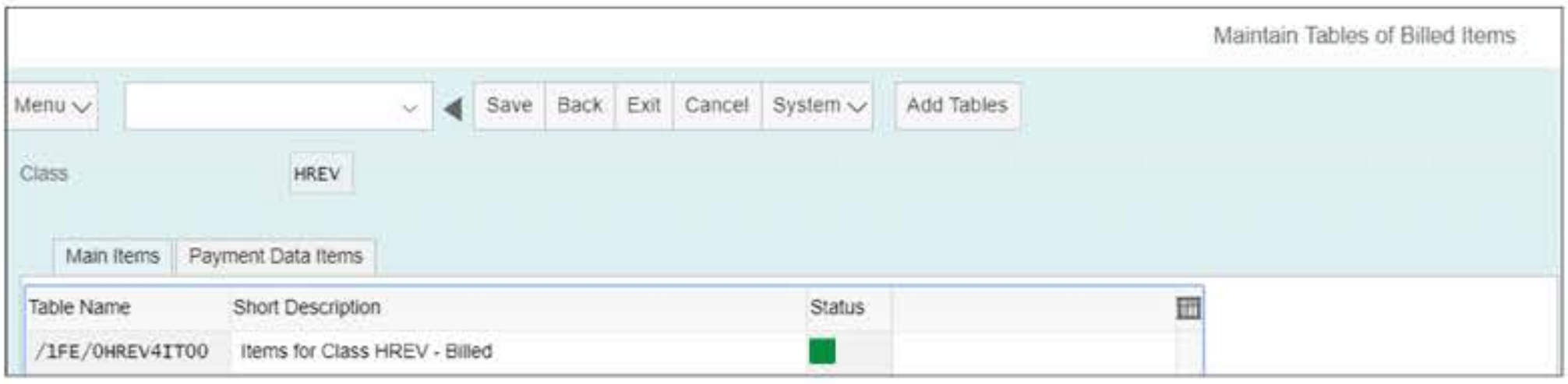

Figure 4.26 Additional Table Creation for Billable Item Class

Using the **Add Tables** button, you can add further tables to the class definition maintained at the node.

Additional tables created for the billable item class are set for data storage under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Item Classes • Billable Item Management • Make Setting for Data Storage** (see Figure 4.27).

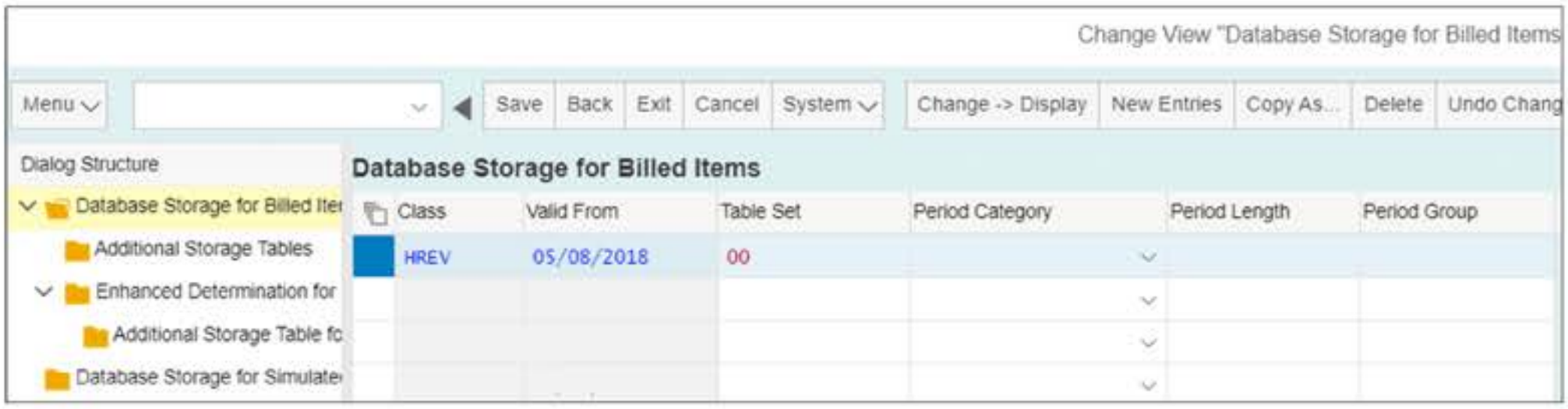

Figure 4.27 Additional Data Storage for Billed and Simulated Items

The default table set is 00 for billed item tables and S0 for simulated items. If additional tables are added, they follow the sequence 01 and S1 for billed and simulated items, respectively. The date available on the storage screen marks the beginning validity date of the table set.

Billable Item Transfer

Billable items originating from SAP Convergent Charging or another external system populate the billable item fields with most of the details required for the processing, but certain fields are required to be obtained within SAP Convergent Invoicing during the transfer of the billable items from the various systems. Various fields for account assignment, such as company code, main transactions and sub-transactions for credit and debit posting, and the payment method assignment for the payment records types

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of the billable item, are set during the billable items transfer. The configuration required for the account assignments is described in the following sections.

Account Assignments

Account assignment is the process of obtaining information such as company code, business area, profit center, main transactions and sub-transactions, and payment method, which will eventually help in the general ledger accounts determination during the invoicing process for posting of receivables in contract accounting. The account determination is essential for recording the sales, expenses, taxes, and payments in the SAP system.

Values for company code, division, business area, segment, and profit center are derived based on the value of the subprocess, billable item type, and set of four external key fields present on the billable items input to SAP Convergent Invoicing. The billable items received contain information about the subprocess and billable item types, based on which the configuration maintained in the posting areas will determine the account assignment values for those billable items.

The **Assignment of External/Internal Account Assignment or Receivable/Payables** interface component in the billable item class is responsible for the account assignment. **Posting area 8120** is maintained for the account assignment. In this posting area, you maintain the selection values for subprocess, billable item type, and, if required, four additional keys to derive account assignment values such as company code, division, business area, segment, and profit center (see Figure 4.28).

The posting area can be accessed using Transaction FQC0 or from menu path **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Items • Billable Item Transfer • Define Account Assignments** (see Figure 4.29).

Billable items during the transfer from an external system to SAP systems will fetch the value of account assignment fields from posting area 8120 based on the subprocess and billable item types. The values obtained will be used during the receivables posting in invoicing process execution.

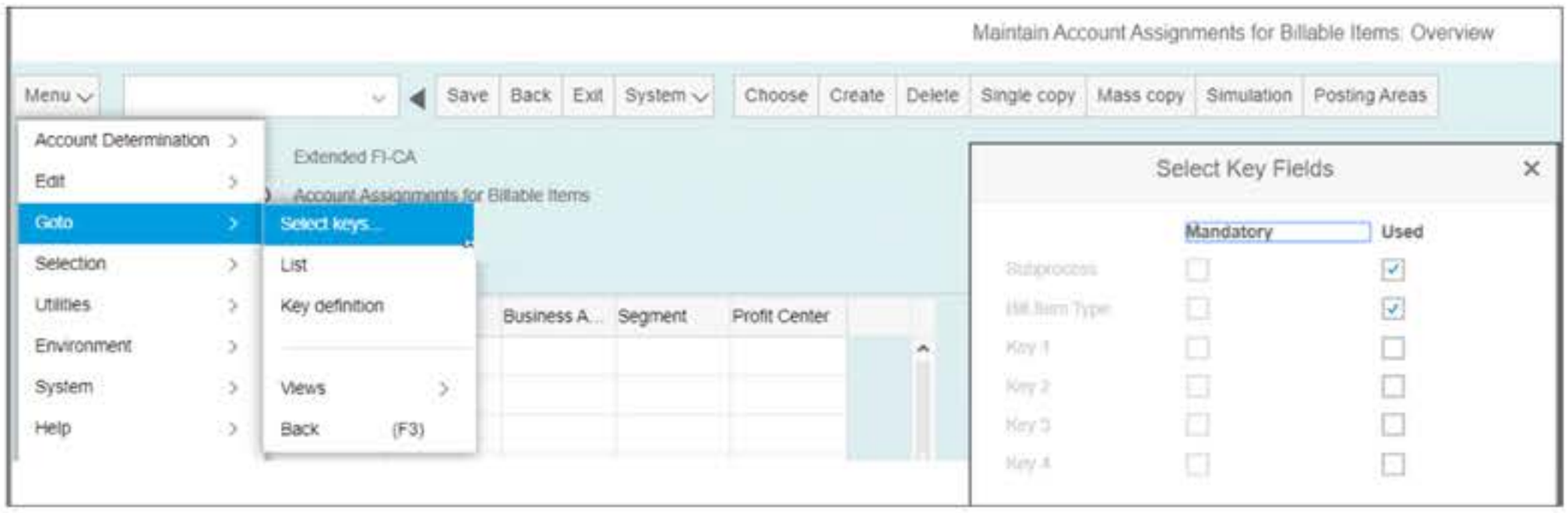

Figure 4.28 Selection Keys for Posting Area 8120

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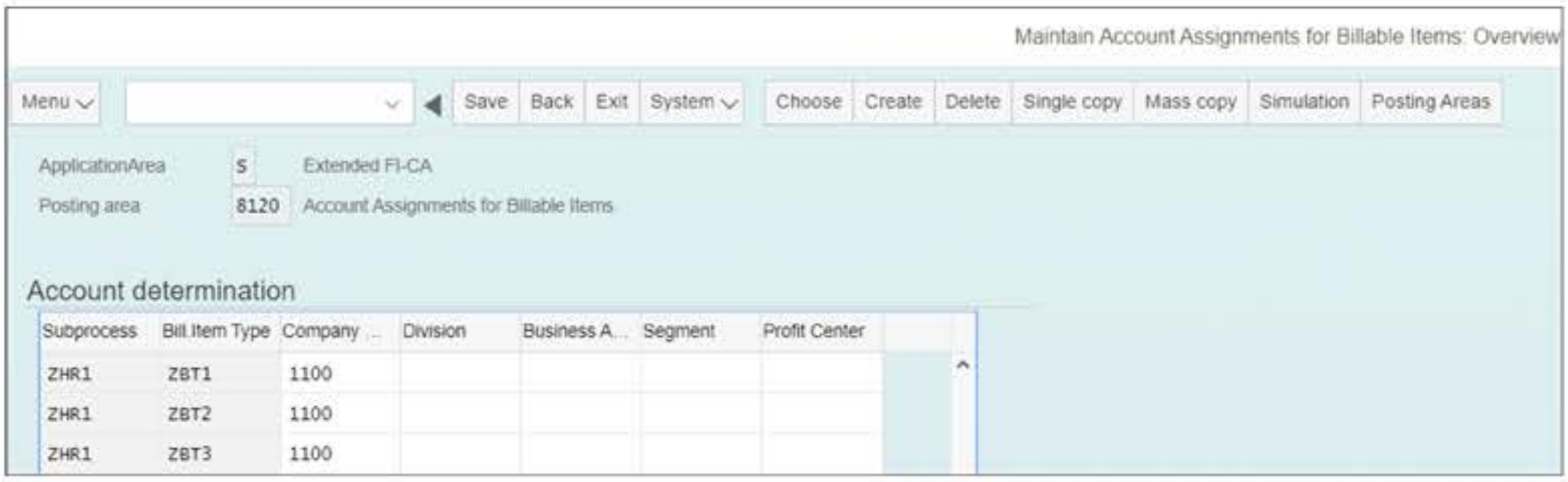

Figure 4.29 Posting Area 8120 for Account Assignment

Main Transactions and Sub-transactions

Main transactions and sub-transactions are used to identify the general ledger accounts. The main transactions identify the subledger account, and main and sub-transactions together identify the revenue general ledger account for the revenue posting in contract accounting.

Business Example

For the different charges for use of service, the billable item types ZBT1, ZBT2, and ZBT3 may be sent. The amount associated with the billable item types needs to be posted to the general ledger account for accounting. The subprocess ZHR1 and the billable item types will determine the main transactions and sub-transactions from posting 8121, will identify the general ledger accounts during invoicing, and will post the receivables with the general ledger account entries.

Main transactions and sub-transactions identify a transaction as a credit or debit posting, which also associates it with the general ledger account determination for financial postings. The general ledger accounts determined by the main transactions and sub-transactions are posted with the amount for the revenue and tax if applicable by the posting document created during the invoicing process. The posting document is an accounting document in the contract accounting component of SAP Billing and Revenue Innovation Management.

The main transaction is maintained under **Contract Account Receivables & Payables • Basic Functions • Postings and Documents • Document • Maintain Document Assignment • Maintain Main Transaction** (see Figure 4.30).

Figure 4.30 Main Transactions Creation Screen

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The sub-transactions are maintained under **Contract Account Receivables & Payables • Basic Functions • Postings and Documents • Document • Maintain Document Assignment • Maintain Sub Transaction** (see Figure 4.31).

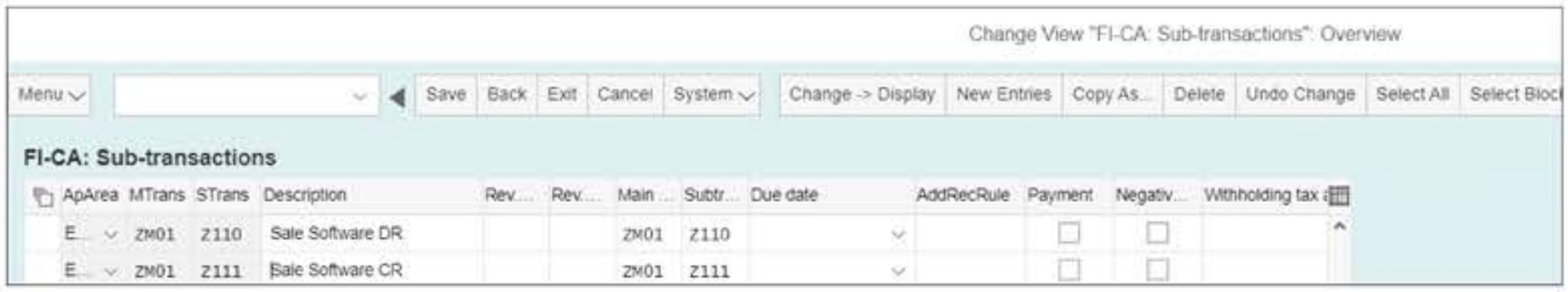

Figure 4.31 Subtransaction Creation Screen for Main Transaction

The main transactions and sub-transactions created are then parameterized to indicate a debit or credit transaction. The configuration is maintained under **Contract Account Receivables & Payables • Basic Functions • Postings and Documents • Document • Maintain Document Assignment • Maintain Transactions for Non-Ind Contract Account Receivables and Payables • Define and Parameterize External Transactions** (see Figure 4.32).

The **Debit/Credit** indicator can be left blank, and the credit/debit posting can be controlled based on the sign of the amount received in the billable items. A negative sign (-) before an amount indicates a credit posting.

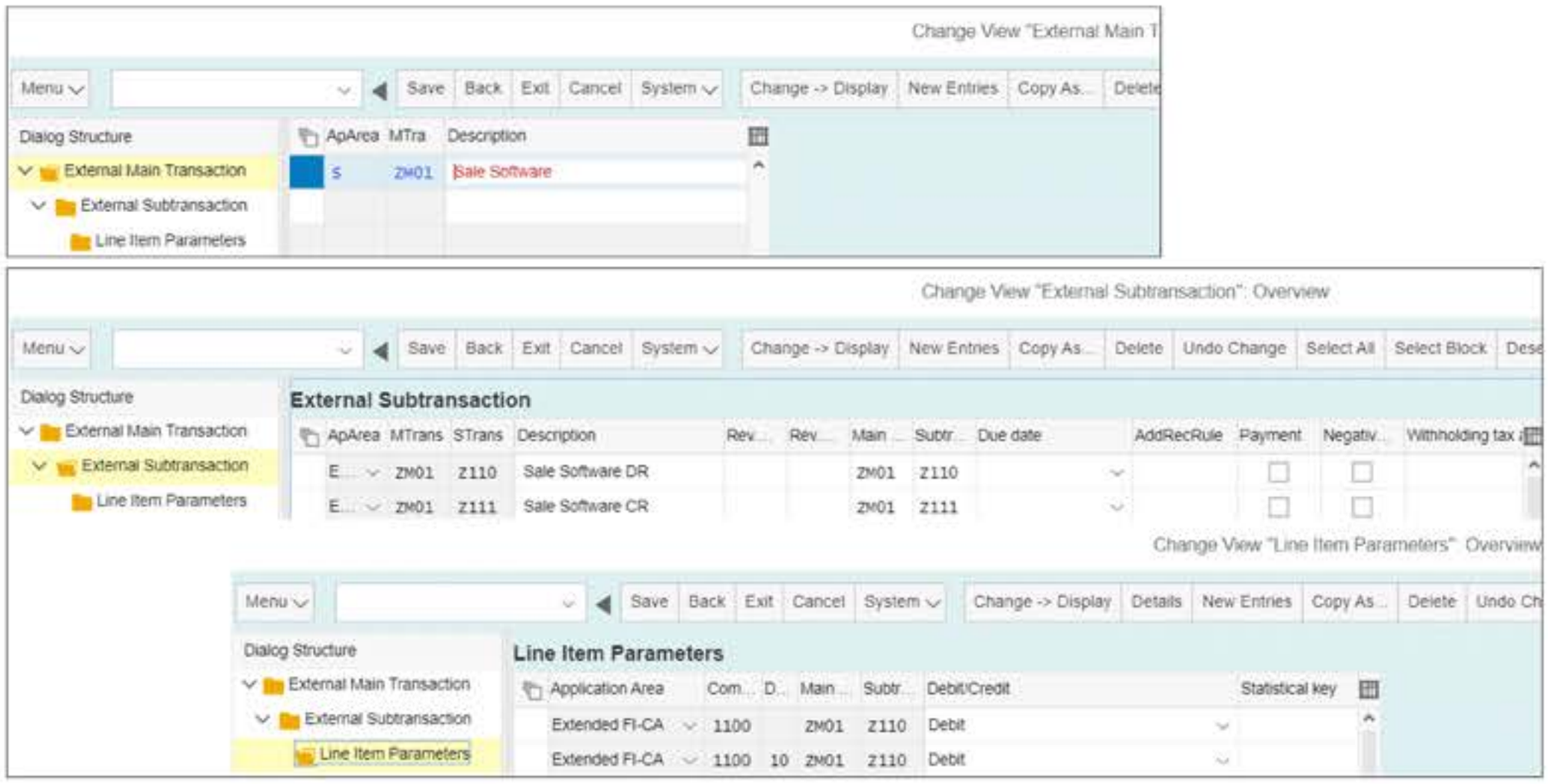

Figure 4.32 Parameterize Main Transactions and Sub-transactions

The main transactions and sub-transactions thus created are then maintained in posting areas 8122 and 8121. These posting areas are maintained with and without product details, respectively, for determination of the main transactions and sub-transactions. The subprocess, billable item type, and four external key fields help derive the main transaction and subtransaction details.

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Posting areas 8122 and 8121 can be accessed using Transaction FQCO or at the following menu paths. The associated configuration is maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Items • Billable Item Transfer • Account Assignment Derivation • Define Main & Sub Transactions for Items with Product Account Assignment** (see Figure 4.33).

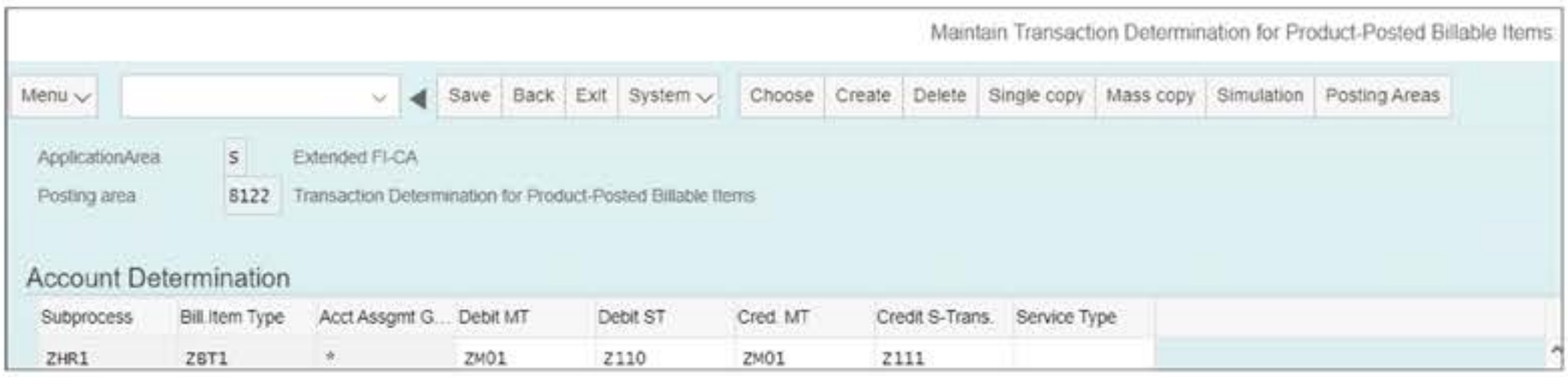

Figure 4.33 Posting Area 8122

To define the main and sub-transactions for posting area 8121, navigate to **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Items • Billable Item Transfer • Account Assignment Derivation • Define Main & Sub Transactions** (see Figure 4.34).

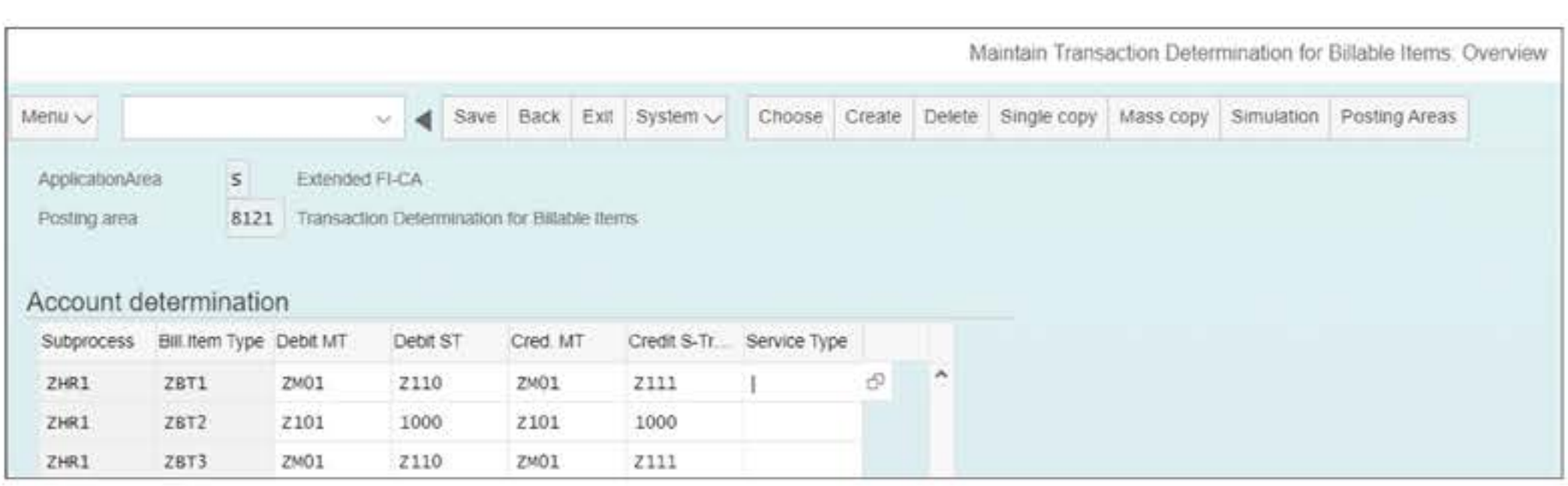

Figure 4.34 Posting Area 8121 for Main Transaction and Subtransaction Determination

Payment Method for Payment Processing

Billable items with record type payment items along with main items contain payment details such as authorization number, card number, card type, and so on. These payment items require a payment method with a corresponding processing type to be updated in the fields of the payment items record type. This information is required at the time of invoicing to post the payment against the receivables posted in contract accounting.

Configuration is maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Items • Billable Item Transfer • Define Payment Method for Payment Processing** (see Figure 4.35).

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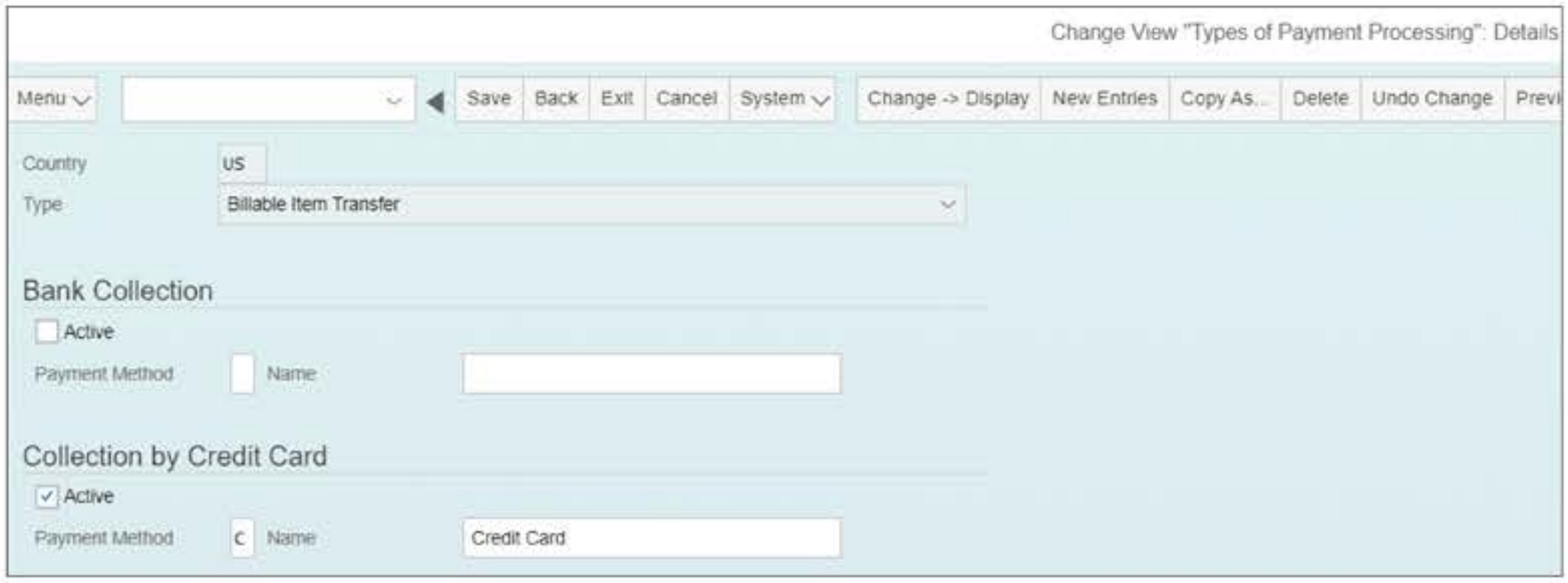

Figure 4.35 Payment Method Settings for Billable Item Type Payment Items

4.3.2 Consumption Items

Consumption items contain the details of the usage of the service, such as type of service, units, duration of usage, and amount per unit, which are processed in the rating process in SAP Convergent Charging. Usage records are also called consumption data records, which are processed to form the billable items that are then billed and invoiced.

Consumption data records are captured either by third-party software or using SAP Convergent Mediation by DigitalRoute. These consumption data records capture detailed information about the usage of services, which is required in rating and charging processes.

The consumption data records are triggered by the business events or usage of the service and are then either passed into the SAP Convergent Charging system and stored as consumption items in SAP Convergent Invoicing or can directly be entered into SAP Convergent Invoicing using the associated APIs and stored as consumption items. Then the rating process is performed to obtain the billable items.

In addition, in a partner settlement functionality scenario in SAP Convergent Invoicing, the consumption items for vendors are created automatically in SAP Convergent Invoicing using details of the billable items. This functionality will be discussed in detail in Section 4.3.7.

Consumption Item Class

Like billable items, consumption items also follow the class concept. To store the consumption items in SAP Convergent Invoicing, a consumption item class is created. SAP provides several interface components that are used according to business requirements to make fields available to store information about the consumption records. A consumption item class specifies the technical attributes of the consumption items.

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Configuration for consumption item classes is maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Consumption Items • Consumption Item Classes • Maintain Consumption Item Classes** (see Figure 4.36).

Upon activation and generation of the class, interfaces create the structures, tables, function modules, and fields for the consumption item class. Standard fields for the consumption items are obtained based on the interfaces selected for the class, and custom fields can be assigned to the class. Creation, activation, and generation of class interfaces are explained in Section 4.3.1.

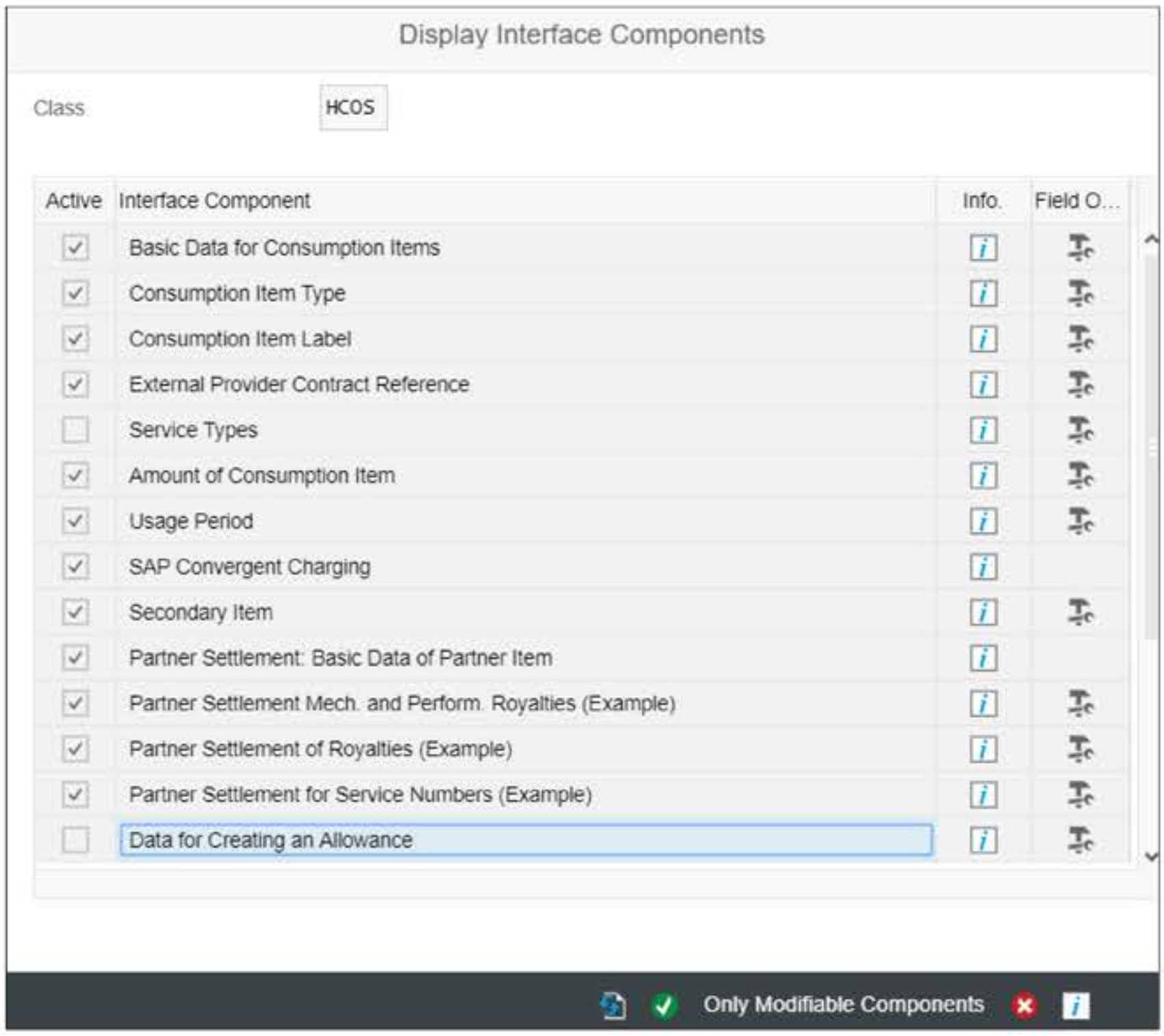

Figure 4.36 Interface Components for Consumption Item Class HCOS

Separate tables are created for each month for the rated items. Each status of the consumption items has separate tables for storing it (see Figure 4.37 and Figure 4.38).

Business Example

For Martex Corp., consumption item class REVE is created for consumptions items that will be used to store the usage records consumption data records, and consumption item class HCOS is created to store the consumption records for the cost of the products obtained from vendors. The HCOS class will be used in the partner settlement functionality to bill and invoice vendors.

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Display Generated Objects

Menu | Back | Exit | Cancel | System | Sort in Ascending Order | Sort in Descending Order | Display Object | Display All Objects

Class HCOS

Generated Objects

Short text	Object Name	ShortTxt
Function Module	/1FC/HCOS_CIT_CREATE_API	Generation of Consumption Items of Class HCOS
Table	/1FC/1HCOS_API	Class HCOS Structure Type API
Function Module	/1FC/HCOS_CIT_CREATE_PROXY	Generation of Consumption Items of Class HCOS
Table	/1FC/1HCOS_PRX	Class HCOS Structure Type PRX
View	/1FC/24HCOS	Consumption View - All Consumption Items of a Class HCOS
Table	/1FC/0HCOS0	Items for Class HCOS - Raw
View	/1FC/20HCOS0	Basis View for Consumption Items of a Class HCOS Status 0
View	/1FC/21HCOS0	Private Basis View for Consumption Item Class HCOS Status 0
Table	/1FC/0HCOS1	Items for Class HCOS - Raw Excepted
Table	/1FC/0HCOS2	Items for Class HCOS - Rateable
View	/1FC/20HCOS2	Basis View for Consumption Items of a Class HCOS Status 2
View	/1FC/21HCOS2	Private Basis View for Consumption Item Class HCOS Status 2
Table	/1FC/0HCOS3	Items for Class HCOS - Rateable Excepted
Table	/1FC/0HCOS400	Items for Class HCOS - Rated
Table	/1FC/0HCOS48C	Items for Class HCOS - Rated MAR 8
Table	/1FC/0HCOS48D	Items for Class HCOS - Rated APR 8
Table	/1FC/0HCOS48E	Items for Class HCOS - Rated MAY 8
Table	/1FC/0HCOS48F	Items for Class HCOS - Rated JUN 8

Not all objects are displayed

Figure 4.37 Objects Generated for HCOS Consumption Item Class

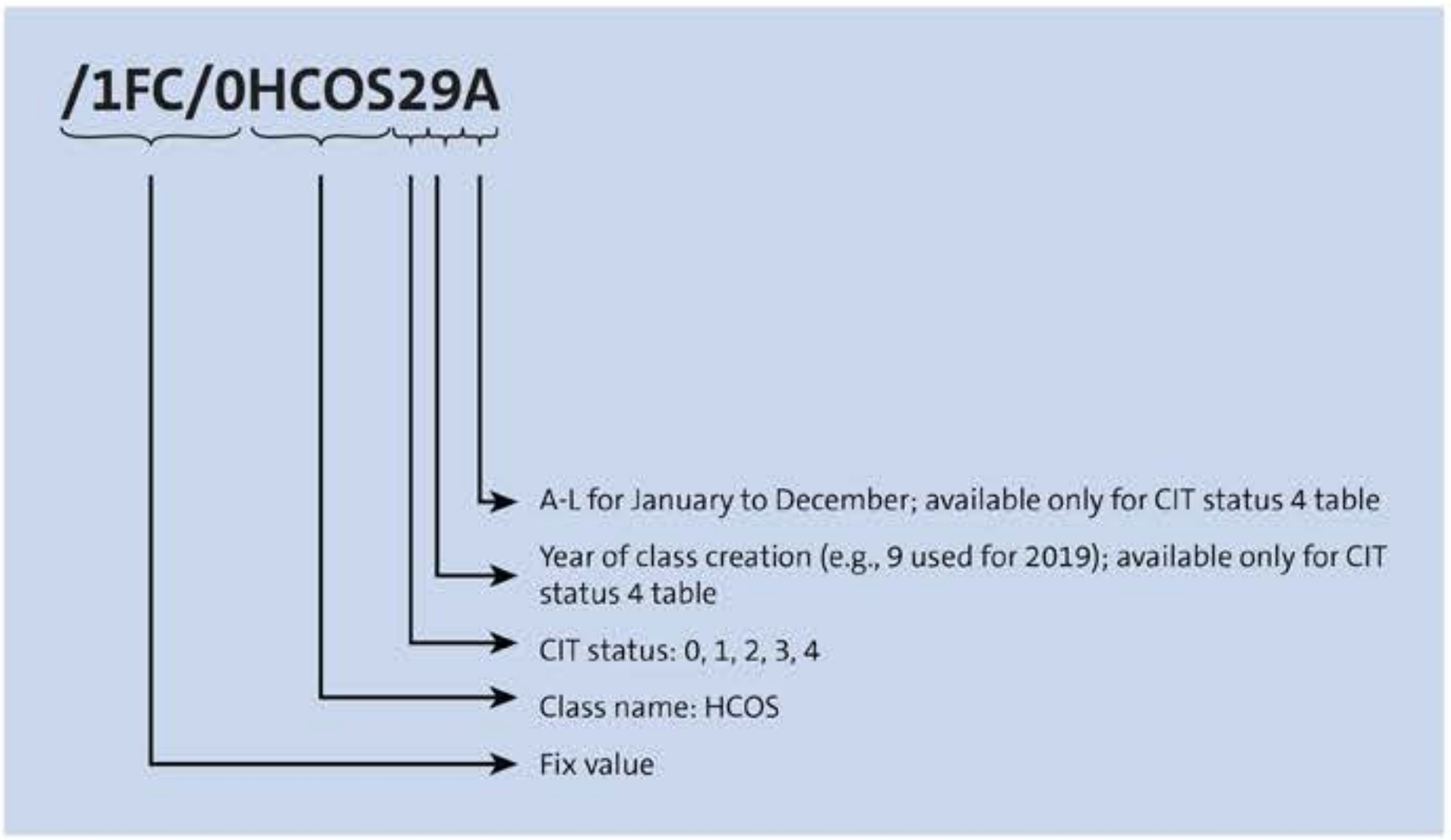

Figure 4.38 Naming Convention for Consumption Item Class Tables

Consumption items obtain various statuses during their lifecycle in the SAP Convergent Invoicing system (see Table 4.3).

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Consumption Item Status	Description
0	Raw
1	Raw-Excepted
2	Unrated
3	Rated-Excepted
4	Rated

Table 4.3 Statuses of Consumption Items

The **Raw** status indicates that the consumption item doesn't contain complete information and values in the fields can be changed. Raw consumption items aren't eligible for the rating process. Raw consumption items should be transferred to status **2** (**Unrated**) to be considered for the rating process. Let's take a look at these statuses:

- **Raw**
  Excepted consumption items aren't considered during transfer or rating. Such items should be restored and then transferred to **Unrated** status to be considered for further processing.
- **Raw-Excepted**
  The consumption items are missing some fields value or a value is incorrect. For such items to be considered for the rating, the status should be restored to **Unrated**.
- **Unrated**
  These items only are considered for the rating process. **Unrated** status indicates the information contained in the consumption items is complete and consistent.
- **Rated-Excepted**
  Consumption items generally get this status when the rating is reversed. Consumption items then should be restored if needed to be considered in the rating process.
- **Rated**
  Consumption items get this status when the rating process is performed successfully on them.

Consumption Item Management

Consumption items contain the consumption item ID type and consumption item ID, which together uniquely identify the consumption items in the system. The consumption item ID type specifies the business significance of the consumption item from the rating perspective.

To store the consumption items in the SAP Convergent Invoicing system, it's essential to store a provider contract in the consumption item. The provider contract contains contract details such as duration, associated business partner, contract account, product details, and billing cycle. The consumption date of the items is checked with the provider contract dates.

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Processing Rules and Program Enhancements

Consumption items transferring from the SAP Convergent Charging system or the external usage recording system to SAP Convergent Invoicing based on business requirements can be enriched with certain values for the consumption item fields. The process of populating certain standard fields or custom fields based on certain conditions during the transfer of consumption items to SAP Convergent Invoicing is called consumption item enrichment. SAP provides sample function modules for each status of the consumption items, which can be modified according to your business requirements.

In addition, SAP provides the facility to process the consumption items received in SAP Convergent Invoicing. The configuration is maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Consumption Items • Consumption Item Management • Define Processing Rules and Enhancements** (see Figure 4.39).

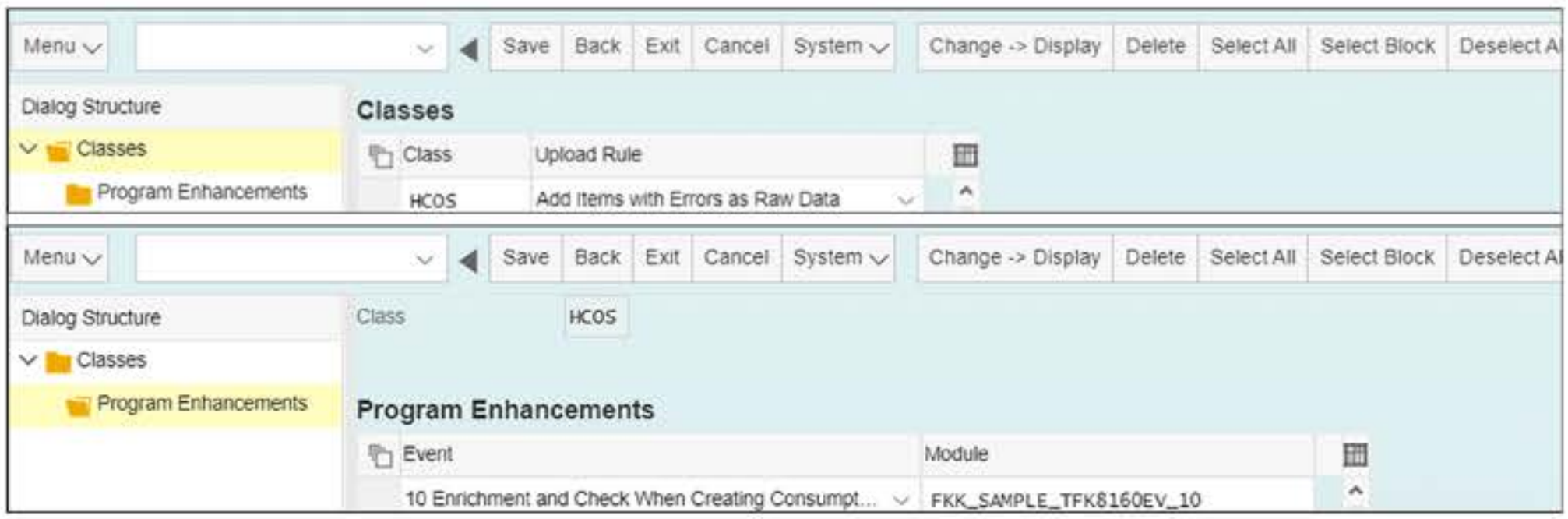

Figure 4.39 Processing Rule and Enhancement for the HCOS Consumption Item Class

Reversal of Consumption items

Consumption items transferred to SAP Convergent Invoicing may sometimes have incorrect data and may require that such incorrect data not be processed. Such consumption item records can be reversed; reversed consumption items aren't considered for the rating and are moved to an **Excepted** status table. The reversal of consumption items requires an exception reason, which provides the following options for reversal of the consumption items:

- **Reverse item (not possible to restore it)**
  The first option reverses the consumption item, and the reversed item can't be restored to rated status and can't be processed further.
- **Except item (possible to restore it)**
  The second option lets you except the consumption items, fields of the items can be edited, the consumption item can be restored to **Unrated** status by providing a restoration reason, and can be further processed in rating.

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Configuration for the reversal, exception reason, and restoration reason is maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Consumption Items • Consumption Item Management**. Here you'll find the following options:

- **Activate Classes for Consumption Items**
- **Define Processing Rules and Program Enhancements**
- **Define Schema for Duplicate check**
- **Assign Schema for Duplicate Check**
- **Define Field Configuration for Consumption Items**
- **Define Changeable Fields for Consumption Items**
- **Enter Exception Reasons**
- **Enter Reasons for Restoration**
- **Make Standard Settings for Reversal**

Consumption items can be created directly in SAP Convergent Invoicing to test the system configuration using Transaction FKKBIXCIT_CWB. This transaction can also be accessed in the SAP Easy Access menu under **Accounting • Financial Accounting • Contract Account Receivables & Payables • Convergent Invoicing • Consumption Items • Enter & Change Manually** (see Figure 4.40).

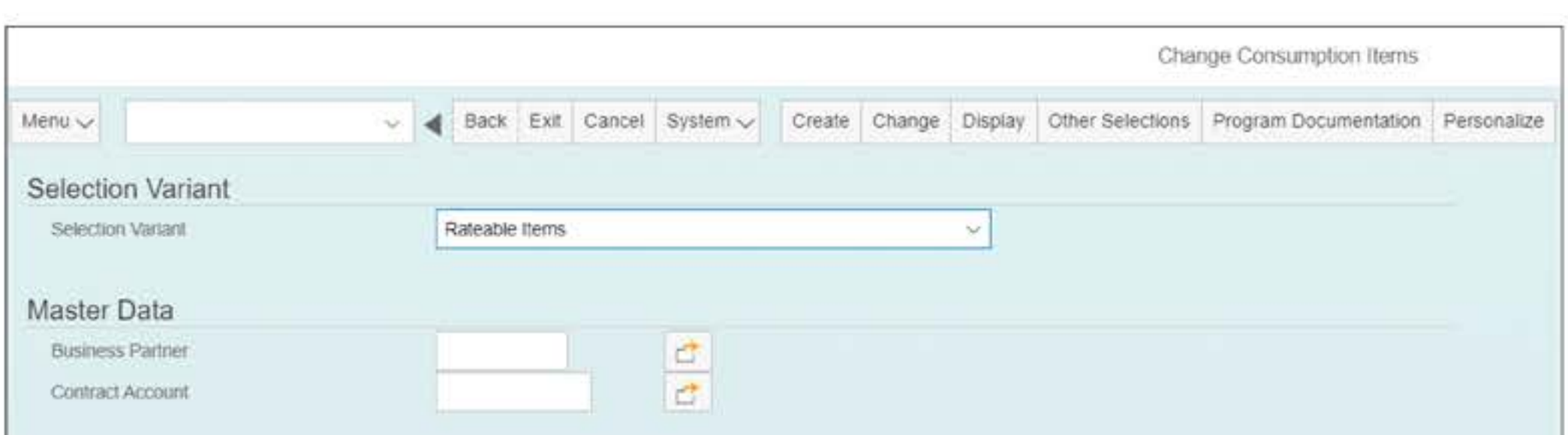

Figure 4.40 Consumption Item Creation Screen

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Transaction FKKBIXCIT_CWB provides the option to create, change, and display consumption items. Using the **Create** option, you can directly create consumption items within SAP Convergent Invoicing. In addition, if certain consumption items present in the system require any field value to be changed, that also can be done using this transaction if a field is made modifiable for the **Unrated** status. For display of consumption items, various selection screen parameters are provided. By selecting a variant for the status, all the consumption items can be displayed at once for a particular status or for all statuses.

The created consumption items are stored in the tables based on the status in which they are created. These items can be viewed using Transaction FKKBIXCIT_MON, which also can be accessed in SAP Easy Access menu under **Accounting • Financial Accounting • Contract Account Receivables & Payables • Convergent Invoicing • Consumption Items • Display** (see Figure 4.41).

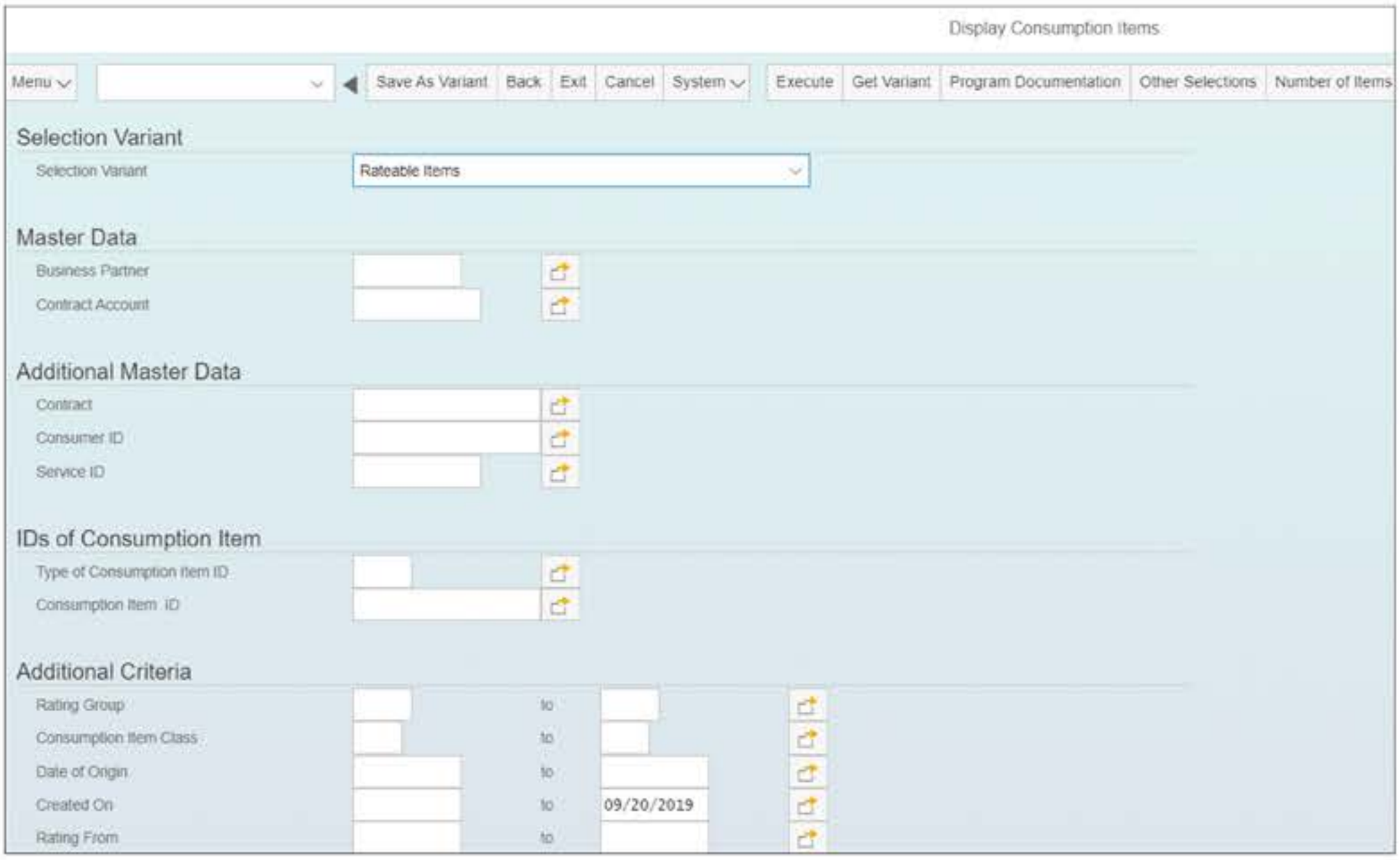

Figure 4.41 Consumption Item Display Screen

Consumption items can be viewed using Transaction FKKBIXCIT_MON. There are several selection parameters provided with which you can view consumption items. A selection variant can be used to view all the consumption items at once based on their status. The other options available on the **Display Consumption Items** screen are rating, restore, reverse, and except (see Figure 4.42).

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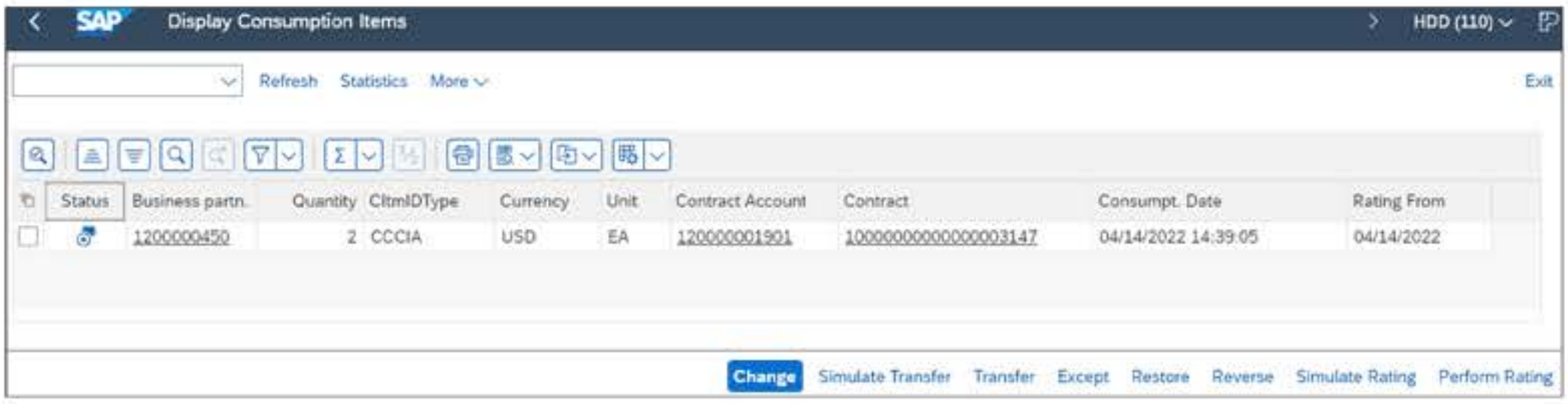

Figure 4.42 Consumption Items Processing Options

4.3.3 Rating and Rating Groups

Rating is a process that takes in the unrated consumption items, applies the associated calculations rule to obtain the amount for the usage, associates it to the contract, and creates the billable items for billing and invoicing. The rating process is triggered in SAP Convergent Invoicing, which sends the consumption items to SAP Convergent Charging for rating and charging and receives the billable items. The rating process performs the following tasks:

- Selection, grouping, and aggregation of consumption items
- Creating billable items for the rated and charged consumption items from SAP Convergent Charging

Rating and charging processes are executed in SAP Convergent Charging. To trigger the rating process from SAP Convergent Invoicing, the following configurations should be maintained:

- Rating group
- Grouping variant
- Aggregation variant
- Billing process

The configuration for setting up the rating process is maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Rating**.

Rating Group

A rating group is essential for the rating process to trigger. The rating group is used to set the parameters required for the rating execution. A rating group is used to group together the consumption items based on business requirements.

Rating groups specify parameters like the following:

- Parallel processing criteria for mass execution of the rating job
- Grouping variants for the creating the rating units
- Assignment of billing process to billable items from rating
- Whether aggregation of billable item should be performed

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The configuration to maintain the rating group can be maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Rating • Define Rating Groups** (see Figure 4.43).

The rating group also provides the enhancement options for the rating process. In the rating group, based on business requirement rating units, billable items can be updated. You can also specify whether a rating is to be executed in SAP Convergent Charging or within SAP Convergent Invoicing. For a rating to be performed in SAP Convergent Charging, an event rating is selected in the program enhancement, and function module `FKK_BIX_RATING_EVENT_30_SAPCC` is assigned (see Figure 4.44).

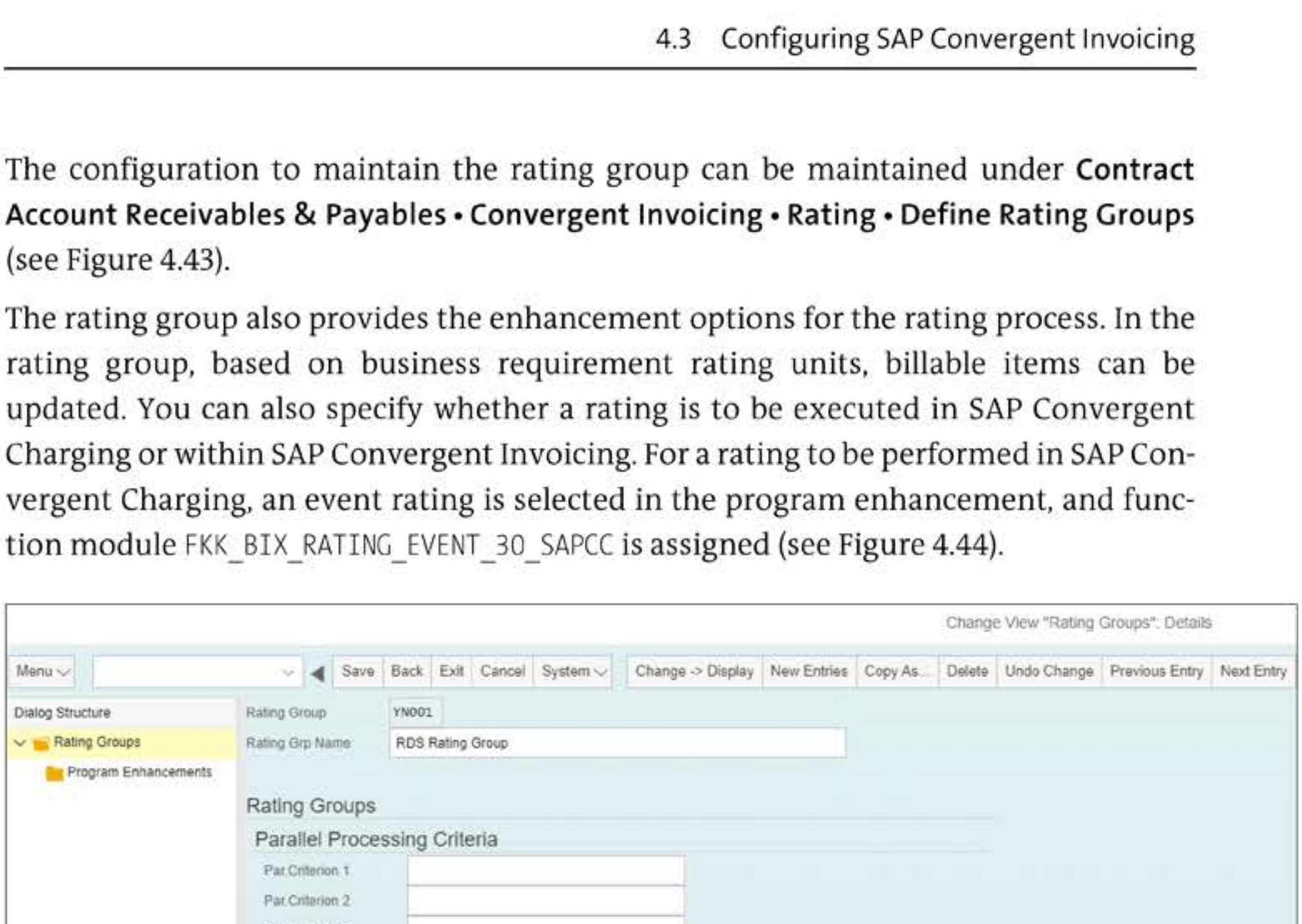

Figure 4.43 Rating Group Creation Screen

Figure 4.44 Program Enhancement for Rating Group

Rating processing in SAP Convergent Invoicing can be performed if function module `FKK_BIX_RATING_EVENT_30` is used for the rating in the program enhancement. Only

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simpler rating logic can be performed in SAP Convergent Invoicing; for more complex logic, SAP Convergent Charging is used. Because we're using SAP Convergent Charging for rating and charging, we'll use function module FKK_BIX_RATING_EVENT_30_SAPCC.

Grouping and aggregation play a very important role in the rating process. Grouping is used for grouping the consumption items together, which are sent as a rating unit to SAP Convergent Charging for rating. An aggregation variant summarizes the consumption items within a group to further consolidate them and sends them for rating, thus reducing the number of individual consumption items sent for rating. The consolidated consumption items contain a total quantity in the consumption item **Quantity** field. Both grouping and aggregation variants are created and assigned with characteristics, based on which the grouping and aggregation are performed:

- **Grouping variant**
  The grouping variant assigned to the rating group is used to group together the consumption items that are required to be rated together based on business requirements. A grouping variant specifies the technical fields or attributes of the consumption item class that will be used to group together the items. Attributes such as rate first, date, product, usage period, and so on are used to group together the consumption items to be rated together. Items grouped together form the rating unit.
- **Aggregation variant**
  The aggregation variant is used to consolidate the consumption items within the rating unit based on the consumption item class attributes. This reduces the number of consumption items sent to SAP Convergent Charging for rating and subsequently reduces the billable items received from SAP Convergent Charging.

Business Example

Martex sells service A. A customer who has subscribed to the service with a reserved instance will generate consumption data or usage records for the use of the service for a month. These usages will be stored as consumption items. Martex's business requirement is to group those consumption items of service A together that have same target rating date (RATE_FROM) and start of consumption period date (CIT_FROM), as well as aggregate the consumption items that have the same consumption item creation date (CITCRDATE).

Rating group YN001 is created for Martex. The grouping variant ZGRP is used with the RATING_FIRST and CIT FROM fields as the grouping criteria. Aggregation variant ZAGV is created with the CITCRDATE parameter for aggregation.

Before creating the grouping variant, the grouping characteristics are maintained. Grouping variants for the rating are maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Rating • Define Grouping Variants** (see Figure 4.45 and Figure 4.46).

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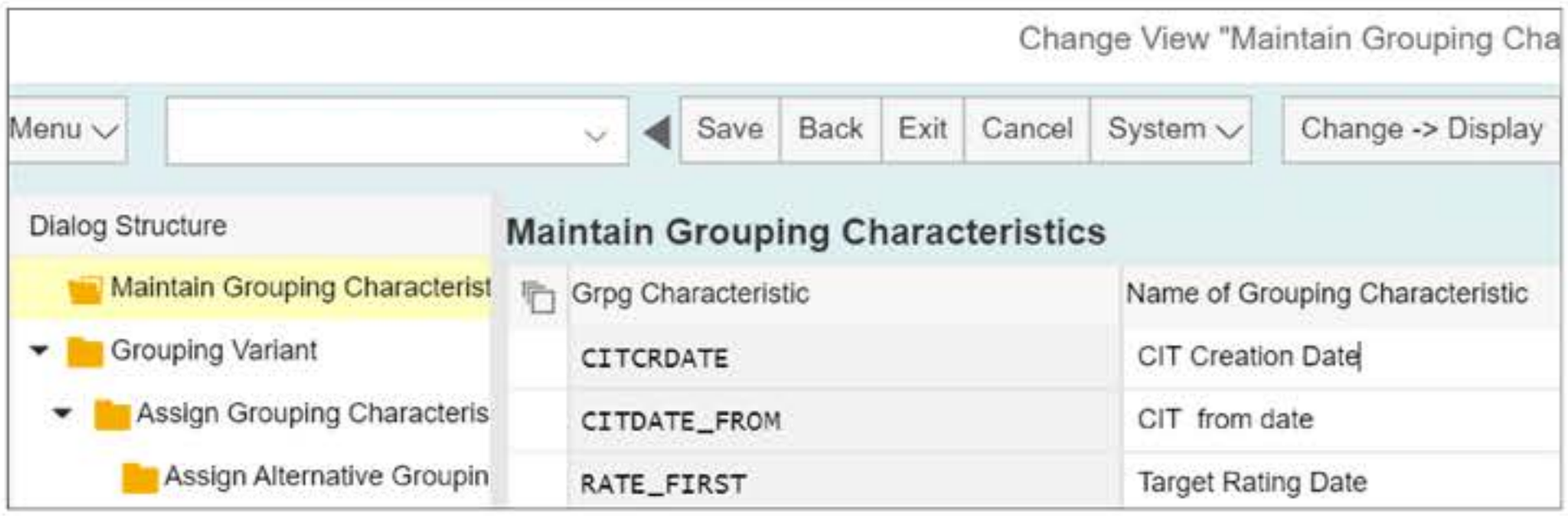

Figure 4.45 Grouping Characteristics

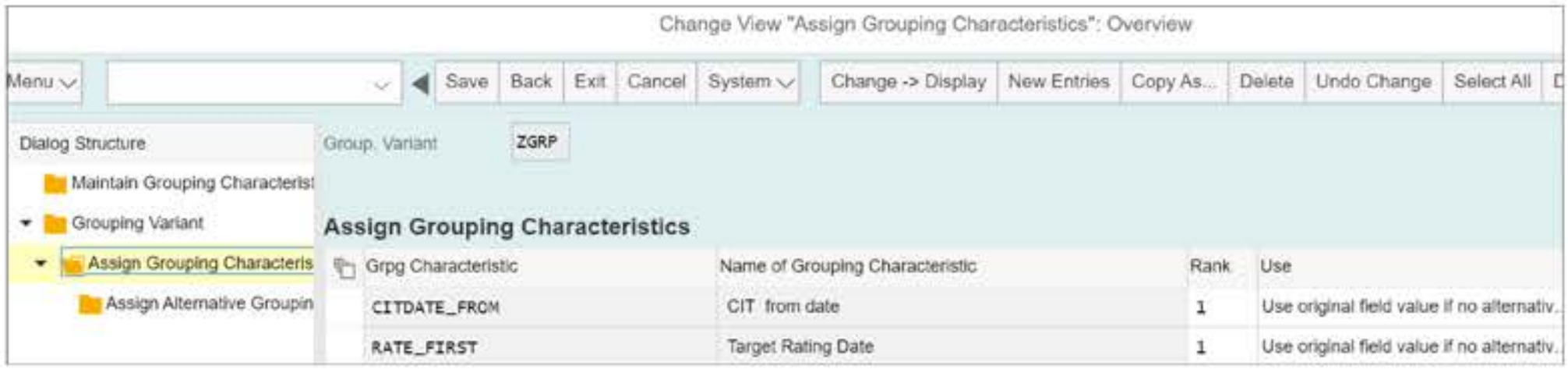

Figure 4.46 Grouping Variant for Rating Group

Aggregation variants are maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Rating • Define Aggregation Variants**. In the configuration node, the first aggregation variant is created, then the consumption item class is assigned, and finally the aggregation characteristics are assigned (see Figure 4.47).

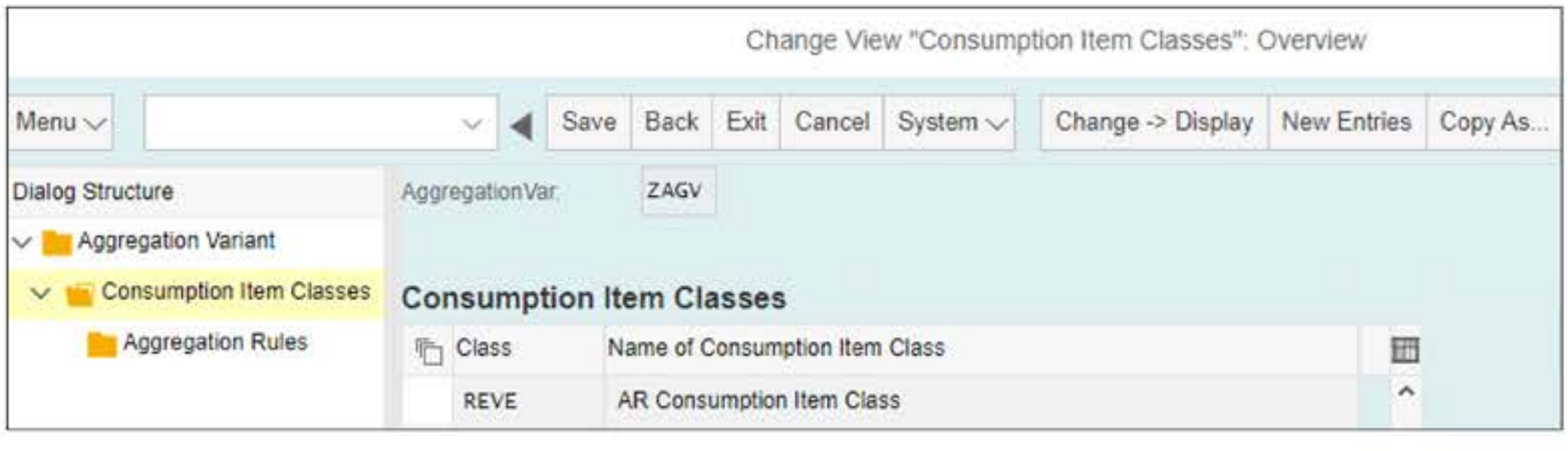

Figure 4.47 Aggregation Variant for Rating Group

The grouping variant and the aggregation variant created are assigned to the rating group. Posting area 8177 is maintained for the assignment. In the posting area, a new entry is created with the rating group as the selection key, and output values are assigned as grouping and aggregation variants. Three additional selection keys are also available and can be used based on business requirements. Selection keys can be obtained via **Menu • Goto**.

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The configuration to maintain posting area 8177 is done under **Contract Account Receivables & Payables • Convergent Invoicing • Rating • Define Aggregation for Consumption Items** (see Figure 4.48).

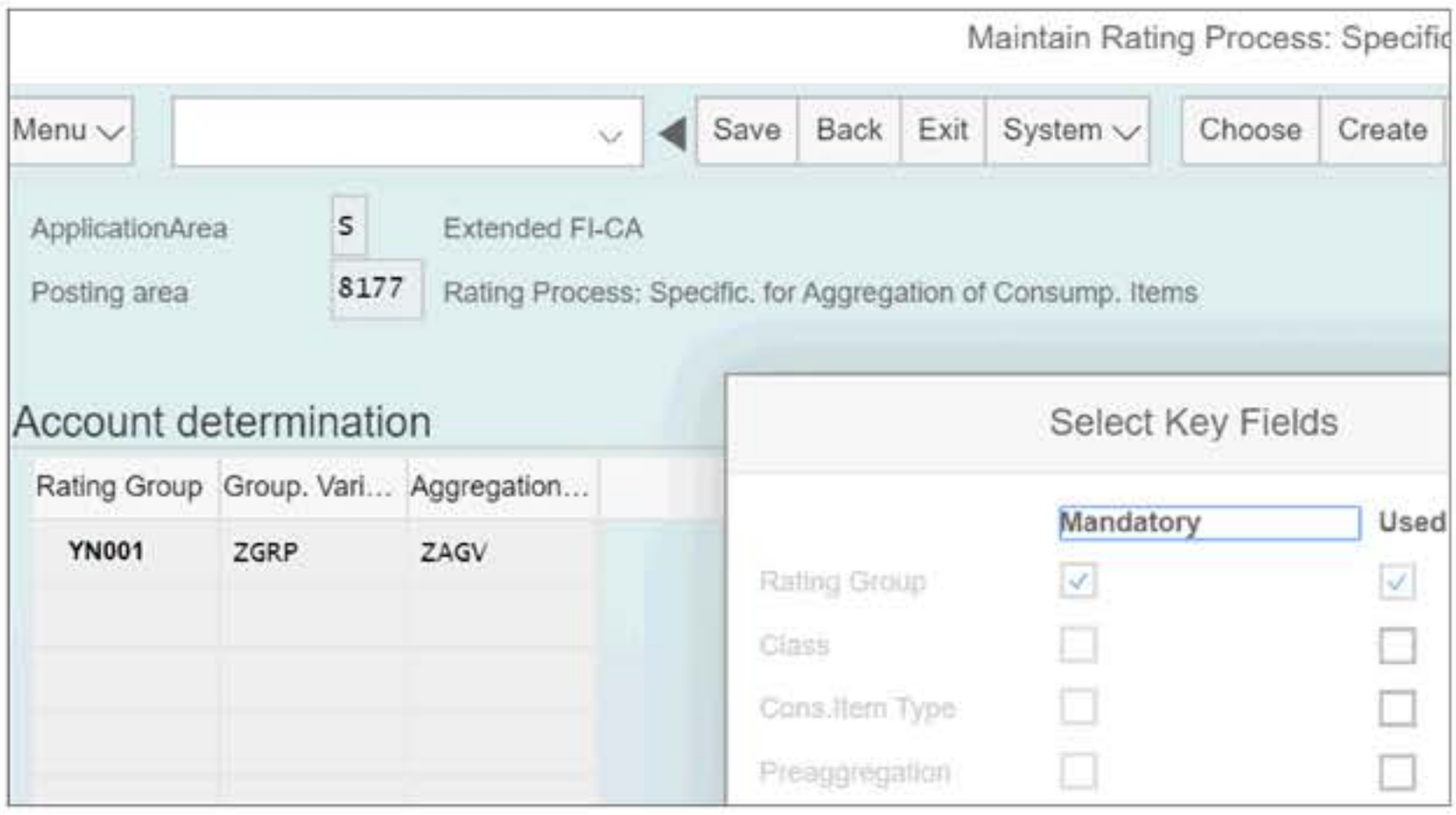

Figure 4.48 Grouping and Aggregation Variant Assignment for Rating Group

Note the following key points of the rating process:

- Consumption items that have the same business partner in the rating process are grouped together by default.
- Consumption items for which the rating first/target rating date isn't reached won't be considered in the rating process.
- If an aggregation variant isn't assigned to the rating group, then each consumption item selected will be rated individually.

Rating process maintenance is shown in Figure 4.49.

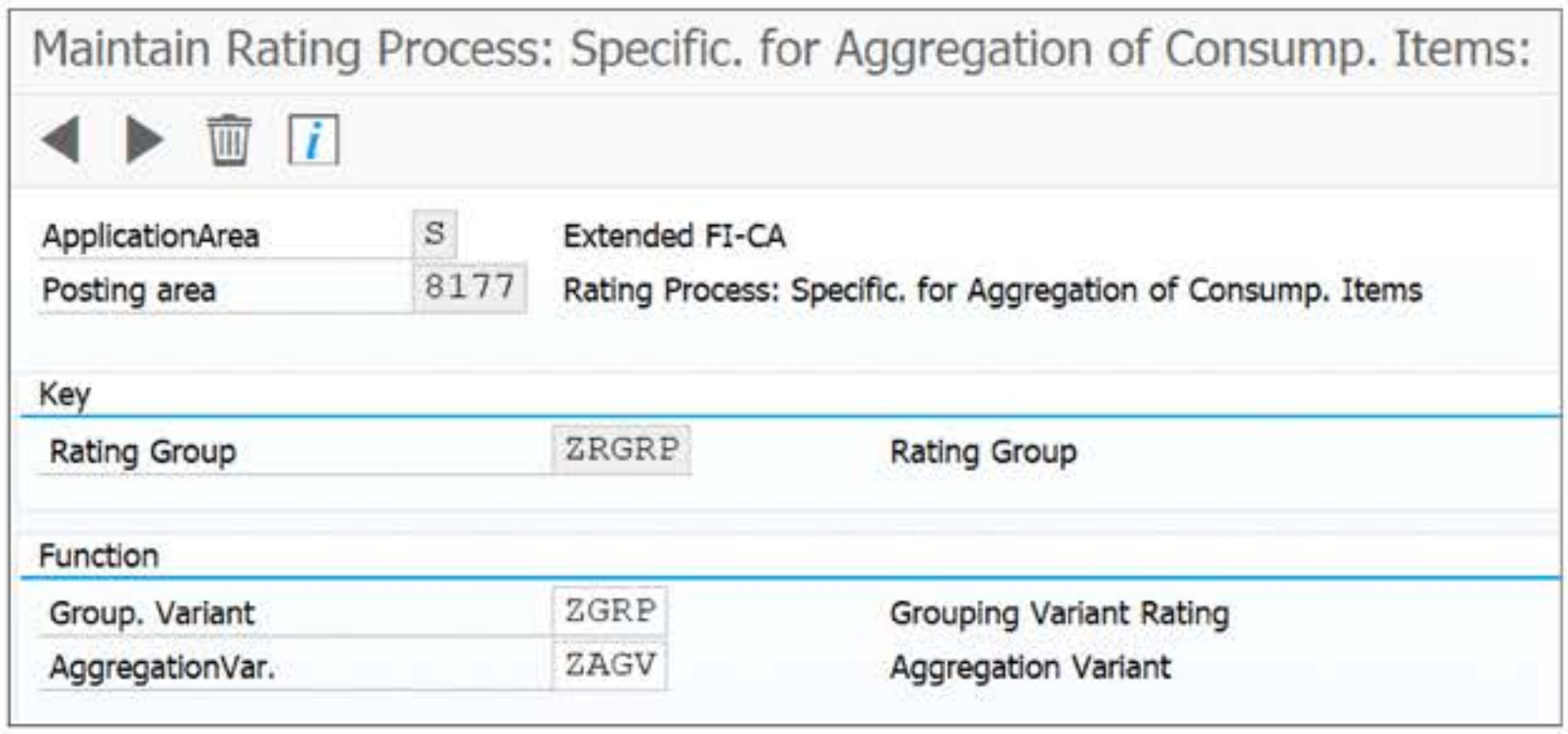

Figure 4.49 Maintain Rating Process

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Ratings can be triggered from SAP Convergent Invoicing individually or en masse. For individual runs, use Transaction FKKBIX_RATE_S, and for mass runs, use Transaction FKKBIX_RATE_MA. These transactions can also be accessed through SAP Easy Access menu **Accounting • Financial Accounting • Contract Account Receivables & Payables • Convergent Invoicing • Periodic Processing • Rating** (see Figure 4.50).

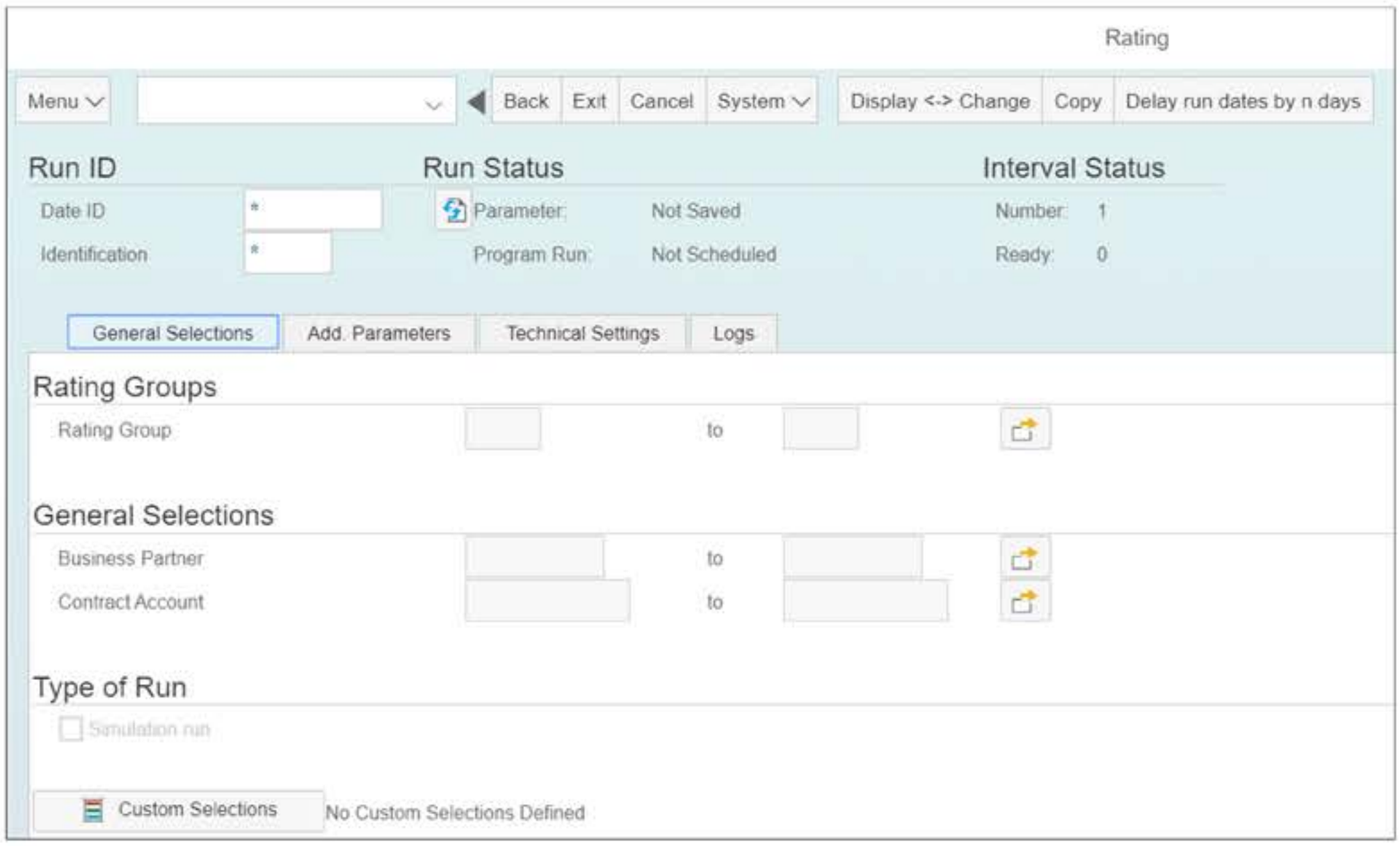

Figure 4.50 Transaction FKKBIX_RATE_MA: Mass Rating

On the execution screen, the run **Date ID** and **Identification** are entered for the rating process. The **General Selections** tab is maintained for the data selection, and the **Rating Group** is entered for which the rating process is executed. The **Add. Parameters** screen contains the target rating date for which rating will be performed. The **Technical Settings** tab contains the parallel processing object. Once the parameters are set, the job can be executed immediately in the background or can be scheduled to run at a later time based on business requirements.

Business Example

Martex uses consumption item class REVE created for usage records. The usage records from the customer for service A will be stored in the consumption item class REVE. Then the rating process will be executed using rating group YN001. The usage records will be sent to SAP Convergent Charging, which will rate and charge the consumption items and send them back to SAP Convergent Invoicing as a billable item of class REVE and billable item type ZBT3.

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Once the rating process is executed, it will send the consumption items to the SAP Convergent Charging system. SAP Convergent Charging will rate the usage records received from SAP Convergent Invoicing. Rated and charged consumption items in SAP Convergent Charging are sent to SAP Convergent Invoicing as billable items. The billable items received from SAP Convergent Charging assign a source transaction ID to the billable items, and the same is updated in the consumption items; this links the consumption item to the corresponding billable items.

Transaction FKKBIXCIT_MON, which is used to view the consumption items, also shows the link to the billable items. In addition, the same transaction provides the option to trigger the rating process for the unrated consumption items. For more details, Section 4.3.2.

The transaction can be accessed in SAP Easy Access menu **Accounting • Financial Accounting • Contract Account Receivables & Payables • Convergent Invoicing • Consumption Items** (see Figure 4.51).

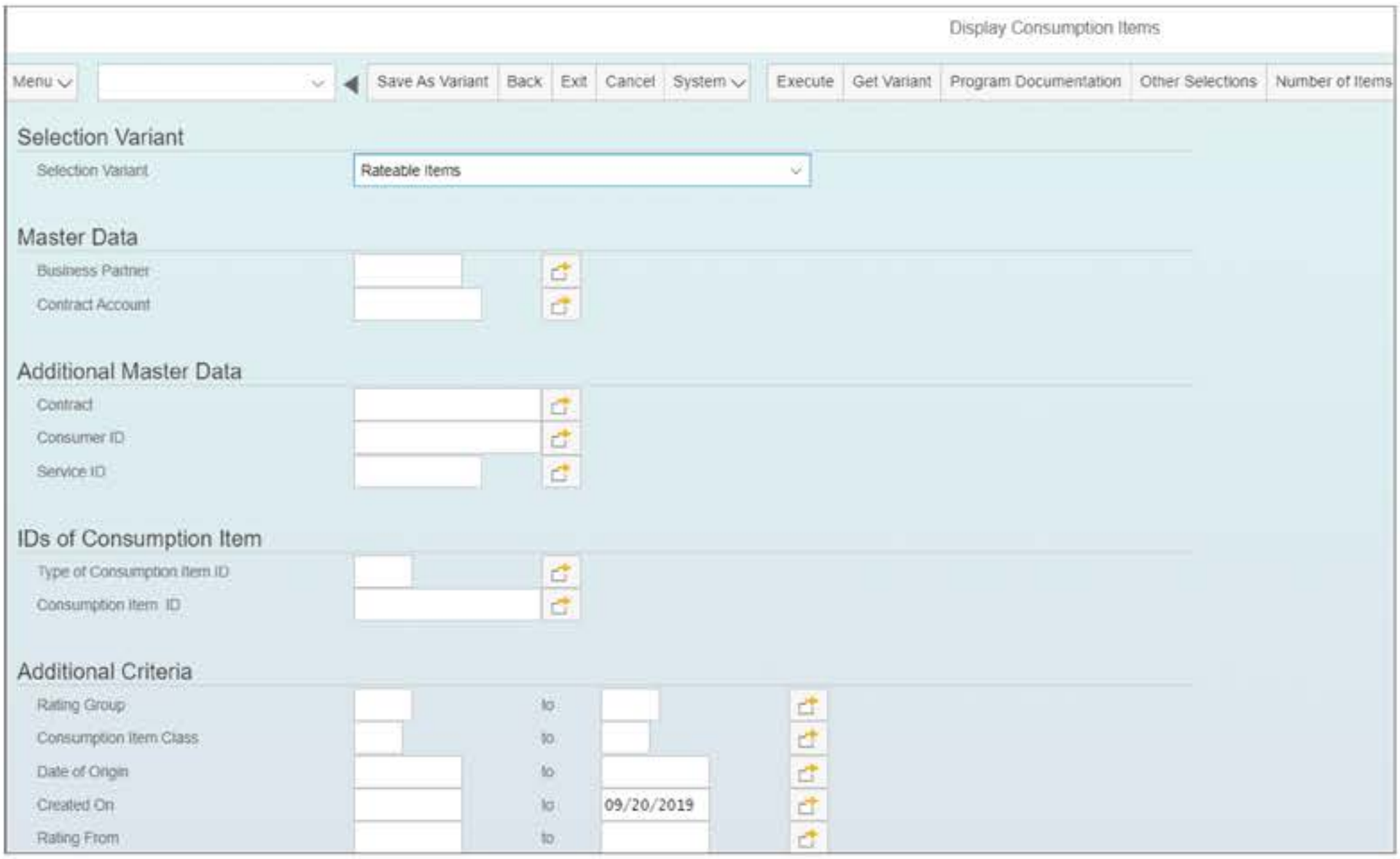

Figure 4.51 Consumption Items Display Screen

4.3.4 Billing

Billable items obtained from the sale of service, usage of service, refunds, cancellations, returns, and so on are processed in the billing process to create the billing document. The billing process creates the billing document by processing the billable items that have the same business significance. Billing groups aggregate the billable items.

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Billing Process

Billing is performed using the billing process configured in the system. A billing process is a collection of billing subprocesses that are responsible for selecting, grouping, and aggregating the billable items (see Figure 4.52).

Business Example

Martex has three different billable item types, ZBT1, ZBT2, and ZBT3, for up-front charges, monthly recurring fees, and usage charges, respectively. Billable items received for the different charges are processed by the billing process. For Martex's billing process, ZBP1 will be created, which will have subprocess ZHR1 assigned.

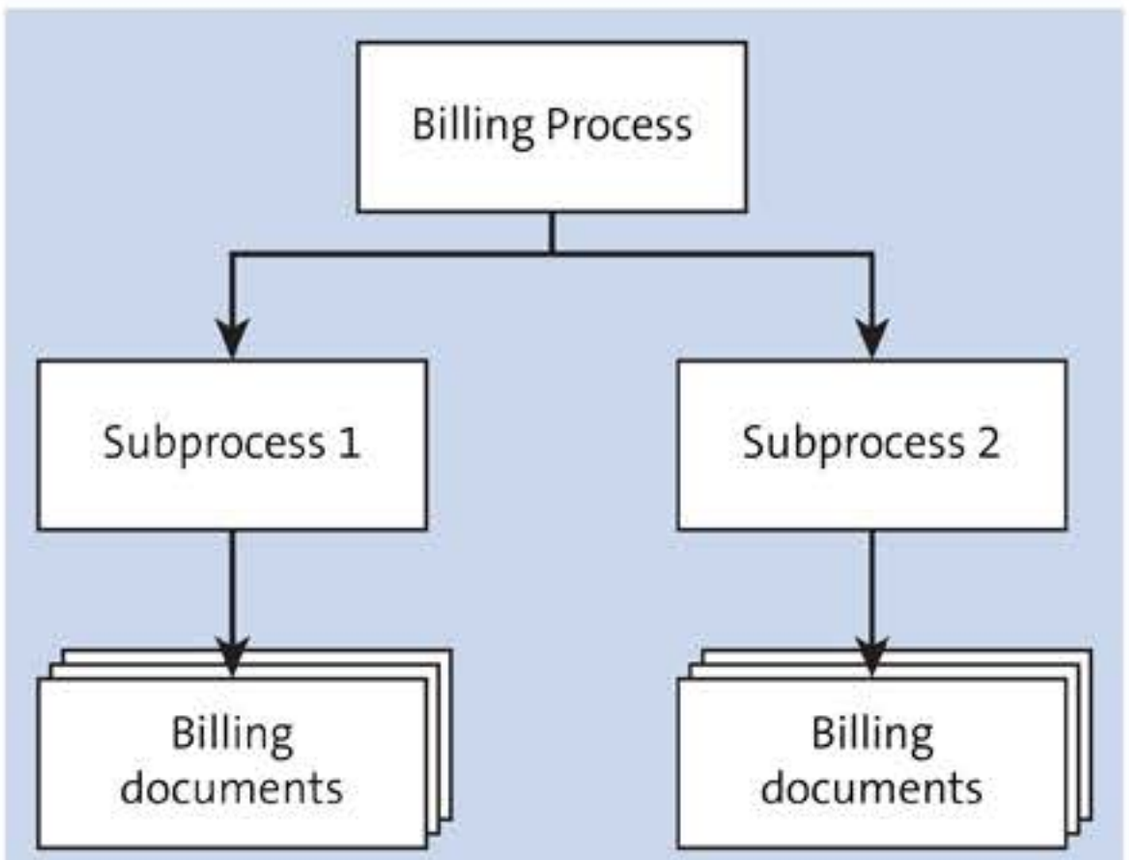

Figure 4.52 Billing Process and Subprocess

A billing process can have one or more subprocesses assigned to it. Billing subprocesses are assigned to the billable item class, and the billable item types of the class are assigned to the subprocesses, which will be considered during the billing process execution. A billing process contains the subprocess, selection variant, grouping variant, and billing type, and aggregation variants are assigned (see Figure 4.53).

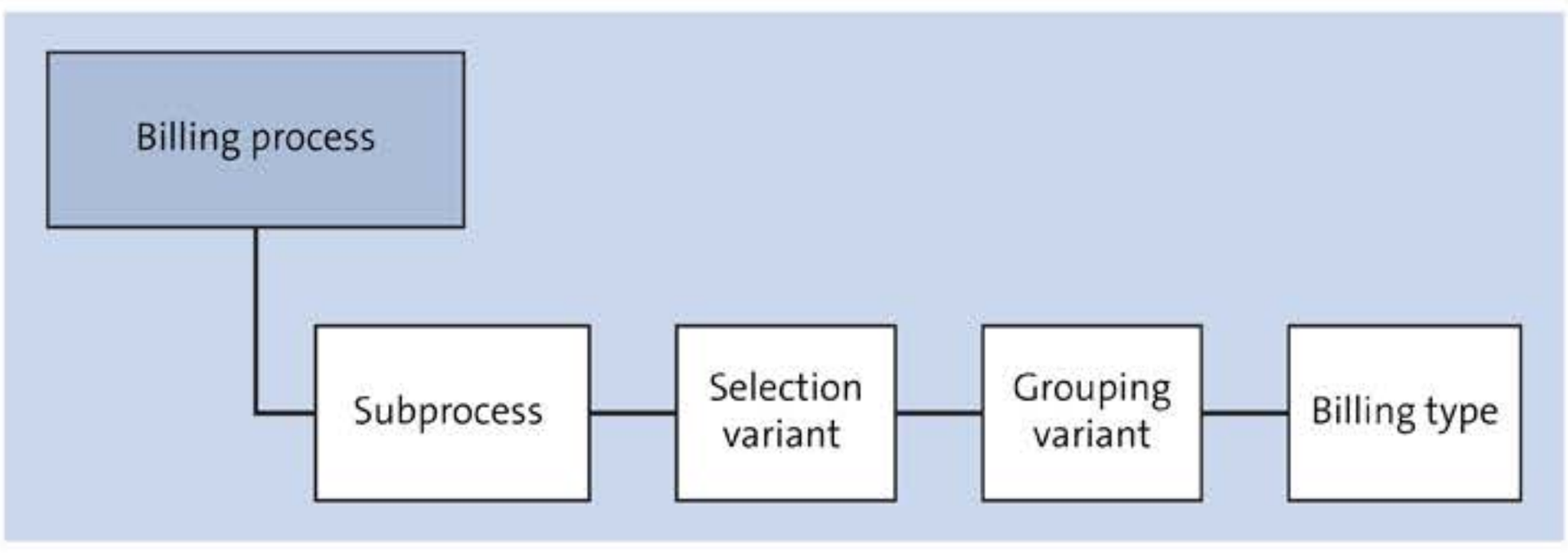

Figure 4.53 Billing Process

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A billing process is created under **Contract Account Receivables & Payables • Convergent Invoicing • Billing • Billing Processes • Define Billing Processes** (see Figure 4.54).

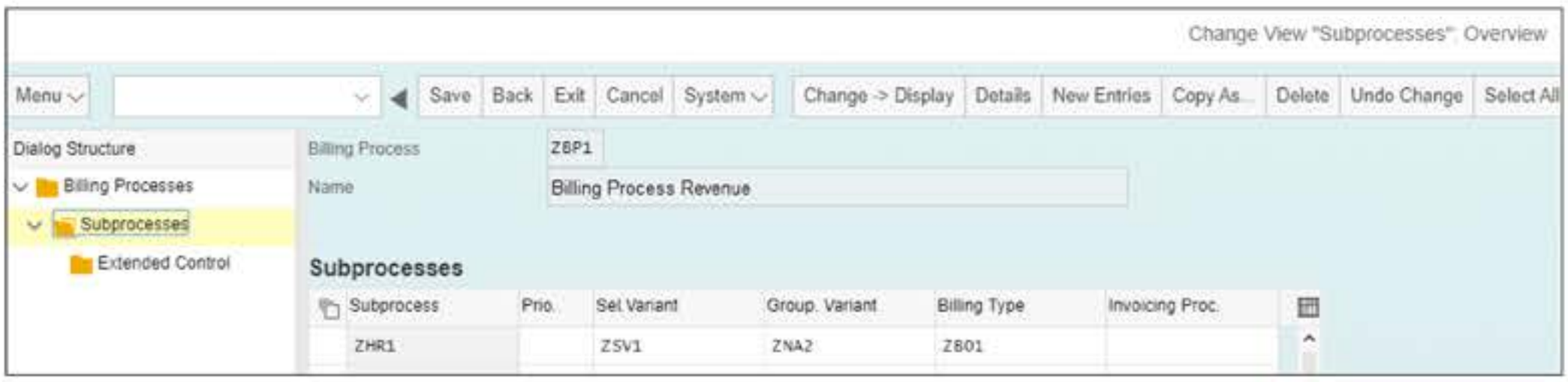

Figure 4.54 Billing Process Creation Screen

A billing subprocess is mandatory for a billing process to execute. Separate billing documents are created based on business partner, contract account, and billing subprocess. Each subprocess will lead to creation of a separate billing document. Billable items belonging to separate subprocesses can't be processed together in a common billing document. The billing subprocess controls selecting billable items, grouping into billing units, and specifying billing types. Billing process assignment to a subprocess and billable item class was shown earlier in Figure 4.18.

The selection variant specifies which billable items of which billable item class will be selected on the execution of the billing process. Billable items belonging to different subprocesses are processed separately in different billing documents.

Business Example

For Martex, there are three charges or billable item types generated. To bill and invoice the customer, these charges should be selected and processed. To select these charges in the billing process, a selection variant `ZSV1` will be created then assigned to the billable item class `HREV`, and the billable item types `ZBT1`, `ZBT2`, and `ZBT3` will be assigned. When the billing process `ZBP1` is executed, all the billable items of type `ZBT1`, `ZBT2`, and `ZBT3` will be selected (see Figure 4.55).

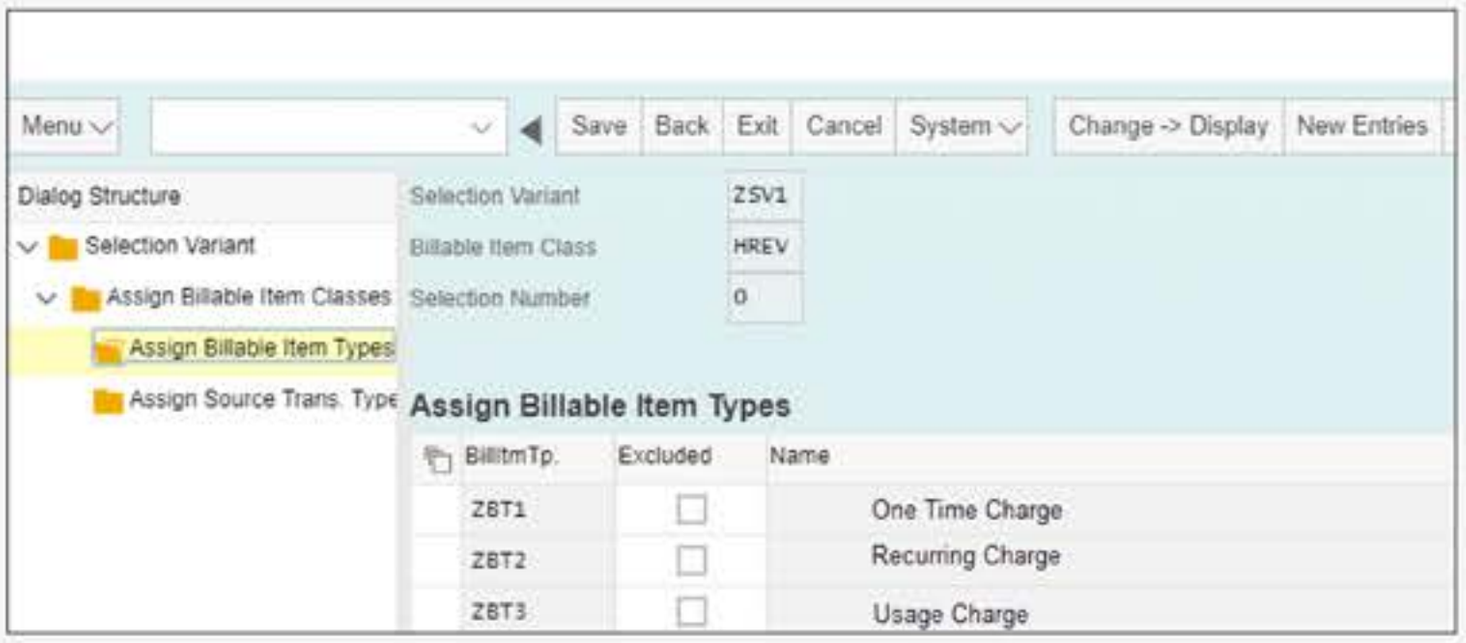

Figure 4.55 Selection Variant for HREV Class

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Business Example

Martex has created billing selection variants for the charges to be selected during the billing process. Now it needs to group the similar charges together and aggregate the selected charges within a group. To achieve this, grouping variant ZNA2 will be created with grouping characteristic as the billable item type, and aggregation variant ZNA1 will be created with aggregation characteristic as the billable item type.

A grouping variant is used for grouping selected billable items based on the grouping criteria. The billable items belonging to the same group will form the billing unit, which will create a billing document. A grouping variant for a billing process is created under **Contract Account Receivables & Payables • Convergent Invoicing • Billing • Billing Processes • Define Grouping Variants** (see Figure 4.56).

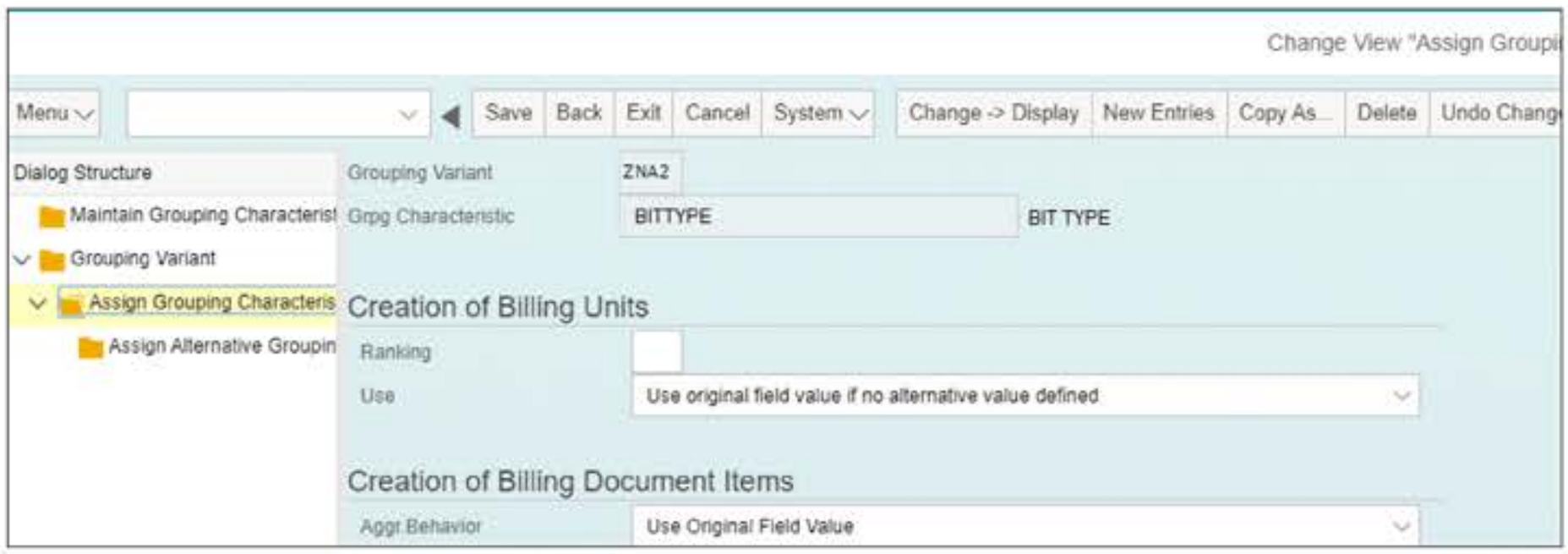

Figure 4.56 Grouping Variant for the Billing Process

For creation of a grouping variant, first grouping characteristics are maintained, which can be assigned to the grouping variant once it's created. Next a grouping variant is created, and then the grouping characteristics are assigned. In addition, alternate grouping characteristics can also be maintained.

Creation of the billing process also requires a billing assignment. Billing types specify the type of the billing process—for example, whether the billing process is for the termination of the service or a periodic billing run. For Martex, billing type ZB01 will be created.

The billing type is maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Billing • Billing Processes • Define Billing Item Types** (see Figure 4.57).

Figure 4.57 Billing Item Type

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Assignment of a billing type to a billing process is shown in Figure 4.57.

An aggregation variant is used for aggregation of the billable items within the same billing unit. The aggregation variant is created in the same node in which a grouping variant is created, but the assignment of the variant happens as an aggregation variant in posting area 8115. The creation of an aggregation is done under **Contract Account Receivables & Payables • Convergent Invoicing • Billing • Billing Processes • Define Grouping Variants** (see Figure 4.58).

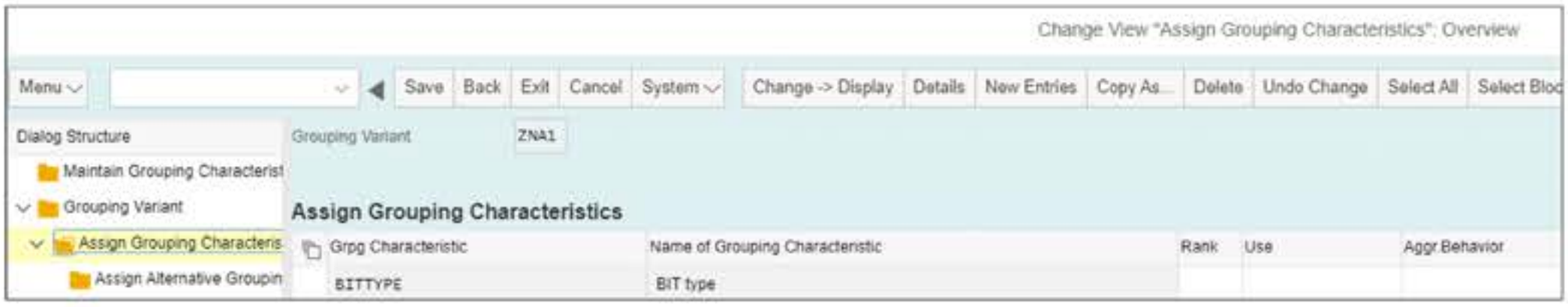

Figure 4.58 Grouping Variant Created for Aggregation for Billable Items

An aggregation variant can be assigned to the combination of the billing process, subprocess, class, and billing type selection keys in posting area 8115. The assignment is performed under **Contract Account Receivables & Payables • Convergent Invoicing • Billing • Billing Processes • Define Aggregation for Billable Items** (see Figure 4.59).

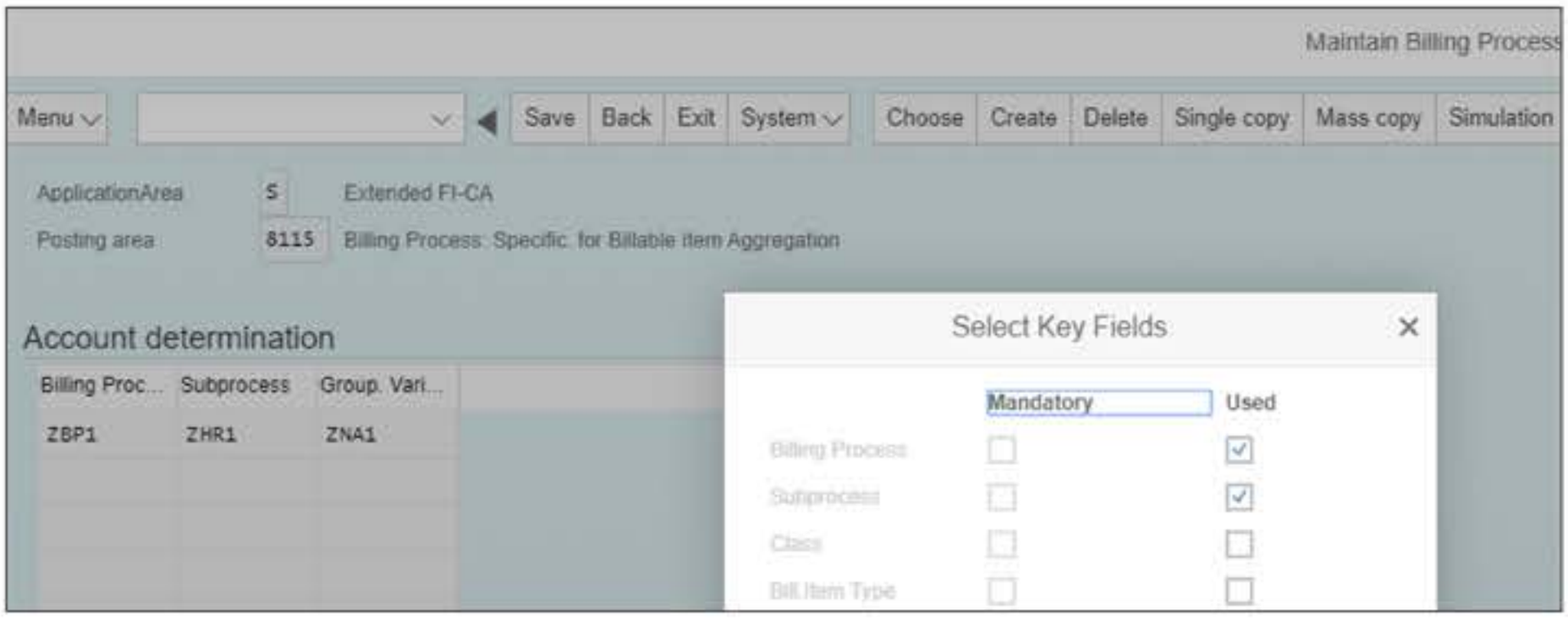

Figure 4.59 Assignment of Variant for Aggregation to Billing Process

Billable Item Types

Billable item types are used for storing business transaction details specific to transaction events such as sales, refunds, recurring charges, and so on. This section covers the different billable item types available.

Billable item types created are assigned to the subprocess, and the subprocess is assigned to a billable item class; therefore, each billable item of the respective class holds the same properties/fields as the class. Billable items are maintained under

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Contract Account Receivables & Payables • Convergent Invoicing • Billable Items • Billable Item Management • Define Item Types.

Billable item types are created per the business requirements associated with the different kinds of charges that will be recorded in SAP Convergent Invoicing as part of day-to-day business. It's essential to map the different kinds of charges originating from the various business transactions to the billable item types; this will help in processing the charges as required and also will be useful if reports must be generated based on billable item types for business transactions.

One Time

One-time billable items are created for charges that are levied once. These are generally for one-time sales of products or service activation charges. One-time charges can be received directly from sales order management. When a sale of a product is done through SAP CRM or SAP Billing and Revenue Innovation Management, subscription order management, the one-time price of the product is transferred to SAP Convergent Invoicing.

The configuration for one-time charges originating from SAP Billing and Revenue Innovation Management, subscription order management, is maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Integration • Customer Relationship Management • Transfer of One-Off Charges • Define Billable item Classes.**

In this node, the billable item class, subprocess, and billable item types are assigned for the one-time charges, which will flow from SAP Billing and Revenue Innovation Management, subscription order management, to SAP Convergent Invoicing (see Figure 4.60).

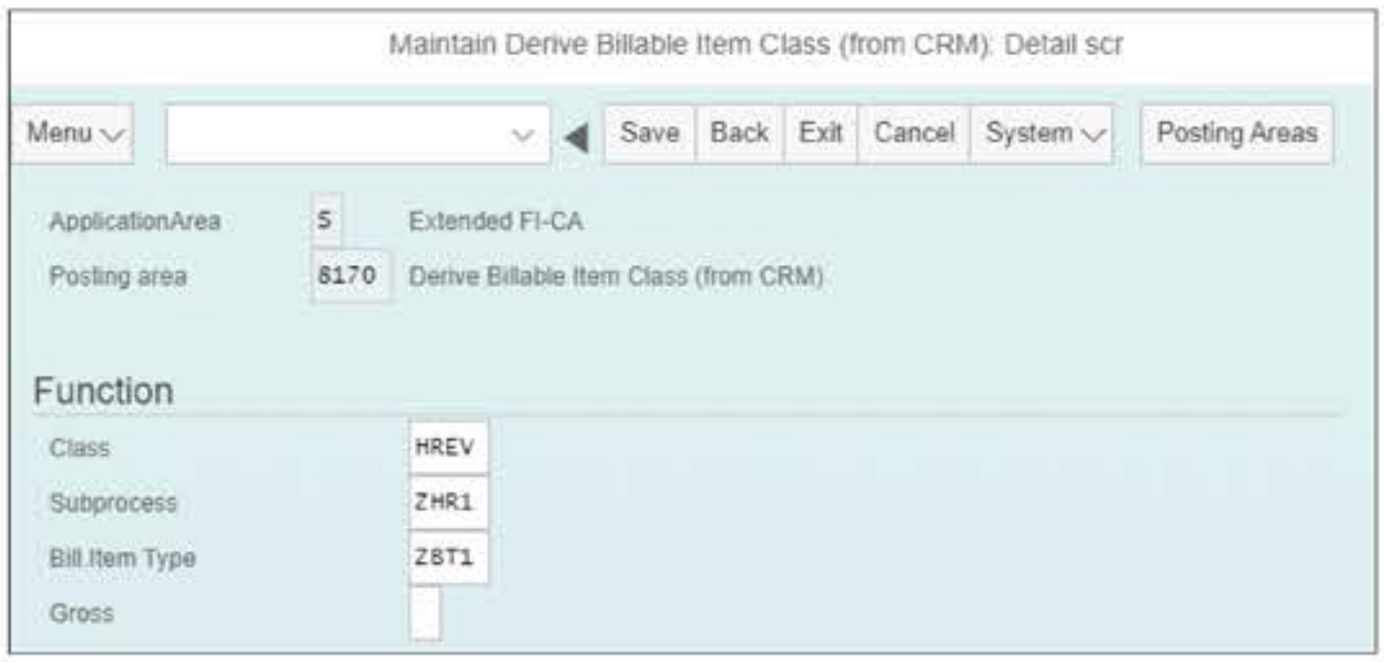

Figure 4.60 Billable Item Class, Subprocess, and Billable Item Type Assignment for One-Time Charge

The next step for the integration is the configuration of account assignment settings for the billable items for the one-time charge. The account assignment objects include company code, division, business area, and segment.

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Account assignment configuration is maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Integration • Customer Relationship Management • Transfer of One-Off Charges • Define Account Assignments** (see Figure 4.61).

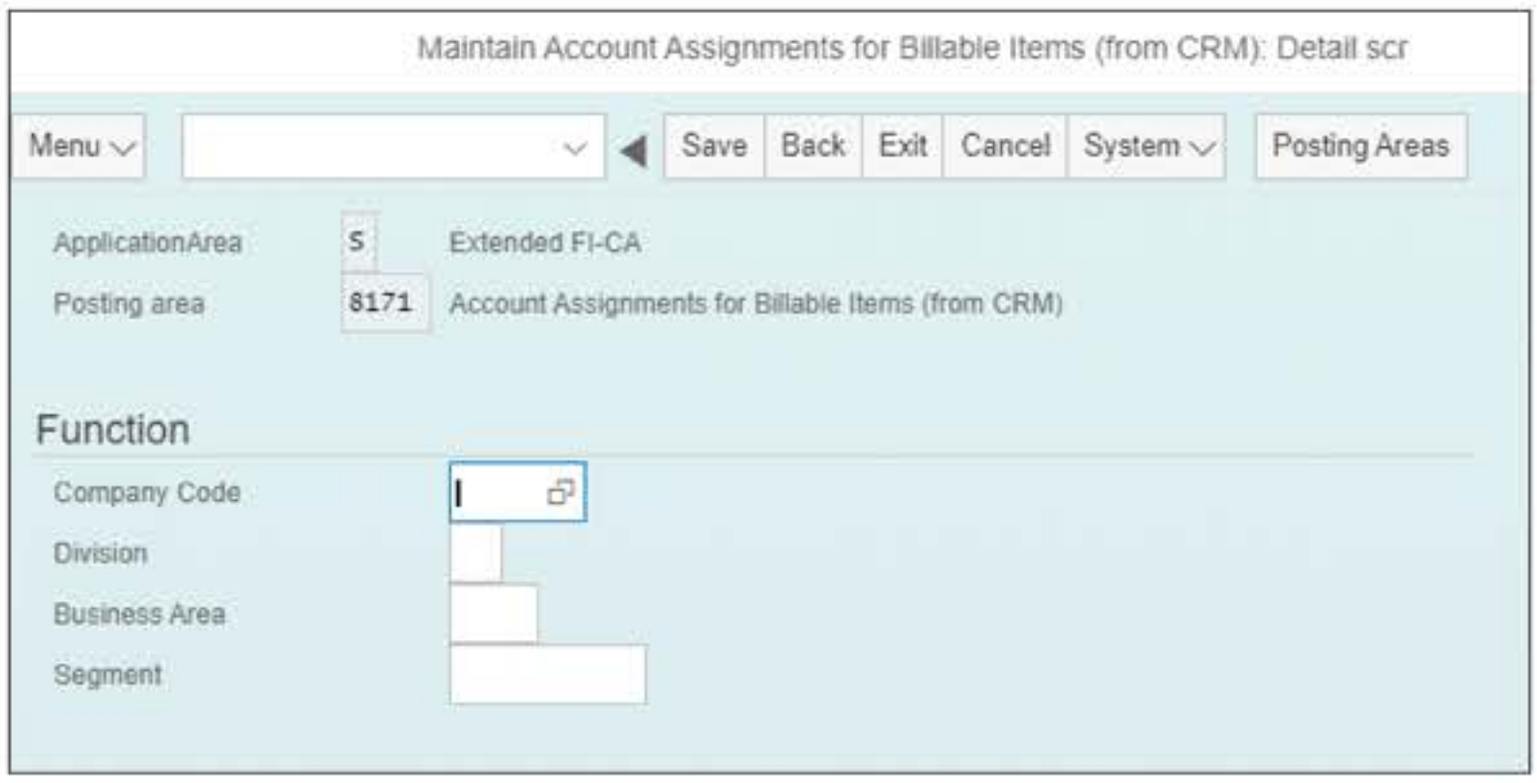

Figure 4.61 Account Assignment for One-Time Charges from SAP Billing and Revenue Innovation Management, Subscription Order Management/SAP CRM

The next step for the configuration setting for the one-time charges from SAP Billing and Revenue Innovation Management, subscription order management, is defining the main transactions and sub-transactions required to post the accounting entries in contract accounting during the invoicing process. Both main transactions and sub-transactions for credit and debit postings are assigned in posting area 8172 for one-time charges.

The configuration is maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Integration • Customer Relationship Management • Transfer of One-Off Charges • Define Main Transactions and Sub Transactions** (see Figure 4.62).

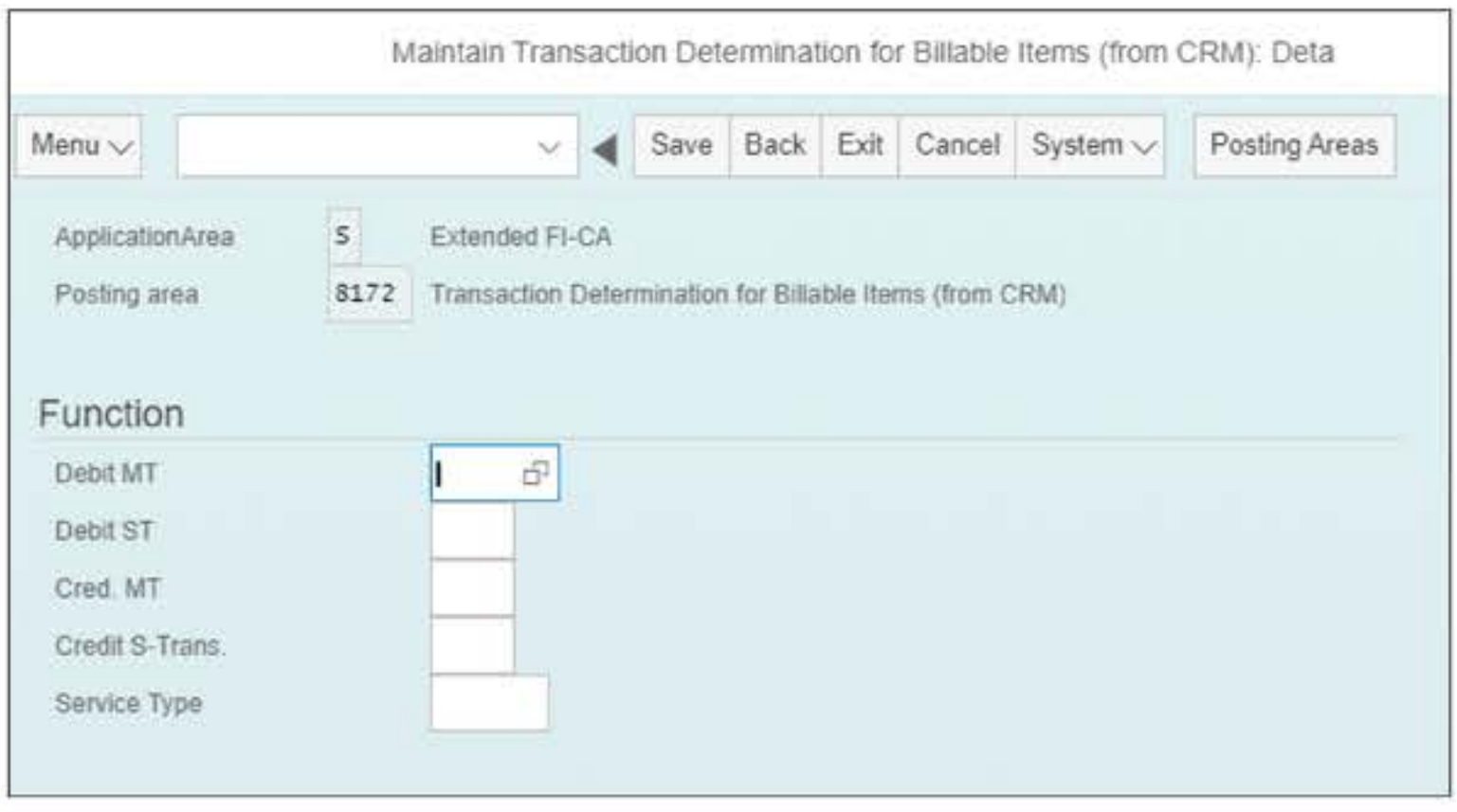

Figure 4.62 Main Transaction and Subtransaction Assignment for One-Time Charges from SAP Billing and Revenue Innovation Management, Subscription Order Management

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Recurring

Charges that are levied for the subscription term recursively for maintaining the subscription are referred to as recurring charges. Recurring charges are posted based on the subscription type; if it's a monthly subscription, then the recurring charges are posted monthly. These charges are usually posted either at the start of the subscription month/period or toward the end of the month/period. When an invoice is generated for the month/period, these charges along with the usage charge and other charges are invoiced and posted on the customer's account.

For Martex, billable item type `ZBT2` is used for posting the recurring subscription charges.

Usage-Based

Usage-based charges are the charges levied for the use of services in the subscription plan by the customer. The usage-based charges are recorded in SAP Convergent Invoicing over a period—monthly, for example—and at the period end are sent for rating to SAP Convergent Charging, which sends the rated and charged items as billable items that are billed and invoiced in the regular periodic invoicing cycle. Receivables are posted to the customer's account.

For Martex, billable item type `ZBT3` is used for the usage-based charges.

Pay-per-Use

These are the charges levied on the customer's account only for when they use the services. These kinds of charges are billed and invoiced as soon as they are received in the SAP Convergent Invoicing system. For example, highway tolls are levied on commuters who use the highway, and these toll charges are made per the use of the highway. A commuter can opt for a monthly pass/subscription instead and can receive the subsidized rate for tolls and be charged at the month end for usage throughout the month.

Refund

These are the charges posted in the system to refund an amount the customer has already paid in advance for purchasing the product or the service subscription. Considering a scenario in which a customer has paid up front for a monthly subscription, as in the case of Martex, this is a reserved instance amount. If the customer later decides not to go ahead with the product or service purchased and asks for the contract to end, the prorated amount is generally refunded from the up-front charge paid if the contract allows. Separate billable item types are generally created to record the refund amount.

Trial

These are generally zero-amount charges or a certain amount of charges that aren't billed and invoiced to the customer. These kinds of charges are used in scenarios of product trials or free subscription trial periods.

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Cost

These kinds of charges are generally the cost of buying a product from a vendor or can be revenue sharing for selling services on behalf of a vendor. For Martex, which purchases the products it sells from different vendors, it uses partner settlement to post the purchases as outgoing charges to the vendors. The partner settlement is described in detail in Section 4.3.6.

Cancellation

Billable item type cancellation is used to create the billable items for the order/subscription cancellation. Cancellation billable items are triggered upon the cancellation of service. Upon the service cancellation, the contract with the service provider ends. The customer account is then invoiced with the final invoice, and any charges applicable are posted to the customer's account upon invoicing.

Billing Documents

The billing process execution results in the creation of the billing documents. Billing documents contain the grouped and aggregated billable items. Billing process execution can generate several billing documents for the same business partner and contract account depending on the subprocess and grouping criteria.

Billing documents can be generated individually or in a mass run of the billing process. Transactions FKKBIX_S and FKKBIX_MA are used for billing process execution individually and en masse, respectively. These transactions can also be accessed through SAP Easy Access menu **Accounting • Financial Accounting • Contract Account Receivables & Payables • Convergent Invoicing • Periodic Processing • Billing** (see Figure 4.63).

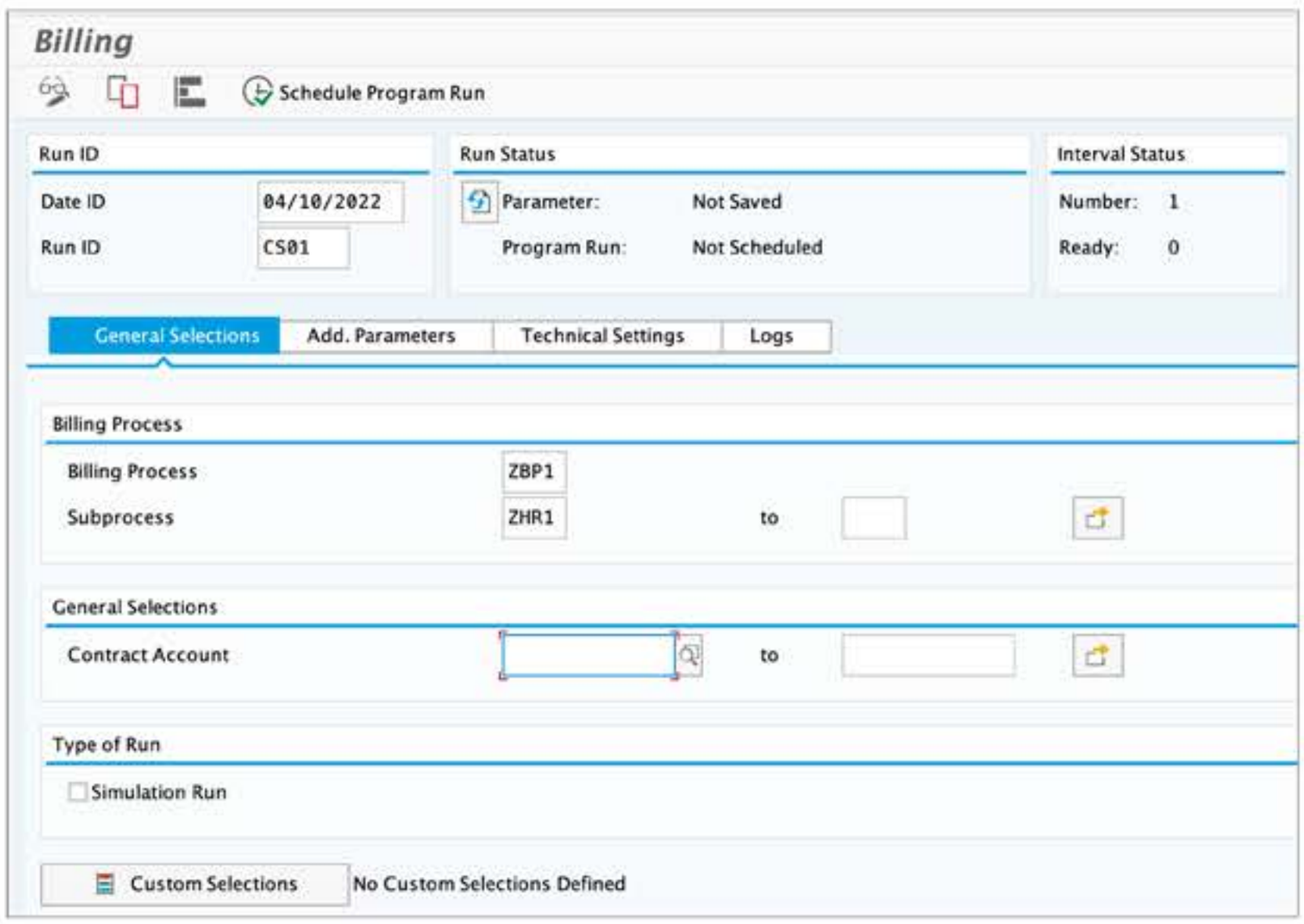

Figure 4.63 Billing Process Mass Run

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Business Example

Billable items created for ZBT1 (one-time charge), ZBT2 (recurring charge), and ZBT3 (usage charge) need to be billed on the billing cycle of the customer. To execute the billing, billing process ZBP1 will be executed. The execution will create the billing documents, which will then be invoiced.

The billing documents created by the billing process have the following sections:

- **Header data details**
  The header section of the billing documents contains details like posting date, customer master data details, start and end dates of the billing period, billing process, billing type, and, most importantly, the invoice from date, which indicates on what date the created billing documents can be invoiced.
- **Line item details**
  The line items section contains details such as quantity, amount, and main transactions and sub-transactions. In the case of aggregated billable items, the **Quantity** field contains the total quantity, and the **Amount** contains the total amount of all the aggregated billable items.
- **Tax items**
  If a tax is calculated on the billable items, then the tax amount and other tax-related information is displayed in this section.
- **Additional items**
  During the billing process, if additional billable items are created and billed together, then this section displays the additional billable items that are created and billed.
- **Payment data items**
  If the customer has paid up front for the purchase, the details of payment are received in the PY record type along with sale billable item record type IT. During the billing process, the payment-related details are populated in the **Payment Data** section, which will be used during the invoicing of the billing documents.
- **Invoicing history**
  This section contains the details of the invoicing document when the billing documents are invoiced.
- **Source items**
  This section contains all the billable item links, which are grouped and billed together; the database table names in which the billed items are stored; and the field billing item group, which you can click to reach a list of all the billable items that are grouped and aggregated together.

Figure 4.64 shows a sample billing document.

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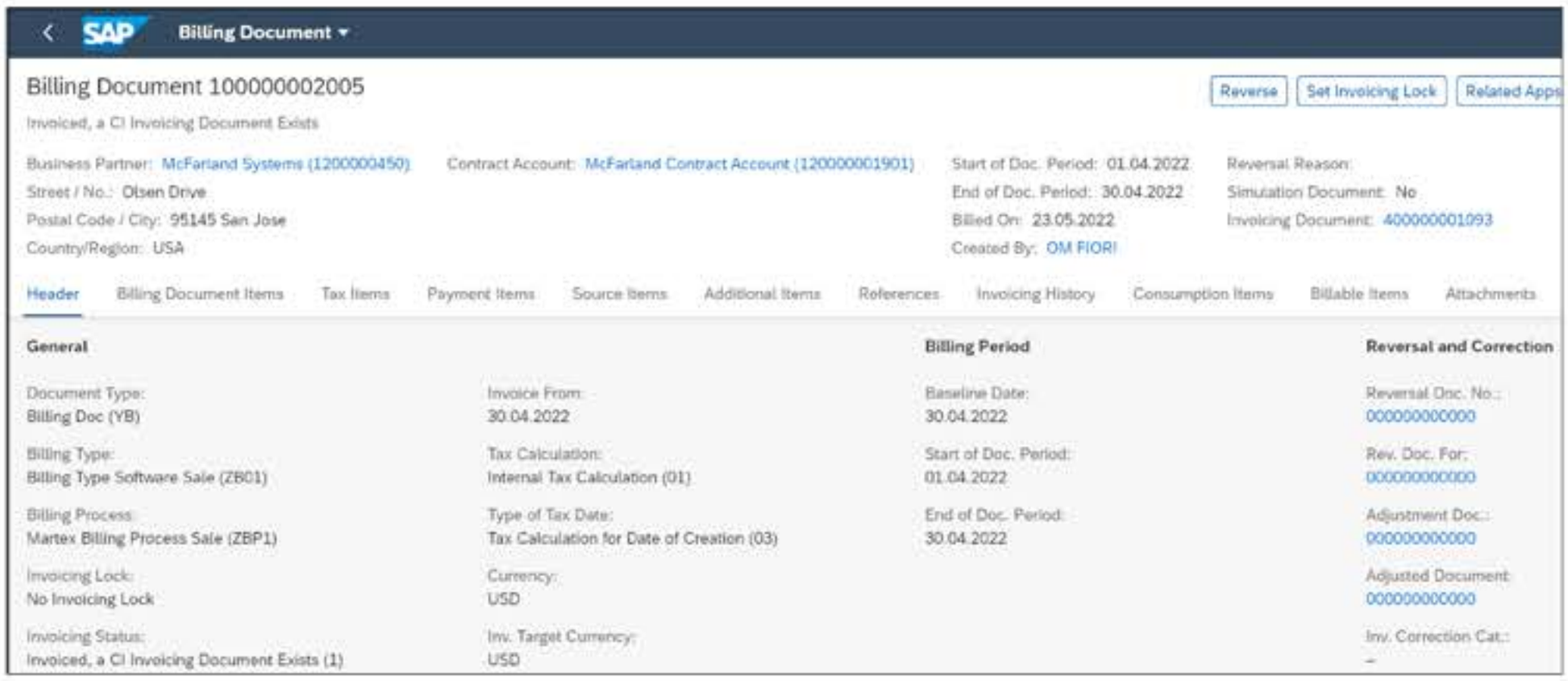

Figure 4.64 Sample Billing Document

We've now discussed billing process and generation of the billing document in detail. In the next section, the invoicing process will be explained in detail, including how billing documents are processed during invoicing.

4.3.5 Invoicing

The invoicing process is used to invoice the billing documents obtained from the billing process. These billing documents can be from the external system, sales and distribution, or from within SAP Convergent Invoicing. The billing documents obtained from the different systems can be invoiced together to provide a consolidated invoice to customers. During the invoicing process, discounts can also be provided to the customers on the basis of a discount key maintained in the contract account. Invoicing also provides options to calculate interest on open items and cash security deposits, plus account maintenance based on the invoicing functions. This section covers each step of invoicing in detail.

Invoicing Process: Revenue/Cost

The invoicing process is set up for processing the billing documents created for the billable item to invoice the customer. The invoicing process creation leads to the creation of receivable posting documents in a scenario for revenue posting from a sale to a customer or a payable document in a scenario for cost posting for obtaining the product from the vendor.

Business Example

Martex generates revenue from the sale of services to its customers, that is, selling services obtained from its vendors. To post the receivables for the revenue, an invoicing process `ZI` will be created.

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The invoicing process has similar steps as the billing process, with the difference in the selection of billing documents instead of billable items. The following steps are required for setting up the invoicing process:

1. Set up the grouping variant.
2. Create the invoicing process.
3. Maintain selection control.
4. Assign invoicing functions.

Each of these steps are described in the following sections in detail.

Grouping Variant

A grouping variant is used to group together the selected billing documents based on grouping characteristics that can be selected from structure `FKKINV_TRIG` for the invoicing orders. Some of the fields include source document category, company code, and contract. In addition, customer-specific fields can be used with a function module that can be assigned to the grouping characteristics.

Grouping variants created are assigned to invoicing processes while creating the invoicing processes. They are maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Invoicing • Invoicing Process • Maintain Grouping Variants** (see Figure 4.65).

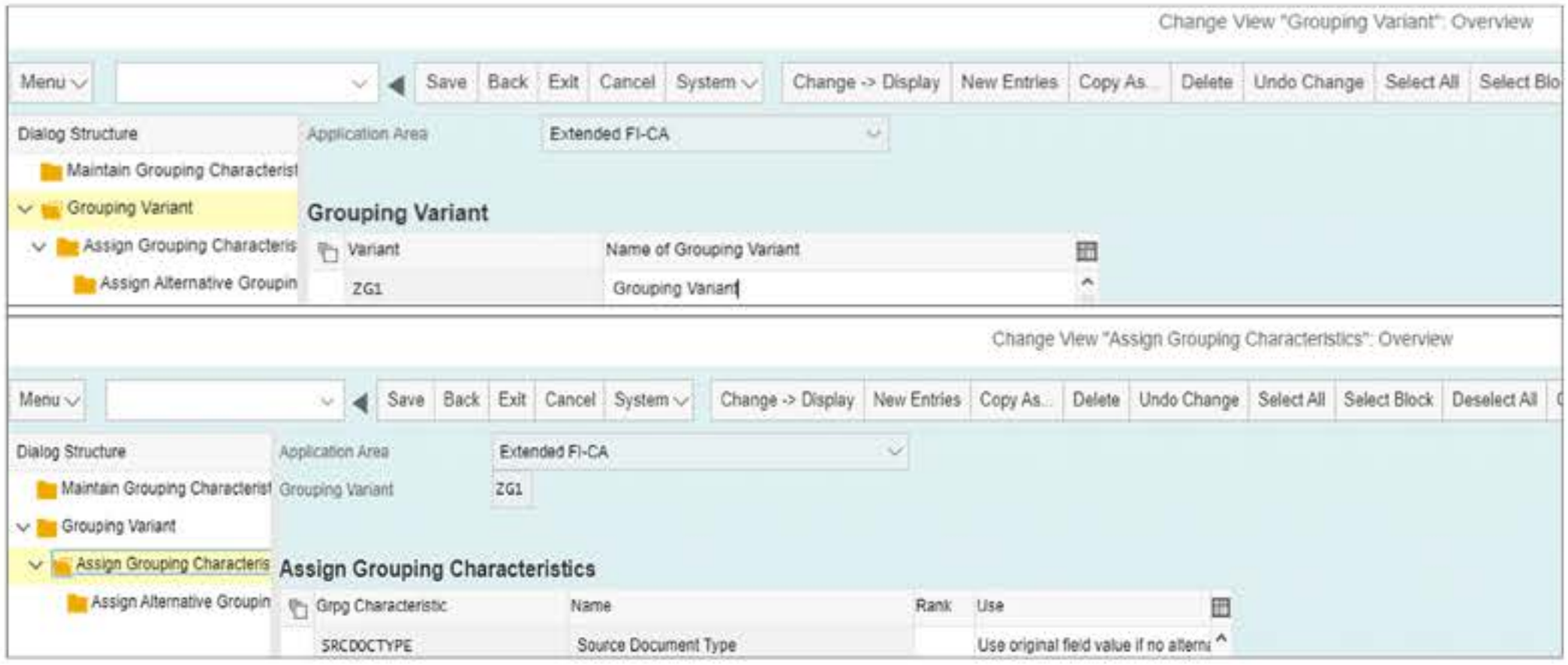

Figure 4.65 Invoicing Grouping Variant

Invoicing Process

To perform invoicing of the billing documents, an invoicing process is created. In the invoicing process, selection conditions are maintained that select the billing documents. A grouping variant is assigned to the invoicing process, which groups the selected billing documents for creation of invoicing units. The invoicing process is assigned invoicing functions that are used for performing the invoicing of billing

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documents. The invoicing process is configured under **Contract Account Receivables & Payables • Convergent Invoicing • Invoicing • Invoicing Process • Define Invoicing Processes** (see Figure 4.66).

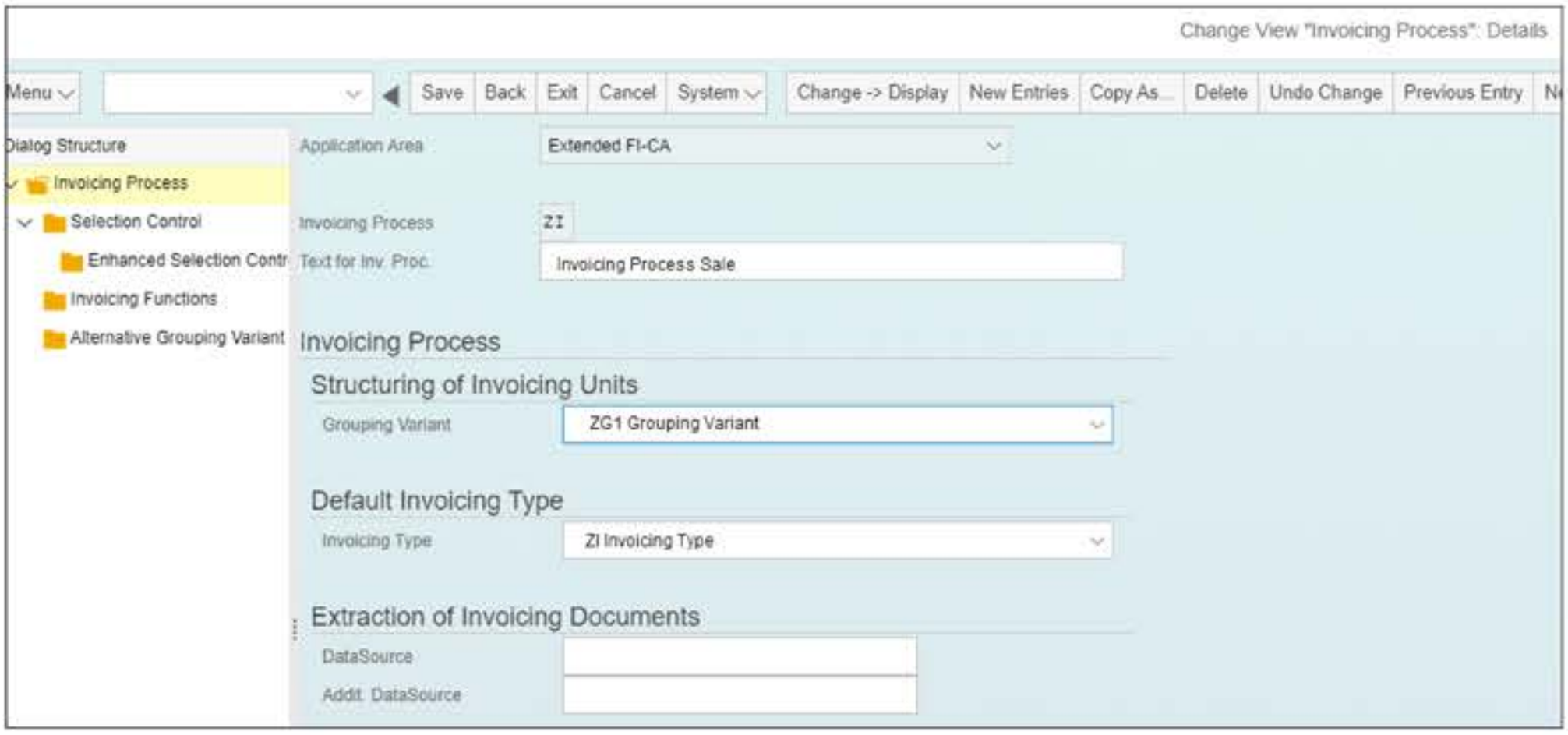

Figure 4.66 Invoicing Process Creation Screen

Selection Control

Selection control is maintained within the invoicing process creation screens. Selection parameters maintained in the selection control are used to control the selection of the billing documents for processing (see Figure 4.67).

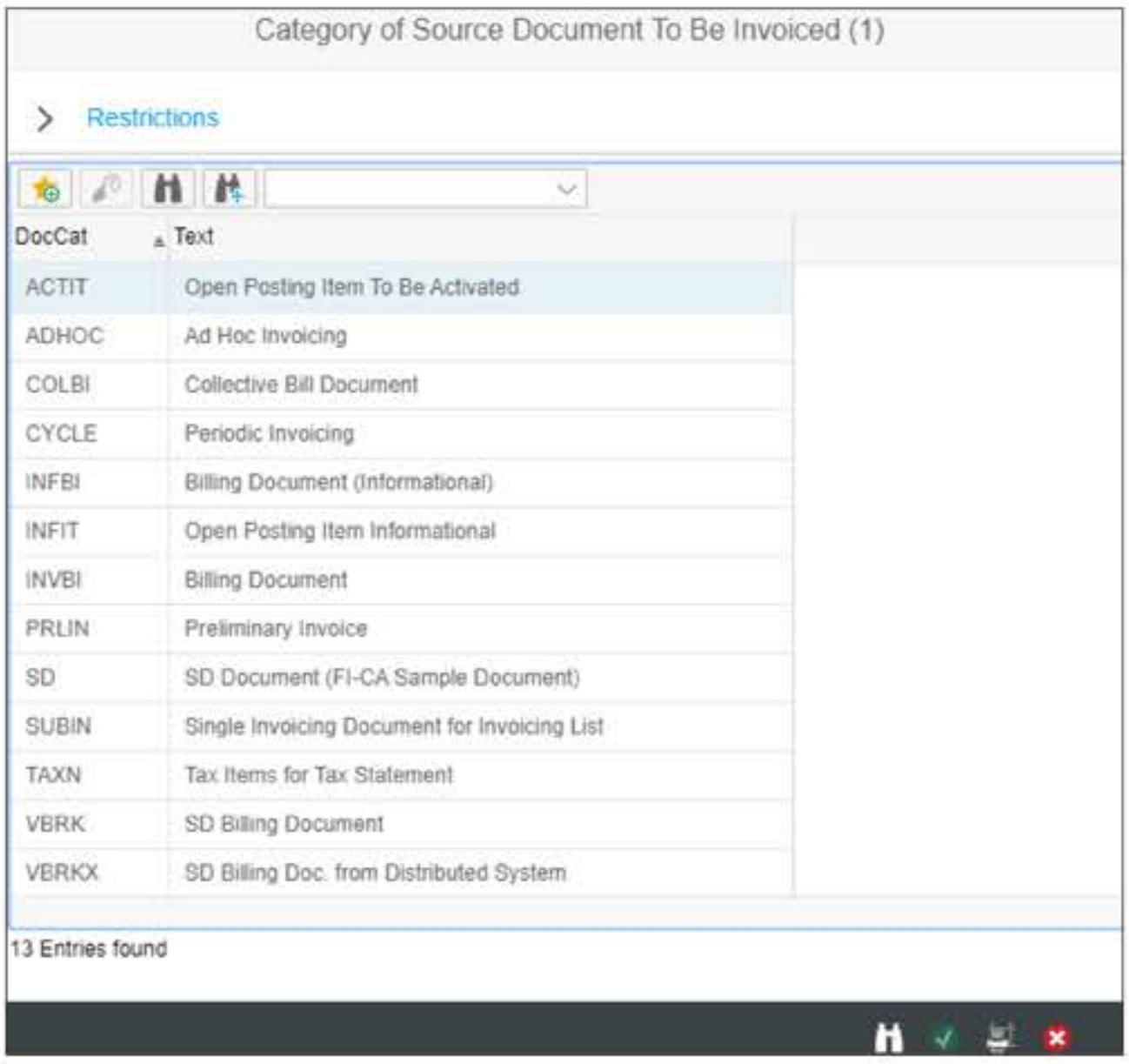

Figure 4.67 SAP Standard Source Document Categories

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The source document category maintained in the selection control specifies which kind of billing document needs to be picked for the invoicing process. Source document categories are the standard values provided by SAP based on the origin of the billing documents.

Figure 4.68 shows the invoicing process selection control screen.

Figure 4.68 Invoicing Process Selection Control Screen

INVBI is the source document category for the billing documents generated within SAP Convergent Invoicing by the billing process executed on the billable items.

The VBRK source document category is used for sales and distribution billing documents, which are pushed to contract accounting for invoicing in SAP Convergent Invoicing.

Invoicing Functions

Invoicing functions indicate the various supported functionalities within an invoicing process. There is a set of standard invoicing functions provided by SAP that is assigned to the invoicing process configuration. Invoicing functions are assigned to the invoicing process at the time of its creation (see Figure 4.69).

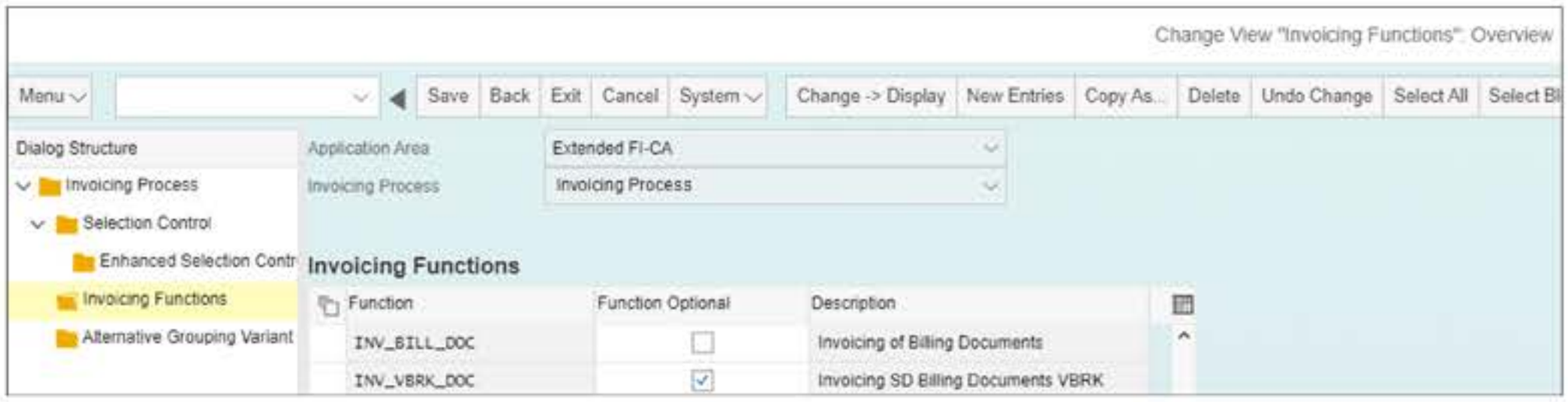

Figure 4.69 Invoicing Function Assignment to Invoicing Process

SAP provides 51 standard functions for invoicing (see Table 4.4).

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Invoicing Function	Description
ACCT_MAINT	Automatic Account Maintenance
ACCT_MAINT_D	Manual Account Maintenance (Dialog Call)
ACCT_MAINTX	Enhanced Automatic Account Maintenance
ACCT_MAINTXD	Enhanced Manual Account Maintenance
ACTIVATE_OI	Activation of Open Items
ACTIVATE_RDI	Activate Documents from Revenue Distribution
ADD_DOC	Creation of Additional (Customer-/Industry-Specific) Docs
BOLLO_CALC	Stamp Tax
BW_INTEGR	Integration in BW Sales Statistics
CHARGE_DISC	Charges and Discounts
CHARGE_DISC2	Individual Charges and Discounts
DEBIT_STAT_I	Debit Entry for Statistical Documents
DEBIT_REV_BY_E	Deferred Revenues - Event-Based
DEF_REVENUES	Deferred Revenues
DUNNING_PROP	Dunning
HIST_ITEMS	Subitems
INSTPLAN_DUE	Set Installment Plan to Due
INTEREST_DEB	Calculate Interest on Open Items
INTEREST_SEC	Calculation of Interest on Cash Security Deposit
INV_ADHOC	Ad Hoc Invoicing
INV_BILL_DOC	Invoicing of Billing Documents
INV_COLLBILL	Creation of Collective Invoicing Document
INV_CYCLE	Periodic Invoicing
INV_OFFSET	Offsetting in Invoicing
INV_SD_DOC	Invoicing of SD Documents
INV_VBRK_DOC	Invoicing SD Billing Documents VBRK
OFFIC_DOC_NO	Official Document Number

Table 4.4 List of Invoicing Functions

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Invoicing Function	Description
PAYFORM	Payment Form
PAYM_HISTORY	Payment History
PAYM_METHOD	Payment Method Determination
PAYM_REF	Payment Reference
PCARD_PAYMET	Card Payment (Payment Method)
PCARD_PAYMNT	Card Payment with Immediate Clearing
PPACC_MAINT	Account Maintenance for Prepaid Account
SUBINV_LIST Create	Invoicing List
SUBINV_TRIG	Preselect Invoicing Document for Invoicing List
PPACC_MAINTD	Manual Account Maintenance for Prepaid Account (Dialog Call)
PPACC_REFILL	Balance Change of Prepaid Account
PRELIM_INV	Creation of Preliminary Invoices
REV_STAT_IT	Reversal of Statistical Line Items
REVERSAL	Invoicing Reversal
RND_INV_DOC	Rounding per Invoicing Document
RND_POST_DOC	Rounding per Posting Document
SECURITY_ADJ	Release/Adjust Cash Security Deposit
SECURITY_CRT	Create Cash Security Deposit
SEPA_PRENOT	SEPA: Direct Debit Pre-Notification
TAX_NOTIF	Create Tax Notification
TRANSFER_IT	Transfer Open Item
VALIDAT_INV	Invoicing Document Plausibility Checks
VALIDAT_SRC	Source Document Plausibility Checks
WRITEOFF	Write Off

Table 4.4 List of Invoicing Functions (Cont.)

The invoicing functions are classified further into main/mandatory and additional/optional invoicing functions. The main invoicing functions are important for the generation of the invoicing documents. The additional invoicing functions are used for

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additional functions that could be performed along with invoice generation. Following are some of the mandatory and additional invoicing functions:

- **Main invoicing functions**
  - Invoicing of Billing Documents (INV_BILL_DOC)
  - Invoicing of SD Billing Documents (INV_VBRK_DOC)
  - Creation of Collective Invoicing Documents (INV_COLLBILL)
  - Creation of Ad Hoc Invoice (INV_ADHOC)
  - Periodic Invoice (INV_CYCLE)
  - Invoicing of SD Documents (INV_SD_DOC)
- **Additional invoicing functions**
  - Card Payment with Immediate Clearing (PCARD_PAYMNT)
  - Card Payment (Payment Method) (PCARD_PAYMET)
  - Payment History (PAYM_HISTORY)
  - Invoicing Reversal (REVERSAL)
  - Automatic Account Maintenance (ACCT_MAINT)

All main and additional functions were listed in Table 4.4.

Invoicing functions should be included for the document categories to be invoiced; for example, invoicing function INVBI should be used for invoicing billing documents created within SAP Convergent Invoicing by the billing process.

Invoicing function PCARD_PAYMNT is used in the scenario in which payment data details are received along with the main billable item record type. During the invoicing process, the receivable is posted in contract accounting, and immediate clearing is performed for the receivable posted using the payment details received in the payment data of billable items.

Invoicing Type

Invoicing types are used to distinguish the invoicing process, such as periodic invoices or final invoices for the customer. Invoicing types can be used to differentiate invoicing units. It's essential to assign an invoicing type to each invoicing process.

During the creation of the invoicing process, invoicing functions are assigned as shown in Figure 4.70, and the assigned functions can be marked as optional. Optional invoicing functions in the invoicing process can be performed based on the activation status maintained in the invoicing type.

There are three type of activation statuses available for the optional invoicing functions, which are available in the invoicing types. Based on the status, an invoicing function can be activated or deactivated when invoicing is performed in dialog mode. Those three activation statuses are as follows:

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- **Active, Cannot Be Deactivated in Dialog**
  An invoicing function with this status would be executed and can't be deactivated.
- **Active, Can Be Deactivated in Dialog**
  Such invoicing functions can be deactivated when invoicing is performed in dialog mode.
- **Not Active, Can Be Activated in Dialog**
  Such invoicing functions aren't active until activated during the dialog mode of invoice process execution.

Invoicing types are maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Invoicing • Invoicing Process • Define Invoicing Types** (see Figure 4.70).

The created invoicing type is assigned to the invoicing process. For Martex, invoicing type **ZI** is assigned to invoicing process **ZI** in the subnode of the invoice type creation node (see Figure 4.71).

Figure 4.70 Invoicing Type Creation Screen

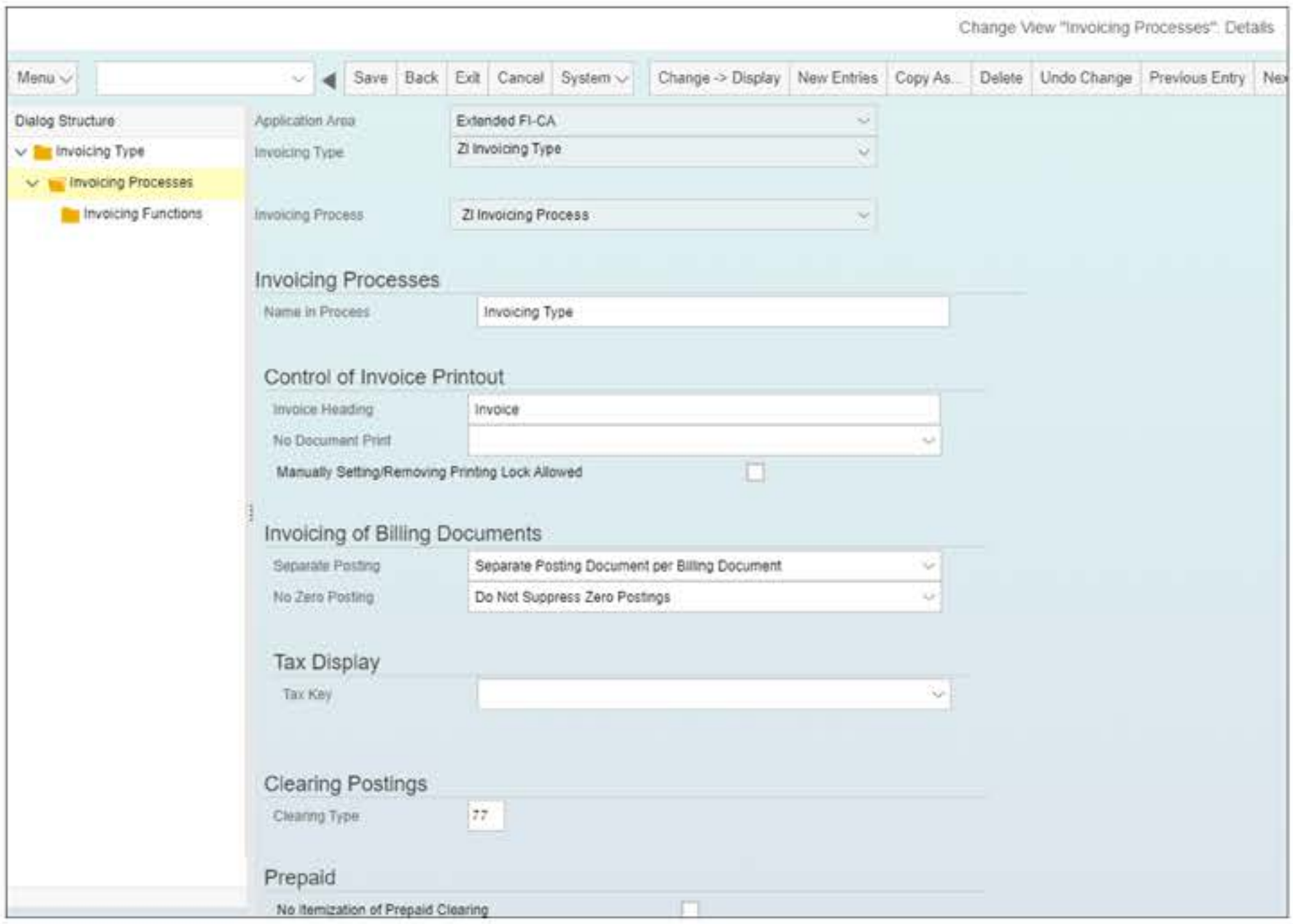

Figure 4.71 Assignment of Invoicing Type to Invoicing Process

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The invoicing functions marked optional in the invoicing process, as shown in Figure 4.72, are assigned to the invoicing type and an activation type is mentioned.

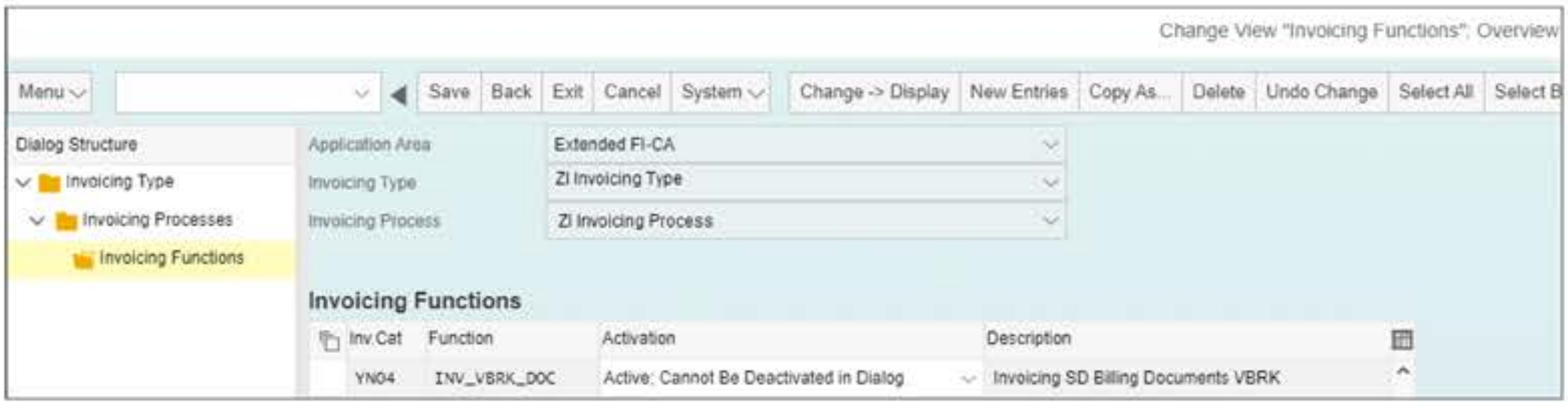

Figure 4.72 Optional Invoicing Function Assignment to Invoicing Type

Invoicing Category

An invoicing category is used for finer control of invoicing. It's assigned to the contract account under the **Convergent Invoicing** tab and is maintained under the invoicing type along with the optional invoicing functions. During the execution of the invoicing process, only the contract accounts with an invoicing category can have the optional invoicing functions executed.

The invoicing category is configured under **Contract Account Receivables & Payables • Convergent Invoicing • Invoicing • Invoicing Process • Define Invoicing Categories**. The assignment of an invoicing category to an invoicing type and optional invoicing function is shown in Figure 4.73.

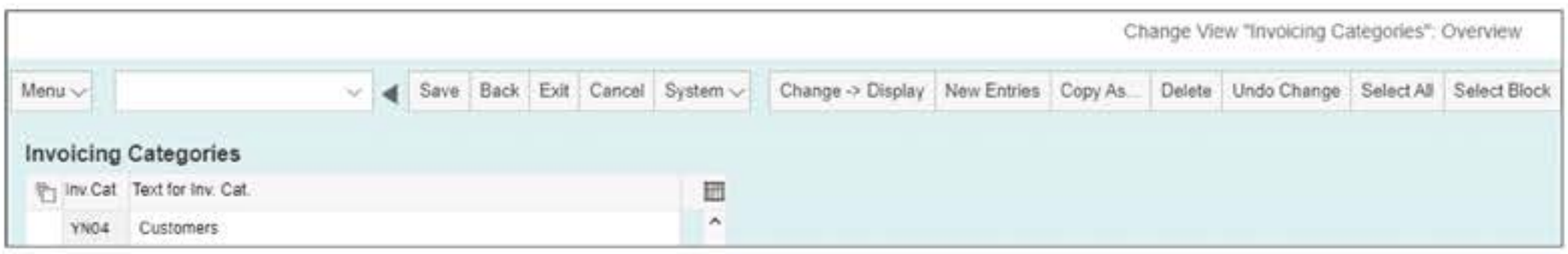

Figure 4.73 Invoicing Categories Creation Screen

Invoice Process Execution

The invoicing process can be executed in individual or mass processing. Its execution requires selection parameters to be filled in. The business partner and contract are general selection parameters and can be used for parallel job execution. An invoicing run execution also requires an invoicing process to be maintained on the selection screen for execution. Based on the selection parameters and any custom selections, the billing documents are selected. Selected billing documents whose target invoicing dates are reached are processed in the invoicing process. Invoice process execution generates an invoicing document and posts the receivables documents in contract accounting on the customer account.

Transactions for execution are FKKINV_S and FKKINV_MA, and the same can be accessed from SAP Easy Access menu **Accounting • Financial Accounting • Contract Account**

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Receivables & Payables • Convergent Invoicing • Invoicing • Documents • Create and from **Accounting • Financial Accounting • Contract Account Receivables & Payables • Convergent Invoicing • Invoicing • Periodic Processing • Invoicing • Execute Invoicing** (see Figure 4.74).

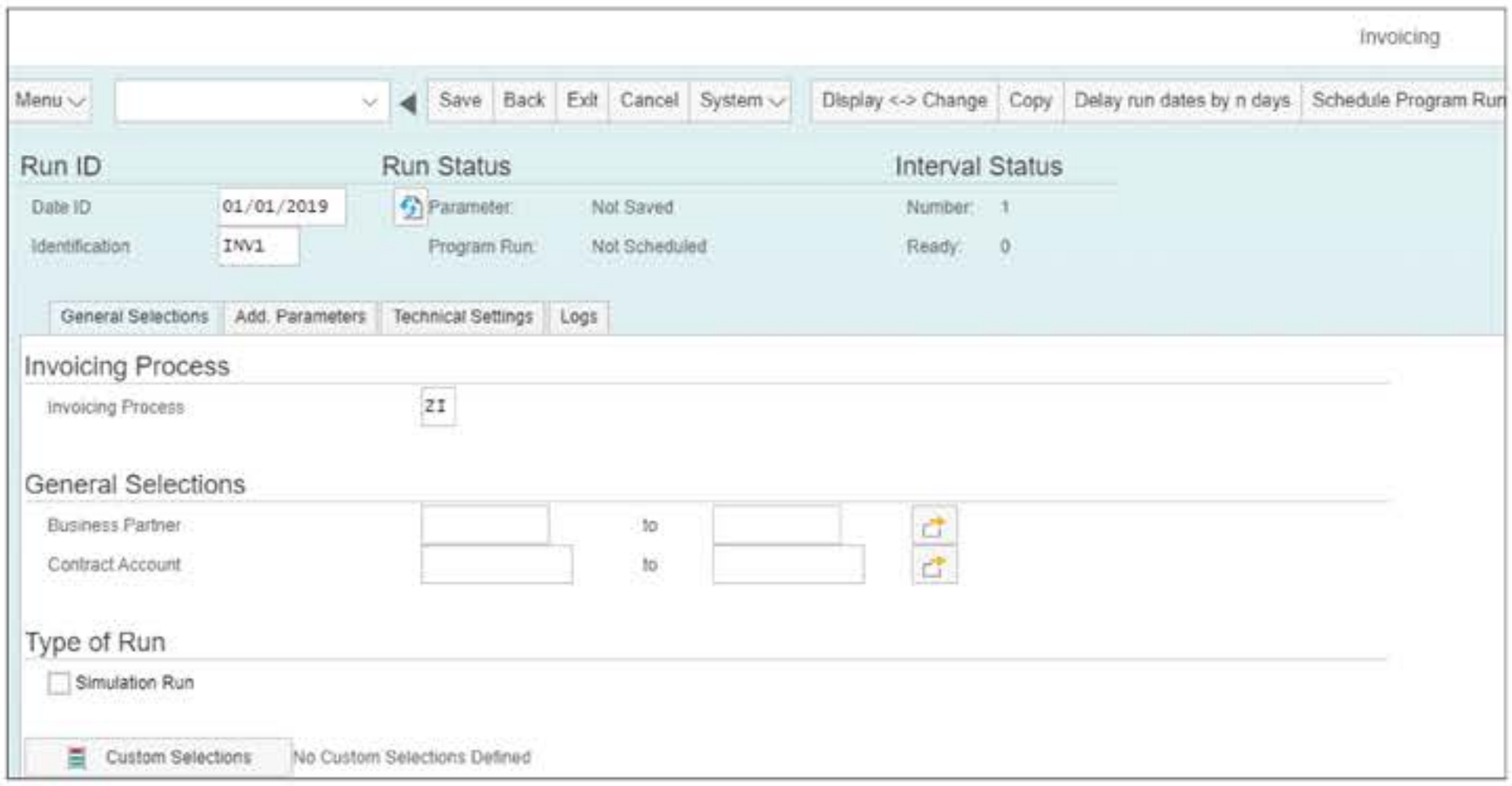

Figure 4.74 Mass Invoicing Run Screen

Invoice Documents

An invoice document is the invoice sent to customers for their use of a service. It contains the details of the amount, usage, previous unpaid receivables, payment details, and so on. Invoicing documents hold financial entries and are generated upon the successful execution of the invoicing process. The invoicing process generates the invoicing document along with the contract accounting posting documents, which post the accounting entries (see Figure 4.75).

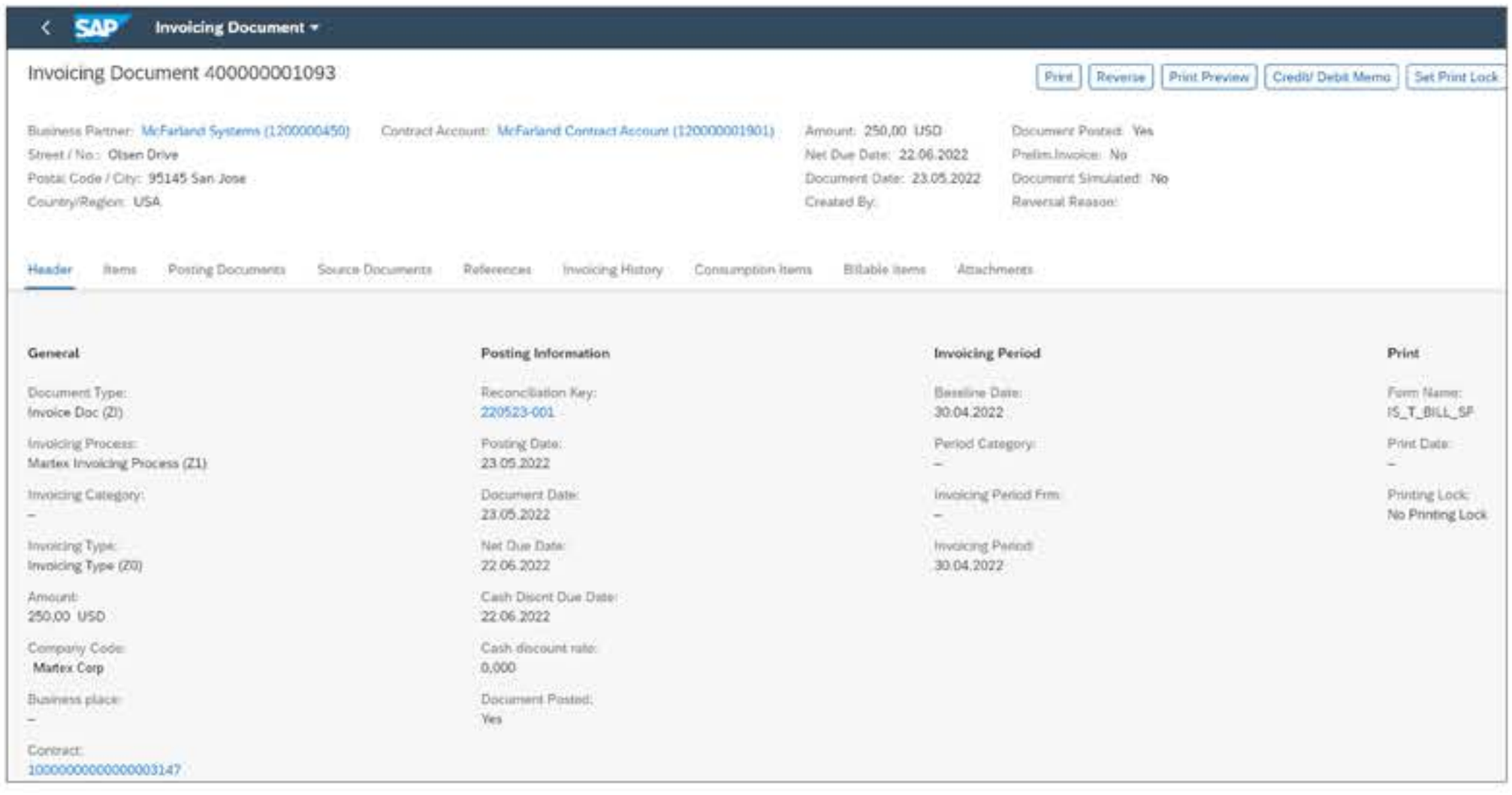

Figure 4.75 Sample Invoicing Document

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An invoicing document contains several pieces of information, structured as follows:

- **Header section**
  Contains information such as posting dates, due dates, master data, total invoiced amount, document type, invoicing process, invoicing type, reconciliation key, and reversal document information.
- **Items**
  Contains the details of invoice line items, including amounts, invoicing types, and account assignment details.
- **Posting documents**
  Contains the details of the contract accounting posting document, which navigates to the document view in contract accounting upon clicking.
- **Source documents**
  Contains the details of the billing documents invoiced together in the invoicing process. Upon selection, you'll be taken to the billing document view.
- **Billable items**
  Contains details of the billable items associated with the billing documents invoiced.
- **References**
  Contains the reference information for the provider contract or provider agreement.
- **Invoicing document type**
  Used to identify the postings. Document types are associated with number ranges for individual and mass processing. A posted document uses these ranges to assign document numbers to the document types. Document types, together with document numbers, uniquely identify the invoicing document posted by the invoicing process.

Customizing for the document type can be maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Invoicing • Documents • Invoicing Documents • Maintain Document Types & Assign Number Ranges** (see Figure 4.76).

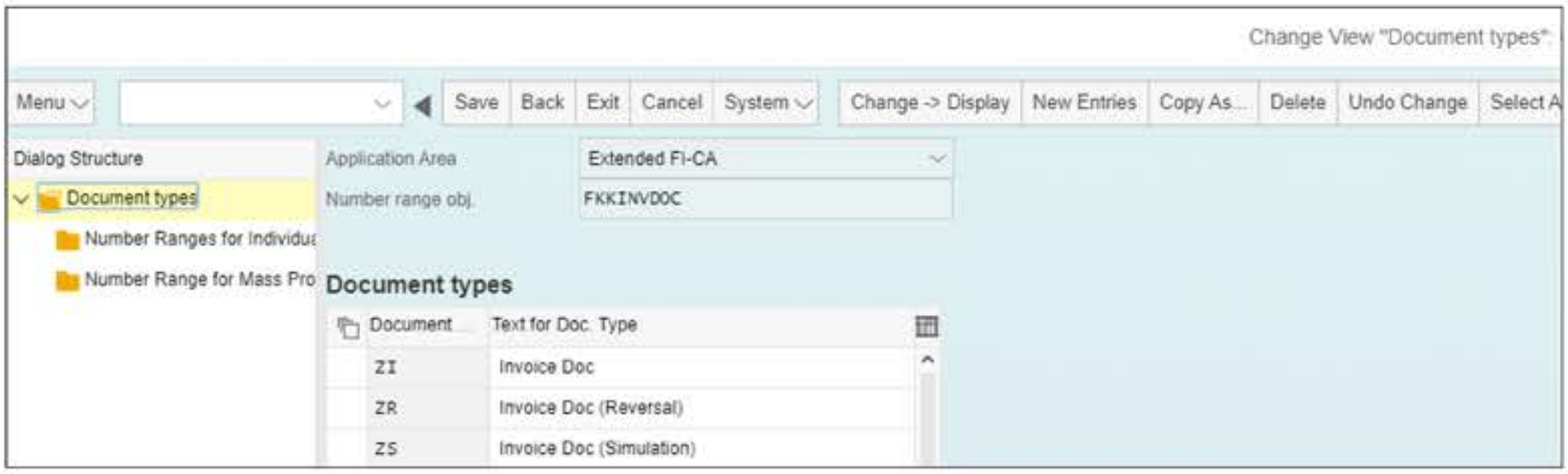

Figure 4.76 Document Type for Invoicing Documents

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Separate document types can be assigned to the invoicing process. Simulation and reversal of invoicing also requires document number ranges to be maintained (see Figure 4.77). The associated **Posting area** is **2600**.

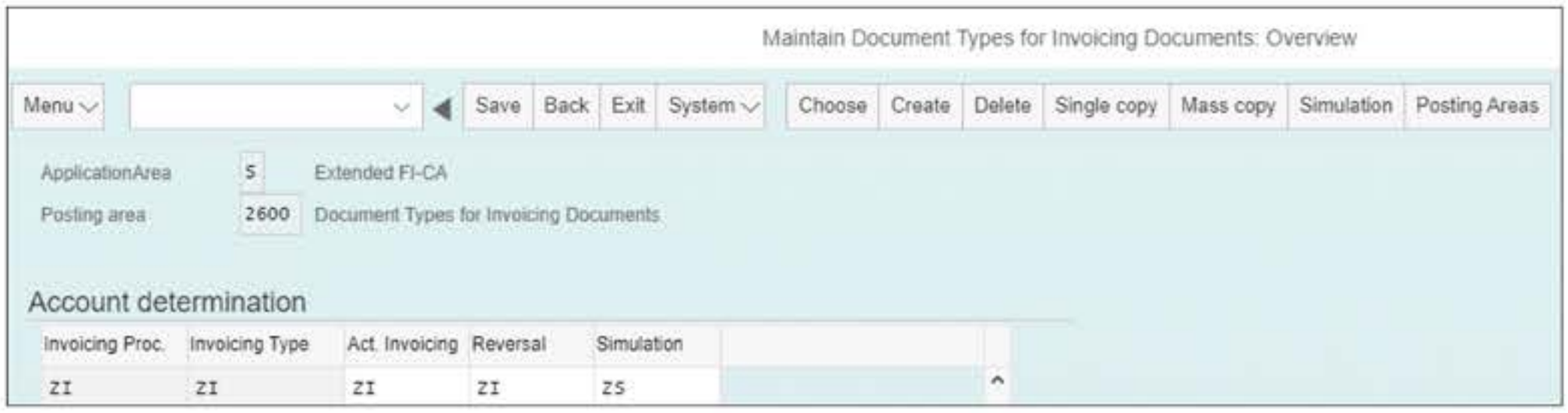

Figure 4.77 Document Type Assignment to Invoicing Process and Invoicing Type

Posting Documents

The invoicing process posts invoicing documents along with the posting documents in contract accounting. A document type also is assigned to the posting documents against each invoicing process and invoicing type. The associated posting area is 2605.

The associated settings can be maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Invoicing • Documents • Posting Documents • Define Document Types for Posting Document for Invoicing** (see Figure 4.78).

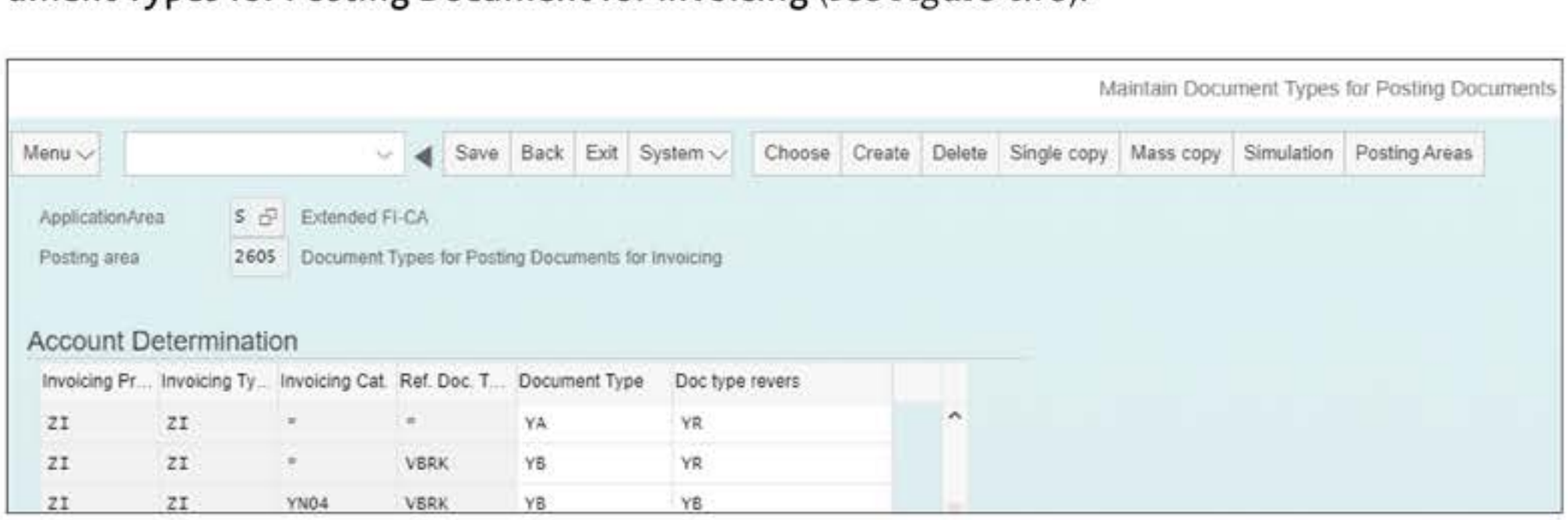

Figure 4.78 Posting Document Types Assignment for Invoicing

Account Assignment for Posting Documents

Posting documents associated with the invoicing document are posted in contract accounting for receivables management. The revenue, expense, tax, and subledger postings are required to be posted in the relevant general ledger accounts. The determination of the general ledger accounts is achieved by using posting areas.

Posting Area 2611 for Business Partner Items

Posting area 2611 contains the subledger/reconciliation accounts for the receivables/payables. General ledger accounts are assigned against each main transaction with a

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combination of key fields such as company codes, account determination IDs, and divisions. The settings can be maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Invoicing • Documents • Posting Documents • Define Account Assignment for Business Partner Items** (see Figure 4.79).

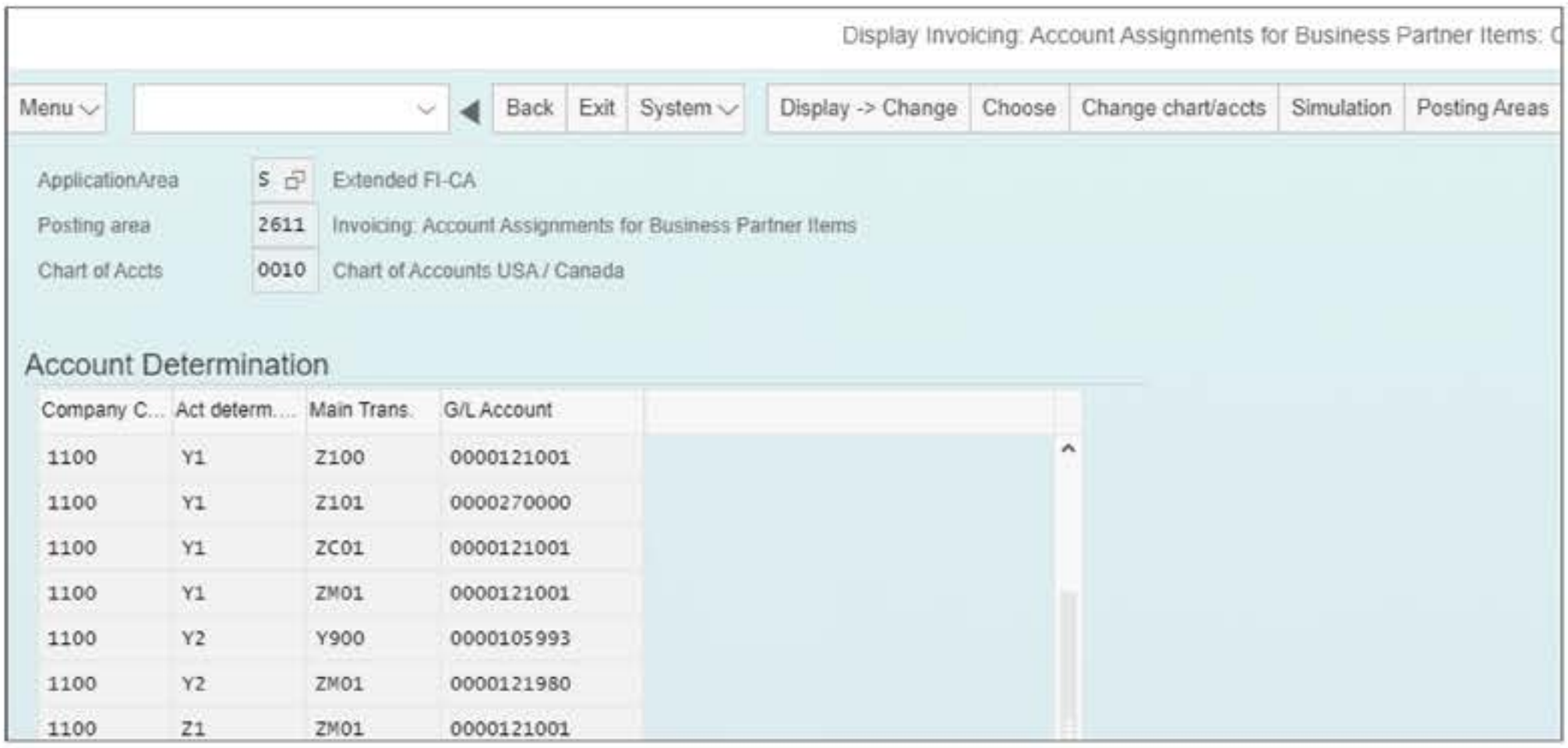

Figure 4.79 Posting Area 2611 for General Ledger Determination for Business Partner Items

Posting Area 2610 for General Ledger Items

Posting area 2610 contains the general ledger accounts for the revenue, expense, and tax determination. The general ledger accounts are maintained against each main transaction and subtransaction combination using key parameters such as company code, division, and account determination ID.

The settings can be maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Invoicing • Documents • Posting Documents • Define Account Assignment for General Ledger Items** (see Figure 4.80).

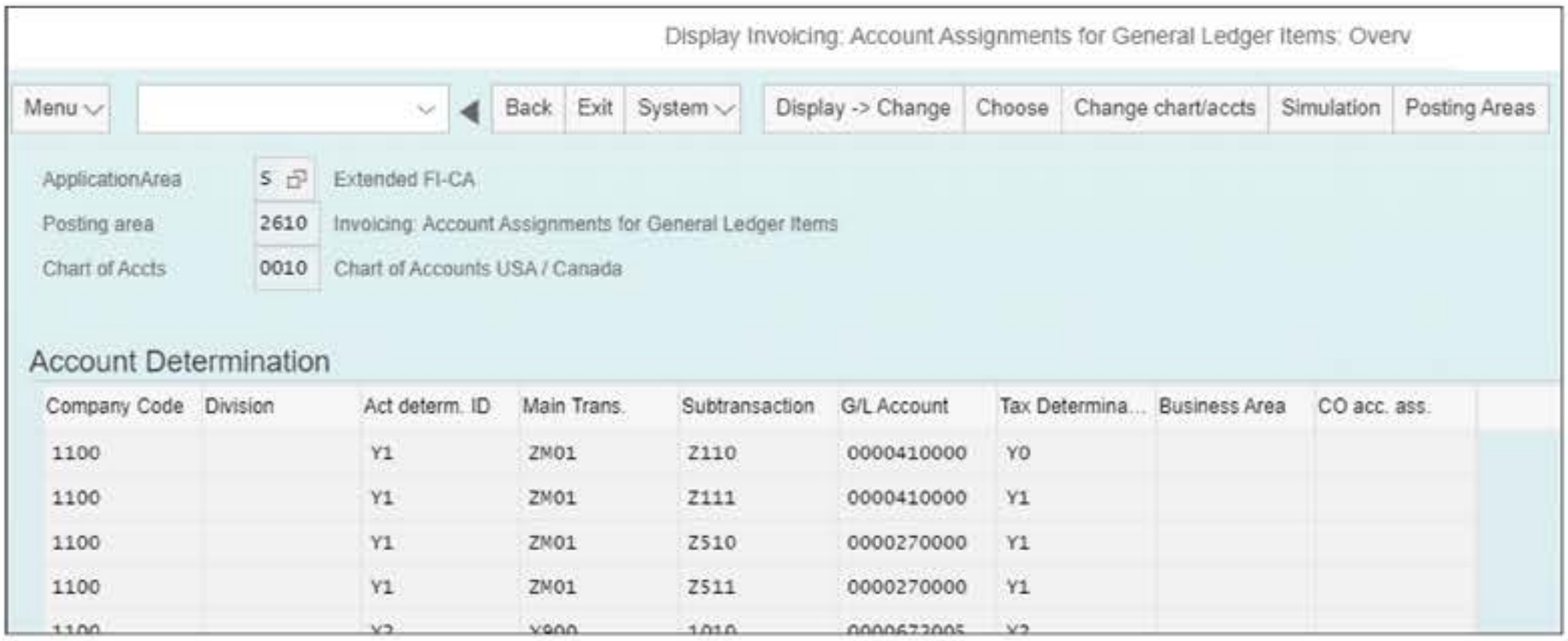

Figure 4.80 Posting Area 2610 for General Ledger Items

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As of SAP S/4HANA 1909, we're able to process intercompany settlement in SAP Convergent Invoicing by handling dual rates and tax account keys. Generating an invoice with the other processes in SAP Convergent Invoicing can significantly reduce delays caused by intercompany reconciliation with sales and distribution processing.

4.3.6 Dependent Items

The dependent items capability enables systematic creation of follow-on related (or) dependent transactions based on a primary transaction. There are many business scenarios that drive a need to create a dependent charge such as a discount to the customer or to generate a payable royalty line item to the vendor. This capability in SAP Billing and Revenue Innovation Management allows automated creation of such dependent transactions based on business rules that can be configured. The dependent item can be either of the following:

- General item (additional charge or discount)
- Partner settlement item

The dependent items themselves could be either of type consumption item or a billable item, as shown in Figure 4.81.

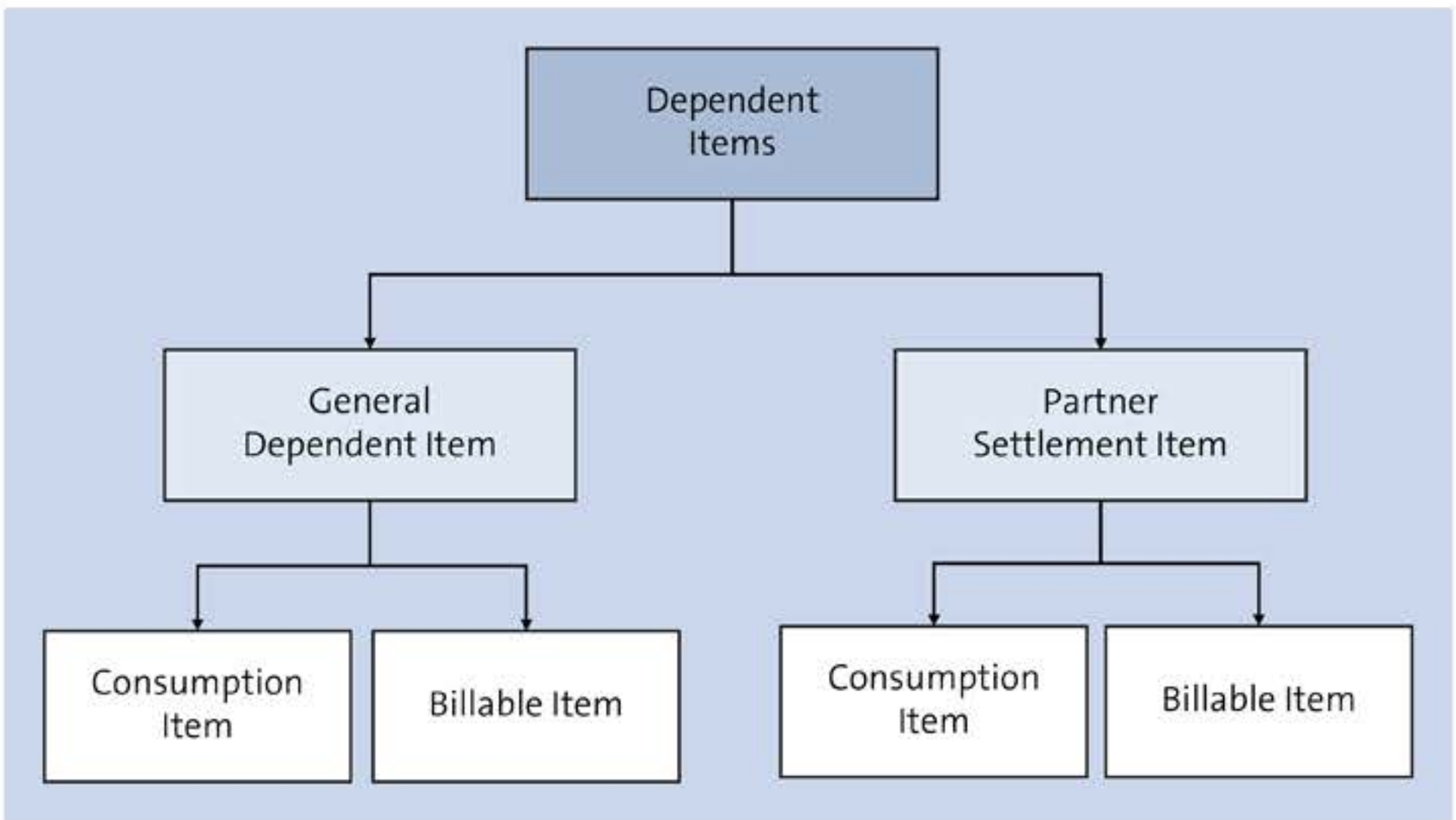

Figure 4.81 Dependent Items with SAP Convergent Invoicing

For the Martex use case, the dependent items capability isn't demonstrated. However, we'll review the concepts and the necessary configurations involving dependent items.

You've seen what a dependent item is and the types of dependent items. A primary transaction or primary item created for a customer drives the creation of a **Dependent**

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Item based on a **Dependent Reason Type**, as shown in Figure 4.82. Examples of a dependent item reason include a dependent charge, discount, or revenue.

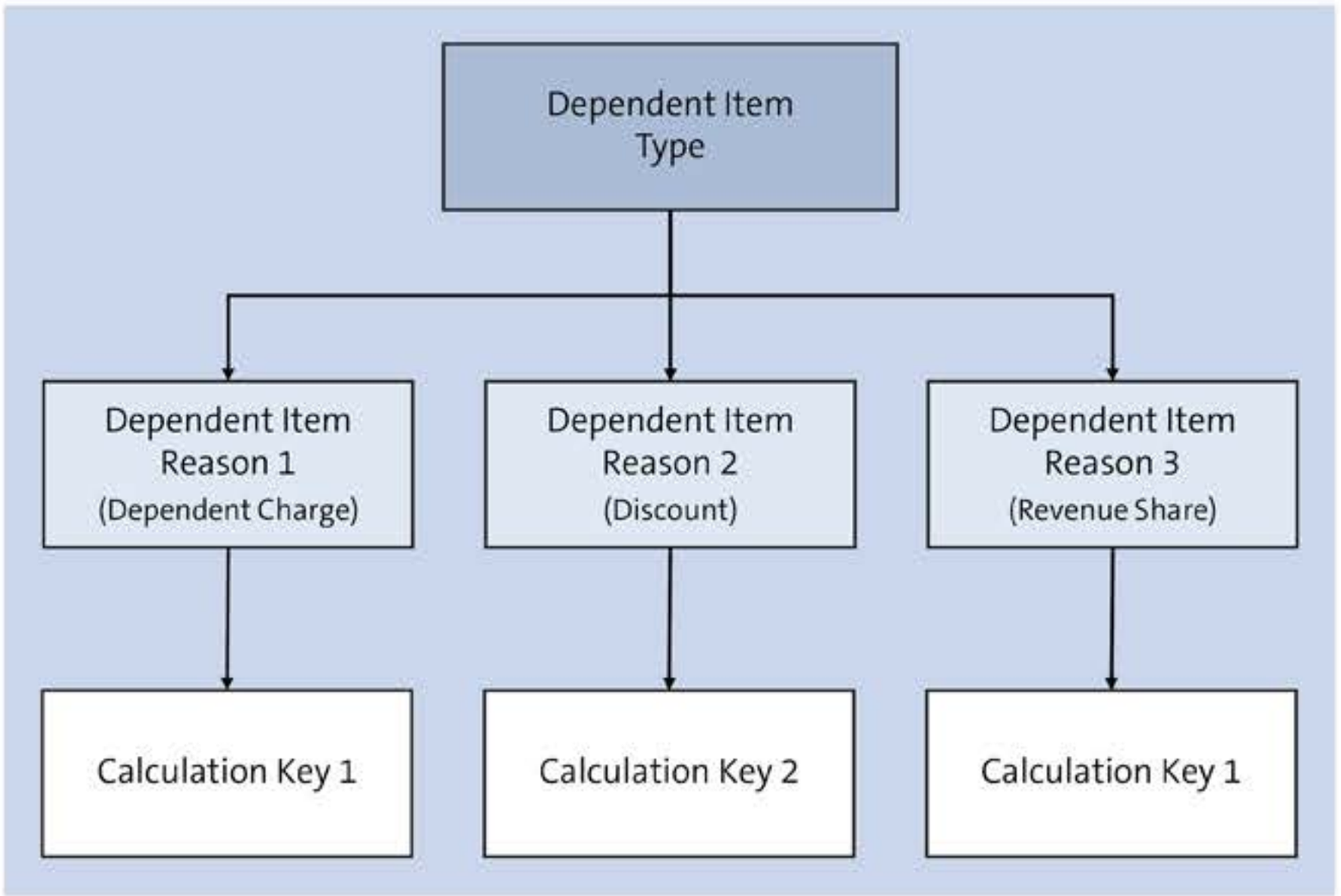

Figure 4.82 Dependent Item Type and Its Relationship with Dependent Item Reasons

In the following sections, we'll cover the configuration setup required to activate the dependent items functionality.

Billable Item Class and Interface Components

As with any feature introduction, SAP provides interface components at the billable item class level to be activated. With the activation of the interface components, additional fields related to the dependent items feature being enabled are now visible. The following interface components will need to be activated in the billable item class under Transaction SPRO path **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Items • Billable Item Classes • Define Interface Components**:

- Activate **DIT_PARTNER** if the generated dependent items are required to be created as a partner settlement item.
- Activate the **DIT_PRIMARY** interface component for the primary item.
- Activate the **DIT_SECONDARY** interface component for the generated dependent items.

As shown in Figure 4.83, the interface components for dependent items are relevant.

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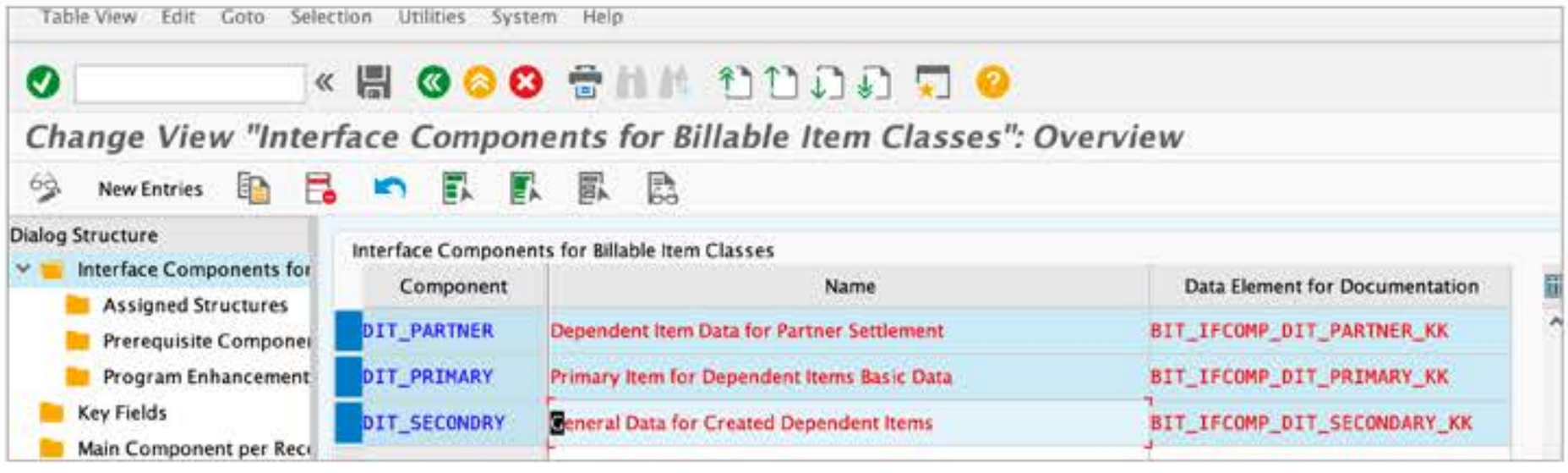

Figure 4.83 Interface Components for Dependent Items

If the dependent items are to be created as a consumption item, the consumption item interface component **SECONDRY_IT** needs to be activated, as shown in Figure 4.84.

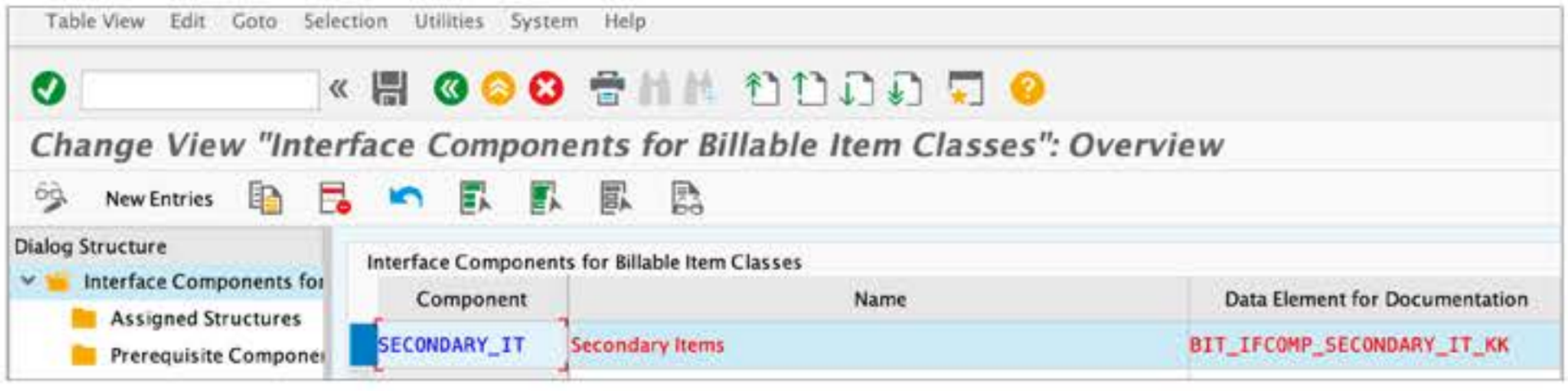

Figure 4.84 Interface Component to Create a Dependent Item as a Consumption Item

As a next step, the interface components for dependent items will need to be activated within the billable item class. To start with, activate the interface component **DIT_PRIMARY** for the primary item, as shown in Figure 4.85. As a reminder, the dependent items are created for the primary item.

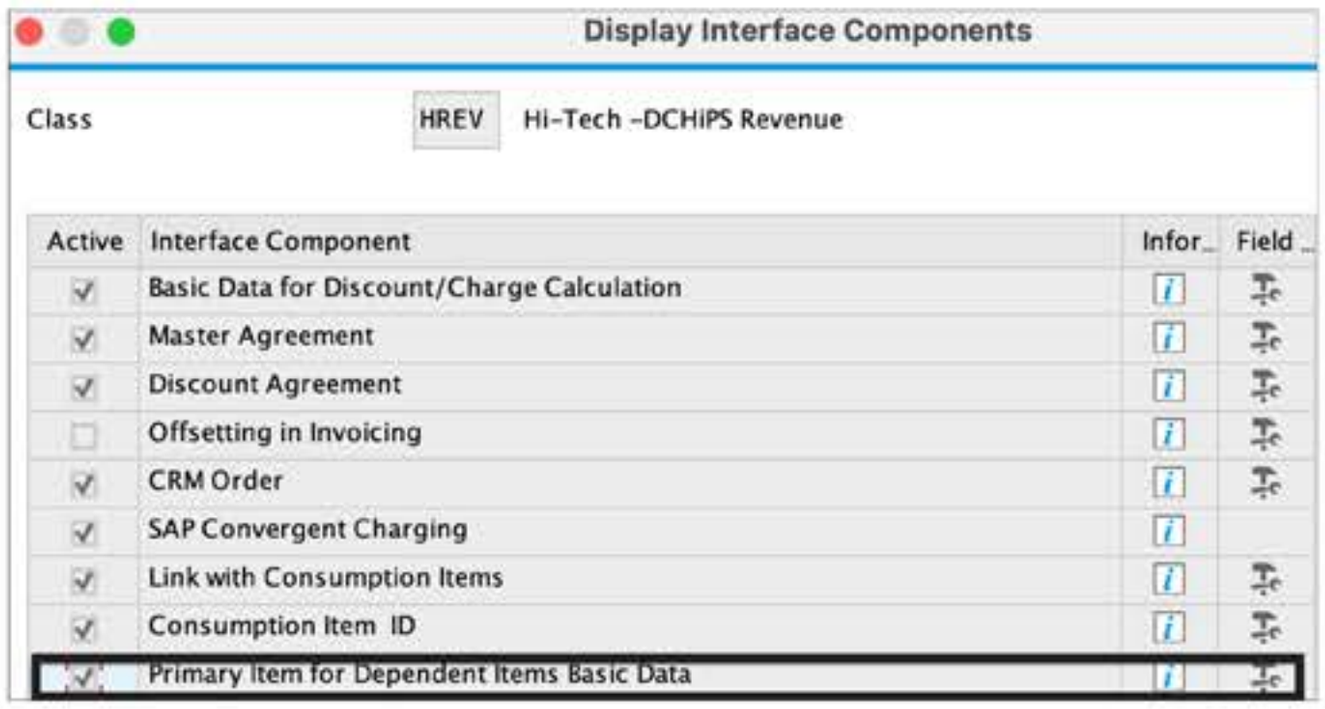

Figure 4.85 Activate Billable Item Transfer Component for DIT_PRIMARY

Next, activate the billable item interface component for **DIT_SECONDARY** and **DIT_PARTNER**. These interface components are related to the dependent items, which are created from or with reference to the primary items, as shown in Figure 4.86. In

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addition, activate the billable item interface component for **SECONDRY_IT** for the consumption item class of the dependent item, as shown in Figure 4.87.

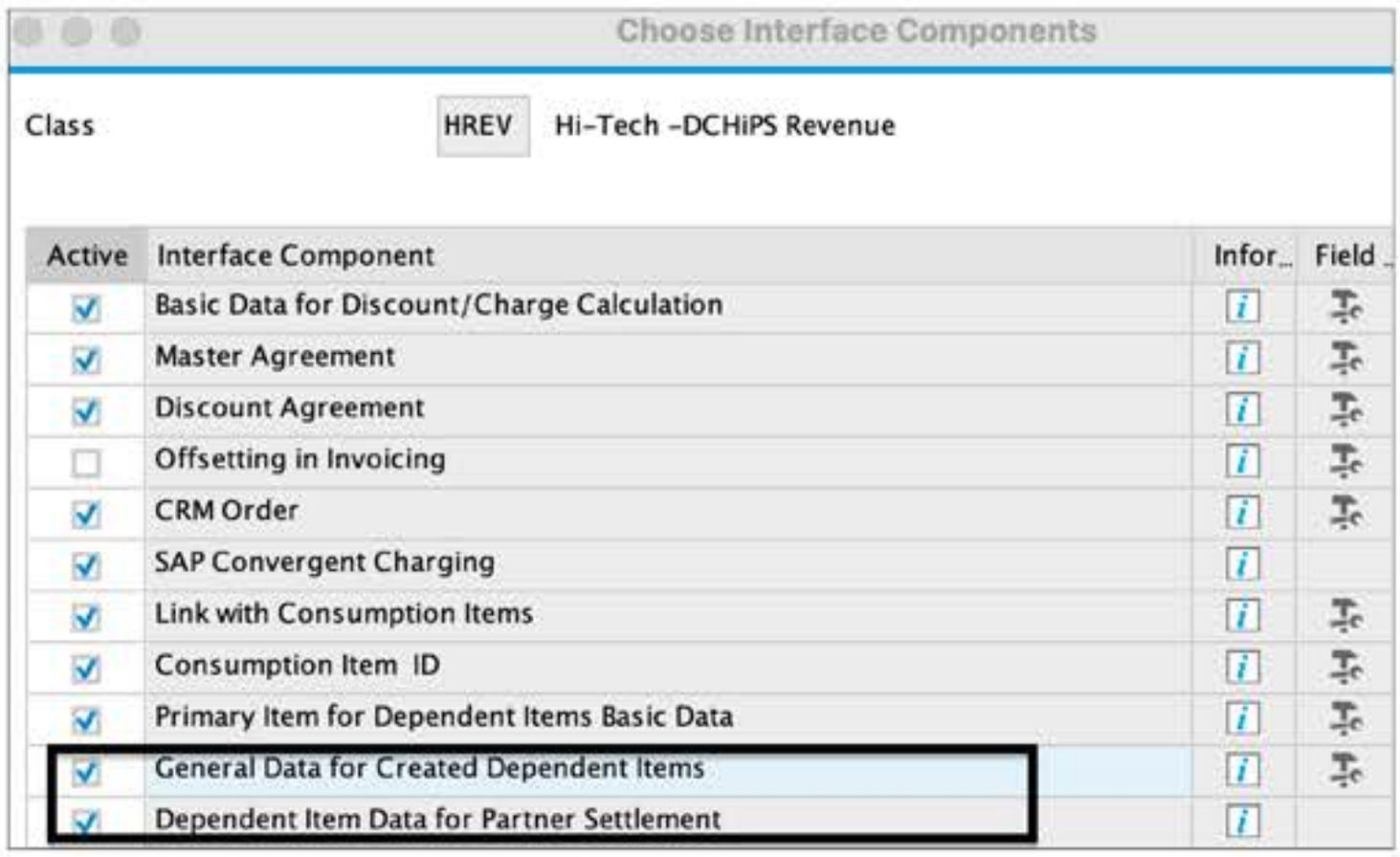

Figure 4.86 Activate Billable Item Interface Component for DIT_SECONDARY and DIT_PARTNER

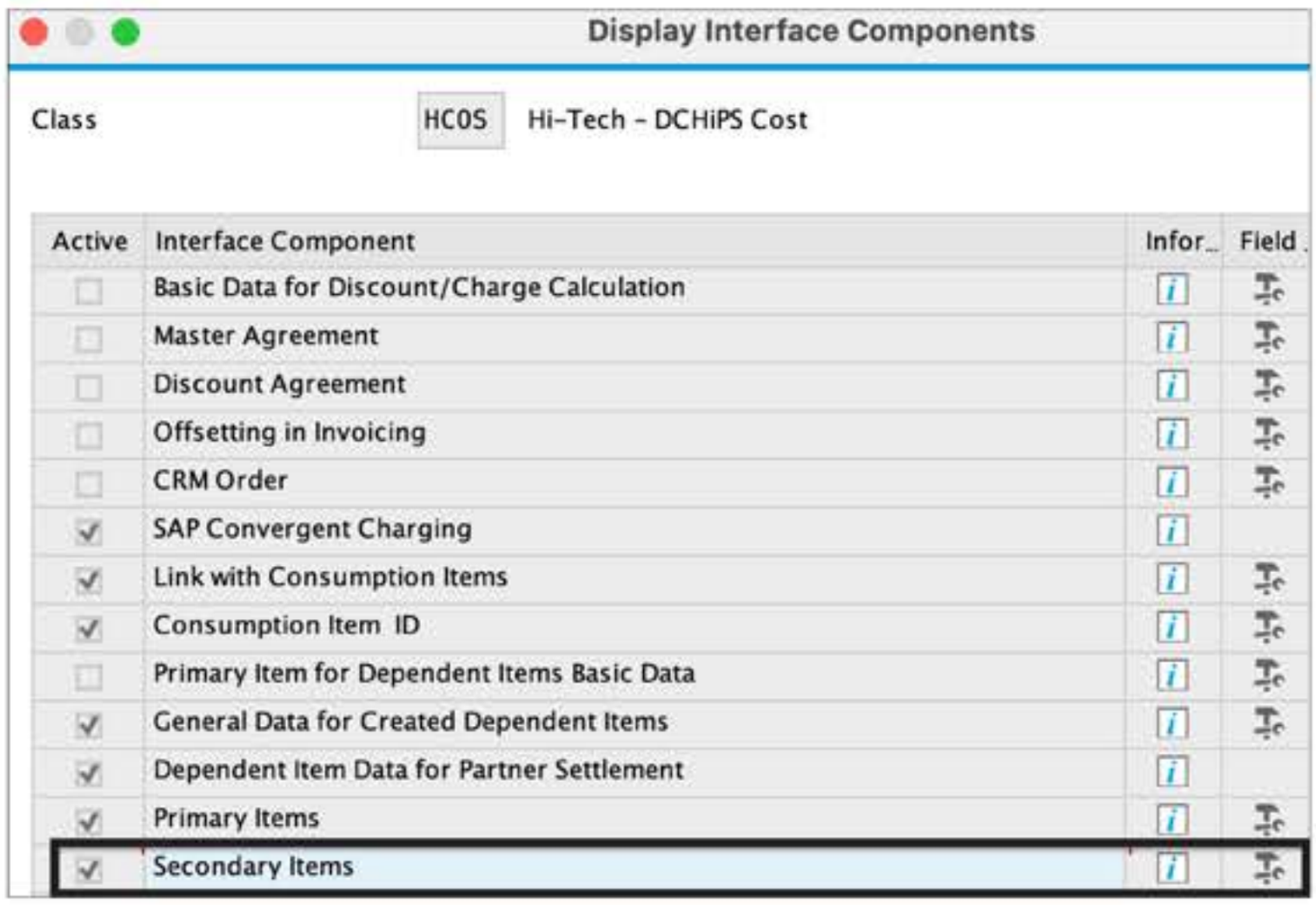

Figure 4.87 Activate Consumption Item Interface Component for SECONDARY_IT

As a refresher, dependent items aren't demonstrated for the Martex case study, so the concept is only discussed here. The screenshots for configurations can be referred to as a guide for any implementation as needed.

Dependent Item Type

Once the interface components are defined and activated, the next steps include defining the dependent item type (**DepItmType**) and the associated calculation key. As

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shown in Figure 4.88, the dependent item types have been configured under Transaction SPRO path **Contract Account Receivables & Payables • Convergent Invoicing • Enhanced Functions • Dependent Items • Define Dependent Item Types**:

- **ZDEPPAY: Payable Dependent Item**
- **ZDEPPYC: Payable Dependent Cons Item**
- **ZDEPREC: Receivable Dependent Item**

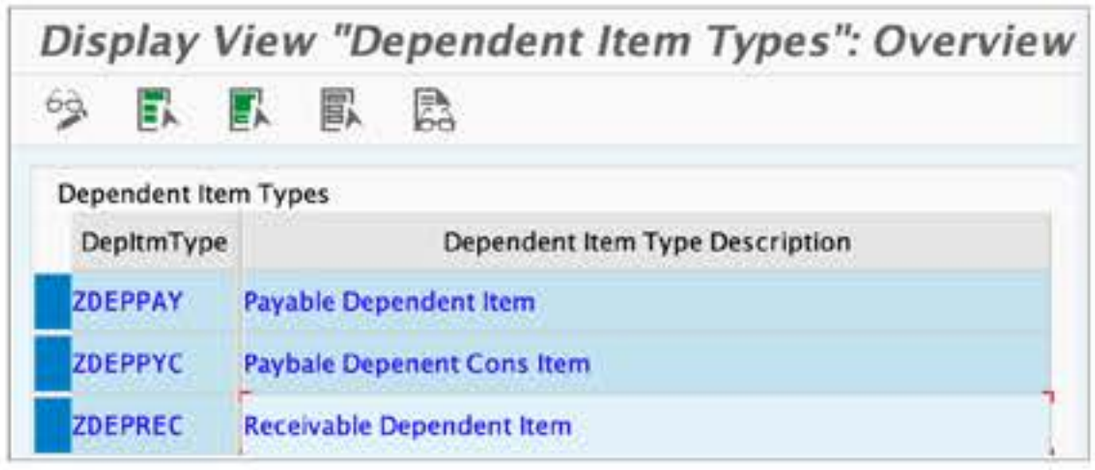

Figure 4.88 Dependent Item Setup for Receivable, Payable, and Payable Consumption Items

Following the definition of dependent items, the next step is to define the **Amount Calculation Key** and the dependent item reasons. The amount calculation key defines the rules and calculations for the dependent item, such as the percentage to calculate the charge or discount, currency, and valid from date, as shown in Figure 4.89 for **ZRV01** and Figure 4.90 for **ZPY01**.

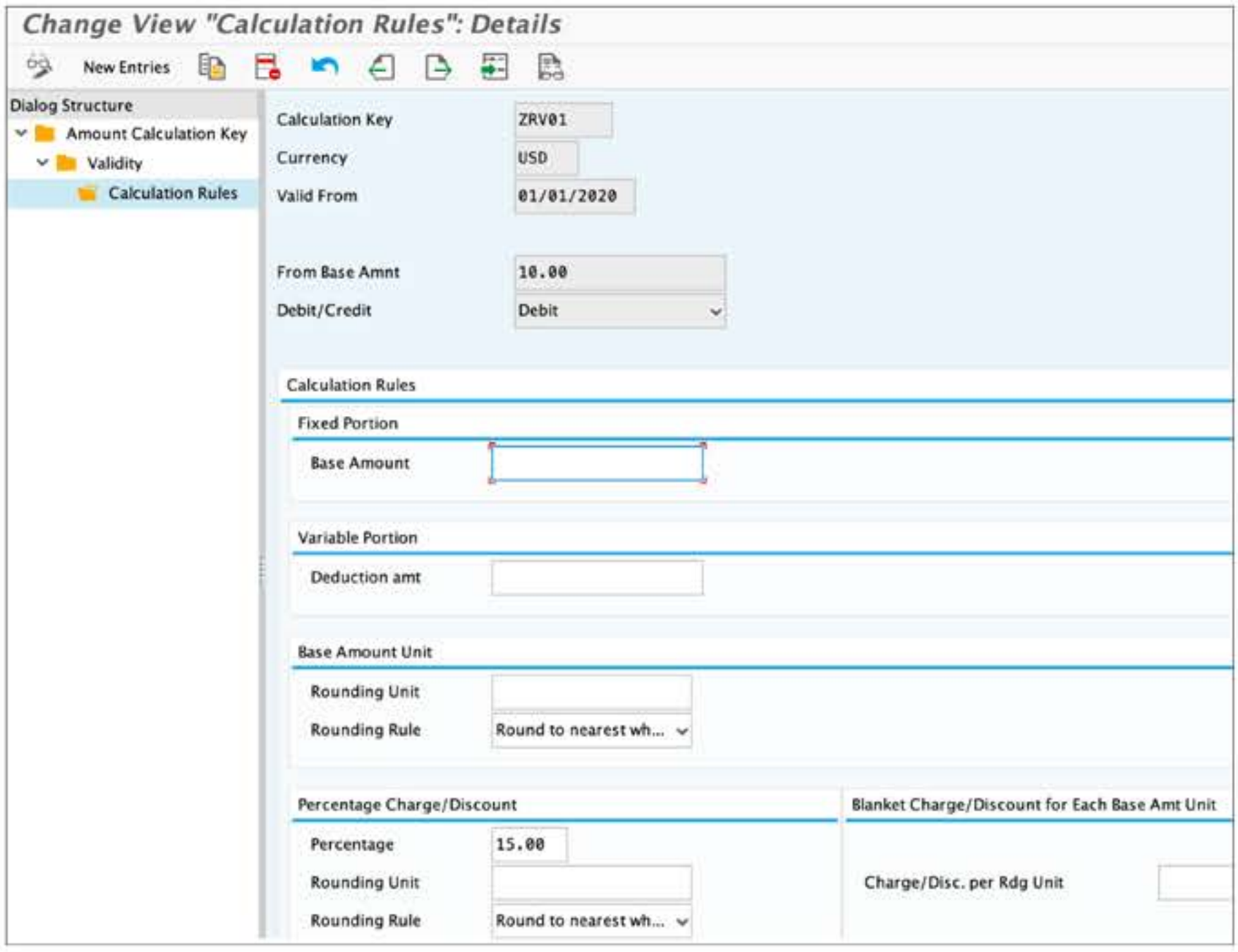

Figure 4.89 Dependent Items: Amount Calculation Key ZRV

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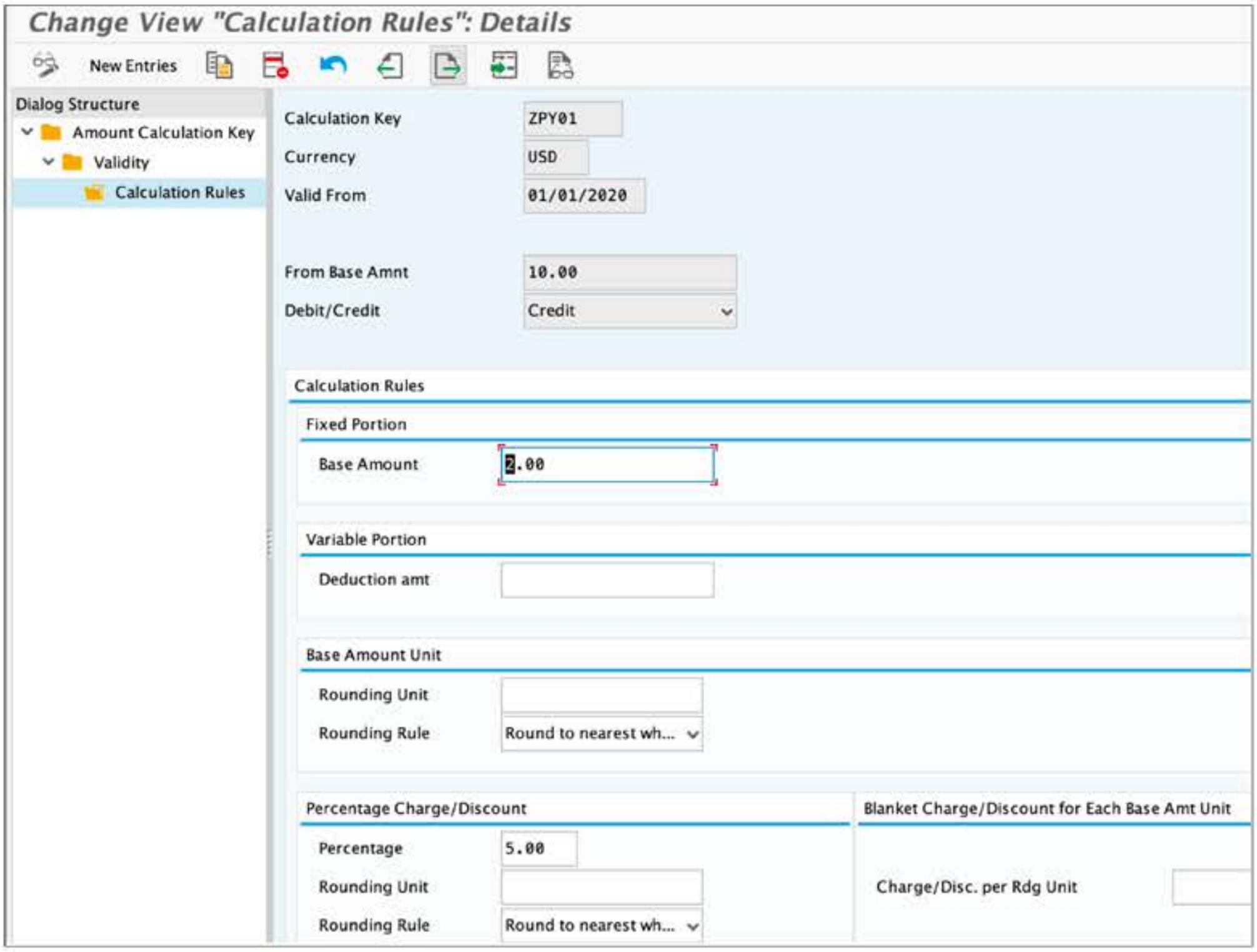

Figure 4.90 Dependent Items: Amount Calculation Key ZPY01

Dependent Item Reason

The dependent item reason defines the reasons why the dependent item of a particular type has to be created. The reasons typically are statutory or contractual in nature, as shown in Figure 4.91. The reasons are then assigned to the amount calculation key, as shown in Figure 4.92, and program extensions, as shown in Figure 4.93. The following program extensions are available for dependent item creation.

- **10 - General key field enrichment**
  Additional logic to determine the generic key fields for master data determination. Sample function module used is `FKK_BIX_DIT_RSN_EVENT_10`.
- **15 - Master data determination**
  Overwrite the master data determined by the system by using the master data ID determined. Sample function module used is `FKK_BIX_DIT_RSN_EVENT_15`.
- **20 - Amount determination**
  Overwrite the amount determined for a dependent item. Sample function module used is `FKK_BIX_DIT_RSN_EVENT_20`.

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- **30 - Adjust dependent consumption item**
  Access all dependent consumption items created by the system based on Customizing, and overwrite these items. Sample function module used is FKK_BIX_DIT_RSN_EVENT_30.
- **35 - Adjust dependent billable item**
  Access all dependent billable items created by the system based on Customizing, and overwrite these items. Sample function module used is FKK_BIX_DIT_RSN_EVENT_35.

Figure 4.91 Dependent Item Reason

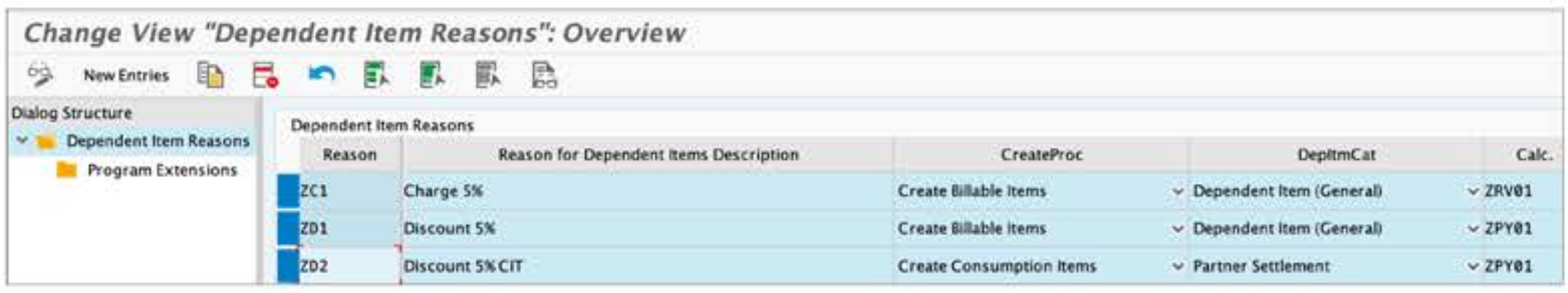

Figure 4.92 Dependent Item Reason Assigned to Amount Calculation Key

Figure 4.93 Dependent Reason Assignment to Program Extensions

Master Data ID

Now that you've seen the definitions of dependent item types, dependent item reasons, and amount calculation key, let's look at master data ID. The **Master Data ID** defines the **Business Partner** and the **Contract Account** related to the dependent item, as shown in Figure 4.94. This master data ID is assigned to the **Dependent Item Type** and **Dep. Item Reason** combination, as shown in Figure 4.95. As a result, the dependent item is created for the business partner and contract account based on the mapping maintained for the dependent item type and dependent item reason combination.

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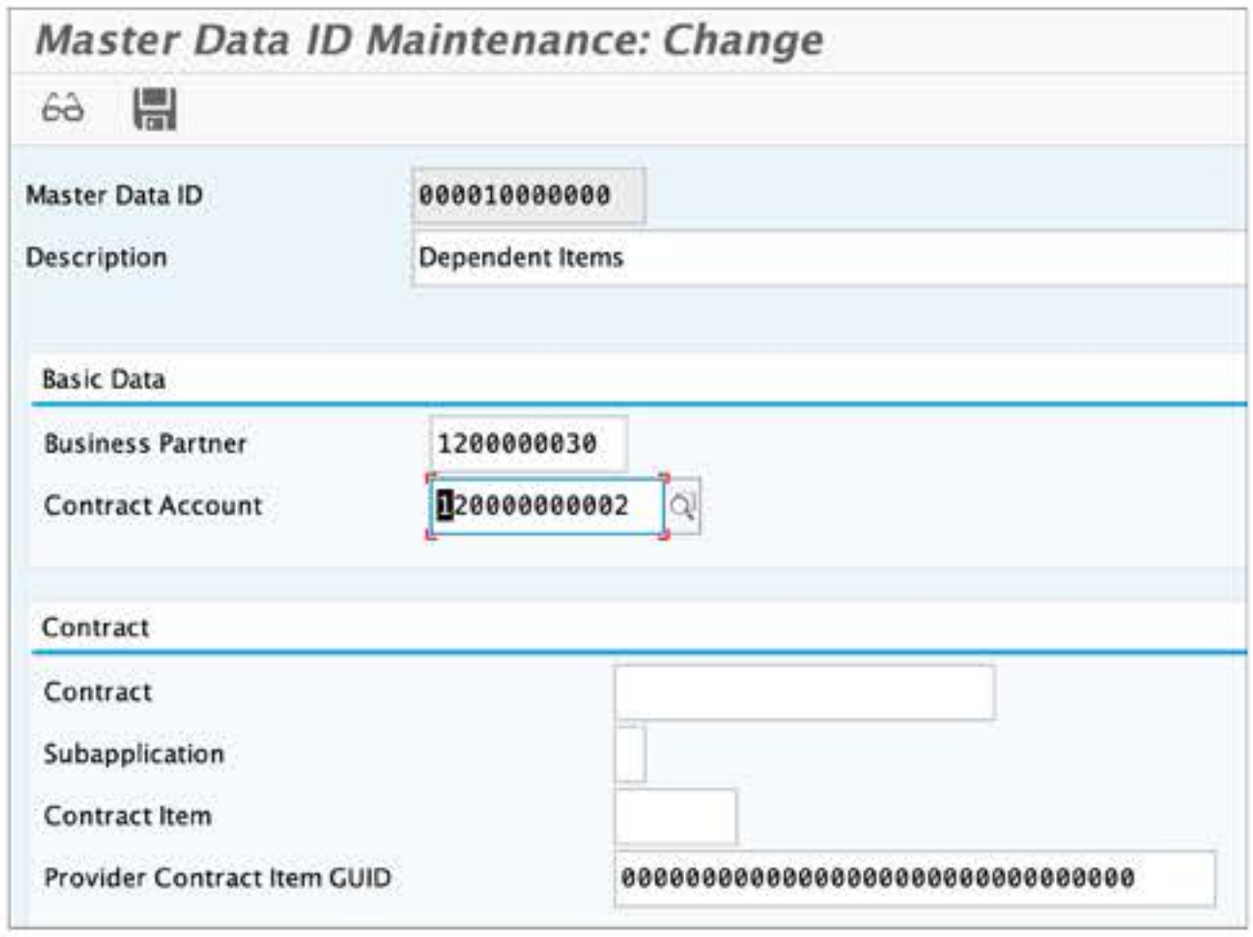

Figure 4.94 Dependent Item: Master Data ID Setup

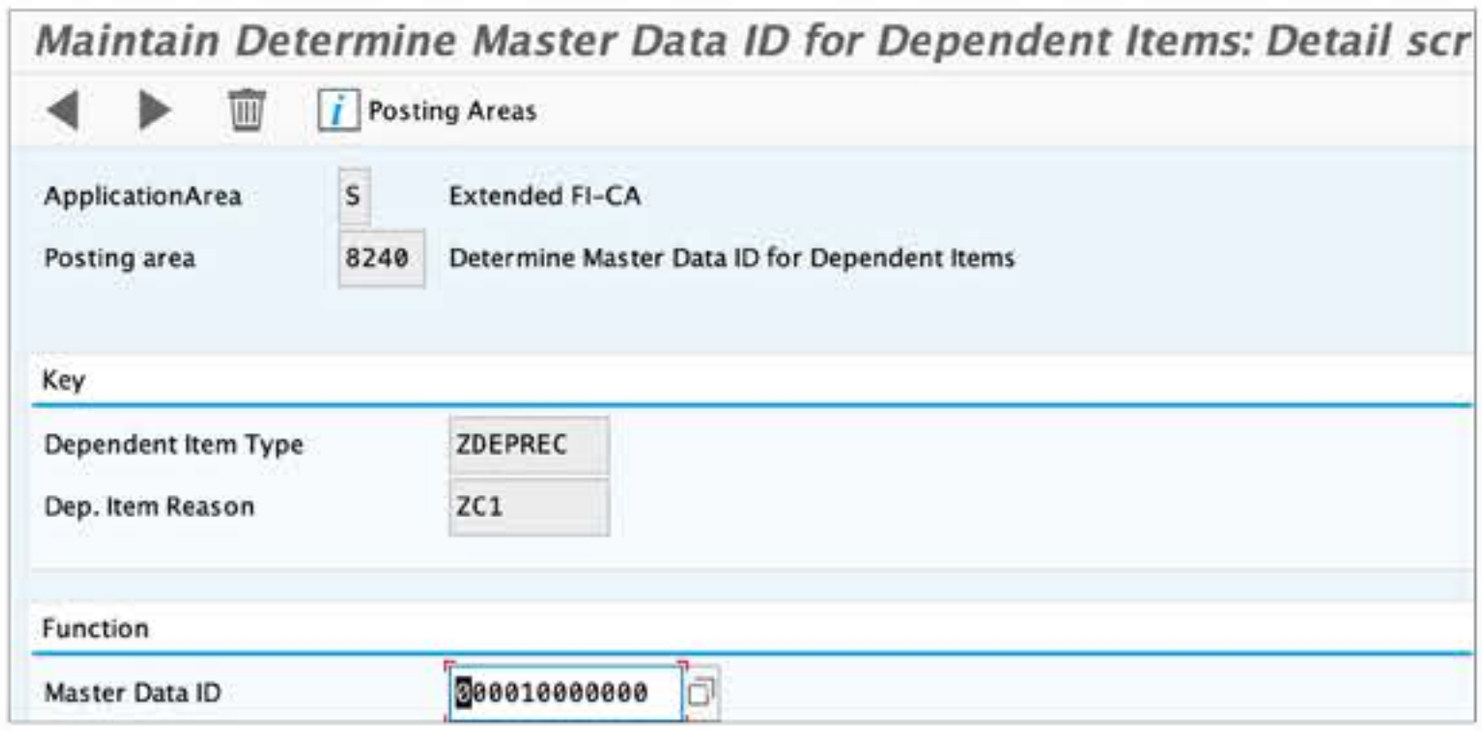

Figure 4.95 Dependent Item Type and Reason Assignment to Master Data ID

As a demonstration, Figure 4.96 shows the creation of the dependent item with dependent reason as **ZC1**. Per the amount calculation key shown previously in Figure 4.89, the dependent item is a payable transaction calculated at 15% of the primary item. The primary item against business partner 1200000030 is $100, and thus the calculated value at 15% for the dependent item is $15.15. You can also see that the dependent item refers to the primary source transaction (**PrmSrcTrns**) as **DEPITEM_CS** as the linkage to the primary item.

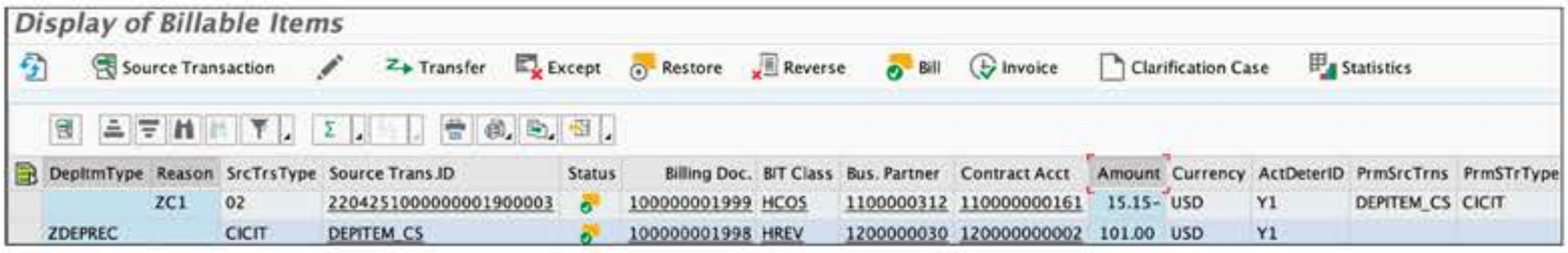

Figure 4.96 Payable Dependent Item Created for the Primary Item

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4.3.7 Partner Settlement

The process of sharing revenue with the vendor, generated from the sale of a product to a customer by the company for the product acquired from a vendor company, is called partner settlement. In scenarios in which products and subscriptions that are acquired from a vendor that owns the rights to the same are sold to end customers, vendors are entitled to revenue sharing, and in SAP Convergent Invoicing, this is achieved via partner settlement. Vendors in such scenarios are called partners with whom partner agreements are set up.

Business Example

Martex sells cloud products to customers on behalf of vendors from which it obtains products. For each sale of a product, Martex pays the vendor the cost of the product via partner settlement, as described in this section.

The details of the sales of a product are recorded in SAP Convergent Invoicing in the form of billable items. When billable items are created, based on the partner settlement category (PSCAT), type of user reference ID (PT_USERID_REFTY), and partner reference ID (PT_USERID_REF) fields, the consumption items are created for the vendor/partner in SAP Convergent Invoicing. The consumption items thus created contain the reference to the partner agreement and are sent for rating and charging to SAP Convergent Charging by the rating process. SAP Convergent Charging then sends billable items for the partner to be billed and invoiced. Upon invoicing, the payables are posted to the partner account in contract accounting.

The partner settlement functionality is activated in SAP Convergent Invoicing by selecting the associated interface component of the class. Upon activation and generation of the class, the fields for the partner settlement become available in the class structure.

The interface components for the class that are used for partner settlement functionality are as follows:

- **Partner settlement: basic data of customer items**
  This interface makes the PSCAT field (partner settlement) category available in the billable item class for the revenue posting for the customer items.
- **Partner settlement: basic data for partner items**
  This interface makes the PTSRL (partner settlement rule) and PTAMOUNT_ADJ (adjustment amount for partner settlement) fields available.
- **Partner settlement of royalties (example)**
  This interface makes the following fields available:
  - PT_USERID_REFTY: Type of reference for partner ID of consumption item
    A value of I means the value in PT_USERID_REF can be directly used as ACCESS_USERID. A value of E means that the value needs to be mapped using a table.

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- PT_USERID_REF: Partner ID of consumption item
  This is used for determining the consumer ID of the consumption item (partner ID) for the partner item.
- PT_PRODUCT_CAT: Product category from partner viewpoint
  This is used for determining the price.

4.3.8 Partner Settlement Categories

The partner settlement category is used to categorize the different revenue-sharing scenarios for the generation of the partner settlement consumption items. In conjunction with the settlement reason, the partner settlement category creates the consumption items. To activate the functionality, first the interface components are selected for the class. Then the partner settlement category is maintained. Categories basically describe the type of service that is used by the customer for partner settlement.

Configuration of partner settlement categories is maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Partner Settlement • Define Categories of Transactions to be Billed** (see Figure 4.97).

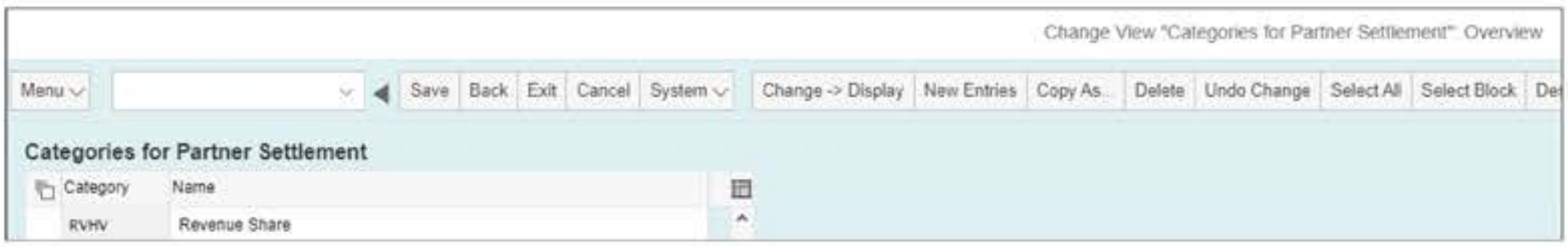

Figure 4.97 Partner Settlement Category

Reason for Partner Settlement

Settlement reasons are contractual agreements based on which the partner amount is decided. Settlement reasons are assigned with function modules, which creates the consumption items automatically when billable items are received with partner settlement categories. After creation of the settlement category, the settlement reason is set up. SAP doesn't provide any standard reason. A settlement reason is created and assigned with a function module, which is used for creation of the consumption items for the partner settlement. SAP provides a sample function module that can be used and can be modified as needed for creation of the consumption items.

Configuration of partner settlement reasons is maintained under **Contract Account Receivables & Payables • Convergent Invoicing • Partner Settlement • Define Reason for Partner Settlement** (see Figure 4.98).

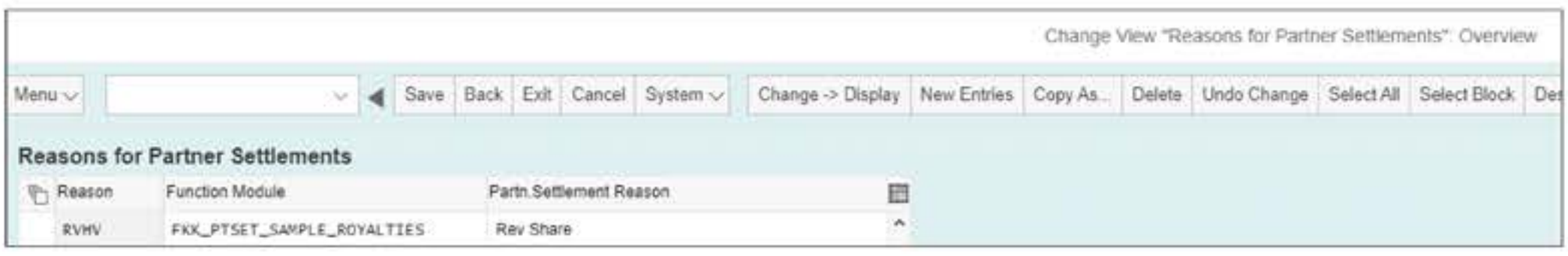

Figure 4.98 Partner Settlement Reason

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Reasons and Categories Assignment

The partner settlement category and settlement reason created are assigned to the consumption item class for the partner settlement. The partner settlement category populated on the billable item field triggers the function module associated with the reason assigned to the category, which creates the consumption items of the assigned consumption item class.

Assignment of reasons and settlement categories to consumption item classes is done under **Contract Account Receivables & Payables • Convergent Invoicing • Partner Settlement • Assign Reasons to Categories** (see Figure 4.99).

Figure 4.99 Consumption Item Class Assignment to Category and Reason

4.3.9 Integration with SAP Revenue Accounting and Reporting

In this section we'll go over the basic elements required to activate the integration between SAP Convergent Invoicing/contract accounting and the SAP Revenue Accounting and Reporting module. SAP Billing and Revenue Innovation Management integrates with SAP Revenue Accounting and Reporting using contract accounting. Note that this section covers the integration concepts between SAP Convergent Invoicing/contract accounting and SAP Revenue Accounting and Reporting, but it won't dive deep into SAP Revenue Accounting and Reporting concepts. In addition, we won't be using the Martex case study to discuss the concepts in this section. They key integration configuration nodes are present in the **SPRO** node **Contract Account Receivables & Payables • Integration • Revenue Accounting**.

SAP Convergent Invoicing/Contract Accounting and SAP Revenue Accounting and Reporting Integration

To enable the integration, the first setup required is shown in Figure 4.100 where the **RA Active** flag is checked. Along with this, the condition types for standalone selling price (SSP) and transaction price are defined. The condition type entered for SSP will be used as a default for order item conditions for SSP. The condition type entered for **Transaction Price** will be used as a default for order item conditions for transaction price. Service types within SAP Revenue Accounting and Reporting define the type of service to be billed for a contract item. It defines the action required for performance obligations (POBs) and the corresponding fulfillment. The **Service Type SALE** will be the default in order items for recurring charges.

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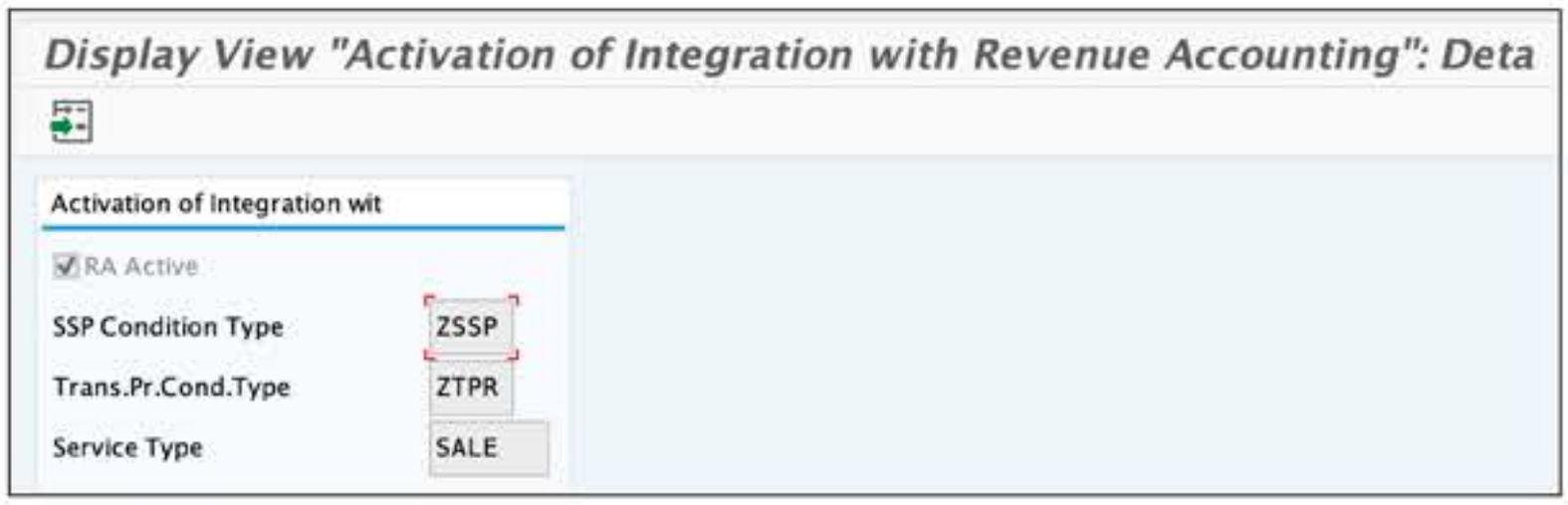

Figure 4.100 Activate Integration with Revenue Accounting

The next step is to define the remote function call (RFC) destination where SAP Revenue Accounting and Recoporting is running (see Figure 4.101). RFC is invoked on the revenue accounting item transfer to SAP Revenue Accounting and Reporting. A two-way connection between the systems where contract accounting and SAP Revenue Accounting and Reporting are running is required, which can also be maintained in the configurations for RFC connections using Transaction SM59.

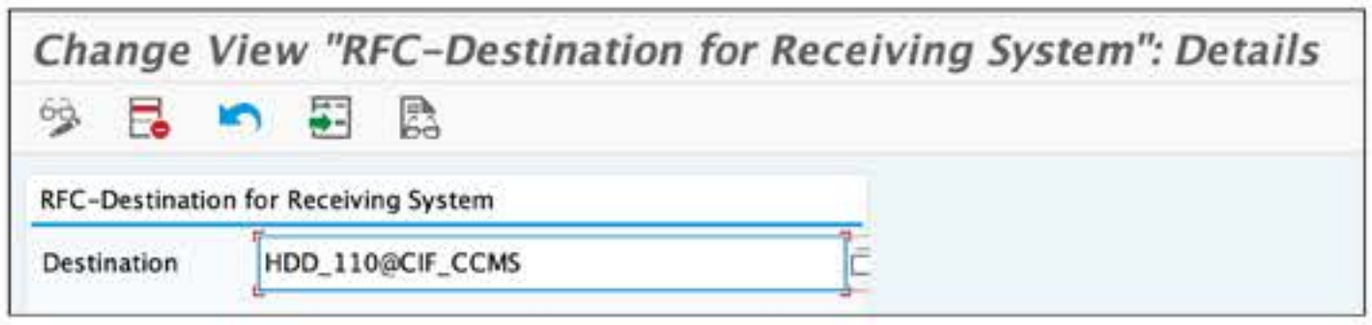

Figure 4.101 Configure an RFC Destination for the SAP Revenue Accounting and Reporting System

Provider Contract and Service Type Activation

The provider contract serves as a key link between the core modules of SAP Billing and Revenue Innovation Management and SAP Revenue Accounting and Reporting (Figure 4.102). The Transaction SPRO node **Contract Account Receivables & Payables • Integration • Revenue Accounting • Activate Provider Contract Items for Revenue Accounting** establishes the integration between the provider contract within SAP Convergent Invoicing/contract accounting and SAP Revenue Accounting and Reporting.

Figure 4.102 Provider Contract Relevance for Revenue Accounting

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The next step is to define the extent to which service types will be relevant. As shown in Figure 4.103, the service type **SALE** can be an **Order Item**, **Fulfillment Item**, and **Invoice Item**. For the purposes of this discussion, we'll consider service type SALE as being relevant for recurring charges.

Figure 4.103 Service Types Definition for Revenue Accounting Items

When the provider contract is created in SAP Convergent Invoicing/contract accounting from SAP Billing and Revenue Innovation Management, subscription order management, service types can be transferred for a provider contract. For certain types of services, such as usage-based services, service IDs are defined in SAP Billing and Revenue Innovation Management, subscription order management, to uniquely identify the type and the instance of the usage service. As shown in Figure 4.104, the service type **USAG** has been assigned to **Service ID TEST1** for illustration purposes. The telemetry feed of type service ID along with a unique ID will be used to track telemetry. So far, we've discussed two types of service IDs, SALE (recurring charge) and USAG (usage charge).

When transferred from SAP Convergent Invoicing/contract accounting to SAP Revenue Accounting and Reporting, each revenue accounting item needs to reference the origin or a reference document/transaction. As shown in Figure 4.105, the reference document for a revenue accounting item in this illustration is a solution quotation.

Figure 4.104 Assign Service IDs to Service Types

Figure 4.105 Define Reference Types for Revenue Account Items

Figure 4.106 shows a sample billable item with the **RA Relevant** flag (revenue recognition flag) and the associated service type for the recurring charge. With these mandatory

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fields present on the billable item, the revenue transfer steps will transfer the revenue accounting item to SAP Revenue Accounting and Reporting for further processing.

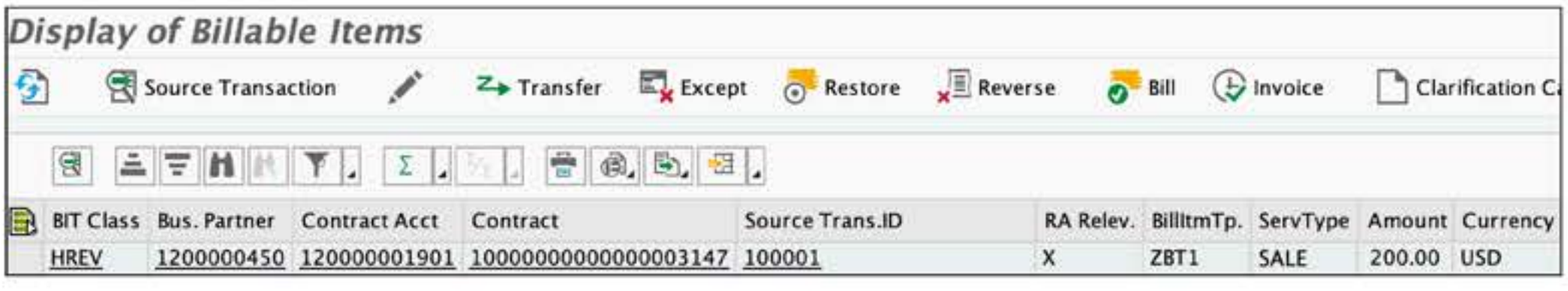

Figure 4.106 Sample Billable Item with Revenue Recognition Relevance

4.4 Using SAP Convergent Invoicing

So far, we've covered the various configurations required for setting up the SAP Convergent Invoicing objects and how to associate them with the business requirements. After setting up the configurations, you need to test them. To test the configurations for SAP Convergent Invoicing alone when a connection to a different SAP system isn't in place, there are different ways that consumption and billable items can be created in SAP Convergent Invoicing directly. Let's discuss them in detail.

4.4.1 Creating a Consumption Item or Billable Item

Consumption items can be created using the following mentioned options via Transaction FKKBIXCIT_CWB. This transaction lets you display, change, and create consumption items. Consumption items can be displayed by selecting the variant for the consumption item and providing the selection conditions (see Figure 4.107). The displayed items can be rated, reversed, excepted, restored, and changed based on the action that needs to be performed.

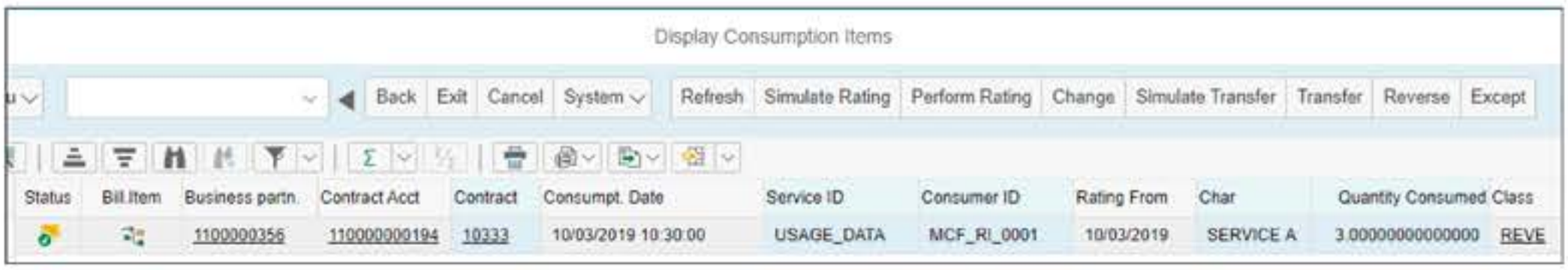

Figure 4.107 Consumption Items Display

Consumption items can be created directly by selecting the **Create** option from within Transaction FKKBIXCIT_CWB. You can enter the **Business Partner**, **Contract**, and consumption item class (**Consumpt. Item Class**) in the **Create Consumption Item** screen, and then select the **Create** option at the bottom (see Figure 4.108).

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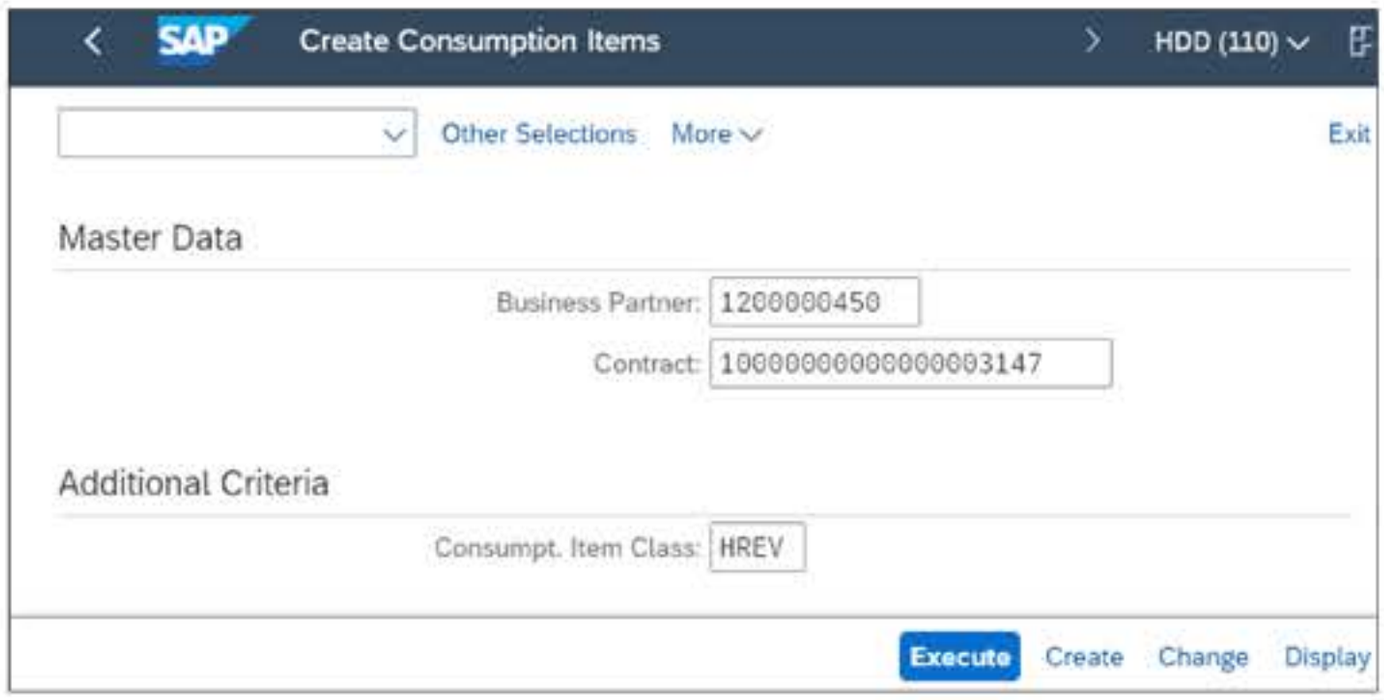

Figure 4.108 Consumption Item Creation Initial Screen

After the **Create** option is selected, select the **Add Item** option. Once details are entered, the consumption item can be saved (see Figure 4.109).

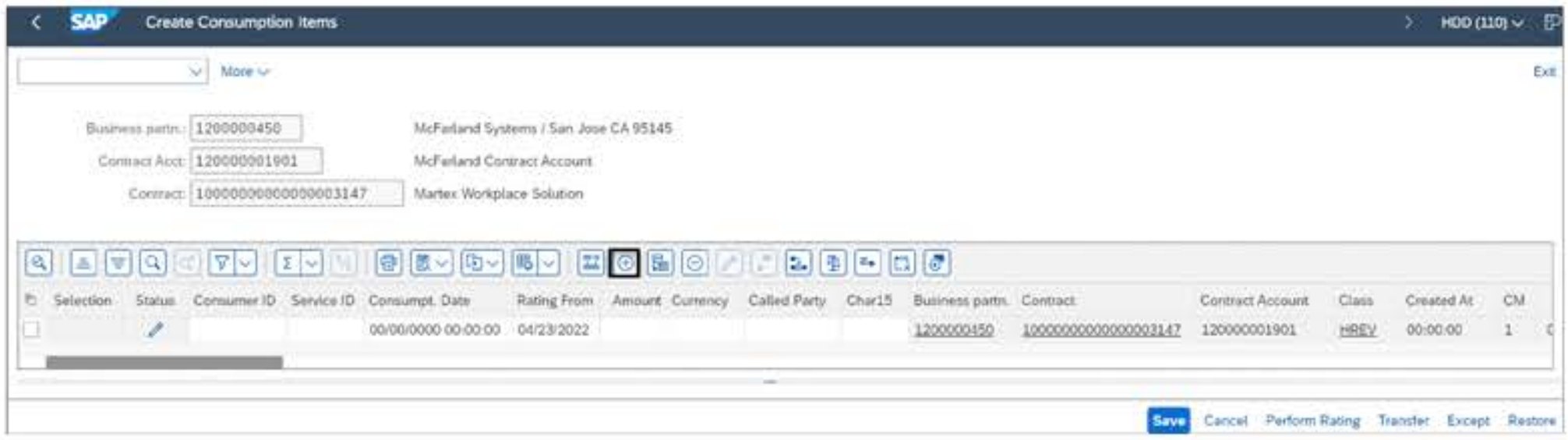

Figure 4.109 Create Consumption Items Screen for Fields Entry

Report RFKKBIXCITSAMPLE

Report RFKKBIXCITSAMPLE is used for creation of consumption items for test data (see Figure 4.110). The report is executed using Transaction SE38.

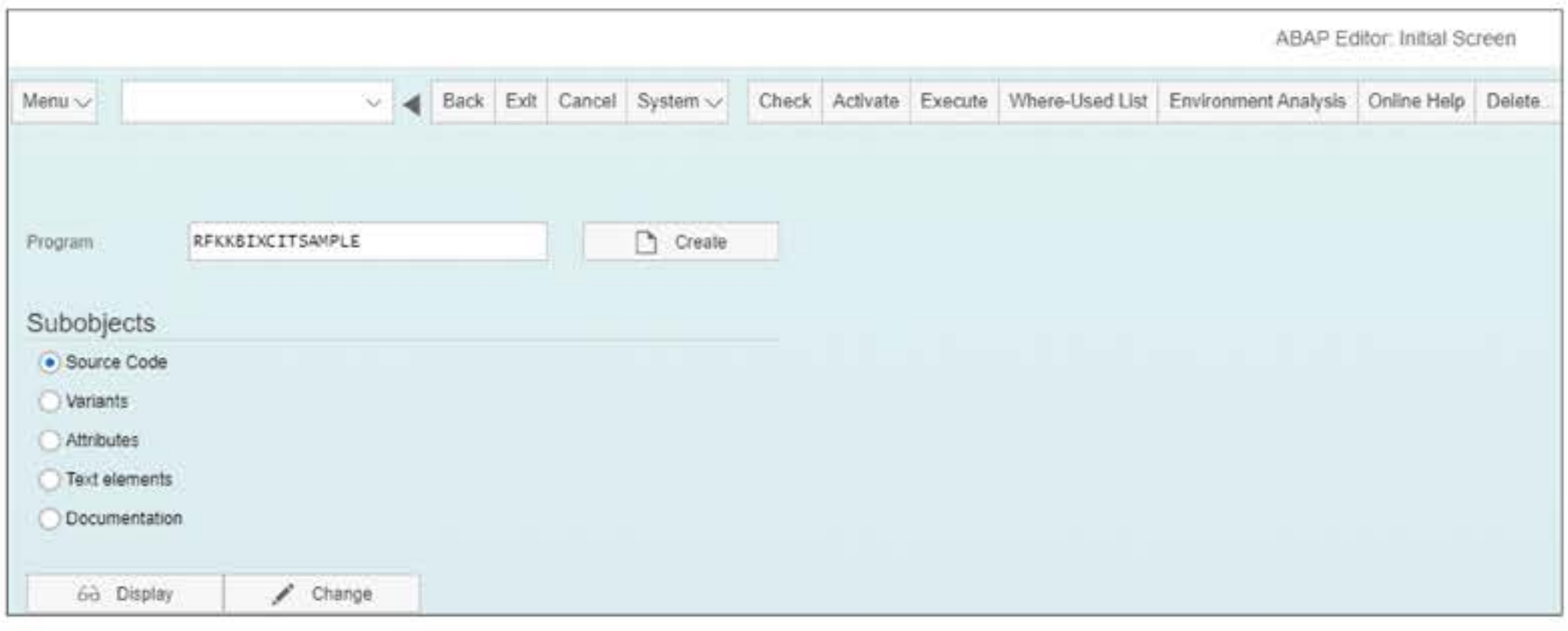

Figure 4.110 Report Program Execution Screen

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In this report, enter consumption item details, and then select the **Execute** option to store the consumption item (see Figure 4.111).

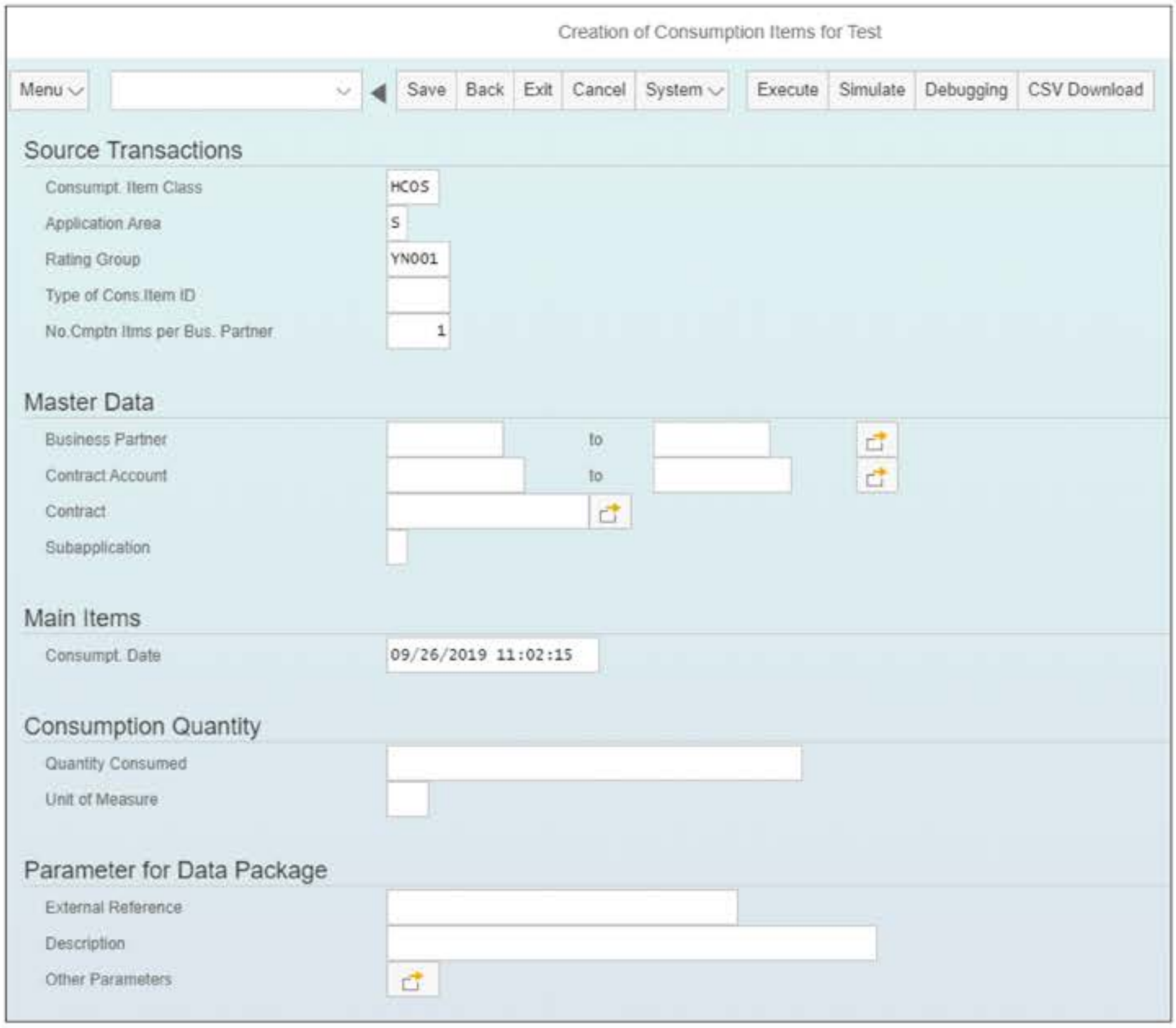

Figure 4.111 Consumption Item Creation Screen

This report also provides the option to simulate the consumption item creation before saving the consumption items. You can also debug the consumption item creation if required.

Using Application Programming Interfaces Generated on Class Generation

Interfaces are assigned during the class creation, and once class interfaces are generated, the tables, structures, and function modules are created for the classes. The function module generated for the consumption item class can be used for creating the consumption items.

The created function module can be obtained by selecting the consumption item class and then selecting the **Objects** option under menu path **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Consumption Items • Consumption Item Classes • Generate Interface for the Consumption Items Classes** (see Figure 4.112).

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Display Generated Objects

Menu | Back | Exit | Cancel | System | Sort in Ascending Order | Sort in Descending Order | Display Object | Display All Objects

Class HCOS

Generated Objects

Short text	Object Name	ShortTxt
Function Module	/1FC/HCOS_CIT_CREATE_API	Generation of Consumption Items of Class HCOS
Table	/1FC/1HCOS_API	Class HCOS Structure Type API
Function Module	/1FC/HCOS_CIT_CREATE_PROXY	Generation of Consumption Items of Class HCOS
Table	/1FC/1HCOS_PRX	Class HCOS Structure Type PRX
View	/1FC/24HCOS	Consumption View - All Consumption Items of a Class HCOS
Table	/1FC/0HCOS0	Items for Class HCOS - Raw
View	/1FC/20HCOS0	Basis View for Consumption Items of a Class HCOS Status 0
View	/1FC/21HCOS0	Private Basis View for Consumption Item Class HCOS Status 0
Table	/1FC/0HCOS1	Items for Class HCOS - Raw Excepted
Table	/1FC/0HCOS2	Items for Class HCOS - Rateable
View	/1FC/20HCOS2	Basis View for Consumption Items of a Class HCOS Status 2

Figure 4.112 Objects for Consumption Item Classes

The function module can be executed in Transaction SE37, and you can enter details for the consumption items to be created. Once the function module is executed, the entries for the consumption items are created (see Figure 4.113).

Test Function Module: Initial Screen

Menu | Save data record | Back | Exit | Cancel | System | Execute | Debugging | Choose | Test data directory

Test for function group /1FC/HCOS
Function module /1FC/HCOS_CIT_CREATE_API
Uppercase/Lowercase ☐

RFC target sys:

Import parameters	Value
I_MANDT	110
I_UNRATED	X
I_TEST	
I_NO_COMMIT_WORK	
I_EXT_LOGNUMBER	
I_EXTERNAL_LOG	
I_MAX_PROBCL	2
I_NO_SUMMARY_MSG	
I_FIRST_CALL	X
I_LAST_CALL	X
IT_ADD_PARAMS	0 Entries

Tables	Value
IT_CIT	1 Entry
ET_CIT0_RETURN	0 Entries
ET_CIT2_RETURN	0 Entries
ET_CIT4_RETURN	0 Entries
ET_CIT_ERROR	0 Entries
ET_MESSAGES	0 Entries

Figure 4.113 Function Module Screen for Consumption Item Creation

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Billable Item Creation

Billable items for test data can be directly created in SAP Convergent Invoicing. Billable items can be created using Transaction FKKBIXBIT_CWB. This transaction provides options to display, change, and create billable items. Billable items can be displayed by selecting the variant for the billable items and providing the selection conditions. The displayed items can be billed, invoiced, reversed, excepted, restored, and changed based on the action to be performed (see Figure 4.114).

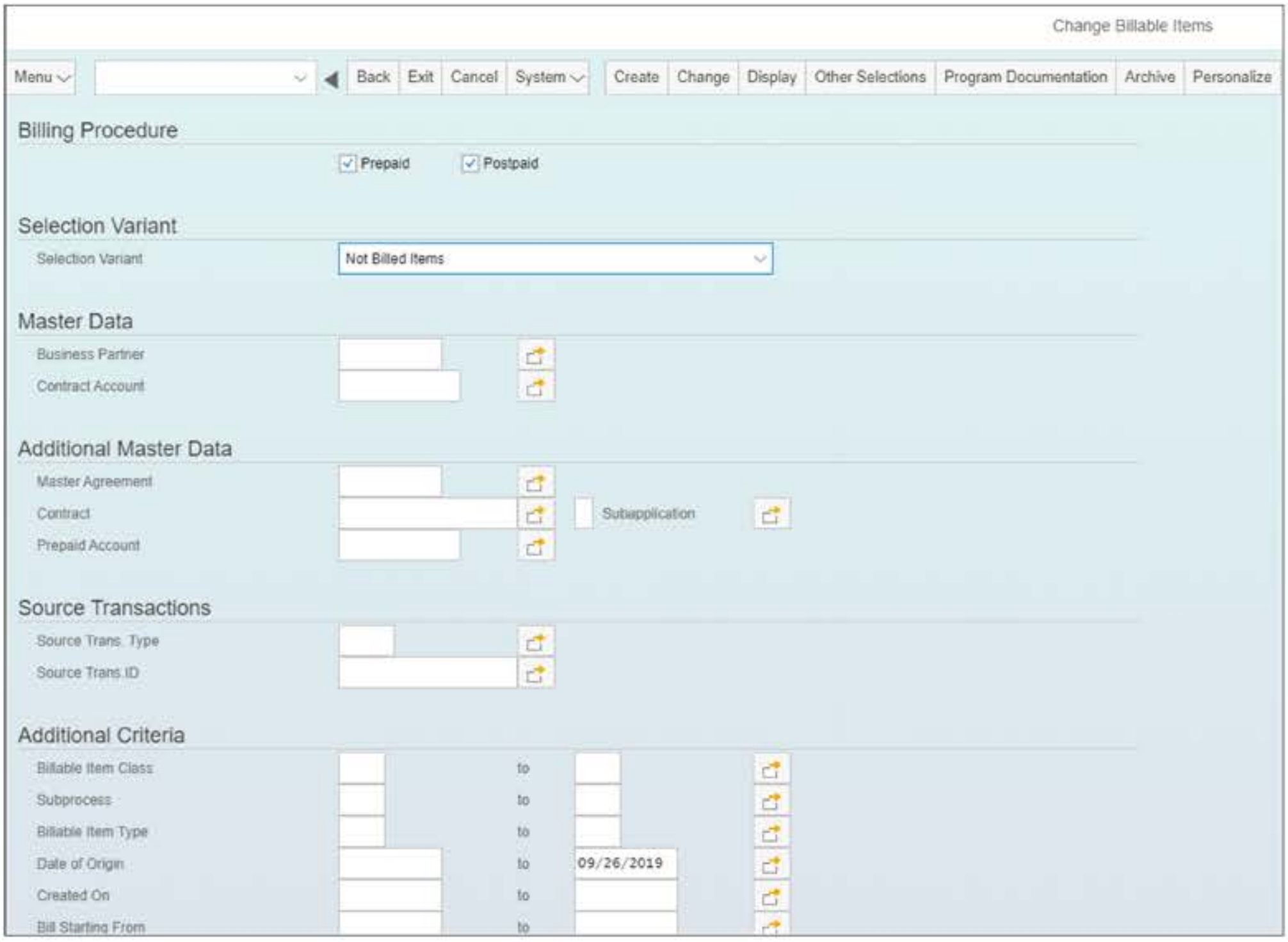

Figure 4.114 Billable Item Display

Billable items can be created directly by selecting the **Create** option from Transaction FKKBIXBIT_CWB. Enter the **Business Partner**, **Contract Account**, **Contract**, and **Billable Item Class** on the **Create Billable Items** screen, and then select the **Create** option (see Figure 4.115).

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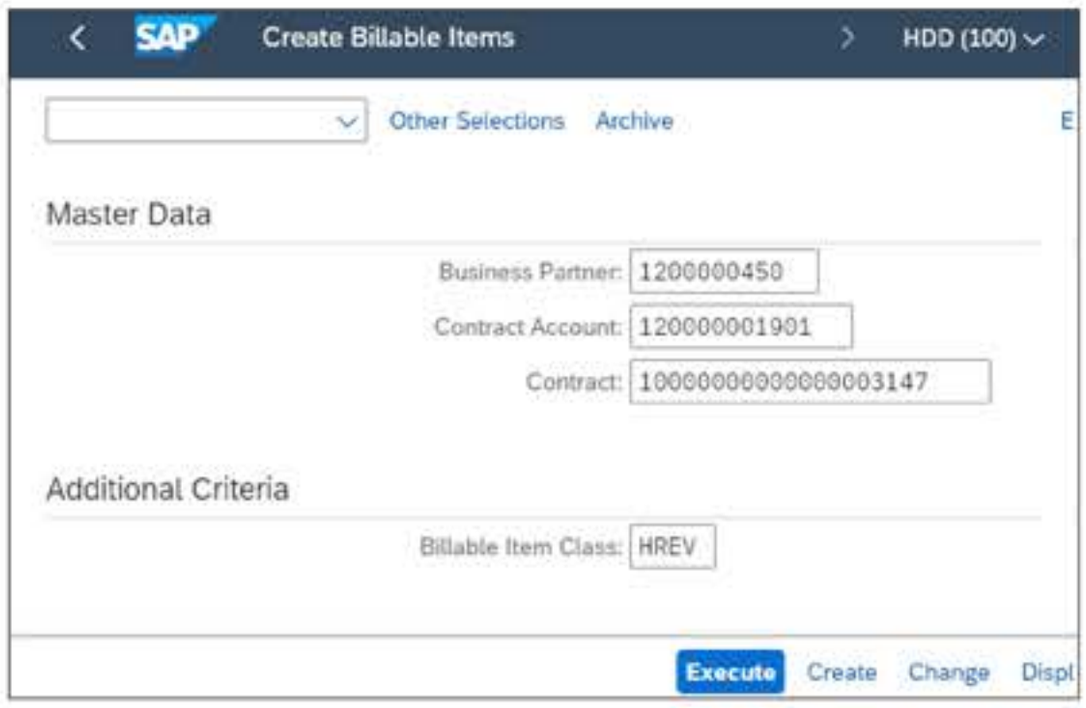

Figure 4.115 Create Billable Items: Initial Screen

After the **Create** option is selected, select the **Add Item** option. Once details are entered, click on **Apply New Values**. The billable item can then be saved (see Figure 4.116).

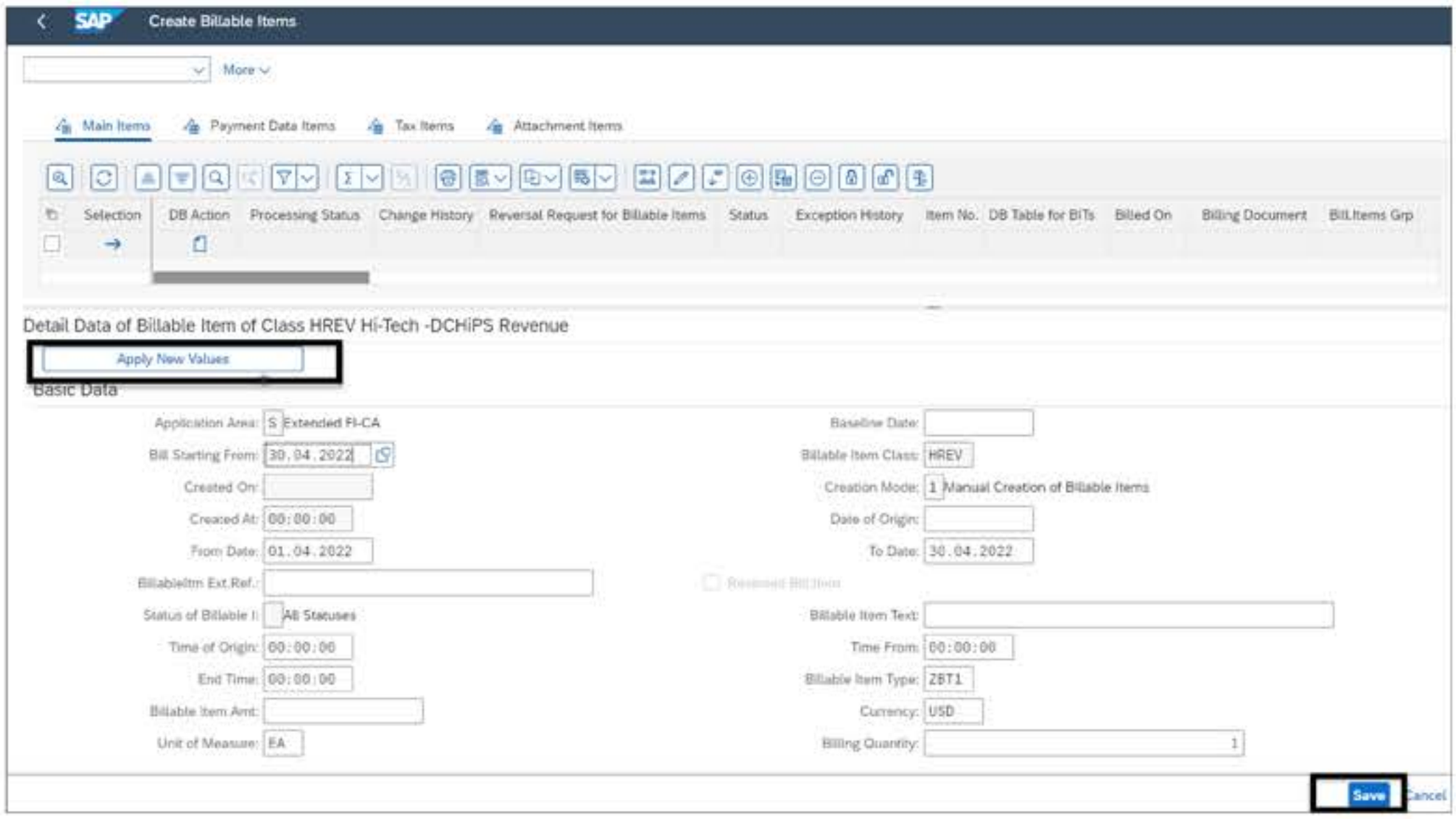

Figure 4.116 Billable Item Creation Screen for Fields Entry

Report RFKKBIXBITSAMPLE

Report RFKKBIXBITSAMPLE is used for creation of billable items for test data (see Figure 4.117). The report is executed using Transaction SE38.

After executing the report, enter billable item details, and then select the **Save** option to store the billable item (see Figure 4.118). This report also provides the option to simulate the billable item creation before executing the billable item creation. The **Execute** option will create the billable item, but the limitation is that it doesn't provide the option to input the value for the custom fields.

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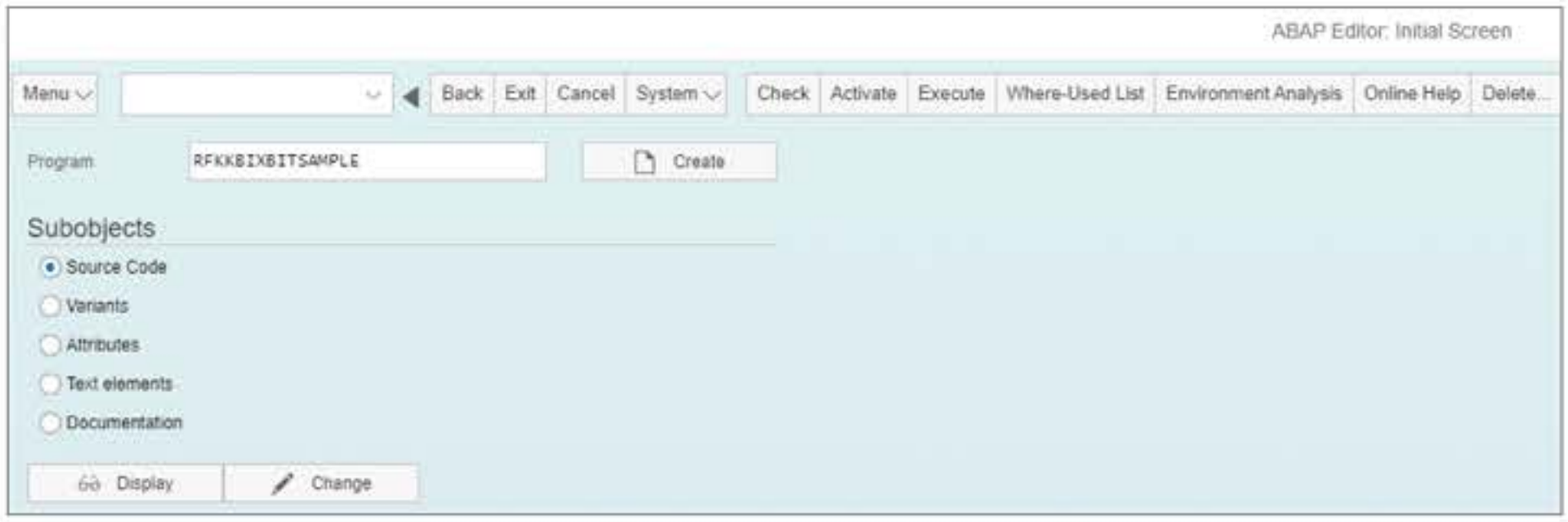

Figure 4.117 Report Program Execution Screen

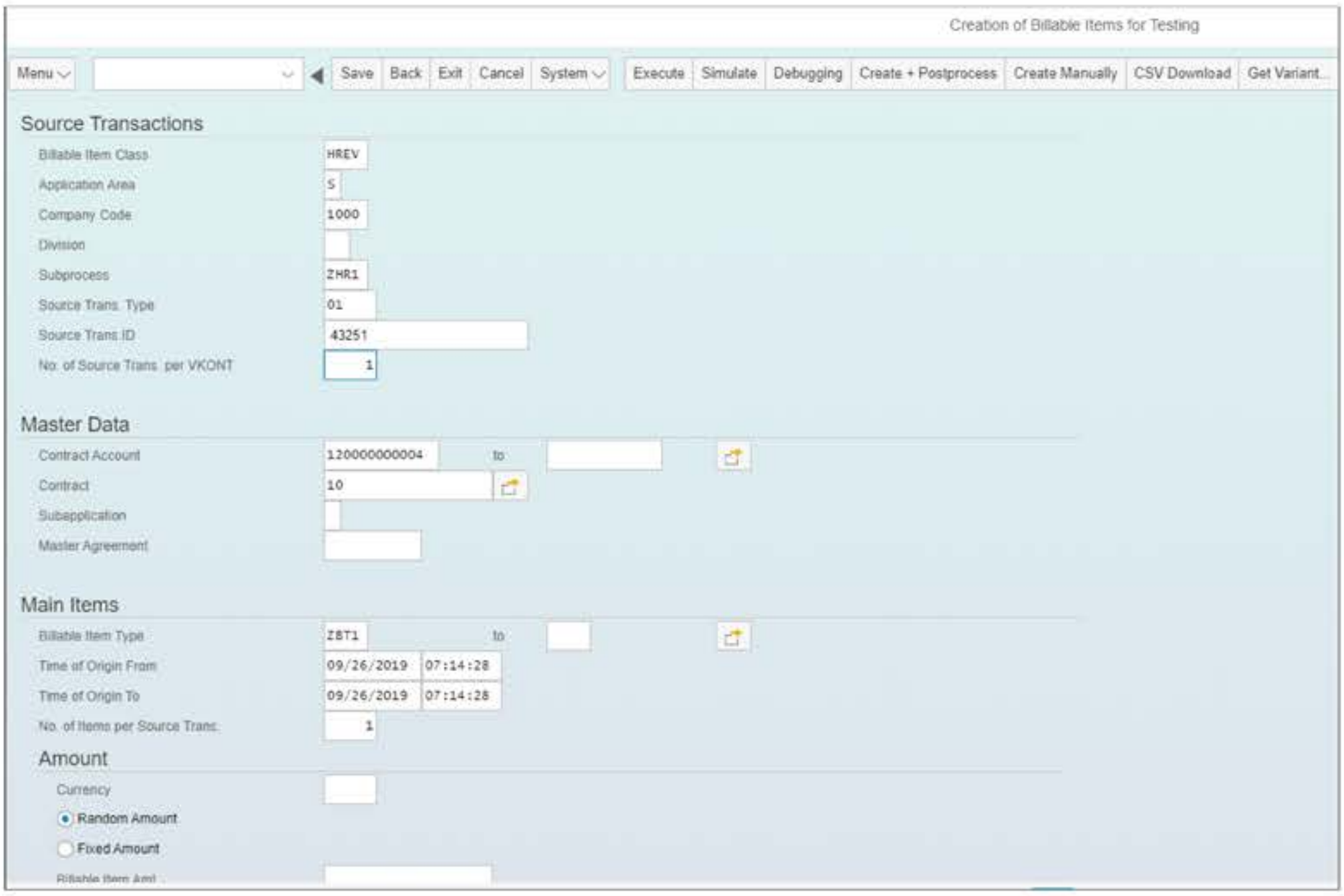

Figure 4.118 Billable Item Creation Screen

The **Create Manually** option lets you enter custom fields. The **Create + Postprocess** option creates the billable items and also sets the billable items in change mode for any field modifications (see Figure 4.119).

Figure 4.119 Create + Postprocess Option Screen for Billable Item Change

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Sample File Download and Uploading Updated Sample File

Report RFKKBIXBITUPLOADSAMPLEFILE (Sample File Download) and report RFKKBIXBITUPLOAD (Uploading Updated Sample File) together are used for billable item creation by uploading the billable item creation file (see Figure 4.120).

Report RFKKBIXBITUPLOADSAMPLEFILE lets you download the sample file that can be updated with the values of fields for the billable item creation. On the selection screen, enter the **Billable Item Class** for which the sample/test file needs to be generated. On execution, the program asks for the storage location. When you provide the same, a comma-separated values (CSV) file will be downloaded, which is used for the billable item data creation.

Figure 4.120 Billable Item Test File Creation Screen

The updated file for billable item creation is then uploaded to create the billable item items (see Figure 4.121). Report RFKKBIXBITUPLOAD is used to upload the file. On execution, it creates the billable items in the SAP Convergent Invoicing system. The **Load Local File** option loads a file from the local system.

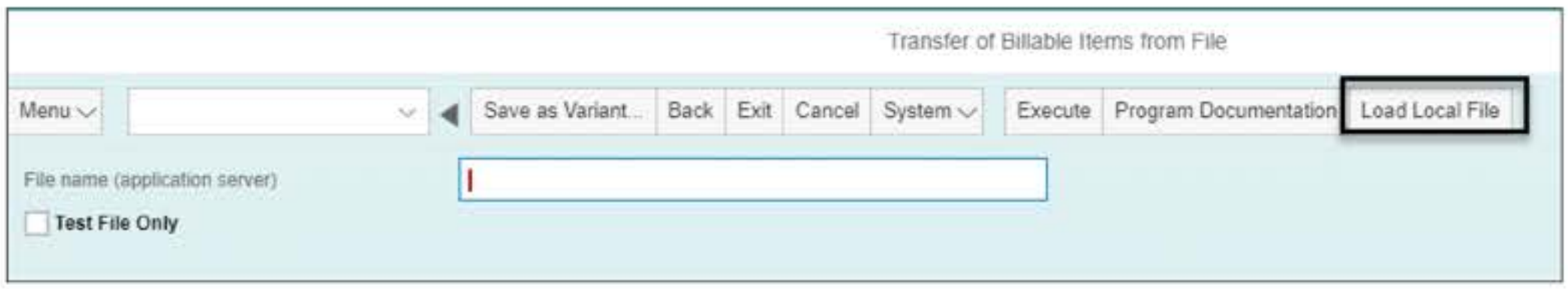

Figure 4.121 Billable Item File Upload Screen

Using Application Programming Interface Created on Class Generation

Interfaces are assigned during the class creation, and once class interfaces are generated, the tables, structures, and function modules are created for the classes. The function module generated for the billable item class can be used for creating billable items. Sample view of billable items is shown in Figure 4.122.

The created function module can be obtained by selecting the billable item class and then the **Objects** option under the path **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Items • Billable Item Classes • Generate Interface for the Billable Items Classes** (see Figure 4.123).

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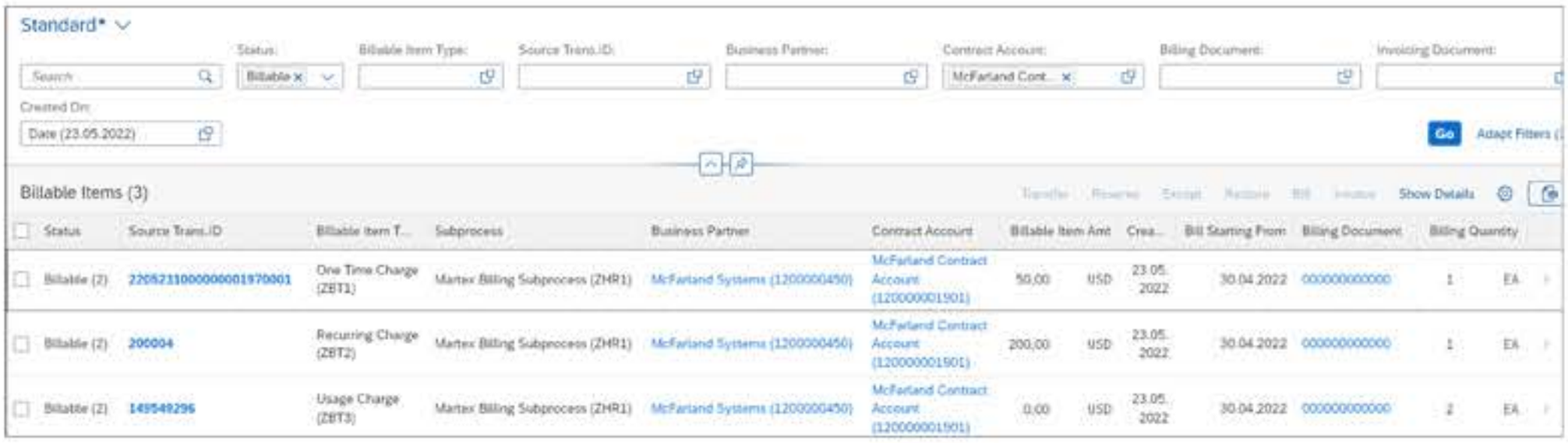

Figure 4.122 Billable Item View

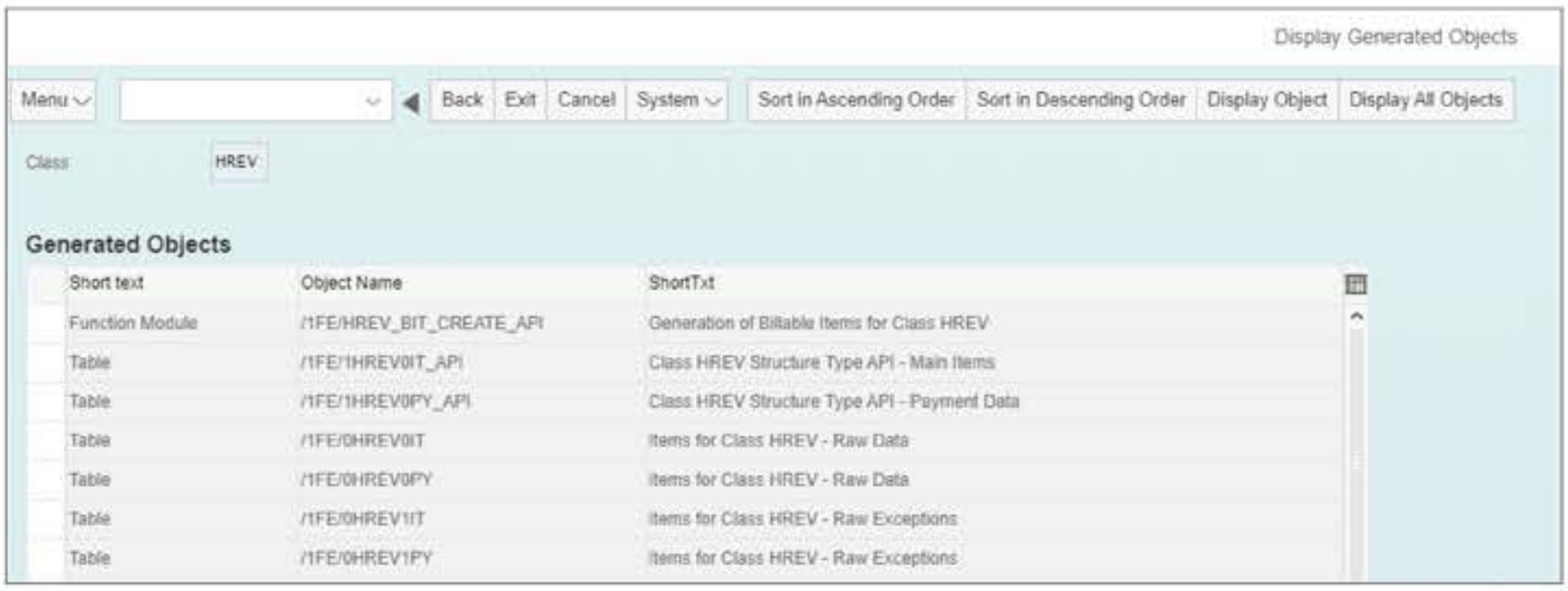

Figure 4.123 Billable Item Class Objects

The function module can be executed in Transaction SE37, where details for the billable items to be created are entered (see Figure 4.124). Once the function module is executed, the entries for the billable items are created.

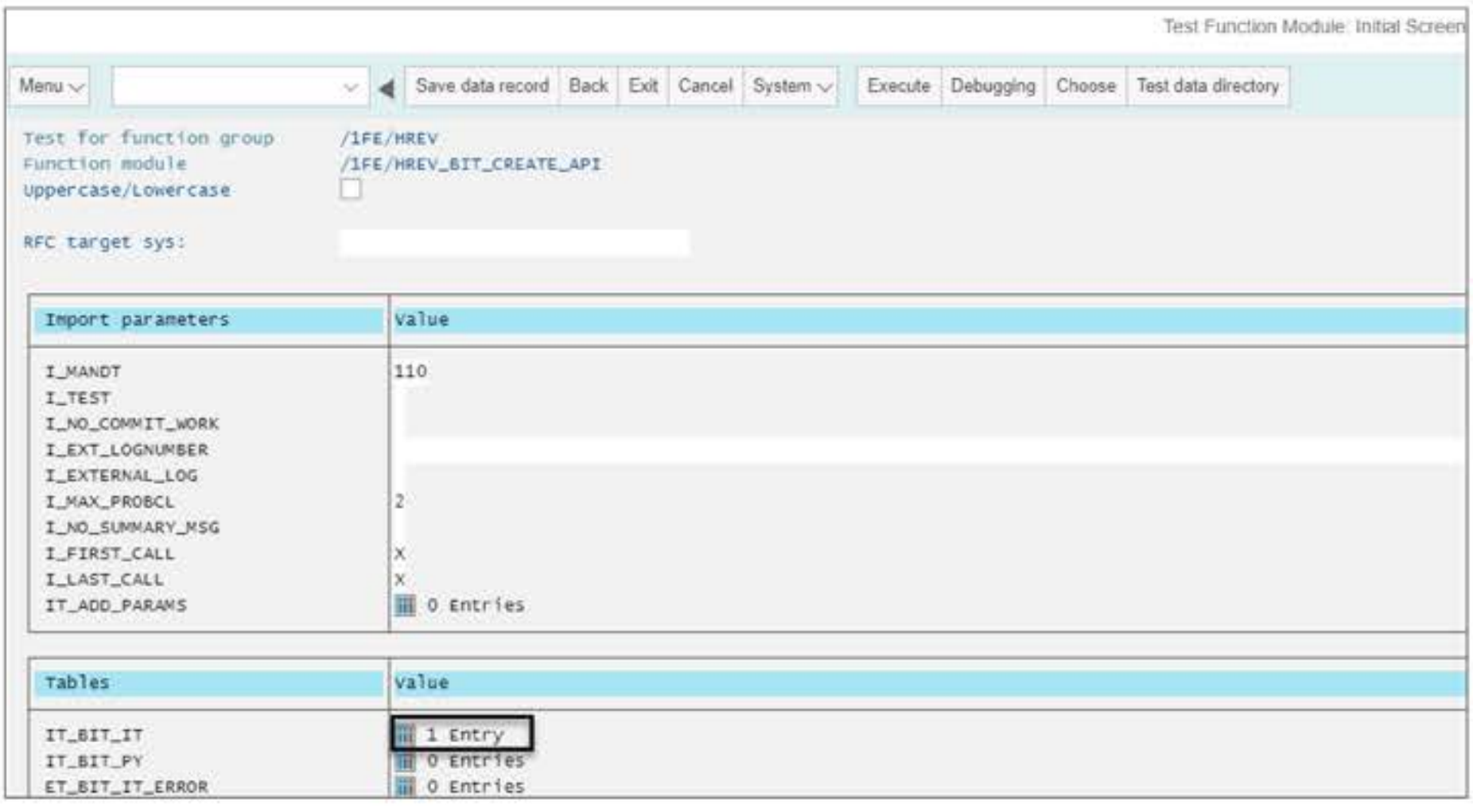

Figure 4.124 Function Module Screen for Billable Item Creation

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4.4.2 Modifying Consumption Items or Billable Items

This section will describe the options available in SAP Convergent Invoicing for modification of consumption and billable items created in the system.

Consumption Item Changes

Consumption item fields created in the system can be changed. The fields that need to be changed must be made modifiable in the system. The consumption item fields can be made modifiable for each status type. The standard fields available for consumption item status **4** (**Rated**) can't be changed.

The fields can be made modifiable under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Consumption Items • Consumption Item Management • Define Changeable Fields for Consumption Items** (see Figure 4.125).

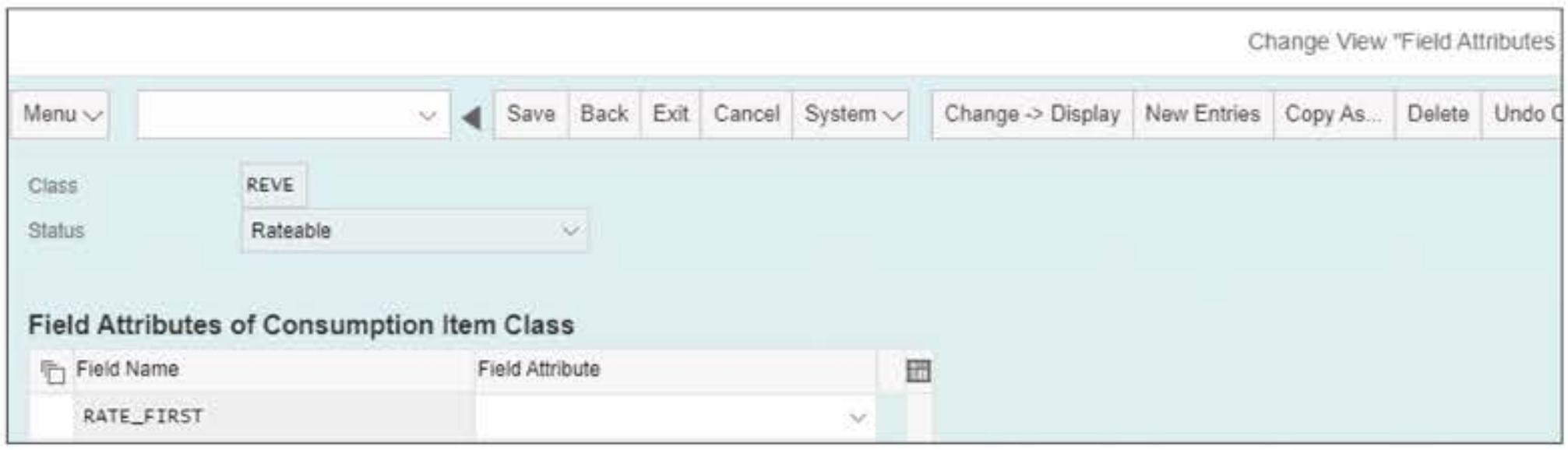

Figure 4.125 Modifiable Consumption Item Field

The consumption item class and status for which the consumption item field is to be assigned are input on creation. Using the **New Entries** option, the field that is to be made modifiable can be entered on the screen.

Transaction FKKBIXCIT_CWB or FKKBIXCIT_MON can be used to modify the consumption item fields. Select the consumption item whose field is to be modified, and select the **Process Item** option (see Figure 4.126).

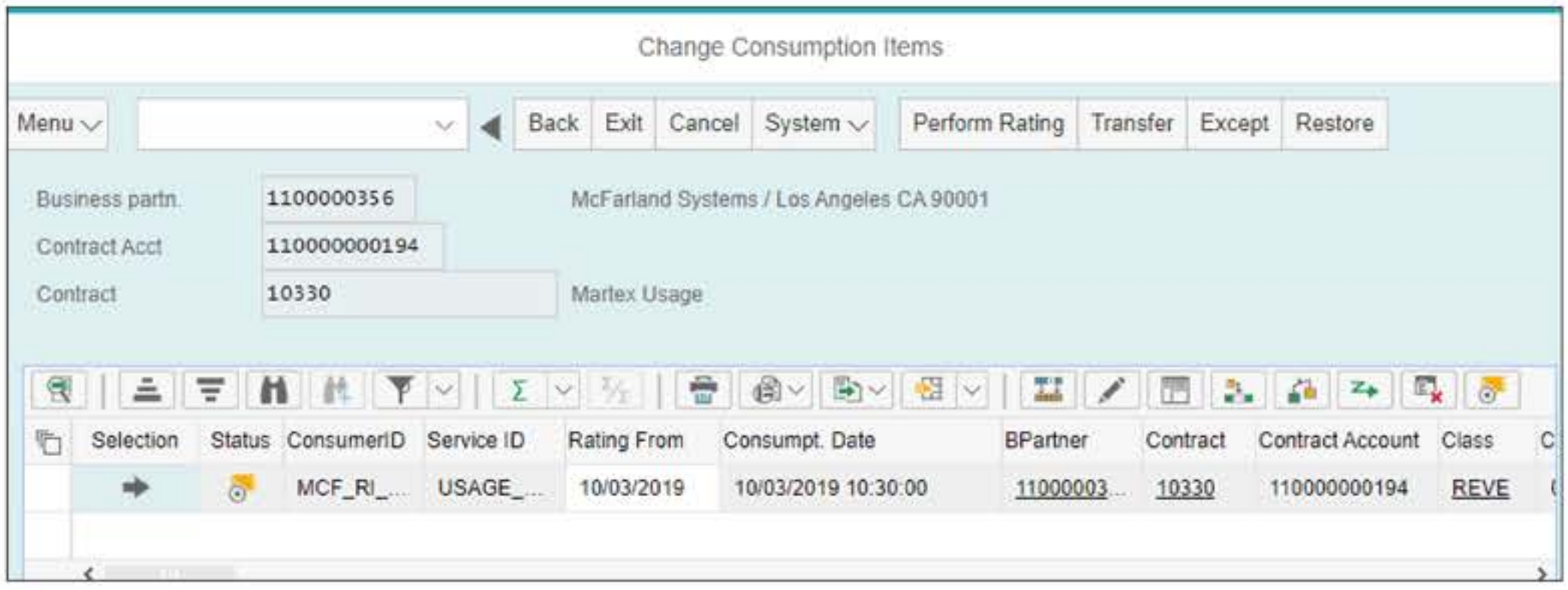

Figure 4.126 Selection of Consumption Item for Modification

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After changing the field value, you select the **Apply New Values** option, which updates the field. On selecting the **Save** option, the new value for the consumption item is saved in the system (see Figure 4.127).

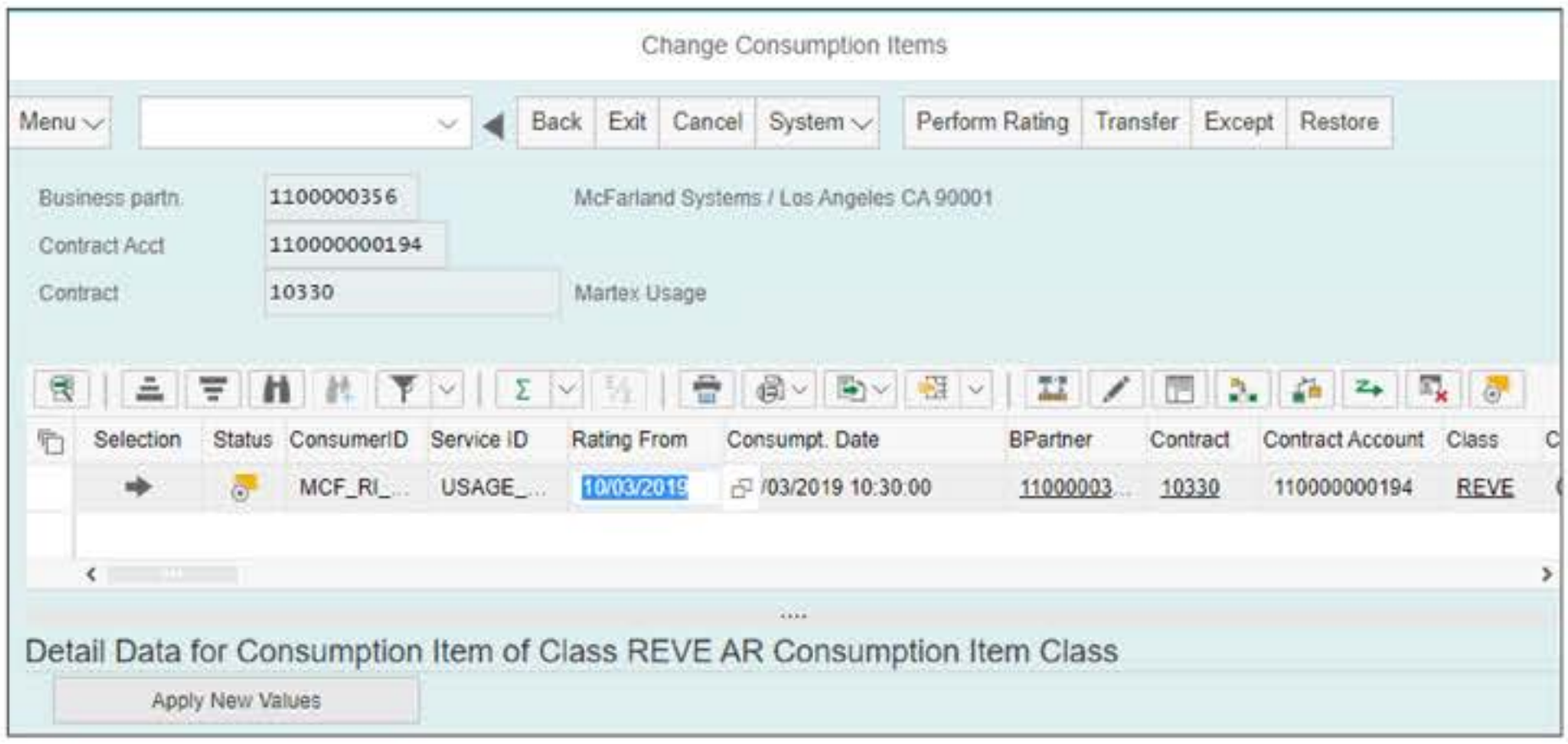

Figure 4.127 Field Modification for Consumption Items

Billable Items Change

Fields of billable items created in the system can be changed. The fields that need to be changed need be made modifiable in the system. The billable item fields can be made modifiable for each status and record type. The standard fields available for billable item status **4** (**Billed**) can't be changed. The fields can be made modifiable under **Contract Account Receivables & Payables • Convergent Invoicing • Basic Functions • Billable Items • Billable Item Management • Define Changeable Fields for Billable Items** (see Figure 4.128).

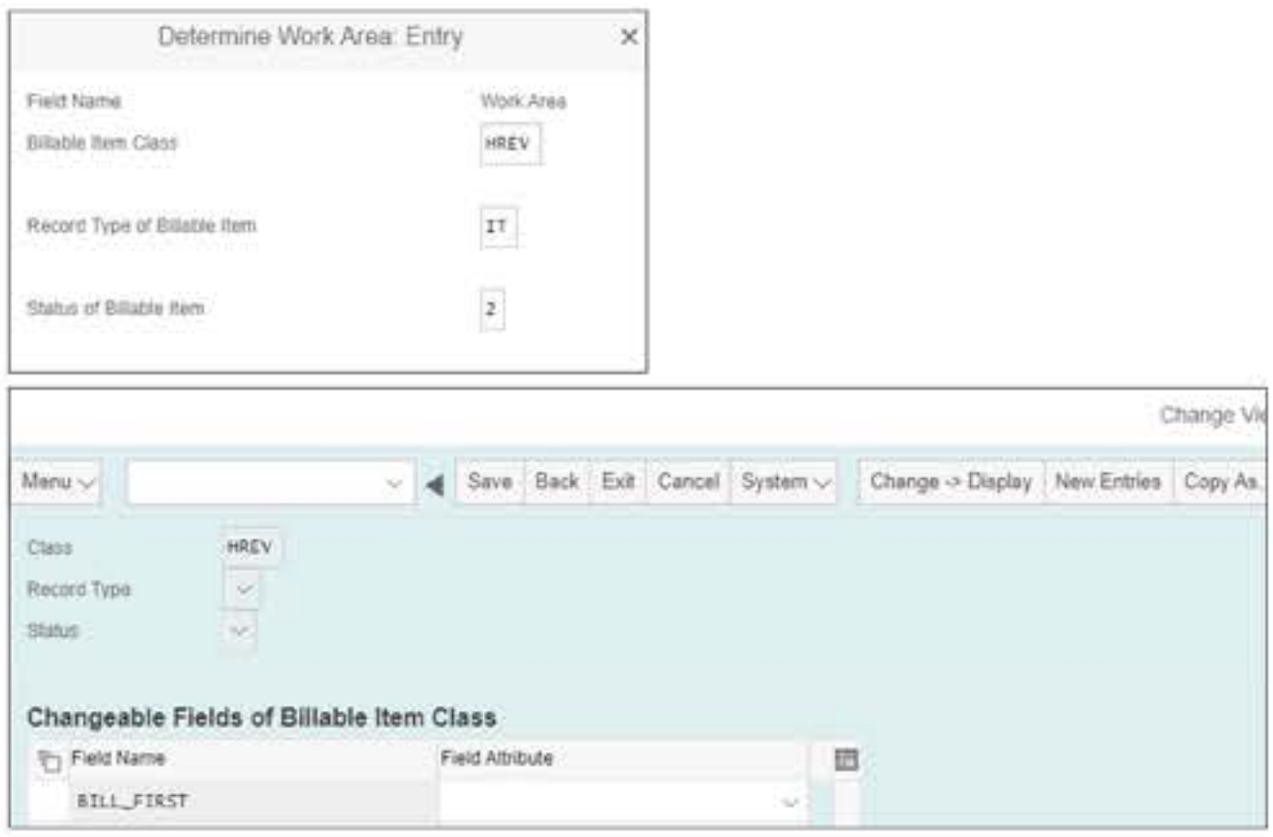

Figure 4.128 Modifiable Field for Billable Item Class Record Type IT

When the field is created, the billable item class, record type, and status for which the billable item field is to be assigned are input. Using the **New Entries** option, you can enter the field to be made modifiable.

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Transaction FKKBIXBIT_CWB or Transaction FKKBIXBIT_MON can be used to modify the billable items fields (see Figure 4.129).

Figure 4.129 Billable Item Selection for Field Modification

You select the billable item whose field is to be modified and then click the **Process Item** option. After changing the field value, click **Save** to save the new value for the billable item in the system (see Figure 4.130).

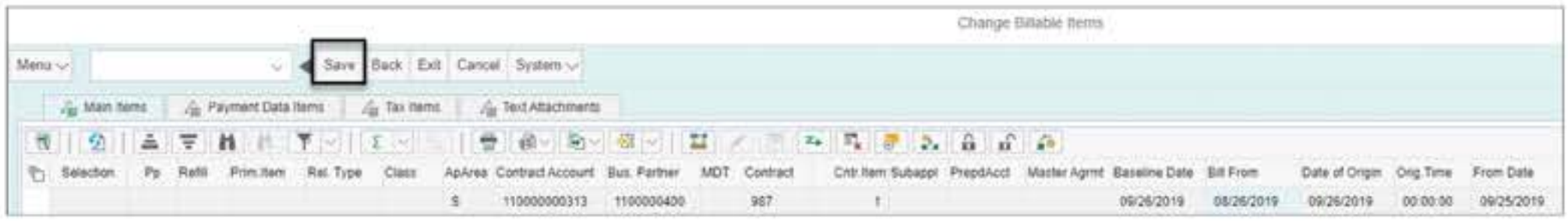

Figure 4.130 Field Modification for Billable Items

4.4.3 Viewing Provider Contract

A provider contract represents the agreement or contractual details with the service provider company. A provider contract is created in contract accounting when a provider order is released in SAP CRM/SAP Billing and Revenue Innovation Management, subscription order management. Provider contracts can be viewed using Transaction FP_VT3 (see Figure 4.131).

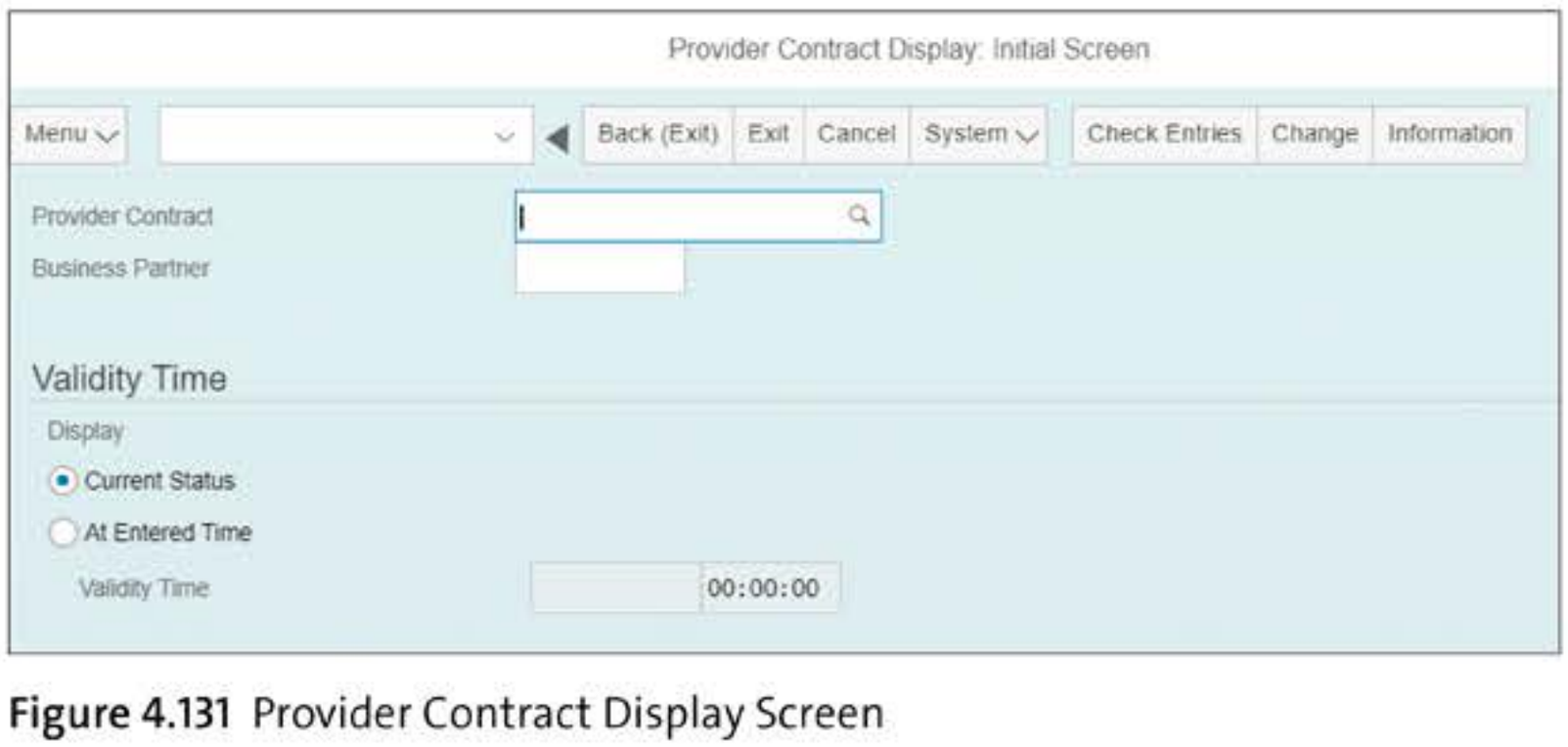

Figure 4.131 Provider Contract Display Screen

4.4.4 Posting a Billing Document, Invoicing Document, and Posting Document

A billing document can be posted using the transactions discussed in the following sections.

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Using Billable Item Monitoring Screen

Transaction FKKBIXBIT_MON is used for billable items monitoring. This transaction also provides the options to bill and invoice the billable items. When a billable item is selected and the **Bill** option is selected, the billing process is triggered for the billable items, thus creating a billing document (see Figure 4.132).

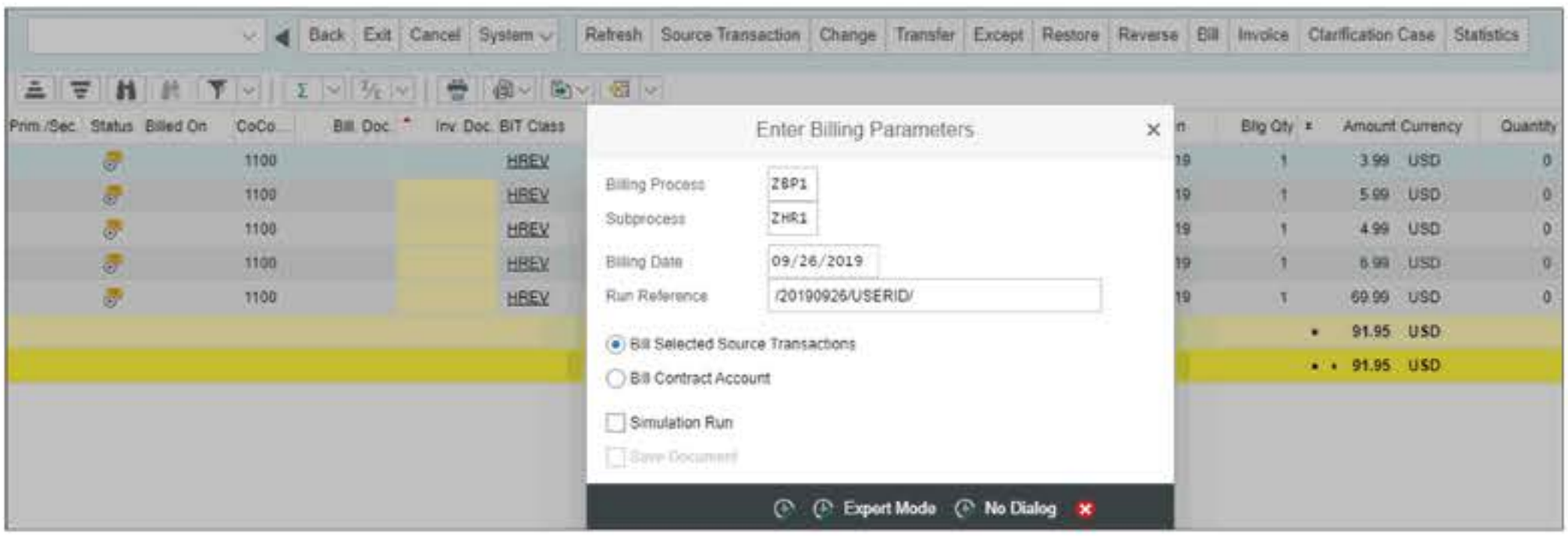

Figure 4.132 Billing Process Execution from Transaction FKKBIXBIT_MON

Individual Billing

The billing process can directly be triggered with Transaction FKKBIX_S for the individual business partner and contract account. You enter the **Billing Process**, **Subprocess**, **Business Partner**, and **Contract Account** on the selection screen for the billing process execution. Billing document simulation is also provided on the screen (see Figure 4.133).

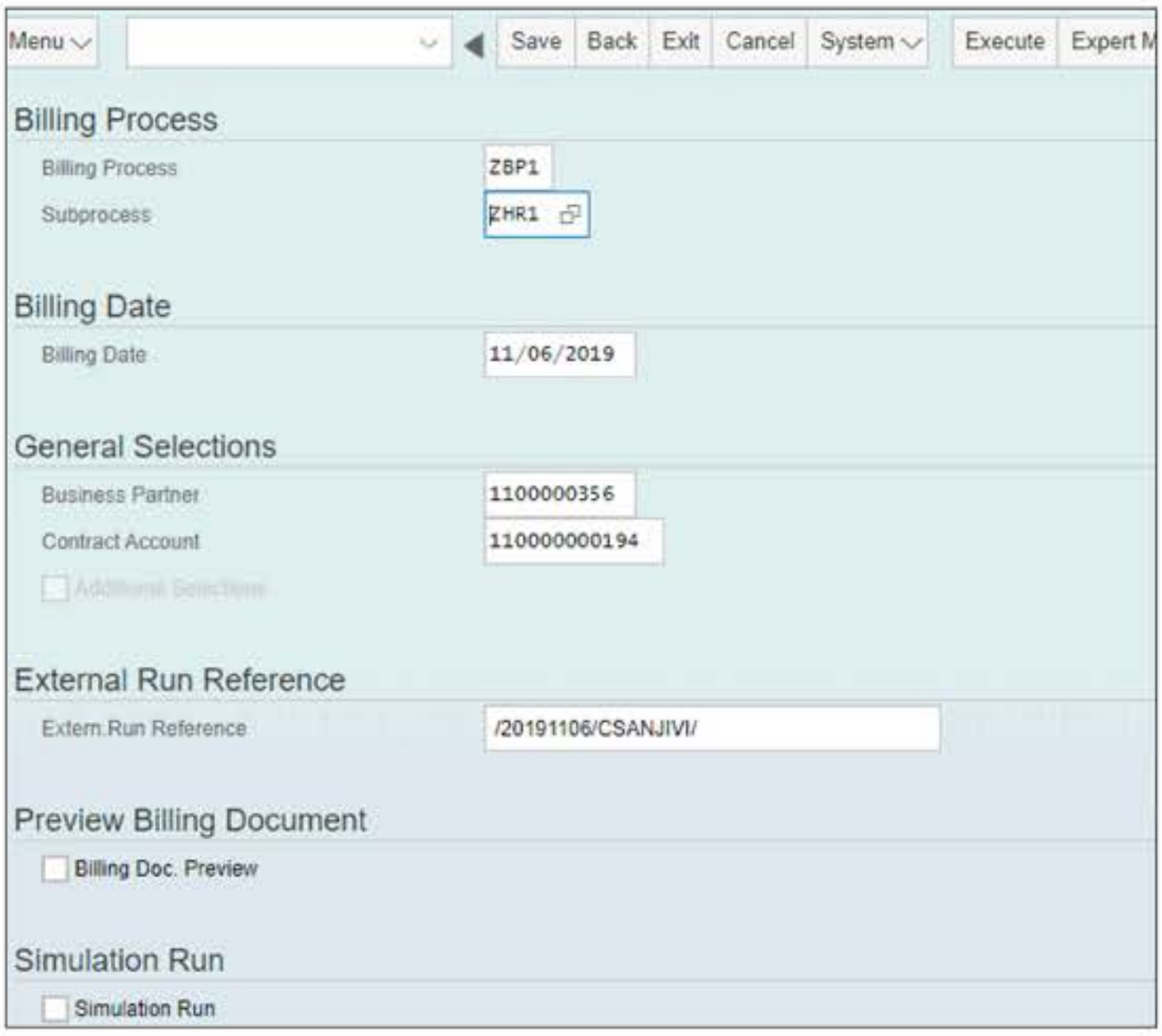

Figure 4.133 Individual Billing Process Execution

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Mass Billing Parallel Run

The billing process in an actual production environment is executed as a mass job via Transaction FKKBIX_MA. Mass jobs are generally set up as batch jobs that execute daily at a desired time and generate the billing documents. Mass jobs are executed in parallel processing with the contract account as a parallel processing parameter. The parallel processing parameter is given in the **Technical Settings** tab of the mass processing screen. The details of the billing process, subprocess, and contract account range are maintained in the **General Selections** tab (see Figure 4.134).

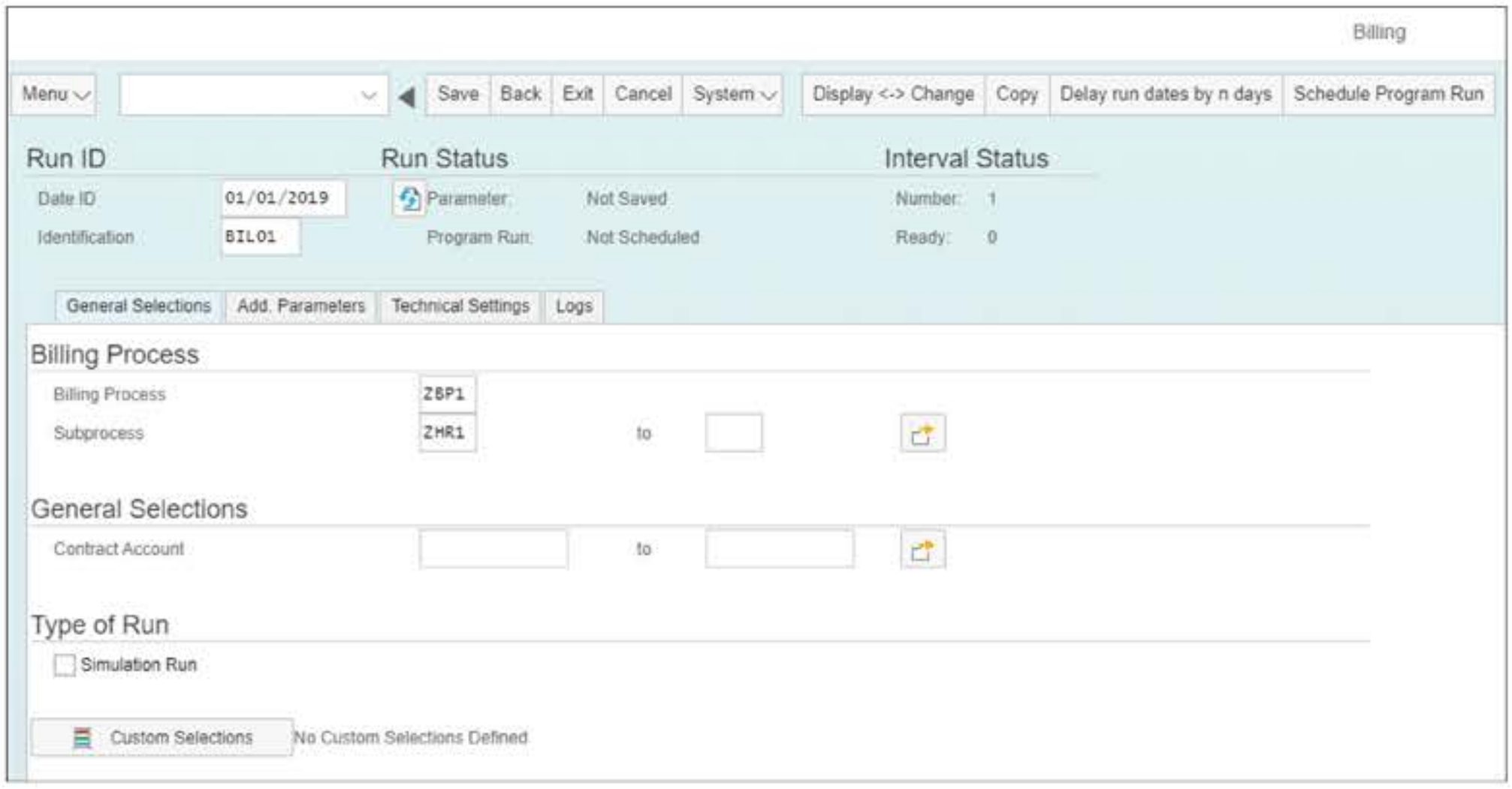

Figure 4.134 Mass Billing Parallel Run Screen

The mass job is scheduled in the background for execution. The job logs of the execution can be seen in the **Logs** tab of the screen.

Mass Billing

Transaction FKKBIX_M also creates the billing documents in mass, but the sequential run is performed and not the parallel run. This job's execution time for similar data is greater than that of Transaction FKKBIX_MA (see Figure 4.135).

An invoicing document can be posted using the transactions discussed in the following sections.

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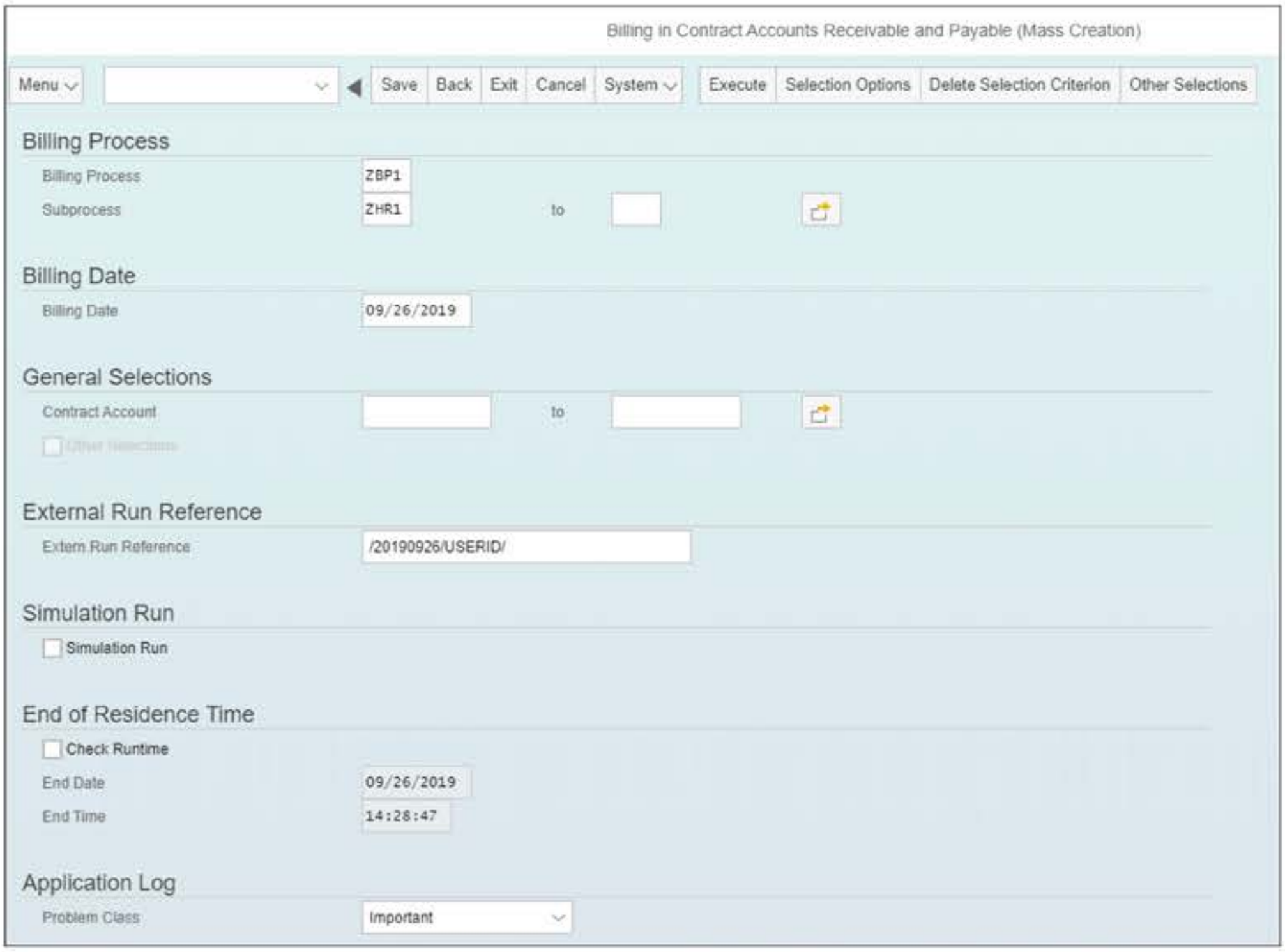

Figure 4.135 Mass Execution Screen for Billing Process

Using Billable Item Monitoring Screen

Transaction FKKBIXBIT_MON is used for billable items monitoring. This transaction also provides an option to invoice the billable items that are already billed. When billable items for which a billing document is posted are selected and the **Invoice** option is selected, the invoicing process is triggered, creating an invoicing document (see Figure 4.136).

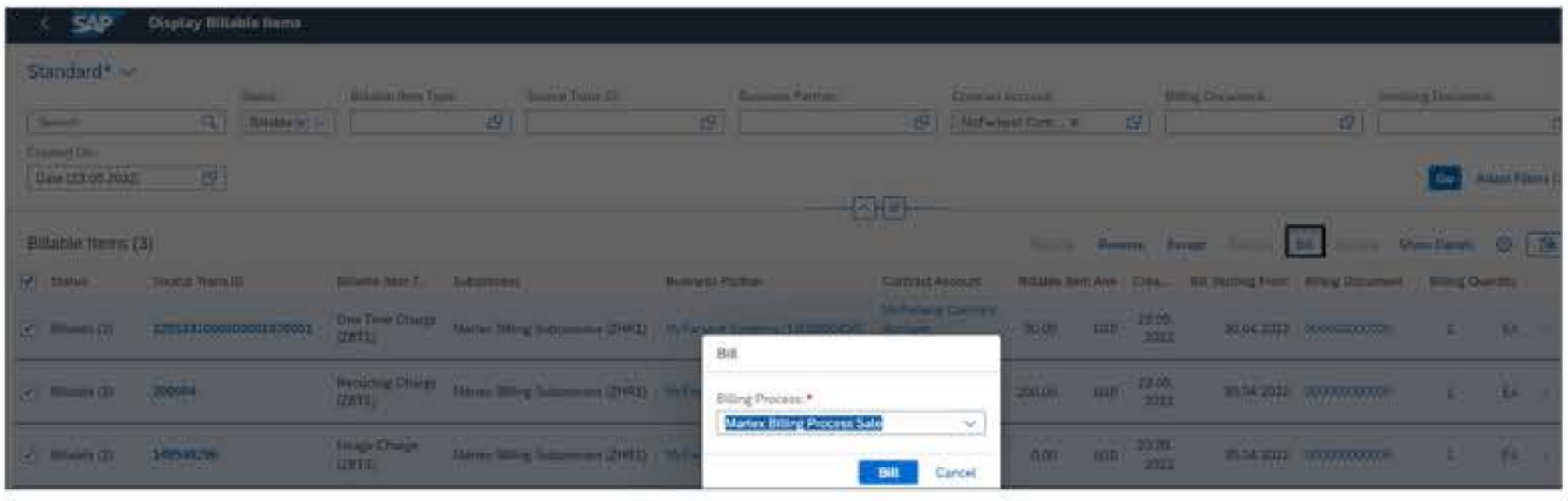

Figure 4.136 Billable Item Monitoring Screen

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Individual Invoicing

The invoicing process can be triggered directly for an individual business partner and contract account or via Transaction FKKINV_S. The invoicing process, business partner, and contract account are entered on the selection screen for the invoicing process execution. Invoicing document simulation is also provided on the screen. The **Expert Mode** option available on the screen provides the flexibility to assign the invoicing type to the invoicing units at the time of execution (see Figure 4.137).

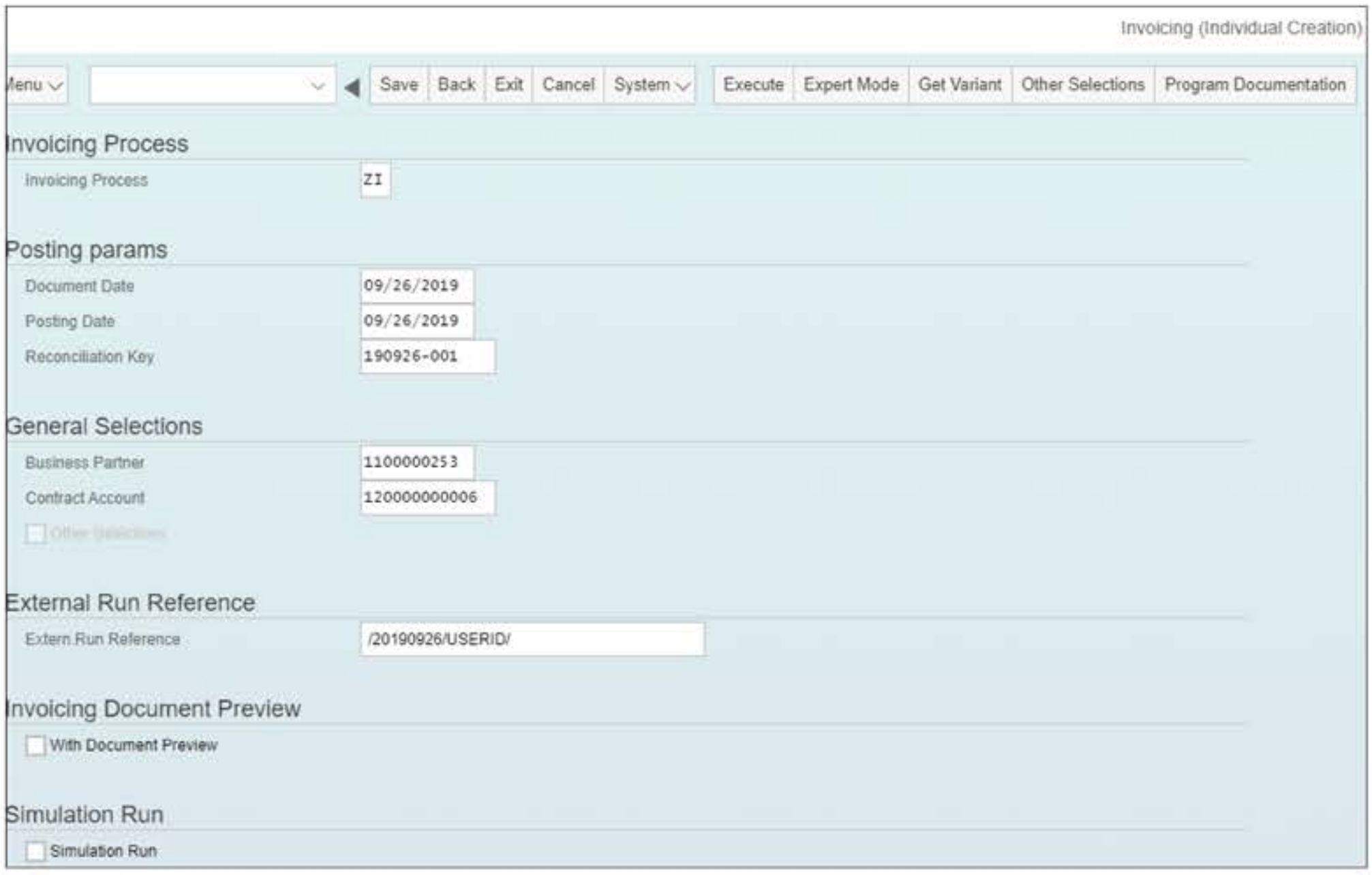

Figure 4.137 Individual Invoicing Run

Mass Invoicing Parallel Run

The invoicing process in an actual production environment is executed as a mass job via Transaction FKKINV_MA. Mass jobs are generally set up as batch jobs that execute daily at a desired time and generate the invoicing documents. Mass jobs are executed in parallel processing with the business partner or contract account as a parallel processing parameter. The parallel processing parameter is given in the **Technical Settings** tab of the mass processing screen. The details of the invoicing process, business process, and contract account range are maintained in the **General Selections** tab (see Figure 4.138).

The mass job is scheduled in the background for execution. The job logs of the execution can be seen in the **Logs** tab of the screen.

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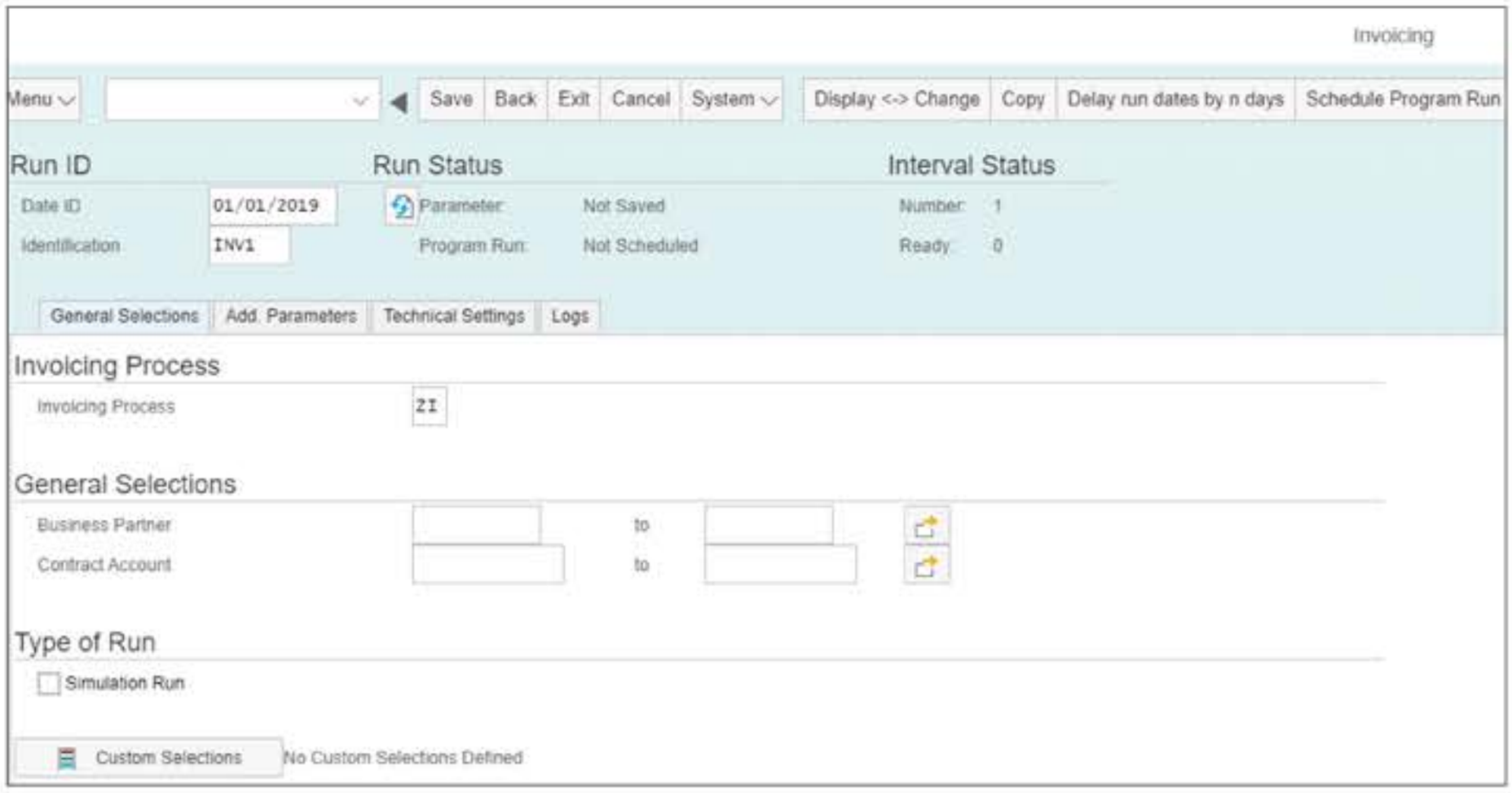

Figure 4.138 Mass Invoicing Run

Mass Invoicing

Transaction FKKINV_M creates the invoicing documents in mass, but the sequential run is performed and not the parallel run. This job's execution time for similar data is greater than that of Transaction FKKINV_MA (see Figure 4.139).

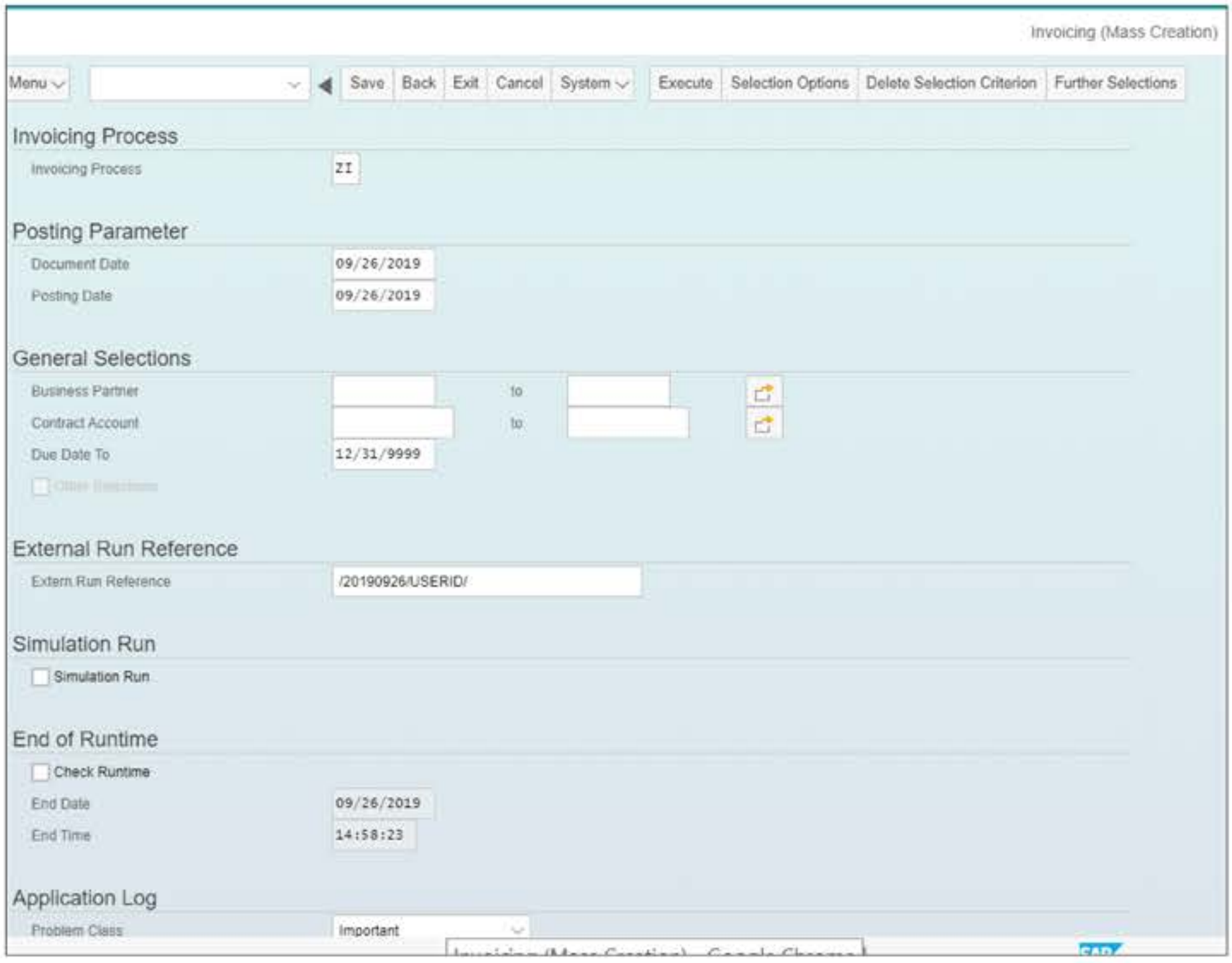

Figure 4.139 Mass Invoicing Sequential Run

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Posting documents are posted using the invoicing functions. When invoicing is performed, the invoicing function posts the posting document or contract accounting receivables document automatically. Posting documents can be viewed by selecting the document number under the **Posting Document** tab of the invoicing document display (see Figure 4.140).

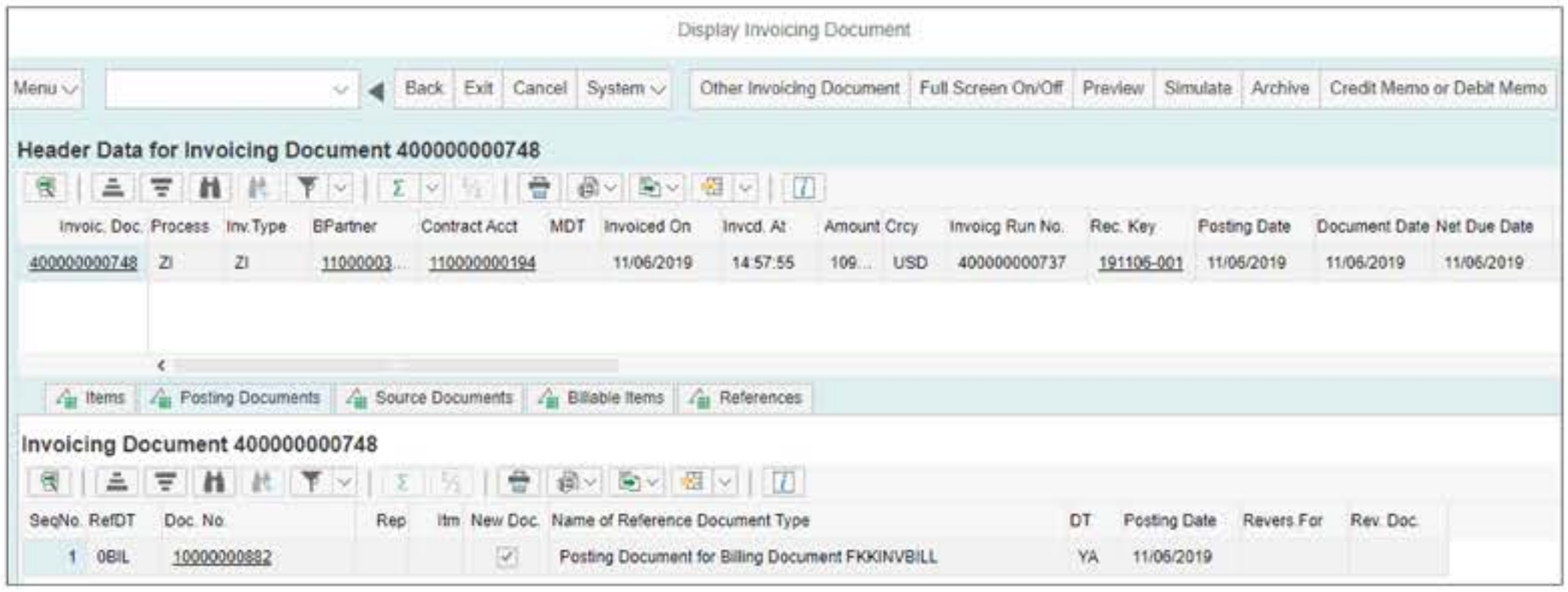

Figure 4.140 Posting Document Number under Posting Documents Tab in the Invoicing Document

In addition, if a posting document number is known, then it can be viewed directly using Transaction FPE3 (see Figure 4.141).

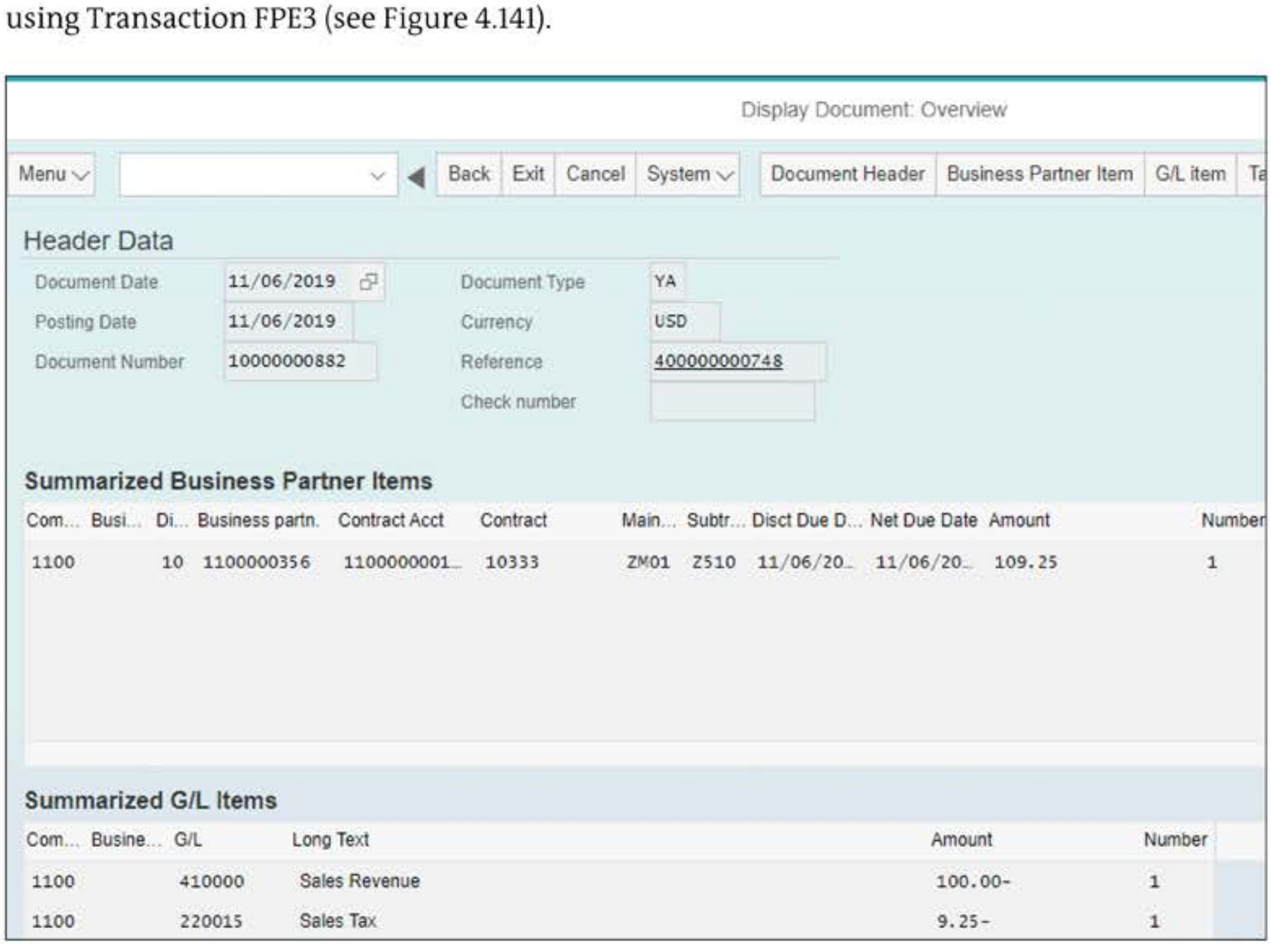

Figure 4.141 Posting Document/Contract Accounting Document Display

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Now that you've seen how the billable items can be created directly within SAP Convergent Invoicing and understand the concepts of provider contract, billing, and invoicing, let's now look at how the billable items have been created for the Martex Corp. use case within the integrated SAP Billing and Revenue Innovation Management solution.

For the use case, we've created the provider contract 10000000000000003147 for the reserved instance. We have also created a partner agreement that references the contract with the vendor because the vendor doesn't sell its software directly to customers, but rather through multiple distributors in different regions. Distributors such as Martex Corp. invoice the customers. In return, the vendor gives Martex Corp. a 10% discount off the MSRP based off partner agreement 10337.

For the first scenario, provider contract 10000000000000003147 (reserved instance), we'll walk through the following transactions:

- **Recurring billable item**
  A monthly recurring billable item for $200 for the sale for customer 1200000450, McFarland Systems, will be triggered from SAP Convergent Charging. A billable item will be created in SAP Convergent Invoicing for this recurring charge.
- **Usage-based charges**
  Usage records are created in SAP Convergent Invoicing as consumption items. They will be rated at a discounted price and converted to billable items. The customers are eligible for a better rate per unit of consumption, as they are on a reserved instance.
- **Cancellation (prorated refund)**
  Contract cancellation will result in a prorated refund to the customer.

Each of these transactions will result in billable items in SAP Convergent Invoicing, indicated in Table 4.5 with a unique billable item type configured to indicate the type of the transaction.

Billable Item	Use for Martex Corp.
ZBT2	Recurring charges
ZBT3	Usage-based charges
ZBT6	Cancellation (prorated refund)
ZBC3	Partner charges (partner settlement)

Table 4.5 Billable Items for Martex Corp.

For the second scenario, provider contract 10335 (nonreserved instance), we'll walk through the following transactions:

- **Usage-based charges**
  Usage records are created in SAP Convergent Invoicing as consumption items. They will be rated at a nondiscounted price and converted to billable items. The customers

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aren't eligible for a discounted rate per unit of consumption, as they are on a nonreserved instance.

- **Partner settlement**
  Vendor payable components will be calculated by SAP Convergent Charging, and a billable item for the vendor payable amount will be created in SAP Convergent Invoicing.

So, let's examine at the transaction flow for provider contract 10333, beginning with the recurring billable item. As shown in Figure 4.142, the billable item for the recurring charge for product MRTX_RI referenced in the provider contract 1000000000000 0003147 is triggered from SAP Convergent Charging on the first day of every month.

Figure 4.142 Billable Item for Recurring Charge for Provider Contract 10333

Once created successfully in SAP Convergent Invoicing, the billable item is now eligible to be billed and invoiced. Figure 4.143 shows the billing document created for the recurring charge for provider contract 10333.

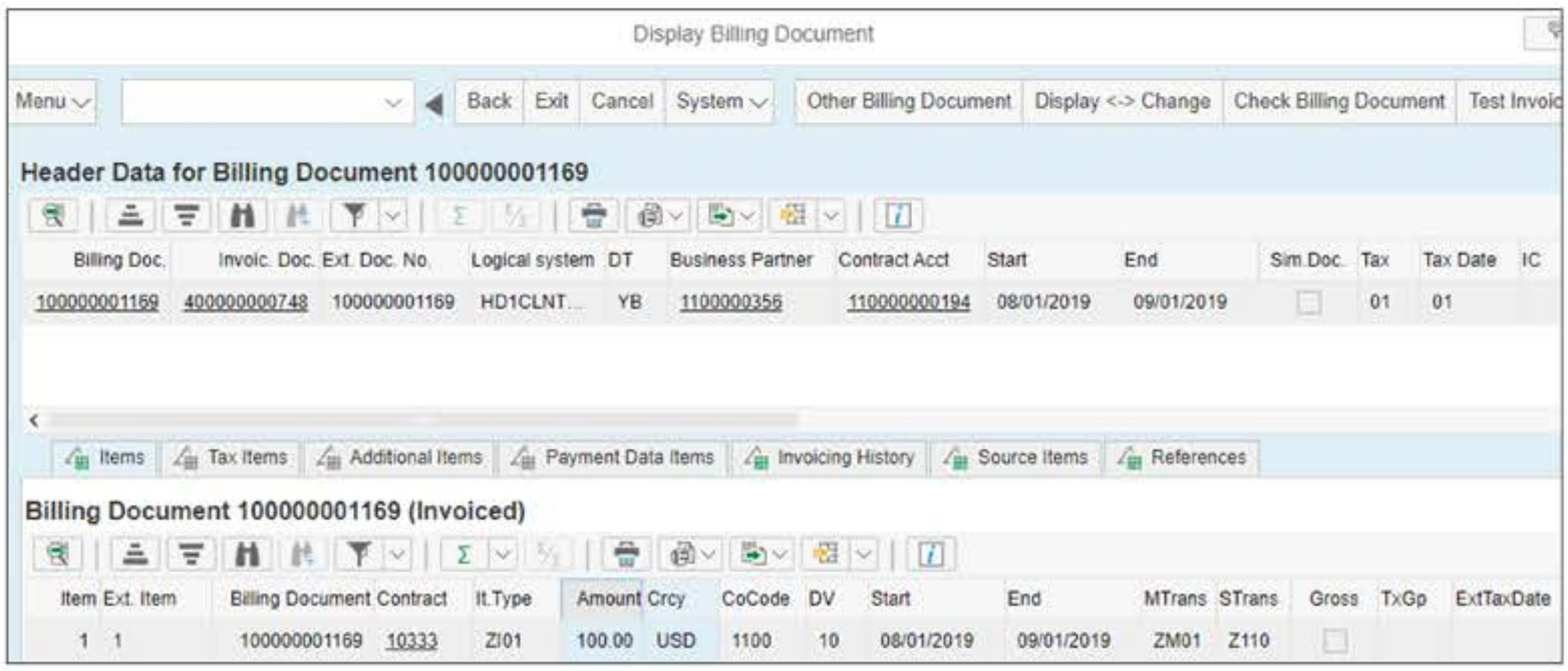

Figure 4.143 Billing Document for the Recurring Charge on Provider Contract 10333

Once the invoicing run for the billing document is performed, the invoicing document shown in Figure 4.144 and the corresponding contract accounting posting document are created, as shown in Figure 4.145 (along with any applicable taxes).

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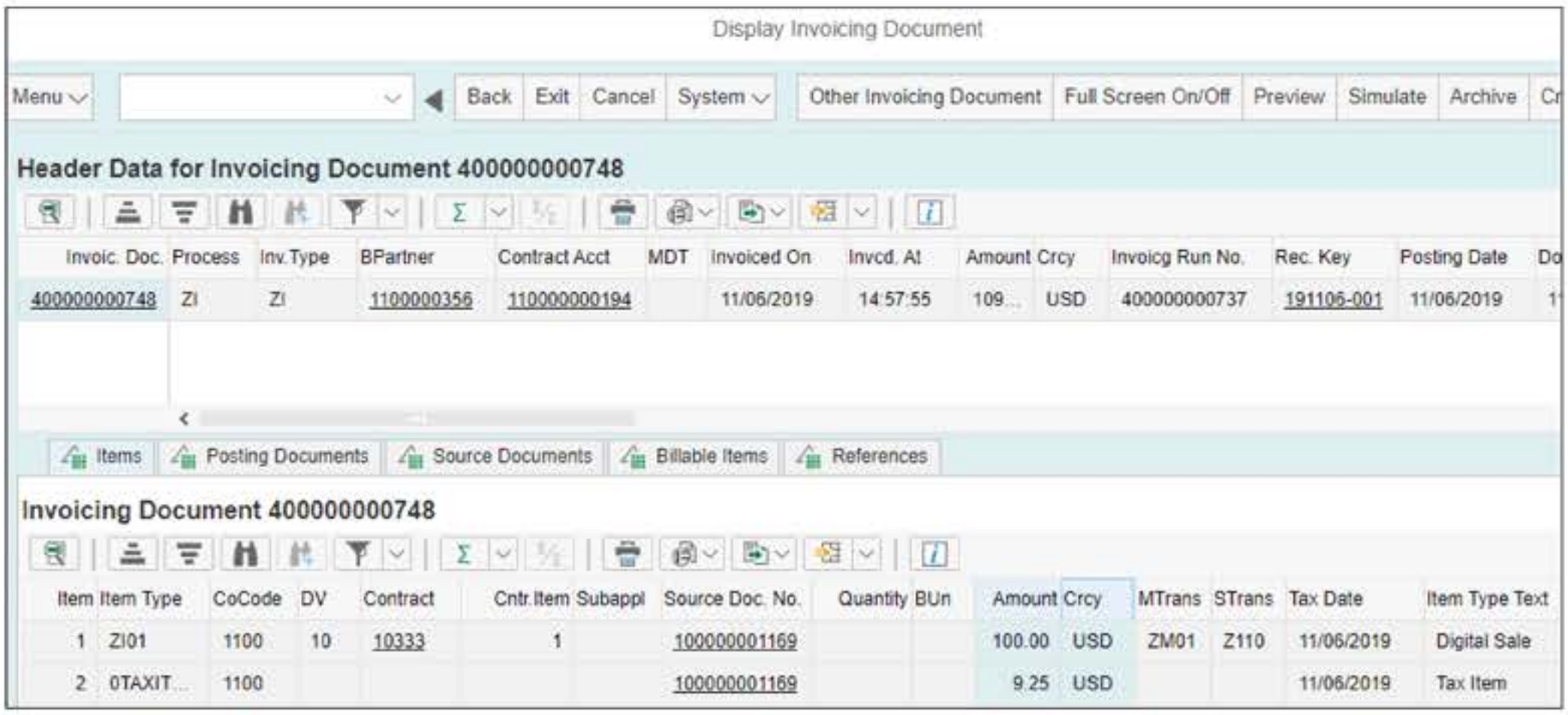

Figure 4.144 Invoicing Document for the Recurring Charge on Provider Contract 10333

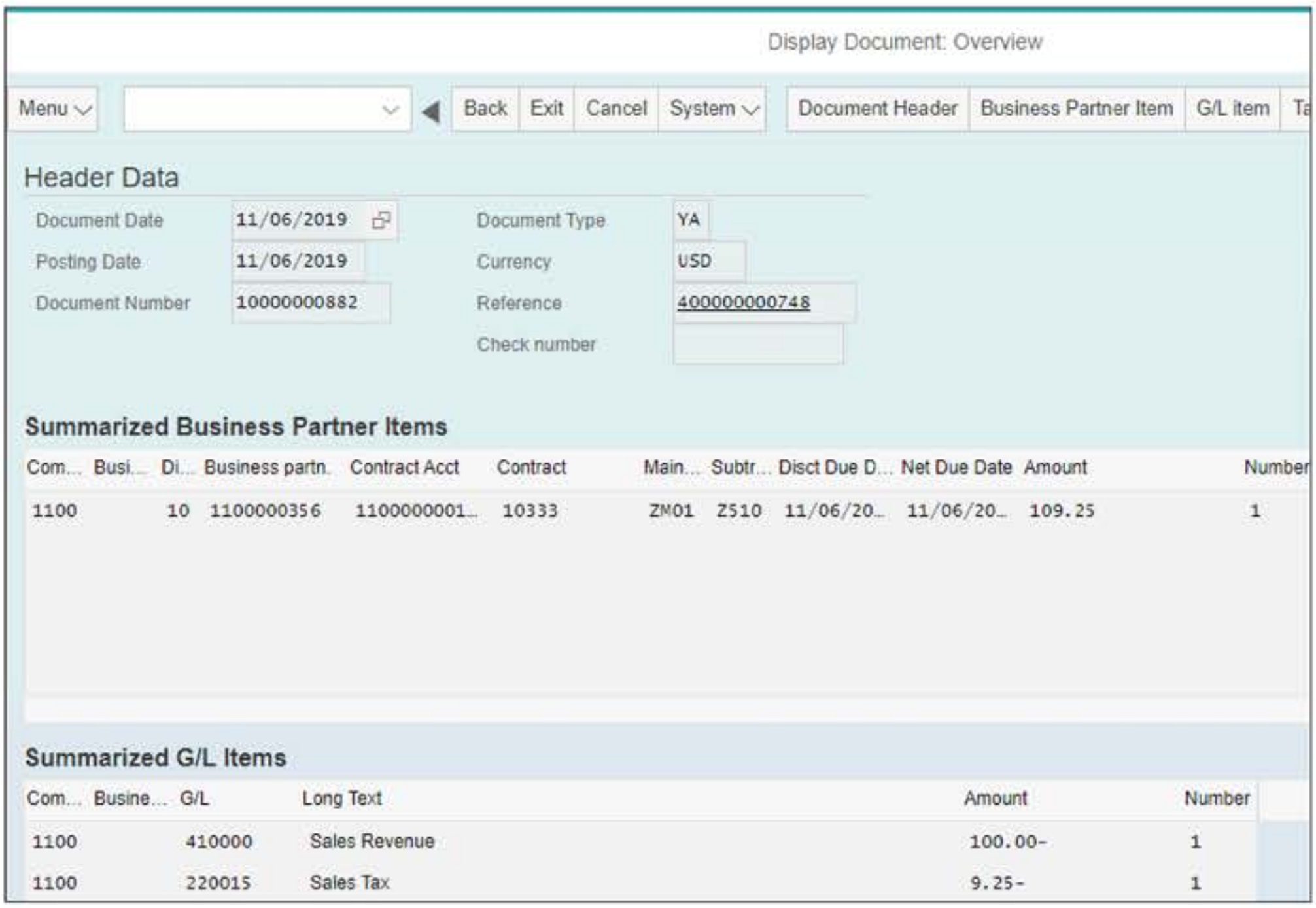

Figure 4.145 Posting Document: Contract Accounting Document Display on Provider Contract 10333

Let's move to provider contract 10333's usage-based charges. The metered usage records are created in SAP Convergent Invoicing as consumption items are shown in Figure 4.146. Next, the rating request for these usage records is initiated from SAP

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Convergent Invoicing and rated in SAP Convergent Charging. As a result, a billable item is created in SAP Convergent Invoicing for this transaction.

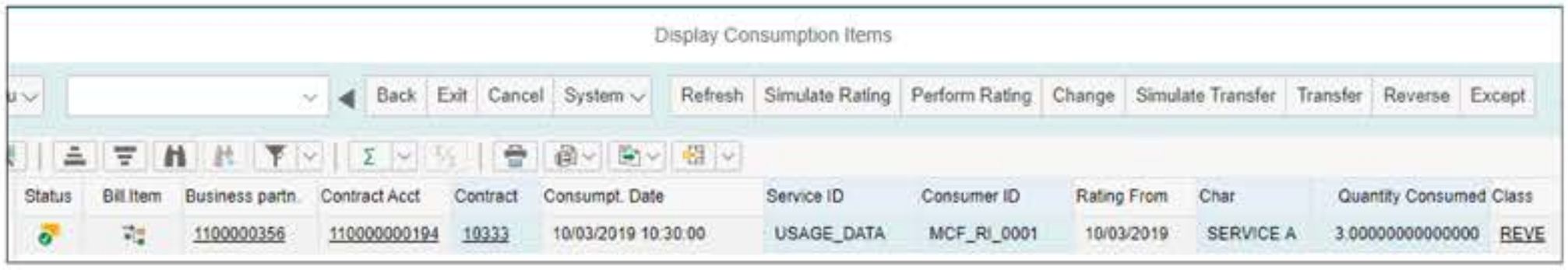

Figure 4.146 Usage Records Created as Consumption Items in SAP Convergent Invoicing on Provider Contract 10333

The billable item resulting from the rating process is shown in Figure 4.147. The total quantity of **3** units has been rated at a rate of $250 per unit (**$750** total), as defined in SAP Convergent Charging; this is a discounted rate because the customer has a reserved instance.

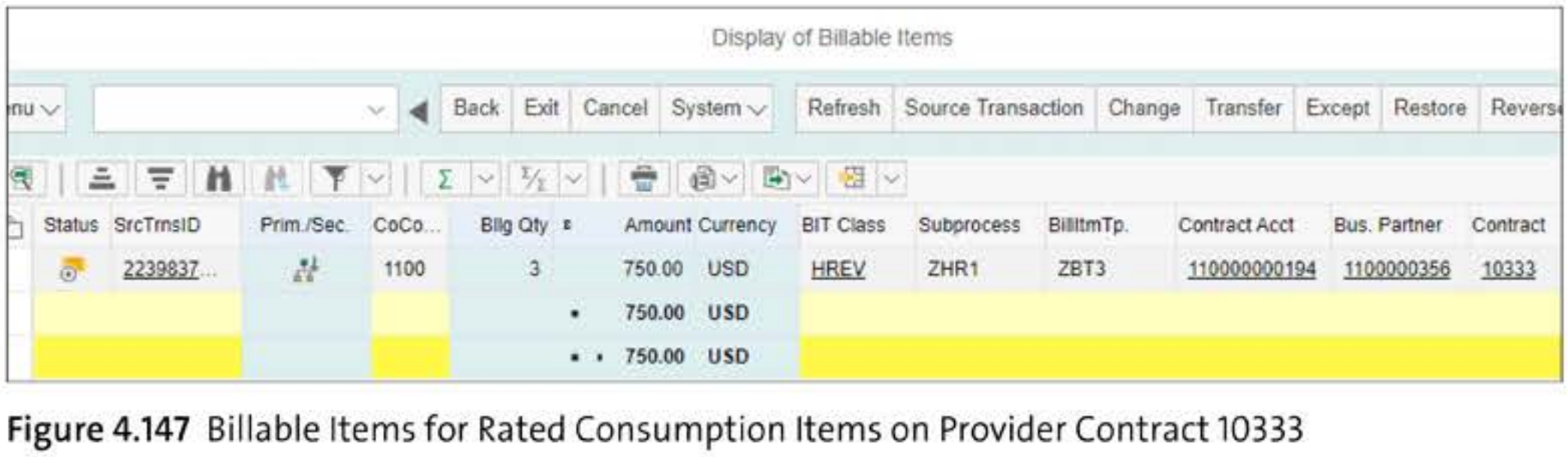

Figure 4.147 Billable Items for Rated Consumption Items on Provider Contract 10333

Once created successfully in SAP Convergent Invoicing, the billable item is now eligible to be billed and invoiced. Figure 4.148 shows the billing document created for the usage charge for provider contract 10333.

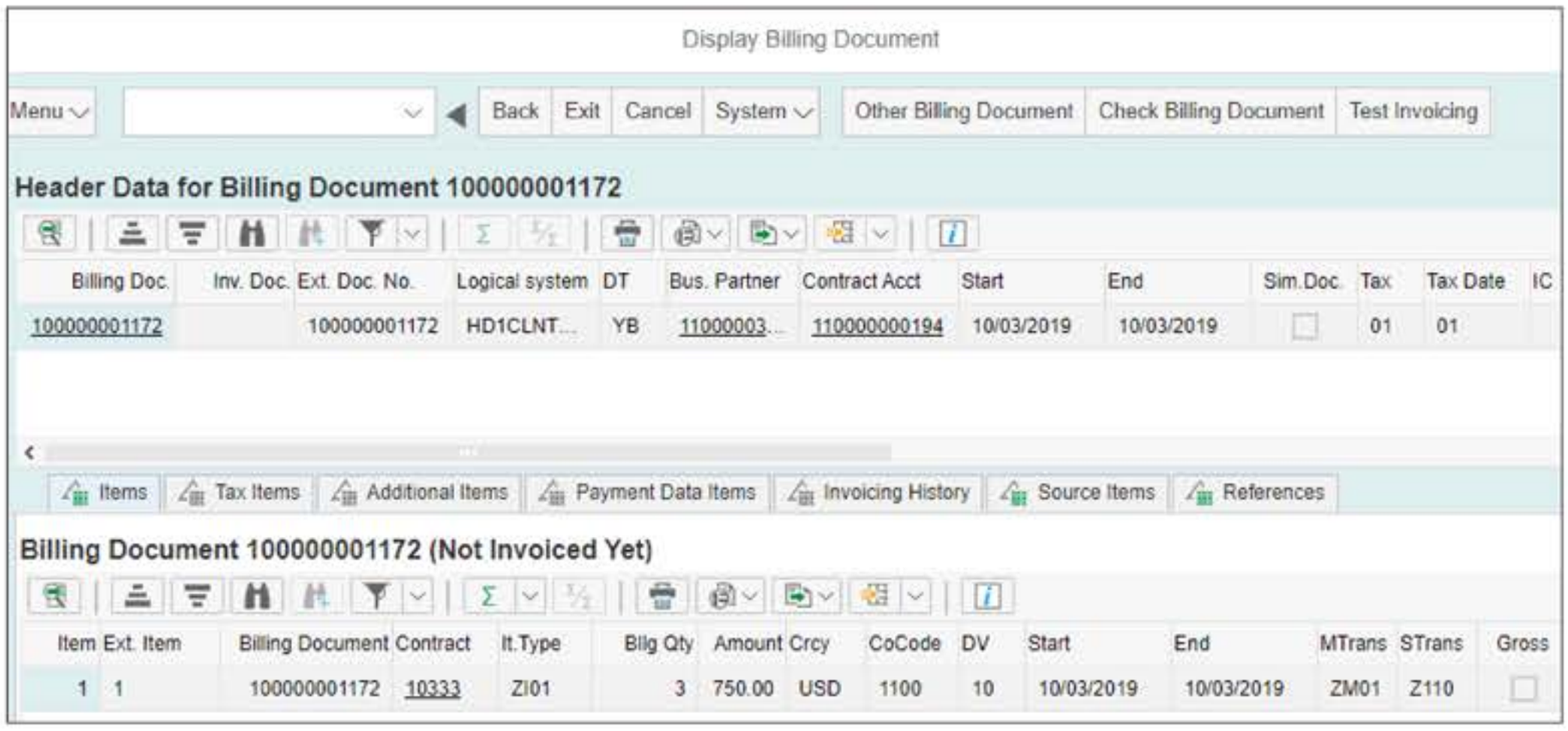

Figure 4.148 Billing Document for Usage Charges on Provider Contract 10333

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Subsequently, once the invoicing run for the billing document is performed, the invoicing document shown in Figure 4.149 and the corresponding contract accounting posting document shown in Figure 4.150 are created (along with any applicable taxes).

Display Invoicing Document

Menu | Back | Exit | Cancel | System | Other Invoicing Document | Full Screen On/Off | Preview | Simulate | Archive

Header Data for Invoicing Document 400000000750

Invoic. Doc.	Process	Inv.Type	BPartner	Contract Acct	MDT	Invoiced On	Invcd. At	Amount	Crcy	Invoicg Run No.	Rec. Key	Posting Date
400000000750	ZI	ZI	1100000356	110000000194		11/07/2019	00:40:40	819.38	USD	400000000750	191107-001	11/07/2019

Items | Posting Documents | Source Documents | Billable Items | References

Invoicing Document 400000000750

Item	Item Type	CoCode	DV	Contract	Cntr.Item	Subappl	Source Doc. No.	Quantity	BUn	Amount	Crcy	MTrans	STrans	Tax Date	Item Type Text
1	ZI01	1100	10	10333	1		100000001172	3	EA	750.00	USD	ZM01	Z110	11/07/2019	Digital Sale
2	0TAXIT...	1100					100000001172			69.38	USD			11/07/2019	Tax Item

Figure 4.149 Invoicing Document for Usage Charges on Provider Contract 10333

Display Document: Overview

Menu | Back | Exit | Cancel | System | Document Header | Business Partner Item | G/L Item | Taxes | Cl

Header Data

Document Date	11/07/2019	Document Type	YA
Posting Date	11/07/2019	Currency	USD
Document Number	10000000884	Reference	400000000750
		Check number	

Summarized Business Partner Items

Com...	Busi...	Di...	Business partn.	Contract Acct	Contract	Main...	Subtr...	Disct Due D...	Net Due Date	Amount	Number
1100		10	1100000356	1100000001...	10333	ZM01	Z510	11/07/20...	11/07/20...	819.38	1

Summarized G/L Items

Com...	Busine...	G/L	Long Text	Amount	Number
1100		410000	Sales Revenue	750.00-	1
1100		220015	Sales Tax	69.38-	1

Figure 4.150 Posting Document Contract Accounting: Document for Usage Charge on Provider Contract 10333

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Finally, when a customer decides to cancel the contract, the contract cancellation process is initiated in SAP Billing and Revenue Innovation Management, subscription order management, as a result of a change order process. A prorated refund amount for the recurring charge is calculated in SAP Convergent Charging, and a prorated refund billable item is sent to SAP Convergent Invoicing, as shown in Figure 4.151.

Let's now look at the transaction flow for the second scenario (provider contract 10335), beginning with usage-based charges. Similar to the previous scenario for the reserved instance, the metered usage records for the nonreserved instance are created in SAP Convergent Invoicing as consumption items, as shown in Figure 4.152.

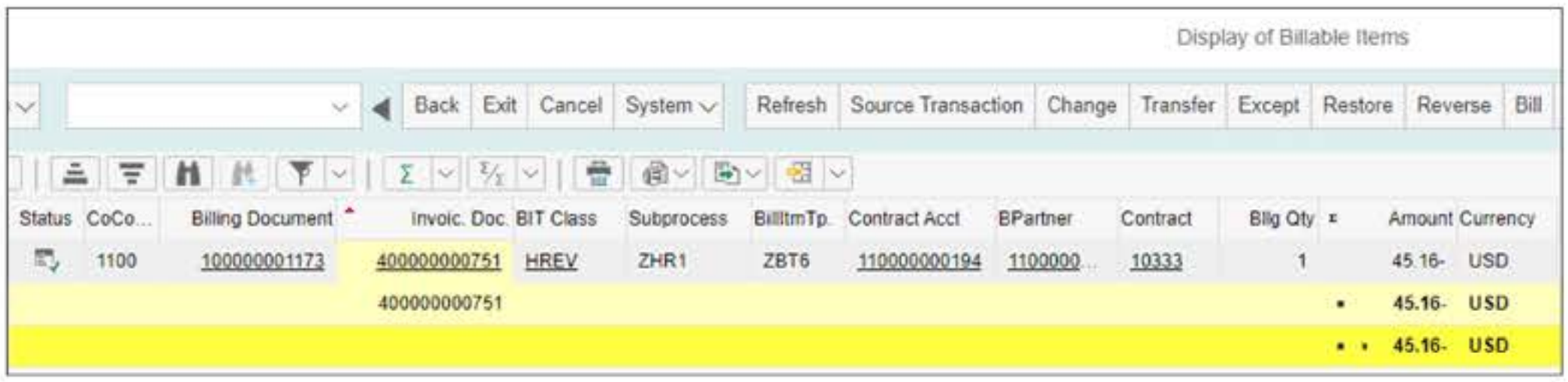

Figure 4.151 Billable Item for Prorated Refund for Contract Cancellation on Provider Contract 10333

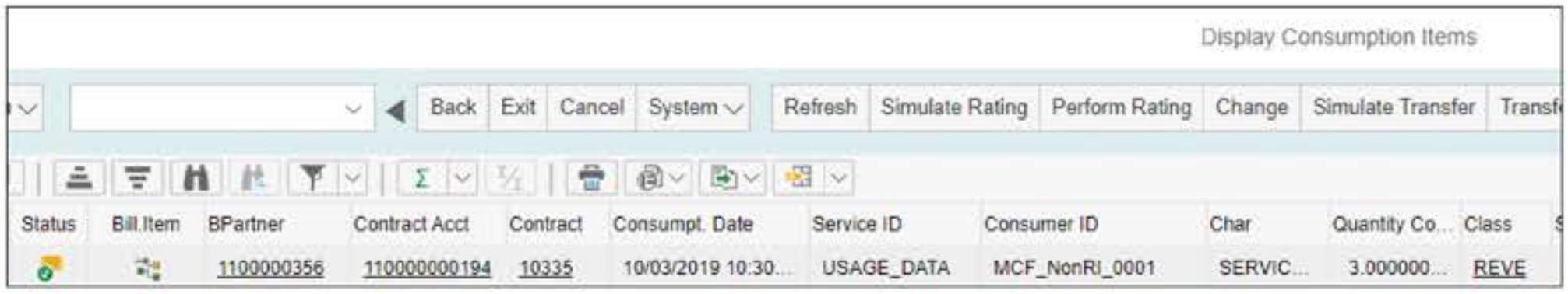

Figure 4.152 Consumption Items Display for Provider Contract 10335

Let's consider the partner settlement. The rating request for the usage record is initiated from SAP Convergent Invoicing and is rated in SAP Convergent Charging. As a result, the following items are created in SAP Convergent Invoicing for this scenario, which is relevant for partner settlement:

- An accounts receivable billable item
- An accounts payable consumption item associated with the usage record

In our example, the accounts receivable billable item accounts receivable for $900 resulted from the rating process shown in Figure 4.153. (The total quantity of **3** units has been rated at a rate of $300 per unit, as defined in SAP Convergent Charging; this is a nondiscounted rate because the customer has a nonreserved instance.) Clicking the icon in the **Prim./Sec.** column navigates to the accounts payable partner consumption item, which is shown in Figure 4.154.

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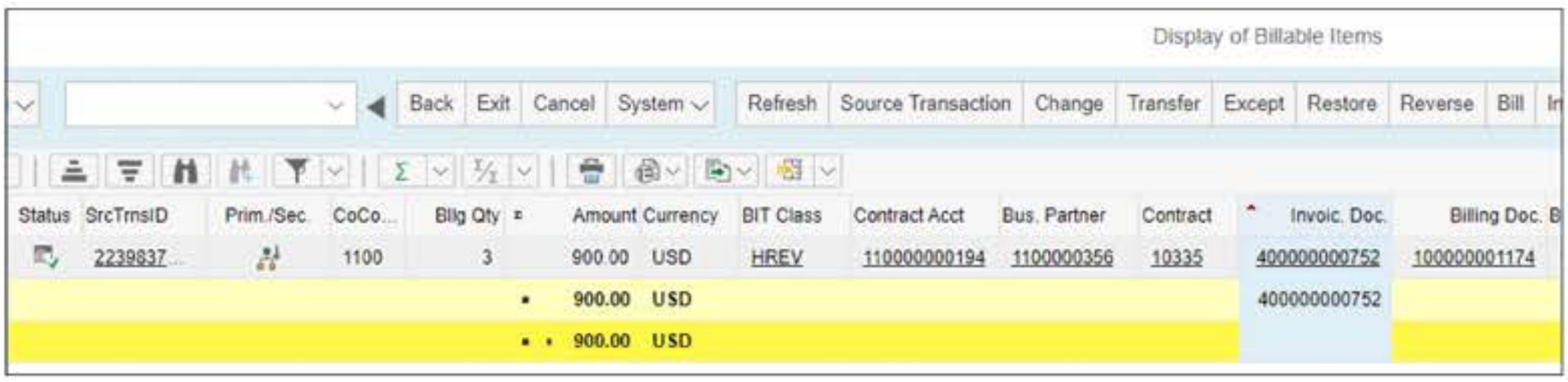

Figure 4.153 Billable Items for Rated Consumption Items for Provider Contract 10335

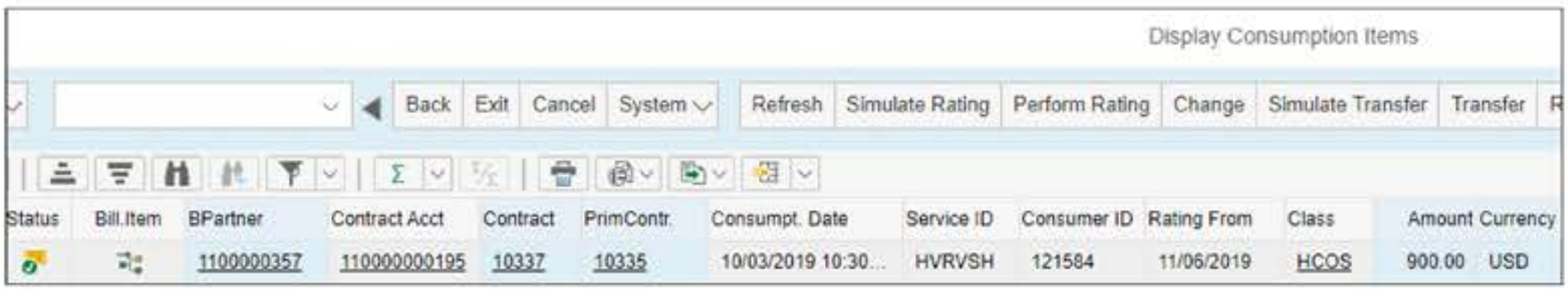

Figure 4.154 Consumption Items Display for Partner Agreement 10337

As shown in Figure 4.155, the vendor payable amount calculated for the vendor Loren Systems is **$675**, as the vendor provides a 10% discount off MSRP for Martex Corp.

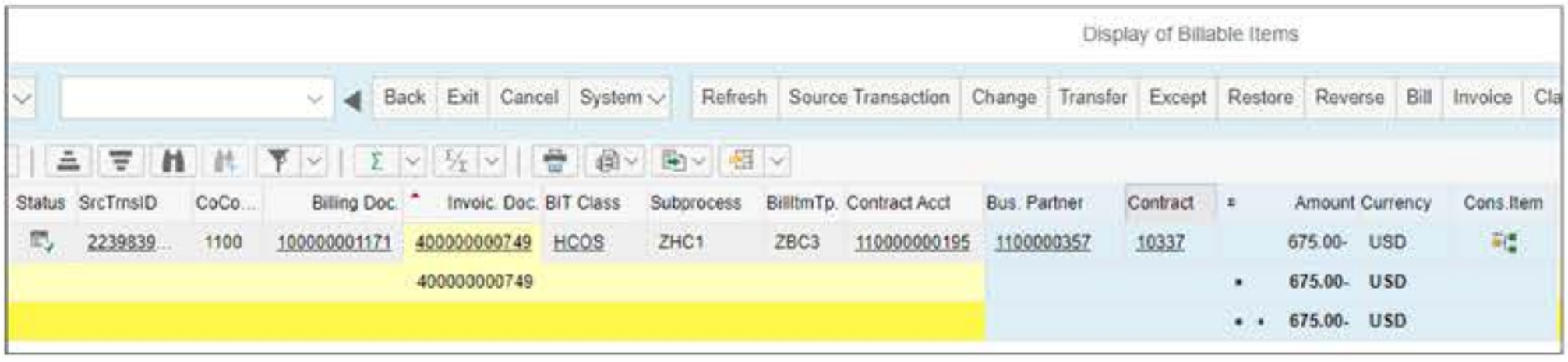

Figure 4.155 Billable Items for Rated Consumption Items for Partner Agreement 10337

4.4.5 SAP Fiori Apps

SAP Fiori apps provide users with a flexible and user-friendly experience to access business transactions on various devices such as laptops, desktops, tablets, mobile phones, and so on. The SAP Fiori user experience model is based on five pillars, which we'll review further in Chapter 5:

- **Role based**
  Users can access the apps according to the roles assigned to complete their tasks specific to their roles
- **Responsive**
  The apps can adapt according to the size and type of device used by the user to access the apps.

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- **Simple**
  The scope of application is simple and follows the 1-1-3 model, which means one user, one use case, and a maximum of three screens per application.
- **Coherent**
  Apps are developed with clear and easy structure. All apps have the same language and can be deployed to multiple landscapes and environments.
- **Instant value**
  Apps are easy to install and use, and the SAP Fiori apps introduced with the SAP S/4HANA 2021 release for SAP Convergent Invoicing are discussed in the rest of this section.

The section outlines the description, functionality, and usage of SAP Fiori apps introduced with the SAP 2021 release and also some commonly used SAP Fiori apps for SAP Convergent Invoicing.

Upload Consumption Items

Consumption items are usage records that are stored in the database tables in SAP Convergent Invoicing in the form of consumption items. The upload of consumption items earlier in the system was performed using Transaction FKKBIXCIT_UPLOAD, as shown in Figure 4.156. Using this transaction, consumptions items are imported in SAP Convergent Invoicing from an external source file.

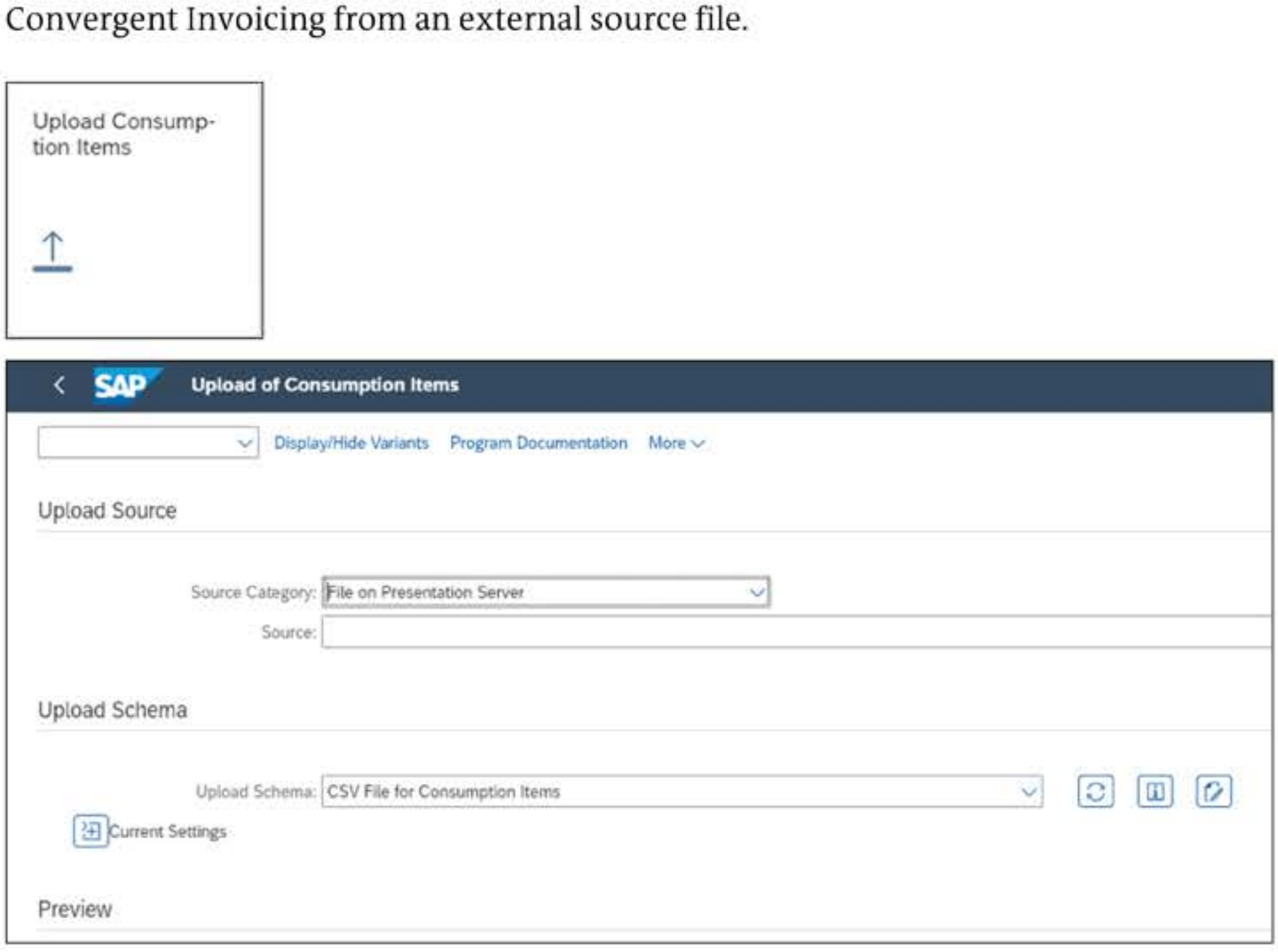

Figure 4.156 Upload Consumption Items Using Transaction FKKBIXCIT_UPLOAD

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With the recent release, SAP has now provided the Upload Consumption Items app to perform the same task. You can perform the following activities via the Upload Consumption Items app:

- Consumption items upload using data form the local file on the computer
- Use of different separators, such as commas or semicolons, to match the source file format
- Simulation of consumption items upload
- Template creation for consumption items upload

Display Error Logs - Consumption Items

Various errors may occur during the processing of consumption items. Errors can originate during the consumption item creation, transfer from **Raw** to **Ratable** status, and also during the rating of the consumption items.

Error analysis and resolution has always been a crucial activity. Proper view of logs is always needed to analyze the errors. SAP has provided the Display Error Logs - Consumption Items app, as shown in Figure 4.157, to view and analyze the error during various steps of consumption items processing.

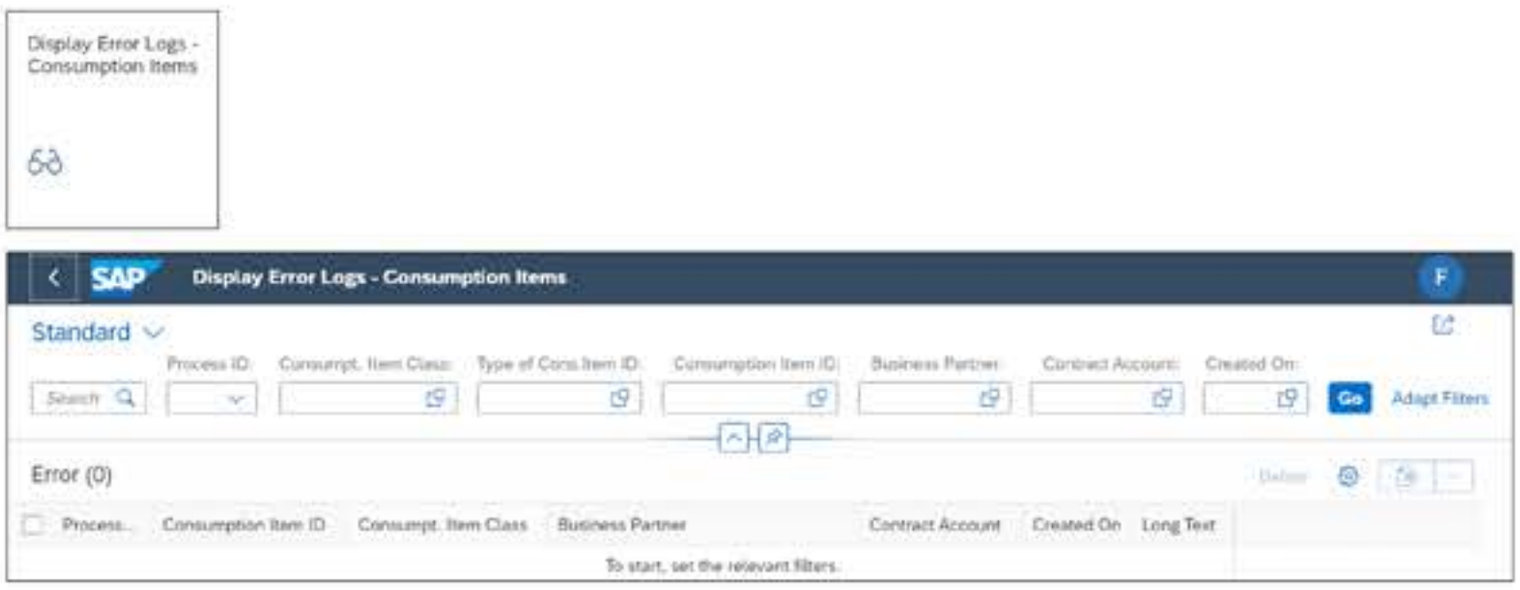

Figure 4.157 Fiori Apps for Consumption Items Error Log

Using this app, you can perform the following activities:

- Display error logs during the following:
  - Consumption items creation
  - Transfer of consumptions items from **Raw** to **Rateable** status
  - Rating of consumption items
- Navigate to application log.
- Delete the error log entry.

To enable the recording of error logs in the system for the consumption items, the configuration settings need to be maintained in Transaction SPRO path **Contract Account Receivables & Payables • Convergent Invoicing • Error Logging • Activate Error Logging for Consumption Items**, as shown in Figure 4.158.

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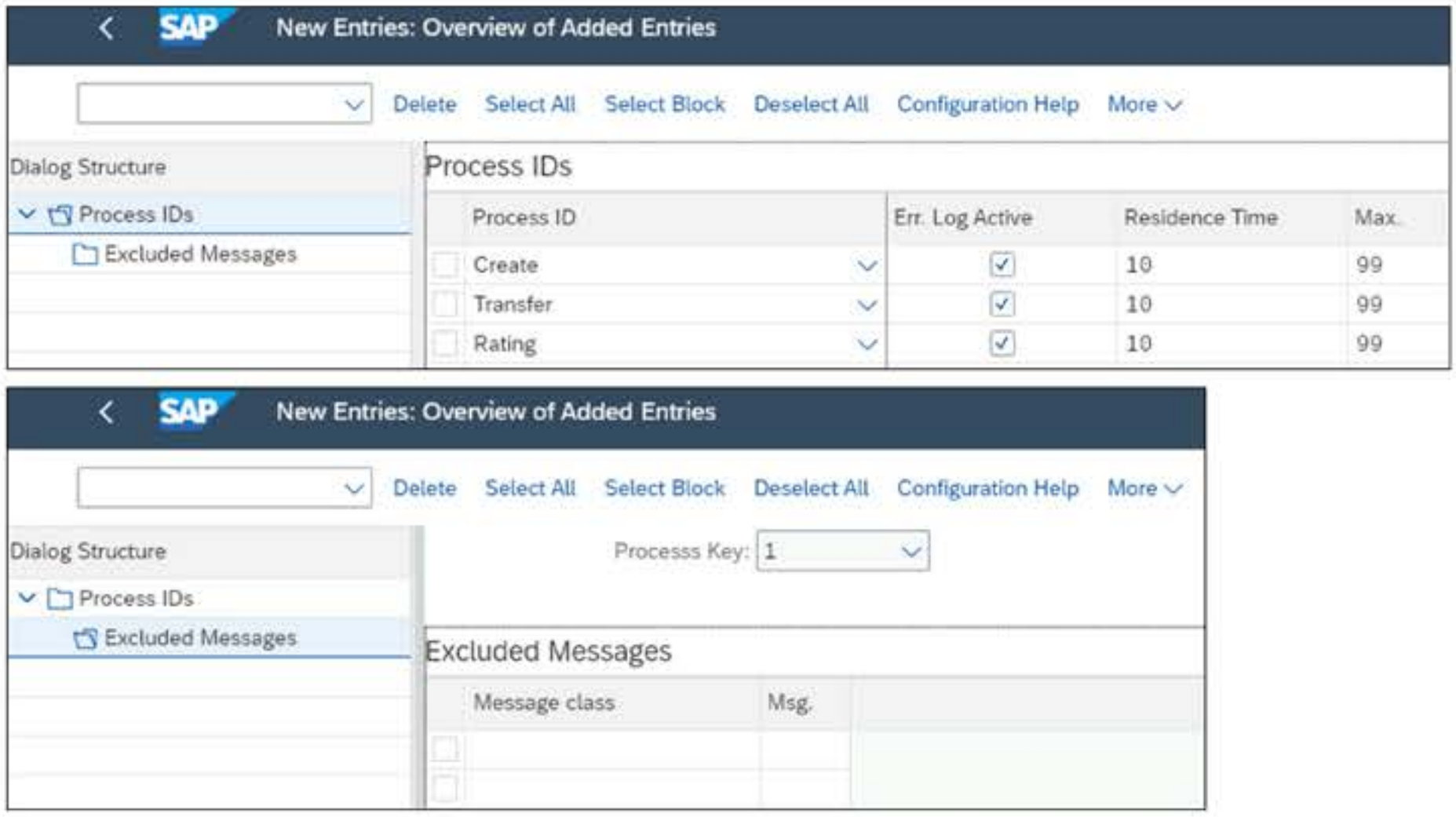

Figure 4.158 Configuration for Enabling Error Logs for Consumption Items

The configuration enables a choice of which error log needs to be captured along with the number of days the errors reside in the system and the maximum number of error messages per execution.

Error messages that aren't required to be captured can be excluded using the **Excluded Messages** subnode.

Display Error Logs - Billable Items

Similar to consumption items, billable items can encounter various errors during billable item processing. Errors can originate during the billable items creation, transfer from **Raw** to **Billable** status, and any other action performed on the billable item.

Error analysis and resolution have always been a needed and important step. Proper view of logs is always needed to analyze the errors. SAP has provided the Display Error Logs - Billable Items app, as shown in Figure 4.159 to view and analyze errors during various steps of billable items processing.

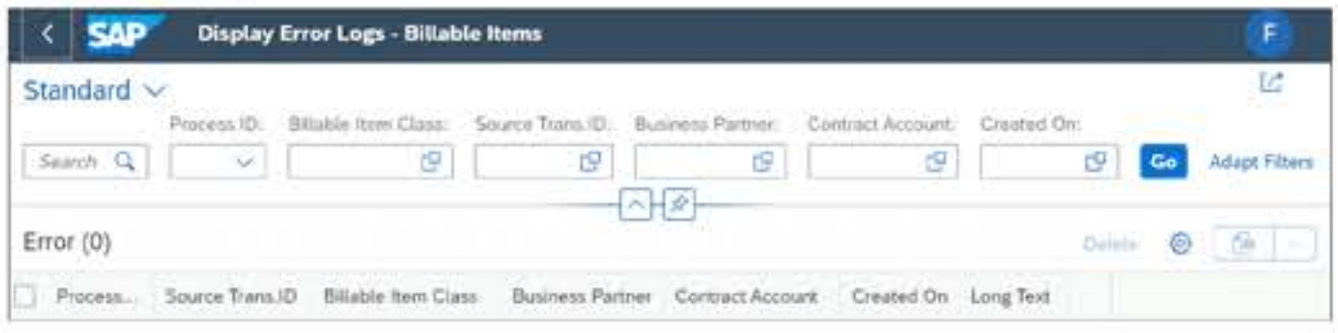

Figure 4.159 Error Log Display for Billable Items

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Using this app, you can perform the following activities:

- Display error logs during billable items creation.
- Transfer billable items from **Raw** to **Billable** status.
- Navigate to the application log.
- Delete the error log entry.

To enable the recording of error logs in the system for the billable items, configuration settings are needed. To do this, navigate to Transaction SPRO path **Contract Account Receivables & Payables • Convergent Invoicing • Error Logging • Activate Error Logging for Billable Items**, as shown in Figure 4.160.

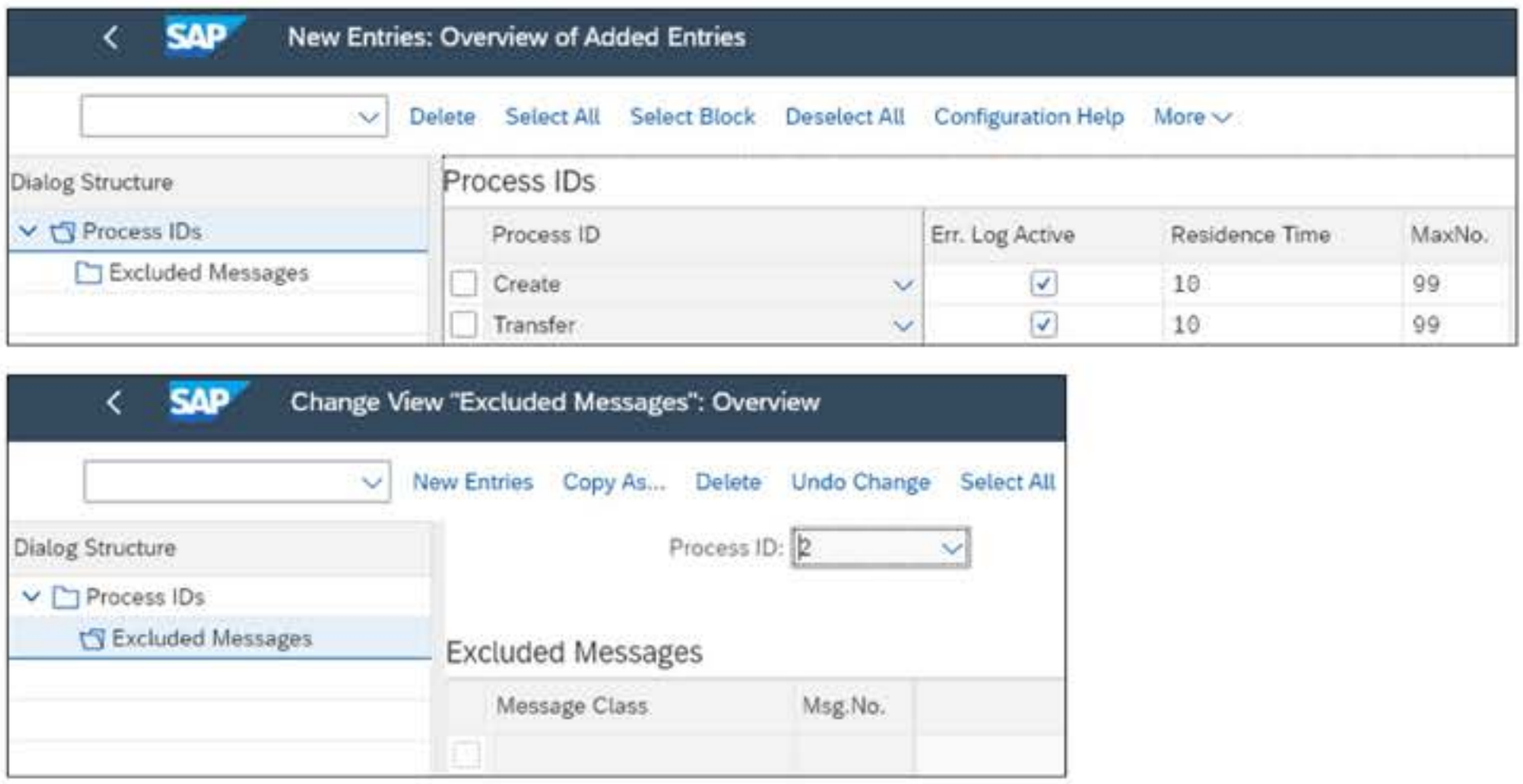

Figure 4.160 Configuration for Enabling the Error Log for Billable Items

The configuration enables a choice of which error log needs to be captured along with the number of days the errors reside in the system and the maximum number of error messages per execution. Error messages that aren't required to be captured can be excluded using the **Excluded Messages** subnode.

There are various other frequently used SAP Fiori apps in SAP Convergent Invoicing, which are discussed next.

Display Billable Items

Business transactions for sales, discounts, and charges are records in the SAP Convergent Invoicing system in the form of billable items. These billable items are then under processing for billing and invoicing.

The display and management of billable items can be performed under the same Display Billable Items app via the following operations, as shown in Figure 4.161 and Figure 4.162:

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- Search billable items using various filters available by status, master data, billable item types, and so on.
- Sort and format the output of search results.
- Display main items, payment items, tax items, and text items for billable items.
- Reverse billable items.
- Except billable Items.
- Transfer billable items from **Raw** to **Billable** status.
- Navigate from billable to consumption items.
- Perform billing and invoicing.
- Navigate to billing and invoicing documents.

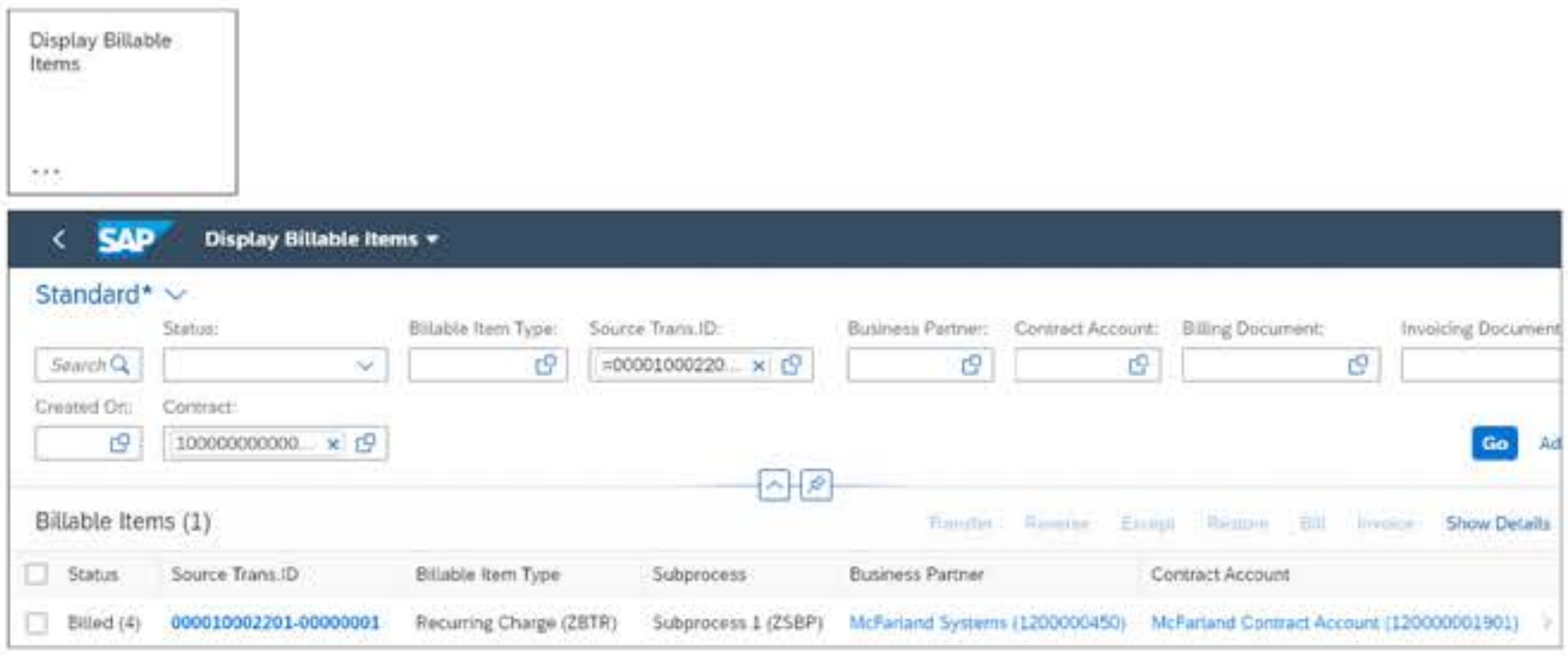

Figure 4.161 Display Billable Items View

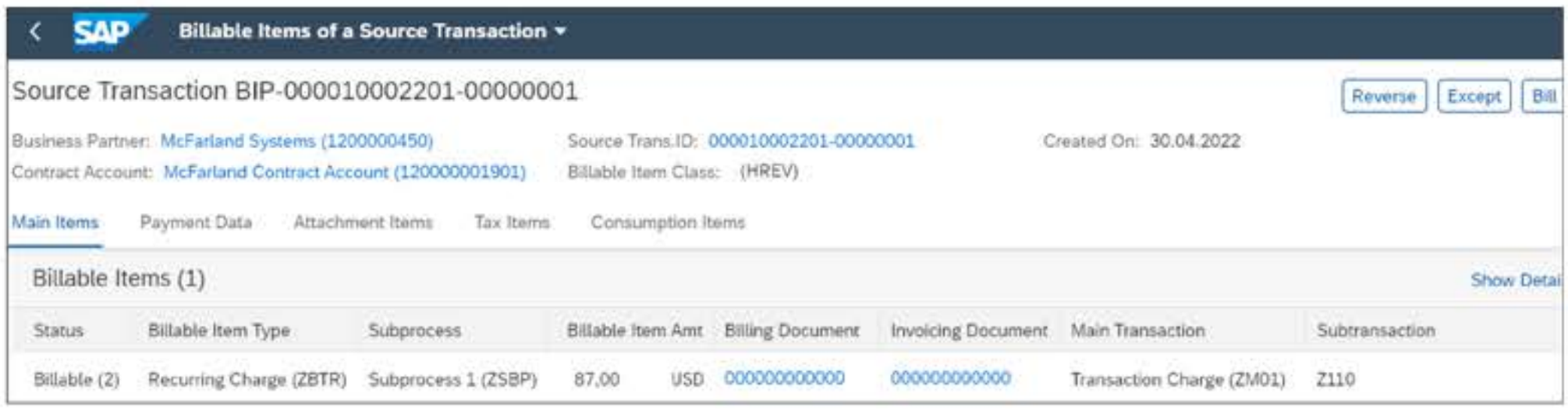

Figure 4.162 Display Billable Item: Detail View

Display Billing Document

Billing is the process that selects, groups, and aggregates billable items to create billing documents. The billing documents created can also be viewed using the Display Billing Documents app. The various sections of billing documents are displayed, and user-defined layouts can be maintained as needed.

Following are the various activities available via the Display Billing Documents app, as shown in Figure 4.163 and Figure 4.164:

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- Search billing documents using various fields for master data, billing dates, invoicing, and billing documents.
- Display reversed/non-reversed billing documents.
- Display a list of billing documents.
- Navigate to billable items and associated consumption items.
- View associated invoicing documents.
- Reverse billing documents.
- Set an invoicing lock.
- View attachments to a billing document.
- Invoice billing documents.

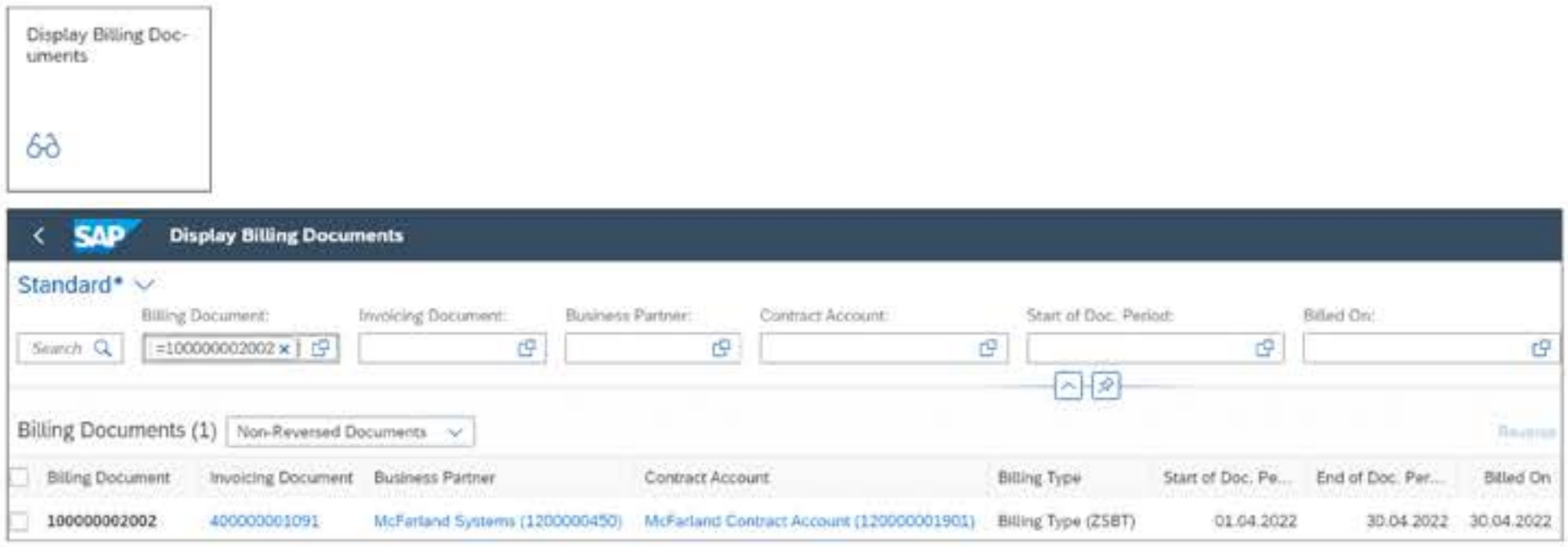

Figure 4.163 Display Billing Documents App

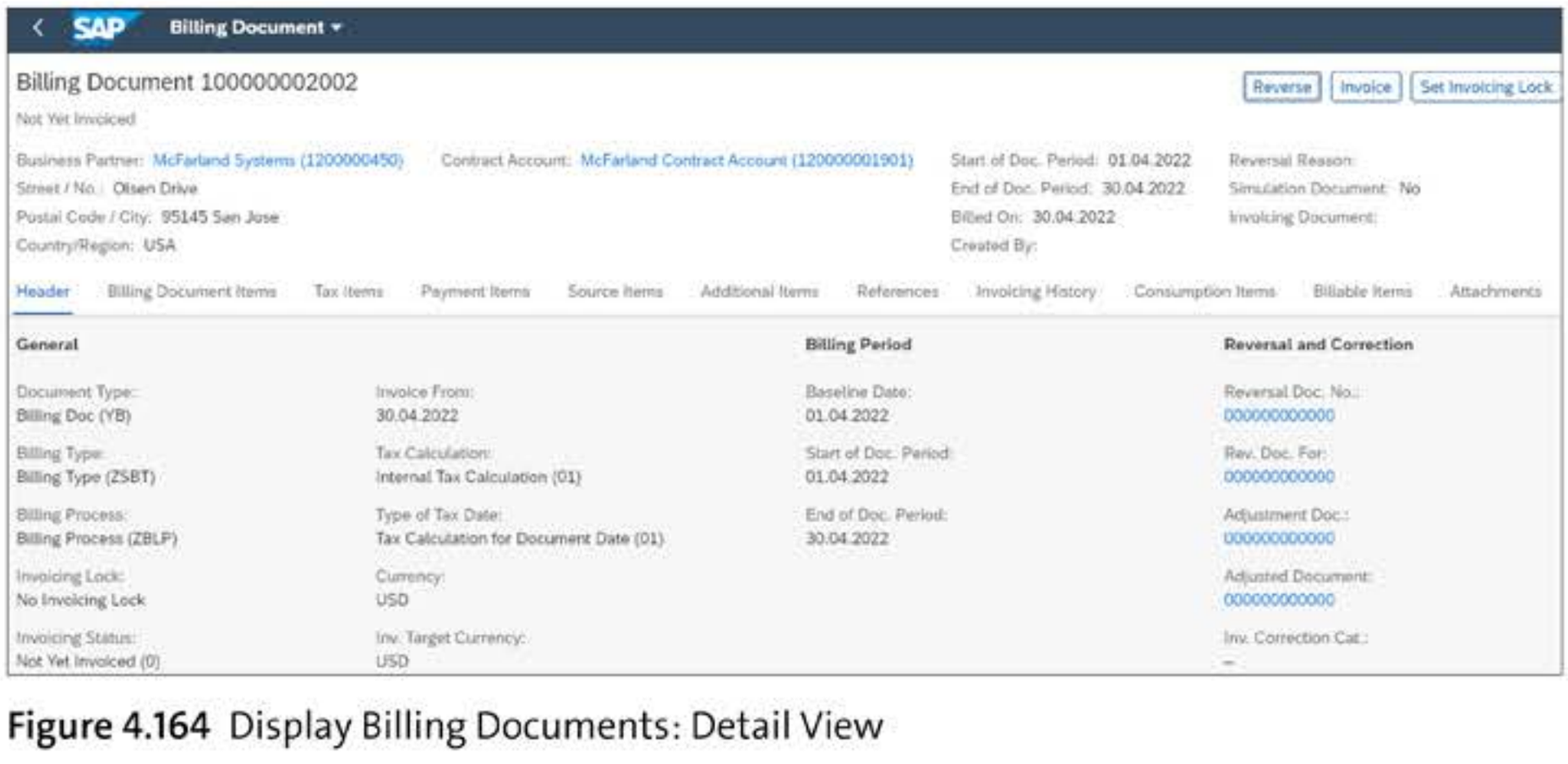

Figure 4.164 Display Billing Documents: Detail View

Display Invoicing Documents

Invoice documents contain the information that is sent to the customer about the sales, tax, and discounts for the purchase or service used by the customer. Invoices are the legal documents against which customers make a payment. Invoicing documents

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in SAP Convergent Invoicing can also be viewed using the Display Invoicing Documents app. Following are various activities available via the Display Invoicing Documents app, as shown in Figure 4.165 and Figure 4.166

- Search invoicing documents using various search fields.
- Display invoicing documents, preliminary invoices, reversed invoices, and non-reversed invoices.
- Navigate to billing document, billable items, and consumption items.
- Navigate to contract accounting posting documents.
- View invoicing history.
- View attachments to invoicing documents.
- Print and print preview invoicing documents.
- Reverse invoicing documents.
- Create credit/debit memos.
- Set print lock.

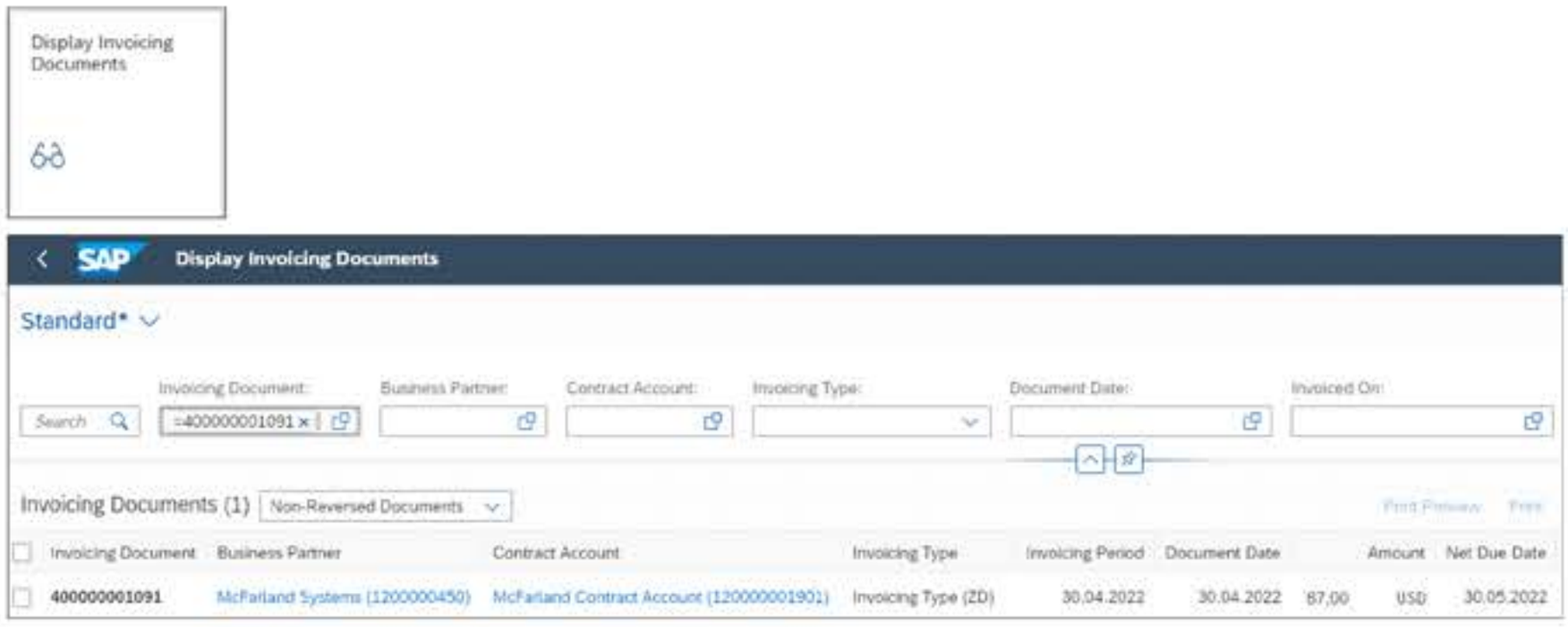

Figure 4.165 Display Invoicing Document App

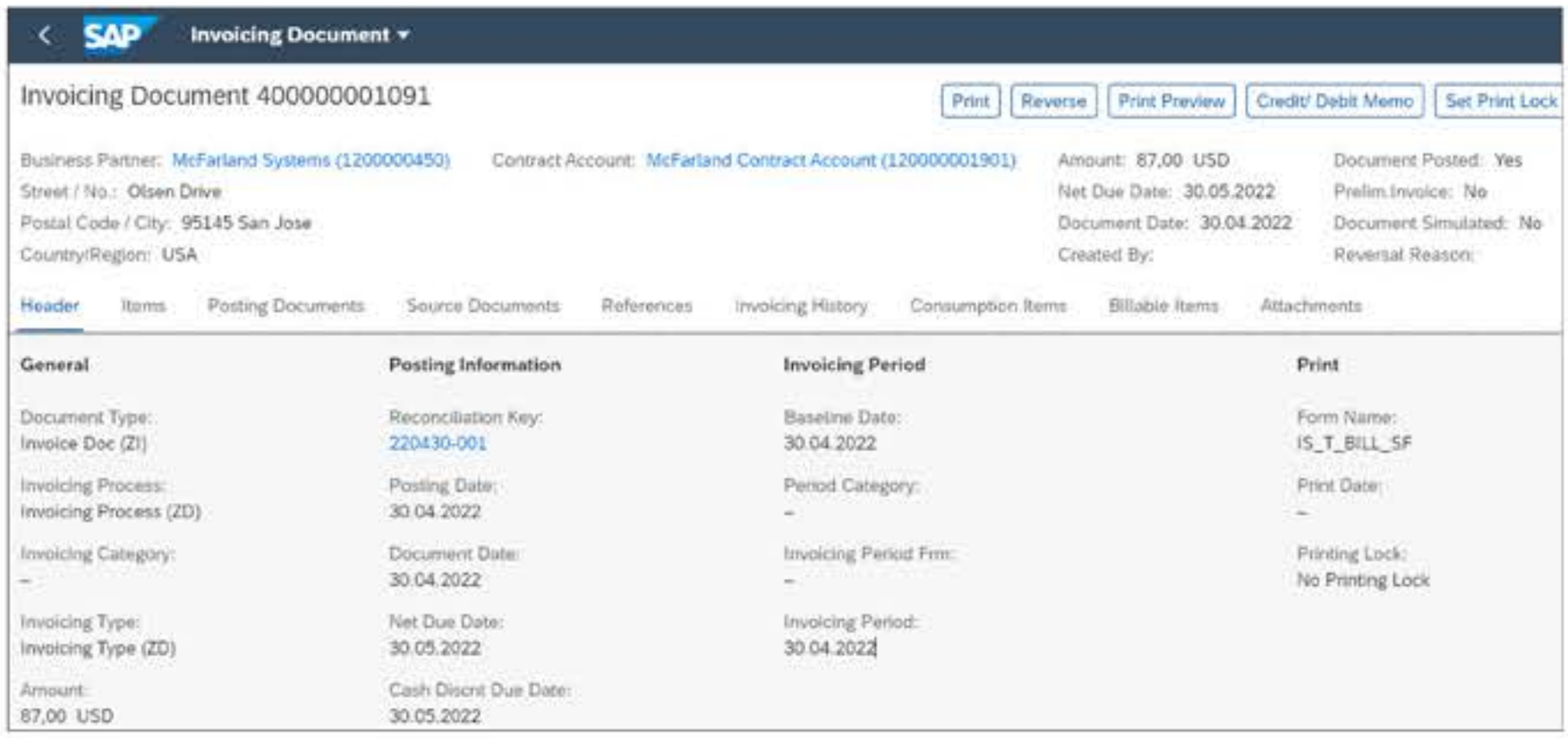

Figure 4.166 Detail View of Invoicing Document App

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4.5 Intercompany Settlement Process

With SAP S/4HANA 1909, the intercompany settlement process has been enabled in SAP Convergent Invoicing. The intercompany process involves a business transaction between two companies within the same enterprise or the same corporate group. Company A receives an order and invoices the customer, the customer makes a payment to company code A, but the fulfillment of the services is done by another Company B. Company A receives an intercompany invoice from Company B and settles the charges owed to Company B. This process can be summarized as follows (see Figure 4.167):

1. Order placed against: Company A
2. Customer invoiced from: Company A
3. Customer pays: Company A (receiving company code)
4. Services delivered from: Company B (supplying company)
5. Intercompany invoice: Company B invoices Company A

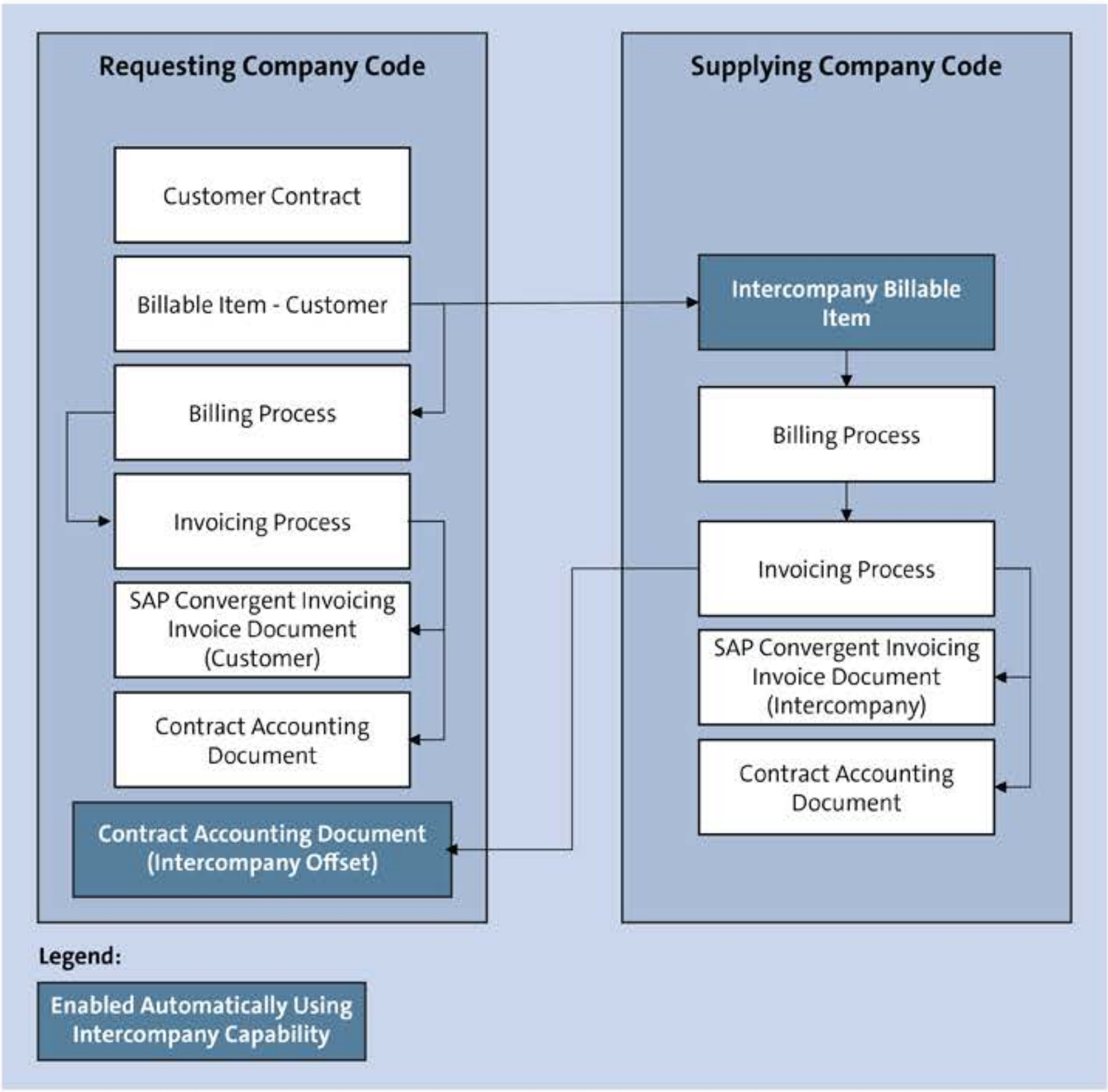

Figure 4.167 Intercompany Settlement Process: Process Flow

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SAP Convergent Invoicing provides a standard capability of generating an intercompany invoice when the transaction is identified as intercompany relevant. A debit memo request is created in the supplying company code, while a payable invoice is created in the receiving company code. As part of this section, we'll take a look at the intercompany sales process, as illustrated in Figure 4.167, and the standard capabilities that the system provides, along with the necessary configuration needed.

Note that the intercompany process isn't demonstrated for the Martex use case. The intercompany settlement process can be enabled in a synchronous and asynchronous setting. Synchronous settlement is enabled using the invoicing process and will be our focus for the demonstration. Asynchronous settlement is enabled using billable items.

4.5.1 Company Code Group Assignment

To use the intercompany settlement capability, both the receiving company code (1100) and the supplying company code (3000) should be assigned to the same **Company Code Group (1100)**, as shown in Figure 4.168, within Transaction SPRO path **Contract Account Receivables & Payables • Organizational Units • Assign Company Codes to Company Code Groups**.

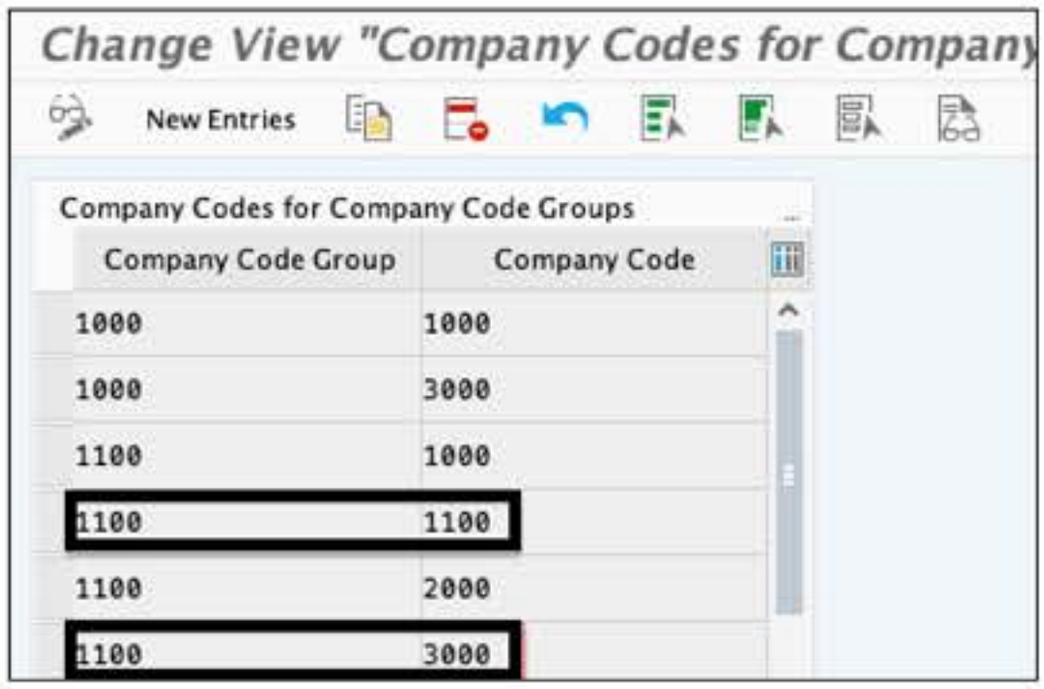

Figure 4.168 Intercompany Setup Requiring Company Code Group Assignment

4.5.2 Intercompany Settlement Types

Once the company code group assignment is complete, the next step is to enable the intercompany settlement configuration. To do this, navigate to Transaction SPRO path **Contract Account Receivables & Payables • Convergent Invoicing • Intercompany Settlement**.

First, **Intercompany Settlement Type** has to be configured. It defines the category or the classification of the billable item in reference to the intercompany settlement posting. The settlement processing type (**IntSettlmntProc**), whether it's **Synchronous Offsetting Entry** (using the invoicing process) or **Asynchronous Offsetting Entry** (using the billable

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item process), is defined as shown in Figure 4.169. You can also define the billable item **Class**, **Subprocess**, and **BIT Type** identifying the requesting company code and the billable item type referencing the supplying company code. There is an option (**Invoicing In**) to also define if the invoicing is expected to be in the supplying company code or the requesting company code.

Figure 4.169 Definition of Intercompany Settlement Type

4.5.3 Settlement Type Determination

Once the intercompany settlement type is defined, it has to be assigned to the requesting company code (**ReqCCode**) and supplying company code (**Supplying CCode**) combination, as shown in Figure 4.170.

Figure 4.170 Intercompany Settlement Type Determination

4.5.4 Master Data Determination

Once the intercompany settlement types are defined, the requesting and supplying company code master data combination needs to be configured to determine the actual and valid master data for the automated intercompany settlement postings, as shown in Figure 4.171.

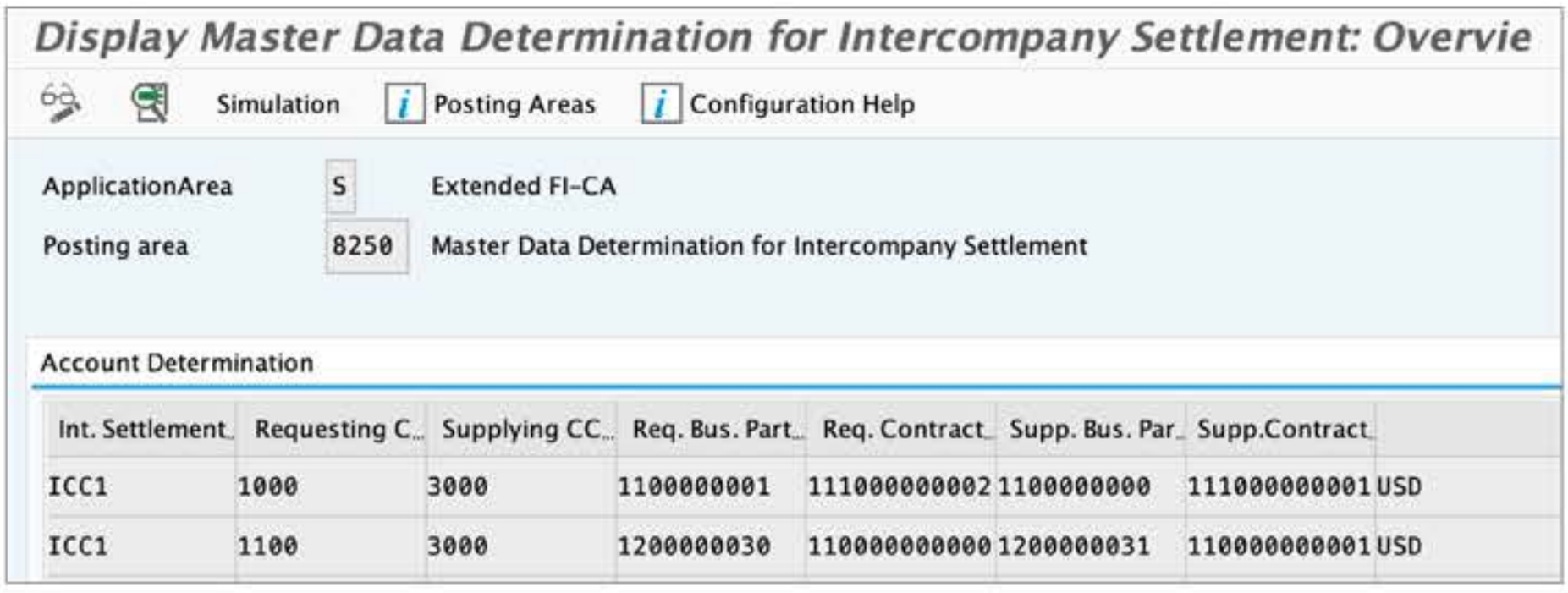

Figure 4.171 Intercompany Settlement: Master Data Determination

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4.5.5 Account Determination and Document Type Settings

The following configurations are related to main transactions and sub-transactions determination for the intercompany postings (see Figure 4.172) and offsetting postings (see Figure 4.173).

Display Trans. Determination for Clearing Within a Corporate Group: Ov

Simulation | Posting Areas | Configuration Help

ApplicationArea S Extended FI-CA

Posting area 8251 Trans. Determination for Clearing Within a Corporate Group

Account Determination

Int. Settlement..	Billable Item T..	Requesting C..	Supplying CC..	Debit Main Tra..	Debit Subtran..	Credit Main Tr..	Credit Subtra..
ICC1	ZBT2	1000	3000	INTM	INDR	INTM	INCR
ICC1	ZBT2	1100	3000	INTM	INDR	INTM	INCR
ICC1	ZBT2	3000	1000	INTM	INDR	INTM	INCR
ICC1	ZBT2	3000	1100	INTM	INDR	INTM	INCR

Figure 4.172 Intercompany Settlement: Main Transactions and Sub-transactions

Display Trans. Determination f. Clearing Within a Corp.Grp.OffstgItm:

Simulation | Posting Areas | Configuration Help

ApplicationArea S Extended FI-CA

Posting area 8252 Trans. Determination f. Clearing Within a Corp.Grp.OffstgItm

Account Determination

Int. Settlement..	Requesting C..	Supplying CC..	Main Transact..	Subtransaction	MTrans. (Offs..	Subtrans.(Off..
ICC1	1000	3000	INTM	INCR	INTM	INCR
ICC1	1000	3000	INTM	INDR	INTM	INDR
ICC1	1100	3000	INTM	INCR	INTM	INCR
ICC1	1100	3000	INTM	INDR	INTM	INDR
ICC1	3000	1100	INTM	INCR	INTM	INCR
ICC1	3000	1100	INTM	INDR	INTM	INDR

Figure 4.173 Intercompany Settlement: Main Transactions and Sub-transactions for Offsetting Entries

As with any new functionality, SAP provides invoicing functions that need to be activated and assigned to the billable item classes. For the intercompany settlement functionality, the interface components are **ICO_DATA** and **INV_OFFSET**, as shown in Figure

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4.174. The associated invoice document type in contract accounting also needs to carry the **Cross-Company** posting flag, as shown in Figure 4.175.

Figure 4.174 Intercompany Settlement: Invoicing Functions

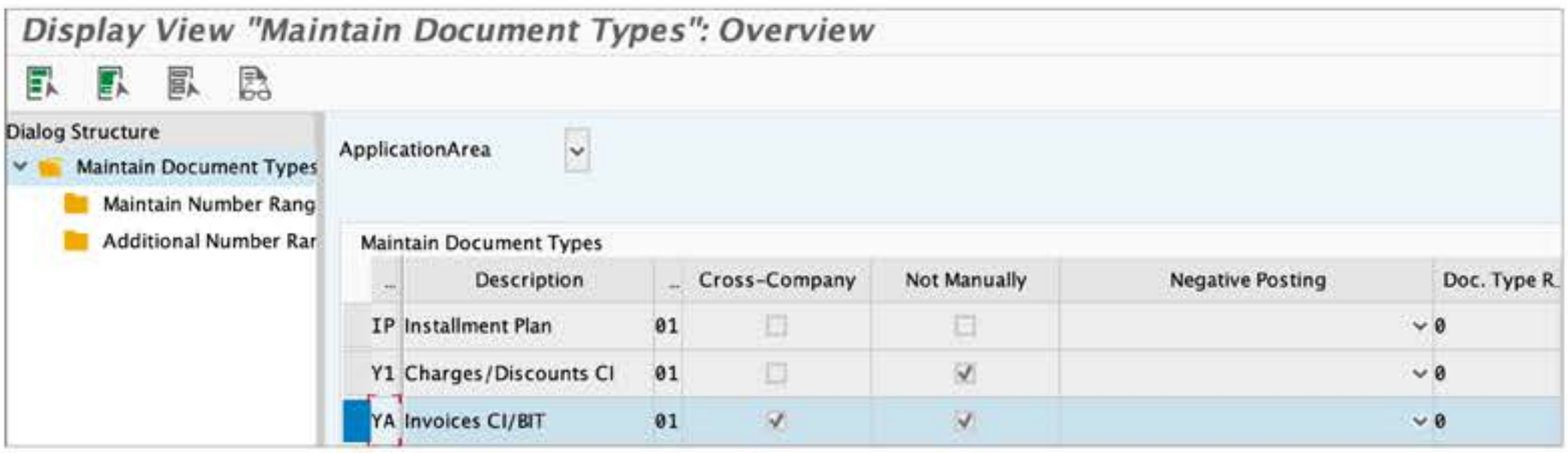

Figure 4.175 Document Type Settings for Intercompany Postings

4.5.6 Intercompany Billable Items and Document Postings

Once the interface components are assigned to the billable item class, intercompany-relevant fields get enabled in the billable item interface. As shown in Figure 4.176, source transaction **ICOSETTLE_CS** represents the billable item for sale transaction.

The intercompany settlement billable item within **CoCode 3000** with source transaction **22042810000000001910001** was generated automatically based on the intercompany settlement configuration. Invoice and posting documents are listed in Table 4.6.

Figure 4.176 Billable Items for the End Customer Sale and Intercompany Settlement

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Billable Item Type	Source Transaction	SAP Convergent Invoicing Invoice	Contract Accounting Document
Sale Posting	ICOSETTLE_CS	400000001089	10000001238
Intercompany (supplying company code)	220428100000000191 0001	400000001090	10000001239 (supplying comany code)
			10000001240 (requesting com-pany code)

Table 4.6 Invoice and Posting Document (Contract Accounting) Details for Requesting and Supplying Company Codes

Figure 4.177 indicates the accounting posting for the sale transaction. In addition, as shown in Figure 4.178, multiple accounting documents (source company code and requesting company code) were posted upon invoicing the intercompany billable Item.

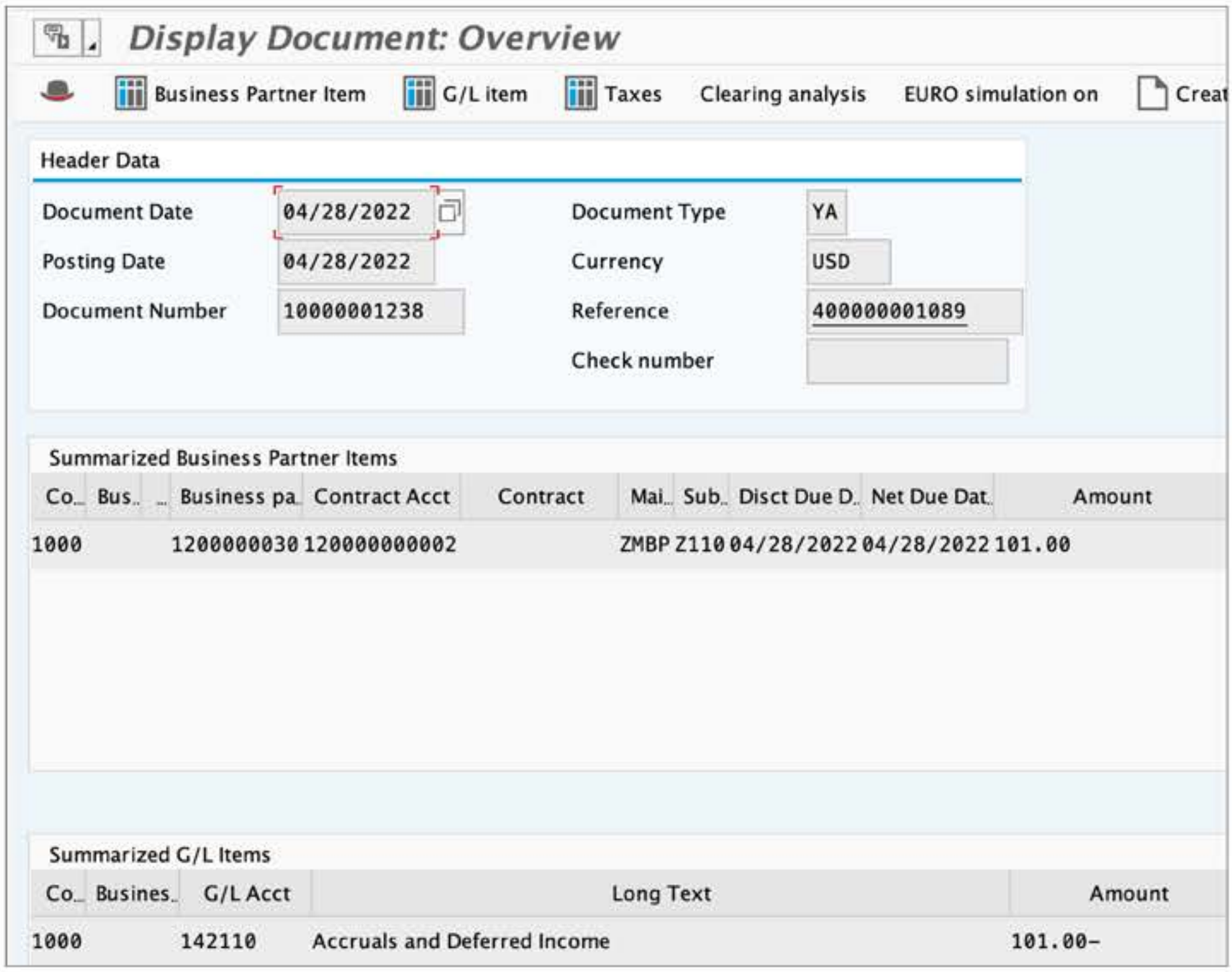

Figure 4.177 Accounting Document for the Sales Transaction

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Display Invoicing Document

Preview | Simulate | Archive | Reverse Document | Create Invoice | Credit Memo or Debit Memo

Header Data for Invoicing Document 400000001090

Invoic. Doc.	BPartner	Contract Acct	Process	Inv.Type	Posting Date	Author	Invoiced On	Invoiced At	Amount	Crcy	Invoicg Run No.
400000001090	1200000030	110000000000	ZI	ZI	04/28/2022	CSANJIVI	04/28/2022	06:15:06	101.00	USD	400000001089

Items | Posting Documents | Source Documents | Billable Items | References

Invoicing Document 400000001090

SeqNo.	RefDT	Doc. No.	Rep	Itm	New Doc.	Name of Reference Document Type	DT	Posting Date	Revers.For	Rev. Doc.
1	0ICO	10000001239			☑	Posting Intercompany Settlement	YA	04/28/2022		
2	0ICS	10000001240			☑	Posting Intercompany Settlement (Offsetting Doc)	YA	04/28/2022		

Figure 4.178 Settlement Invoice with Multiple Accounting Documents

Finally, accounting documents for the supplying company code (see Figure 4.179) and the requesting company code (see Figure 4.180) indicate the resultant accounting postings from the automated intercompany settlement process.

Display Document: Overview

Business Partner Item | G/L item | Taxes | Clearing analysis | EURO simulation on | Crea

Header Data

Document Date	04/28/2022	Document Type	YA
Posting Date	04/28/2022	Currency	USD
Document Number	10000001239	Reference	400000001090
		Check number	

Summarized Business Partner Items

Co..	Bus..	..	Business pa..	Contract Acct	Contract	Mai..	Sub..	Disct Due D..	Net Due Dat..	Amount
3000			1200000030	110000000000		INTM	INDR	04/28/2022	04/28/2022	101.00

Summarized G/L Items

Co..	Busines..	G/L Acct	Long Text	Amount
3000		410000	Sales Revenue	101.00-

Figure 4.179 Settlement Accounting Document in the Supplying Company Code

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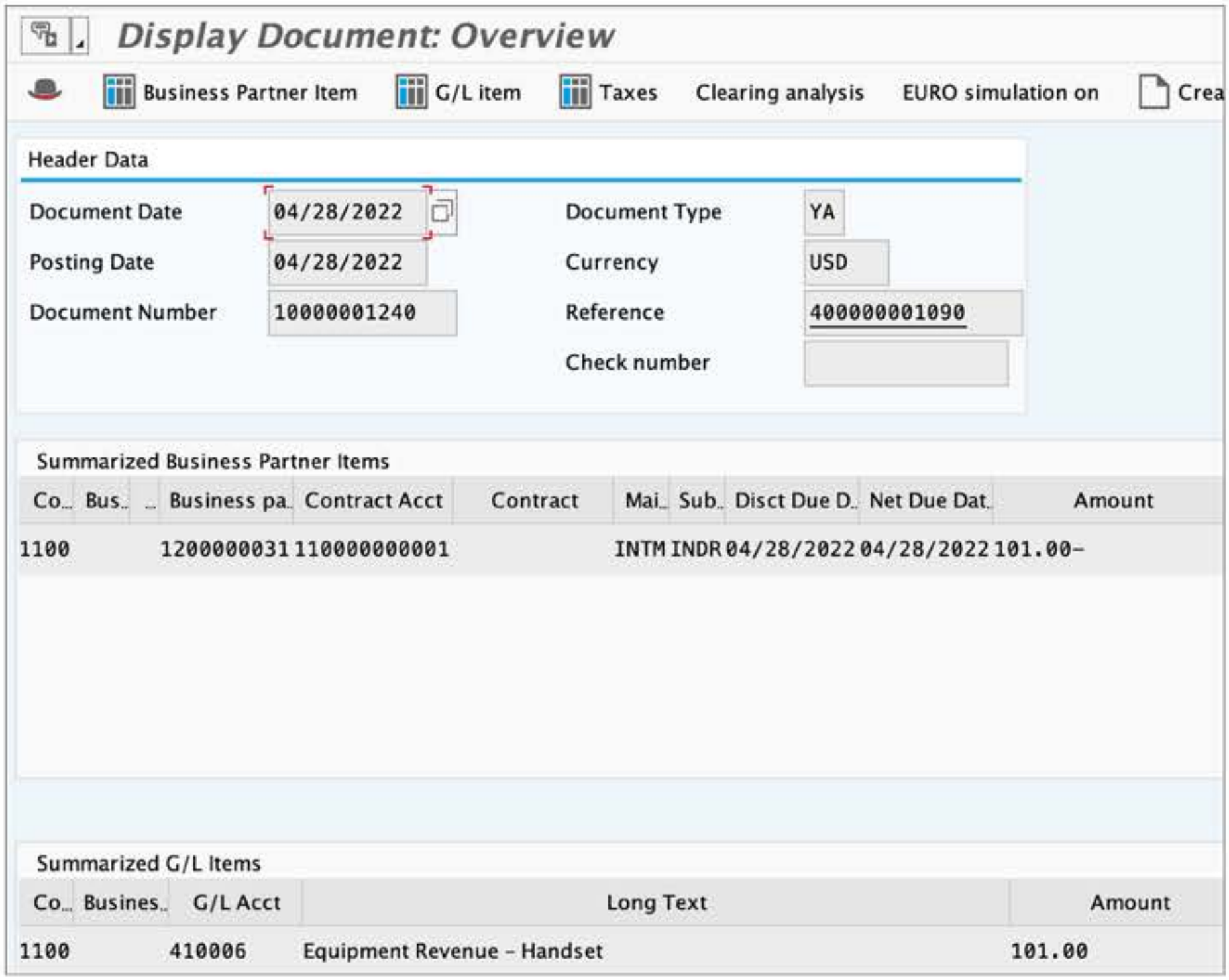

Figure 4.180 Settlement Accounting Document in the Receiving Company Code

4.6 Reporting and Analytics

Now we'll explore the reporting and analytics functionality with SAP Convergent Invoicing, from viewing and processing billable items and consumption items to document analysis.

4.6.1 All Billable Items View/Processing

Billable items created in the system can be viewed using Transaction FKKBIXBIT_MON or using the SAP Easy Access menu path **Accounting • Financial Accounting • Contract Account Receivables & Payables • Convergent Invoicing • Billable Items** (see Figure 4.181).

This transaction provides a view of billable items based on their statuses, among other selection parameters. The **ALL STATUSES** status is used to display billable items of all statuses on a single screen (see Figure 4.182).

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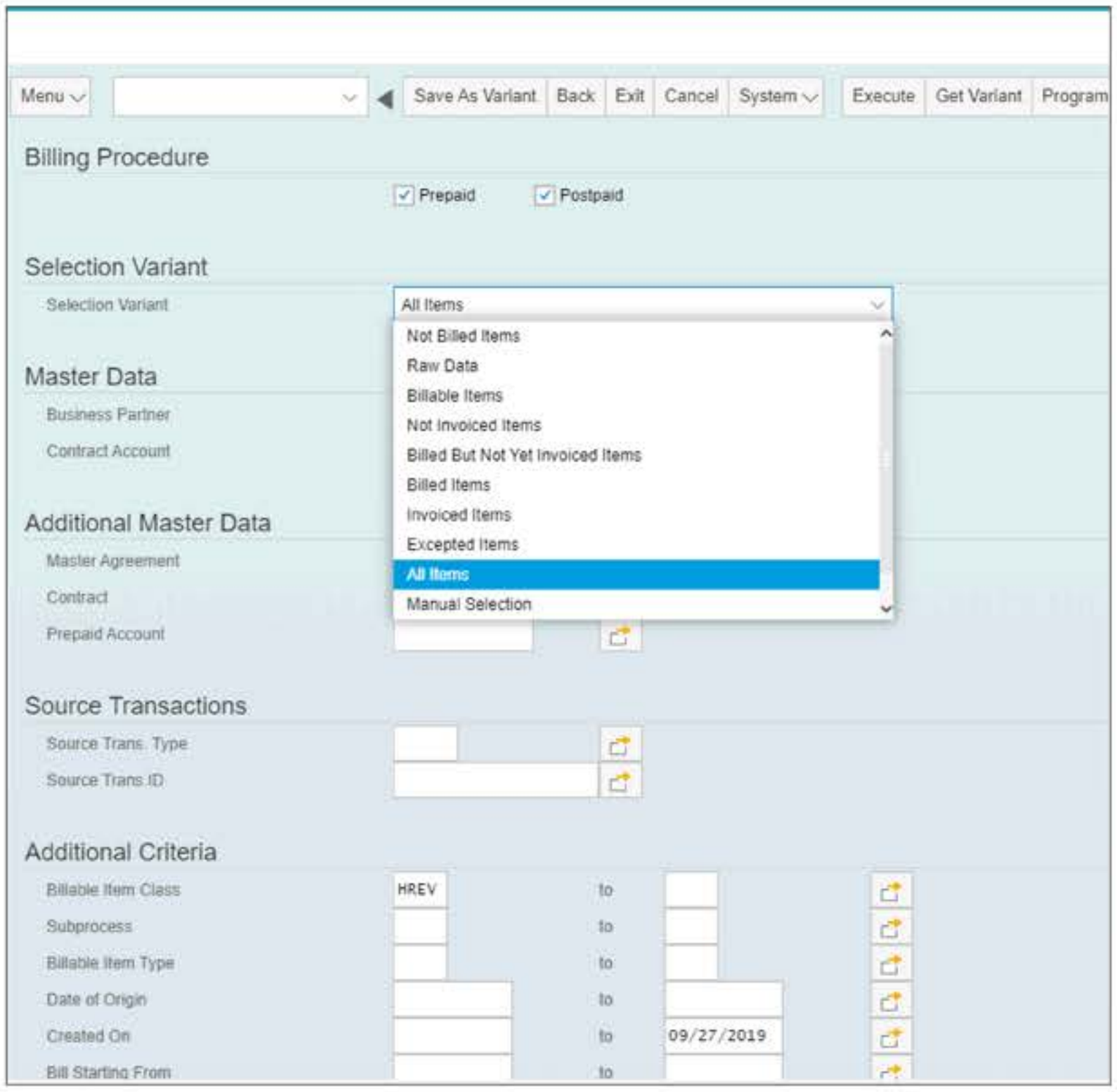

Figure 4.181 Billable Item Display Selection Screen

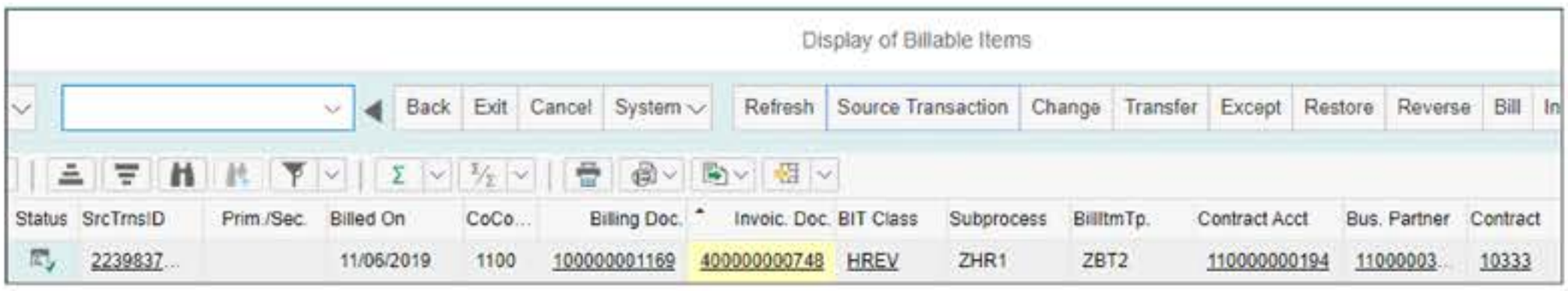

Figure 4.182 Billable Item Display

The options available on the screen for processing billable items are as follows:

- **Change**
  Used for modifying the fields of the billable items.
- **Transfer**
  Used for transferring **Raw** status billable items to **Billable** status.
- **Except**
  Used for changing the status of billable items from **Billable** to **Billable-Excepted** to exclude them from the billing process. Excepting billable items requires an exception reason.

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- **Restore**
  Used for restoring a **Billable-Excepted** status billable item to **Billable** status to be considered for billing. Restoration of billable items requires a restoration reason.
- **Reverse**
  Used for reversing the billable item moving to **Excepted** status and reversing the rating of associated consumption items, changing them to either **Unrated** or **Excepted** status based on exception reason settings.
- **Bill**
  Billable items in status **Billable** are considered for billing. The billing process can be executed on the selected billable items.
- **Invoice**
  Billed billable items can be processed in invoicing using this option.
- **Clarification Case**
  Billing documents and even invoice documents of the billable items can be put in **Clarification Case** status for further processing based on clarification rules regarding amount or quantity.

4.6.2 Consumption Items View/Processing

Consumption items in the system can be viewed using Transaction FKKBIXCIT_MON, which also can be accessed under SAP Easy Access menu path **Accounting • Financial Accounting • Contract Account Receivables & Payables • Convergent Invoicing • Consumption Items** (see Figure 4.183).

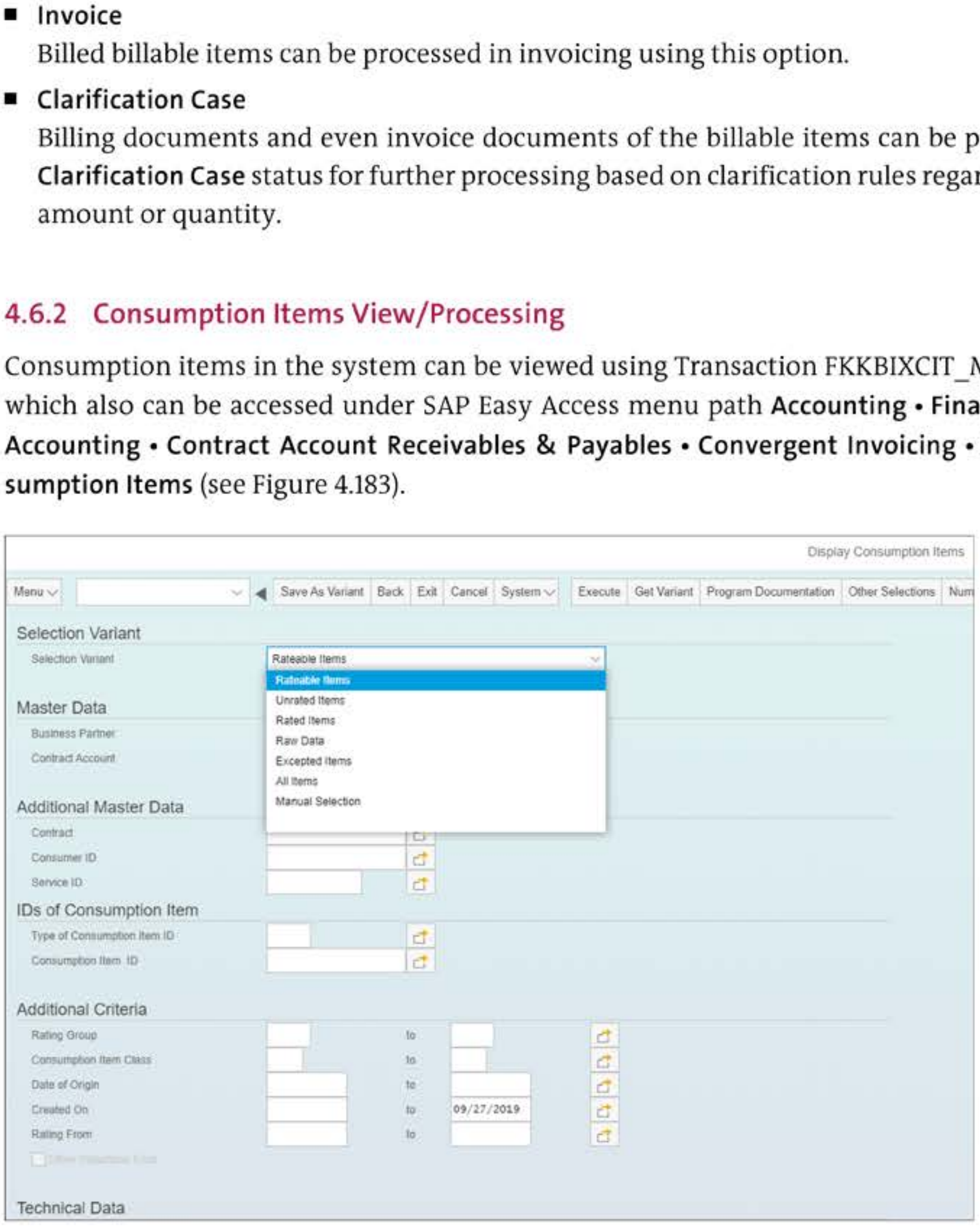

Figure 4.183 Consumption Item Display Selection Screen

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The transaction provides the option to view the consumption items sorted by status or all together. This is controlled by the selection variant provided on the screen as one of the selection parameters (see Figure 4.184).

Figure 4.184 Consumption Item Display Screen

Several processing options are available on the display screen that can be performed on consumption items based on requirements:

- **Perform Rating**
  The rating process is described in detail in Section 4.3.3. The unrated consumption items can be sent to the SAP Convergent Charging system for rating through this option.
- **Change**
  The consumption item fields can be modified using this option. The consumption item fields that are made modifiable in the configuration can be modified. The configuration was shown earlier in Figure 4.125.
- **Reverse**
  The consumption items that aren't to be used for rating or are created in error can be reversed using this option. The reversing will require a reversal reason.
- **Except**
  Erroneous unrated consumption items can be moved to an **Excepted** status using this option by providing an exception reason.
- **Restore**
  The excepted unrated consumption items can be moved to an unrated status using this option. Erroneous consumption items that are modified can be restored.

4.6.3 Raw Data View of Billable Items

Billable items of **Raw** status can be viewed using Transaction FKKBIXBIT_MON by selecting **Raw Data** as the **Selection Variant** (see Figure 4.185).

In addition, Transaction FKKBIXBITR_MON can be used directly to view **Raw** and **Raw Excepted** status billable items (see Figure 4.186).

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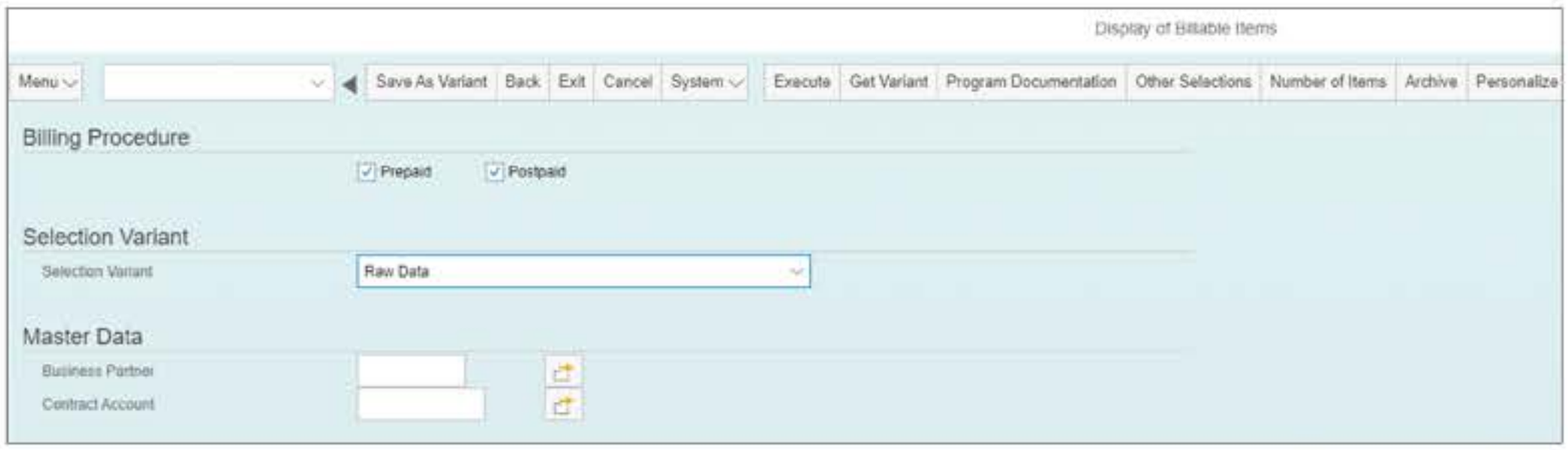

Figure 4.185 Raw Data Selection for Billable Items

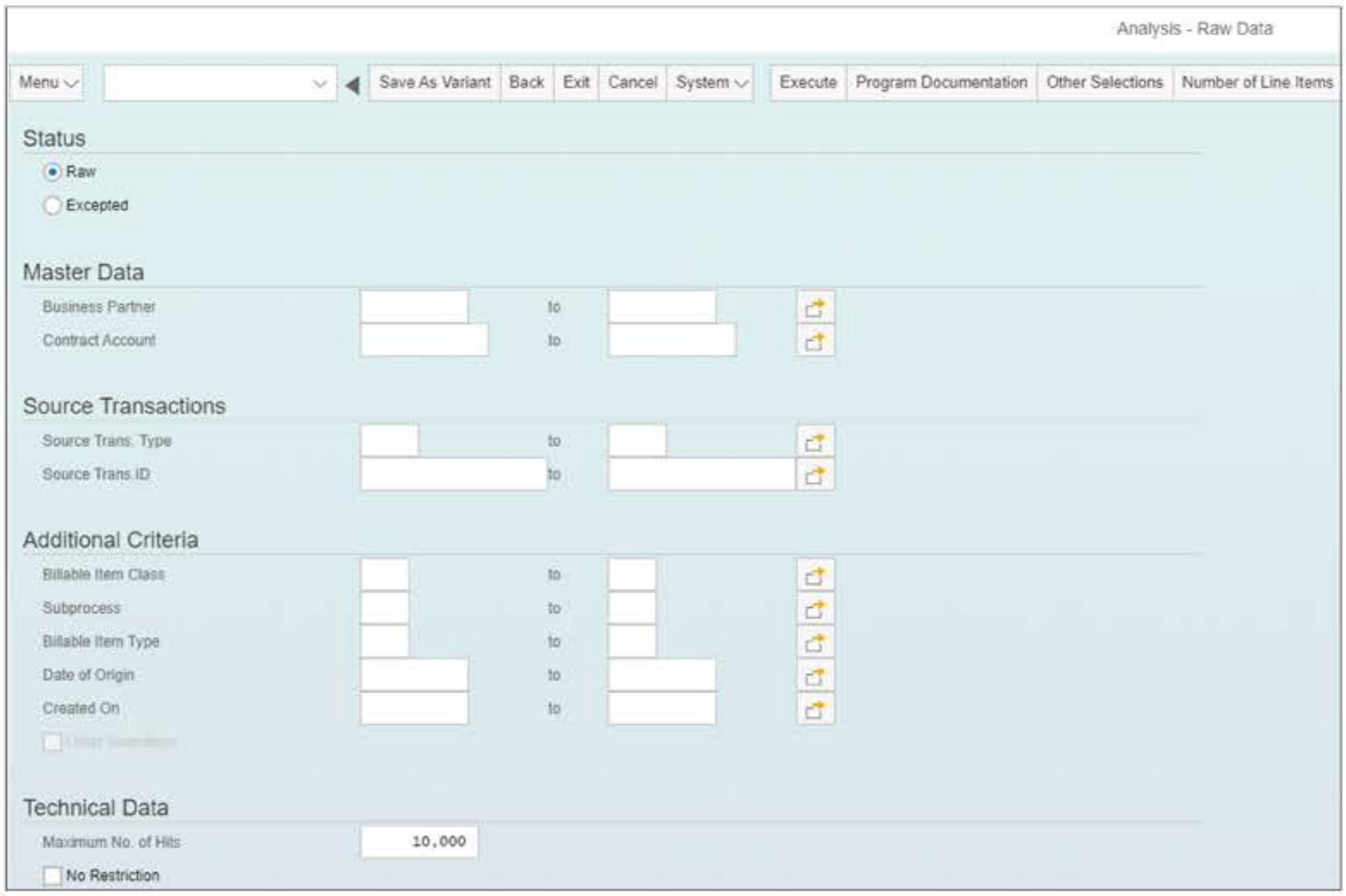

Figure 4.186 Raw Data Billable Items Analysis

4.6.4 Display Invoicing Orders

Billing documents that are eligible for invoicing or for which invoicing orders are present in table DFKKINV_TRIG can be viewed using Transaction FKKINV_MON, which also can be accessed under SAP Easy Access menu path **Accounting • Financial Accounting • Contract Account Receivables & Payables • Convergent Invoicing • Invoicing Documents** (see Figure 4.187).

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The billing documents that originate from different processes or billing systems are differentiated by the source document category. To view the invoicing order, the source document category is provided as the selection criterion, among other selection parameters (see Figure 4.188).

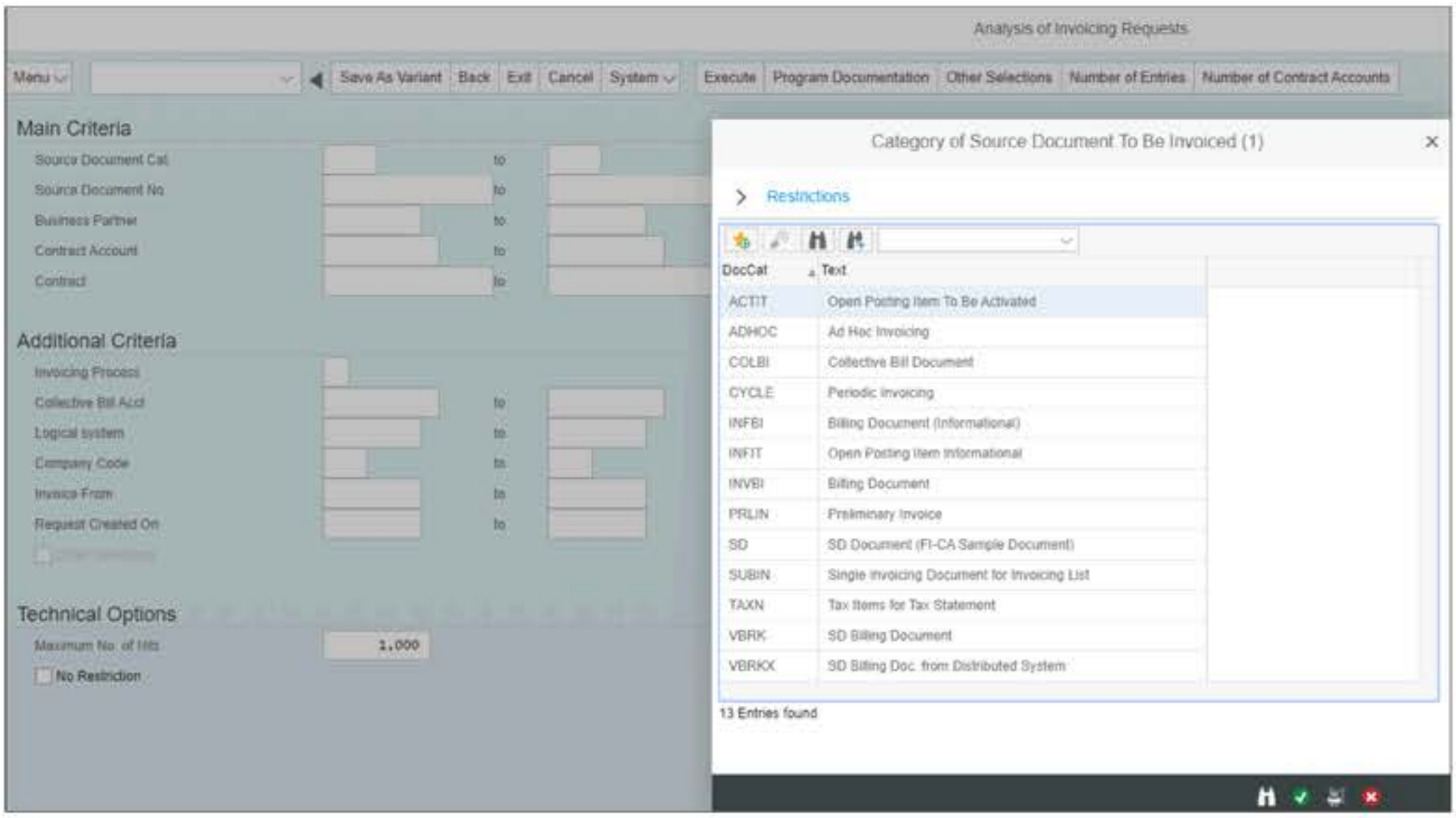

Figure 4.187 Invoice Order Monitoring Screen

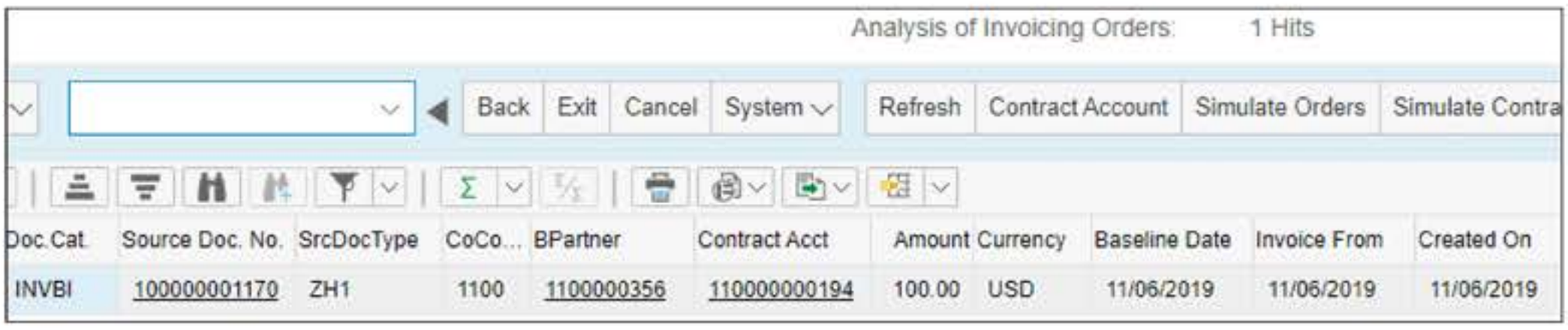

Figure 4.188 Invoicing Order Display Screen

4.6.5 Display Accrual/Deferral Items

With revenue recognition in the standard, the revenue posting takes place at the time of invoicing itself. Revenue accrual/deferral in SAP Convergent Invoicing is related to the revenue recognition of the billable items that aren't invoiced. For a class to have billable items for the accrual or deferral, the interface components for accrual/deferral should be selected, and during the creation/transfer of billable items, the accrual/deferral posting type should be assigned.

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The accrual/deferral of billable items created in SAP Convergent Invoicing can be viewed using Transaction FKKBIX_REVREC_MON (see Figure 4.189).

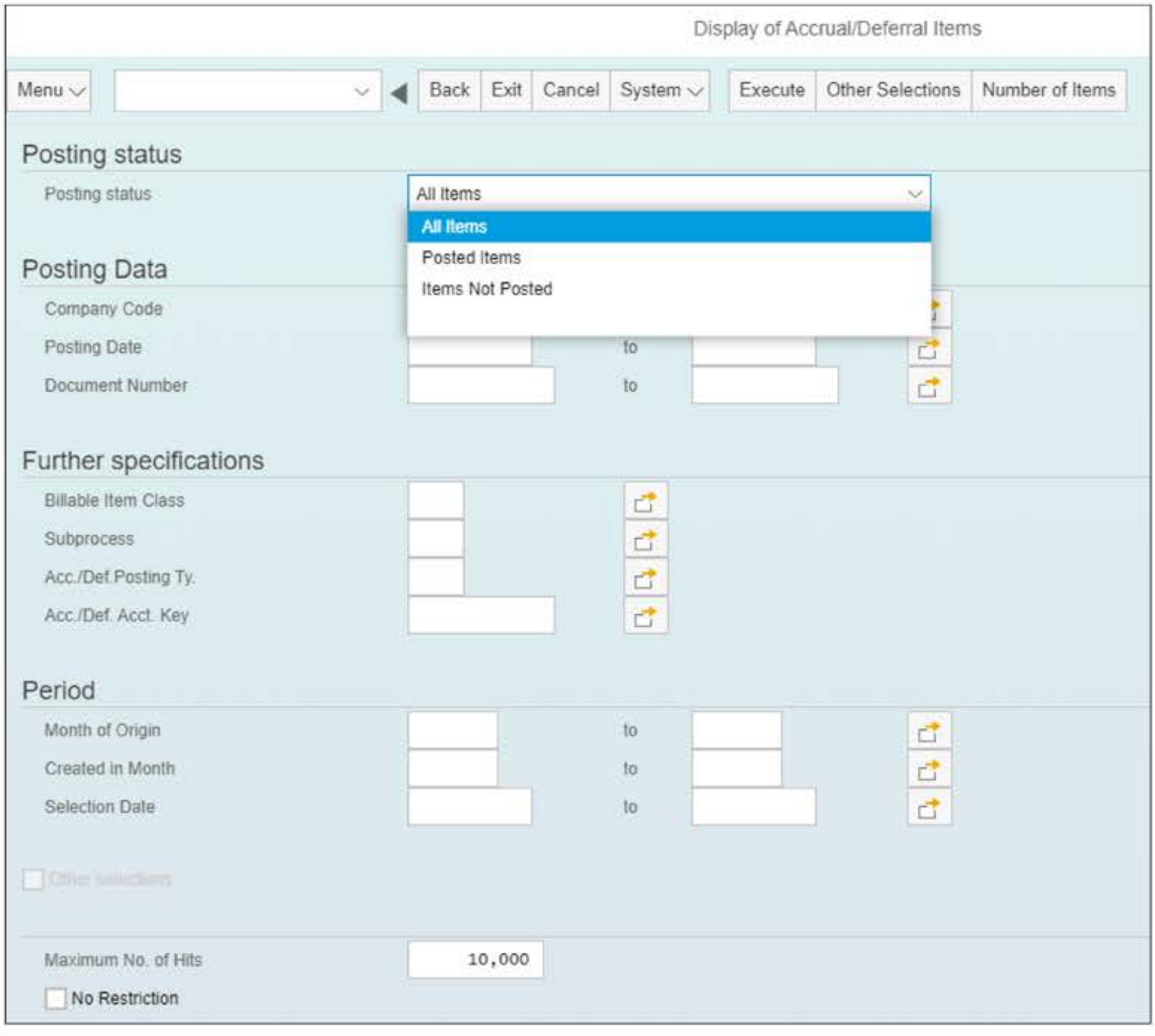

Figure 4.189 Accrual/Deferral Display Screen

4.6.6 Billable Items View/Processing

The billable items can be viewed using Transaction FKKBIXBIT_MON by selecting the **Billable Items** variant. The view and processing details are described in Section 4.6.1. In addition, the billable items also can be viewed using Transaction FKKBIXBITB_MON, which also can be accessed under SAP Easy Access menu path **Accounting • Financial Accounting • Contract Account Receivables & Payables • Convergent Invoicing • Billing Documents** (see Figure 4.190).

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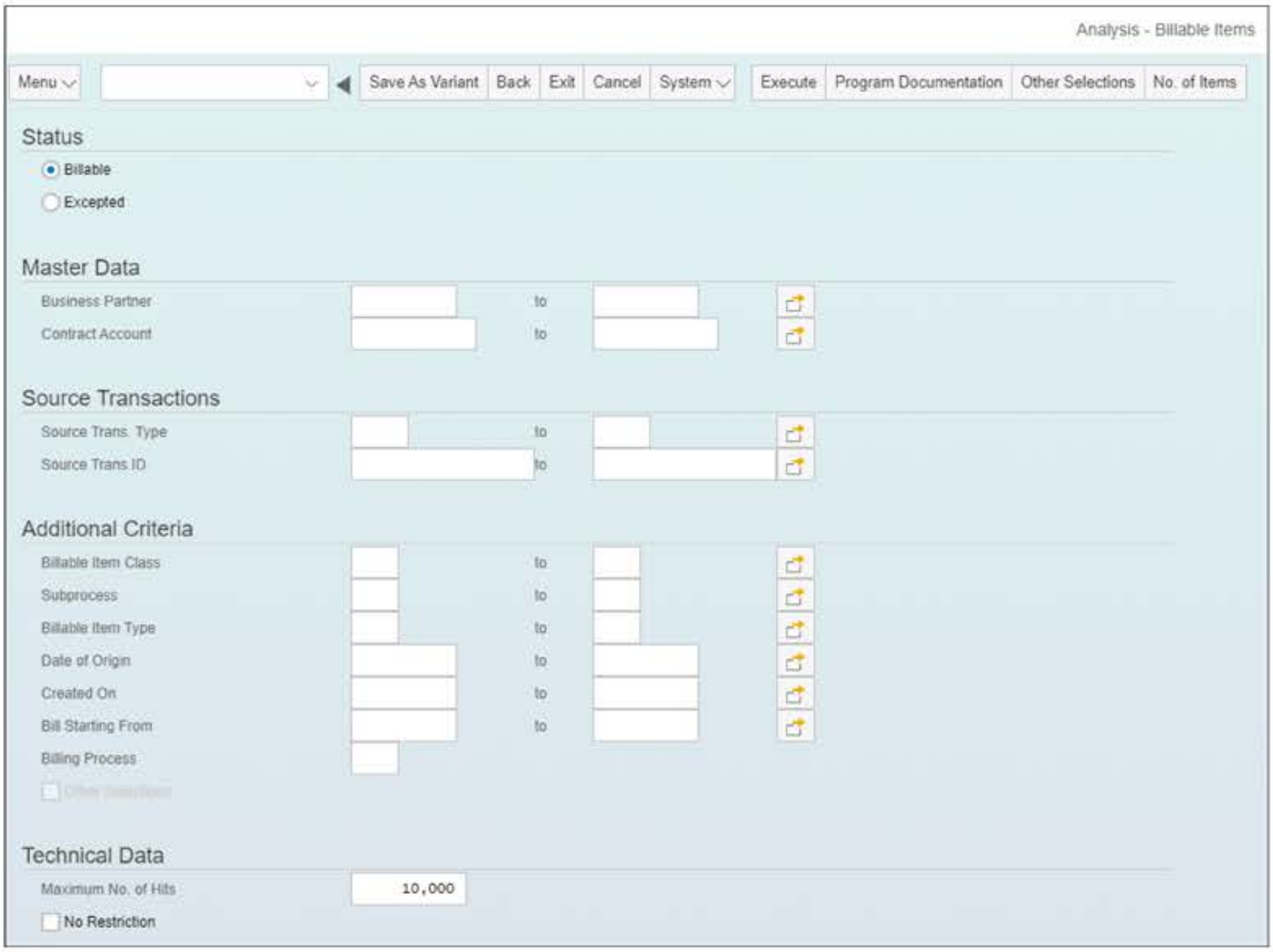

Figure 4.190 Billable Items Display Screen for Status Billable

4.6.7 Reversal Request for Invoicing Documents Analysis

The invoicing documents posted in the system either can be reversed directly or a reversal request can be created for the document reversal. Reversal of an invoicing document can be due to incorrect invoicing data or a customer request. For reversal to happen, the reversal request/task is created.

The reversal request for the invoicing documents can be viewed using Transaction FKKINV_REV_MON, which also can be accessed under the SAP Easy Access menu: **Accounting • Financial Accounting • Contract Account Receivables & Payables • Convergent Invoicing • Invoicing Documents** (see Figure 4.191).

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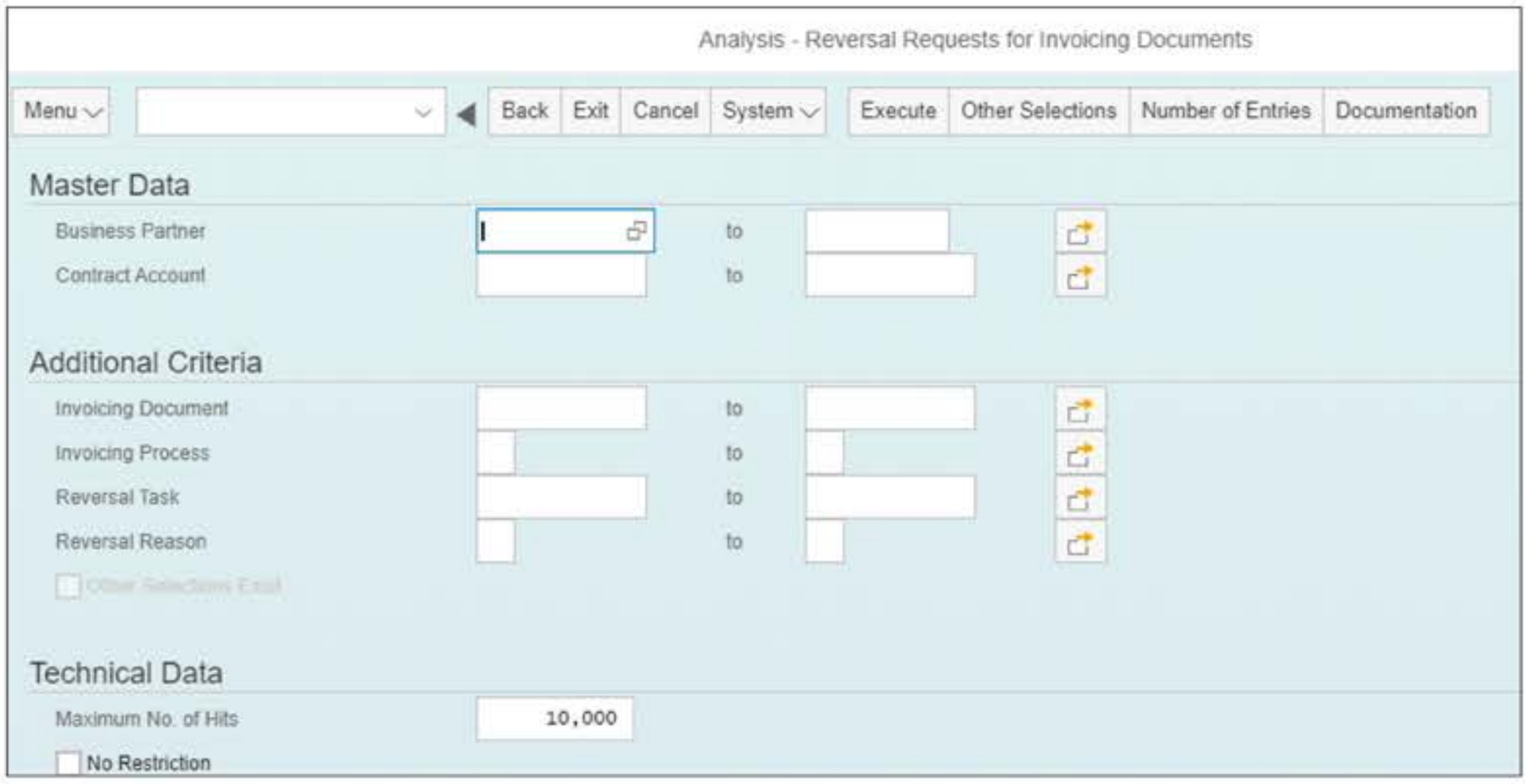

Figure 4.191 Invoice Document Reversal Request Display Screen

4.6.8 Reversal Request for Billing Documents Analysis

Like invoicing document reversal requests, the billing documents posted in the system either can be reversed directly or a reversal request can be created for the document reversal. Reversal of a billing document can be due to incorrect billing data. For reversal to happen, the reversal request/task is created (see Figure 4.192).

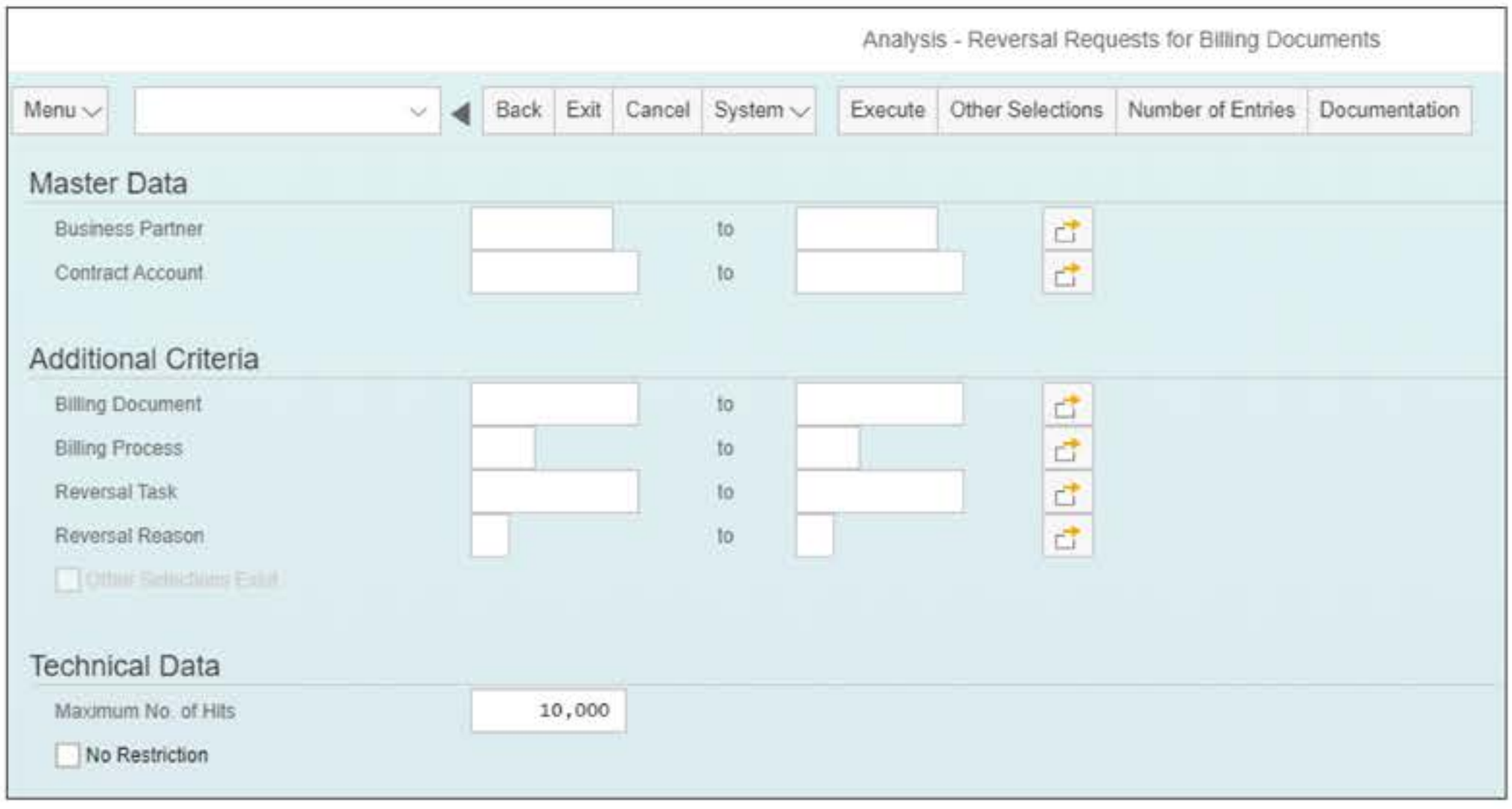

Figure 4.192 Billing Document Request Display Screen

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The reversal request for the billing documents can be viewed using Transaction FKKINVBILL_REV_MON, which also can be accessed under SAP Easy Access menu path **Accounting • Financial Accounting • Contract Account Receivables & Payables • Convergent Invoicing • Billing Documents** (see Figure 4.192).

4.7 Solving Common Billing and Invoicing Issues

This chapter highlights some of the frequent issues encountered in the billing and invoicing process. As you've learned so far, the input to the billing and invoicing process is either from SAP Billing and Revenue Innovation Management, subscription order management (as a one-off charge) from SAP Convergent Charging (typically a usage-based charge), or from SAP Convergent Invoicing (as a recurring charge using a billing plan).

Following are some of the common issues encountered in the billing and invoicing process:

- **Billable Item not picked up during billing**
  This example is more of a standard system behavior and not a system issue. However, beginners typically spend a considerable amount of time figuring out the root cause. There are instances where billable items aren't picked up during the billing process. While all the attributes of the billable item may seem to contain all required fields/valid data, the bill from date may be in the future. Billing transactions always pick up billable items with a bill from date that is current or in the past. All billable items with a bill from date in the future aren't picked up by the billing transaction.
- **Invoicing failure due to a failed tax call**
  In a scenario where an external tax engine integration is enabled, it's also crucial that the integration is active and working during the invoicing run. Any issues with the tax call may result in an invoicing run failure. The billing document remains in the system until the tax system is up, and the next invoicing job processes the billing document to create the invoicing document.
- **Manual lock placed on the provider contract**
  Billing and invoicing runs for billable items for which the associated provider contract is locked aren't picked up during the billing and invoicing runs. It's suggested to review the status of the provider contract and look for any such locks that may prevent further processing
- **Invoicing failure due to missing general ledger account**
  There are instances where the general ledger accounts configured for invoicing and contract accounting postings aren't extended to the appropriate company code. In such instances, the invoicing run fails and doesn't post a corresponding SAP Convergent Invoicing document and contract accounting document.

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- **Reversal of a SAP Convergent Invoicing invoice document**
  The SAP Convergent Invoicing invoice document is created only once a billable item is successfully created and billed. The SAP Convergent Invoicing invoice step then creates a SAP Convergent Invoicing invoice document and a corresponding contract accounting posting. In the event that certain attributes on the billable item are missing or incorrect after the invoicing document is posted, the billable item can't be directly reversed. The SAP Convergent Invoicing invoice document will need to be reversed first, which also reverses the contract accounting posting. Once this is complete, the SAP Convergent Invoicing billing document can be reversed. This will allow the billable item to be corrected or excepted.
- **Units of measure on a billable item**
  In a business scenario where there are multiple units of measure, there is a need to configure these additional units as a custom unit of measure. Unit of measure by standard has a length of three units. It's also displayed on correspondences and is typically interfaced to external non-SAP applications, but there may be inconsistencies in the definition. It's a key field to watch out for and needs to be kept consistent in terms of the decimal places and the unit of measure key.

4.8 Summary

You should now understand the consume-to-cash process and the system architecture and various terminology used in SAP Convergent Invoicing. You also should be able to set up the system with SAP Convergent Invoicing configurations and be able to map your business requirements.

The following chapter will cover the basic settings, postings, reporting, and accounting aspects of contract accounting. You'll learn about the SAP Convergent Invoicing and contract accounting integration and see how the invoicing document from SAP Convergent Invoicing flows into contract accounting and various receivables management functionalities.

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Chapter 5

Contract Accounting

This chapter explains how to set up and use contract accounting. It provides a brief overview of master data and transactional data, discusses some key concepts (e.g., account determination, open item management, account balance display, and integration with general ledger accounting), and delves into some of the standard out-of-the-box reporting capabilities and relevant transaction codes for contract accounting.

Contract accounting enables subledger accounting for receivables and payables, typically in industries that require processing of huge volumes of business transactions and business partners. It also enables necessary capabilities to manage accounts receivable and accounts payable (e.g., open item management and account balance display), along with capabilities such as dunning and interest calculation. Figure 5.1 shows some of the relevant industry applications for the contract accounting solution.

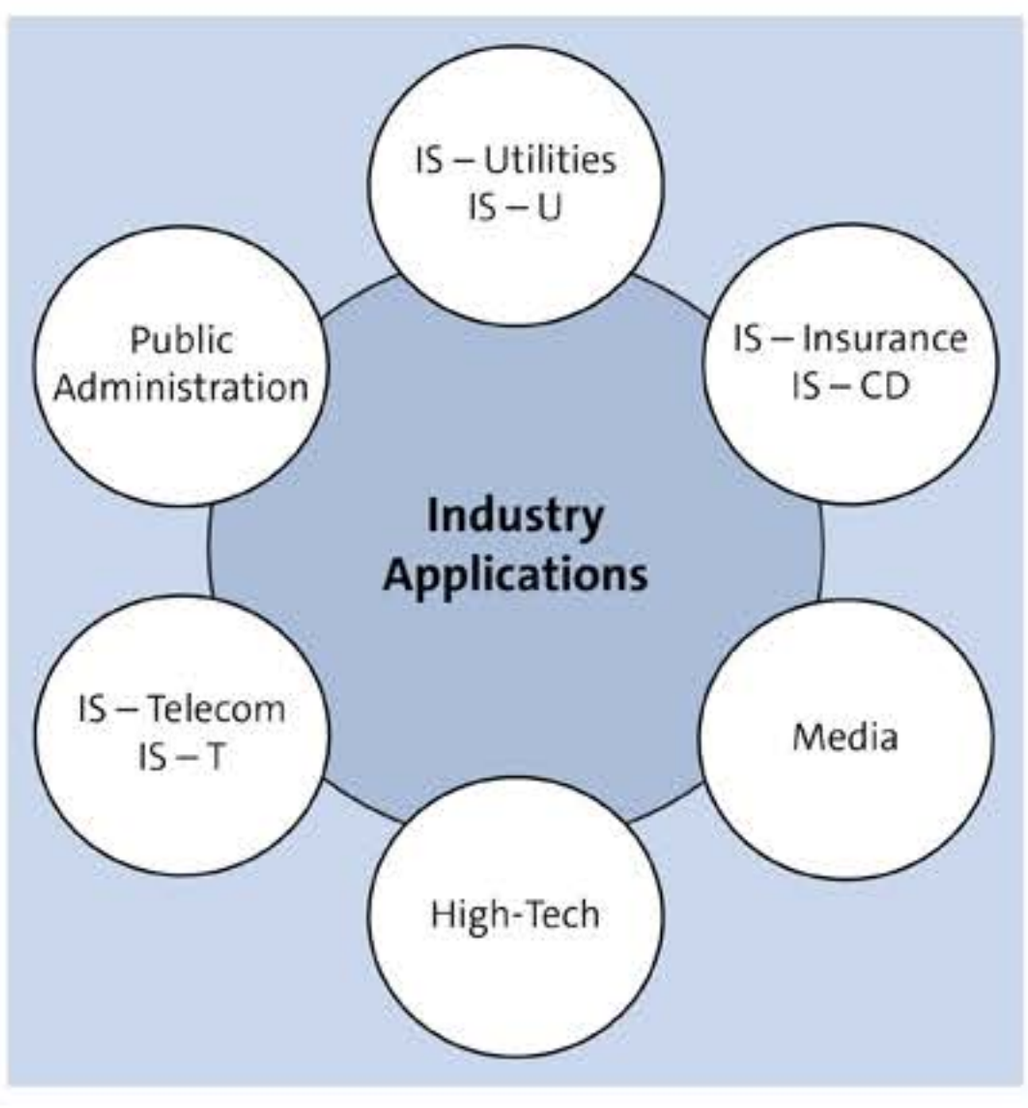

Figure 5.1 Relevant Industry Applications for Contract Accounting

This chapter describes the relevant master data for posting documents (e.g., business partners, contract accounts, and general ledger accounts) and then shows how to configure and use contract accounting with step-by-step instructions. The chapter concludes with a description of core contract accounting reporting and analytics capabilities.

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Martex Corp.

As a quick refresher of our example case study, Martex Corp., a company that distributes software, has transformed itself into a company that distributes cloud products with infrastructure as a service (IaaS) and software as a service (SaaS) offerings. With its new strategy, Martex obtains products from multiple vendors and sells them to its customers on behalf of those vendors. Each product is sold in different types of offerings, such as recurring, usage consumption, one-off, and pay per use. There are two use cases, which were explained in Chapter 1, related to SaaS combo charges and SaaS prorated charges. In this chapter, we'll work through the setup and concepts primarily related to account determination, open item management, and account balance display.

SAP Convergent Invoicing, which we discussed in Chapter 4, is a submodule of contract accounting. All the necessary master data attributes that SAP Convergent Invoicing uses for transaction processing is primarily defined in contract accounting. Let's take a look at the following key functionalities carried out in each of the modules within the SAP Billing and Revenue Innovation Management solution:

- **Non-SAP system**
  Captures network data usages and sends the payload.
- **SAP Convergent Mediation by DigitalRoute**
  Transforms raw consumption data into an SAP Convergent Charging–readable comma-separated values (CSV) file.
- **SAP Convergent Charging**
  Performs rating and charging on the usage items and creates billable items in SAP Convergent Invoicing.
- **SAP Convergent Invoicing**
  Stores rated usage items and performs billing and invoicing.
- **Contract accounting**
  Posts receivables from invoicing, process payments, collections, dunning, and more (see Figure 5.2 for the configuration node).

The steps of the consume-to-cash process are as follows:

1. **Capture consumption data.**
   The usage of a service is captured in a non-SAP system. The usage is the consumption of the service provided by the service provider. The consumption data records details such as the time of usage, quantity consumed, and master data of the customer for which consumption is tracked. This consumption data is in a raw format and needs to be formatted before rating can be performed on it.
2. **Transform consumption data.**
   The consumption data recorded outside of SAP needs to be transformed into an SAP format before it can be used for the rating process. SAP provides SAP Convergent

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Mediation by DigitalRoute to transform the raw data captured in the non-SAP system and transform it into a format that can be fed into SAP Convergent Charging.

3. **Perform the rating and charging process.**
   The consumption data record transformed by SAP Convergent Mediation by DigitalRoute is fed into SAP Convergent Charging for the rating and charging process. The rating and charging process calculates the amount of the consumed quantity based on the pricing rules maintained in the SAP Convergent Charging system. The calculated amount is then associated with the business partner account that has consumed the service in the charging process. The rated and charged consumption items, which are known as billable items, are sent to the SAP Convergent Invoicing system for the billing and invoicing process.
4. **Bill and invoice.**
   The billable items received from SAP Convergent Charging act as an input to the billing process. The billing process groups and aggregates the billable items and creates the billing documents, which are used as an input to the invoicing process. The invoicing process groups and aggregates the billing documents and creates the invoicing document that is sent to the customer.
5. **Manage receivables.**
   The invoicing process creates a posting document for the receivables posting in contract accounting. The posting document posts the accounting entries to the customer account, against which payment is received from the customer.

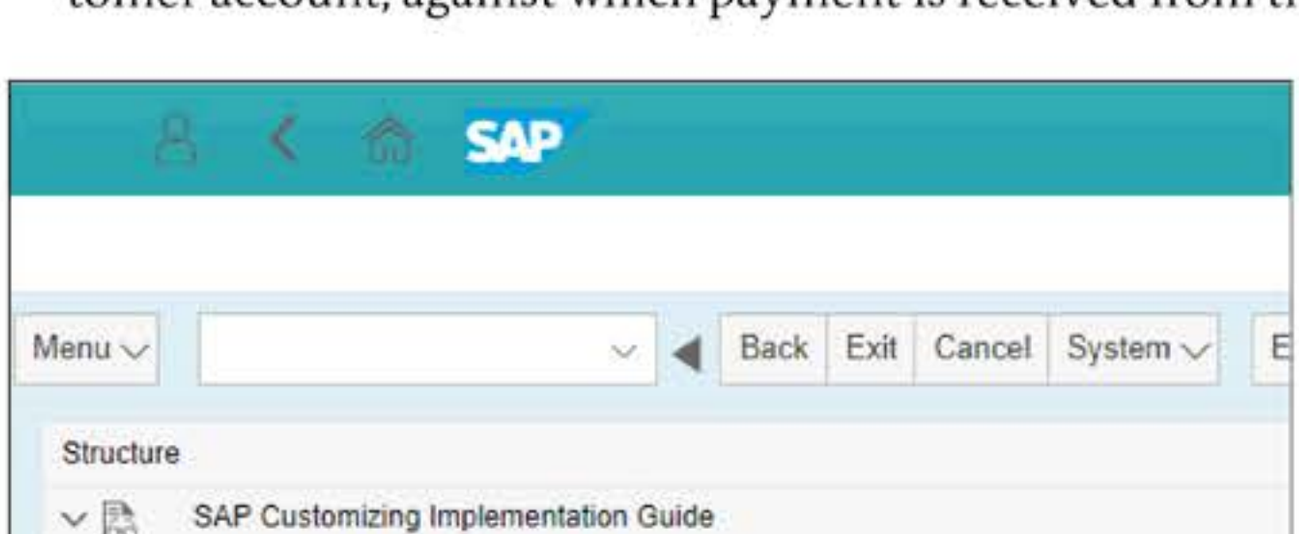

Figure 5.2 Configuration Node for Contract Accounting

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As described in Chapter 4, the receivables created by the invoicing process finally get posted in contract accounting as a subledger posting. The payment is later realized in contract accounting, and the customer open items and the account balance is managed. The general ledger postings are transferred to financials as a result of general ledger integration from contract accounting.

5.1 Master Data

In this section, we'll look at the master data elements in contract accounting relevant to the Martex case study established in Chapter 1. The master data elements will be required to create and process transactions using contract accounting. We'll discuss business partners, business agreements, products, contract accounts, provider contracts, and partner agreements.

The following master data objects will be explained in detail in this chapter:

- Business partner
- Contract account
- Provider contract
- Partner agreement

The following master data objects have been explained in detail in other chapters, so we'll only call out their significance for transaction postings from a contract accounting perspective in this chapter:

- Business agreement (SAP Billing and Revenue Innovation Management, subscription order management)
- Product (SAP Billing and Revenue Innovation Management, subscription order management)

5.1.1 Business Partner

This section will explain the concept of the business partner master data object from a contract accounting perspective. We'll go through the configuration setup for the business partner in contract accounting and discuss why we need a business partner object as a prerequisite for account determination, open item management, and account balance display.

In this chapter, we'll be referencing a customer or a vendor in contract accounting as a business partner. A business partner can be an organization, a person, or a group of persons or organizations that have a business relationship with a company. A business partner can also be extended to various partner roles (e.g., general business partner, contract partner, customer, or vendor).

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Figure 5.3 shows the business partner view in contract accounting under the business partner general role.

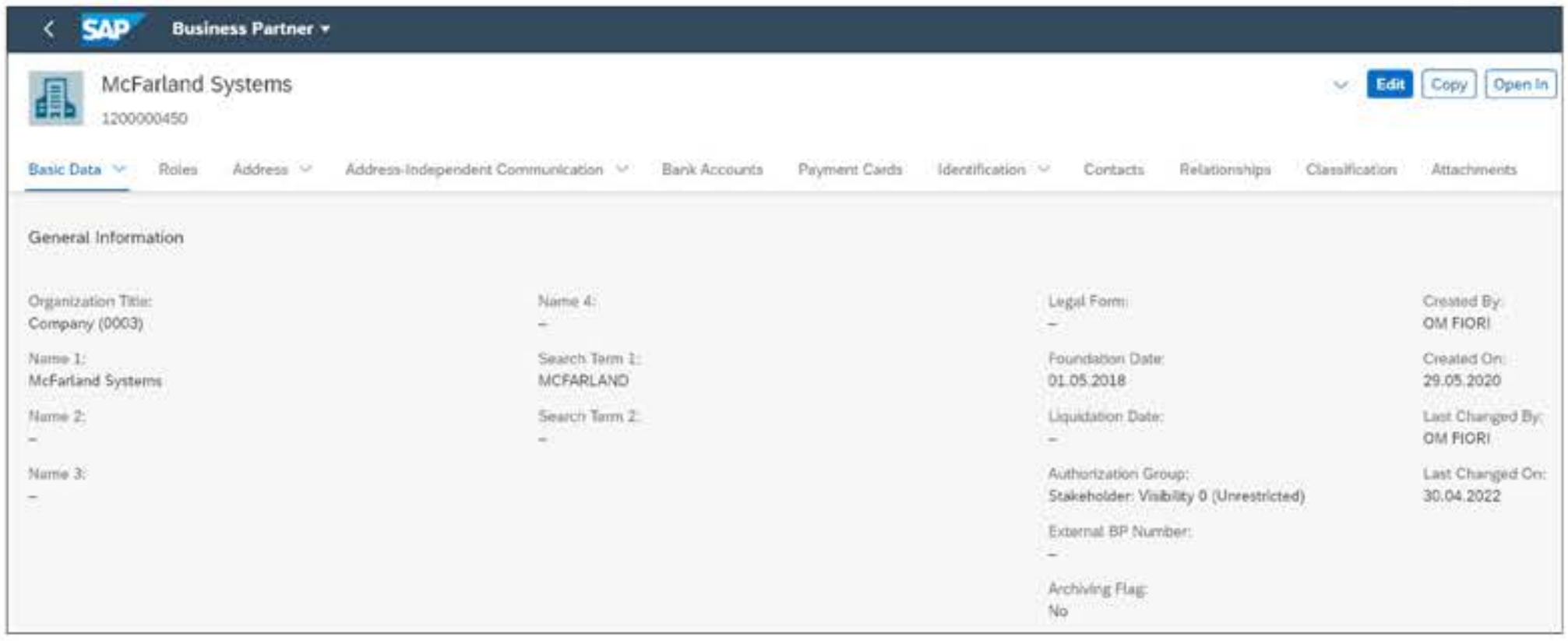

Figure 5.3 Business Partner View in Contract Accounting

You use Transaction BP to view a business partner in contract accounting.

Next, let's extend the business partner to the contract partner role to ensure that a corresponding contract account is created for the business partner. Figure 5.4 shows some of the business partner roles in the **Roles** tab that the business partner can be extended to, including **Business Partner (Gen.)** (the general business partner) which is displayed as **Sol-to-Party** and **Contract Partner** roles, which are particularly relevant for our case study.

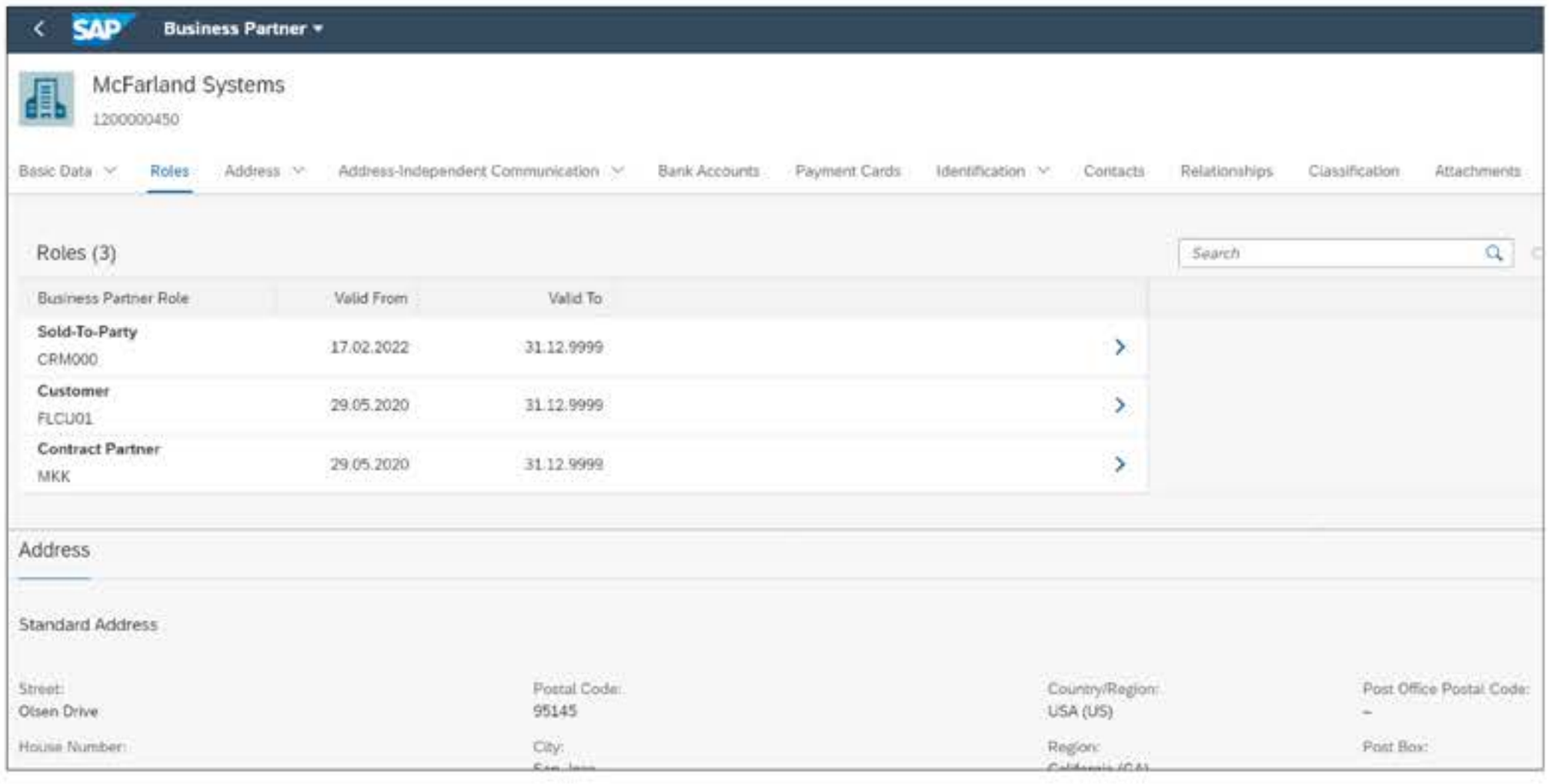

Figure 5.4 Business Partner Roles

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The general business partner role enables maintenance of business partner general data such as name, address, and payment data (bank details, payment card information, etc.). In contract accounting, it's necessary to create a business partner and a contract account to be able to record a transaction in the system.

We'll walk through the key integration points and setup needed to create a business partner in contract accounting. Similar setup is also needed in SAP Customer Relationship Management (SAP CRM) to enable business partner setup and replication to contract accounting.

Figure 5.5 shows the master data flow across SAP systems.

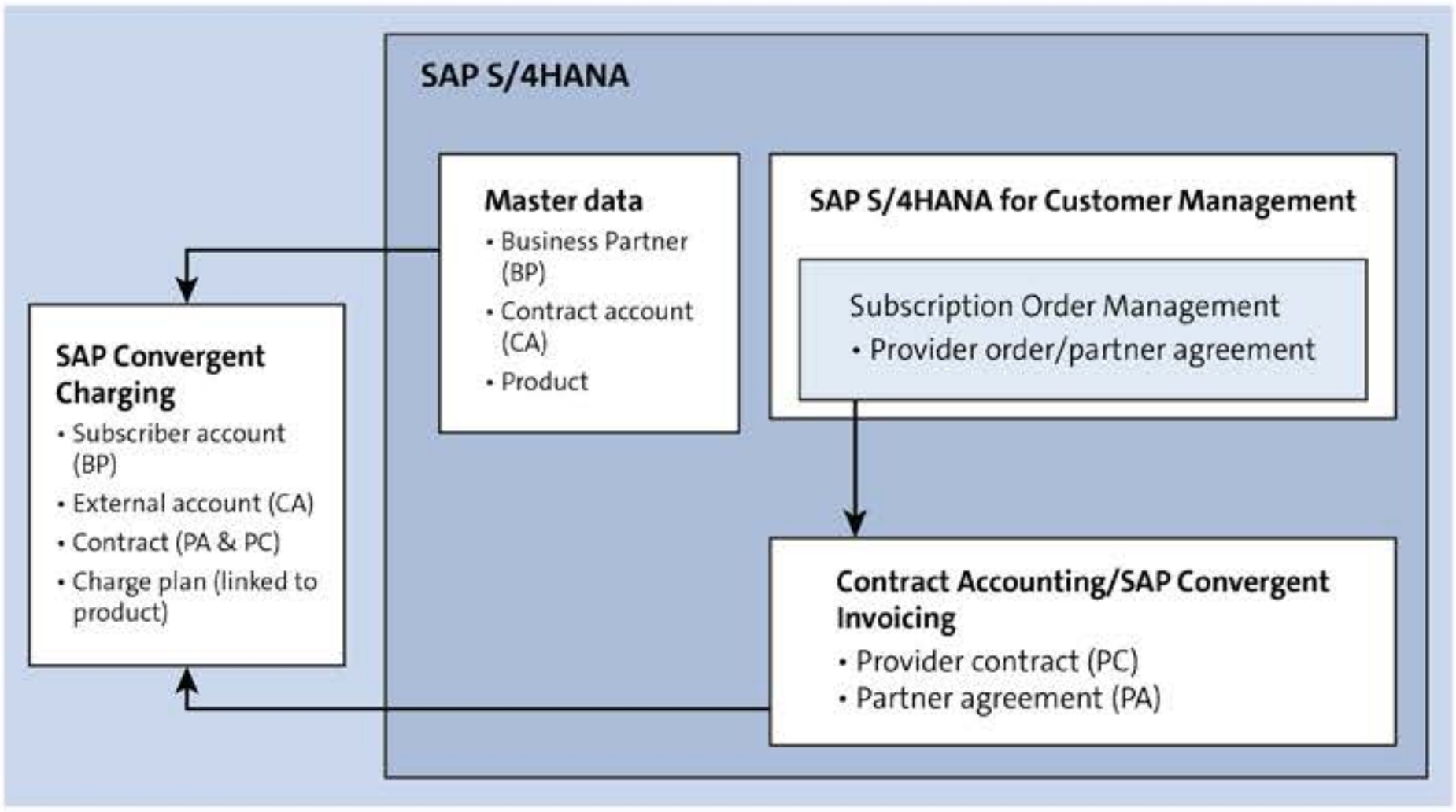

Figure 5.5 Master Data Flow across SAP Systems

Now let's configure a business partner. To configure a business partner in SAP ERP or SAP S/4HANA, navigate to **SAP Customizing Implementation Guide • Cross Application Components • SAP Business Partner • Activation Status for Functions**.

Figure 5.6 shows the functions necessary to be activated for business partners. Activate the functions as shown. There are quite a few other functions that are available and can be activated as needed.

Several standard business partner roles are delivered by the application, and each role determines the functions that can be carried out with the business partner. The business partner is the highest level, equivalent to a customer in accounts receivable. The contract account is the next level. A contract account can't be created without a corresponding business partner.

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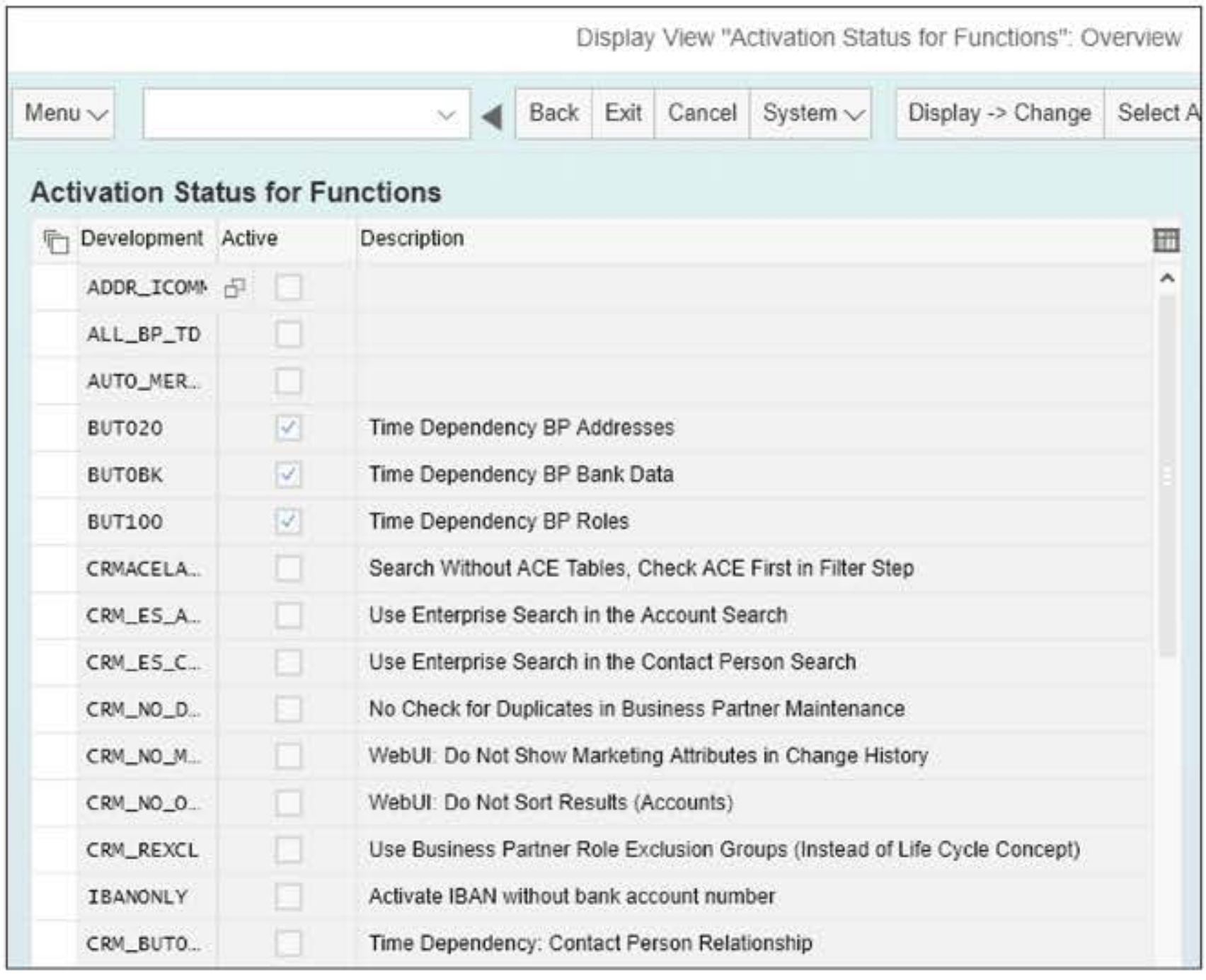

Figure 5.6 Activation Status for Functions

To create a contract account for a business partner, the prerequisite is to extend the business partner to an `MKK` contract partner role. To create the contract partner role, navigate to **SAP Customizing Implementation Guide • Cross Application Components • SAP Business Partner • Business Partner • Basic Settings • Business Partner Roles • Define BP Roles**. Then, click **New Entries** and create the **MKK** role. Once the **MKK** role has been created and saved, the new entry is visible under the list of business partner roles, as shown in Figure 5.7.

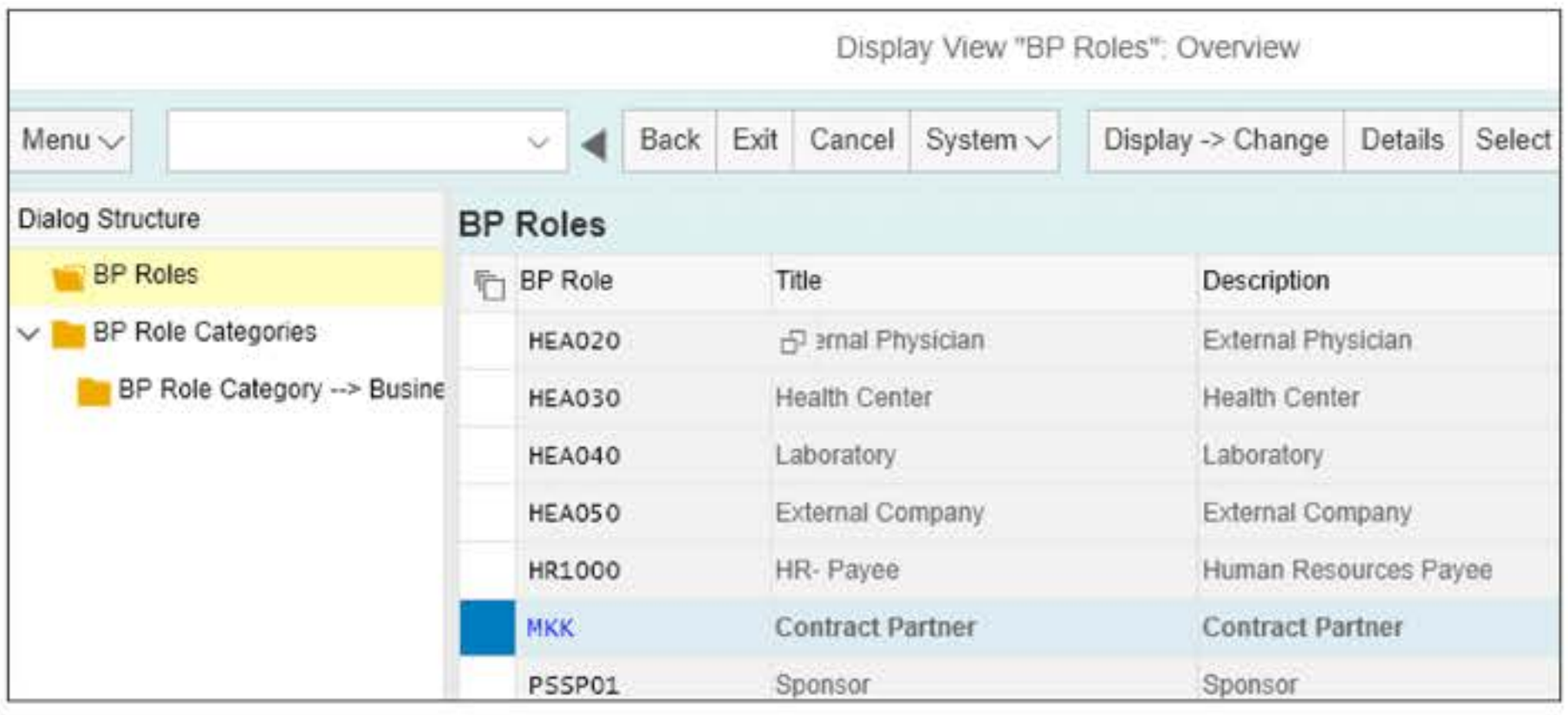

Figure 5.7 MKK Role for Contract Partner Created as New Business Partner Role

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Now, let's assign the business transactions, such as Business Partner Usage (BPUS) and Change Business Partner (CHAN), which are functions needed for the MKK role. Click **New Entries** as shown in Figure 5.8, and then assign **BPUS** and **CHAN**, as shown in Figure 5.9.

Figure 5.8 Business Partner Role Category to Business Transaction Assignment

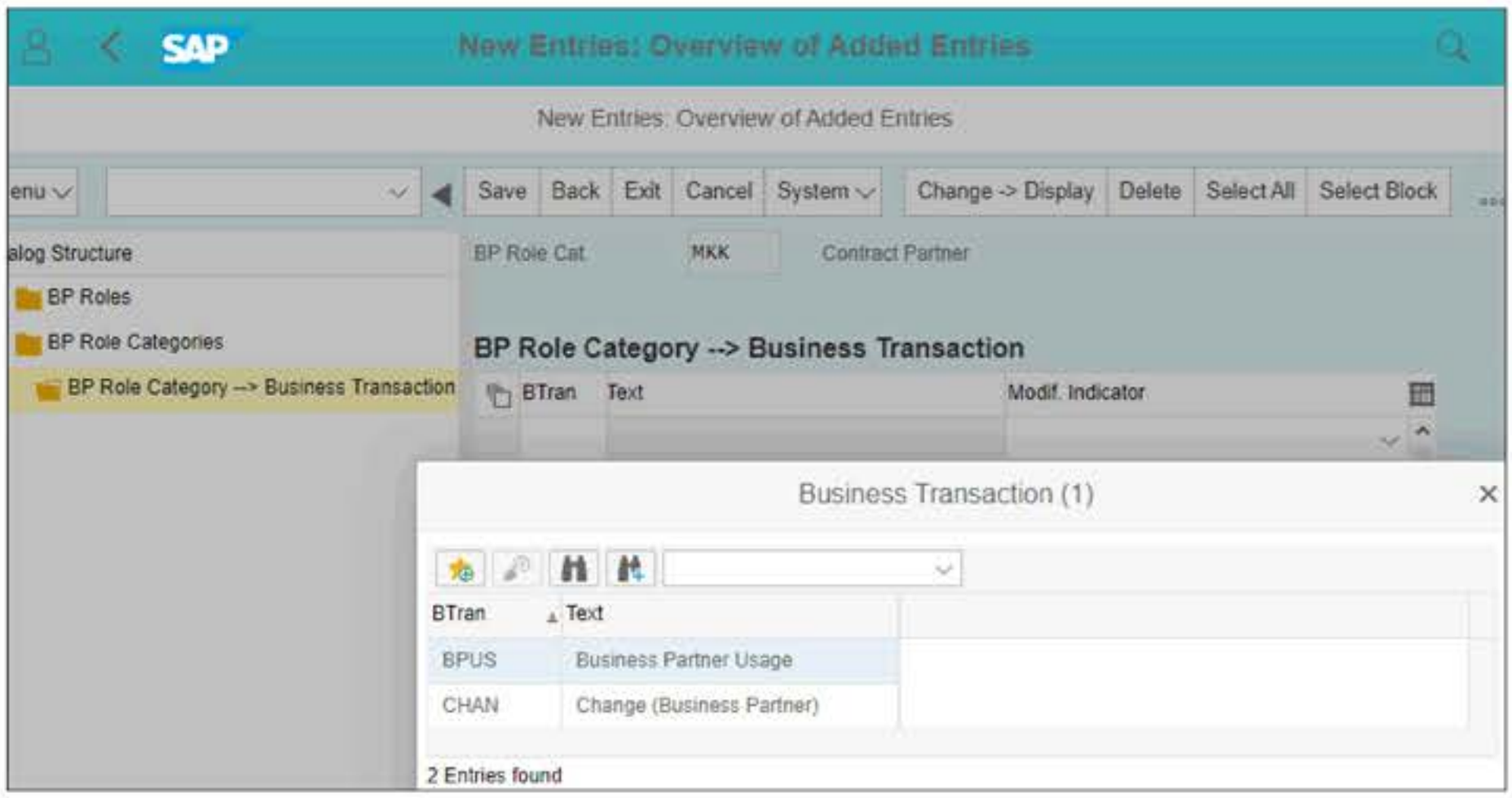

Figure 5.9 Business Transaction Selection

What we've seen thus far, is creation of the **MKK Contract Partner** role relevant for contract accounting, as shown in Figure 5.10.

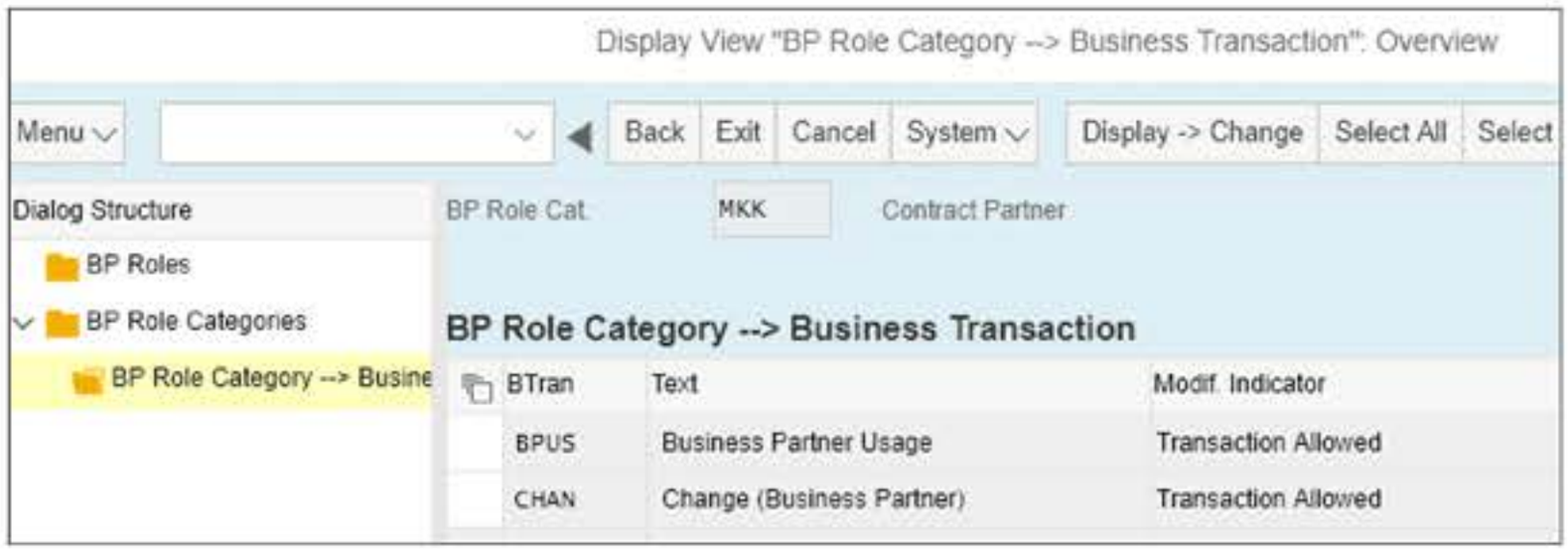

Figure 5.10 Business Transactions Assigned to MKK Contract Partner Business Partner Role

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5.1.2 Business Agreement

A business agreement is a master data object that is the equivalent of a contract account in contract accounting. One business partner can have several business agreements associated with it. Business agreements house data that controls billing and invoicing processes in SAP Convergent Invoicing, payments, correspondences, and tax calculations.

A business agreement created in SAP Billing and Revenue Innovation Management, subscription order management, is replicated to contract accounting as a contract account with SAP standard replications.

Navigate to the SAP Customizing Implementation Guide menu path **Financial Accounting • Contract Accounts Receivable and Payable • Integration • Customer Relationship Management • Business Agreement • Determine Template for Contract Account for Replication**. Next, as shown in Figure 5.11, enter a reference contract account, which will be used as a template when a business agreement is replicated as a contract account in contract accounting. All the attributes available in the template will be copied over to the new contract account.

Figure 5.11 Contract Accounting Integration with SAP CRM for Business Agreement Replication

A business partner and business agreement are required to create a provider order in SAP Billing and Revenue Innovation Management, subscription order management. For more information about business agreements, see Chapter 2, Section 2.1.2.

With the SAP S/4HANA release 1809 release, SAP has moved SAP Billing and Revenue Innovation Management, subscription order management, to customer management within the SAP S/4HANA system. With SAP Billing and Revenue Innovation Management, subscription order management, moved to SAP S/4HANA, the contract account creation can be done directly in contract accounting.

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5.1.3 Product

A product is another component necessary to create a provider order. The product can either be a tangible product defined within materials management that contains all the relevant information required for purchasing, sales, accounting, and so on, or a subscription product defined within contract accounting as a contract accounting product that contains a lean product structure required for subscription processing (as explained in Chapter 4, Section 4.1.2). As illustrated in Figure 5.12, the contract accounting subscription product is a lean structure containing the description of the product and the sales area data. Once created in SAP Billing and Revenue Innovation Management, subscription order management, the product will be replicated to contract accounting. Use Transaction FPPRD3 to view the contract accounting product in contract accounting.

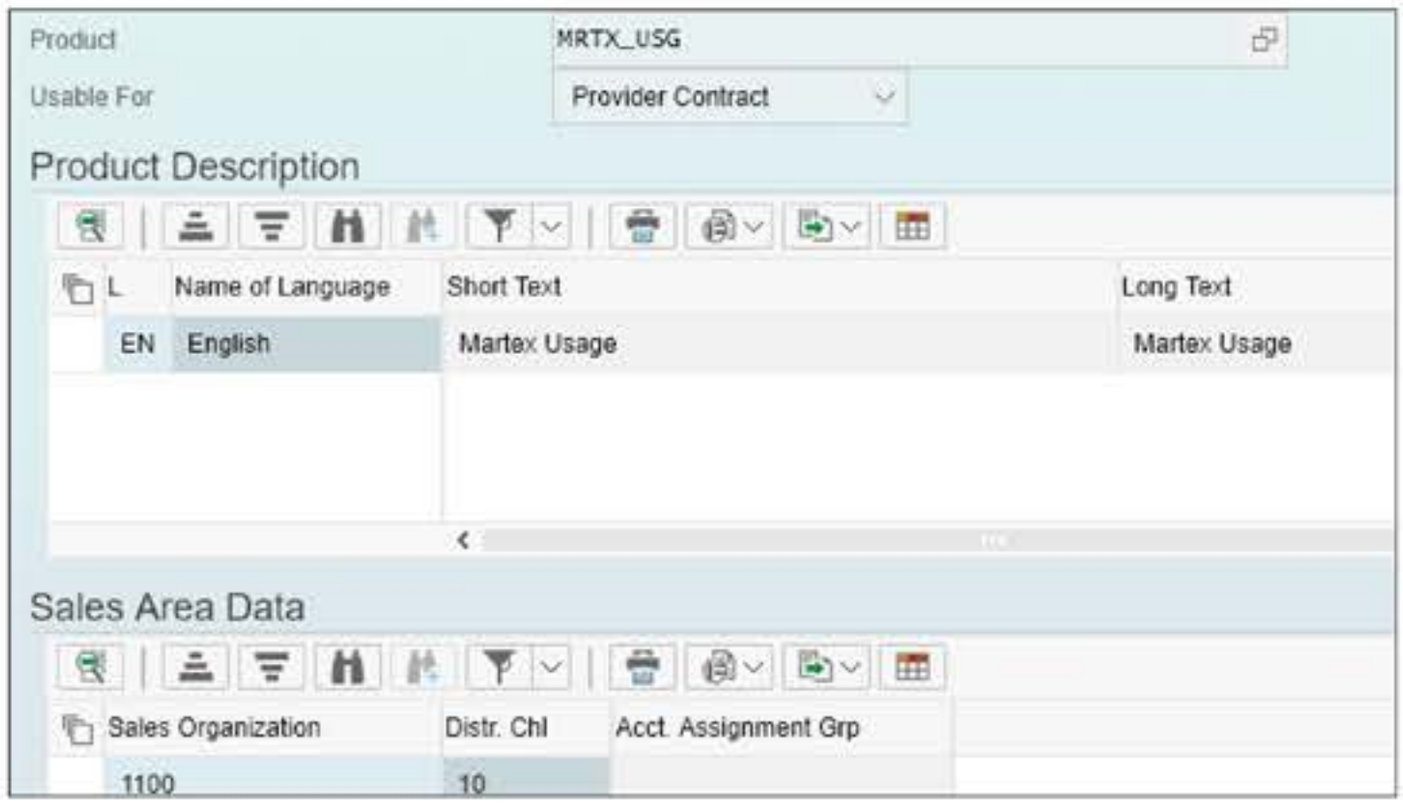

Figure 5.12 Contract Accounting Product in Contract Accounting

The next step is to define the type of product to be used for a provider contract.

5.1.4 Contract Account

A contract account is an essential master data object in contract accounting that stores payment/tax, dunning/correspondence, and SAP Convergent Invoicing attributes that will be applicable for the contract items created for the contract account. It's a mandatory object required to create billable items in SAP Convergent Invoicing. The billable items are created for a business partner and contract account combination. A contract account will be extended to a company code, and all the attributes stored in the contract account will be at the company code level.

Figure 5.13 illustrates the relationship between a business partner and a contract account. As shown, the business partner is extended to one or more sales organizations. A contract account is extended to a company code. A 1:N relationship exists between a business partner and a contract account.

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For Martex Corp., a business partner and a contract account will need to be created to create billable items for the business partner and contract account combination.

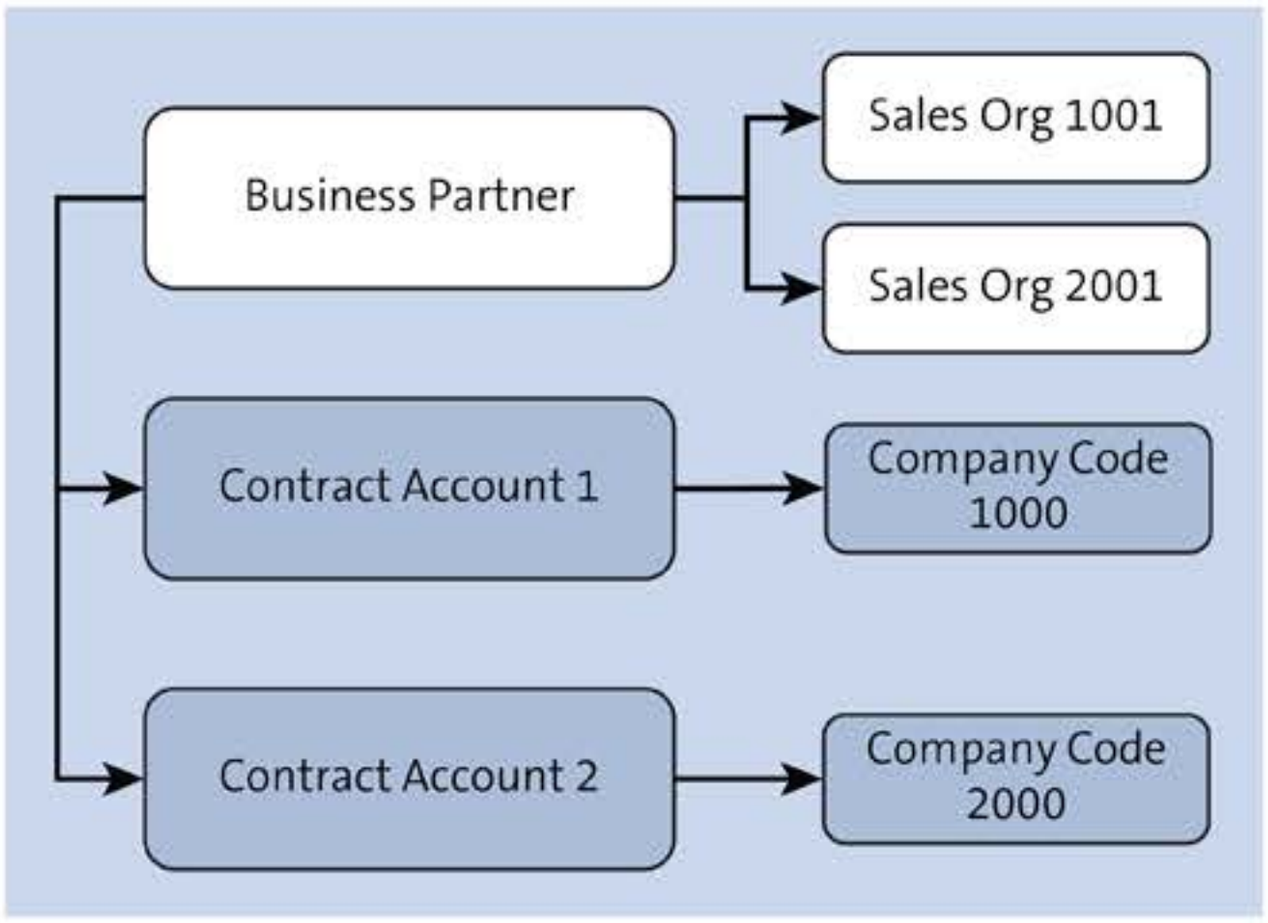

Figure 5.13 Business Partner and Contract Account Relationship

A contract account category categorizes contract accounts that require similar controls or reporting for a specific business. Now that you understand what a contract account is and its relationship with the business partner, let's configure the number ranges and the contract account categories.

First, navigate to menu path **SAP Customizing Implementation Guide • Financial Accounting • Contract Accounts Receivable and Payable • Basic Functions • Contract Accounts • Number Ranges and Contract Account Categories • Define Number Ranges**. Here you can define the number ranges for the contract account, as shown in Figure 5.14.

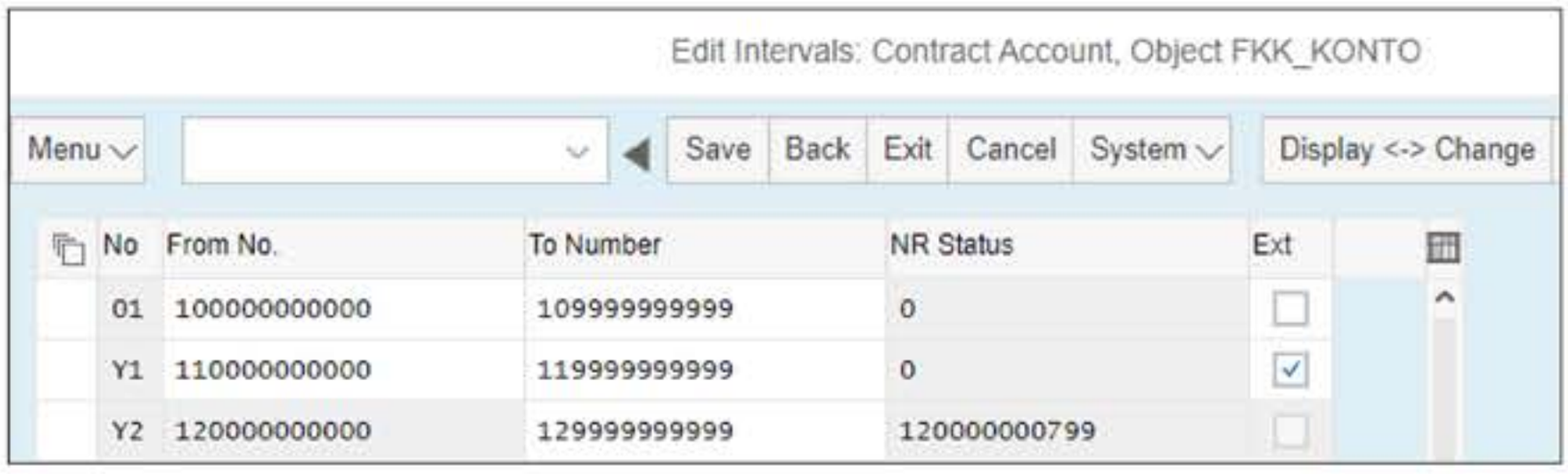

Figure 5.14 Number Ranges for the Contract Account

Next, navigate to menu path **SAP Customizing Implementation Guide • Financial Accounting • Contract Accounts Receivable and Payable • Basic Functions • Contract Accounts • Number Ranges and Contract Account Categories • Configure Contract Acct Categories and Assign Number Ranges**. Here you can configure the contract account categories and assign the number ranges.

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Figure 5.15 illustrates the definition of contract account categories and number range assignments for contract account creation.

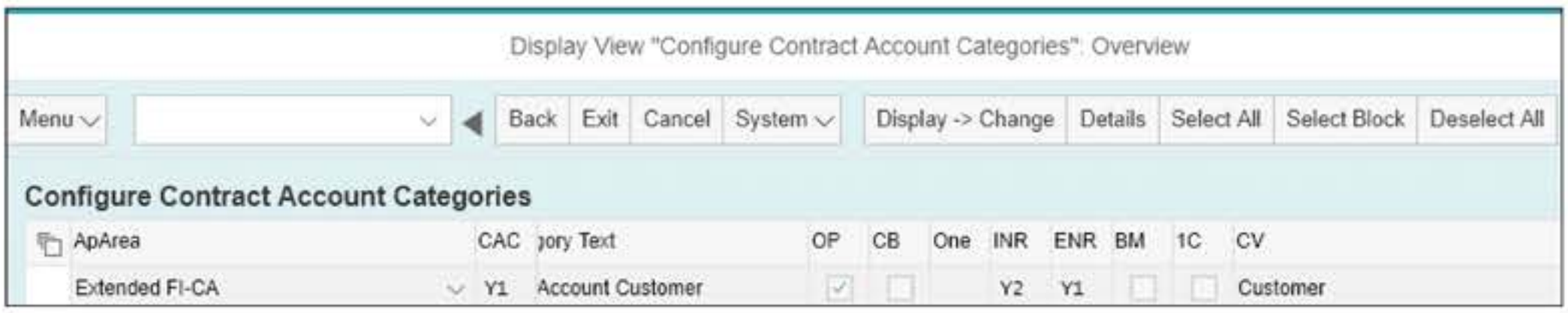

Figure 5.15 Configure Contract Account Categories

Contract account relationships are required to determine the relationship that the business partner holds with the contract account. If there are multiple business partners for a contract account, the account relationship differentiates the relations that each business partner holds for a contract account.

Navigate to menu path **SAP Customizing Implementation Guide • Financial Accounting • Contract Accounts Receivable and Payable • Basic Functions • Contract Accounts • Contract Account Relationships • Define Contract Account/Business Partner Relationships**. Here, assign **Account Holder** as the account relationship (**Acct Rel.**). As shown in Figure 5.16, the account relationship is defined to determine the relation between the business partner and the contract account.

Figure 5.16 Definition of Partner Account Relationship

Now, navigate to **SAP Customizing Implementation Guide • Financial Accounting • Contract Accounts Receivable and Payable • Basic Functions • Contract Accounts • Define Account Determination Characteristics**. Here you can configure the account determination ID, a key characteristic that will be used in subsequent configurations to automatically determine the general ledger accounts for posting a document in contract accounting (see Figure 5.17).

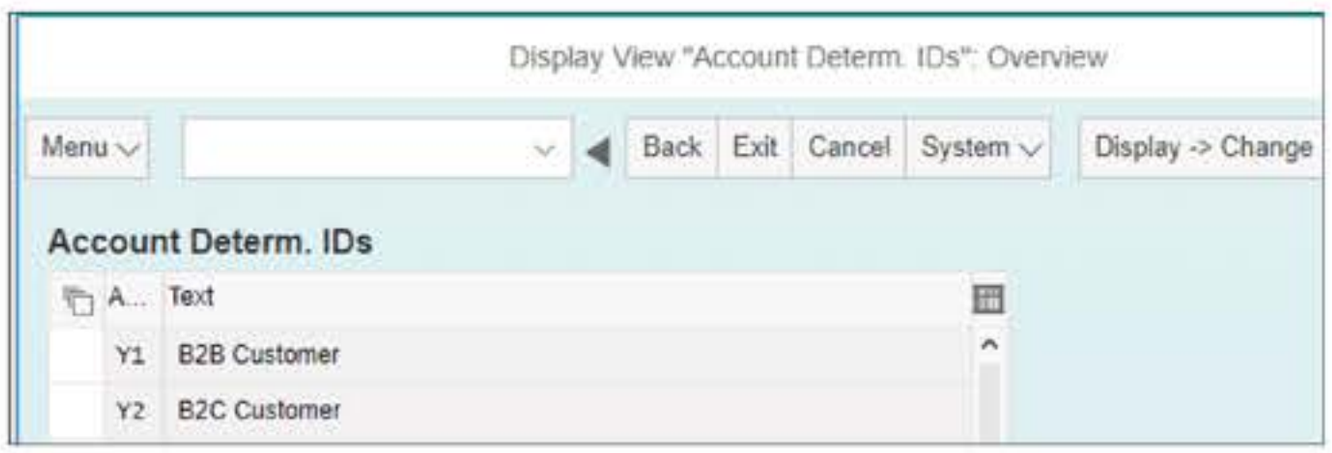

Figure 5.17 Contract Account: Account Determination IDs

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Figure 5.18 shows the various tabs available for a contract account.

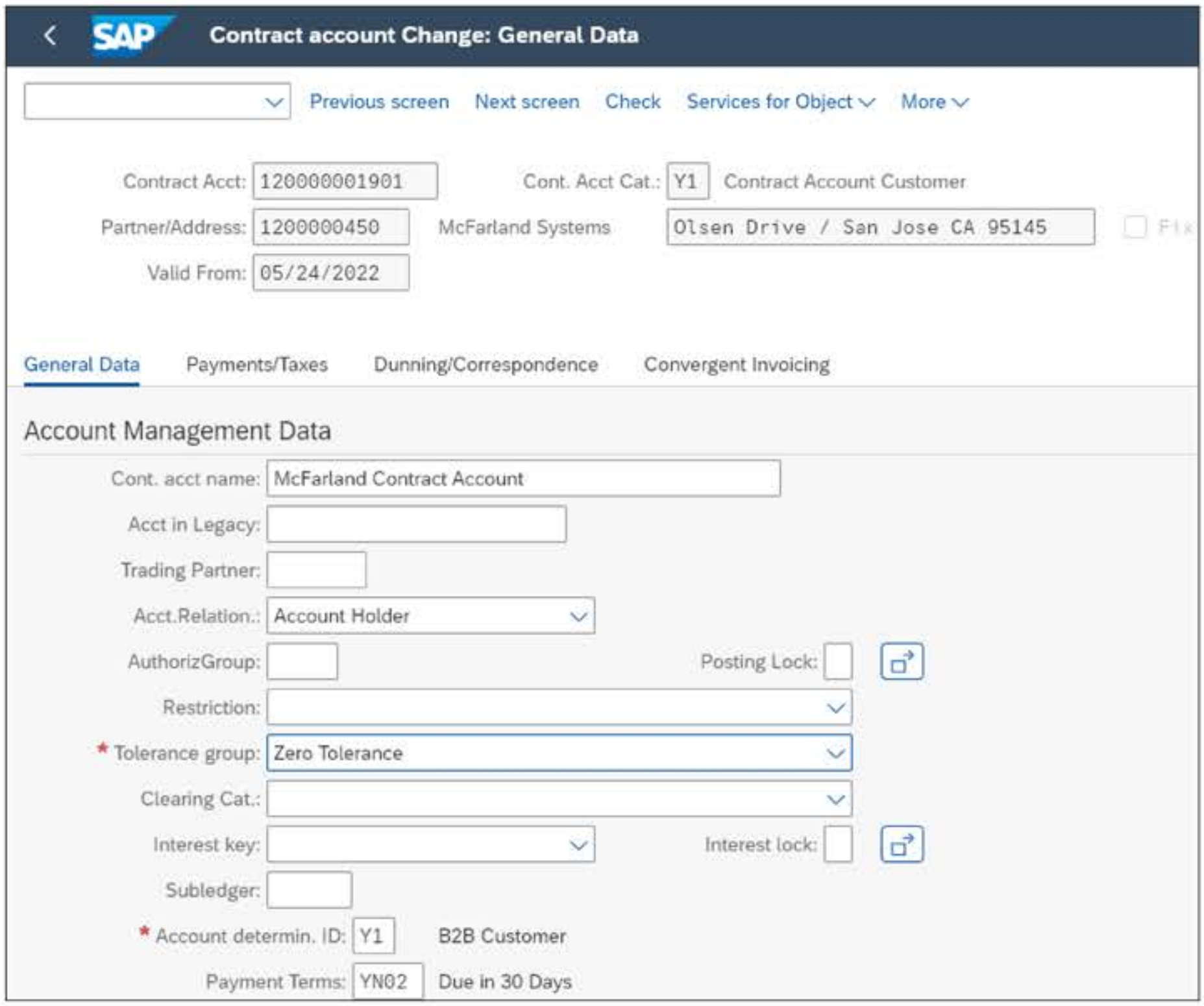

Figure 5.18 Contract Account General Data

The business agreement 120000001901 created in SAP Billing and Revenue Innovation Management, subscription order management, for the Martex Corp. case study has been replicated as a contract account. The tabs for the contract account are as follows:

- **General Data**

  As shown in Figure 5.18, the **General Data** tab contains some key information, such as contract account name, account reference from legacy, account relationship with the business partner, tolerance group, account determination ID, payment terms, and so on.

  The values updated in this object will be common for all the billable items created for this contract account.

- **Payments/Taxes**

  As shown in Figure 5.19, the **Payments/Taxes** tab contains some key information related to the group: standard company code information, incoming and outgoing payment methods, and more.

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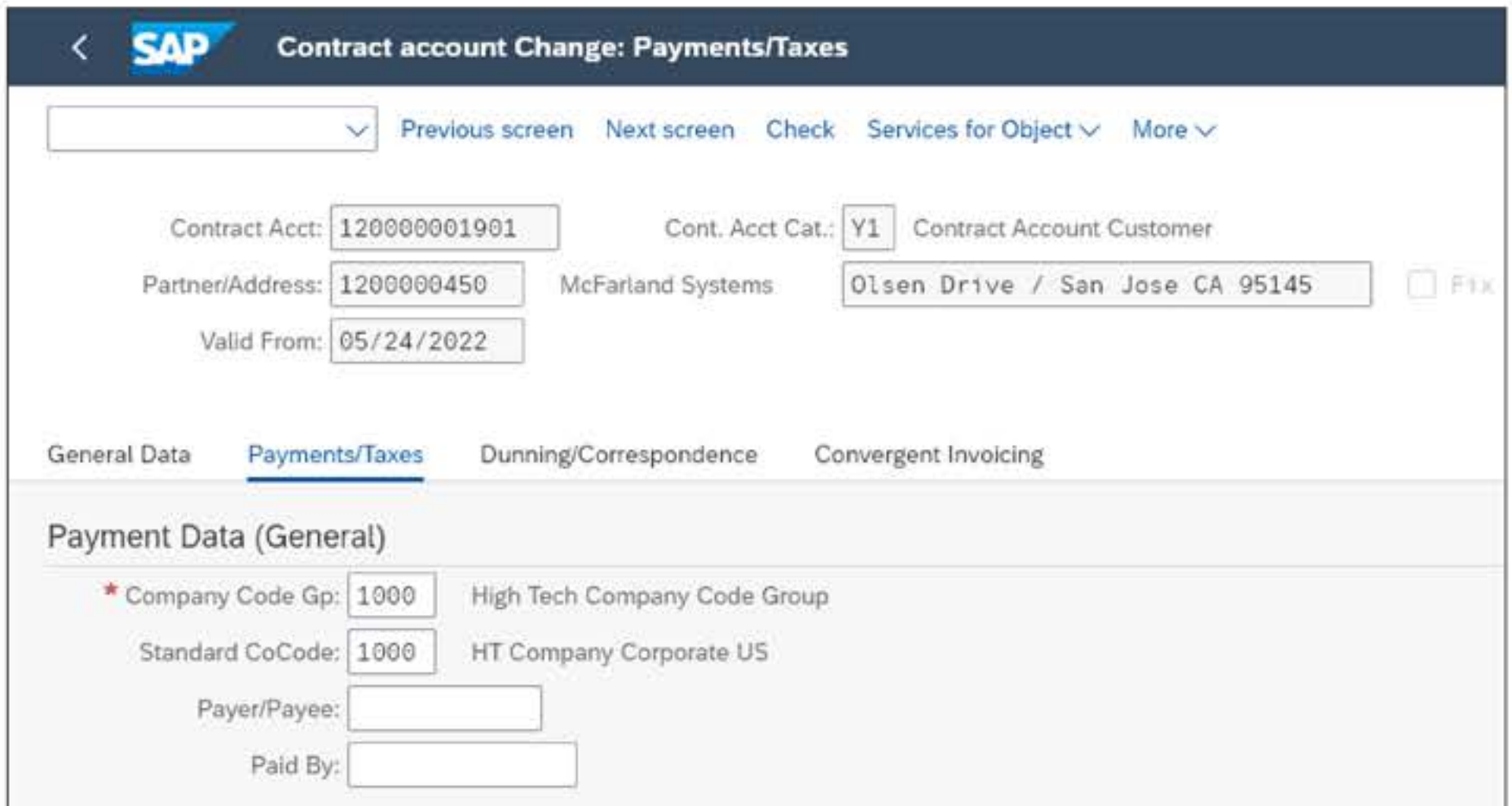

Figure 5.19 Contract Account Payments/Taxes Tab

- **Dunning/Correspondence**
  As shown in Figure 5.20, the **Dunning/Correspondence** tab contains key information such as alternative dunning recipient, dunning procedure, dunning lock, correspondence controls, and so on. We won't be using this tab for the Martex case study.

- **Convergent Invoicing**
  As shown in Figure 5.21, the **Convergent Invoicing** tab contains some of the key billing/invoicing attributes that will be used to determine the selection criteria for the accounts for billing/invoicing, as well as for determination of the billing cycles for the periodic bills.

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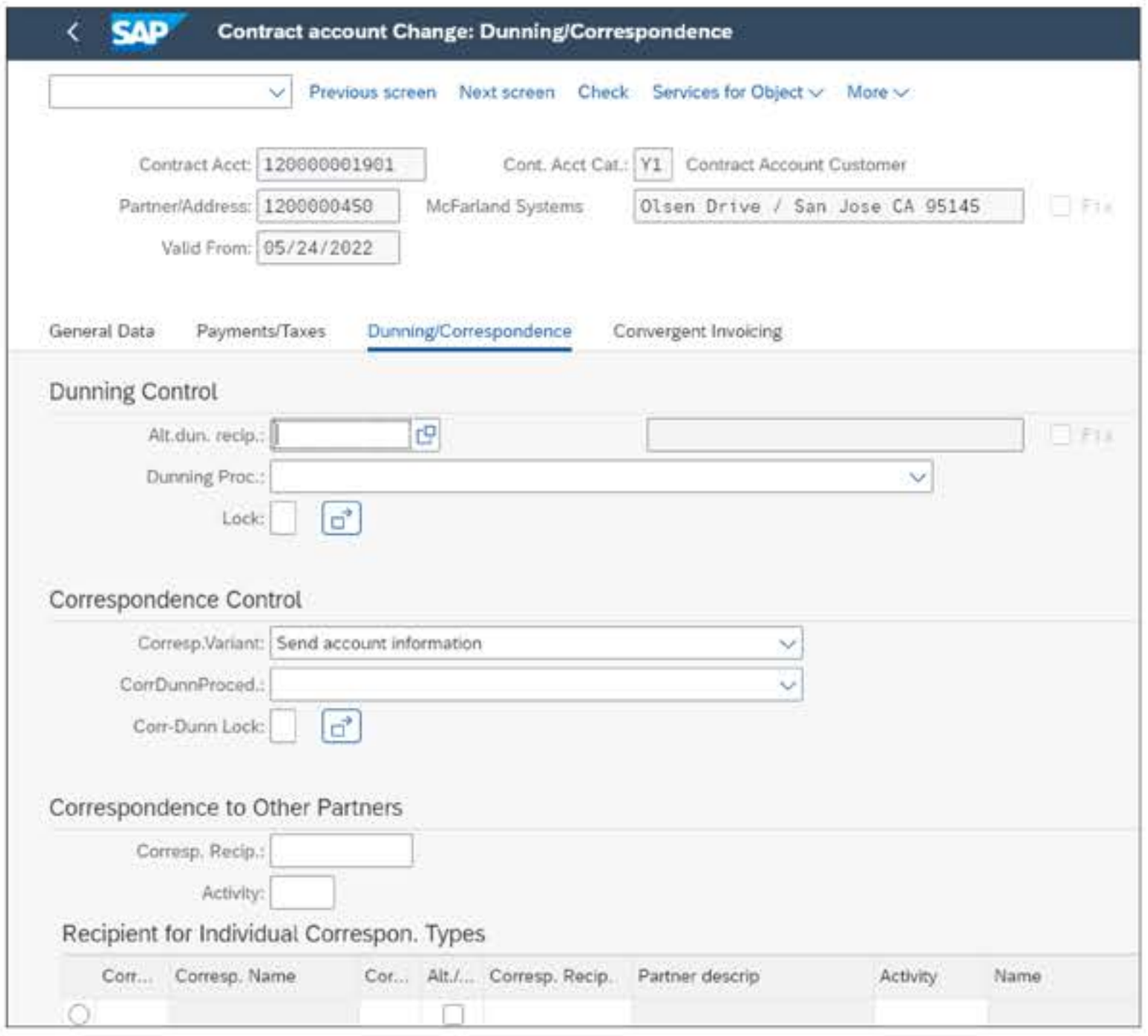

Figure 5.20 Contract Account Dunning Data Tab

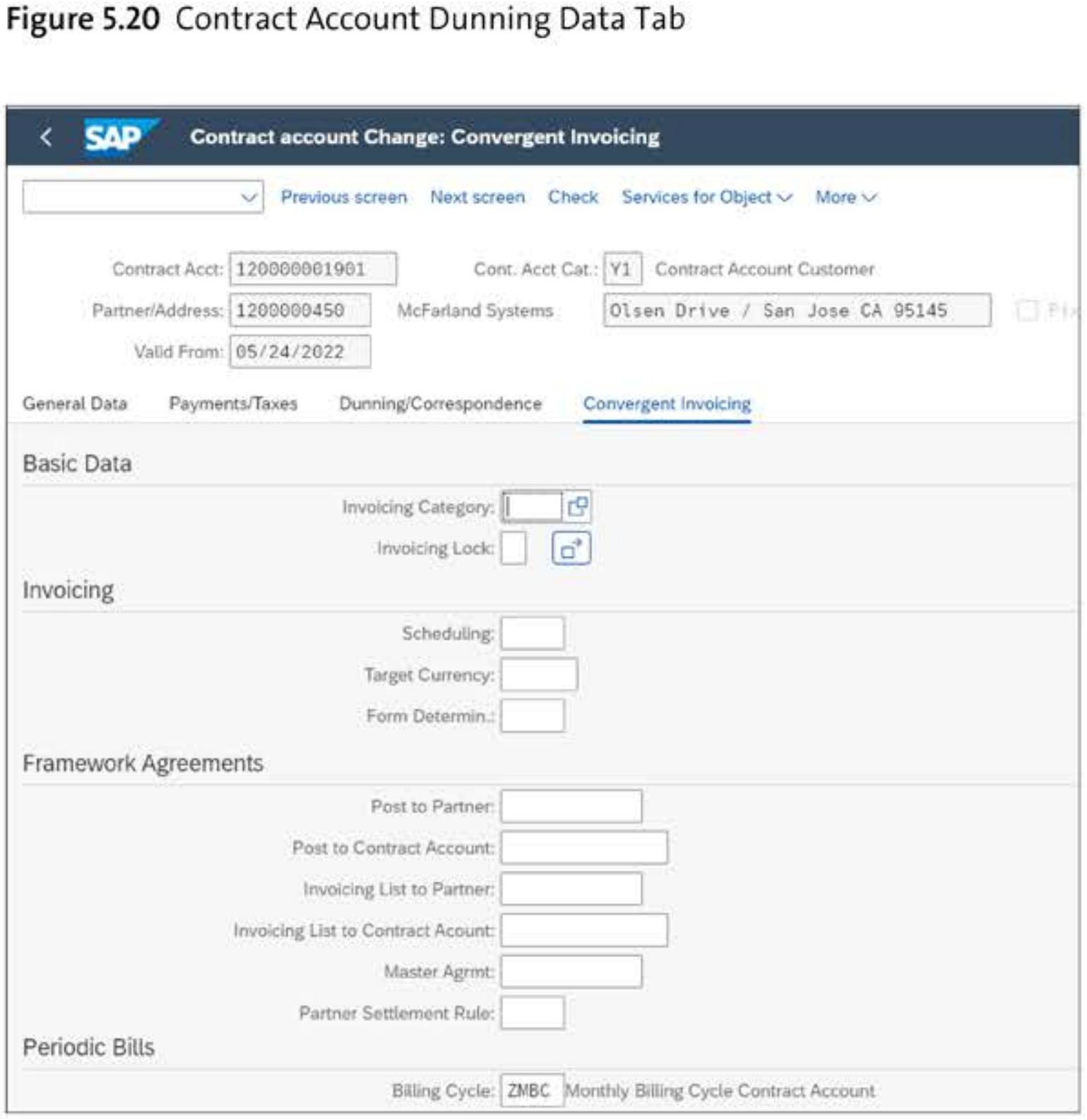

Figure 5.21 Contract Account Convergent Invoicing Tab

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The business agreement created in SAP Billing and Revenue Innovation Management, subscription order management, is replicated in contract accounting as a contract account. This is defined in contract accounting as part of the contract accounting integration with SAP CRM. See Section 5.1.2 for contract accounting integration with SAP CRM to replicate the business agreement as a contract account in contract accounting.

5.1.5 Provider Contract

A provider contract is the contractual agreement between the business partner and the provider offering the contract. The provider contract contains a header section to capture general data such as name, business partner, and contract header validity dates. It also contains an item section in which some key item-level details are captured, such as item text; contract account; master agreement (see Chapter 2, Section 2.1.4); organizational data determining which sales organization, distribution channel, and division combination the contract belongs to; product information; and billing cycle and discount information. There can be multiple provider contracts created for a business partner and contract account combination.

Figure 5.22 illustrates the relationship between the business partner, contract account, and provider contract. As mentioned, the provider contract is a lower-level item that represents the contract. A provider contract can also contain multiple items pertaining to different contract accounts.

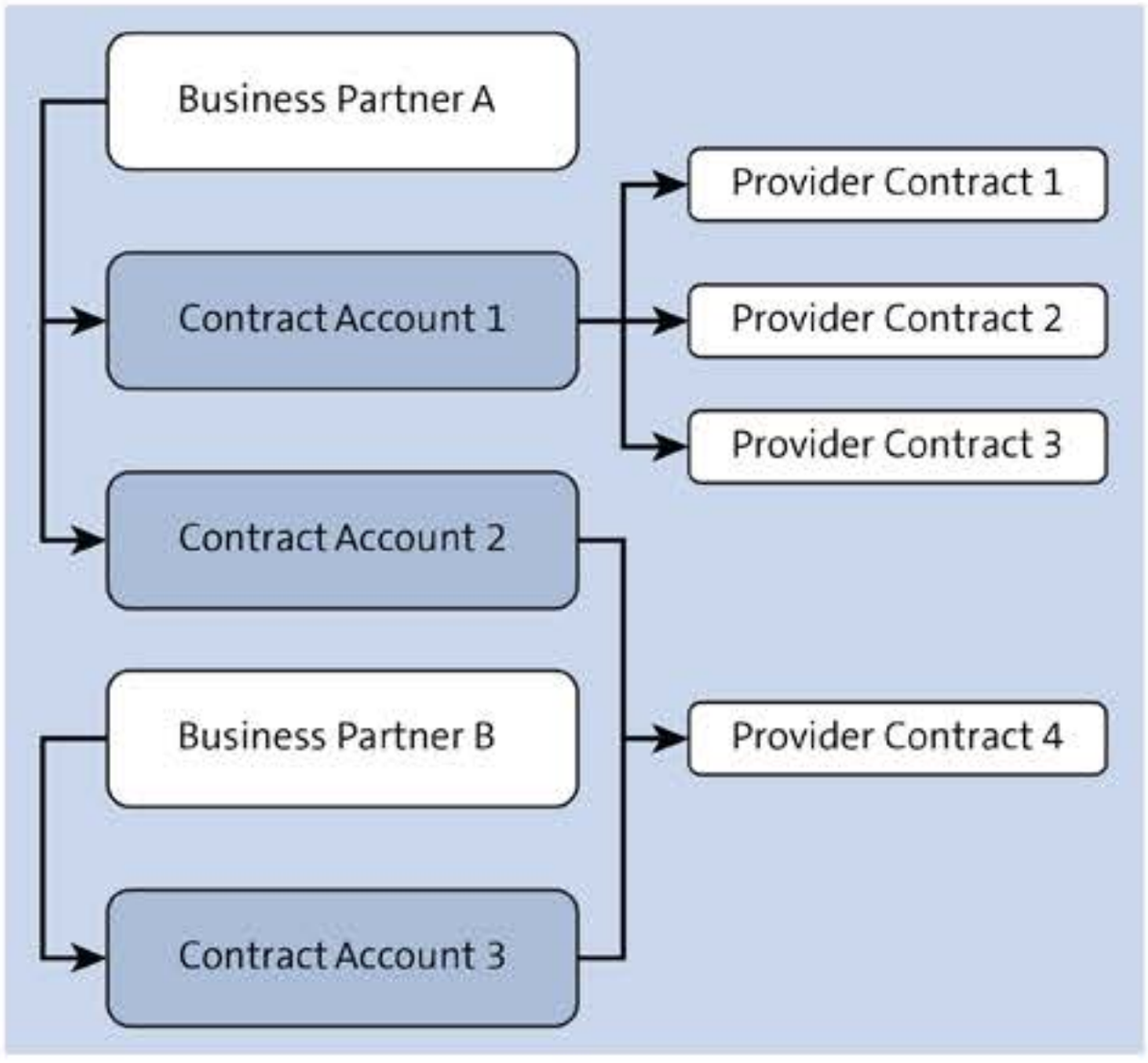

Figure 5.22 Business Partner, Contract Account, and Provider Contract Relationship

The configuration for provider contract setup in contract accounting is found under **Contract Account Receivables & Payable • Basic Functions • Provider Contract • Activate Provider Contract**.

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Now let's define the central settings for the provider contract, as shown in Figure 5.23.

In addition to this setup, it's necessary to define identification types for the provider contract. For the Martex example, a customer ID has been defined to uniquely identify the provider contract in contract accounting.

The configuration to define types of IDs for the provider contract setup in contract accounting is found under **Contract Account Receivables & Payables • Basic Functions • Provider Contract • Define the Type of ID**.

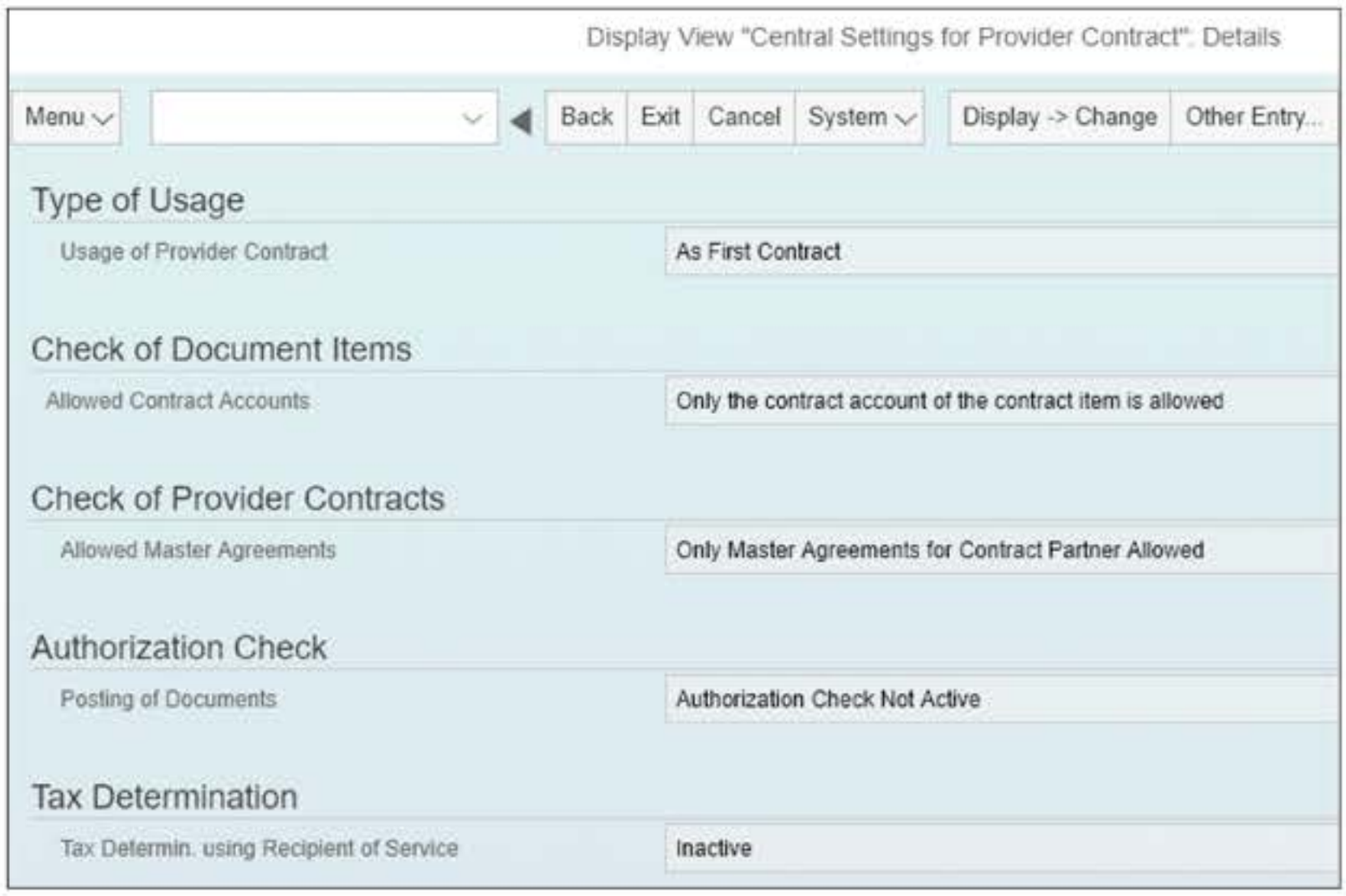

Figure 5.23 Defining the Provider Contract in Contract Accounting

As shown in Figure 5.24, **Booking ID BK** has been configured as a unique identifier for the contract. A contract created for a customer will have a unique identifier (e.g., a mobile number).

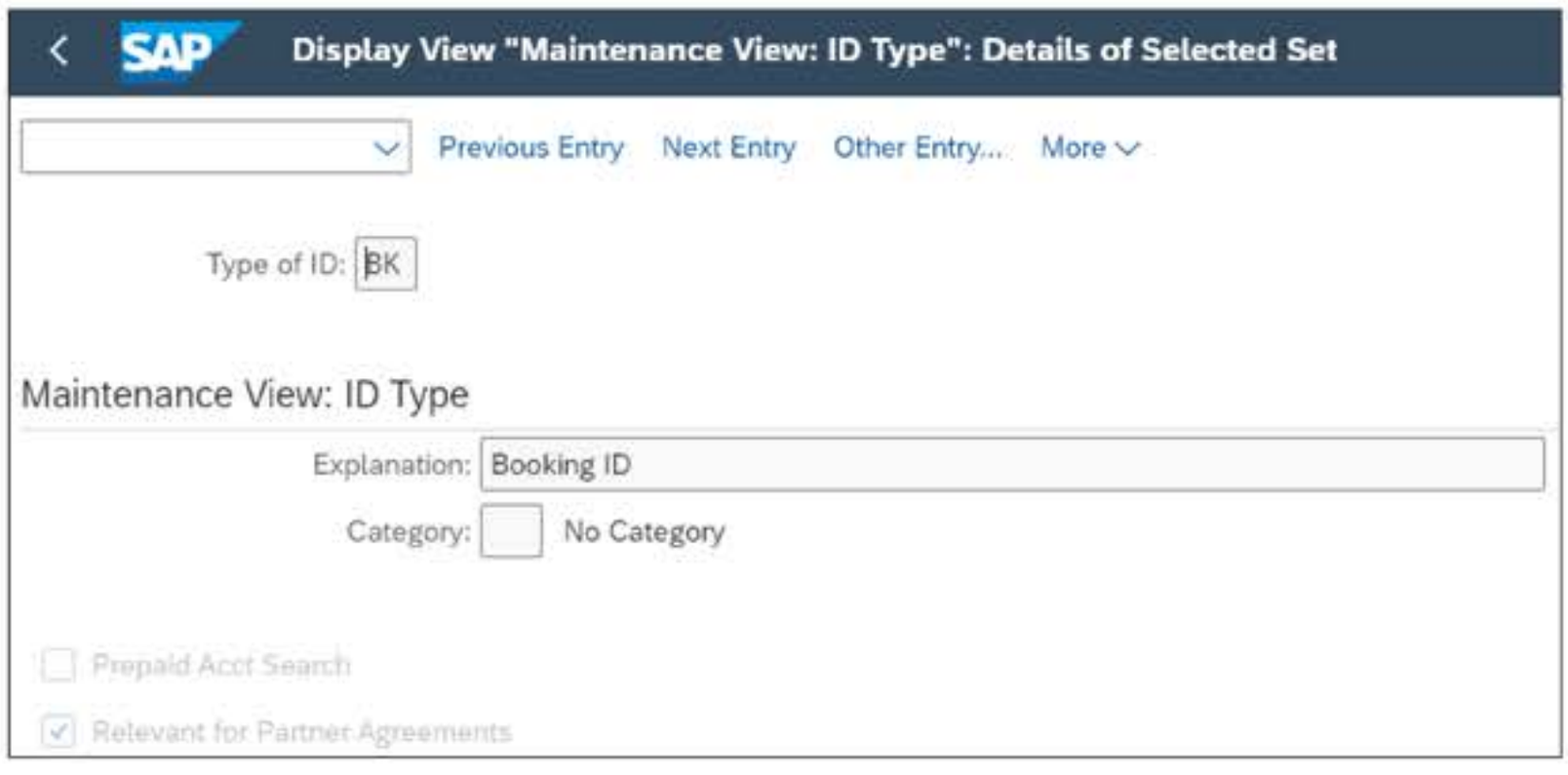

Figure 5.24 Type of ID for Customer Identification

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Once the type of identification has been configured, select the **Relevant for Partner Agreements** checkbox. This will ensure that the corresponding partner agreement (for any partner payouts) is looked up automatically with reference to this unique ID.

As explained in Section 5.1.3, the subscription product is defined as a contract accounting product, which is assigned to the provider contract in contract accounting. Now, as shown in Figure 5.25, define the type of product assigned to the provider contract. The contract accounting product is a lean product structure that contains the product description and sales area data. The menu path to maintain the central settings for a product is **SAP Customizing Implementation Guide • Financial Accounting • Contract Accounts Receivable and Payable • Basic Functions • Provider Contract • Make Central Settings for Products**.

Figure 5.25 Product Storage Definition Indicating Contract Accounting Product as Product Used for Provider Contract

The setup explained here was focused on the key configuration elements required to create and replicate a provider contract in contract accounting.

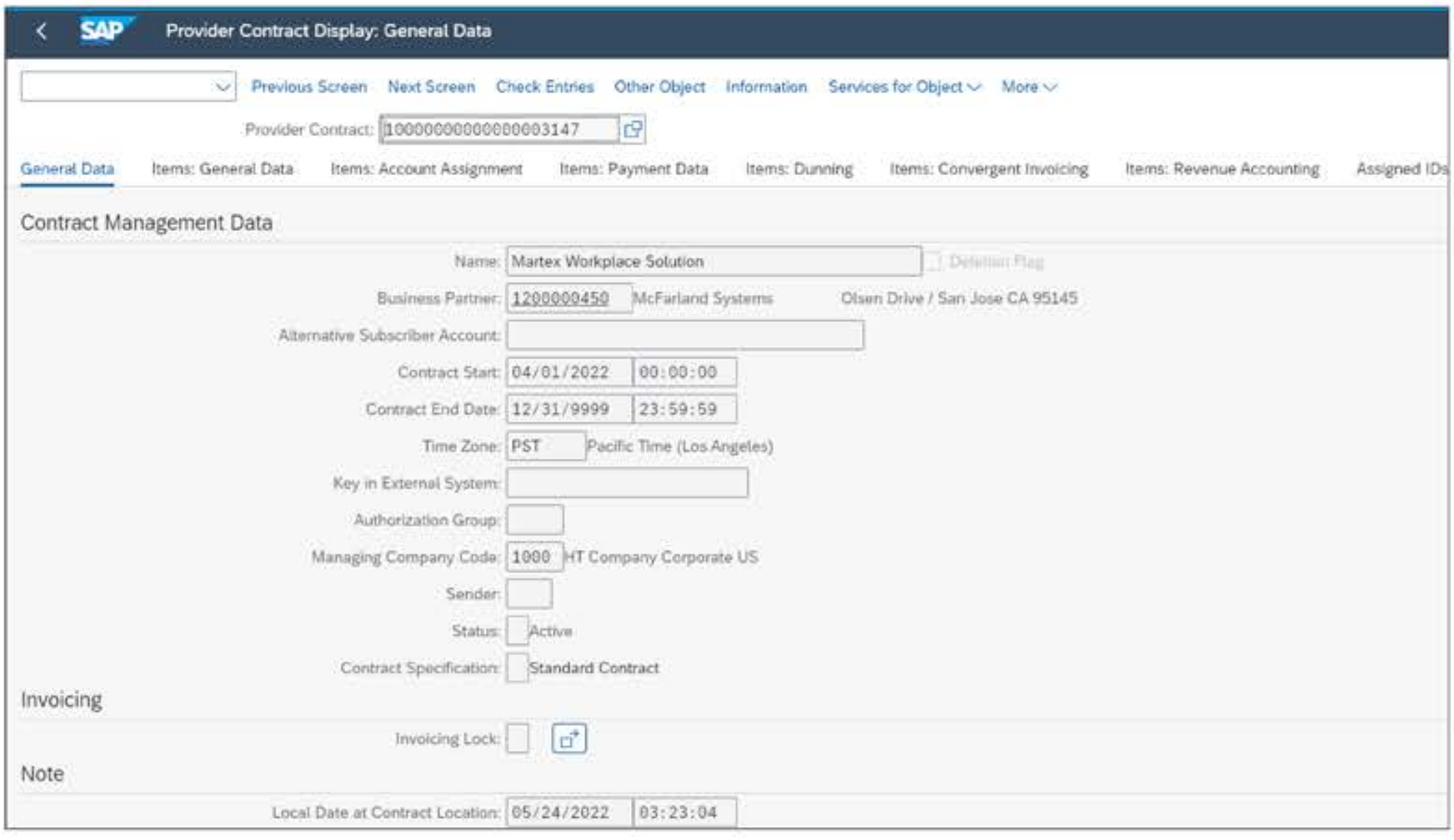

Figure 5.26 Provider Contract General Data Tab

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Now, use Transaction FP_VT3 to view the provider contract in contract accounting, which has been created for the Martex example case study. Figure 5.26 shows the details of the general data stored in the provider contract, which represents the header section of the contract. Here you see key attributes such as **Contract Start**, **Contract End Date**, and **Managing Company Code**.

Figure 5.27 and Figure 5.28 show the contract item view of the provider contract. The item-level details—**Valid From**, **Valid To**, **Contract Account**, **Sales Organization Data**, **Company Code**, and **Product**—are stored in this section.

As shown in Figure 5.29, the **Booking ID** configured to uniquely identify the contract, **BK**, is stored in this section of the provider contract.

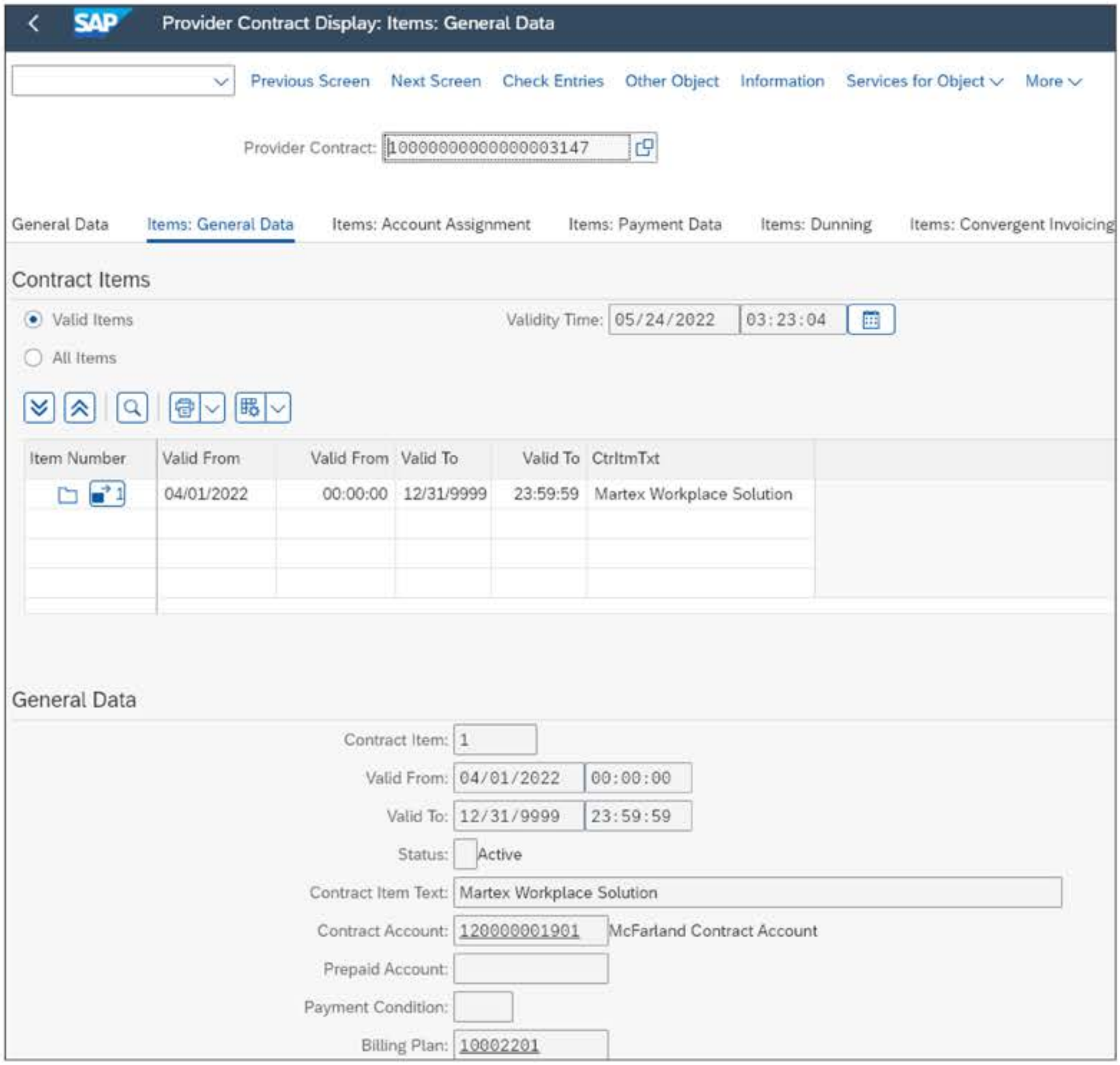

Figure 5.27 Provider Contract General Data Item View

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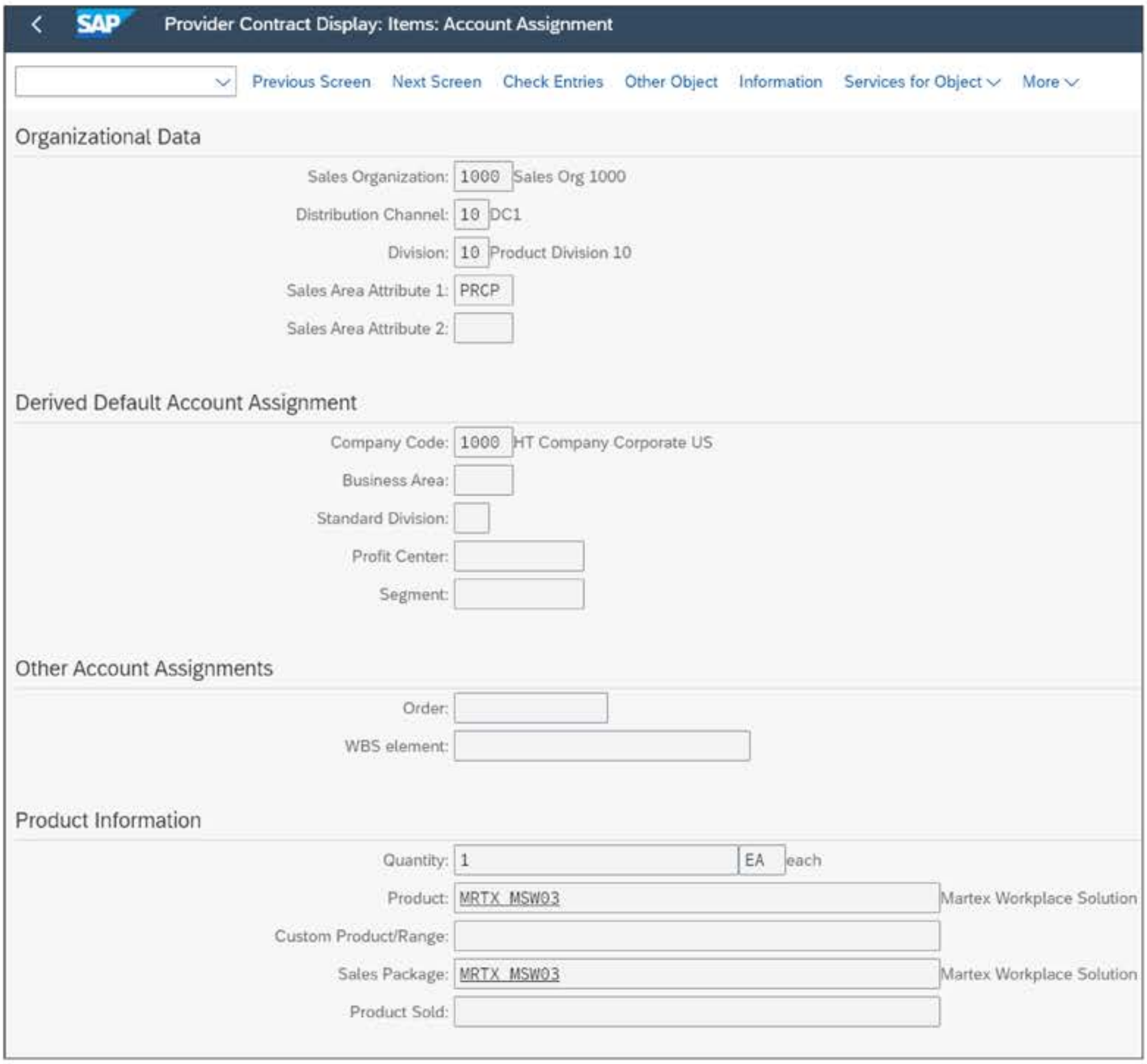

Figure 5.28 Provider Contract Account Assignment View

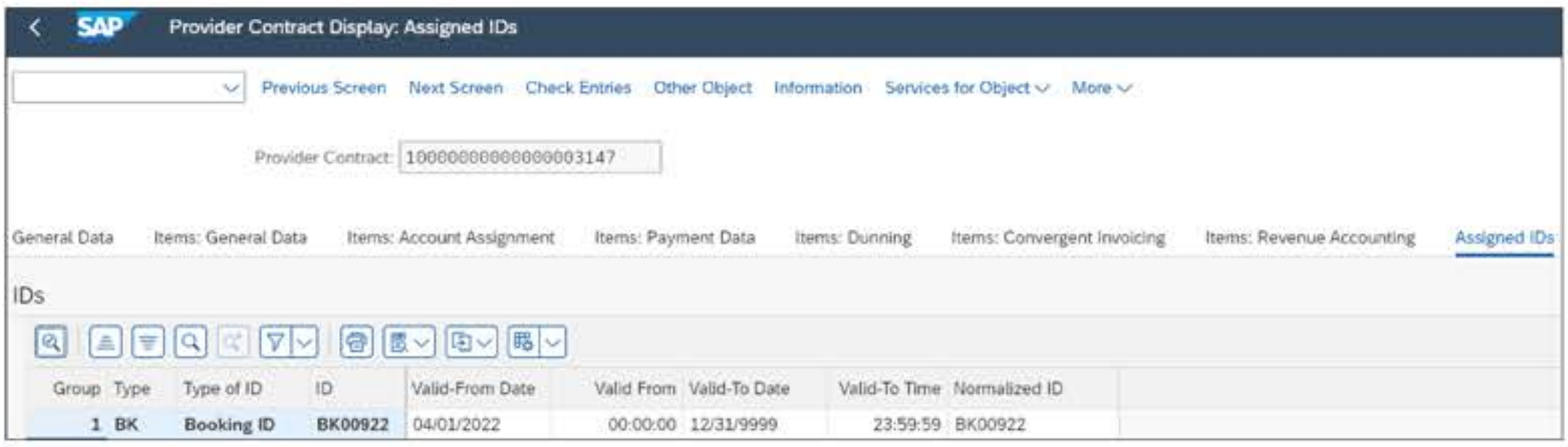

Figure 5.29 Provider Contract Assigned IDs

Billable items created for the provider contract inherit the attributes stored in the provider contract. For example, the billing cycle stored in the provider contract will be used to determine the billing date for the billable item. If the billing cycle isn't stored in

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the provider contract, the billing cycle at the contract account level will be picked up for processing. The values for the attributes maintained at the provider contract level precede the values maintained for the attributes at the contract account level.

5.1.6 Partner Agreement

A partner agreement is the contractual agreement between the business partner and the provider offering the contract. The partner agreement contains a header section to capture general data such as name, business partner, and contract header validity dates. It also contains an item section in which some key item-level details are captured, such as item text; contract account; master agreement (see Chapter 2, Section 2.1.4); organizational data determining which sales organization, distribution channel, and division combination the contract belongs to; product information; and billing cycle and discount information. There can be multiple provider contracts created for a business partner and contract account combination.

Now use Transaction FP_PV3 to view the partner agreement object in contract accounting. Figure 5.30 illustrates the partner agreement object in contract accounting representing the vendor agreement.

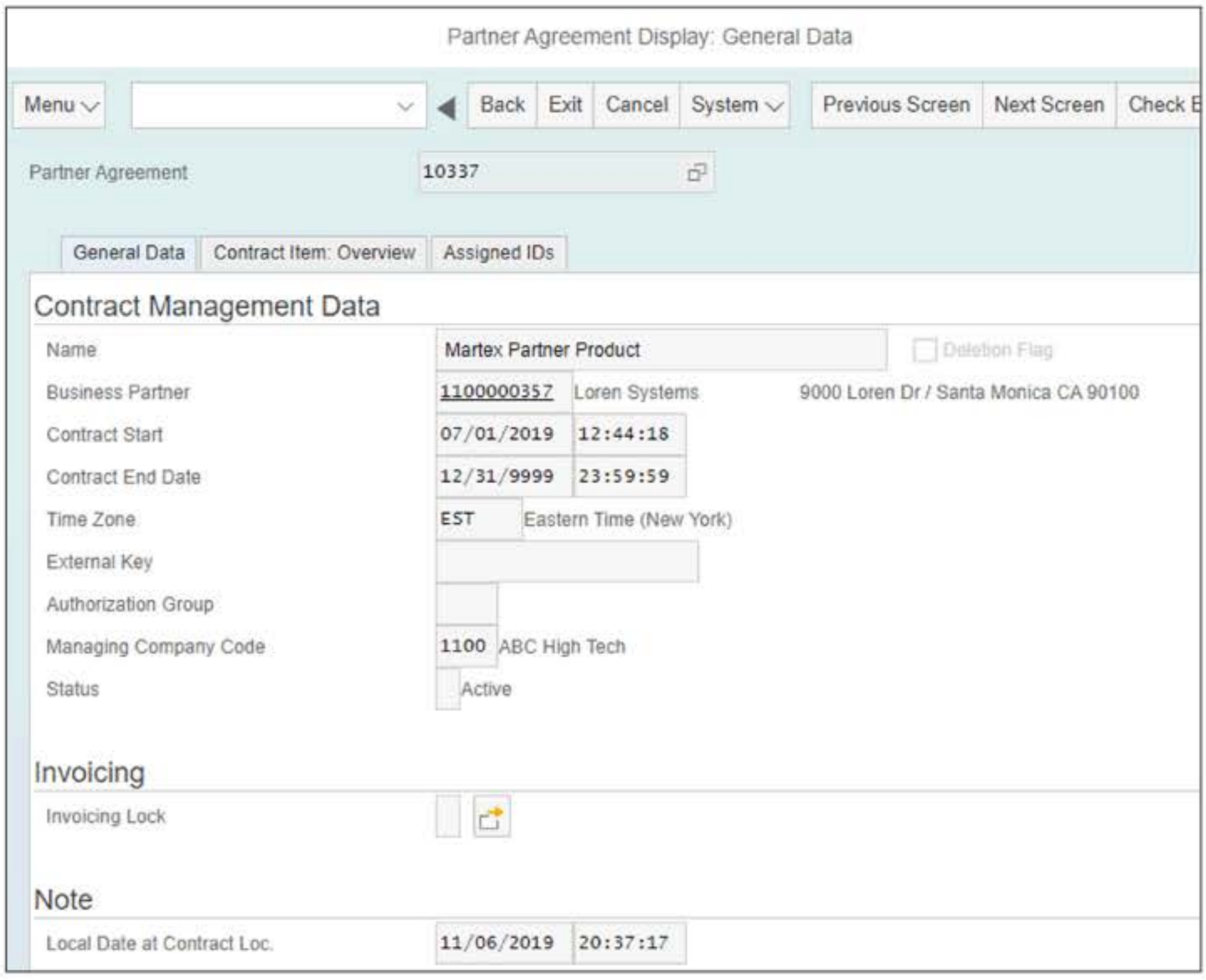

Figure 5.30 Partner Agreement in Contract Accounting: General Data

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This object has been replicated in contract accounting from SAP Billing and Revenue Innovation Management, subscription order management. The **General Data** tab contains some key attributes such as the business partner (representing a vendor, Loren Systems), contract start date, contract end date, and company code.

Figure 5.31 shows the contract item view of the partner agreement. Note the key fields such as **Valid From**, **Valid To**, **Contract Account**, and **Partner Settlement Rule** that are stored in this section.

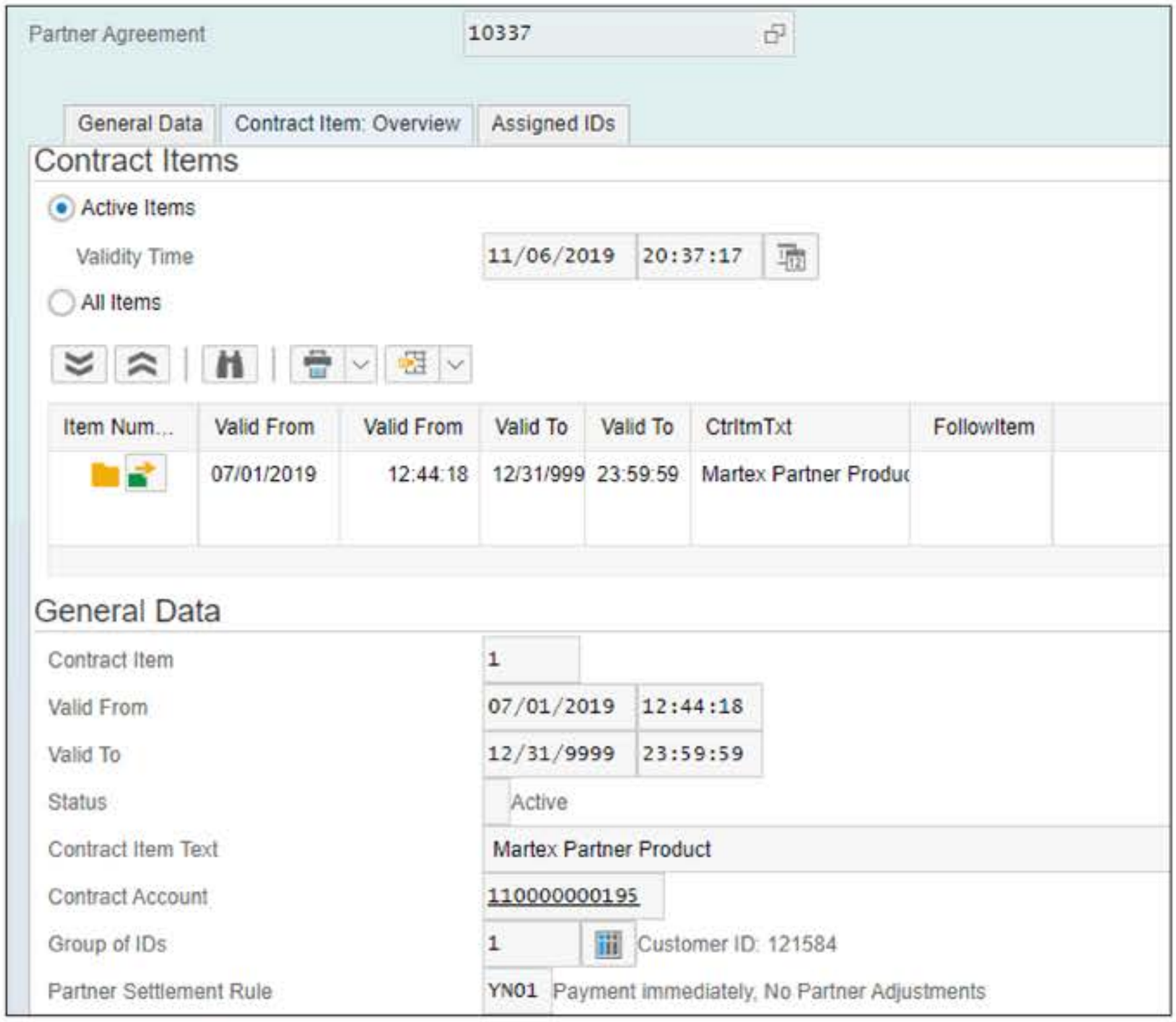

Figure 5.31 Partner Agreement in Contract Accounting: Contract Item Overview

Chapter 4, Section 4.2.6 , explains the partner settlement setup and the partner settlement rule definition in detail. The partner settlement rule is referenced in the partner agreement.

Figure 5.32 shows customer ID **CN**, which is referenced in the partner agreement.

As explained in Chapter 4, Section 4.2.6 , this ID will be referenced in the revenue billable item, based on which, the partner agreement will automatically be determined for any given transaction relevant for partner payouts.

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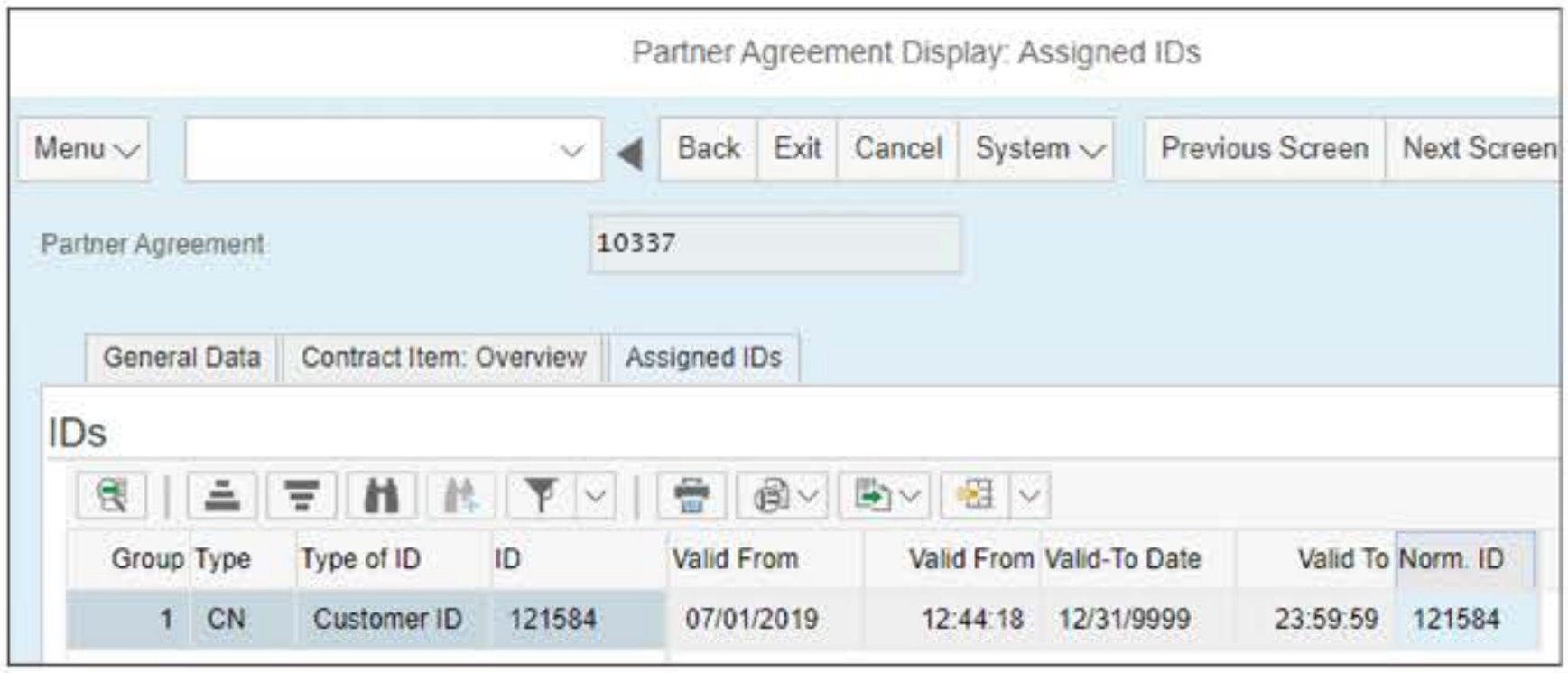

Figure 5.32 Partner Agreement in Contract Accounting: Assigned IDs

5.2 Configuring Contract Accounting

In this section, we'll continue to look at some key configuration elements for posting a document, viewing the account balance display, and performing open item management in contract accounting. We'll explain key concepts such as main transactions and subtransactions needed to determine accounts for posting a contract accounting document. Figure 5.33 shows how the general ledger account for receivables, revenue, and expense is derived based on the combination of key transaction attributes such as company code, division, account determination, main transaction, and subtransaction.

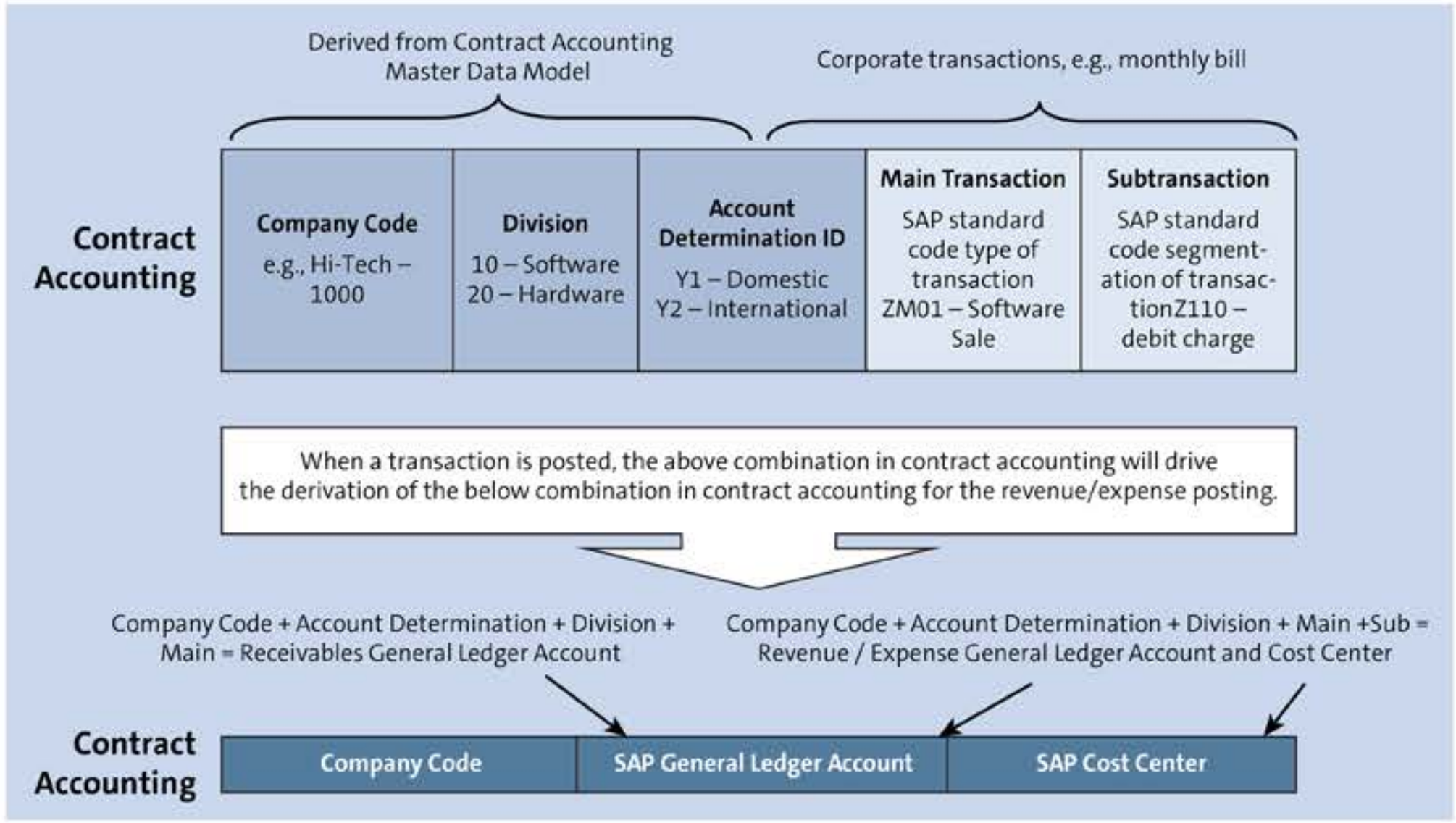

Figure 5.33 General Ledger Account Determination for Financial Posting

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5.2.1 Account Determination

In this section, we'll explain basic concepts such as the account determination ID, main transactions, and subtransactions related to contract accounting document postings, and then we'll configure the same in contract accounting. The account determination ID helps automatically determine the general ledger account for a business transaction. See Section 5.1.4 for more details on account determination ID setup. When an invoicing document in SAP Convergent Invoicing is posted, a financial posting can be automatically enabled in contract accounting. The subledger posting and a general ledger posting is possible with a combination of company code, account determination, division, and main transactions and subtransactions.

Main Transaction

The main transaction is a key element used to determine the receivables and payables account, along with a combination of other document posting characteristics, as shown in Figure 5.34. Each main transaction has a description associated with the definition, which is printed on the customer's invoice.

The configuration setup for main transactions is found under **Contract Account Receivables & Payables • Basic Functions • Postings and Documents • Document • Maintain Document Assignments • Maintain Main Transaction**. Now, create a main transaction, as shown in Figure 5.34. This will be the main transaction used to determine the account assignment for the Martex example case study.

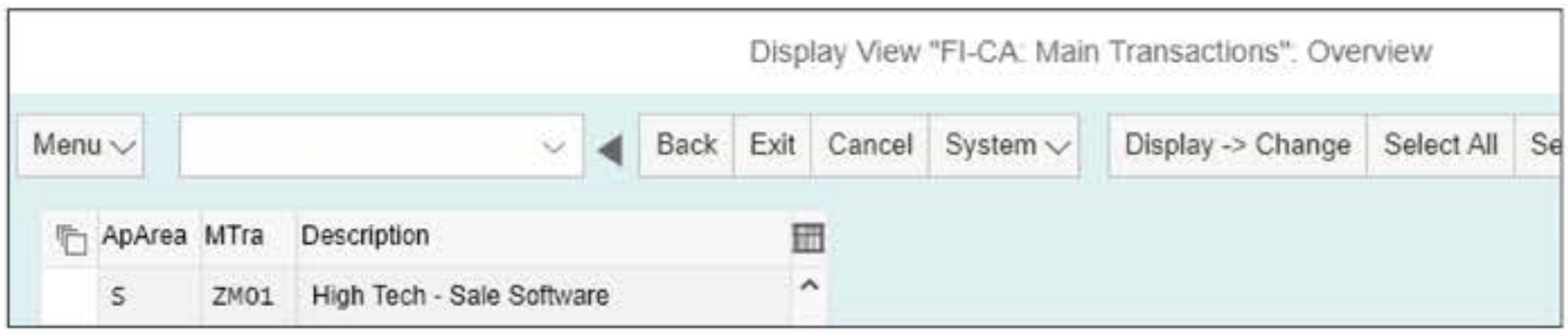

Figure 5.34 Contract Accounting: Main Transaction

Subtransactions

The subtransactions are key elements used to determine the revenue and expense accounts, along with a combination of other document posting characteristics, as shown earlier in Figure 5.33. See Section 5.2.5 for further details. Subtransactions also control the following:

- Tax determination
- Debit/credit posting for a transaction
- Additional account assignments (company account assignment and business area)

The configuration setup for subtransactions is found under **Contract Account Receivables & Payables • Basic Functions • Postings and Documents • Document • Maintain**

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Document Assignments • Maintain Sub Transaction. Now, create a subtransaction as shown in Figure 5.35 to determine the account assignment for the Martex example.

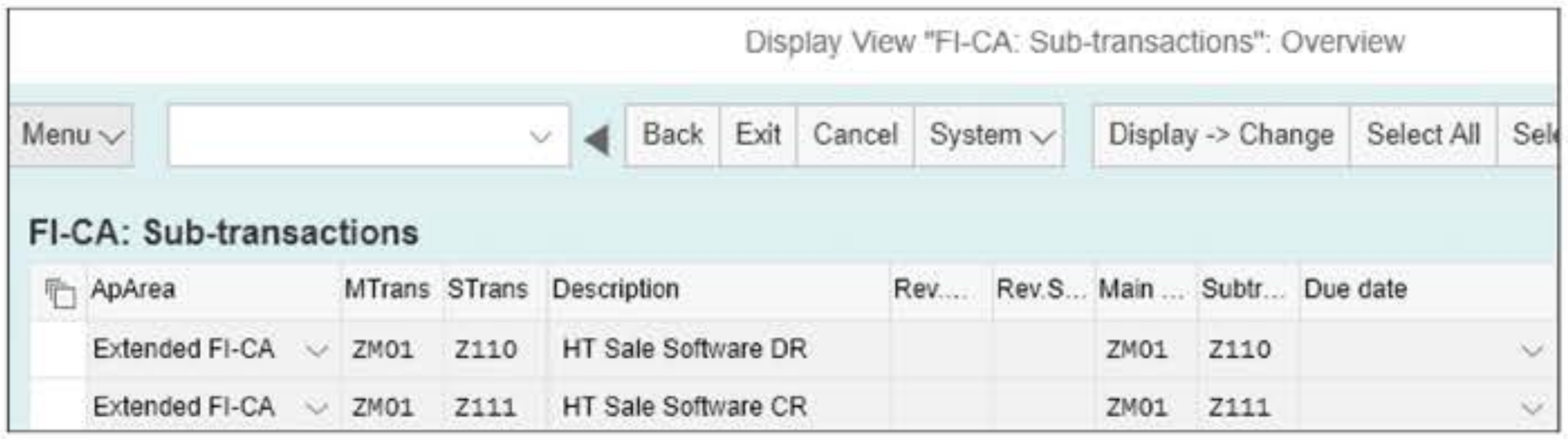

Figure 5.35 Contract Accounting: Subtransactions

Once the main and subtransactions are defined, the combination of these can be configured to determine the financial postings for any given transaction. See Section 5.2.5 for more on posting area 2610 (account assignment for general ledger items) and posting area 2611 (general ledger determination for business partner items).

5.2.2 Open Item Management

Every financial subledger posting as a result of a business transaction remains as an open item until it has been completely paid. An accounts receivable or accounts payable invoice posting to a contract account has to be paid and cleared to appear as a closed transaction. Similar to accounts receivable, contract accounting provides a capability for users to manage such open accounts receivable or accounts payable transactions.

Every open subledger posting can be tracked under the contract accounting subledger as an open item against which an incoming payment can be received or an outgoing payment can be posted. Various screen layouts can be configured to ensure that select fields relevant to the business process are available for display as part of an open item report. The report provides a view of open invoices as well as unapplied payments (if any), which aids accounts receivable in carrying out cash application.

Let's now look at some of the key steps involved in configuring open item management in contract accounting. To start, navigate to menu path **Contract Account Receivables & Payables • Basic Functions • Open Item Management • Define Line Layout Variants for Open Item Processing**.

Next, configure **Line Layout YN1** with the variant fields, as shown in Figure 5.36. Some of the technical fields configured are `GPART` (Business Partner), `VKONT` (Contract Account), `OBPEL` (Document Number), `FAEDN` (Net Due Date), and `TXTU1` (Text). You can choose any field available in the contract accounting open item management table `FKKCLIT` as required.

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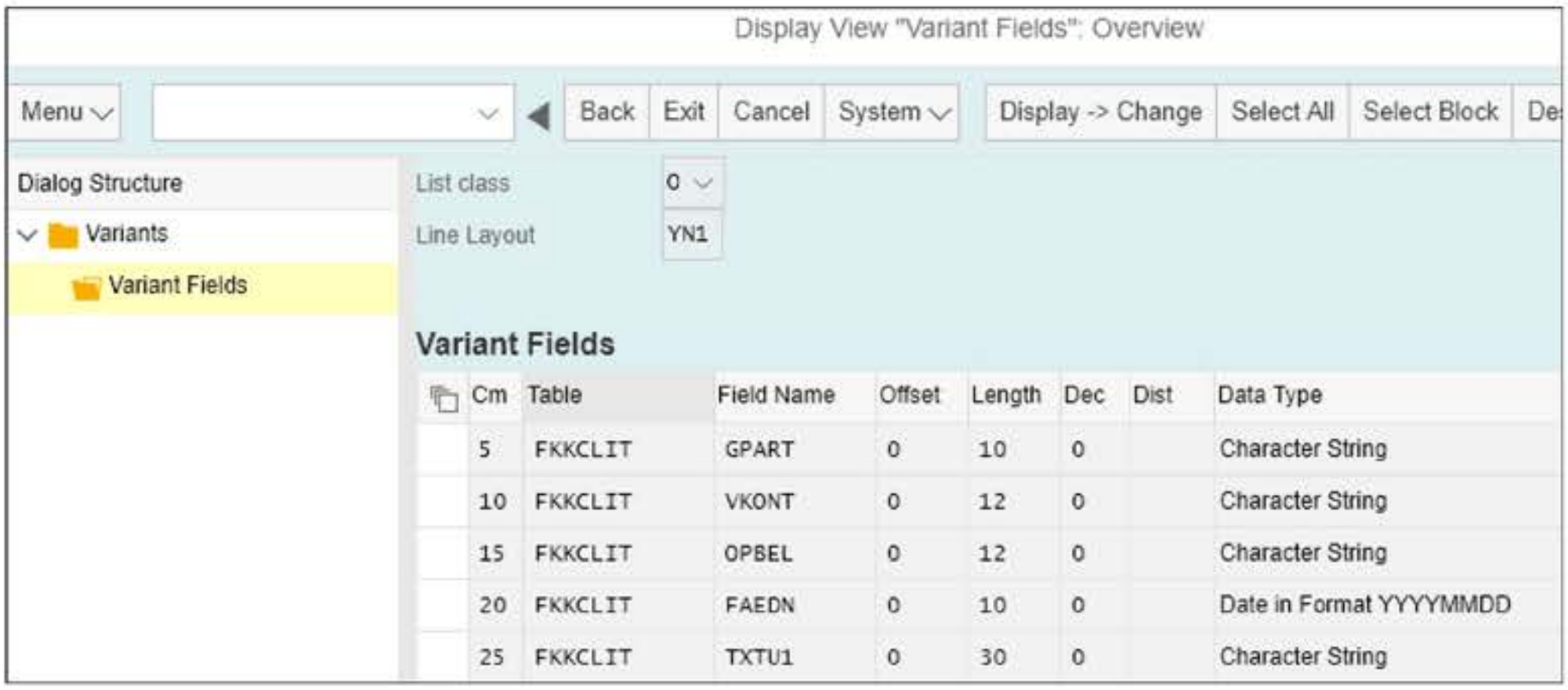

Figure 5.36 Line Layout Variants for Open Item Processing

Next, use Transaction FP06 (Account Maintenance: Process Open Items) to view the layout you've defined. As shown in Figure 5.37, enter the **Business Partner** ID, and then press Enter.

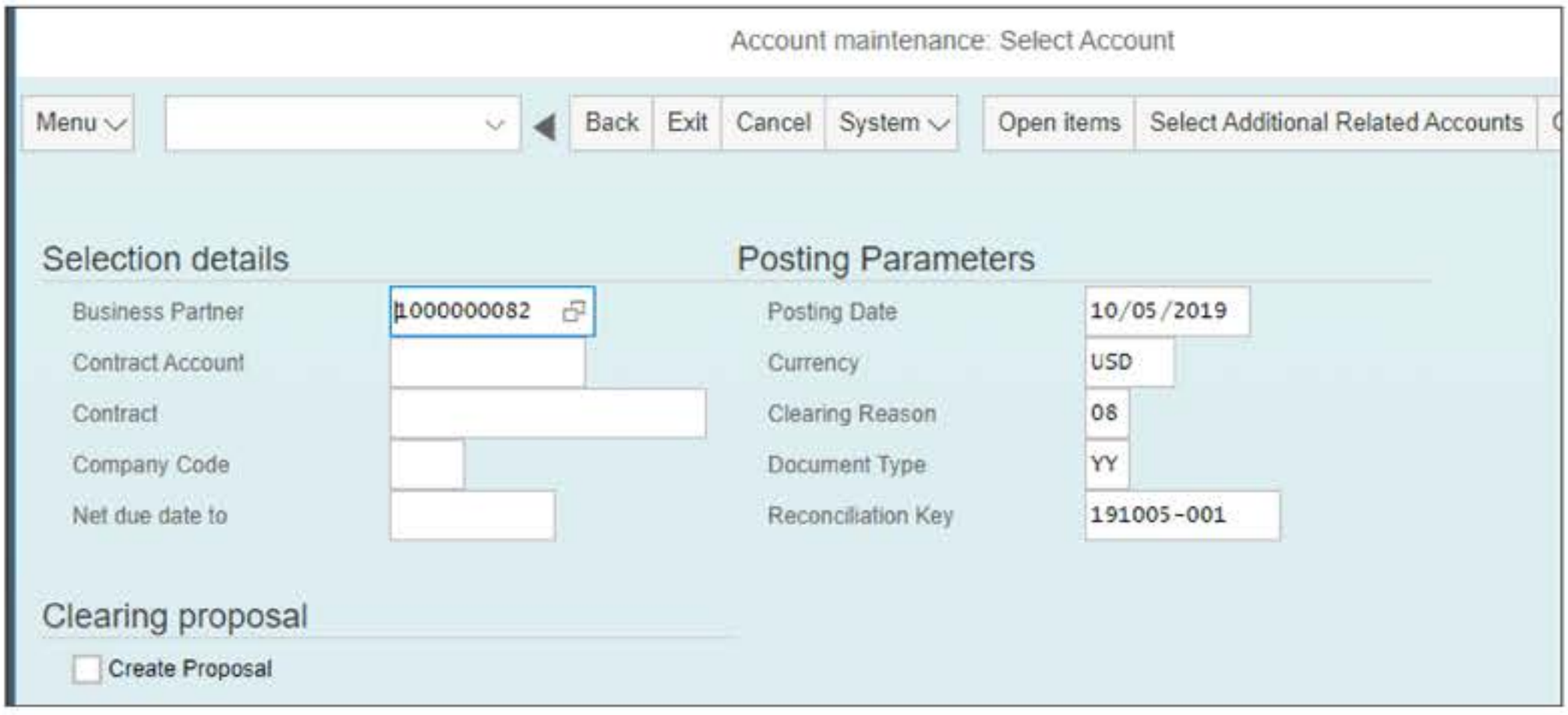

Figure 5.37 Open Item Management: Transaction FP06

As shown in Figure 5.38, the visible fields are configured like the line layout variants for open item management.

Appropriate tolerance groups can also be defined if needed to accept any minor payment differences. The tolerance group is primarily defined in accounts receivable and can be invoked as a configuration item within contract accounting.

The configuration setup for open item management is found under **Contract Account Receivables & Payables • Basic Functions • Open Item Management • Maintain Tolerance Groups**. The next step is to configure the tolerance group, as shown in Figure 5.39.

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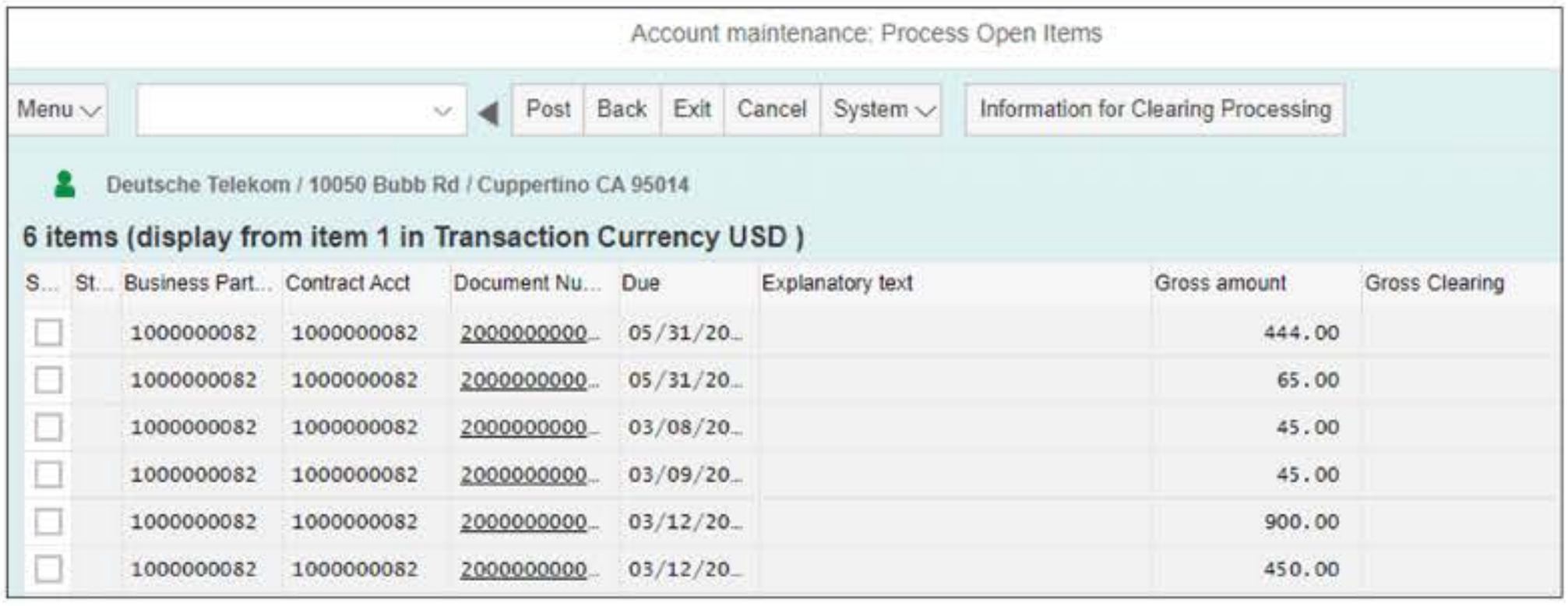

Figure 5.38 Account Maintenance: Process Open Items

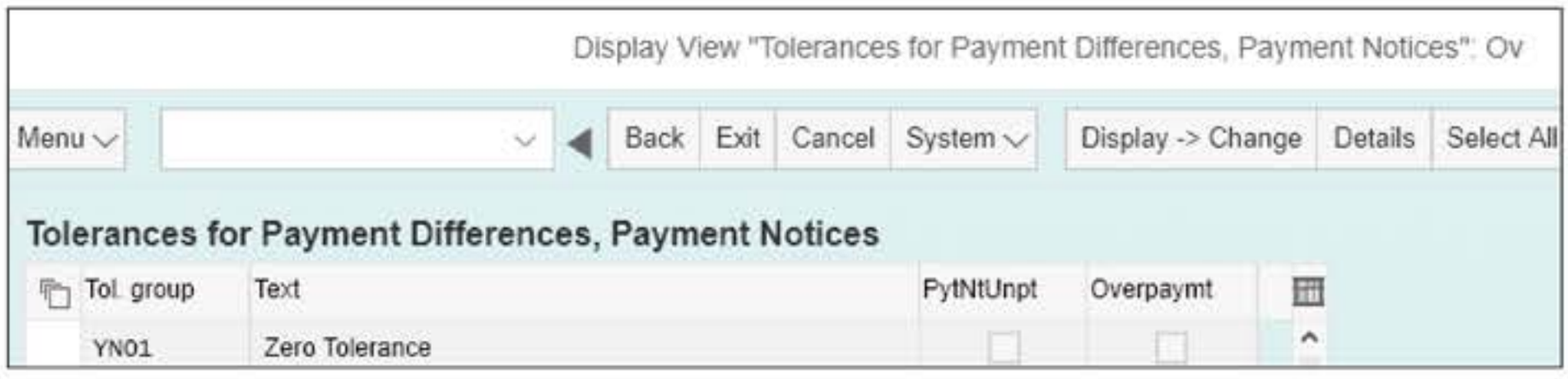

Figure 5.39 Tolerance Group for Payment Differences in Contract Accounting

Once the tolerance group has been defined, the next step is to determine the dollar or percentage of tolerance allowed, which has been set to zero (**Zero Tolerance**) in this case, as shown in Figure 5.40.

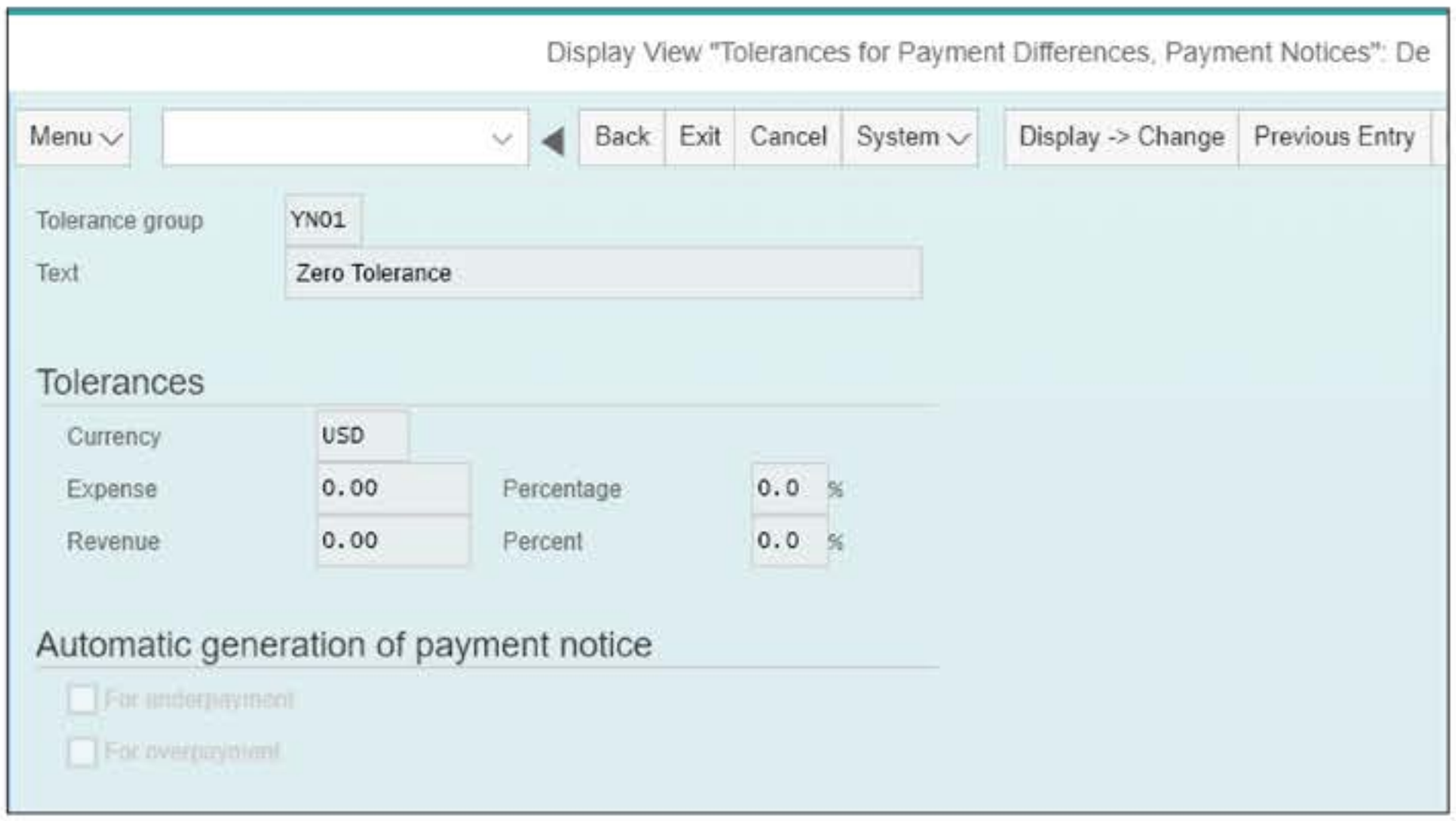

Figure 5.40 Tolerance Amount for Payment Differences during Open Item Clearing

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5.2.3 Account Balance Display

The account balance display provides a detailed view of the account transactional information, including any open, cleared, or statistical transactions for a business partner. The account balance can be viewed for a given business partner and contract account relation or for all the contract accounts for a business partner. The report also can be executed for a group of business partners. Various screen layouts can be configured to ensure that selection fields relevant to the business process are available for display as part of the report output. The configuration setup for account balance display is found under **Contract Account Receivables & Payables • Basic Functions • Account Balance Display • Define List Category**. Now, create an account balance list type as the first step. As shown in Figure 5.41, create your own **List Type** (in this case, enter "ALLES" for *all items*) definition for the account balance.

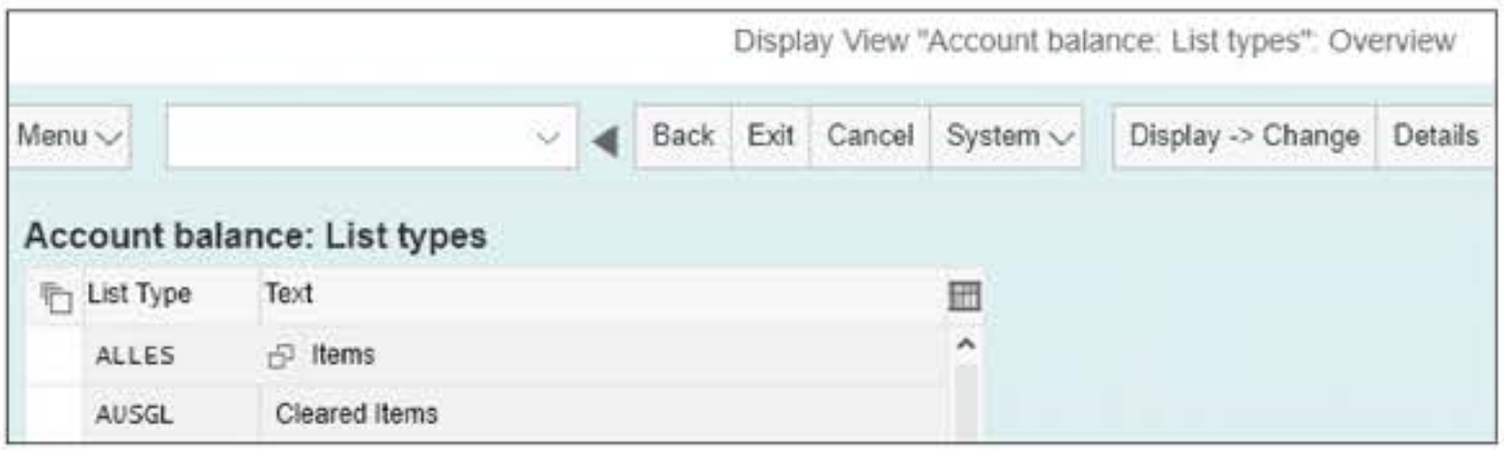

Figure 5.41 Account Balance: List Type

Once the list type has been created, refer to Figure 5.42 to configure further selections to display **Openitems**, **Cleared items**, and **Non-stat. items** (nonstatistical items). The list type will need to be selected on the account balance display screen to execute the report.

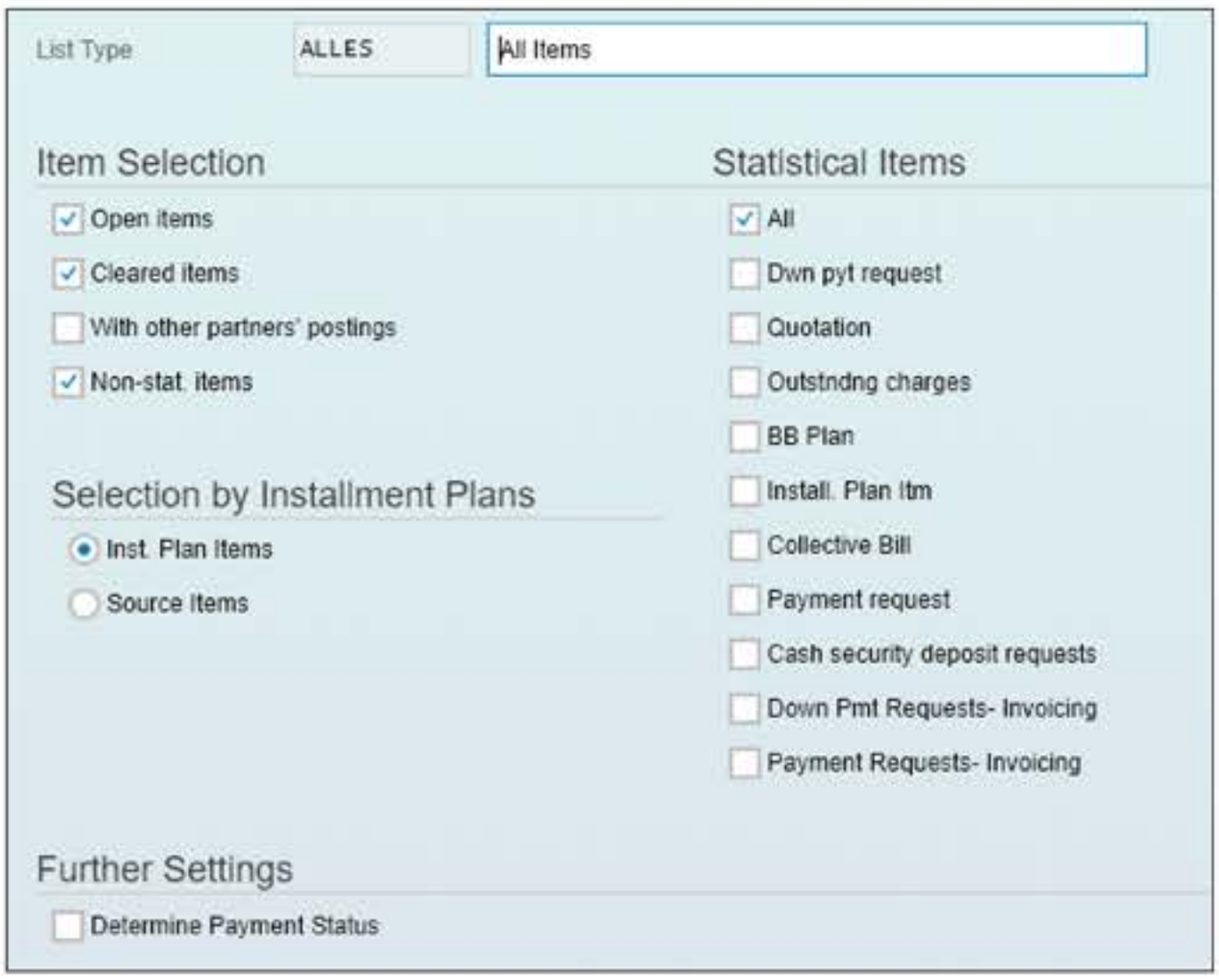

Figure 5.42 Account Balance: List Type Selection

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Once the account balance list type has been configured, the next step is to configure the line layout variants for the account balance display. Navigate to menu path **Contract Account Receivables & Payables • Basic Functions • Account Balance Display • Define Line Layout Variants for Account Balance**. Now configure the line layout as shown in Figure 5.43. contract accounting enables you to define your own line layout for the account balance display. This configuration screen contains two main sections. Enter the header of the layout with the desired spacing in the **Text** section.

Next, select the **Fields of the variant** that need to be displayed on the account balance report. See Figure 5.43 for the field selections.

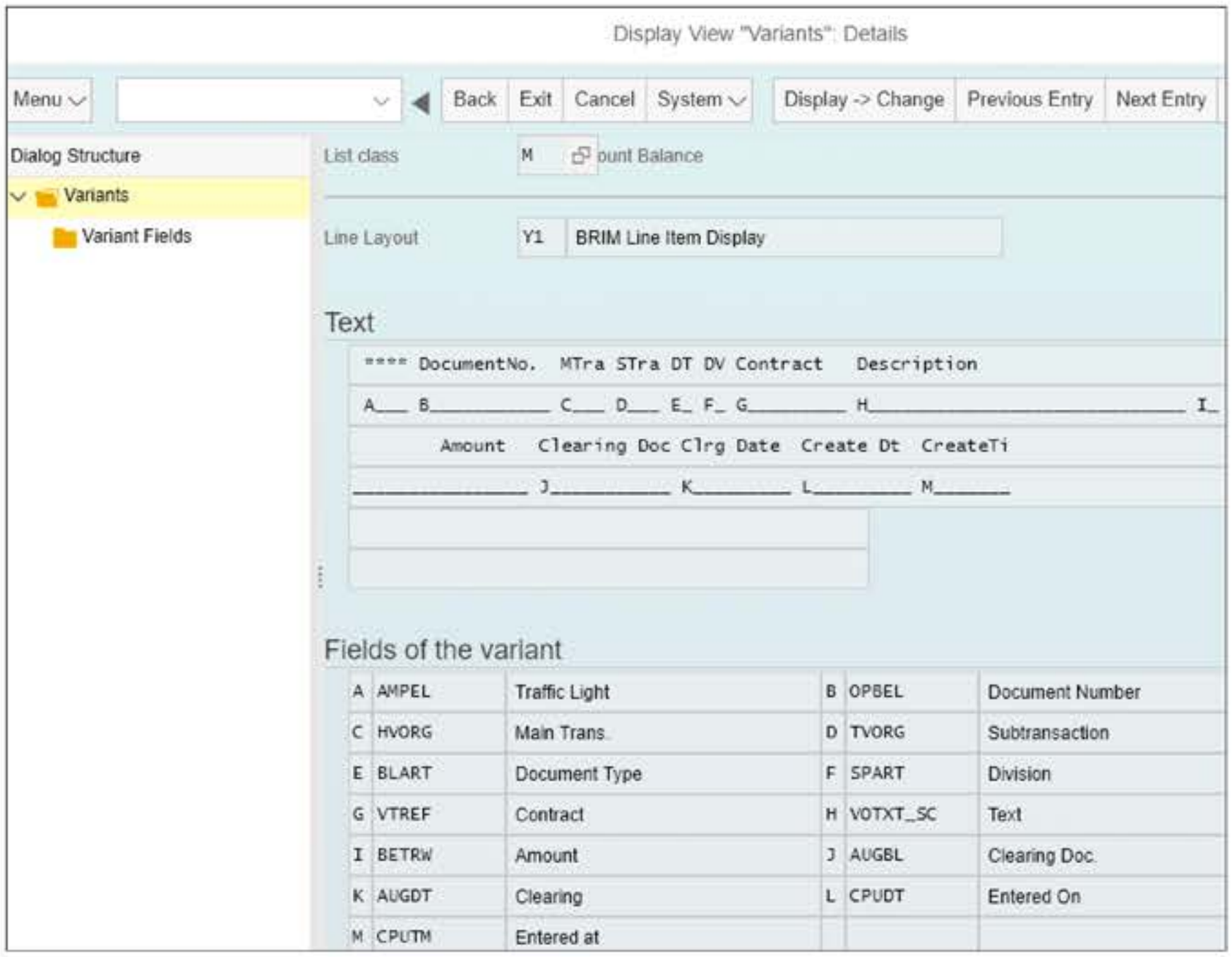

Figure 5.43 Account Balance Display: Line Layout Variant

For further details on how to use the account balance display, Section 5.4.5.

5.2.4 Payments and Returns

Similar to accounts receivable, contract accounting also provides standard out-of-the-box capabilities to handle incoming payments to clear open accounts receivable and outgoing payments to clear open accounts payable invoices. Contract accounting also supports the returns process in case of payment returns from the bank.

This section highlights the capabilities of incoming and outgoing payments that can be enabled within contract accounting, including payment lot, check lot, credit card lot, and more.

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Clarification account postings, which are a key feature of contract accounting, enable posting to a clarification account in case a corresponding open item or contract account isn't determined automatically for payment posting.

This book doesn't dwell on the specifics of enabling incoming payments and clearing the open accounts receivable posted for our running example. Our primary focus is on the integration aspects of contract accounting within the SAP Billing and Revenue Innovation Management solution.

The automatic payment program within contract accounting enables payment postings and clearing of open accounts receivable and accounts payable transactions, in addition to creating a payment medium output with payment instructions to carry out bank transfers, check payments, and so on.

Electronic bank statement postings within contract accounting enable bank statement reconciliation and clearing of open accounts receivable postings in case of bank transfers. The bank statement setup includes integration aspects of the electronic bank statement setup between financial accounting and contract accounting.

As of SAP S/4HANA 1909, contract accounting also provides a feature to receive and automate payment clearing by receiving bank files that contain a customer's check information.

5.2.5 Integration with SAP Convergent Charging

Replication of business partners (as external accounts in SAP Convergent Charging) and contract accounts (as subscriber accounts in SAP Convergent Charging) is enabled via integration between contract accounting and SAP Convergent Charging. Similarly, provider contracts are also replicated from SAP Billing and Revenue Innovation Management, subscription order management, to SAP Convergent Invoicing/contract accounting and then to SAP Convergent Charging.

Some of the key integration aspects of contract accounting with SAP Convergent Charging are as follows:

- **Activate joint replication of all master data**
  This option ensures that the business partner, contract account, and provider contract are replicated from contract accounting to SAP Convergent Charing at the same time. The business partner is created first, followed by the contract account, and finally the provider contract. If this option is selected, the business partner and the contract account remain in contract accounting and aren't replicated to SAP Convergent Charging until the provider contract is created in contract accounting.

  Navigate to menu path **Contract Accounts Receivable and Payable • Integration • Convergent Charging • Activate Joint Replication of All Master Data**. Select the **Joint Replication of All Master Data to SAP CC** checkbox, as shown in Figure 5.44.

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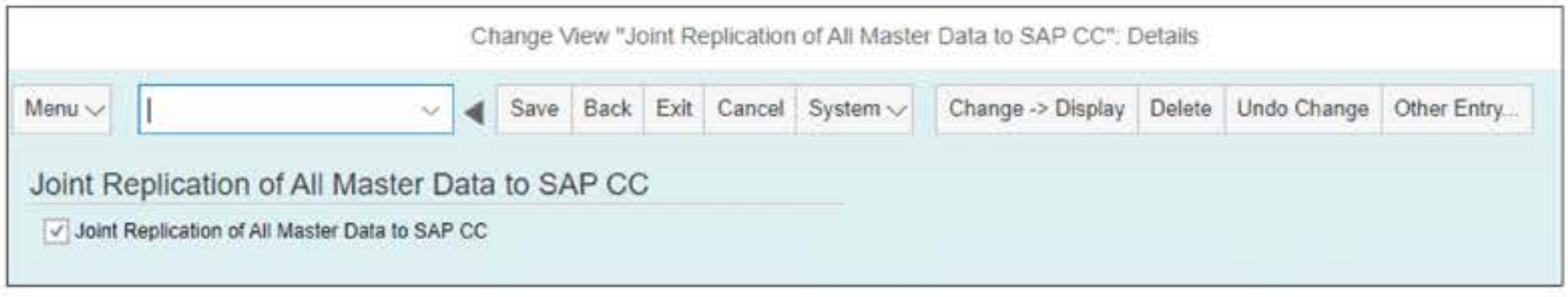

Figure 5.44 Contract Accounting Integration with SAP Convergent Charging: Joint Replication of Master Data

- **Make specifications for connecting multiple contract accounting systems to an SAP Convergent Charging system**
  It's possible to connect multiple contract accounting systems to an SAP Convergent Charging System. This a feature provided by SAP Billing and Revenue Innovation Management. We aren't using this feature for our example case study, so we won't activate it.
- **Specifications for connecting multiple SAP Convergent Charging systems to a contract accounting system**
  It's also possible to connect multiple SAP Convergent Charging systems to a contract accounting system. We aren't using this feature for our example case study, so we won't activate it.

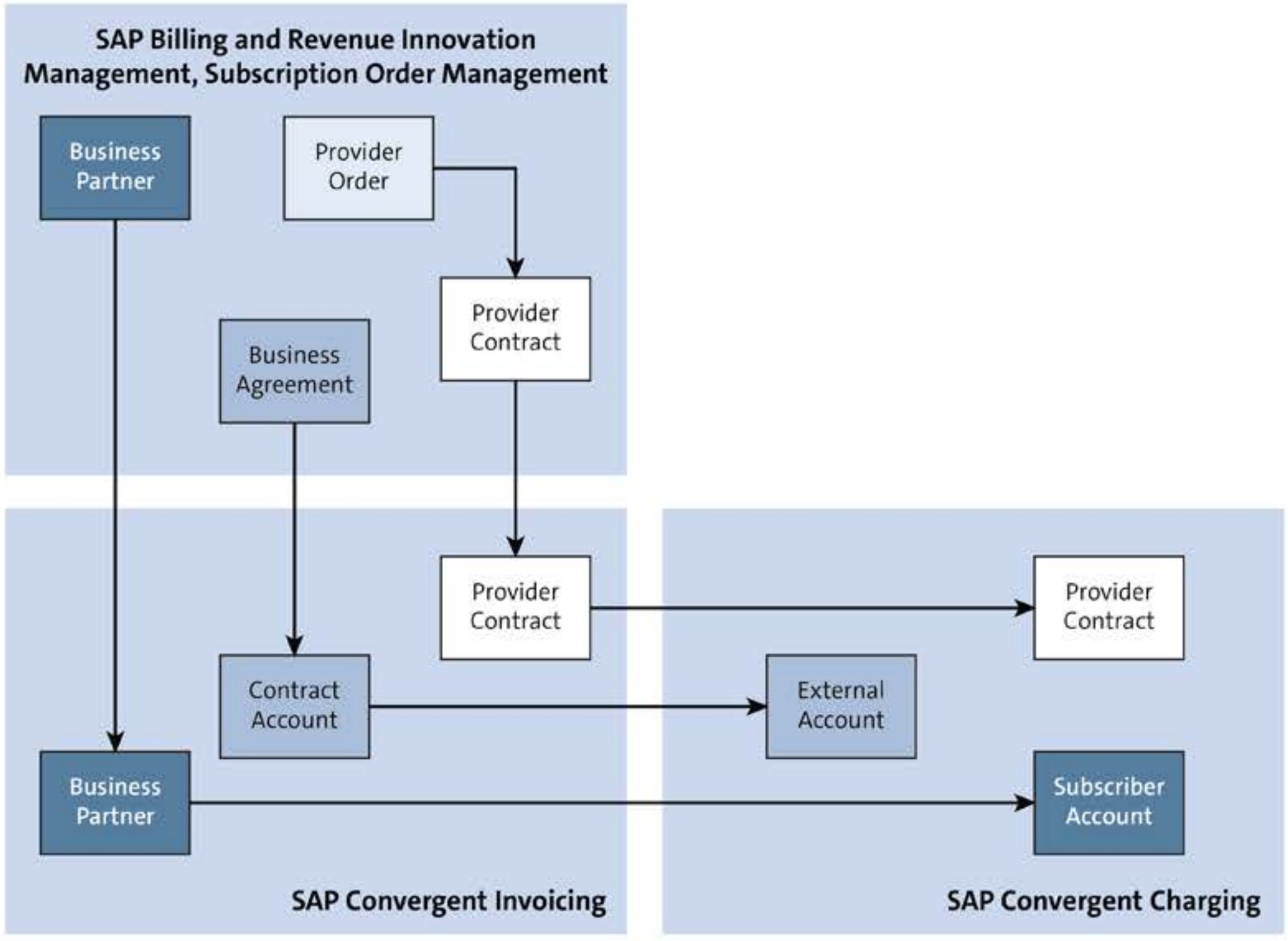

Figure 5.45 Master Data Replication

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Master data replication between SAP Billing and Revenue Innovation Management, subscription order management; SAP Convergent Charging; and SAP Convergent Invoicing can be seen in Figure 5.45.

In addition to this integration setup, Transaction SOAMANAGER settings are needed to enable replication of data and transactions between SAP Billing and Revenue Innovation Management, subscription order management; SAP Convergent Charging; and SAP Convergent Invoicing. As discussed in Chapter 3, SAP Convergent Charging is a Java-based application, and web service settings are required for SAP Billing and Revenue Innovation Management, subscription order management, and for SAP Convergent Invoicing to communicate with SAP Convergent Charging.

5.2.6 Integration with the General Ledger

This section explains the capability of transferring general ledger summary records from contract accounting to the general ledger. All the financial documents posted in contract accounting are eventually transferred to the general ledger for reporting.

Let's discuss the reconciliation key (as in Figure 5.46). This key stores a summary of all the line items resulting from the contract accounting posting. Stored in the **Document Headers** section of the documents transferred, the reconciliation key is used to reconcile the contract accounting document postings with the corresponding general ledger postings. The reconciliation key can be determined automatically by background system batch runs, as well as manually during any foreground execution of a transaction.

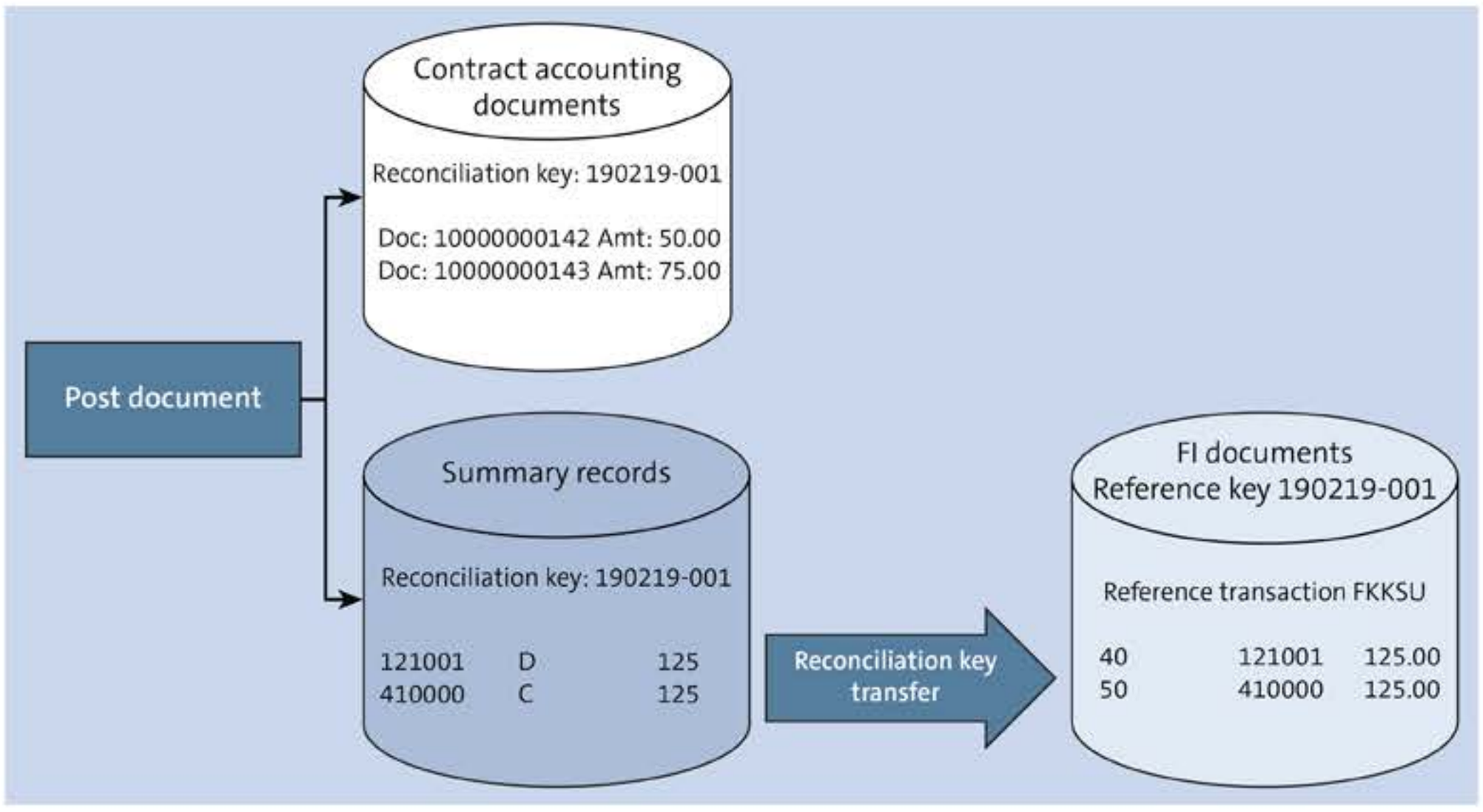

Figure 5.46 Reconciliation Key

Reconciliation key statuses are described in Table 5.1.

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Status	Description
Created/Open	Postings can be made under the key.
Closed	No further postings can be made under the key, and the key is ready for transfer.
Transferred	The key is transferred to the general ledger and profitability analysis.

Table 5.1 Reconciliation Key Statuses

Now that you've learned about the reconciliation key, let's use Transaction FPF3 to display the reconciliation key in contract accounting (as in Figure 5.47).

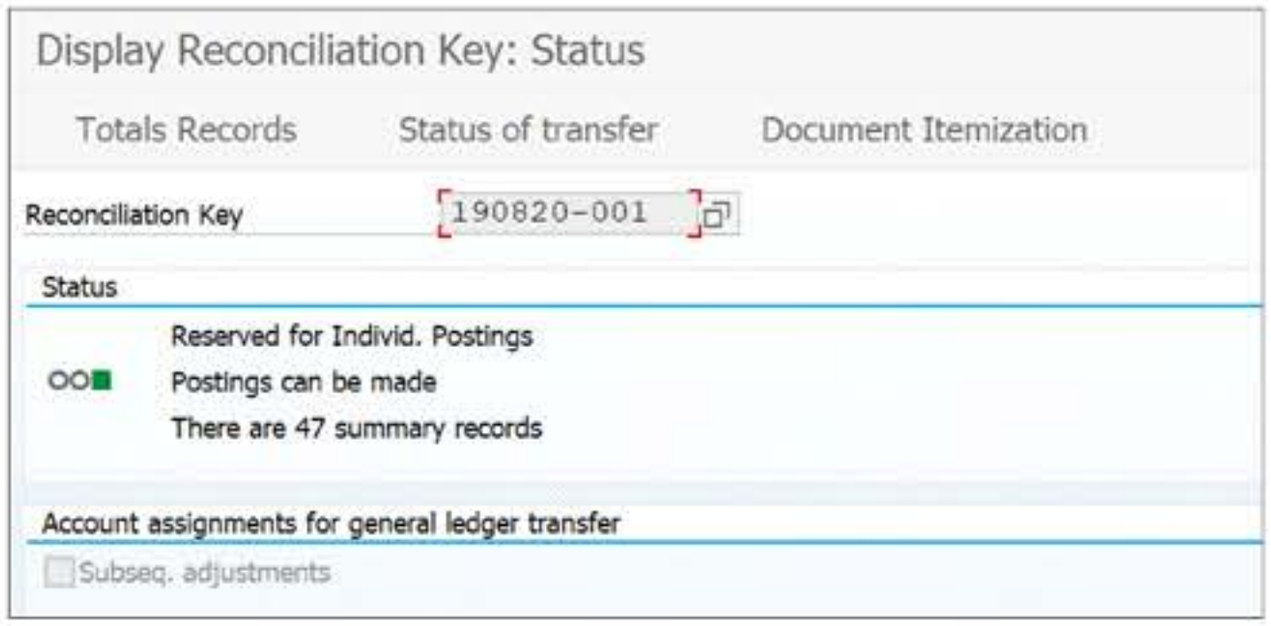
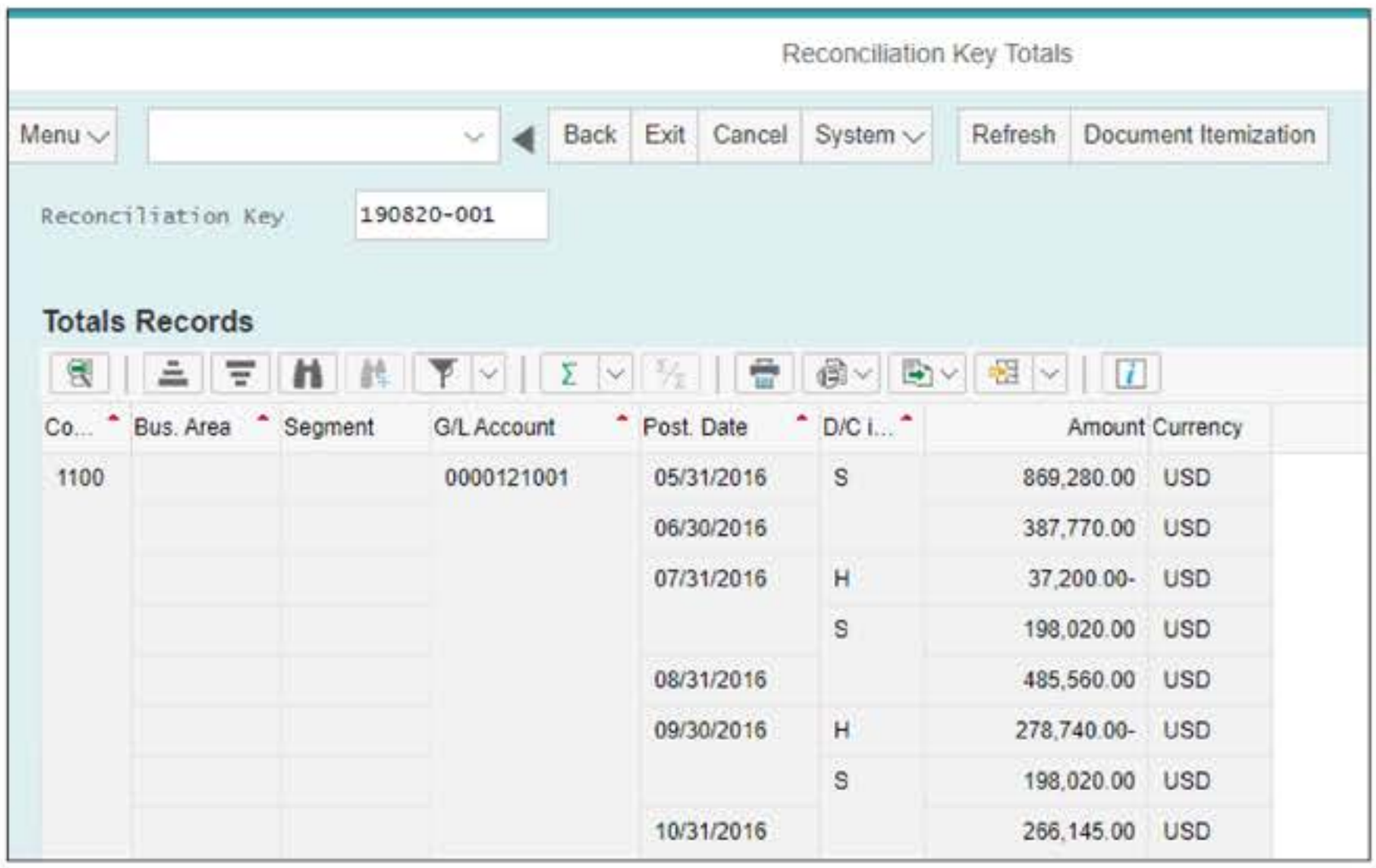


Figure 5.47 Reconciliation Key Display Screen

Click on **Totals Records** to view all the documents tied to the reconciliation key that are related to the general ledger account. As shown in Figure 5.48, a detailed list of multiple records tied to the reconciliation key is displayed.



Co...	Bus. Area	Segment	G/L Account	Post. Date	D/C i...	Amount	Currency
1100			0000121001	05/31/2016	S	869,280.00	USD
				06/30/2016		387,770.00	USD
				07/31/2016	H	37,200.00-	USD
					S	198,020.00	USD
				08/31/2016		485,560.00	USD
				09/30/2016	H	278,740.00-	USD
					S	198,020.00	USD
				10/31/2016		266,145.00	USD

Figure 5.48 Reconciliation Key: Totals

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Double-click a line item to view the documents tied to that line item. As shown in Figure 5.49, here you can view the document itemization of the posting totals.

Itemization for Posting Totals

Menu | Back | Exit | Cancel | System | First Column | Column Left | Column Right | Last Column | Choose Detail

Rec. Key / Document Number	Rec. cnter / Item	Clrd	CoCode / Doc.	Bus. Area / BPartner	G/L acct / Cont.Account	Posting Date	Document Date	Amount / Contract	Tax / Tax Base Amou
190820-001	19		1100		121001	09/30/2016	09/30/2016	278,740.00-	
10000000377	2		1100000253		120000000006	10305		253,400.00-	0.0
10000000391	1		1100000253		120000000006	10303		25,340.00-	0.0
190820-001	18		1100		121001	09/30/2016	09/30/2016	198,020.00	
10000000377	1		1100000253		120000000006	10305		99,010.00	0.0
10000000391	2		1100000253		120000000006	10303		99,010.00	0.0

Figure 5.49 Reconciliation Key: Itemization for Posting Totals

Next, use Transaction FPE3 to view a contract accounting document and the reconciliation key, which is present in the document header of every contract accounting document regardless of its origin. As shown in Figure 5.50, reconciliation key (**Reconciliation. Key**) **160511-001** is listed in the document header.

Display Document: Header

Menu | Back | Exit | Cancel | System | Overview | Business Partner Item | G/L item

Document Date	05/11/2016	Document Type	YB	Currency	USD
Posting Date	05/11/2016			Translation dte	05/11/2016
Document Number	200000000001	Branch		Exchange rate	
Reference	400000000001	Single Doc		Reconcil. Key	160511-001
				Tfr Doc. Type	

Administrative Data

Entered On	05/11/2016	Ref. procedure	VBRK
Entered at	14:56:38	Object key	0090000131
Created by	FIORI	Document Class	
Origin	77 CI Invoicing		
Clearing Info	Document did not clear any items		

Figure 5.50 Document Header: Reconciliation Key

The reconciliation key holds information about all the postings made to the general ledger accounts. Once closed and transferred to financial accounting, it updates the general ledger accounts.

Reconciliation keys for most of the mass processes are closed automatically as soon as the job is completed, but certain processes in which individual postings are made

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require closing reconciliation keys manually. A reconciliation key can be closed via Transaction FPG4 (as in Figure 5.51).

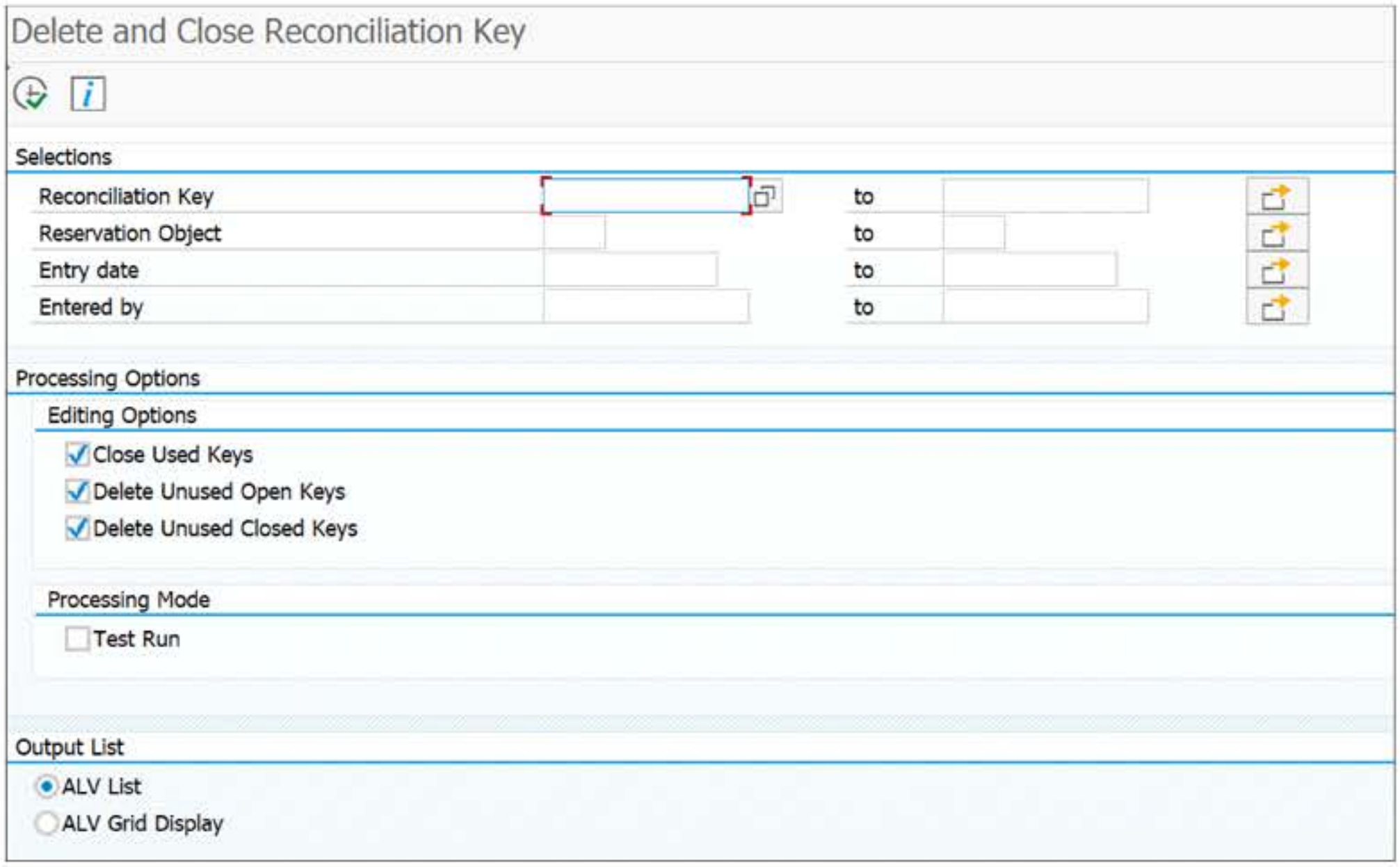

Figure 5.51 Reconciliation Key Close Transaction Screen

The closed reconciliation keys are then transferred using Transaction FPG1M for mass transfer (as in Figure 5.52).

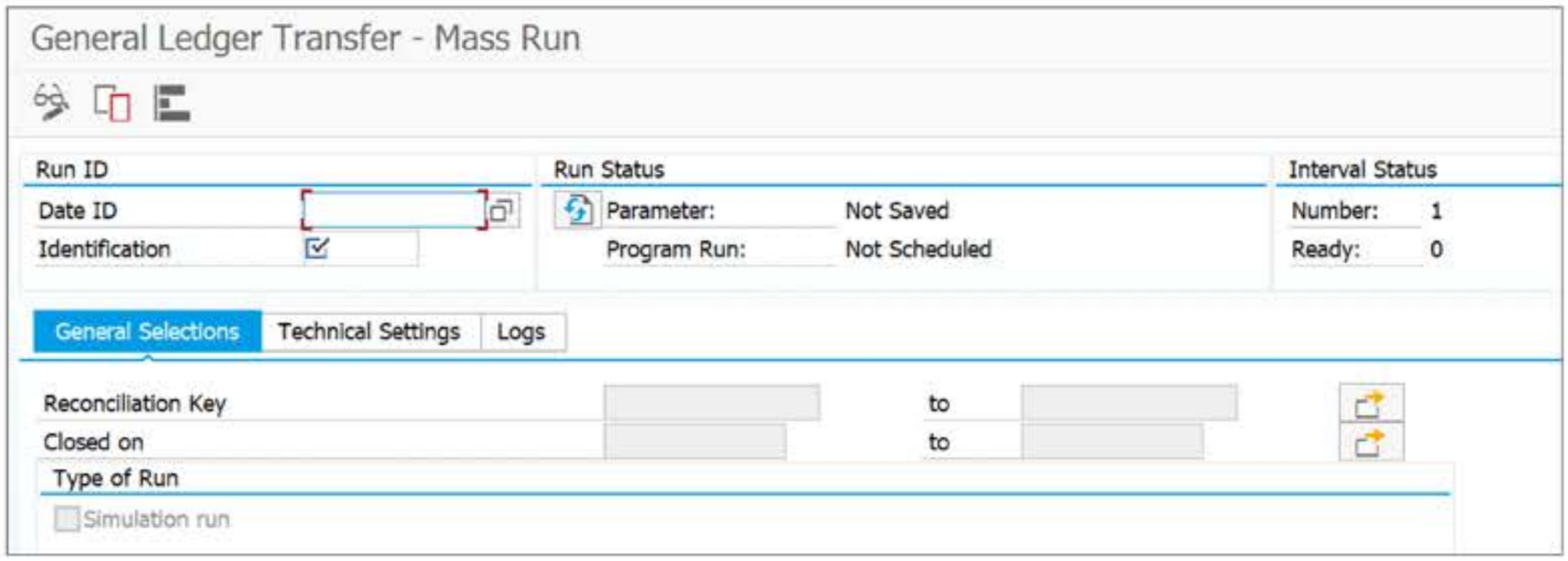

Figure 5.52 Reconciliation Key Mass Transfer Screen

As of SAP S/4HANA 1909, manually posted and reversed contract accounting documents can be updated directly to the general ledger.

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5.2.7 Financial Customer Care

Financial Customer Care (FCC) provides various customer account–related information to the interaction center agent/call center agent, which gives a wholistic view of receivable/payables and the various transactions, correspondence, and dispute cases on the account.

FCC is part of the Interaction Center web client, which integrates contract accounting, displays financial information, and allows you to post certain payments and manage dispute cases.

Several pieces of financial information can be displayed in FCC:

- **Customer account balance details**
  Information displayed under customer account balance includes all items, open items, cleared items, dispute case, installment plan items etc.
- **Credit memo**
  Credit memos created on the accounts are displayed under this section. Credit memos can also be posted on the customer account using the Interaction center Financial Customer Care
- **Reversal document**
  Details of the reversal document posted on the customer accounts are displayed under this section
- **Installment plan types**
  Installment plans created on the original documents are displayed under this section.
- **Promise to Pay**
  Information pertaining to Promise to Pay items are displayed
- **Dunning notices**
  Dunning notices issued to the customer are displayed under the **Dunning history** tab of the **Financial Customer Care Fact sheet**. Display of dunning notices issues, reversed, notices with open or cleared items can also be controlled.
- **Correspondence**
  Various correspondences issued to the customer be it installment plan letter, dunning notices, invoices pdf or account statement are displayed under the correspondence section of the Financial Customer Care fact sheet. Account statement can also be issued for the FCC
- **Collection history**
  Items which are marked by dunning to be sent to collection agency are displayed
- **Refills**
  Prepaid account balances display, refill, amount transfer are some activities which can be performed using FCC.

Each piece of information mentioned can be displayed on the FCC fact sheet. The information is controlled by the contract accounting profiles, which are configured under

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the node **SAP IMG • Services • Interaction Center Web Client • Industry Specific Functions • Integration with Contract Account Receivables and Payables (FI-CA) • Define FI-CA Profiles** (see Figure 5.53).

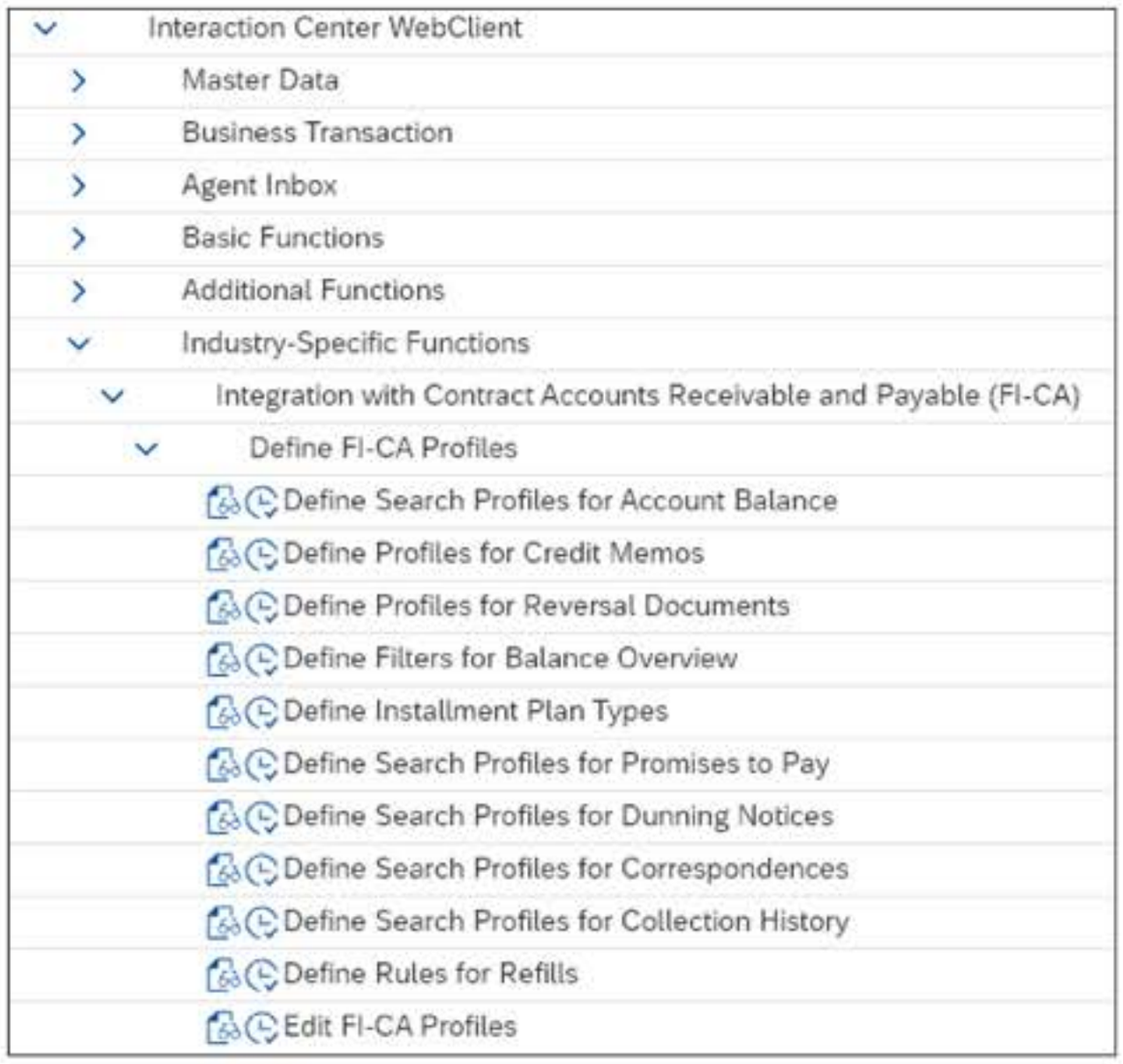

Figure 5.53 Contract Accounting Profile Maintenance for Information Display in the CRM Interaction Center Web Client for FCC

A sample contract accounting profile provided by SAP for account balance searches is displayed in Figure 5.54.

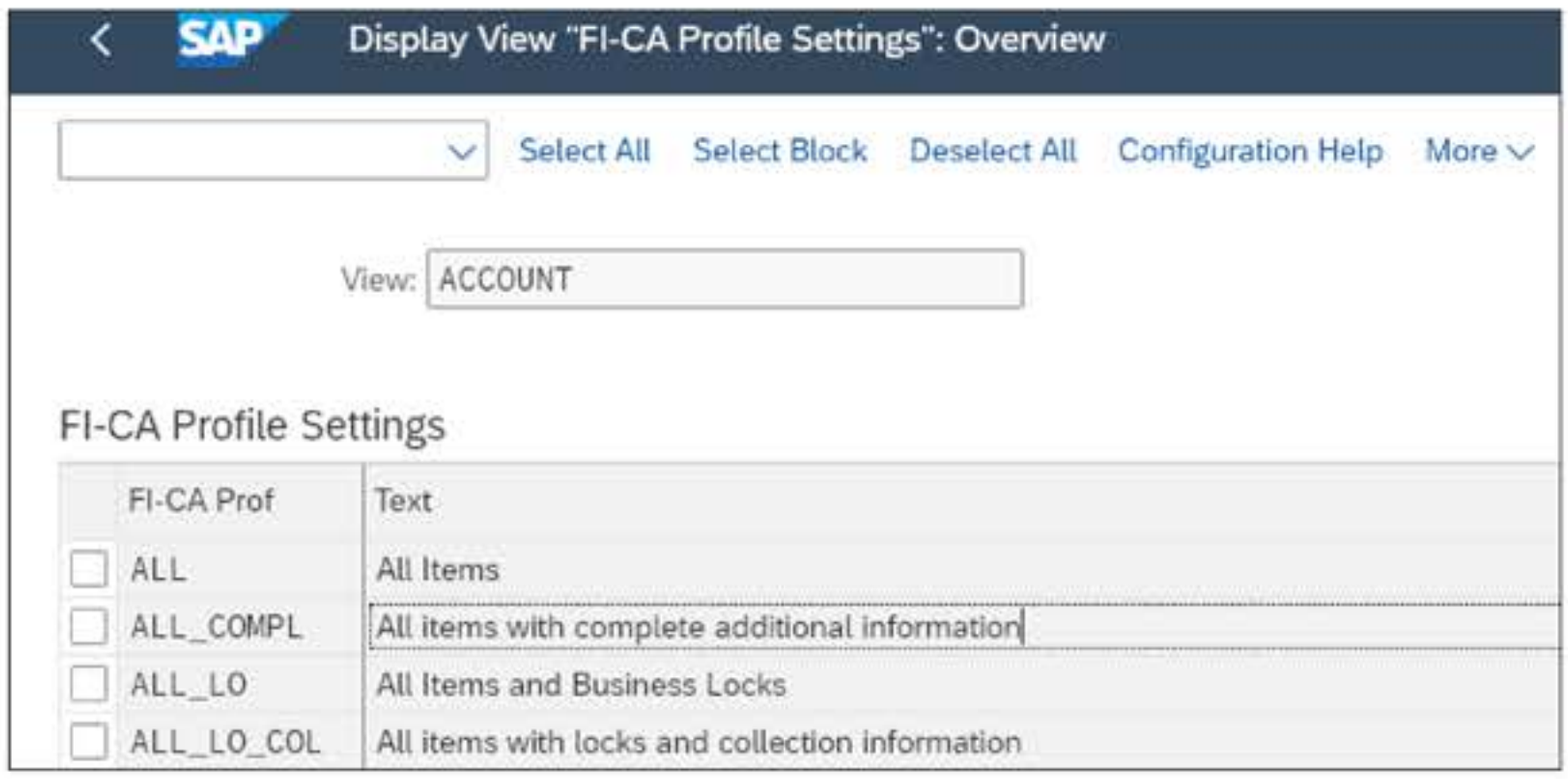

Figure 5.54 Contract Accounting Profile for Account Balance Search

Each of the profiles created for information display is under the common FCC profile provided by SAP standard. Settings for contract accounting profiles are maintained under the node **SAP IMG • Services • Interaction Center Web Client • Industry Specific**

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Functions • Integration with Contract Account Receivables and Payables (FI-CA) • Define FI-CA Profiles • Edit FI-CA Profiles, as shown in Figure 5.55.

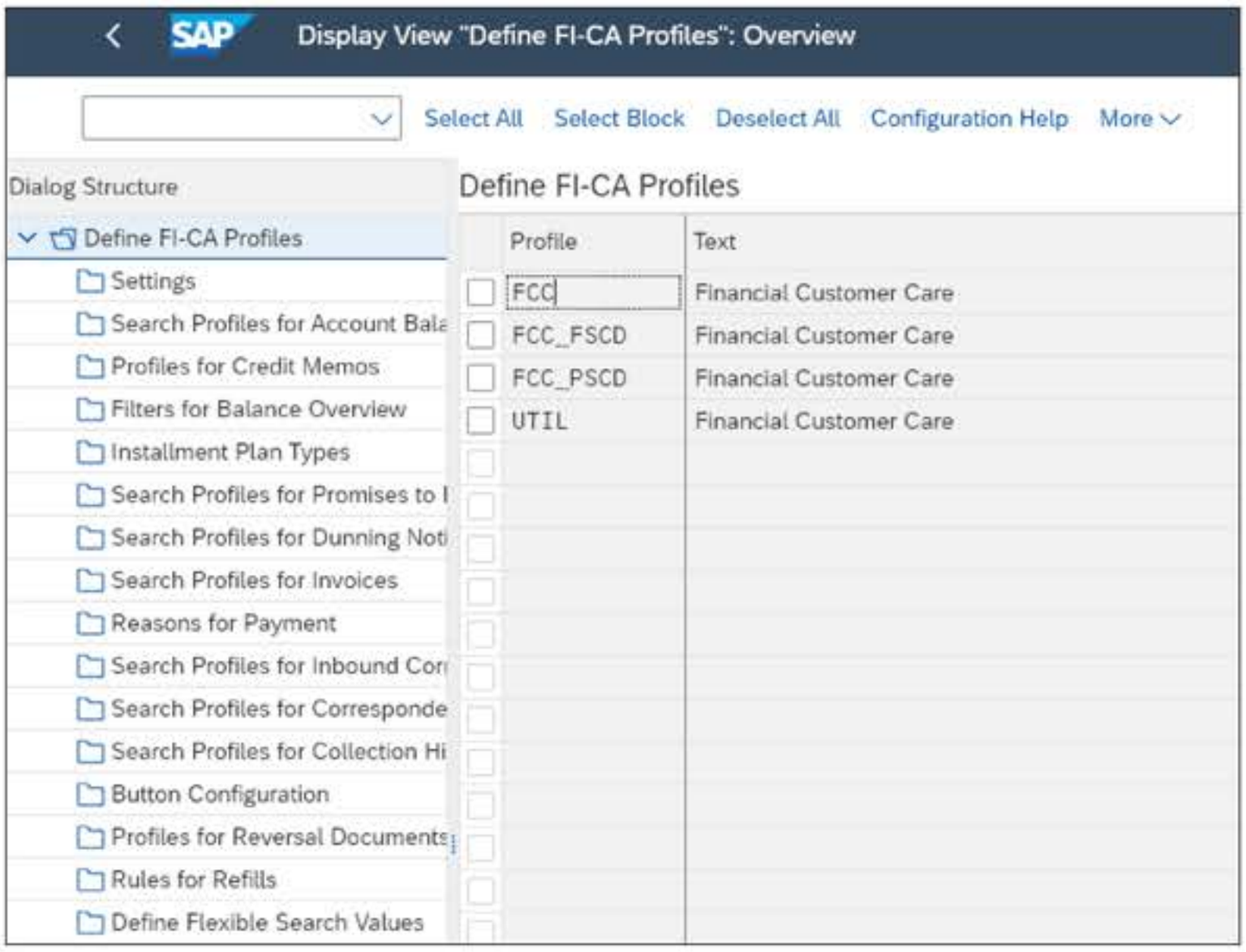

Figure 5.55 Contract Accounting Profiles Maintenance

FCC profiles are provided by SAP standard. These profiles contain the settings specific to the various information segments, as shown in Figure 5.56.

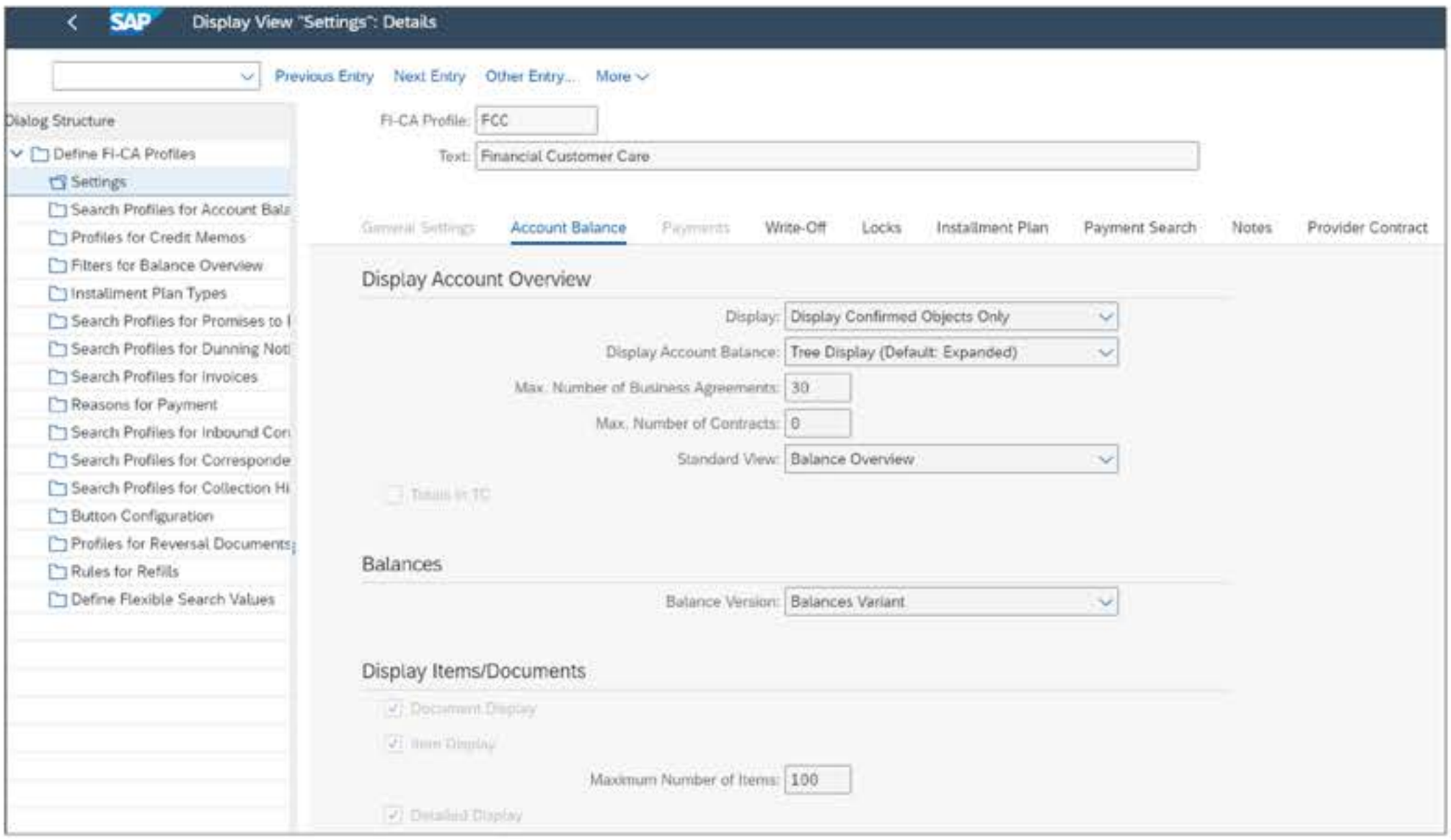

Figure 5.56 SAP Standard Profiles for FCC

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The contract accounting search profiles are then assigned to the FCC contract accounting profile for each information segment. The contract accounting search profiles are assigned to the default FCC by selecting the **Search Profiles for Account Balance** subnode. Under the subnode, you can click on New Entries and select the contract accounting profile from F4 help. Assign the contract accounting search profiles and mark one of them **Default** (see Figure 5.57).

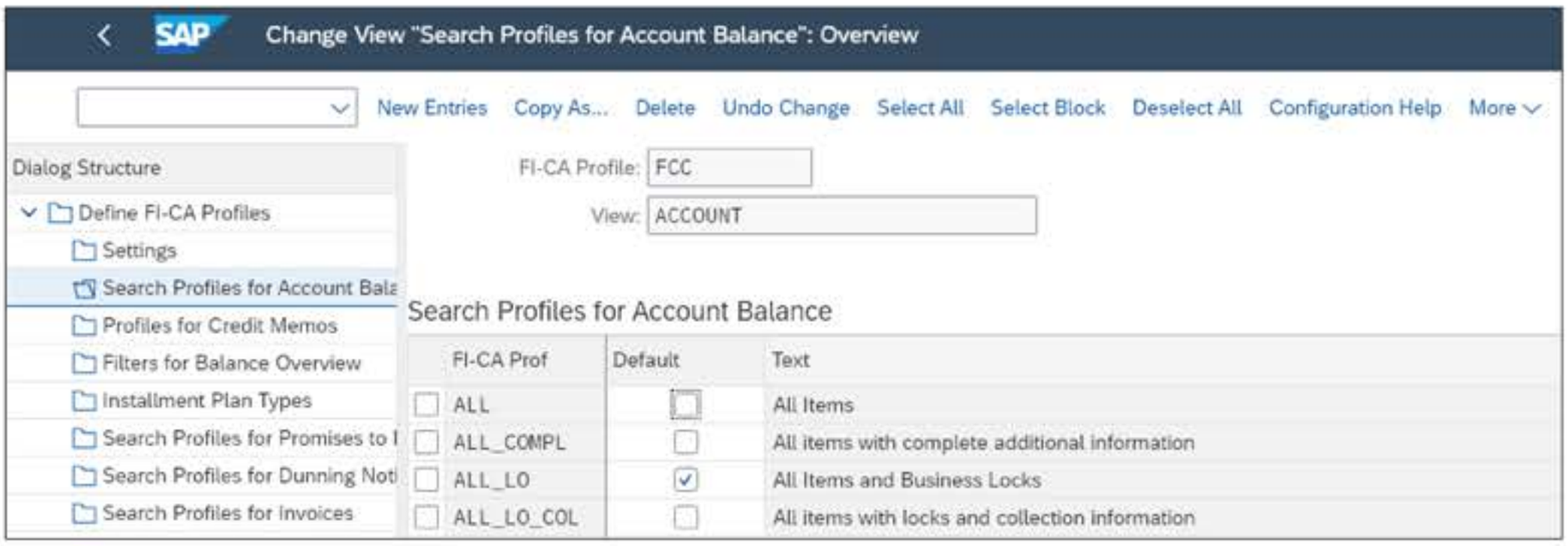

Figure 5.57 Search Profile Assignment to FCC Profile

Settings for each search profile are also maintained under the same configuration node. Figure 5.58 shows the setting for contract accounting **Search Profiles for Invoices**, which has settings maintained for the duration of invoice details to be displayed.

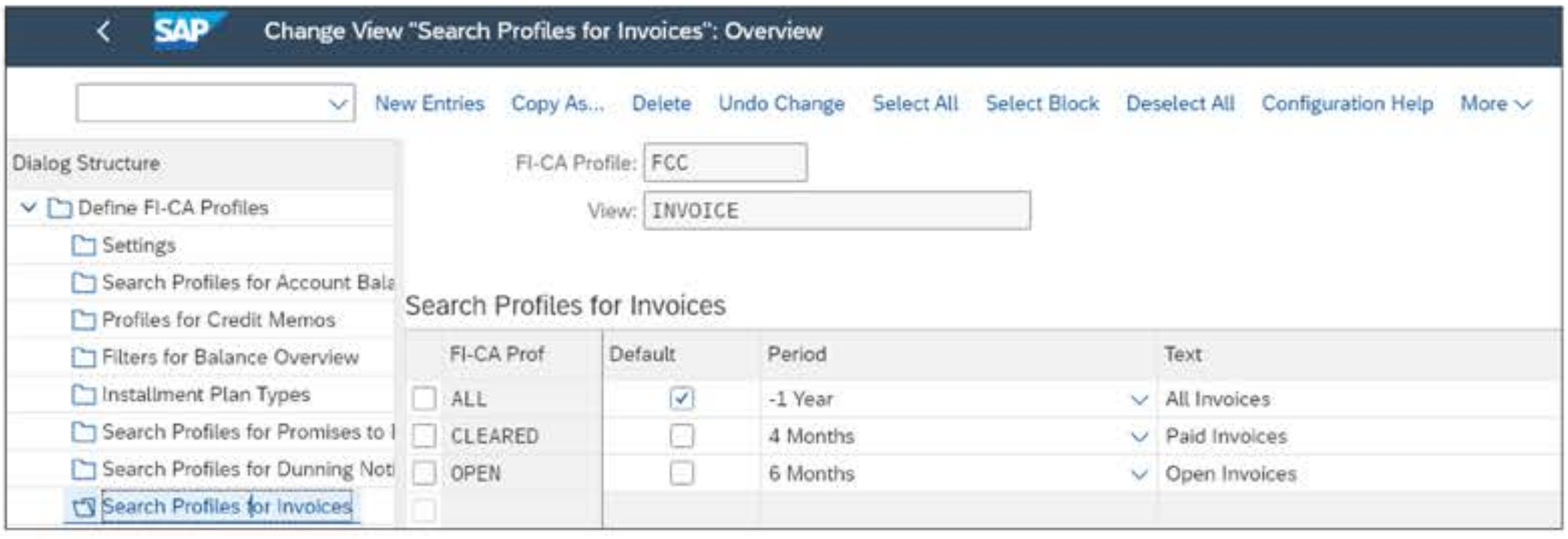

Figure 5.58 Contract Accounting: Search Profiles for Invoices

The contract accounting profiles set up under the **Services** nodes are then assigned with function modules to derive the information per the profiles/functionality. The function module is assigned in two steps:

1. **Define the key for each process, and assign the function module to it.**
   The configuration is maintained under the node **Contract Accounts Receivables and Payables • Integration • Customer Relationship Management • Additional Information in IC Web Client • Register Function Modules**. Figure 5.59 shows the keys defined

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for contract accounting processes and the corresponding function modules assignment.

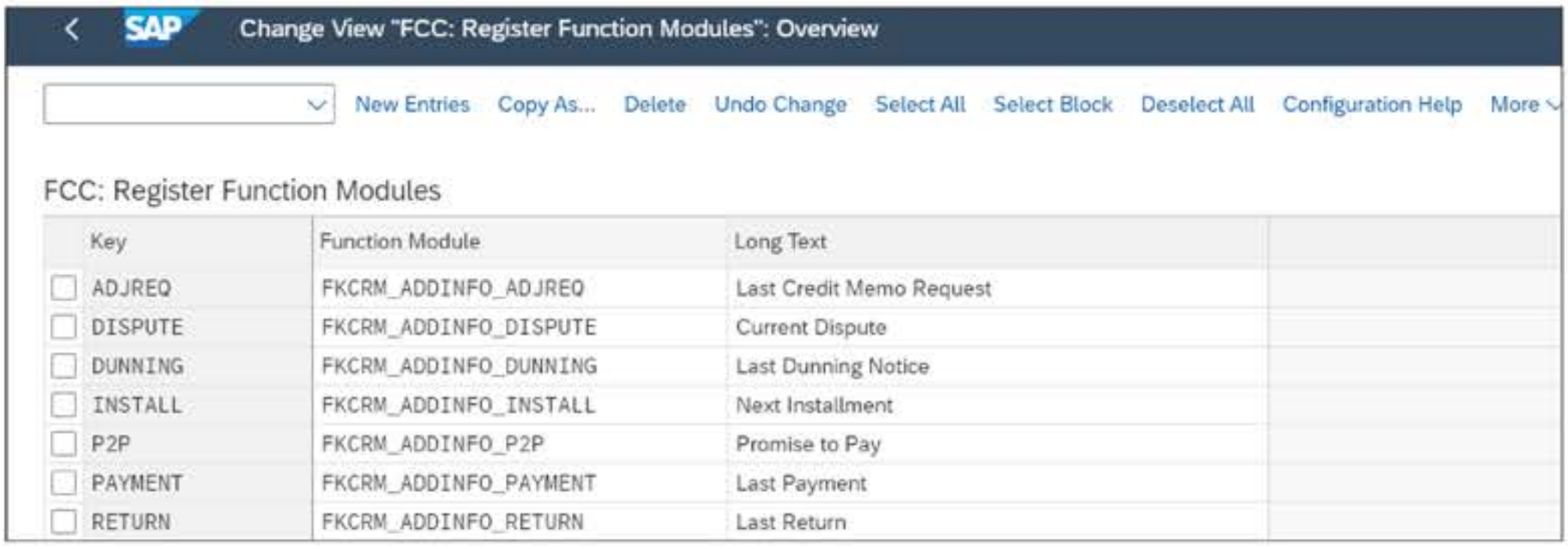

Figure 5.59 Key Creation and Function Module Assignment for Contract Accounting Search Profile Assignment

2. **Assign the keys to the contract accounting profiles created under the Services node.**
   In this node, contract accounting search profiles are assigned to the keys created under the integration node. Figure 5.60 shows the contract accounting search profiles assigned to the SAP CRM profile.

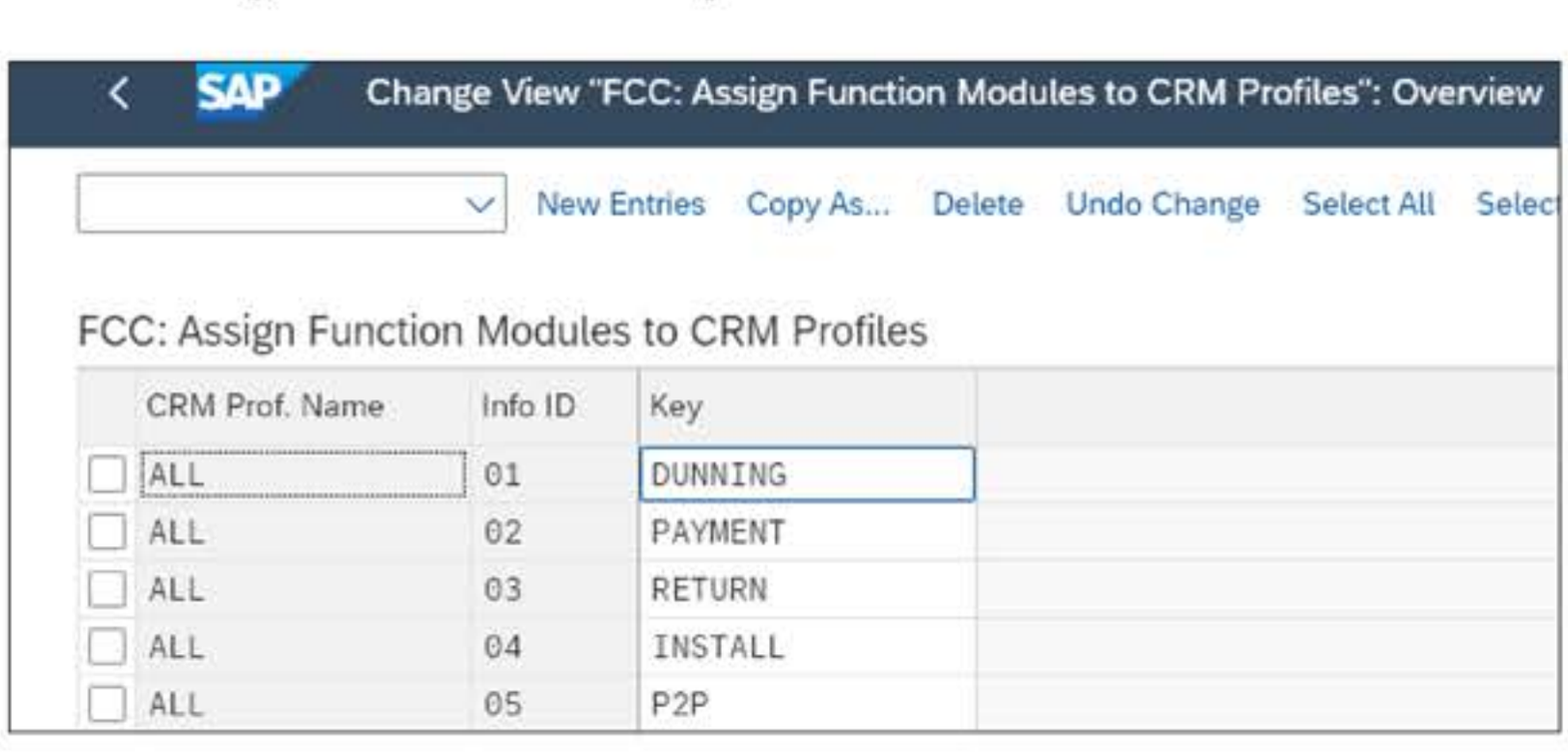

Figure 5.60 Contract Accounting Search Profiles for Account Balance Assignment to Contract Accounting Keys for FCC

SAP has provided certain standard function modules for each type of information to be displayed. Custom function modules can also be created by taking the standard reference and information retrieval, and display can be controlled per the requirement.

In the Interaction Center web client screen shown in Figure 5.61, the customer/business partner is searched using the available search parameters. **Account ID** is the SAP

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Business Partner number. Once the account is identified, the business partner fact sheet can be viewed to see the various transaction details on the account.

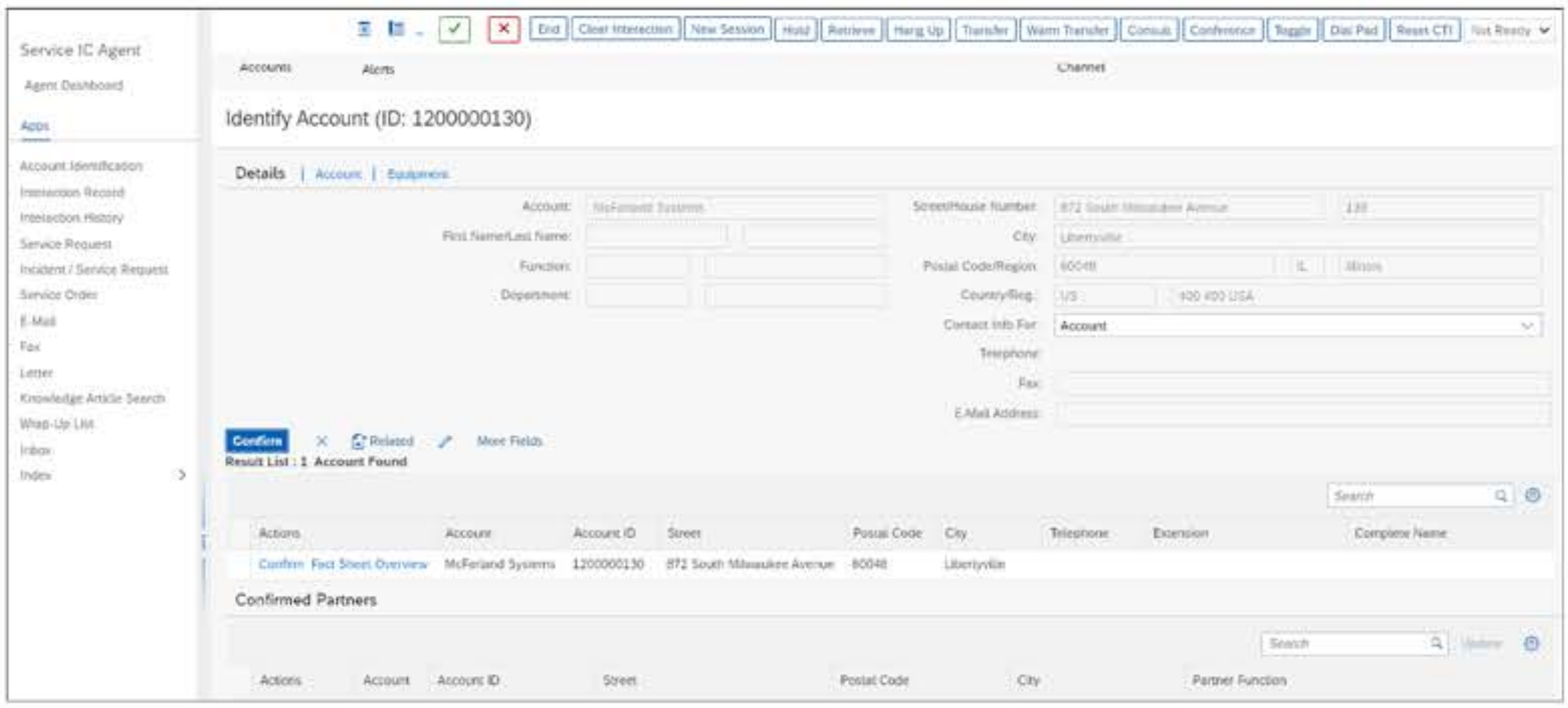

Figure 5.61 Interaction Center Web Client Screen

The account/business partner fact sheet shows the details of the transactions on the associated contract accounts (see Figure 5.62). Information such as open items, correspondence, dunning, and other details can be seen in the relevant tabs of the fact sheet.

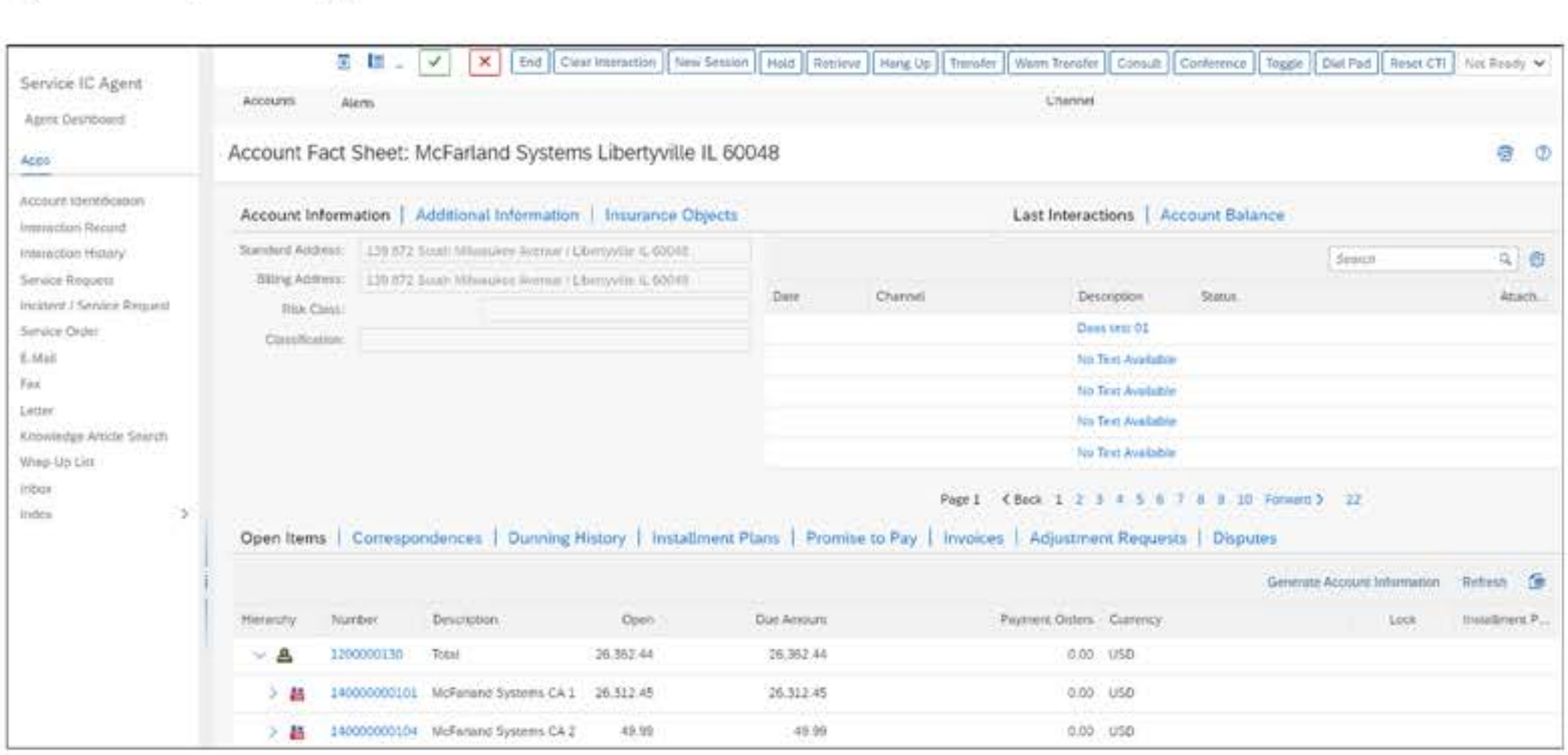

Figure 5.62 Fact Sheet View of the Business Partner

The overview of the account in Figure 5.63 shows the general information, such as name, address, and correspondence details maintained at the business partner.

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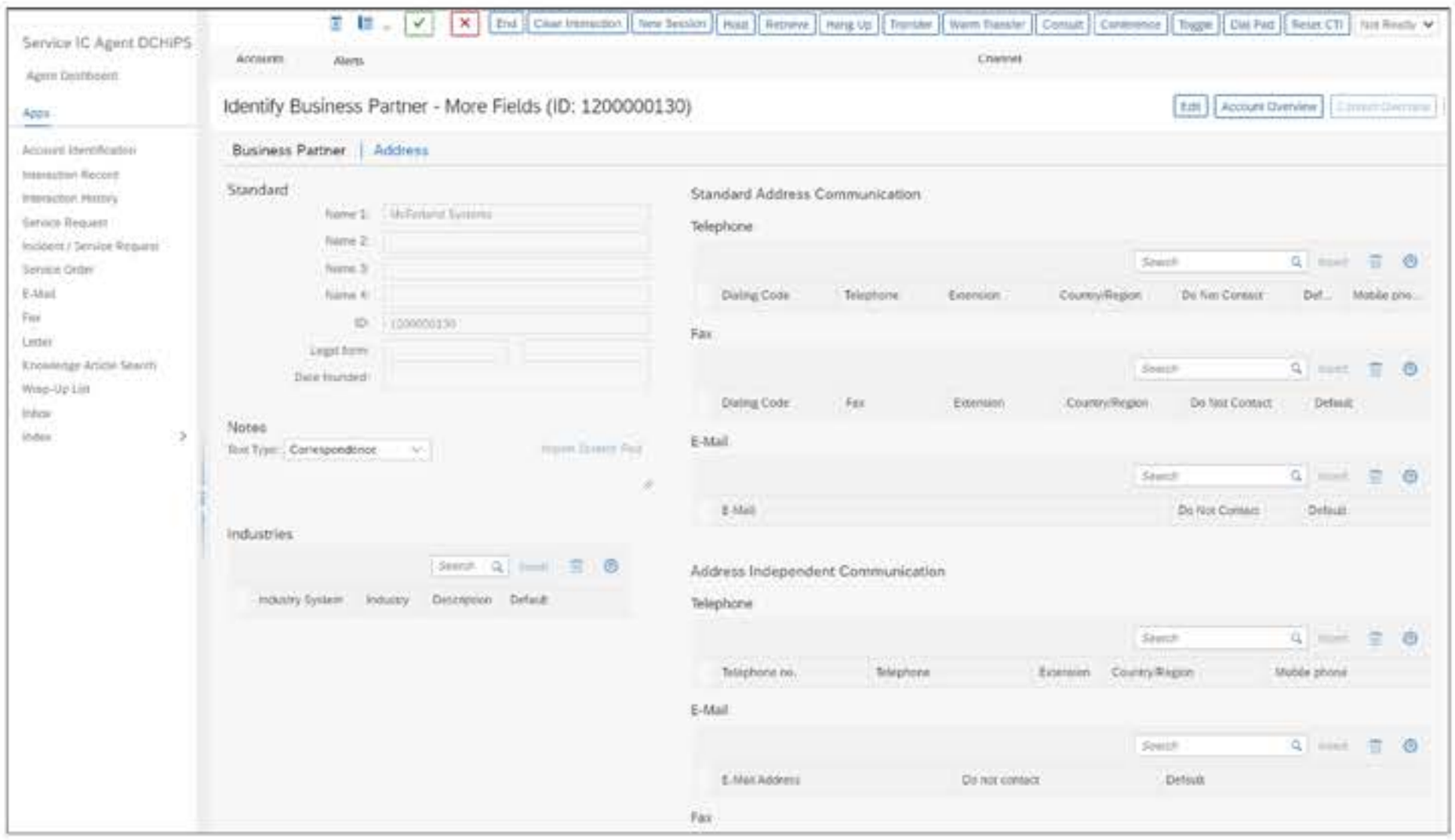

Figure 5.63 Account Overview in CRM Interaction Center Web Client

5.2.8 SAP Cash Application in Contract Accounting

Businesses with huge volumes of transactions generate a significant number of invoices for the services used by their customers. The payments for these invoices are also in significant numbers, which makes it important to clear the invoices against which payment is made or post on the customer account if the exact invoice can't be determined to avoid late payment, dunning, and bad credit ratings for customers.

Customer payments can be in different ways and can be against single or multiple invoices. They can also be full or partial payments that may or may not contain proper notes to the payee to identify the correct invoices or accounts to which payment can be applied.

Payments not applied to the correct invoice or account get posted to a clarifications account and then must be clarified manually. The manual clearing of each clarification for large numbers of customers or payments is very time consuming and leads to an increase in effort required of the accounts team. The additional effort needed for clearing the payments in clarification requires more accounts team members, increasing costs further to the company. Figure 5.64 shows the payment posting upon bank statement processing.

Contract accounting includes settings to make the payment posting and clearing process automated as much as possible, but that can't address the various challenges faced in payment/cash application to invoices. In addition, the rules can't be modified very frequently to adapt to the various scenario faced in cash application.

SAP Cash Application is an additional add-on provided by SAP that is based on machine learning, which can be used to automate the clearing of clarification cases created after posting the payment lot.

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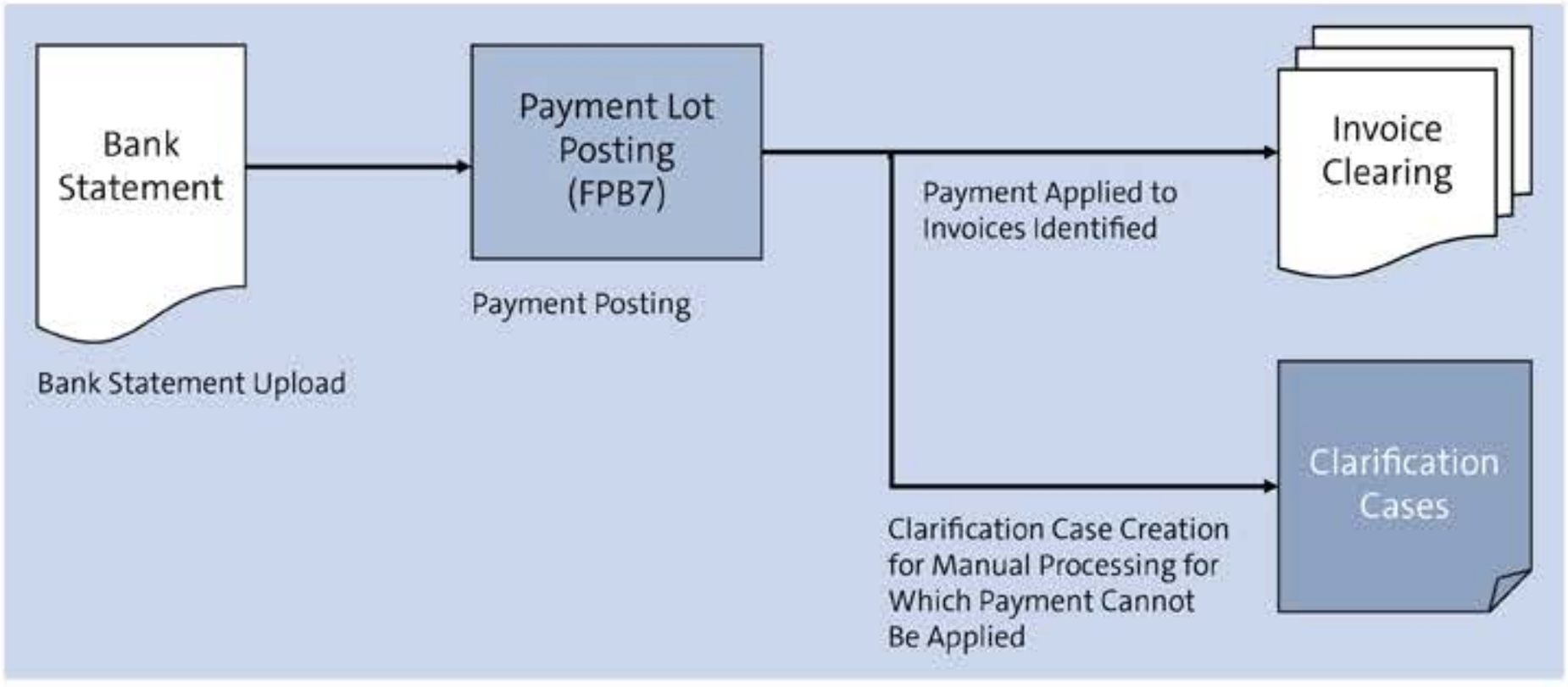

Figure 5.64 Bank Statement Processing for Payment Posting

SAP Cash Application uses the learnings accumulated from the past records to help identify the invoices/accounts of customers to which payments can be applied and to provide the proposal with a confidence level. Based on the processing set for the confidence level, the payments are applied on the invoices/accounts, and the clarifications with the lowest confidence level are released for manual processing. This reduces the manual efforts significantly and reduces the days sales outstanding. Figure 5.65 shows the payment processing with SAP Cash Application active.

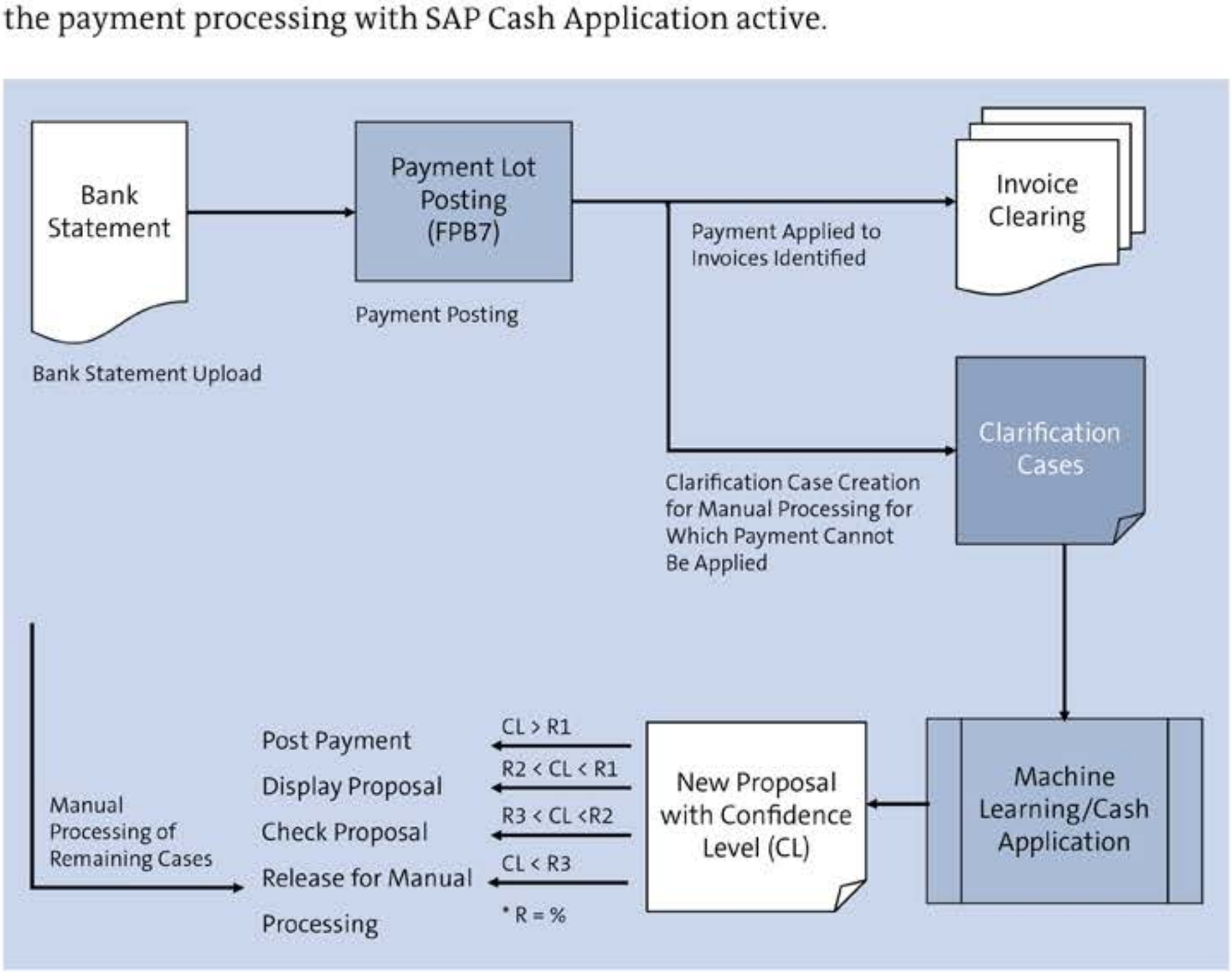

Figure 5.65 Payment Processing with SAP Cash Application

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SAP Cash Application can be set up using the following steps:

1. **Perform prerequisites and configuration.**
   The following prerequisites are required for the SAP Cash Application, add-on for contract accounting:
   - Activated SAP Business Technology Platform (SAP BTP) account
   - Subaccount for SAP Cash Application
   - Service key containing technical details of outbound services and OAuth 2.0 client creation

Note

Go to *http://s-prs.co/v554204* to view the administrative guide and get more details on the prerequisites.

Configuration steps are as follows:

- The HTTP connection must be set up to the external server.
- The remote function call (RFC) destination to external server needs to be set up via Transaction SM59. Figure 5.66 shows the RFC destination setup details, which are explained in more detail in this list. Different RFCs are provided for the various data that needs to be shared with the cash application.

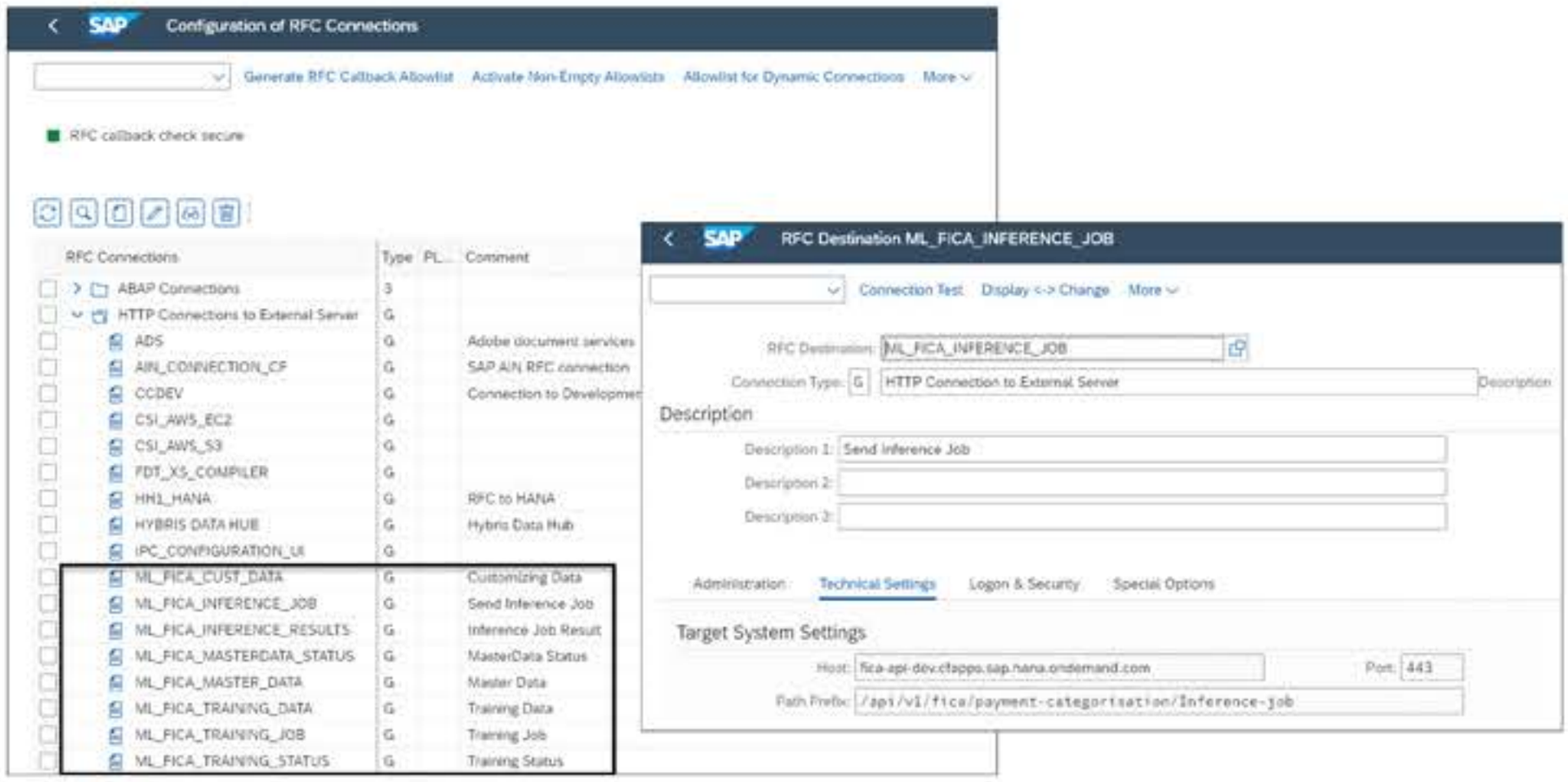

Figure 5.66 Transaction SM59: RFC Destination Setup

- The RFC destination must be assigned under the contract accounting node.

  RFC created destinations are assigned in the configuration node under menu path **Contract Account Receivables & Payables • Business Transactions • Payments • Processing Incoming & Outgoing Payments • Incoming Payment Clarification with Machine Learning • Define RFC Destinations**. Figure 5.67 shows the RFC destinations assigned in contract accounting node.

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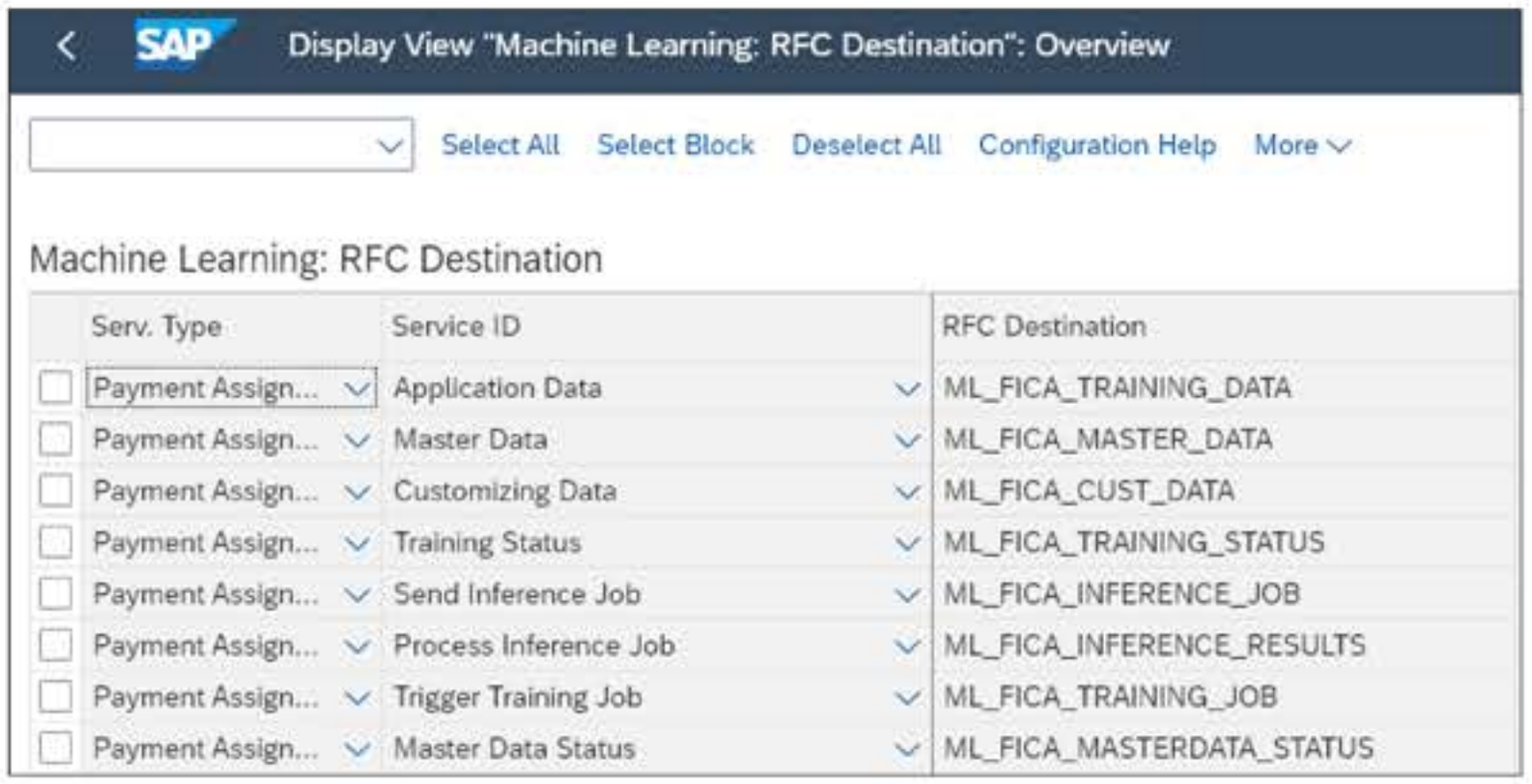

Figure 5.67 RFC Destination Assignment in the Contract Accounting Node

RFC destinations are maintained for each processing step involved in setting up SAP Cash Application and processing payment data.

– Activate the **Machine Learning Active for Payment Assignment** checkbox.

  In the company code settings with which SAP Cash Application will be used, the **Machine Learning Active for Payment Assignment** checkbox must be checked via contract accounting menu path **Contract Accounts Receivable & Payables • Organizational Units • Set Up Company Codes for Contract Accounts Receivables and Payables**. Figure 5.68 shows the configuration maintained for the company code in contract accounting for machine learning for payment application activation.

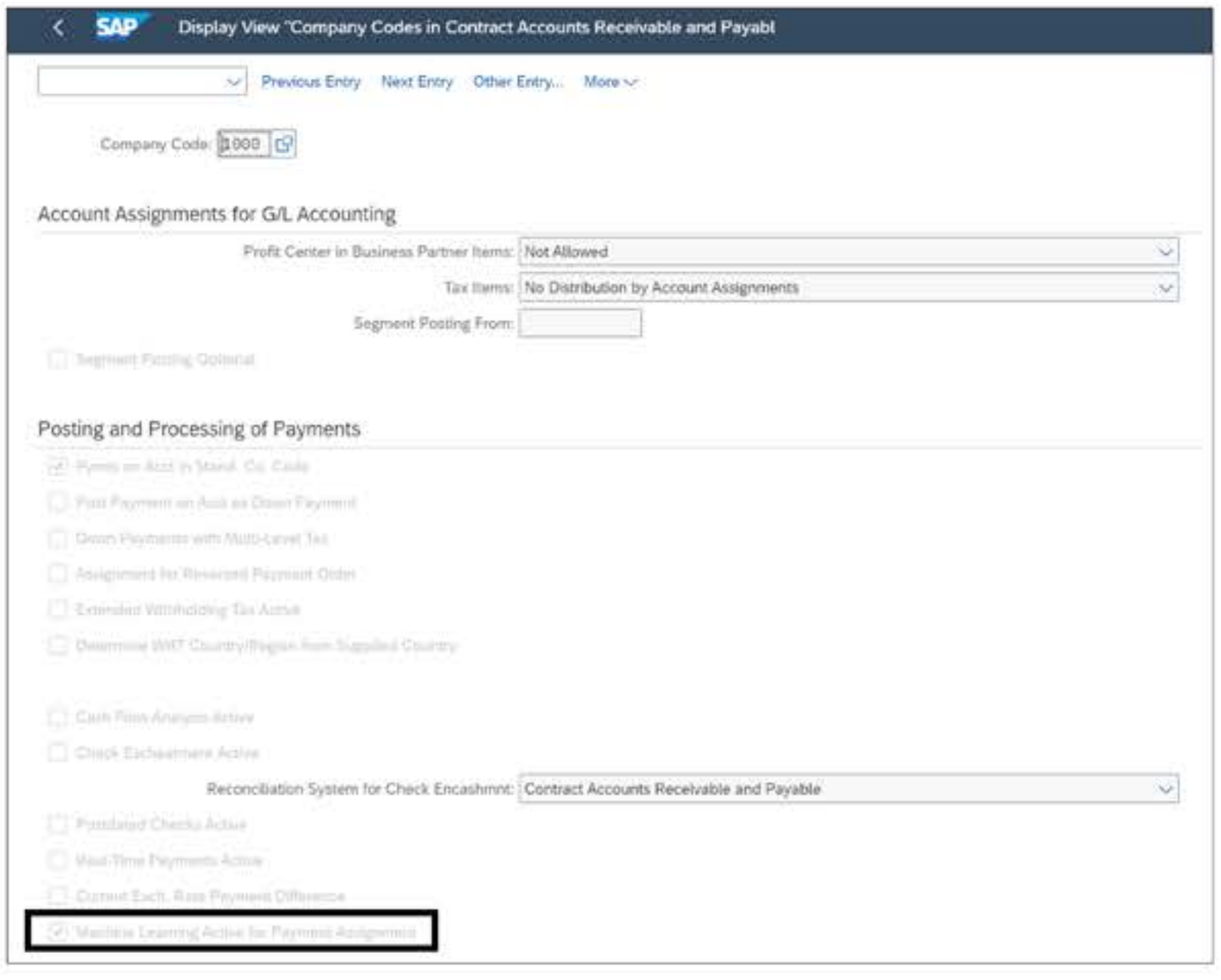

Figure 5.68 Machine Learning Active for Payment Assignment Setting

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– Perform the settings for SAP Cash Application in contract accounting.

  Under this configuration, settings are maintained for the following:

  - Days after which master data is transferred to SAP Cash Application
  - Frequency in months by which data is transferred to SAP Cash Application for training
  - Confidence level percentage maintenance

  You can find these settings under menu path **Contract Accounts Receivables and Payables • Business Transactions • Payments • Processing Incoming and Outgoing Payments • Incoming Payment Clarification with Machine Learning • Make Setting for SAP Cash Application for Contract accounting**. Figure 5.69 shows the sample values maintained for the SAP Cash Application settings.

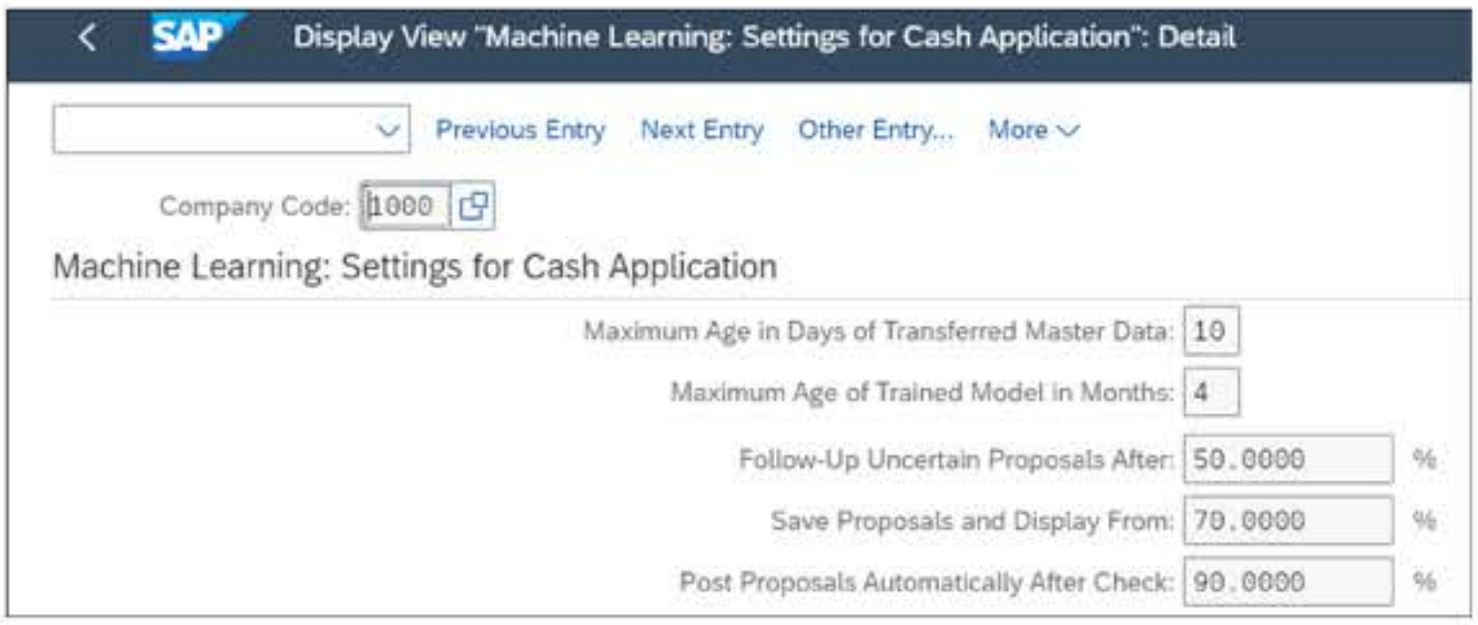

Figure 5.69 Settings for SAP Cash Application under Contract Accounting

2. **Perform data migration.**

   The following two actions are performed as part of historical payment data migration:

   – Determination and storing of objects for parallel processing in the payment lot items table and in the clarification cases table

   – Selection of historical payment data by company code and posting date of the payment lot

     This is done using Transaction FPML_CASHAPP_MIG. Figure 5.70 shows the selection screen for the program used for historical data transfer.

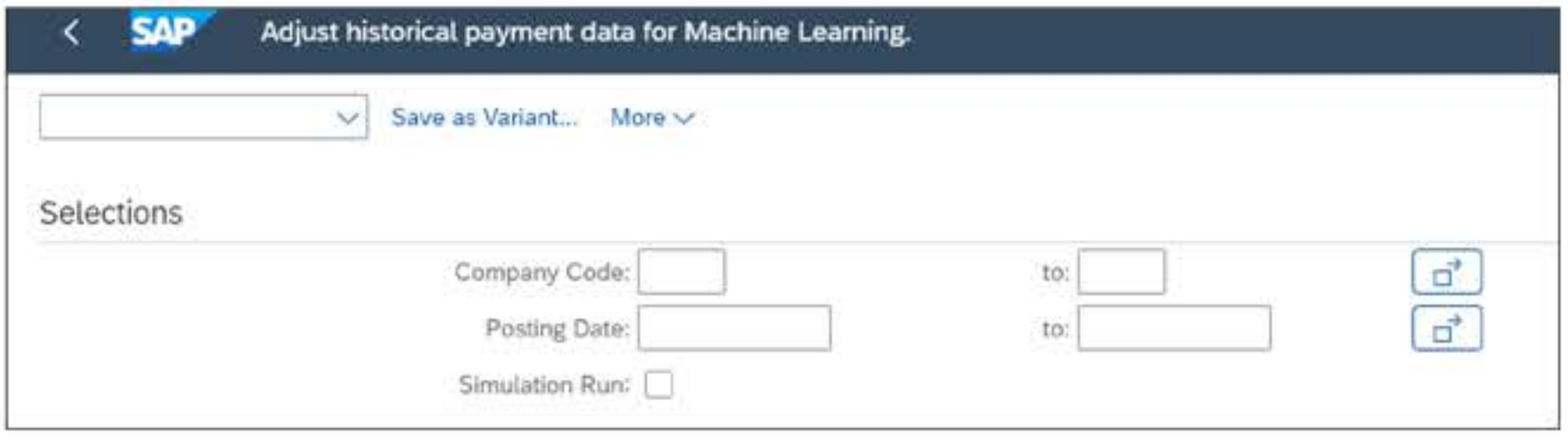

Figure 5.70 Historical Data Migration to SAP Cash Application Program

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3. **Train SAP Cash Application.**
   SAP Cash Application is trained using the historical data from the SAP system. Training processes comprise five substeps:
   - Transfer selection categories.

     In this substep, the selection categories configured in the SAP system are extracted and sent to the machine learning server in CSV format. This is a one-time activity that only needs to be repeated if the selection category configuration is changed in the SAP system.

     The activity is performed using Transaction FPML_CASHAPP_CUSTOM. Figure 5.71 shows the selection screen of the program.

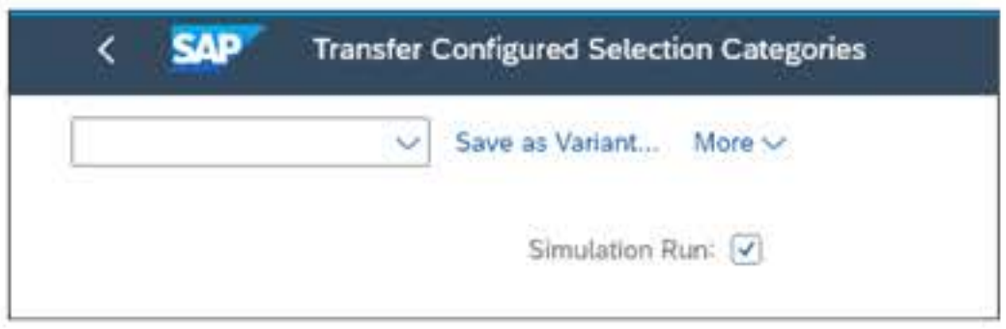

Figure 5.71 Program for Transferring Selection Categories to SAP Cash Application

   - Transfer master data.

     Master data such as business partner, contract accounts, contracts, and bank details are sent to SAP Cash Application to keep the master data up to date for use in training.

     The master data transfer is performed using Transaction FPML_CASHAPP_MASTER. Figure 5.72 shows the program screen for master data transfers.

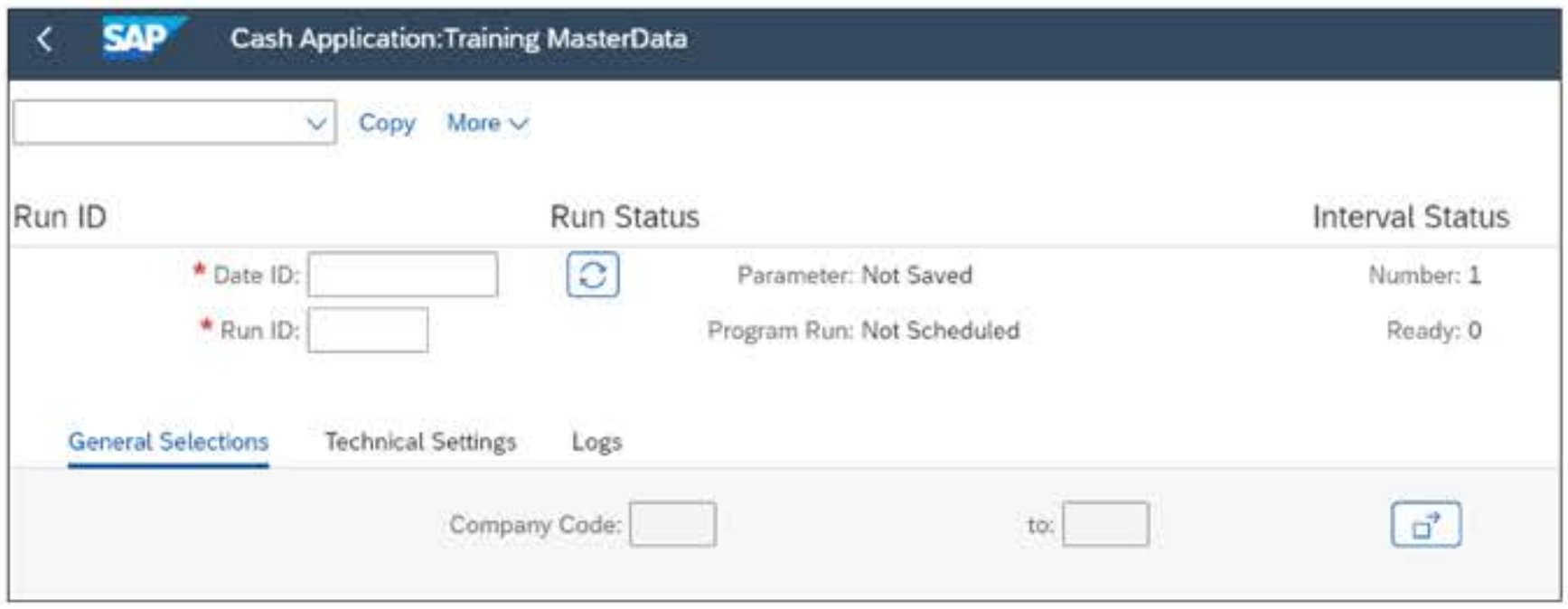

Figure 5.72 Master Data Transfer Program to SAP Cash Application

   - Query master data status.

     The program is used to query the SAP system and SAP Cash Application to check for master data status synchronization.

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Master data status can be queried using Transaction FPML_CASHAPP_MDSTAT. Figure 5.73 shows the program screen for master data query.

Figure 5.73 Master Data Query Program for Status Sync Check

– Transfer payment lot items and trigger training.

  At this step, historical payment lot data is sent to the SAP Cash Application training data set to trigger the training process. If data is huge it will be split into several CSV files for each run date and execution ID. This step is executed using Transaction FPML_CASHAPP_TRAIN. Figure 5.74 shows the program screen for the training program.

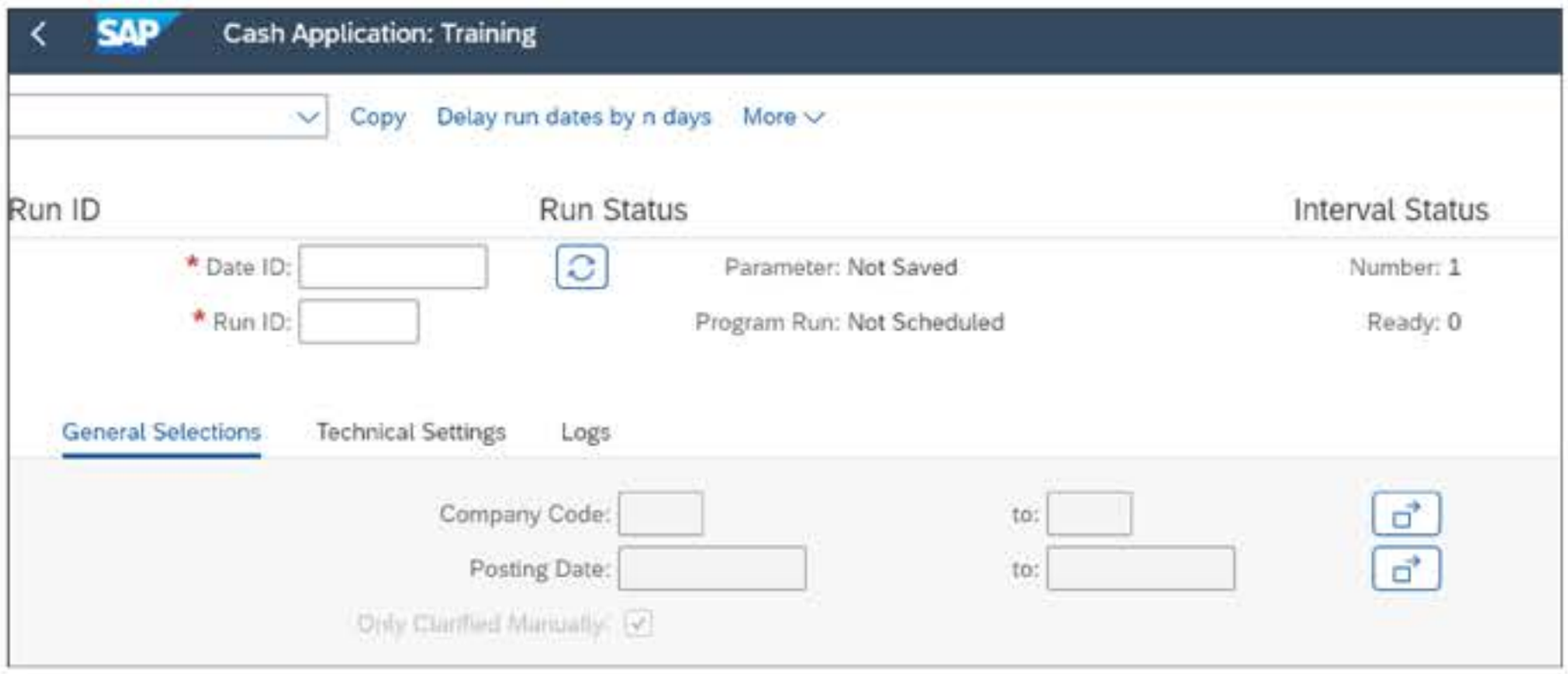

Figure 5.74 Program for Transferring Payment Lot Data and Triggering Training

– Query the training status.

  In this step, a status query is made regarding the training triggered for the training program. The run date and ID for the training program are used to query the status of the training. This step is performed using Transaction FPML_CASHAPP_TSTAT. Figure 5.75 shows the program screen for the training status query.

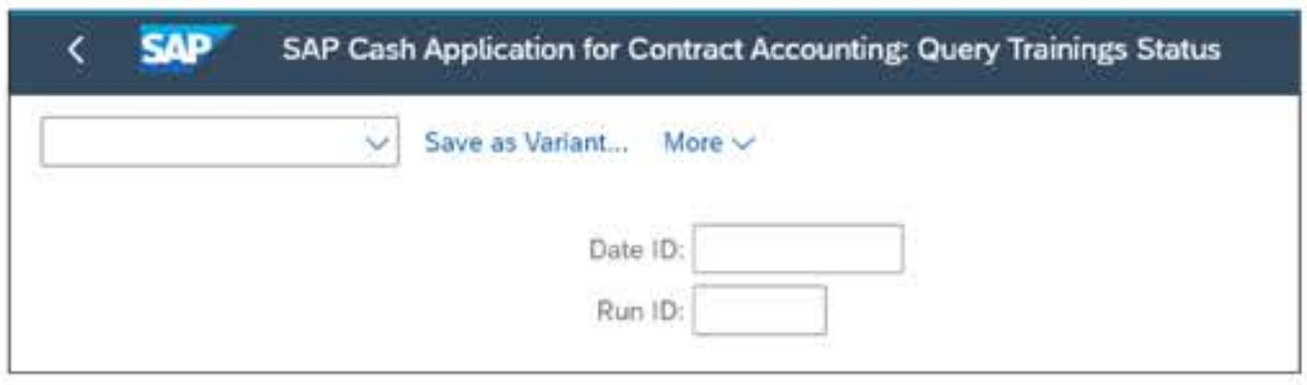

Figure 5.75 Training Status Query Program

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4. **Inference**

   This step marks the actual execution of the clarification cases created upon posting the payment lot. Before clarification cases can be transferred to SAP Cash Application, the prerequisite transactions need to be executed, including the bank statement import and payment posting using Transactions FF_5 and FPB7, respectively. The payment posted can be seen using Transaction FPO5.

   The payments that can't be applied to the invoice or account are posted to the clarification account, and clarification cases are created. Once SAP Cash Application is made active, the clarification cases for that company code can't be processed directly manually.

   Inference is performed in two steps:

   – Clarification cases data transfer to SAP Cash Application: The clarification cases created upon the payment lot posting are sent to the SAP Cash application using Transaction FPML_CASHAPP_INF. Figure 5.76 shows the program for clarification cases data transfer to SAP Cash Application.

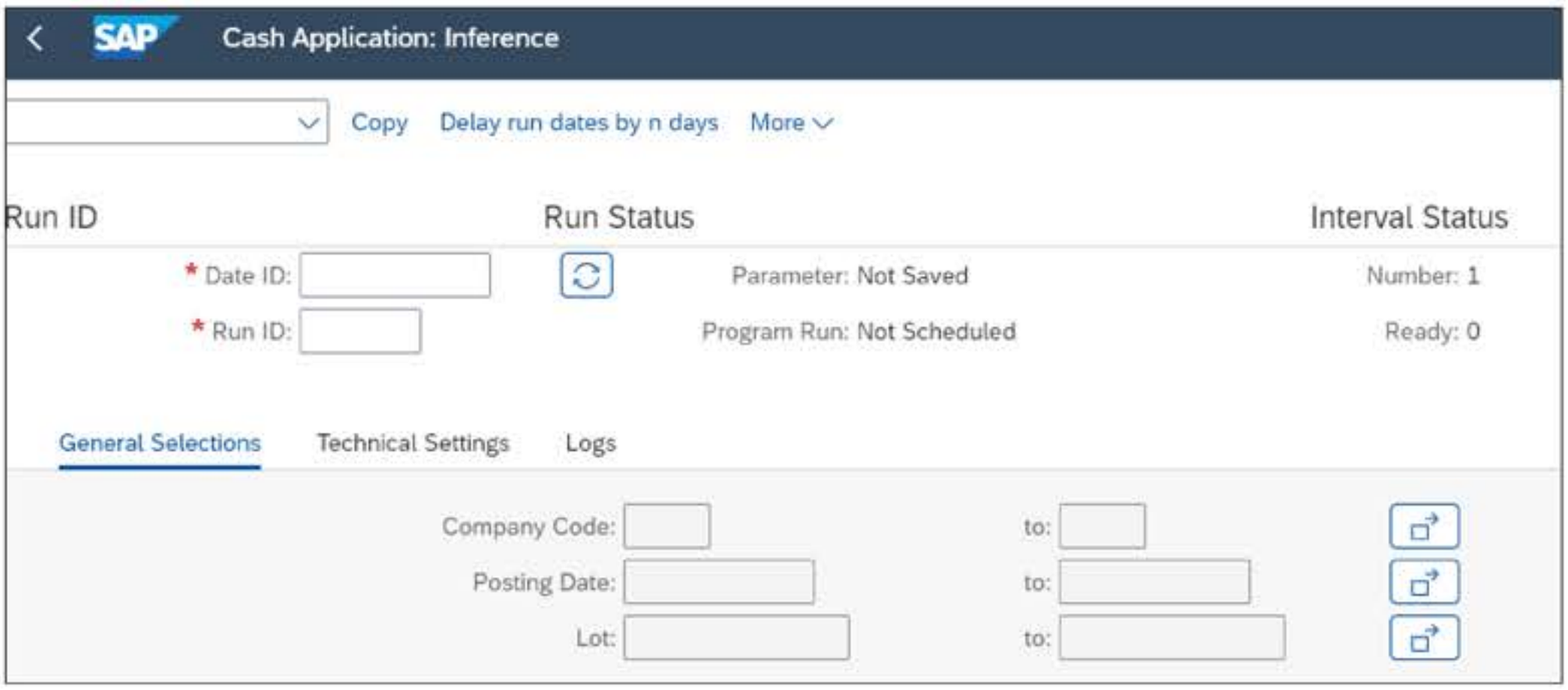

Figure 5.76 Inference Program for Clarification Cases Data Transfer to SAP Cash Application

   – Processing for returned and reversed clarification cases: In this step, the inference jobs generated by the inference program when transferring clarification cases data to SAP Cash Application continue processing for posting lot for which the confidence level is met and/or release the clarification cases that can be processed automatically.

   Transaction FPMP_CASHAPP_PROP is used to process the inference jobs for payment posting of clarification cases and releasing them for further manual intervention if required. Figure 5.77 shows the screen for the returned and reversed clarification cases processing program.

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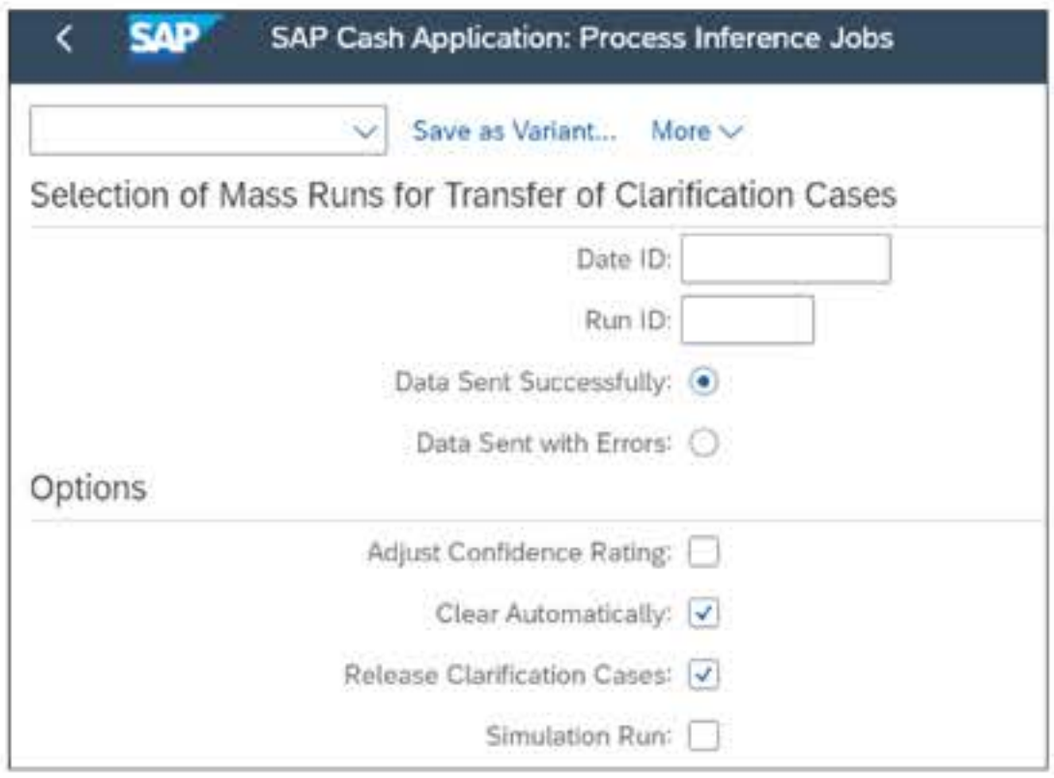

Figure 5.77 Program for Processing Returned and Reserved Clarification Cases

You've now learned about the necessity of SAP Cash Application and how you can set up this add-on and can automate the processing of the clarification cases resulting from the payment lot posting for which invoices and accounts can't be identified for clearing and posting.

5.3 Using Contract Accounting

This section explains how to use key transaction codes in contract accounting, including creating a business partner or contract account, viewing or editing a document, viewing the customer's balance, and using open item management. After learning to configure account determination, account balance display, and open item management in the previous sections, it's now time to run a few transaction codes and view the output.

5.3.1 Creating a Business Partner

The business partner can be created directly in SAP Billing and Revenue Innovation Management, subscription order management, and can then be replicated to contract accounting or directly created in contract accounting. For our example, the business partner is created in SAP Billing and Revenue Innovation Management, subscription order management, and then replicated to contract accounting and to SAP Convergent Charging.

The main definition of a business partner within the SAP ERP or SAP S/4HANA system resides at menu path **SAP Customizing Implementation Guide • Cross-Application Components • SAP Business Partner**. The definition of business partner roles, number ranges, business partner types, and so on is performed in this node.

In this section, we'll focus on the master data elements needed for the case study. Business partner 1200000450 (McFarland Systems) has been created for the Martex Corp. example. First, use Transaction BP to view business partner 1200000450. As shown in

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Figure 5.78, enter the **Business Partner** number, and click **Start**. This will return the business partner replicated in contract accounting from SAP Billing and Revenue Innovation Management, subscription order management.

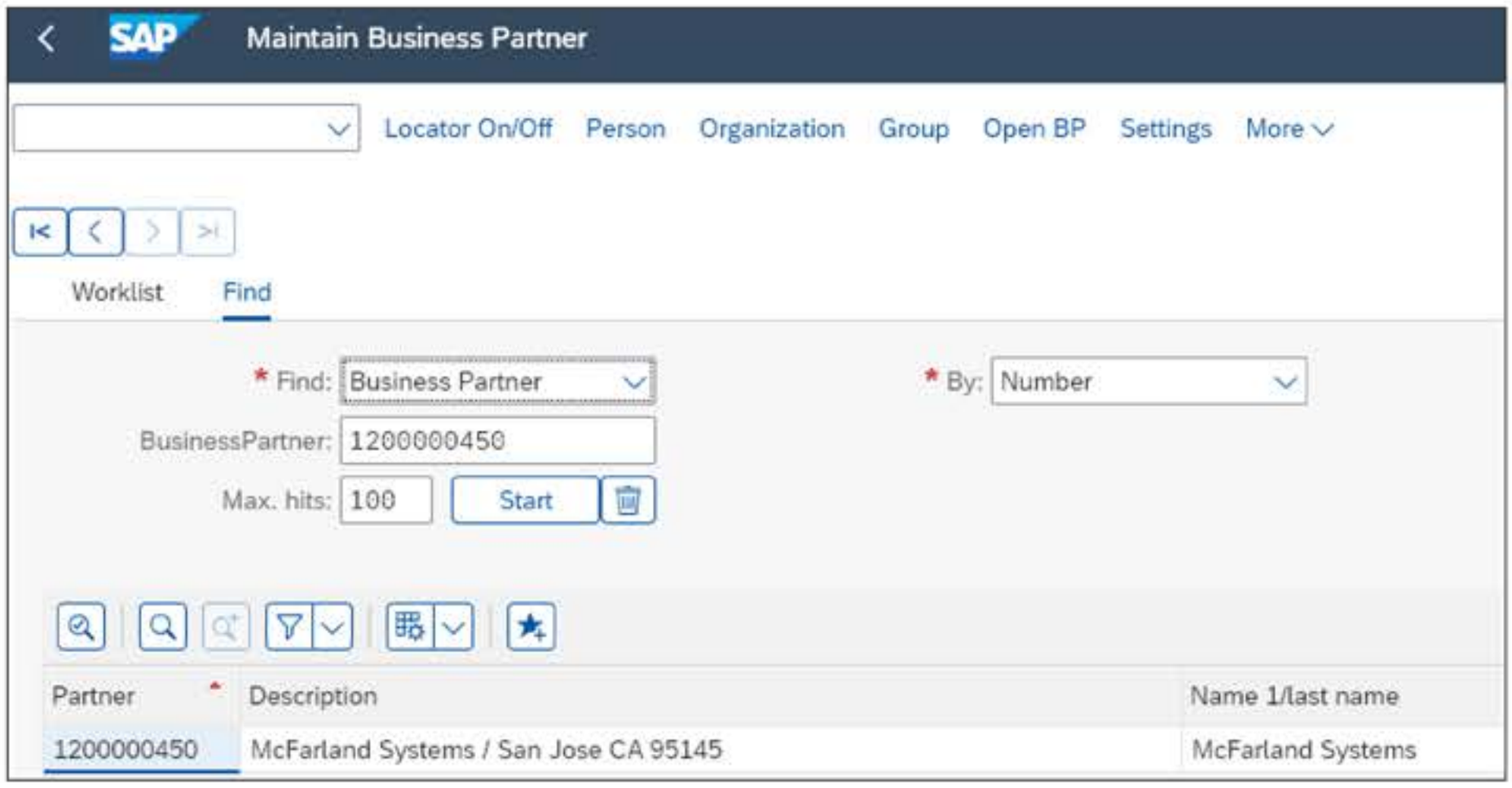

Figure 5.78 Transaction BP: Business Partner View

Next, double-click the business partner. As shown in Figure 5.79, the **General Data** view of the business partner contains key attributes such as the name, search term, and address information of the customer.

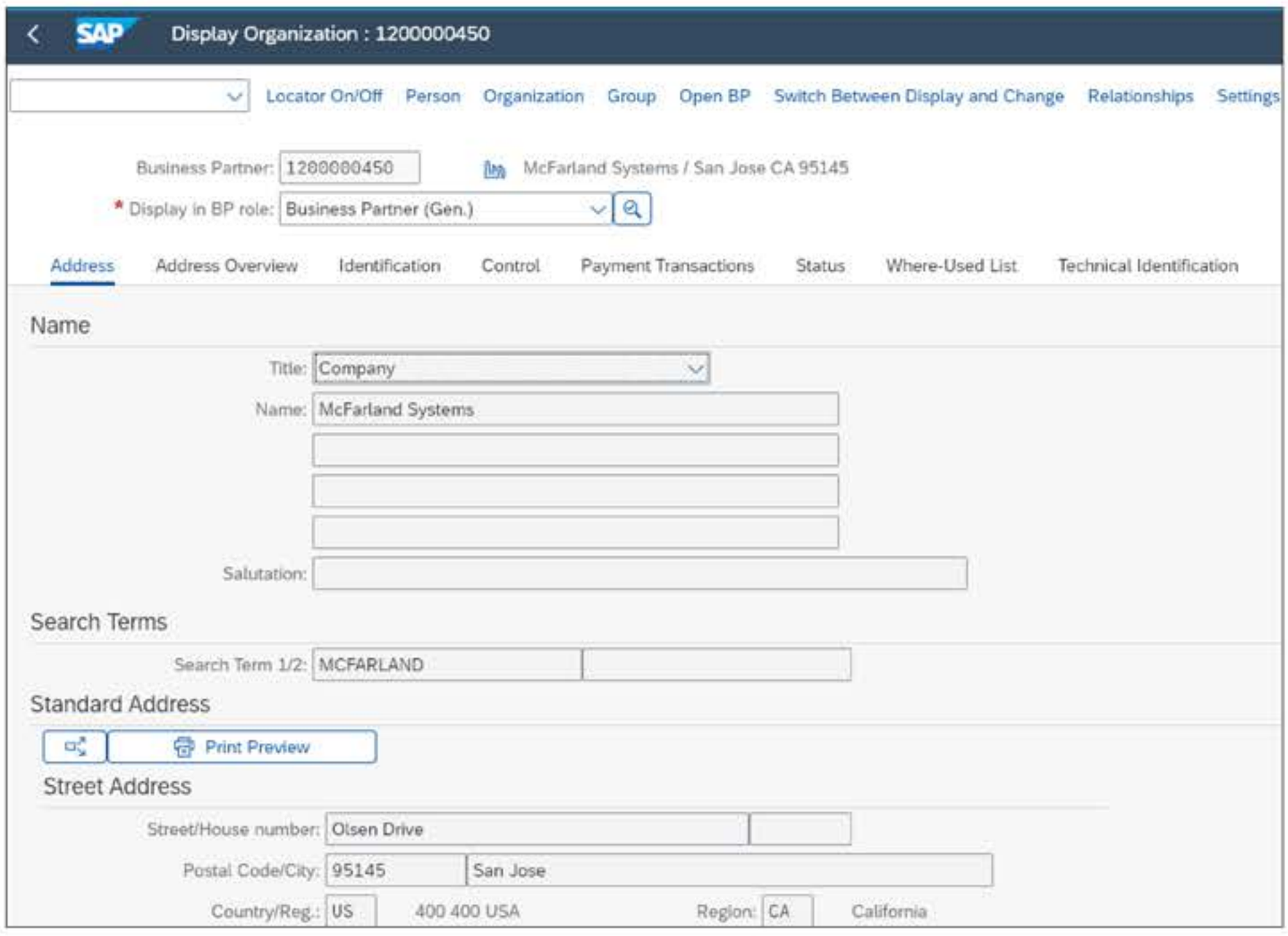

Figure 5.79 Business Partner: General Data View

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Now, click the **Where-Used List** tab for the business partner, as shown in Figure 5.80. You'll be able to see the references of the contract accounts and the provider contracts created for the business partner. Double-clicking any of the references will take you to the related screen; for example, double-clicking **Contract Account** will lead you to the contract account screen. This feature is very useful for navigating from one screen to another and is prevalent across all contract accounting screens.

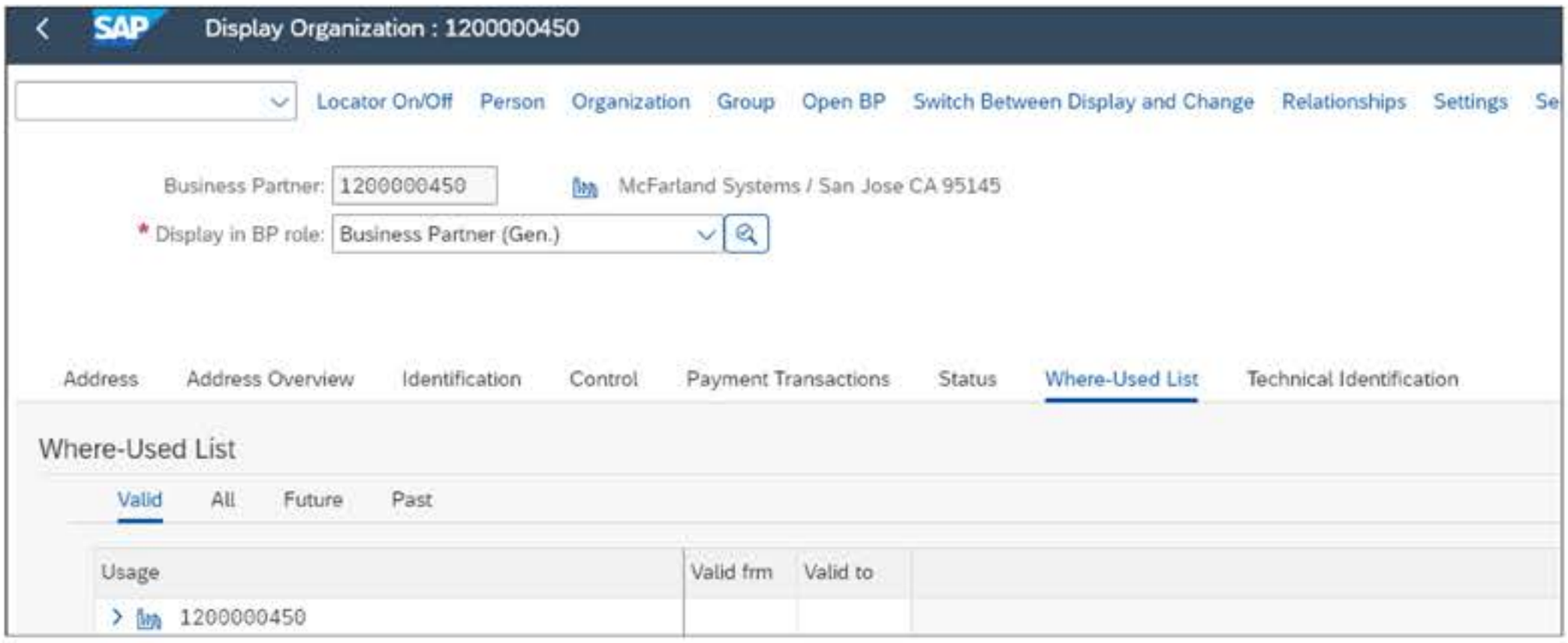

Figure 5.80 Business Partner: Where-Used List

To create a business partner directly in contract accounting, use Transaction BP. The business partner will need to be extended to the `MKK` contract partner role to create a corresponding contract account.

5.3.2 Creating a Contract Account

A contract account holds the transactional data for the business partner. All contract accounting document postings done for a business partner, whether payments, receivables, or interest, are recorded under the contract account. The contract account master data contains various details associated with mode payments, dunning, account determination, and billing cycles. This information is available under various tabs on the contract account display screen.

The contract account can be created directly in SAP Billing and Revenue Innovation Management, subscription order management (as a business agreement linked to the business partner), and replicated to contract accounting, or it can be created directly in contract accounting and replicated to SAP Convergent Charging (as a subscriber account).

Note

The basic settings required for the creation of the contract account are described in detail in Chapter 4, Section 4.2.3.

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As shown in Figure 5.81, contract account can be directly created in contract accounting using Transaction CAA1, which can also be accessed via the SAP Easy Access menu: **Accounting • Financial Accounting • Contract Account Receivables & Payables • Master Data • Contract Account**.

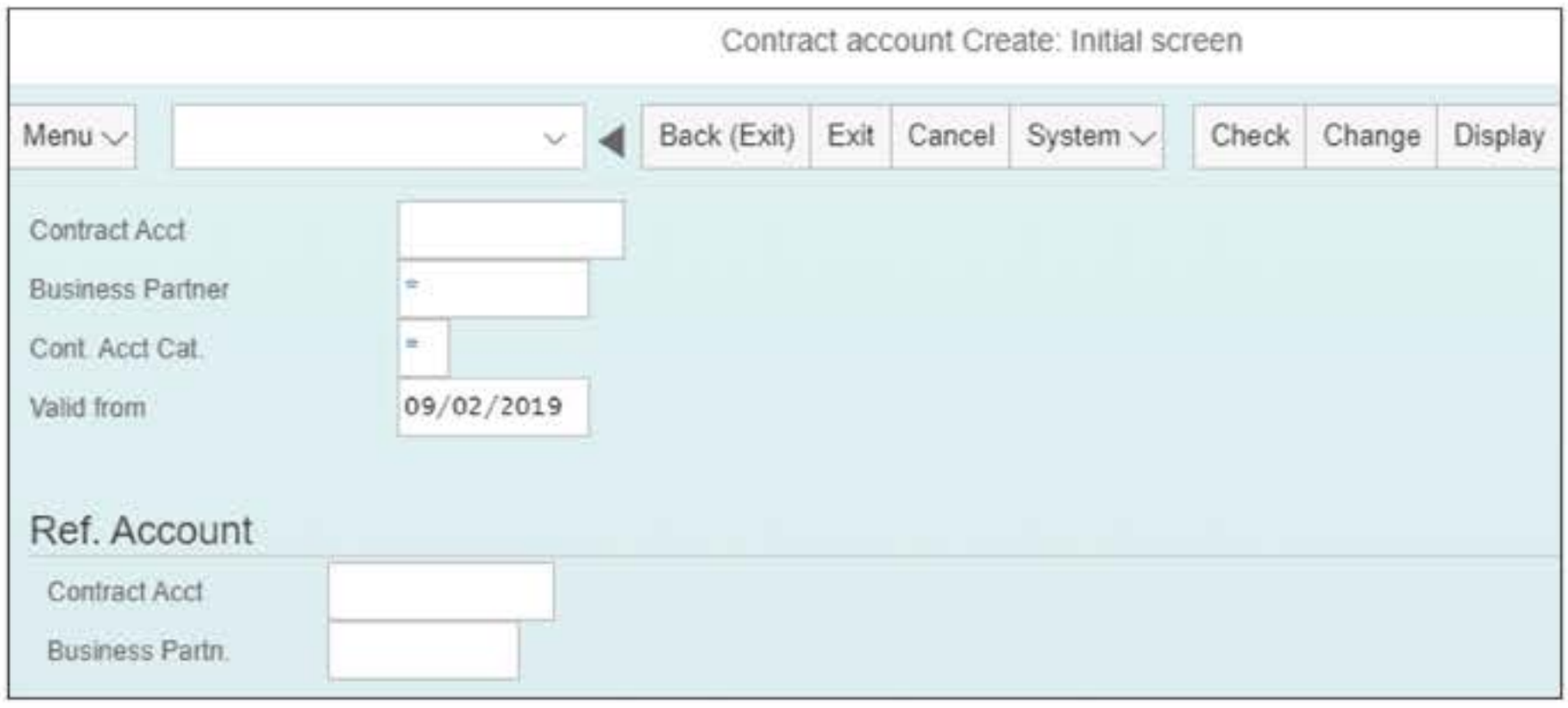

Figure 5.81 Contract Account Creation: Initial Screen

The contract account can be created in reference to the existing contract account by providing the reference contract account on the creation screen. Certain details such as contract account category, account relationship, tolerance group, clearing category, account determination ID, payment terms, and company code are mandatory values for creating a contract account.

Contract Account 110000000194 for McFarland Systems has been created for the Martex example.

Now let's use Transaction CAA3 to view the contract account (see Figure 5.82). It can also be accessed via the SAP Easy Access menu: **Accounting • Financial Accounting • Contract Account Receivables & Payables • Master Data • Contract Account**.

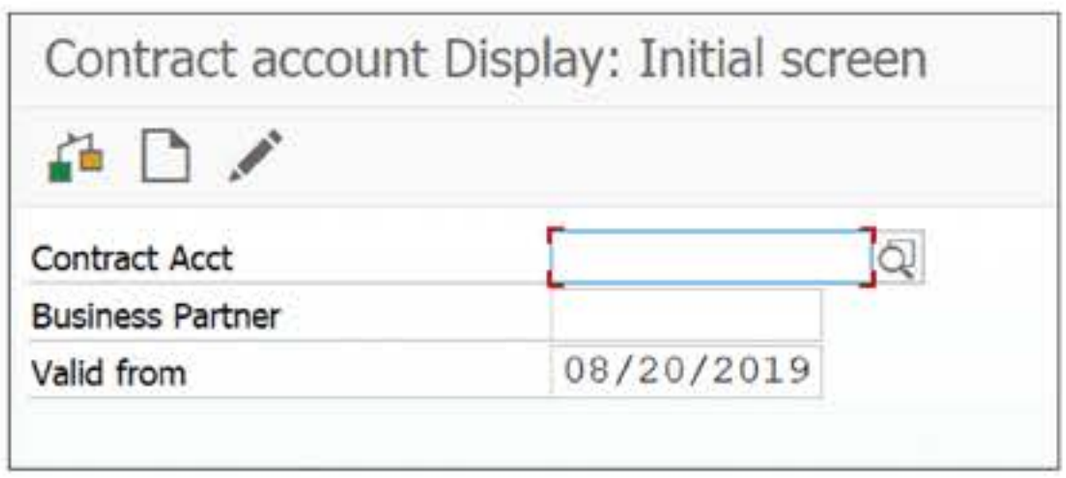

Figure 5.82 Contract Account Display Screen

5.3.3 Viewing or Editing a Posting Document

A posting document in contract accounting is an accounting document that gets posted during the invoicing process execution. The posting document contains both

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master data details of the customer, such as business partner, contract account, and contract, and also the transaction data for the business transaction, such as sale of service, tax data, and other payment-related details. Contract accounting documents can originate directly within contract accounting via processes such as interest calculation, dunning charges, installment plans, and payment lot postings.

The documents in contract accounting can be viewed using Transaction FPE3, which also can be accessed via the SAP Easy Access menu: **Accounting • Financial Accounting • Contract Account Receivables & Payables • Document** (see Figure 5.83).

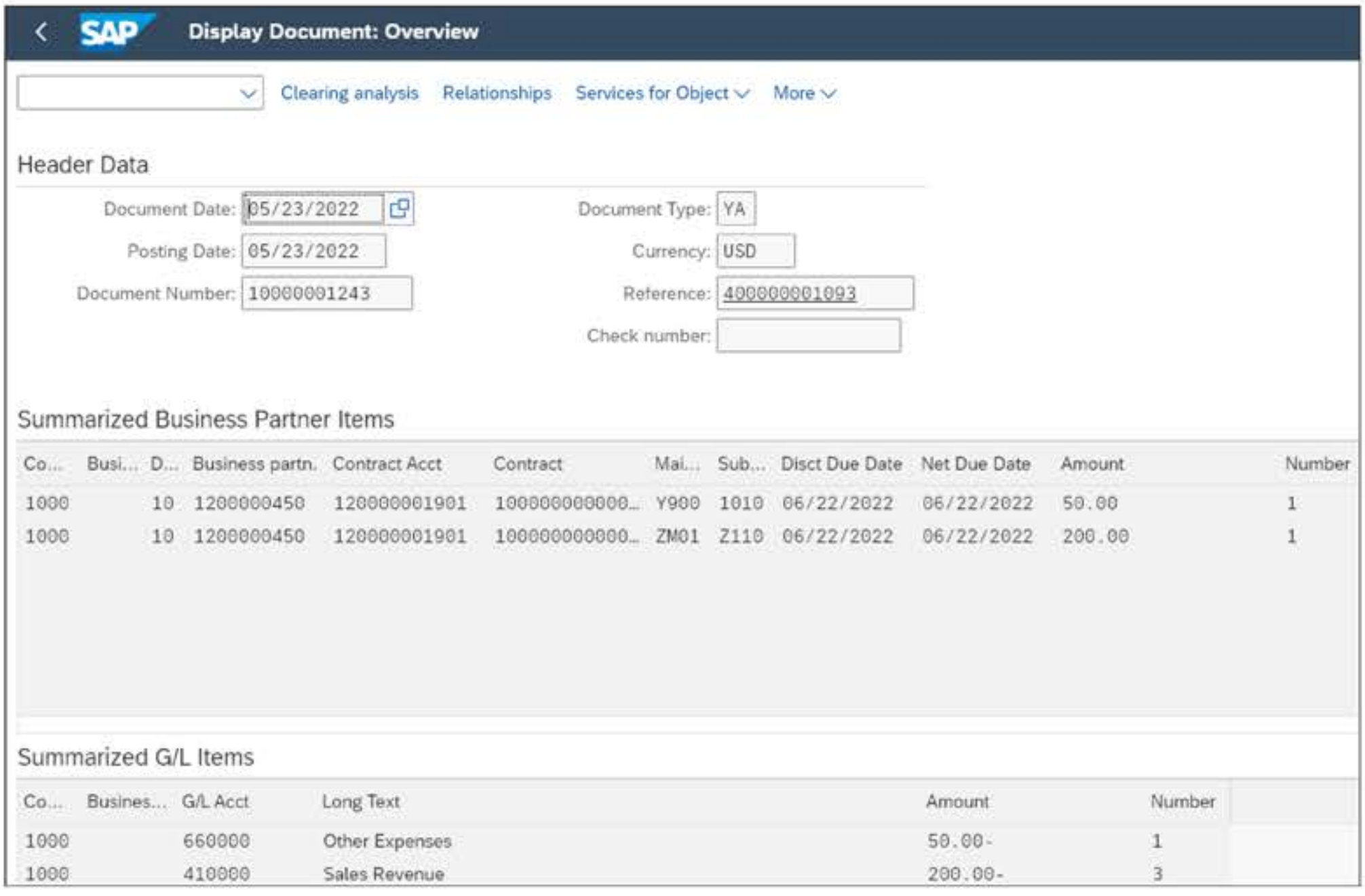

Figure 5.83 Contract Accounting Document View

The screen provides the option to directly view the business partner items, general ledger items, and document header details. Upon execution, the header, summarized business partner items, and summarized general ledger items are displayed (see Figure 5.84).

Certain posting document fields can be modified using the **Display -> Change** document option available under the **More** menu on the display screen (see Figure 5.85).

A document can also be modified using Transaction FPE2, which can be accessed via the SAP Easy Access menu: **Accounting • Financial Accounting • Contract Account Receivables & Payables • Document** (see Figure 5.86).

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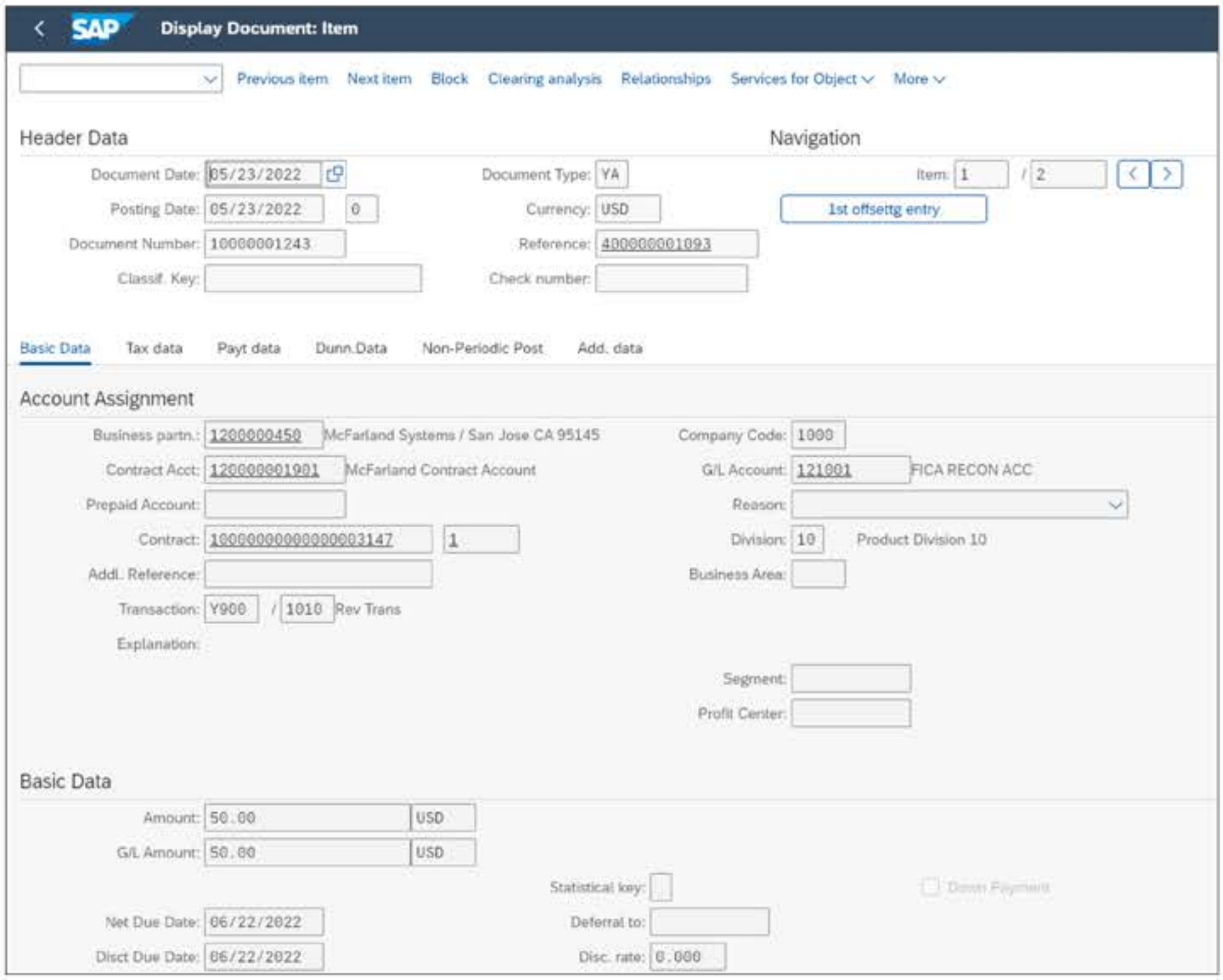

Figure 5.84 Contract Accounting Document Item View

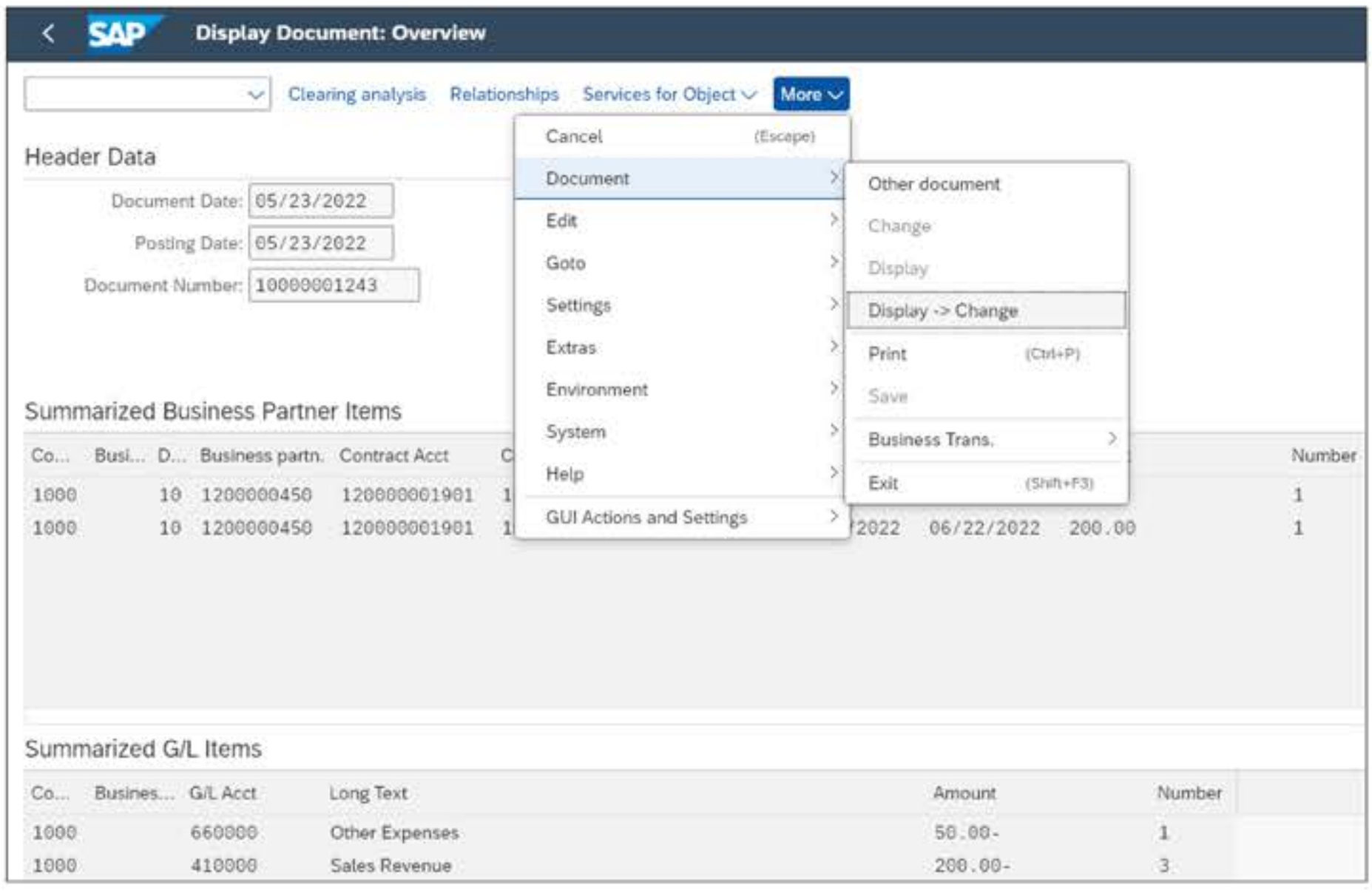

Figure 5.85 Menu Path for Document Change

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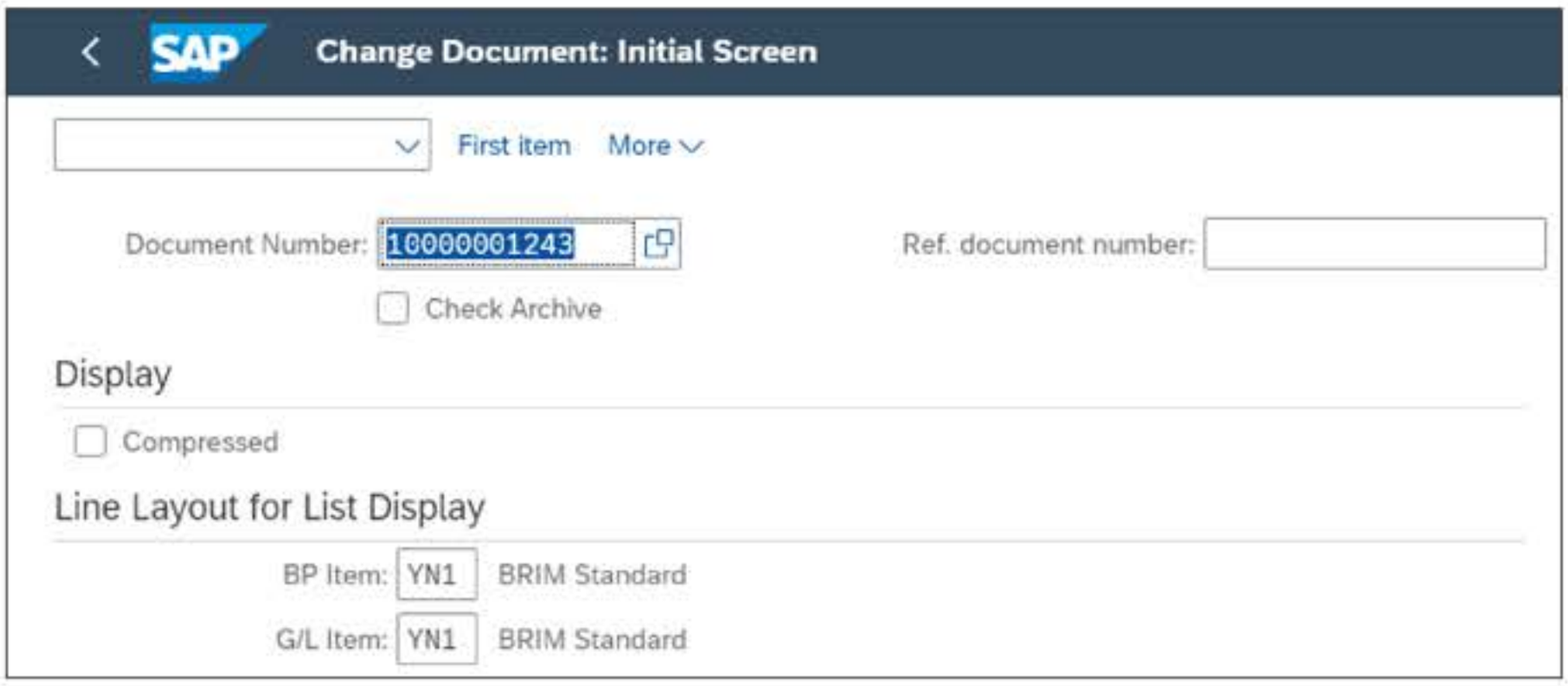

Figure 5.86 Transaction FPE2 for Document Change

The fields that are made changeable via configuration are available to change in the **Change Document: Initial Screen**.

Fields can be made changeable under the Transaction SPRO node at **Contract Account Receivables & Payables • Basic Functions • Postings & Documents • Basic Settings • Define Authorizations for Field Changes** (see Figure 5.87).

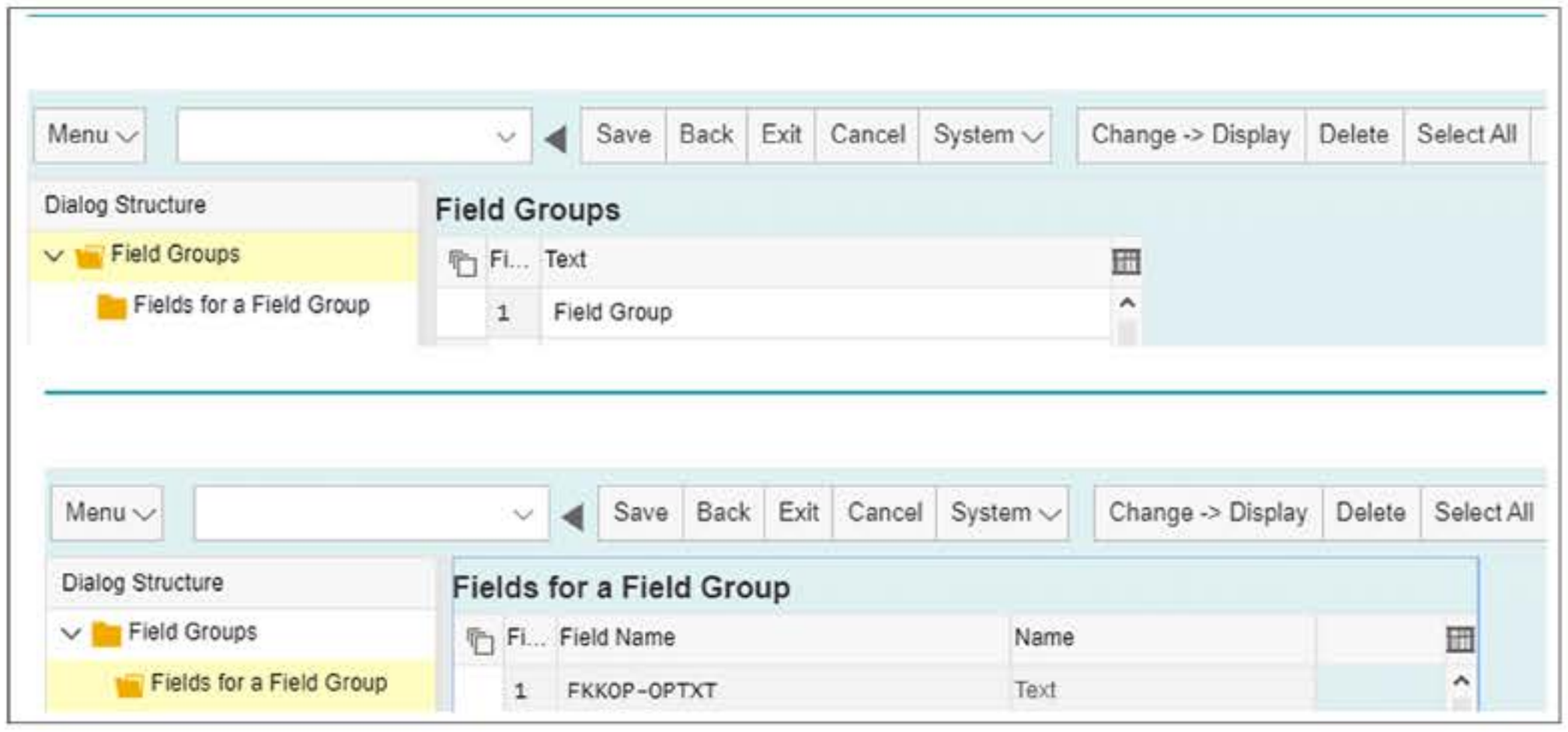

Figure 5.87 Field Group for Modifiable Document Fields

For the created group, change authorization can be given to users.

5.3.4 Viewing a Customer's Balance on Account

The customer's account balance display provides a view of all the transactions or accounting entries posted to the customer's account and even provides other details, such as receivable item status (**Open**, **Cleared**, or **Overdue**) or dunning level.

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The customer account balance can be viewed in contract accounting, as shown in as shown in Figure 5.88 and Figure 5.89, using Transaction FPL9, which also can be accessed via the SAP Easy Access menu: **Accounting • Financial Accounting • Contract Account Receivables & Payables • Account**.

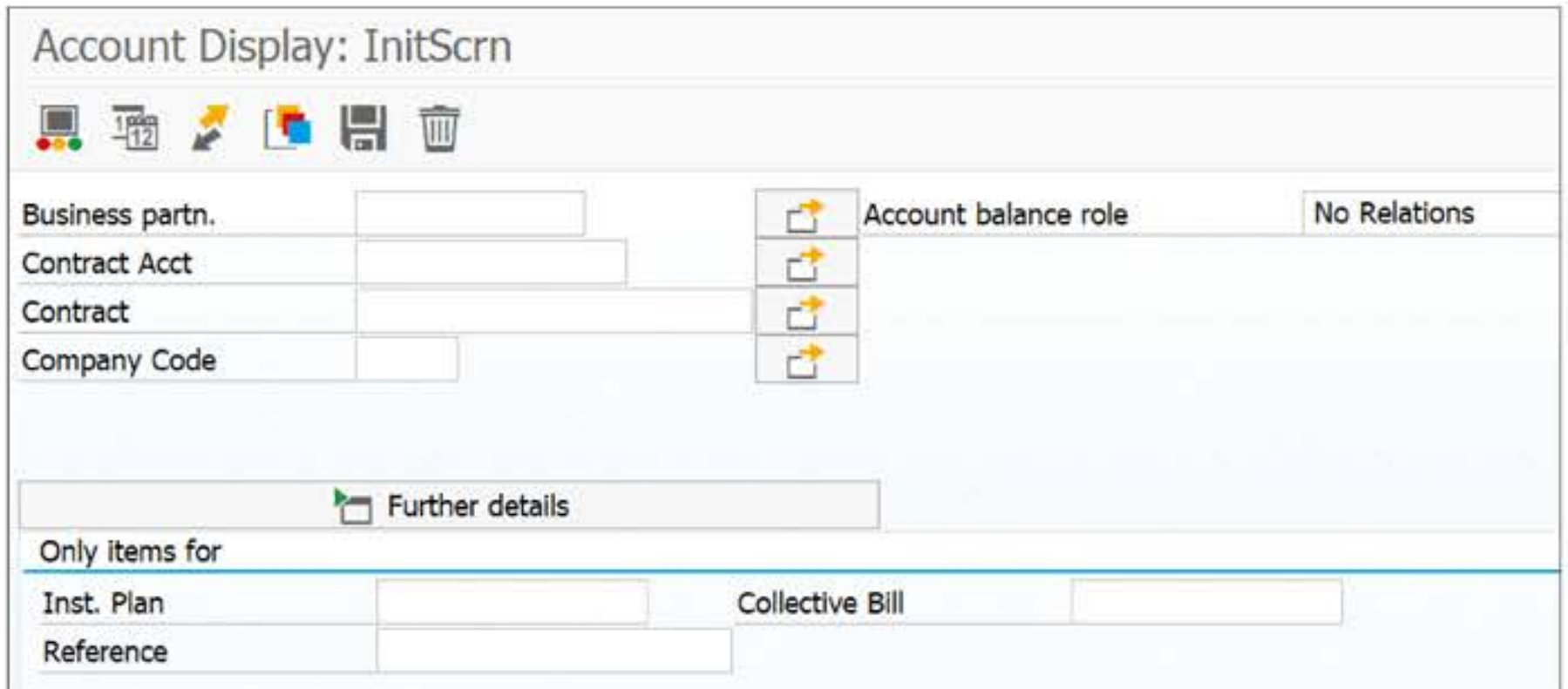

Figure 5.88 Account Balance Initial Screen

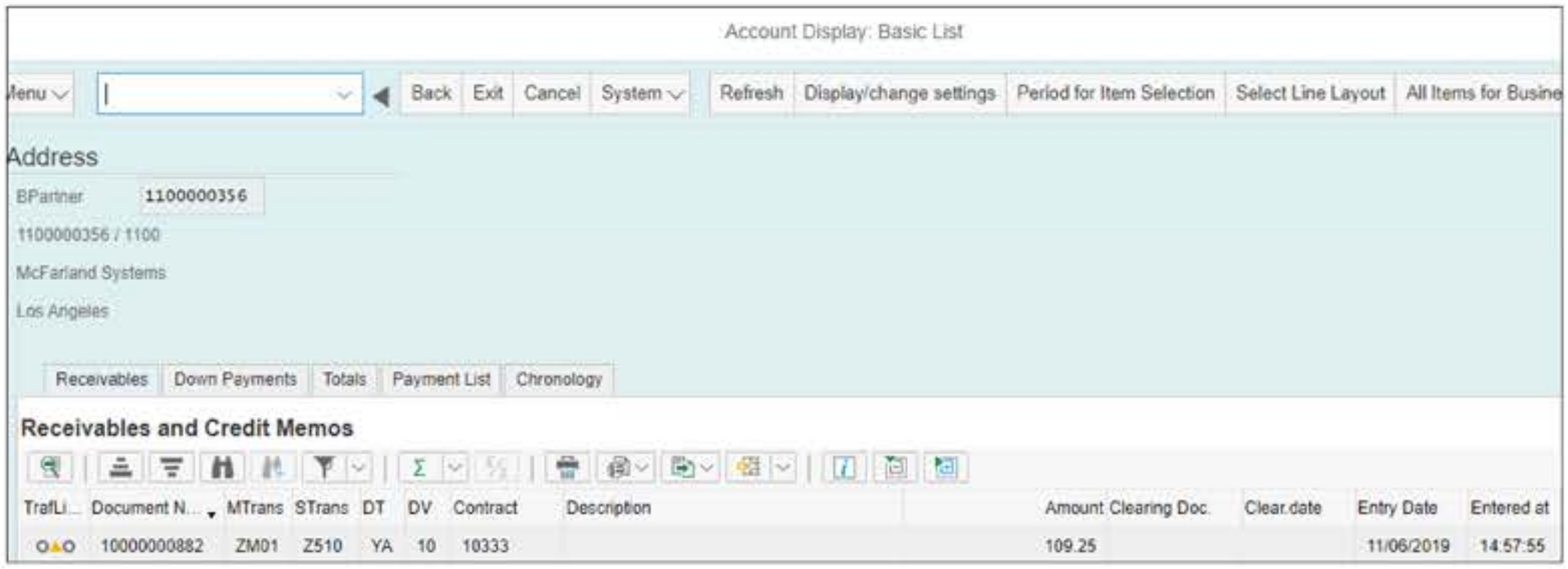

Figure 5.89 Account Balance Screen

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The receivables and payment postings are displayed under the **Receivables** tab. The **Down Payments** tab contains the released security deposit payment or down payment items, the **Totals** tab shows the total amount, the **Payment List** tab shows only the payments posted on the account, and the **Chronology** tab displays the receivables and associated posted payments.

Further details of each posted document can be seen by double-clicking the document. The various columns shown in the account balance screen are controlled using the line-item layout. The account balance display is described in detail in Section 5.2.3.

5.3.5 Using Open Item Management

This section explains the various contract accounting functionalities for managing the open items on the customer/vendor account. Functionalities such as clearing, account maintenance, clearing control, and clearing restrictions are covered here.

Account Maintenance

Account maintenance is used to set off a credit balance with a debit balance present on a customer's account. This is key in scenarios in which there is a credit balance originating from a credit posting or the release of a security deposit and the credit amount needs to be adjusted with the monthly or the final invoice.

Account maintenance can be performed explicitly by manually executing the account maintenance transaction or in a mass batch job each day at the end of business hours, or it can be done implicitly during the process such as clearing during payments or invoicing.

To execute manual account maintenance (see Figure 5.90), use Transaction FPO6, which also can be accessed through the SAP Easy Access menu: **Accounting • Financial Accounting • Contract Account Receivables & Payables • Account**.

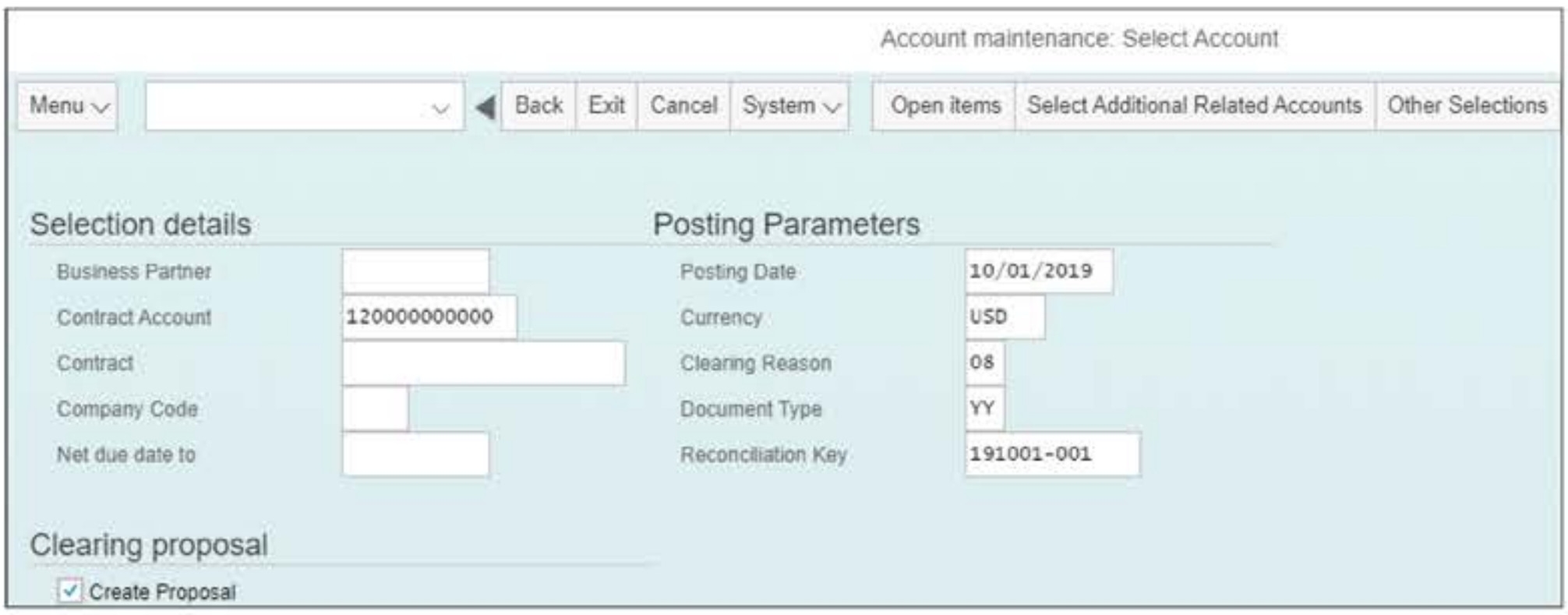

Figure 5.90 Account Maintenance Screen

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On the **Account maintenance** screen, the selection parameters are entered to determine the account for which account maintenance needs to be performed. On execution, the list of open items and payment or credit on an account is displayed. The following steps should be performed for account maintenance:

1. **Select and activate items.**
   The items that are to be set off are selected and activated. Upon activation, the items are marked for account maintenance. Under the processing status, the difference becomes green if the difference amount is zero; otherwise, it's red. If the difference amount isn't zero, then a debit balance amount is made equal to the credit item in the **Gross Clearing** field, as shown in Figure 5.91.

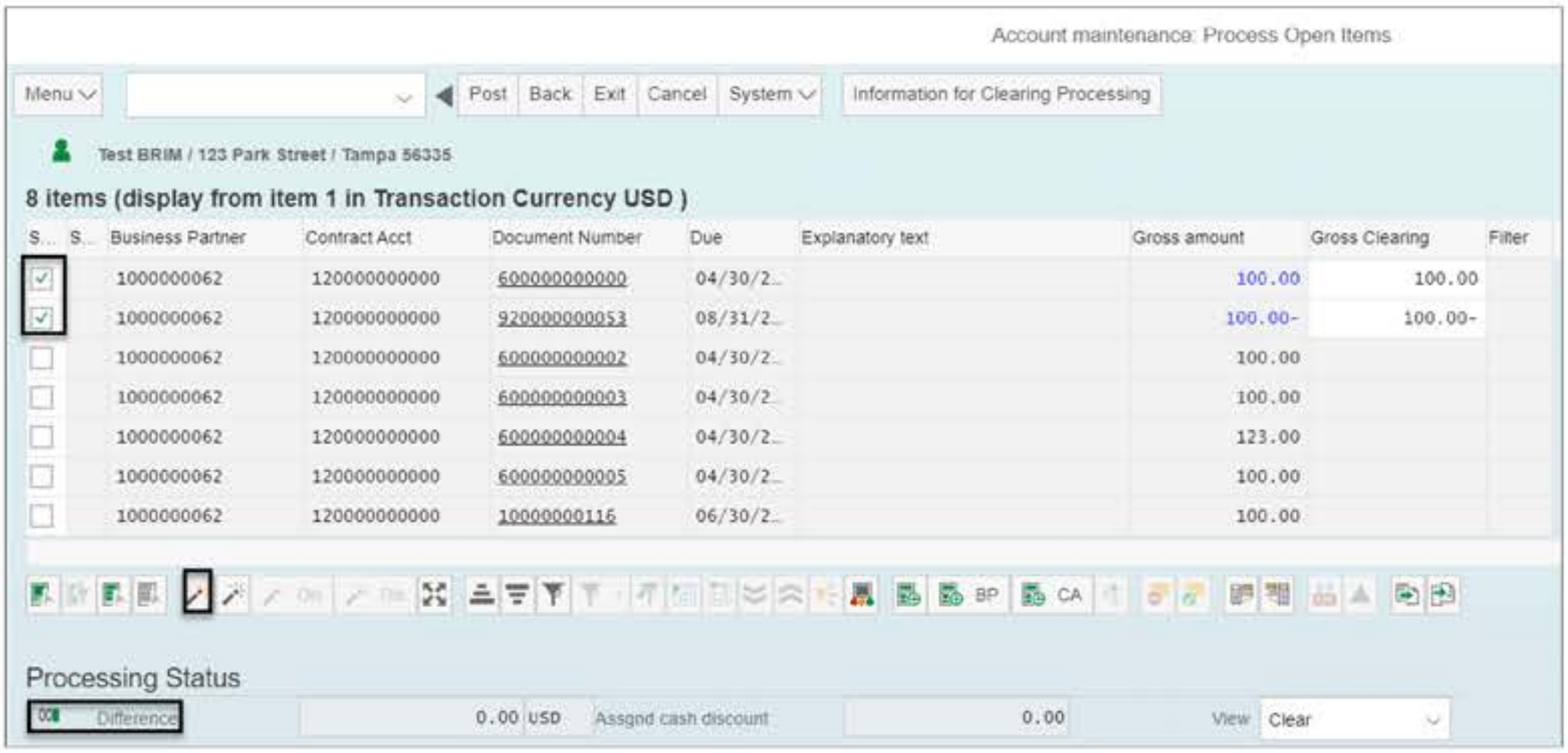

Figure 5.91 Selection and Activation of Account Items

2. **Post the clearing document.**
   Selecting the **Post** option posts the clearing document, and the credit and debit items are set off as shown in Figure 5.92.

The account maintenance in the production system is done using a mass batch job. When account maintenance must be done on multiple accounts and in parallel, the mass run is executed to set off the credit balance on the customer's account with the debit balance. Use Transaction FPMA for this task, which also can be accessed through the SAP Easy Access menu: **Accounting • Financial Accounting • Contract Account Receivables & Payables • Periodic Processing • For Contract Accounts**.

When the run date and identification are entered and the selection parameters maintained, the corresponding open credit and debit balances are picked up and cleared on execution based on the clearing rules maintained for the automatic clearing.

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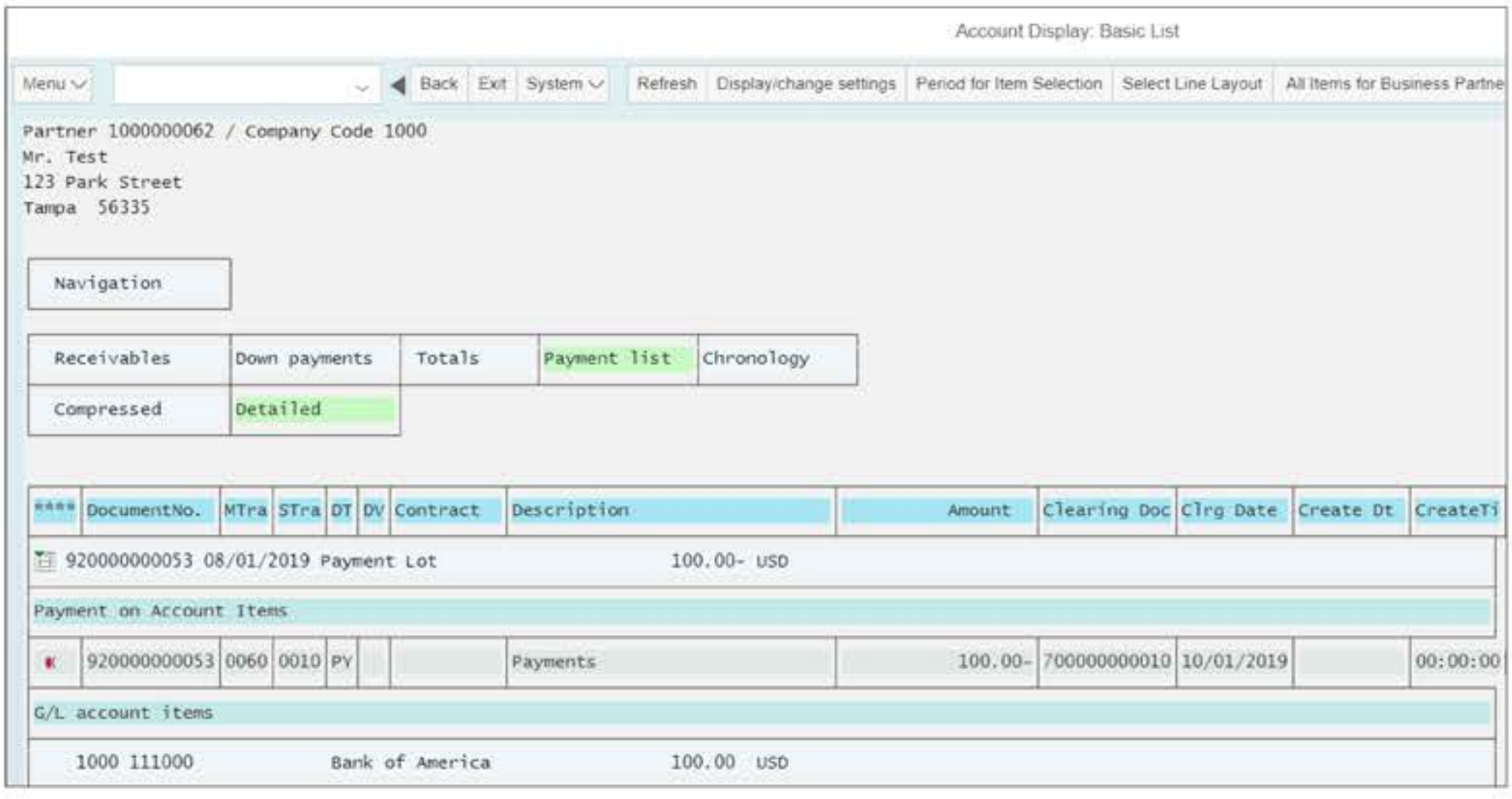

Figure 5.92 Transaction FPL9: Sample Clearing Document Posted by Account Maintenance

Certain configurations are required for the account maintenance to determine the document type clearing reason while the transaction is executed for posting the clearing document. Posting area 1020 is maintained for these entries.

Posting Area 1020

Default settings associated with the account maintenance such as document type, clearing reason, and whether a clearing proposal should be created during account maintenance are maintained in posting area 1020, as shown in Figure 5.93.

SAP provides a set of standard clearing reasons for the type of process the clearing is originating from. Clearing reason 08 is used for account maintenance performed using Transaction FP06 or done manually.

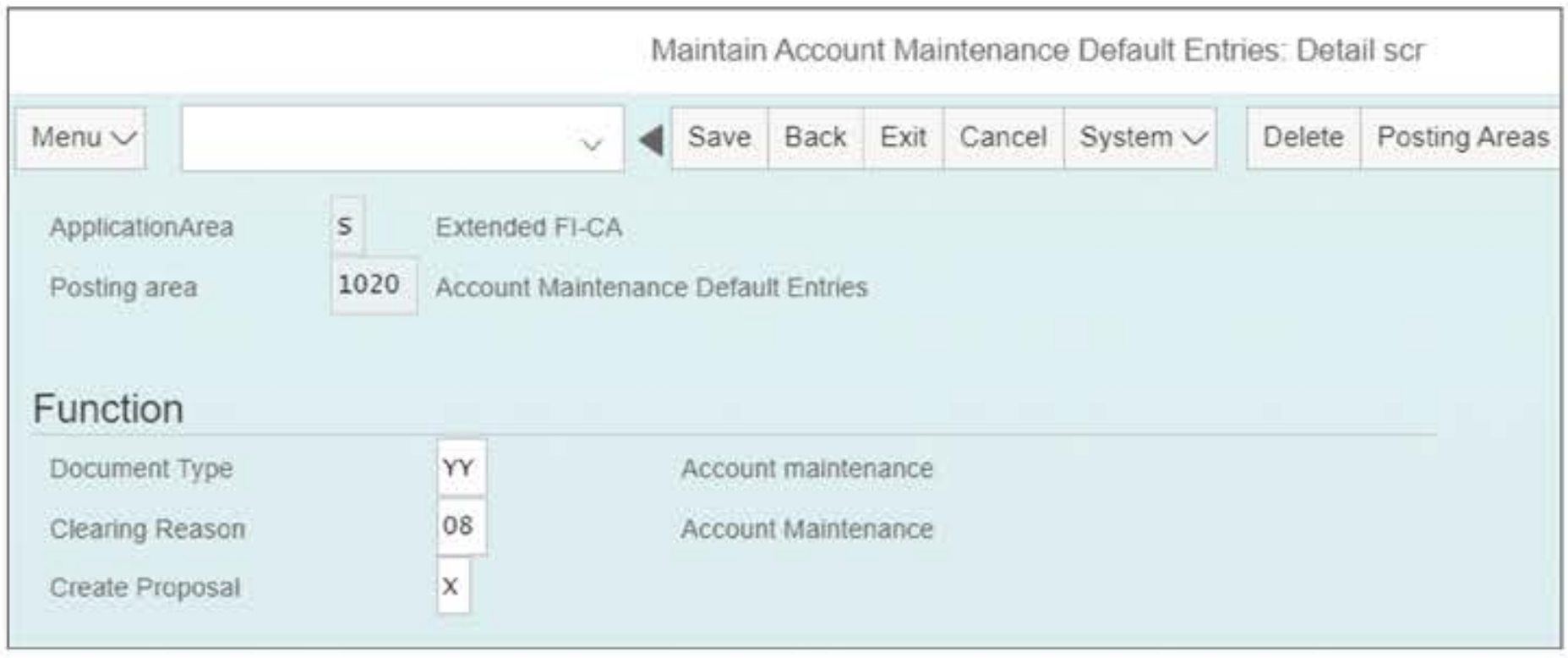

Figure 5.93 Posting Area 1020

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Posting Area 1025

Automatic clearing also requires values such as **Document Type** and **Clearing Reason** to be determined during the posting of the automatic clearing or mass clearing. These parameters are maintained in posting area 1025, as shown in Figure 5.94.

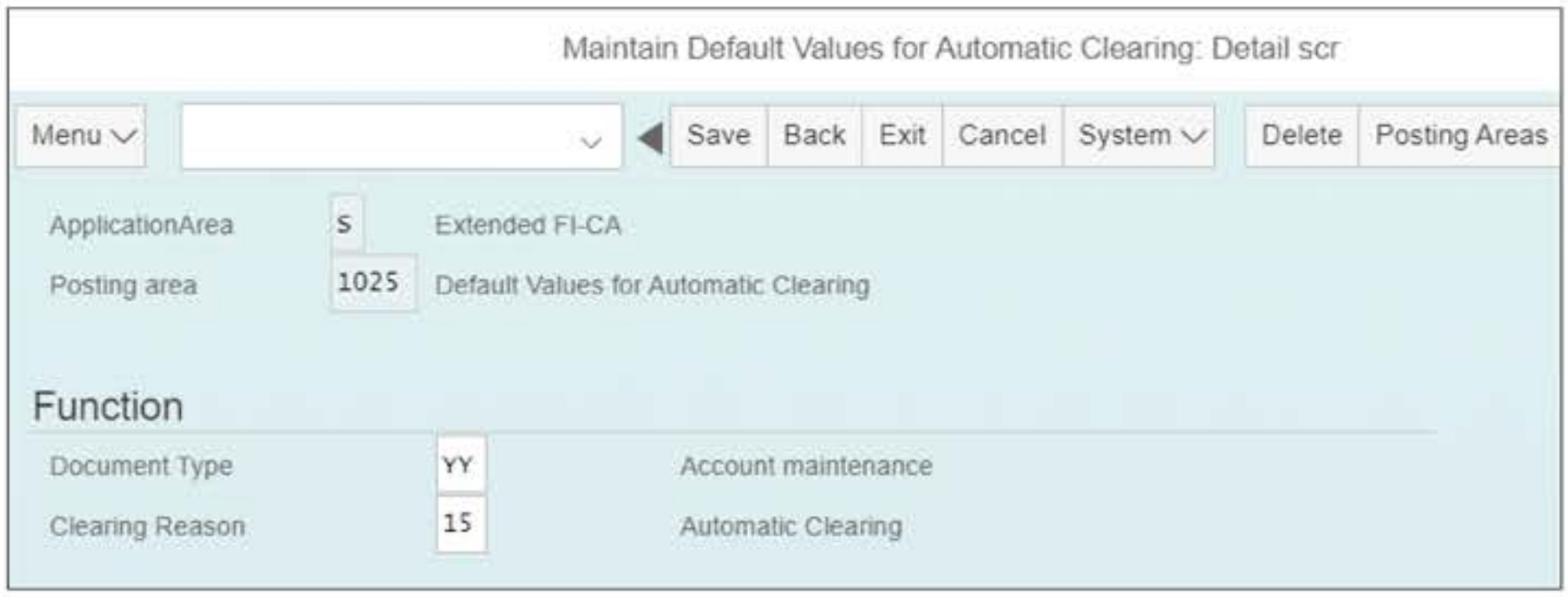

Figure 5.94 Posting Area 1025: Default Values for Automatic Clearing

Reset Clearing

A document can be cleared by the incoming payment, account maintenance, or invoicing process. When there is a need to separate the clearing document from the cleared item to either reverse the receivable item or make the incoming payment a payment on account, the clearing document is reset. Resetting the clearing document (see Figure 5.95) makes the receivable document an open item again, and the payment document becomes the payment on account.

Transaction FPO7 is used to perform the clearing reset (see Figure 5.95) and can also be accessed through the SAP Easy Access menu: **Accounting • Financial Accounting • Contract Account Receivables & Payables • Periodic Processing • For Contract Accounts • Documents**.

You enter the clearing document that needs to be reset in the **Clearing Doc.** field of the **Reset Cleared Items** screen and then select the scope of the reset. If a full reset of the clearing document is required, which makes the receivable item fully open (not partially cleared), you select the **Whole clearing** option.

Next, you select the setting for the posting clearing amount. To retain the original accounting entries of the receivable item, select the **Retain distribution to accounts** option. If you select **Create a new open item**, the original receivable remains cleared, and the new receivable is posted. In this scenario, the original receivable can't be reversed. On selecting the **Post** option, you reset the clearing and the payment, and the receivable item becomes open on the customer account.

On the **Reset Cleared Items** screen, parameters such as **Document Type** and **Clearing Reason** are required for posting the reset clearing document.

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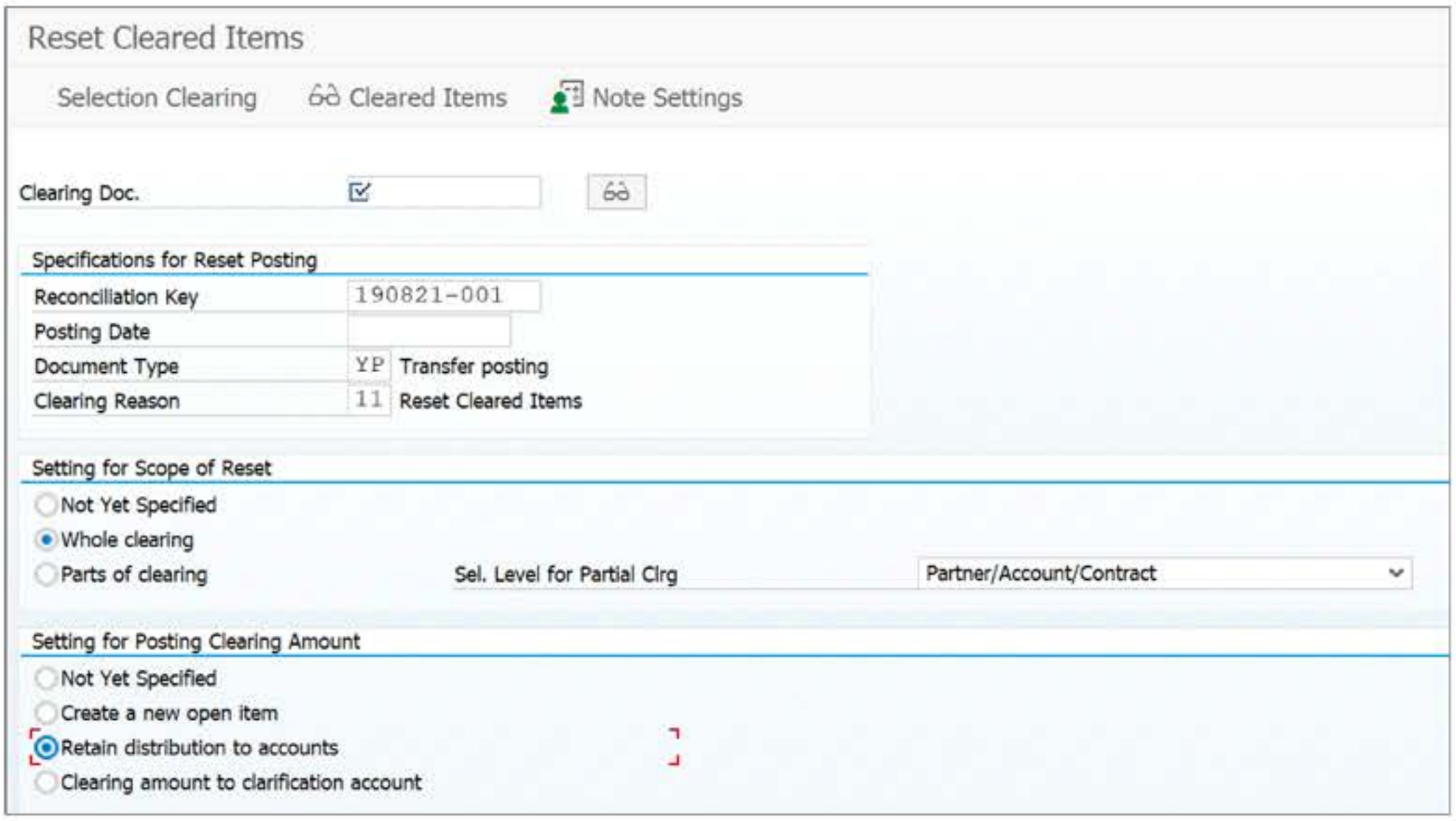

Figure 5.95 Reset Cleared Items Screen

Posting Area 1060

A clearing reason and document type for the reset clearing document are maintained in posting area 1060. These values are automatically populated on the reset clearing screen and are used for posting the reset clearing document, as shown in Figure 5.96.

Figure 5.96 Default Parameters for Reset Clearing

Posting Area 1090

While posting the reset clearing, if the **Create new open items** option is selected in the settings for the posting clearing amount, a new receivable item is created. The creation of a new receivable item requires the main transactions and subtransactions for the

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posting. Posting area 1090 is used to assign the main transactions and subtransactions for creation of new items during the reset clearing, as shown in Figure 5.97.

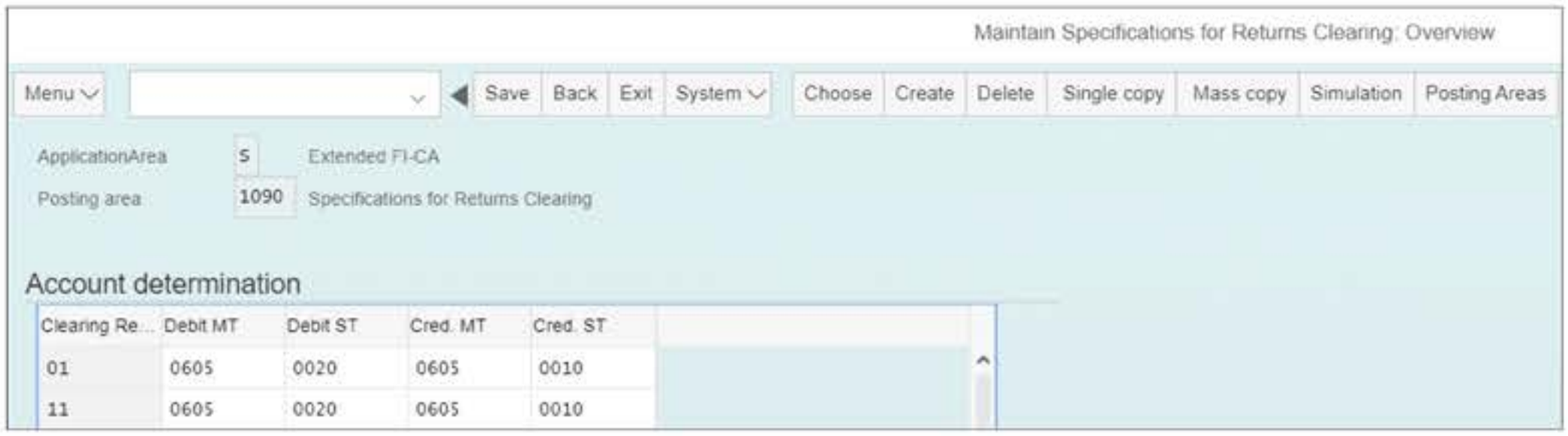

Figure 5.97 Main Transactions and Subtransactions Assignment to Clearing Reason for New Item Creation

Clearing Restrictions

Clearing restrictions are generally applied on the open items that should be ignored during the automatic clearing process. Based on business requirements, these restrictions can be removed, and the account items can be considered for clearing.

Clearing restrictions can be created (see Figure 5.98) under menu path **Contract Account Receivables & Payables • Basic Functions • Open Item Management • Define Modifiable Clearing Restrictions**.

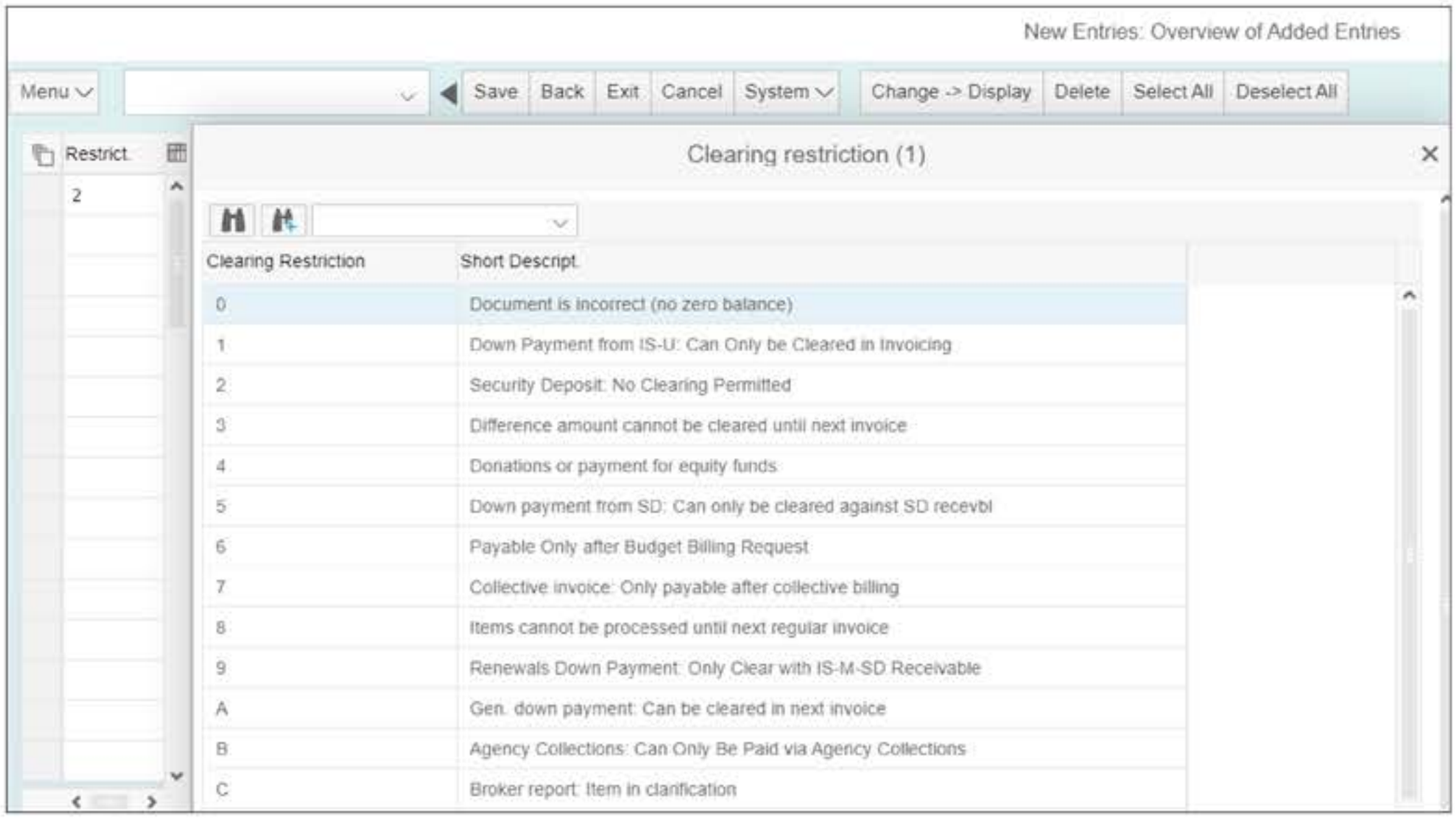

Figure 5.98 Modifiable Clearing Restrictions

The clearing restrictions applied on the documents can be removed manually using the change document functionality described in Section 5.3.3.

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Clearing Control

The clearing control is a very important functionality when it comes to the clearing of the open items. Product or service provider companies have different rules or laws based on which different kinds of receivables should be cleared.

Clearing control provides a set of configuration nodes in which you can set up rules for how the clearing of open items takes place. Different rules can be assigned to different payment originating processes, such as payment lot, payment run, account maintenance, invoicing, and so on. These rules guide the selection, grouping, and sorting of the open items for the payment or clearing to be applied.

The configuration for the clearing control is maintained under **Contract Account Receivables & Payables • Basic Functions • Open Item Management • Clearing Control • Clearing Variants**.

> **Clearing Variant**
>
> A clearing variant is created for different processes, and the steps are maintained, which are basically selecting, grouping, and sorting rules for the clearing process. These variants are assigned to the standard processes.

The various steps involved in setting up clearing variants are as follows:

1. **Define the grouping and sorting features.**
   SAP provides a set of standard fields that can be used for grouping and sorting the open items. These fields are from the FKKOP structure. Standard function module FSC_SAMPLE_TFK116 can also be used to create a custom function module based on requirements for the grouping fields, as shown in Figure 5.99.

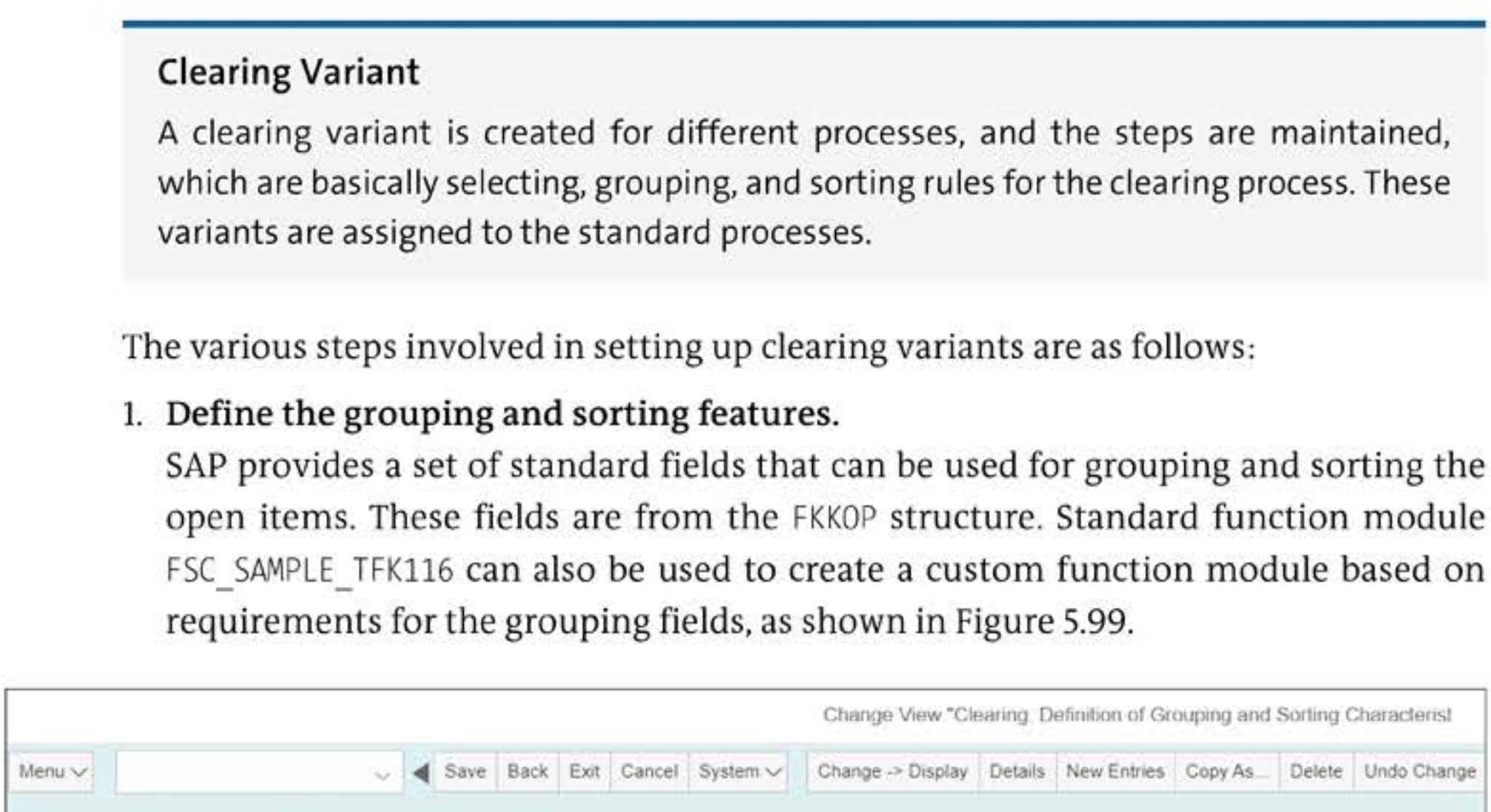

Figure 5.99 Grouping and Sorting Fields

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2. **Define a clearing variant and the clearing steps.**
   A clearing variant is created that is specific to the processes through which the clearing process is executed, such as account maintenance, invoicing, and incoming payments, as shown in Figure 5.100.

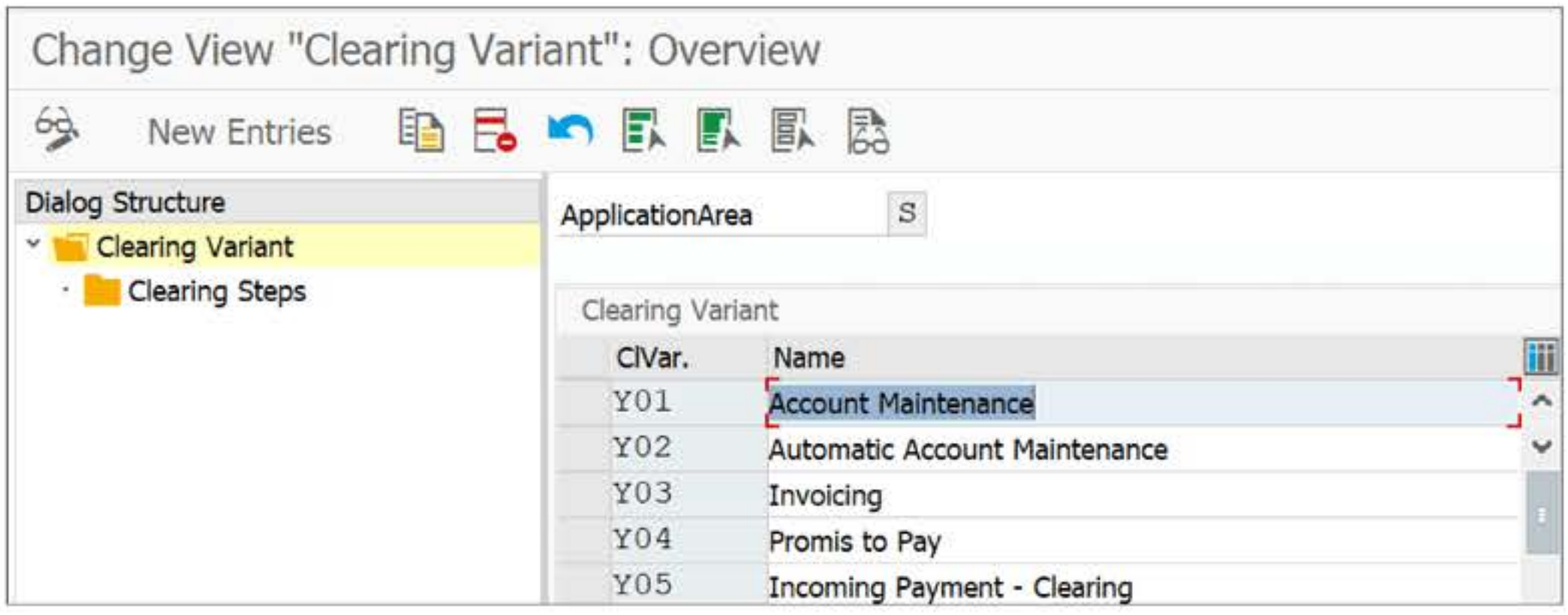

Figure 5.100 Clearing Variants

Clearing steps are created next for the clearing variants created. Clearing steps use the grouping and sorting fields defined for grouping the items and sorting them, as shown in Figure 5.101.

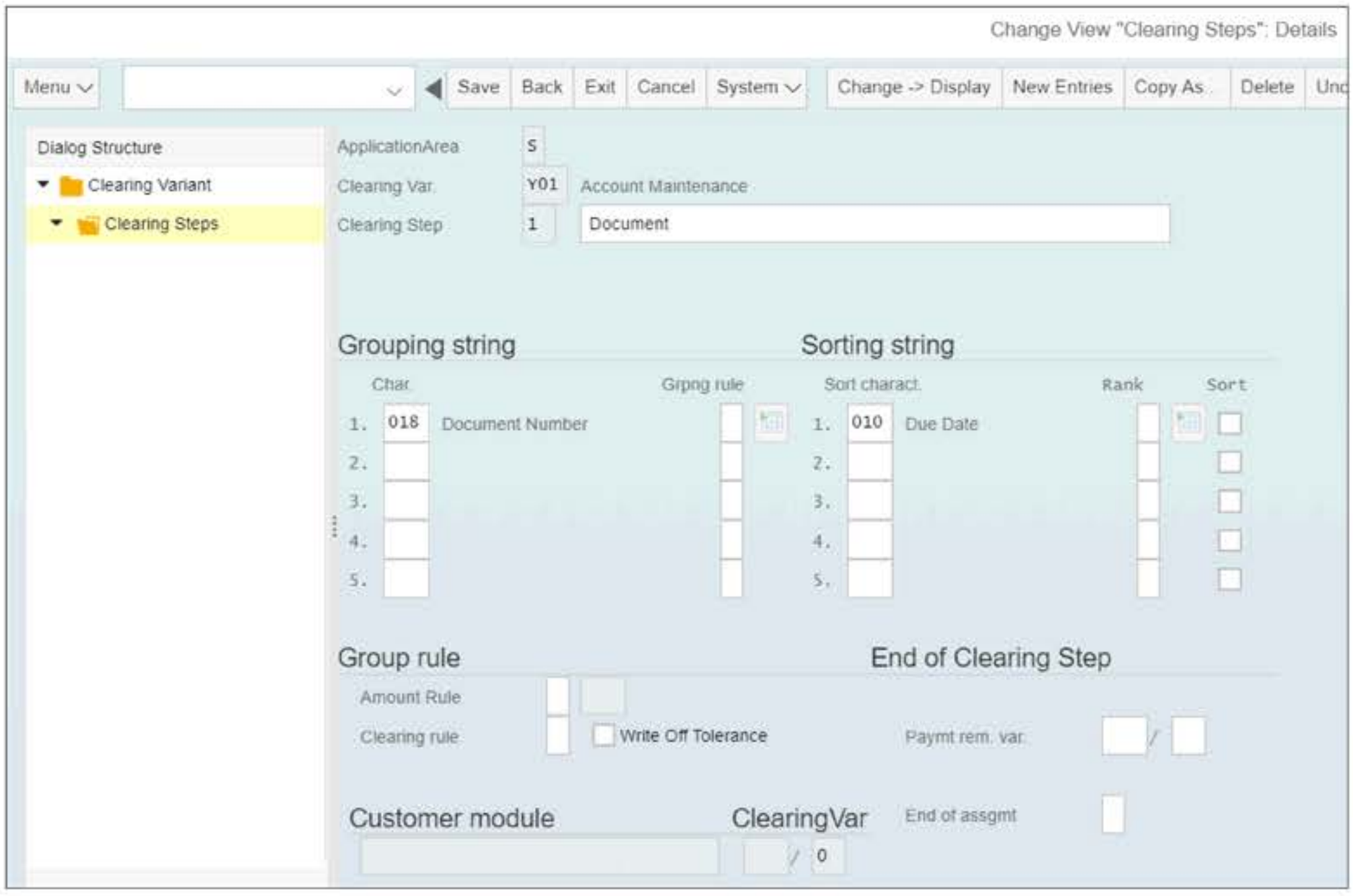

Figure 5.101 Clearing Steps

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After the clearing variants are created and clearing steps are defined, the clearing variants are assigned to the clearing types. The clearing types are the processes through which clearing is triggered upon execution. The assignment of the clearing variant to the clearing type is done under **Contract Account Receivables & Payables • Basic Functions • Open Item Management • Clearing Control • Define Clearing Specifications for Clearing Types**. Figure 5.102 shows the assignment.

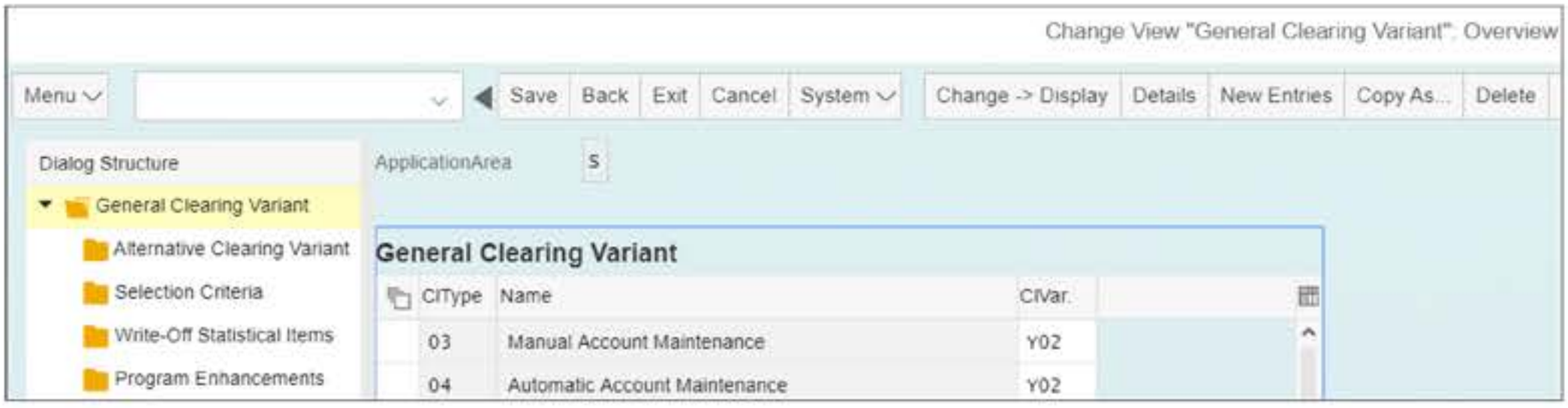

Figure 5.102 Clearing Variant Assignment to Clearing Type

5.3.6 SAP Fiori Apps

To provide the best user experience (UX) while interacting with SAP Business Suite, SAP has provided SAP Fiori apps. With SAP Fiori apps, users can access critical business applications on any device easily.

The SAP Fiori UX model is based on five pillars, as shown in Figure 5.103:

- **Role-Based**
  Users can access the apps according to the roles assigned to complete tasks specific to their roles.
- **Responsive**
  The apps can adapt according to the size and type of device used by the user to access the apps.
- **Simple**
  The scope of application is simple and follows a 1-1-3 model, which means 1 user, 1 use case, and maximum of 3 screens per application.
- **Coherent**
  Apps are developed with a clear and easy structure. All apps have the same language and can be deployed to multiple landscapes and environments.
- **Instant Value**
  Apps are easy to install and use.

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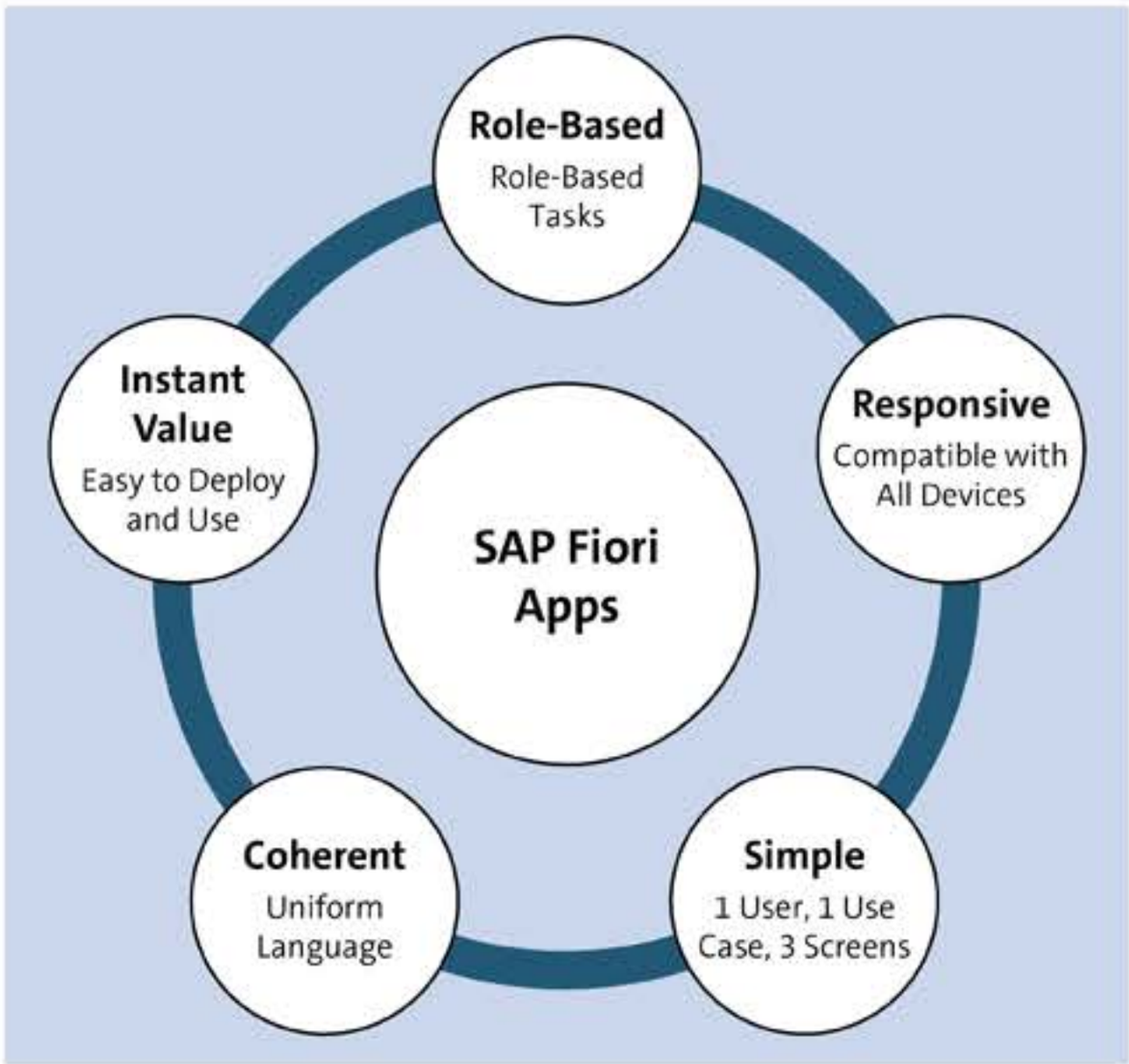

Figure 5.103 Pillars of SAP Fiori Apps

The commonly used SAP Fiori apps for contract accounting are described in the following list:

- **Manage Business Partner Master Data**
  This app is used to create, change, and display the business partner. Business partner role, business relationship, business partner general data, company code data, and sales and distribution data maintenance are performed using this app, as shown in Figure 5.104.

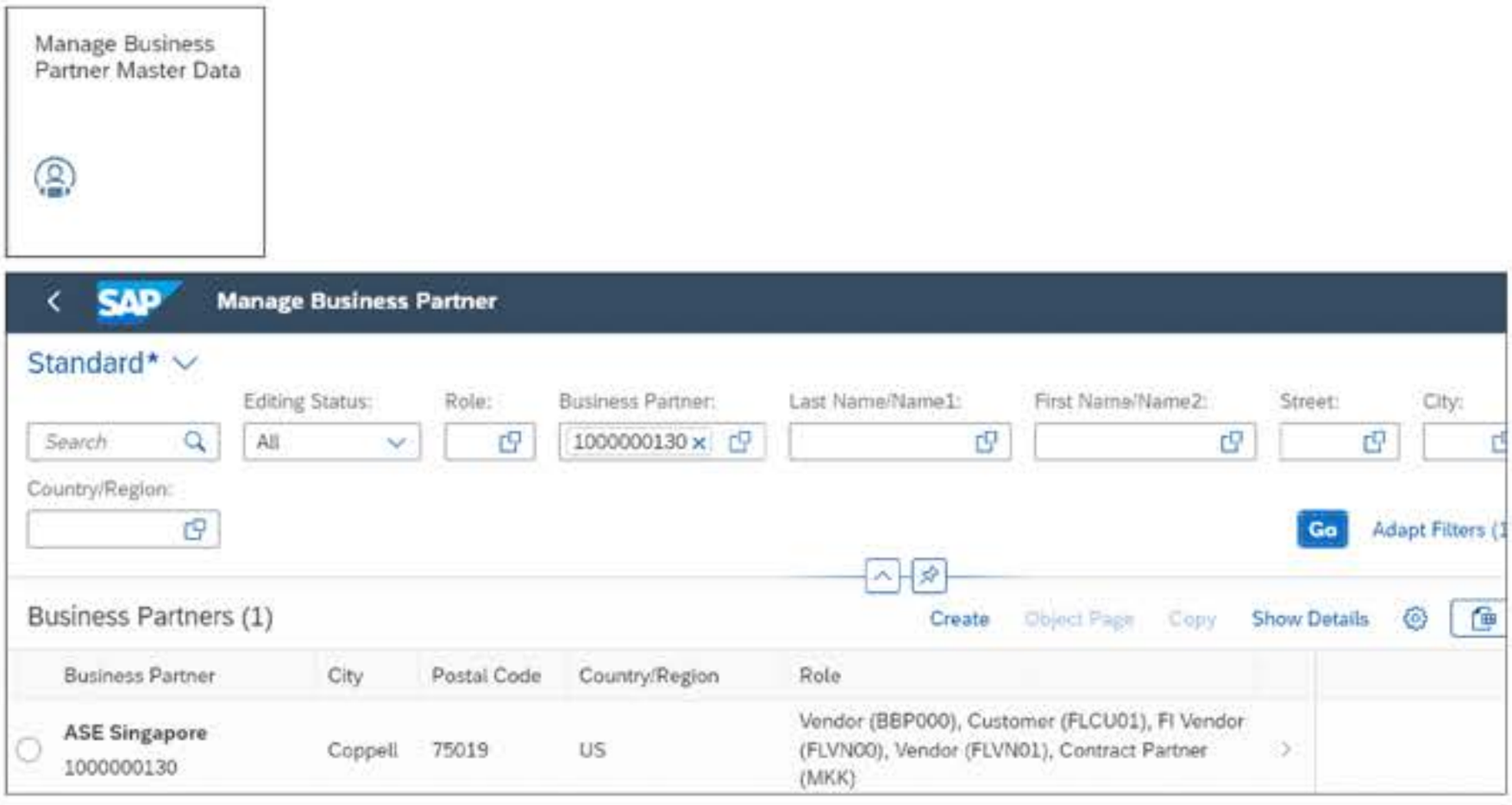

Figure 5.104 Manage Business Partner Master Data App

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- **Manage Contract Accounts**

  This app is used to create, change, and display contract accounts. Contract accounts contain various information such as payment terms, account determination ID, clearing category, and other payment- and dunning-related details that can be easily modified using the app. Figure 5.105 shows the screen and options for actions that can be performed on the contract account.

  To create the contract account, the business partner and contract account category are required.

  There are several mandatory fields as well, such as **Account Determination ID**, **Clearing Category**, **Tolerance Group**, **Account Relationship**, **Company Code**, and **Group Company Code**, which are required to be maintained to create the contract account. These fields can be found under the **General Data** and **Payment/Taxes** tabs of the contract account creation screen.

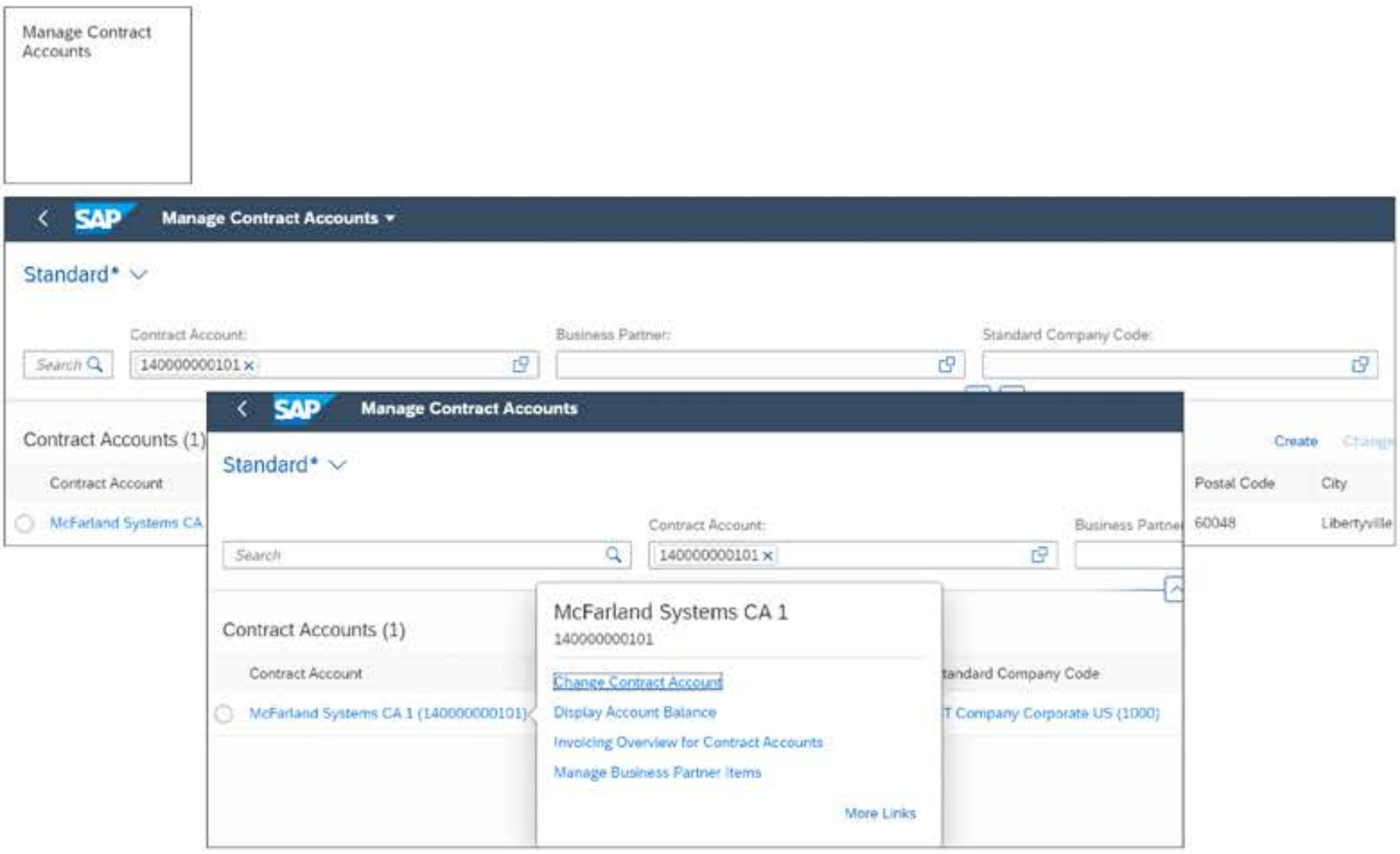

Figure 5.105 Manage Contract Accounts App

- **Manage Business Partner Items**

  This app is used to view and process documents posted on contract accounts under the given business partners. You can perform activities such as reversal, write-off, change, reset clearing, and clearing. Figure 5.106 shows the app name ❶ and the item view of the documents posted on the contract accounts ❷.

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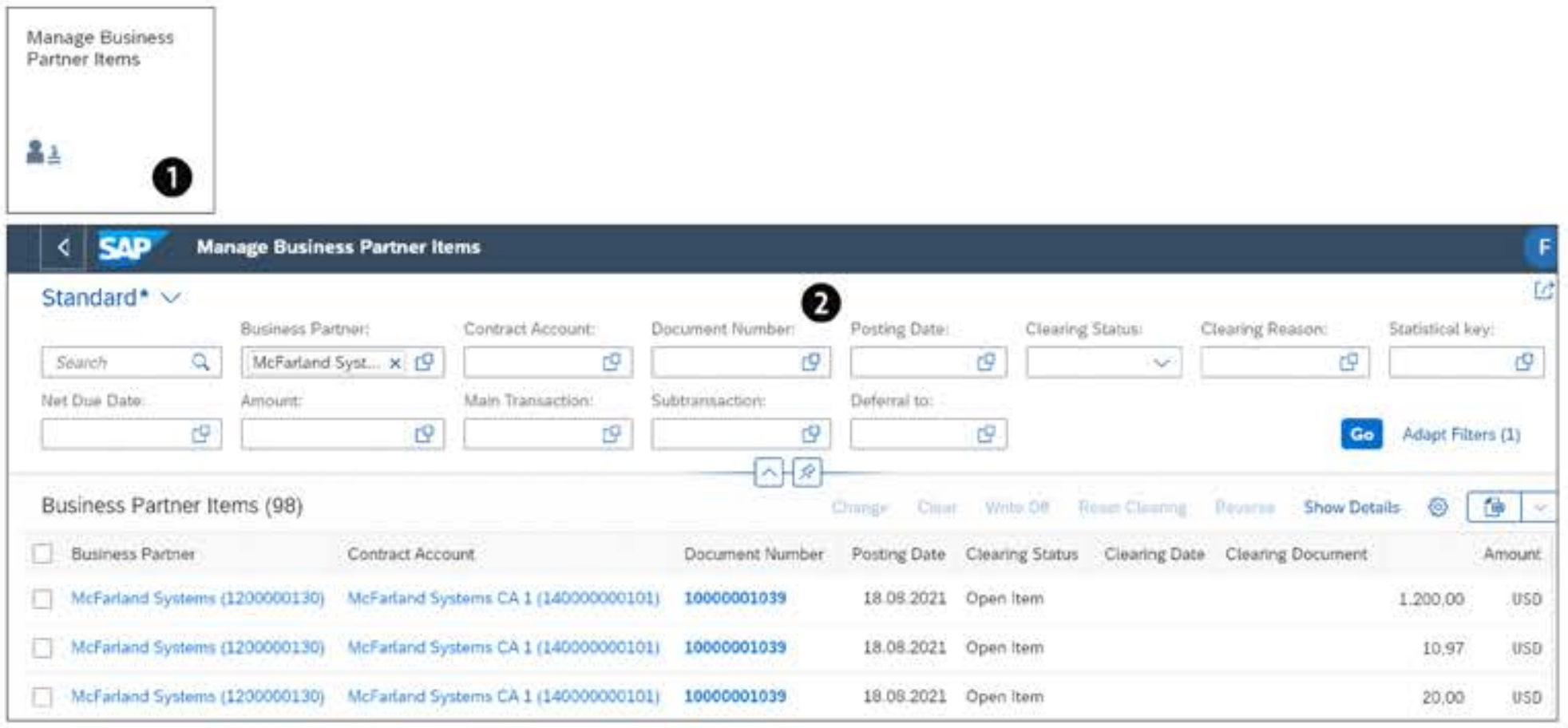

Figure 5.106 Manage Business Partner Items App

Various selection criteria are available to control the output display. Clicking on the individual document number opens the list of actions that can be performed on the document, as shown in Figure 5.107. When you select the action, the document will open for further processing.

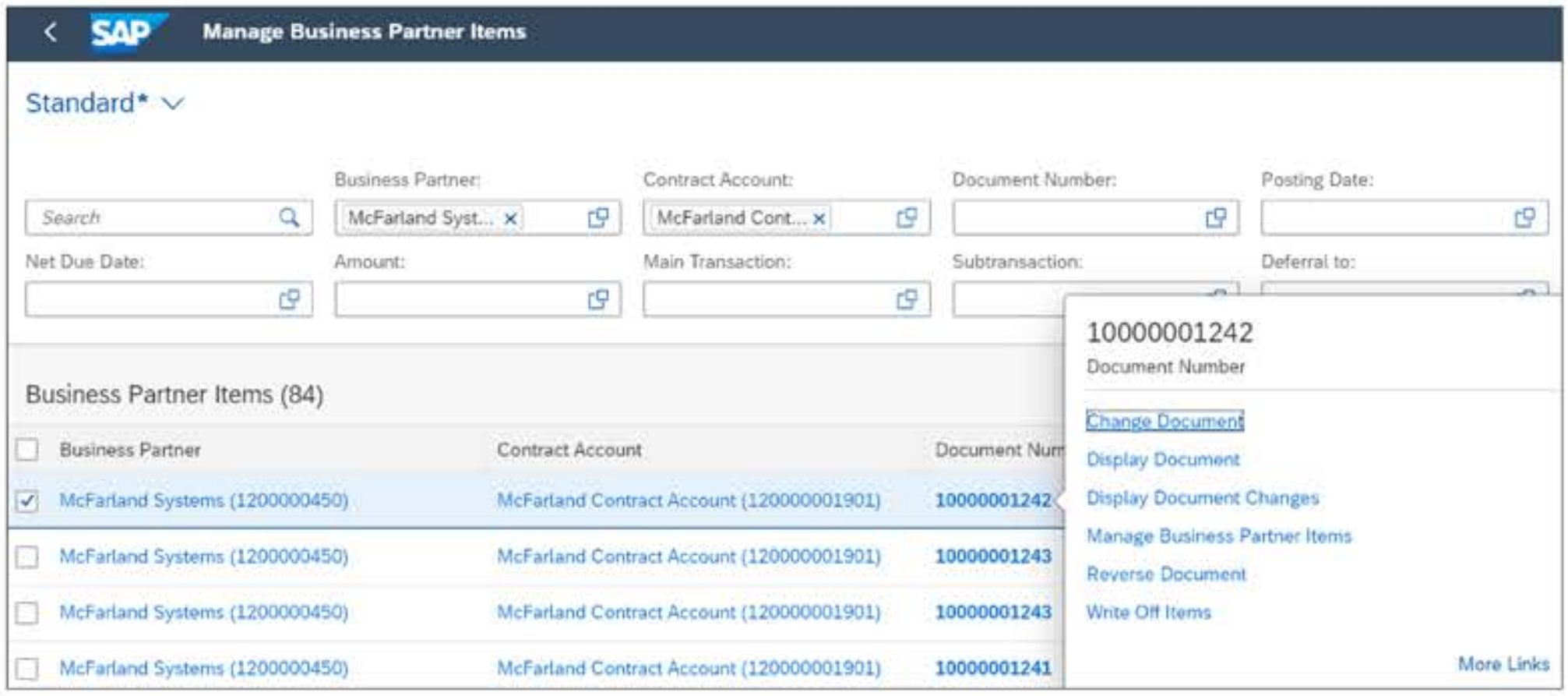

Figure 5.107 Manage Business Partner Items App: Actions That Can Be Performed on the Document

- **Display Account Balance**
  This app is like the Manage Business Partner Items app with the difference that only one contract account can be viewed and processed at a time. Figure 5.108 shows the initial screen ❶ and balance/items view of documents on the contract account ❷.

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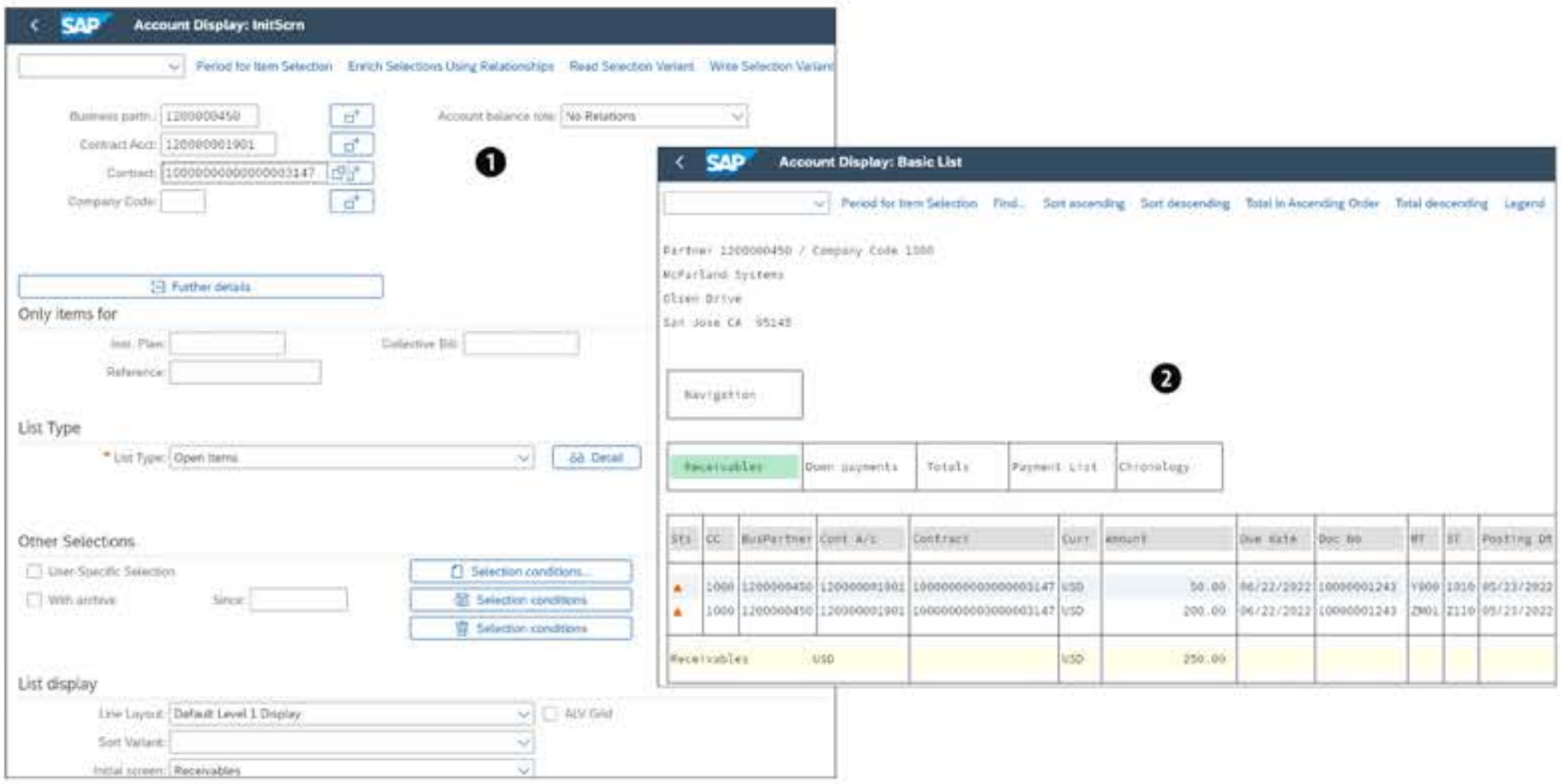

Figure 5.108 Display Account Balance App: Account Balance Initial and Item View

This is basically a GUI display of Transaction code FPL9. The various open item activities, such as account maintenance, payment reversal, reset clearing, dunning items, installment plan view and so on, can be performed on the items on the contract account.

- **Manage Repayment Requests**
  This app is used for management of repayment requests. Figure 5.109 shows the search screen for Manage Repayment Requests. Using the app, the following activities can be performed:
  - Search and sort the repayment request.
  - Set the filters to add or remove columns from the display.
  - Change the due date and usage text.
  - Reverse the repayment request.
  - View list and detailed view of repayment requests.

Manage Repayment Requests

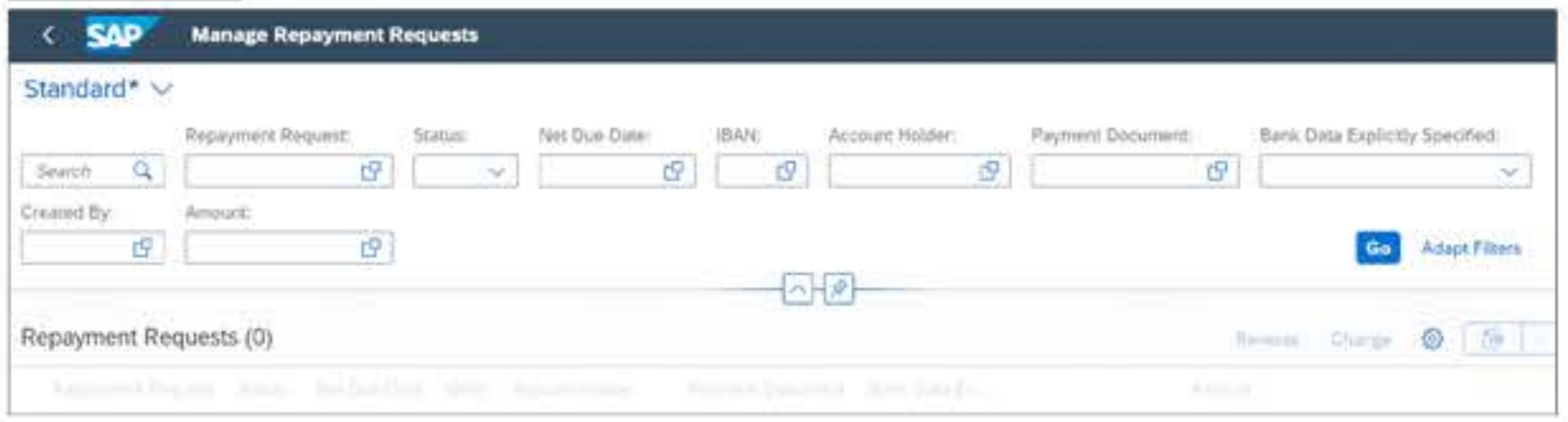

Figure 5.109 Manage Repayment Requests App

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- **Manage Installment Plans**
  An installment plan is created to allow the customer to pay for due items in installments rather than paying the whole amount at once. In scenarios where the customer is unable to pay the whole invoice amount at once due to various reasons such as a large invoice amount, financial difficulties, and so on, an installment plan can be created monthly, quarterly, half yearly, or annually based on the business guidelines. The Manage Installment Plans app shown in Figure 5.110 is used for creating installment plan. Using this app, the following activities can be performed:
  - Create, change, and display installment plans.
  - Display the installment plan history.
  - Analyze installment plans.

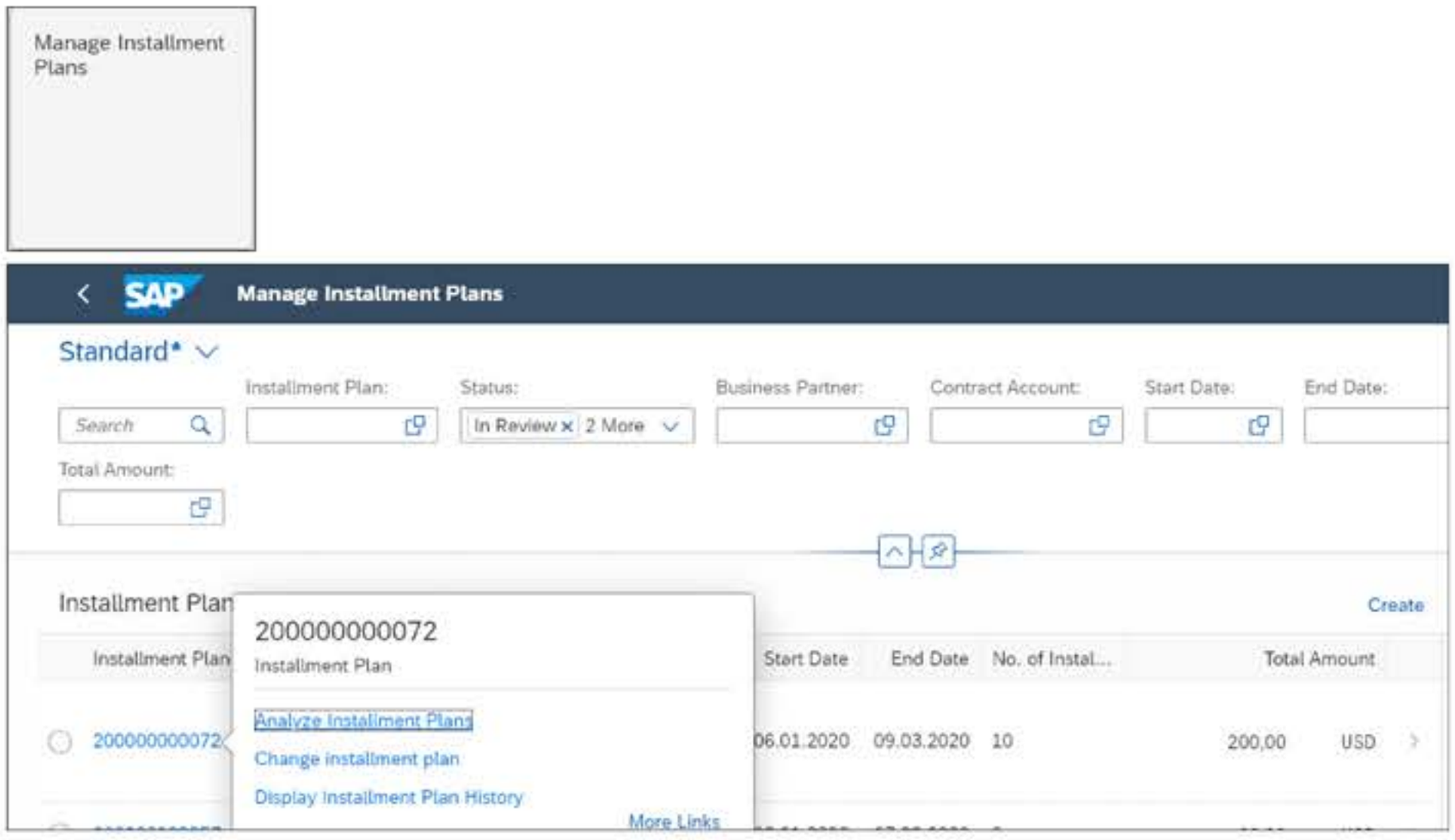

Figure 5.110 Manage Installment Plans App

- **Analyze Installment Plans**
  Installment plans created in the system can be analyzed for items that are coming due soon, deactivated, and waiting for approval. Using this app, the following activities can be performed:
  - See a diagram view of installment plans available in the system with their statuses.
  - View the amount due in various currencies.
  - See a diagram view of the installment plan by status, due date, start date, end date, deactivation reason, and so on.

Figure 5.111 shows the app name ❶ and diagram view ❷ of installment plans based on the search criteria and displayed according to status, due date, and amount.

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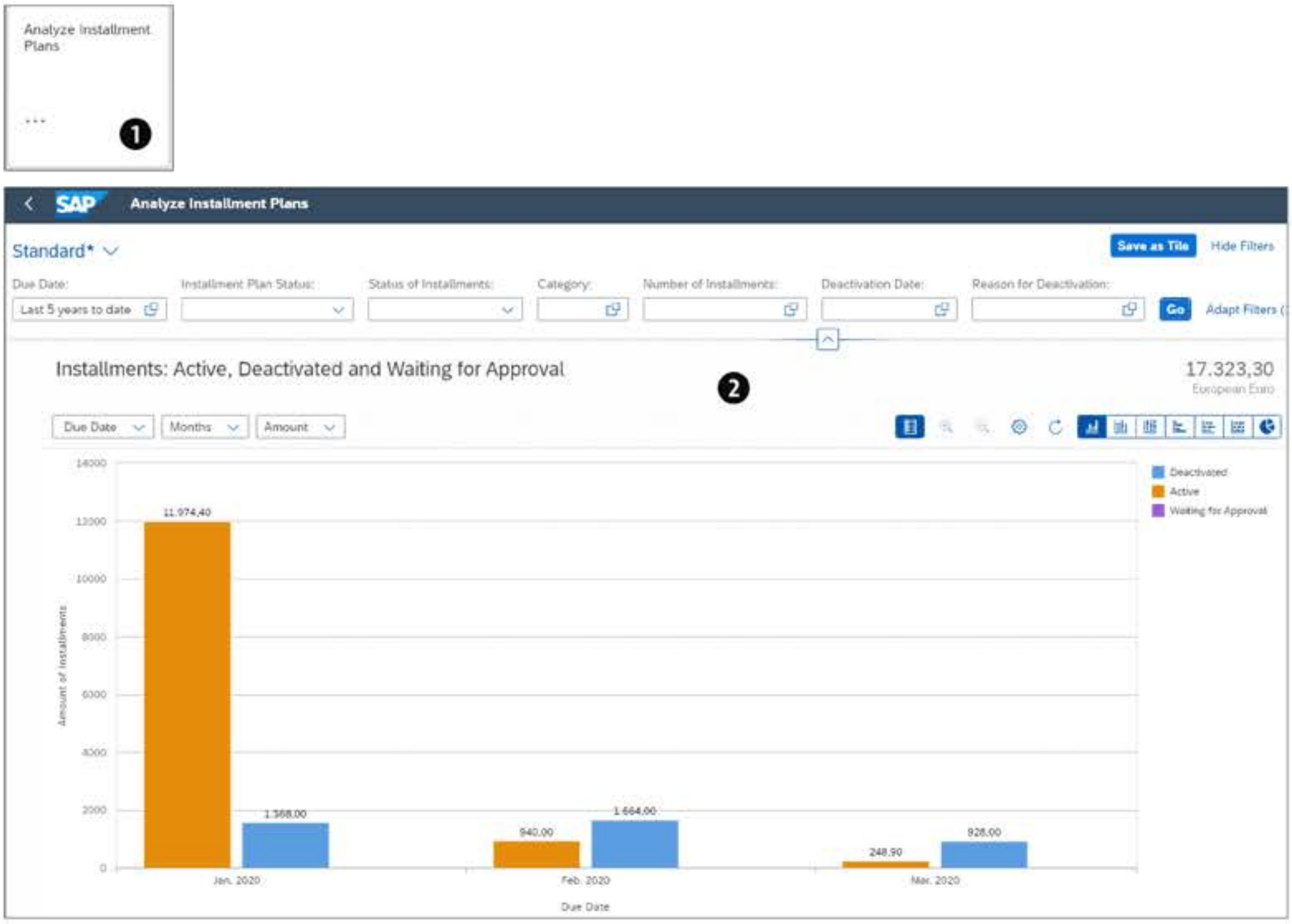

Figure 5.111 Analyze Installment Plans App

5.4 Reporting and Analytics

SAP provides several reporting options for the account postings performed in contract accounting. The reports provided are useful for extracting the details of the business partner/customer items based on their status—that is, whether items are cleared or open—using several parameters such as dunning, document type, and so on. Reports can also be used to obtain the entries for the general ledger account postings in contract accounting. The reports obtained provide a view of the data and help in analysis. This section will cover several report options available in contract accounting.

5.4.1 Contract Accounting Items List: Open/All Items

Transaction FPO4 is used to obtain the details or a report of the receivable and payable items on an account. This report can be extracted based on the master data details and even the general ledger accounts. These reports specifically can get the open items on a key date and extract the details of all items, as shown in Figure 5.112.

In the **Output Control** tab, the output of the report can be controlled and obtained in the desired format.

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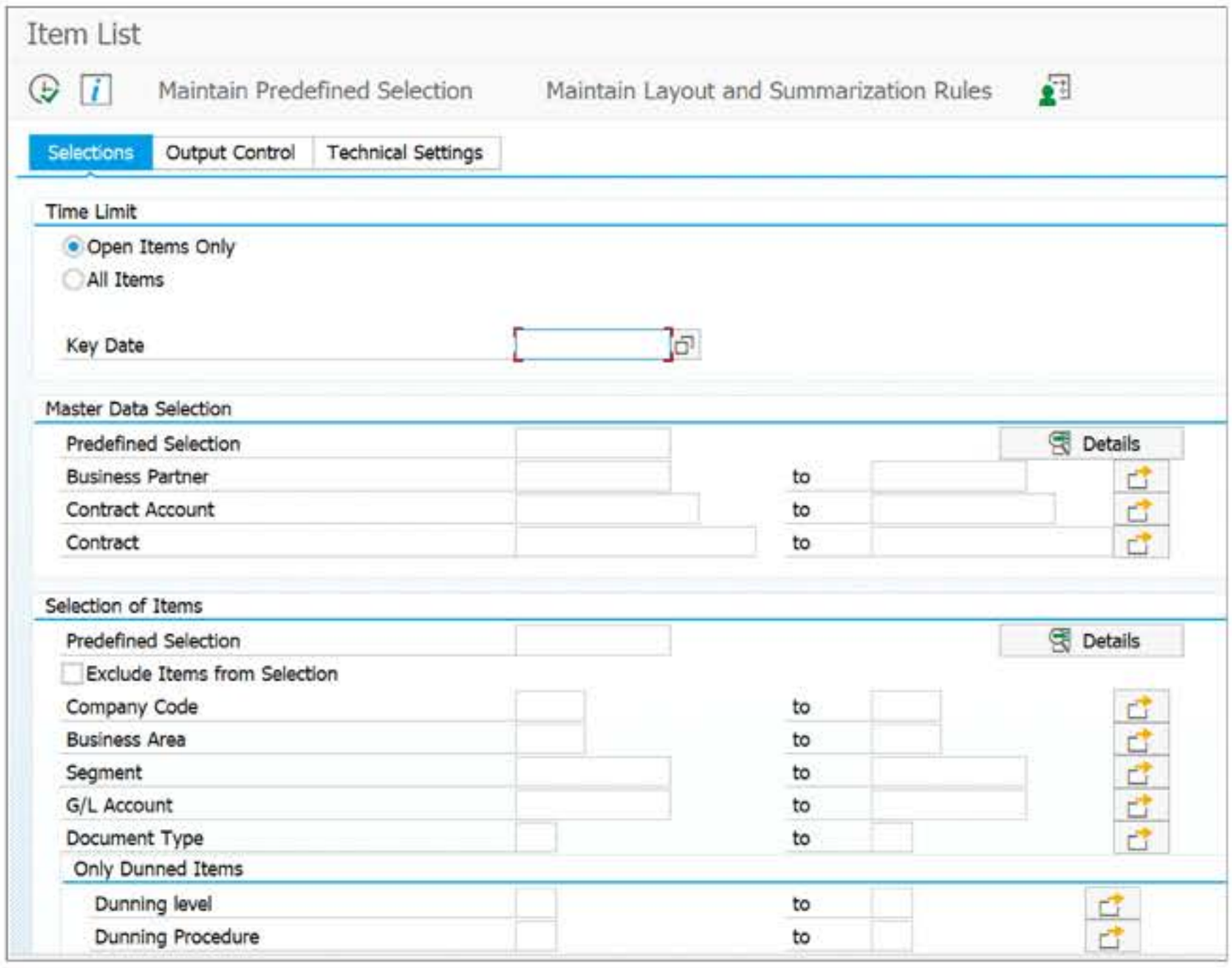

Figure 5.112 Report for Open/All Items

5.4.2 List of Open Items at Key Date

The contract accounting documents that aren't cleared in the system by invoicing, account maintenance, reversal, or payments are called open items. Transaction FPO1P is used to obtain the list of open items on a key date (see Figure 5.113).

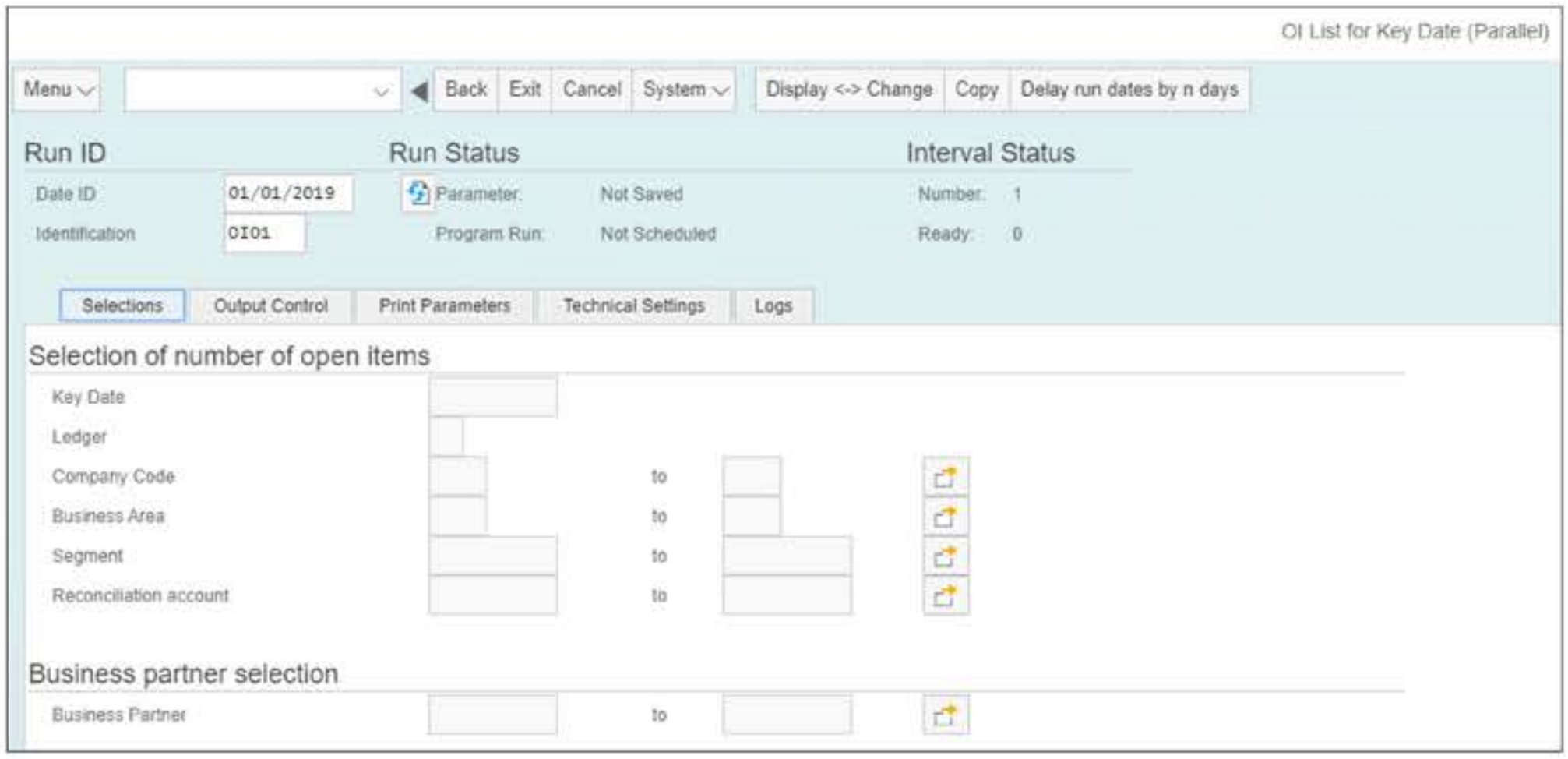

Figure 5.113 Report to Obtain Open Items at Key Date

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This transaction can also be accessed through the SAP Easy Access menu: **Accounting • Financial Accounting • Contract Account Receivables & Payables • Periodic Processing • Closing Preparation**. This transaction is a parallel mass job that generates the spool, and the display of the spool is controlled by the parameters maintained in the output control of the job settings.

5.4.3 Reconciliation of Open Items and General Ledger

All contract accounting documents are posted under a reconciliation key. The reconciliation key at close of business hours for the day is then closed and transferred to the financial accounting system. The reconciliation key transfer posts a finance document and updates the general ledger accounts with the accounting entries. To reconcile the accounting entries transferred to the financial accounting system and those that have yet to be transferred, Transaction FPO2 is used for an individual run to obtain the details of the amounts present in the general ledger accounts, the amount under the reconciliation key, and the total amount of open items (see Figure 5.114). The report provides the data per reconciliation general ledger account. Transaction FPO2P is used for a mass run (see Figure 5.115).

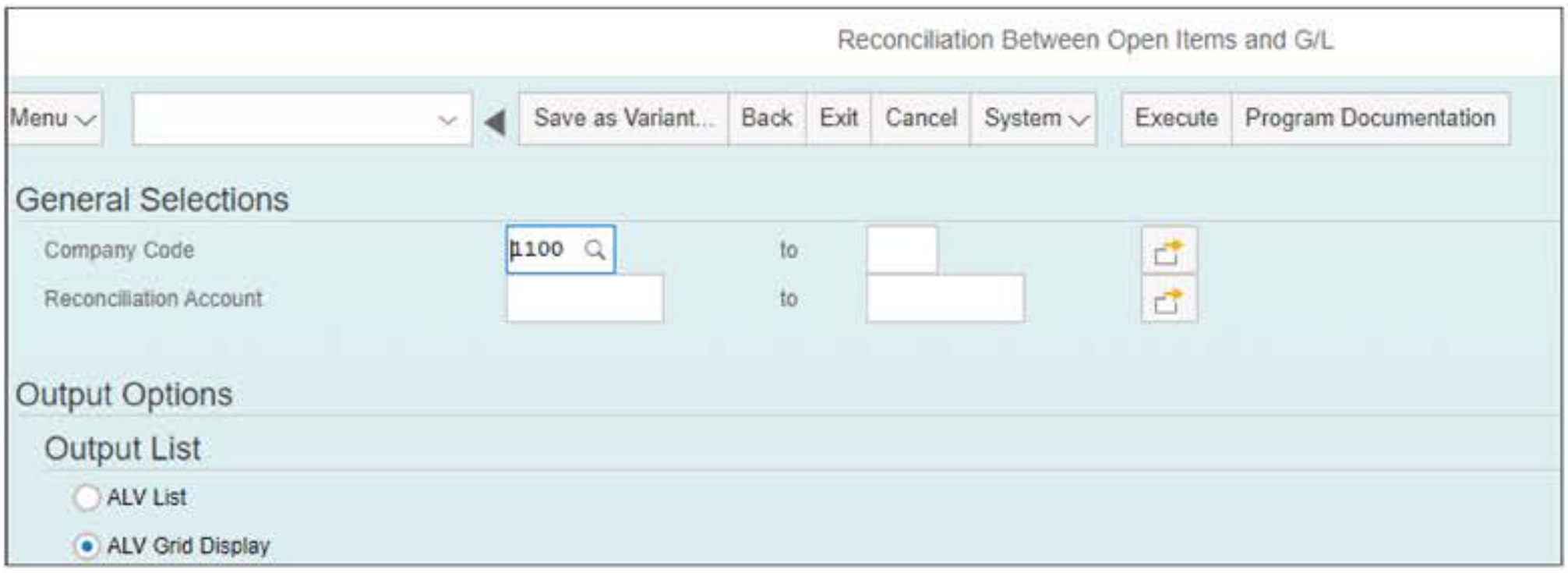

Figure 5.114 Transaction FPO2 Reconciliation Report: Initial Screen

Figure 5.115 Transaction FPO2 Reconciliation Report Data

The report provides a status for the reconciliation. The red status indicates that the reconciliation has a difference. The yellow status indicates that there is no difference

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amount: the amounts in the general ledger accounts and the reconciliation key match with the open item balance, but the reconciliation key isn't yet transferred. The green status indicates that the reconciliation keys are transferred and the reconciliation is successful.

5.4.4 List of Open Items for Business Partner on Key Date

As mentioned earlier, the posting documents that aren't cleared are called open items. A business partner can have several contract accounts, and each contract account can have multiple open items posted on the account. This report provides a list of all the open items on a key date for the business partner, extracting open items from all the contract accounts under the business partner.

Transaction FPO1_ACC is available to extract the details of the open items on a key date specific to the business partners (see Figure 5.116).

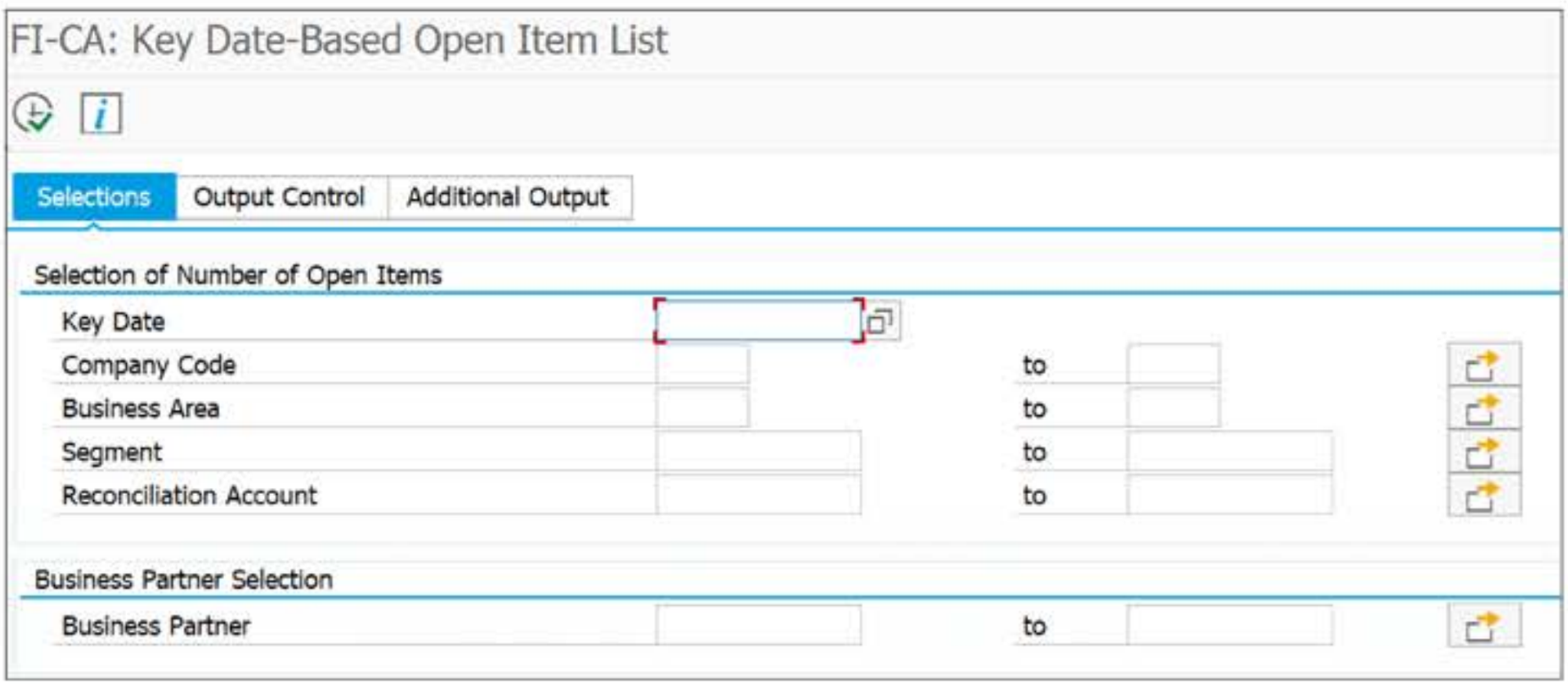

Figure 5.116 Report for Extracting Open Items for Business Partner on Key Date

In the **Selections** tab, input is maintained for the data selection; the **Output Control** tab controls the output data display; and the **Additional Output** tab controls the data format.

5.4.5 Customer Account Balance Display

All receivables, payables, payments, charges, interest, and several other transactions are posted on the contract accounts. The customer account balance display provides the detailed information about the various documents posted on the contract account, as well as the details of the amount of each posting and the total amount for the complete account.

Transaction FPL9 is available to display the customer account balance details (see Figure 5.117). The output is controlled by the list type and line layout, discussed in Section 5.2.3. The items/documents—receivables, cleared items, open items, overdue items,

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statistical postings such as installment plans, security deposit requests, down payments, and so on—can be displayed here. The layout contains several fields that can be set up easily based on business needs.

The transaction can also be accessed via SAP Easy Access menu: **Accounting • Financial Accounting • Contract Account Receivables & Payables • Account.**

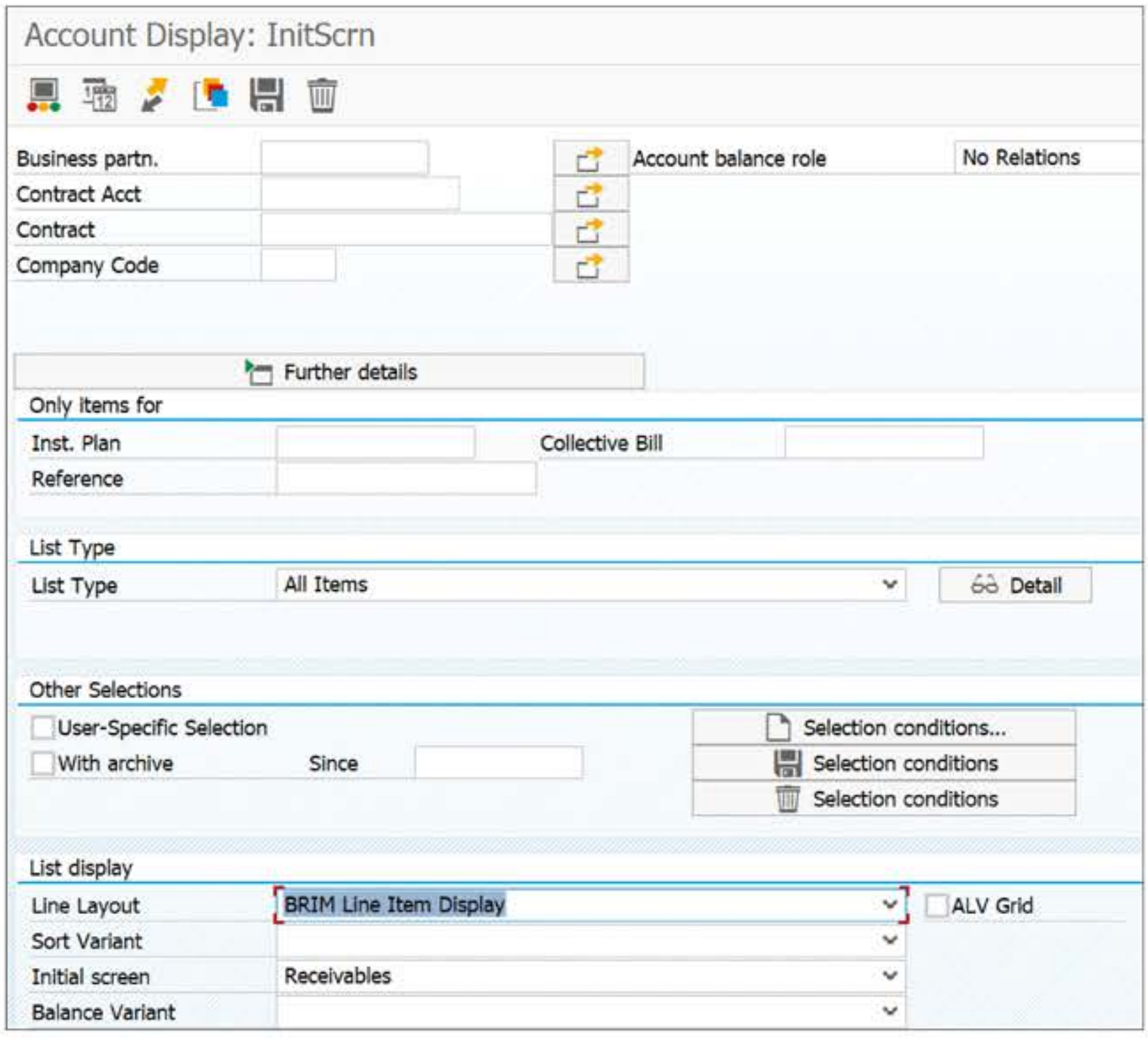

Figure 5.117 Account Balance Display Screen

The account items can be obtained by master data such as the business partner, contract account, and contract. Information such as clearing documents, reversal documents, details related to a document, posting dates, and so on can be seen easily in a single place in the account balance display.

5.5 Solving Common Contract Accounting Requirements

In this section, we'll discuss the most common challenge faced in business requirements where the receivables management system used is accounts receivable/accounts payable. Contract accounting is also used as part of the SAP Billing and Revenue Innovation Management solution for posting receivables, but all open item management has to be handled in the accounts receivable/accounts payable system.

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Until SAP release 1909, this has been a challenge as there was no standard solution available to address this requirement. Several companies have already invested huge amounts in setting up their accounts receivable/accounts payable system for receivable and payables management and wanted to use the same with SAP Billing and Revenue Innovation Management. With release 2020, SAP has provided a standard solution to move the contract accounting open items to accounts receivable open items and marking contract accounting open items as technically cleared.

After the movement of contract accounting open items to accounts receivable or accounts payable, all open item activities can be performed on the documents.

Figure 5.118 shows the view of a contract accounting document transferred to accounts receivable as an open item.

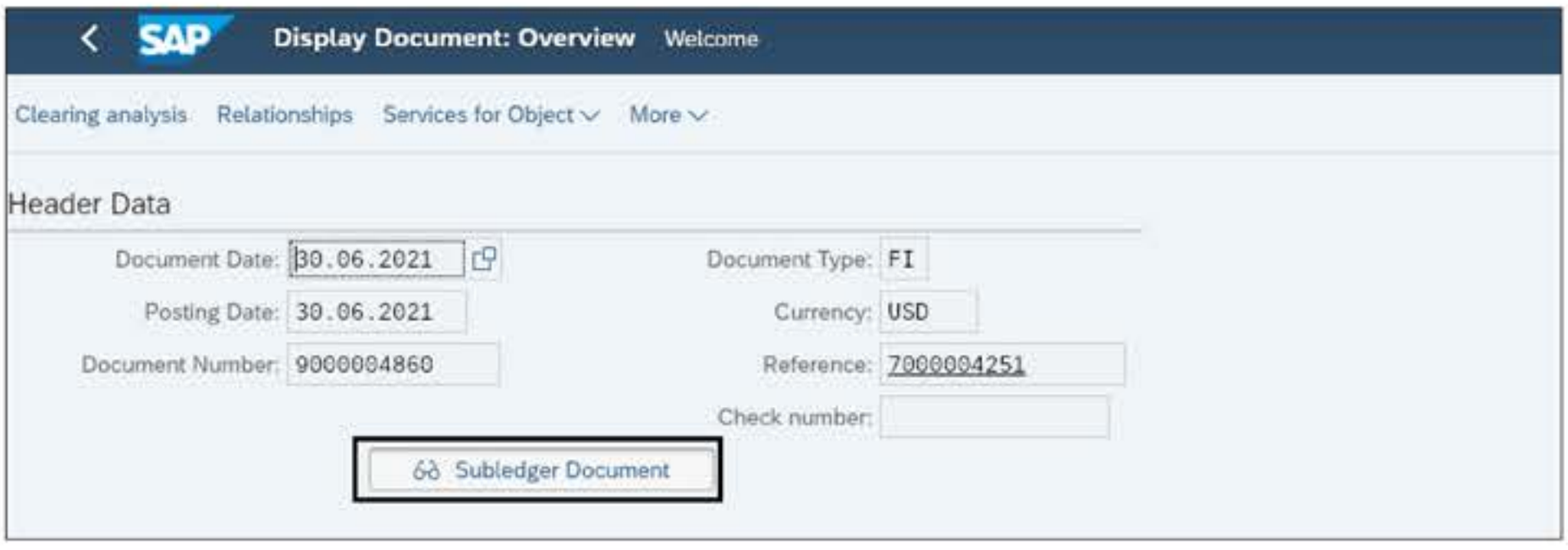

Figure 5.118 Contract Accounting Document Transferred to Accounts Receivable via the Subledger Document Button

Contract accounting documents that are supposed to be transferred to accounts receivable as open items are updated with **Restriction L Subledger Transfer to FI-AR/AP**, as shown in Figure 5.119.

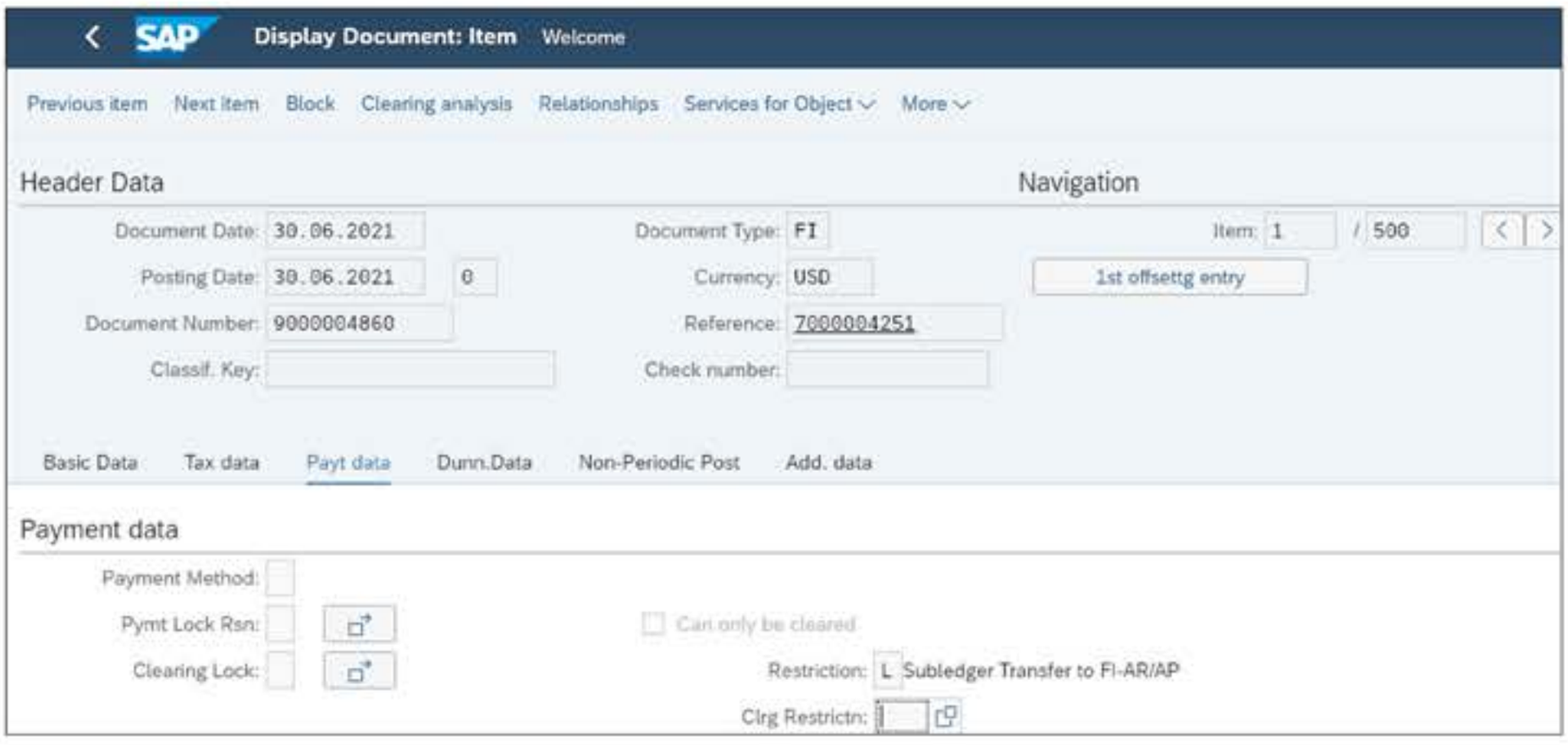

Figure 5.119 Restriction L-Subledger Transfer to Accounts Receivable/Accounts Payable Updated on the Contract Accounting Document

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Contract accounting documents that are marked for accounts receivable/accounts payable transfer are transferred to accounts receivable/accounts payable using the batch job via Transaction FPSL, as shown in Figure 5.120.

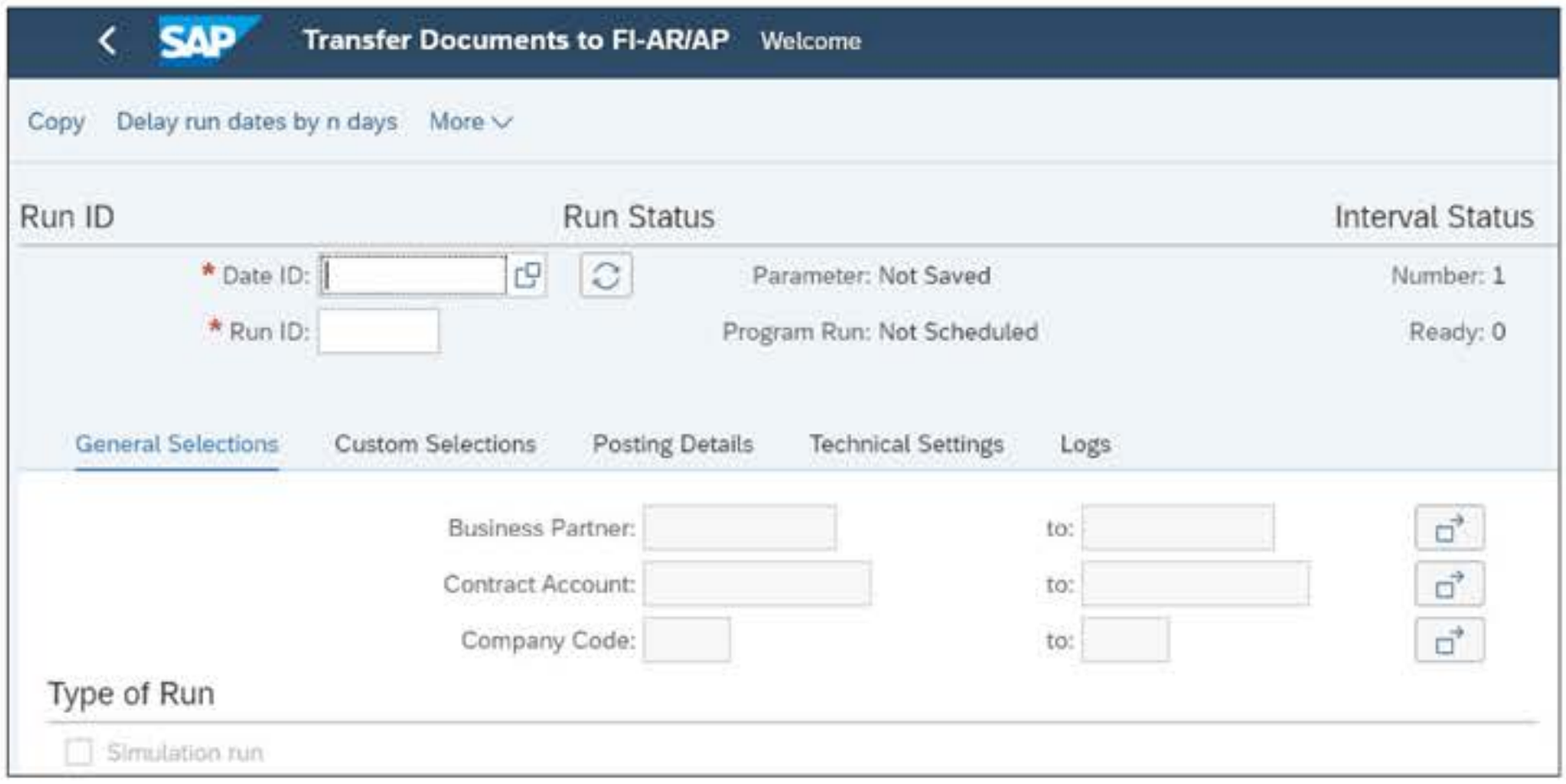

Figure 5.120 Transaction FPSL screen for Contract Accounting Document Transfer to Accounts Receivable/Accounts Payable

The following contract accounting configuration nodes must be maintained:

- **Activate Document Transfer to FI-AR/AP**
  Follow menu path **Contract Account Receivable and Payables • Integration • Account Receivables/Account Payables • Activate Document Transfer to Account Receivables/Account Payables**. Figure 5.121 shows the settings maintained to active the contract accounting to accounts receivable/accounts payable transfer.

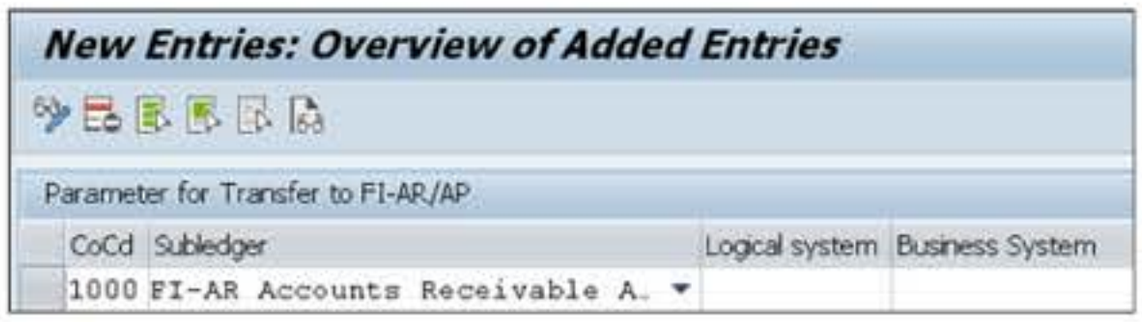

Figure 5.121 Company Code Activation for Accounts Receivable/Accounts Payable Transfer

- **Document Types for Contract Accounting Clearing and FI-AR/AP Posting**
  Follow menu path **Contract Account Receivable and Payables • Integration • Account Receivables/Account Payables • Define Document Types for the Document Transfer.** Figure 5.122 shows the document types applicable for contract accounting document clearing and accounts receivable and accounts payable posting, which are to be used when transferring the contract accounting document.

- **Maintain Sub-transaction Applicable for FI-AR/AP Transfer**
  Follow menu path **Contract Account Receivable and Payables • Basic Functions • Postings and Documents • Documents • Documents • Maintain document Assignments •**

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Maintain Sub Transactions. Figure 5.123 shows the subledger type and posting key to be used for accounts receivable/accounts payable transfer for the associated main and subtransactions.

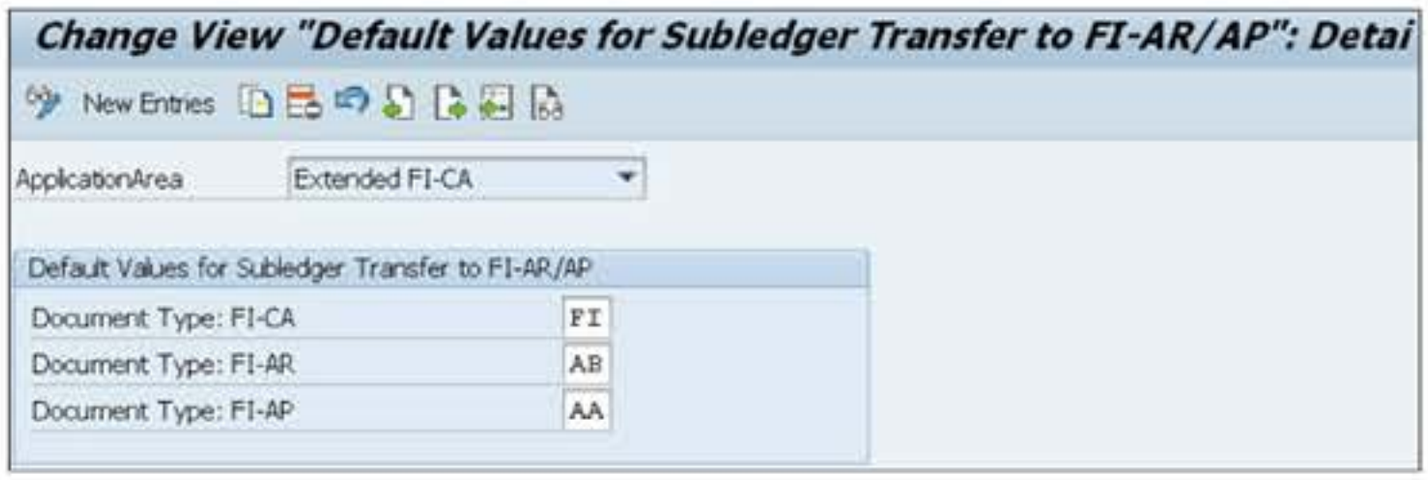

Figure 5.122 Document Types Used for Contract Accounting Clearing and Accounts Receivable and Accounts Payable Transfer Posting

Figure 5.123 Main and Subtransaction for Accounts Receivable/Accounts Payable Transfer

- **Contract Account Subledger Field**

  For contract accounts for which contract accounting to accounts payable/accounts receivable transfer must be executed, the **Subledger** field must be updated with **FI**, as shown in Figure 5.124.

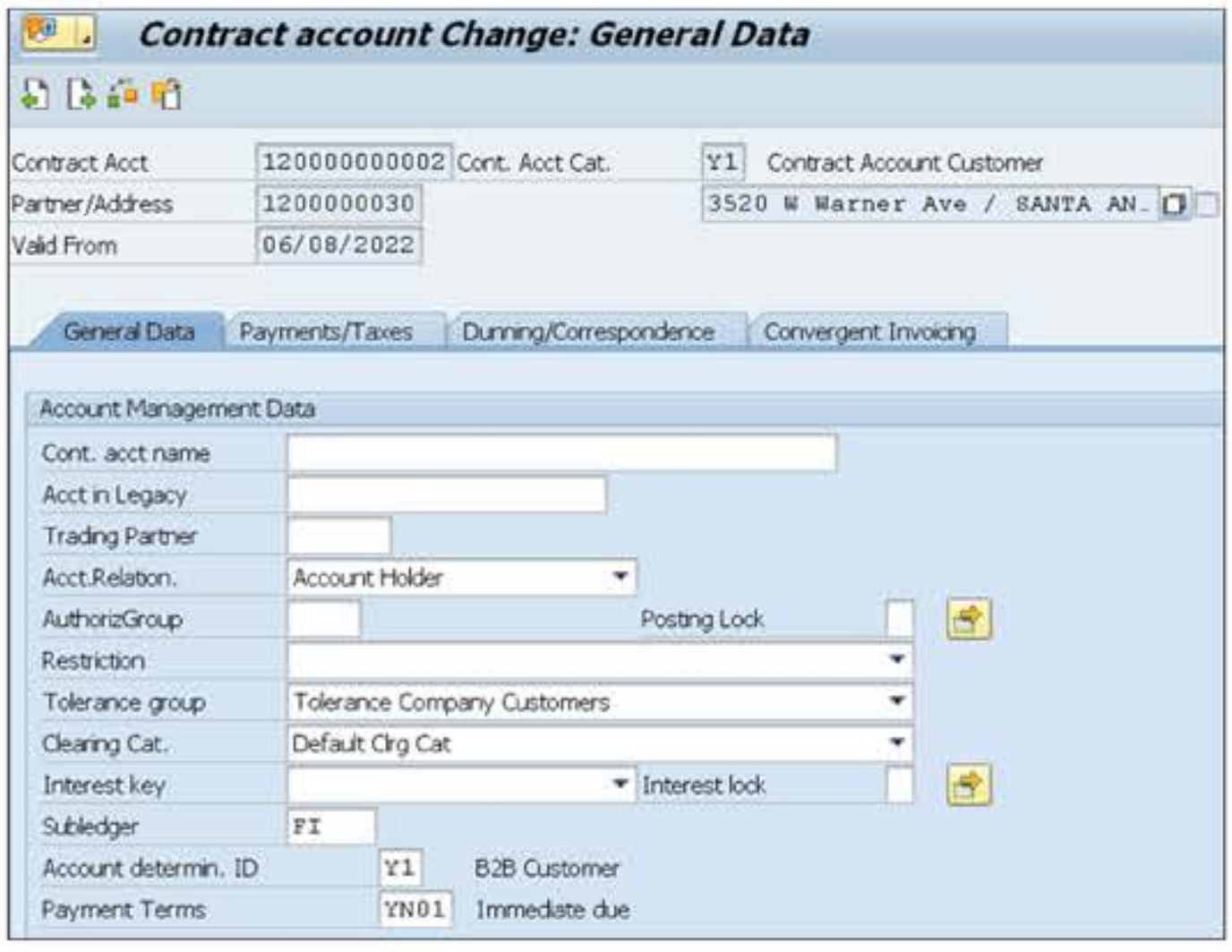

Figure 5.124 Subledger Indicator FI on Contract Account for Contract Accounting to Accounts Receivable Transfer

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- **Posting Area 1421 for Revenue Account Determination in Another Subledger**
  Figure 5.125 shows the **Posting area** entry for revenue account determination in subledger accounts receivable/accounts payable. In this posting, the revenue account appears that needs to be mapped to an accounts receivable document upon transfer from contract accounting to accounts receivable. This can be the same or a new document based on the requirement.

Figure 5.125 Posting Area 1421 for Revenue Account Determination

5.6 Summary

This chapter has described in detail the various contract accounting objects, configurations required for setting up the contract accounting processes, and the reports available for data reconciliation and analysis. You should now be able to configure your system to meet your business requirements and perform various contract accounting processes.

In the next chapter, you'll read more about subscription billing, which will help you understand the subscription master and transactional data concepts.

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Chapter 6

SAP Subscription Billing

In this chapter, we'll introduce a relatively new cloud native solution called SAP Subscription Billing, which is used to manage subscription and usage-based business models. We'll briefly discuss the similarities and differences between SAP Subscription Billing and the traditional SAP Billing and Revenue Innovation Management solution, and we'll cover some of the key features of SAP Subscription Billing. Then we'll describe and execute the necessary configuration steps to build a sample use case, along with discussing the integration with SAP S/4HANA and SAP Convergent Mediation (by DigitalRoute).

SAP Subscription Billing is a multitenant public cloud solution that is primarily built on SAP Cloud Platform. The solution can integrate with front-office applications and SAP Customer Relationship Management (SAP CRM) applications for customer engagement such as commerce platforms, sales, service, and marketing applications. It can also integrate with the backend enterprise resource planning (ERP) system and digital core solution to provide a centralized and easy-to-manage revenue management application in the cloud. SAP Subscription Billing allows you to manage your business models with subscriptions, one-time fees, and usage-based charges, and it offers advanced pricing features such as block, volume, and tiered pricing to help design your business model.

6.1 Overview of SAP Subscription Billing

SAP Subscription Billing is a relatively new solution in comparison with SAP Billing and Revenue Innovation Management. Though it supports traditional business use cases around the as-a-service model, it lacks the flexibility of SAP Billing and Revenue Innovation Management when working with client-specific requirements. You can customize the application based on the different configuration options provided by SAP to handle business use cases based on your requirements; however, to implement any feature that isn't supported out of the box, you need to rely on integrated applications or workarounds. Usually, SAP Convergent Mediation by DigitalRoute can be used with SAP Subscription Billing to handle such requirements. In addition, SAP Subscription Billing provides a quarterly update where new features are introduced to make the solution more robust and mature.

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Because SAP Subscription Billing is built on SAP Business Technology Platform, it allows you to integrate with legacy or customized provisioning services or applications easily in your process flow for the activation of services through the order fulfillment or subscription creation. SAP Subscription Billing also provides you an option to monitor the order orchestration process from order creation to billing with different stages, and it provides option to handle errors and to reinitiate the process when errors are fixed.

The SAP Subscription Billing solution is designed for small- to mid-size subscription-driven clients to not only increase customer satisfaction but also to provide a business-friendly application to manage recurring and usage-driven revenue models. This automated solution simplifies both recurring and usage billing processes, thereby making it more efficient and transparent, which helps in reducing customer disputes. The solution is filled with preconfigured components to enable business users and IT configurators to create new and innovative business models based on industry-specific requirements. The application is easy to use and supports generating offers and services easily to manage usage, one-time, and subscription-based charges The SAP Subscription Billing application helps to streamline master data management, sales, billing, and payment processes.

In this chapter, we'll guide you through how to configure SAP Subscription Billing. We'll discuss how you can set up the required business configuration, perform master data management (e.g., markets, business partners, rate plans, and products), and configure recurring and usage-based transactions. We'll also highlight the options available to set up integration with SAP S/4HANA and SAP Convergent Mediation by DigitalRoute.

6.2 Business Configuration

To set up the SAP Subscription Billing application with a client-specific use case, you need to start with business configuration first. To access the application, use your tenant URL along with your user credentials. Once you're logged in to the system, you can navigate to **Master Data and Configuration • Business Configuration**.

The application consists of different configuration options that must be set up prior to setting up any master data or transactional data. In this section, we'll discuss some of the core settings, such as markets, units of measure, and rate elements, to enable your application to function.

6.2.1 Markets

In SAP Subscription Billing, a *market* is defined as an entity that offers digital product and services and has various legally relevant attributes. Each subscription to a digital product is associated with a single market. A market represents the organizational unit that offers a product, setting the legal framework and reflecting various business-relevant attributes, including a reference to a sales organization and further aspects, such

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as currency, price type, and time zone. Note that it's not possible to delete an existing market or to change the ID, currency, time zone, or price type of a market once saved.

For our use case, you need to create "MARTEX_USA" where the products are managed and sold. To navigate in SAP Subscription Billing, navigate to **Master Data and Configuration • Business Configuration • Markets**.

As shown in Figure 6.1, while setting up markets, select **Currency** and **Price Type** values in SAP Subscription Billing. Depending on the **Price Type** selected, either gross or net prices can be entered in the Products app. The default price type is **Net**.

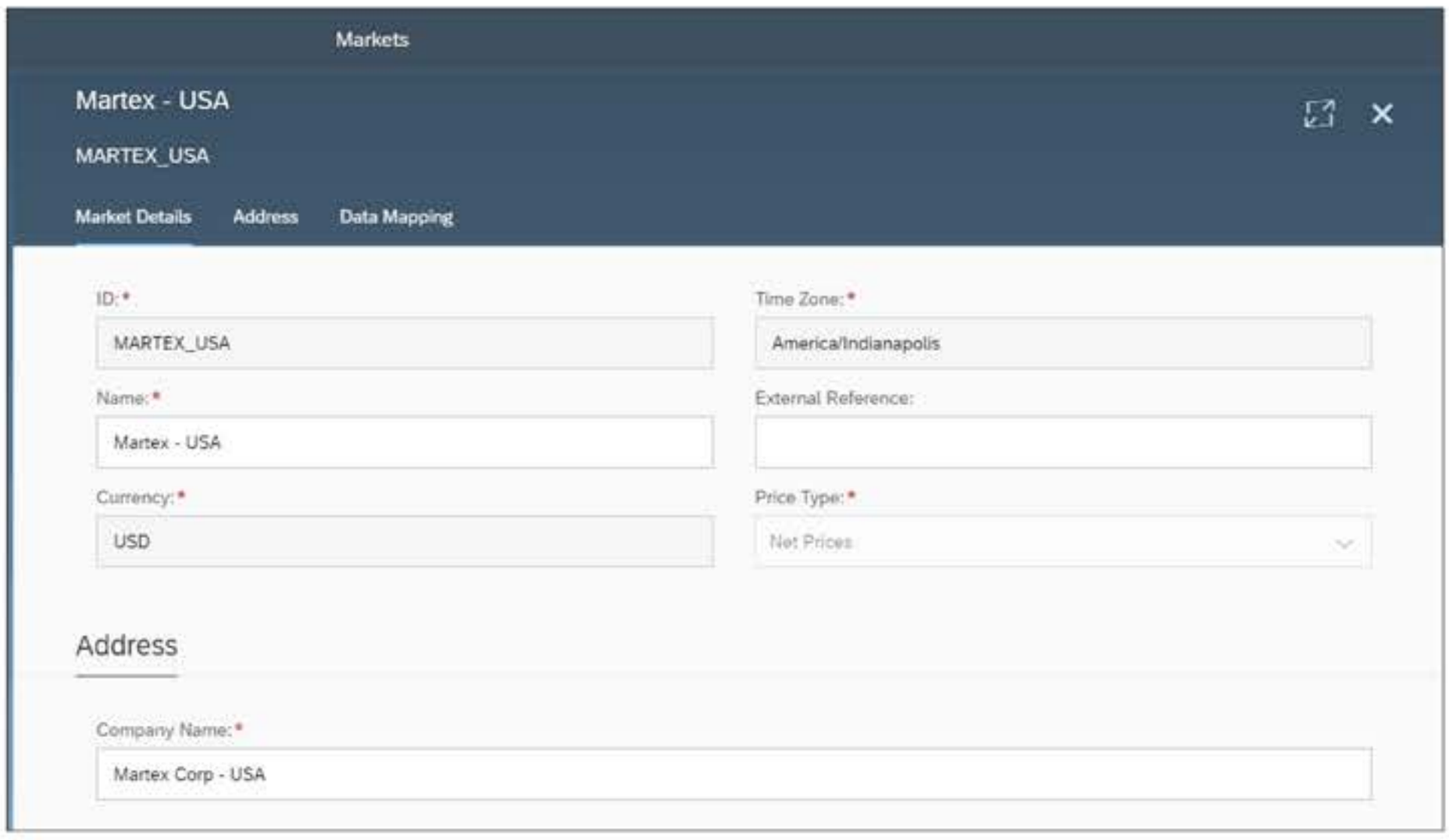

Figure 6.1 Markets in SAP Subscription Billing

In addition, as shown in Figure 6.2, there is also a Sales Area section for related information, such as **Sales Organization**, **Distribution Channel**, and **Division**, in markets.

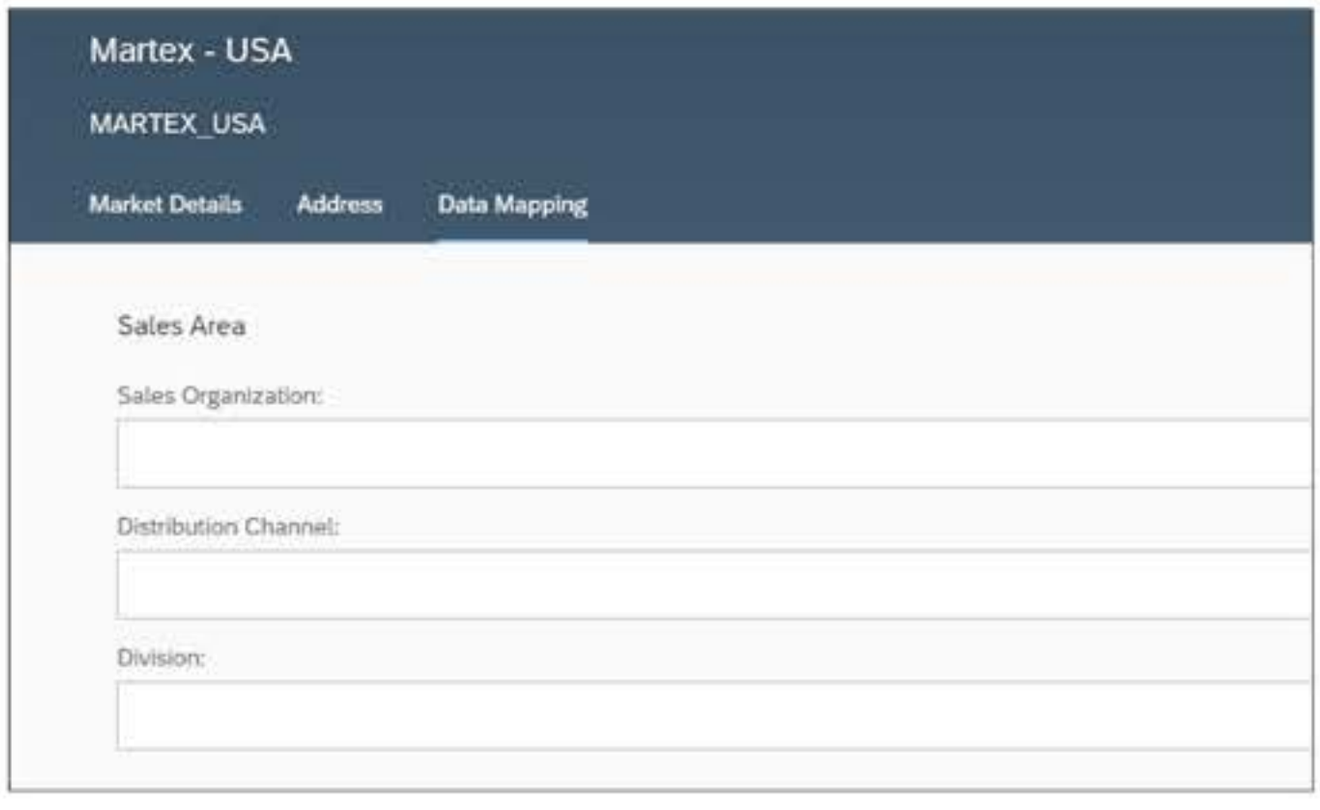

Figure 6.2 Define Sales Area for SAP Subscription Billing Markets

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6.2.2 Units of Measure

In this section, you can maintain different units of measure that you'll need for your billing and pricing. The recommendation is to define **Each (EA)** for one-time charges and **Each Per Period (EPP)** for fixed recurring charges.

6.2.3 Rate Elements

Rate elements are used to define properties usually when setting up products and billing. SAP Subscription Billing offers three types of rate elements.

- Recurring
- Usage
- One-time/initial fee

As part of our Martex use case, define a recurring rate element by entering "Martex Recurring Charge" in the **Label** field, as shown in Figure 6.3.

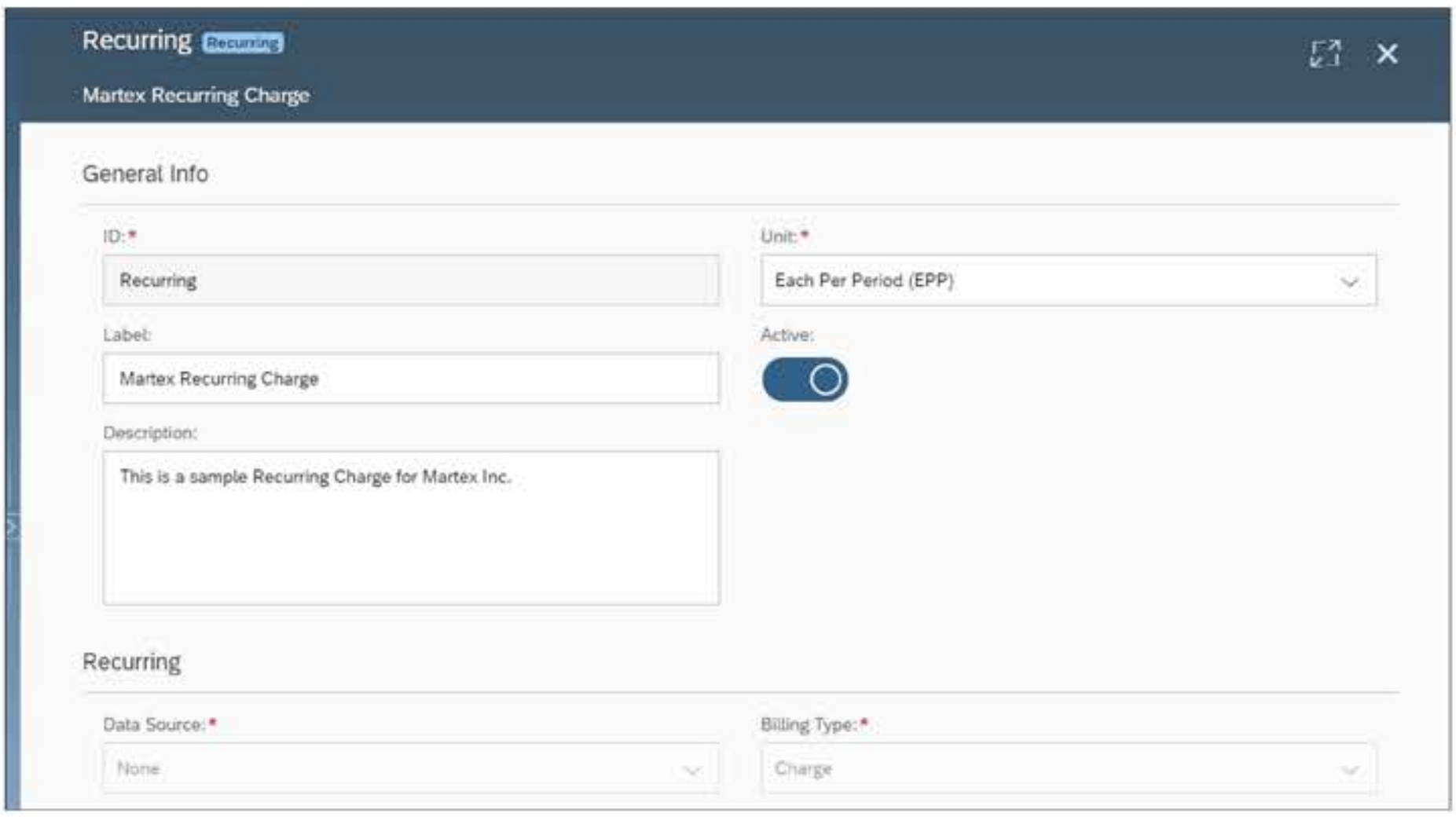

Figure 6.3 Martex Recurring Rate Element

The **Data Source** field for the rate element refers to the source of the quantity information that can be used to calculate the billing amount. Because fixed price will be used for this use case, select **None**. However, if you want to factor quantity while calculating amount, you would select the **Data Source** as **Subscription**. The **Billing Type** field is used to specify if the amount is to be charged to a customer or to be credited.

The preceding business configuration will allow you to start with the master data setup. We'll refer to business configuration as needed once we move to the next settings.

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6.3 Master Data

Master data in SAP Subscription Billing is any data that plays an anchoring role in any of the business processes in SAP Subscription Billing. It includes data for markets, customers, subscription products, rate plans, and rate plan templates that don't change frequently across any given business process. We'll discuss setting up the following master data in detail in the context of SAP Subscription Billing implemented for Martex.

6.3.1 Customer Master Data

In this section, we'll discuss the customer concept and describe the configuration steps to set up a customer in SAP Subscription Billing. In SAP Subscription Billing, you can create and edit master data for corporate (i.e., business-to-business [B2B]) and individual (i.e., business-to-consumer [B2C]) customers using the Customers app.

Customer (business partner) data can be replicated between SAP Subscription Billing and SAP S/4HANA Cloud in both directions, along with address information. In SAP Subscription Billing, each customer has only one address; however, SAP S/4HANA supports multiple addresses. During replication, all nonstandard addresses in SAP S/4HANA Cloud are deleted, and the standard address is overwritten by the default address from SAP Subscription Billing. It's recommended to use one system as the leading system (or SAP Master Data Governance [SAP MDG] system) to keep the data consistent, and only allow your users to edit the customer data in that system.

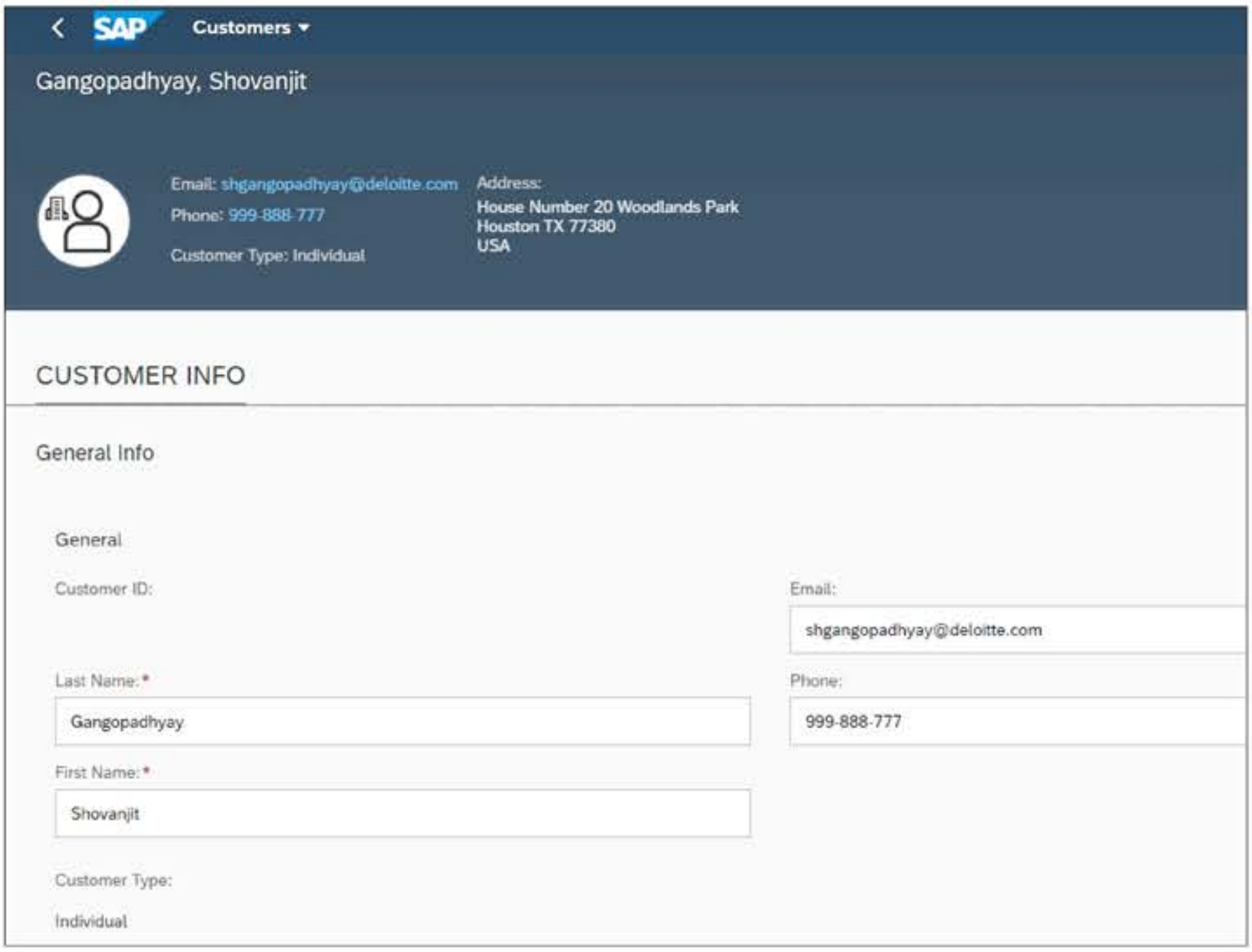

Figure 6.4 Creating a Customer in SAP Subscription Billing

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To manage the requirement for Martex Corp., you need to create an individual customer in SAP Subscription Billing and extend the customer to the existing market called MARTEX_USA. To create a customer in SAP Subscription Billing, navigate to **Master Data and Configuration • Customers • Create**.

In the **Customer Type** dropdown, select the new customer type as either **Individual** or **Corporate**. For this use case, select **Individual**. Figure 6.4 represents the create customer transaction in SAP Subscription Billing. You also need to extend the customer for applicable markets to enable transaction authority, as shown in Figure 6.5. One customer can be extended to one or multiple markets.

Figure 6.5 Extending the Customer for Existing SAP Subscription Billing Markets

You can also define custom references specific to "Customer" to store client-specific information, as shown in Figure 6.6. To define custom references, navigate to **Master Data and Configuration • Business Configuration • Customer References**. Because the existing ERP solution used to manage Martex Corp. was a non-SAP portal, create a new **Reference Type** as **Legacy Customer ID** for the customer object where you'll store the existing ERP customer identification number.

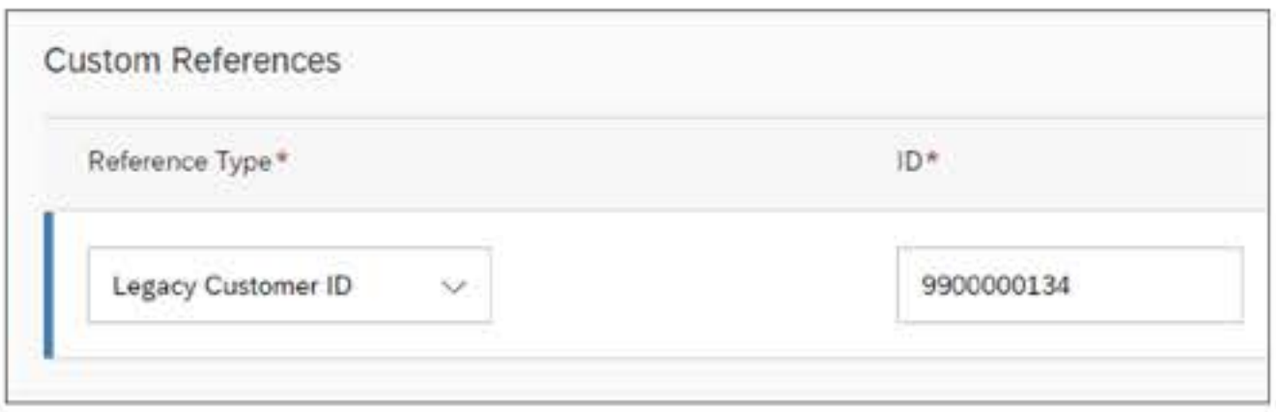

Figure 6.6 Defining the Custom Reference for the Customer Object

The custom reference will now be available for use and can be updated manually, as shown in Figure 6.7. The custom reference can also be updated using custom predefined application programming interfaces (APIs) if integrated with SAP or non-SAP systems for customer replication.

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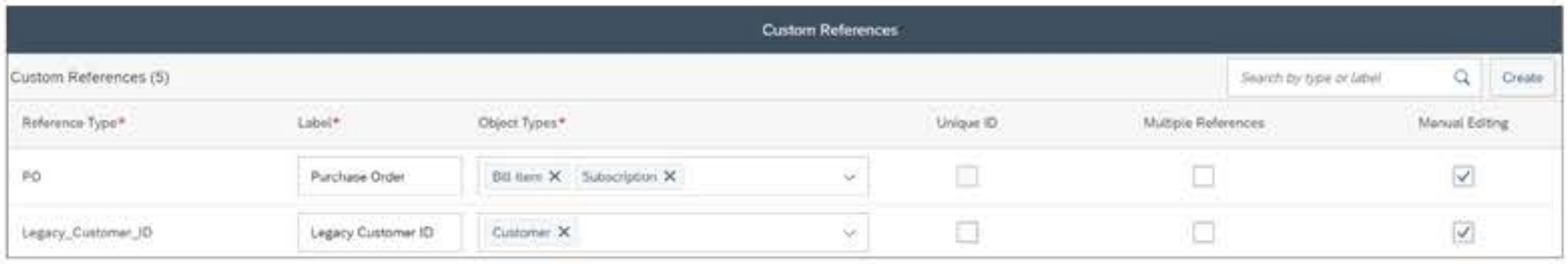

Figure 6.7 Defining Custom References for Customers

6.3.2 Rate Plan Templates

In this section, we'll discuss the relevance of rate plan templates in SAP Subscription Billing. Rate plans and rate plan templates are core components to manage pricing and billing. They are used to define the details of subscription charges and credits in SAP Subscription Billing. Rate plan templates are the basis for creating rate plans. Rate plan templates resemble a reusable model that includes billing- and pricing-related settings as well as the charges or credits used for the rating process. To define rate plan templates, navigate to **Pricing • Rate Plan Templates**. Figure 6.8 is a reference of a Rate Plan app in SAP Subscription Billing.

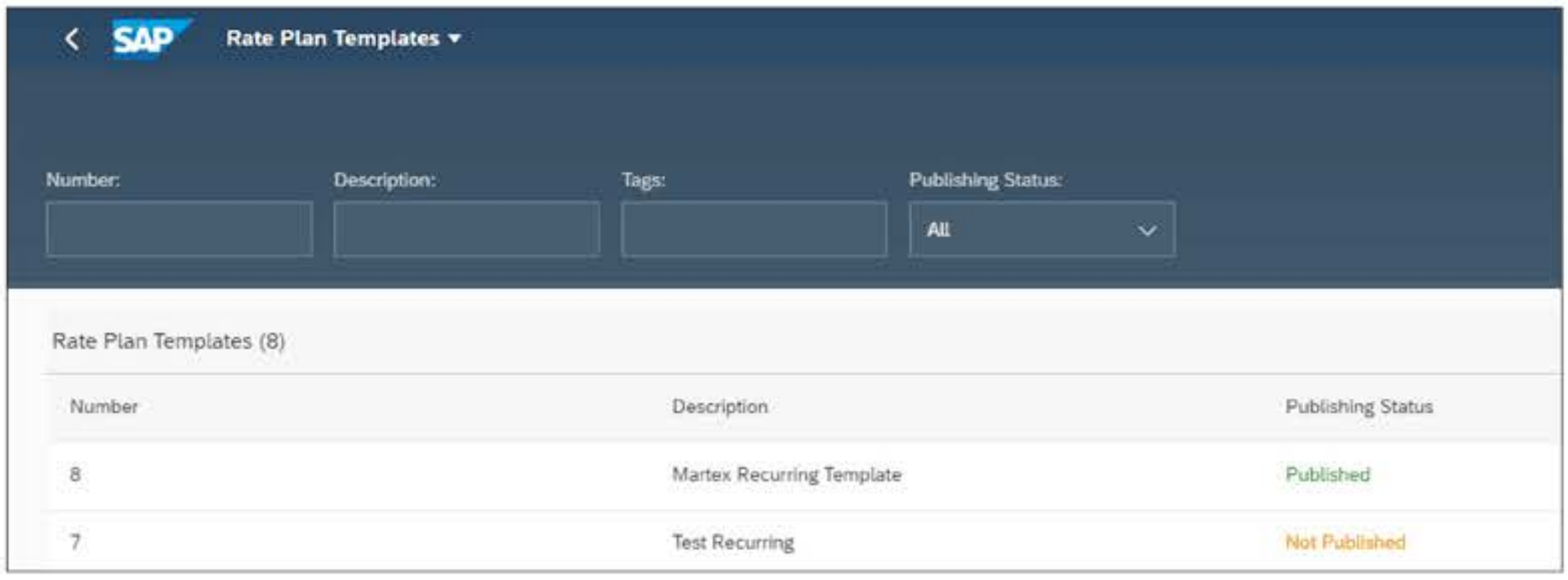

Figure 6.8 Rate Plan App in SAP Subscription Billing

While setting up a new rate plan template, there are two predefined options available in SAP Subscription Billing:

- **Rate plan templates without a pricing scheme**
  In this option, the pricing logic ID is defined in SAP Subscription Billing using rate plan templates.
- **Rate plan templates with a pricing scheme**
  In this option, the pricing logic is defined using a pricing scheme and related entities such as the price element specification.

For the Martex use case, as shown in Figure 6.9, use rate plan templates without a pricing schema.

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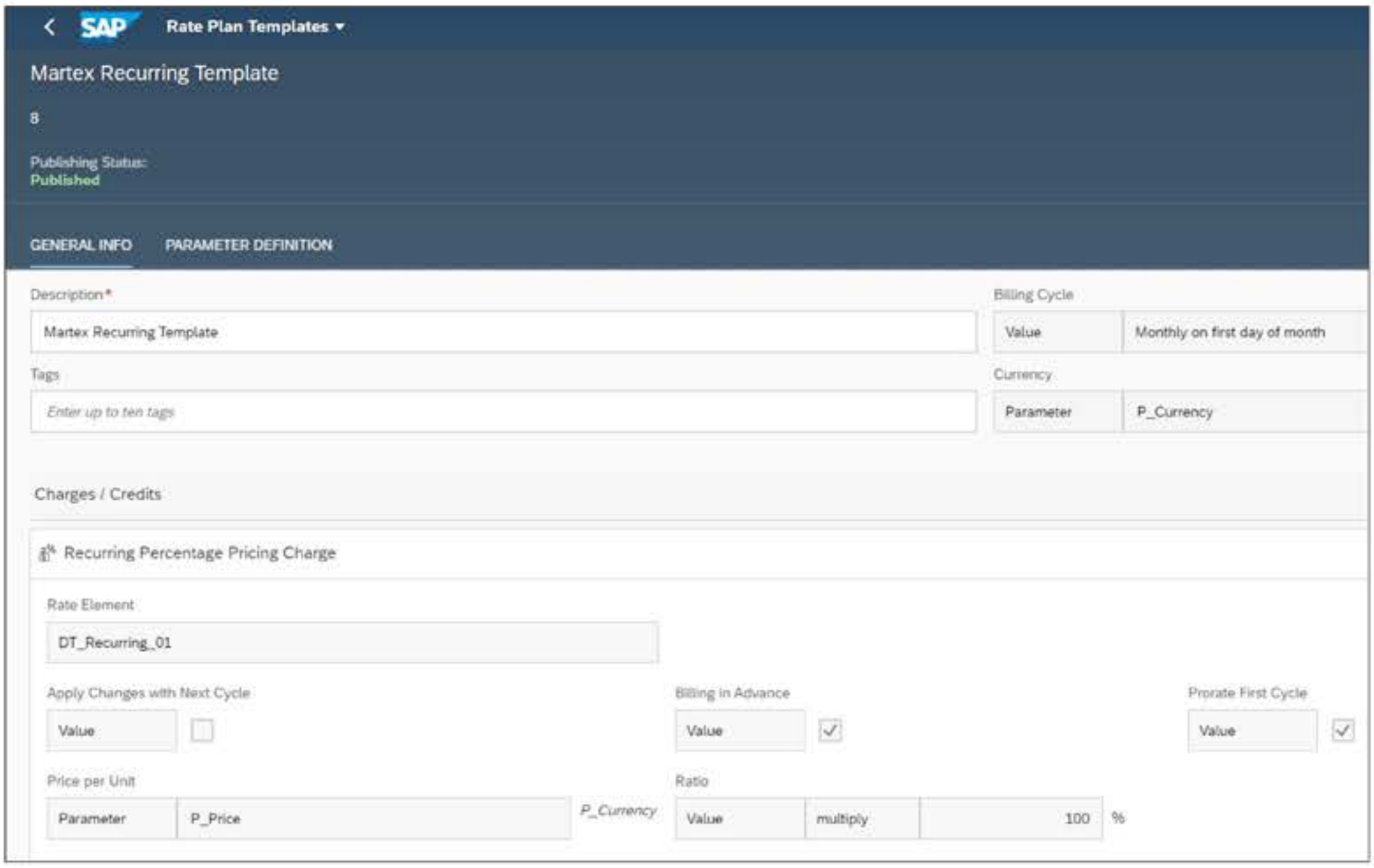

Figure 6.9 Setting Up Martex Recurring Template in SAP Subscription Billing

You can define a **Description** and **Tags** for the newly created rate plan template in SAP Subscription Billing. Rate plan templates consist of different components. Let's discuss and understand each of the components in detail next:

- **Billing Cycle**
  The billing cycle determines the frequency of billing for a service or a product. SAP Subscription Billing provides standard billing cycles, which are monthly, quarterly, half-yearly, or yearly. The start of the billing cycle can be the first day of the calendar period or the billing cycle reference date (i.e., the start of the bill date) of the subscription. The billing cycle reference date is always defaulted to the start of subscription (or contract start date) and most used to manage the start of an anniversary billing cycle.
- **Currency**
  This defines the currency of the rate plan template. Note that in SAP Subscription Billing, the values for **Billing Cycle** and **Currency** can be defined as either a fixed **Value** or a **Parameter**. If the values are defined as parameters, the information must be specified when a rate plan is created based on the template.
- **Charges/Credits**
  Based on the template definition, you can set up charges/credits to manage the rating type and pricing model. There are three types of predefined rating models available in SAP Subscription Billing:
  - **One-Time**
    The one-time rating model is used to manage fixed charges that are billed only

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once during the subscription lifecycle, such as an activation fee or installation fee. One-time charges are always billed in advance.

- **Recurring**
  The recurring rating model is used to manage periodic bills in each rating period.
- **Usage-Based**
  The usage rating model is used to define usage-based charges that are billed in each rating period. Usage-based charges are always billed in arrears.

For each of the rating types, there are predefined pricing models available. You would need to select one pricing model based on the rating type to configure further, as shown in Figure 6.10.

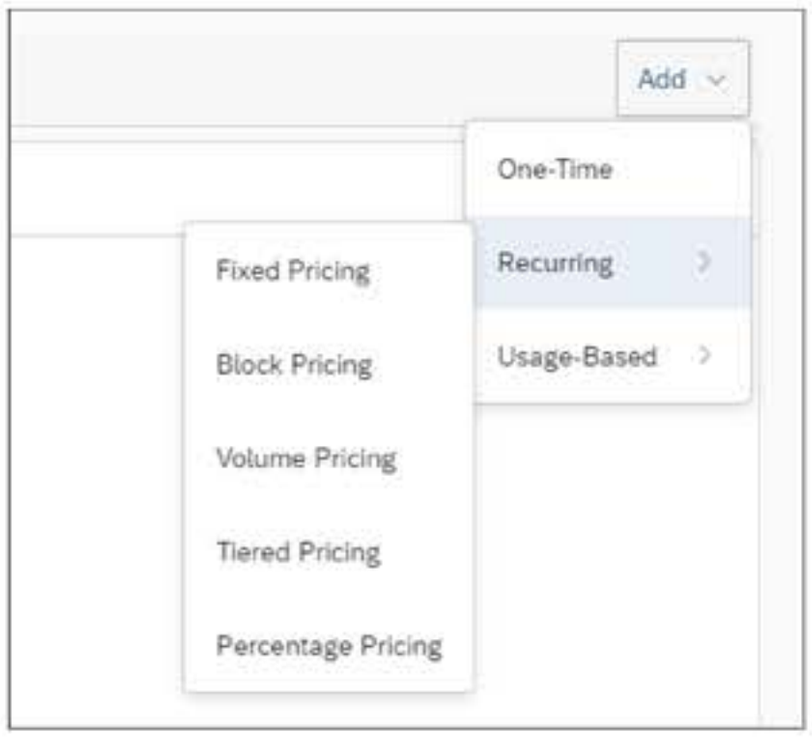

Figure 6.10 Pricing Models for Recurring Rating

Table 6.1 illustrates the various pricing models available for each rating type.

Pricing Model	One-Time	Recurring	Usage-Based
Fixed pricing	Yes	Yes	-
Block pricing	-	Yes	Yes
Volume pricing	-	Yes	Yes
Tiered pricing	-	Yes	Yes
Percentage pricing	-	Yes	Yes

Table 6.1 Pricing Models in SAP Subscription Billing

For the Martex Corp. use case, the customer will be billed with a fixed recurring charge on a monthly basis. Now, let's discuss each of the pricing models to check which model will be applicable to the use case:

- **Fixed pricing**
  This pricing model is used to define a fixed amount to be billed to a customer each period, as shown in Figure 6.11.

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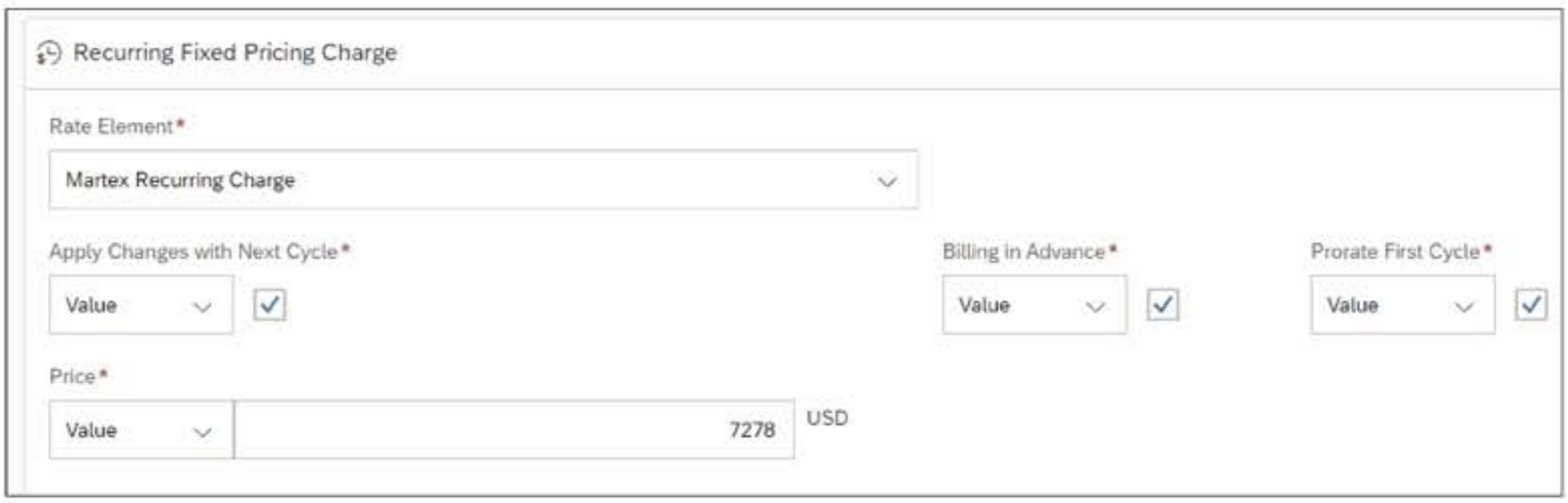

Figure 6.11 Fixed Pricing for Recurring Rating

- **Block pricing**
  This model enables you to define the price for blocks of usage, as shown in Figure 6.12. In SAP Subscription Billing, a block is a quantity of usage units.

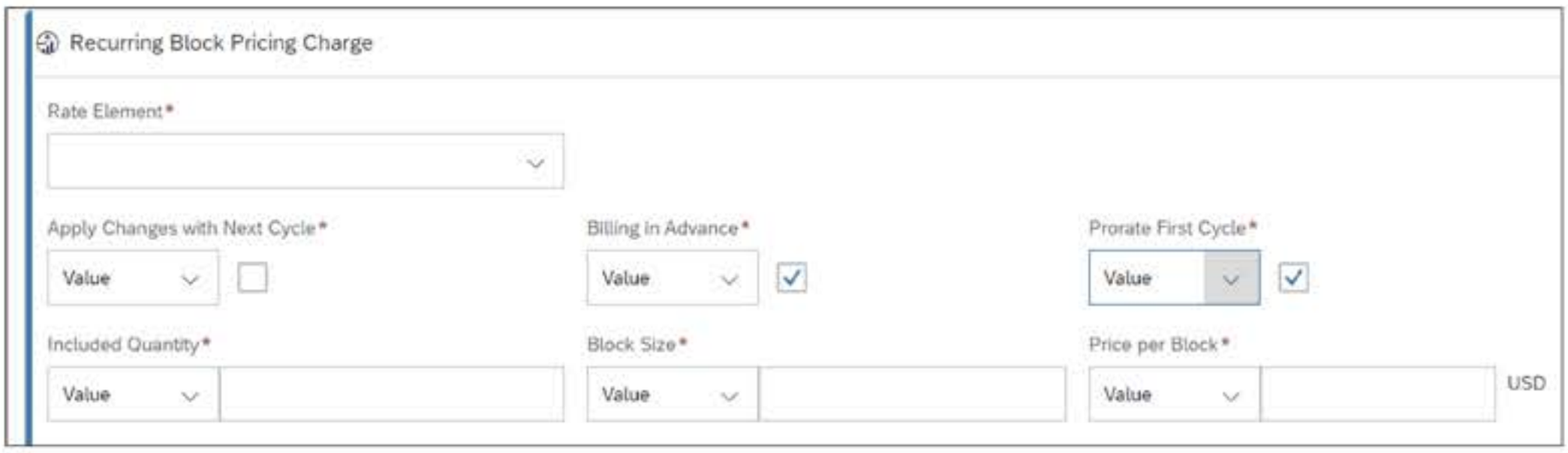

Figure 6.12 Block Pricing for Recurring Rating

- **Volume pricing**
  This pricing model allows you to define the price in pricing tiers, as shown in Figure 6.13. The applicable price per block and any fixed price are determined from the tier. This is most used for usage scenarios and typically defines decreasing prices per block for increasing usage. The pricing block can be defined as following:
  - **Up To**
    This is used to define the upper limit of usage blocks in each tier. The uppermost limit is **Unlimited**.
  - **Price per Block**
    This is used to manage the price of each block. For usage-based pricing, the amount is calculated by multiplying the number of blocks by the price per block of the relevant tier of usage volume.
  - **Fixed Price**
    This is used to manage any fixed charge for usage (fixed monthly service plan), and this price is in addition to the usage-based price.

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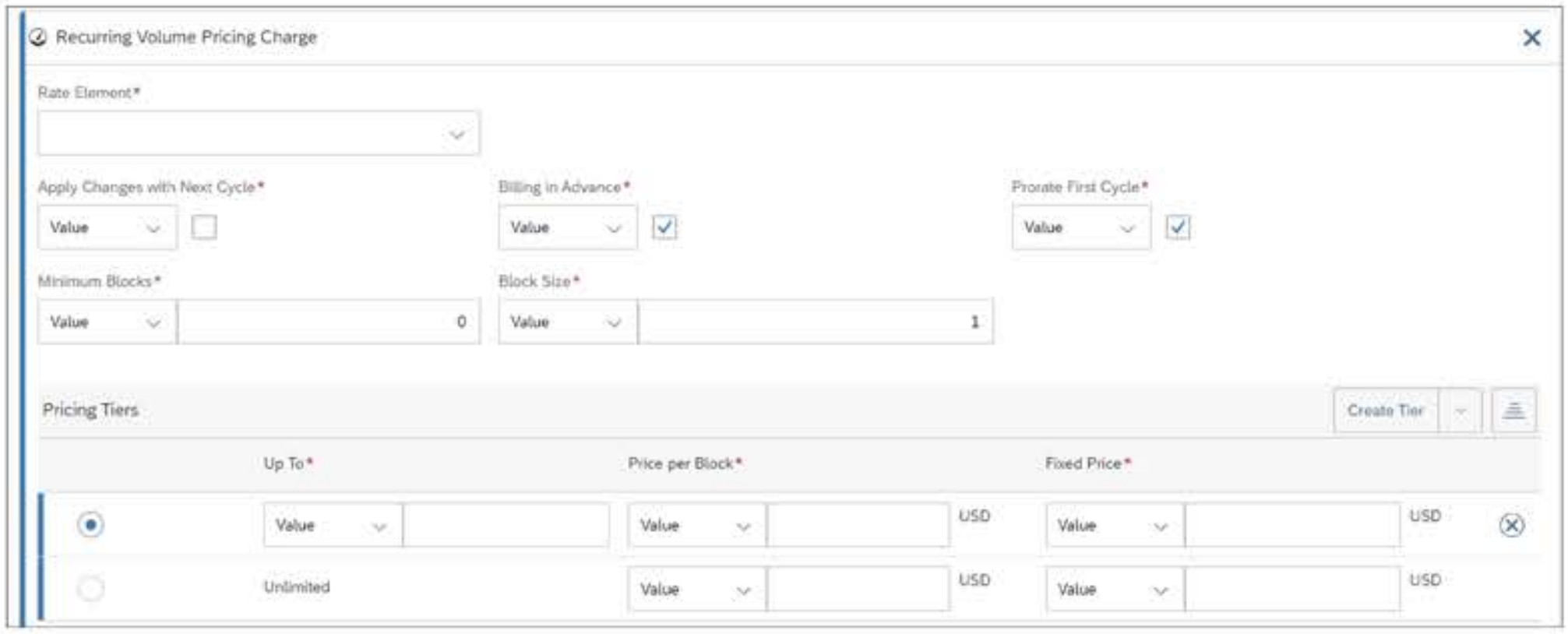

Figure 6.13 Volume Pricing for Recurring Rating

- **Tiered pricing**
  This pricing model allows you to define the price of usage in pricing tiers, as shown in Figure 6.14. The applicable price per block that is defined for each usage tier is applied to the usage volume within that tier. The prices for each relevant tier are added to calculate the price for the entire usage volume. The pricing tier can be defined as following:
  - **Up To**
    Used to define the number of usage blocks in each tier. The uppermost limit is **Unlimited**.
  - **Price per Block**
    This is used to manage the price of each usage block. The price for all blocks is calculated according to the number of blocks per relevant tier of usage volume. Each calculated amount is added together to form the price for all blocks.

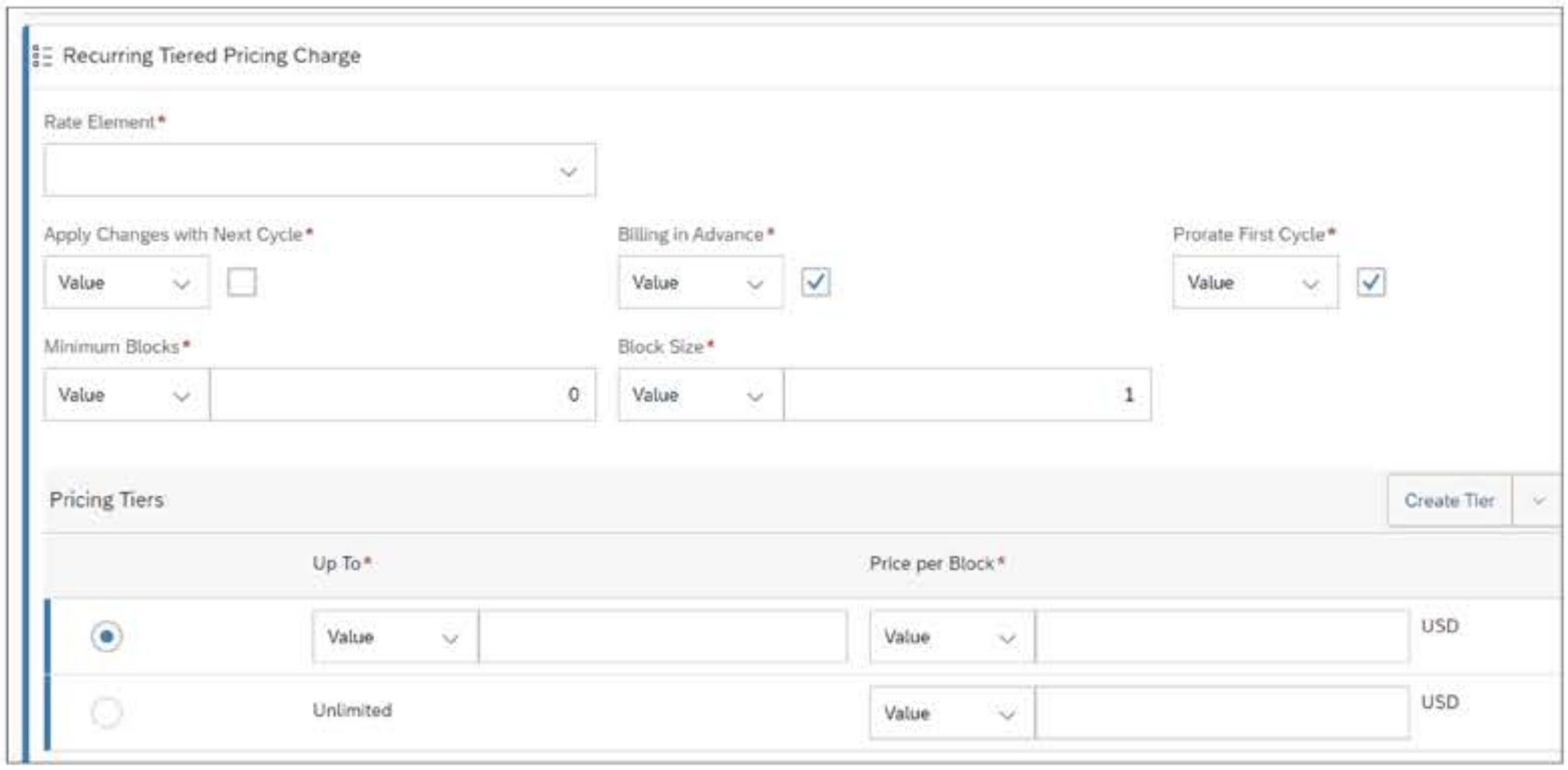

Figure 6.14 Tiered Pricing model for Recurring Charge

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- **Percentage pricing**
  By using this pricing model, as shown in Figure 6.15, you can define the price of usage quantities by specifying a percentage by which the price is required to be multiplied, added, or subtracted. To define a percentage charge, you need to specify the price per unit and then define the ratio by specifying a percentage and operation:
  - **multiply**
    Calculates a percentage of the price of the usage quantity.
  - **add**
    Calculates a base price for the usage quantity and adds a percentage of this price on top.
  - **subtract**
    Calculates a base price for the usage quantity and subtracts a percentage of this price.

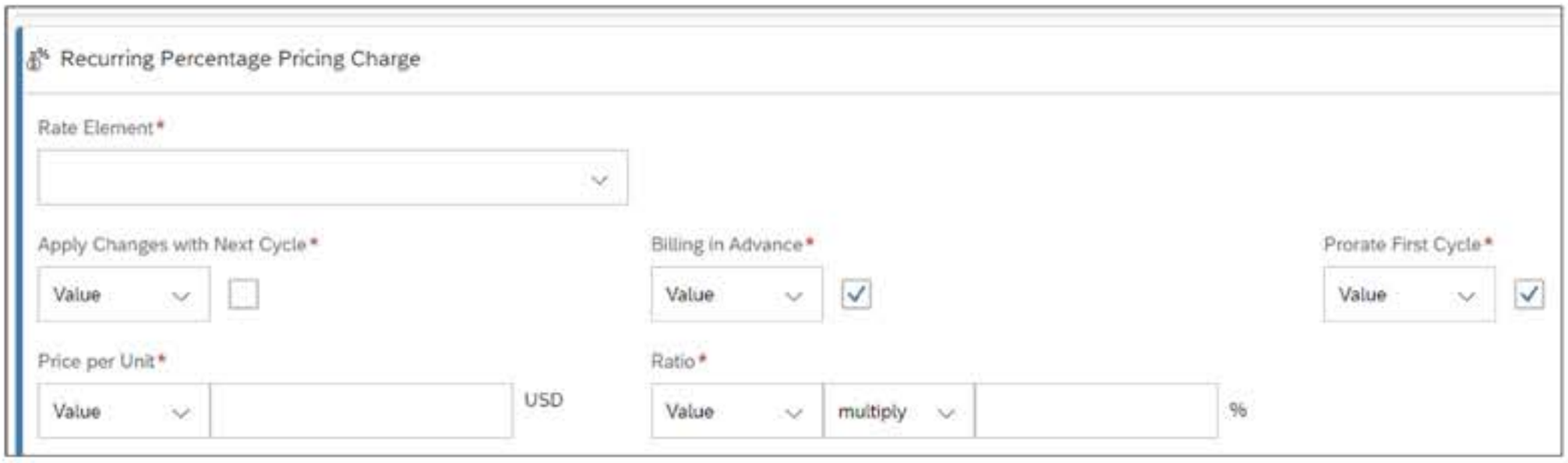

Figure 6.15 Percentage Pricing Model for Recurring Charge.

The following list describes other standard billing options for a recurring model:

- **Apply Changes with Next Cycle**
  This option manages changes to a rate plan in the form of a new rate plan snapshot or to the pricing data of the subscription in the form of a new subscription snapshot from the next billing cycle of a subscription. With this setting, you can keep the prices stable for the complete billing cycle.
- **Billing in Advance**
  SAP Subscription Billing offers options to bill customer either in arrears or in advance. This use case requires charging the customer up front, so select the **Billing in Advance** option.
- **Prorate First Cycle**
  This option allows auto-determining charges based on the actual number of days the subscription was active during the first rating period. This option is available out of the box and can't be customized. To see the prorated option, select **Prorate First Cycle** for **Martex Recurring Template**.
- **Custom amounts (for all charge types except one-time allowance charges)**
  Custom amounts are displayed if they have been defined for a charge. These are further amounts related to charges, for example, the cost of a service. Custom amounts are used for reporting purposes only and aren't billed to customers.

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For this use case, define a recurring rate plan template named "Martex Recurring Template". In addition, both **Price** and **Currency** will be using as parameters ("P_Currency" and "P_Price") as shown in the **PARAMETER DEFINITION** tab for the Martex rate plan template in Figure 6.16.

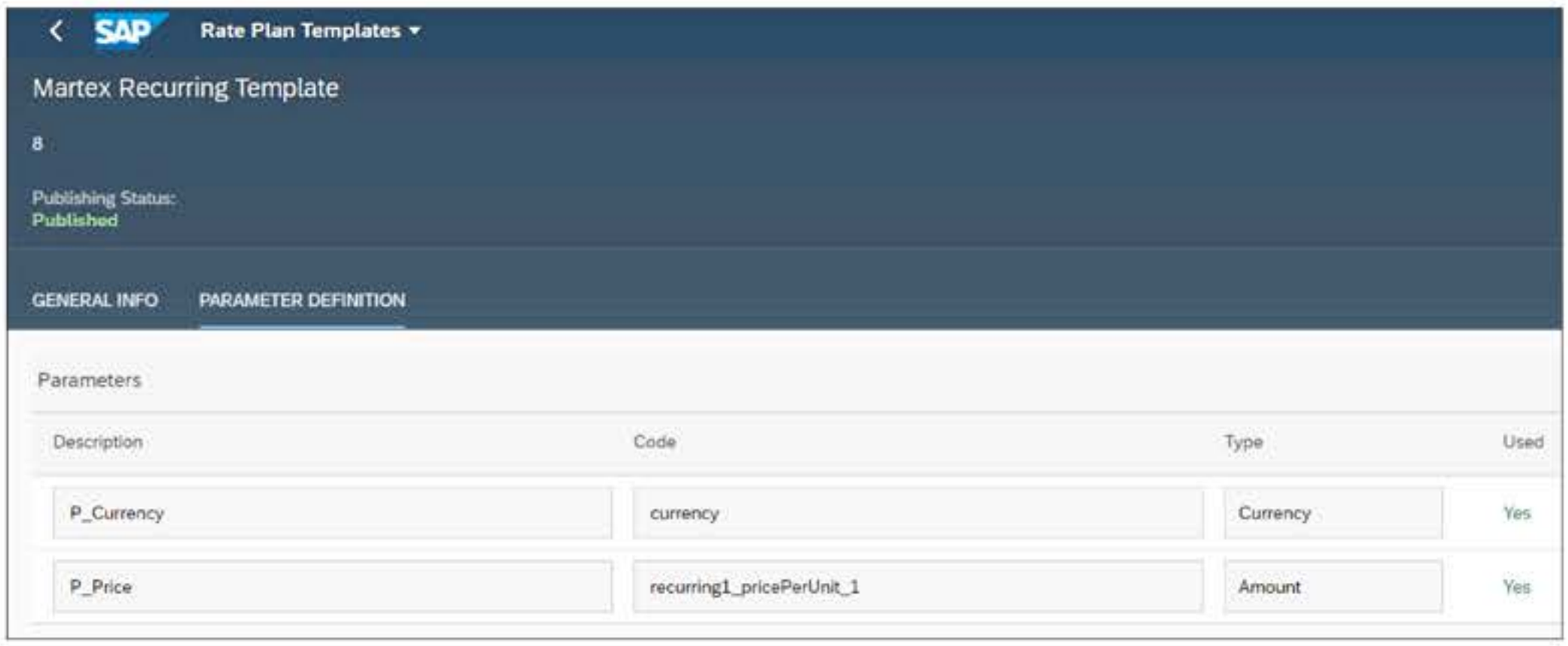

Figure 6.16 Parameter Definition for the Martex Rate Plan Template

6.3.3 Rate Plans

In this section, we'll discuss rate plans in SAP Subscription Billing. Rate plans are used to define the details of subscription charges and credits in SAP Subscription Billing. Rate plans are always created based on rate plan templates. To define rate plans, navigate to **Pricing • Rate Plans**. Figure 6.17 shows a view of the rate plans in SAP Subscription Billing. You can also create rate plans in the Products app, in which case, they are automatically assigned to a product.

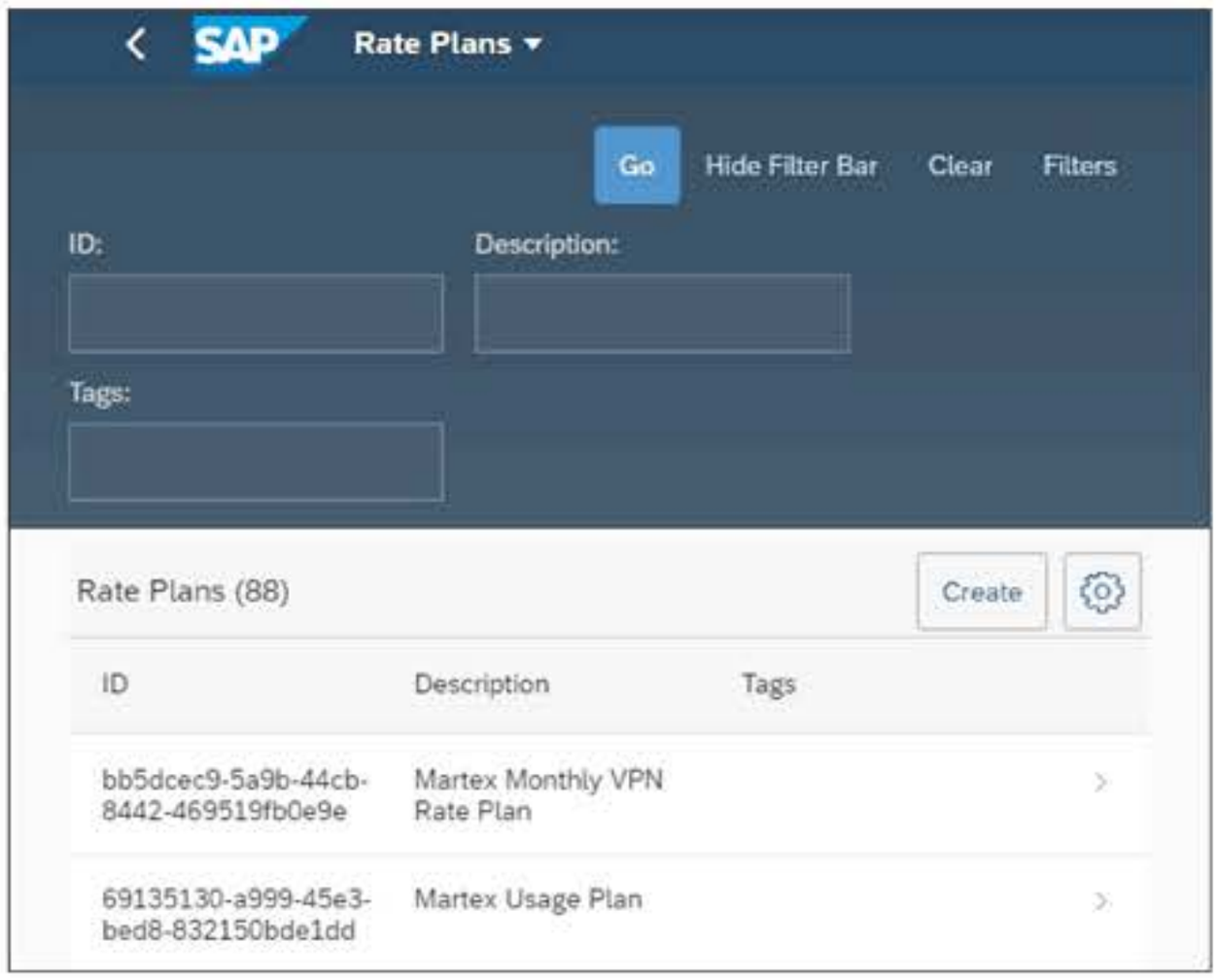

Figure 6.17 Rate Plans in SAP Subscription Billing

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To manage the requirement for Martex Corp., as shown in Figure 6.18, create **Martex Monthly VPN Rate Plan** with the previously created Martex Recurring Template.

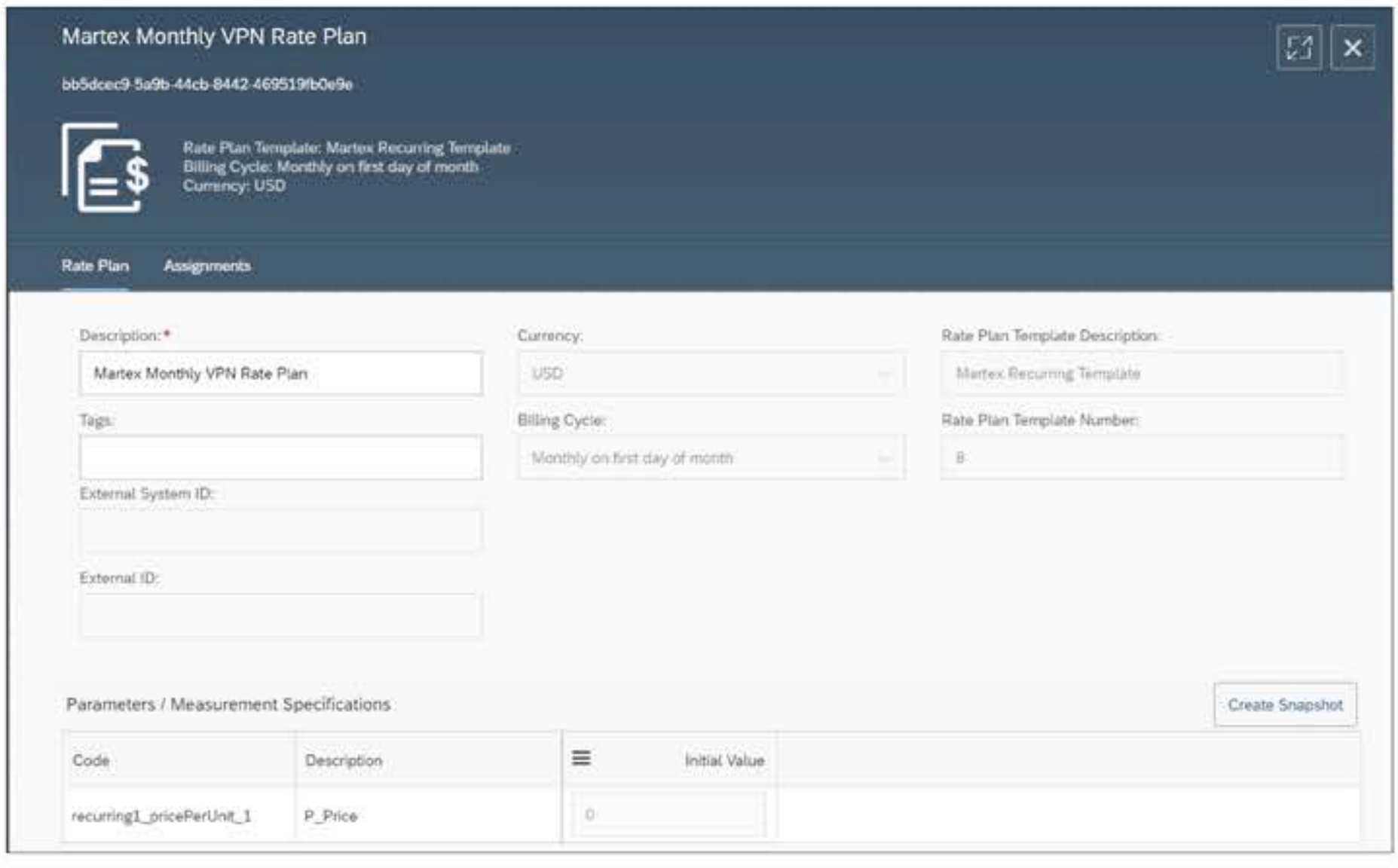

Figure 6.18 Martex Monthly VPN Rate Plan

6.3.4 Subscription Products

In this section, we'll discuss SAP Subscription Billing products that help business users set up and manage digital services and map these products with rate plans. The products are required to be extended for at least one market and rate plan template. In addition, subscription products can only be sold in markets for which a rate plan is defined and active in the product master data. To define **Products**, navigate to **Master Data and Configuration • Products**. In this section, you'll create a monthly digital service called "Martex Monthly VPN Subscription" for Martex Corp., as shown in Figure 6.19.

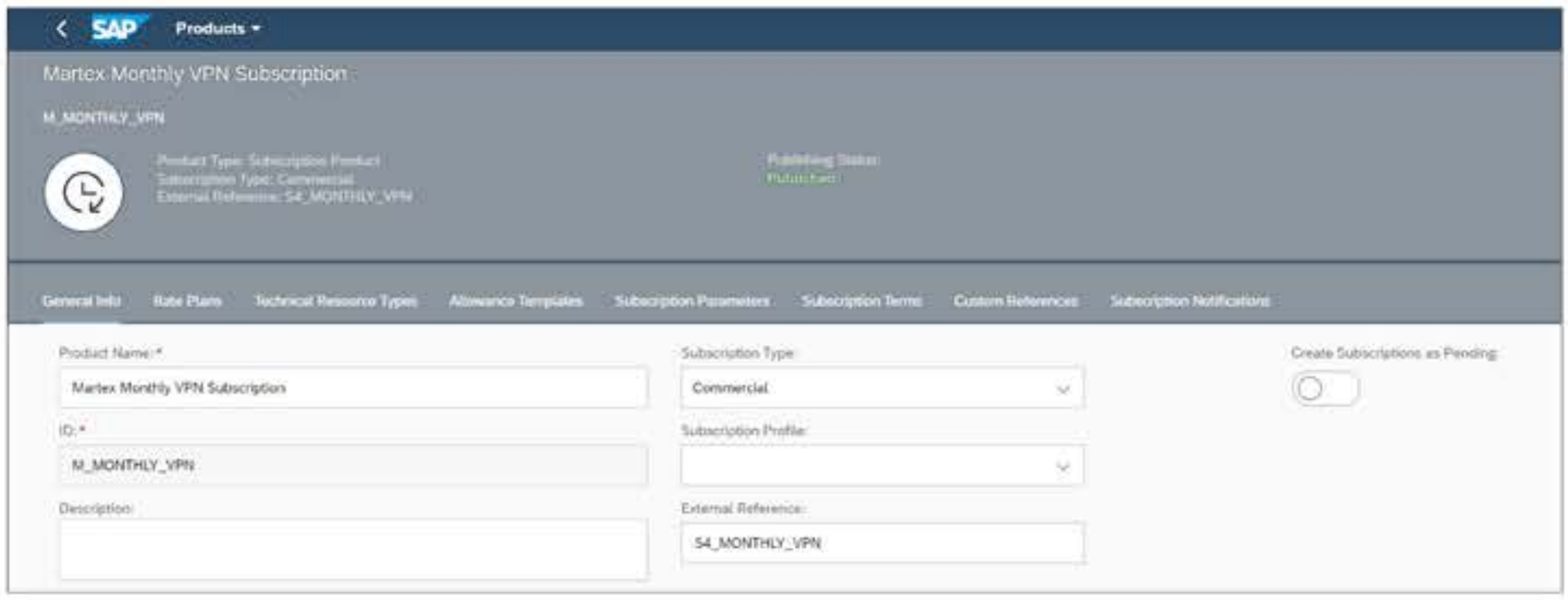

Figure 6.19 Creating Subscription Products in SAP Subscription Billing

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Following are some of the important components in SAP Subscription Billing products:

- **Subscription Type**
  Subscription Type defines the purpose of the product or service use. Following are the available subscription types available while setting up products.
  - **Commercial**
    Used for products for which bills are created and relevant for invoicing. For this use case, set the **Subscription Type** as **Commercial** for the Martex Monthly VPN subscription.
  - **Internal Billing**
    Used to keep track of internal cost or bills; this isn't relevant for invoice documents.
  - **Trial/Beta**
    Used for product trials, specifically for usage products to test usage data collection, but bills and invoice documents aren't created.
  - **Test**
    Used purely for testing purposes; no bills are created.
- **Subscription Profile**
  A subscription profile specifies a series of settings to distinguish different kinds of subscriptions.
- **External Reference**
  You can enter an identifier from an external system, for example, a material number from SAP S/4HANA, and so on.
- **Create Subscriptions as Pending**
  This is a very useful option if you want to implement deferred billing scenarios. By now, you're already aware of device as a service (DaaS) feature in SAP Billing and Revenue Innovation Management, which allows you to activate a subscription based on hardware fulfillment. In SAP Subscription Billing, you can model such a requirement using this option.

  This setting will allow you to specify that all subscriptions for the product are created with the status **Pending**. Typically, until the hardware is shipped to a customer base, you should not activate the subscription to generate customer bills. This setting will keep your subscription in pending state until you collect technical resources (e.g., hardware serial number, domain ID, etc.) or custom references. Until the **Pending** status is removed or subscription data is completed, no rating and billing takes place in the SAP Subscription Billing system.
- **Rate Plans**
  This option enables you to assign a rate plan template. Use the rate plan template for Martex Corp. created as "Martex Recurring Template" as a model for this rate plan. Figure 6.20 shows the assignment of rate plans in subscription products.

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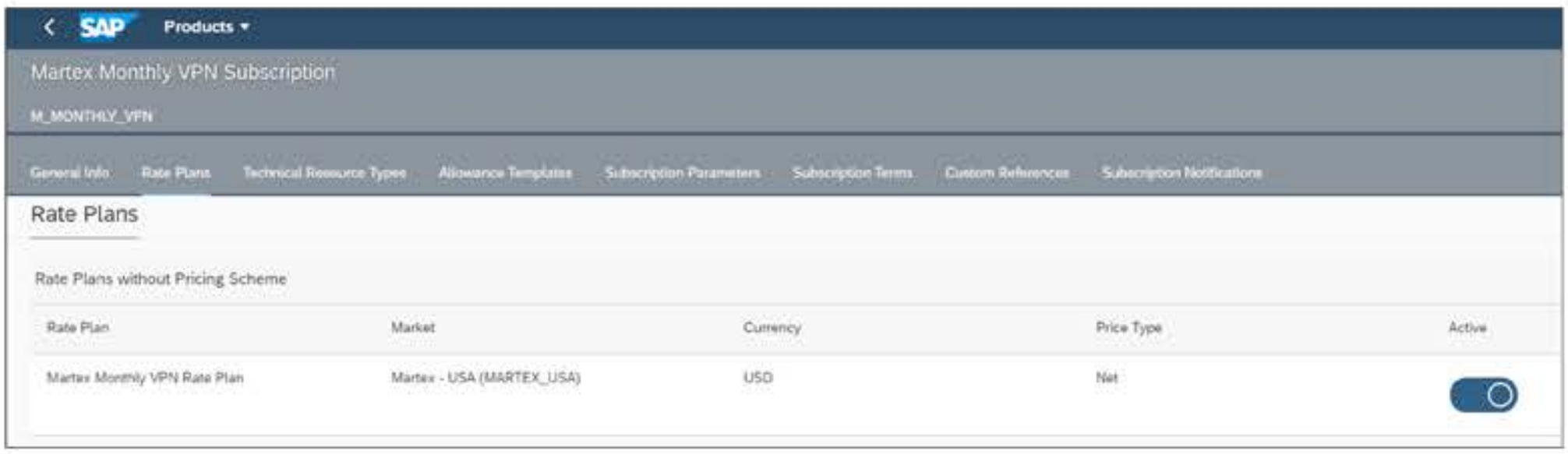

Figure 6.20 Assigning Rate Plans in Subscription Products

- **Technical Resource Types**
  For usage-based services, you need technical resources to bill your customers based on the usage of a service. SAP Subscription Billing is usually integrated with SAP Convergent Mediation by DigitalRoute to capture usage data records. The technical resource is used to assign the usage record to the subscriber and the subscribed service. When you model a pay-per-use product, you assign the technical resource types that are valid for your product. This assignment determines which rate elements you can use for usage charges in the rate plan of the product.

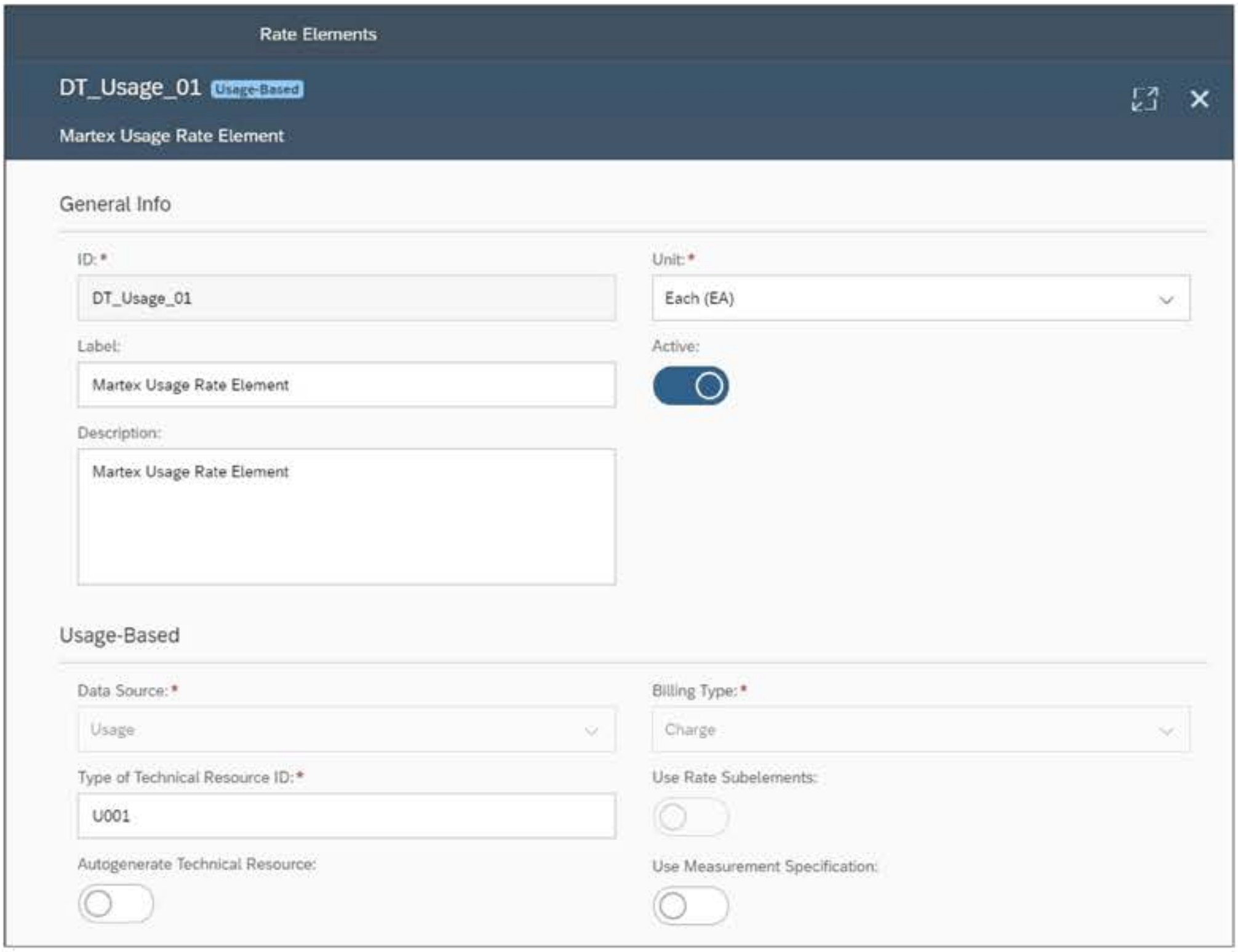

Figure 6.21 Defining Usage Rate Elements in SAP Subscription Billing

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To define **Technical Resource Types**, navigate to **Master Data and Configuration • Business Configuration • Rate Elements**. When you define a rate element, you can specify a type of technical resource ID to describe what type of ID is expected for your resource, as shown in Figure 6.21. You also have an option to auto-generate technical resources in SAP Subscription Billing.

Once the usage rate element is defined, you can map the same in the **Technical Resource Types** area of the product, as shown in Figure 6.22.

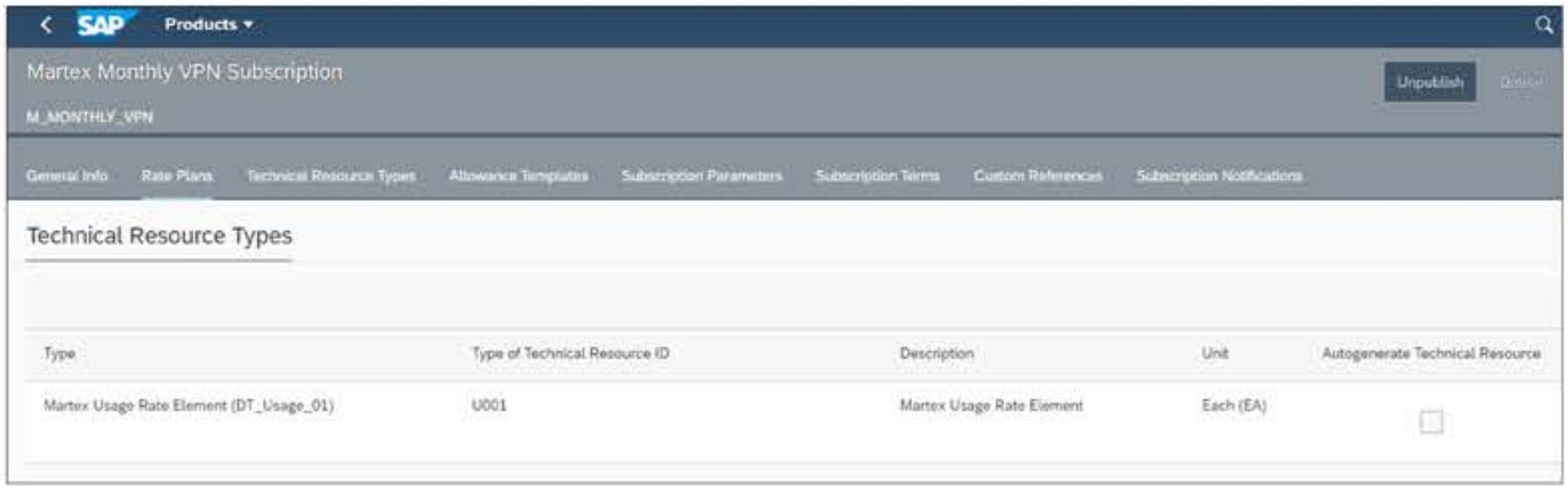

Figure 6.22 Technical Resource Types in Products

- **Subscription Terms**
  This option enables you to define a list of subscription options when the product is used in transactions. Let's discuss a few of the components
  - **Withdrawal Period**
    This period allows cancellation of a subscription without penalty. This is like a trial period, where the customer can make use of the service and then decide for continuation or cancellation. If the customer cancels the subscription before the end of the withdrawal period, all charges incurred are fully refunded.
  - **Term of Notice**
    This refers to the cancellation opt-out period. The period is defined in the number of days between the receipt of a cancellation notification from the customer or sales representative and the actual subscription cancellation. Note that when defining your cancellation policy, also consider whether you want to allow the cancellation to take effect during a billing cycle (if so, select **Allow Mid-Billing Cycle Expiration**).
  - **Minimum Term**
    The minimum duration of the contract in months. The customer can't cancel the subscription during the minimum term. Note that the end date of the minimum term is calculated based on the defined length, the start date of the subscription, and the billing cycle.
  - **Initial Fixed Term**
    The default length of the subscription in months. In the product, you can define a default fixed term for all the subscriptions from that product.

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- **Expected Term**
  The projected duration of the subscription in months.
- **Renewal Term**
  The period in months by which the subscription extends automatically after the minimum contract term.

Note that, if needed, you can later overwrite the default periods for individual subscriptions in the Subscriptions app. Figure 6.23 is a reference of **Subscription Terms** parameters in a product.

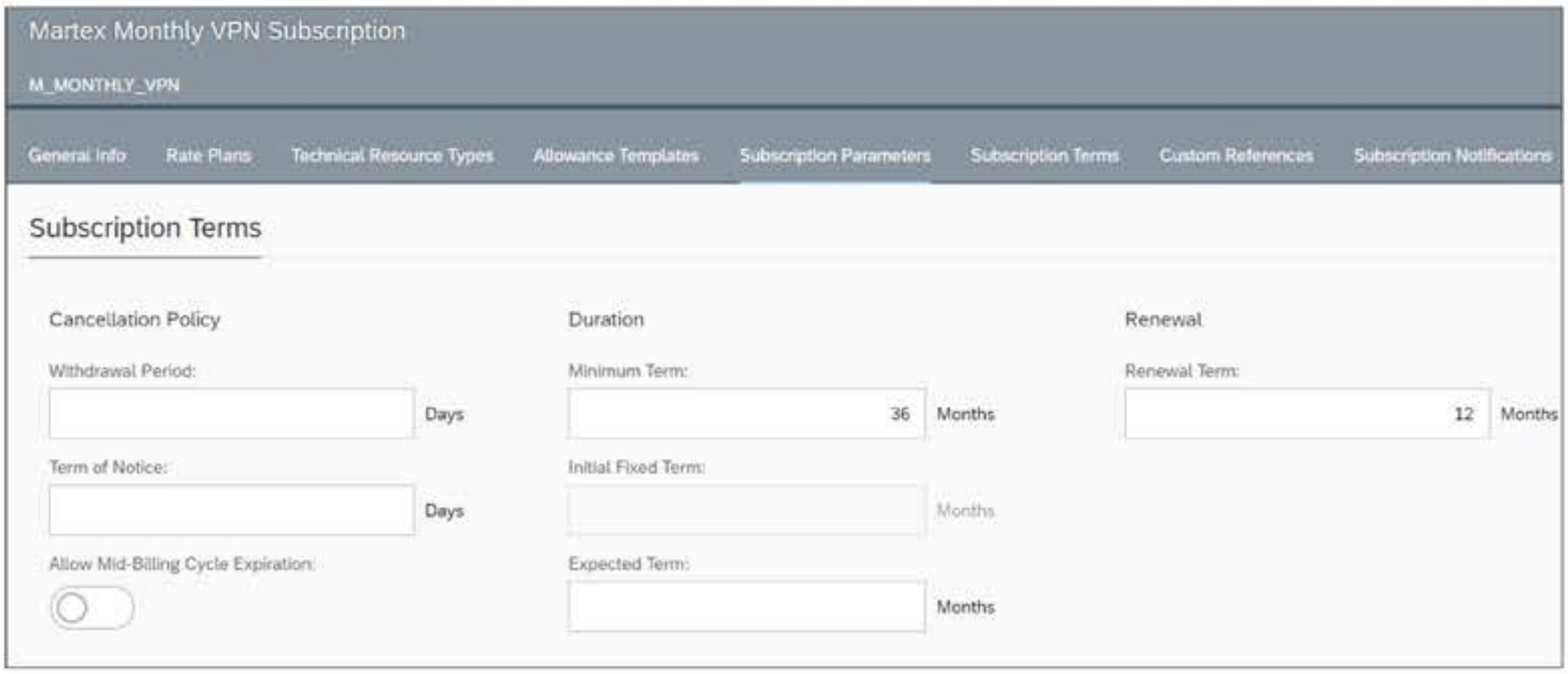

Figure 6.23 Subscription Terms Parameters in Products

Once the product is set up, you can publish the subscription product so that you can use the product for subscription transactions. The app also allows you to search for exiting products in the application, as shown in Figure 6.24.

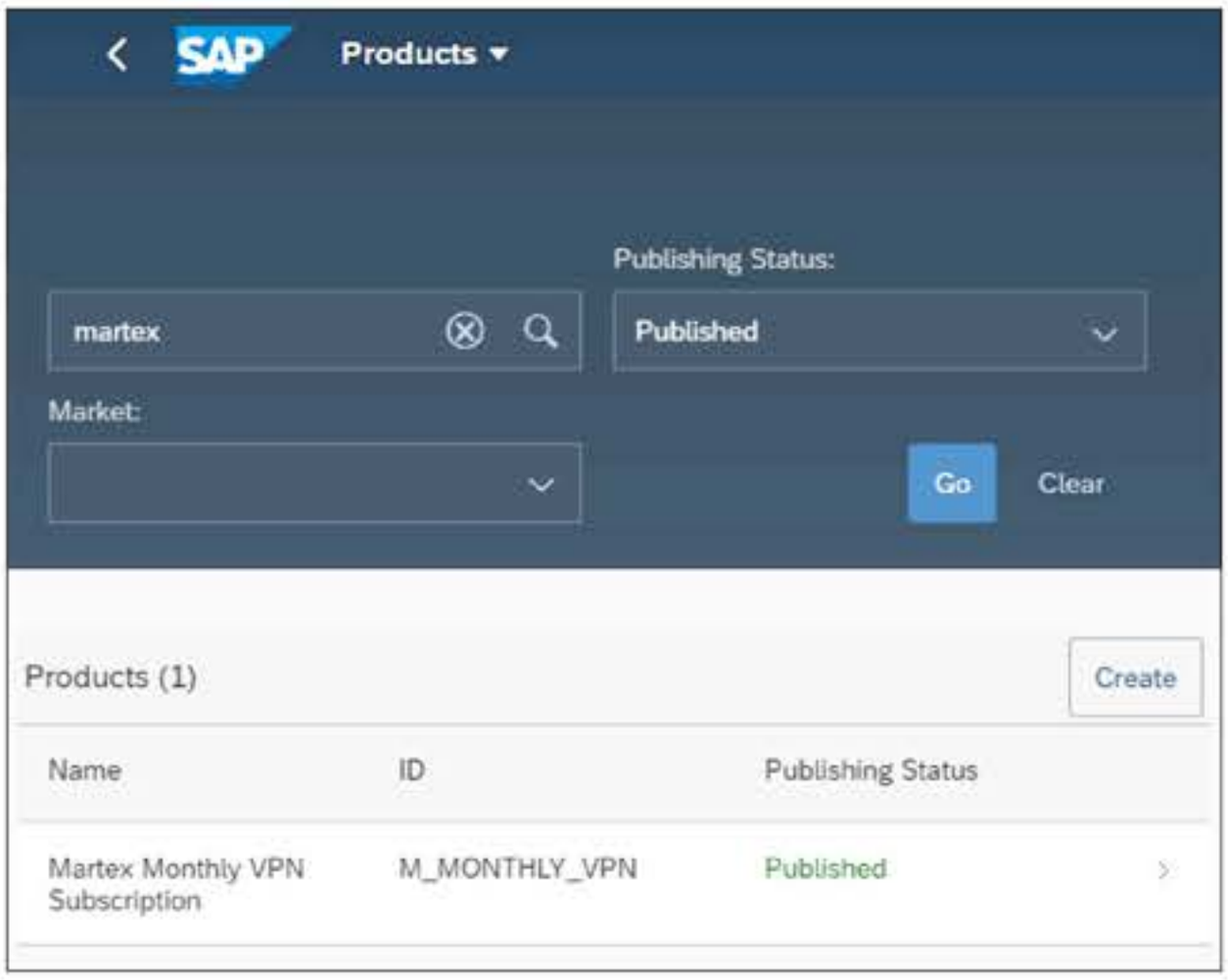

Figure 6.24 Searching Subscription Products in SAP Subscription Billing

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6.4 Provisioning Subscription Transactions

Once the master data is set up, you can proceed with setting up transactions in the SAP Subscription Billing application. In SAP Subscription Billing, *provisioning* refers to all activities needed to set up a subscription. Typically, in most implementations, you'll find that the creation of a subscription and its provisioning are triggered by an integrated upstream or front-office system. SAP Subscription Billing offers APIs that can be used to create transactions from different systems. You also have options to create pending subscriptions that can later be activated upon fulfillment. The application also provides you with apps that can be used to create transactions directly in SAP Subscription Billing.

To create a provisioning transaction manually in SAP Subscription Billing, navigate to **Subscriptions and Billing • Provisioning Request**. Now, you can follow these steps to create a provisioning request:

- **Select the customer.**
  The first object you need to create a subscription will be the customer. You can search existing customers available in SAP Subscription Billing. Once you've identified your customer, select the same for the provisioning request, as shown in Figure 6.25.

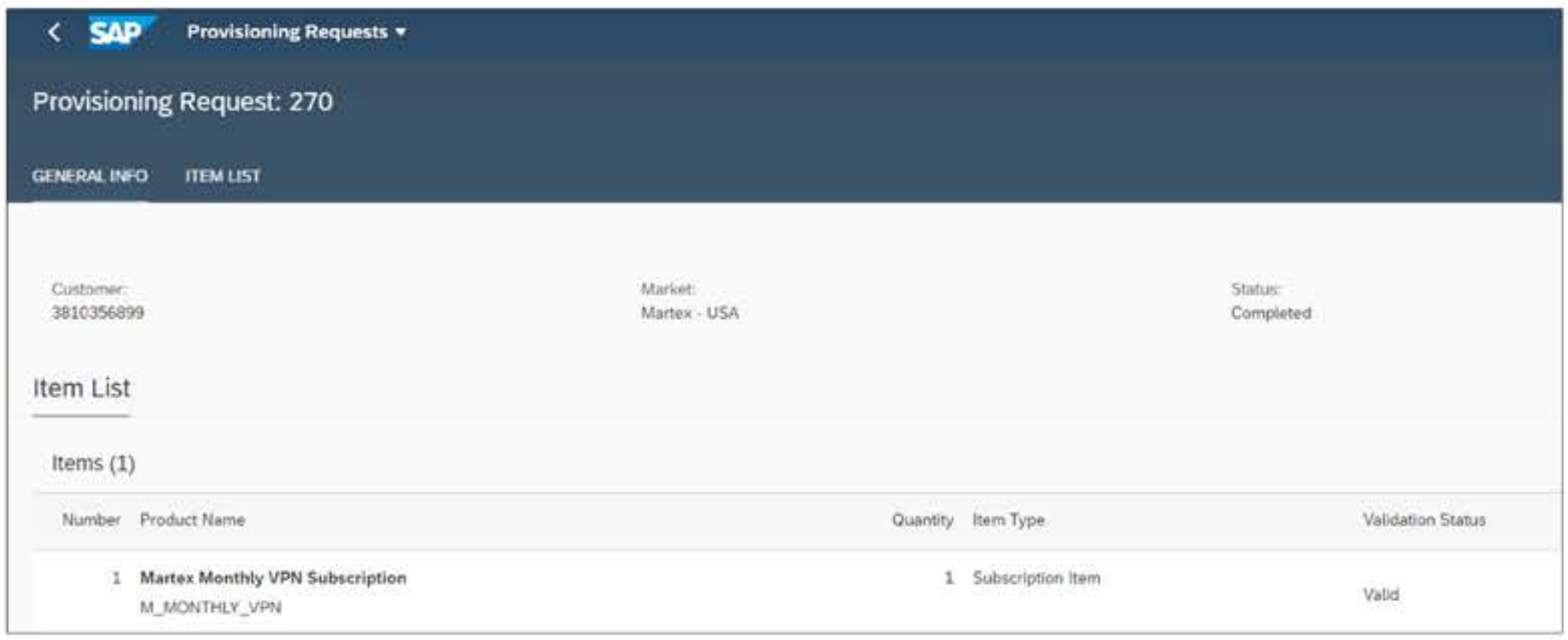

Figure 6.25 Selecting the Customer for the Provisioning Request

- **Add items.**
  The next object you need is the service or products offered to the customer. You can add a subscription item for each subscription that you want to create and complete the required information under the **GENERAL INFO** tab. To create a valid subscription item, you need to complete the mandatory information, such as an existing product, a quantity, and the subscription validity dates. Figure 6.26 shows the **Pricing** reference.

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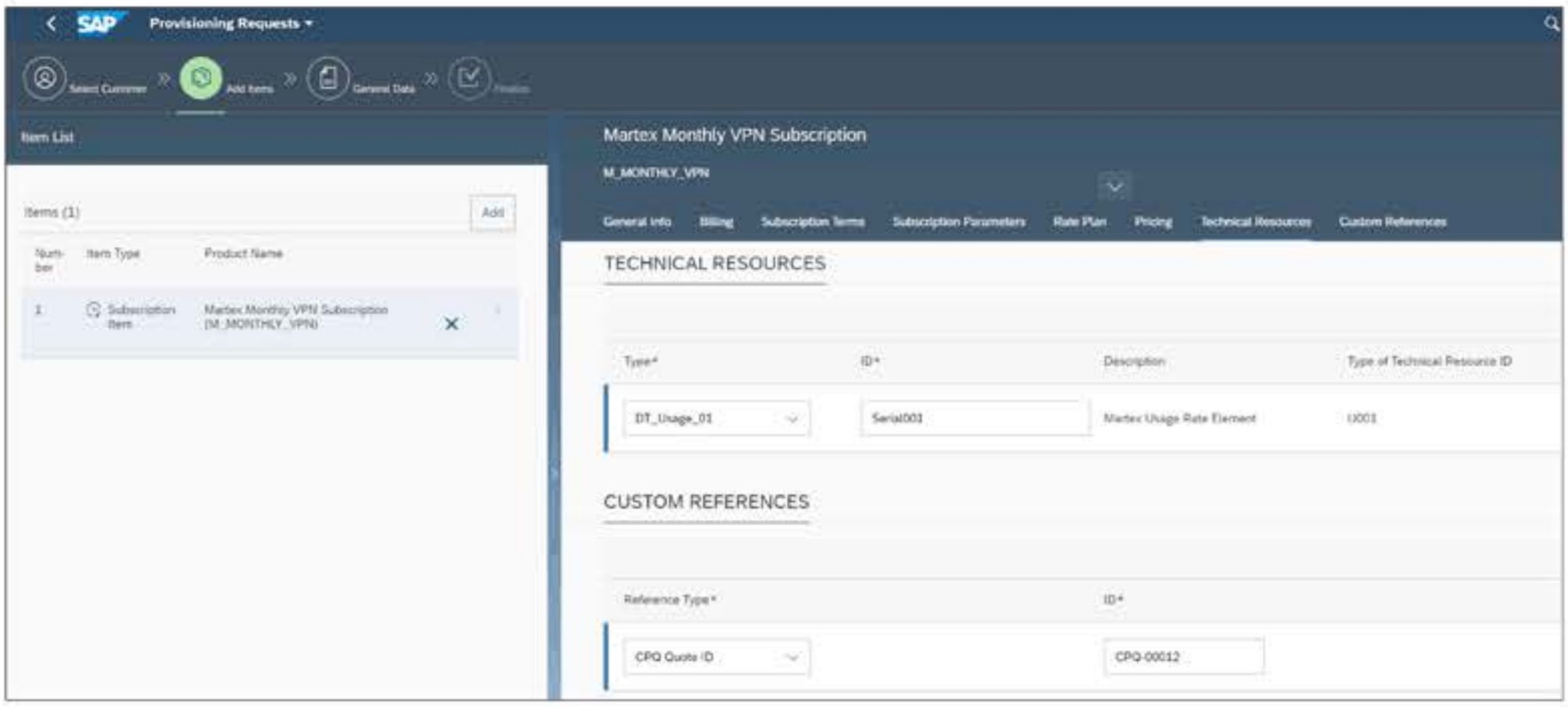

Figure 6.26 Adding Price and Technical Resource IDs

- **Define the general data.**
  You need to define general data such as the payment method and custom references for your provisioning request, as shown in Figure 6.27.

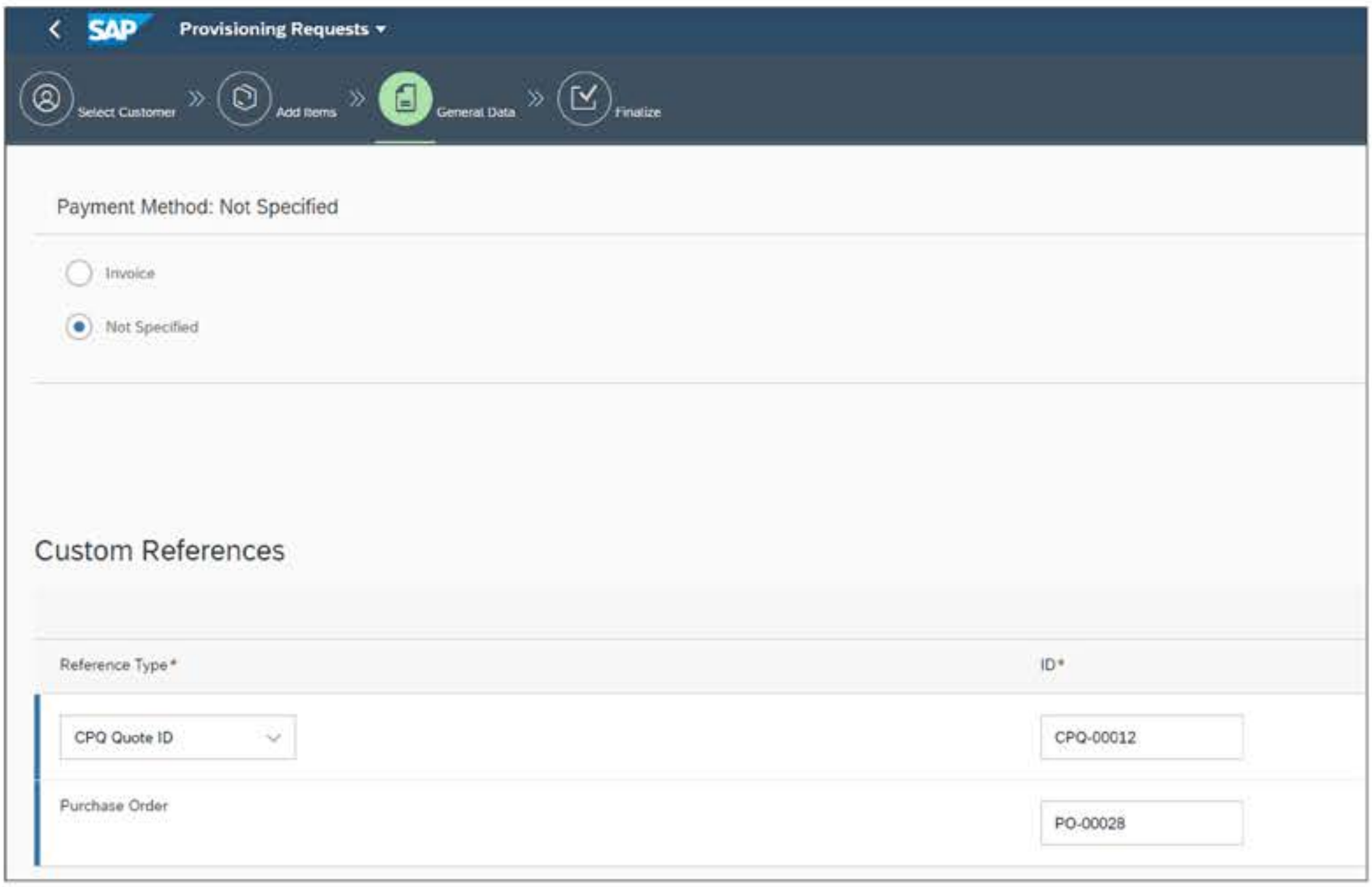

Figure 6.27 General Data in the Provisioning Request

- **Finalize the provisioning request.**
  This is the final step. Before you create the provisioning request, use the summary of

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the provisioning request data to make sure that everything is correct. Figure 6.28 shows the specifics regarding finalizing the provisioning request. You can change your entries in the previous steps if necessary.

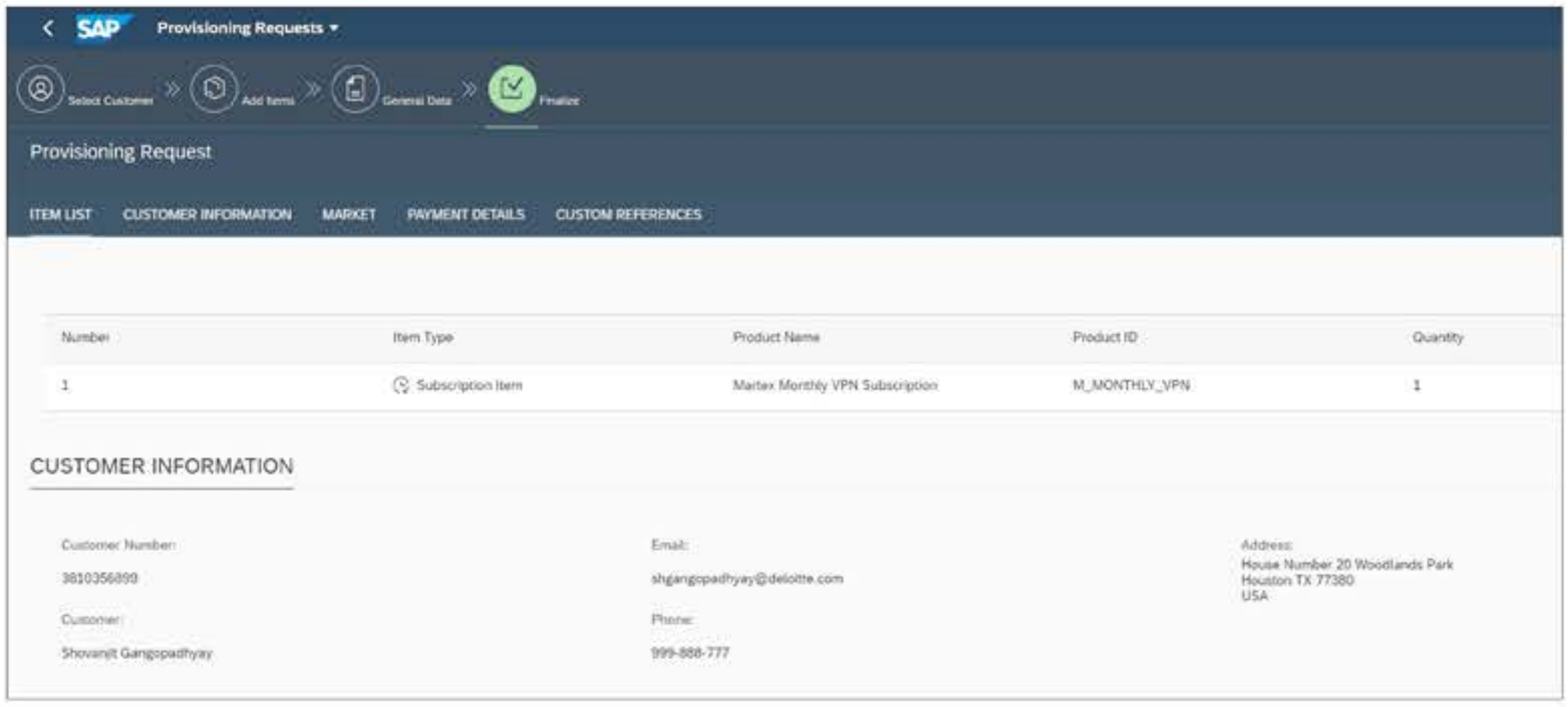

Figure 6.28 Finalizing the Provisioning Request

Once the provisioning request is created, you can track the progress of the subscription creation in the Provisioning Requests app. You can display each step with its detailed business document status and progress information, such as processing time, success, and troubleshooting information, in the Progress Tracker.

When a provisioning request is released, SAP Subscription Billing automatically generates subscriptions for each subscription item of the provisioning request. During the generation process, the technical resources that are required for a subscription to a digital product are determined. Note that the Provisioning Request app is planned to be deprecated in the upcoming release.

If you've used the Pending Subscription feature, you can complete your pending subscriptions in the Subscriptions app by following these steps:

1. Open the pending subscription, and select **Complete**.
2. Follow the wizard to provide technical resources and custom references, and to set the start date of the subscription.
3. Once you've fully provisioned the subscription, select **Save as Complete**.

Once you've saved the subscription, the status will change from **Pending** to one of the following statuses:

- **Active**: If the chosen start date is today or a date in the past.
- **Not Started**: If the chosen start date is a date in the future.

To view the subscription, navigate to **Subscriptions and Billing • Subscriptions**. Figure 6.29 is a view of a subscription transaction.

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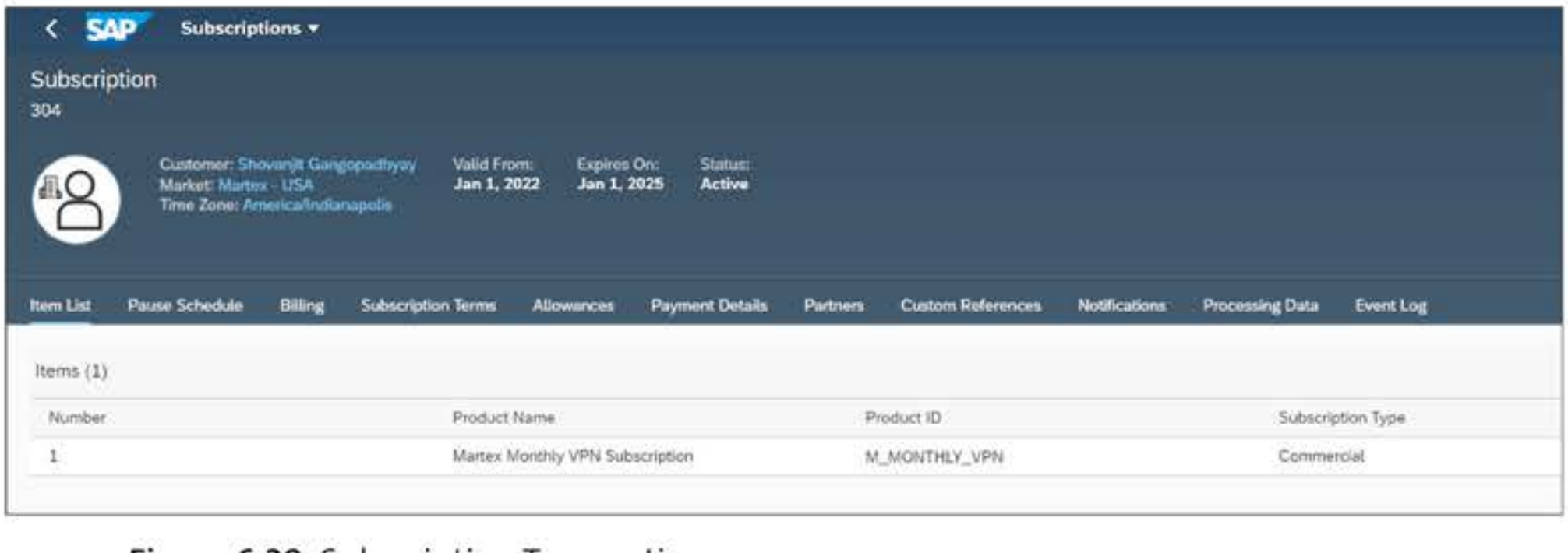

Figure 6.29 Subscription Transaction

In the Subscription app, you see the items for the current effective period. You can select another period in the list using **Items Effective From** to view the state of the items in the past or in the future. The subscription items are displayed according to their effective period. Every time you make any change on the item level, such as a change in the pricing parameters or in the custom references, you need to define as of which date this change becomes effective. This is similar to change order management in SAP Billing and Revenue Innovation Management.

You can also define custom references, such as external documents, while setting up your subscription. For example, you might want to store an SAP CRM application number for a subscription so that you can use the same for any downstream reporting requirement. You can define custom references either on the header of the subscription or for individual subscription items.

In the application, all changes to a subscription are tracked in the event log.

6.5 Billing Process in SAP Subscription Billing

Once your transactions are created and subscription is activated, the next step is to generate customer billing. In SAP Subscription Billing, you can't manually trigger billing by yourself. The application generates billing data continuously, based on data from the rating process. Billing data is stored in billing documents. Note that the billing data isn't auto transferred to integrated systems. The billing data needs to be queried periodically so that the information can then be transferred to invoicing processes in an integrated system. SAP Subscription Billing also supports proration as a standard feature.

In SAP Subscription Billing, to view bills, navigate to **Subscriptions and Billing • Bills**. Bills provide an initial statement of payable or credited amounts and must be transferred to an invoicing system to create customer invoices. Each bill contains charges or credits that share certain attributes such as customer and billing date. Charges or

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credits for multiple subscriptions can be included in the same bill. Each bill contains the following key information at the header and item levels:

- **Header level**
  - References to the relevant customer and market
  - Billing date and total bill amount
  - Billing date and due date
  - Bill status
  - Information about any split element defined in the relevant subscriptions, which caused the bill to be split
  - Transfer status and successor document information if bills are transferred to an integrated system
  - Payment details or SAP S/4HANA contract account
  - Any custom references that are inherited from the customer
- **Item level**
  - Reference to the subscription and product
  - Bill item amount
  - Detailed information about the charges or credits, such as the rate element, rating period, pricing and usage information, and allowance consumption
  - Any custom references that are inherited from the subscription, subscription item, or product

Figure 6.30 shows the charge details for a monthly subscription bill charged between the rating period May 5, 2022, and Jun 4, 2022, with net amount $165.

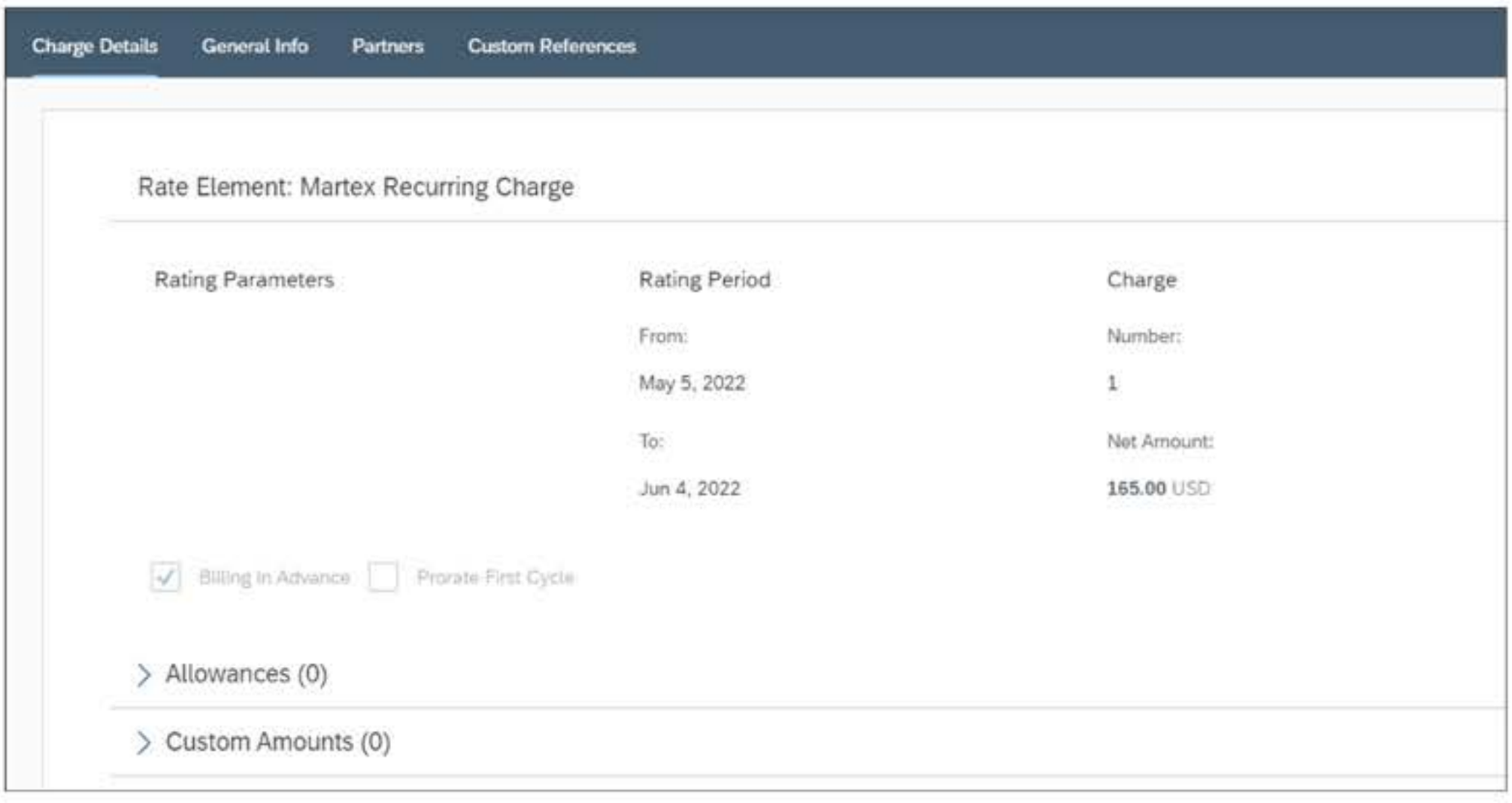

Figure 6.30 Sample Bill: Charge Details

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SAP Subscription Billing provides a standard API to query billing data from application and send it back to integrated invoice management systems. You need to set up a periodic job that can handle this process. Typically, SAP Cloud Platform Integration is used to query such records and set the bill status once the transfer to the downstream system is successful.

SAP Subscription Billing also offers additional features related to billing. To define additional billing settings, you can navigate to **Master Data and Configuration • Business Configuration • Billing Settings**.

6.6 Integration with SAP Convergent Invoicing

Being a cloud solution, SAP Subscription Billing does provide various predefined integration packages to enable data transfer between SAP Subscription Billing and other SAP and non-SAP ERP applications. From an SAP Billing and Revenue Innovation Management standpoint, SAP Subscription Billing offers native integration with SAP Convergent Invoicing. To enable the integration, the following settings are required in **Master Data and Configuration • Business Configuration • Subscription Settings**:

- **Replicate Subscriptions to Provider Contracts**
  This setting enables integration of subscription objects from SAP Subscription Billing with SAP Convergent Invoicing. Once you activate the setting, the subscriptions that use the profile are replicated to SAP S/4HANA as provider contracts. The replicated provider contracts have the same IDs as the corresponding subscriptions. You can then use the provider contracts to manage subsequent process and reporting in SAP S/4HANA.
- **Require Contract Account in Subscriptions**
  As you know, one of the mandatory master data items required for SAP Billing and Revenue Innovation Management is contract accounts. However, in SAP Subscription Billing, subscriptions can be created without contract account information. But if you activate the setting, it's mandatory to provide a contract account in the items of a subscription. The contract account is used to manage the payment details for the subscriptions that use the subscription profile. The contract account also controls other aspects of payments, such as the dunning procedure and tolerances.

In addition to SAP Convergent Invoicing, there are various out-of-the-box integrations available in SAP Subscription Billing. It offers native integration with SAP CPQ, the SAP Digital Payments Add-On, SAP Commerce, and so on.

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6.7 Additional Features in SAP Subscription Billing

SAP Subscription Billing offers additional features that you can explore based on project requirements. For example, you can manage payment details with SAP Subscription Billing to enable credit cards as a payment method by integrating with the SAP Digital Payments Add-On. It also allows you to integrate with SAP Event Mesh to manage different types of events.

One of the core features of SAP Subscription Billing is its native integration with SAP Convergent Mediation by DigitalRoute. The Usage Records API enables you to create and update usage records to store consumption data from an external metering system. Each usage record is valid for a given aggregation period. A usage record is defined as follows:

- Should contain quantity information corresponding to the number of units measured. The quantity can be a positive number or zero.
- Should have the consumption start and end date of the usage (range of dates during which the quantity has been aggregated).
- Should have a technical user identifier from which the consumer will be derived.

If the usage data consists of additional client-specific information, then custom references can be configured using the Business Configuration app. You can navigate to **Subscriptions and Billing • Usage** to view usage details.

6.8 Summary

SAP Subscription Billing provides businesses of all sizes with the flexibility to offer services on subscription and meet changing business needs with speed and agility. Because it's a preconfigured application, it also reduces time to market as well as infrastructure requirements. However, in comparison to SAP Billing and Revenue Innovation Management, it lacks in flexibility and features. Still, SAP Subscription Billing can simplify anything as a service (XaaS) business transformation. The nature of this unique, flexible, and easy-to-use platform makes it one of the most scalable solutions for a subscription-based business model.

In the next chapter, we'll discuss some of the core integrations with SAP Billing and Revenue Innovation Management, including SAP Revenue Accounting and Reporting, SAP Convergent Mediation (by DigitalRoute), and integrating tax engines and payment gateways.

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Chapter 7

Integration with Other Solutions

This chapter will provide an overall picture of the integration points for all SAP Billing and Revenue Innovation Management components. These points include the integrations between each of the applications within SAP Billing and Revenue Innovation Management and those between SAP Billing and Revenue Innovation Management and other solutions.

SAP Billing and Revenue Innovation Management is a full solution suite, but there are rare cases in which companies will only implement some SAP Billing and Revenue Innovation Management components. As the best practice across industries, SAP Billing and Revenue Innovation Management works best when all or most of the components are implemented together and with other additional applications. As a result, companies typically implement SAP Billing and Revenue Innovation Management in conjunction with other applications to create a quote-to-cash (or consume-to-cash) ecosystem that provides an end-to-end solution.

This chapter will first highlight the integrations between core SAP Billing and Revenue Innovation Management components and how they interact with each other. The chapter won't go in depth into the step-by-step setup of the technical activities, but rather it will describe how data and information flow between these components. Once this is done, we'll branch out to additional applications. This is where we'll explore how SAP Billing and Revenue Innovation Management is integrated with other auxiliary applications, such as SAP Convergent Mediation by DigitalRoute, SAP Customer Financial Management, and more.

7.1 Integration between SAP Billing and Revenue Innovation Management Core Components

SAP Billing and Revenue Innovation Management is implemented to streamline high-volume revenue management processes. SAP Billing and Revenue Innovation Management has solutions based on the following industries: SAP for Telecommunications, SAP Utilities Customer Engagement, and SAP Public Sector Collection and Disbursement (PSCD). SAP Billing and Revenue Innovation Management is modular, so you have the option of deploying one or more elements depending on your requirements.

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A key step in SAP Billing and Revenue Innovation Management integration is activation of the business functions in your SAP S/4HANA system. The set of business functions is based on your industry. If you want to use any additional integration of the business processes in SAP S/4HANA, you'll need to activate them as well.

For SAP Billing and Revenue Innovation Management integration, products are defined as the item for which you're building the revenue management process. They are divided into three logical views: commercial, charging, and technical. Let's discuss each view:

- **Commercial**
  A commercial view of products is provided in SAP Billing and Revenue Innovation Management, subscription order management. In fact, it contains what can be defined as the master commercial view of all products, including the definition of the available options for customer selection, the bundling of different services, and the one-off charges.
- **Charging**
  The charging view is provided in SAP Convergent Charging, which is where you define the charge plans for the consumption of services. This system receives chargeable items or consumption data records. Chargeable items are real-time consumption data, whereas consumption data records are in batch mode. The charge plans and the refill plans are defined here and used to perform the rating, charging, and refilling processes for consumption and refill data.
- **Technical**
  The technical view is where you activate the services to which your customers subscribe.

In SAP Billing and Revenue Innovation Management, the product prices are maintained accordingly:

- You maintain usage prices/tiers in SAP Convergent Charging or in the subscription contract parameter.
- You maintain one-off prices in SAP S/4HANA as pricing condition records.
- You maintain recurring prices in SAP S/4HANA as pricing condition records.

There are different ways of implementing recurring pricing and charging scenarios in SAP Billing and Revenue Innovation Management. If the recurring pricing logic is complex, it's advised to use SAP Convergent Charging to avoid custom enhancements in SAP S/4HANA. If the recurring pricing is straightforward and SAP Convergent Charging isn't integrated, it's possible to use an SAP Convergent Invoicing billing plan to charge periodically. Based on the recurring pricing scenarios implemented, the price is used in an SAP Convergent Invoicing billing plan to generate billable items during each billing cycle. If the recurring prices are managed using SAP Convergent Charging, the price conditions are then replicated to SAP Convergent Charging. Replicating the prices to SAP Convergent Charging applies the recurring charges during each billing cycle. SAP

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Convergent Charging can prorate prices if services have only been consumed during part of a billing cycle. To distribute prices from SAP Billing and Revenue Innovation Management, subscription order management, to SAP Convergent Charging, you must configure the consumer proxy. This configuration is done in SAP Customer Relationship Management (SAP CRM) via Transaction SOAMANAGER.

The product definition is shown in Figure 7.1.

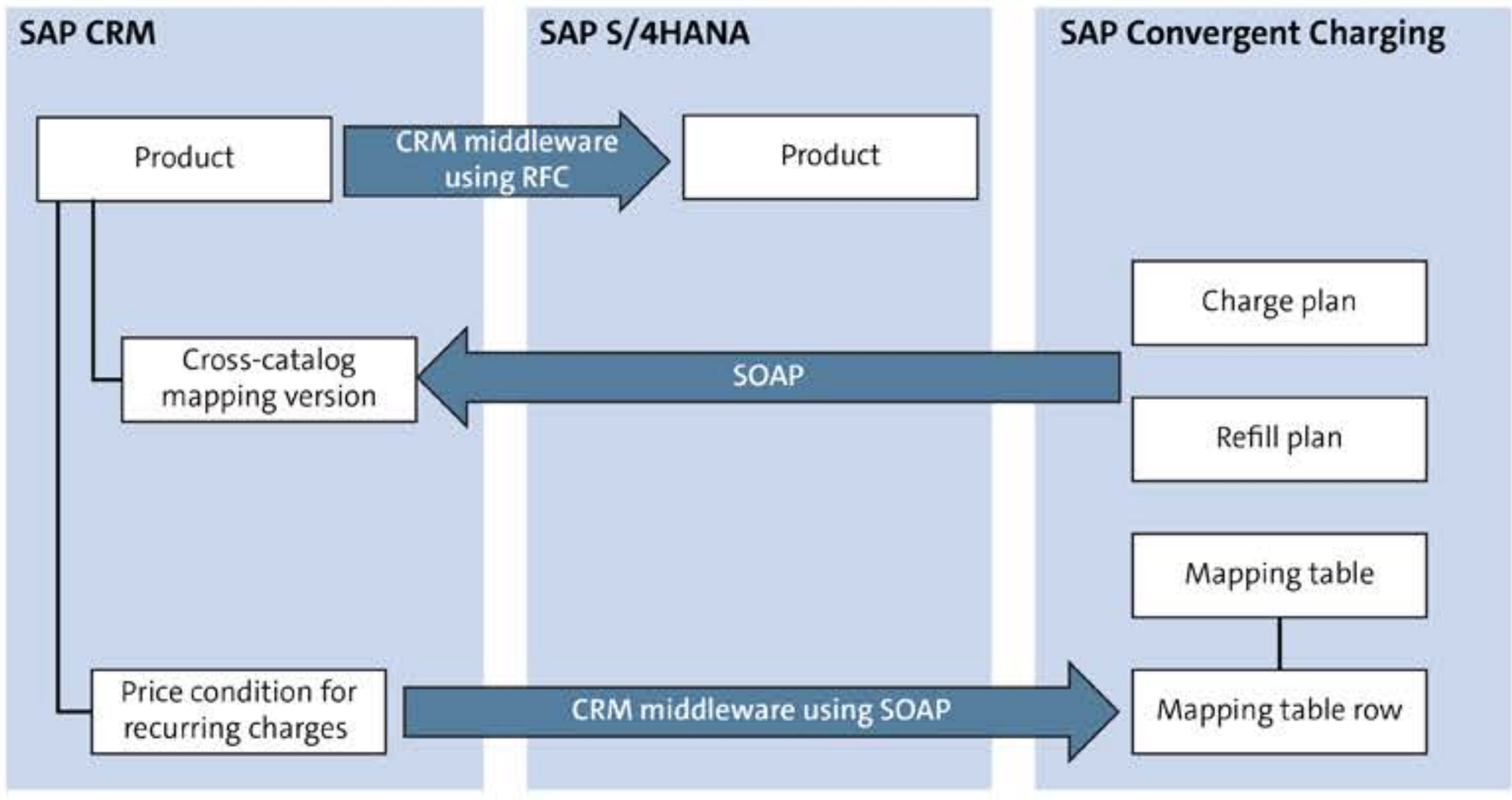

Figure 7.1 Product Definitions

Business partner master data is usually managed in SAP S/4HANA or SAP Master Data Governance (SAP MDG). The SAP S/4HANA system contains the contract accounting functions and is usually the master data system where customer and contract account data is entered and managed. The master data is also replicated to SAP Convergent Charging via web services using SOAP. Based on the replication of business partner, contract account and subscription contract, the corresponding subscriber account, external account, and provider contracts are created in SAP Convergent Charging.

When the subscription contract is created in SAP S/4HANA or SAP Convergent Charging, each component only stores those attributes relevant for further processing within that component. The business partner, contract accounts, and prepaid accounts are distributed to the SAP Convergent Charging system from the SAP S/4HANA system when the data of the subscription contract they belong to is received in SAP S/4HANA.

The transactional flow of data begins in SAP Billing and Revenue Innovation Management, subscription order management, which is used as an orchestration system where the order or quote type is created. SAP Convergent Invoicing contains the billing and invoicing. SAP Convergent Charging maintains pricing and charging logic, which receives the chargeable item and sends it to SAP Convergent Invoicing as billable items for billing and invoicing processes (see Figure 7.2).

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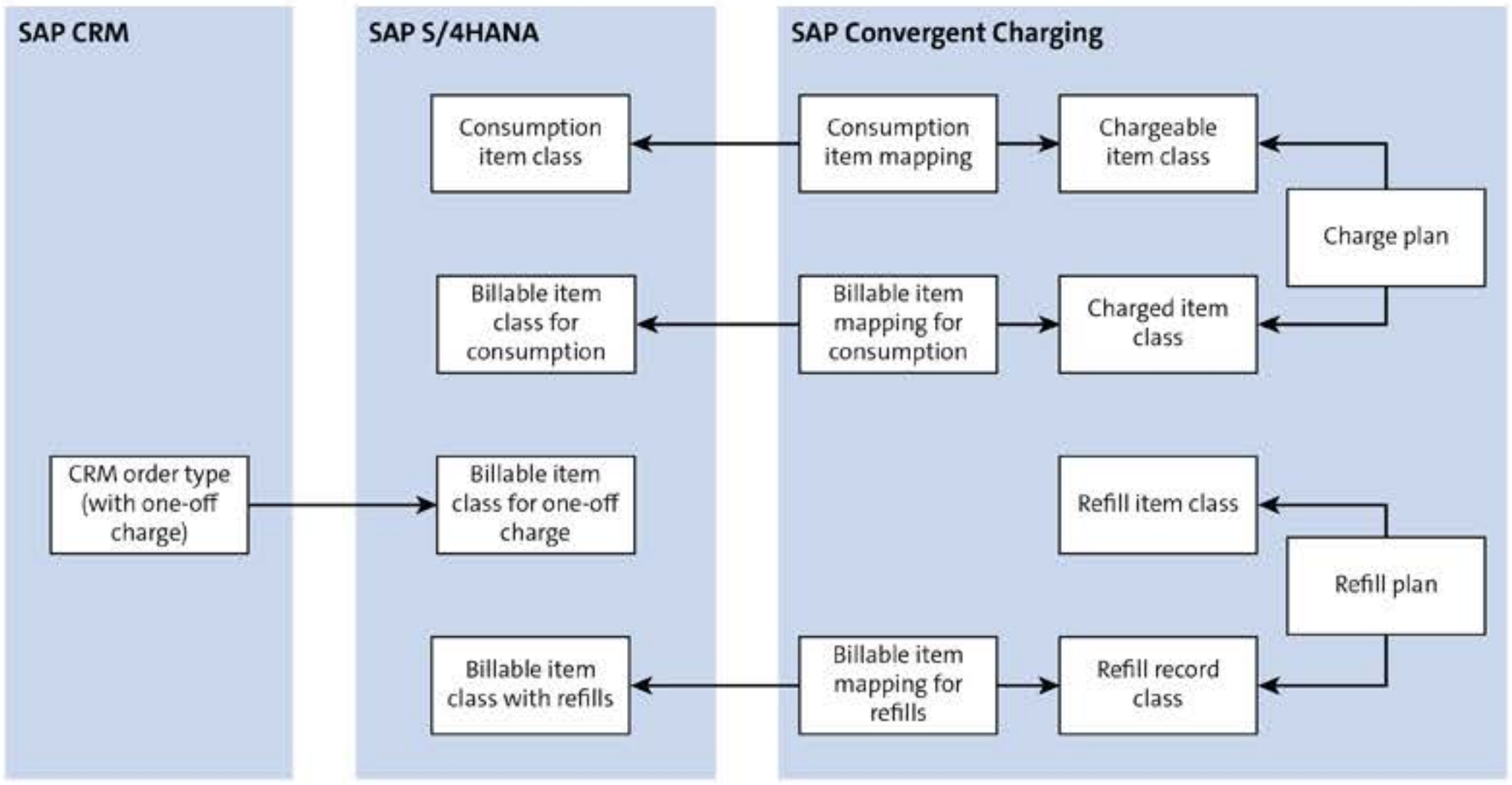

Figure 7.2 Transactional Data Flow from the Master Guide

7.1.1 Consumption Data

The original consumption data is stored in SAP Convergent Invoicing in the SAP S/4HANA system in the form of consumption items both before and after rating. The benefits of storing consumption data in SAP S/4HANA are as follows:

- You can work with scaled prices.
- Rerating is triggered in SAP Convergent Invoicing.
- You can use partner settlement.
- You can correct consumption items in SAP S/4HANA.
- You can determine the source transaction for a billable item.

Charging data, such as charged items, refill records, account-based marketing (ABM) notifications, and unrated chargeable items, is created from rated or unrated chargeable items or refill requests in SAP Convergent Charging. SAP Convergent Charging then reads and maps these files to the format of the associated billable item class or consumption item class in SAP S/4HANA. Then the SAP Convergent Charging bulkloader loads the data as billable items and consumption items into SAP S/4HANA.

SAP Convergent Charging receives unrated consumption items and enriches them with information related to provider contracts or subscriptions. SAP Convergent Charging then transfers the unrated consumption items to SAP Convergent Invoicing, which stores and manages the unrated consumption items. SAP Convergent Invoicing then sends them back to SAP Convergent Charging for rating.

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7.1.2 One-Off Charges

SAP Billing and Revenue Innovation Management, subscription order management, transfers the one-off charges to SAP S/4HANA automatically using standard, predefined order item type configuration. One-off charges are processed in SAP S/4HANA using a billable item class that is configured for one-off charges and using the CRM Order interface component.

7.1.3 Prepaid Data

Billing in SAP Convergent Invoicing receives prepaid data for refills as billable items from SAP Convergent Charging. It stores, manages, and groups the data together with the existing consumption of the customer. SAP Convergent Invoicing then summarizes the data into a billing document. One or more billing documents are merged into an invoice. SAP Convergent Invoicing is a module in the SAP S/4HANA system.

7.1.4 Billing and Invoicing

Billable items originate in SAP Convergent Charging as consumption and refill data. SAP Convergent Charging triggers periodic or event-based charges and creates billable items in the SAP S/4HANA system.

The billing component in SAP S/4HANA receives billable items from SAP Convergent Charging. The billable items are charged from consumption and refill data processed in SAP Convergent Charging and charges from SAP Billing and Revenue Innovation Management, subscription order management (one-off charges) and bill plans (recurring charges) in SAP S/4HANA. During invoicing, one or more billing documents can then be merged into one invoice.

Invoicing creates an invoice for the customer based on the billing document and on additional source documents from other systems if available. Invoicing posts this total amount to the contract account of the business partner as a receivable.

7.1.5 Contract Data

When configuring SAP Billing and Revenue Innovation Management, you work with the following types of contracts and agreements:

- **Subscription contract**
  This comprises all legally binding agreements regarding the provision and billing of services that are entered by a customer and a company for a specified period.
- **Master agreement**
  This is a contractual agreement between a service provider and a business customer.
- **Partner agreements**
  These are used to share revenue with a content provider or to pay royalties. With end

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customers, you have provider contracts; with partners, you have partner agreements.

- **Sharing contract**
  This specifies services, such as shared minutes or data on a phone plan. The contract applies to all individual contracts assigned to a sharing contract and can be used by all contract partners.

7.1.6 Disputes

A dispute is created from the bill stored in SAP Convergent Invoicing. It's replicated to contract accounting in a contract accounting dispute case, which contains information on the receivables related to the bill and the disputed object. A dispute can be created for bills, bill line items, or billing detail records. All data relating to and used for processing the dispute is stored in the SAP S/4HANA contract accounting system. Adjustment requests can be created from a bill or as a subitem of a dispute. Adjustments are posted directly in SAP S/4HANA contract accounting as credit items.

7.1.7 Communication between Systems

SAP S/4HANA communicates with SAP Convergent Charging via web services. These web services use SOAP and are configured in each respective system using Transaction SOMANAGER. SAP Convergent Charging communicates with SAP S/4HANA via an SAP Java connector communication.

SAP provides preconfigured application programming interfaces (APIs) that can be used to communicate with non-SAP or legacy platforms to manage overall business processes as required for the customer. Some of the more common integrations involve integrating with SAP Convergent Mediation by DigitalRoute; SAP SAP Configure, Price, and Quote (SAP CPQ); and SAP Revenue Accounting and Reporting.

Configuring these integrations is core to implementing the SAP Billing and Revenue Innovation Management solution.

7.2 SAP Convergent Mediation by DigitalRoute

As discussed in previous chapters, consumption data can be passed directly into SAP Billing and Revenue Innovation Management systems as consumption items. However, there are some scenarios in which a mediation layer is required to aggregate consumption data or to enrich consumption data with additional information. SAP Convergent Mediation by DigitalRoute is a separate platform (requires additional license, etc.) which helps to manage extract, transform, and load (ETL) jobs.

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SAP Convergent Mediation by DigitalRoute provides native integration with SAP Convergent Charging and SAP Convergent Invoicing. SAP Convergent Mediation by DigitalRoute is a platform that collects and processes any type of data from any type of system, with zero data loss. It transforms usage data to create accurate, chargeable items for business systems.

The SAP Convergent Mediation by DigitalRoute architecture consists of three zones:

- **Access zone**
  Access zone provides GUIs (known as SAP Convergent Mediation by DigitalRoute desktop) to view/update/configure SAP Convergent Mediation by DigitalRoute systems for users. In the recent versions, a web-based version is also available to manage SAP Convergent Mediation by DigitalRoute.
- **Control zone**
  Also known as platform, the control zone oversees the entire system. This also consists of a system database (e.g., Oracle, SAP HANA, Postgres, etc.) that holds configuration information.
- **Execution zone**
  Also known as execution context, this zone consists of one or more servers (based on the system configuration and infrastructure) to execute workflows. The execution zone can be distributed across multiple servers and locations.

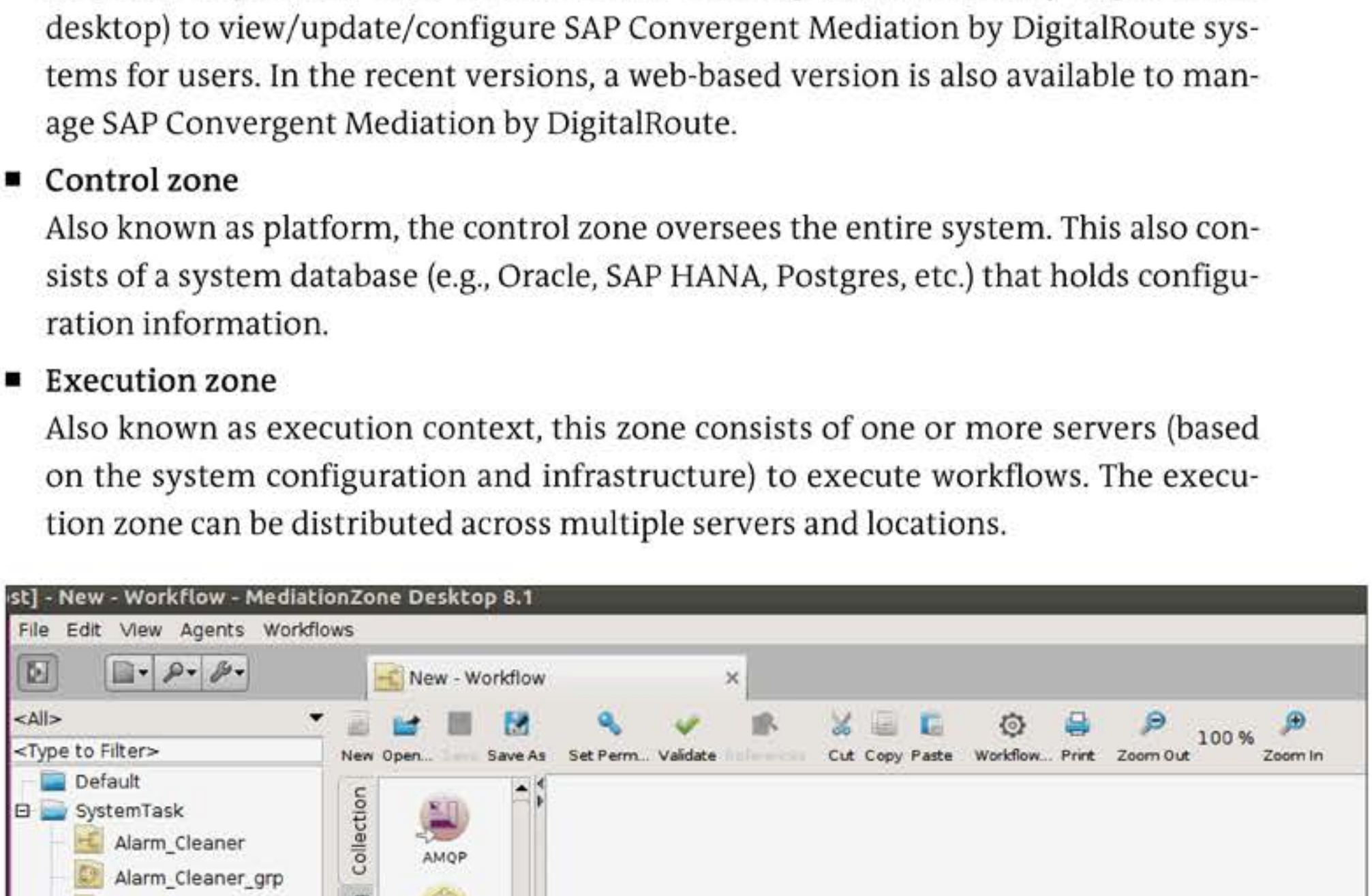

Figure 7.3 Desktop Launcher

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As mentioned earlier, SAP Convergent Mediation by DigitalRoute can be accessed using the desktop launcher provided as part of the installation file. You need to have a compatible Java version (Java 1.8) installed. Once installed, you can add the host name or IP with port number to the control zone to start accessing the system.

As part of this section, we'll highlight the native integration components available in SAP Convergent Mediation by DigitalRoute for SAP Convergent Charging and SAP Convergent Invoicing. This section, however, won't include in-depth information about installation and setup, such as platform and execution containers.

Mediation logic or rules for SAP Convergent Mediation by DigitalRoute are defined in the workflow object. Figure 7.3 illustrates a sample of how the desktop launcher is used for modeling workflows.

7.2.1 Integration with SAP Convergent Charging

SAP Convergent Mediation by DigitalRoute supports connectivity with any rating and charging APIs, but it has out-of-the-box integration with SAP Convergent Charging. In Figure 7.4, in the **Processing** tab, you can see two agents for SAP Convergent Charging: **SAP_CC_Batch** and **SAP_CC_Online**.

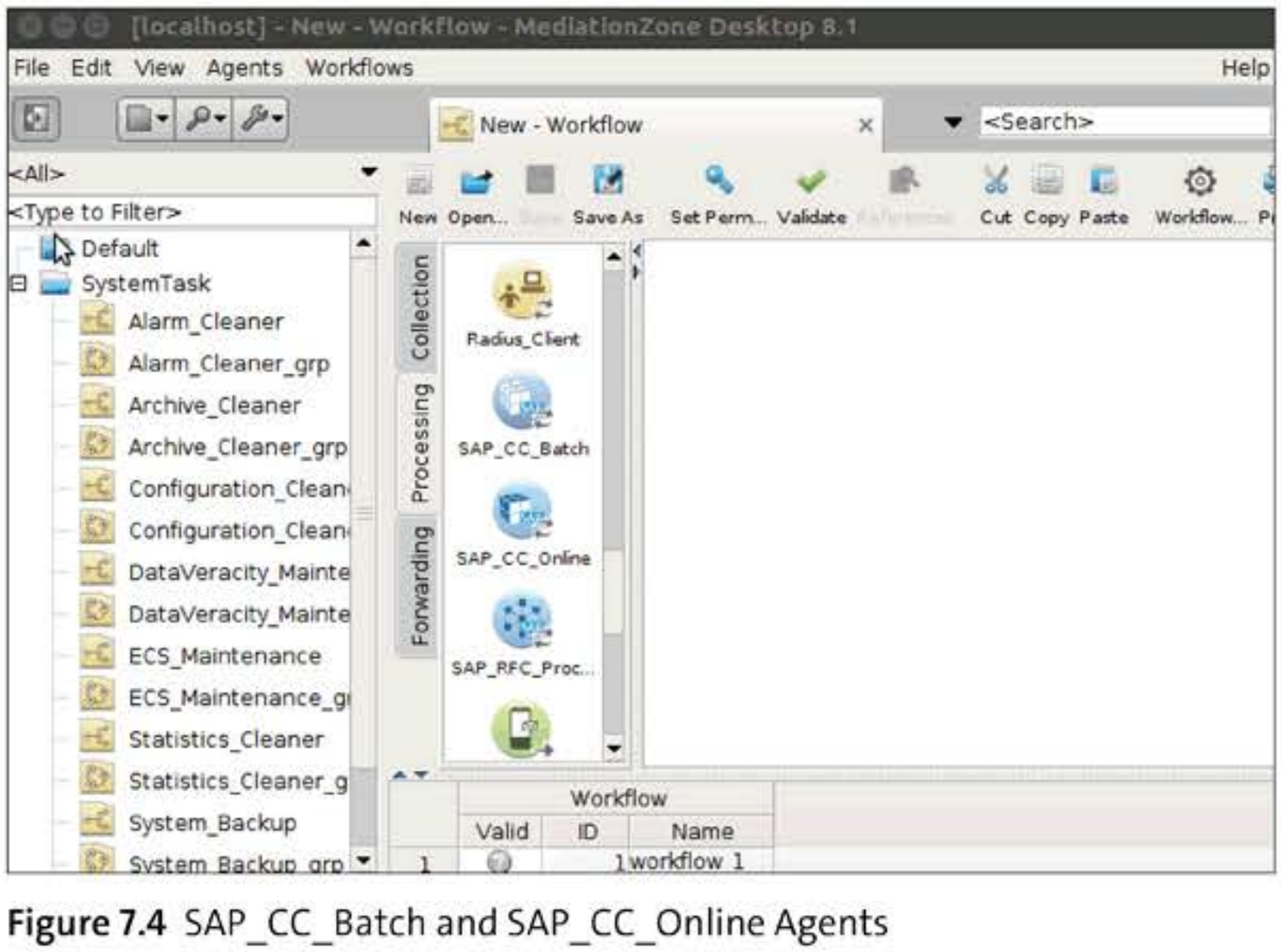

Figure 7.4 SAP_CC_Batch and SAP_CC_Online Agents

SAP Convergent Mediation by DigitalRoute has three types of agents: collection, processing, and forwarding. When integrated with SAP Convergent Charging, the collection agent is typically used to read files or incoming streams from a source location. SAP Convergent Mediation supports receiving data (also known as usage data records [UDR]) from multiple source types, such as File Transfer Protocol (FTP), direct disk reads,

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or message queues. In Figure 7.5, a disk agent is being used as the collection agent. In the collection agent, there are fields to specify the directory and filename to process.

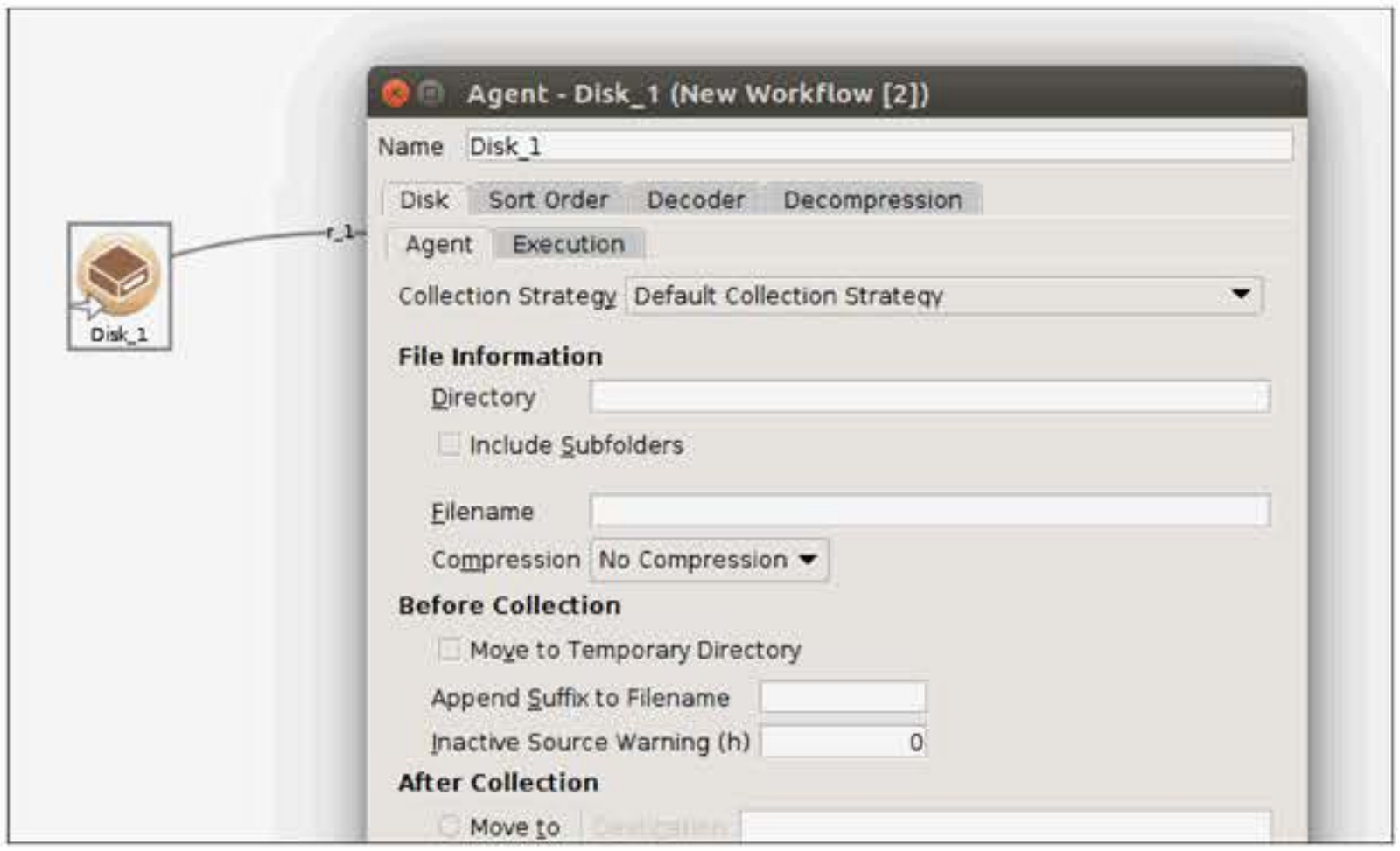

Figure 7.5 Disk Agent

To pass the correct data to the processing agent, the UDR format needs to be defined within the collection agent. SAP Convergent Mediation by DigitalRoute provides a pre-defined UDR format for communication with SAP Convergent Charging. This setting can be accessed from the **Decoder** tab of the collection agent (Figure 7.6).

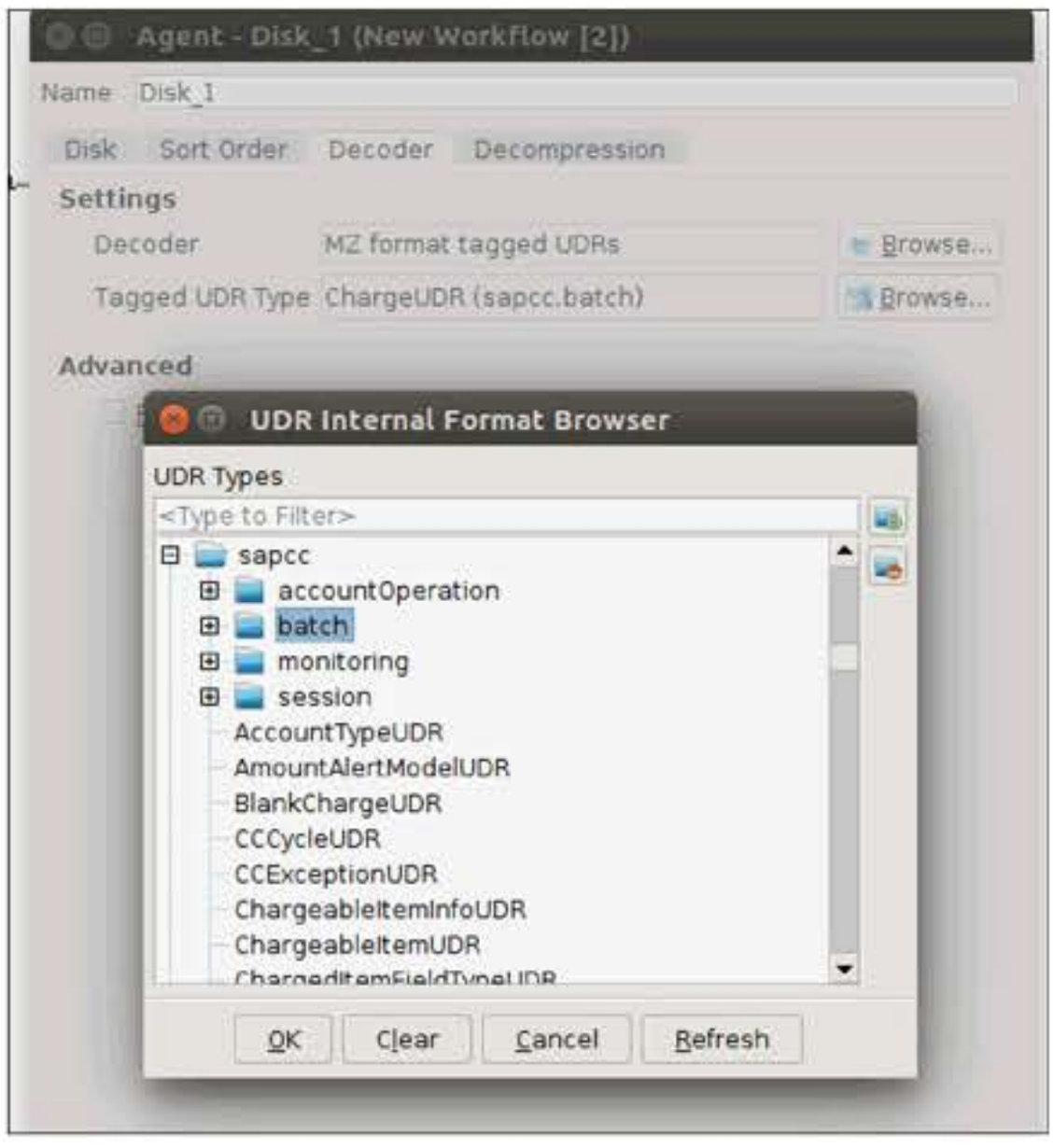

Figure 7.6 Decoder

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Next, let's consider the processing agent. SAP_CC_Online is an example of a processing agent via which a rating can be triggered within SAP Convergent Charging if it's correctly configured within the agent (Figure 7.7). The dispatcher of the SAP Convergent Charging component needs to be added to the host name/port.

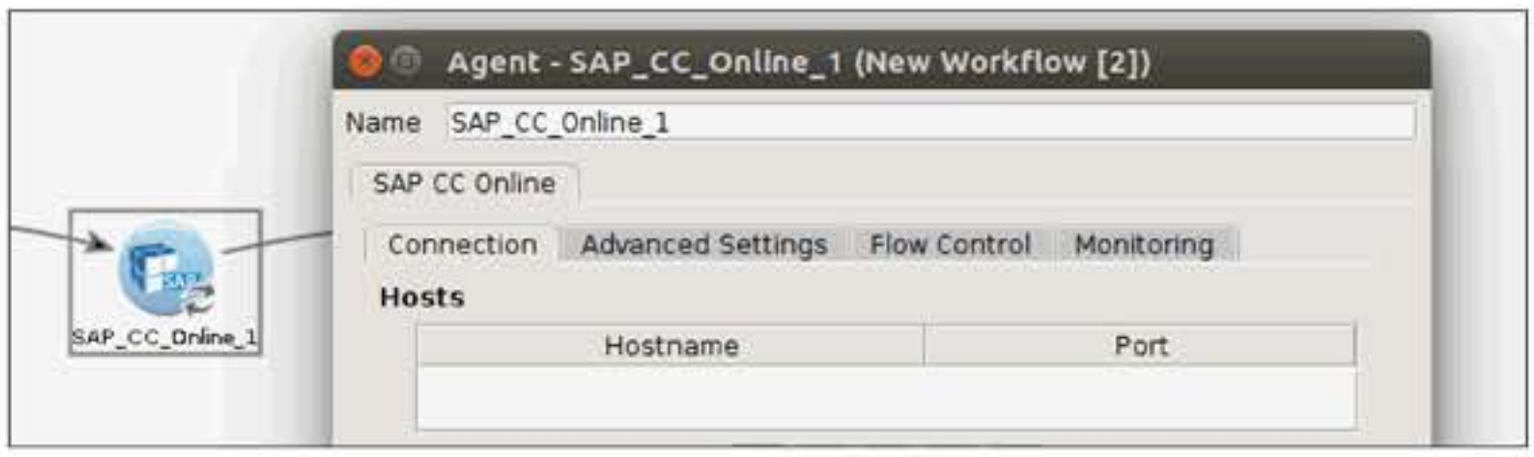

Figure 7.7 SAP_CC_Online Agent

7.2.2 Integration with SAP Convergent Invoicing

In addition to SAP Convergent Charging, SAP Convergent Mediation by DigitalRoute can also directly connect to SAP Convergent Invoicing. This scenario is common in implementations in which other rating and charging systems are being used or an external rating engine integrates directly with SAP Convergent Invoicing via SAP Convergent Mediation by DigitalRoute (Figure 7.8). SAP Convergent Mediation by DigitalRoute can be used to read output from the rating system, enrich rating results, and create billable items directly in SAP Convergent Invoicing. The remote function call (RFC) processor can be used in the workflow after setting up the RFC profile (Figure 7.9).

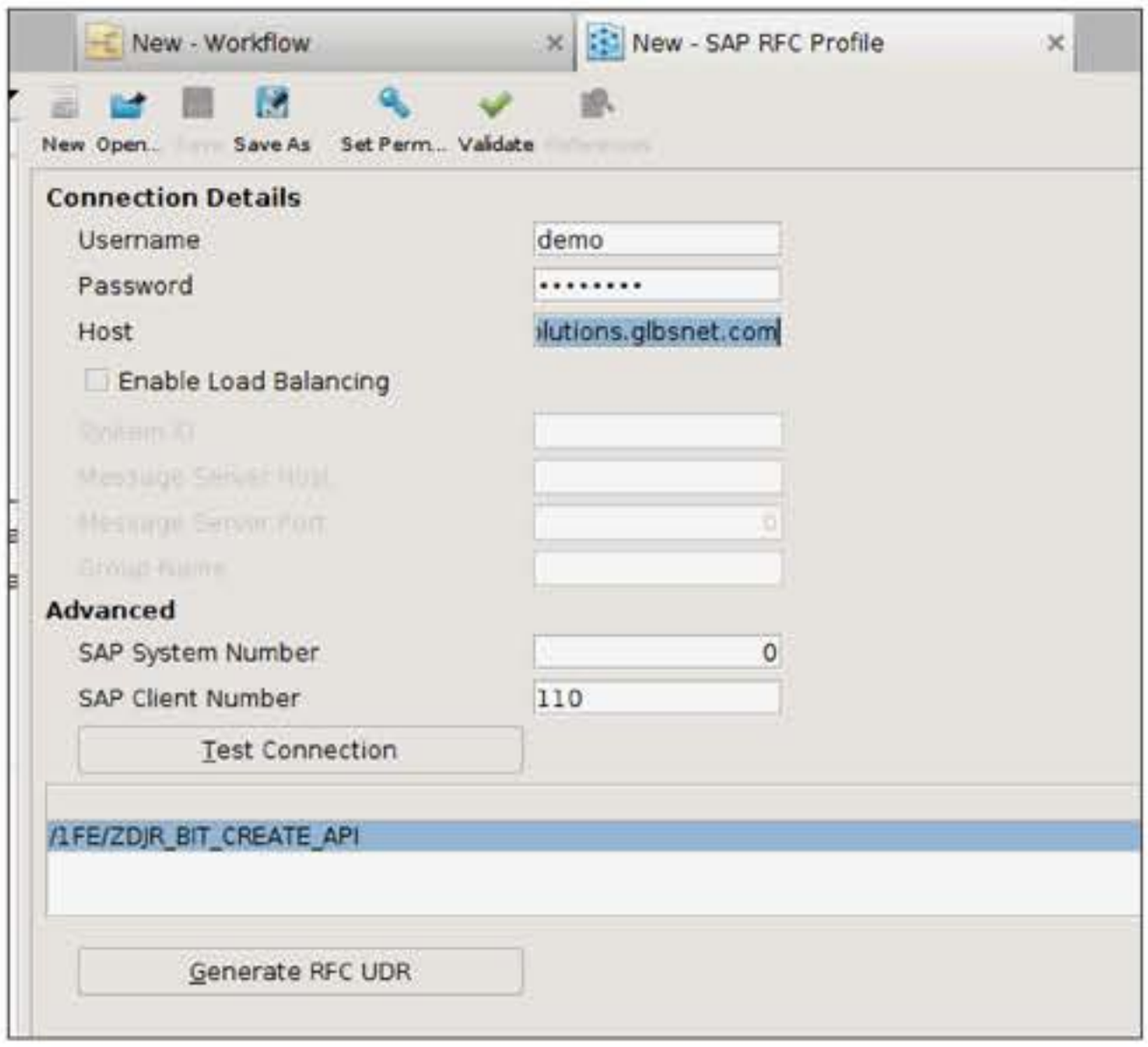

Figure 7.8 New SAP RFC Profile

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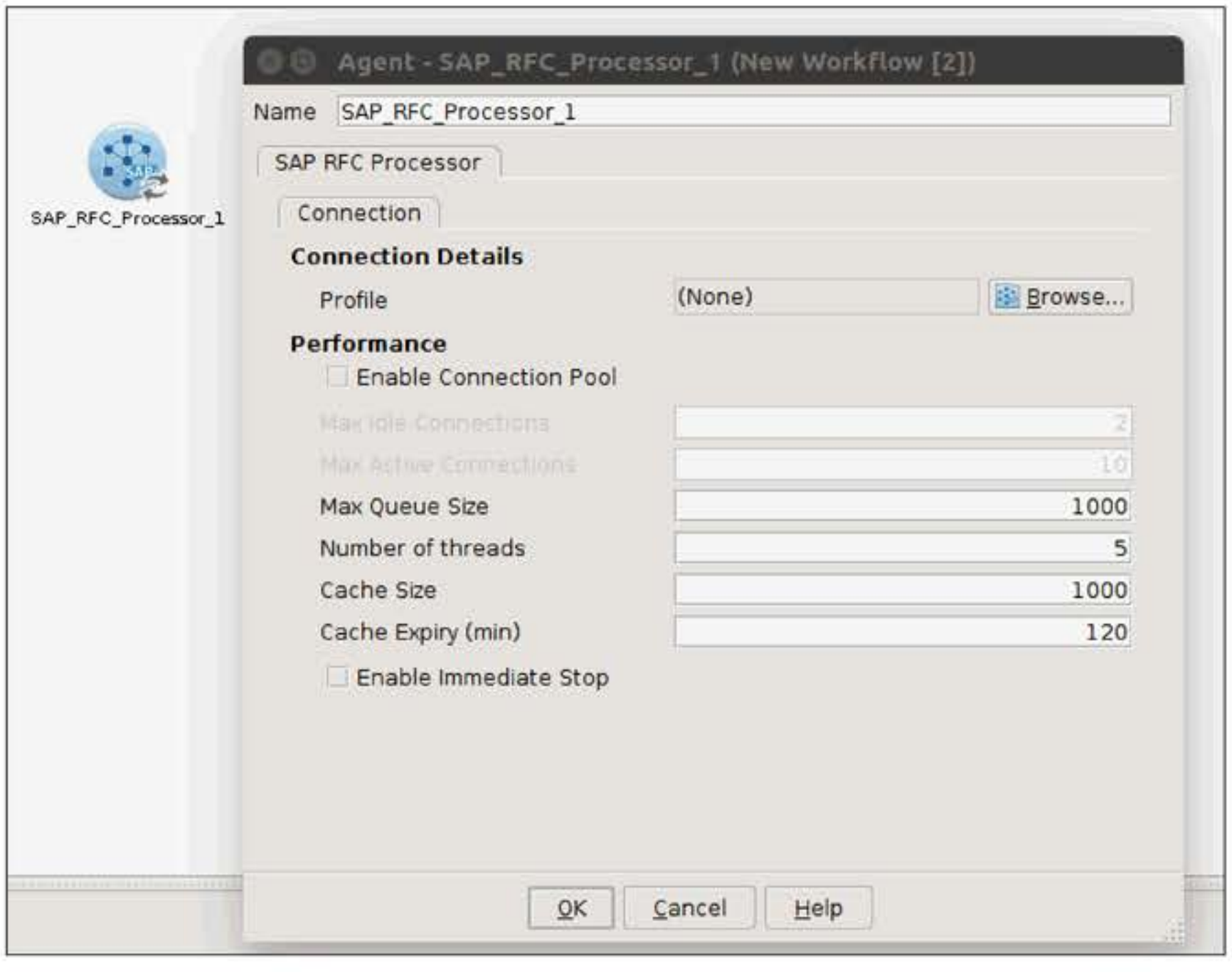

Figure 7.9 SAP RFC Processor

As mentioned earlier, SAP Convergent Mediation by DigitalRoute has many other features that are out of scope for this chapter. Documents for other advanced capabilities, such as aggregation or data enrichment of UDRs, are available on the DigitalRoute wiki website.

7.3 SAP Billing and Revenue Innovation Management and SAP Customer Experience Platform

The current SAP Customer Experience is the fourth generation of the suite that can help shape lead-to-cash processes with SAP Billing and Revenue Innovation Management. The lead-to-cash process is an agile, integrated business process, combining multiple SAP solutions, enriched with artificial intelligence and experience management scenarios. It covers the entire business process, from initial contact with a prospective customer, to order fulfillment and service delivery. Five subprocesses are used to structure the overall process: contact-to-lead, lead-to-opportunity, opportunity-to-quote, quote-to-order, and order-to-cash. With these subprocesses, the lead-to-cash process expands across the entire SAP Customer Experience solution portfolio and SAP S/4HANA, along with SAP Billing and Revenue Innovation Management.

The SAP Customer Experience portfolio offers various solutions:

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- **SAP Marketing Cloud**
  Used for campaign automation, segmentation, and marketing-related activities.
- **SAP Commerce Cloud**
  Helps to improve the customer's shopping experiences, with configurable functionality, through seamless, personalized, and integrated web shops.
- **SAP Customer Data Cloud**
  Manages customer identity, profile management, and customer consent management.
- **SAP Sales Cloud**
  Offers sales and quoting automation.
- **SAP Service Cloud**
  Offers customer service functionality, service ticket management, and interaction management.

SAP also uses the functions of the Qualtrics XM platform for improved predictive intelligence to gain customer insights through presale to post-sale customer experiences.

SAP Customer Experience primarily manages the demand chain portion of the process, which is contact-to-lead through quote-to-order. SAP S/4HANA with SAP Billing and Revenue Innovation Management manages the service, supply, and billing processes, which is order-to-cash.

Because lead-to-cash comprises multiple subprocesses, let's go through each of the submodules to envision a journey and see which modules are integrated to support the process:

- **Contact-to-lead stage**
  In this phase, the marketing team creates a campaign by targeting existing customers through email or targeting new contacts through a website. One of the goals of the campaign is to achieve the customer's consent to share his information and interact through promotional offers. If the customer gives consent, all interactions are organized and scored. Once a contact reaches a certain score threshold, the contact qualifies as a lead for the sales team.
- **Lead-to-opportunity stage**
  In this stage, the sales team can determine if the lead should be dropped or pursued. Analytics tools can allow sales to directly access the contact and request more information. If the lead has potential, sales will convert the lead into an opportunity.
- **Opportunity-to-quote stage**
  In this stage, sales determines if this opportunity is ready to receive a quote. Note that there are two distinct sales channels—direct sales and e-commerce. With a direct sales channel, a sales rep would perform some of the same activities from the previous stage, such as contacting the customer and analyzing existing data. With an e-commerce or omni-channel scenario, a contact could use a web shop to create

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a quote. The e-commerce option expedites the activities by creating a quote online, thus skipping the lead and opportunity stages. Through the quote, the sales team organizes and delivers the best selection of products, services, subscriptions, and discounts.

- **Quote-to-order stage**
  Once the quote is submitted, it may or may not require approval or acceptance from the sales team based on predefined criteria. If a review request is received, the quote can be altered until an agreement is reached. Once the customer confirms the acceptance of the quote, the order is generated.
- **Order-to-cash stage**
  With the generated order, the order-to-cash stage begins. In this stage, the order is divided into various parts for fulfillment and provisioning. In this stage, physical products are delivered, and one-time and recurring services are planned and dispatched. As order fulfillment is processed, the customer can view the order status and receive notifications about the order. Customer invoicing and subscription billing are offered with the final steps of this stage.

SAP Billing and Revenue Innovation Management and SAP Customer Experience offer various APIs that can be used to enhance and integrate system components. Usually, SAP Billing and Revenue Innovation Management, subscription order management is managed as an order orchestration module, which can distribute different types of products and services for fulfillment based on their types (e.g., hardware to sales and distribution, services for cloud, and recurring and usage products to SAP Convergent Charging, etc.). Though we won't cover individual APIs as part of this section, the following provides a brief understanding of how SAP Billing and Revenue Innovation Management can be integrated with the SAP Commerce platform

SAP Billing and Revenue Innovation Management, subscription order management, provides user interfaces (UIs) in the form of an SAP WebClient UI to manage SAP Billing and Revenue Innovation Management objects such as business partners, contract accounts, provider orders, and contracts. However, for most implementations, companies (especially business-to-consumer [B2C] companies) implementing SAP Billing and Revenue Innovation Management also have e-commerce systems.

SAP has provided several APIs in the form of RFCs to integrate with the SAP Customer Experience commerce systems (Figure 7.10).

When integrating with a commerce system, there are two types of approaches: asynchronous and synchronous. An asynchronous scenario is usually applicable in a B2C scenario, in which transaction volumes are high and real-time pricing isn't required. In this approach, the e-commerce system will pass in the orders to SAP Billing and Revenue Innovation Management, subscription order management, using a message queue service, such as IDocs or any message bus system.

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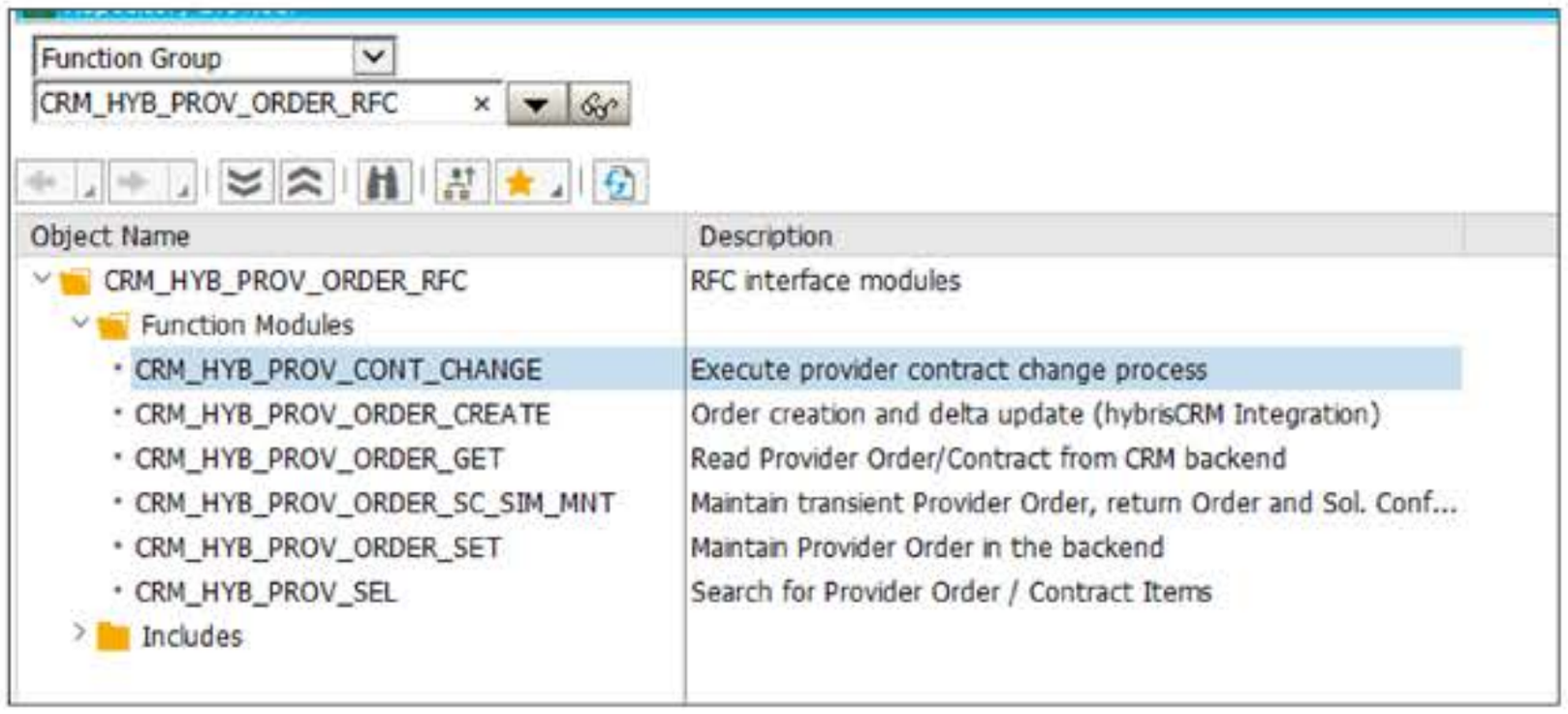

Figure 7.10 APIs for Integration with SAP Customer Experience Commerce Systems

Synchronous scenarios are usually applicable for business-to-business (B2B) scenarios in which transaction volumes are low and real-time pricing information is required. In such a scenario, the RFC discussed earlier can be called directly or wrapped in OData or web services.

7.4 SAP Revenue Accounting and Reporting

The SAP Revenue Accounting and Reporting add-on can be integrated with the offer-to-cash SAP Billing and Revenue Innovation Management integration. SAP Revenue Accounting and Reporting provides functions to manage revenue recognition regulations. The add-on is used for managing regulations in countries subject to both US Generally Accepted Accounting Principles (GAAP) and International Financial Reporting Standards (IFRS).

This section provides key considerations for the integration of SAP Revenue Accounting and Reporting and SAP Billing and Revenue Innovation Management. We'll discuss the nuances of revenue recognition from a process perspective. The other chapters discussed earlier focused on describing the episode-related key components and linking them back to the configuration setup needed for the Martex case study.

7.4.1 Optimized Contract Management

SAP Revenue Accounting and Reporting offers a new inbound processing version to enable integration of operational documents called optimized contract management (OCM). OCM and classic contract management (CCM) have key functionalities that are different, including the following: SAP Fiori apps and analytical apps, elimination of steps to process operational documents, and new tables to store revenue accounting items (RAIs). Inbound processing for SAP Revenue Accounting and Reporting to use OCM must be activated via the Business Add-In (BAdI) Determination of Inbound Processing Version.

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The integration from SAP Revenue Accounting and Reporting to SAP Billing and Revenue Innovation Management remains similar. Processing RAIs from SAP Billing and Revenue Innovation Management to SAP Revenue Accounting and Reporting includes only one step with OCM via Transaction FP_RAI_MON (Revenue Accounting Transfer) and eliminates the step to process additionally in Transaction FARR_RAI_MON.

Additional revenue accounting tables have been introduced in OCM as described next. Generally, to facilitate faster processing and the ability to partially process RAIs and monitor "postponed" RAIs. Table 7.1, Table 7.2, and Table 7.3 describe the database tables for processed RAIs, partially processed RAIs, and postponed RAIs, respectively. Table 7.4 describes the database table where profitability segment data is stored.

Database Table	Description
FARR_D_ORD_MI	Processed Order RAIs - Main Items
FARR_D_ORD_CO	Processed Order RAIs – Condition Items
FARR_D_FLFMT_MI	Processed Fulfillment RAIs- Main Items
FARR_D_INV_MI	Processed Invoice RAIs – Main Items
FARR_D_INV_CO	Process Invoice RAIs – Condition Items
FARR_D_COST_MI	Processed Cost RAIs – Main Items
FARR_D_COST_CO	Processed Cost RAIs – Condition Items

Table 7.1 Database Tables for Processed RAIs

Database Table	Description
FARR_D_ITEM_PROC	Contains RAI and accounting principle for successful processing

Table 7.2 Database Table for Partially Processed RAIs

Database Table	Description
FARR_D_INB_MI	Postponed RAIs – Main items table
FARR_D_INB_CO	Postponed RAIs – Condition items table

Table 7.3 Database Tables for Postponed RAIs

Database Tables	Description
FARR_D_INB_COPA	Profitability segment data of postponed and partially processed RAIs

Table 7.4 Database Table for Profitability Segment Data

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Functionality in OCM and CCM differ in regards to data modeling, account determination, allocation, fulfillments change processes, and migration. These differences won't be covered in the scope of this chapter, but we'll focus on the changes in integration with SAP Billing and Revenue Innovation Management and key functionalities related to subscription type models.

7.4.2 Integration Architecture

SAP S/4HANA offers standard compatibility between SAP Billing and Revenue Innovation Management and SAP Revenue Accounting and Reporting. The architecture provides direct integration to the sales and distribution system, SAP Convergent Invoicing billable items, and SAP Convergent Invoicing provider contracts. In addition—new in SAP S/4HANA 2021—SAP Convergent Invoice billing plans are enabled to integrate to SAP Revenue Accounting and Reporting. For the purposes of SAP Billing and Revenue Innovation Management integration, the focus of this section will be on SAP Convergent Invoicing and SAP Revenue Accounting and Reporting integration.

SAP Revenue Accounting and Reporting is driven by three main data types of RAIs: order items (CAO1), fulfillment items (CAO2), and invoice items (CAO3). SAP Convergent Invoicing supplies SAP Revenue Accounting and Reporting with these RAIs in the SAP Billing and Revenue Innovation Management framework. Initial setup for settings needs to be completed for activating the standard revenue item classes and interfaces for CAO1, CAO2, and CAO3. These general activations, assigning logic systems, BRFplus application maintenance, and other general SAP Revenue Accounting and Reporting–specific configurations, won't be covered in detail here to keep our focus on the SAP Billing and Revenue Innovation Management and SAP Revenue Accounting and Reporting integration.

Figure 7.11 shows the business function that must be activated to enable SAP Revenue Accounting and Reporting integration with SAP Convergent Invoicing.

FICA_EHP7_RA	Integration with Revenue Accounting	Business func. will remain activated		S4CORE	618	FI-CA
FICA_EHP7_RA2	Integration of Revenue Accounting 2 (Reversible)	Business func. will remain deactivate		S4CORE	801	FI-CA

Figure 7.11 Business Functions for SAP Revenue Accounting and Reporting and SAP Convergent Invoicing Integration

Once these business functions are activated, the configuration nodes shown in Figure 7.12 for SAP Convergent Invoicing integration with SAP Revenue Accounting and Reporting will be available under **Financial Accounting • Contract Accounts Receivable and Payable • Integration • Revenue Accounting**.

Once the configuration nodes are available, as a starting step, the interface components relevant to SAP Revenue Accounting and Reporting based on the business use case will need to be activated.

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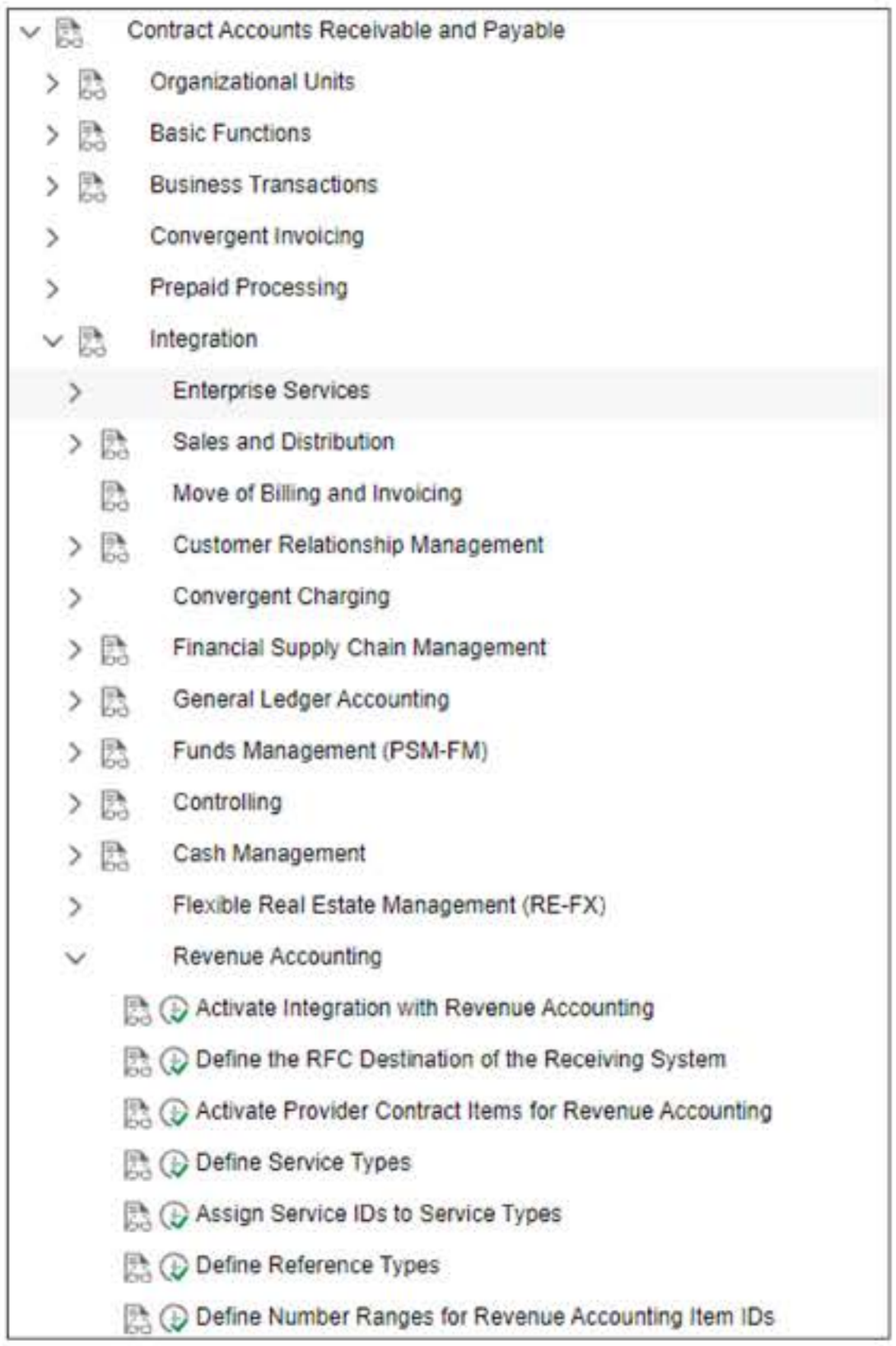

Figure 7.12 Configuration Nodes for SAP Revenue Accounting and Reporting Integration with SAP Convergent Invoicing

7.4.3 Five-Step Model

Accounting standard ASC606 defines a five-step model to recognize revenue for multielement arrangements, which led to the development of SAP Revenue Accounting and Reporting. This section will break down the ASC606 five-step model with relevant integration with SAP Revenue Accounting and Reporting steps, key configurations, and examples from the Martex case study.

The ASC606 revenue recognition five-step model is as follows:

1. Identify a contract with a customer.
2. Identify the separate performance obligations in the contract.
3. Determine the transaction price.
4. Allocate the transaction price to the performance obligations of the contract.
5. Recognize revenue when (or as) the entity satisfies a performance obligation.

Let's look at each step in more detail.

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Step 1: Identify a Contract with a Customer

In the SAP Billing and Revenue Innovation Management integration with SAP Revenue Accounting and Reporting, a provider contract or multiple provider contracts can be combined to create a single holistic SAP Revenue Accounting and Reporting contract for a business partner. In the Martex case study, there is a one-to-one relationship between the provider contract and SAP Revenue Accounting and Reporting contract. Once the SAP Revenue Accounting and Reporting–relevant integration business functions and interface components are activated, nominating provider contracts and service types for SAP Revenue Accounting and Reporting processing can be configured.

The RFC destination and activating provider contract items for SAP Revenue Accounting and Reporting relevance are configured with selections and indicators (Figure 7.13 and Figure 7.14).

Figure 7.13 RFC Destination for Receiving System

Figure 7.14 Determine Provider Contract Relevance for SAP Revenue Accounting and Reporting

These configurations will make available a separate SAP Revenue Accounting and Reporting–specific tab in the provider contract. This enrichment will include line items with service types that will be described in step 2 of the five-step model. Transaction price and standalone selling price (SSP) details will also be available to receive relevant details from upstream applications. The various attributes in this configuration (sales organization, distribution channel, division, sales area 1/2, charge type, service type) will determine if a provider contract line item is relevant to be transferred to SAP Revenue Accounting and Reporting. Additionally with SAP S/4HANA 2021, the origin type attribute is available to determine if the data from the SAP Convergent Invoicing provider contract or billing plan line items will be transferred to SAP Revenue Accounting and Reporting for the creation of RAI order items. SAP Convergent Invoicing billing plan line items can be used for both recurring and one-off charges. In addition, this new integration allows for the passing of pricing conditions to SAP Revenue Accounting

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and Reporting to have granularity between gross, discount, and other relevant pricing for enhanced reporting and accounting views.

Step 2: Identify the Separate Performance Obligations in the Contract

The performance obligation (POB) data object in SAP Revenue Accounting and Reporting holds attributes in relation to revenue recognition, including but not limited to general data, revenue recognition fulfillment attributes, allocation data, status, and account assignments. POB attributes are defined by a combination of the operational document (provider contract) configuration and BRFplus decision tables (Figure 7.15).

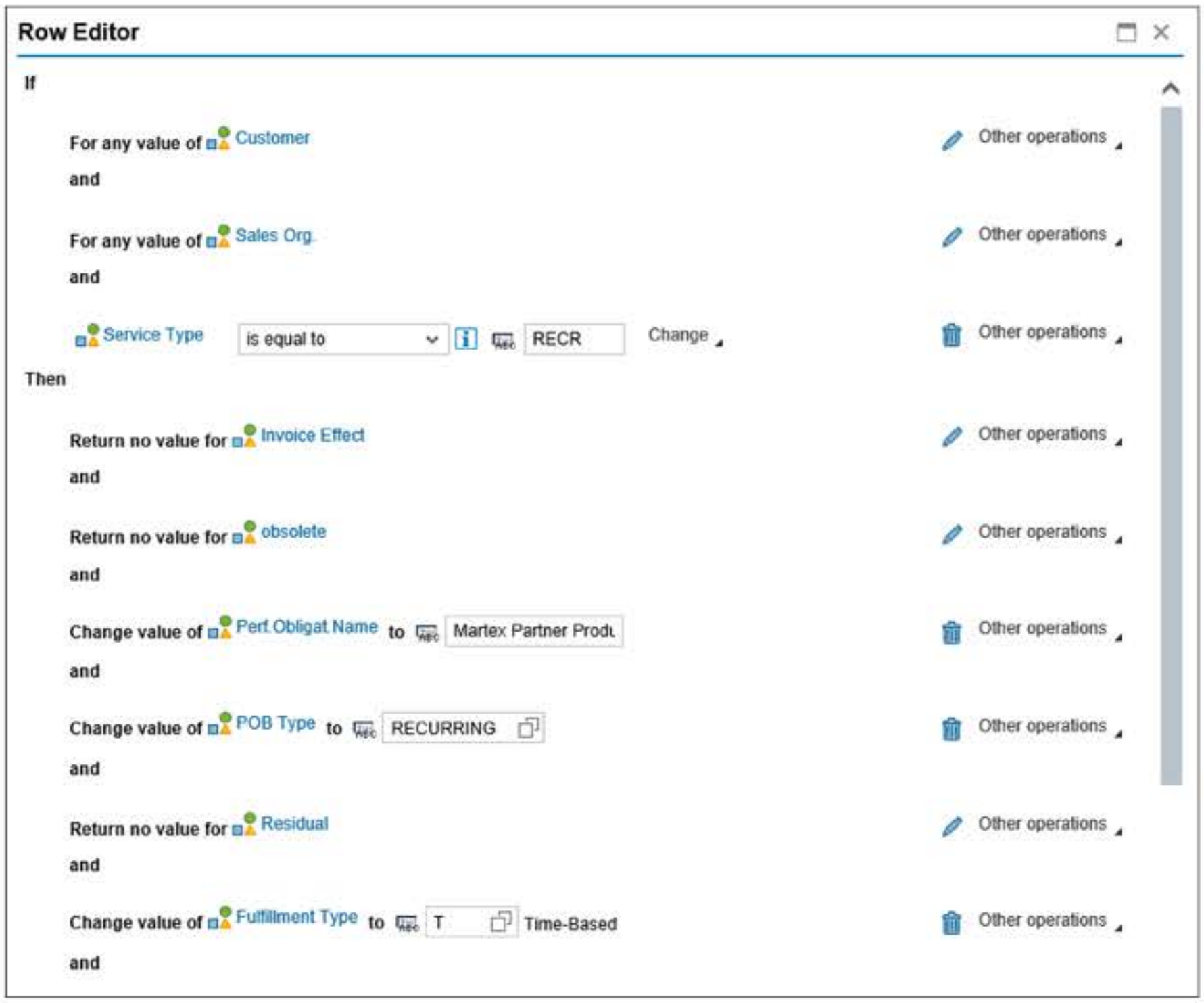

Figure 7.15 Example of Decision Table Entry for Performance Obligation Determination via BRFplus

For Martex, the POB is related to line items in the Martex usage and/or Martex partner product provider contracts. The definition of service types drives the creation of SAP Revenue Accounting and Reporting order items, fulfillment items, invoice items, and one-off charges. Via configuration, service types are mapped to SAP Billing and Revenue Innovation Management, subscription order management, service IDs, save for the recurring charge service type. After a service type is indicated as an order item, a provider contract creation with a recurring charge (Figure 7.16) and/or a usage service type will generate an order item RAI to subsequently create an SAP Revenue Accounting and

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Reporting contract and relevant POBs. POB determination configuration in SAP Revenue Accounting and Reporting–specific BRFplus tables needs to be appropriately maintained, along with relevant account determination and contract categories.

Service Types for Revenue Accounting Items

Service Type	Service Type Text	Order Item	One-Time Charge	Fulfil. Item	Invoice Item
RECR	Martex Partner Product	☑	☐	☐	☑
ZUSG	Martex Usage	☑	☐	☑	☑

Figure 7.16 Define Service Types

The integration of SAP Convergent Invoicing billable items and provider contracts to create order RAIs sets the table for SAP Billing and Revenue Innovation Management transactions to supply SAP Revenue Accounting and Reporting with fulfillment items and invoice items. The decoupling of the ordering and billing systems supports transactional integrity, auditable end-to-end document referencing, automated adherence to the timing of revenue, and accuracy of balance sheet postings.

Step 3: Determine the Transaction Price

The transaction price in an SAP Revenue Accounting and Reporting contract is relevant for the amount to be invoiced and fulfilled for the duration of an agreement with a customer. As a part of the business functions shown earlier in Figure 7.11, SAP delivers extensions to the provider contract replication from SAP CRM to SAP ERP. The enhanced structure includes a selection of the relevant fields described in Table 7.5.

Field	Contents	Example
REVACC_REFID	Revenue accounting reference ID	Default provider order number
REVACC_REFTYPE	Revenue accounting reference type	Standard value PO
TRANS_PRICE	Revenue accounting transactional price	100.00
TP_CURRENCY	Transactional price currency	USD
TP_REC_DURATION	Duration of a recurrence period	1
TP_REC_UNIT	Time unit of a recurrence period	Month
STANDALONE_PRICE	Revenue accounting SSP	85.00
SP_CURRENCY	SSP currency	USD
SP_REC_DURATION	Duration of a recurrence period	1
SP_REC_UNIT	Time unit of a recurrence period	Month
VALIDTO_RT	Valid to time stamp	End of duration of contract: 06/30/2021

Table 7.5 Provider Contract Revenue Accounting Key Fields

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SAP Revenue Accounting and Reporting calculates the total transaction price for a recurring charge as follows:

Revenue accounting transactional price ÷ Days in recurrence period × Days in valid from date to end of duration

In the case of Martex, the revenue accounting transactional price is 100.00 USD (as noted in Table 7.6). The number of days in the recurrence period is set to 30 as the recurring charge is once a month; although days in a month can vary, SAP uses 30 to derive a per-day transactional price. The valid from date is 07/01/2021. An end of duration date will be provided in a provider contract to calculate the SAP Revenue Accounting and Reporting transaction price, although this agreement may not have a specified end date. For Martex, the end of duration is 06/30/2020, for a total of 365 days between the valid from and end of duration dates. The total transaction price of the POB is calculated as $1,216.67 (see Table 7.7). Similarly, the SSP can be sent as an optional field from the provider contract replication and would derive a total SSP for the POB of $1,034.17. The SSP can also be determined via BRFplus decision tables.

SAP Revenue Accounting and Reporting Contract					
Performance Obligation	Performance Obligation Type	Contractual Price	SSP	Frequency	Quantity
Martex partner product subscription	Subscription	$100.00	$85.00	Per month	1
Martex usage	Usage	$10.00	$10.00	Per unit	200

Table 7.6 Martex SAP Revenue Accounting and Reporting Contract 1

SAP Revenue Accounting and Reporting Contract					
Performance Obligation	Start Date	End Date	Fulfillment Type	TP Total	SSP Total
Martex partner product subscription	7/1/2021	6/30/2020	Time-based	$1,216.67	$1,034.17
Martex usage			Event-based consumption	$2,000.00	$2,000.00

Table 7.7 Martex SAP Revenue Accounting and Reporting Contract 2

Default condition types for SSP and transaction price must be defined, as well as a default service type for recurring charges. The default service type is defined for SAP Revenue Accounting and Reporting because service IDs from SAP Billing and Revenue

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Innovation Management, subscription order management, aren't maintained for recurring charges (see Figure 7.17).

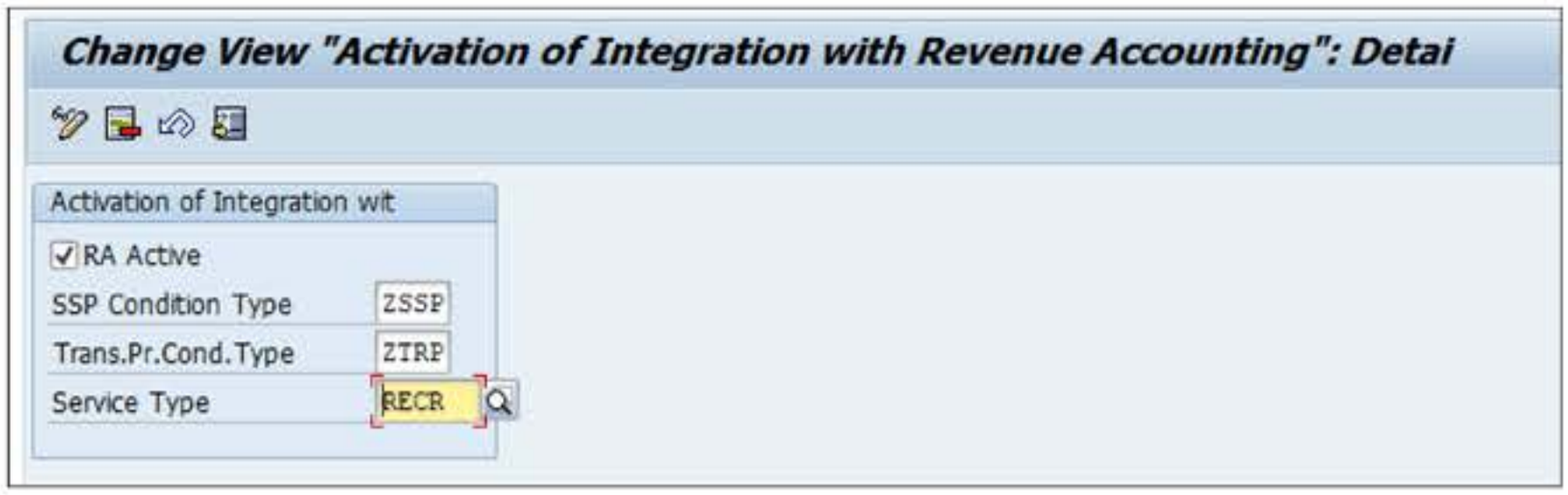

Figure 7.17 Activation of Integration with SAP Revenue Accounting and Reporting

At the time of provider order creation, service type ZUSG for usage/consumption-based charges won't be available for POB creation; at this time, there is no quantity or transaction price for these items. It may be deemed valuable to estimate consumption at the usage-based POB inception. For this, BRFplus decision tables are available to provide SAP Revenue Accounting and Reporting estimated usage quantity, transactional price, and SSP per unit. Such estimates are based on quantitative evaluations that are calculated outside the framework of SAP Billing and Revenue Innovation Management and SAP Revenue Accounting and Reporting. An **Estimate Quantity** indicator in the order item RAI (usage-based) must be set to use BRFplus for estimating consumption. For Martex, a usage estimation has been determined for this contract of 200 units at a $10 transaction and SSP per unit.

This SAP Revenue Accounting and Reporting contract now has two POBs, one recurring and one usage-based with transaction price and SSPs maintained. We'll see how allocation, fulfillment, and invoicing affect the revenue recognition in the next steps.

Step 4: Allocate the Transaction Price to the Performance Obligations of the Contract

Allocating the transaction price is done based on the SSP of the POBs in the contract. The overall objective is to obtain an allocation of the amount of consideration that the organization considers to be entitled to pass the good or service promised. The calculation of the allocated transaction price is handled by SAP Revenue Accounting and Reporting and is calculated as follows:

(Total SSP of the POB ÷ Total SSP of the SAP Revenue Accounting and Reporting contract) × (Total transaction price of the contract) = Allocated transaction price of the POB

For Martex, the allocated amounts are $1,096.37 and $2,120.30 for the two POBs, as noted in Table 7.8.

SAP Revenue Accounting and Reporting automatically calculates the allocation of POBs at SAP Revenue Accounting and Reporting contract creation. In the next step, you'll see how the fulfillments and invoices from SAP Billing and Revenue Innovation

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Management allow SAP Revenue Accounting and Reporting to calculate revenue, contract asset/unbilled receivables and contract liabilities, and deferred revenue.

Revenue Allocation			
Performance Obligation	Transaction Price Total	SSP Total	Allocation to Performance Obligations
Martex partner product subscription	$1,216.67	$1,034.17	$1,096.37
Martex usage	$2,000.00	$2,000.00	$2,120.30
Total contract value	**$3,216.67**	**$3,034.17**	**$3,216.67**

Table 7.8 Revenue Allocation

Step 5: Recognize Revenue When (or as) the Entity Satisfies a Performance Obligation

In the final step of the five-step model, SAP Revenue Accounting and Reporting will receive fulfillments and invoice data from SAP Billing and Revenue Innovation Management to subsequently adjust the general ledger based on calculations for revenue and balance sheet items. In SAP Revenue Accounting and Reporting, a *fulfillment* is defined as the point in time to recognize revenue and any related cost of goods sold. In this step, we'll expand on consumption fulfillments, time-based fulfillments, and invoice transfers from SAP Convergent Invoicing. Briefly, we'll touch on general ledger account determination and SAP Revenue Accounting and Reporting subledger general ledger transfer.

SAP Revenue Accounting and Reporting defines and provides various standard fulfillment methods for revenue recognition, including but not limited to event-based customer invoice, shipping, or consumption; percentage of completion; and time-based with various deferral methods. In the SAP Billing and Revenue Innovation Management integration with SAP Revenue Accounting and Reporting, the fulfillment methods generally used are time-based, event-consumption, and event-customer invoice.

Referring to service types, an indicator with a fulfillment item is set for billable items related to usage in configuration, as shown in Figure 7.18. This configuration enables billable items to be evaluated and transferred periodically to be processed by SAP Revenue Accounting and Reporting to fulfill usage-based performance.

Service Types for Revenue Accounting Items

Service Type	Service Type Text	Order Item	One-Time Charge	Fulfill. Item	Invoice Item
RECR	Martex Partner Product	☑	☐	☐	☑
ZUSG	Martex Usage	☑	☐	☑	☑

Figure 7.18 Define Service Types

SAP Revenue Accounting and Reporting–specific fields are enabled in the billable items from SAP Convergent Invoicing to support the consumption values to SAP Revenue

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Accounting and Reporting. Usage billable items can rise in data volume because organizational units of consumption can be at a low level, such as minutes used or gigabytes consumed. An aggregation occurs of the billable items as fulfillments in batches based on organizational need of frequency and corresponding revenue are realized. Transaction FP_RAI_TRANSF transfers fulfillment RAIs from SAP Convergent Invoicing to SAP Revenue Accounting and Reporting en masse and can be set up as a recurring batch process.

Because SAP Revenue Accounting and Reporting relies on SAP Billing and Revenue Innovation Management to provide consumption events via billable items on a periodic basis to appropriately recognize revenue, the relevant fulfillment details for recurring charges are available at provider contract inception. For recurring charges or subscription-based billing, related POBs receive the transaction price and the start and end dates from the provider contract. The start and end dates are received from the **Valid To** and **End of Duration** fields in the provider contract. SAP Revenue Accounting and Reporting will automatically generate a revenue schedule of amounts to be recognized over financial periods based on the start and end dates and deferral method. A deferral method defines how revenue will be amortized; there are several to choose from in standard SAP Revenue Accounting and Reporting based on business requirements. For the purposes of Martex, we'll be using *1: Linear Distribution, Day-Specific, 365/366 Basis*. SAP Revenue Accounting and Reporting will calculate and spread the allocated transaction price for the POBs over financial periods based on how many days are in the period.

Figure 7.18 also displays the **Invoice Item** indicator, allowing the transfer of invoicing from SAP Convergent Invoicing to SAP Revenue Accounting and Reporting. As invoices are processed in SAP Convergent Invoicing, the relevant details are transferred to SAP Revenue Accounting and Reporting in batches periodically via Transaction FP_RAI_TRANSF as invoicing RAIs. Invoicing RAIs contain main and condition items in parallel to order RAIs. These attributes, as well as the source item ID of the original order item, will support referencing the POB to enrich invoicing values. The difference between the recognized revenue and invoiced amount for the POB is calculated and posted to the balance sheet. This adjustment will post to contract asset/unbilled receivable and contract liability/deferred revenue, respectively, based on account determination and whether the difference is positive or negative.

For the Martex case study, we'll consider consumption for usage charges (Martex usage) and time-based for recurring charges (Martex partner product subscription) scenarios.

In Table 7.9 and Table 7.10, a depiction of the SAP Revenue Accounting and Reporting contract allocation and revenue recognition for the first period of contract inception is displayed. As discussed in step 4, the allocation to the POBs has occurred. The usage- and time-based fulfillments occur for July 2021 for these POBs, and revenue is recognized accordingly. With SAP Revenue Accounting and Reporting 2020 and subsequent

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releases, the ability to fulfill and recognize revenue based on the consumption value at the time of billable item transfer is available, to allow for unbilled revenue recognition in billing in arrears scenarios. For this configuration in BRFplus, fulfillment type is equal to consumption, and value-based fulfillments are active. Estimated quantity and values in BRFplus remain required in this configuration. The revenue calculated for the usage-based POB is calculated as 20 units at $2,120.30/200 = $10.6 each, for a total of $212.03 recognized revenue for the period. The Martex partner product subscription is calculated for 31 days in July 2021. The allocated transaction price ($1,096.37) is recognized for 31 days out of 365 for a total of $93.12 in recognized revenue for the July financial period. As a result of receiving a total invoice amount of $300 for the contract in the July period, there is a difference between the total recognized amount and invoiced amount. Because this difference realizes a recognized amount greater than the invoiced/billed amount, a net amount of $5.15 will be posted to the unbilled receivables balance sheet general ledger account.

SAP Revenue Accounting and Reporting Revenue Recognition for July 2021			
Performance Obligation	Performance Obligation Type	Transaction Price Total	Allocation to Performance Obligations
Martex partner product subscription	Subscription	$1,216.67	$1,096.37
Martex usage	Usage	$2,000.00	$ 2,120.30

Table 7.9 Martex SAP Revenue Accounting and Reporting Contract July 2021: Part 1

SAP Revenue Accounting and Reporting Revenue Recognition for July, 2021				
Performance Obligation	Effective Quantity	Fulfilled Quantity	Recognized Amount	Invoiced Amount
Martex partner product subscription	1	N/A	$93.12	$100.00
Martex usage	200	20	$212.03	$200.00
			$305.15	$300.00

Table 7.10 Martex SAP Revenue Accounting and Reporting Contract July 2021: Part 2

There are two ledgers SAP Revenue Accounting and Reporting interacts with: the revenue accounting subledger and the financial accounting general ledger. The SAP Revenue Accounting and Reporting subledger defines more details to support external disclosure reporting, analysis, and reconciliation. The transfer from the SAP Revenue

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Accounting and Reporting subledger to the financial accounting general ledger occurs in batches and generally has summarized details, based on configuration. The postings aren't in the scope of this documentation, but understand that SAP Revenue Accounting and Reporting derives accounts to be posted to via BRFplus configuration. SAP Revenue Accounting and Reporting account determination can derive the general ledger account to post to if a reference general ledger account is provided. A reference general ledger account is provided by the sender system. In the case of SAP Billing and Revenue Innovation Management, the provider contract won't provide reference accounts. BRFplus has the capability to use other attributes from the provider contract, such as service type, company code, business partner, and others, to determine the general ledger accounts for SAP Revenue Accounting and Reporting to post to. Revenue, deferred revenue, unbilled receivables, contract assets, and contract liabilities, among others outside the scope of this chapter, can be determined.

7.4.4 Freeze and Unfreeze Performance Obligations

New capabilities were released in SAP S/4HANA 2020 in SAP Revenue Accounting and Reporting. A relevant capability to support as-a-service type business models is the ability to freeze/unfreeze POBs. In time-based or subscription-based business models, there are cases where service will be frozen, paused, or suspended due to seasonality of service, fraud-related suspensions, or nonpayment among others. In these freeze periods, revenue and cost should not be recognized in the ledger, as the service isn't being provided to the related party.

SAP Revenue Accounting and Reporting requires the operational document to provide the periods in which a freeze period start and ends. Once these freeze periods are provided, all revenue accounting contracts and POBs related to the order item operational document will be applied freeze dates. There is no change to the transaction price of the SAP Revenue Accounting and Reporting contract or related POBs when a freeze is applied. A recalculation of days to recognize revenue occurs when SAP Revenue Accounting and Reporting is notified of freeze and unfreeze dates. Subsequently, the revenue recognition deferral is updated to reflect the freeze periods. When a freeze period is created, the end date of the revenue accounting contract extends by the number of days in the freeze period. Likewise, on deletion of a freeze period, the end date is reduced as shown in Table 7.11.

It's recommended to use *Deferral Method 1: Linear Distribution, Day-Specific, 365/366 Basis* for time-based POBs eligible for freeze periods due to the day-based calculation needed for appropriate extension of contracts. For a successful application of freeze periods, end date can't be earlier than the unfreeze date similar to how the start date can't be later than the freeze date. The start date of the relevant POB can't be empty and the freeze periods may not coincide.

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Action	Performance Obligation	Start Date	Freeze Start	Freeze End	End Date
SAP Revenue Accounting and Reporting contract creation	123	02/01/2022	N/A	N/A	1/31/2023
Application of freeze period	123	02/01/2022	3/01/2022	03/10/2022	2/10/2023

Table 7.11 Freeze and Unfreeze Performance Obligations

The asynchronous inbound service API available for managing freeze periods is `RevenueAccountingContractSetFreezePeriodsIn`. The fields described in Table 7.12 are used to enable applying freeze and unfreeze periods to POBs.

Field	Description	Mandatory/Optional
`DocumentItemID`	Order item in operational document. In SAP Billing and Revenue Innovation Management integration with SAP Revenue Accounting and Reporting this is the provider contract and item ID.	Mandatory
`DocumentTypeCode`	Type of operational document. In SAP Billing and Revenue Innovation Management integration `CAOI` can be used.	Mandatory
`SendingComponentID`	ID of sending component. In SAP Billing and Revenue Innovation Management integration `CA` can be used.	Mandatory
`LogicalSystemID`	ID of the logical system with client and client number.	Mandatory
`DocumentItemEndDate`	End date after freeze period is applied to POB.	Mandatory
`FreezeDate`	Start of freeze period.	Mandatory
`UnfreezeDate`	End of freeze period plus one day.	Optional

Table 7.12 API Fields to Manage Freeze and Unfreeze Periods to POBs

7.5 SAP Customer Financial Management

This section explains some of the key capabilities of SAP Customer Financial Management, which include SAP Collections and Dispute Management, SAP Credit Management, and payments. SAP Customer Financial Management is an integrated solution

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within SAP Billing and Revenue Innovation Management that provides the capability of tracking, monitoring, and taking proactive action to get ahead of the financials of your customers. SAP Customer Financial Management is an integrated solution suite alongside SAP Billing and Revenue Innovation Management. SAP Customer Financial Management is also available as part of the separate FSCM suite or the full SAP S/4HANA solution suite. With SAP Customer Financial Management, companies can effectively record and manage their accounts receivable data for all customers, which leads to an increase in automation and reduction in manual effort and eventual costs. Additional benefits also include the availability of integrated data for dispute, collections, and credit management applications.

7.5.1 Collection Management

The collection management component integrates with the SAP Convergent Invoicing and Contract Account Receivable and Payable to support the processes of receivables management. As part of SAP S/4HANA, the following business functions are delivered by SAP as activated:

- FSCM, Enablement of Financial Shared Services (`FIN_FSCM_SSC_AIC_1`)
- FSCM Functions 3 (`FIN_FSCM_CCD_3`)
- FSCM Functions 2 (`FIN_FSCM_CCD_2`)
- FSCM Function (`FIN_FSCM_CCD`)

Collection management works in conjunction with accounts receivable to proactively collect outstanding payments from customers and allow companies to effectively manage their collections operations. There are three main processes supported by collection management:

- Collecting receivables
- Controlling the collection of receivables
- Data synchronization and creation of worklist

These processes are supported by various configurations, organizational data and master data.

7.5.2 Organizational Structure

There are three components of an ERP solution: organizational structure, master data, and transactional data. The organizational structure is the foundation of the ERP solution and reflects the structure of your company or business group.

Collection management's foundational structure is represented by collection profiles, collection segments, and collection groups. The collection profile is the highest-level node in the organizational structure. Collection profiles are assigned with collection segments, and each collection segment can be assigned to multiple company codes to

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group the receivables from the respective company codes. Collection segments assigned to company codes represent lines of business.

In some firms, the accounts receivable team and collections team are the same; in others, they are separate. Depending on the client requirements and collections team structure, you can define the equivalent node in collection management, called a collection group. A collection group represents the collections team. You can assign a collection group to one or multiple collection segments depending on which company codes these collection groups will be performing receivables collection activities for.

7.5.3 Master Data

Collection activities are performed by collection segments and for business partners. Business partners need to be set up for collection management. A collection profile, collection segment, and collection group need to be assigned to a business partner along with the collection specialist. When the collection specialist or collection group is assigned to a business partner, all the receivables related to that business partner go to the worklist of the specialist that's assigned to the business partner. If the business partner isn't assigned with a specialist, then the receivables for all the business partners should be randomly distributed among specialists that are present in the collection group in the business partner master data. Based on who will handle the collection activity for the business partner, collection specialists are assigned to collection groups, and the collection groups are assigned to business partners in the business partner master data. One more important component of the master data/configuration is the collection strategy that allows a supervisor to define the rules for your worklist prioritization and display of items.

For a look at the organizational structure for collection management, see Figure 7.19.

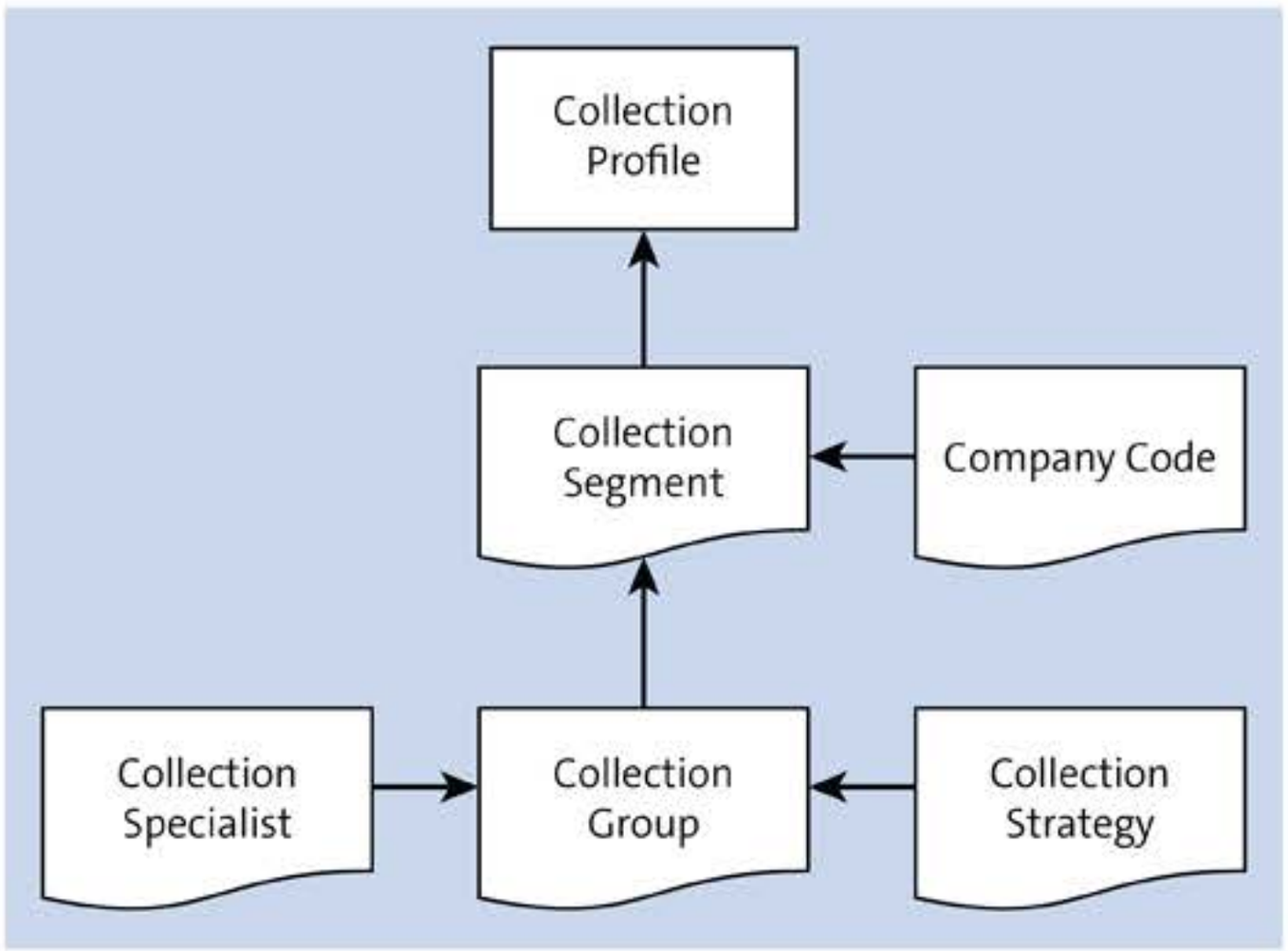

Figure 7.19 Organizational Structure for Collection Management

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7.5.4 Collection Strategies

The worklist of a collection specialist is prepared automatically based on the settings maintained in the collection strategy. The collection strategy allows you to define the currency for the worklist, and usually it should be consistent with the document currency.

The collection strategy allows you to configure the following aspects of the worklist:

- Overdue receivables aging bucket
- Due receivables aging bucket
- Prioritizing business partners for collection activity based on the collection rules
- Receivables processing before due date
- Excluding items that are legally dunned or defining tolerance days after dunning

SAP delivers standard basic rules that serve as conditions for prioritization of worklist items for each business partner. It also determines if the business partner is supposed to be included in the worklist or not. For example, Rule BR00000016 Customer isn't a direct payer. If this rule is part of the strategy as an exit rule, then any business partner that satisfies this rule won't be included in the worklist.

One or more basic rules are assigned to a collection rule, and along with preconditions defined in the collection rule for the basic rules, they allow you to prioritize and filter the worklist based on business partners. At the time of worklist generation, the business partners go through the check of collection rules and, in turn, basic rules. Collection rules are configured in collection strategies, a collection strategy is assigned to a collection group, and a collection group is assigned to a business partner.

7.5.5 Promise to Pay

One of the main activities of collection specialists is to keep track of customers who have promised to pay certain invoices/amounts by certain dates and make sure they are being followed through with in a timely manner. The promise-to-pay functionality of collection management allows you to execute this function of a collection specialist by allowing you to create a promise to pay for each open item for a business partner.

The promise-to-pay functionality allows you to keep track of the promises to pay that are created for an invoice:

- Open (no payment exists)
- Open (payment exists, not applied)
- Broken
- Partially kept
- Kept

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Withdrawal of the promise to pay is allowed if a customer gives another promise to pay with a new date. Each promise to pay received for a business partner receives a level so that collection specialists can keep track of the number of promises to pay issued.

The installment plan function of the promise to pay allows a customer to define the number of installments for the amount that is being promised.

7.5.6 Customer Contact

In day-to-day collection activity performed by collection specialists, they will record multiple pieces of information based on follow-ups with customers. These follow-ups, whether successful or unsuccessful, can be recorded in a customer contact with a reason, date/time, contact type, result, user, and contact person. Actions such as promise to pay, dispute cases, resubmissions, or notes create a customer contact automatically.

7.5.7 Resubmission and Note

Resubmission works as a reminder for the collection specialist to carry out any activity for a customer or invoice. Resubmission can also be used as a criterion in generating the worklist.

7.5.8 SAP Collections and Dispute Management

The FSCM suite in SAP S/4HANA provides a comprehensive solution for receivables management by seamlessly integrating collection management, SAP Collections and Dispute Management, and credit management.

In collection management, you can activate the integration component of SAP Collections and Dispute Management. You can also define a dispute case status code that you want to consider not collectible.

In the world of providing products or services to a customer and charging them for those products and services, there will always be scenarios in which customers dispute a charge for various reasons. SAP provides the SAP Collections and Dispute Management component to address this process and make sure that the complaints lodged by customers are recorded properly until they have reached their logical conclusions.

The SAP Collections and Dispute Management component can be configured to integrate with the SAP Convergent Invoicing component of SAP Billing and Revenue Innovation Management, as discussed in the following section. In most SAP Convergent Invoicing processes, a customer is on a recurring auto debit, so a customer usually raises a complaint rather than an automatic dispute case being created based on an underpayment. We'll also cover configuration of SAP Collections and Dispute Management.

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Integration of SAP Collections and Dispute Management and SAP Convergent Invoicing

SAP Collections and Dispute Management can be integrated with SAP Convergent Invoicing by activating the `FICA_FSCM_CRM_DISPUTE` business function. In addition, the `FIN_FSCM_CCD` and `FIN_FSCM_INTEGRATION` business functions also need to be activated to ensure the prerequisites are complete. Once these business functions are activated, they all are available for integration and configuration.

Configuring SAP Collections and Dispute Management

The SAP Collections and Dispute Management component is integrated with the SAP Convergent Invoicing system at the company code level to determine which legal entity's receivables and customers are going to be part of the SAP Collections and Dispute Management process. Figure 7.20 and the following configuration path show how to set up this configuration: **Financial Accounting • Contract Accounts Receivable and Payable • Organizational Units • Set Up Company Code for Contract Accounts Receivable and Payable**.

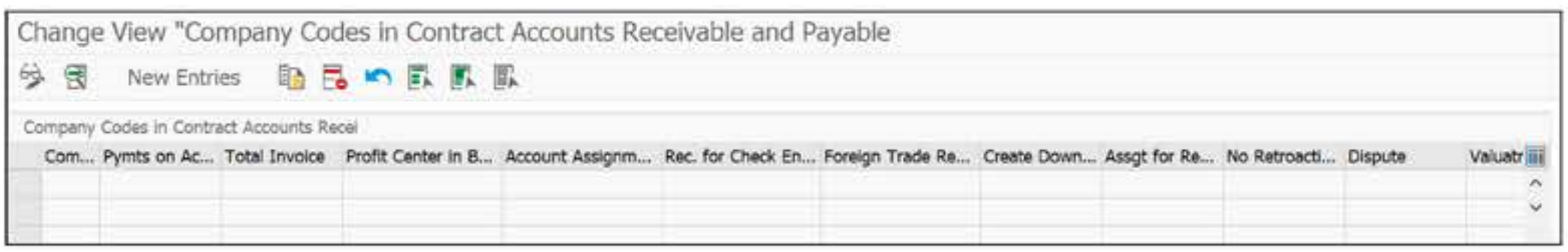

Figure 7.20 Company Codes in SAP Convergent Invoicing

Assuming the company code for SAP Convergent Invoicing in contract accounting is already done, you need to select the company for dispute cases in SAP Collections and Dispute Management.

The next step in the process is to configure the integration of both components (see Figure 7.21). Go to **Financial Accounting • Contract Accounts Receivable and Payable • Integration • Financial Supply Chain Management • Dispute Management • Make Basic Settings for Dispute Management**.

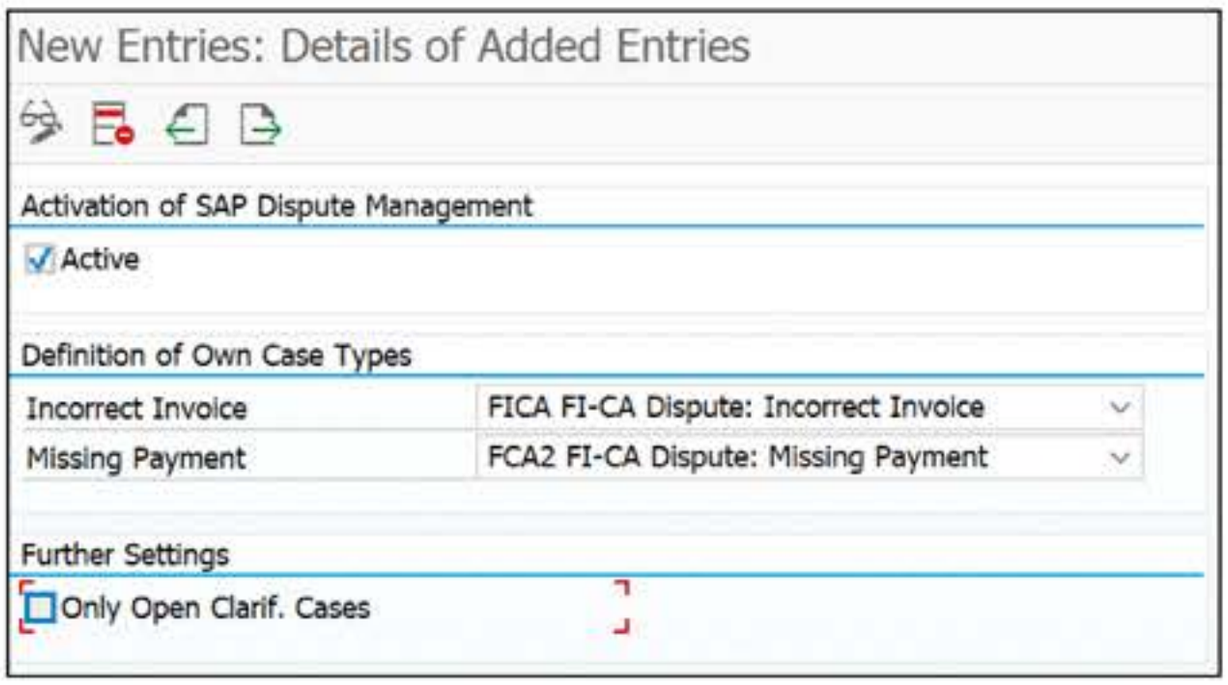

Figure 7.21 Make Basic Settings for SAP Collections and Dispute Management

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As you can see, you need to set the **Active** checkbox to tell the SAP Convergent Invoicing system that, in this instance, SAP Collections and Dispute Management is active.

Dispute cases are created by case types that act as containers for the dispute case that holds all the attributes for the dispute case. There are two scenarios—(1) incorrect invoices, incorrect credits, or missing credits; and (2) missing payments—for which SAP provides two standard case types: FICA and FCA2. You can create your own case types if there is a client-specific requirement to add more fields or change the case attribute profile, search profile, and so on. These standard case types contain attributes such as priority, disputed amount, reason for the complaint, clerk responsible for resolution, and contact data of the business partner. Let's take a deeper look:

- **Missing payment**
  Customer reaches out to raise a complaint that he has paid but is still receiving a dunning notice
- **Incorrect invoice**
  Customer complains about an incorrect amount on the invoice or an incorrect invoice in its entirety.

7.5.9 SAP Collections and Dispute Management for SAP Convergent Invoicing

In this section, we'll cover SAP Collections and Dispute Management. First, we'll start with the records management system.

Records Management System

The records management system is the instance in which you want to create dispute cases and store them. You can create multiple record management systems or use the standard one provided by SAP: `UDM_DISPUTE`. Change the record management system ID to "UDM_FICA_DISPUTE" to process cases from SAP Convergent Invoicing (see Figure 7.22).

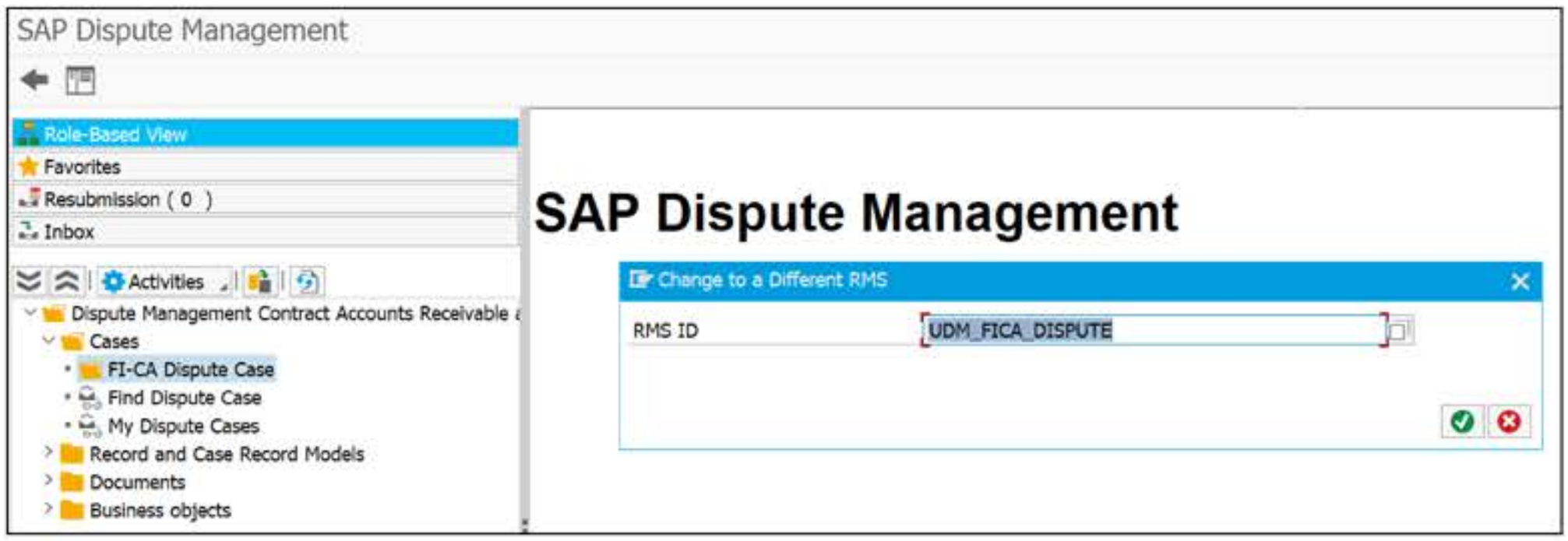

Figure 7.22 Select Record Management System ID

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The UDM_DISPUTE record management system ID delivers standard element types in the dispute case processing system. Element types such as dispute cases, business objects with contact account/business partner, invoice, credits, and more (see Figure 7.23).

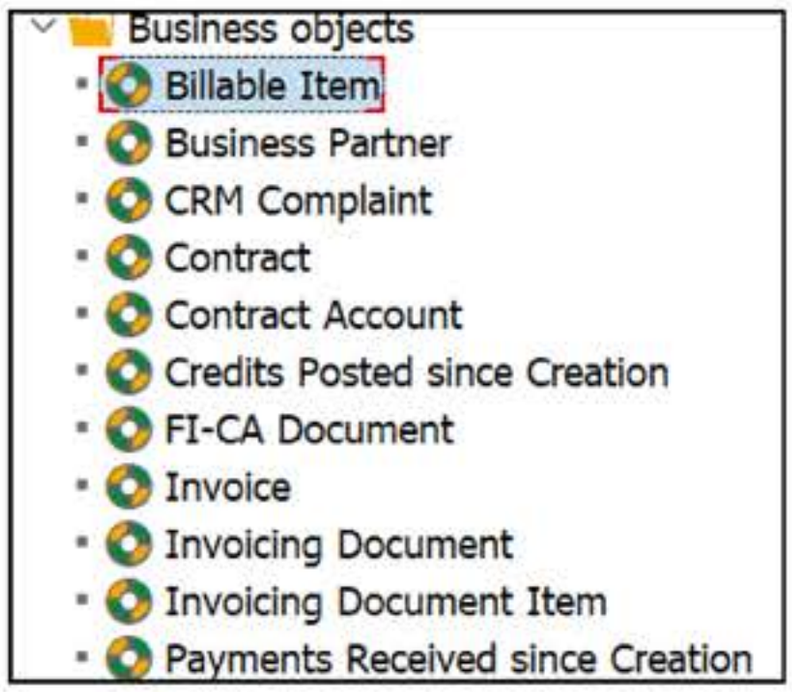

Figure 7.23 Element Types in Dispute Cases

Following are some important fields related to the dispute case lifecycle:

- **Total Amount**
  This is the total of all items for a business partner within a dispute case. This field isn't updated automatically.
- **Disputed Amount**
  This is the amount in dispute raised by the customer. This doesn't change automatically if postings are made.
- **Amount Credited**
  This is updated automatically if the credit is posted against an invoice for an open dispute case using Transaction FPE1. Credits that are not invoice related are handled by the system to link them with the dispute cases based on the missing credit case type, business partner, and event 1251.
- **Amount Paid**
  Based on the missing payment category, the system determines the total amount paid for the total payment received for dispute cases.
- **Amount Written Off**
  If you write off a certain amount from a dispute case, the system will move that amount to this field.

Note that the creation of a dispute case isn't supported from SAP Collections and Dispute Management. Dispute cases can only be created in the SAP Convergent Invoicing system.

Account Balance Display

There are two scenarios in which you can create a dispute case:

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- **Dispute case with reference to an invoice**
  In the SAP Easy Access menu, choose **Account • Account Balance**. Display the balances for a business partner (contract account) and then select one of the open documents to create a dispute case in reference to it.

  The dispute case type will be either incorrect invoice or incorrect credit. The scenario here shows that the customer has determined that the invoice in question is open incorrectly even though he has made the payment or a credit should be applied.
- **Dispute case without reference to an invoice**
  Inside the **Account Balance** menu, choose **Environment • Account • Dispute Cases • Create**.

  If you create a dispute case without selecting any document, then the system creates a dispute case without reference to an invoice or credit. This type of dispute case is defined as a missing payment case type. The scenario is that the customer has called in to report that he has made the payment, but it's not applied to an invoice.

Dispute Cases in SAP Collections and Dispute Management

Once you access Transaction UDM_DISPUTE, it calls dispute case processing. You'll need to expand the **Case** folder to find the dispute case or select **My Dispute Case** to find the dispute cases that are assigned to you as a person responsible or coordinator.

You can change the dispute case in Transaction UDM_DISPUTE by going to dispute case change mode. Only lot payments are assigned to the dispute case in the case of a missing payment because none of the payment-program-related payments are missing. Statistical amount, down payment, payment posted in the payment program, and payment posted in posting area 0113 aren't considered as amounts of credits.

Attribute Profile

SAP delivers standard attribute profiles that are assigned to the standard case types that we talked about earlier (FICA, FCA2), as shown in Figure 7.24.

FICADIS2	FI-CA Dispute: Missed Payment	UDMCASEATTR10
FICADISP	FI-CA Dispute: Incorrect Invoice	UDMCASEATTR10

Figure 7.24 Standard Case Types

The attribute profile determines the field in the dispute case; if the standard attribute is insufficient or is too detailed, you can change the profile.

Status Profile

The status determines the lifecycle of the dispute case and allows you to identify the stage the dispute case currently is in. SAP delivers standard status profiles that provide statuses such as **New**, **Being Processed**, **Closed**, **Confirmed**, and **Voided**.

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If you want to implement additional statuses by connecting to central status management, then you need to use BAdI SCMG_SET_STATUS_S for the initial status and BAdI SCMG_INIT_STATUS_S for a status change. To change the status, use the CRM_STATUS_CHANGE_FOR_ACTIVITY function module.

For the list of system statuses, see Table 7.13.

Status code	Description
003	External Processing
007	Completed
008	Confirmed
009	Canceled

Table 7.13 System Statuses

Transfer to External Application

There are scenarios in which your SAP Convergent Invoicing system and SAP Collections and Dispute Management system are in separate systems. For that, you can configure the external application that you want to transfer the dispute case to.

7.5.10 Credit Management

Managing credit in today's complicated marketplace is very important, including managing the risk associated with the customer. The credit management component is seamlessly integrated with SAP Convergent Invoicing in SAP Billing and Revenue Innovation Management and allows you to manage credit for contract accounts. Activate credit management under SAP Convergent Invoicing in the integration section (see Figure 7.25).

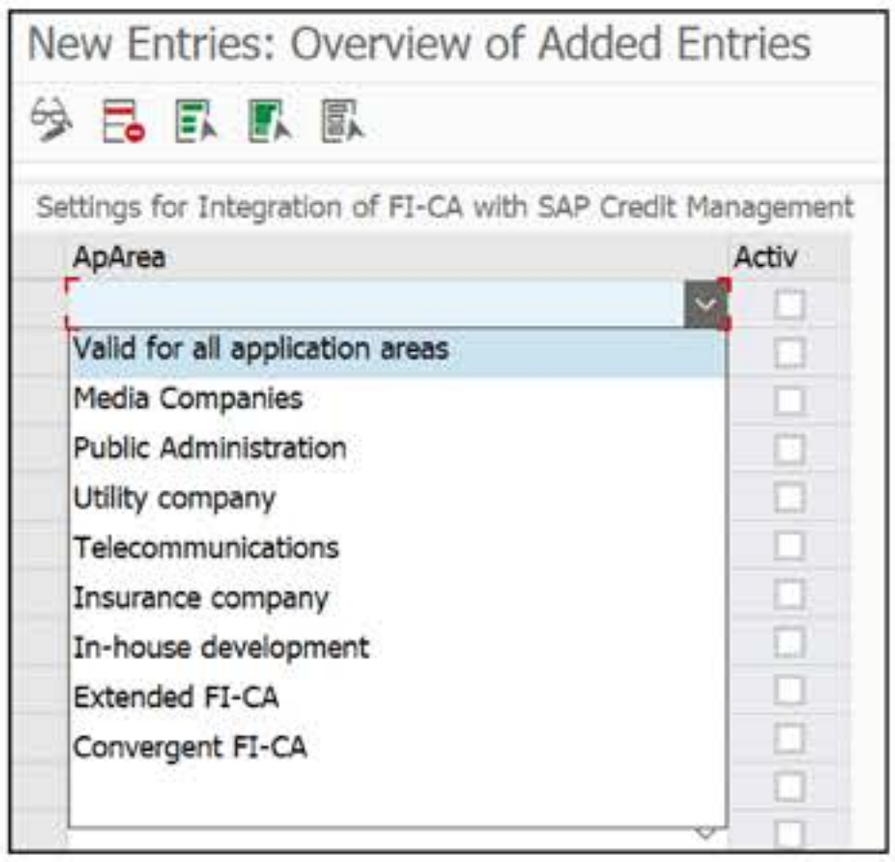

Figure 7.25 Activate Credit Management

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Navigate to **Financial Accounting • Contract Accounts Receivable and Payable • Integration • Financial Supply Chain Management • Credit Management • Activate Credit Management.**

This configuration allows you to configure the previously listed application area. You need to activate credit management for the application areas that you'll be using.

Maintain Credit Segment

Accounts receivable credit management or sales and distribution credit management uses a credit control area for maintaining the credit limit at the credit control area level, but in credit management, you can define a credit segment that allows you to store more than one credit limit and exposure for a business partner. You can also define the currency for the credit segment, which behaves like document currency for the credit segment.

Assign a credit segment to a posting area that uses the parameters, posting area, division, account determination ID, and business area.

Credit Worthiness

In SAP Convergent Invoicing, integrated credit management is stored in an external credit rating procedure. This creditworthiness information is included in calculating the score for the business partner (contract account). The rating procedure for SAP Convergent Invoicing with the function rating in the formula is configured in the following location: **Financial Supply Chain Management • Credit Management • Credit Risk Monitoring • Master Data • Define Rating Procedure**. To check credit limits, navigate to **Financial Supply Chain Management • Credit Management • Credit Risk Monitoring • Credit Limit Check**.

To determine credit worthiness, consider the following formula:

Score = SAP Convergent Invoicing creditworthiness + Further items from SAP Credit Management

For a commitment query to work, you must define the check rule at the following location: **Financial Supply Chain Management • Credit Management • Credit Risk Monitoring • Credit Limit Check • Define Check Rules**.

At the credit exposure update from contract accounting, all the business partners are transferred from SAP Credit Management. There are some mass activities for SAP Convergent Invoicing:

- Business partner exposure data is sent to SAP Credit Management.
- Business partner payment behavior is sent to SAP Credit Management.
- The creditworthiness value is replicated.

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7.6 Tax Engine

Countries such as the United States and Canada have specific tax calculation requirements that are generally addressed by third-party tax engines. These tax engines provide a tax calculation outside of SAP systems, but they are flexible and seamlessly integrated with SAP systems.

Indirect tax calculation for the sale of hardware or software is addressed by using external tax engines. Indirect taxes are levied by the service provider company against the customers, and then the tax collected is paid to the government by the service provider company. During the sale of a product, the estimated tax is provided to the customers; upon invoicing, the actual tax is calculated and posted to the customer's account (see Figure 7.26).

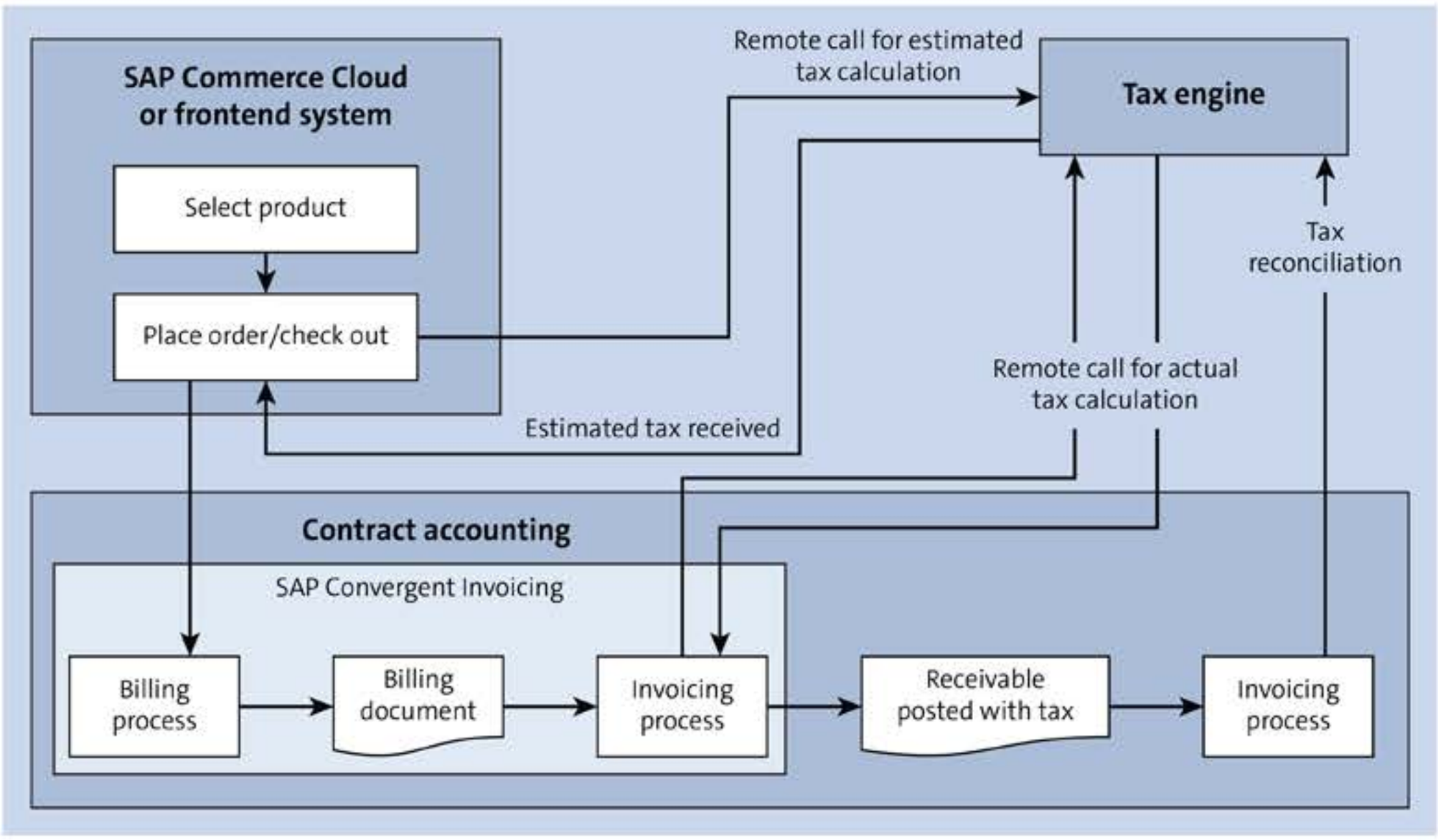

Figure 7.26 Tax Calculation Process

When SAP Commerce is used for frontend sales of a service, an RFC is made to the external tax engine for the estimated tax calculation. Following this, the billable items for the sale of the service are received in SAP Convergent Invoicing, which undergoes a billing process and subsequently an invoicing process.

During the invoicing process, an RFC is made to an external tax engine, and the data for tax calculation is passed to the tax system in which the actual tax is calculated. The tax system then passes the tax data back to the SAP Convergent Invoicing system, and receivables are posted with the tax amounts posted to the tax account.

SAP provides `FQEVENTS` as enhancement points, which can be used for coding custom logic if required. `FQEVENT 1110` is used for enhancing the invoicing process and for

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making an RFC for actual tax calculation. The data associated with the tax calculated externally is stored in database table DFKKEXTTAX.

Transaction FPTX is used for a tax audit for the tax calculation made by the external tax engine. Successful execution of this transaction updates the flag in table DFKKEXTTAX (see Figure 7.27).

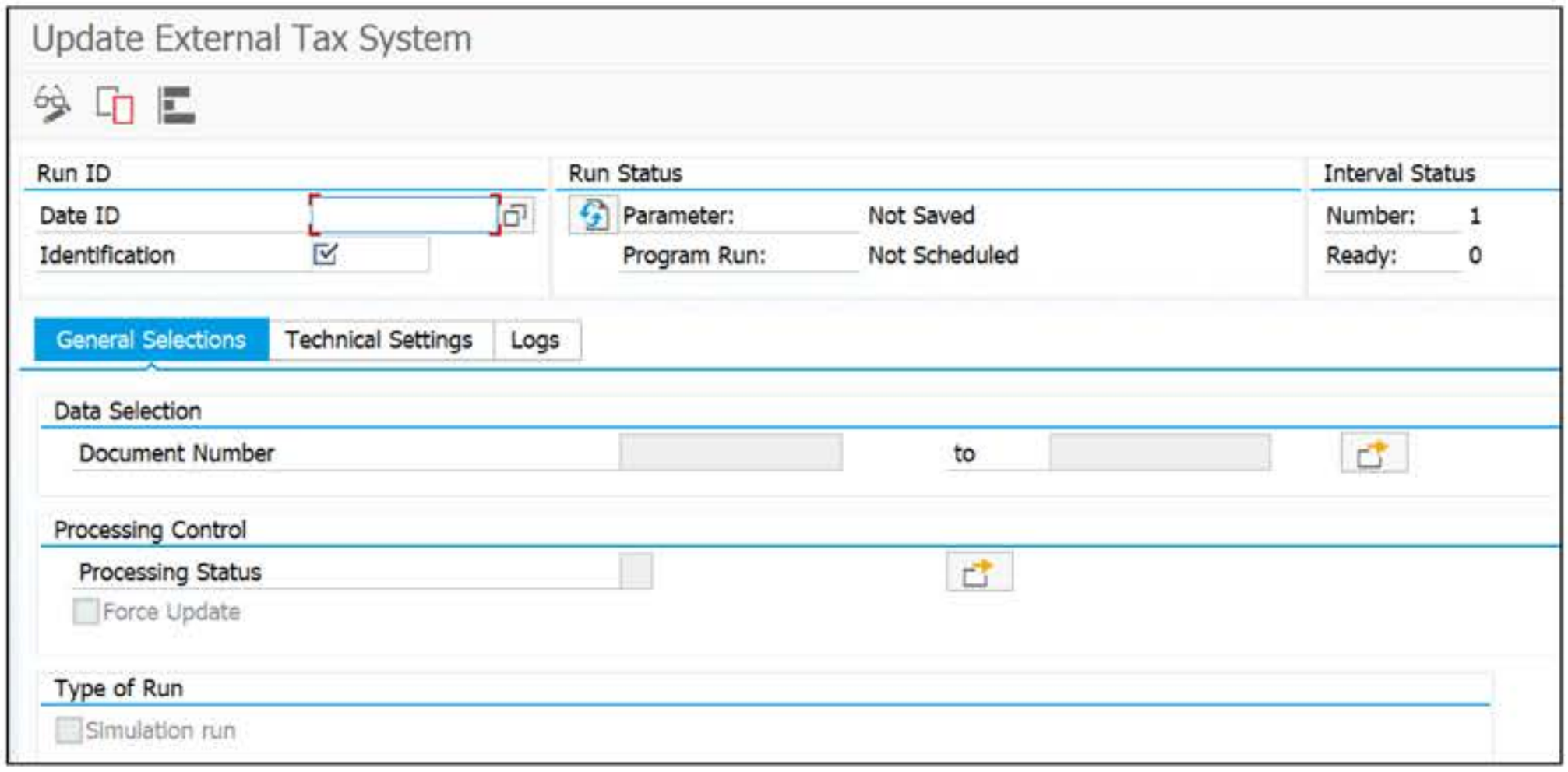

Figure 7.27 Transaction FPTX: Update External Tax System

In SAP Convergent Invoicing, during the invoicing process, the RFC is made to the tax engine for calculation of the final tax amount. The calculated tax is represented as a separate line item in the invoicing document, and the total amount is available in the invoice header.

There are different SAP-standard invoice item types that represent the line item types within the invoicing document. OTAXITEM is an item type for tax items. Tax items form a separate line item within the invoice (see Figure 7.28).

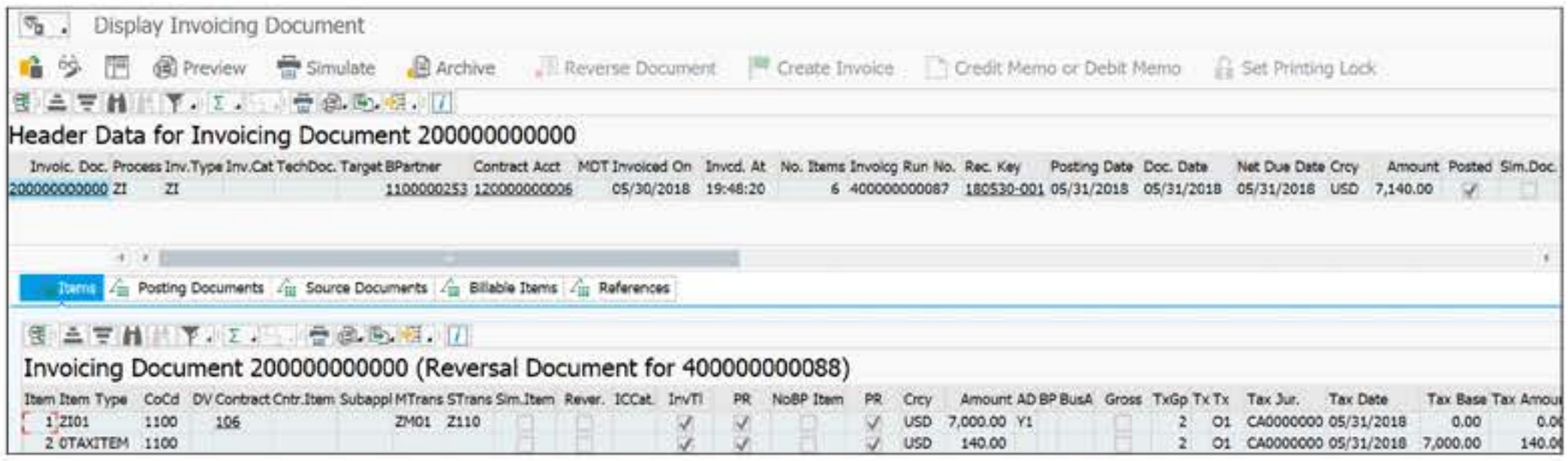

Figure 7.28 Invoice Document with Tax Item

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The general ledger account determination for the tax posting is based on the tax determination code.

Posting area 2610 can be maintained at Transaction SPRO node **Contract Account Receivables and Payables • Convergent Invoicing • Invoicing • Documents • Posting Documents • Define Account Assignment for General Ledger Items**, and it contains the details of the revenue account determination and the tax determination code. This tax determination code linked to a tax code leads to the determination of a tax general ledger account for tax amount posting.

During the invoicing process, the tax calculated is shown in the line item of the invoicing document and the posting document, which is the SAP Convergent Invoicing receivable document that contains the posting details of both the revenue and tax items (see Figure 7.29).

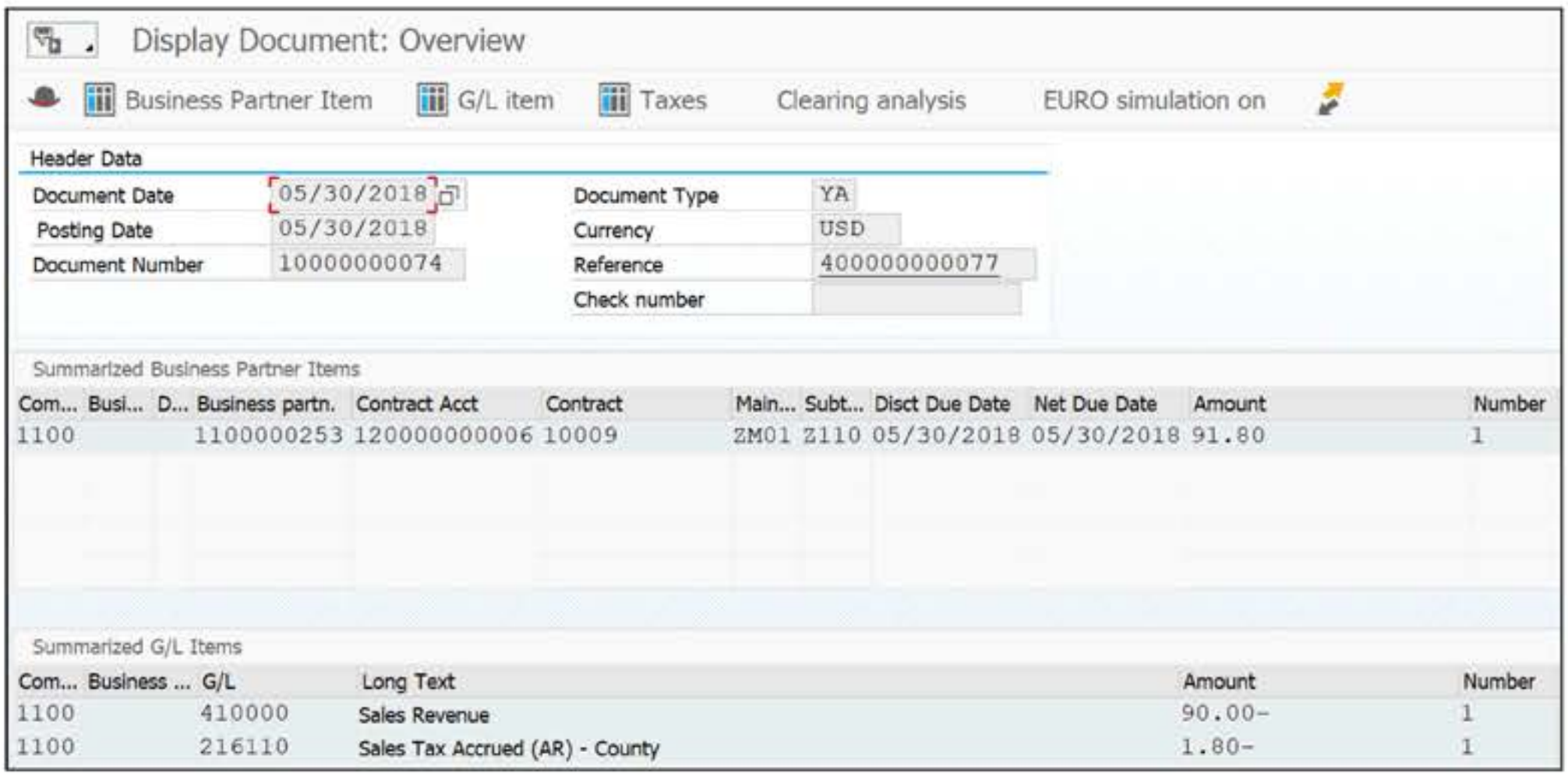

Figure 7.29 Sample SAP Convergent Invoicing Document with Tax Line Item

7.7 Payment Gateway

In this section, we'll explain the integration aspects of SAP Billing and Revenue Innovation Management with external payment gateways. This section generally focuses on options available within SAP Billing and Revenue Innovation Management, in which SAP Convergent Invoicing can be integrated with the payment gateway for credit card settlements.

Let's start with some key terminology:

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- **Payment gateway**
  Enables a payment transaction by transferring payment information, such as credit card details, transaction amount, and so on, securely between the merchant and the payment processor.
- **Payment processor**
  Processes the payment transactions from various channels, such as credit/debit cards, to the acquiring bank.
- **Acquiring bank**
  Processes the credit/debit card payments on behalf of the merchant. The payments from the issuing bank are transferred eventually to the acquiring bank.

In a scenario in which a customer makes a purchase with their credit card from any customer portal, the credit card details are validated and processed by the payment gateway. Once the payment is authorized, the credit card is charged, and the payment details are sent to SAP Convergent Invoicing as a payment billable item, along with the revenue transaction in an integrated SAP Billing and Revenue Innovation Management solution. As explained in Chapter 4, every transaction is passed to SAP Convergent Invoicing as a billable item—in this case, a revenue billable item and a payment billable item. The accounting entries after billing and invoicing are as follows:

- Revenue Posting (Revenue Billable Item):
  - Dr. Customer
  - Cr. Sales
- Payment Posting (Payment Billable Item):
  - Dr. Credit Card Clr. A/C
  - Cr. Customer

The payment received as part of the credit card charge has been captured in the credit card clearing A/C, and the customer account has been cleared; that is, no open items remain on the customer account.

The next step is for the payment processor to settle the funds with the bank of the company providing the services. This can be automated from SAP Convergent Invoicing as part of a credit card settlement run. As a result of the credit card settlement run, the following accounting entries are posted:

- Dr. Bank
- Cr. Credit Card Clr. A/C

The credit card settlement run is executed by running Transaction FPCS in SAP Convergent Invoicing. As part of the settlement run, an integration can be built with the payment gateway to help enable the automated credit card settlement.

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7.8 Summary

The SAP Billing and Revenue Innovation Management integration with SAP Revenue Accounting and Reporting has been important for organizations that have been affected by accounting standard ASC606. Many of the same organizations that leverage SAP Billing and Revenue Innovation Management for subscription and usage-based models are in the technology, media, and entertainment industries. These industries are the most affected by the transition to ASC606 due to the multielement nature of their agreements with customers.

A standard integration to handle large volumes of transactions from SAP Billing and Revenue Innovation Management in SAP Revenue Accounting and Reporting provides benefits for automated revenue recognition, flexible disclosure, and adherence accounting standards. For accounting organizations, the automation and reconciliation functionality provided by SAP Revenue Accounting and Reporting will reduce critical month-end close activities, as well as provide external auditors with a clear end-to-end order-to-revenue view to provide detailed support for reported financials.

Now that we've discussed each component and their integrations, the next chapter will focus on legacy data migration applicable in SAP Billing and Revenue Innovation Management. We'll discuss different migration objects and their dependencies in detail.

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Chapter 8

Extension and Connectivity

Various extension and interface components are delivered as part of SAP Billing and Revenue Innovation Management. Integration is offered in different forms—SOAP and REST (OData)—and can be used based on the compatibility of SAP or non-SAP platforms. We'll limit the discussions to the standard, out-of-the-box integration options available as part of SAP Billing and Revenue Innovation Management 2021 only.

In an integrated enterprise resource planning (ERP) solution, SAP Billing and Revenue Innovation Management provides connection protocols to enable integrations with various platforms in an echo system. In a scenario where non-SAP platforms are part of the solutions to support the end-to-end business processes and data interchange, such as master data and transactional data, interfaces are critical components for systems integrations. As of now, SAP has defined the SAP API Business Hub (*https://api.sap.com*) as the central place for publishing all external application programming interfaces (APIs). The publishing of APIs for SAP S/4HANA has started and is updated frequently.

In SAP S/4HANA, you must perform some configurations to enable the APIs to be called or consumed by the external systems. APIs are exposed in SAP S/4HANA in several ways, such as web services and remote function calls (RFCs). SAP supports two types of web services: SOAP and OData services. For OData, you need to enable the OData gateway by calling Transaction /IWFND/MAINT_SERVICE. If the service is available in the backend system, it's possible to configure and test the API in SAP S/4HANA. You also need to define a role and a technical user to use the OData service. For the SOAP API, you also must perform the necessary steps to enable it. Typically, in implementations involving SAP and non-SAP systems, you'll also have a middleware system, such as SAP Process Integration/SAP Process Orchestration or SAP Cloud Platform Interface to manage the integration and application security requirements.

Figure 8.1 showcases a baseline overview of API integration with SAP S/4HANA and SAP S/4HANA Cloud with the boundary as SAP or non-SAP applications.

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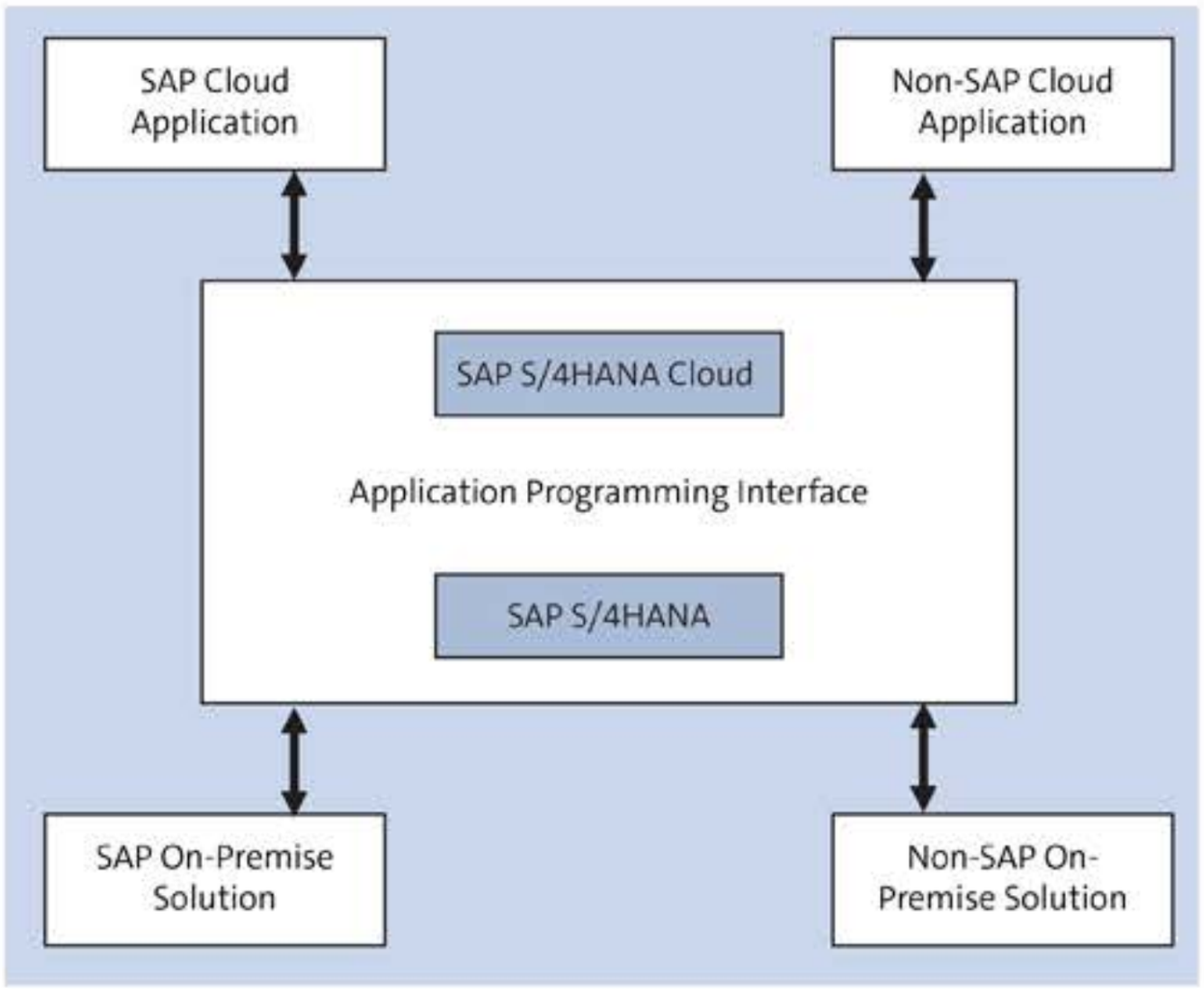

Figure 8.1 Overview of API Integration with SAP and Non-SAP Applications

In SAP S/4HANA, you must do some configuration steps to test the capability of APIs. The methods are also dependent on the type of APIs. If the API is of type OData, you need to use an OData gateway by calling GUI Transaction /IWFND/MAINT_SERVICE. If the service is available in the backend system, it's possible to configure and test the API in SAP S/4HANA. You also need to define a role and a technical user to use the OData service. If the API is a SOAP API, then you must perform the steps to configure SOAP APIs. In the following sections, we'll briefly discuss some of the most-used standard APIs available for each module of SAP Billing and Revenue Innovation Management. To explore the full list of APIs applicable to SAP Billing and Revenue Innovation Management, we recommend visiting the SAP API Business Hub.

8.1 Extension for SAP Billing and Revenue Innovation Management, Subscription Order Management

SAP Billing and Revenue Innovation Management, subscription order management, provides different APIs to support the creation and maintenance of different types of transactions, such as master agreements, solution quotations, subscription orders, partner agreements, and so on. All the APIs provided by SAP can be enhanced to meet the business requirements of your company. SAP Billing and Revenue Innovation Management, subscription order management, also provides web-based user interfaces (UIs) such as SAP Fiori to perform the maintenance of the transactions mentioned earlier. SAP Application Interface Framework is used to monitor interfaces as well as execute error handling. The SOAP APIs can be monitored under the `/CMSOM` namespace. The following sections cover some of the commonly used objects in SAP Billing and

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Revenue Innovation Management, subscription order management, and related API and extension information.

8.1.1 Business Partners

The business partner is one of the core master data objects used in SAP Billing and Revenue Innovation Management. In enterprise solutions, business partner or customer data is one of the master data objects required to perform transactions such as creating solution quotations or subscription orders. It's important that the business partner or customer master is always in sync between SAP and the boundary systems to avoid incorrect transactions that will result in incorrect billing and financial postings. Additionally, in an integrated systems environment, it's always recommended to use a one-direction flow to create and change business partner information from SAP S/4HANA to the front office (e.g., Salesforce.com, an ecommerce platform, etc.) to avoid data inconsistency issues. Even though SAP S/4HANA can be used to manage or to govern business partner/customer data, SAP S/4HANA can also be used to integrate with non-SAP application within the ecosystems. SAP S/4HANA offers various integration options and extensions. The native business partner API in SAP S/4HANA enables you to create, read, update, and delete (CRUD) data related to customers and suppliers.

If you leverage OData services, you can use the standard synchronous OData-based inbound API `API_BUSINESS_PARTNER`. To create a business partner record, you need to provide values for mandatory fields or attributes such as business partner roles, name, address, and so on. Additionally, you need to provide values to populate optional/custom attributes in the payload before calling the API. These steps are applicable for all API calls. If the API is successfully executed, you should receive a response with code "201" with the corresponding SAP S/4HANA business partner number. In case of any failure due to validation or incorrect namespace, you'll a receive standard error message with error code to troubleshoot further.

In addition, you can use the standard function module `BAPI_BUPA_CREATE_FROM_DATA` to create a business partner with the required role and information in SAP S/4HANA from external sources. The standard function module can be used to generate a custom web service that can be consumed by non-SAP platforms to create business partner records.

8.1.2 Contract Accounts

The contract account is another primary master data object managed as part of the SAP Billing and Revenue Innovation Management solution. Contract accounts typically hold payment-related information for a customer (e.g., payment terms, methods, bank information, etc.). SAP offers a set of APIs for contract accounts that can be used to create or read records from SAP S/4HANA.

To create, retrieve, or update contract accounts from external non-SAP system, SAP provides a synchronous SOAP service: `FKCMANAGECONTRACTACCOUNT`. Alternatively, you

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can use the standard SAP S/4HANA function module BAPI_CTRACCONTRACTACCOUNT_CR1. The external system should provide mandatary parameter details such as business partner ID, payment terms and conditions, and so on to maintain contract accounts in SAP S/4HANA.

Typically, an order-to-cash solution requires a validation of or information about payment-related input from SAP S/4HANA while creating a transaction. The read native contract account API in SAP S/4HANA enables you to retrieve a list of contract accounts, read contract account details and contract account-business partner relationships, and retrieve the account balance for a contract account.

8.1.3 Subscription Products

A product is another set of master data that is required in SAP Billing and Revenue Innovation Management implementations. Like other master data objects, the product master should be one-directional. In an integrated ERP solution where a non-SAP application also requires master data, we need to identify the primary source system for product maintenance. Once the source system is identified, you can then develop a one-directional flow to manage creation and update of products in integrated systems.

Let's consider an example where a non-SAP application is managed as a source system for product maintenance and data governance. To replicate product information from non-SAP application to SAP S/4HANA, you can use the API_PRODUCT_SRV synchronous OData service. This API can be used to create subscription products in SAP S/4HANA with required material attributes such as basic product data, sales organization data, and units of measure. The API can also be used to read and update products in the SAP S/4HANA system. Note that the API doesn't support cross-catalog mapping or managing subscription product-specific data. You can either update subscription-specific settings manually once the product is created in the SAP S/4HANA system or develop a custom program to automate the process.

You can also use standard Business Application Programming Interface (BAPI) BAPI_MATERIAL_SAVEDATA to create a new material master record in SAP S/4HANA or to change existing material master data.

If SAP S/4HANA is used as the primary system for product maintenance, then product information should be shared from SAP S/4HANA to non-SAP applications. SAP also provides standard SOAP web service SubscriptionProductReplicationRequest_Out to replicate subscription product-specific data from an SAP S/4HANA system to an external system.

8.1.4 Subscription Transactional Data

Now that we've covered the master data integration required to implement SAP Billing and Revenue Innovation Management, this section discusses transactional data. SAP

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Billing and Revenue Innovation Management has several sets of transaction types such as master agreements, subscription orders, solution quotes, and so on. Even though SAP Billing and Revenue Innovation Management, subscription order management, comes with the standard SAP Fiori UI to create and maintain these transactions, SAP provides multiple APIs out of the box to enable the integration with third-party systems to maintain (create and update) these types of transactions. Depending on the solution design and business framework, transactions can be initiated in SAP S/4HANA in SAP Billing and Revenue Innovation Management, subscription order management, or in non-SAP applications. If the transactions are initiated in a non-SAP application, you must replicate the transaction details into SAP Billing and Revenue Innovation Management, subscription order management, in SAP S/4HANA. This will ensure that the process of subsequent fulfillment and billing managed in SAP Billing and Revenue Innovation Management can be fulfilled and processed. To maintain the consistency of the subscription contract between the source and the target systems, it's important to capture and process any change events (e.g., extension, cancellation, etc.). In this case, it's from a non-SAP application to SAP S/4HANA.

For transactions that are initiated from non-SAP systems or e-commerce platform, you can use the standard asynchronous SOAP web service `SubscriptionOrderRequest_In` (Technical Name - `II_CRMS4_PRVO_CREATE_IN`) to replicate the subscription order into SAP Billing and Revenue Innovation Management, subscription order management. The API also allows you to create subscription orders and subscription contracts in a single step. SAP provides standard API `CRM_ISX_BTX_API_ORDER_CREATE` for creation of subscription transactions in SAP Billing and Revenue Innovation Management. Because SAP Billing and Revenue Innovation Management transactions follow one order framework, you can use the same API with different transaction types (`PROCESS_TYPE`) to create various subscription transactions.

The solution quotation is another common transaction type used in SAP Billing and Revenue Innovation Management implementations. SAP provides standard OData API `API_BUS_SOLUTION_QUOTATION_SRV` (available as of SAP S/4HANA 1909) to create, read, and update solution quotation objects. In addition, you can use the service to accept solution quotations in a single step, which triggers the automatic creation of follow-up transactions.

The OData service can also be used by external applications to read existing solution quotations in SAP Billing and Revenue Innovation Management. For example, to read an existing solution quotation with Quotation ID – "1025", the sample request URL is */sap/opu/odata/sap/API_BUS_SOLUTION_QUOTATION_SRV/A_BusinessSolutionQuotation('1025')*.

It's also possible to read a specific quotation line item that exists in SAP S/4HANA, as shown in Figure 8.2.

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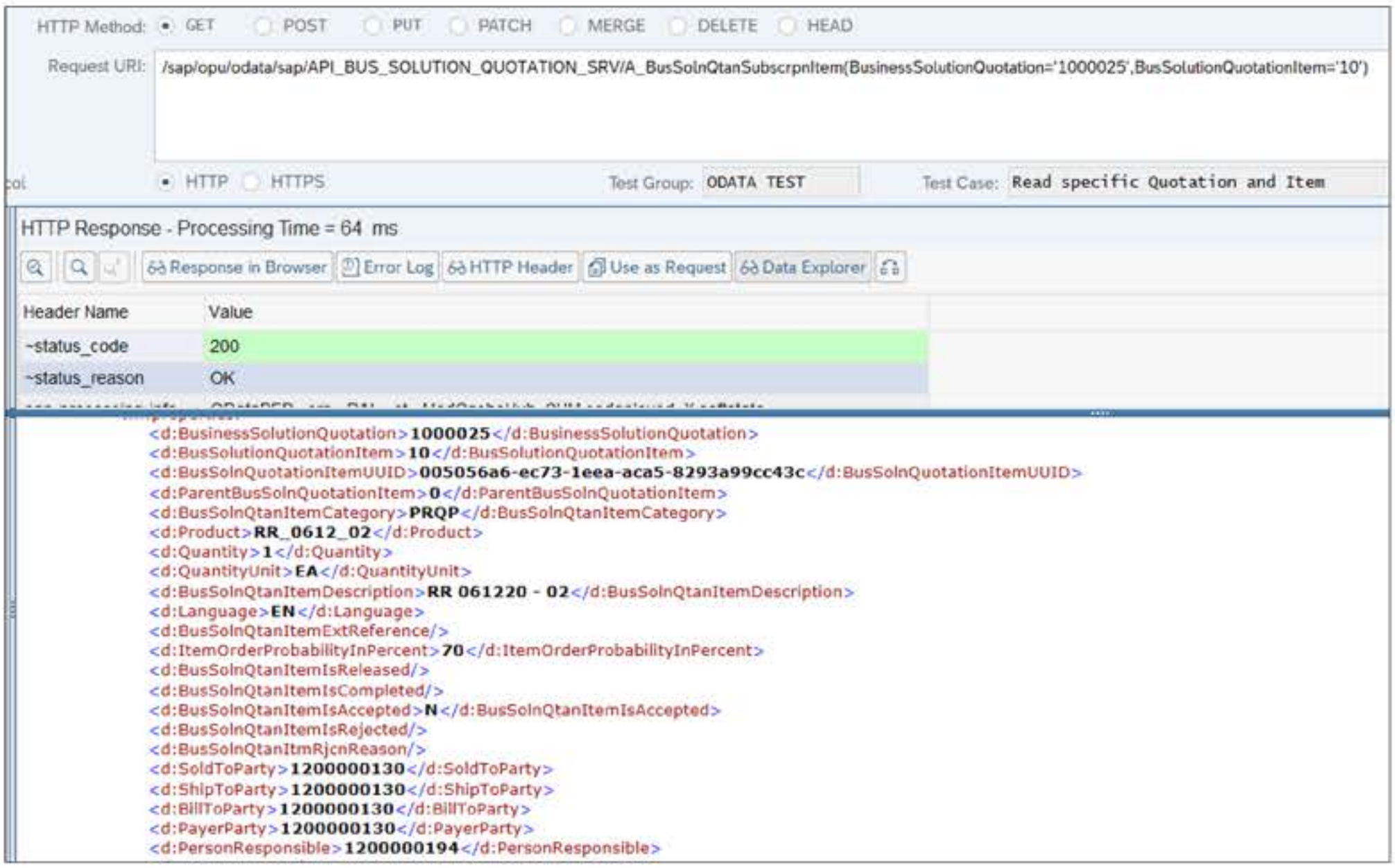

Figure 8.2 Read Solution Quote with Quote ID and Line-Item Number

Most SAP Billing and Revenue Innovation Management implementations are integrated with front-office applications. For that reason, SAP Billing and Revenue Innovation Management provides a standard SOAP web service `SubscriptionContractChangeProcessRequest_In` to manage subscription contract changes. The standard API `CRM_ISX_BTX_API_CONT_CHANGE` is used to manage contract changes. Based on your change scenarios, you can define the process type (`PROCESS`) and related process attributes (`PROCESS_ATTRIBUTES`) to manage change order scenarios. For example, if the external system requests an extension, you can define the process type value as "EXTENSION" along with extension terms (`DURATION`, `UNIT`) information. If the external system requests cancellation, you can define the process type value as "CANCEL" along with cancellation request data and cancellation reason (`CANC_REQUEST_DATE`, `CANC_REASON`) information.

8.2 Extension for SAP Convergent Charging

SAP Convergent Charging is one of the core components of the SAP Billing and Revenue Innovation Management solution. SAP Convergent Charging and its functionalities are discussed at great length in Chapter 3 of this book. SAP Convergent Charging isn't part of the SAP S/4HANA suite; rather, it's a standalone Java application. SAP Convergent Charging consists of two main components: Core Server and Core Tool. Both require SAP's Java Virtual Machine (JVM) to run. The integrations between SAP Convergent

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Charging and other modules of SAP Billing and Revenue Innovation Management (e.g., SAP Billing and Revenue Innovation Management, subscription order management; and SAP Convergent Invoicing) are provided by SAP. The integration doesn't require a separate middleware application.

The following sections provide an overview of how SAP Convergent Charging works and then walks you through some of the SAP Convergent Charging services, which can be consumed by SAP or non-SAP applications.

8.2.1 Overview

SAP Convergent Charging provides web services to perform rating, charging, and other operations. These technical interfaces are hosted by the SAP Convergent Charging Core Server system. Any non-SAP applications that require integrating directly with SAP Convergent Charging must consume these business services via SOAP web services. These web services can be used to manage master data, perform contract provisioning, rate customer usage consumptions, refill prepaid accounts, and so on. Figure 8.3 illustrates an overview of SAP Convergent Charging web services and APIs.

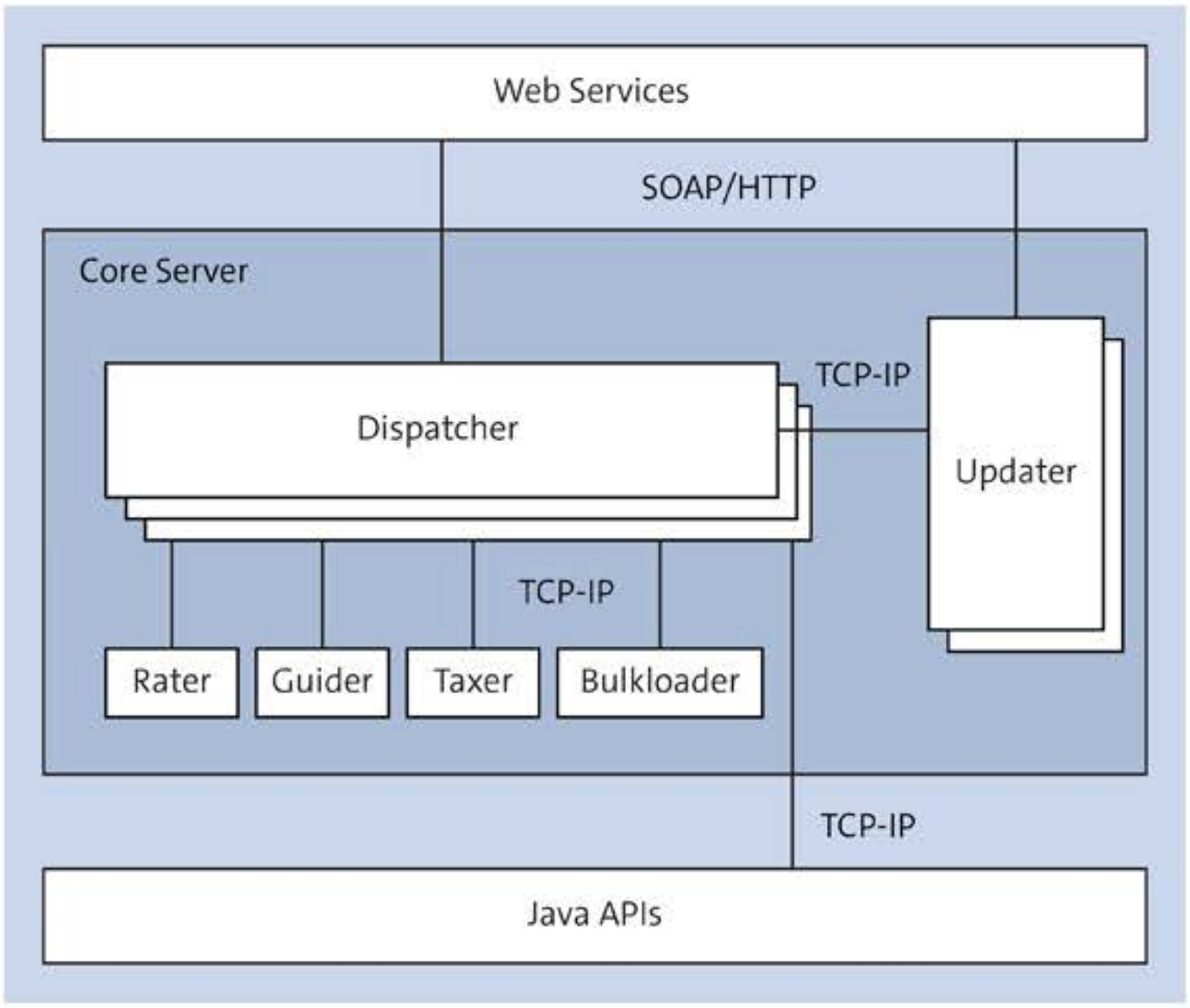

Figure 8.3 SAP Convergent Charging Web Service and Java APIs

SAP Convergent Charging provides several endpoints for accessing web services, which are defined by a Universal Resource Identifier (URI). The URI includes a network address of a specific instance of the SAP Convergent Charging system and a port number dedicated to these communications. Refer to Chapter 3 for detailed information. A sample URL structure of a URI is given as a reference here:

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https://<INSTANCE_HOST_ADDRESS>:<INSTANCE_WS_PORT_NB>/<WS_VERSION>/<WS_TECH_NAME>

The following is a brief description of the components of the URI:

- *INSTANCE_HOST_ADDRESS*
  Network address (domain name server [DNS] name or IP address) of the host server of the active updater instance or the dispatcher instance of SAP Convergent Charging.
- *INSTANCE_WS_PORT_NB*
  The port number used for web services communications for the target instance.
- *WS_TECH_NAME*
  Technical name of the web service.
- *WS_VERSION*
  Version of the web service.

Following are two sample endpoints:

- *http://Your_CC_Upadater_Instance_HostName:9081/v2/contractProvisioning?wsdl*
- *http:// Your_CC_Dispatcher_Instance_HostName:9180/activation?wsdl*

For a complete list of all available services, refer to the **Web Services Documentation** section at the following URL: *http://s-prs.co/v554201*.

Note that the port numbers for updater and dispatcher instances for SAP Convergent Charging are usually 9081 and 9080, respectively. These port numbers are default values from SAP, but you can change the port numbers while setting up your SAP Convergent Charging application.

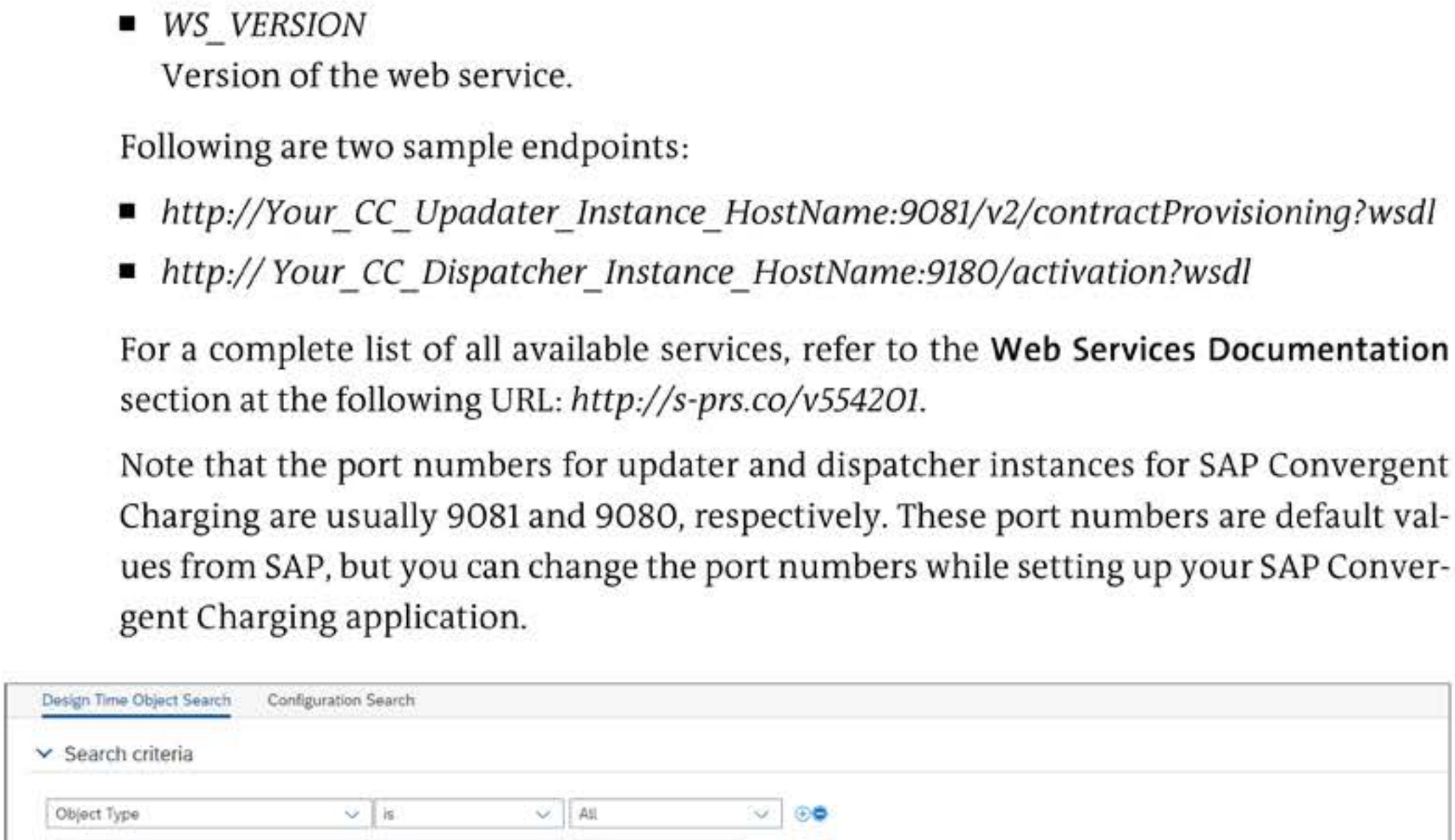

Figure 8.4 Configure SAP Convergent Charging Web Service in Transaction SOAMANAGER

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All the standard services to integrate SAP Convergent Charging with SAP Billing and Revenue Innovation Management, subscription order management, and SAP Convergent Invoicing are provided by SAP. You can search and configure different integration objects between SAP Convergent Charging and SAP S/4HANA, as shown in Figure 8.4.

8.2.2 Managing the Catalog

As discussed earlier, subscription products are maintained in SAP Billing and Revenue Innovation Management, subscription order management. SAP Convergent Charging doesn't store any product information separately. However, SAP Convergent Charging provides services that can be used to extend product master data with pricing information. This process component enables you to assign a released charge plan and refill plan in SAP Convergent Charging with the commercial products stored in SAP Billing and Revenue Innovation Management, subscription order management. The assignment between a product in SAP Billing and Revenue Innovation Management, subscription order management, and a charge plan or refill plan in SAP Convergent Charging is done using cross-catalog mapping. This service also manages operations related to SAP Convergent Charging mapping table and range table maintenance. Following are the operations managed using the services:

- `chargePlanClassFind`
- `refillPlanClassFind`
- `mappingTableCreate`
- `mappingTableMaintain`
- `mappingTableRowMaintain`
- `rangeTableMaintain`
- `rangeTableCancel`

The endpoint is *http://Your_CC_Upadater_Instance_HostName:9081/v1/catalog*.

8.2.3 Manage Contract Provisioning

SAP Convergent Charging will always integrate with external customer relationship management (CRM) or order management applications to capture contract information. SAP Billing and Revenue Innovation Management, subscription order management, is used to create provider contracts with master data information (business partner, contract account, etc.), and then the contract is replicated to SAP Convergent Charging. SAP Convergent Charging offers a standard `contractProvisioning` service that provides the necessary operations to manage provisioning of provider contracts. A provider contract will have the pricing and charging data that is then used by SAP Convergent Charging to generate billable items based on predefined contract terms and conditions. This service also manages changes in provider contracts (change of

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contract attributes, cancellations), and so on. Following are the operations managed using the service:

- `chargingContractMaintain`
- `chargingContractCancel`
- `chargingContractFind`
- `chargingContractMaintainBundle`
- `chargingContractMaintainMass`

The endpoint is *http://Your_CC_Upadater_Instance_HostName:9081/v2/contractProvisioning*.

8.2.4 Managing Contract Activation

One of the prerequisites for SAP Convergent Charging to generate charges based on contract terms is to activate the provider contract. The contract activation service enables external applications to request activation for a provider contract in SAP Convergent Charging. This is an essential step before you plan to run the billing and invoicing process in SAP Convergent Invoicing. However, note that you can only activate charging contracts with an operational status set as **Active**. The operation provided by the service is called `chargingContractActivate`. Once the contract is activated, SAP Convergent Charging can generate the charges based on the billing reference date and process billable items to SAP Convergent Invoicing. The endpoint of the service is *http://Your_CC_Dispatcher_Instance_HostName:9180/activation*.

8.2.5 Managing Chargeable Items

SAP Convergent Charging is capable of processing a high volume of usage data records or consumption data records sent from external source systems. In most implementations, you'll need a mediation system or middleware to integrate with SAP Convergent Charging for usage rating and charging processes. SAP Convergent Charging provides a standard service that can be used to capture and process chargeable items from external sources. SAP Convergent Charging can manage both online and offline charging. The mandatory information required to process the service is `userTechnicalId` (i.e., technical reference ID of the provider contract), `serviceId` (SAP Convergent Charging service ID/type), and `consumptionDate` (date of service consumption). The API is flexible to support custom attributes as key-value pairs within `userProperties`. In SAP Billing and Revenue Innovation Management, the data points are processed in SAP Convergent Charging to generate billable items for SAP Convergent Invoicing.

Following are the different services offered:

- `chargeableItemCharge`
- `chargeableItemChargeMass`

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- chargeableItemChargeBundle
- chargeableItemCheckCharge

The charging operation generates multiple output data files for billing or invoicing purposes. If your billing system is SAP Billing and Revenue Innovation Management, the output files are transferred as billable items into SAP Convergent Invoicing. The transfer is performed by one of SAP Convergent Charging instances, which is called Bulkloader.

The endpoint will be *http://Your_CC_Dispatcher_Instance_HostName:9180/chargeableItemCharging*.

8.3 Extension for SAP Convergent Invoicing

Like the SAP Billing and Revenue Innovation Management components discussed previously, SAP Convergent Invoicing offers standard APIs to integrate with third-party systems. SAP Convergent Invoicing also natively integrates with SAP Convergent Charging. Primarily, SAP Convergent Invoicing acts as the system of record in SAP Billing and Revenue Innovation Management to store consumption items and billable items. In addition, as previously discussed in Chapter 4, SAP Convergent Invoicing is the billing and invoicing module. Based on business needs, SAP Convergent Invoicing is required to integrate with various applications or portals to interface invoice data, integrate with third-party tax solutions, and integrate with other ERP solutions to ingest or export billable transactions. In the following sections, we'll discuss some of the commonly used APIs for SAP Convergent Invoicing.

8.3.1 Application Programming Interfaces for Consumption Items

As discussed in Chapter 4, consumption items refer to usage records that are stored in SAP Billing and Revenue Innovation Management. A metering system tracks the usage consumed by a customer for a given period and interfaces the usage data to SAP Billing and Revenue Innovation Management for billing. These usage records can be fed into SAP Convergent Invoicing using standard API `FKK_BIX_CIT_CREATE_API`. This asynchronous service enables you to create consumption items in SAP Convergent Invoicing with the data provided in a payload of the API call.

Configuration of the consumption item class generates a unique RFC function module for every consumption item class. You use this RFC function module to transfer consumption items for the consumption item class. The naming convention for the function modules is `/1FC/xxxx_CIT_CREATE_API`, where *xxxx* stands for the four-character name of the consumption item class. The generated RFC function module adapts the structure of the consumption item class along with its parameters.

The following parameters return the result of the consumption item transfer:

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- ET_CIT_ERROR: Items with errors.
- ET_CIT0: Items added as raw data.
- ET_CIT2: Items added as ratable.
- ET_CIT4: Items added as rated.

In addition, the system returns the source transactions for the rated items in the corresponding parameters. You can also use FKK_BIX_CITCAT_LIST_GET_API to return a list of the productively usable consumption item classes.

8.3.2 Application Programming Instances for Billable Items

As discussed in Chapter 4, billable items refer to transactional data of a service consumed. A billable item contains information such as quantity consumed, units, master data, amount, and other key information required to further process a billable transaction. In SAP Convergent Invoicing, an out-of-the-box, asynchronous inbound function module FKK_BIX_BIT_CREATE_API can be used to create billable items. It contains details of service consumed, amount, account assignment, and other information required to bill and invoice the billable item. Once the billable item has been created, the source transaction ID is sent in the response with the data included in the billable item. If there are any issues when the billable item is created, the system displays error messages in the response.

Configuration of the billable item class generates a unique RFC function module for every billable item class. The naming convention for the function modules is /1FE/xxxx_BIT_CREATE_API, where *xxxx* stands for the four-character name of the class.

You can also use FKK_BIX_BITCAT_LIST_GET_API to return a list of the productive and usable billable item classes.

8.3.3 Application Programming Interfaces for Customer Invoicing

Once the billable items are processed in SAP Convergent Invoicing, the next step is to process billing, followed by generating a customer invoice. The customer invoice is a crucial transactional document, and customers reference the invoice to pay for the services consumed. Based on business needs, the invoice documents generated in SAP Convergent Invoicing are required to be interfaced to customer-facing portals or payment portals for reference. SAP Convergent Invoicing provides standard APIs to share invoicing document information to external portals. Function module FKK_INVDOC_SELECT_SINGLE_API can be used to interface invoice details for a given invoice document. For a list of invoices, function module FKK_INVDOC_SELECT_API can be used.

You can also use the standard OData API API_CABUSPARTINVOICE to read invoice items for a business partner and read the corresponding invoice headers with this service. For each invoice, you can read correspondence data, including the printed document.

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8.4 Extension for Contract Accounting

Contract accounting offers standard APIs and extension points that can be used to manage business scenarios required to interface with external systems. The most common scenarios of external integrations are payment gateways where the payment information is captured and processed in SAP S/4HANA. SAP Customer Financial Management solution caters to capabilities such as credit management, payment processing, and SAP Collection and Dispute Management. Contract accounting addresses a majority of the aforesaid capabilities. Various standard APIs are available to aid external integrations and to reduce customizations. Receivables management and payment handling within contract accounting primarily deal with large volumes created by billing and invoicing processes, and the APIs are designed to deal with huge volumes. The following list covers some of the common integration scenarios encountered while working with contract accounting:

- **APIs for contract accounting document posting**
  SAP S/4HANA provides standard APIs to post documents in contract accounting from external, non-SAP applications. You can use `BAPI_CTRACDOCUMENT_CREATE` to post a contract accounting document with document attribute values based on document types. During posting of a contract accounting document, general ledger account determination automatically occurs based on the configurations maintained.

  If you have an existing contract accounting document in an SAP S/4HANA system, you can use `BAPI_CTRACDOCUMENT_CHANGE` to update the document. Note that you can only change business partner items using the API.

  If an external application requires information of existing contract accounting documents, you can use standard API `BAPI_CTRACDOCUMENT_GETDETAIL`. The API can be used to read data from a contract accounting document. SAP also provides inbound OData service `CADocumentManage`, which enables external systems and other consumers to read contract accounting document data. The service makes the document data available through its document header, document business partner line item, and document general ledger line-item entities.

- **APIs for open item clearing**
  As you may already know, an open item refers to an invoice item that isn't cleared or paid. Once the invoice is issued to a customer, an open item is created on the customer account until the invoice is settled or paid. Open items in contract accounting can be cleared with an offsetting amount posted on the account.

  SAP Billing and Revenue Innovation Management can be integrated to a non-SAP application or portals that require customer open item information to be displayed for a given period. You can use the standard API `BAPI_AR_ACC_GETOPENITEMS` to interface open items for an account based on selection criteria. The API can provide a customer's open items on a given key date.

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You can also use standard API BAPI_CTRACCONTRACTACCOUNT_CLR to clear open items in contract accounting.

- **APIs for customer account balance**
  SAP provides standard API BAPI_CTRACCONTRACTACCOUNT_GBAL to read account balances for existing customers for a key date. The API can make a query based on item type (open, cleared, statistical), date as well as customer information (business partner, contract account, contract). If you've enabled SAP Biller Direct, you can also use the standard inbound synchronous service ContractAccountAccountBalanceByCustomerQueryResponse_In. This inbound operation returns the current account balances of a specified business partner's contract accounts. The response message returns the following balance items:
  - Total amount for open items
  - Total amount for due items
  - Total amounts for items that are 30, 60, or 90 days overdue
- **APIs for business partner payments**
  There are various inbound and outbound payment processes depending on the type of business, size of business, and country. In a typical implementation, you'll need to integrate with non-SAP applications to execute an end-to-end payment process. SAP Billing and Revenue Innovation Management offers standard APIs such as API_CABUSPARTPAYMENT, which can be used to retrieve a list of payments and make and release payments. The API also helps to read payment items and corresponding payment headers for a business partner. Payment items are either document items that are cleared by payments, payments on account, or released payments. Note that a released payment can also be withdrawn unless the corresponding items are not cleared.
- **APIs for credit management**
  One of the core modules of SAP Customer Financial Management is credit management, which helps companies efficiently determine the risk of losses on receivables from customers by assessing customer credit information. There are also provisions to integrate third-party credit agencies for credit decisions. Usually, the credit check is a standard process that happens before a sales transaction is committed, and it's important to synchronize a customer's credit information from SAP S/4HANA to external or internal applications to make an informed decision.
  SAP S/4HANA provides a standard SOAP-based synchronous API CREDITMANAGEMENTACCOUNTBYIDQU1 to request credit-relevant data, such as the credit limit or the credit exposure of a business partner. The message interface consists of two message types: CreditManagementAccountByIDQuery, which represents a message for requesting credit data, and CreditManagementAccountByIDResponse, which represents a response to the request for credit data. This service contains header, selection, language code, credit management account, and log nodes.

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In addition, inbound OData service `CACreditWorthiness` can be used to enable external systems and other consumers to retrieve creditworthiness data of business partners. This data includes information such as the creditworthiness score, the creditworthiness date, and whether the creditworthiness value is frozen for a business partner. The OData service transfers a list of creditworthiness records for individual business partners.

- **APIs for managing disputes**
  Disputes can be managed effectively using SAP Collections and Dispute Management on a dispute case created for a receivable item in contract accounting. SAP Billing and Revenue Innovation Management offers standard APIs that can be used to create, read, and update a dispute case. It also enables creation or deletion of disputed objects from a dispute case. A disputed object could be a business partner, contract account, contract, or contract accounting documents. Notes and attachments also can be created and read. SAP Billing and Revenue Innovation Management provides a standard inbound OData service `CADisputeCase`.

Now that we've discussed some of the available APIs and integration points, these may not cover all use-cases during your solution implementation. If you require additional or detailed references of SAP Billing and Revenue Innovation Management APIs, go to *https://api.sap.com*, and search with "SAP Billing and Revenue Innovation Management" or an object name to check all available standard API options, as shown in Figure 8.5.

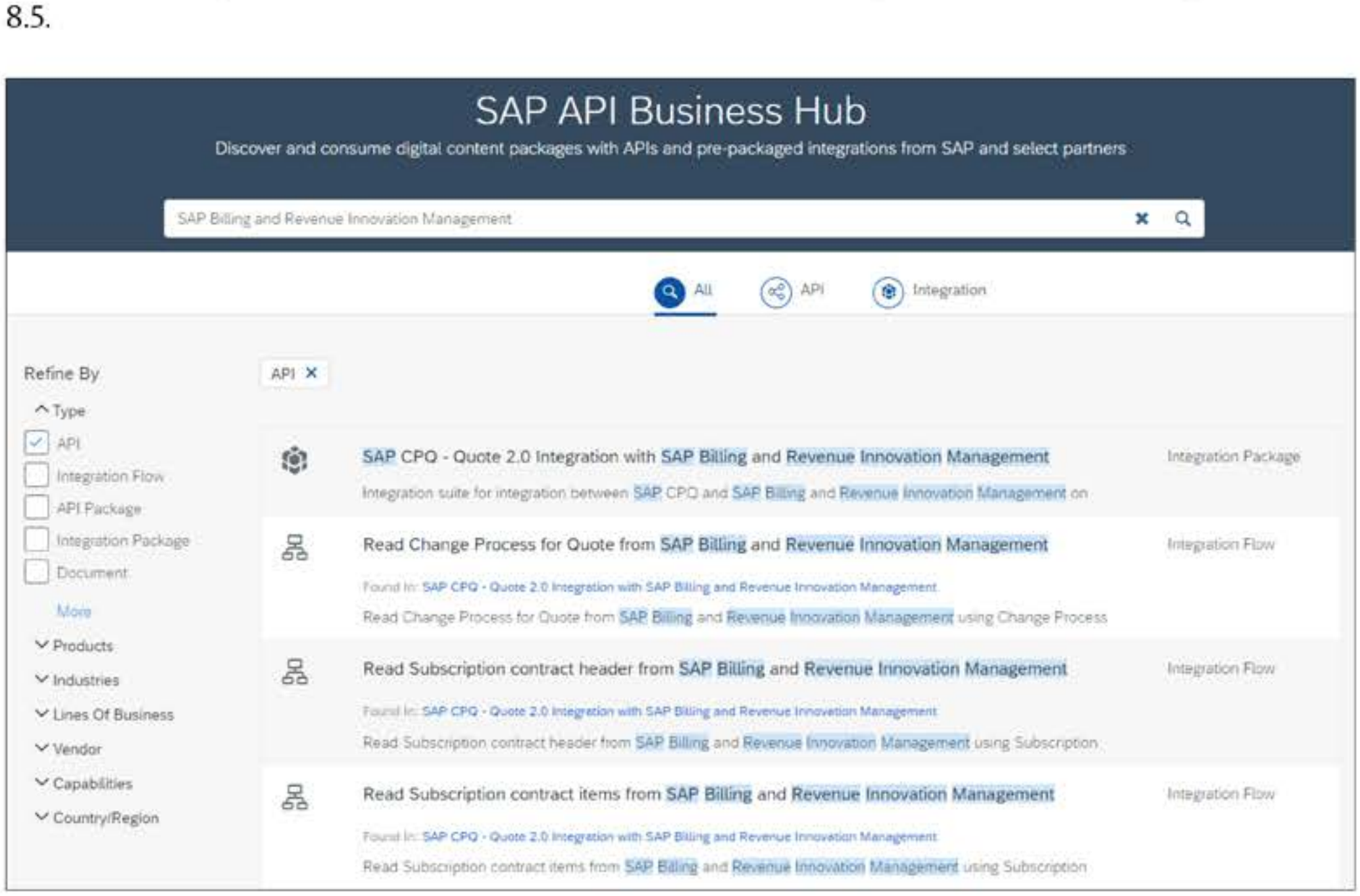

Figure 8.5 SAP API Business Hub

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8.5 Summary

By now, you know that SAP Billing and Revenue Innovation Management uses sets of APIs with different authentication protocols to build applications, integrations with legacy and non-SAP platforms and extensions easily. You've learned to set up, create, and test all the necessary API and integrated objects, and you understand the purpose and functions of these objects and how they play their part in this process. SAP Billing and Revenue Innovation Management is an integrated solution. We therefore recommend working through the other chapters in this book as well, which may prove helpful for reaching your goals.

In the next chapter, we'll discuss how to handle data migration processes while working with SAP Billing and Revenue Innovation Management, including different associated objects and their dependencies.

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Chapter 9

Data Migration

In this chapter, we'll introduce the data migration strategy and various methods you can use to migrate legacy data into SAP Billing and Revenue Innovation Management. We'll use Martex Corp. as an example where we have a traditional non-SAP ERP solution and see how each type of master data and transactional data can be migrated to SAP Billing and Revenue Innovation Management. We'll also describe and execute the necessary configuration steps to manage integrations with SAP Convergent Charging and SAP Convergent Invoicing.

Data migration relates to the process of migrating master data and transactional data along with business process information from one system to another. The migration may be required while transforming a business from a non-SAP ERP solution to SAP, migrating from a hardware infrastructure to a cloud-based system, or upgrading projects. Because each company is unique and may operate with centralized or fragmented data management tools, it's very important to strategize migration based on different parameters such as data volume, applications involved, data quality, and validation process. Note that based on project scope, there could be different objects or methods involved. But this chapter will help you get insight on some of the most used data migration strategies while working with SAP Billing and Revenue Innovation Management. As you're aware now, SAP Billing and Revenue Innovation Management, subscription order management, is included in SAP S/4HANA starting with version 1809, so we'll also cover a quick guide on how to manage migration from SAP Billing and Revenue Innovation Management, subscription order management, based on SAP Customer Relationship Management (SAP CRM) to SAP Billing and Revenue Innovation Management, subscription order management, based on SAP S/4HANA.

In this chapter, we'll primarily focus on the use case of Martex Corp., which is currently using a non-SAP ERP solution that was developed in-house and has a decentralized nature. Martex is facing a lot of issues to consolidate their revenue and manage their audit as it involves manual reconciliation between different systems. Because they have decided to adapt SAP S/4HANA as their desired ERP solution with SAP Billing and Revenue Innovation Management, we'll focus on the different steps involved in data migration and the core data migration objects for an SAP Billing and Revenue Innovation Management implementation.

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9.1 Data Migration Process

Before we start with our data migration execution, let's look at the different phases involved while planning for data migration. There are different ways to plan for data migration, and we can divide different phases of the data migration process to have a better understanding on the planning and execution. The phases can be overlapped with an implementation timeline, and they don't necessarily depend on completion of implementation phases. However, it's important to finalize the end-to-end design to evaluate dependencies of different migration objects. Following are the different phases involved with brief overviews when planning for an enterprise data migration:

- **Discover**
  In this phase, the implementation team should explore different migration tools available to support data migration. SAP provides various migration cockpits and workbenches as part of the standard offering. You can also evaluate the use of SAP Data Services or any other non-SAP extract, transform, and load (ETL) tools for data integration, quality, profiling, and processing. The tool selection also drives the discussion on data validation. Based on requirements, it's also common to develop custom migration programs or reports using the standard SAP function modules and application programming interfaces (APIs) available.
- **Prepare**
  This phase is used to prepare the data migration approach and strategy. This phase primarily involves business process owners as well as stakeholders from legacy platforms along with the implementation and data migration team. Before planning for migration, it's important to plan the data migration approach and strategy. In this phase, business processes are mapped based on underlying relations and dependencies. The data migration team, along with the implementation team, will start preparation for the data migration workshop with the business process and legacy application team. This also involves conducting legacy data health checks to identify potential risks and mitigate using data validation plans. Some of the deliverables of this phase include the following:
  - Data migration strategy document
  - Business process review document
  - Documentation of key business stakeholders
- **Explore**
  In this phase, the team will explore different options to manage the design of the legacy data migration. This design is important to identify object dependency and resourcing. This phase will also evaluate the time taken by each of the different migration scenarios with advantages and dis-advantages. In this phase, you'll also conduct data mapping workshops, data quality assessments, migration approaches, manual processes, and so on.
- **Realize**
  In this phase, the team will execute the data migration. The migration cockpit will be set up so that the same can be used to load into a QA system. In addition, cutover

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planning will also start from this phase. Usually, data is extracted from legacy platforms and then processed through a set of validations to eliminate any bad data and avoid corrections after data load. The data is initially loaded into the quality environment to improve migration performance, identify object and configuration dependencies, and so on. You should also plan for a mock data load in the preproduction environment and compare load reports to ensure that the migration process is working as expected. The primary deliverable from this phase is the cutover plan and data migration mapping document.

- **Deploy**
  In this phase, the team will migrate the data to the production environment. Because we're dealing with the production environment, it's advisable to perform a simulation run prior to start loading all at once. Make sure to conduct a data quality check as well as evaluate volume. For large implementations, it's often a best practice to break the large volume of data in batches.

  Before planning for the data load, make sure that all the production changes and transports are already in the system. Specifically for SAP Billing and Revenue Innovation Management, as it includes integrated systems, it's necessary to ensure that all changes related to SAP Billing and Revenue Innovation Management, subscription order management; SAP Convergent Charging; SAP Convergent Invoicing; and contract accounting are live in the production instance. If there are manual cutover activities, make sure to complete that, too, prior to loading production data. Once the data load is complete, you should generate an error report with all the failed data information and create a generation file for reconciliation. This is an important deliverable from a business standpoint to even start transacting in the new production environment. The failed records, as available in the error report, can be reprocessed after resolving the error messages.

- **Live system**
  In this phase, the new application is ready to be used by business users and customers to run. This phase might require a few weeks of IT support, which is typically called Hypercare.

As we discussed the different phases involved in data migration, the timeline for each phase is dependent on the project scope, volume, and complexity involved. The planning for data migration should involve a specialized team of data migration experts to mitigate any risk and planning for the mitigation plan. Though the timeline can be finalized based on different parameters as discussed previously, it's very important to execute each phase to avoid any unforeseen trouble later.

9.2 Data Migration Objects

As part of supporting data migration for Martex Corp. with the SAP Billing and Revenue Innovation Management implementation in SAP S/4HANA, we'll manage the following master data and transactional data objects in this chapter:

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- Master data objects:
  - Customers (or business partners)
  - Contract accounts
  - Products
- Transactional data objects
  - Subscription order/subscription contract
  - Open accounts receivable
  - Payments

To start with the SAP Billing and Revenue Innovation Management transactional data, master data for customers, products, and contract accounts is a must. Existing business contracts can be migrated, along with open account receivables, if any, depending on business scenarios. The order of sequence for data migration should be business partner, products master data, subscription-specific product master data, existing business contracts, and then accounts receivables, if any, as there are dependencies among other objects.

Figure 9.1 shows a high-level object dependency as well as sequence of object loads for SAP Billing and Revenue Innovation Management implementations.

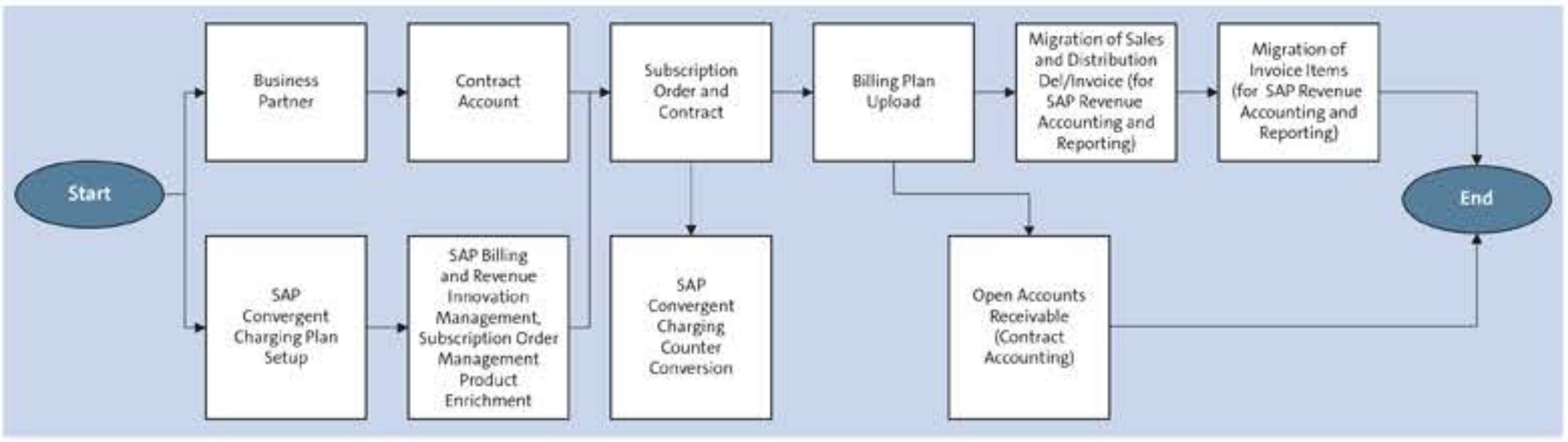

Figure 9.1 SAP Billing and Revenue Innovation Management Data Migration Objects with High-Level Dependencies

9.3 Data Migration Types

There are different ways of handling data migration while implementing SAP Billing and Revenue Innovation Management, and the migration strategy is primarily dependent on business use cases, dependencies, project scope, volume, and other attributes. In some cases, the implementation team can make use of SAP BODS, which is enabled with an additional license; use custom programs developed using standard APIs; or use the Industry Solutions Migration Workbench (ISMW).

In this section, we'll primarily describe the migration plan using ISMW as a standard migration tool. We'll discuss the standard migration APIs that can also be used to

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develop a custom program to manage data migration. Migration using SAP BODS or detailed ISMW execution isn't in the scope of this chapter.

9.3.1 Creating a Company Using Industry Solutions Migration Workbench

To use ISMW, the first thing you must do is create a company. A company can be referred to as a migration project scope. A default company called SAP is provided that contains all standard migration objects. However, for our Martex migration, we'll create a new company.

To create a new migration company or project, you can navigate to **IS-U Migration • Company Maintenance** and then maintain the details for **Company** "MRTX" (Martex), as shown in Figure 9.2 and Figure 9.3.

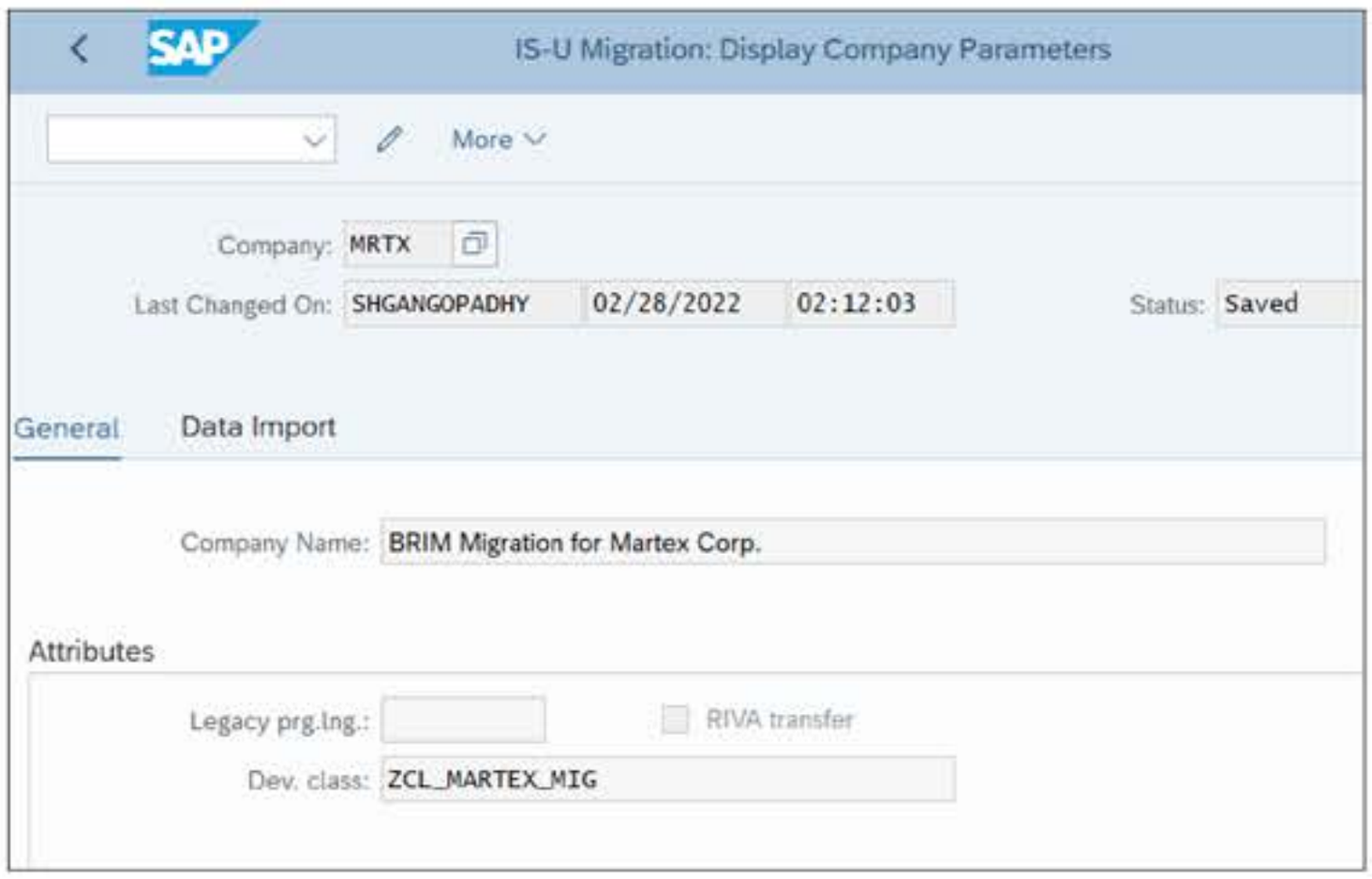

Figure 9.2 Maintenance of Martex Company: General View

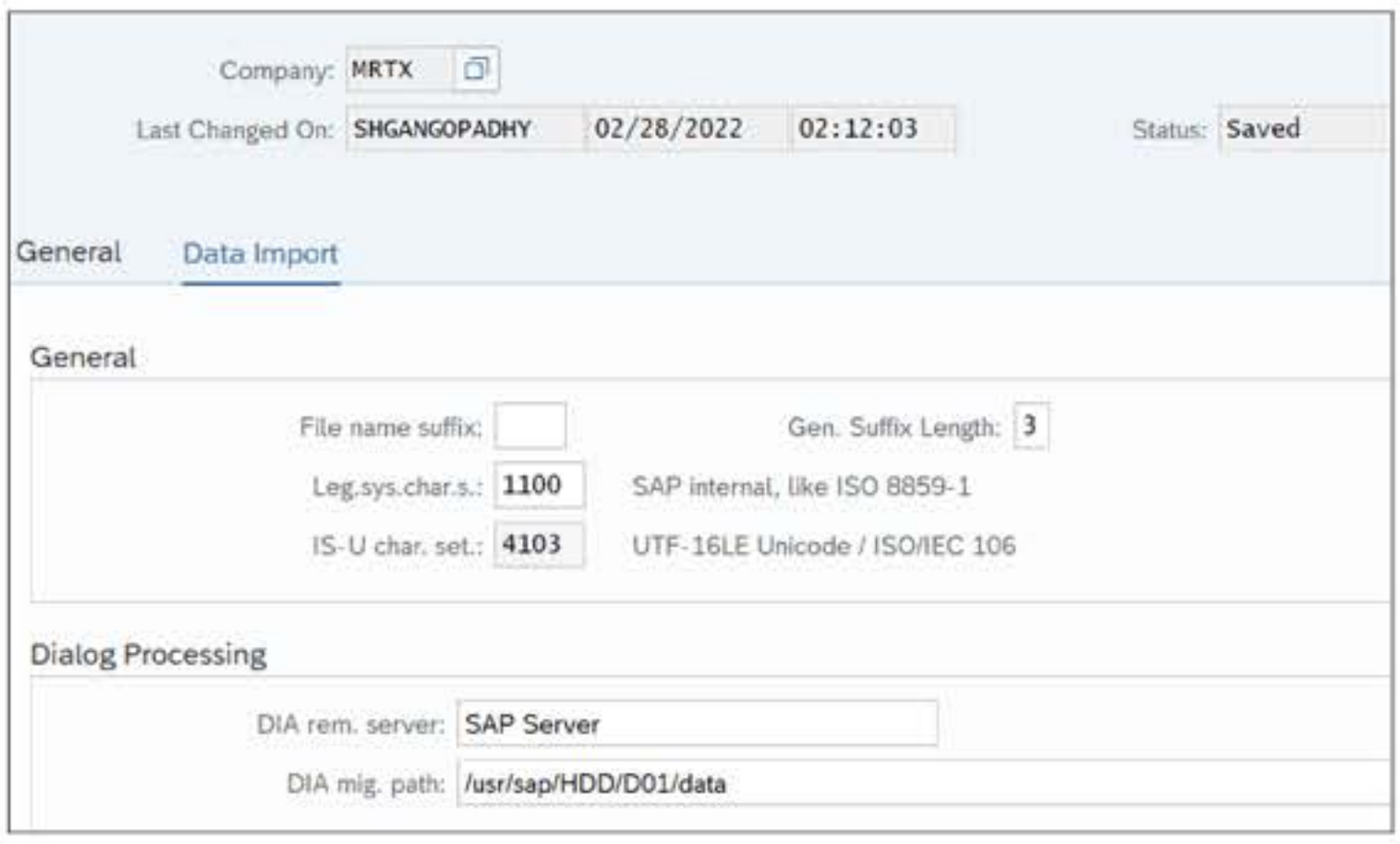

Figure 9.3 Maintenance of Martex Company: Data Import View

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Now, re-execute Transaction EMIGALL, and, this time, select **MRTX** as **Company**. In your workbench, you won't find any migration objects because you haven't yet added migration objects based on your needs.

9.3.2 Customer Migration from Legacy Systems

Business partners represent all the parties involved in a business transaction, such as persons, groups of persons, and organizations. Every company will have a set of customer records that are required to be loaded into the system as part of migration. Before you start the migration, be sure to extract all active customers from the legacy platform to perform data validation. Some clients may choose to also migrate archived customers to have a historical reference. As part of the SAP Billing and Revenue Innovation Management migration, the following roles are required to be assigned during data migration

- FLCU01
  Customer role for all business partners for which transactions will be created/migrated.
- FLCU00
  Role to manage partners for sales and distribution sales processes.
- MKK
  Contract partner role, specifically for SAP Billing and Revenue Innovation Management implementations. Without this role, business partners can't be used in contract accounting.
- FLVN00
  Vendor role for partners, typically used in partner settlement scenarios.

Now, because you're trying to migrate business partners, you need to copy the standard migration object for business partner available in the SAP company to our Martex company. To proceed, select **IS-U Migration**, and then select **Other Company**. Now, based on the standard list of migration objects, select the desired migration object (in this case, **PARTNER**), then select **Migration Object** from the menu bar, and select **Copy**. Here, choose to migrate to your **Target Company** (in this case, **MRTX**), and then select **Copy**. Now, navigate again to your company **MRTX**, and you'll see that the **PARTNER** object is added as shown in Figure 9.4.

Following are some of the prerequisites for business partner migration:

- Company code and sales area data should be maintained in SAP S/4HANA.
- Settings related to business partner are completed, and the number range is maintained in the production environment.

Business partner information is usually uploaded in SAP S/4HANA first and then it's replicated to SAP Convergent Charging as a subscriber account. Figure 9.5 shows the data replication scenario while migrating business partner information.

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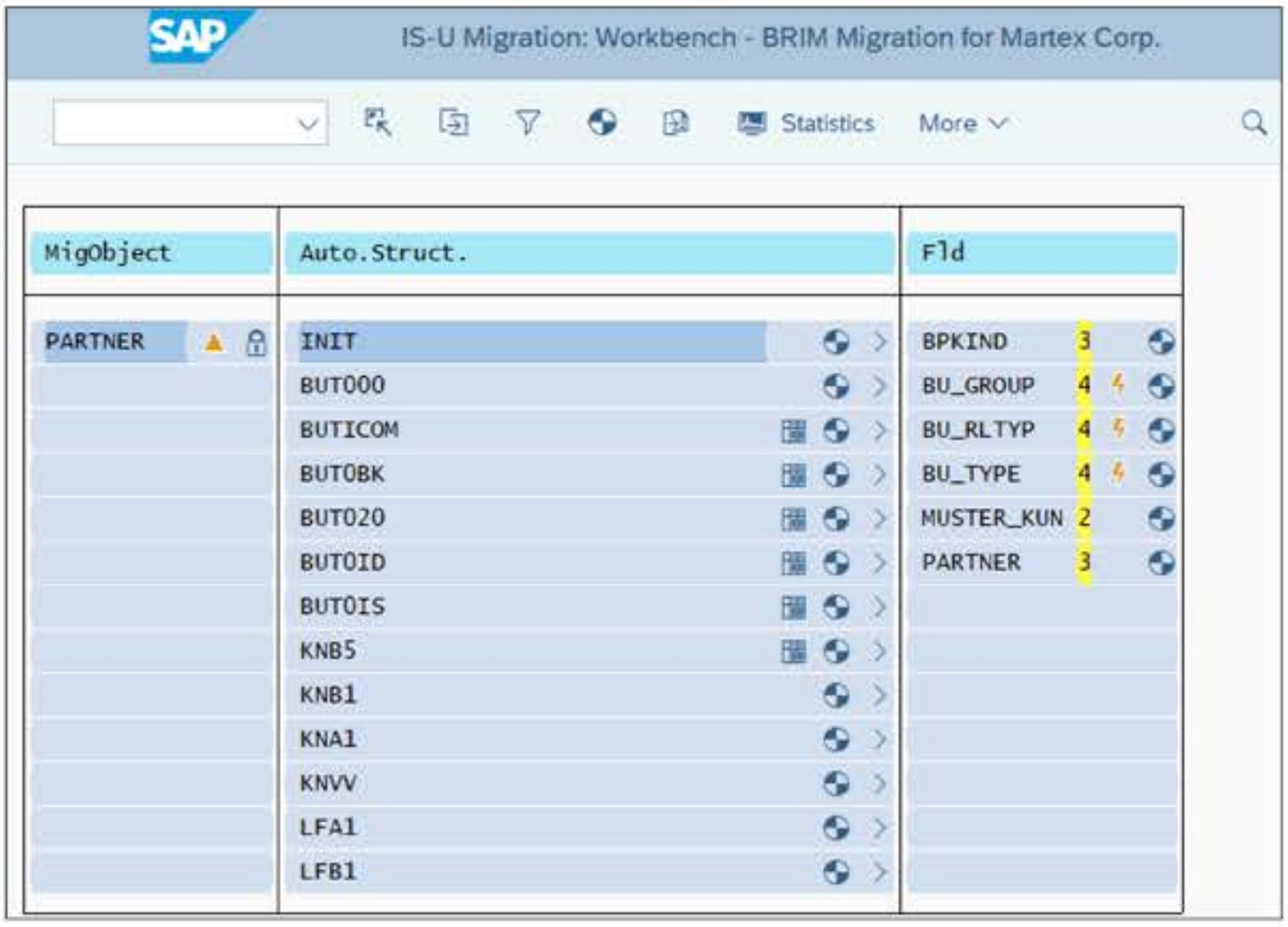

Figure 9.4 PARTNER Object for Company Martex (MRTX)

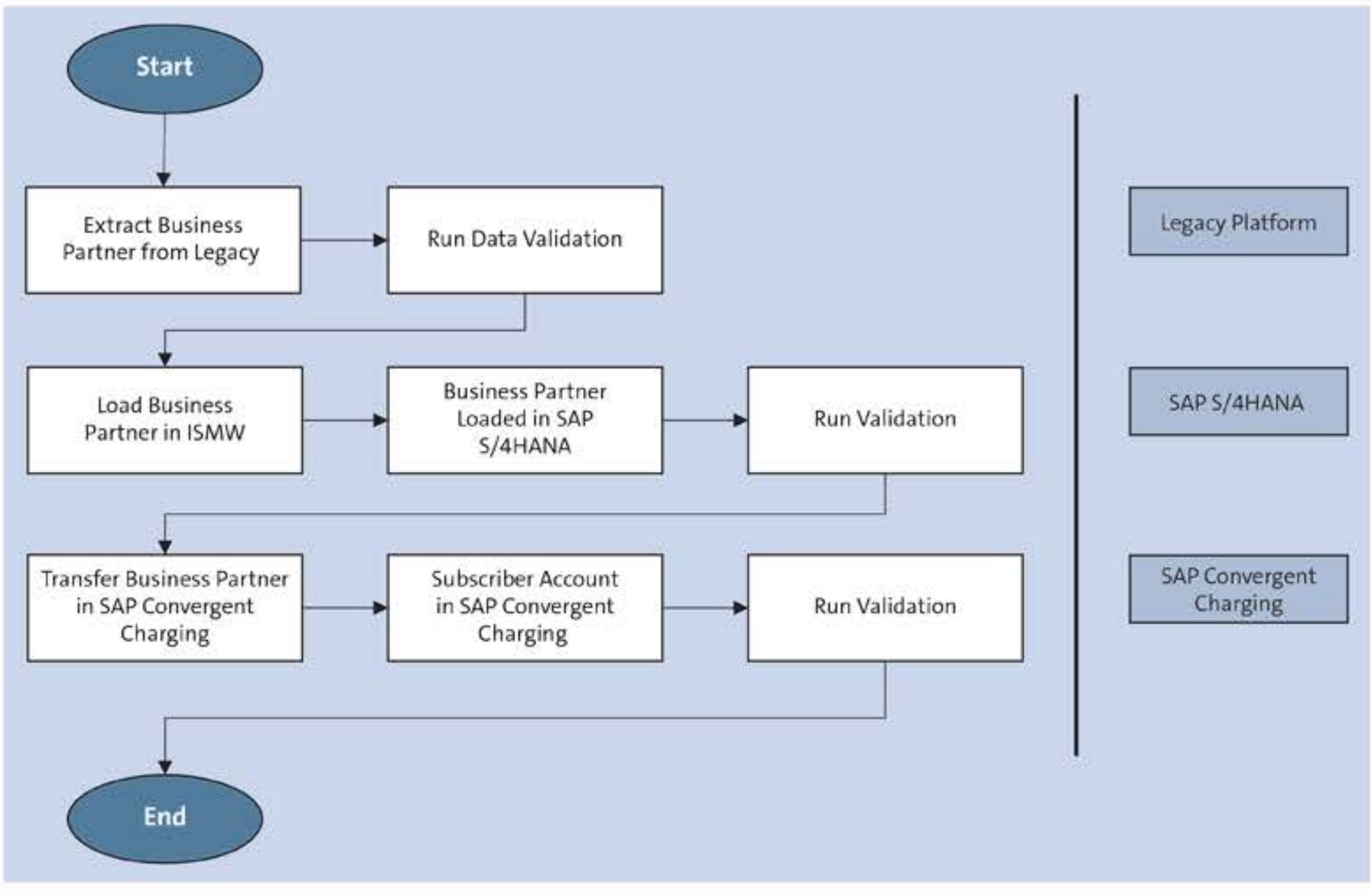

Figure 9.5 Migration of Business Partner

During the business partner migration to SAP S/4HANA using ISMW, the `CC_REPL_OFF` parameter ensures that all migrated business partners are stored in database table `DFKK_CC_MD_MIG`. Once your data migration in SAP S/4HANA is complete for both business

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partners and related contract accounts, you can then use Transaction FKKCC_MD_MIG, which, based on the user selection criteria, can migrate the information in SAP Convergent Charging. The transaction internally queries table DFKK_CC_MD_MIG to migrate master data information in SAP Convergent Charging. To turn off the migration, execute Transaction EMIGALL, and then navigate to **IS Migration • Settings • SAP Settings**. You can set the control parameter CC_REPL_OFF to the value X. This, however, only restricts replication for migration by using ISMW. For manual order creation, using SAP Fiori or APIs, the parameter will have no effect.

Note that while performing migration with SAP Convergent Charging, it's important to migrate rating area information. However, it's also possible to migrate rating areas for master data (business partners, contract accounts) during migration of provider contracts by completing the migration of all SAP Convergent Charging master data objects in SAP S/4HANA (business partners, contract accounts, and provider contracts) and then starting with SAP Convergent Charging migration. There is no predefined object to manage rating area migration, so you'll need to create a new rating area object for partner by using function module FKK_MIGRATION_RATING_AREA. While setting up the migration object, if your project landscape includes a single SAP Convergent Charging system, you can mark a SPACE for the field RATING AREA. You can also generate a custom SAP S/4HANA report with standard function module BAPI_BUPA_CREATE_FROM_DATA for business partner migration. Ensure that you develop simulation, log file, and reprocessing mechanisms in case you're planning to develop a custom report for migration.

9.3.3 Contract Account Migration

The contract account is a critical master data element, and it's mandatory to have a contract account corresponding to the business partner for customers that need billing and invoicing for financial transactions while using the SAP Billing and Revenue Innovation Management solution in SAP S/4HANA. One business partner can have multiple contract accounts based on the requirements, such as different company codes, business units, alternate payers, and so on. If the legacy platform doesn't have contract account information, it's important to design contract account management in the SAP S/4HANA system. Based on different scenarios, you can decide to create contract accounts for individual business partners. Usually, contract accounts are loaded with all posting receivables and payables.

The prerequisites for contract account migration include the following:

- Business partner has been created with the business partner general role and contract partner role.
- Contract account–related settings are completed in production environment.
- The number range for the contract account is maintained.

Now, because you're migrating a contract account, you need to copy a standard migration account. To proceed, select **IS-U Migration • Other Company • SAP**. Now, based on the standard list of migration objects, select the desired migration object (here,

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ACCOUNT), select **Migration Object** from the menu bar, and then select **Copy**. Choose the option to migrate to your **Target Company** (here, **MRTX**), and then select **Copy**. Now, navigate again to your company **MRTX**, and you'll see that the **ACCOUNT** object is added, as shown in Figure 9.6.

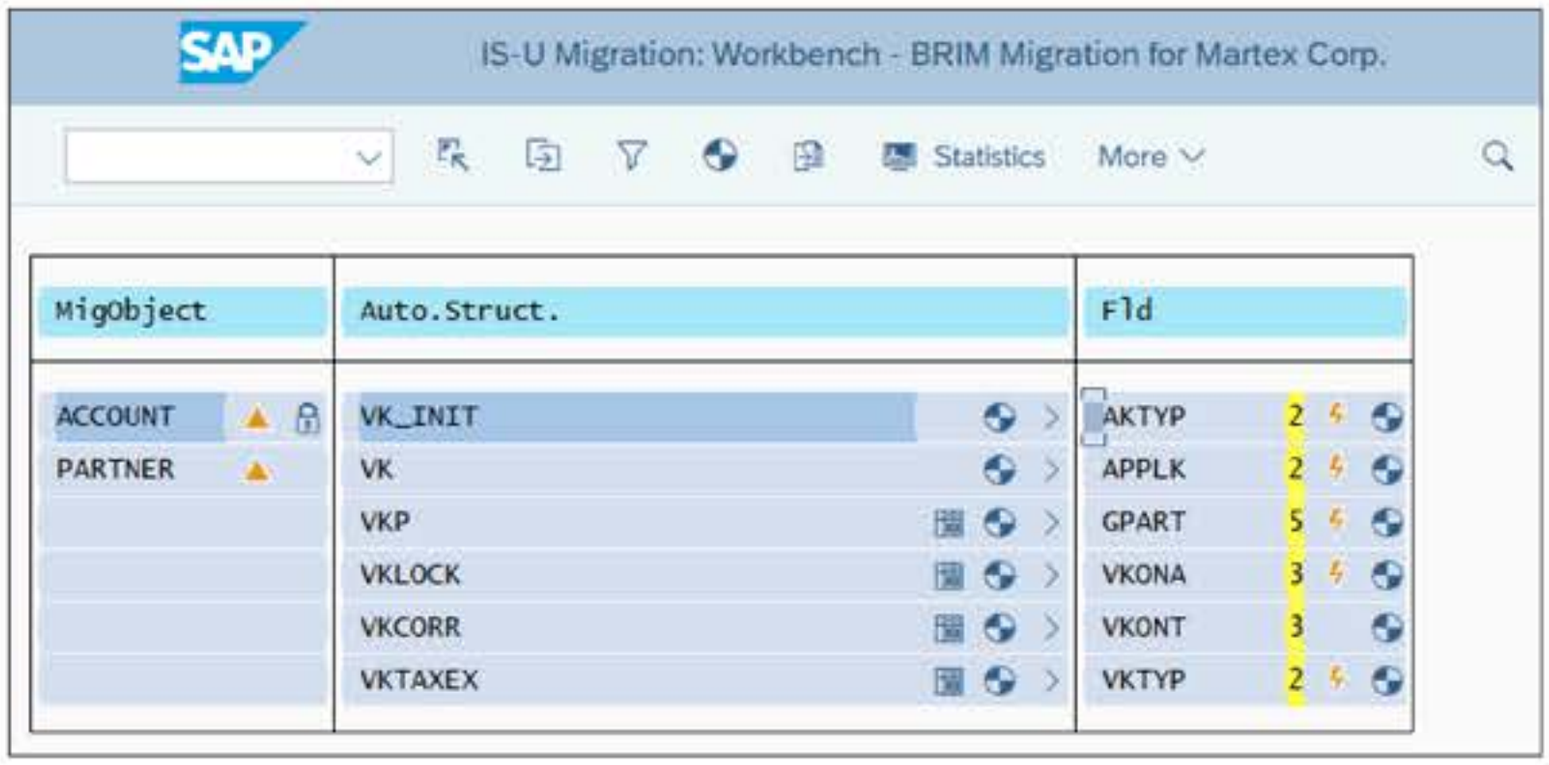

Figure 9.6 ACCOUNT Object for company Martex

To save the legacy identification of contract accounts, you can use the standard **Account in Legacy** field. Contract account information is usually uploaded in SAP S/4HANA first and then it's replicated to SAP Convergent Charging as an external account. Figure 9.7 shows the data replication scenario while migrating contract account information.

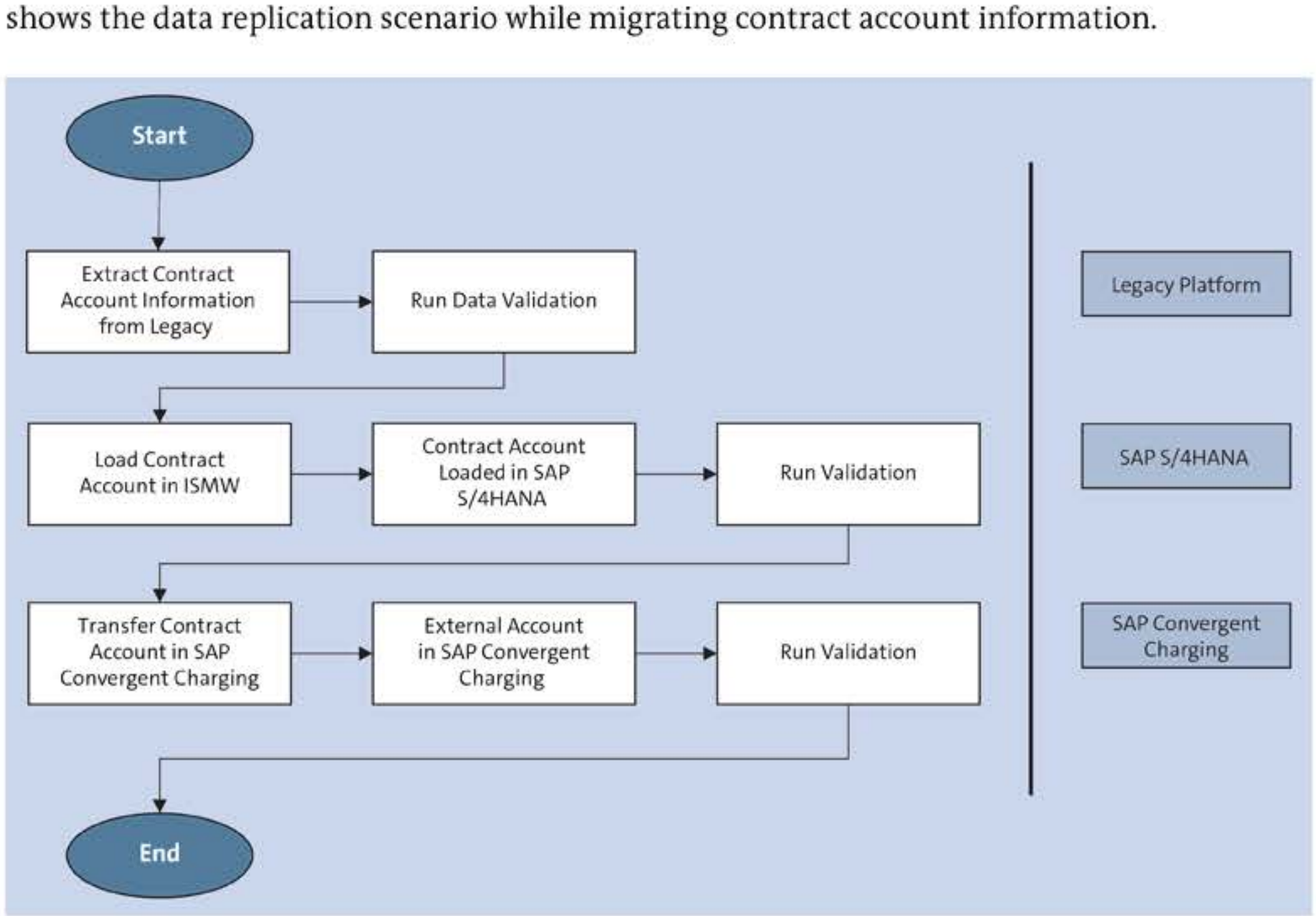

Figure 9.7 Migration of Contract Account

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As an alternative, it's also possible to develop a custom migration program using the standard contract account creation API BAPI_CTRAACCONTRACTACCOUNT. While generating a custom report, make sure to enable a trace report and log for data corrections and validations.

Once the business partner and contract are migrated and validated into SAP S/4HANA successfully, you can now distribute the master data–related objects to SAP Convergent Charging using Transaction FKKCC_PC_MIG. The transaction makes use of mass-capable web service operations to replicate multiple objects into SAP Convergent Charging. Note that this transaction should not be executed until both business partners and contract accounts are migrated to SAP S/4HANA. Using this transaction, you can even replicate various ranges of business partners in parallel jobs. It's possible to define the ranges using the business partner number and custom selections, as shown in Figure 9.8.

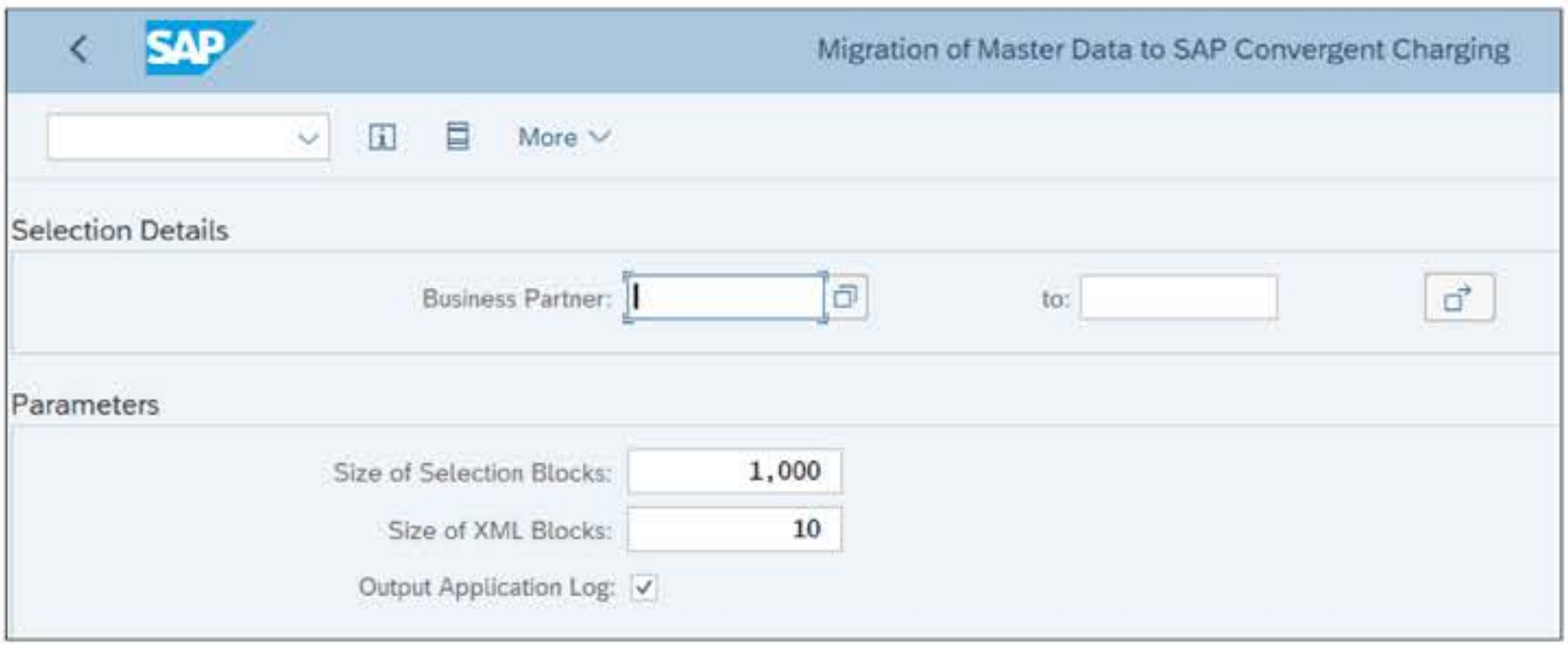

Figure 9.8 Migration of Master Data to SAP Convergent Charging

9.3.4 Subscription Product Migration

Now that business partners and contract accounts are loaded, the next master data object will be products. While using SAP Billing and Revenue Innovation Management in SAP S/4HANA, if you're working with recurring, usage, or partner settlement scenarios, you can use subscription products to manage your business models. Depending on the business scenarios, you can create different types of products. Because SAP Billing and Revenue Innovation Management, subscription order management, is now integrated in SAP S/4HANA, the solution uses traditional product/material governance processes in SAP S/4HANA. The product creation for subscription products is very similar to hardware materials with additional attributes supporting subscription-specific use cases. For example, contract duration and contract extension terms are specifically provided to support transactions using subscription scenarios, which won't be applicable for traditional hardware products. Similarly, subscription products are usually nontangible, so plant-specific information (essential for setting up hardware products) isn't applicable while setting up or migrating subscription products.

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The prerequisites for subscription product setup include the following:

- Configuration related to subscription products should be available in production. If the settings define using SAP S/4HANA number ranges for product IDs, then the product number range should be maintained in the system.
- Enterprise structure and related configuration should be available in production.
- If you're using SAP Convergent Charging, then the latest version of the charge plan should exist in SAP Convergent Charging.
- Maintain the technical resource type for the assignment schema (if you're using usage business models).

Because subscription products require additional steps for enrichment, you must perform multiple dependent steps to complete the migration. Note that unlike business partner and contract account, SAP Convergent Charging doesn't store subscription product details. Figure 9.9 shows the data replication scenario while migrating subscription products.

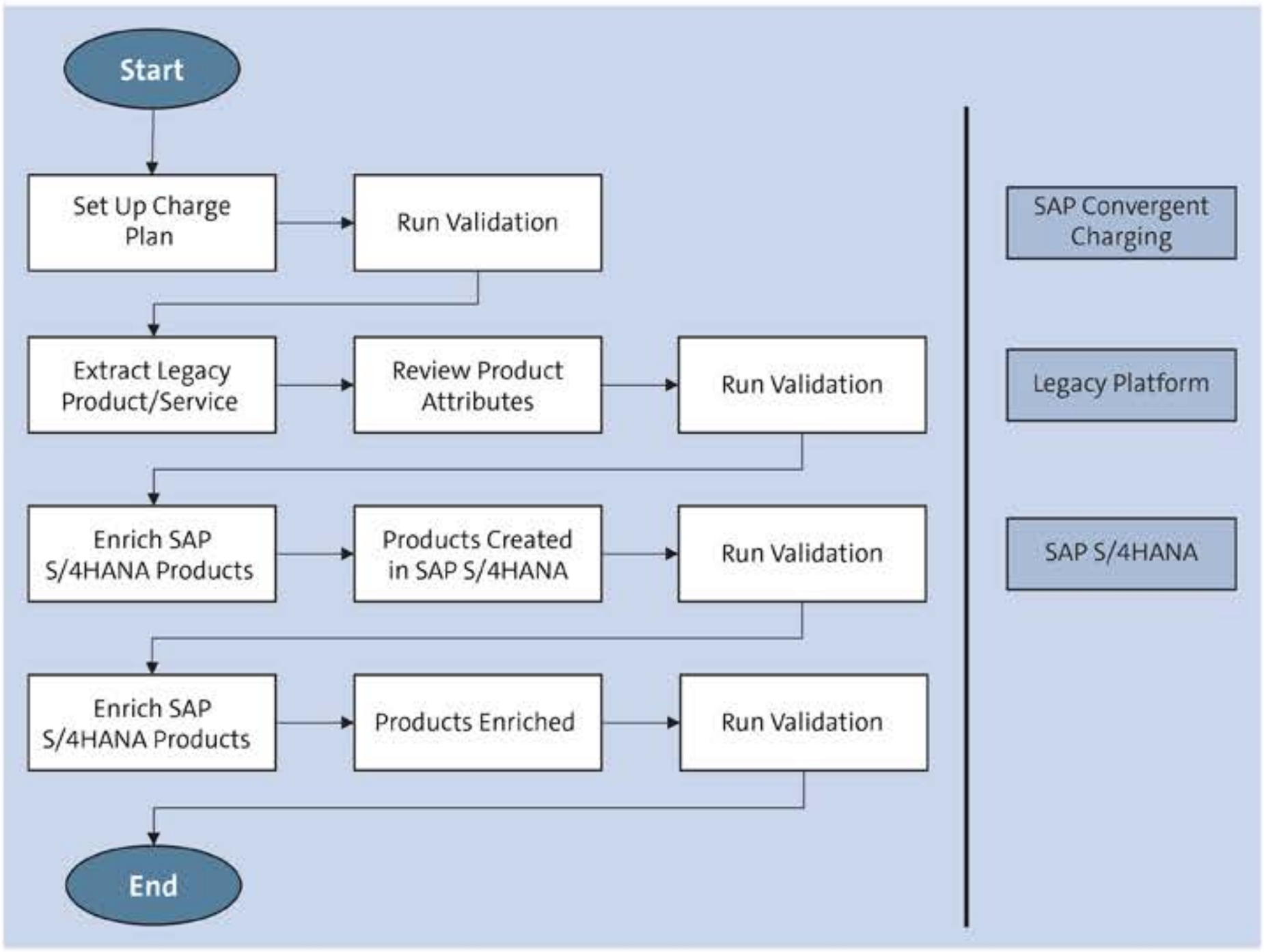

Figure 9.9 SAP Billing and Revenue Innovation Management Subscription Product Migration

There is no existing migration object available for subscription product migration so you'll need to create a migration object in Transaction EMIGALL. Follow these steps in sequence to do so:

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1. **Create subscription products in SAP S/4HANA.**
   In this step, we'll focus on creating the subscription materials without subscription-specific attributes in SAP S/4HANA. You should create a migration object "PRODUCT1" using the standard function module `BAPI_MATERIAL_SAVEREPLICA`. The function module can migrate all standard product data (e.g., material master) in SAP S/4HANA. The function module calls `MATERIAL_MAINTAIN_DARK`, which creates the material in SAP S/4HANA. The commit control takes place in ISMW. Once the products are created using the function module, you should get the list of subscription products for your next migration steps.
2. **Update the subscription-specific attributes in SAP S/4HANA.**
   Once subscription materials are generated and reference IDs are available, you need to update the existing materials to update subscription-specific attributes (e.g., billing cycle, contract terms, extension terms, etc.). To perform this step, you need to create a new migration object "PRODUCT2" using the function module `CRMS4_SOM_MATERIAL_UPDATE`. This function module can update table `MARA` (Billing Cycle Information) and table `MVKE` (Contract and Extension Periods). The function module calls the class method `CL_CRMS4_SOM_PRODUCT_API -> UPDATE_PRODUCT`, which updates the material with subscription-specific attributes.
3. **Perform product enrichment (cross-catalog mapping, etc.).**
   Once subscription materials are generated and subscription-specific attribute data is maintained, you need to perform the next steps for subscription-specific enrichment. This step is essential if you're using SAP Convergent Charging. Although you can use the standard SAP Billing and Revenue Innovation Management, subscription order management, app in SAP Fiori, if the volume is high, you can follow this process to enrich the products. Again, because there is no standard migration object, create a new migration object "PRODUCT3" using the function module `CRMS4_SOM_SPEC_PROD_UPDATE`. This function module is used to update the SAP Billing and Revenue Innovation Management, subscription order management–specific product data (cross-catalog mapping, charge plans, parameters, external objects, and discounts/charges). The function module internally calls the class method `CL_CRMS4_SOM_PROD_SPEC_API->CREATE_PRODUCT_SPECIFIC`, which updates the material with SAP Convergent Charging data (classic contract management [CCM] data). The commit control takes place in ISMW.

Once these steps are complete, you can start using the products to load subscription orders or contracts. The APIs mentioned can also be used to generate custom programs instead of using ISMW. Because the steps are interdependent, it's very important that the execution is in sequence and validated after each migration step.

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9.3.5 Subscription Order and Contract Migration from Legacy Systems

When migrating from legacy platforms to SAP S/4HANA, it's very important to migrate all open transactions/subscriptions from the legacy platform.. Because subscription lifecycles span across months or years, you must refer to the original terms and conditions with which the subscription was created in the legacy platform. In addition, there could be a number of changes (e.g., change in price, change in billing cycle, etc.) that might have been recorded on the subscription in the legacy platforms, which should be retained once the transactional data is migrated into SAP S/4HANA. To maintain business continuity, legacy contracts need to migrate to the SAP Billing and Revenue Innovation Management system. Depending on the business use cases, various types of subscription order management documents can be leveraged. The most-used transactions in SAP Billing and Revenue Innovation Management implementations are subscription orders and contracts, master agreements, and so on. In this case study, you're migrating the subscription orders and contracts for Martex Corp.

Note that during the migration of subscription contracts or provider contracts, only the active and most recent status of contracts are considered. Changes to the contract in the past, which led to multiple change orders for the lifecycle of the provider contract in the legacy system, aren't considered for transfer. You can opt for custom enhancements if there is a requirement to store historical contract changes.

The prerequisites for migration of open accounts receivable items include the following:

- Baseline configuration related to SAP Billing and Revenue Innovation Management, subscription order management; SAP Convergent Invoicing; and SAP Convergent Charging is completed.
- Number ranges for subscription order/contract and any other transaction type in scope for migration should be maintained in the production environment.
- Master data (i.e., business partner, contract account, product) should be available in the system.
- Integration should be tested successfully between SAP Billing and Revenue Innovation Management, subscription order management; SAP Convergent Invoicing; and SAP Convergent Charging.
- If migration includes configurable products, make sure to migrate the variant pricing conditions together with the variant configuration characteristics. Variant conditions are stored as one of the characteristics used during the price determination.

Figure 9.10 provides high-level details of subscription order and subscription contract migration.

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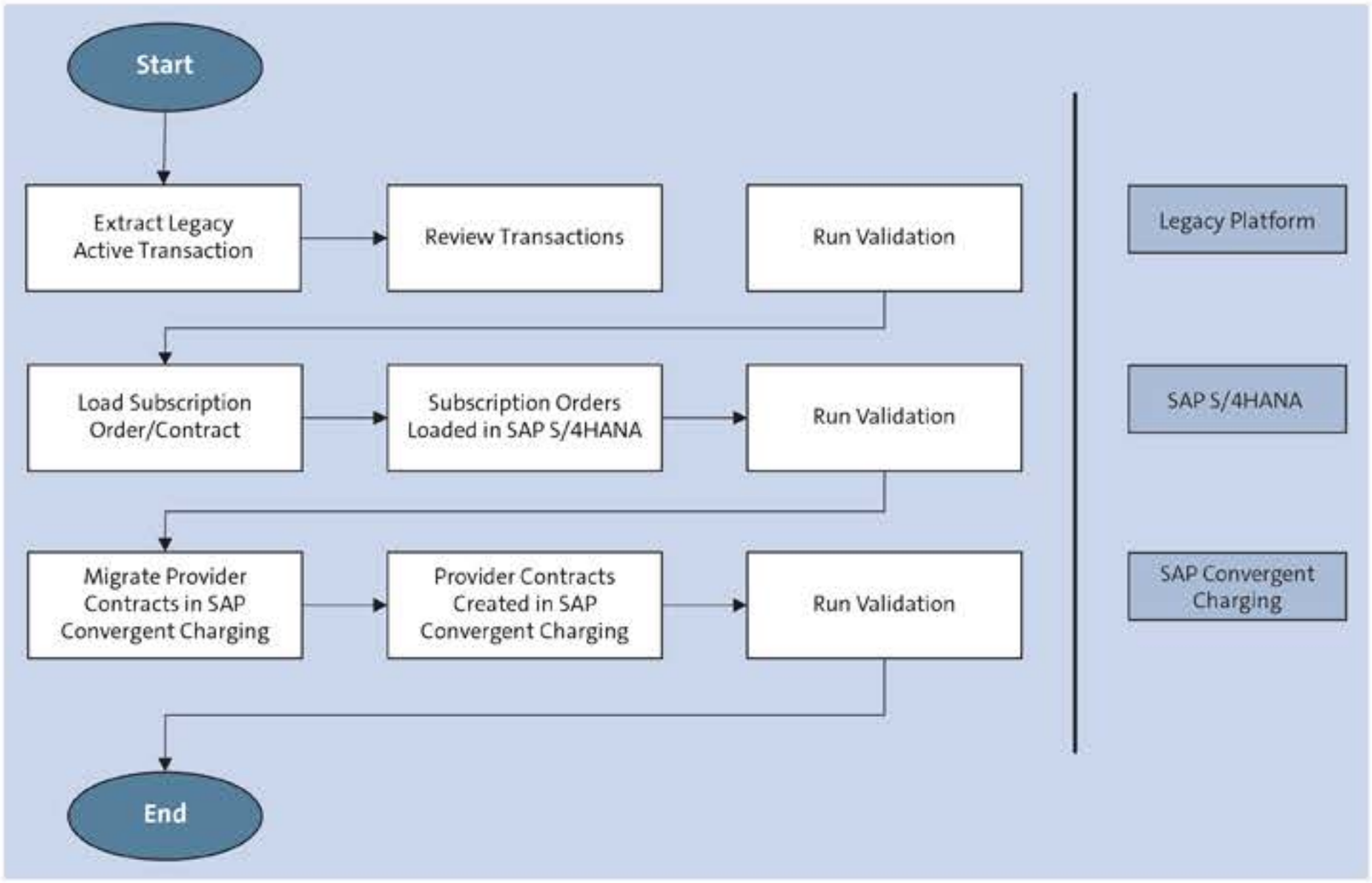

Figure 9.10 Subscription Transaction Migration

There are different ways of migrating subscription orders. The most common way is to reuse the standard API `CRM ISX BTX API ORDER CREATE` for order creation. This API is flexible to support various methods, which helps to migrate the documents easily. The API contains importing parameter `IS_CONTROL`, which contains control indicators to control the order/transaction creation process. Some of the important indicators are listed here:

- **`SUBMIT_ORDER`**
  This indicator helps to set the corresponding status of the subscription order to create the subscription contract. This functions the same as creating subscription orders manually using the **Submit** button in the Manage Subscription Orders application.
- **`SAVE_ORDER`**
  You can use this option to prevalidate the subscription order before loading the transactions in batches. If you only check this indicator and uncheck `SUBMIT_ORDER`, it won't release the subscription contract to SAP Convergent Charging or SAP Convergent Invoicing. Note that once subscription orders are submitted, there is no option to revert released contracts if there are data issues. The only available option is to use change process cancellation or mass cancellation.
- **`SAVE_WITH_ERRORS`**
  This indicator helps to record transactions with errors, so that you can troubleshoot the order/transaction further.

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If the integrations with SAP Convergent Charging and SAP Convergent Invoicing are established, then the subscription contract will auto-distribute to SAP Convergent Charging and SAP Convergent Invoicing. You don't have to migrate provider contracts separately into SAP Convergent Charging.

You can also migrate subscription contracts using ISMW. There is no standard migration object available for subscription orders, so you need to create a new migration object (**Mig. object** field) called "SUBSCRIPT" using Transaction EMIGALL, as shown in Figure 9.11 and Figure 9.12. The function module to use for the migration is CRMS4_SOM_SUBSCRPN_CONTR_MIG.

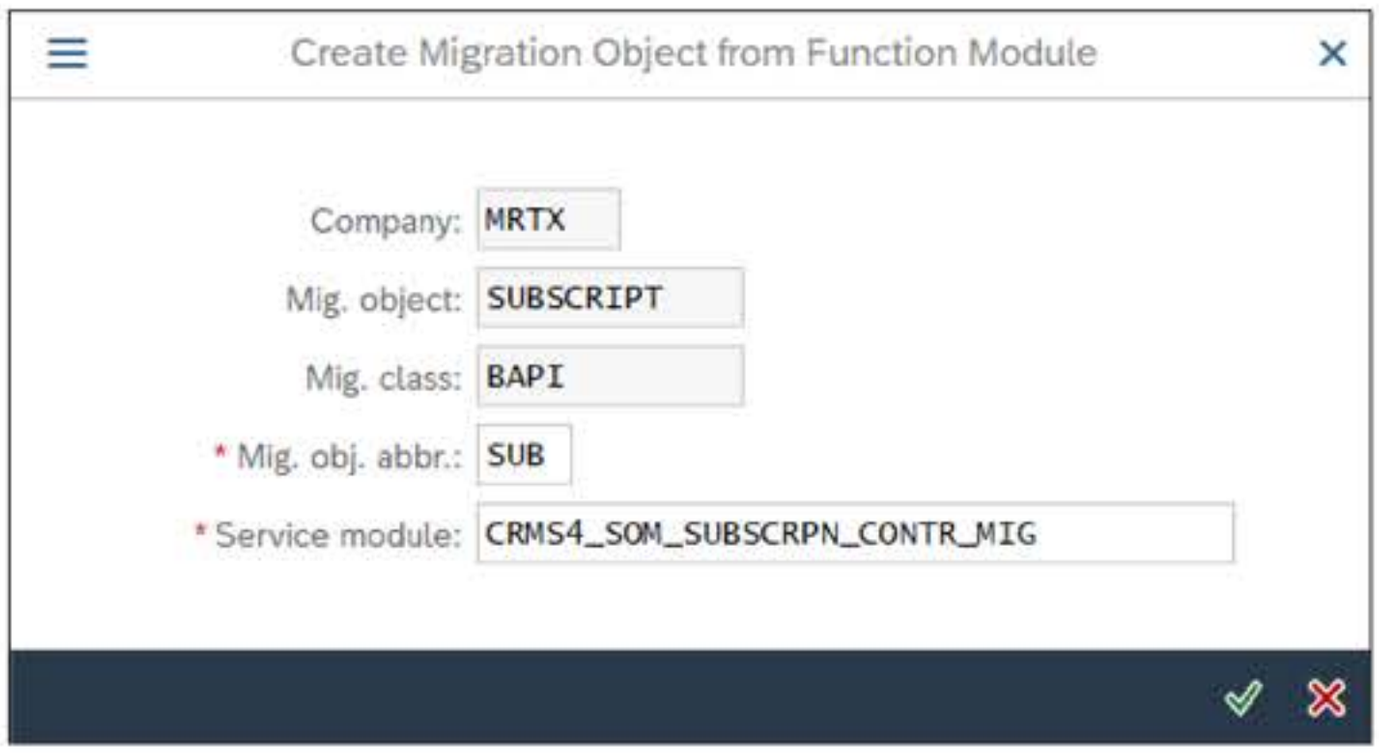

Figure 9.11 Creation of Migration Object SUBSCRIPT to Migrate Subscription Contracts

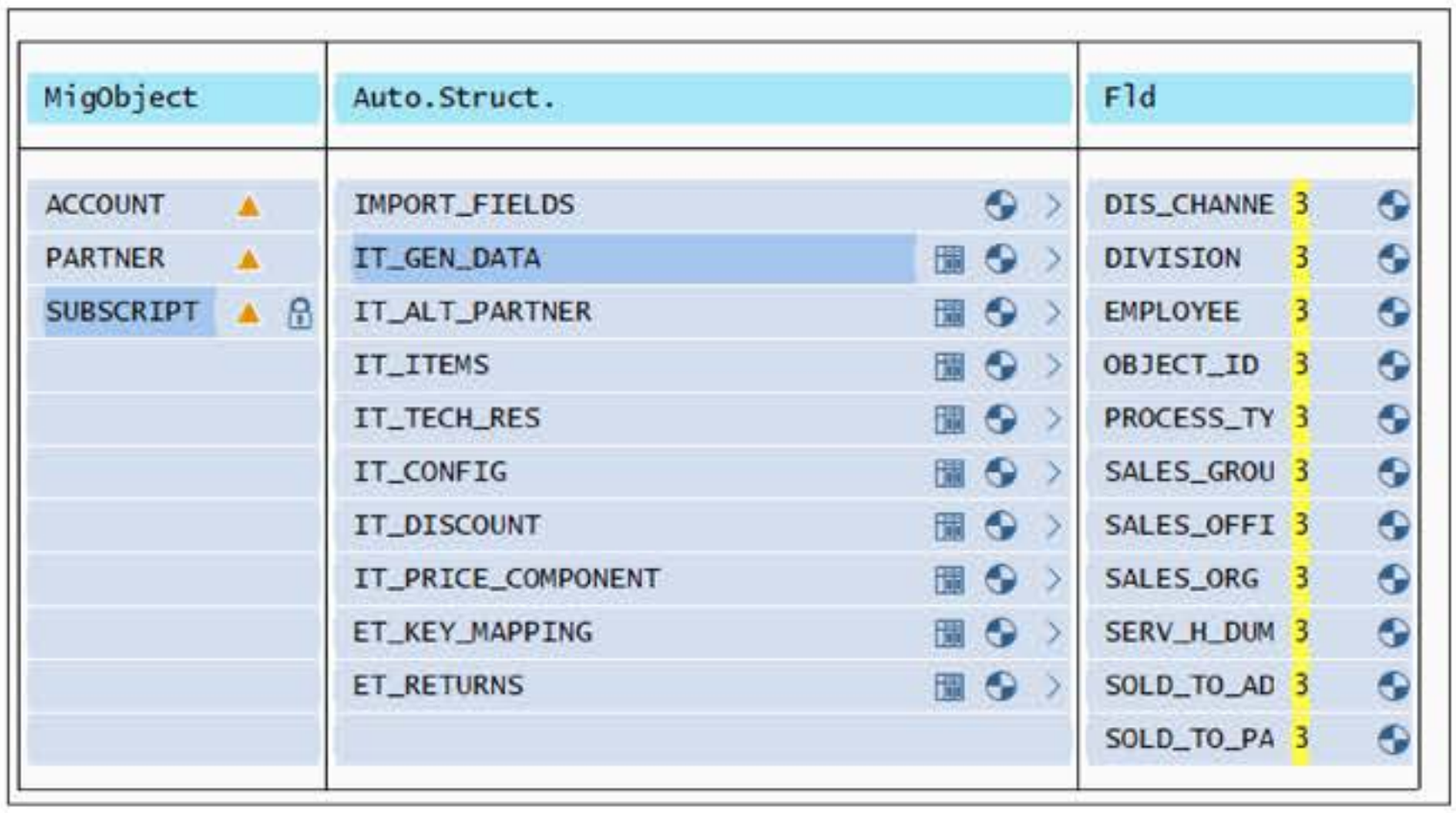

Figure 9.12 Migration Object for Subscription Orders

ISMW calls the function module CRMS4_SOM_SUBSCRPN_CONTR_MIG, which in turn calls method MAP_DATA of the CL_CRMS4_SOM_SUB_CONTR_MIG_MAP class, which generates template data to be migrated. Then function module CRMS4_SOM_SUBSCRPN_CONTR_MIG calls

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method COPY_ORDER_TO_CONTRACT of the CL_CRMS4_SOM_MIG_COPY_CONT_HND class by passing the template generated in the previous step of the process flow. Commit control takes place in ISMW.

As you know by now, a subscription contract is an actual document that has the agreed-on terms, duration, and pricing, and it can't be edited manually or directly. This document gets replicated to SAP Convergent Charging and SAP Convergent Invoicing. After successful creation of subscription contracts, be sure to validate some of the attributes of subscription contracts.

If your solution is integrated with SAP Convergent Charging, then you might need a way to restrict the generation of billable items from the previous billing cycles. Typically, the bills should have been generated in the legacy platform, which doesn't require being included in SAP S/4HANA again. However, if the contract was created in the past and was never charged in legacy, then you can use SAP Convergent Charging to auto-generate charges for the current billing cycle as well as for previous billing cycles.

To restrict duplicate billing, if the contract was created in the past and was billed in legacy, you can restrict generation of duplicate billable items by one of the following options:

- You can set the **Contract Start Date** as the migration date. This will prevent generation of billable items from SAP Convergent Charging for past dates, but it's generally not recommended as it will cause issues for revenue forecasting, SAP Revenue Accounting and Reporting integration, and reporting.
- Once the contracts are migrated, you can use the web service chargingContract-StateManagement. Using this web service, you can set values for the counters **Last Activation Date** and **Last Activation Status**. These two values are used by the charging logic to determine if a charged item must be created. If the **Last Activation Date** is set as the migration date, then SAP Convergent Charging will generate the billable items considering the migration date only. This will ensure that duplicate bills aren't generated that are already available in the legacy system.
- You can make use of characteristics with information related to the **Legacy** indicator and **Last Bill Date** information. This can then be used in SAP Convergent Charging to set up a rule to consider billable item generation only after the last bill date.

9.3.6 Billable Items and Consumption Items

In the event that consumption items or billable items are required to be preloaded in the system as part of data migration, you can use the standard function modules FKK_BIX_CIT_CREATE_API and FKK_BIX_BIT_CREATE_API, respectively. A common use case occurs where you need to refer to historical data to manage reporting or future sale conditions. You can also make use of item status to ensure that the migrated charges

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are never billed or considered for any financial posting based on your business requirement.

Let's discuss a typical business use case in which you want to store the billable items information migrated from a legacy system. Let's say you're migrating a yearly contract, which was initially valid for five years, into SAP Billing and Revenue Innovation Management. The contract was created in a legacy platform and was billed for two years in the legacy platform. Now the contract is being migrated and will be billed using SAP Billing and Revenue Innovation Management in the SAP S/4HANA landscape. Now, if there is a need for a revenue projection on total contract value with additional invoiced parameters, then you can use billable items to store relevant details and use that as one of your reporting sources. In addition, if you need to build logic in which the historical billable details should determine the adjustment amounts, then you can also make use of migrated billable items.

As a prerequisite for migration of billable items and consumption items, the baseline configuration related to SAP Convergent Invoicing as well as consumption item and billing item classes should be generated successfully.

9.3.7 Open Accounts Receivable Conversion from a Legacy Platform

Open items are the items for which payment is due from the customer. Open item transfer is an important activity that needs to be performed by moving to the SAP Billing and Revenue Innovation Management system from the legacy system. The transaction posting leading to the open item's creation needs to be frozen before the migration activity can be performed. The migration of open items is required for open item management in the new SAP system.

The open items can be posted to a reconciliation account per the business requirement. The open items in the legacy system needs to be written off in the legacy system or be moved to the new system. All posting receivables and payables are assigned to one contract account.

The starting point for the migration of a subsidiary ledger such as contract accounting must always be a legacy system that is reconciled with the general ledger. All postings from the legacy system must have been transferred to the general ledger.

The prerequisite for the migration of open items to contract accounting in the SAP S/4HANA system is the migration of the master data objects: business partner, contract account, prepaid account, and provider contract.

Various procedures are available for the migration of open items to contract accounting. All these procedures share the following: those open items are always migrated individually, and all open items are always migrated. The open items must in total explain the balances of the related reconciliation accounts.

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At the time of the migration of open items, the migration of the general ledger accounts and balances has already taken place. Through the transfer of the migrated open items to the general ledger, therefore, the general ledger will no longer be correct. Therefore, you must make a posting to the general ledger before the transfer of the items to be migrated. This posting must set the balance of the reconciliation account to zero. This measure is necessary because now the general ledger account can still be posted directly. As soon as you use the account as a reconciliation account for contract accounting, the account can no longer be posted to directly. Don't use the general ledger account as a reconciliation account (account type V for contract accounting) until after the receivables account has a balance of zero.

For the posting, you use an anonymous transfer account for the revenues; this account must be a revenue account and not a balance sheet account. Normally, revenues aren't considered during a migration, for the following reasons:

- Items cleared in the legacy system could have already been archived. Archived documents aren't considered during a migration. The balance of the revenue account in financial accounting can therefore not be explained by the sum of all postings in the legacy system.
- If the revenues were transferred again to the general ledger, then controlling would also be updated again. It's necessary to avoid secondary effects of this kind.

If the original posting is from the previous year, and if account assignments that are no longer valid in the year of the transfer were used in this posting, then you need to exchange the account assignments accordingly.

The events `3910` (Migration: Document Transfer - Convert Document Data) and `3920` (Migration: Document Transfer- Test Document Data) are processed in contract accounting for the migration of documents. See the documentation of the function modules. These events are called from function module `ISU_M_DOCUMENT_CREATE`. If you're using ISMW, you need to create a new object as "DOCUMENT" and use the function module `ISU_M_DOCUMENT_CREATE`, which internally calls function module `FKK_CREATE_DOC_MASS`.

Based on the migration status of revenue and tax items, there are two possible scenarios for open accounts receivable migration:

- **Revenue and tax items aren't migrated.**
  The OLD receivables account can be posted to manually. The NEW receivables account becomes the reconciliation account for contract accounting. Every receivable item is migrated with a tax code (`FKKOP-MWSKZ`); tax items (`OPKs`) aren't migrated. The offsetting item of the migrated document corresponds to the gross amount of the invoice (see Figure 9.13).

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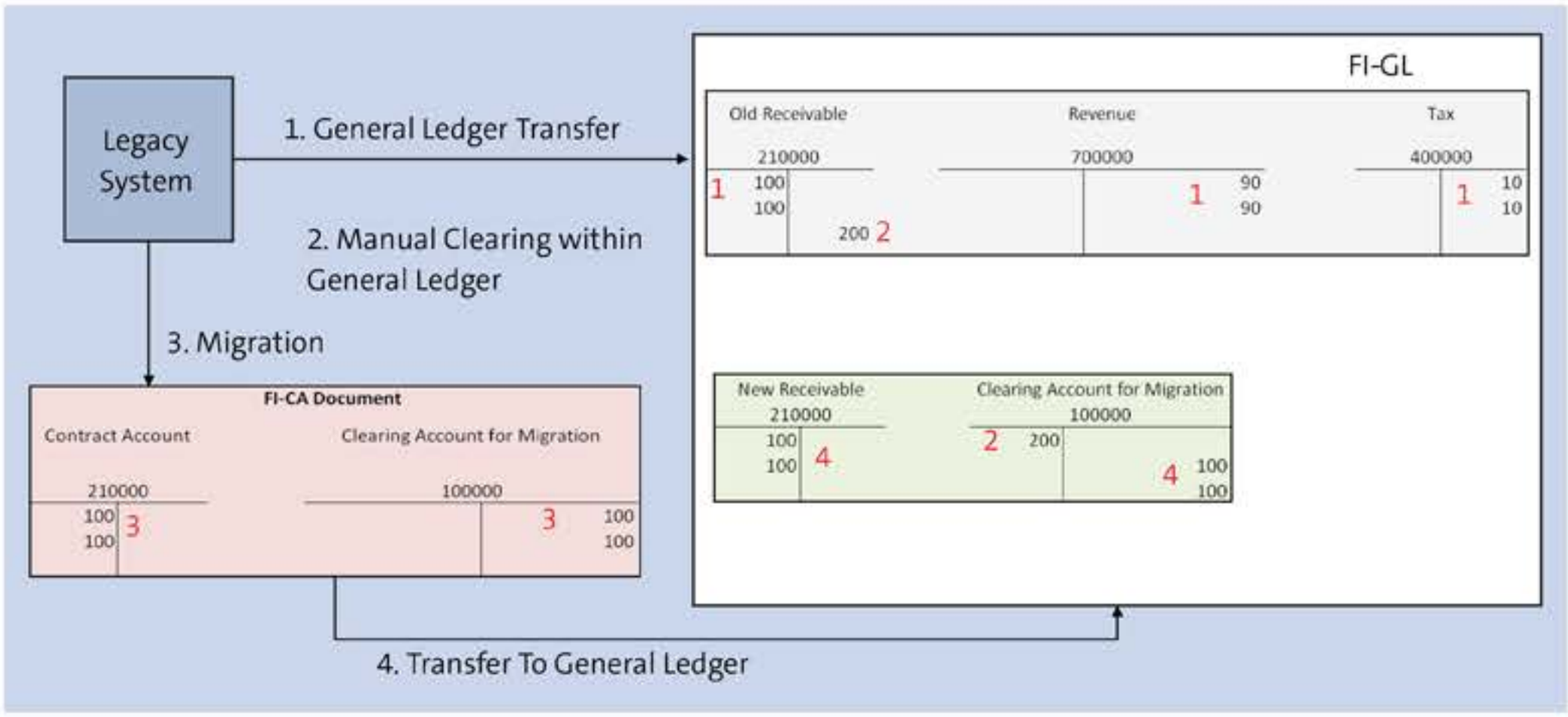

Figure 9.13 Open Item Transfer without Revenue and Tax Items

The disadvantage of this option is that the documents migrated in this way aren't allowed to be reversed because the tax amounts aren't known. However, you can use write-off or manual posting of credits, which you can offset against the existing open items.

- **Revenue and tax items are migrated.**
  The OLD receivables account can be posted to manually. The NEW receivables account becomes the reconciliation account for contract accounting (see Figure 9.14).

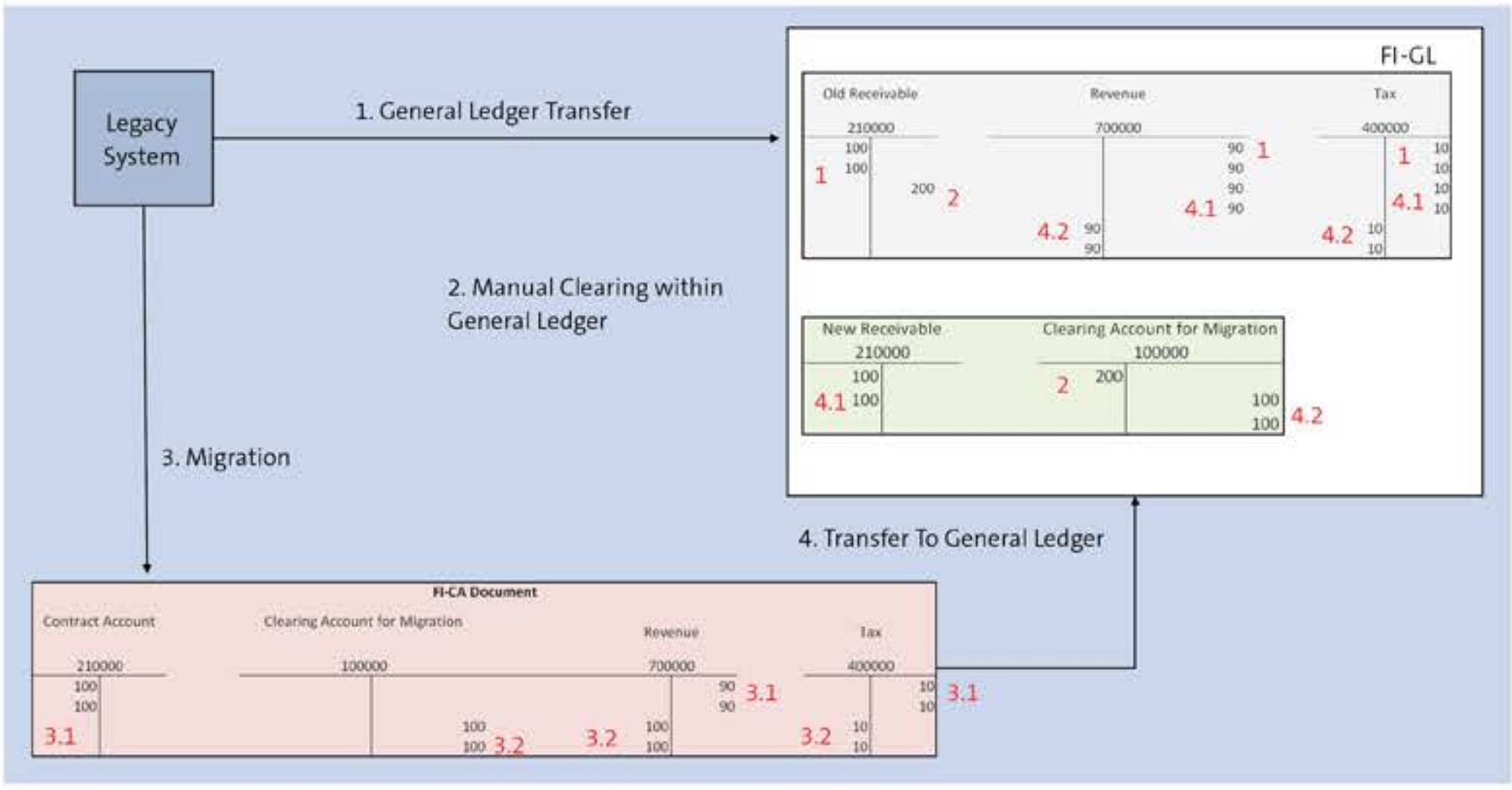

Figure 9.14 Open Item Transfer with Revenue and Tax Items

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The advantage of migrating revenue items and tax items is that the migrated receivables can be reversed. For each open item, two postings are migrated to contract accounting.

9.3.8 Migration from SAP CRM-Based SAP Billing and Revenue Innovation Management, Subscription Order Management, to SAP S/4HANA

The initial release of SAP Billing and Revenue Innovation Management was hosting SAP Billing and Revenue Innovation Management, subscription order management, in SAP CRM prior to SAP S/4HANA 1809. There are quite a few projects implemented using the SAP CRM provider order management process to integrate with SAP S/4HANA and the SAP Convergent Charging system. Because SAP Billing and Revenue Innovation Management, subscription order management, is now included in SAP S/4HANA, it's a good opportunity to migrate existing SAP CRM-based SAP Billing and Revenue Innovation Management, subscription order management, functionality into SAP S/4HANA. This not only helps projects avoid middleware-related issues between SAP S/4HANA and SAP CRM but also helps to adapt newer features that are released keep up with the business model transformation in the subscription- and usage-based economy.

We'll discuss briefly a few of the considerations to keep in mind while planning for migration from SAP CRM provider order management to SAP S/4HANA-based SAP Billing and Revenue Innovation Management, subscription order management. We'll assume that the initial implementation of SAP Billing and Revenue Innovation Management using SAP CRM is already up and running with production business transactions. In addition, base customization of SAP Billing and Revenue Innovation Management, subscription order management, is completed in SAP S/4HANA and ready for data migration.

Handling Master Data

The following master data is handled during migration:

- **Organization structure**
  Though SAP S/4HANA will continue to have an enterprise structure, you'll need to set up and manage organization data as well. This is similar to the implementation of SAP CRM–based SAP Billing and Revenue Innovation Management, and you can use the report to manage conversion of the SAP S/4HANA sales structure into the organization structure.
- **Business partner**
  Conversion of business partners isn't required because the roles and customer-specific information will already be available in SAP S/4HANA and SAP Convergent Charging.
- **Contract account**
  Like business partners, we don't need to manage the conversion of contract accounts.

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- **Subscription product**
  To support SAP Billing and Revenue Innovation Management, subscription order management, using SAP S/4HANA Service, you'll need to adapt the SAP S/4HANA material master. During the SAP CRM–based implementation, the products might have been distributed to SAP S/4HANA using middleware as the material master or only as contract accounting products. In addition, the product modeling and cross-catalog mapping is also required to be adapted based on the SAP S/4HANA Service scenario.

  Following are additional checks and steps you can do depending on the SAP Billing and Revenue Innovation Management landscape:

  - If your initial implementation was only managed using contract accounting products, you must create the products as the material master with the same name. Additionally, you need to change your Customizing in contract accounting.
  - If you're already working with the material master, you must change the material type for your products, for example, using the relevant transaction (e.g., Transaction MMAM).
  - You must ensure that the item category group (`MTPOS`) on the sales organization level (`MVKE`) is adapted, for example, via Transaction MM17, to the item category group you need for your subscription processes, for example, `PRRP` (Subscription Rate Plan).

  You can also add subscription-specific fields to product master data using the Manage Subscription Product-Specific Data application.

Handling Transactional Data

For migrating subscription orders and contracts, you can refer to Section 9.3.5.

For more details, you can also refer to SAP Note 2843936 if you're migrating from SAP CRM to SAP S/4HANA Service or SAP Note 2770484 if you're migrating from SAP CRM to SAP S/4HANA for customer management.

9.4 Summary

Data migration is an integral part of any implementation. This is an important task to ensure that all historical data is migrated to the new landscape to ensure business continuity. Details analysis needs to be done on migration objects and sequences to be followed. This chapter provided a high-level view of the migration objects to work with for any SAP Billing and Revenue Innovation Management implementation. This chapter will also help you if you're planning to migrate from an SAP CRM–based Hybris billing implementation to an SAP S/4HANA–based SAP Billing and Revenue Innovation Management solution.

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In the next chapter, we'll discuss different deployment options available while implementing SAP Billing and Revenue Innovation Management and related infrastructure requirements for different deployment types.

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Chapter 10

First Steps to SAP Billing and Revenue Innovation Management

Like other software implementation projects, getting SAP Billing and Revenue Innovation Management up and running requires planning, evaluation, and thoughtful consideration of current and future states. This chapter offers landscape and infrastructure details as well as best practices for planning and considering SAP Billing and Revenue Innovation Management as a solution. You'll also learn how to avoid common pitfalls and gain insights into key decision points.

In this chapter, we'll evaluate the leading thought principles you should consider for implementation of the SAP Billing and Revenue Innovation Management solution. There are many key decision points to consider when implementing SAP Billing and Revenue Innovation Management, including but certainly not limited to deployment options, business processes, different type of products and services offered, and master data governance and considerations. Throughout this chapter, we'll highlight key concepts to consider when implementing SAP Billing and Revenue Innovation Management and provide insights into how to successfully deploy this solution.

Each SAP implementation is filled with unique experiences driven by many factors, such as business scenarios, customer requirements, system landscapes, scope of the project, transactional volume, and company policy regarding security and governance—to name a few. Other factors that impact the implementation are the types of deployment, such as on-premises versus cloud. These factors drive the implementation approach, size of hosting platform, type of software to install, and project duration.

In every SAP project, or any IT project, for that matter, initial planning is required. The planning phase is a critical step that can take weeks or even months. Depending on the project, it may require a full-time commitment from different groups such as IT and business stakeholders, as well as time committed in multiple working sessions.

In this phase, team members gather information about current systems and business processes, discuss the deliverables of the project, and determine any dependencies to identify gaps.

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10.1 Why Move into Subscription-Based Business?

In today's digital economy, most businesses transform their operating models into subscription-based business models. This move is driven by several factors, such as the following:

- The need to grow the business by expanding their offering portfolio horizontally and vertically via ambition and acquisition.
- The need for a new path to market found by transforming the current product offerings from delivery-based into service-based ones.

These business models also let customers access, consume, and pay for services or products based on need and usage. From a financial standpoint, subscription-based business enables companies to have a more predictable and continuous revenue stream.

There are multiple flavors of subscription-based business models to support different types of revenue streams. The following provides some of the common ones:

- **Recurring and on-demand subscription**
  This offer allows customer to consume services with or without limits, according to the contract (e.g., video streaming, on-demand, and fixed monthly fee, such as cable TV and internet services)
- **Usage-based subscription**
  This offer is based on services consumed by customers such as telecommunications data plan, cloud storage, and so on.
- **One-off charge fee**
  This type of charge is usually triggered by new contract activation, contract renewal, and early termination of a contract.
- **Predefined subscription**
  This offer provides access to specific resources (e.g., SaaS, PaaS, XaaS).
- **Brokerage/transaction fees**
  This type of subscription is based on a per transaction fee (e.g., Ameritrade and E-Trade).
- **Bundled subscription or package**
  This type of service includes everything in a box model, which may include subscription, service, or devices, along with a warranty, and so on.

10.2 Evaluating Your Current Landscape

There are several different scenarios to be considered when implementing the SAP Billing and Revenue Innovation Management solution. Before starting the project, you need to evaluate your current system landscape to determine what components of SAP

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Billing and Revenue Innovation Management need to be installed. In addition, because SAP Billing and Revenue Innovation Management consists of different core components, you need to evaluate your business model to decide if you'll require the full suite of components, involving SAP S/4HANA, SAP Convergent Charging, SAP Convergent Mediation (by DigitalRoute), and so on, or a few specific components based on your project goal.

It should be clear by now that the SAP Billing and Revenue Innovation Management solution consists of four different application components: SAP Billing and Revenue Innovation Management, subscription order management; SAP Convergent Charging; SAP Convergent Invoicing; and contract accounting. Prior to SAP S/4HANA 1809, SAP Billing and Revenue Innovation Management, subscription order management (which was also known as provider order management) was part of the SAP Customer Relationship Management (SAP CRM) system. As of SAP S/4HANA 1809, we now have SAP Billing and Revenue Innovation Management, subscription order management. Now, let's take a look at some of the key considerations to start with while implementing the SAP Billing and Revenue Innovation Management solution.

10.2.1 Company Currently Running SAP

If your company is already using SAP and the business model requires the full suite SAP Billing and Revenue Innovation Management implementation, use the following checks to help you determine what existing components can be used for SAP Billing and Revenue Innovation Management:

- **Your company only uses SAP ERP or SAP S/4HANA prior to 1809 with no upgrade planned.**
  - If no upgrade is planned, then SAP S/4HANA version 1709, 1610, or 1511 can be used.
  - You'll need to install SAP CRM 7.0, EHP 4, to manage SAP Billing and Revenue Innovation Management, subscription order management, as well as SAP Convergent Charging. The first edition of the book extensively covers SAP Billing and Revenue Innovation Management using the SAP CRM system.
  - You'll need to install SAP Convergent Charging.
- **Your company plans to upgrade to SAP S/4HANA 1809 or higher.**
  - You'll only require an upgrade to SAP S/4HANA 1809.
  - You'll need to install SAP Convergent Charging.
- **Your company plans to install SAP Billing and Revenue Innovation Management from the ground up (greenfield implementation).**
  - It would be best to install SAP S/4HANA 2020 or later.
  - You'll need to install SAP Convergent Charging with the latest support pack.

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The only supported database for SAP S/4HANA is SAP HANA. The recommended database for the other components such as SAP Convergent Charging is SAP HANA 2.0, but other databases are supported. Check the Product Availability Matrix (PAM) for compatibility. Note that the best practice recommends the installation of the latest support packs for all components.

10.2.2 Set Up SAP Billing and Revenue Innovation Management Infrastructure

Once you've decided on SAP Billing and Revenue Innovation Management components to be installed and evaluated your current IT landscape, you'll need to set up your infrastructure. You have multiple options for setup: on-premises, cloud, or software as a service (SaaS). While there are multiple cloud providers, we'll briefly cover Amazon Web Services (AWS) and Google Cloud. If you're not ready to build your infrastructure, consider the SAP SaaS: SAP S/4HANA Cloud.

Whatever option you choose, you should follow the first steps. Determine the size of your servers by performing a sizing exercise. Use the Maintenance Planner to plan your systems, whether you're upgrading or doing a greenfield installation.

Once installed, you'll need to activate the appropriate business function and perform the post-steps for configuring the SAP Fiori apps. You have the option of installing a separate SAP Fiori frontend server for on-premises or cloud, but, for SAP HANA Cloud, the only option is embedded. There are pros and cons for both such as maintenance and cost. There are some key points for each infrastructure option, which we'll discuss in the following subsections.

SAP Billing and Revenue Innovation Management Deployed On-Premises

On-premises was the standard solution for installing SAP systems whether securing new servers from a hardware vendor or adding new virtual machines (VMs) to existing hardware. You can choose your architecture, either a two-tier where database and application are on the same server, or a three-tier where the application and database are on separate servers. Most production systems run on three-tier because the architecture allows each server to be dedicated to processing either data or application requests, providing a more manageable system and less contention for resources.

SAP Billing and Revenue Innovation Management Deployed on Amazon Web Services

AWS is another means of provisioning infrastructure. This subscription-based solution is usually quicker and cheaper than deploying in the cloud. AWS is an SAP Certified Provider. Like on-premises, you'll need to provide the software and license. The software can be uploaded from the on-premises system to AWS, or you can configure a server to connect to the internet for download from SAP Marketplace.

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While a lot of the considerations are the same as on-premises, there are some things to consider. There are some SAP Notes to consult before venturing into AWS:

- SAP Note 1656099 - SAP Applications on AWS: Supported DB/OS and AWS EC2 Products
- SAP Note 1656250 - SAP on AWS: Support Prerequisites

When deploying on AWS, consider the supported operating systems, network, and storage. The operating systems offered by AWS that are currently supported by SAP are SUSE Linux Enterprise Server (SLES) 11 or higher, Red Hat Enterprise Linux (RHEL) 6 or higher, Oracle Linux (OEL) 6.4 or higher, and Windows Server 2008 R2, 2012 (R2), 2016, and 2019. Check the notes mentioned previously as the supported hardware are updated as new versions become available. Choose an Amazon Machine Image (AMI) that is using hardware virtualized machine (HVM) virtualization instead of paravirtualized (PV) as this makes migration to future AWS instance types simpler.

A common approach for the network is to create and configure an Amazon Virtual Private Cloud (VPC). The VPC is a private, isolated section of AWS Cloud. This extends your internal network into the AWS Cloud.

Storage is needed for your operating system, SAP software, and database. SAP recommends using Elastic Block Store (EBS) Optimized EC2 instances for databases. The EBS volumes should use some software RAID level higher than RAID-0. The log files should be on different EBS volumes than the data files. The operating system uses EBS persistent storage, which is presented as raw block devices. The storage for the application can be any Solid State Drive (SSD) backed EBS volume types as all volume types are supported. Consider S3 for storing backups. For Linux-based SAP systems, you'll need an Elastic File System (EFS), a Network File Service (NFS), for files that are stored on highly available shared file systems. This would be for the */sapmnt* and */usr/sap/trans* file systems.

SAP Billing and Revenue Innovation Management Deployed on Google Cloud Platform

SAP Billing and Revenue Innovation Management is also certified on Google machine engine types. There are several prerequisites to run SAP on GCP, including the type of support agreement and monitoring. The support agreement must be Premium or Enhanced. While Google offers other support plans (Silver, Gold, Platinum, Enterprise Support, and Role-Based Production Support), Premium and Enhanced are the only ones considered valid options. Choose the option that best fits your budget and organization's policies.

SAP monitoring must be activated. The GCP SAP monitoring agent needs to be installed on the VM. This enables the GCP performance counters to be included in SAP enhanced monitoring. SAP enhanced the SAP HOST agent and monitoring Transaction ST06 for Google Cloud monitoring. The minimum SAP Host Agent version for Google

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Cloud Enhanced Monitoring is release 7.21 patch level 26. For more information, check SAP Note 2460297 - SAP on Linux on Google Cloud: Enhanced Monitoring.

GCP offers Compute Engine VMs and Bare Metal Solution bare-metal machines for SAP HANA. Compute Engine VMs are what most people are familiar with. You choose a server from a list of certified machine types based on your application. The supported operating systems are Microsoft Windows Server 2012 R2, 2016, 2019, and 2022; SUSE Linux Enterprise Server 11 SP4 (SLES11) and higher; SUSE SLES for SAP Applications (based on SLES11 and higher); and Red Hat Enterprise Linux 6 (RHEL6) and higher. It's a good practice to check SAP Note 2456432 (SAP Applications on Google Cloud: Supported Products and Google Cloud) when installing in GCP as the latest supported operating systems and databases are listed there.

For bare-metal machines, you're responsible for the operating system and any hypervisor software. You must bring your own license (BYOL). Google installs the initial operating system or hypervisor, and then you're responsible once they are turned over to you. The supported architecture for SAP software on bare-metal is a three-tier architecture, in which SAP NetWeaver and the application servers run on different servers than SAP HANA.

There are two SAP HANA on Bare Metal Solution architectures from which to choose:

- **All-metal application architecture**
  All your servers run in the same Bare Metal Solution environment. This solution is an option for workloads that require a lower latency between SAP HANA and the application servers.
- **Hybrid application architecture**
  SAP HANA runs in the Bare Metal Solution environment, and the SAP application servers run on Google Cloud. This architecture offers lower costs and more flexibility than the all-metal architecture.

SAP Billing and Revenue Innovation Management Deployed on SAP S/4HANA Cloud

If you're new to SAP or want a completely managed cloud solution, there is SAP S/4HANA Cloud. SAP S/4HANA Cloud is a subscription-based software solution hosted on SAP servers and completely managed by SAP.

Another option offered by SAP is RISE with SAP in which you work with SAP and partner with a system integrator and hyperscaler to get a system designed for your needs. Also included with this solution are multiple services and tools to help with your transformation to SAP S/4HANA Cloud, such as deployment assistance, analytics, and intelligent business design.

Whether you choose SAP S/4HANA Cloud or Rise with SAP, you're using a hosted version of SAP S/4HANA, eliminating the need to run the solution on-premises or manage the solution in the cloud.

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10.2.3 Communication between Systems in the SAP Billing and Revenue Innovation Management Landscape

Because the SAP Billing and Revenue Innovation Management solution integrates with different SAP and non-SAP components, you need to consider the communication, specifically between SAP S/4HANA, SAP Convergent Charging, and SAP Convergent Mediation (by DigitalRoute). SAP, however, offers native integration between the different SAP Billing and Revenue Innovation Management components. In addition, projects involving various front-office or back-office applications (SAP or non-SAP) will also require integration as well. Depending on your security policies and application type, you'll need to set up secure remote function calls (RFC) and web service calls. The communication to SAP Convergent Charging is through web service calls configured in Transaction SOAMANAGER. The connection can use an HTTP or HTTPS protocol depending on requirements. Systems spanning networks may require more secure communications such as HTTPS, while systems in the same network may not have that requirement. The communication from SAP Convergent Charging and SAP S/4HANA leverages an SAP Java connector connection. This should also use load balancing if SAP ERP has multiple application servers. RFC and JCo connections must be authenticated with a username and password. This user should be created as a system user. A system user won't be able to log on to your systems directly. You can also configure SAP ERP to support Secure Sockets Layer (SSL). A good practice is to use RFC load balancing to connect ABAP systems.

Once you have an agreement on a deployment scenario, you may need to decide to update or upgrade some of your products.

SAP Billing and Revenue Innovation Management integration usually consists of the following products:

- SAP S/4HANA (SAP Billing and Revenue Innovation Management, subscription order management; SAP Convergent Invoicing, contract accounting, SAP Customer Financial Management)
- SAP Convergent Mediation (by DigitalRoute)
- SAP Convergent Charging

10.2.4 Verify Landscape Selection with the Product Availability Matrix

A good tool to check when deciding on product updates or upgrades is SAP's Product Availability Matrix (PAM). The PAM provides the technical information for all SAP product versions. It has upgrade paths, release dates, and support end dates. It should be the starting point for planning your environment. You also can check the system landscape directory (SLD) and landscape management database (LMDB). This can be done in SAP Solution Manager either directly in Transaction LMDB or with the **Landscape Management** tile in SAP Solution Manager (see Figure 10.1).

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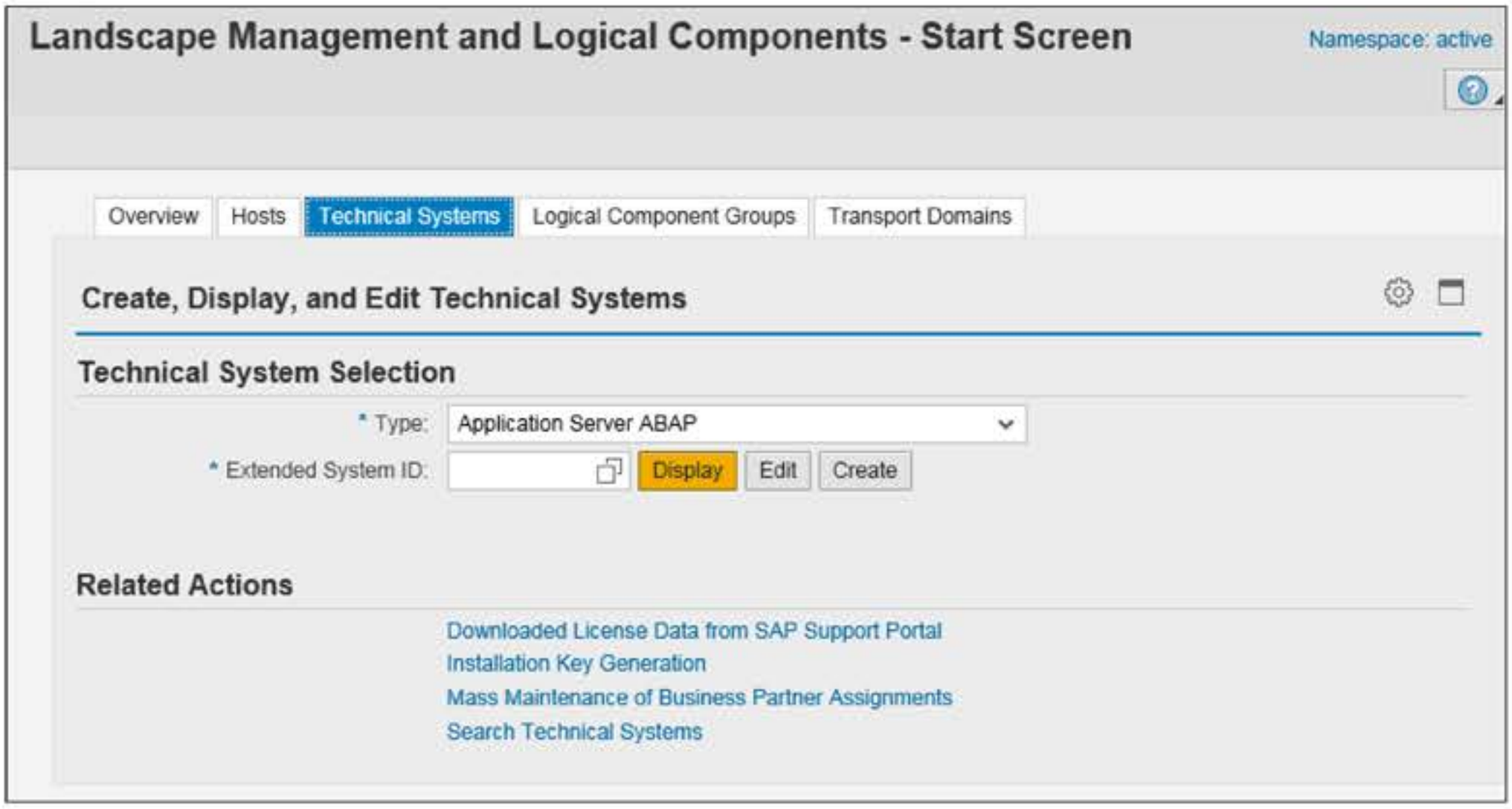

Figure 10.1 LMDB Technical Systems Search

Once you've decided on products and versions, you can plan landscape changes with the Maintenance Planner, another tool offered by SAP for planning (see Figure 10.2). This tool should contain information about all the products in your landscape.

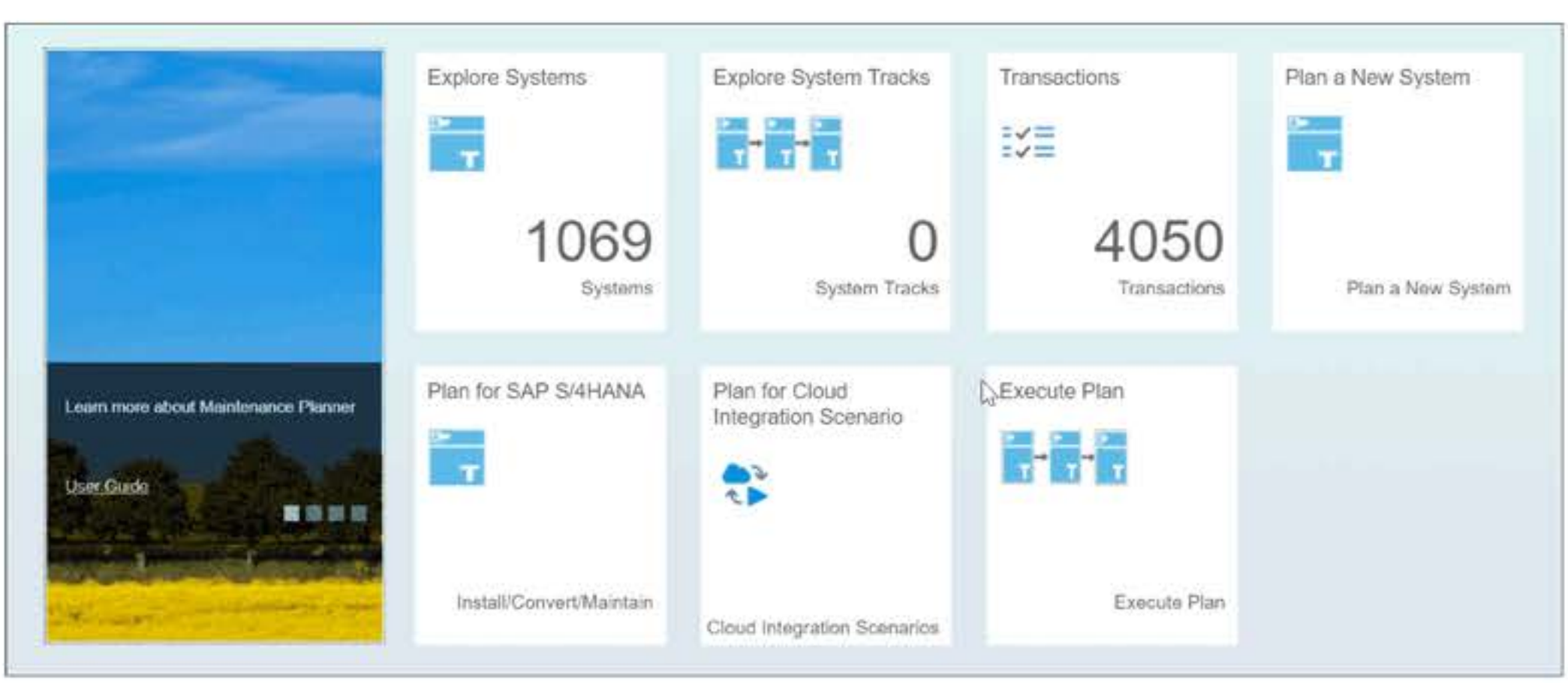

Figure 10.2 Maintenance Planner

From here, you can select a system and the planned upgrade. This will help determine the upgrade path. Follow-up steps include planning and executing the upgrade throughout your landscape.

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10.2.5 Security for Your Connected Systems

Some organizations have a restrictive security policy, which requires securing the systems in your SAP Billing and Revenue Innovation Management landscape. Securing SAP Convergent Charging requires configuring Secure Sockets Layer (SSL)/Transport Layer Security (TLS). When configuring, you need to generate certificates to identify your systems. Depending on your policy, you can use self-signed or signed certificates. Self-signed certificates are generated by open-source tools such as OpenSSL. Signed certificates are signed by a certificate authority. Certificates are imported into the SAP Convergent Charging Core Server and should also be added to the default keystore on the server.

One of the important things to remember when using signed certificates for implementing enhanced security is that the root and intermediate certificates provided with the signed certificate must be imported into the default Java trust-store on the server.

10.2.6 Load Balancing

Performance of the solution is always essential to its success. To increase the responsiveness of the SAP Billing and Revenue Innovation Management solution, consider implementing load balancing. Load balancing distributes the application load across multiple servers, thereby increasing the responsiveness of the solution. Load balancing can be in a round-robin style in which you let the application decide, or you can determine how the load is distributed by design.

One of the great features of SAP Convergent Charging is that it has internal load balancing. When multiple dispatcher instances are installed, one dispatcher instance, the master dispatcher, controls where requests are directed. You can further decide on the robustness of your load balancing solution by the distribution of instances over multiple servers. For example, you can install dispatcher, guider, rater, bulkloader, and updater instances on multiple servers, or you can install some combination of the instances on a dispatcher/guider, rater/bulkloader, and updater on separate servers. The distribution and number of servers can be determined by performing a sizing exercise.

There are multiple options for databases for all systems in the landscape. SAP is encouraging moving to the SAP HANA database for SAP products. SAP HANA is an in-memory database that has faster transaction speeds. When processing, speed is a critical factor. SAP HANA also is scalable. You can scale by adding additional resources on a single hardware element or by adding additional hardware. You can start with a single node and add additional nodes later. With SAP HANA, you can also implement a high-availability solution, which is a requirement for most production systems. The goal for any production system is to *not* have a single point of failure. With SAP HANA 2.0, multitenancy is a default. Multitenancy provides multiple isolated databases per SAP

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HANA system. For instance, there can be one development SAP HANA system with multiple SAP HANA databases with their own system IDs.

For one specific client using SAP S/4HANA 1709, the SAP S/4HANA system was already on-premises, and SAP Convergent Charging was to be installed. Two instances were required to limit latency. The systems were built in the United States and EMEA regions. For using the SAP Convergent Charging client applications, enhanced security was required for communicating across regions. Communication between SAP CRM and SAP Convergent Charging was done through web service calls. Because there was an SAP CRM and SAP Financial Supply Chain Management (FSCM) system in each region, this communication was in the same region, so enhanced security wasn't required.

If hardware isn't available, it's possible to install on temporary hardware and migrate to newer hardware later. This is possible using VMs. This is seamless for the application and can be done by moving a VM from one physical host to another. This can take place during a downtime period and usually takes a few hours.

10.2.7 Migrating Changes

How do you move changes throughout the landscape? For ABAP-based systems, this isn't an issue: you can use a domain controller and add systems and transport routes. Because SAP Convergent Charging is Java-based, enhanced Change and Transport System (CTS+) is an option. For CTS+, you need to agree on the ABAP system to use for attaching SAP Convergent Charging. SAP Solution Manager is most often used because it has an ABAP and a Java stack—CTS+ requires both. Additional roles may need to be built so that developers/configurators can attach the work developed in SAP Convergent Charging to a transport. Usually, with segregation of duties, the creator of a transport can't import a transport. If you've secured your landscape by enabling HTTPS, additional steps must be done in CTS+ configuration to connect to the core server for importing transports. If you choose not to use CTS+ to migrate your changes through the landscape, you can use the client tool, called Core Tool, to export and import your changes.

10.2.8 Monitoring

You'll need to have a monitoring and alerting strategy. SAP Solution Manager is required for ABAP monitoring. There is alerting available, as well as notifications that can be triggered. In addition, there are tools for monitoring SAP HANA (see Figure 10.3). The SAP HANA cockpit lets you monitor all your SAP HANA systems in one location. The cockpit runs on SAP HANA, and there is usually a separate SAP HANA instance solely for it. For SAP Convergent Charging root-cause analysis, uses Wily Introscope Agent, which is part of SAP Solution Manager.

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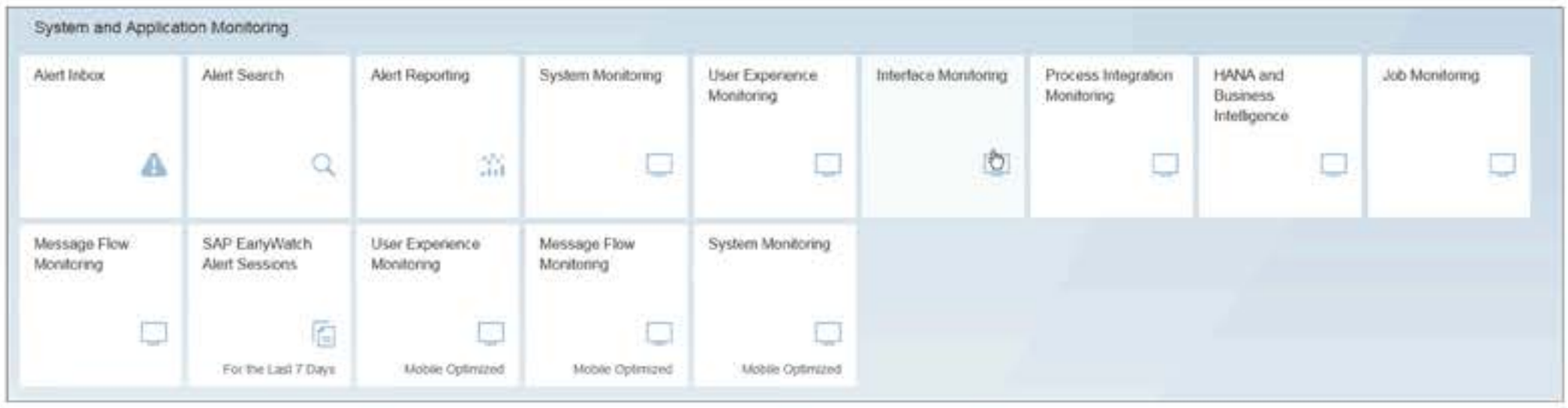


Figure 10.3 SAP Solution Manager Monitoring

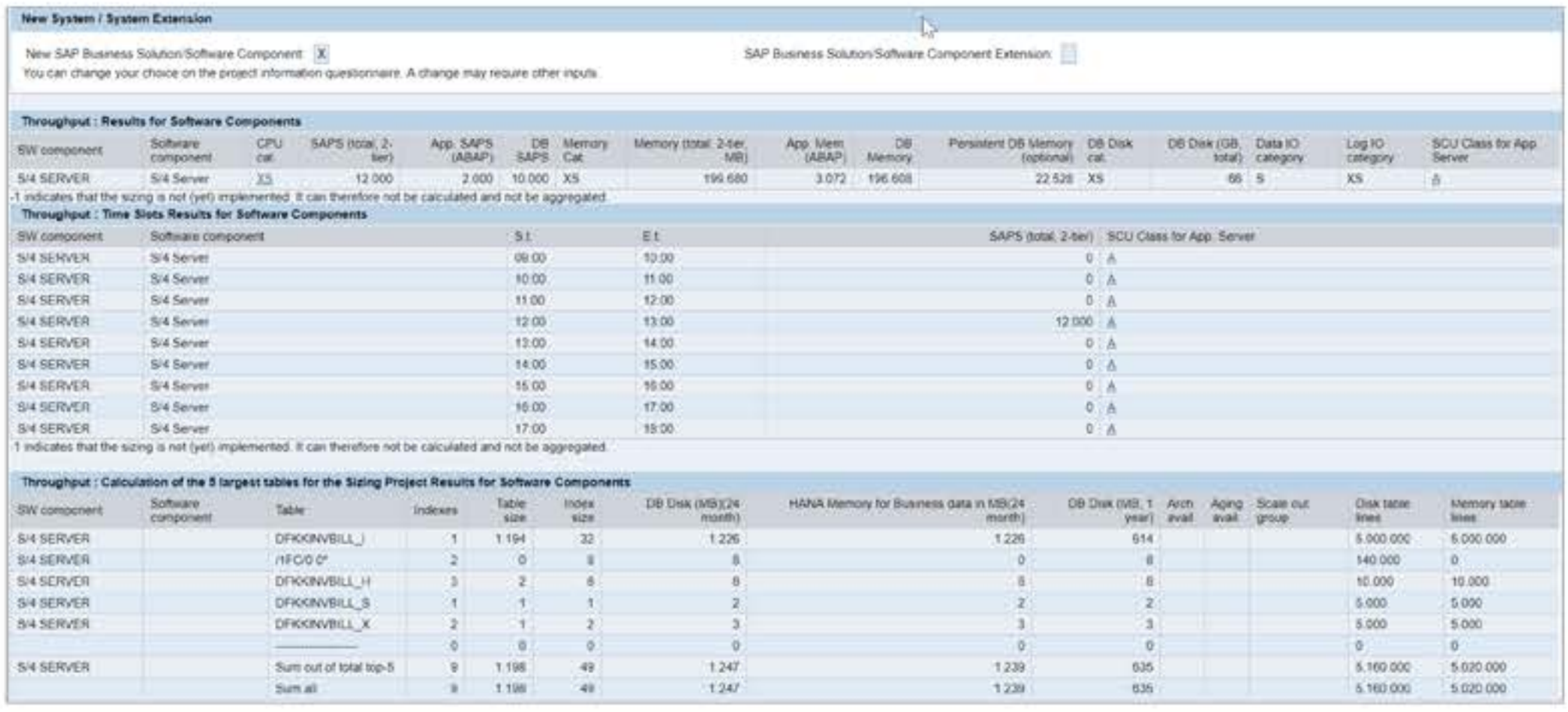
10.2.9 Data Protection

Data protection involves using a backup/recovery strategy. Most clients use a third-party tool such as Tivoli for backing up their entire landscape. Best practice is to have some automated backup strategy. Furthermore, you should decide when to back up and when to have a full backup or a delta backup. For SAP HANA, online backup is possible; other databases have online or offline options. It's possible to back up to disk and then to tape. Best practice is to test the system recovery to ensure your strategy is working.

10.2.10 Sizing

In addition to determining a production version, operating system, and database, you need to decide on hardware to determine if the current hardware is robust enough to handle additional functionality. A sizing exercise is a good approach to answer this question. SAP has defined three types of sizing—greenfield, production, and expert:

- **Greenfield sizing**
  This is usually for new implementations. The tool to use for this is Quick Sizer, which uses several metrics of your environment to produce recommendations for CPU, memory, database space, and disk I/O (see Figure 10.4).



New System / System Extension

New SAP Business Solution/Software Component: X
SAP Business Solution/Software Component Extension:
You can change your choice on the project information questionnaire. A change may require other inputs

Throughput : Results for Software Components

SW component	Software component	CPU cat.	SAPS (total, 2-tier)	App. SAPS (ABAP)	DB SAPS	Memory Cat.	Memory (total, 2-tier, MB)	App. Mem (ABAP)	DB Memory	Persistent DB Memory (optional)	DB Disk cat.	DB Disk (GB, total)	Data IO category	Log IO category	SCU Class for App Server
S/4 SERVER	S/4 Server	XS	12.000	2.000	10.000	XS	196.680	3.072	196.608	22.528	XS	66	S	XS	A

-1 indicates that the sizing is not (yet) implemented. It can therefore not be calculated and not be aggregated

Throughput : Time Slots Results for Software Components

SW component	Software component	S.t.	E.t.	SAPS (total, 2-tier)	SCU Class for App. Server
S/4 SERVER	S/4 Server	09.00	10.00	0	A
S/4 SERVER	S/4 Server	10.00	11.00	0	A
S/4 SERVER	S/4 Server	11.00	12.00	12.000	A
S/4 SERVER	S/4 Server	12.00	13.00	0	A
S/4 SERVER	S/4 Server	13.00	14.00	0	A
S/4 SERVER	S/4 Server	14.00	15.00	0	A
S/4 SERVER	S/4 Server	15.00	16.00	0	A
S/4 SERVER	S/4 Server	16.00	17.00	0	A
S/4 SERVER	S/4 Server	17.00	18.00	0	A

1 indicates that the sizing is not (yet) implemented. It can therefore not be calculated and not be aggregated

Throughput : Calculation of the 5 largest tables for the Sizing Project Results for Software Components

SW component	Software component	Table	Indexes	Table size	Index size	DB Disk (MB)(24 month)	HANA Memory for Business data in MB(24 month)	DB Disk (MB, 1 year)	Arch avail.	Aging avail.	Scale out group	Disk table lines	Memory table lines
S/4 SERVER		DFKKINVBILL_I	1	1.194	32	1.226	1.226	614				5.000.000	5.000.000
S/4 SERVER		/1FCO/0 0*	2	0	8	8	0	8				140.000	0
S/4 SERVER		DFKKINVBILL_H	3	2	8	8	8	8				10.000	10.000
S/4 SERVER		DFKKINVBILL_S	1	1	1	2	2	2				5.000	5.000
S/4 SERVER		DFKKINVBILL_X	2	1	2	3	3	3				5.000	5.000
		----------	0	0	0	0	0	0				0	0
S/4 SERVER		Sum out of total top-5	9	1.198	49	1.247	1.239	635				5.160.000	5.020.000
		Sum all	9	1.198	49	1.247	1.239	635				5.160.000	5.020.000

Figure 10.4 Quick Sizer

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With these results, you can work with your hardware vendor or check the benchmark table—both on the Quick Sizer site.

- **Productive sizing**
  This is done when you've already gone live and want to add new functionality. For this, you can look at past performance and extrapolate. You can check transactions such as Transaction ST03 and Transaction STAD as part of that process. In addition, reports such as SAP EarlyWatch Alert can provide further details. SAP has provided report /SDF/HDB_SIZING to assist with sizing SAP HANA (see Figure 10.5).

```
RESULTS OF SUITE ON HANA SIZING IN GB

Based on the selected table(s),the anticipated memory requirement is:

  Column store data                                                    11,3
+ Row store data                                                       15,0
+ Work space                                                           26,3
+ Hybrid LOB stored on disk                                            13,3
+ Fixed size for code, stack and other services                        50,0
-------------------------------------------------------------------------
= Anticipated total memory requirement considering 20% records stored
    on disk are cached in memory                                      105,3

Sizing report:                                          /SDF/HDB_SIZING
SID                                                                 HTP
NW release:                                                     740 SP 7
Date of analysis:                                           08.07.2019
Version of the report:                                              44
Selected accuracy:                                                   L
Number of work processes used:                                      01
Type of analyzed database:                                       MSSQL

Number of tables successfully analyzed:                        103.674
Number of tables partially analyzed:                                0
Number of tables with error:                                        3
Error estimation in % due to tables with errors:                    0
```

Figure 10.5 Sample Report from Quick Sizer

- **Expert sizing**
  For SAP Convergent Charging, expert sizing is done by SAP; it's not on the Quick Sizer site. SAP will analyze your environment based on data such as number of contracts, growth rate, recurring charges, and charge plan complexity. With that input, SAP will determine the required throughput. A sizing recommendation is provided for the number of servers, number of CPUs, and amount of memory. This is provided to a hardware vendor to provision the required servers.

Expert sizing for the SAP HANA database for SAP Convergent Charging is also not available via the Quick Sizer tool, and there is no official documentation. It's done by SAP based on SAP's internal testing and other clients running SAP Convergent Charging on SAP HANA. To provide a recommendation, several factors are taken into consideration, such as if the application is more geared toward real-time or batch-based processing. In

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addition, SAP requires the hardware be certified to run SAP HANA. A list of certified vendors is provided in the Certified and Supported SAP HANA Hardware Directory.

Let's look at a sample SAP Convergent Charging SAP HANA sizing recommendation report (see Table 10.1):

- Deploy six server hosts for SAP Convergent Charging Core Server instances:
  - Two servers to host dispatcher and guider instances (one each)
  - Two servers to host rater and bulkloader instances (one each)
  - Two servers to host updater instances (one each)
- Deploy a minimum-sized SAP HANA database node (RAM), 192 GB, based on certified hardware and deployment model.
- Load test before go-live to assess suitability and to verify that the CPU sizing is sufficient.

2M Contracts	SAPS	Memory (GB)	Storage (GB)
Server (dispatcher and guider)	1000	10	128
Server (rater and bulkloader)	7500	22	750
Server (updater)	7500	15	128
Database server (SAP HANA)	-	192	711

Table 10.1 Sample SAP Convergent Charging SAP HANA Sizing Recommendation Report

Whether you choose greenfield, production, or expert sizing, a sizing exercise should be performed for all products that make up SAP Billing and Revenue Innovation Management. Your approach may be a mixture of types depending on what is already installed on your landscape.

When evaluating your landscape, there are multiple factors to consider. Consider hardware, software, security, and performance, as well as the other factors discussed in this section. Appraise your landscape and how you want to deploy SAP Billing and Revenue Innovation Management. Defining your landscape and deployment options is a building block for defining your SAP Billing and Revenue Innovation Management solution.

There are multiple factors to evaluate when deciding to implement SAP Billing and Revenue Innovation Management. First, decide what products to implement and determine what needs to be installed, updated, or upgraded. Second, what additional configuration needs to be done on existing products? Determine where servers will be located to reduce latency if you require global access. It's also important to decide on infrastructure—that is, physical versus virtual. In addition, you should consider security, error reporting, and data protection.

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10.3 Evaluating Your Deployment Options

Several deployment options are available according to different business requirements. Every client and engagement will have its own unique requirements and needs to be analyzed carefully while evaluating the deployment options. In one good example of such a scenario, say that after the generation of invoices in SAP Convergent Invoicing, the process is extended for the billing process in sales and distribution. Another business requirement could be related to the billing of bundled products with hardware sales through sales and distribution and digital sales through SAP Billing and Revenue Innovation Management, subscription order management; SAP Convergent Charging; and SAP Convergent Invoicing in a single invoice. There are requirements for contract accounting postings as well, in which invoiced items in SAP Convergent Invoicing can be further processed for account postings either using the standard contract accounting integration or via financial accounting with custom developments. There are also clients who want to feed in rated transactions from legacy portals to SAP Convergent Invoicing directly and invoice the customers. Remember that SAP Billing and Revenue Innovation Management is a modular solution, so the framework is flexible to address specific business needs and issues and doesn't require all modules to be implemented at the same time (if not necessary). Figure 10.6 shows different components of SAP Billing and Revenue Innovation Management and modules available in SAP S/4HANA systems (1809 and onwards).

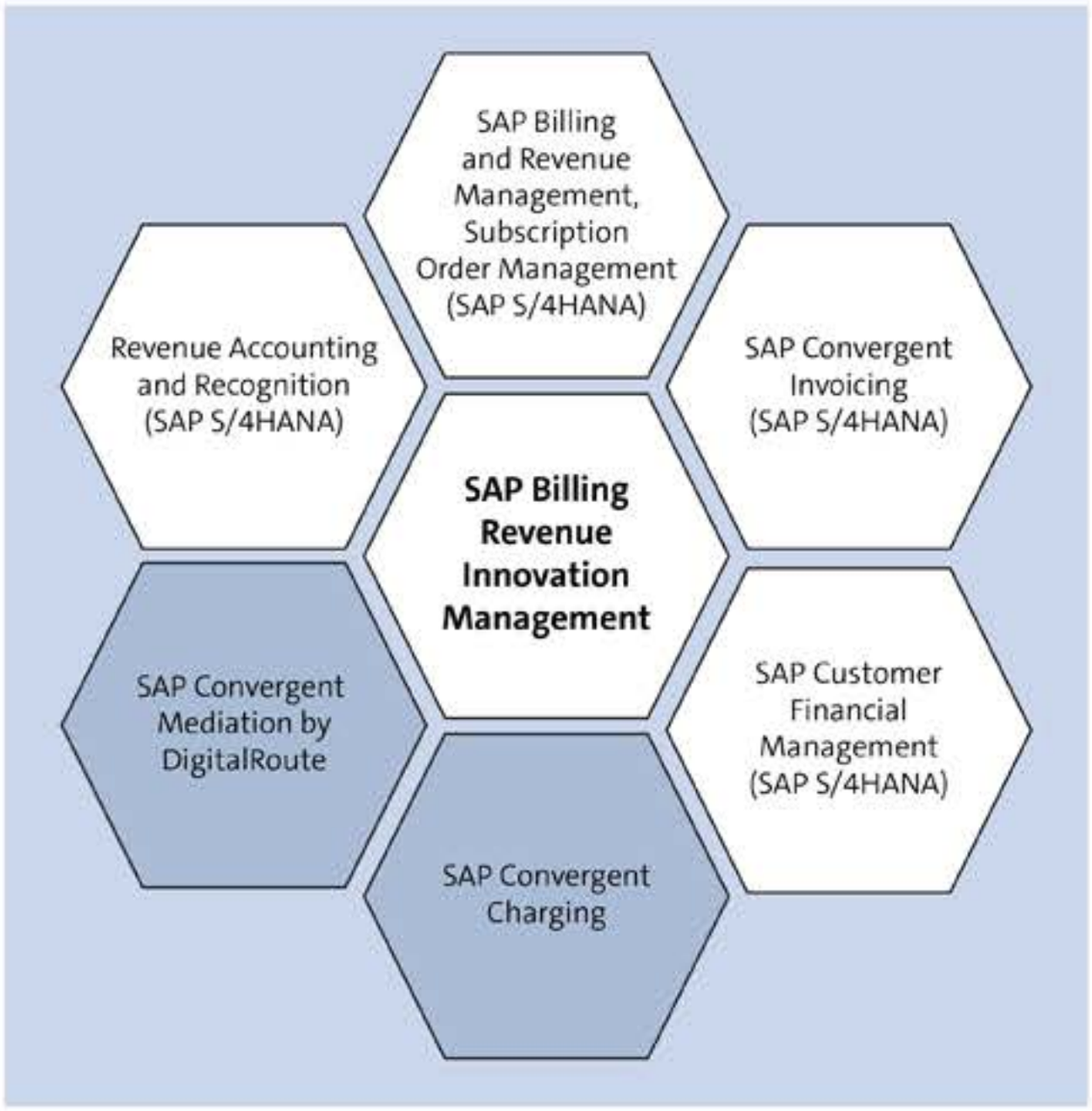

Figure 10.6 SAP Billing and Revenue Innovation Management: Modular components

The following sections discuss a few deployment options.

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10.3.1 SAP Billing and Revenue Innovation Management: Full Suite Deployment

The SAP Billing and Revenue Innovation Management full suite implementation refers to implementing all components of SAP Billing and Revenue Innovation Management (i.e., SAP Subscription Billing, SAP Convergent Charging, SAP Convergent Invoicing, and contract accounting) at the same time. Usually, clients opt for the full suite if they don't have a robust order and product management solution but are looking to adapt an XaaS model or are upgrading from a legacy order-to-cash platform. In addition, implementing the full suite provides flexibility as well as optimization, which helps the client build their subscription and usage business effectively. Because all components of SAP Billing and Revenue Innovation Management are natively integrated, it doesn't require any additional middleware solution between each component.

10.3.2 SAP Billing and Revenue Innovation Management: Phased Deployment Using SAP Convergent Charging, SAP Convergent Invoicing, and Contract Accounting

The deployment option is suitable to manage specific requirements for usage rating and billing. This framework will require additional integration with legacy order and master data management systems. Usually, clients looking forward to streamlining their usage and billing processes typically opt for such a deployment option. Note that this model may require complex integrations and enhancements with existing order management systems based on the type of features to be enabled.

10.3.3 SAP Billing and Revenue Innovation Management: Phased Deployment Using SAP Convergent Invoicing and Contract Accounting

This deployment is typically used to streamline billing-specific requirements and also to adapt to the SAP Billing and Revenue Innovation Management framework. Clients looking for phased release typically start streamlining their billing-specific requirements and issues, and gradually transform the solution model to include SAP Convergent Charging and SAP Billing and Revenue Innovation Management, subscription order management.

10.3.4 SAP Billing and Revenue Innovation Management Deployment Using SAP Convergent Charging Only

SAP Convergent Charging helps to improve and streamline usage handling processes. Clients who are looking to streamline usage rating and charging processes can opt for SAP Convergent Charging and make use of standard web services to integrate with existing order management, mediation, and ERP solutions to process the order-to-cash flow.

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10.3.5 Deployment Option with SAP Flexible Solution Billing

In this option, billing data from SAP Convergent Invoicing and other legacy systems can be integrated in sales and distribution for a unified billing process in SAP Flexible Solution Billing. This option is best suited for billing the bundled products having hardware sales through sales and distribution and digital sales through SAP Billing and Revenue Innovation Management (SAP Billing and Revenue Innovation Management, subscription order management, SAP Convergent Charging, and SAP Convergent Invoicing) in a single invoice.

10.3.6 Deployment Option with Sales and Distribution Integration

As of SAP S/4HANA 1909 and above, SAP Billing and Revenue Innovation Management, subscription order management, offers native integration with the sales and distribution module. In this option, hardware sales managed using sales and distribution can be integrated with SAP Convergent Invoicing to consolidate different streams of revenue. This option is best suited for billing the bundled products with hardware sales through sales and distribution and with digital sales in a single invoice through SAP Billing and Revenue Innovation Management, subscription order management; SAP Convergent Charging; and SAP Convergent Invoicing.

10.3.7 Deployment Options with Contract Accounting Posting

In this option, invoiced items in SAP Convergent Invoicing can be further processed for account postings, either using the standard contract accounting integration or via financial accounting with custom developments. Contract accounting includes all the standard features of financial accounting, such as open item management, payments processing, dunning, and so on. In addition, contract accounting provides more capabilities, such as mass parallel processing, summarized postings, and standard integration with the general ledger.

10.4 SAP Standard Content Activation Service

The SAP standard content activation service is a prepackaged, ready-to-use, end-to-end reference solution for SAP S/4HANA, SAP Customer Experience, or SAP SuccessFactors for a specific industry or line of business. The SAP standard content activation service is planned to decrease the risk during discovery, preparation, exploration, business process changes, and realization activities, and it helps to increase quality and reduce cost by accelerating adoption time.

The SAP standard content activation service is also offered for the SAP Billing and Revenue Innovation Management ready-to-run, comprehensive reference solution built

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on SAP S/4HANA, SAP CRM application (for SAP S/4HANA version prior to 1809), and the SAP Convergent Charging solution. It's delivered with relevant business content, accelerators, and enablement services that help clients reduce costs, decrease risks during implementation, and accelerate adoption during discovery, exploration, and realization activities.

Using the SAP standard content activation service, you can also gain comprehensive control of the entire order-capture and billing chain for finance and billing with automated operations. You can calculate and report on flexible revenue-share terms for partner management in a timely and accurate manner.

Following are some of the benefits of using the SAP standard content activation service with SAP Billing and Revenue Innovation Management:

- Reduced costs and resource effort thanks to a preconfigured, ready-to-run solution
- Elimination of risk through a proven reference-solution architecture and comprehensive process support
- Increased innovation adoption and decreased time to value through agile and lean principles
- Accelerated enablement and increased engagement

10.5 Reviewing Rapid Deployment Solutions and Implementation Best Practices

Rapid deployment solutions are preconfigured applications that allow companies to deploy software in weeks or months for a set price. SAP Rapid Deployment solutions are designed to address one area of an SAP enterprise application and address a specific function or need. They are designed to meet about 60% to 70% of what a customer needs, beyond which the customer can customize and tailor as needed to address unique requirements. SAP Rapid Deployment solutions aren't just toolkits, but solutions that have configuration templates (customized settings) with supporting consulting services. They also provide best practices to customers who use SAP software. Licensed customers get an added advantage because the content is free, and this helps them to reduce implementation costs and speed up deployment, which in turn adds benefits to their investment in SAP products.

SAP Rapid Deployment solutions use the concept of assemble-to-order, which helps to create one project from different SAP Rapid Deployment solutions to address any critical business challenges. With the help of SAP Rapid Deployment solutions, you can innovate and extend your technology to provide business solutions, provide mobile user experiences, and get help from the cloud. SAP Rapid Deployment solutions work with the ASAP methodology, which provides a framework for SAP projects for implementation, enhancements, and customizations needed for postproduction support.

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Deploying SAP Rapid Deployment solutions can be done individually or can be combined. But combining solutions will take extra integration steps and testing. For installing a solution, one or more software products will be needed, and SAP Rapid Deployment solutions come with an SAP Solution Manager template, which can be used individually or combined. A solution can be either implemented or configured using manual configuration or automated configuration. Once the configuration is done and activation is completed, you can test your solutions using the template test scripts provided along with the scope items.

SAP Rapid Deployment solutions provide a complete configuration and master data setup, including business partners, products, pricing, and business agreements/contracts. The following sections provide basic information about master data and how to set it up for an SAP Billing and Revenue Innovation Management implementation.

10.5.1 Master Data

Master data is one of the most important objects in the SAP application framework and is used across almost all processes in every area of SAP and SAP Billing and Revenue Innovation Management. The following master data needs to be considered while implementing SAP Billing and Revenue Innovation Management:

- Business partner
- Contract account
- Product
- Pricing

Business Partner

Business partner is a generic term used for parties involved in a business scenario and can represent anyone involved in the business or who shows interest in the business or project. The business transactions or scenarios they can access depend on the roles assigned to them. The business partner record holds vital information such as name, address, bank details, communication details, and more. It's always best practice to manage business partners in SAP S/4HANA and SAP Convergent Charging as a subscriber account.

Each business partner is identified using a unique business partner number. A business partner is created with a sold-to party and contract partner role. The contract partner role is mandatory for creating transactional data (subscription order and subscription contract) in SAP Convergent Invoicing. The business partner number range can be external or internal depending on the business scenario. For example, if business partner creation is being handled by a third-party system, then the external number range should be configured in SAP to accept the business partner ID. If SAP S/4HANA is the leading system for maintaining master data, then an internal number range needs to

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be configured in SAP and SAP will generate business partner IDs when new business partners are created. An external or internal number range should be assigned to a business partner grouping.

Contract Account

The contract account is an object used to store controlling data for a long-term business relationship with a business partner. A contract account is also used to store financial data details such as the following:

- Payment scheduling
- Payment terms
- Tax
- Invoice creation
- Correspondence
- Dunning control
- Bank details/payment card details

These values control processes in open item accounting and invoicing.

A contract account is used to handle both prepaid and postpaid products. In prepaid products, there is a prepaid account involved, which is a subset of the business agreement. Prepaid accounts have additional details, such as a minimum balance, and are widely used in telecommunication and tollway solutions.

A contract account is always attached to a business partner, and a business partner can have one or more contract accounts. When processing a business transaction (contract creation), you can always make sure that the system determines a suitable contract account.

Products

SAP Billing and Revenue Innovation Management requires a product as part of the master data setup. The type of the product is a subscription product. You create a subscription product in SAP Billing and Revenue Innovation Management, subscription order management. A subscription product provides attributes and settings required to create a subscription order/solution quote and subscription contract. Once the business requirements of the client are obtained, you can start planning what attributes and settings are required to create the products. Product setup starts with a product hierarchy and product category. Product master data is bundled into a product model, which is a container for modeling data. If your implementation requires configurable products, you need to perform additional setup, such as setting up classifications, dependencies, characteristics, and characteristic values.

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For configurable products, a customer can choose from multiple given options. They can contain many characteristics, and a customer can select values for any characteristic while ordering the product. According to the value of each of the characteristics of each attribute/value, product components will be determined automatically by the rules configured during product modeling.

There are different types of products based on business usage:

- Subscription products
- One-off product
- Hardware/physical products
- Partner products

A subscription product must be assigned to a charge plan through cross-catalog mapping, which will be discussed in the next section. You need to create a charge plan in SAP Convergent Charging before you can perform cross-catalog mapping. A charge plan represents the charging part of a service that a service provider offers to its customers or partners via the product catalog and service provisioning. A charge plan must have one or more charge objects in the same catalog. A charge has three types of price plans: recurring, one-time, and usage-based types.

Cross-Catalog Mapping

Cross-catalog mapping is a process to assign a subscription product in SAP Billing and Revenue Innovation Management, subscription order management, to a charge plan in SAP Convergent Charging. Cross-catalog mapping is done during product setup in SAP Billing and Revenue Innovation Management, subscription order management. This was discussed in detail in Chapter 2, and Figure 10.7 and Figure 10.8 shows how cross-catalog mapping is managed in SAP Billing and Revenue Innovation Management, subscription order management.

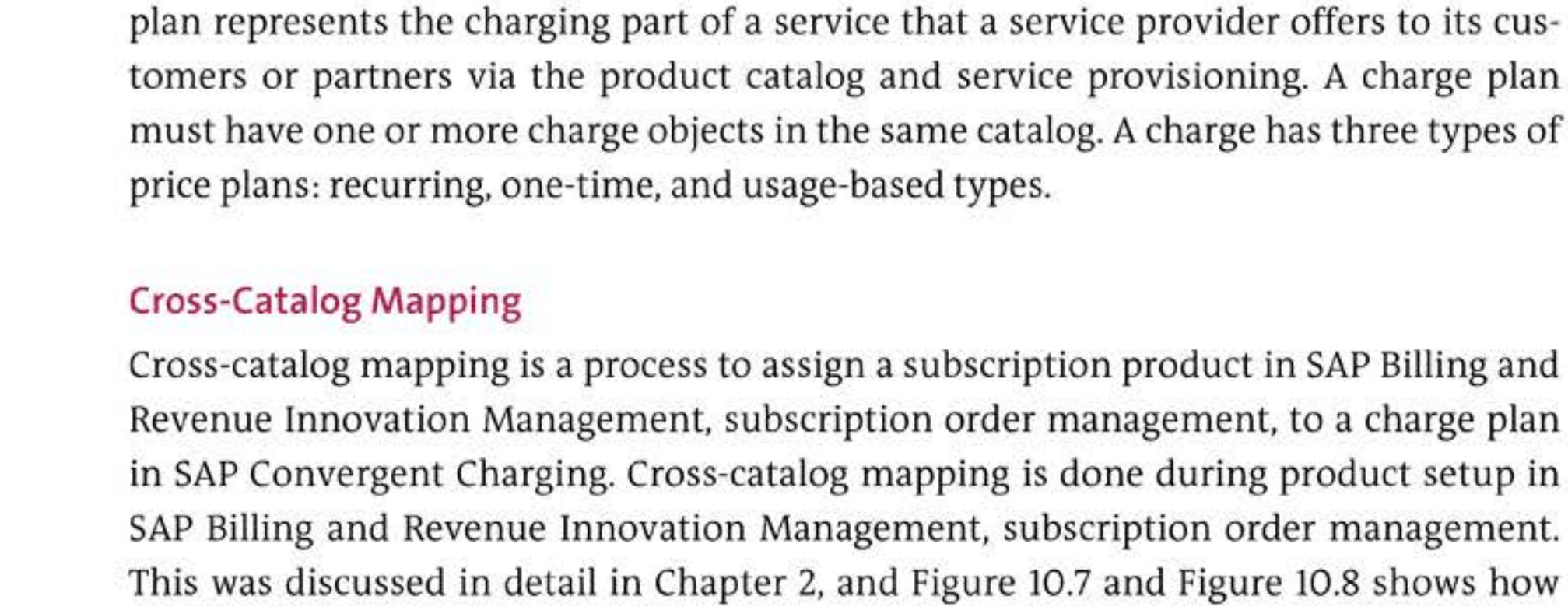

Figure 10.7 Setting up Cross-Catalog Mapping in SAP Billing and Revenue Innovation Management, Subscription Order Management

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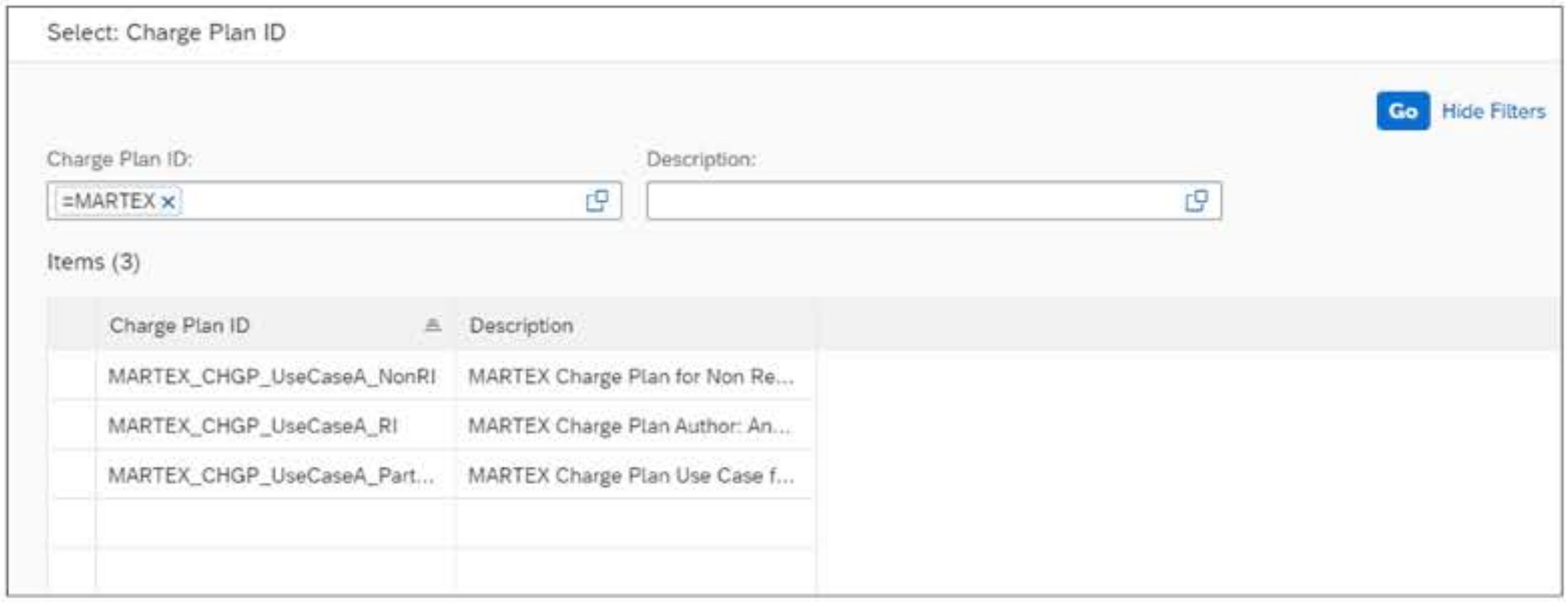

Figure 10.8 Search SAP Convergent Charging Charge Plans

During this process, SAP Billing and Revenue Innovation Management, subscription order management, calls a web service of SAP Convergent Charging to show the list of available charge plans. After the mapping process is completed, a mapping version will be automatically created. The mapping version should be in **Released** status for the cross-catalog mapping to be activated. The information stored in this mapping is then used during replication of contracts to SAP Convergent Invoicing and SAP Convergent Charging. Information needed in billing, provisioning, or other systems is gathered using the mapping rules defined in cross-catalog mapping. Cross-catalog mapping versions and details are available in the **Cross-Catalog Mapping** assignment block in the Subscription Product-Specific app in SAP Fiori.

Once the products are successfully created, they should be uploaded/replicated to contract accounting. There are two different product types available to replicate these products to contract accounting:

- Contract accounting product
- Material master product

A contract accounting product is a lightweight entity that holds only language-dependent texts and sales-area-dependent data. If your implementation only involves either accounts receivable or accounts payable transactions, you can use either a contract accounting product or a material master product because it will be used only in subscription contracts and partner agreements. However, if your implementation involves partner settlements or revenue shares, which will create accounts payable transactions along with the accounts receivable ones, you have the option to use either material master product or contract accounting product. Using a material master product for a partner settlement or revenue-share scenario allows both the subscription contract and partner agreement to use the same product. On the other hand, if you choose to use a contract accounting product in SAP ERP, it only allows you to create a subscription contract. For a partner agreement, you need to create a new product

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because SAP doesn't allow you to use the same product for both the subscription contract (accounts receivable) and partner agreement (accounts payable).

Pricing

In a typical SAP Billing and Revenue Innovation Management implementation, as mentioned previously, there are different types of pricing:

- Pricing for a one-time charge
- Pricing for a recurring charge
- Pricing for a usage-based charge

Customers will be charged a one-time, recurring, or usage-based subscription charge based on the service they subscribe to and consume. This offers more flexibility in settlement of charges. For example, a mobile service provider offers services for customers such as a data plan based on usage consumption and voice and text messaging with a fixed monthly recurring charge; in addition, a one-time charge is usually applied for a new customer to activate services.

10.5.2 Transactional Data

Let's look at some of the transactional data considerations, from the subscription order to subscription contracts.

Subscription Order

In a traditional SAP implementation, sales are usually captured in the sales order object. However, SAP Billing and Revenue Innovation Management has its own version of a sales order, called the subscription order. The subscription order differs from a typical sales order because it has special handling built-in to support both hardware and subscription-based products. For example, consider the technical resource ID, which is used by telecommunication companies to store a phone number that acts as a unique identifier for a customer order. Because SAP Billing and Revenue Innovation Management contains three distinctive modules, the data model of a subscription order is designed to integrate with other SAP Billing and Revenue Innovation Management modules (e.g., placeholders for business agreements for SAP Convergent Invoicing/contract accounting integration and triggering of the order distribution infrastructure).

As part of SAP's standard delivery, a baseline subscription order configuration is provided as transaction type PRVO. While that allows for end-to-end SAP Billing and Revenue Innovation Management integration, business processes specific to certain companies aren't part of the flow (e.g., approvals or integration with different payment gateways/tax engines). Therefore, during SAP Billing and Revenue Innovation Management project implementation, making copies of the standard transaction is recommended.

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Subscription Contracts

A subscription contract is an agreement placed between a customer and a service provider for the services rendered to the customer over an arc of time. A subscription contract is generated as a subsequent document when a subscription order that contains a rate plan product is released. During the release process, contract-relevant data from line items containing the rate plan product will be copied to the subscription contract. Each line item containing a subscription product will generate a new contract; therefore, it's common to see a subscription order generating multiple subscription contracts as a subsequent document.

Over the lifecycle of a subscription contract, changes to the contract (e.g., cancellation or product change) are managed using a change process framework. Figure 10.9 shows some of the standard change processes available for subscription contracts.

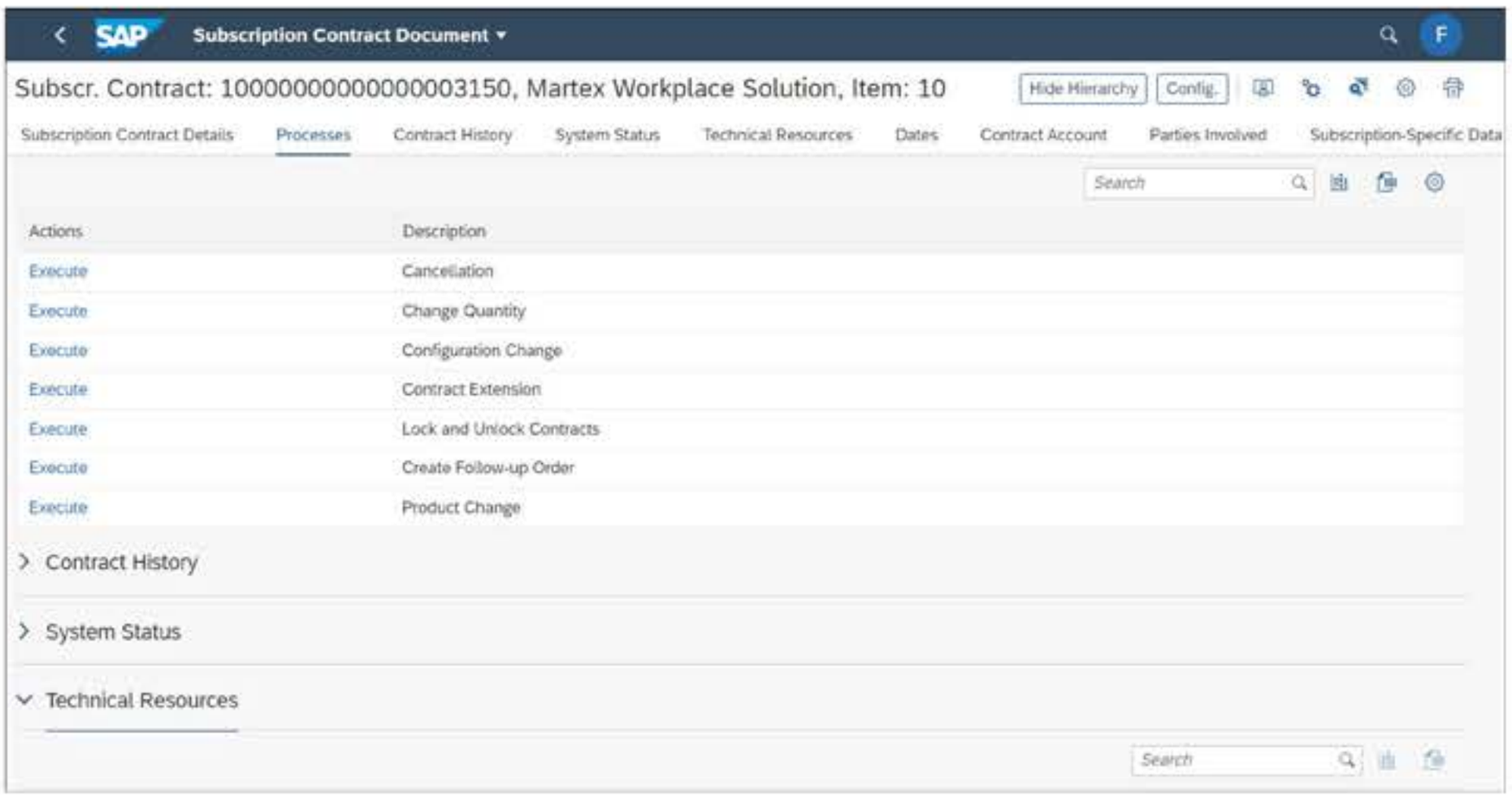

Figure 10.9 Change Process for Subscription Contracts

Changes to a contract will first generate a new object called a change order. Change orders are temporary documents that hold contract changes for review or approval before merging the changes back to the subscription contract.

As part of SAP's standard delivery, a baseline subscription contract configuration is provided as transaction type PRVC, but it's often recommended to make a copy to support company-specific flows during implementation.

10.6 Replication of Subscription Contracts to SAP Convergent Invoicing and SAP Convergent Charging

Once the subscription contract is successfully created in SAP Billing and Revenue Innovation Management, subscription order management, it will be replicated using the

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Order Distribution Infrastructure (ODI). The ODI uses the Common Object Layer framework to distribute the subscription contracts. This distribution of contract data to SAP ERP is applicable for create, change, and cancel scenarios/processes. ODI steps can be configured in SAP Billing and Revenue Innovation Management, subscription order management, where you need to define the step types and schema. The step types will be assigned to the schema based on the replication scenario.

Distribution of these contracts through ODI can be monitored using the distribution monitor. Using this monitor, contract replication can be reprocessed (displayed, rescheduled, and rerun). Once the subscription contract is successfully replicated to SAP Convergent Invoicing and SAP Convergent Charging, the status of the contract in SAP Billing and Revenue Innovation Management, subscription order management, will be **Technically Active**.

10.7 Summary

The first step toward SAP Billing and Revenue Innovation Management is deciding to transition to a subscription-based business model. Team members then need to gather information on the current systems and business processes, as well as understand the deliverables and dependencies. As our customers' needs and requirements rapidly evolve, it's imperative that we continue to demonstrate a willingness to adapt and change our standard business practices. Implementing SAP Billing and Revenue Innovation Management is one of the first steps you can take to make this transition easier and smoother. As a reminder, the key steps to consider in the transition to SAP Billing and Revenue Innovation Management include the following:

- Considering the different subscription-based business models
- Evaluating your landscape to determine the best deployment scenario
- Choosing deployment options based on your product types
- Reviewing SAP Rapid Deployment solutions and implementation best practices
- Considering master data for implementing SAP Billing and Revenue Innovation Management
- Determining standard versus custom transaction types for subscription orders or subscription contracts
- Understanding the replication of subscription contracts to SAP Convergent Invoicing or SAP Convergent Charging
- Considering the integration points between core modules of SAP Billing and Revenue Innovation Management

With the tools and knowledge provided, SAP Billing and Revenue Innovation Management can be leveraged as a solution in your journey toward a subscription-based business model.

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Conclusion

Now, we will quickly revisit the contents of the previous chapters, which include the SAP Billing and Revenue Innovation Management components, the capabilities enabled by each of the components, and how they integrate with each other. We will then discuss the evolving business models and their move to the subscription economy and the need for organizations to offer flexible consumption offerings to have the competitive edge.

In **Chapter 1, Introduction to SAP Billing and Revenue Innovation Management**, we talked about the primary business drivers that motivate companies to follow the new trends in the subscription economy. The chapter explains the opportunities and risk for companies in the new economy. Then, we introduced the key SAP Billing and Revenue Innovation Management components and its supporting add-ons. Finally, to prepare the readers for the subsequent chapters, a case study was outlined. The case study focused on how Martex Corp., a fictitious company, is trying to revamp its billing processes to stay competitive. Martex served as the baseline for the book with the definition of master data and configuration set up.

In **Chapter 2, Subscription Order Management**, we learned how to setup and enable SAP Billing and Revenue Innovation Management, subscription order management, including the configuration steps to build the use case and the integration with SAP Convergent Charging and SAP Convergent Invoicing.

In **Chapter 3, Rating and Charging**, we showed you how to setup and use SAP Convergent Charging and its modeling process based on Martex Corp. We went through the charge creation process by explaining the core tool, the charge types (recurring, one time, usage), and interface objects such as chargeable item classes and charged item classes.

In **Chapter 4, Invoicing**, we learned how to setup and use SAP Convergent Invoicing. We discuss billable items, consumption items, rating, billing, invoicing, and partner settlement, as well as some of the standard out-of-the-box reporting capabilities that SAP Convergent Invoicing offers.

In **Chapter 5, Contract Accounting**, we saw how to setup and use contract accounting, along with key concepts like account determination, open item management, account balance display, and integration with the general ledger, as well as some of the standard out-of-the-box reporting capabilities and relevant transaction codes for contract accounting.

In **Chapter 6, SAP** Subscription Billing, we discussed how SAP Subscription Billing is a cloud-based solution built on SAP Business Technology Platform, integrating the front-office SAP Customer Relationship Management (SAP CRM) applications to the back-

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end digital core solution. We also learned how SAP Subscription Billing can help clients increase customer satisfaction as well as customer retention.

In **Chapter 7, Integration with Other Solutions**, we learned how SAP Billing and Revenue Innovation Management is implemented in a heterogenous SAP CRM landscape and with other ancillary systems such as SAP Convergent Mediation by DigitalRoute, SAP Revenue Accounting and Reporting, SAP Customer Experience, SAP Customer Financial Management, entitlement management systems, tax engines, and payment gateways.

In **Chapter 8, Extension and Connectivity**, we talked about various Extension and interface components (SOAP/REST) delivered as part of SAP Billing and Revenue Innovation Management suite, their extension span across SAP Billing and Revenue Innovation Management, subscription order management, SAP Convergent Charging, SAP Convergent Invoicing, and contract accounting functionalities.

In **Chapter 9, Data Migration**, we looked at various methods for bringing the legacy data into SAP Billing and Revenue Innovation Management. The chapter further outlined types of data and sequencing of objects such as business partner, contract accounts, product, subscription orders/contracts, and accounts receivable, if any.

Chapter 10, First Steps to SAP Billing and Revenue Innovation Management, explores leading practices for planning, evaluating, and considering SAP Billing and Revenue Innovation Management as a solution for the future. This chapter offered best practices for planning and considering SAP Billing and Revenue Innovation Management as a solution. You will learn how to avoid common pitfalls and gain insights into key decision points.

To thrive and succeed, companies should be able to see around corners, make quick course corrections, and harness technology to reimagine business models and reinvent how to deliver value to their customers. These strategies ensure companies maintain their competitive advantage and stay ahead of the curve.

With those principles in mind, many companies are looking beyond traditional infrastructure by migrating up the value chain—from products to platforms, software, and services. At the same time, many customers are demanding more control over what they consume and how they pay for it. Consequently, companies are shifting to the flexible consumption or anything as a service (XaaS) model for their enterprise. Flexible consumption models can present XaaS providers with substantial potential benefits. These range from predictable, recurring revenue streams and improved margins to higher customer retention (driven by stickier, long-term relationships with those customers). Perhaps the biggest reason for some companies to consider transitioning to an XaaS model is that they can't afford not to.

There's a gigantic wave of companies shifting toward this new XaaS model. The first and most noticeable shift to the new flexible consumption model is with the technology industry, and most exciting is with the entertainment industry. The success of

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Netflix and Hulu have motivated many other companies to rethink their content delivery approaches. The traditional model, where the relationship ends with the final sale, is outdated. Many companies are doing everything they can to catch up or to further strengthen their position as the leaders. Technology companies such as HPE are determined to stay ahead of the competition with their flexible consumption offerings and to have a recurring revenue business model. HPE is accelerating its move to deliver the full HPE portfolio in an as-a-service subscription model by the end of 2022.

Let's take another look into the wave of companies are shifting into the new flexible consumption model. If you are reading this book, that means you're familiar with the publisher of this book, SAP PRESS. This could mean two things: you have bought the book directly, or perhaps you have acquired this book using the SAP PRESS subscription service.

This is not meant as a sales pitch for the SAP PRESS subscription service (even though they earn every right to have that). The paragraph serves as a clear example of how the companies across industries are getting into the new paradigm–the subscription economy, where XaaS will be the new norm.

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Appendix A

Certification Preparation

This chapter introduces a few practice questions and answers on the concepts of SAP Billing and Revenue Innovation Management. The intent of this appendix is to allow you to gauge your understanding of SAP Billing and Revenue Innovation Management and assess your preparedness for SAP Billing and Revenue Innovation Management certification and implementation. Though the questions listed here may not appear as identical in the actual SAP Billing and Revenue Innovation Management examinations and should not be used as a practice questionnaire, they will help you identify the exam pattern and revisit key topics to increase your chances to be successful on the exam.

A.1 Objectives of the Test

The SAP Billing and Revenue Innovation Management certification exam validates that the candidate possesses the fundamental knowledge in business processes related to the subscription economy and the high-level configuration required to enable the use cases using SAP Billing and Revenue Innovation Management. This certification helps candidates be better prepared as it does the following:

- Measures configuration and validation skills.
- Reinforces preparedness and the ability to succeed in a real-life SAP Billing and Revenue Innovation Management project.

As of May 2022, SAP offers the following certification exams related to SAP Billing and Revenue Innovation Management:

- **SAP Certified Application Associate - SAP Billing and Revenue Innovation Mgmt. - Subscription Order Management (C_BRSOM_2020)**

To appear for this exam, you should have fundamental and core knowledge in subscription-based business processes and system configuration knowledge in SAP Billing and Revenue Innovation Management, subscription order management. The exam scope covers an overview of SAP Billing and Revenue Innovation Management components and how each of the components play a role to implement the end-to-end solution. However, most of the scope lies with SAP Billing and Revenue Innovation Management, subscription order management, with topics related to master data, subscription transactions, and integrations. The majority of the content is covered in Chapter 2 of this book.

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- **SAP Certified Application Associate - SAP Billing and Revenue Innovation Management - Usage to Cash (C_BRU2C_2020)**
  To appear for this exam, you should have fundamental and core knowledge in subscription-based business processes and high-level configuration knowledge in SAP Convergent Charging, SAP Convergent Invoicing, and contract accounting. You need to have knowledge in foundational components as well as architecture components for each module.

The following sections of this appendix will cover questions from all modules. Note that the questions are not identical to the SAP exam but will help you revisit your learning journey so far and understand your key strength and improvement areas.

Sections A.2 to A.4 provide practice questions and answers. Individual sections allow you to do the following:

- Solidify your skills with tailor-made questions.
- Review and reinforce what you've learned.

A.2 Key Terminology

Following are some of the key terms you might encounter while going through the exam questionnaires:

- Anything as a service (Xaas)
- Business partner
- Contract accounting
- Cross-catalog mapping (CCM)
- Device as a service (Daas)
- Enterprise resource planning (ERP)
- Graphical user interface (GUI)
- Infrastructure as a service (IaaS)
- Implementation Guide (IMG)
- Open Data Protocol (OData)
- Order Distribution Infrastructure (ODI)
- Platform as a service (PaaS)
- Software as a service (Saas)
- Sales and distribution
- Simple Object Access Protocol (SOAP)
- SAP Billing and Revenue Innovation Management, subscription order management
- SAP Convergent Charging

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- SAP Convergent Invoicing
- SAP Convergent Mediation by DigitalRoute
- SAP Customer Financial Management
- SAP Customer Relationship Management (SAP CRM)
- SAP Master Data Governance (SAP MDG)

A.3 Practice Questions

The following sections provide practice questions for different topics that appear on the exam, including SAP Billing and Revenue Innovation Management, subscription order management; SAP Convergent Charging, SAP Convergent Invoicing, and contract accounting.

A.3.1 SAP Billing and Revenue Innovation Management, Subscription Order Management

1. What are the key benefits for XaaS for service providers? (There is more than one correct answer.)
   A. Value acceleration
   B. Revenue predictability
   C. Customer engagement
   D. Improved field service management
2. Identify the key business partner role for setting up contract accounts.
   A. MKK
   B. FLUCOO
   C. DROO
   D. CRMOO
3. True or False – SAP Billing and Revenue Innovation Management, subscription order management, should always be integrated with SAP Convergent Charging to capture pricing information.
   A. True
   B. False
4. Which of the following additional fields are required for subscription products?
   A. **Billing Cycle Determination Rule**
   B. **Contract Terms**
   C. **Payment Method**
   D. **Extension Terms**

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5. Which of the following condition types are delivered in the sales and distribution component for SAP Billing and Revenue Innovation Management, subscription order management?

   A. PSMB

   B. PSPB

   C. PPBS

   D. SOMP

6. You use the subscription order change process to . . .

   A. change contract duration.

   B. correct a usage fee.

   C. change an assigned technical ID.

   D. execute cancellation.

7. Which of the following objects are shared between SAP S/4HANA and SAP Convergent Charging?

   A. Contract account

   B. Price plan

   C. Charge plan

   D. Business partner

8. Select all the tiles that are used for product modeling.

   A. **Manage Solution Quotation**

   B. **Manage Product Bundles**

   C. **Manage Product Master Data**

   D. **Manage Range Tables**

9. For each subscription contract change, a change order is created.

   A. True

   B. False

10. A subscription order once submitted can't be changed.

    A. True

    B. False

11. For every process type, only one document distribution schema can be assigned.

    A. True

    B. False

12. It is possible to group and execute different changes on a subscription contract simultaneously.

    A. True

    B. False

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13. It is possible to extend the end date of the contract . . .
    A. when the contract end date is in the past.
    B. when the contract end date is in the future.
    C. when the contract end date is the current date.
14. While doing cross-catalog mapping for the subscription products, there can be one or more charge plans that can be in **Released** status.
    A. True
    B. False
15. SAP Billing and Revenue Innovation Management, subscription order management, can be activated with which of the following?
    A. SAP CRM provider order management
    B. SAP S/4HANA Service
    C. SAP S/4HANA sales and distribution
    D. SAP Convergent Charging
16. Subscription contracts include which of the following objects?
    A. Technical identifier
    B. Contract account
    C. Business partner
    D. Cross-selling products
17. Master agreements can define . . .
    A. who is allowed to order the service.
    B. which price is defined for a service.
    C. how much the bill is.
    D. what can be ordered.

A.3.2 Charging

1. For which document can you use the counter in a charge plan for all contracts collectively?
    A. Master agreement
    B. Sharing contract
    C. Sharing order
    D. Provider order

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2. Which provider contract objects are shared between SAP CRM, SAP ERP, and SAP Convergent Charging?
   A. Contract account
   B. Price plan
   C. Charge plan
   D. Business partner
3. What is a part of the customer master data managed by SAP Convergent Charging and linked to an end customer of a service provider?
   A. Business partner
   B. General ledger account
   C. Subscriber account
   D. Business partner relationship
4. A subscriber account is linked to the following component of a charge?
   A. Price plan
   B. Usage rate
   C. Charge plan
   D. Charging plan
5. What are the two categories of counters?
   A. Global counters
   B. Universal counters
   C. Persistent counters
   D. Transient counters
6. When a consumption item class is replicated from SAP Convergent Invoicing to SAP Convergent Charging, which object/component is created in SAP Convergent Charging?
   A. Charged item class
   B. Chargeable item class
   C. Charging item class
   D. Billable item class
7. Where do we define the output format of the rating and charging process?
   A. Charged item class
   B. Chargeable item class
   C. Charging item class
   D. Consumption item class

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8. Which of the following statements are true?
   A. We can add multiple charges inside a charge plan.
   B. We can add multiple rates inside a price plan.
   C. We can add multiple price plans inside a charge.
   D. We can add multiple charging components inside a charging plan.
9. Which of the following rates are triggered during suspension or activation of a provider contract?
   A. One-time
   B. Recurring
   C. Usage rate
   D. Consumption rate
10. Which of the following is NOT a charging plan component?
   A. Operators
   B. Charging
   C. Objects
   D. Comparators
11. With the help of which component can we reuse the same logic across multiple price plans?
   A. Prefix switching
   B. Pricing macro
   C. Switch table
   D. Property introducer
12. There is a parameter defined as a string. However, in your logic, this parameter is to be used as a number. Which of these logical components will you use?
   A. Property Type Converter
   B. Property Introducer
   C. Prefix Switching
   D. Allowance Property Introducer
13. In a standard SAP Billing and Revenue Innovation Management landscape, the flow of a provider contract in all three systems of SAP Billing and Revenue Innovation Management is?
   A. SAP Billing and Revenue Innovation Management, subscription order management, to SAP Convergent Invoicing to SAP Convergent Charging
   B. SAP Billing and Revenue Innovation Management, subscription order management, to SAP Convergent Charging to SAP Convergent Invoicing
   C. SAP Convergent Invoicing to SAP Billing and Revenue Innovation Management, subscription order management, to SAP Convergent Charging
   D. SAP Convergent Invoicing to SAP Convergent Charging to SAP Billing and Revenue Innovation Management, subscription order management

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14. In Which of the following is true about SAP Convergent Charging Core Server?
    A. Defines the pricing algorithms and structures
    B. Computes the price that corresponds to a service consumption
    C. Defines the accounts to charge according to the defined charging plans
    D. Manages the batch mode processing
15. Which instance of SAP Convergent Charging Core Server is used to load the files generated by the rating process into SAP Convergent Invoicing?
    A. Dispatcher
    B. Guider
    C. Rater
    D. Bulkloader

A.3.3 Invoicing

1. Billing items creation can be achieved through . . .
    A. a combination of business partner, contract account, and subprocess.
    B. rules defined under the grouping control parameter for grouping variant.
    C. manually using expert mode during the execution of the billing process.
    D. all of the above.
2. Which is the highest node of configuration in SAP Convergent Invoicing?
    A. **Business Partner**
    B. **Account Determination**
    C. **Billable Item Class**
    D. None of the above
3. Which of the following statements are true?

    1. Interface components defined for a billable item class help to determine standard/custom fields required for a business transaction.
    2. The logical sequence of process steps prior to billing constitutes selection, aggregation, and grouping.
    3. Only billable items in **Billed** status can be billed.
    4. The receivables and payables of a business transaction are stored under the Main Items record type.

    A. Only 1
    B. 1 and 4
    C. 1, 2, and 4
    D. All of the above

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4. The billing process in SAP Convergent Invoicing will select the billable items in _____ status for billing.

   A. **Raw** and **Billable**

   B. **Raw**, **Raw-Excepted**, and **Billable**

   C. only **Billable**

   D. **Raw** and **Raw-Excepted**

5. Multiple billing documents cannot be grouped into one invoice document.

   A. True

   B. False

6. What are selection, grouping, and aggregation variants assigned to?

   A. Billing process only

   B. Billing process and subprocess

   C. Subprocess and billable item class

   D. Billing process and billable item type

7. How is the unrated partner consumption item generated in SAP Convergent Invoicing for partner settlement?

   1. The file upload functionality (`FKKBIXCIT_UPLOAD`) is used for uploading partner usages.
   2. The system generates consumption items for partner settlement when the billable item is created for the end customer.

   A. 1

   B. 2

   C. Either 1 or 2

   D. Both 1 and 2

8. The partner settlement process is a solution used for the _____ business model.

   A. revenue-based

   B. promotional

   C. revenue-sharing

   D. cost-based

9. What are the key interface components for partner settlement?

   A. Billable item class - partner settlement: basic data of customer item

   B. Billable item class - partner settlement: basic data of partner item

   C. Consumption item class - partner settlement: basic data of partner item

   D. All of the above

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10. Determine the correct match between posting area and its significance in SAP Convergent Invoicing/contract accounting.

    1. 8120 - Company code derivation
    2. 2610 - General ledger account determination for document posting
    3. 2611 - General ledger account determination for document posting
    4. 8121 - Company code derivation

    A. 1 and 3
    B. 1 and 2
    C. 2 and 4
    D. None of the above

A.3.4 Contract Accounting

1. Which of the following is a summarization criterion for a reconciliation key?

    A. Company code
    B. Currency
    C. Controlling account assignment
    D. All of the above

2. Which of the following statements is true regarding the reconciliation key in contract accounting?

    1. It can be determined both manually and automatically.
    2. Multiple contract accounting documents can be consolidated into summary records under a unique key before closing the reconciliation key.
    3. It is the link between the accounting document and the contract accounting document.
    4. Multiple contract accounting documents can be consolidated into summary records under a unique key after closing the reconciliation key and before transferring to the general ledger.

    A. 1, 2, and 3
    B. 1, 3, and 4
    C. 3 and 4
    D. None of the above

3. Which of the following is not true about the reconciliation key?

    A. Under the reconciliation key, summary records are recorded for transferring contract accounting documents to the general ledger.

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B. Reconciliation keys can be determined automatically.

C. Even if the reconciliation key is closed but not transferred, postings can be made to the reconciliation key.

D. The reconciliation key provides the link between the contract accounting document and the general ledger document.

4. Where can the payment method be updated ?

A. Contract account level

B. Document level

C. Both

D. None of the above

5. Characteristics from master data (contract account and contract) do not have any input in account determination in contract accounting.

A. True

B. False

6. For what reasons do payments go to clarification?

1. Incorrect CA
2. Posting period locked
3. Posting lock on CA
4. Payment amount greater than open receivables amount

A. 1, 2, and 4

B. 2 and 3

C. 1 and 2

D. 1, 2, and 3

7. Can credit documents be written off?

A. Yes

B. No

8. What are the components of contract accounting?

A. Reconciliation key

B. General ledger accounts

C. Profit center and cost center

D. All of the above

9. The main transaction will determine the reconciliation account.

A. True

B. False

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A.3.5 Extension and Connectivity

1. What is the standard API used for subscription order creation?
   A. CRM_ISX_BTX_API_ORD_CREATE
   B. CRM_MSG_BTX_API_ORDER_CREATE
   C. CRM_ISX_BTX_API_ORDER_CREATE
   D. None of the above
2. What is the standard API used to change the solution quotation before releasing?
   A. CRM_ISX_BTX_API_ORD_MAINTAIN
   B. CRM_MSG_BTX_API_ORDER_MAINTAIN
   C. CRM_ORDER_MAINTAIN
   D. None of the above
3. What is the standard API to create solution quotations?
   A. API_BUS_SOLUTION_QUOTATION_SRV
   B. CRM_ISX_BTX_API_ORDER_CREATE
   C. CRM_ISX_BTX_API_ORD_MAINTAIN
   D. None of the above
4. What partner role needs to be created while migrating a business partner to create a contract account?
   A. MKK
   B. FLCU01
   C. FLCU00
   D. None of the above
5. What is the standard API used for the subscription contract change process?
   A. CRM_ISX_BTX_API_CONT_CHANGE
   B. CRM_MSG_BTX_API_CONT_CHANGE
   C. CRM_ISX_BTX_API_CONTRACT_CHANGE
   D. None of the above
6. What is the standard BAdI used to validate subscription contract validity with reference to a master agreement?
   A. CRM_ISX_BTX_DATE_DETERMINE
   B. CRM_PRVMA_DATE_DETERMINE
   C. CL_CRM_PRVMA_DATE_DETERM_DL
   D. None of the above

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7. Which of the following standard SAP Convergent Charging web services are implemented in SAP S/4HANA?
   A. CO_CRM_ISX_CC_ACTIVATION_SRV
   B. CO_CRM_ISX_CC_SUBSCRIBER_MAPPI
   C. CO_ISX_ALLOWANCE_MANAGEMENT_SE
   D. All of the above
8. Which API is used for mass runs in SAP Billing and Revenue Innovation Management, subscription order management?
   A. ISX_CRM_BTX_SOM_MR_CREATE
   B. CRMS4_SOM_MR_CREATE
   C. ISX_CRM_SOM_MR_CREATE
   D. None of the above
9. What API can we leverage to create billable items?
   A. FKK_BIX_CIT_CREATE_API
   B. FKK_BIX_BIT_CREATE_API
   C. FKK_CRM_BIX_BIT_CREATE_API
   D. None of the above
10. What report needs to be triggered to add custom fields in the master agreement header and item table?
    A. Report CRMS4_EXTENSION_POSTPROCESSING
    B. Report ISX_CRM_EXTENSION_POSTPROCESSING
    C. Report ISX_BTX_EXTENSION_POSTPROCESSING
    D. None of the above

A.3.6 Data Migration

1. What prerequisites are required before contract account migration?
   A. Business partner with roles MKK and FLCU01
   B. Product mast data
   C. Subscription contracts
   D. None of the above
2. What is the correct sequence for data migration objects?
   A. Business partner, products, contract accounts, subscription contract
   B. Business partner, contract accounts, subscription contract, products
   C. Subscription contracts, products, contract accounts, business partner
   D. None of the above

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3. What validation needs to be done after subscription contracts are migrated?
   A. Contract has technically active status as **Yes**
   B. Contract account
   C. Contract end date and start date
   D. All of the above

A.4 Practice Answers and Explanations

The following sections provide answers to the practice questions listed previously.

A.4.1 SAP Billing and Revenue Innovation Management, Subscription Order Management

1. Correct answer: **A**, **B**, and **C**
   Value acceleration, revenue predictability, and customer engagement are the key benefits for XaaS for service providers.

2. Correct answer: **A**
   `MKK` is a key business partner role for setting up contract accounts. The `MKK` contract partner role displays the contract accounts linked to the corresponding business partner and is the key link between these two master data entities.

3. Correct answer: **B**
   SAP Billing and Revenue Innovation Management, subscription order management, should not always be integrated with SAP Convergent Charging to capture pricing information because consumption billing isn't always required. SAP Convergent Charging is required for services with consumption billing to enable rating the billing items for the consumption and calculating any overage charges.

4. Correct answer: **A**, **B**, and **D**
   Billing cycle determination rule, contract terms, and extension terms are required to configure subscription products. These important parameters drive key business processes. For example, the billing cycle determines the frequency of payments, whereas the contract term specifies the term of a subscription and length of a customer's commitment.

5. Correct answer: **A** and **B**
   `PSMB` (Bundle Price) and `PSPB` (Monthly Subscription Fee) condition types are delivered in the sales and distribution component for SAP Billing and Revenue Innovation Management, subscription order management. These two conditions are predefined by SAP to price product bundles and subscription services in a solution quotation.

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6. Correct answer: **A**, **C**, and **D**
   You use the subscription order change process to change contract duration, change an assigned technical ID, and execute cancellation. There are several more subscription change order processes predelivered in SAP Billing and Revenue Innovation Management that can be defined in Customizing to trigger contract changes. The three mentioned here are only a few common change processes. Changing a contract duration occurs when the customer requests to change the end date of their contract (i.e., extensions, short-life, co-term.). Changing an assigned technical ID changes the identifier, also referred to as the technical resource on solution quotations, used for rating and charging a subscription. Executing a cancellation can be more than just changing the end date. For example, a customer may be charged a fee to end their contract early, which can be captured on the cancellation change order. It's also possible to execute multiple change orders through change process groups.

7. Correct answer: **A** and **D**
   Contract account and business partner are shared between SAP S/4HANA and SAP Convergent Charging. The contract account represents the billable account for financial postings. The business partners are also replicated to identify the key parties involved, similar to the solution quotation or service order.

8. Correct answer: **B** and **C**
   Manage Product Bundles and **Manage Product Master Data** tiles can be used for product modeling. The Manage Product Bundles app is used to build the product bundle, set required or optional indicators, and configure the validity of the components. The Manage Product Master Data app is used to define the key product master data elements such as the general information (e.g., basic data: division, language, description, etc.), classification, sales data, plants, and so on.

9. Correct answer: **A**
   True. For each subscription contract change, a change order is created that applies the changes to an existing subscription and enables associated billing and invoicing. Submission of the change order applies the change for the downstream process in SAP Convergent Charging and SAP Convergent Invoicing.

10. Correct answer: **A**
    True. Once submitted, a subscription order can't be changed because those changes have already been applied to an existing subscription contract and replicated to the corresponding provider contract in SAP Convergent Invoicing.

11. Correct answer: **B**
    False. More than one document distribution schema can be assigned to support flexibility in the steps defined in each schema and the schema determination based on item category.

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12. Correct answer: **A**
    True; it is possible to group and execute different change processes on a subscription contract simultaneously. This is helpful when more than one subscription change process (e.g., price change and extension of the contract) should be executed at the same time. The Change Process Groups in customization can be used to define the change processes and sequence to be executed within the group.

13. Correct answer: **B** and **C**
    It is possible to extend the end date of the contract when the contract end date is in the future OR when the contract end date is the current date. The contract end date would be extended by triggering a subscription change process and submitting the change order to apply the changes to the existing subscription or contract.

14. Correct answer: **B**
    False. While doing cross-catalog mapping for the subscription products, there can be only one charge plan in **Released** status; otherwise, the system wouldn't know which charge plan to select.

15. Correct answer: **A** and **B**
    SAP Billing and Revenue Innovation Management, subscription order management, can be activated with SAP CRM provider order management and SAP S/4HANA Service. Initially, SAP Billing and Revenue Innovation Management, subscription order management, was called provider order management and was managed outside of SAP S/4HANA using SAP CRM. As of SAP S/4HANA 1809, SAP Billing and Revenue Innovation Management, subscription order management, lies within SAP S/4HANA.

16. Correct answer: **A**, **B**, and **C**
    A subscription contract includes technical identifier, contract account, and business partner. Cross-selling products aren't included as a part of a subscription contract because the contract is for a single sales or service organizational entity.

17. Correct answer: **A**, **B**, and **D**
    A master agreement can define who is allowed to order the service, which price is defined for a service, and what can be ordered. How much the bill is will be determined on the rating and charging of the subscription and the pricing model (consumption *or* fixed fees).

A.4.2 Charging

1. Correct answer: **B**
   For a sharing contract, you can use the counter in a charge plan for all contracts collectively because the sharing contract can access the same counter.

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2. Correct answer: **A** and **D**
   Contract account and business partner objects are shared between SAP CRM, SAP ERP, and SAP Convergent Charging. The contract account is created in the SAP ERP system and used by SAP Billing and Revenue Innovation Management, subscription order management as well as replicated into SAP Convergent Charging as the external account used to determine which account to charge. Like contract accounts, business partners are created in the SAP ERP system and used by SAP Billing and Revenue Innovation Management, subscription order management, as well as replicated into SAP Convergent Charging as a subscriber account.

3. Correct answer: **B**
   The subscriber account is a part of the customer master data managed by SAP Convergent Charging and linked to an end customer of a service provider. The subscriber account is a representation of a business partner in SAP Convergent Charging, which owns the contract.

4. Correct answer: **D**
   A subscriber account is linked to the charging plan.

5. Correct answer: **C** and **D**
   A persistent counters account and transient counters are two categories of counters. A charge object only supports two types of counters persistent and transient counter. Persistent counters are updated and saved to the backend database during the charging process, while transient counters are only stored in internal memory during the charging process.

6. Correct answer: **B**
   When a consumption item class is replicated from SAP Convergent Invoicing to SAP Convergent Charging, a chargeable item class is created in SAP Convergent Charging. The chargeable item class is a representation of the consumption class in SAP Convergent Invoicing.

7. Correct answer: **A**
   In a charged item class, we define the output format of the rating and charging process.

8. Correct answer: **A**, **B**, and **C**
   We can add multiple charges inside a charge plan, we can add multiple rates inside a price plan, and we can add multiple charging components inside a charging plan.

9. Correct answer: **A**
   One-time rates are triggered during suspension or activation of a provider contract. One-off charges can only be triggered once by an external event such as activation or suspension.

10. Correct answer: **C**
    Objects are *not* a charging plan component.

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11. Correct answer: **B**
    With the help of a pricing macro, we can reuse the same logic across multiple price plans. A pricing macro is like a subroutine or function in programing language. The purpose is for reusability.

12. Correct answer: **A**
    Property Type Converter can be used as a string OR a number.

13. Correct answer: **A**
    The flow of a provider contract in all three systems of SAP Billing and Revenue Innovation Management is SAP Billing and Revenue Innovation Management, subscription order management, to SAP Convergent Invoicing to SAP Convergent Charging. Because a contract object is required by the three modules, when a subscription contract is created in SAP Billing and Revenue Innovation Management, subscription order management, it will be automatically replicated to SAP Convergent Invoicing and SAP Convergent Charging as it's needed to process the transactions in each module.

14. Correct answer: **A**, **B**, and **C**
    All three options are true.

15. Correct answer: **D**
    The bulkloader instance of SAP Convergent Charging Core Server is used to load the files generated by the rating process into SAP Convergent Invoicing. Bulkloader is the instance that is responsible to push the output from SAP Convergent Charging to SAP Convergent Invoicing as billable items. It uses the SAP Java connector to call the billable item API.

A.4.3 Invoicing

1. Correct answer: **D**
    Billing items creation can be achieved through a combination of business partner, contract account, and subprocess; rules defined under the grouping control parameter for grouping variant; and manually using expert mode during the execution of the billing process.

2. Correct answer: **C**
    Billable Item Class is the highest node of configuration in SAP Convergent Invoicing.

3. Correct answer: **B**
    1 and 4 are the true statements.

4. Correct answer: **C**
    The billing process in SAP Convergent Invoicing will select the billable items in only **Billing** status.

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5. Correct answer: **B**
   False. Multiple billing documents can be grouped into one invoice document.

6. Correct answer: **B**
   Selection, grouping, and aggregation variants are assigned to the billing process and subprocess.

7. Correct answer: **B**
   The unrated partner consumption item is generated in SAP Convergent Invoicing for partner settlement as follows when the billable item is created for the end customer.

8. Correct answer: **C**
   The partner settlement process is a solution used for the revenue-sharing business model.

9. Correct answer: **D**
   All three choices are key interface components for partner settlement.

10. Correct answer: **B**
    Here are the matches between posting area and its significance in SAP Convergent Invoicing/contract accounting: 8120 - Company code derivation *and* 2610 - General ledger account determination for document posting.

A.4.4 Contract Accounting

1. Correct answer: **D**
   All three options represent summarization criteria for a reconciliation key: company code, currency, and controlling account assignment.

2. Correct answer: **A**
   The first three choices are true. Reconciliation key in contract accounting can be determined both manually and automatically. Multiple contract accounting documents can be consolidated into summary records under a unique key before closing the reconciliation key. Reconciliation key is the link between the accounting document and contract accounting document.

3. Correct answer: **C**
   If the reconciliation key is closed but not transferred, postings can't be made to the reconciliation key.

4. Correct answer: **C**
   Both. Payment method can be updated at the contract account level and the document level.

5. Correct answer: **B**
   False. Characteristics from master data (CA and contract) have an input in account determination in contract accounting.

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6. Correct answer: **D**
   The first three choices are true. Incorrect CA, posting period locked, and posting lock on CA are reasons for payments to go to clarification.

7. Correct answer: **A**
   Yes, we can write off credit documents.

8. Correct answer: **D**
   Key components of contract accounting are reconciliation key, general ledger accounts, and profit center and cost center.

9. Correct answer: **A**
   True. The main transaction will determine the reconciliation account.

A.4.5 Extension and Connectivity

1. Correct answer: **C**
   The `CRM_ISX_BTX_API_ORDER_CREATE` API can be used to create subscription orders, solution quotations, and master agreements. By leveraging this API, the contracts/master agreement can be created from third-party applications such as Salesforce. This API can be leveraged for data migration objects.

2. Correct answer: **C**
   `CRM_ORDER_MAINTAIN` is the standard API used to change the solution quotation before releasing. `CRM_ORDER_MAINTAIN` can be used to change existing solution quotations, as well as master agreements from third-party applications (e.g., SFDC). They provide more options than the create order API. These can be leveraged during data migration.

3. Correct answer: **A**
   `API_BUS_SOLUTION_QUOTATION_SRV` is the standard API to create a solution quotation. By leveraging this API, quotations can be created from third-party applications (e.g., SFDC). Solution quotations can have service items, subscription items, and hardware items. More information on this API is available on *https://api.sap.com*.

4. Correct answer: **A**
   The `MKK` partner role needs to be created while migrating a business partner to create the contract account.

5. Correct answer: **A**
   `CRM_ISX_BTX_API_CONT_CHANGE` has multiple applications for carrying out changes to subscription contracts. This API can be integrated with third-party applications (e.g., SFDC). Using the same API, we can cancel or extend existing subscription contracts.

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6. Correct answer: **B**
   CRM_PRVMA_DATE_DETERMINE is used to check which date will be used for checking the validity of a subscription order/solution quotation with a master agreement item. Generally, the date must be within the validity period defined in the master agreement item referenced. This logic can be changed if required by custom implementation.

7. Correct answer: **D**
   All of the above. All these web services need to be activated in SAP S/4HANA as only then will the order distribution interface execute correctly and the contracts get replicated to SAP Convergent Charging from SAP S/4HANA.There are more than nine web services that need to be activated.

8. Correct answer: **A**
   ISX_CRM_BTX_SOM_MR_CREATE is the mass run API that can be used to trigger contract changes and creation. Using this API, we can create subscription orders and contracts en masse.

9. Correct answer: **B**
   FKK_BIX_BIT_CREATE_API is used to create billable items directly in SAP Convergent Invoicing. We can use this function module during migration, when we want to bring open items from legacy systems. Custom programs can be created using this function module to invoice customers for miscellaneous items.

10. Correct answer: **A**
    When custom fields are added in SAP Fiori for master agreement header and item, we need to run program CRMS4_EXTENSION_POSTPROCESSING, which adds these fields in backend tables. Running the report also needs to be done for other SAP Fiori apps when extensions are done.

A.4.6 Data Migration

1. Correct answer: **A**
   You should have a business partner with roles MKK and FLCU01 before contract account migration.

2. Correct answer: **A**
   The correct sequence is business partner, products, contract accounts, subscription contract.

3. Correct answer: **C**
   All these validations need to be performed.

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A.5 Key Takeaways

As we are at the end of this appendix, it's important to look back and understand how it can help you to make the most in your certification journey. Before appearing for the SAP Billing and Revenue Innovation Management certification examination, keep the following in mind:

- SAP Billing and Revenue Innovation Management certification is recommended as an entry-level qualification and confirms that the practitioner has a foundational understanding of the solution.
- For either of the exams, it's important to understand the concept of the subscription- and usage-driven economy. This is covered as part of Chapter 1 in this book, and you shouldn't skip the chapter to directly proceed with SAP Billing and Revenue Innovation Management, subscription order management, or other modules. Chapter 1 provides you a detailed idea of how the revenue model suits the SAP Billing and Revenue Innovation Management process.
- The exams usually cover various topic areas with corresponding weighting, which can be found in *https://training.sap.com*. For example, questions for C_BRSOM_2020 for SAP Billing and Revenue Innovation Management, subscription order management, are divided into Overview (>12%), Order and Contracts (>12%), Master Data (>12%), and Integration to SAP Convergent Charging/Integration to SAP Convergent Invoicing and SAP Fiori Applications (>12%).
- Some of the questions in the examination are detailed and can only be answered if you have hands-on experience with the solution. We strongly encourage you to develop baseline use cases for Martex Corp. in your hands-on system to get the most benefit from this book.
- In the certification, there could be a few ambiguous questions. The best way to handle such questions is to eliminate the incorrect options first.
- During your certification examination, if you aren't sure about an answer to a question, you can bookmark the question and skip it to move to the next one. You will have time to revisit the question again later.

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Appendix B

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